IHP

SG13G2

SG13G2 DC CV NMOS HV

Rev. 1.2 (190924)

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1 Documentation

1.1 Included Effects in Simulation Model

General	Date, Version	24.09.2019
	Processversion	SG13G2
	Simulator	ADS 2019, Spectre 18.1
	Simulation model	PSP 103.6
	Warning Boundaries	On

Statistic	Corner	x
	Process variations for Monte Carlo Simulation	e
	Mismatch	x
Transistor	DC behavior	x
	Drain diode	x
	Overlap cap.	x
	Junction cap.	x
	1/f Noise	x_1
	HF / Thermal Noise	-
	Substrate resistance (RF)	x_2
	Gate resistance (RF)	x_2
Temperature	-40°C, 25°C, 70°C, 125°C	x

- x measured and modelled
- x_1 parameters taken from SG13
- x_2 measurements are done from a different wafer run than DC model which leads to slight offsets
- e estimated and modelled
- o not included
- - not applicable

1.2 Time range measurement

The verification of the time range measurements is done by the standard IHP verification procedure.

2 Setup

2.1 Measurement Conditions

2.1.1 Measurement Conditions for Transistors

	Sweep	Start[V]	Stop[V]	No. Points	List or f(Vth)
idvg	vg	-500m	3.6	83	
	vb	0	-3.6	5	
	vd	100m	3.3	3	
	vs	0		1	
idvd	vd	0	3.6	37	
	vg	Lin f(Vth)	3.6	5	Ref. Setup: idvg Idref: 100.0n Offset: 100.0m Resolution: 1.000m
	vb	0		1	
	vs	0		1	
idvd_vbmin	vd	0	3.6	37	
	vg	Lin f(Vth)	3.6	5	Ref. Setup: idvg Idref: 100.0n Offset: 100.0m Resolution: 1.000m
	vb	-3.6		1	
	vs	0		1	
idvd_vth	vd	0	3.6	37	
	vg	List f(Vth)		3	Ref. Setup: idvg Idref: 100.0n Offset: -200.0m Resolution: 1.000m Steps: 200.0m 200.0m
	vb	0		1	
	vs	0		1	
idvd_vth_vbmin	vd	0	3.6	37	
	vg	List		3	Ref. Setup: idvg

		f(Vth)			Idref: 100.0n Offset: -200.0m Resolution: 1.000m Steps: 200.0m 200.0m
	vb	-3.6		1	
	vs	0		1	
RF idvg	vg	-500m	3.6	83	
	vd	100m	3.3	3	
RF idvd	vd	0	3.6	37	
	vg	0	3.6	13	
RF sdut	freq	100MEG	47.6G	31	
	vd	0	3.6	13	
	vg	0	3.6	13	

2.1.2 Measurement Conditions for Capacitance

	Sweep	Start[V]	Step[V]	Stop[V]	High Node	Low Node
Junction Bulk-Drain	1 VHI	-5	200m	200m	B	D
Junction Bulk-Source	1 VHI	-1	20m	300m	B	S
Oxide	1 VHI	3.6	-100m	-3.6	B	G
Overlap	1 VHI	-3.6	100m	3.6	G	D

2.2 Devices

2.2.1 Setup of DC Test Devices

Name	W[um]	L[um]	AD[um²]	AS[um²]	PD[um]	PS[um]	NRD	NRS	NF	Comment	Size Category
W10u0_L10u0_S557_3	10	10	3.4	3.4	20.7	20.7	0	0	1		Large

W0u3_L10u0_S547_4	300m	10	102m	102m	1.28	1.28	0	0	1		Narrow
W10u0_L0u5_S387_5	10	500m	3.4	3.4	20.7	20.7	0	0	1		Short
W10u0_L0u35_S556_3	10	350m	3.4	3.4	20.7	20.7	0	0	1		Short
W10u0_L0u5_T356_S387_5	10	500m	0	0	0	0	0	0	1		Short
W0u3_L0u35_S560_3	300m	350m	102m	102m	1.28	1.28	0	0	1		Small
W10u0_L0u5_S556_4	10	500m	3.4	3.4	20.7	20.7	0	0	1		L Scale
W10u0_L3u0_S557_1	10	3	3.4	3.4	20.7	20.7	0	0	1		L Scale
W10u0_L5u0_S557_2	10	5	3.4	3.4	20.7	20.7	0	0	1		L Scale
W10u0_L1u2_S556_5	10	1.2	3.4	3.4	20.7	20.7	0	0	1		L Scale
W05u0_L10u0_S558_1	5	10	1.7	1.7	10.7	10.7	0	0	1		W Scale
W02u0_L10u0_S558_5	2	10	680m	680m	4.68	4.68	0	0	1		W Scale
W0u6_L10u0_S559_5	600m	10	204m	204m	1.88	1.88	0	0	1		W Scale
W02u0_L3u0_S558_4	2	3	680m	680m	4.68	4.68	0	0	1		LW Scale
W0u6_L1u2_S559_4	600m	1.2	204m	204m	1.88	1.88	0	0	1		LW Scale
W02u0_L1u2_S558_3	2	1.2	680m	680m	4.68	4.68	0	0	1		LW Scale
W05u0_L5u0_S557_5	5	5	1.7	1.7	10.7	10.7	0	0	1		LW Scale
W0u3_L1u2_S560_5	300m	1.2	102m	102m	1.28	1.28	0	0	1		LW Scale
W0u6_L0u35_S559_2	600m	350m	204m	204m	1.88	1.88	0	0	1		LW Scale
W0u6_L0u5_S559_3	600m	500m	204m	204m	1.88	1.88	0	0	1		LW Scale
W0u3_L0u5_S560_4	300m	500m	102m	102m	1.28	1.28	0	0	1		LW Scale
W02u0_L0u33_S558_2	2	330	680	680	4.6	4.6	0	0	1		Additio

		m	m	m	8	8					nal
W10u0_L0u33_S556_2	10	330 m	3.4	3.4	20.7	20.7	0	0	1		Additional
W10u0_L0u4_T356_S387_4	10	400 m	0	0	0	0	0	0	1		Additional
W0u3_L0u3_S560_1	300 m	300 m	102 m	102 m	1.28	1.28	0	0	1		Additional
W10u0_L0u45_T356_S384M_1	10	450 m	0	0	0	0	0	0	1		Additional
W0u15_L0u33_S547_5	150 m	330 m	101 m	101 m	1.34	1.34	0	0	1		Additional
W10u0_L0u4_S387_4	10	400 m	3.4	3.4	20.7	20.7	0	0	1		Additional
W10u0_L0u3_S556_1	10	300 m	3.4	3.4	20.7	20.7	0	0	1		Additional
W0u3_L0u45_T356_S384M_3	300 m	450 m	0	0	0	0	0	0	1		Additional
W0u3_L0u33_S560_2	300 m	330 m	102 m	102 m	1.28	1.28	0	0	1		Additional
W0u6_L0u33_S559_1	600 m	330 m	204 m	204 m	1.88	1.88	0	0	1		Additional
W05u0_L0u33_S557_4	5	330 m	1.7	1.7	10.7	10.7	0	0	1		Additional

2.2.2 Setup of CV Test Devices

Name	W[um]	L[um]	NF	AD[um²]	AS[um²]	PD[um]	PS[um]	Comment	Size Category
Junction Capacitance Bulk - Drain									
CJBDarea_S350B_5			1	37.5K		850		B-High, S/D-Low	BD Area
CJBDperim_S350B_6			1	28.6K		102K		B-High, S/D-Low	BD Perim

Junction Capacitance Bulk - Source									
CJBSarea_S350B_5			1	0		0		B-High, S/D-Low	BS Area
CJBSperim_S350B_6			1	0		0		B-High, S/D-Low	BS Perim
Oxide Capacitance									
Cox_S342K_3	160	250	1					B-High, G-Low	Oxide
Overlap Capacitance									
Cgds1_S350B_9	6K	300m	600	1.14K	1.14K	6.25K	6.25K	G-High, S/D-Low, B+Sub-GND	Overlap GDS
Cgdsb1_S350B_9	6K	300m	600	1.14K	1.14K	6.25K	6.25K	G-High, S/D+B-Low	Overlap GDSB
Cgds2_S350B_10	400	10	40	79.8	79.8	436	436	G-High, S/D-Low, B+Sub-GND	Overlap 1 GDS
Cgdsb2_S350B_10	400	10	200	79.8	79.8	436	436	G-High, S/D+B-Low	Overlap 1 GDSB

3 Corner Devices

3.1 Target Values vs. Simulated Values

3.1.1 W = 10.00u, L = 450.0n, NF = 1

	Slow			Typic.			Fast		
	Target	Sim.	Error [%]	Target	Sim.	Error [%]	Target	Sim.	Error [%]
VT	770m	770m	0	700m	700m	0	630m	630m	0
Ron*W	-	-	-	1.42	1.42	0	-	-	-
Ion/W	480	482	-0.31	550	550	0	620	622	-0.33
log(Ioff/W)	-14	-12.5	-3.53	-13	-12.4	-2.5	-12	-12.1	1.12

3.1.2 W = 10.00u, L = 10.00u, NF = 1

	Slow			Typic.			Fast		
	Target	Sim.	Error [%]	Target	Sim.	Error [%]	Target	Sim.	Error [%]
VT	730m	730m	0	690m	690m	0	650m	650m	0
Ion/W	33	33.1	-0.22	37	37	0	42	42	0

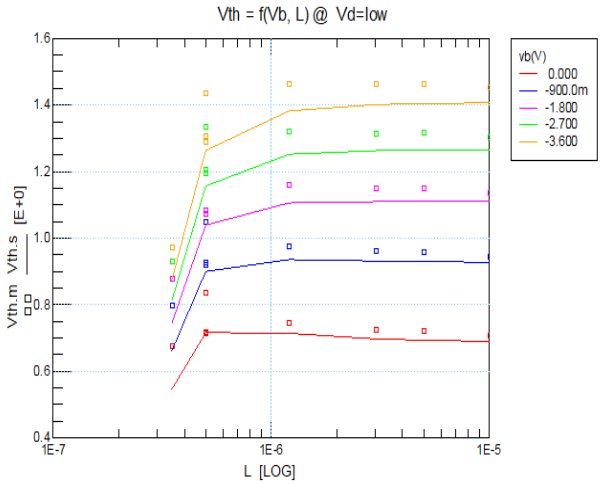
3.1.3 W = 300.0n, L = 450.0n, NF = 1

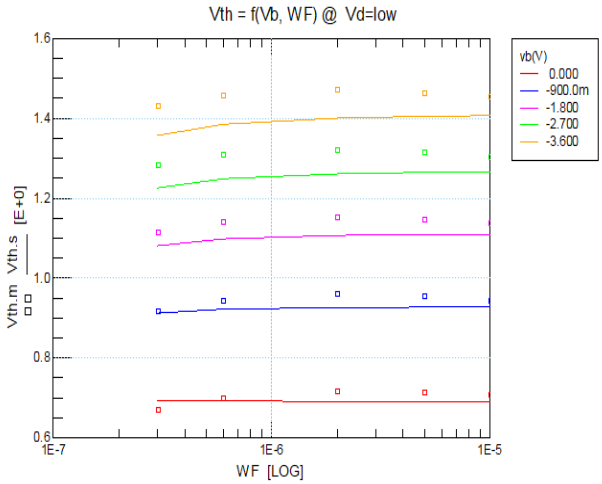
	Slow			Typic.			Fast		
	Target	Sim.	Error [%]	Target	Sim.	Error [%]	Target	Sim.	Error [%]
VT	750m	750m	0	670m	670m	0	590m	590m	0
Ion/W	490	491	-0.21	580	580	0	670	671	-0.21

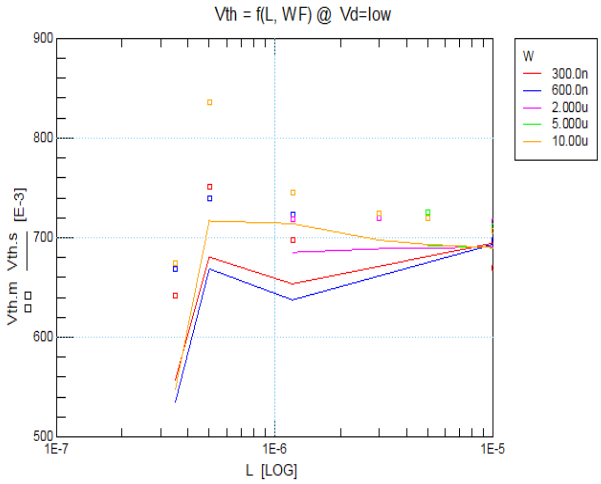
4 DC Transistors

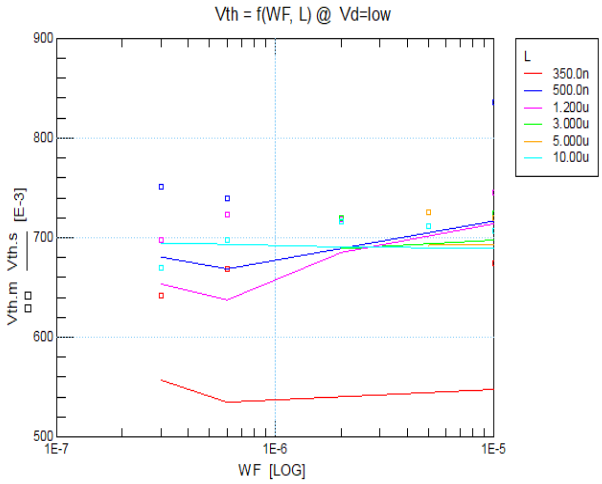
4.1 Summarized Results

4.1.1 Threshold Voltage Linear Region

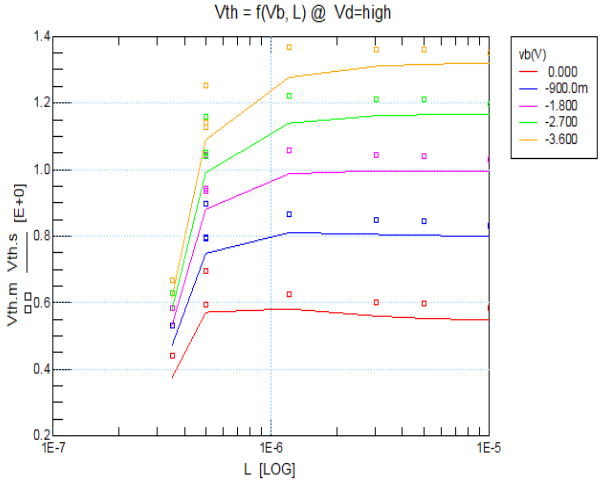


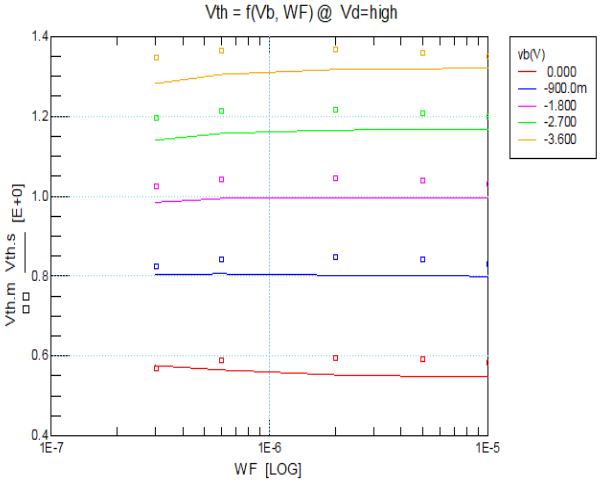


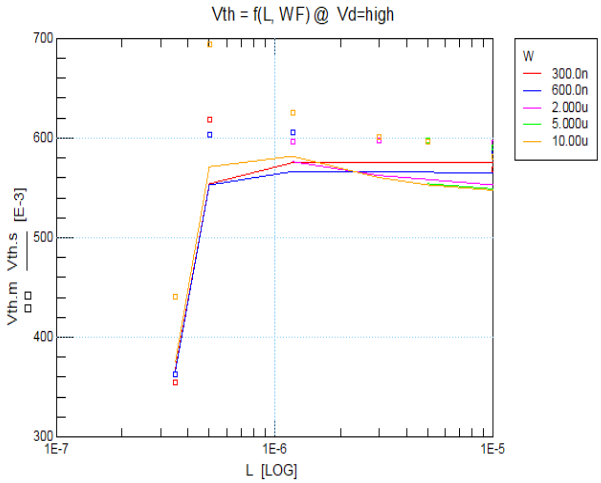


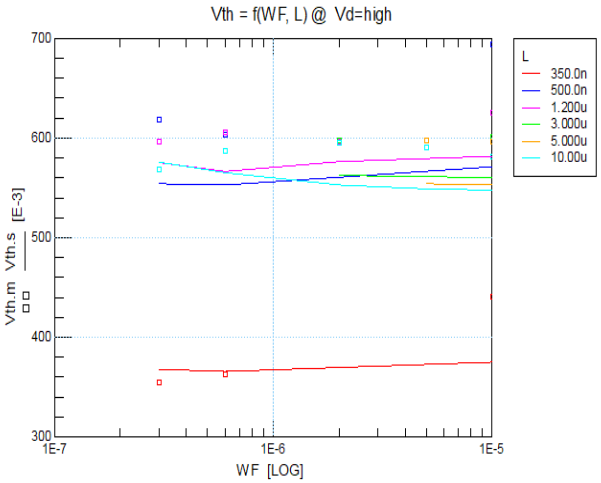


4.1.2 Threshold Voltage in Saturation

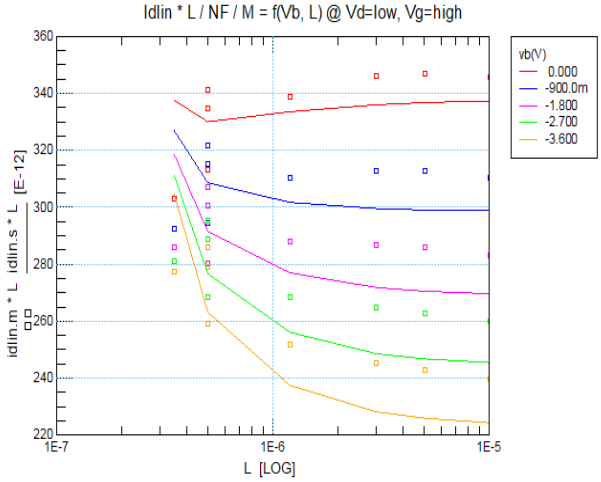


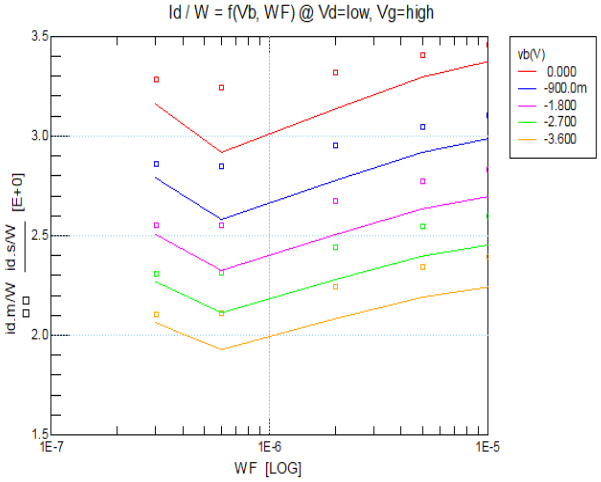


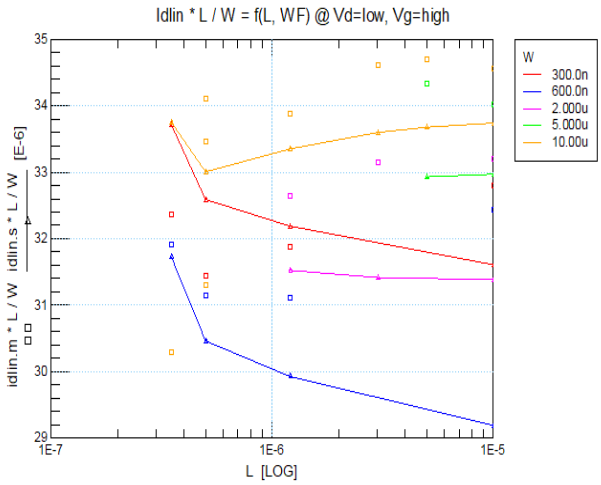


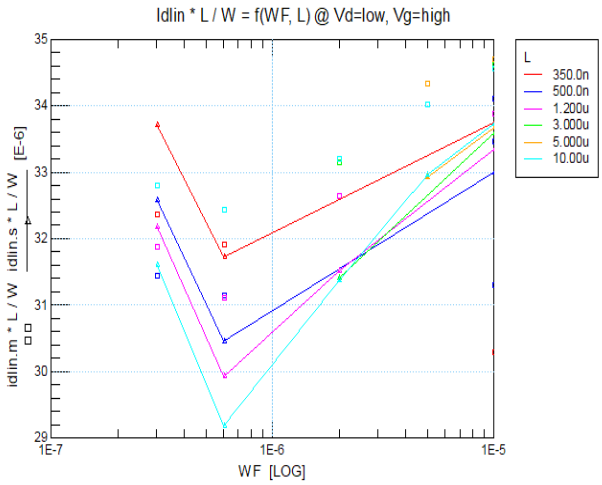


4.1.3 Maximum Drain Current in Linear Region

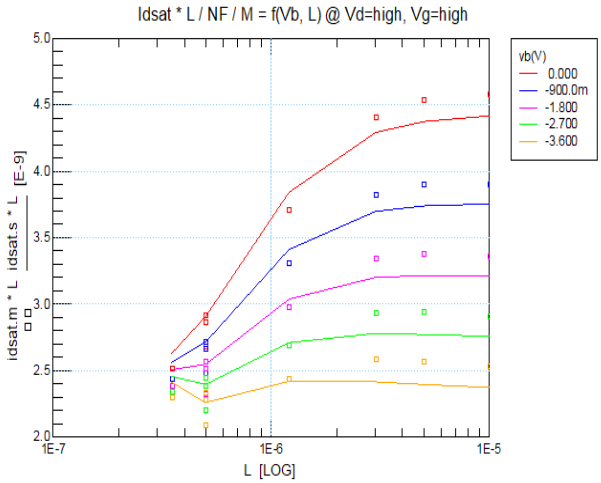


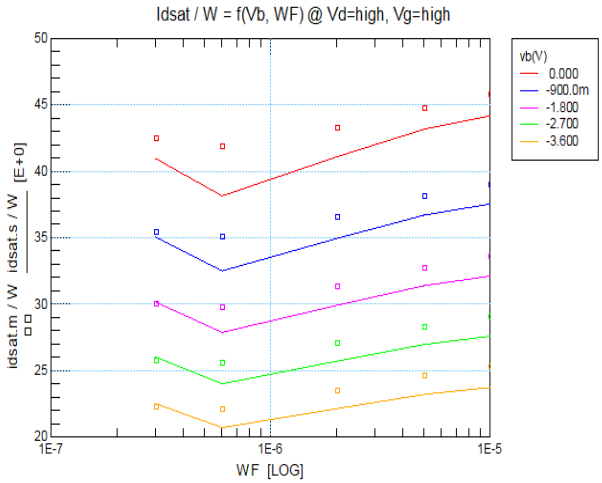


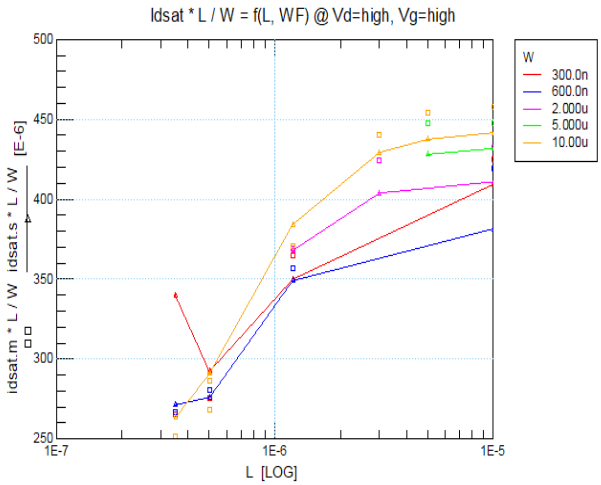


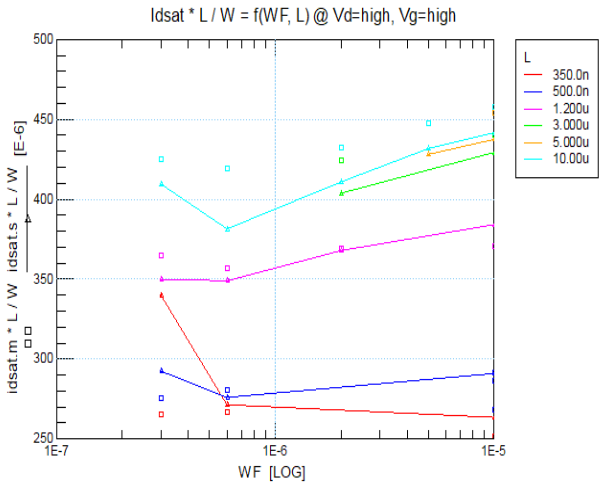


4.1.4 Maximum Drain Current in Saturation Region



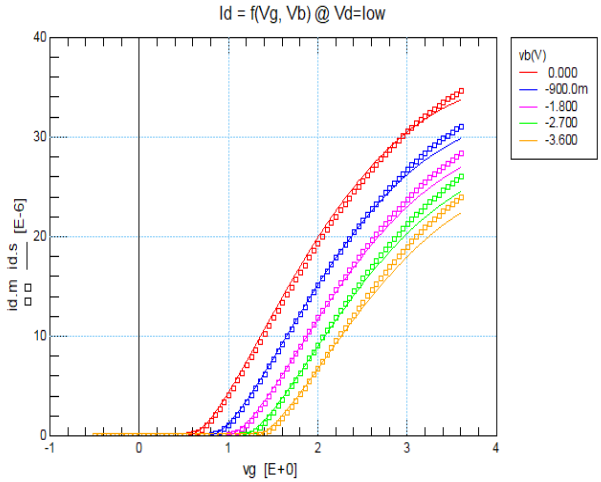


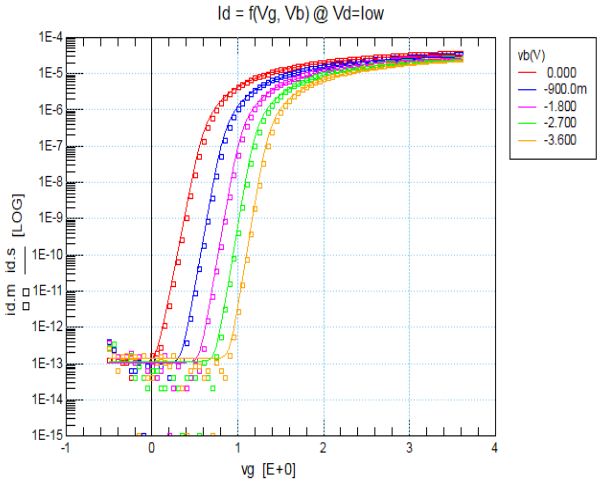


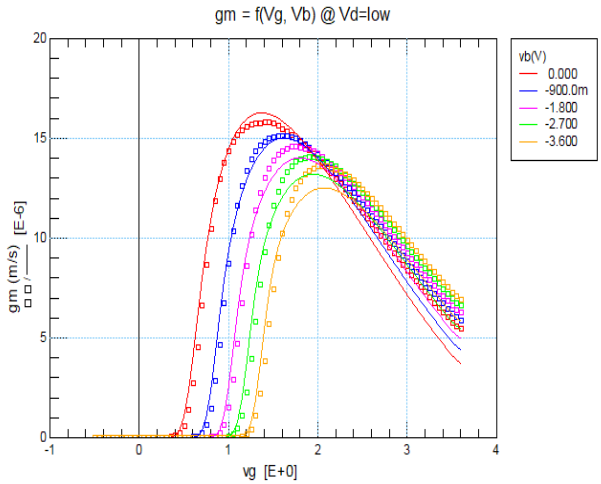


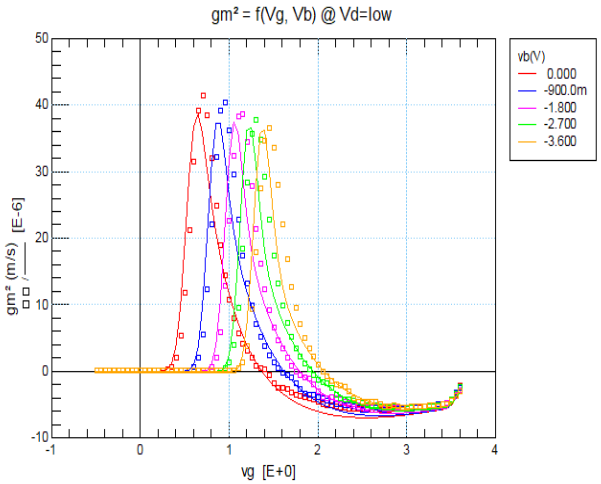
4.2 W10u0_L10u0_S557_3

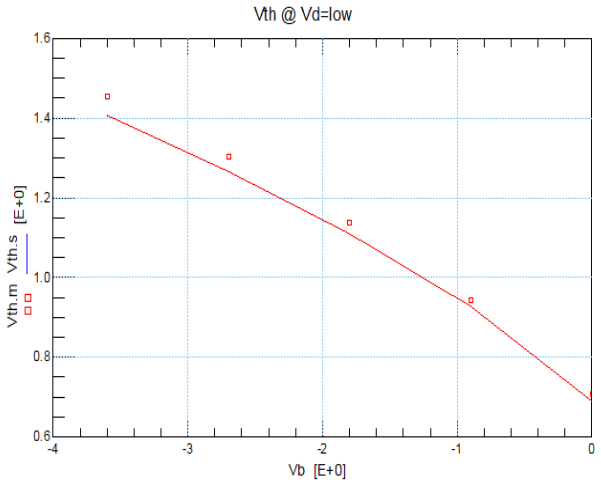
idvg, Vd = 0.1V, T = 27°C

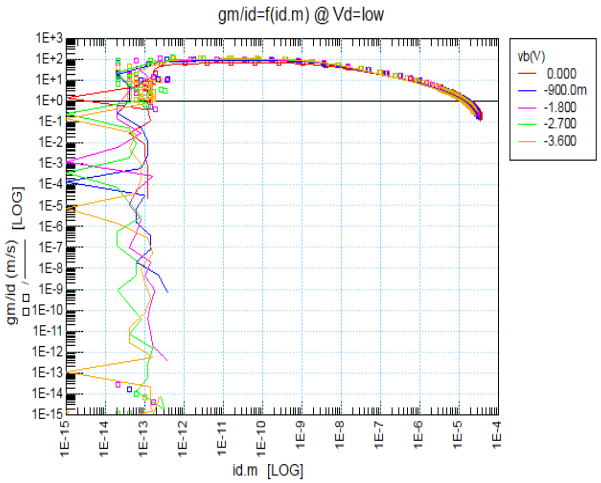


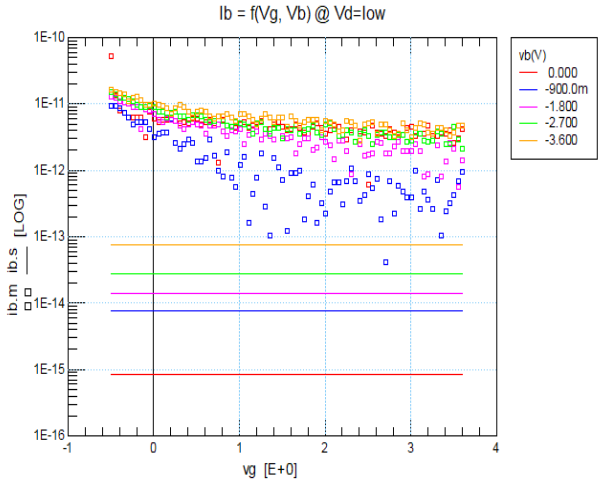


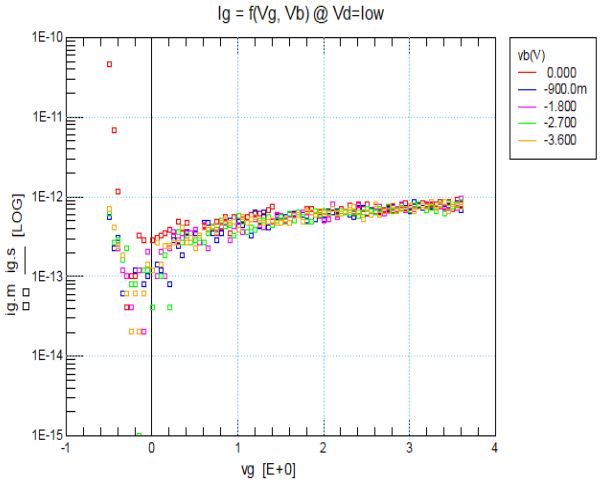




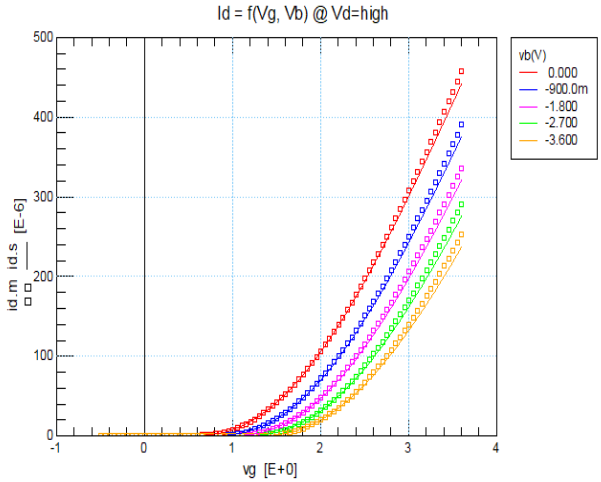


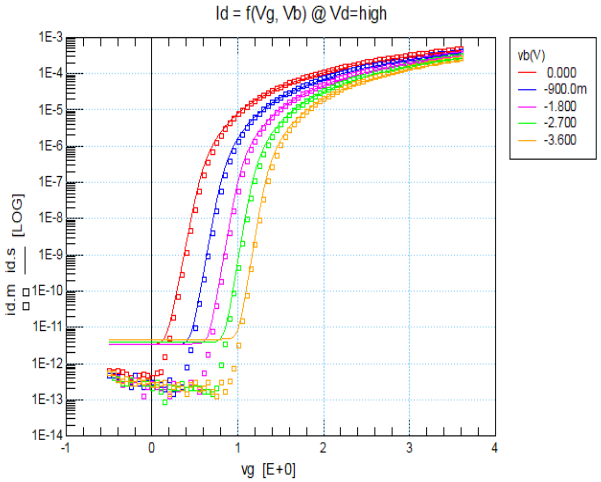


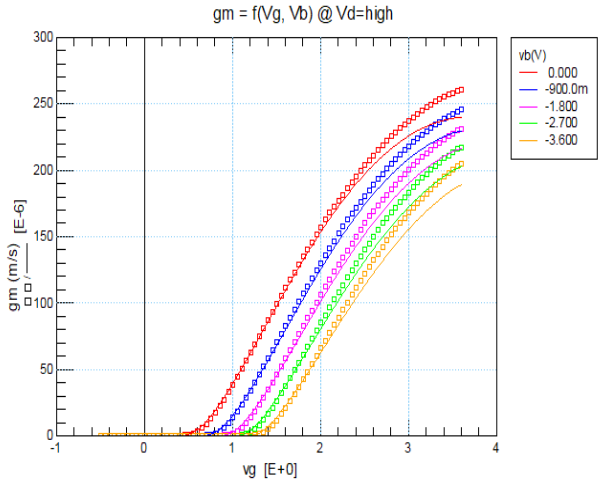


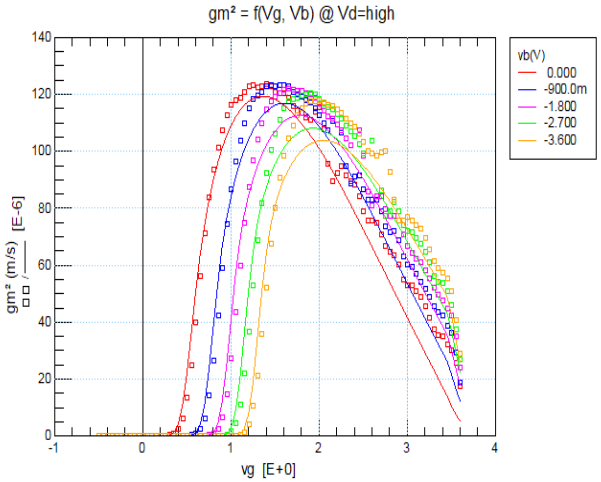


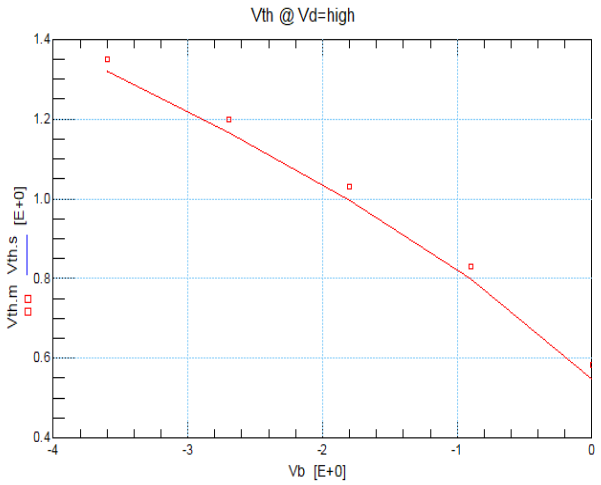
idvg, Vd = 3.3V, T = 27°C

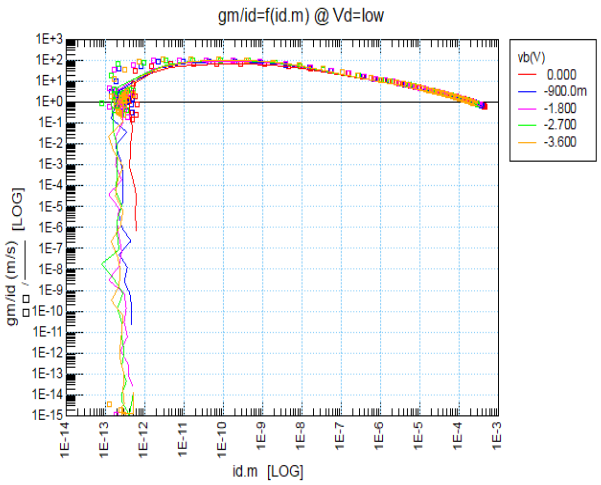


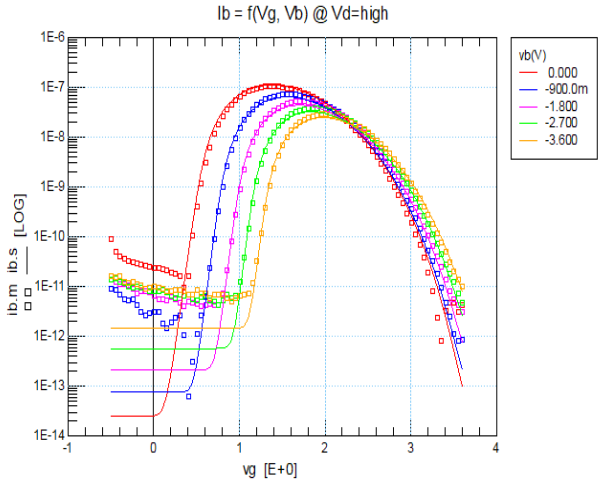


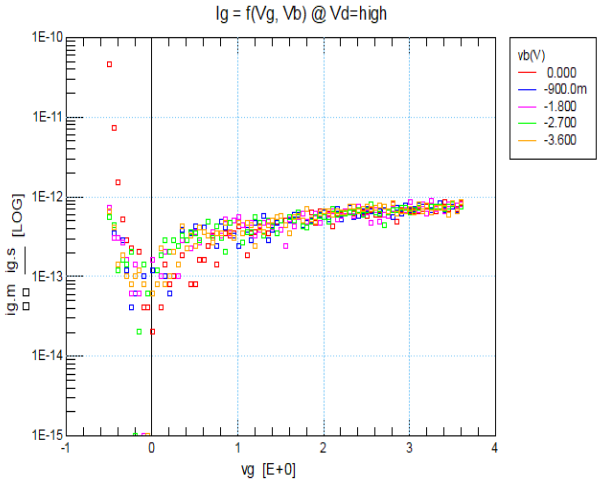




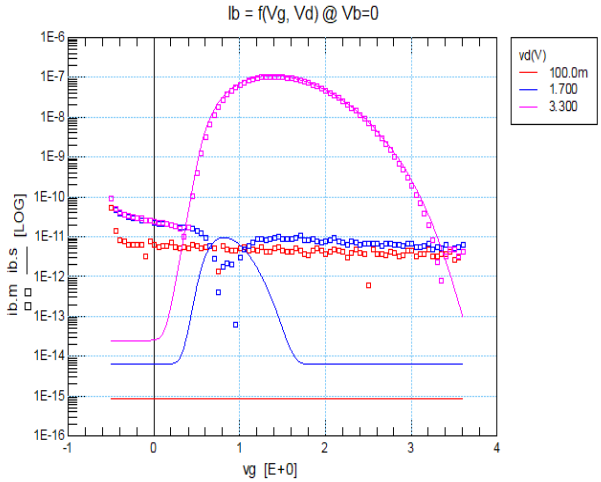


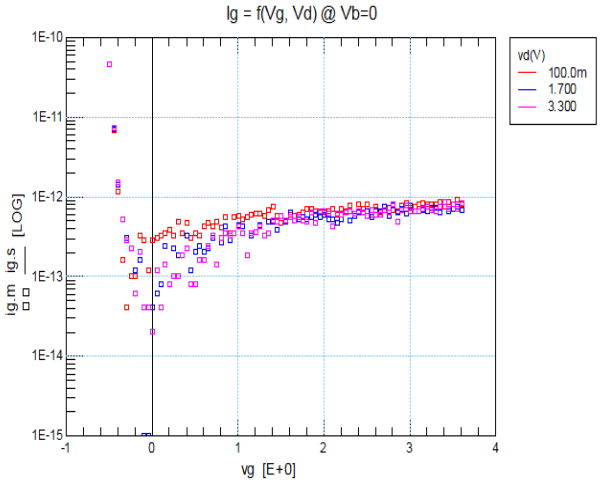




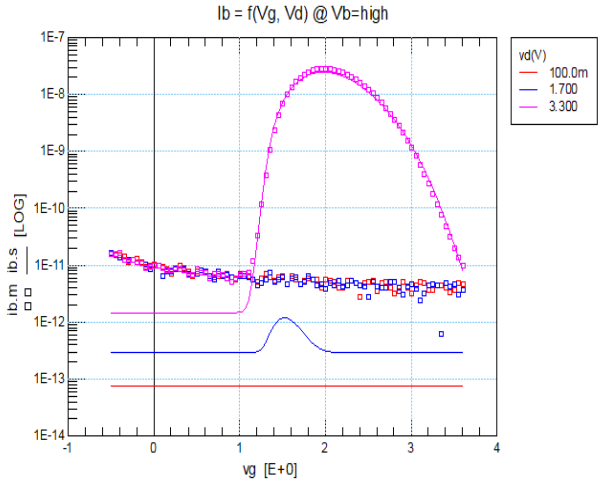


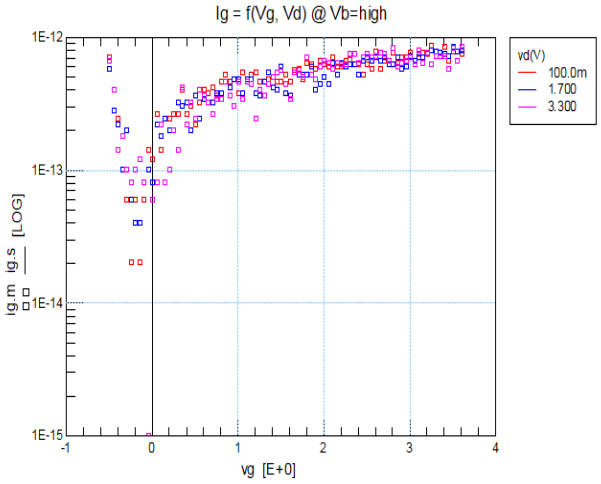
idvg, Ib,Ig, Vb = 0V, T = 27°C



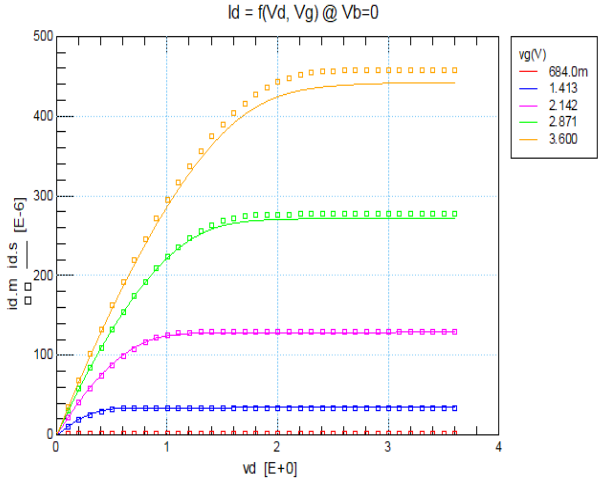


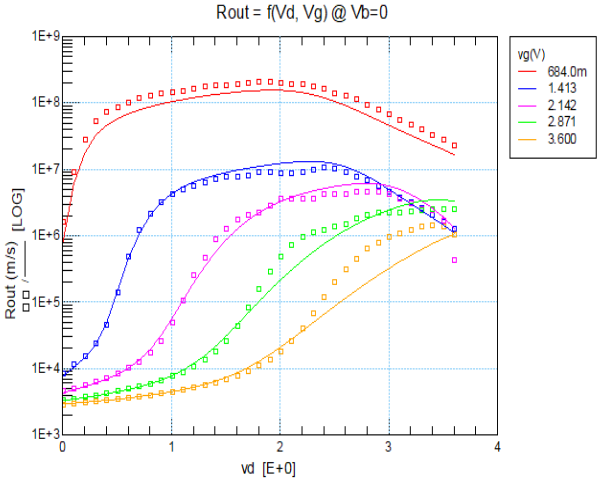
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

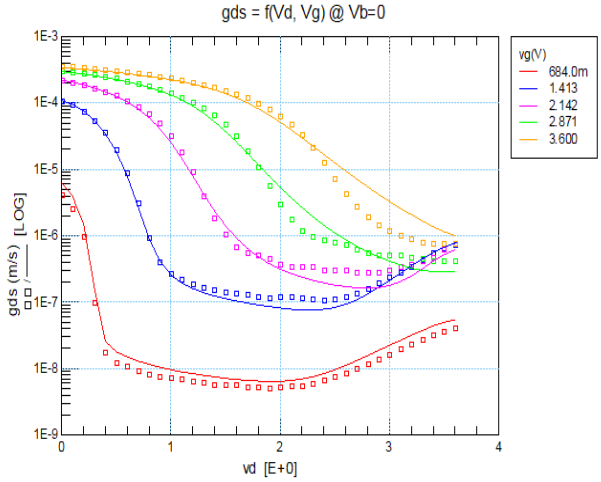




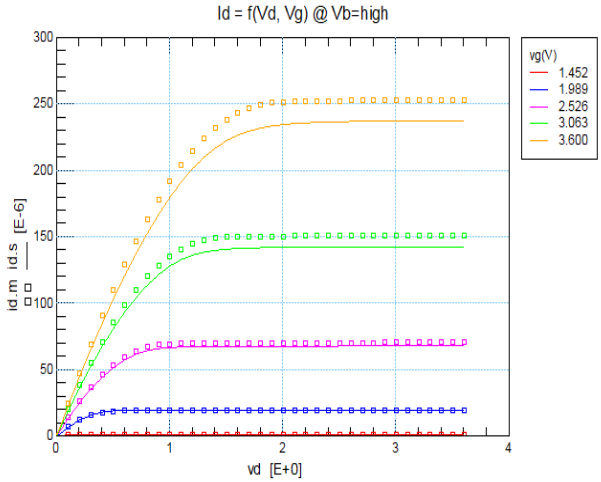
idvd, Vb = 0V, T = 27°C

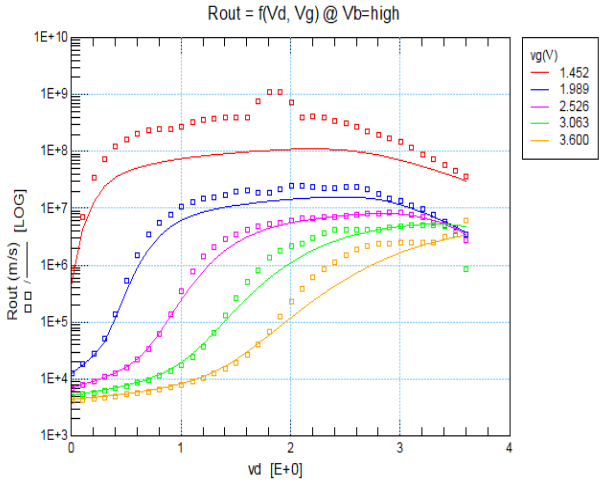


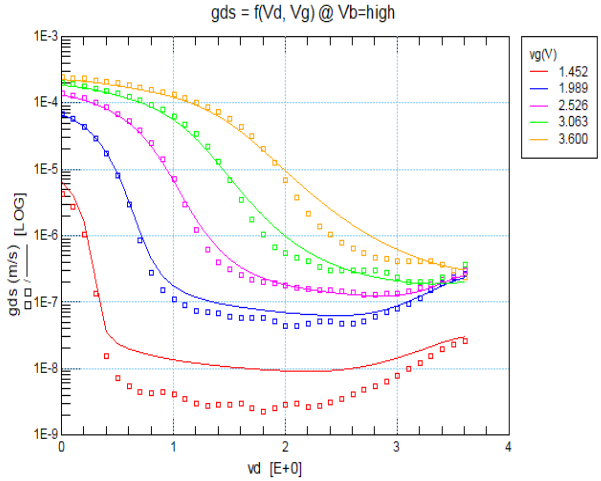




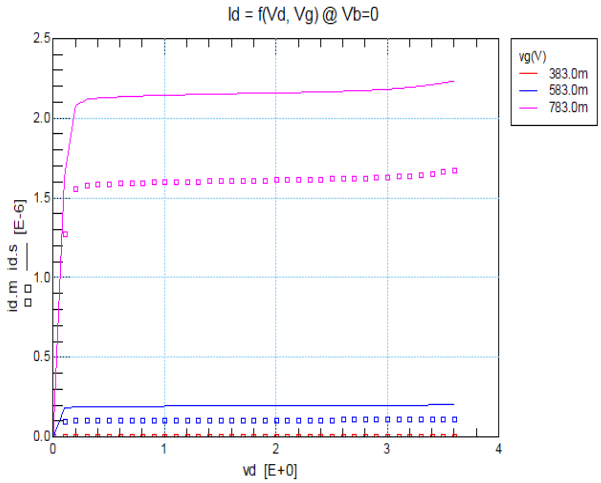
idvd_vbmin, Vb = -3.6V, T = 27°C

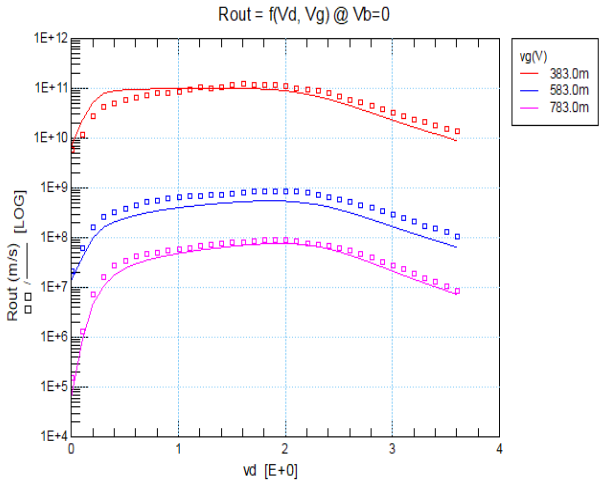


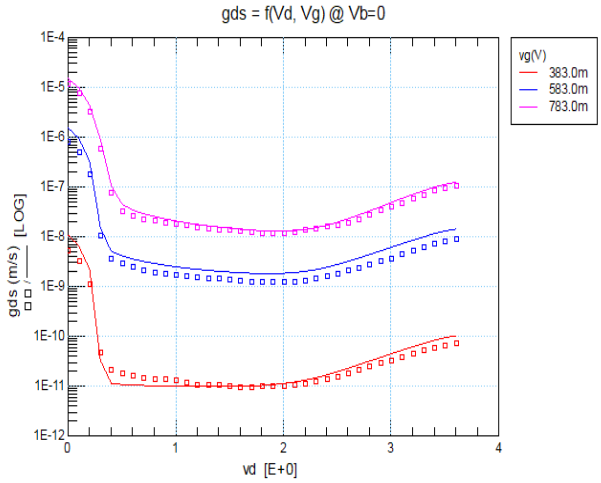




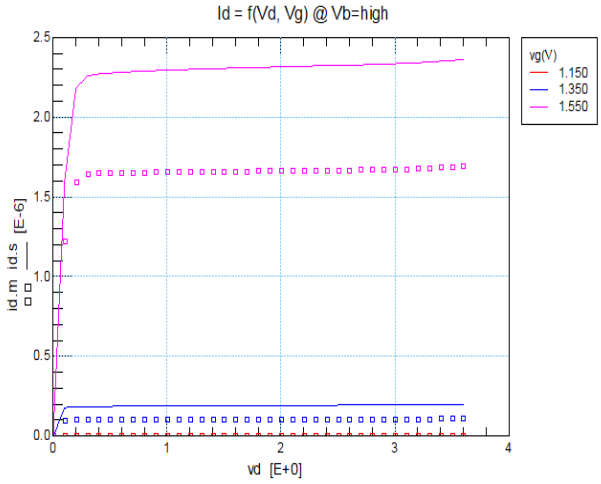
idvd_vth, Vb = 0V, T = 27°C

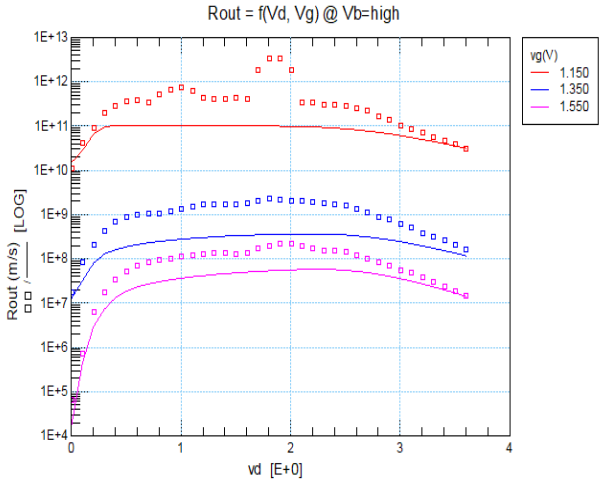


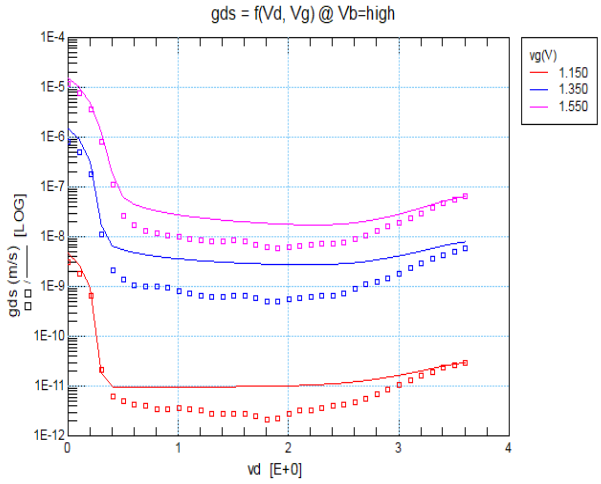




idvd_vth_vbmin, Vb = -3.6V, T = 27°C

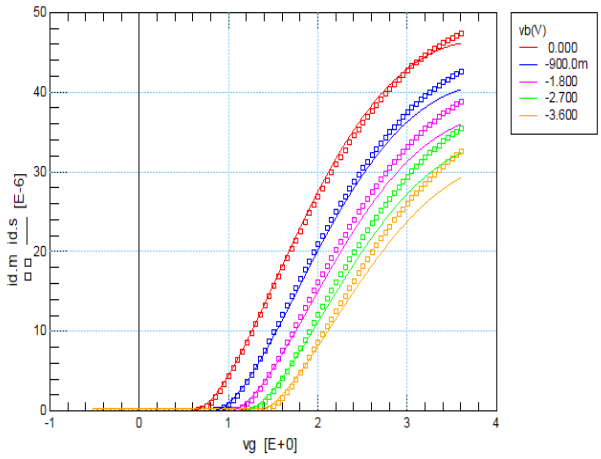




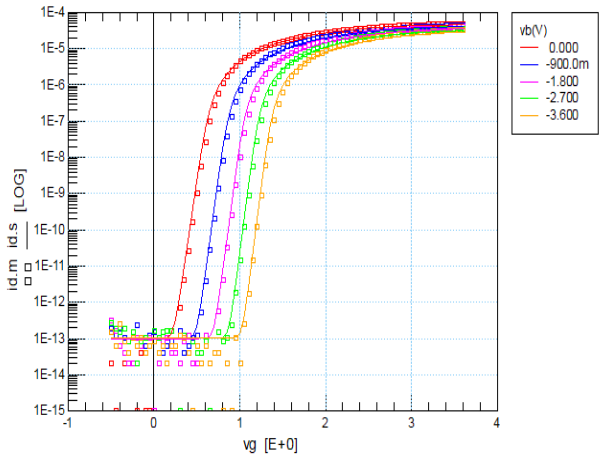


idvg, Vd = 0.1V, T = -40°C

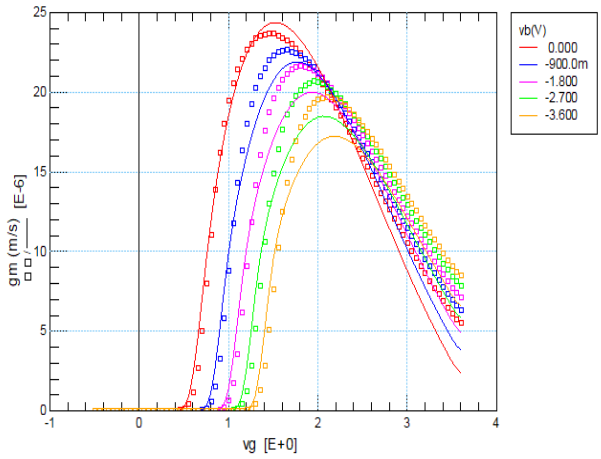
Id = f(Vg, Vb) @ Vd=low



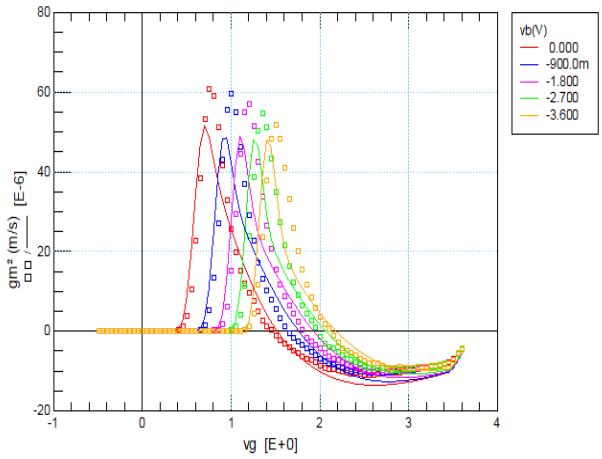
Id = f(Vg, Vb) @ Vd=low



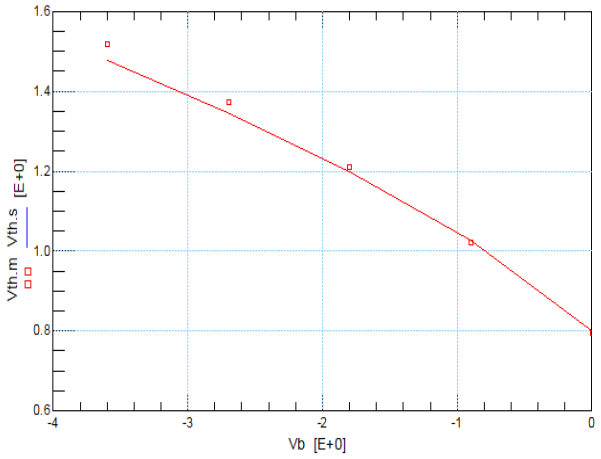
gm = f(Vg, Vb) @ Vd=low



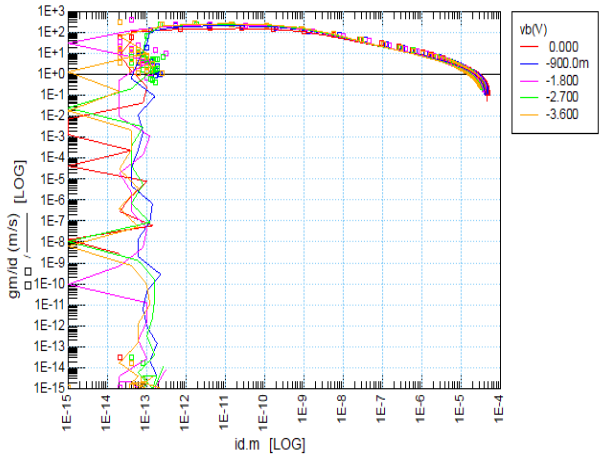
gm² = f(Vg, Vb) @ Vd=low

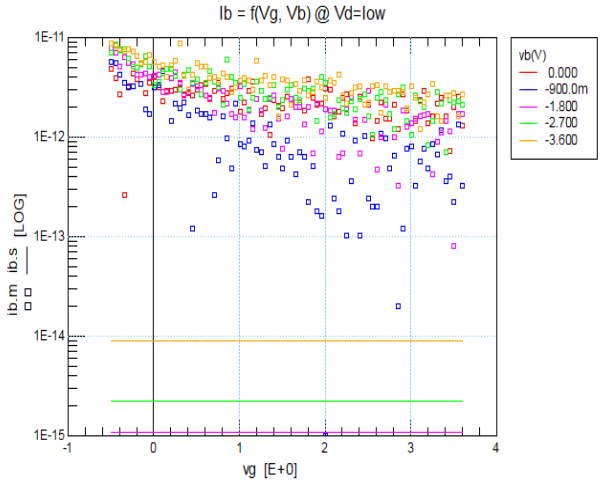


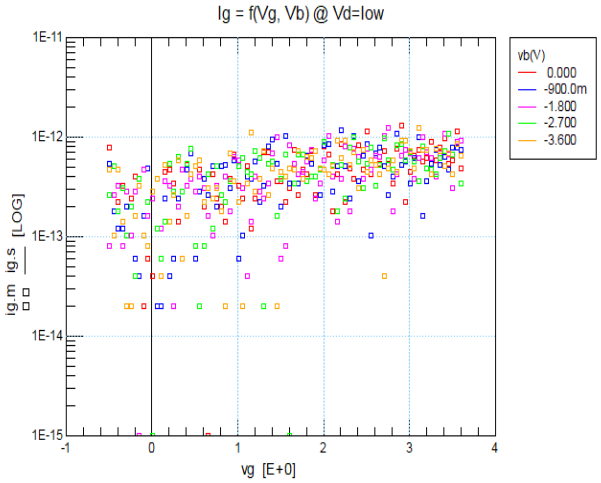
Vth @ Vd=low



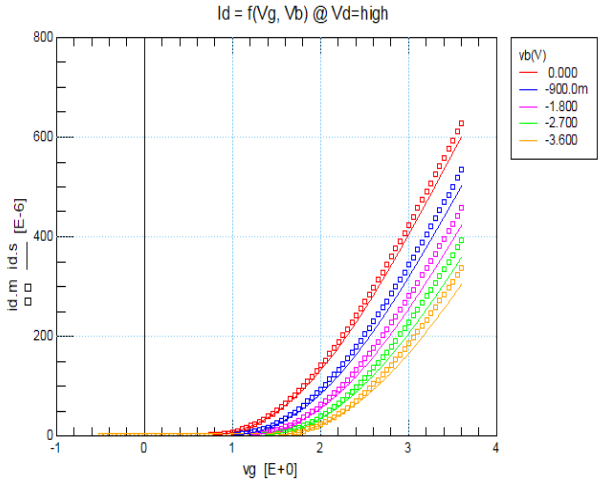
gm/id=f(id.m) @ Vd=low

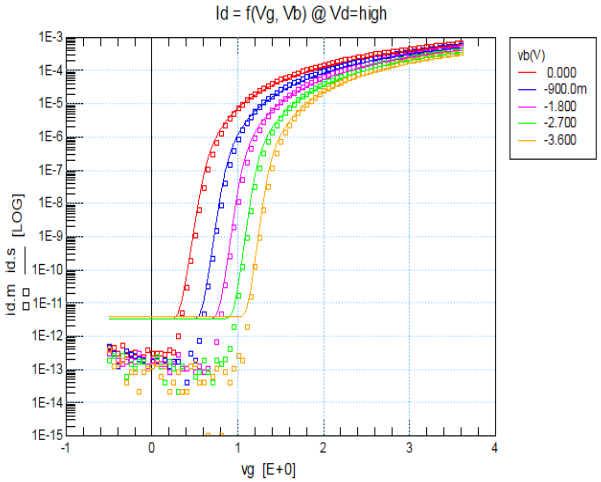


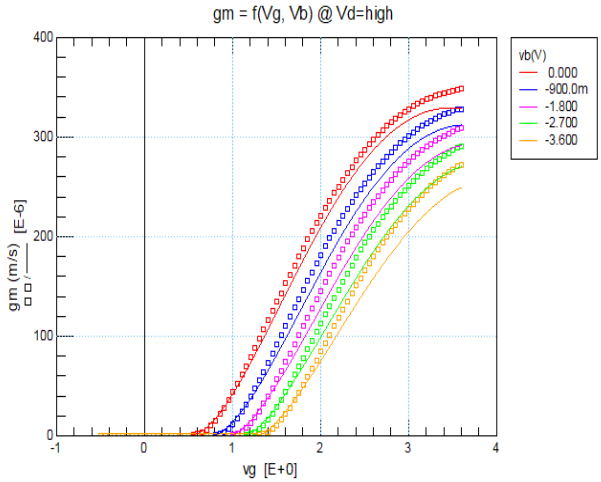


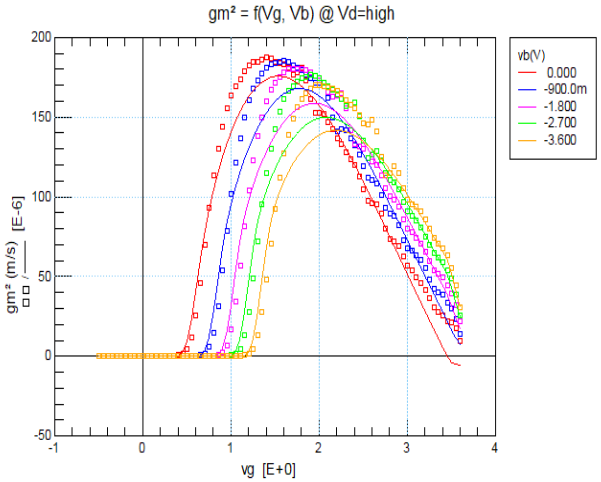


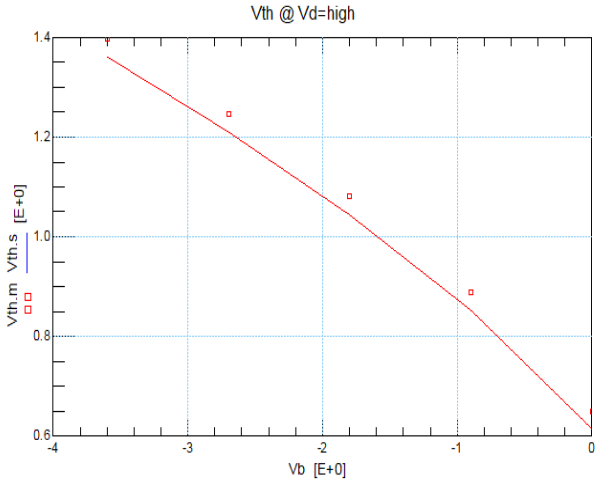
idvg, Vd = 3.3V, T = -40°C

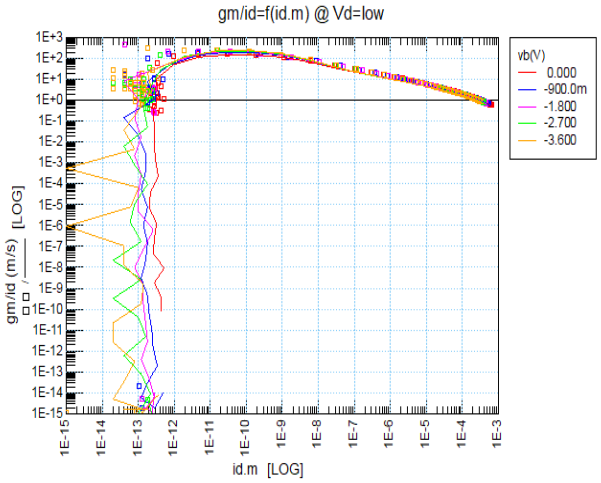


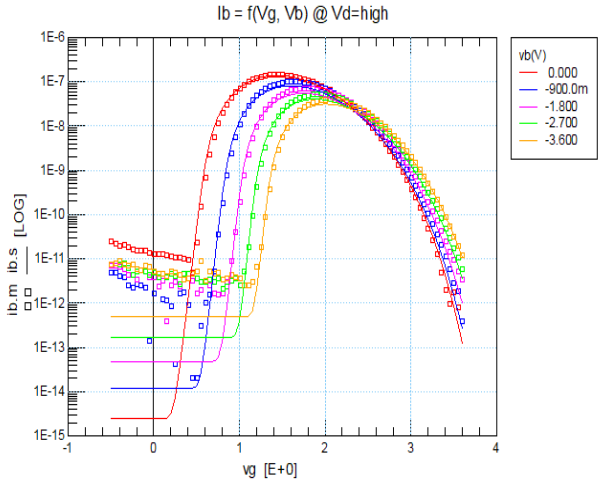


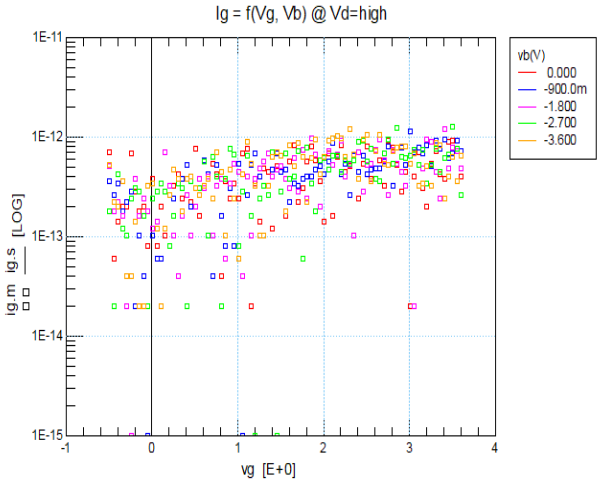




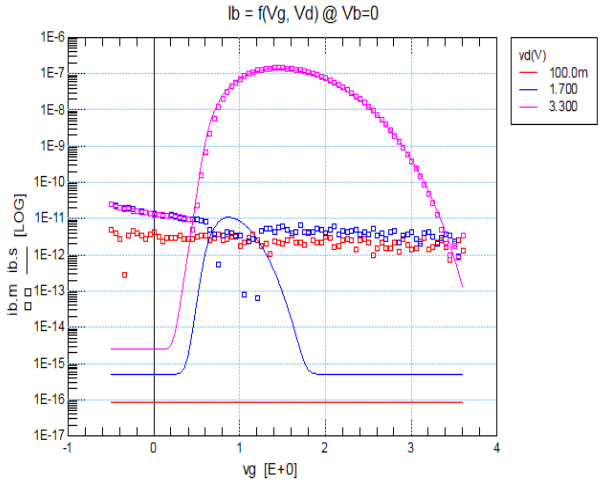


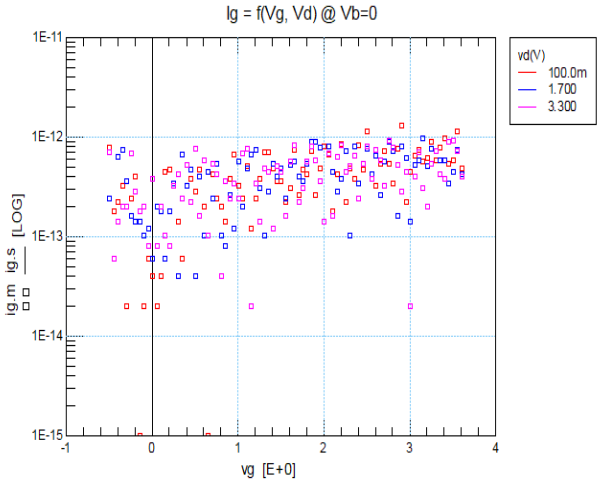




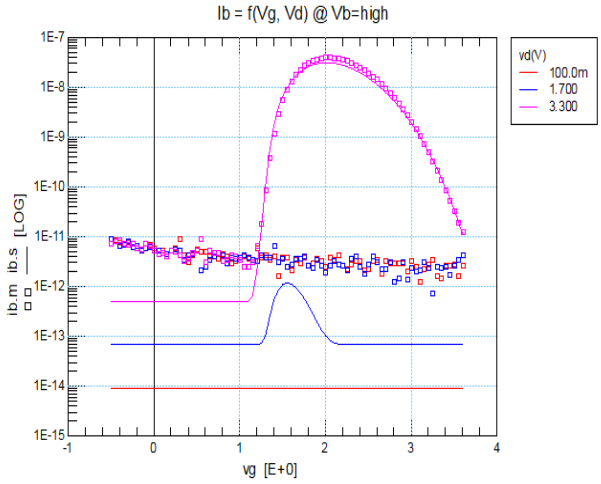


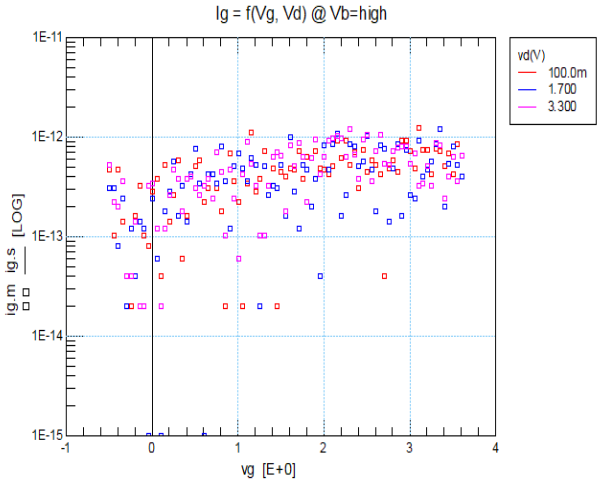
idvg, Ib,Ig, Vb = 0V, T = -40°C



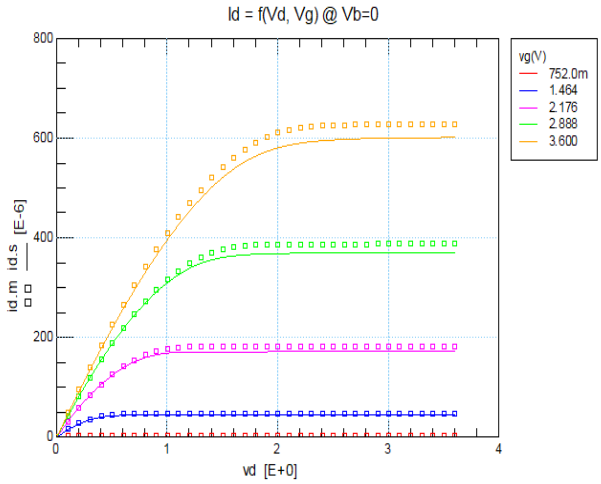


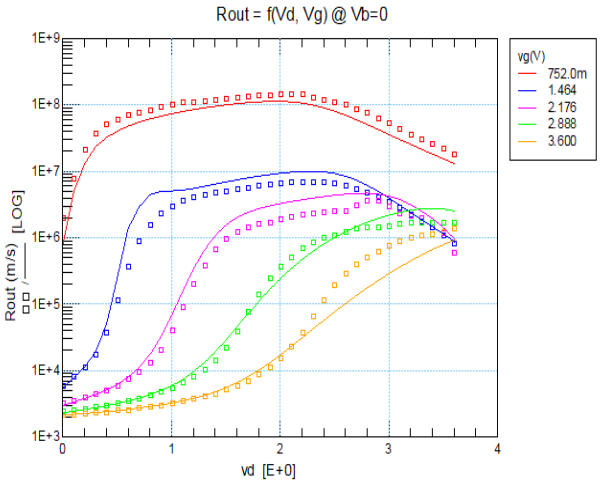
idvg, Ib,Ig, Vb = -3.6V, T = -40°C

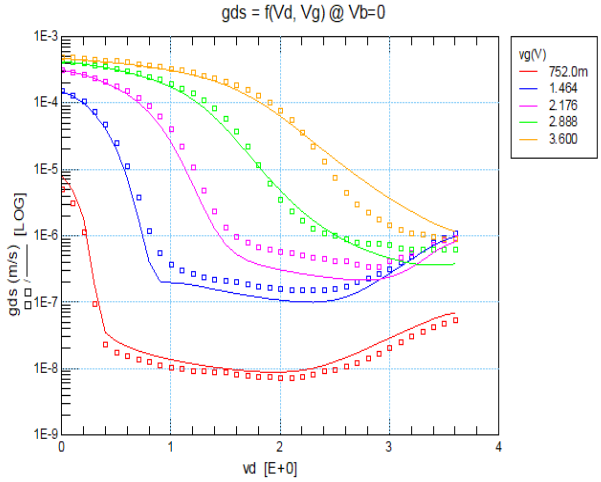




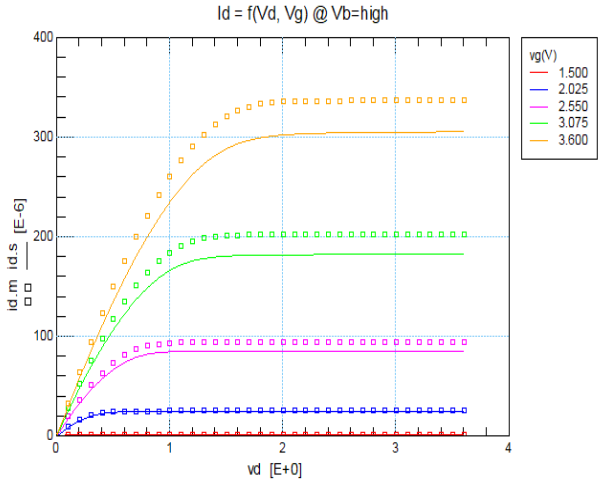
idvd, Vb = 0V, T = -40°C

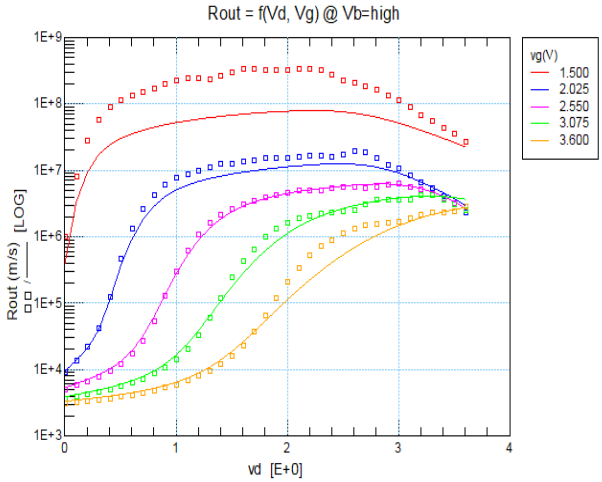


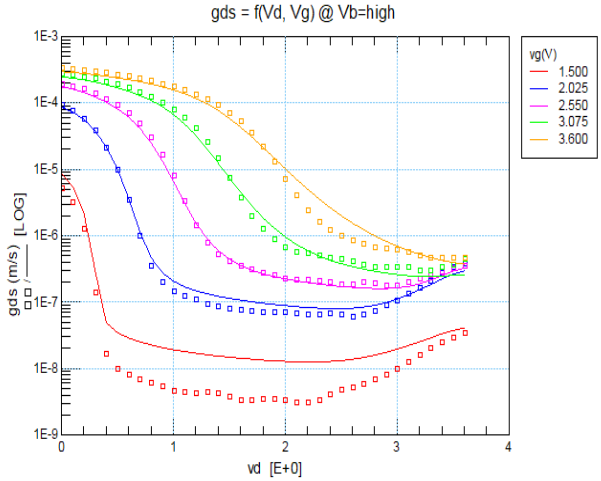




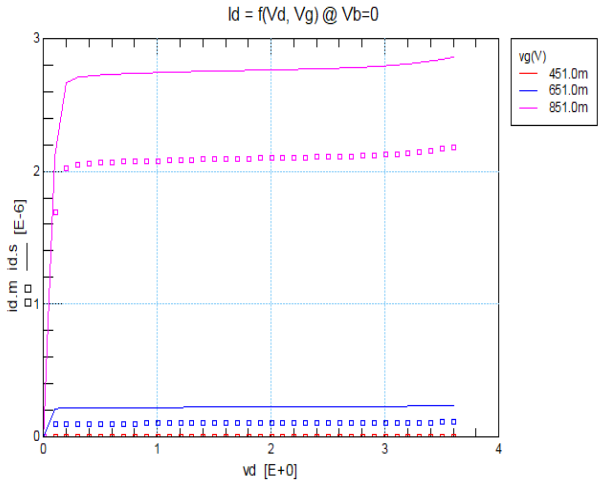
idvd_vbmin, Vb = -3.6V, T = -40°C

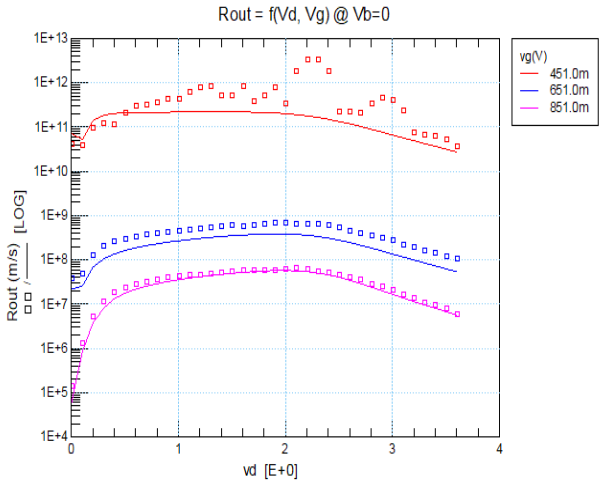


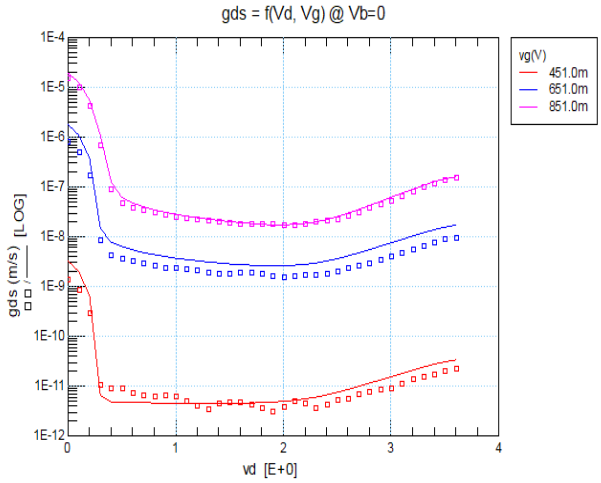




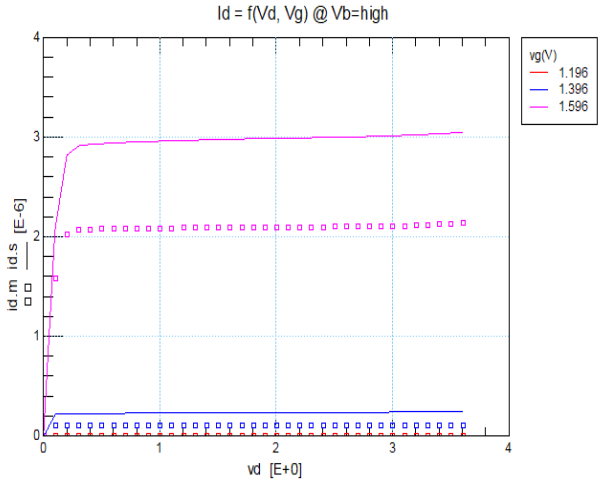
idvd_vth, Vb = 0V, T = -40°C

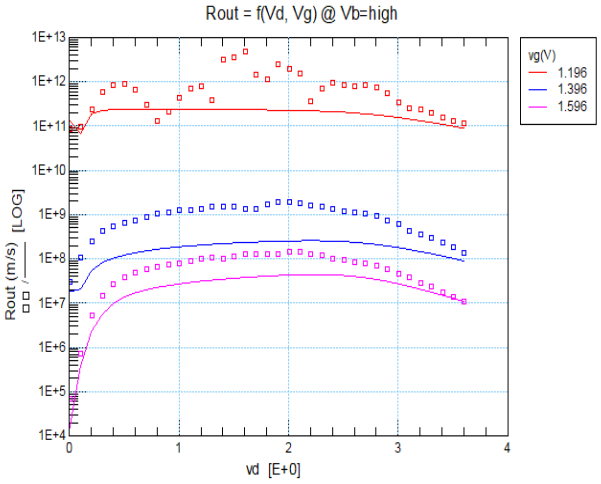


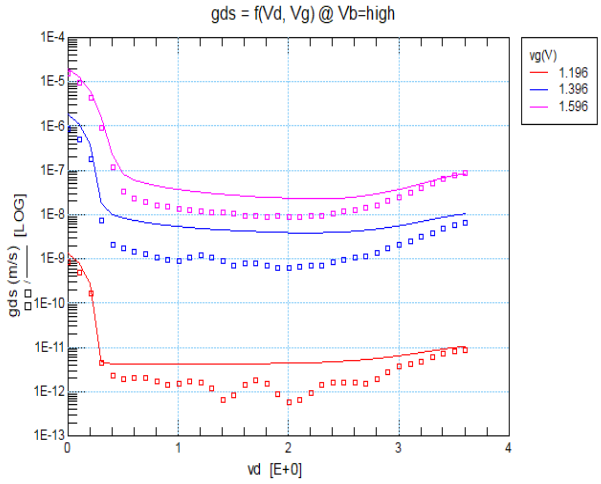




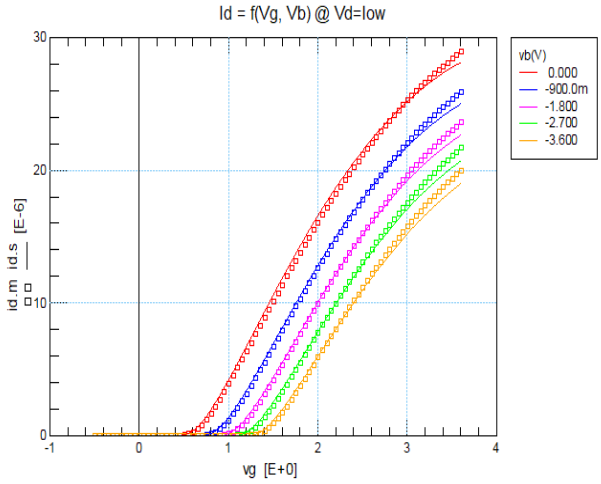
idvd_vth_vbmin, Vb = -3.6V, T = -40°C

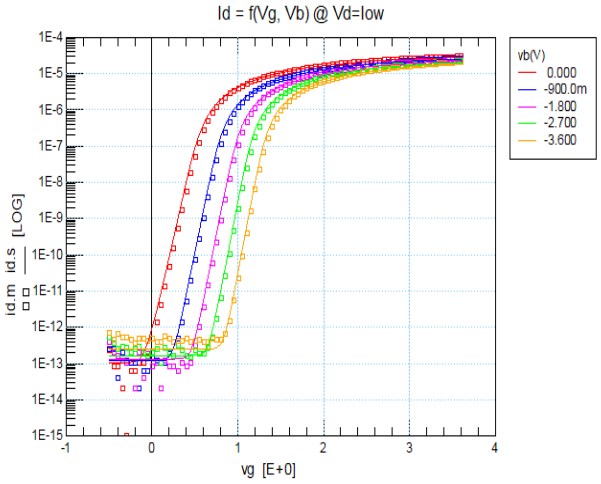


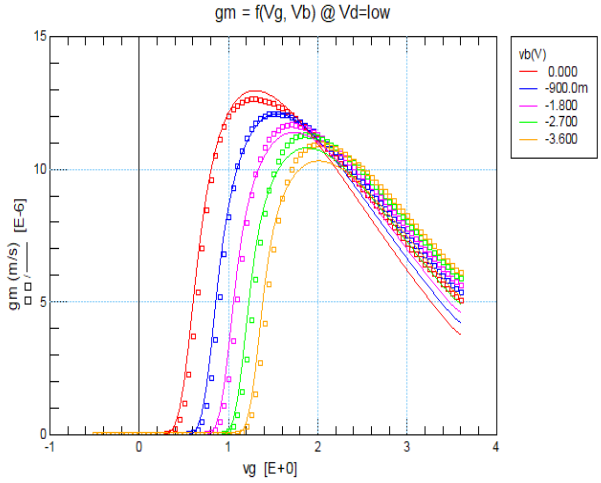


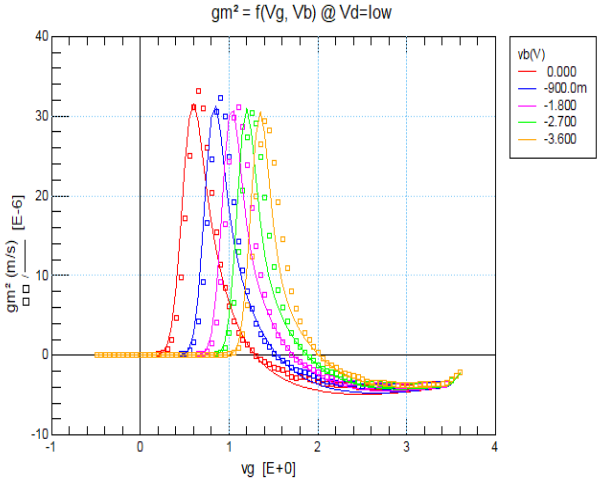


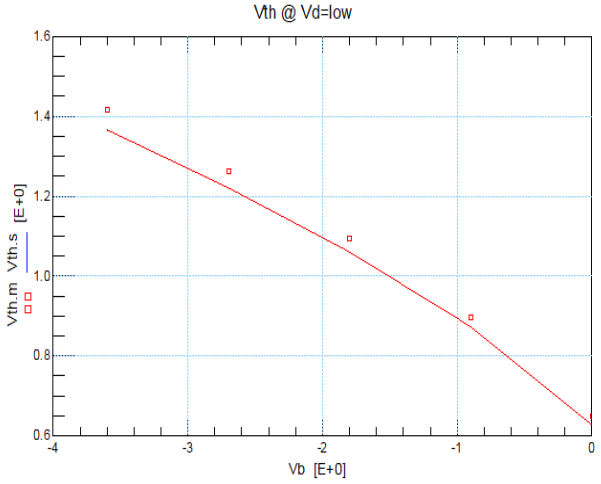
idvg, Vd = 0.1V, T = 70°C

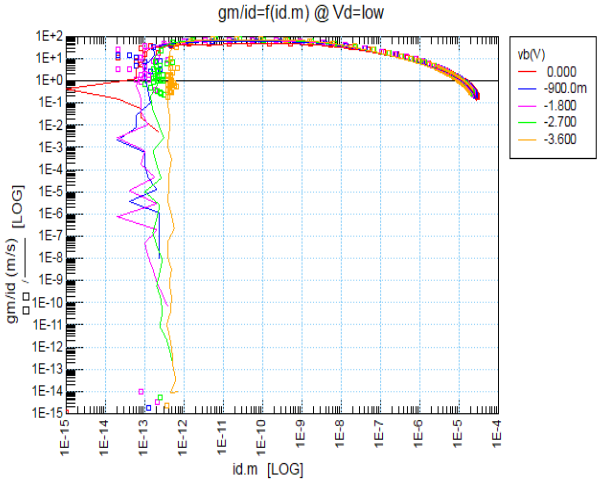


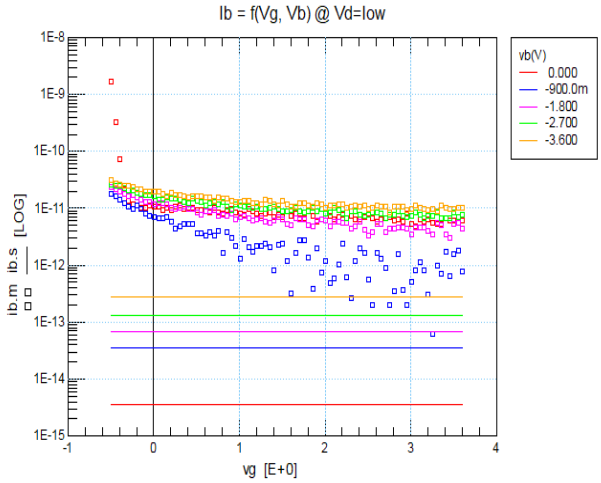


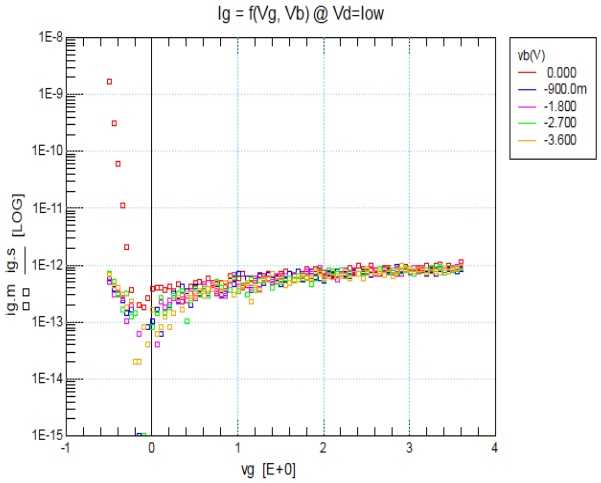




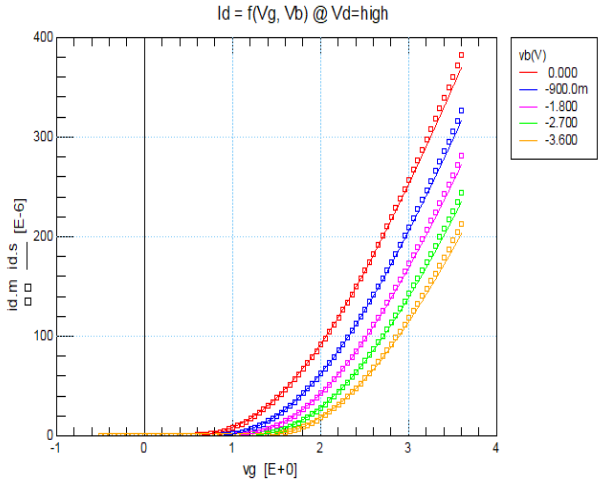


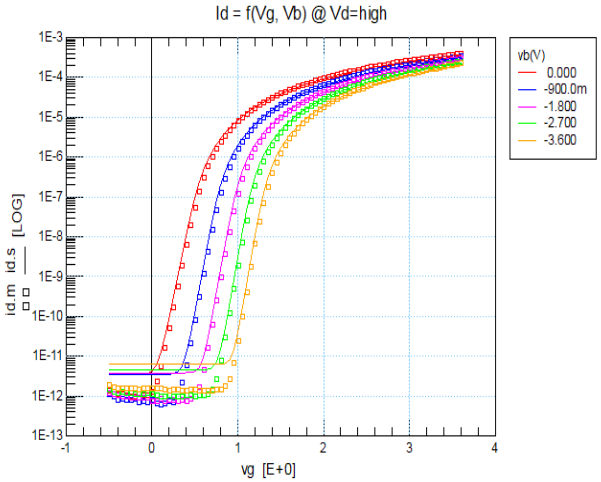


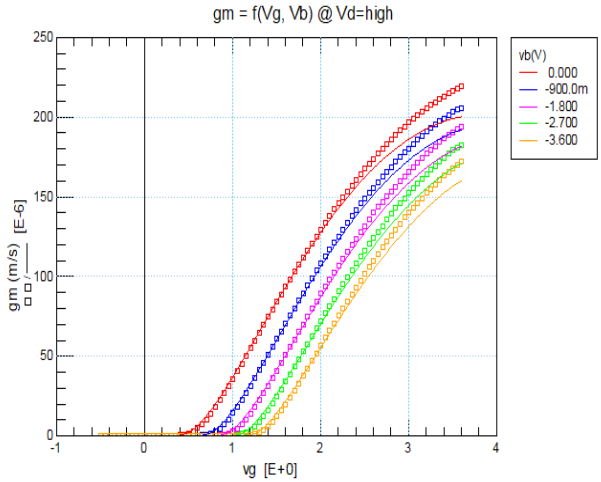


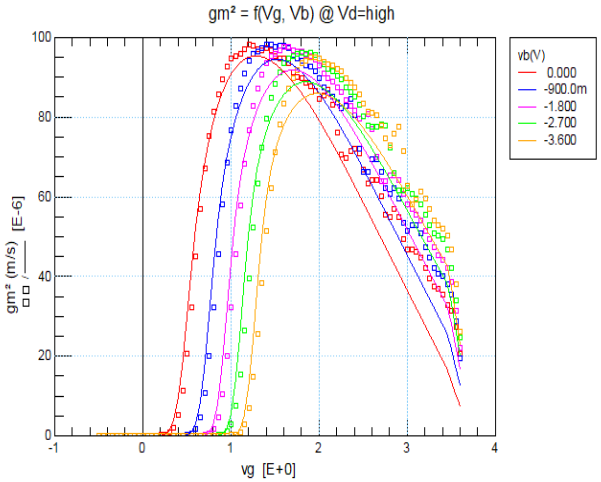


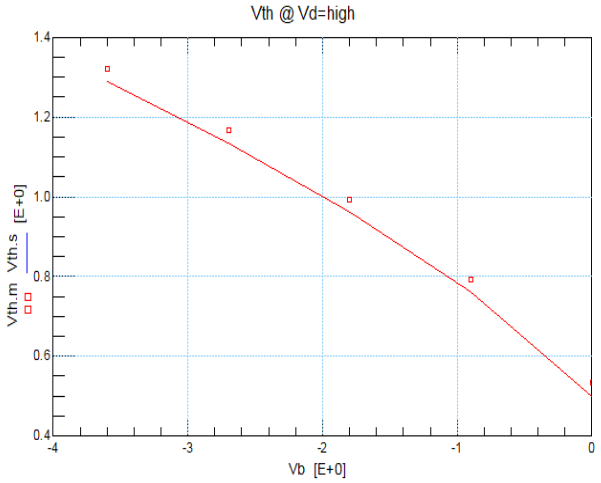
idvg, Vd = 3.3V, T = 70°C

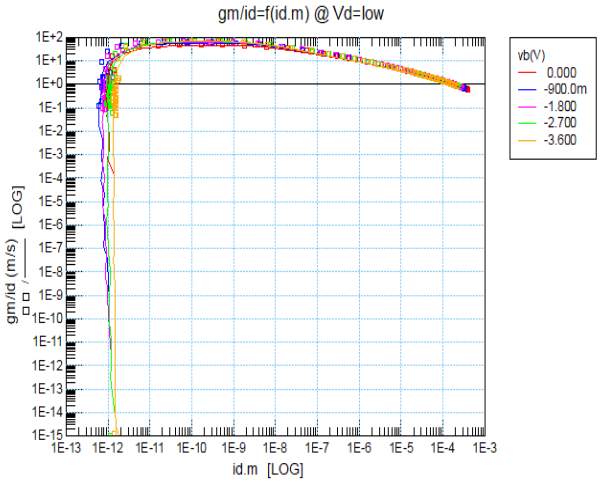


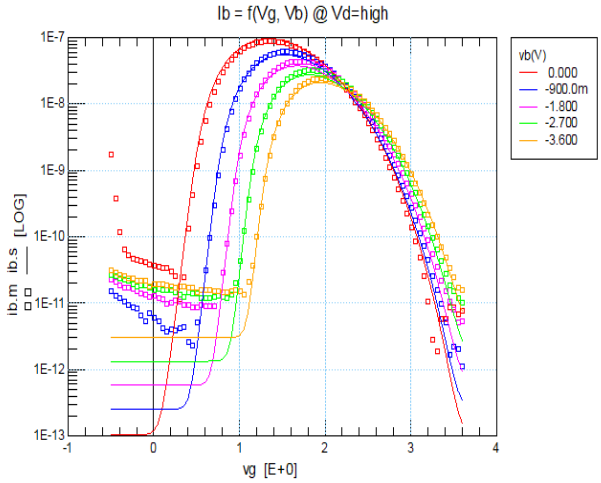


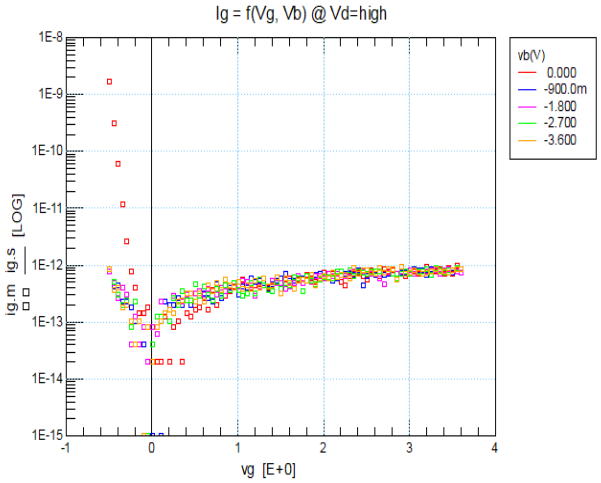




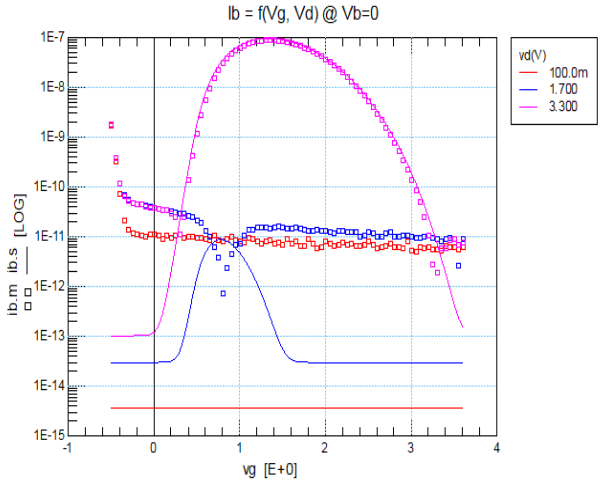


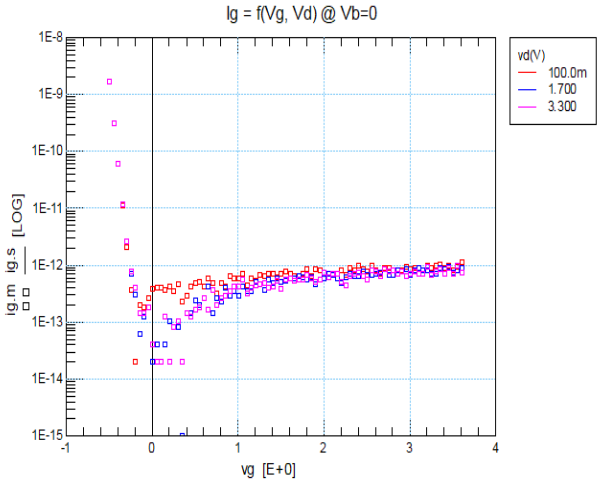




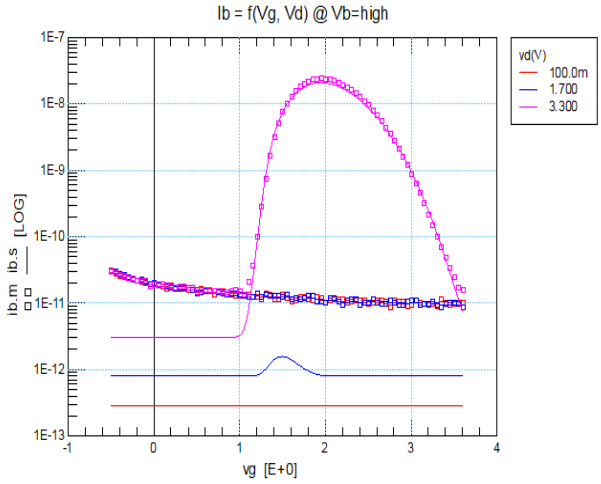


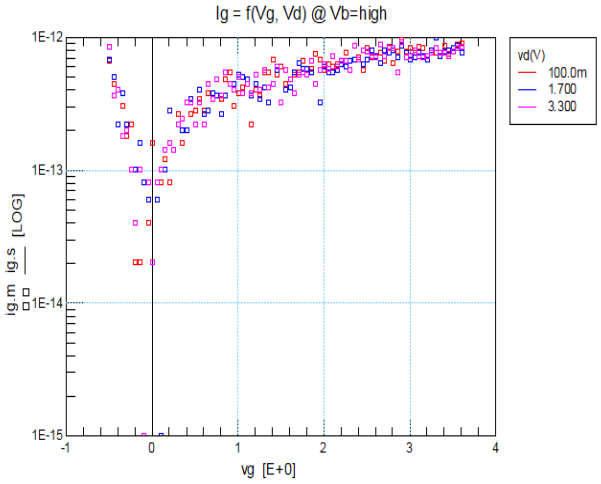
idvg, Ib,Ig, Vb = 0V, T = 70°C



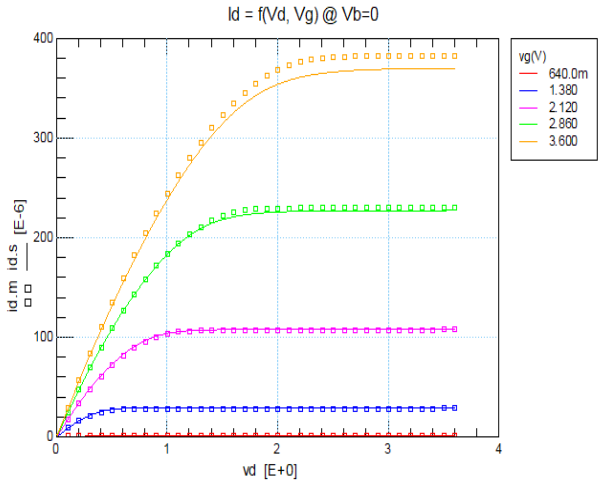


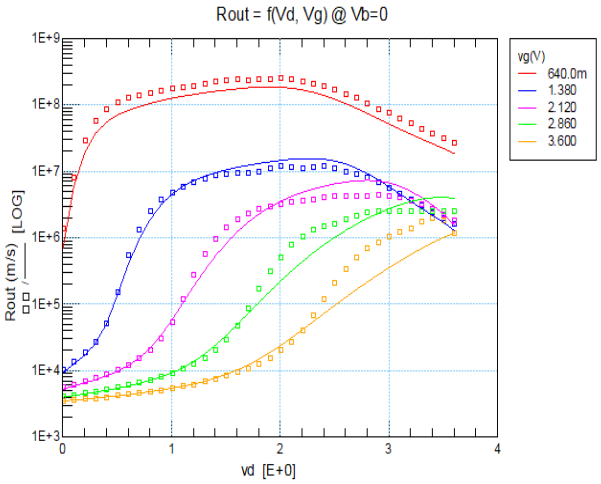
idvg, Ib,Ig, Vb = -3.6V, T = 70°C

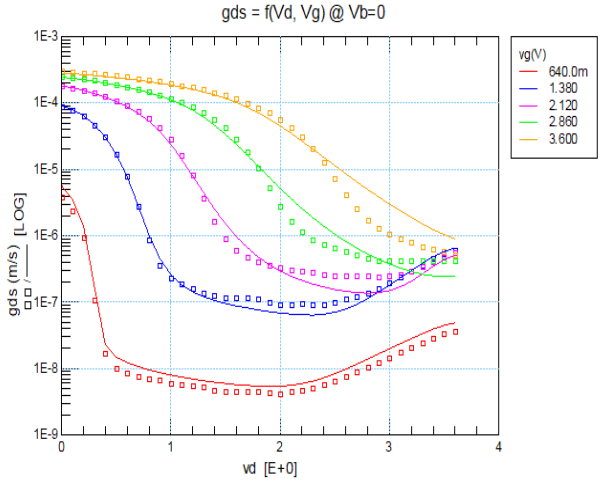




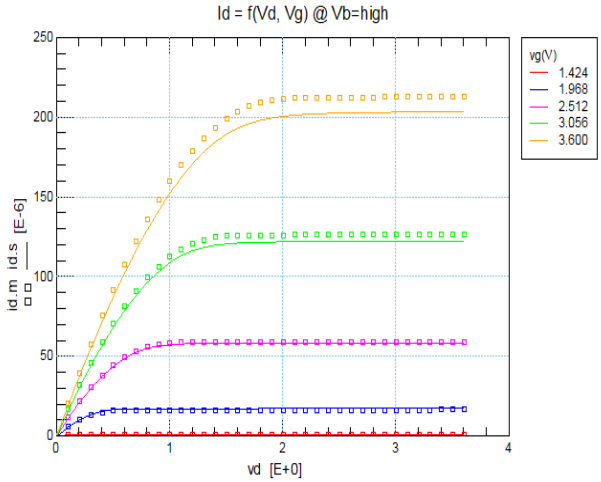
idvd, Vb = 0V, T = 70°C

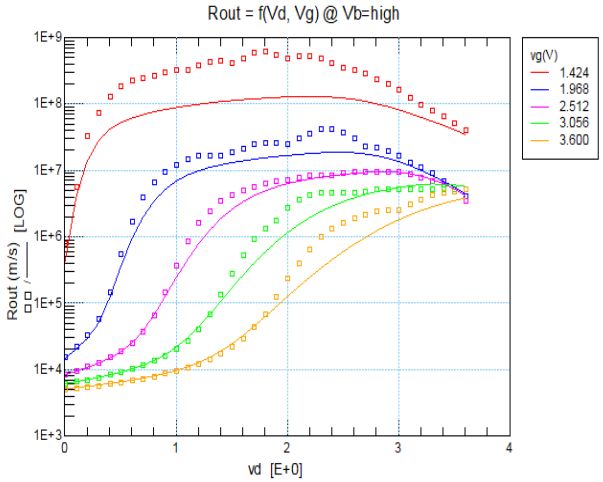


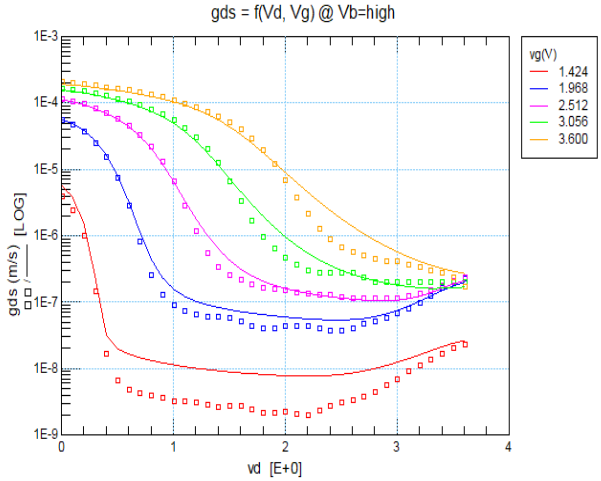




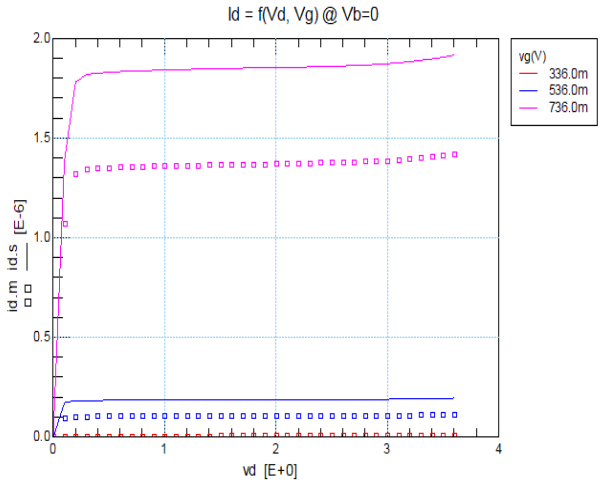
idvd_vbmin, Vb = -3.6V, T = 70°C

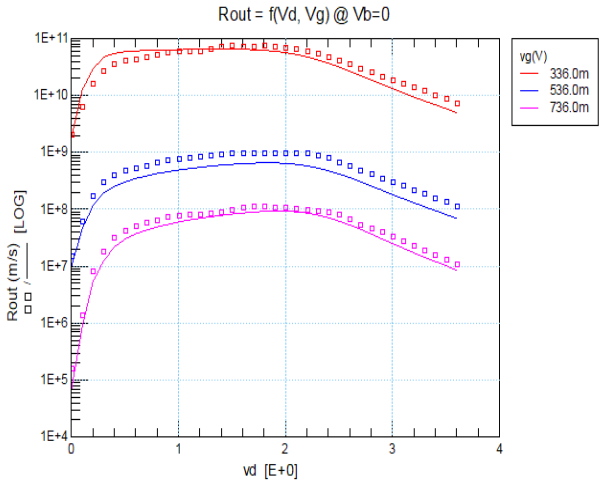


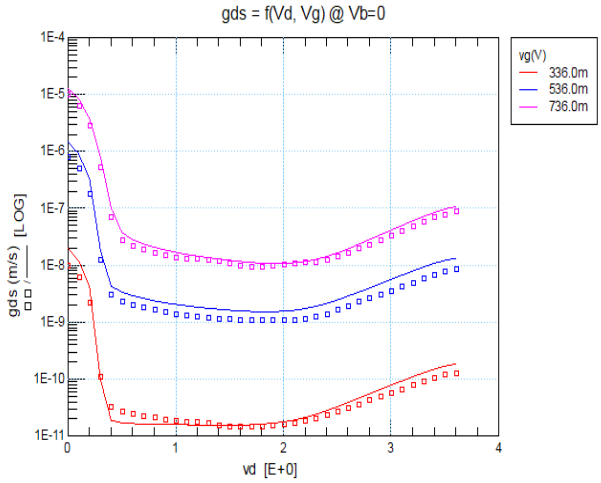




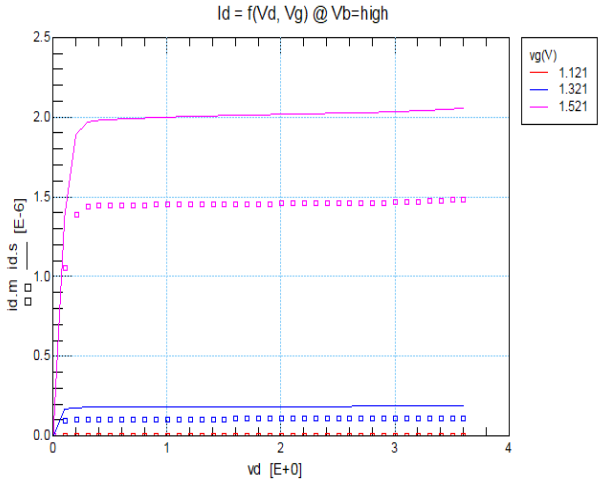
idvd_vth, Vb = 0V, T = 70°C

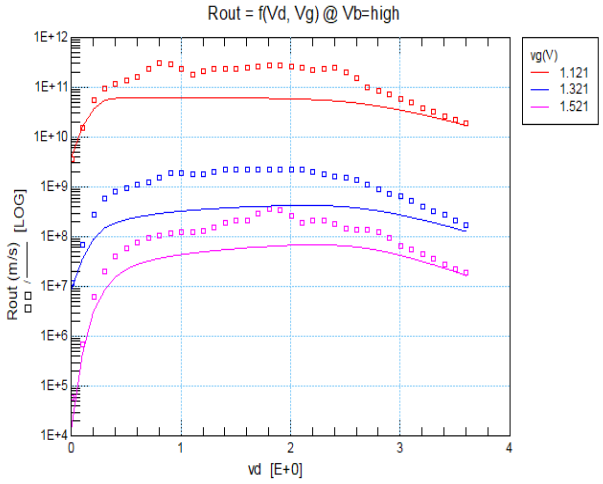


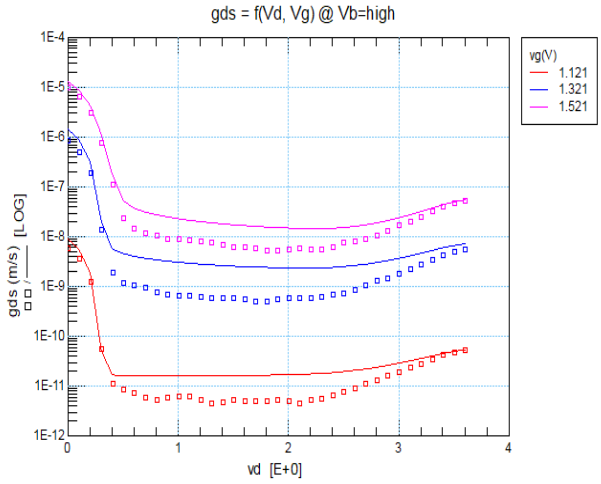




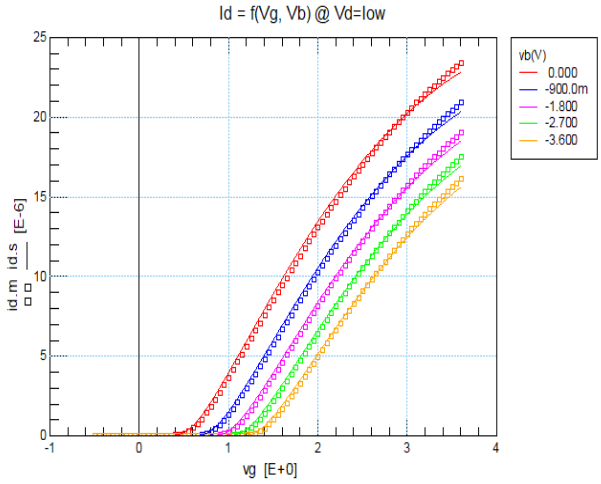
idvd_vth_vbmin, Vb = -3.6V, T = 70°C

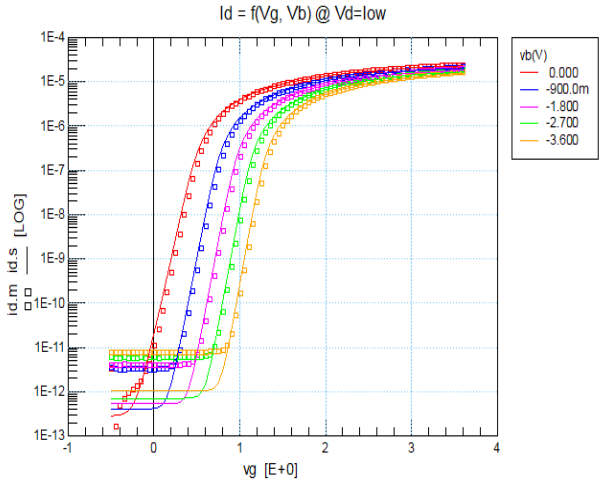


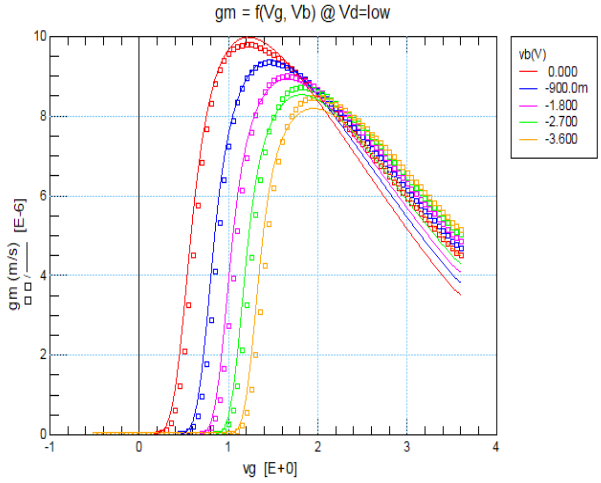


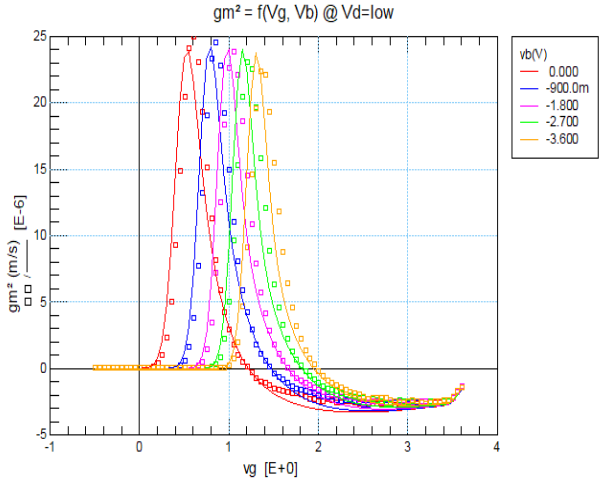


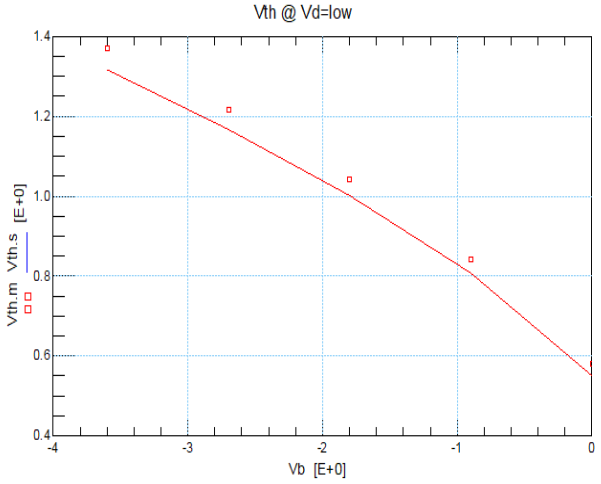
idvg, Vd = 0.1V, T = 125°C

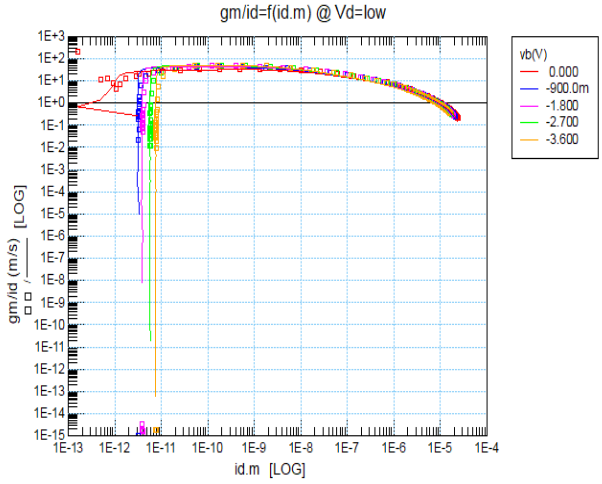


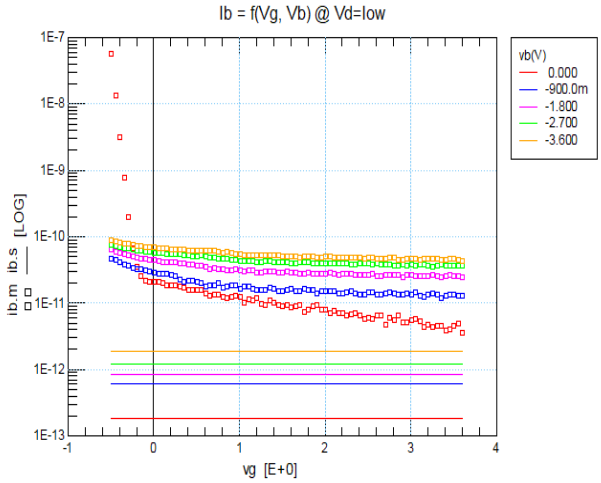


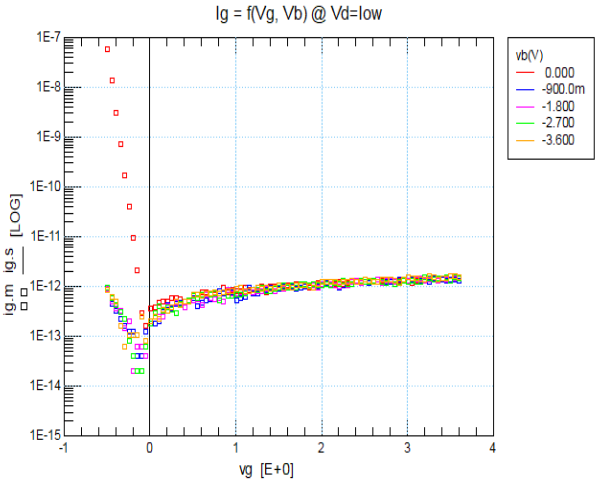




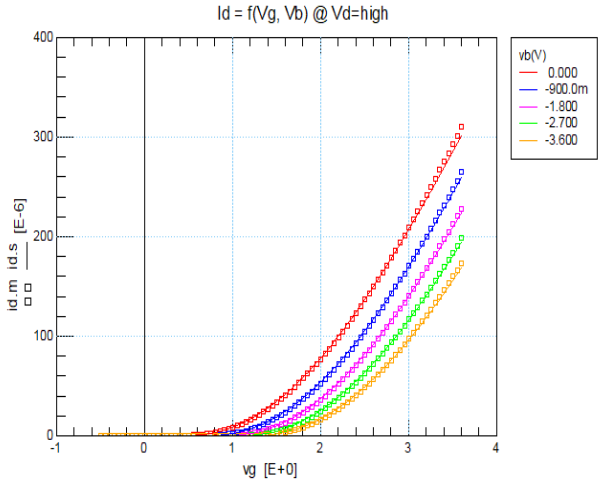


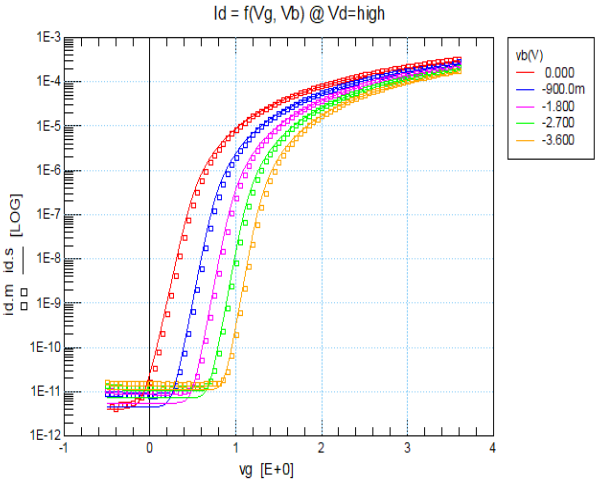


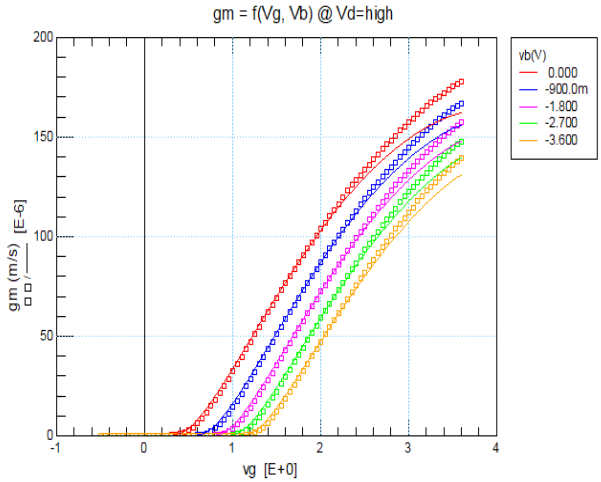


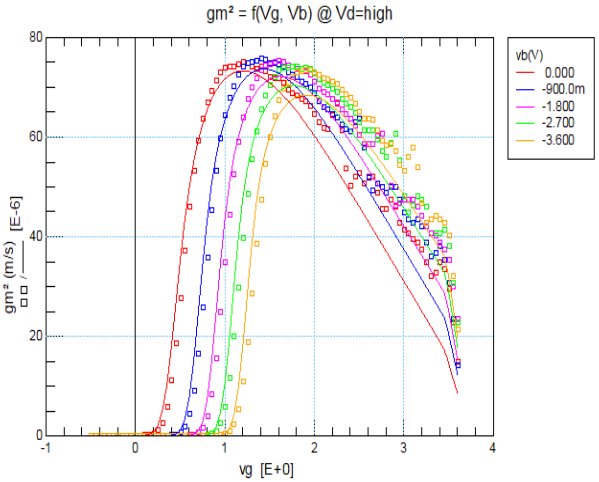


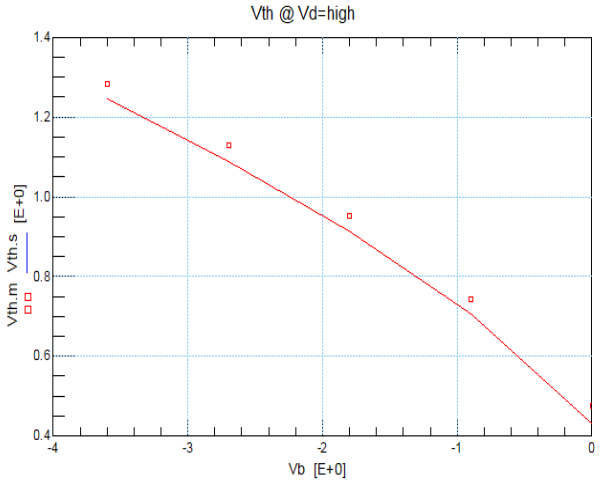
idvg, Vd = 3.3V, T = 125°C

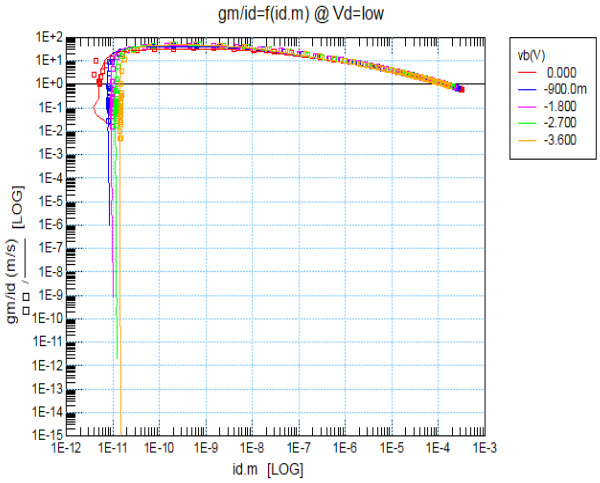


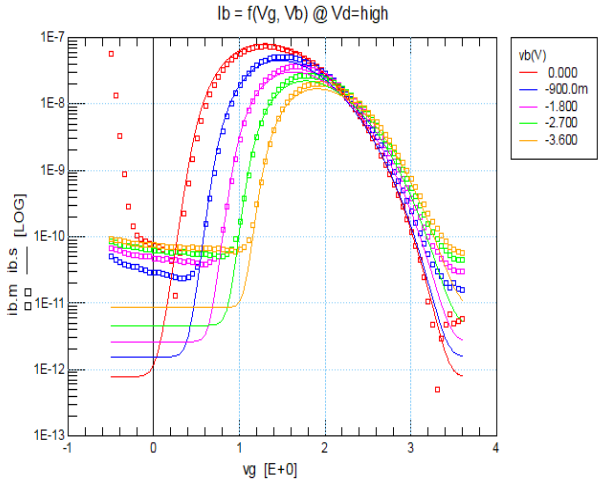


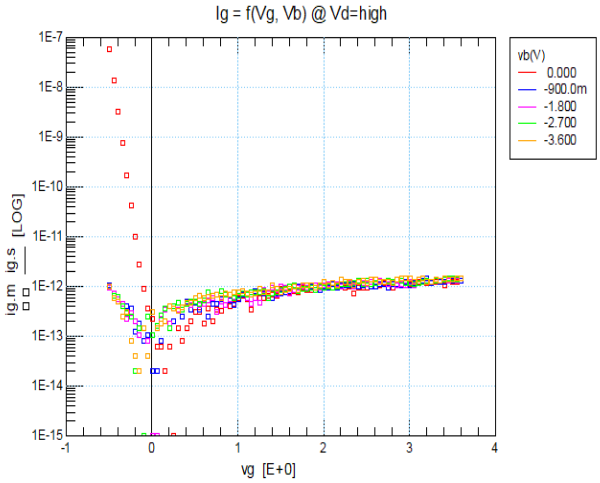




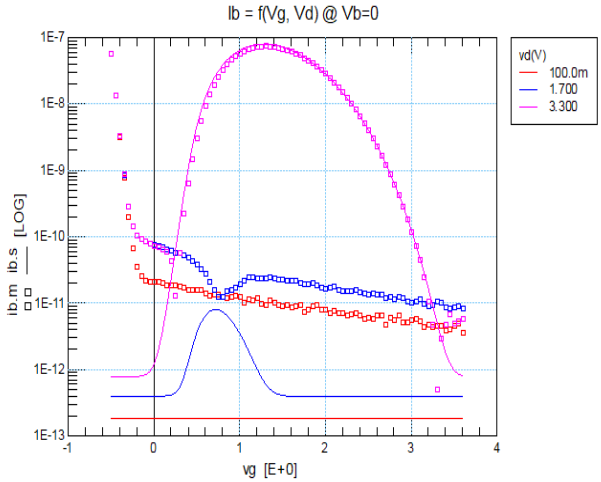


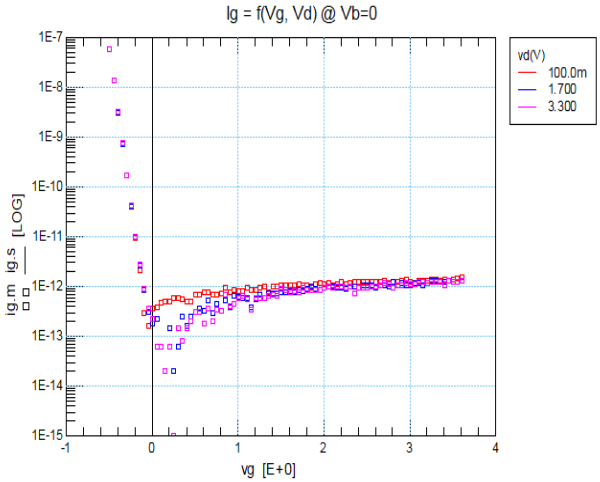




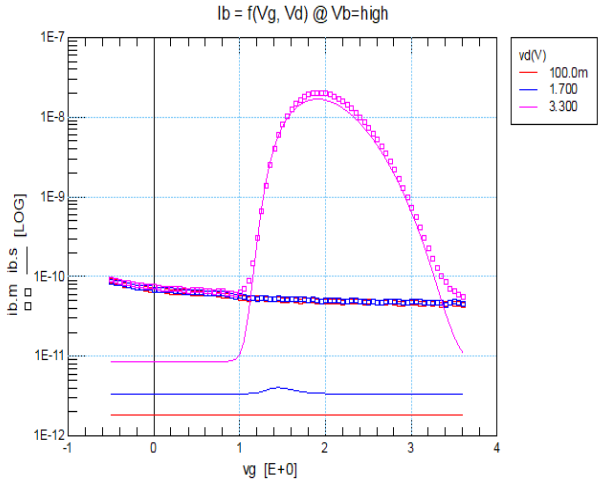


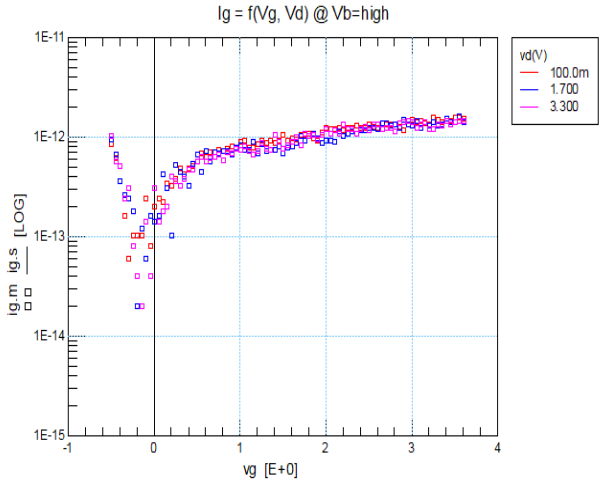
idvg, Ib,Ig, Vb = 0V, T = 125°C



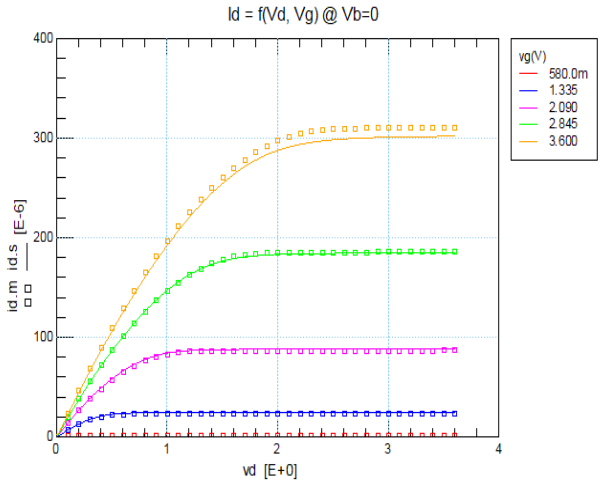


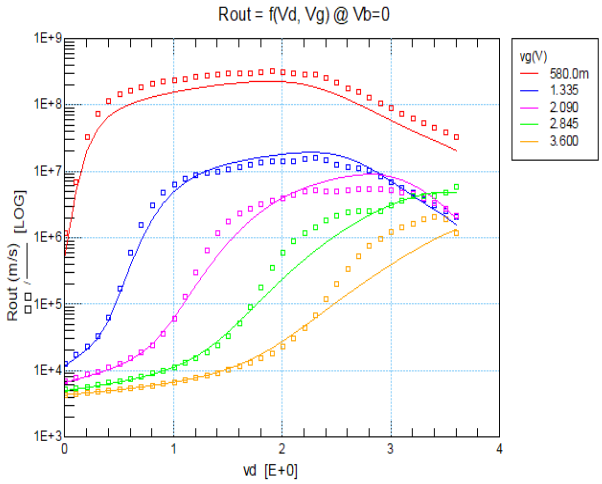
idvg, Ib,Ig, Vb = -3.6V, T = 125°C

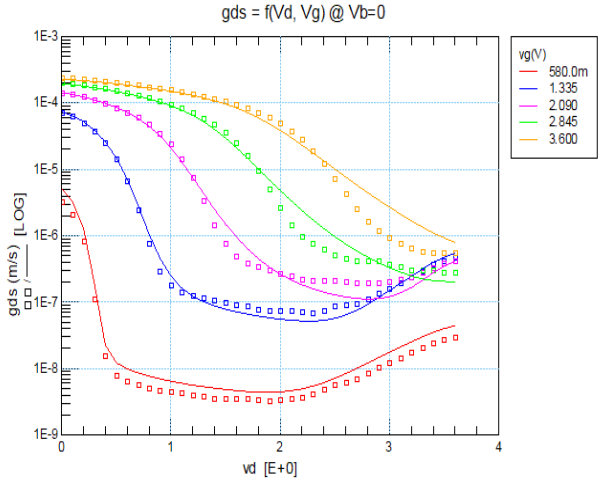




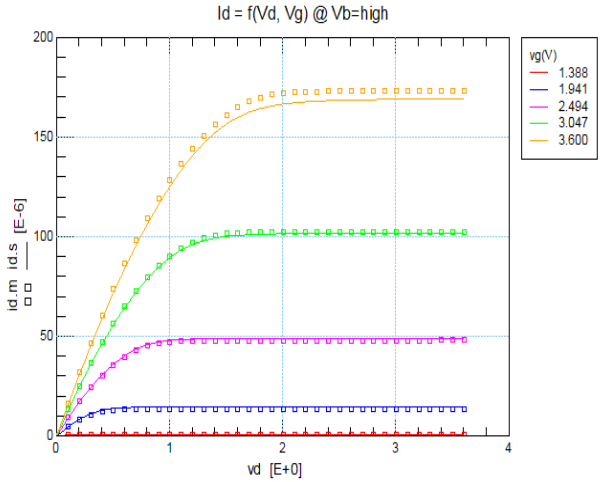
idvd, Vb = 0V, T = 125°C

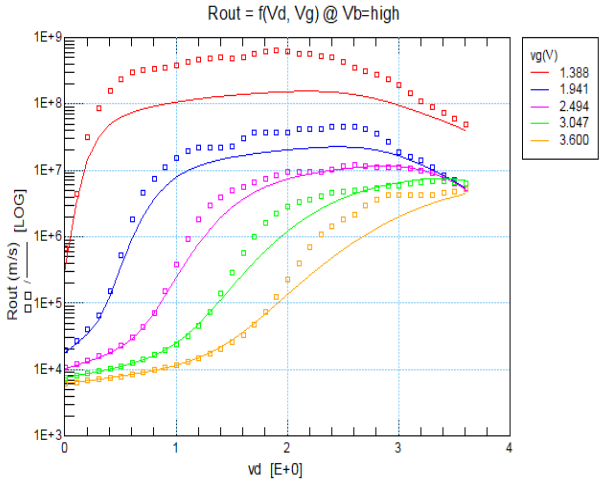


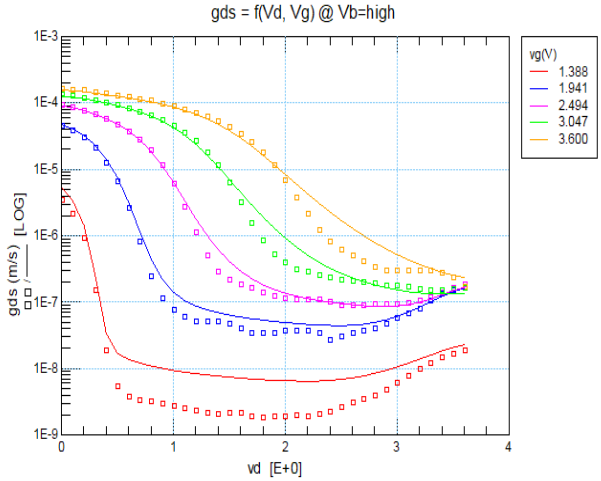




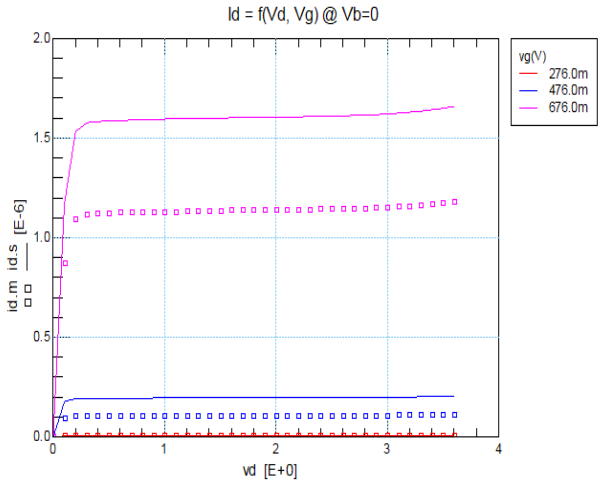
idvd_vbmin, Vb = -3.6V, T = 125°C

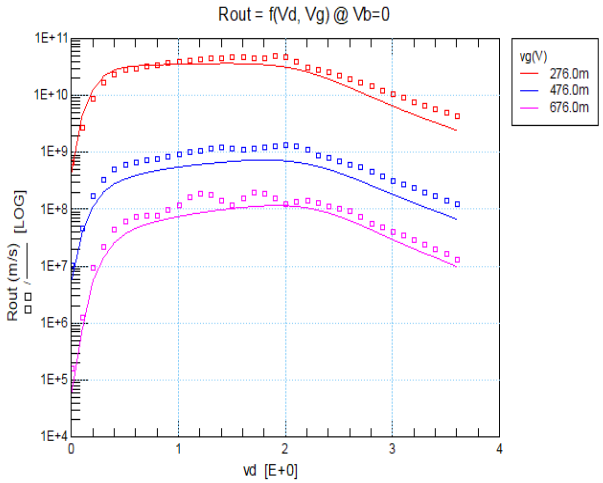


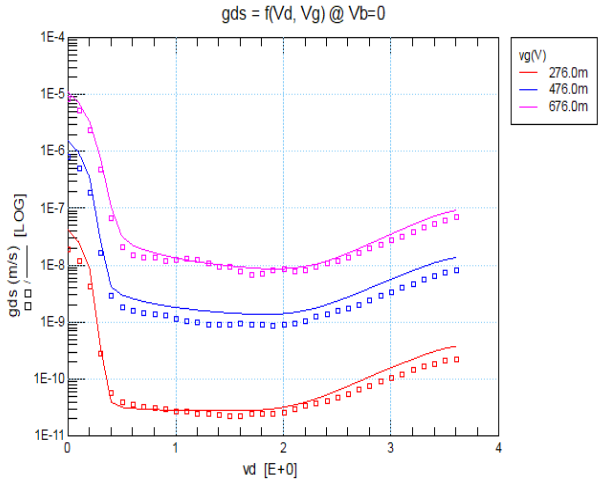




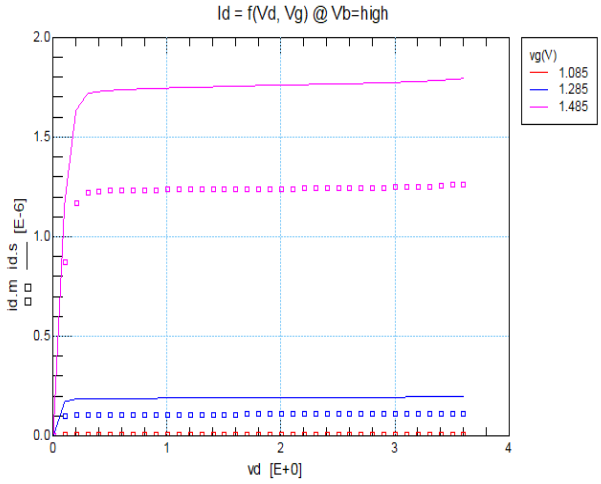
idvd_vth, Vb = 0V, T = 125°C

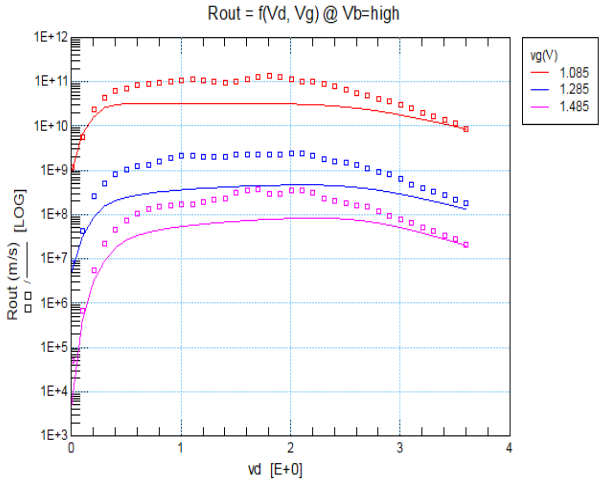


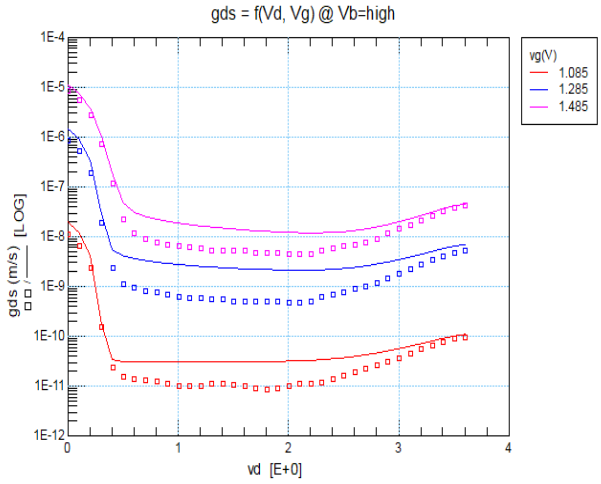




idvd_vth_vbmin, Vb = -3.6V, T = 125°C

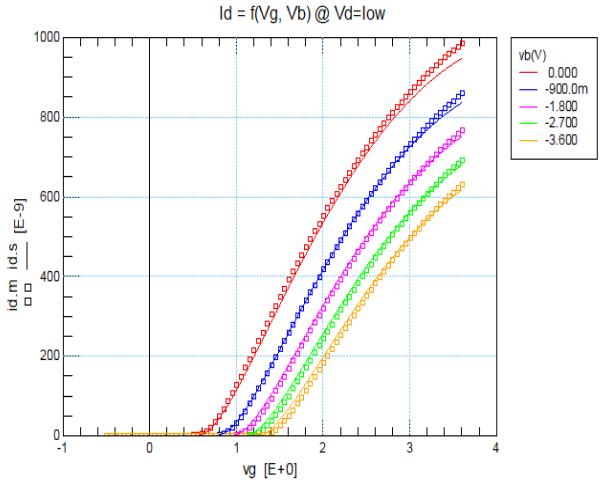


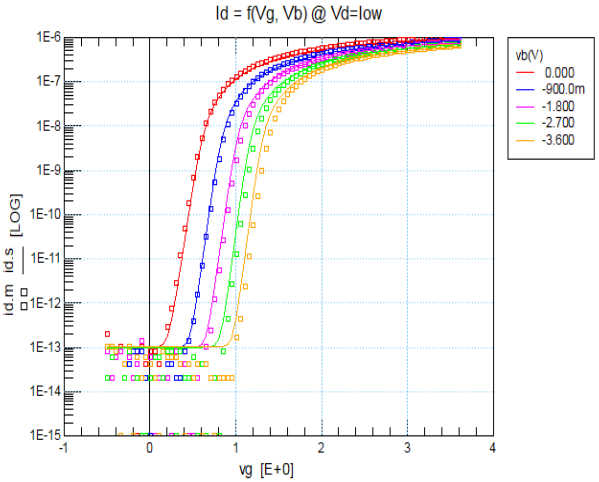


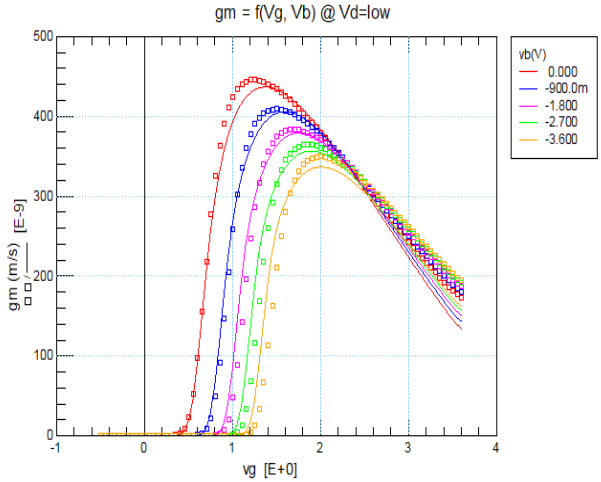


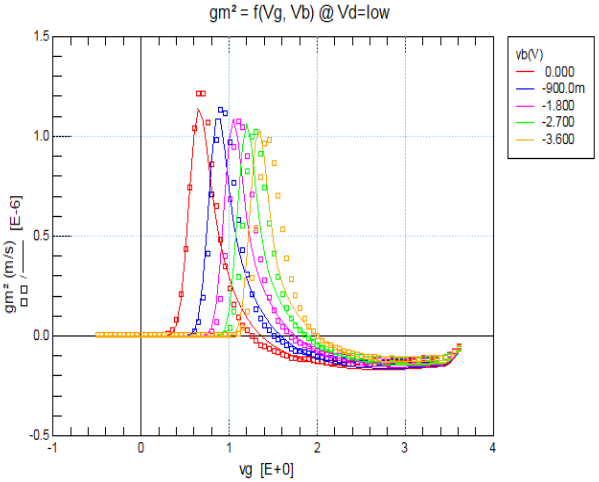
4.3 W0u3_L10u0_S547_4

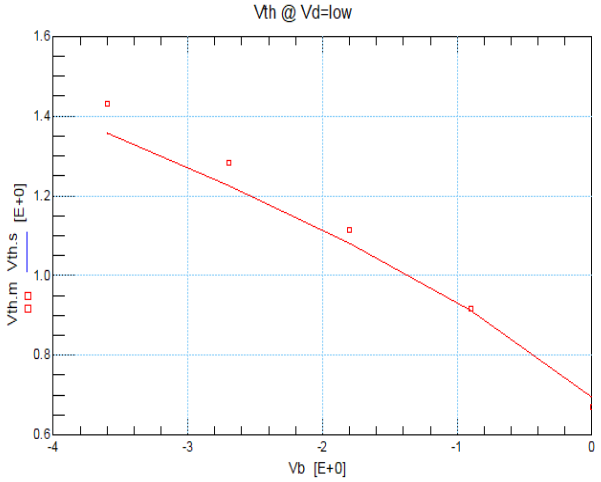
idvg, Vd = 0.1V, T = 27°C

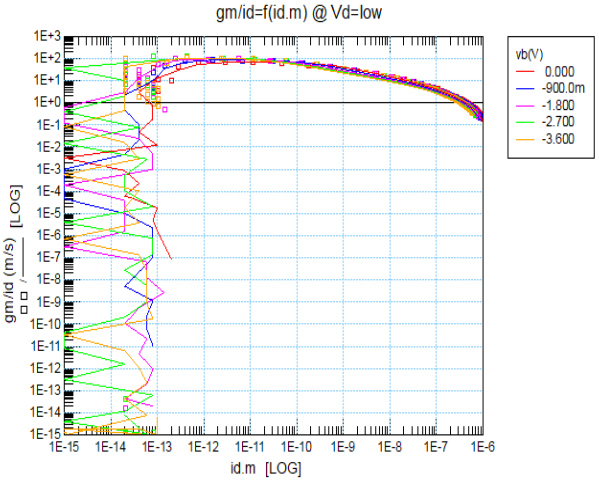


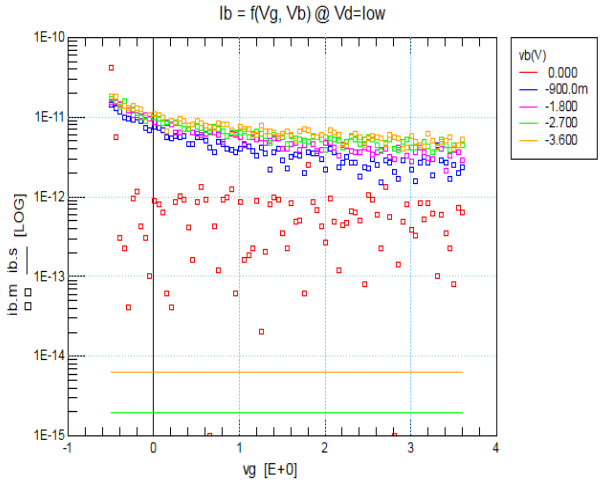


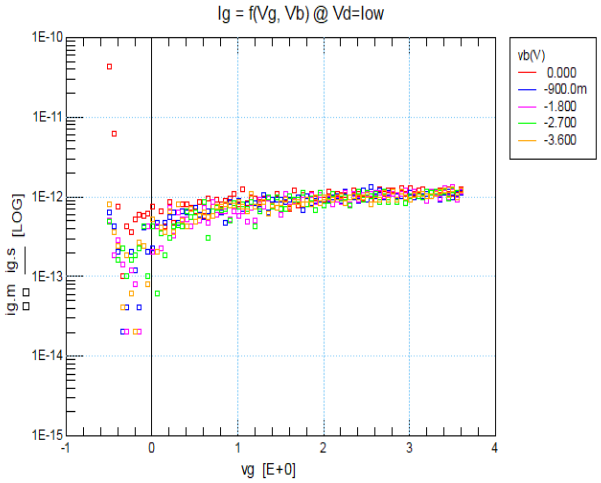




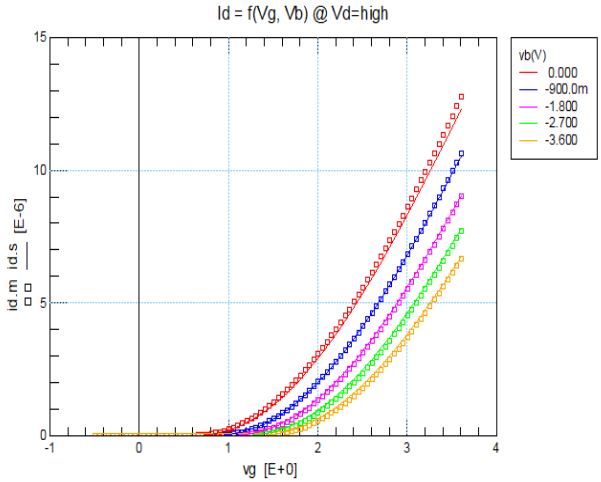


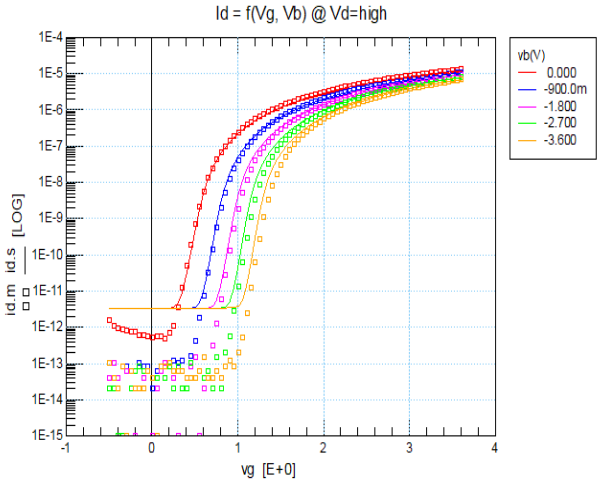


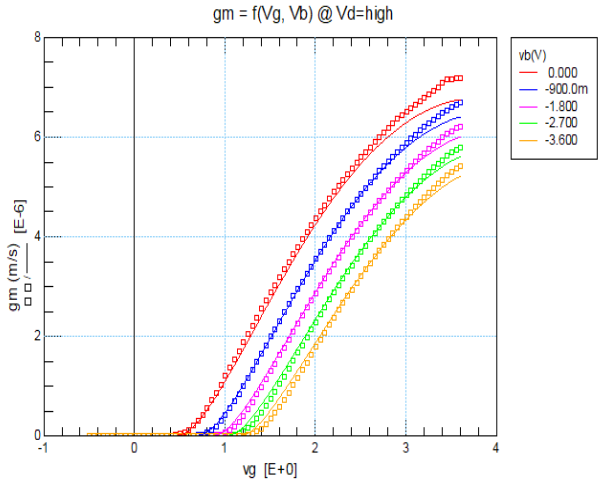


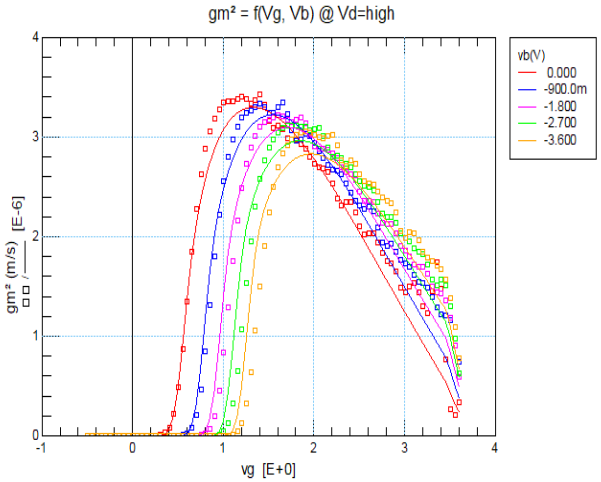


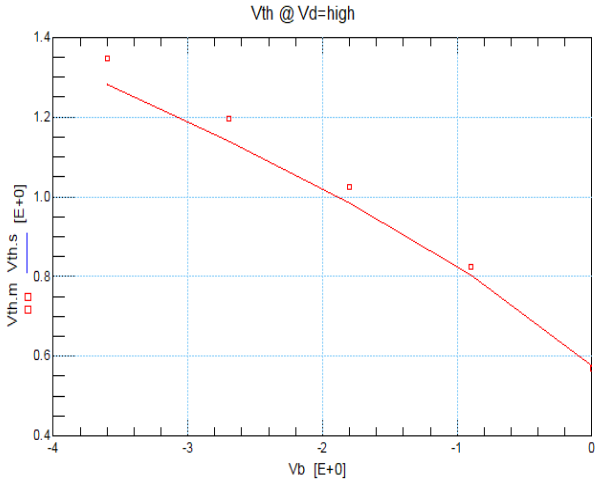
idvg, Vd = 3.3V, T = 27°C

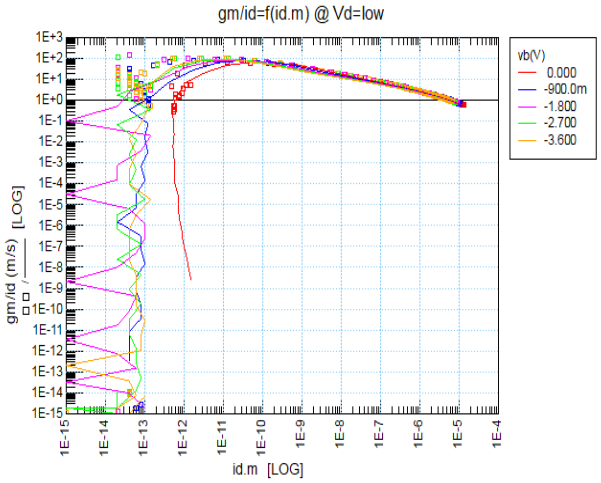


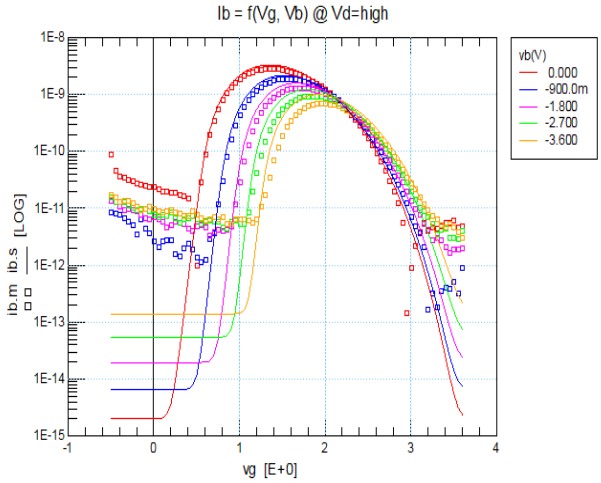


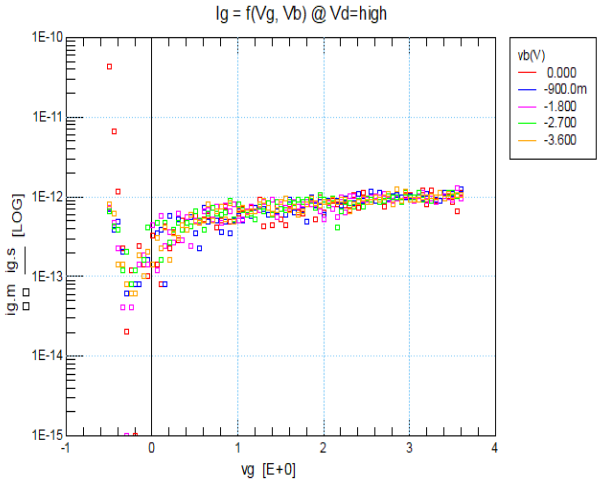




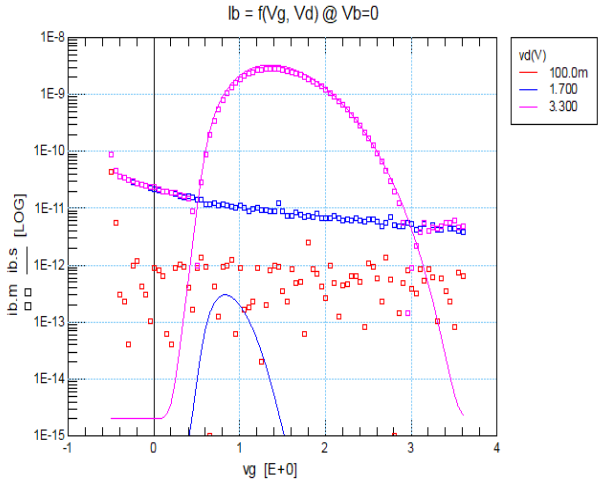


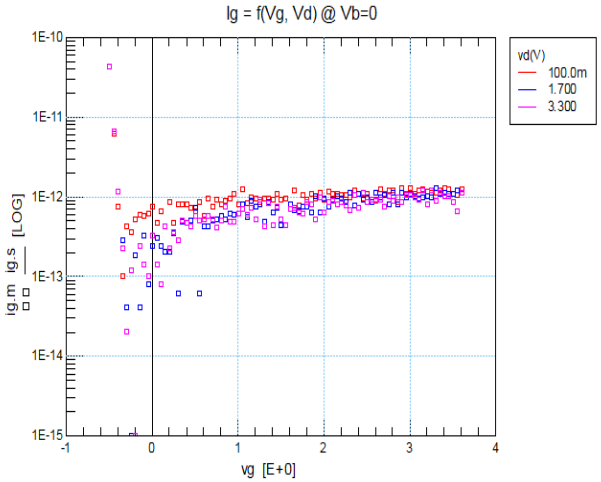




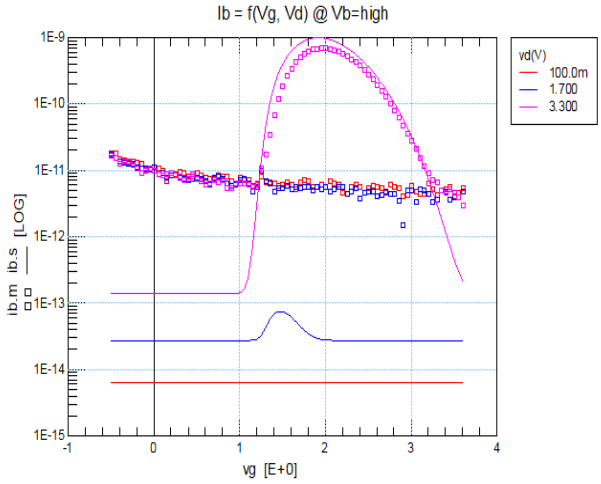


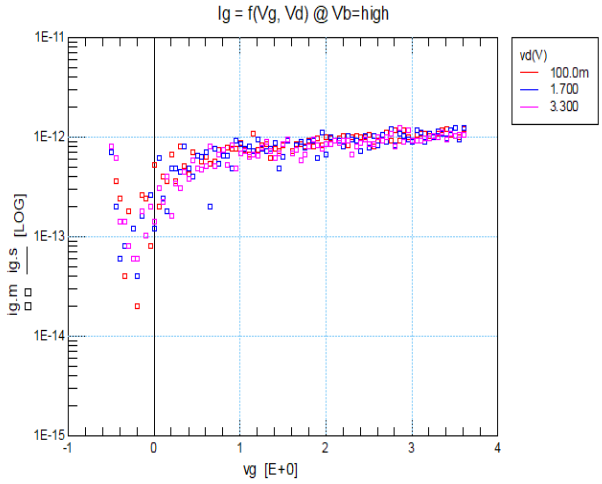
idvg, Ib,Ig, Vb = 0V, T = 27°C



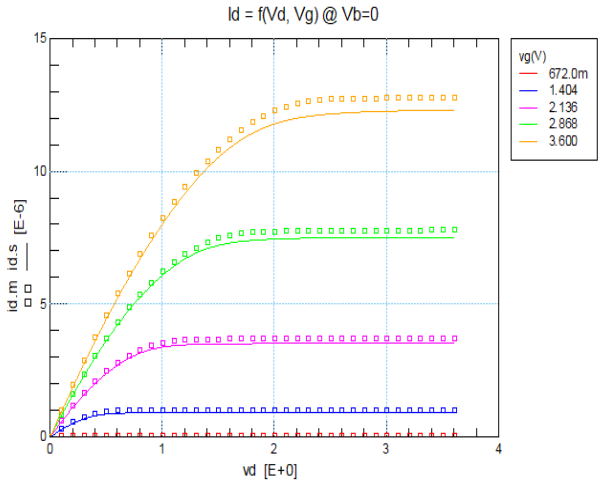


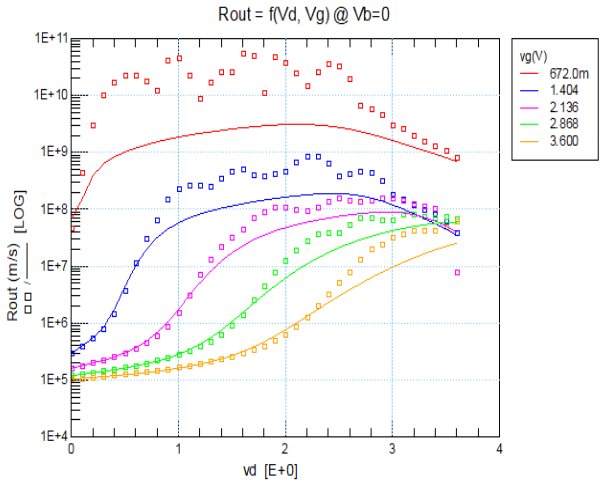
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

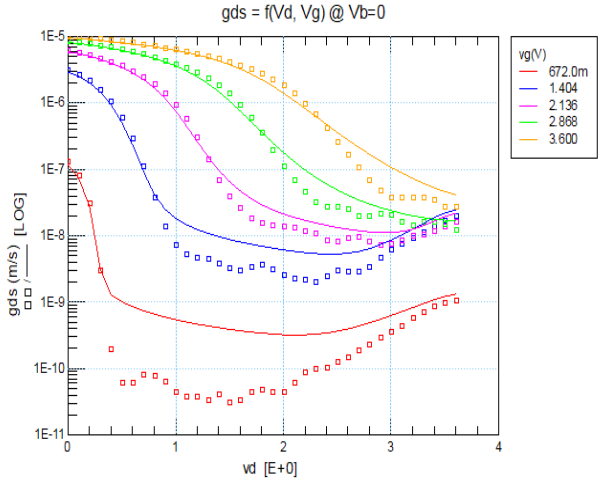




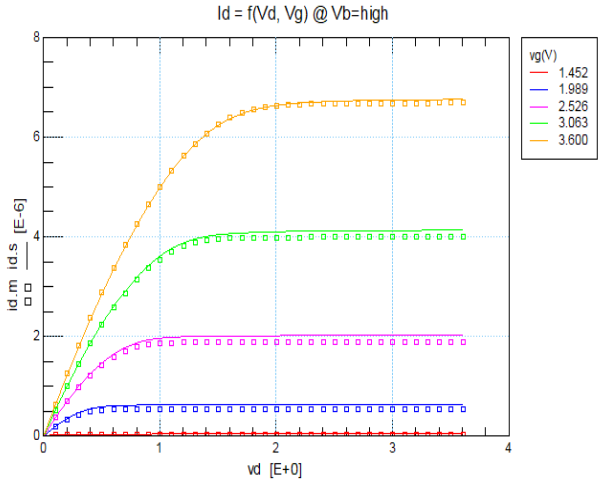
idvd, Vb = 0V, T = 27°C

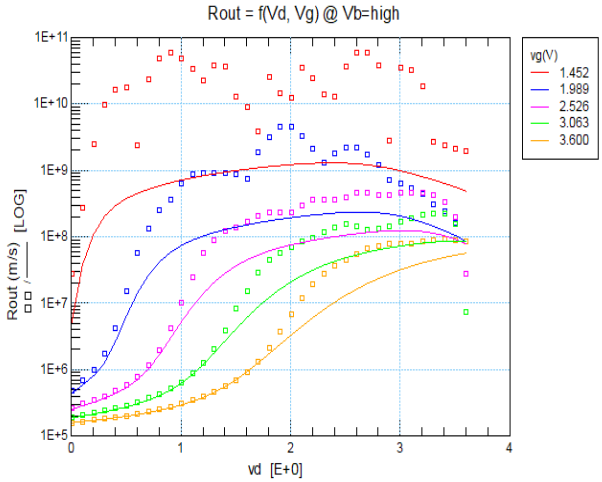


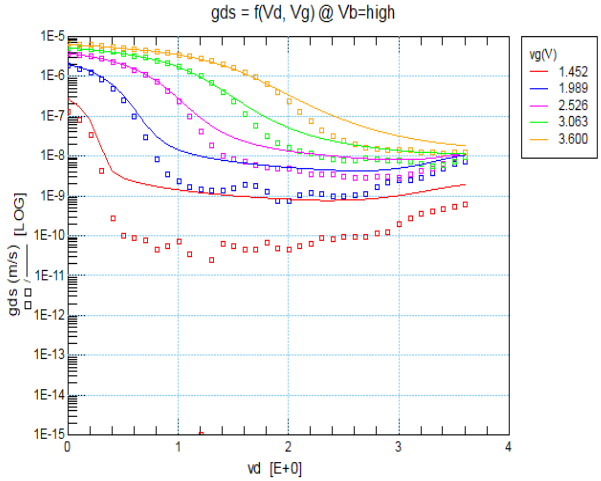




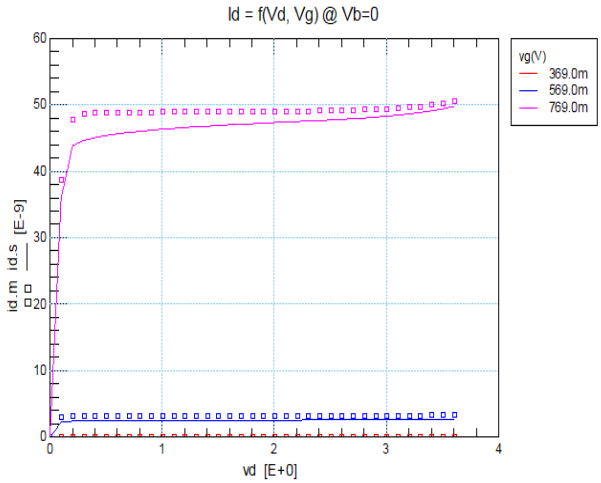
idvd_vbmin, Vb = -3.6V, T = 27°C

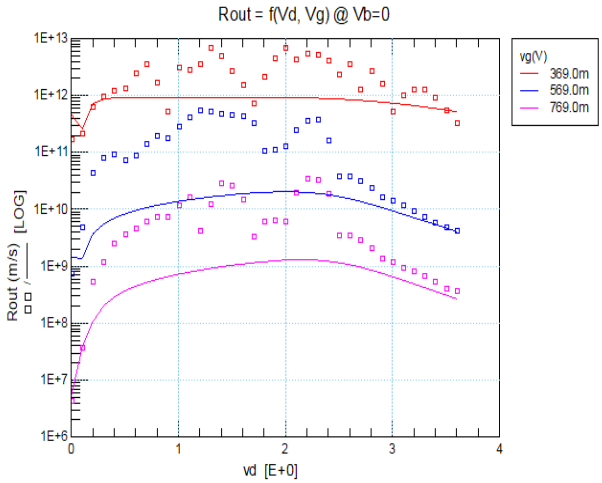


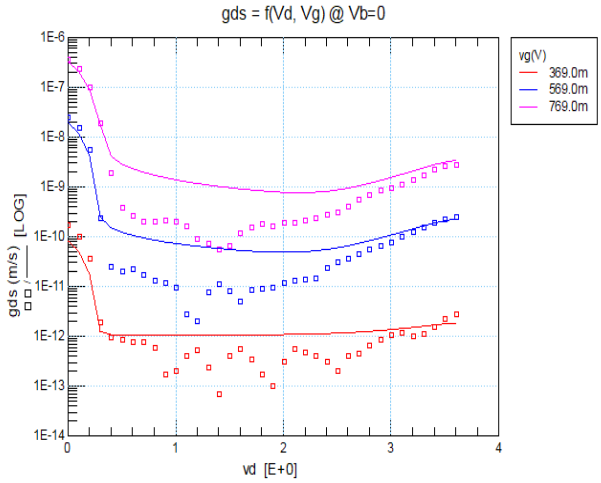




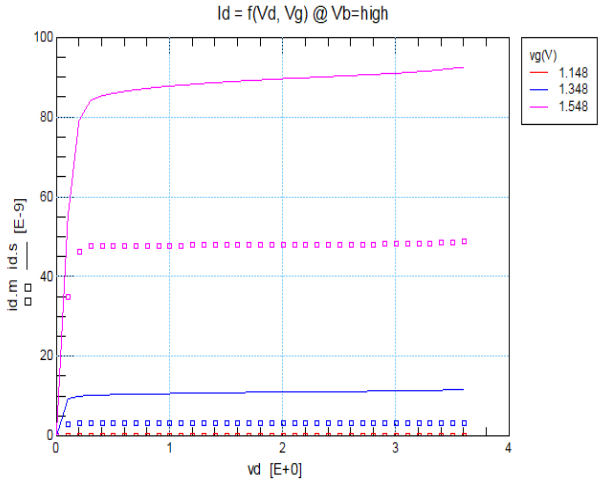
idvd_vth, Vb = 0V, T = 27°C

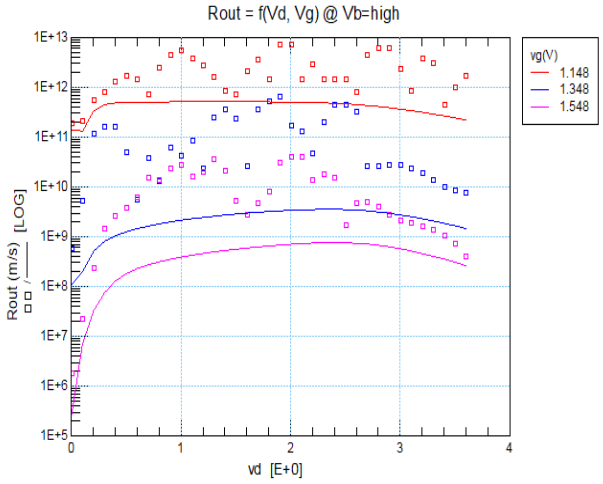


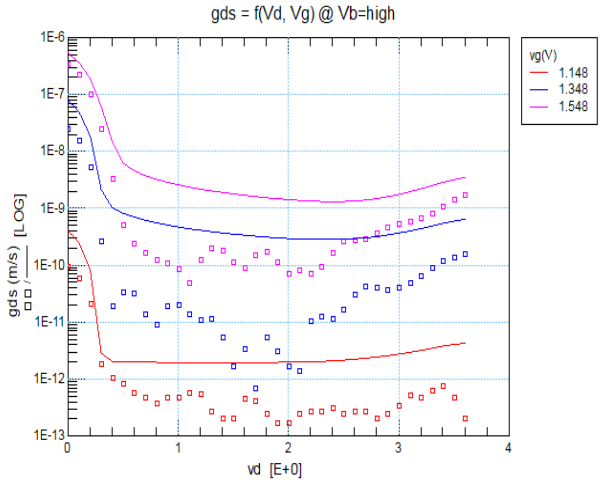




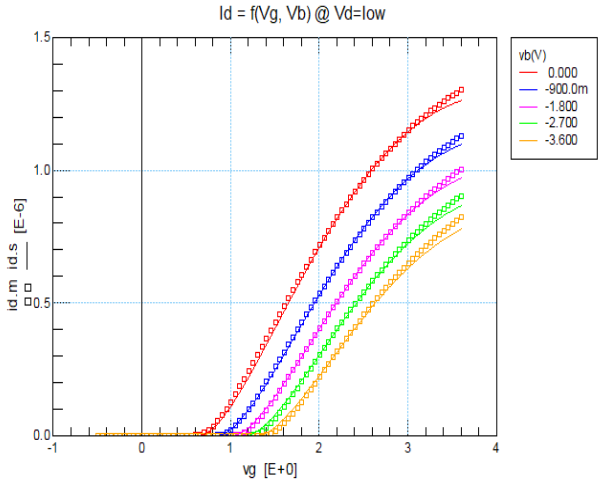
idvd_vth_vbmin, Vb = -3.6V, T = 27°C

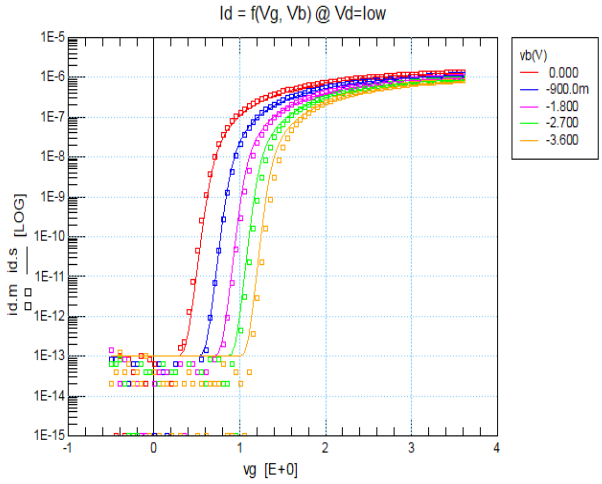


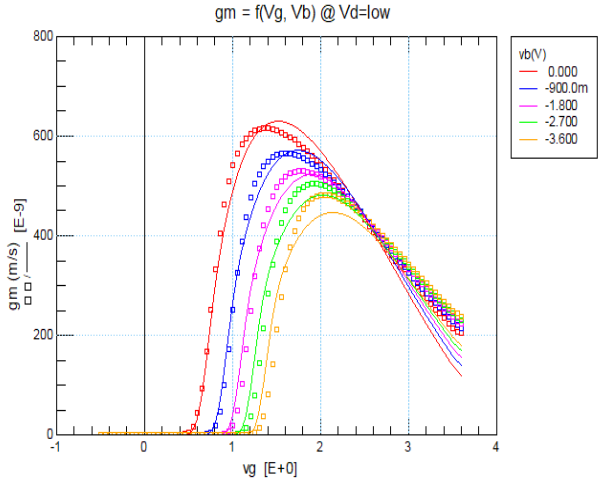


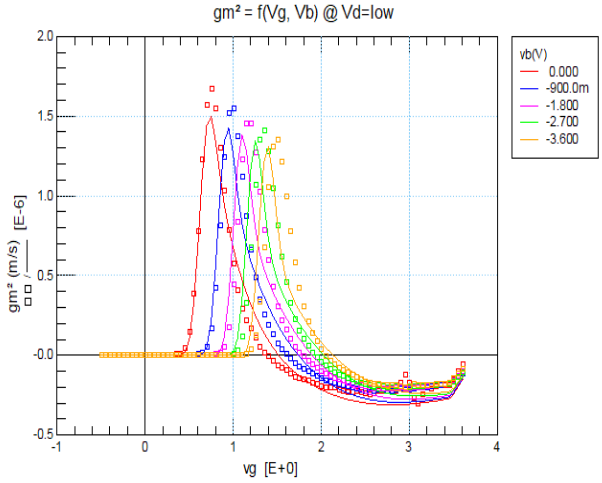


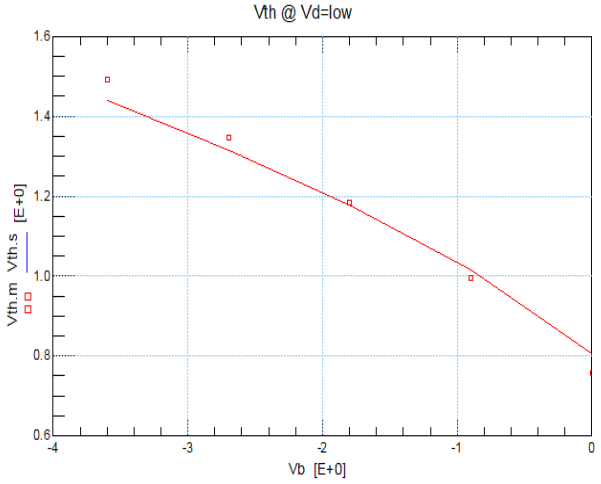
idvg, Vd = 0.1V, T = -40°C

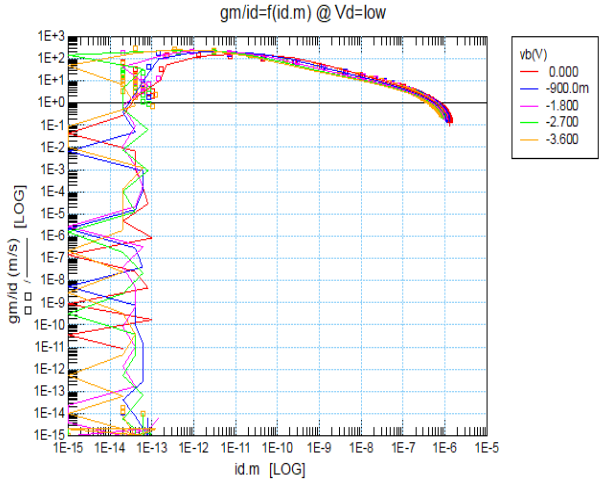


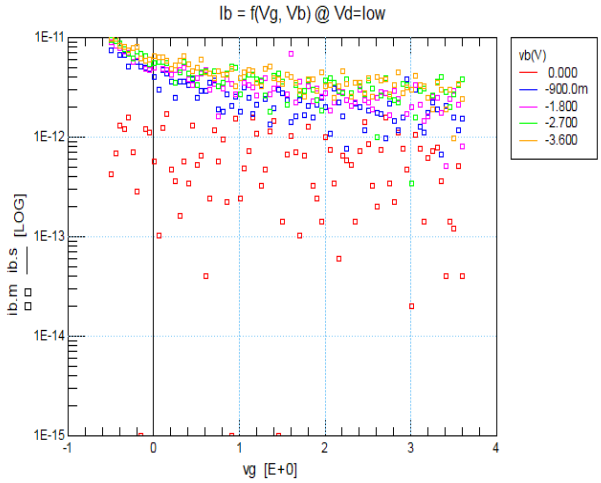


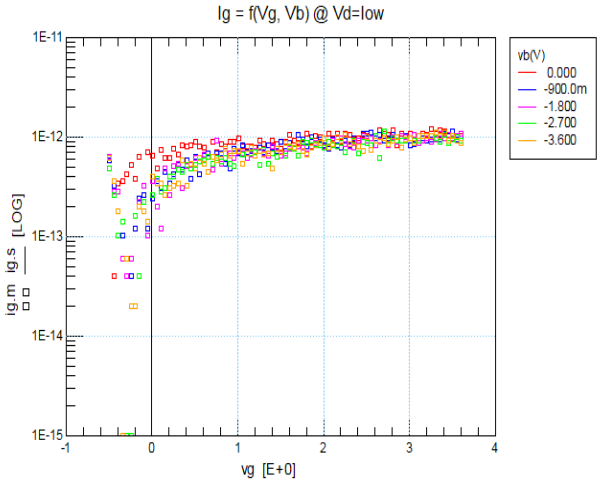




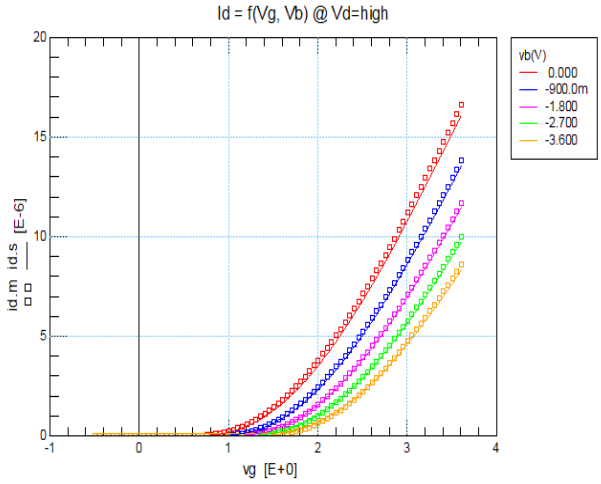


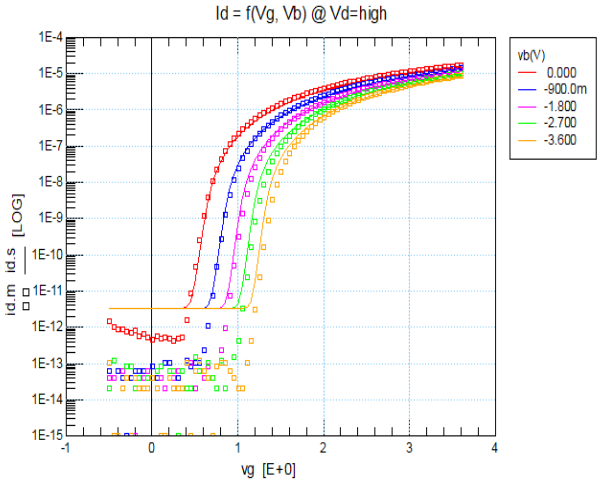


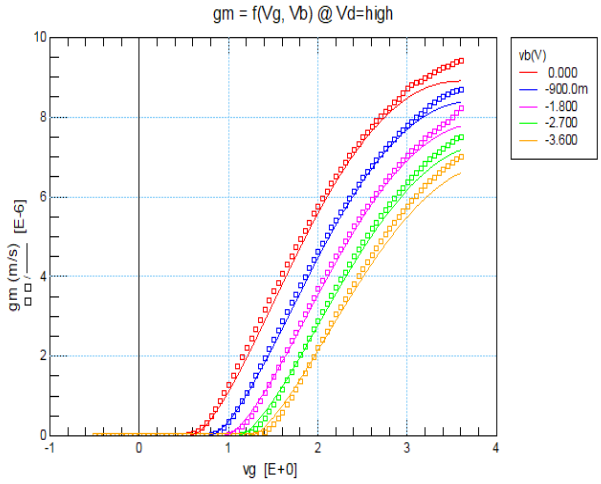


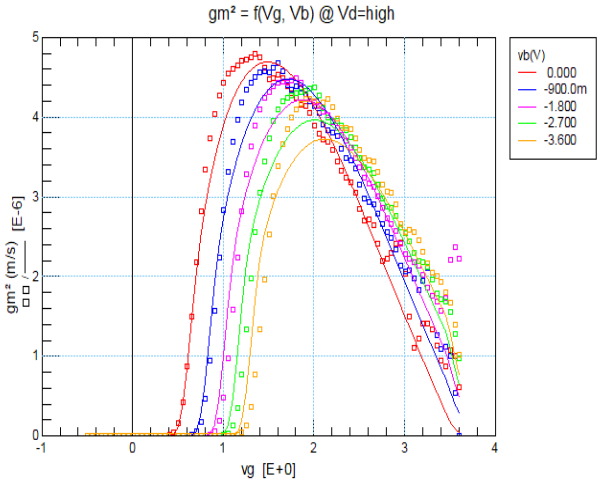


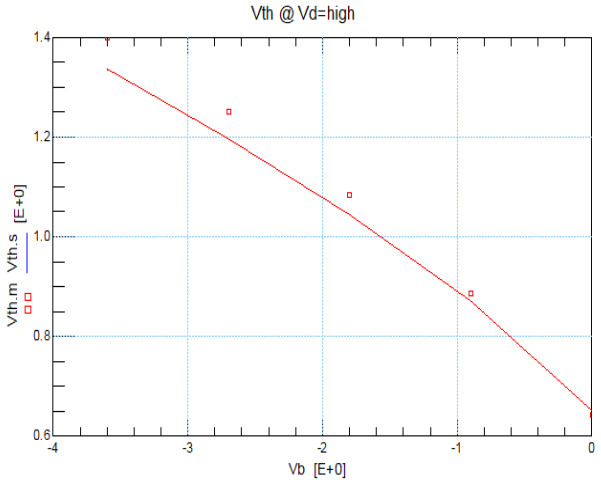
idvg, Vd = 3.3V, T = -40°C

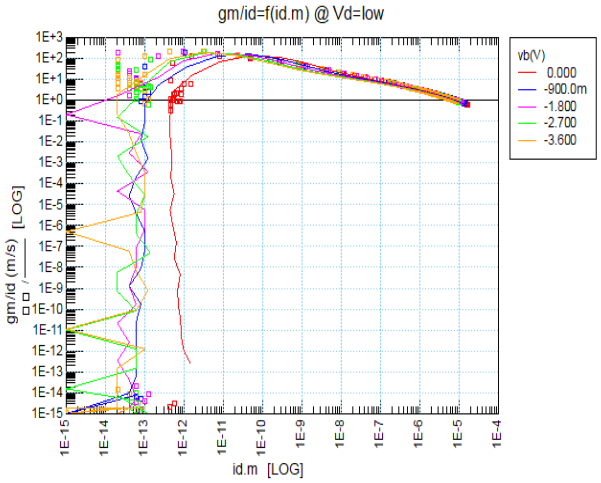


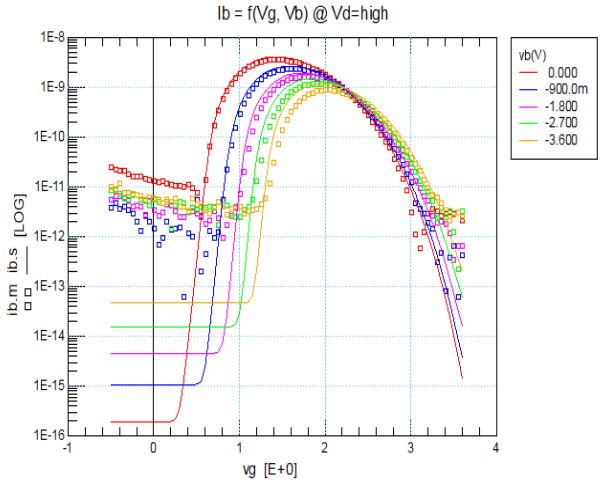


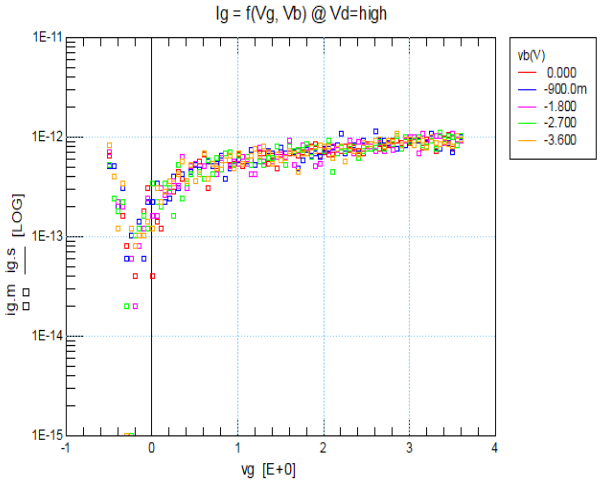




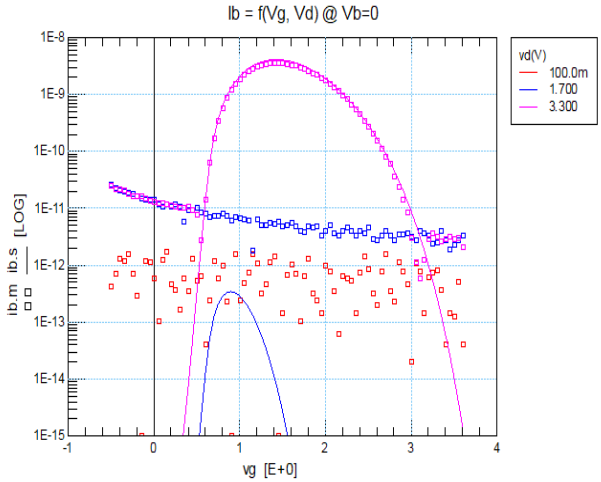


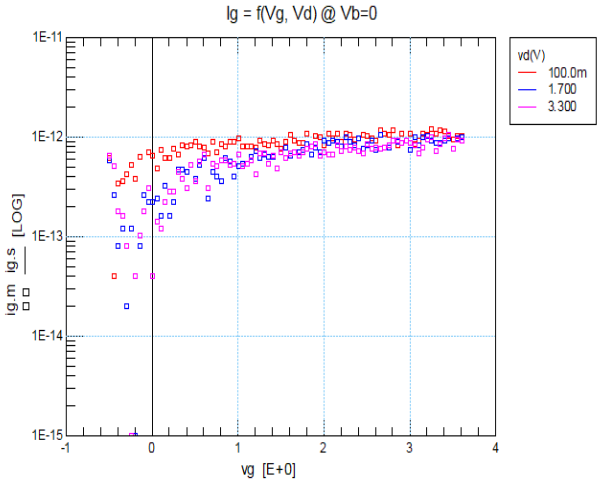




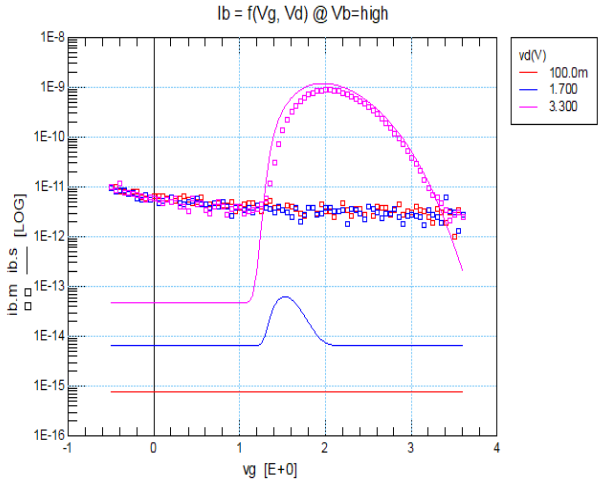


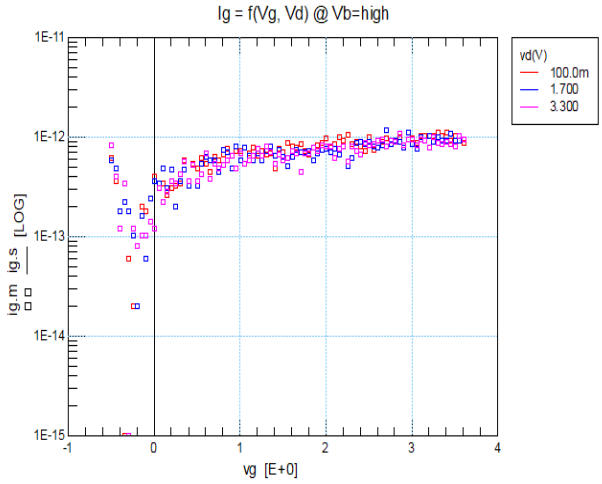
idvg, Ib,Ig, Vb = 0V, T = -40°C



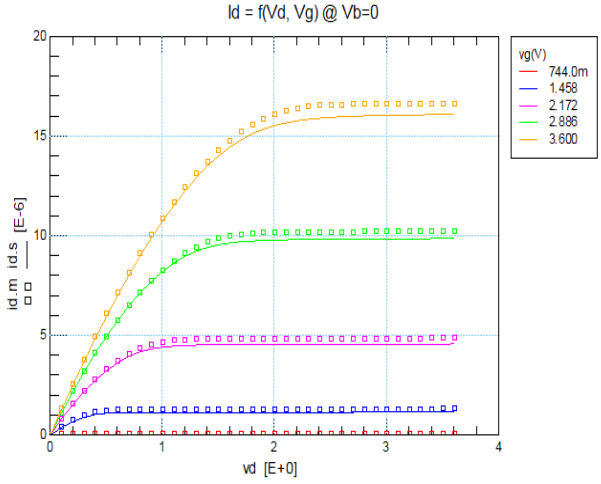


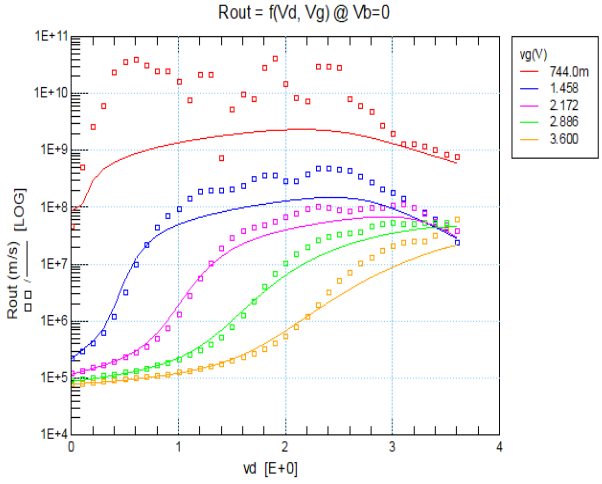
idvg, Ib,Ig, Vb = -3.6V, T = -40°C

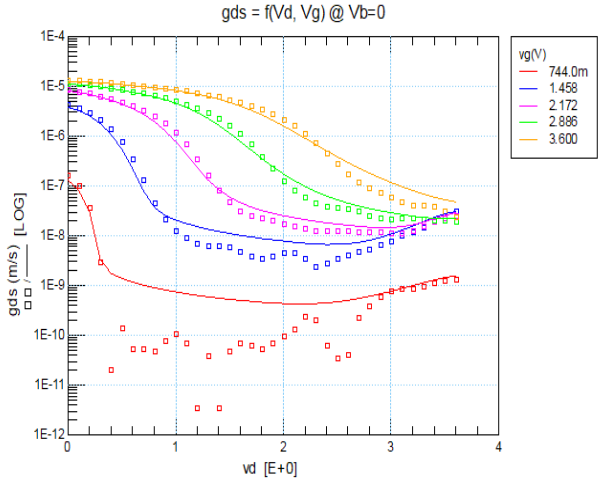




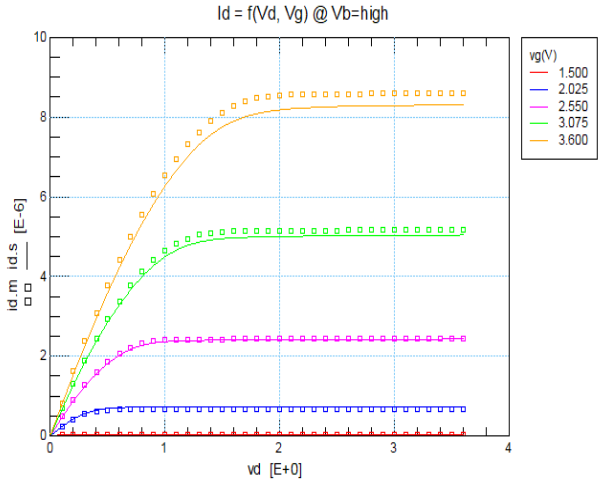
idvd, Vb = 0V, T = -40°C

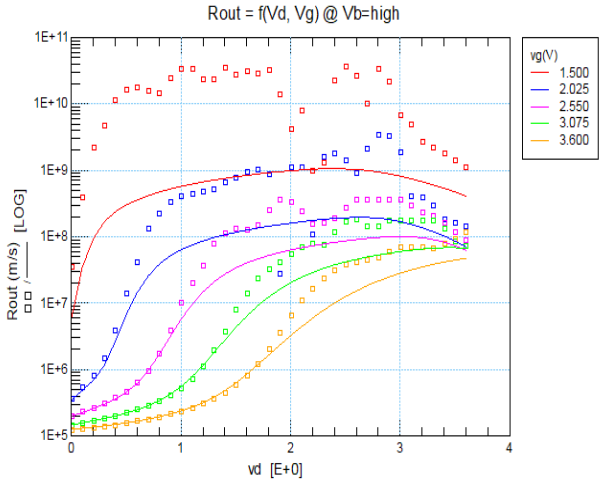


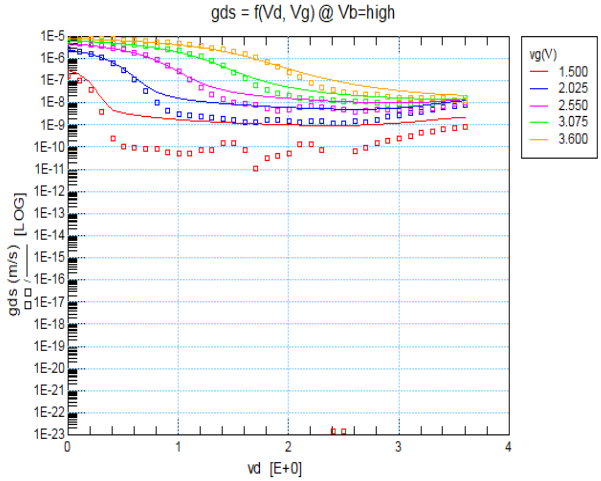




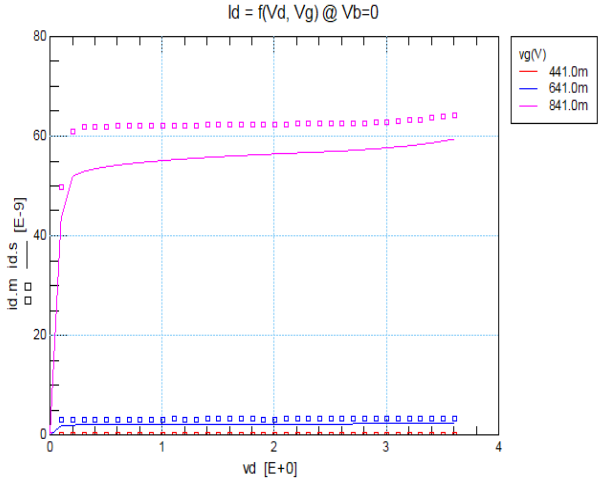
idvd_vbmin, Vb = -3.6V, T = -40°C

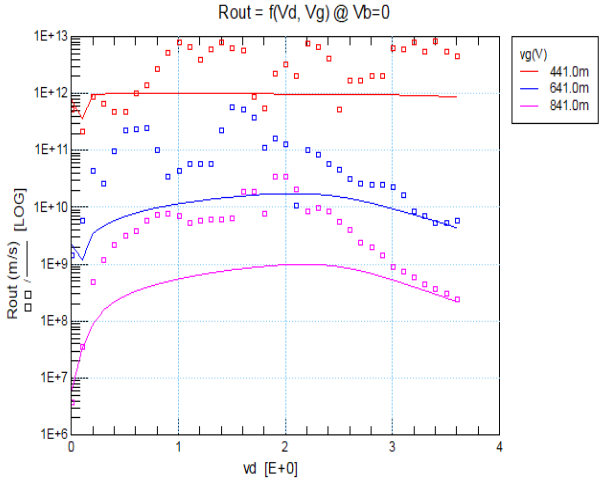


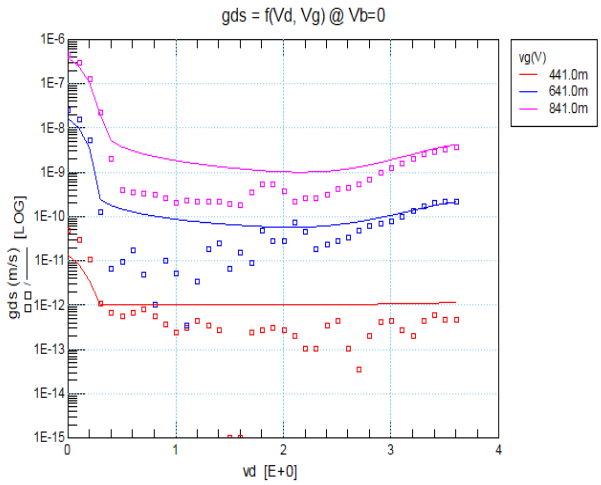




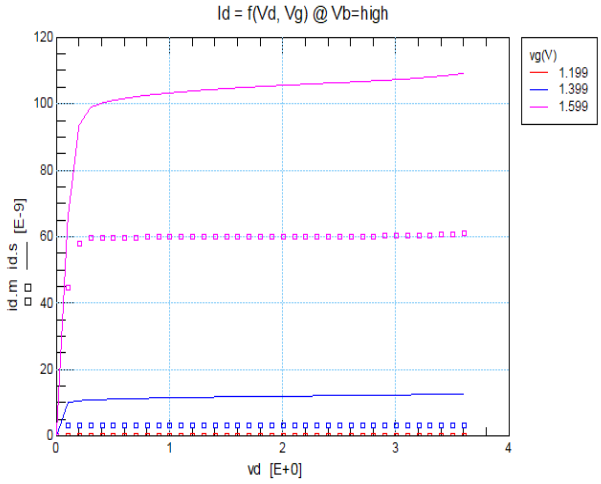
idvd_vth, Vb = 0V, T = -40°C

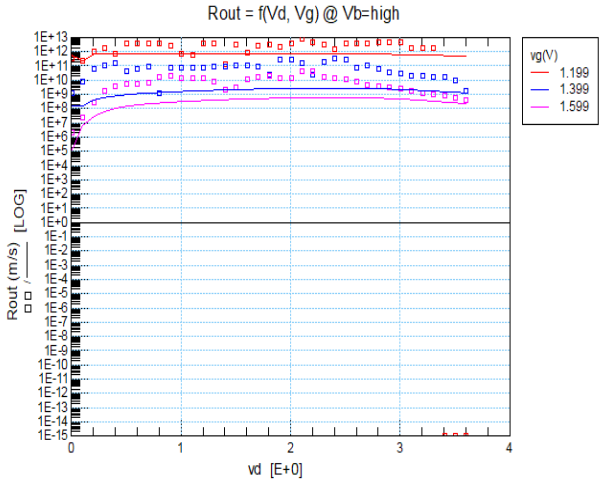


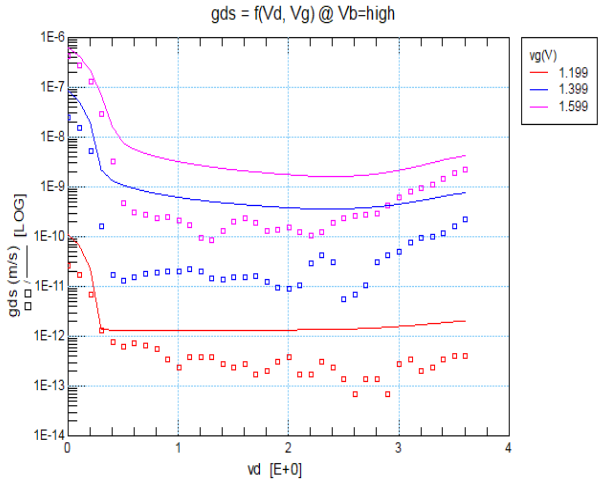




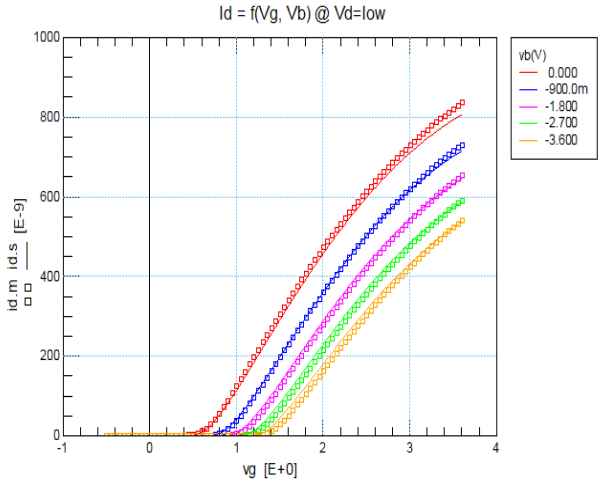
idvd_vth_vbmin, Vb = -3.6V, T = -40°C

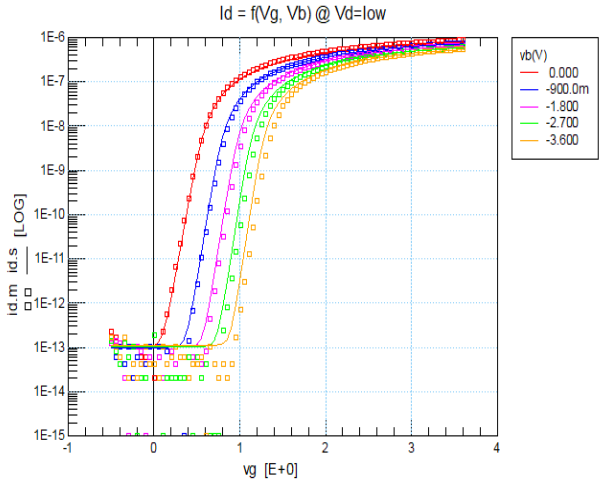


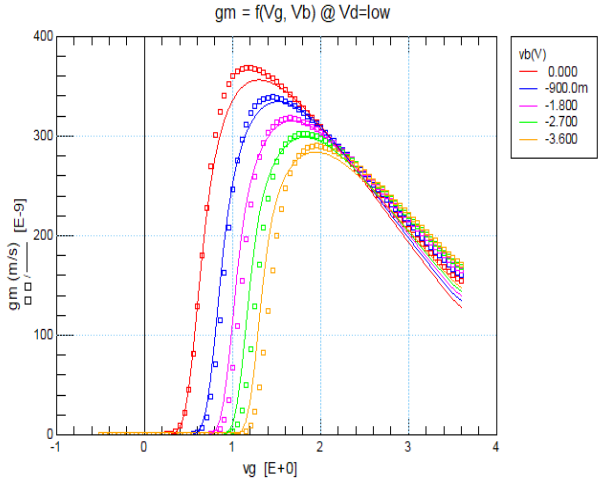


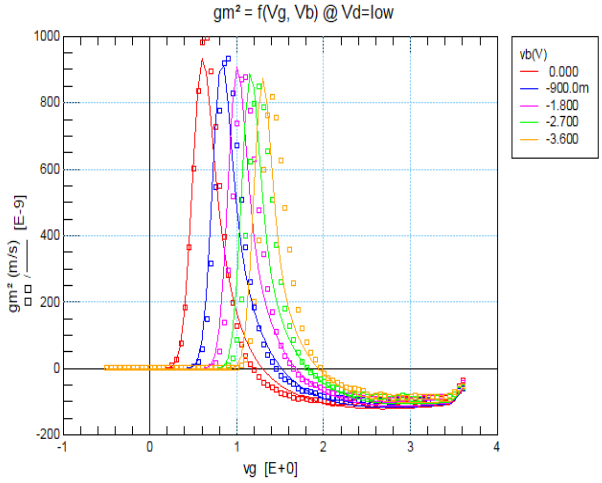


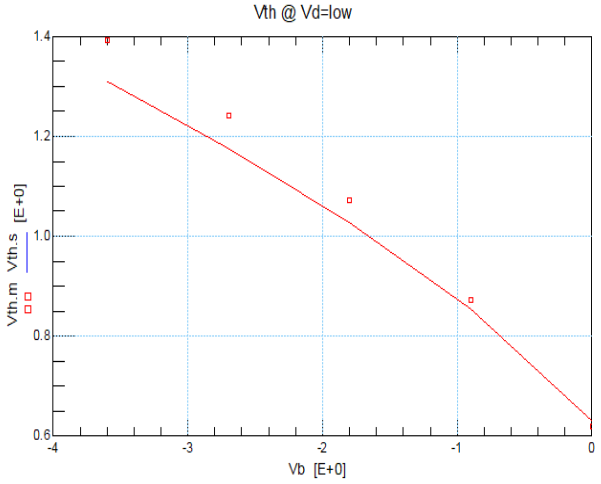
idvg, Vd = 0.1V, T = 70°C

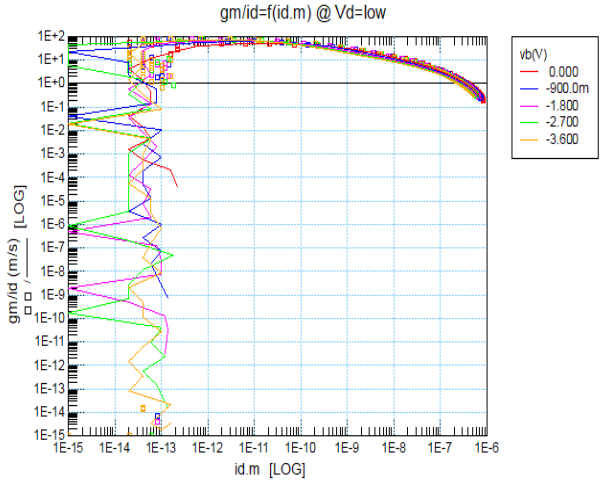


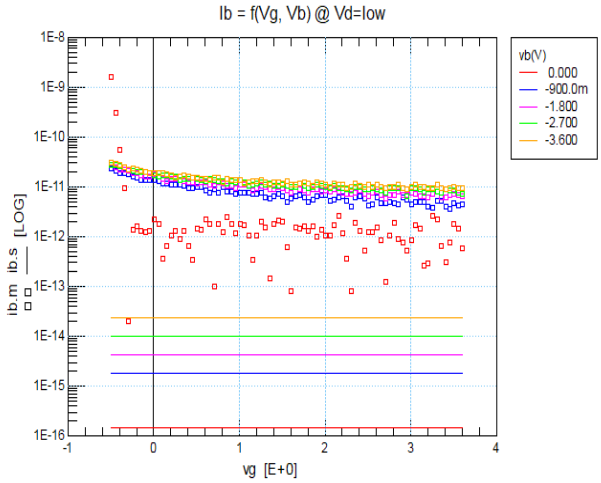


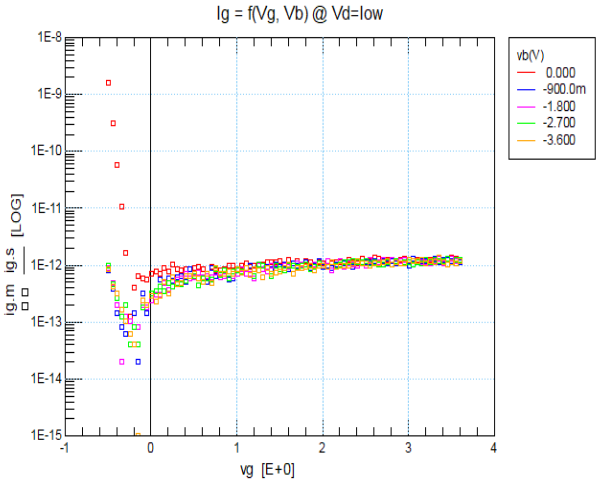




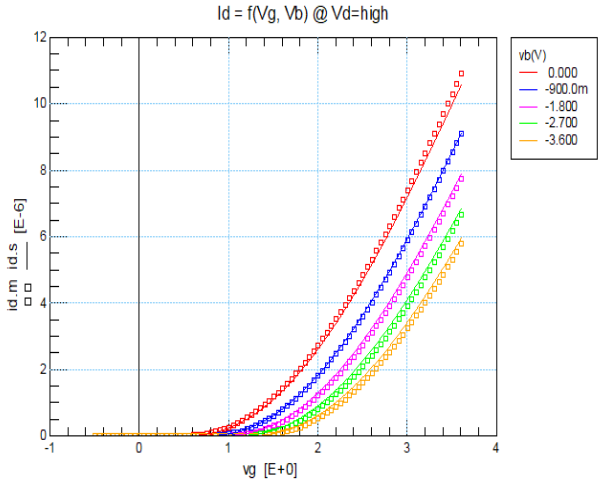


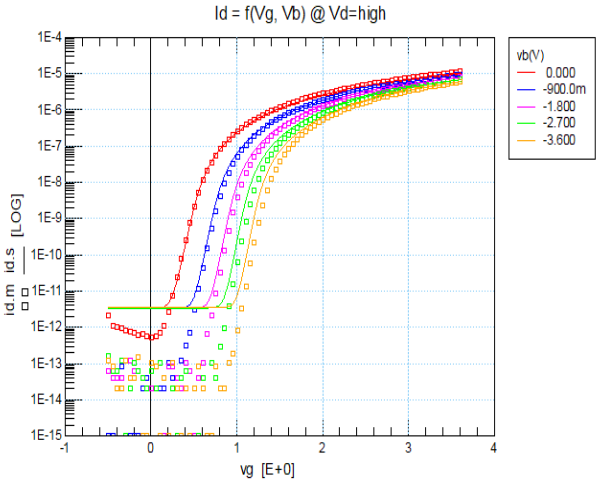


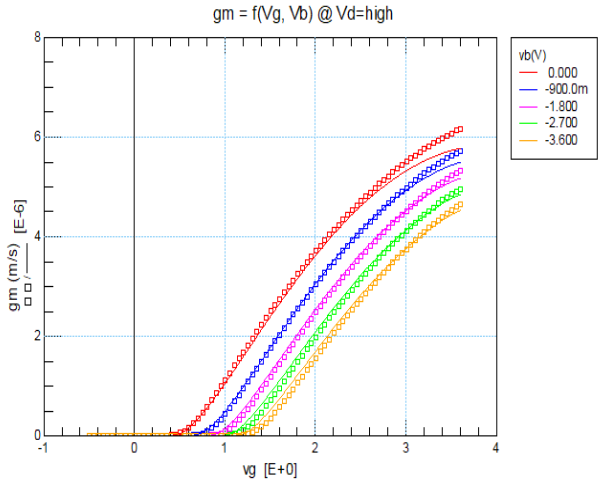


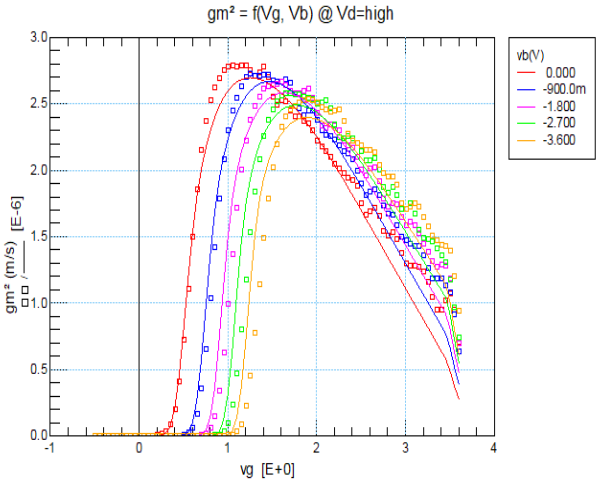


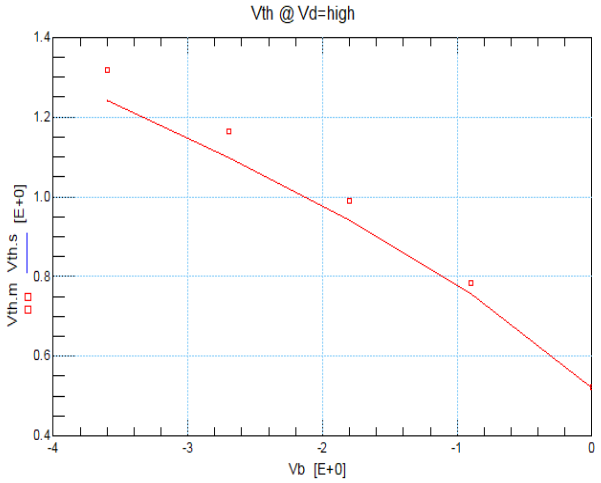
idvg, Vd = 3.3V, T = 70°C

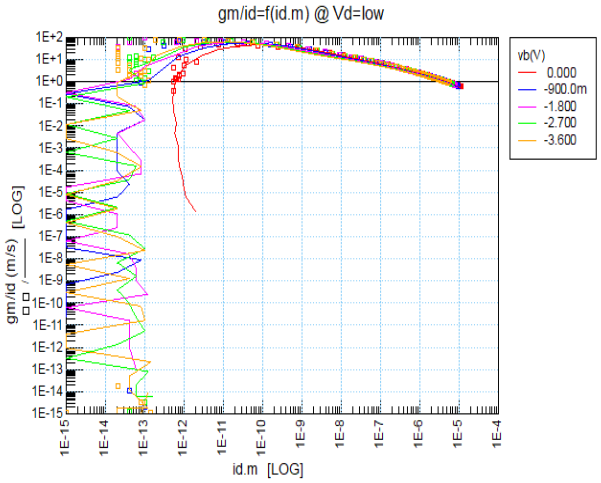


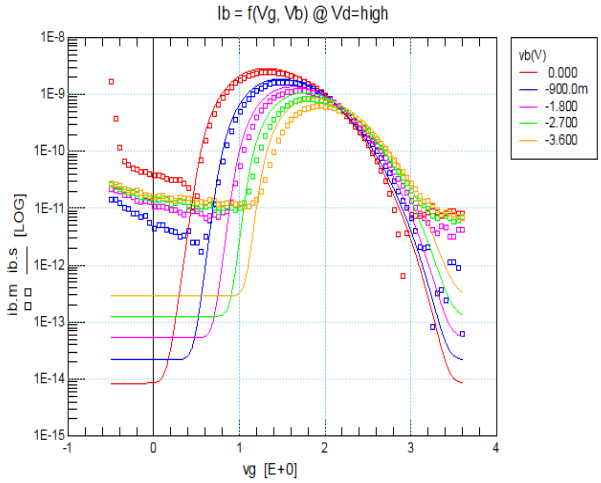


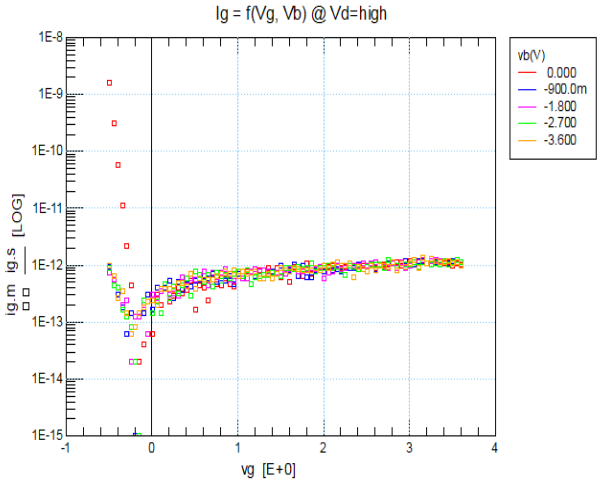




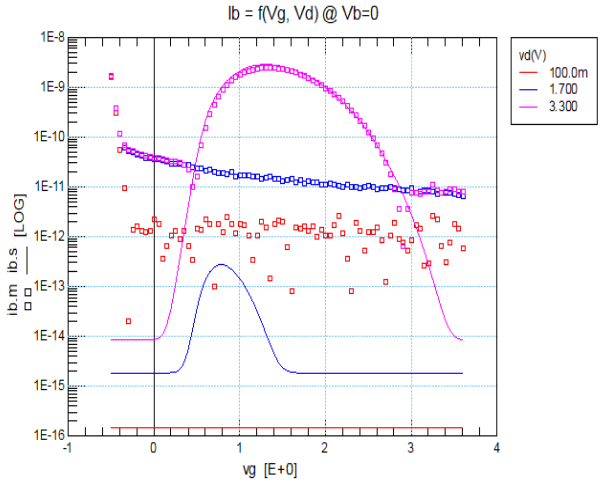


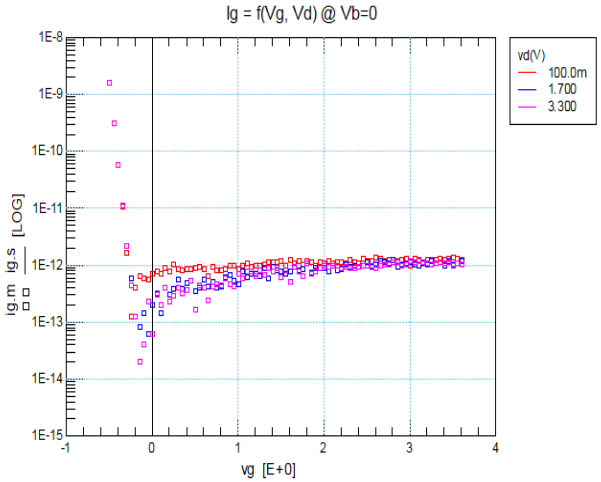




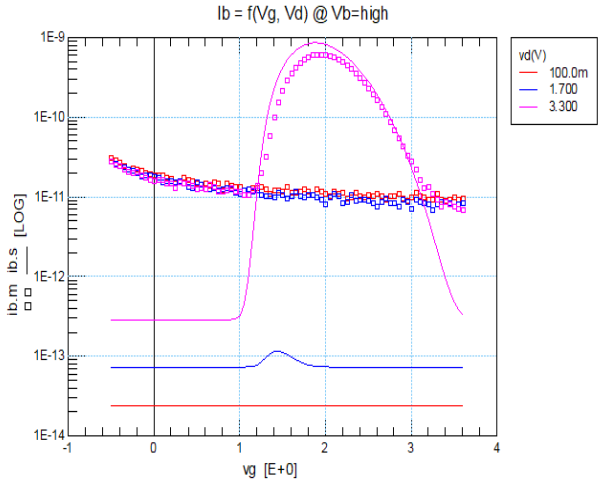


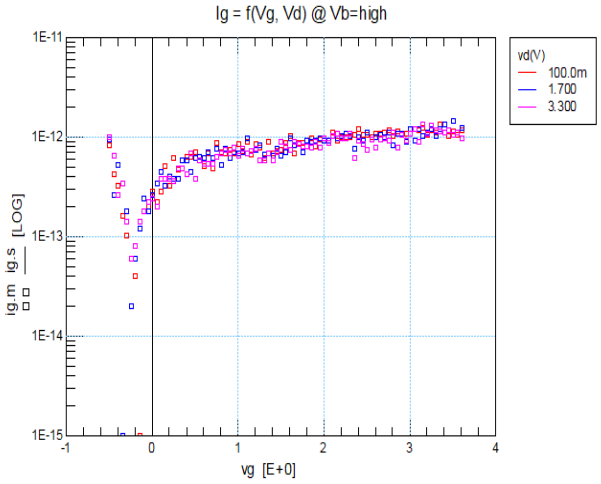
idvg, Ib,Ig, Vb = 0V, T = 70°C



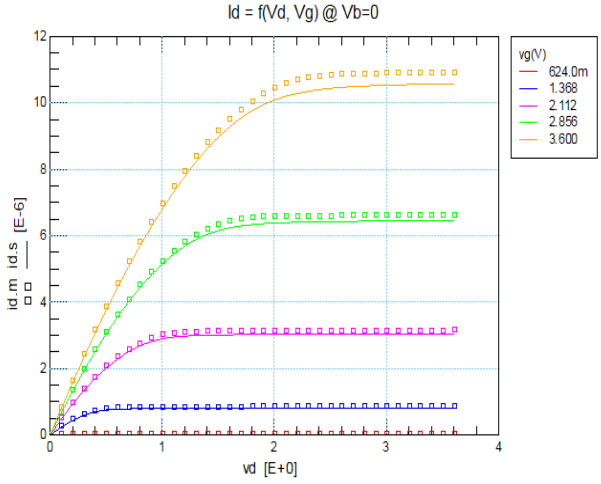


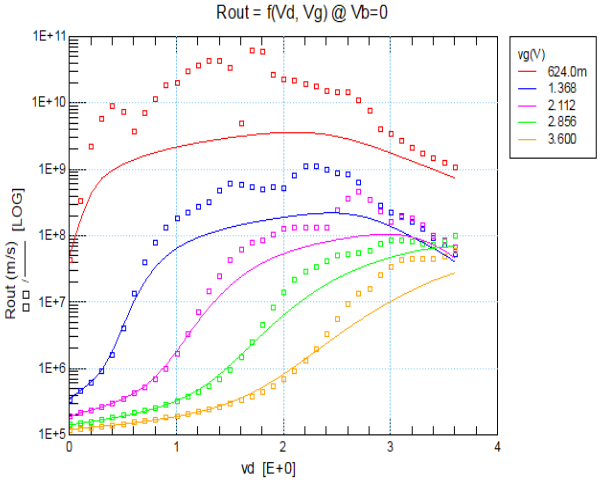
idvg, Ib,Ig, Vb = -3.6V, T = 70°C

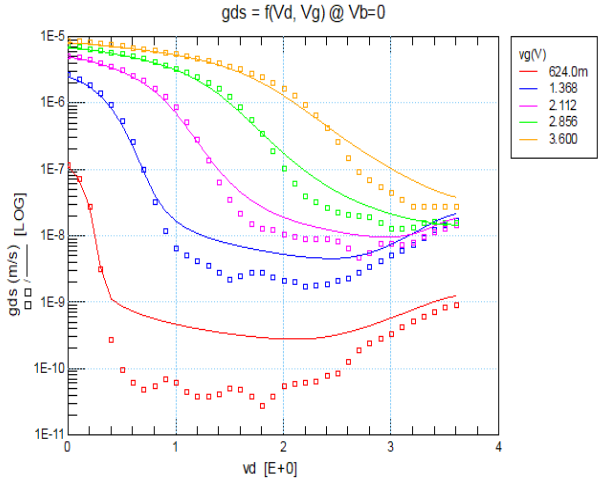




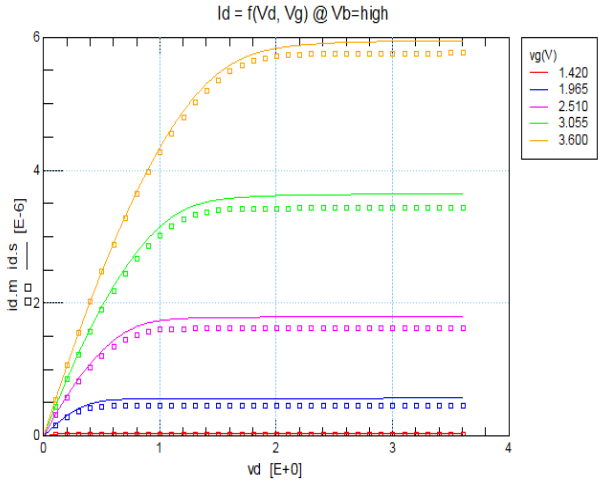
idvd, Vb = 0V, T = 70°C

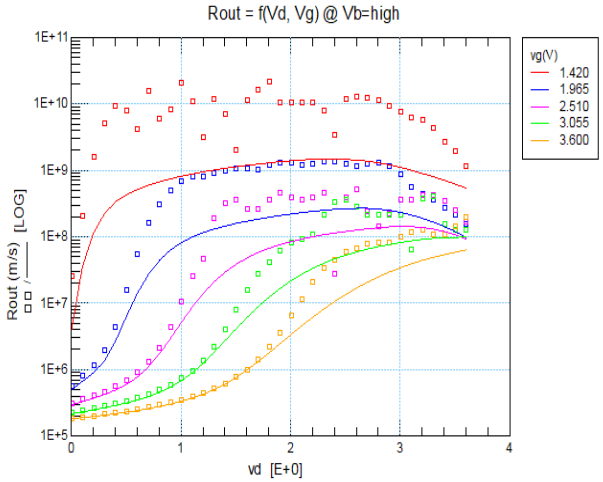


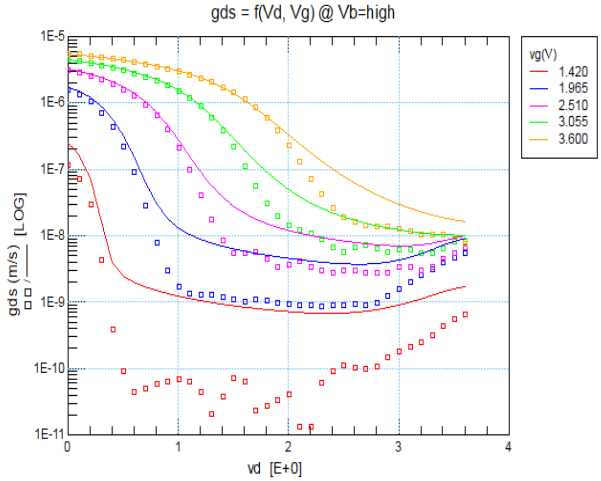




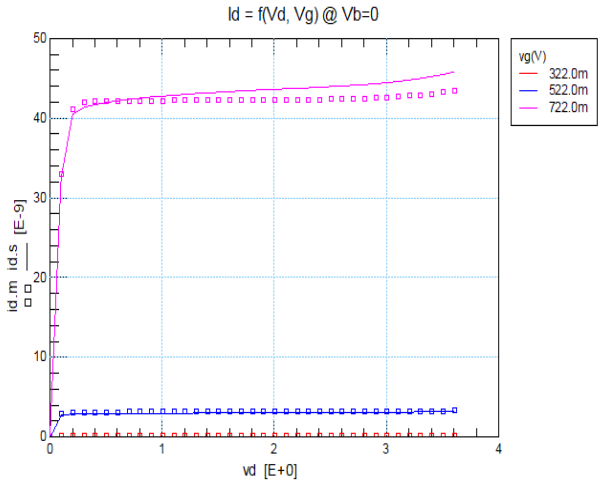
idvd_vbmin, Vb = -3.6V, T = 70°C

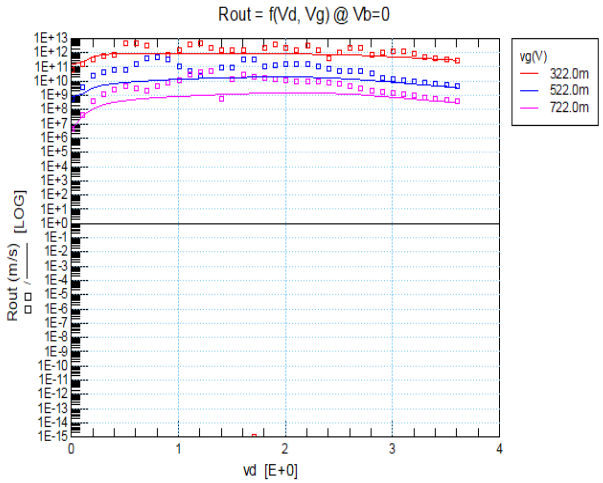


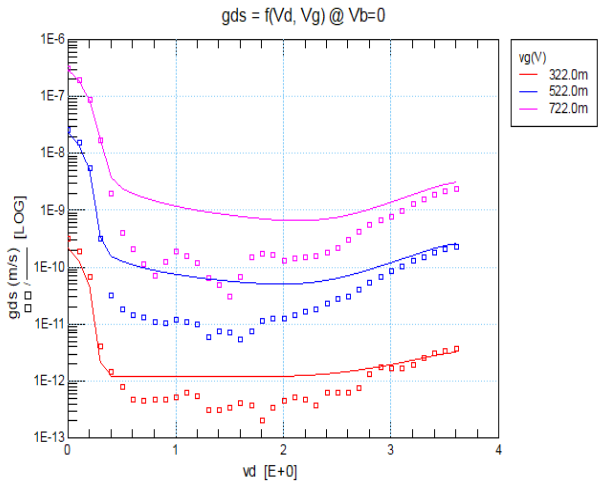




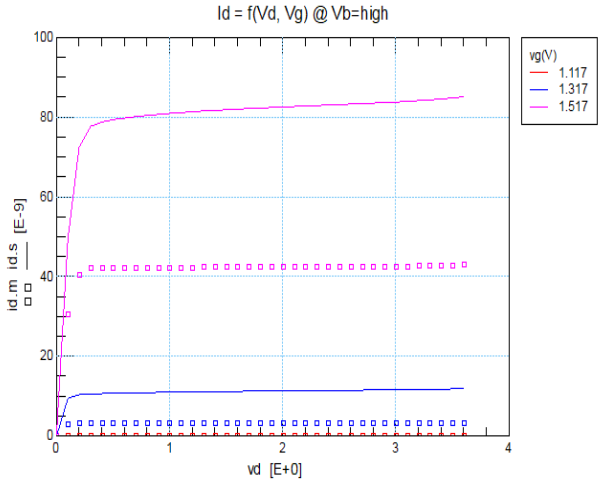
idvd_vth, Vb = 0V, T = 70°C

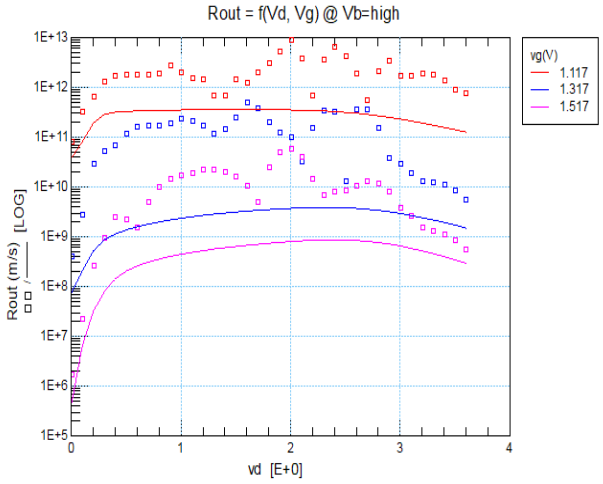


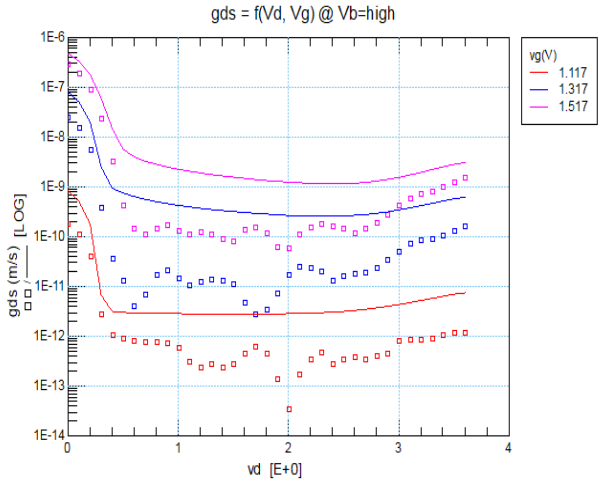




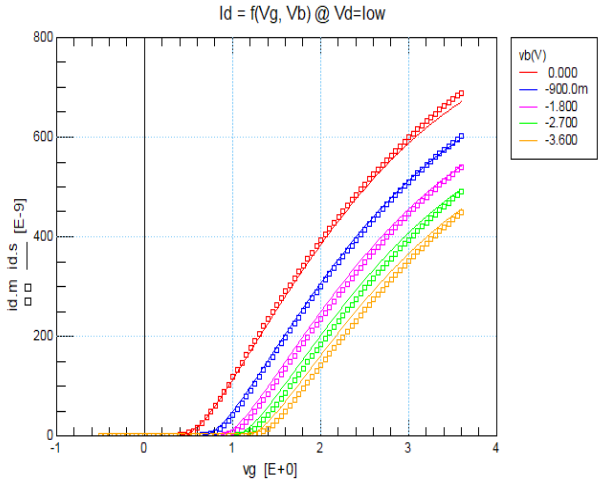
idvd_vth_vbmin, Vb = -3.6V, T = 70°C

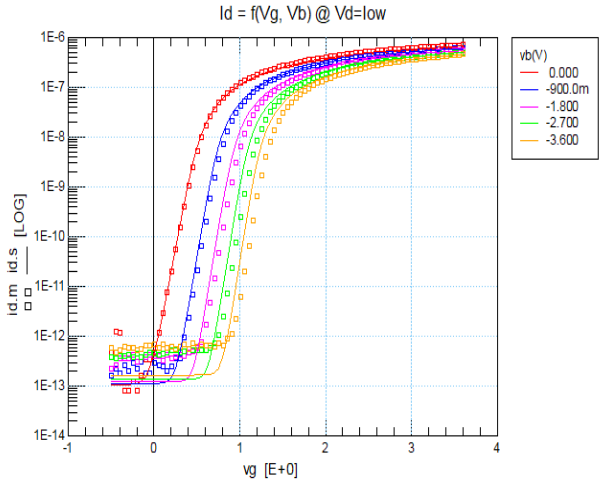


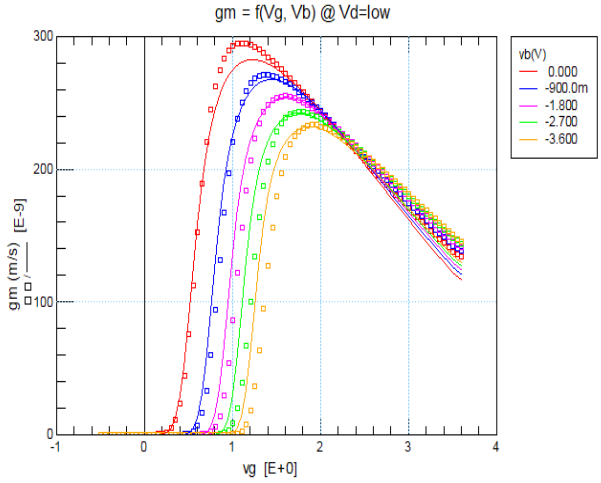


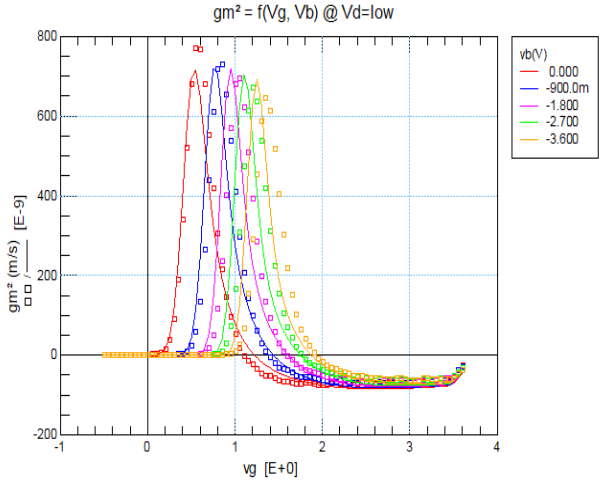


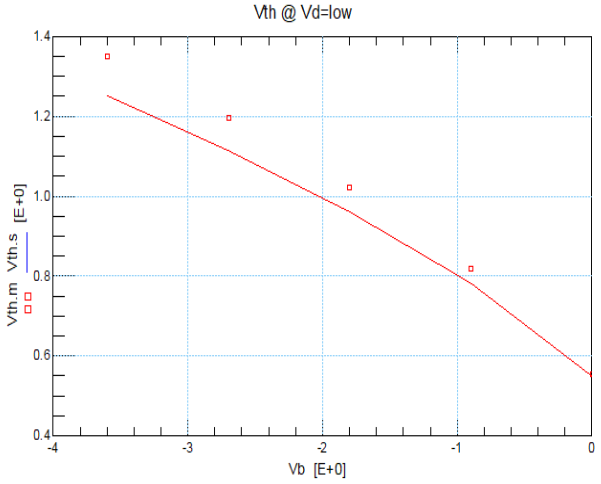
idvg, Vd = 0.1V, T = 125°C

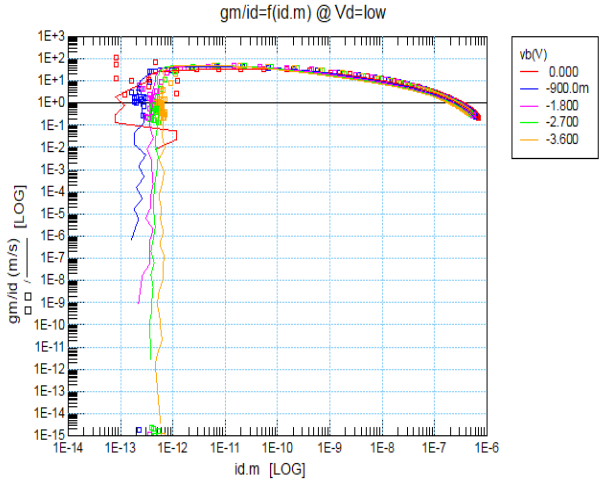


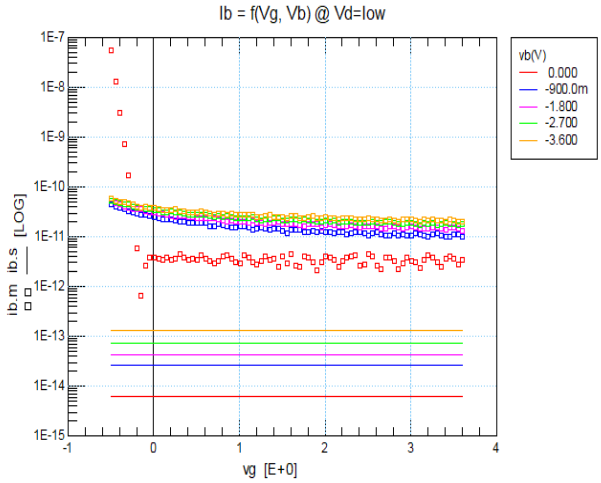


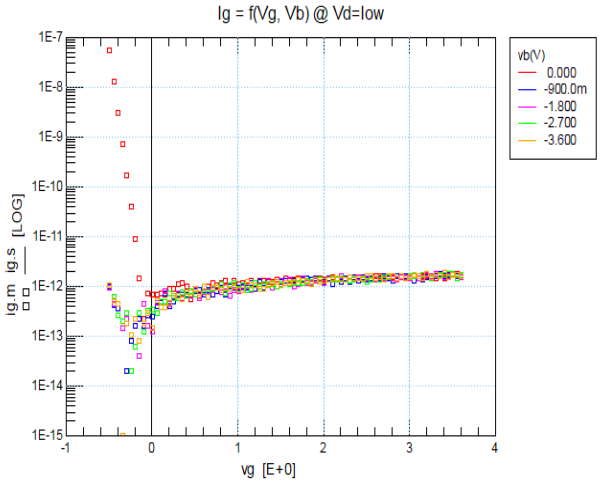




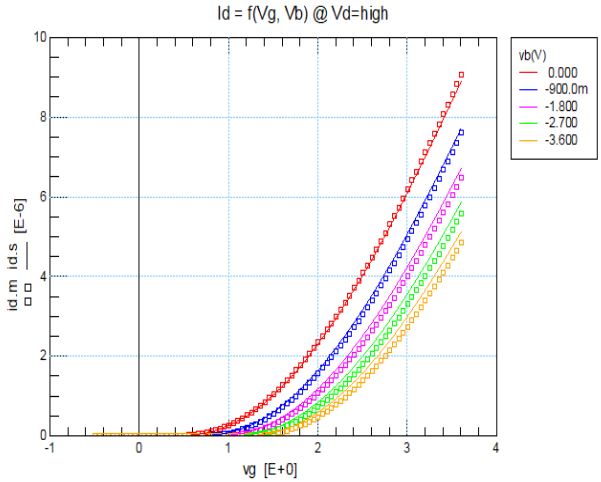


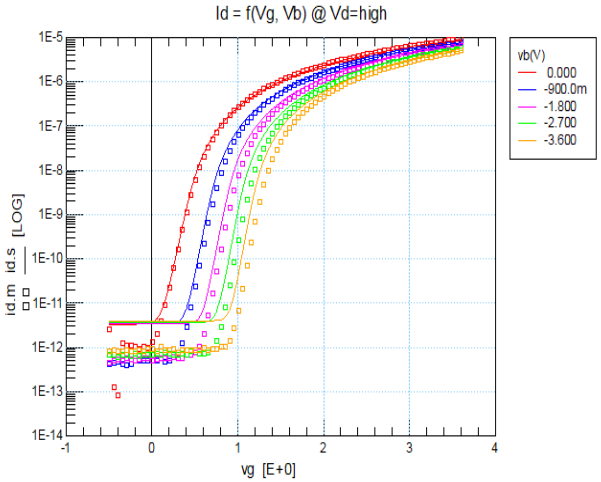


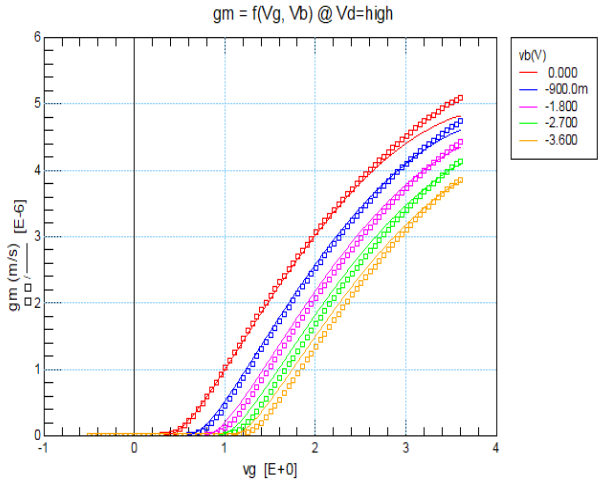


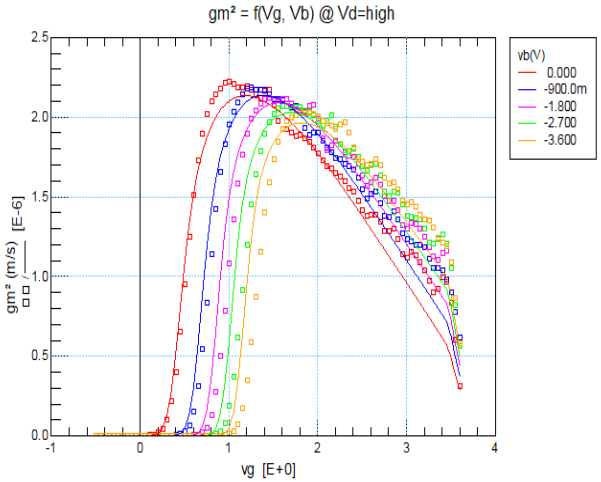


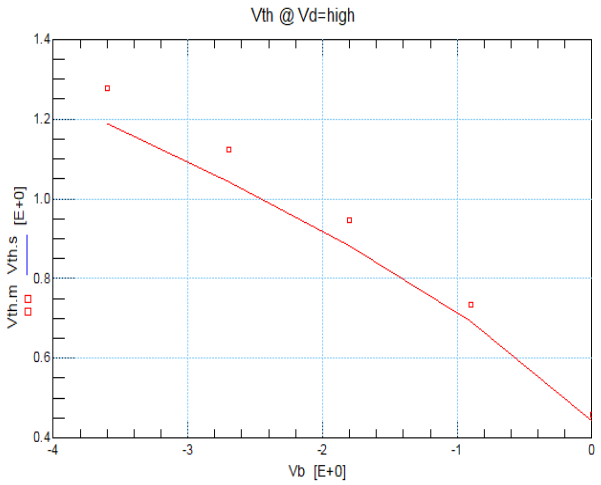
idvg, Vd = 3.3V, T = 125°C

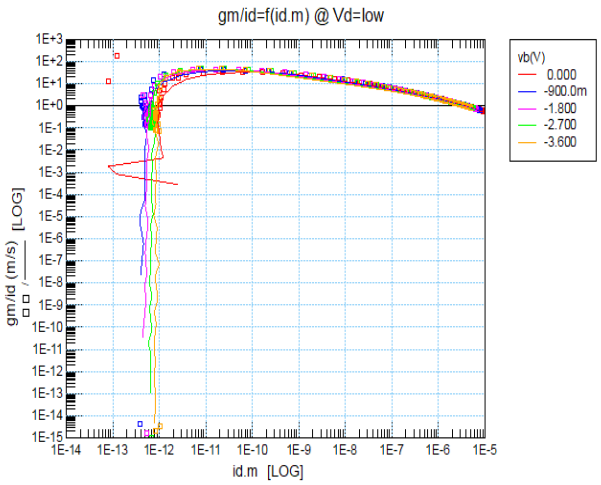


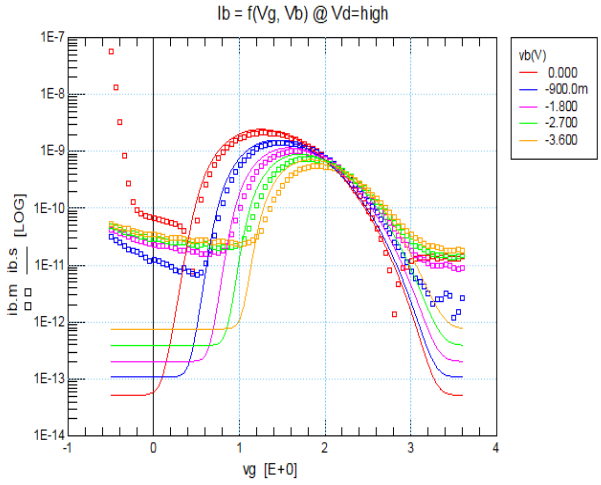


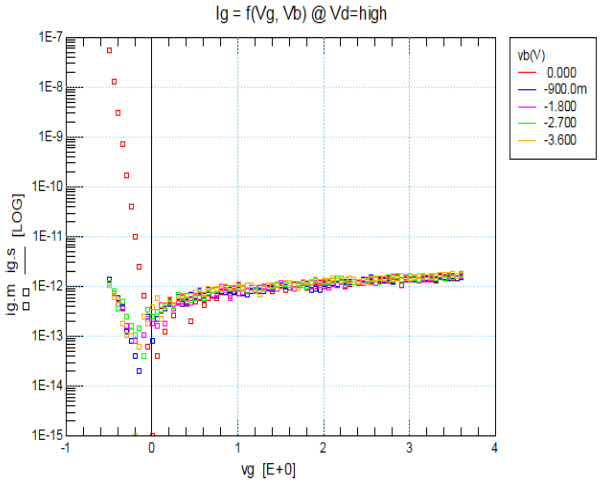




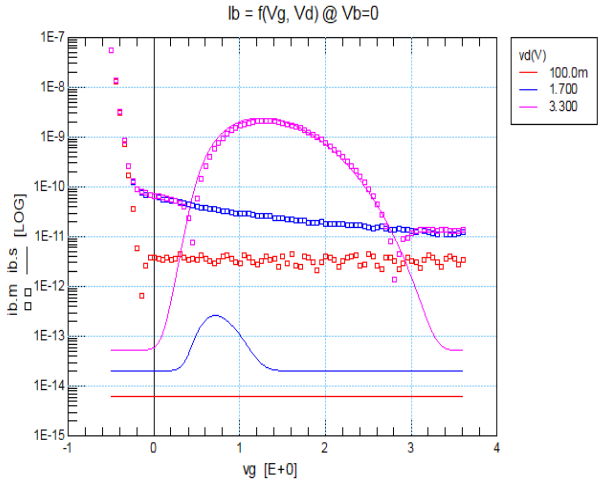


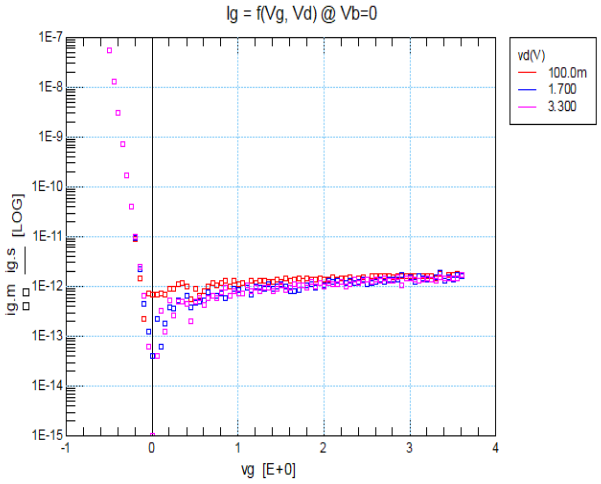




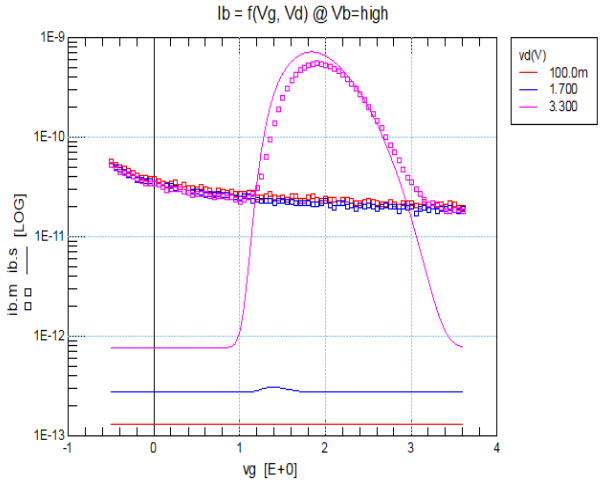


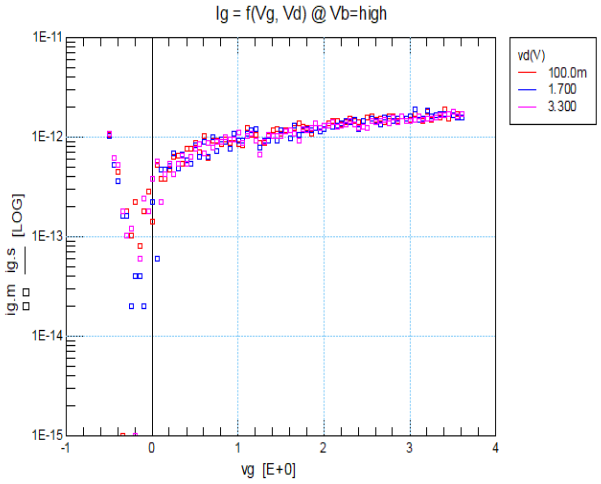
idvg, Ib,Ig, Vb = 0V, T = 125°C



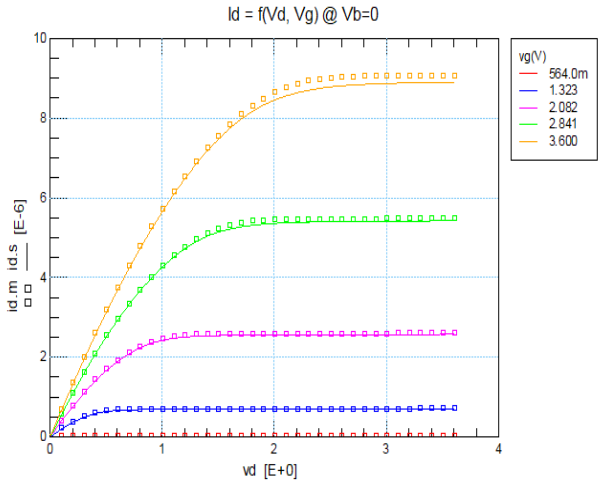


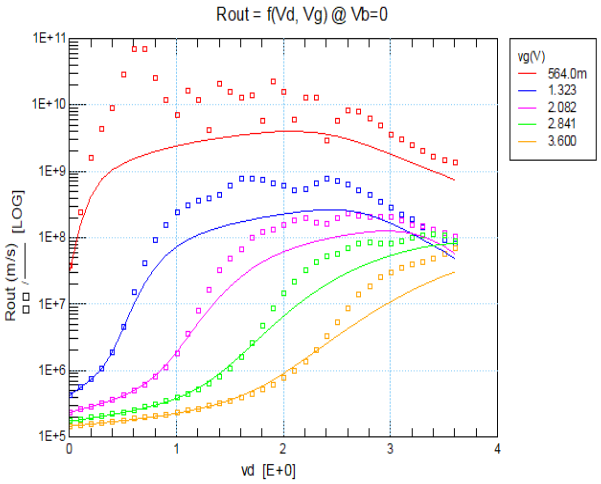
idvg, Ib,Ig, Vb = -3.6V, T = 125°C

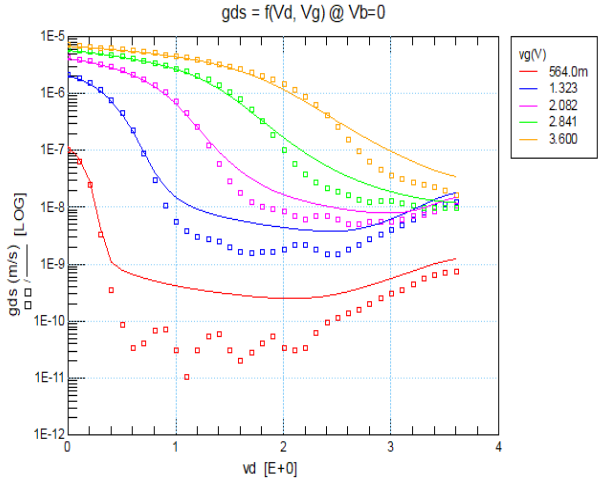




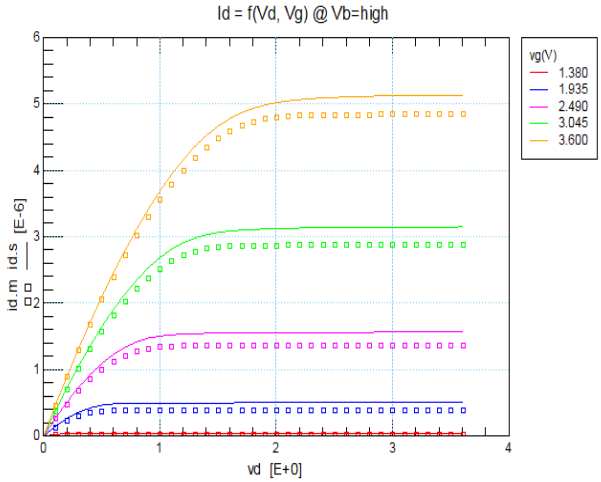
idvd, Vb = 0V, T = 125°C

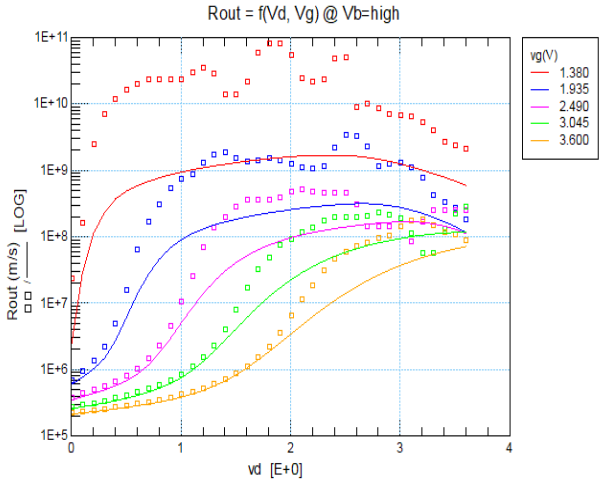


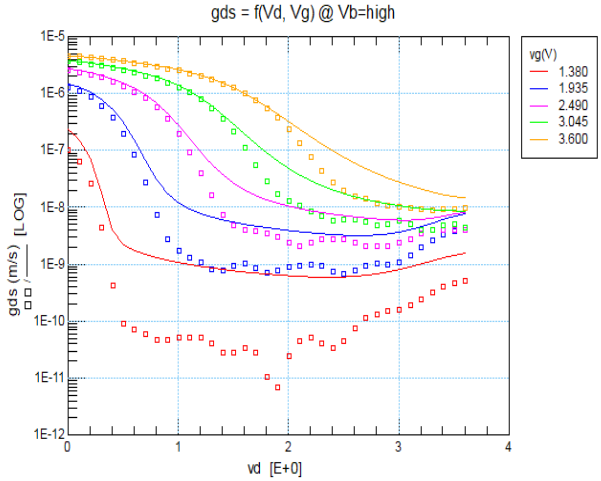




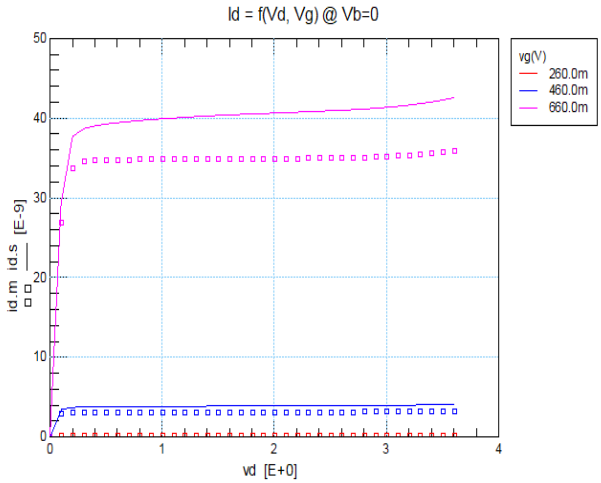
idvd_vbmin, Vb = -3.6V, T = 125°C

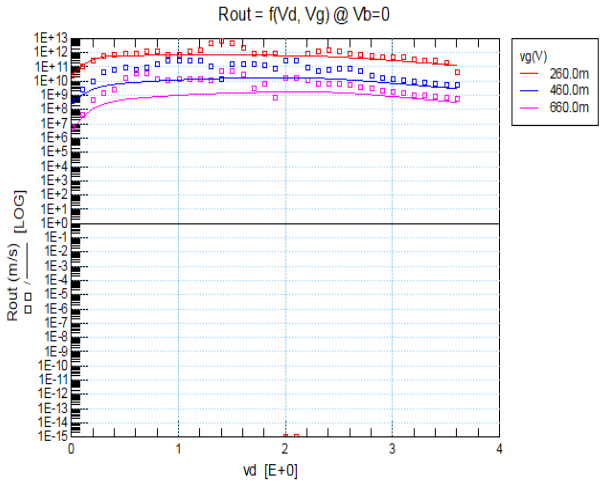


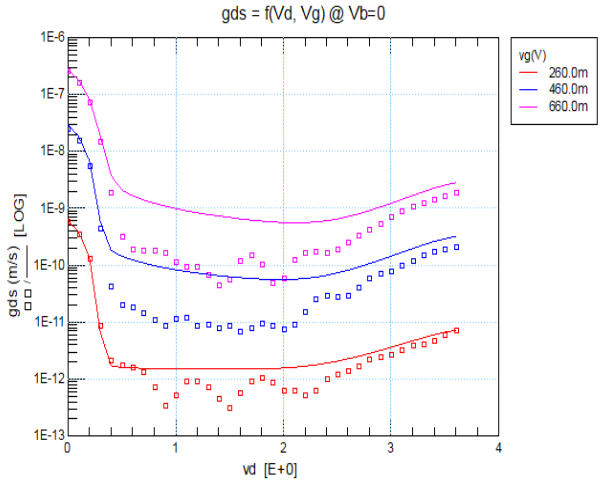




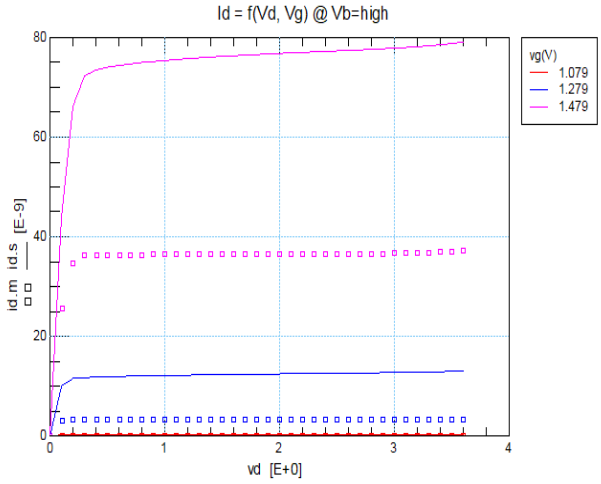
idvd_vth, Vb = 0V, T = 125°C

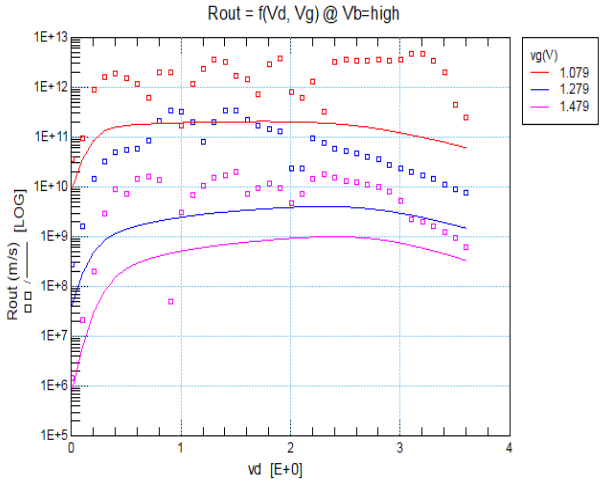


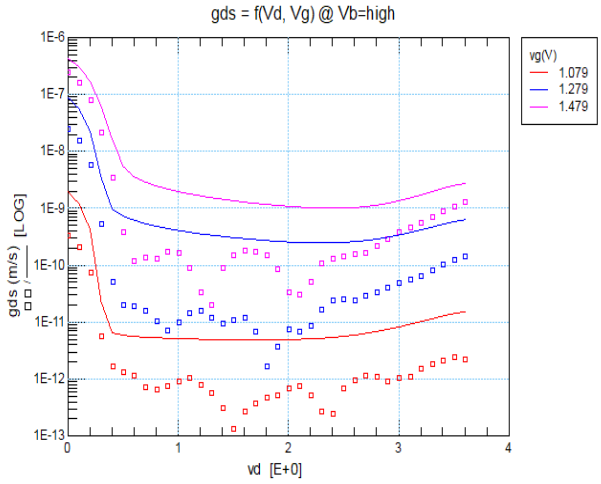




idvd_vth_vbmin, Vb = -3.6V, T = 125°C



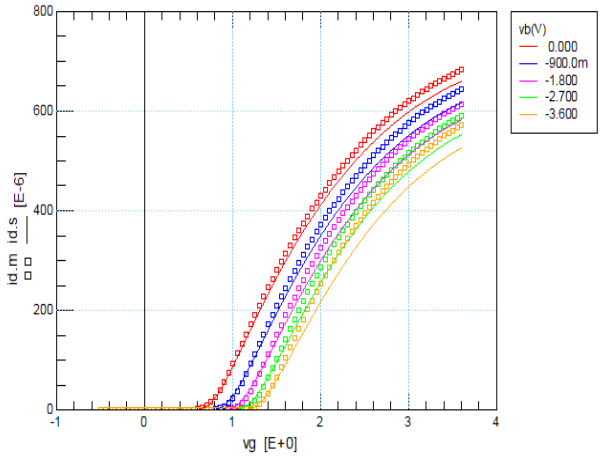




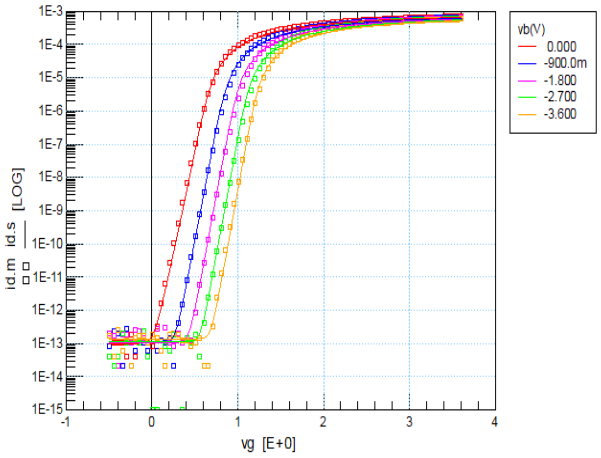
4.4 W10u0_L0u5_S387_5

idvg, Vd = 0.1V, T = 27°C

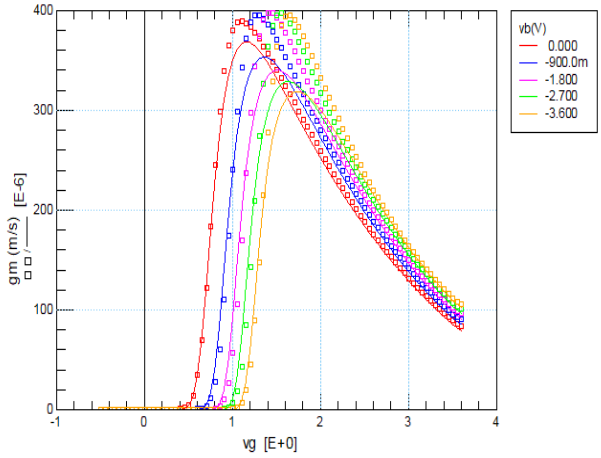
Id = f(Vg, Vb) @ Vd=low



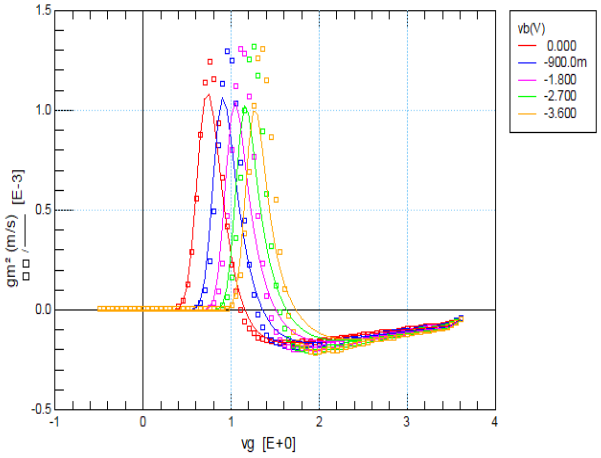
Id = f(Vg, Vb) @ Vd=low



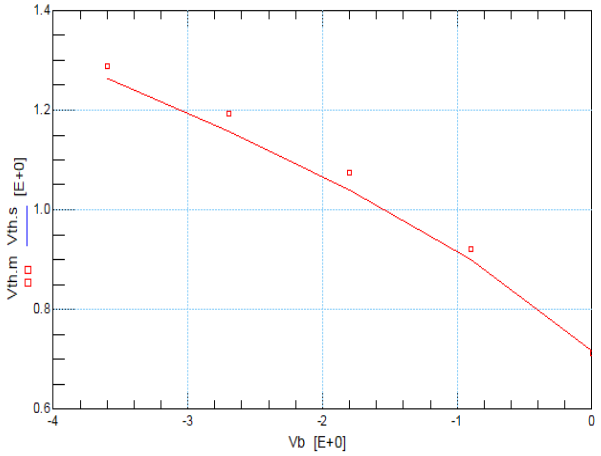
gm = f(Vg, Vb) @ Vd=low



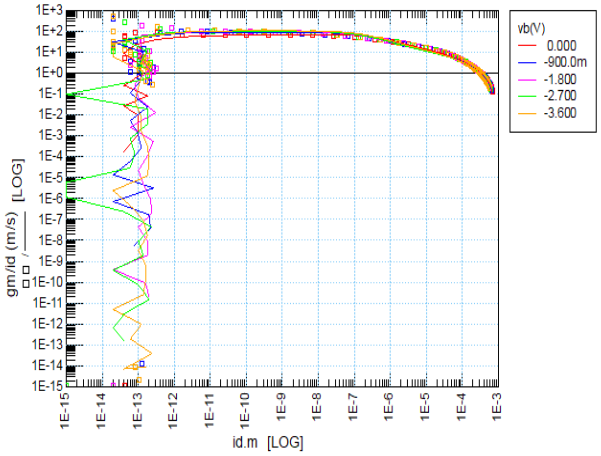
gm² = f(Vg, Vb) @ Vd=low

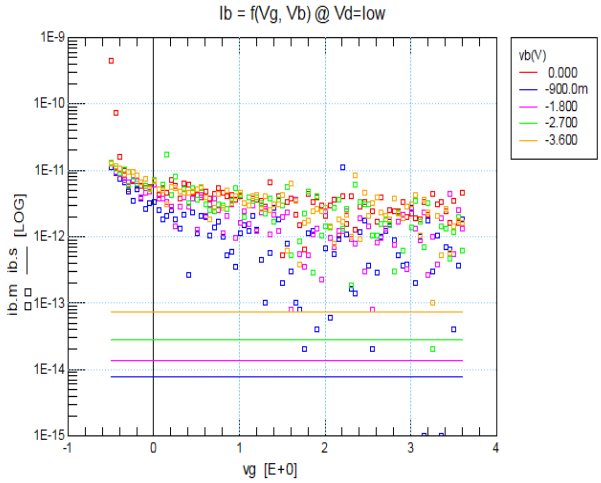


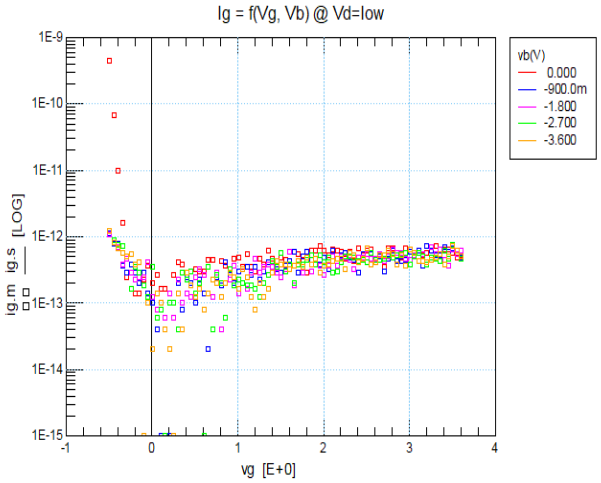
Vth @ Vd=low



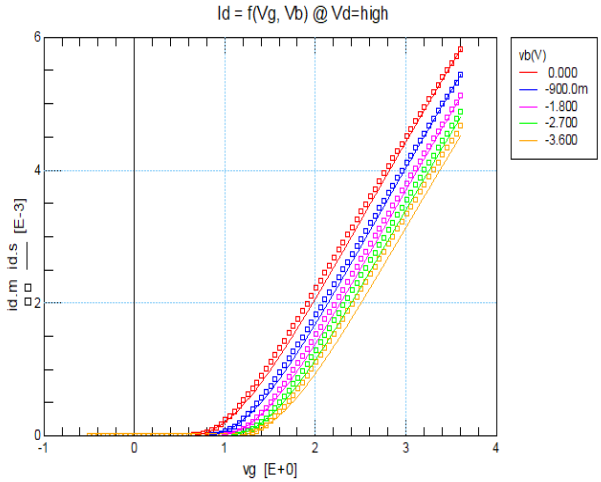
gm/id=f(id.m) @ Vd=low

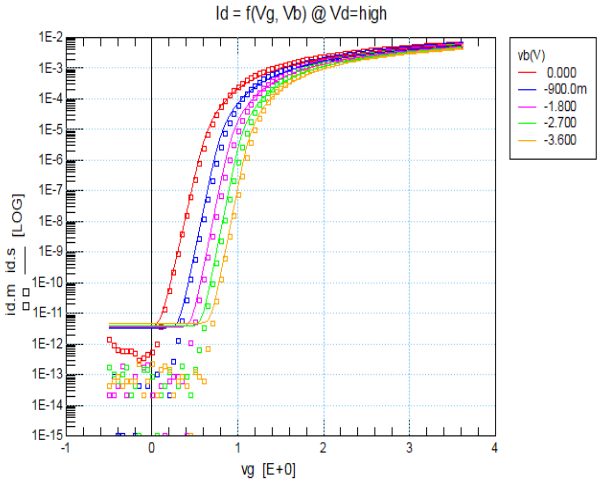


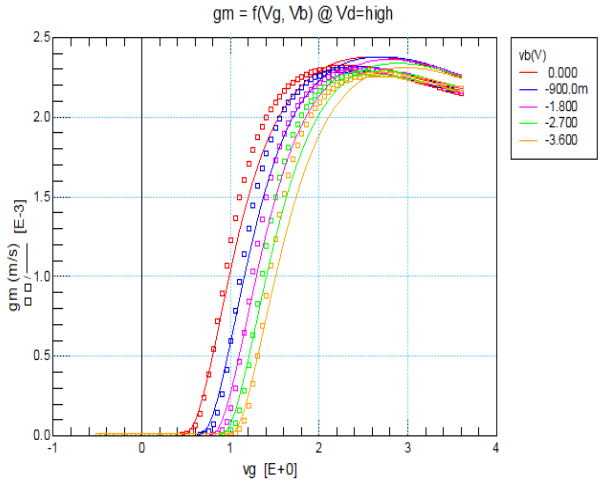


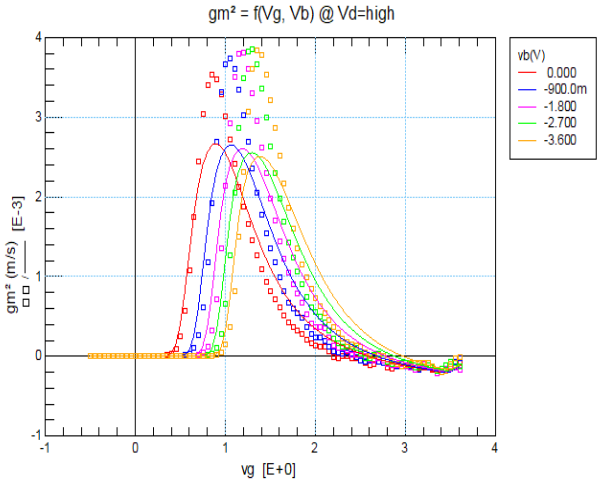


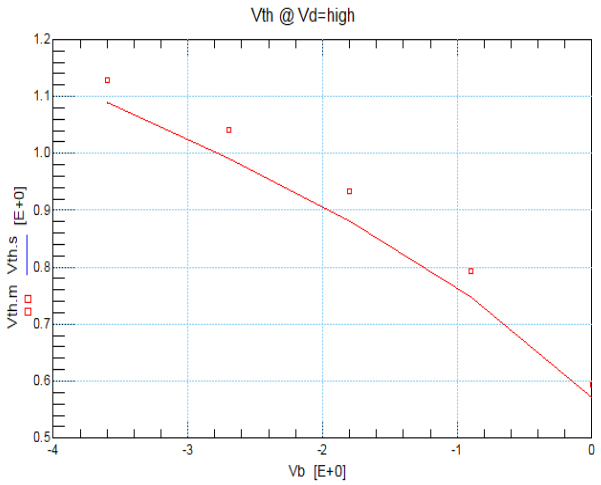
idvg, Vd = 3.3V, T = 27°C

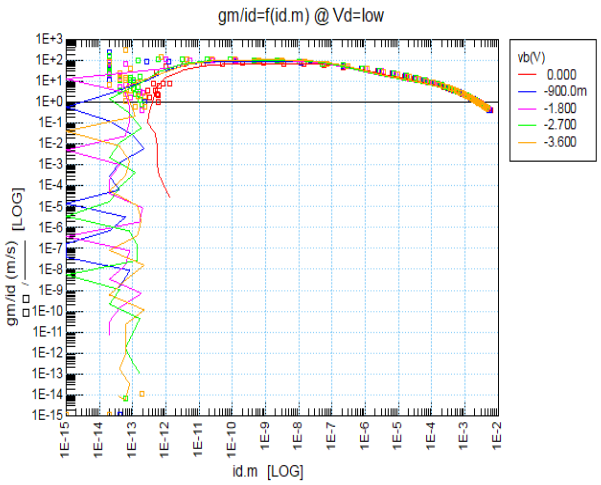


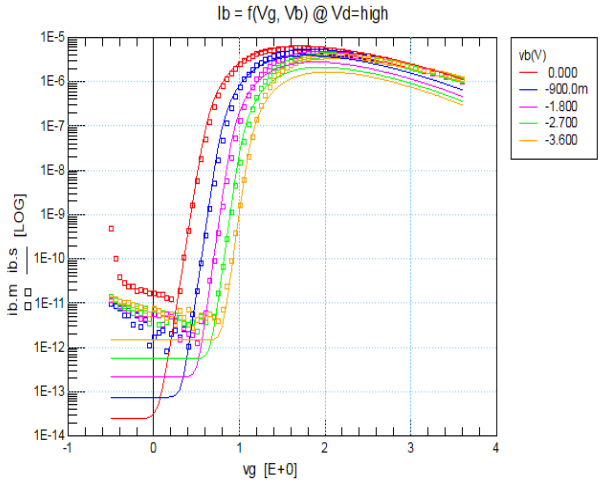


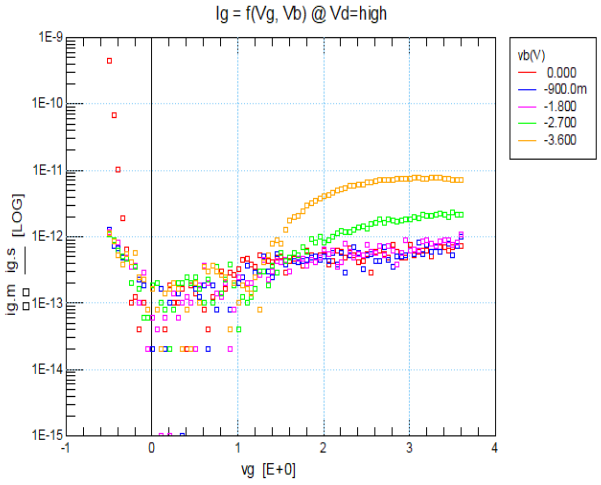




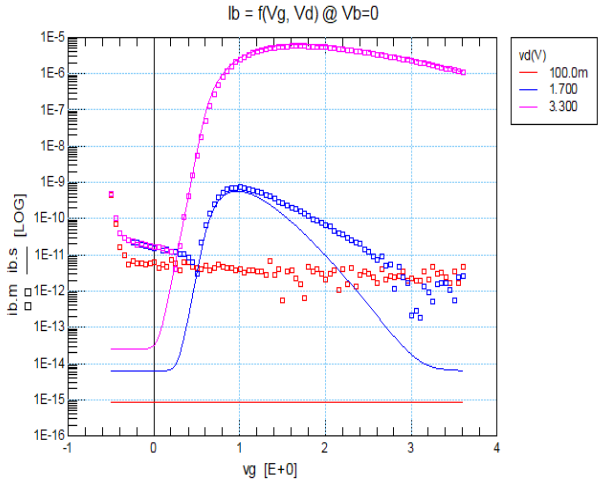


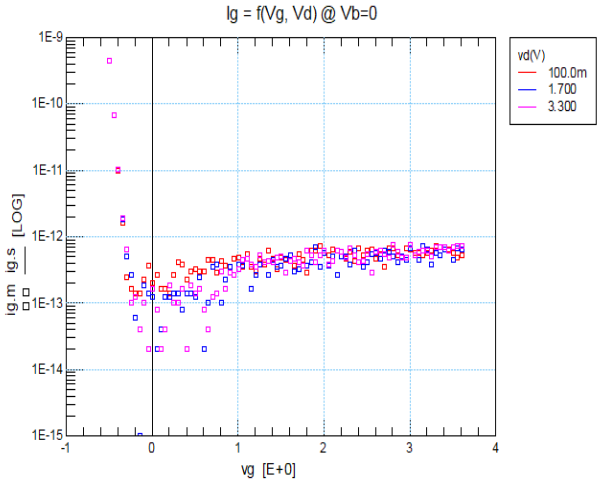




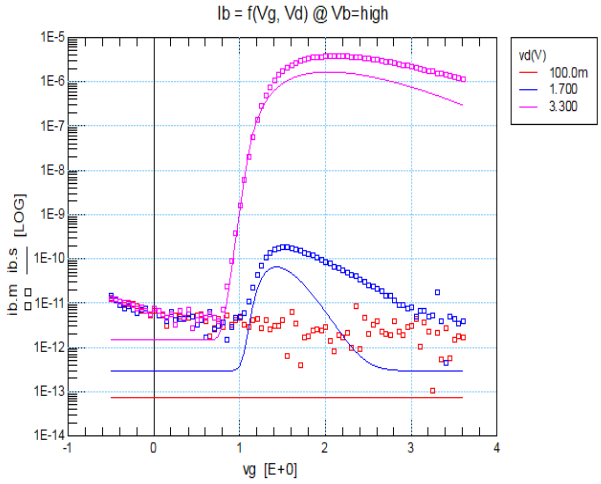


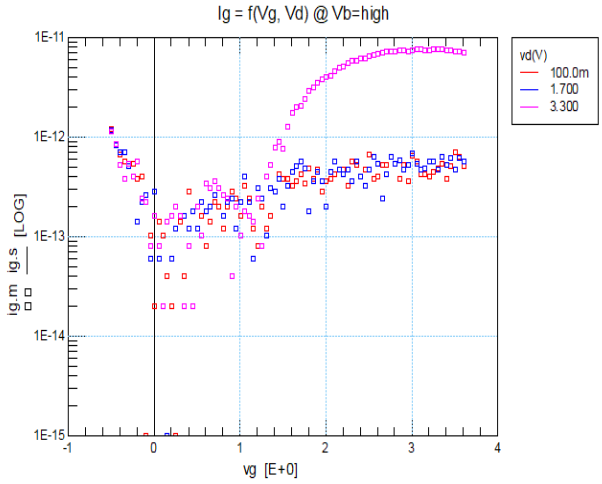
idvg, Ib,Ig, Vb = 0V, T = 27°C



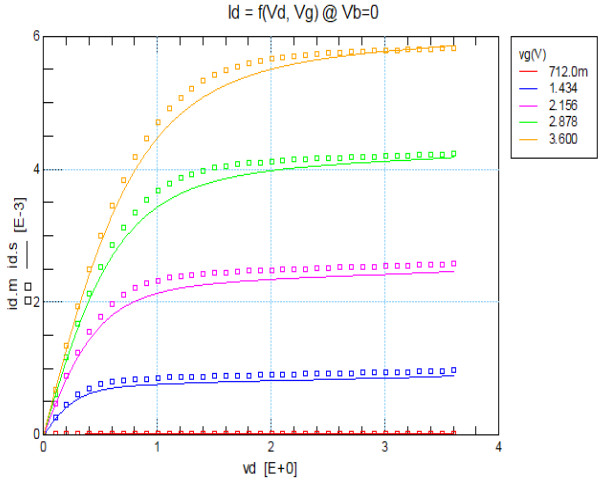


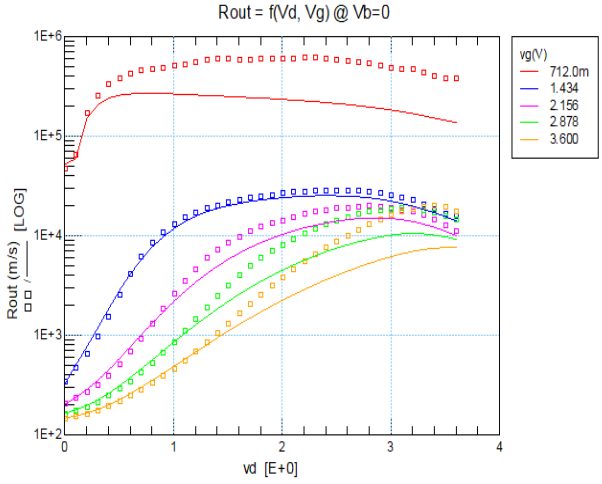
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

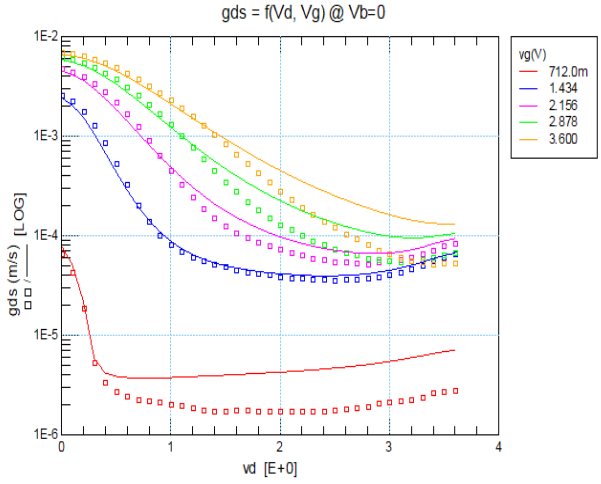




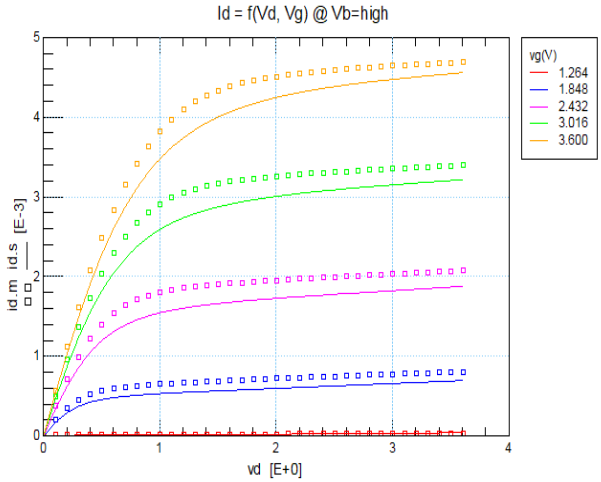
idvd, Vb = 0V, T = 27°C

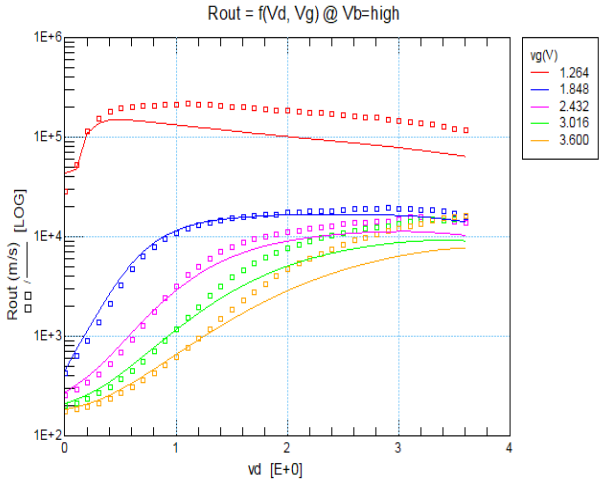


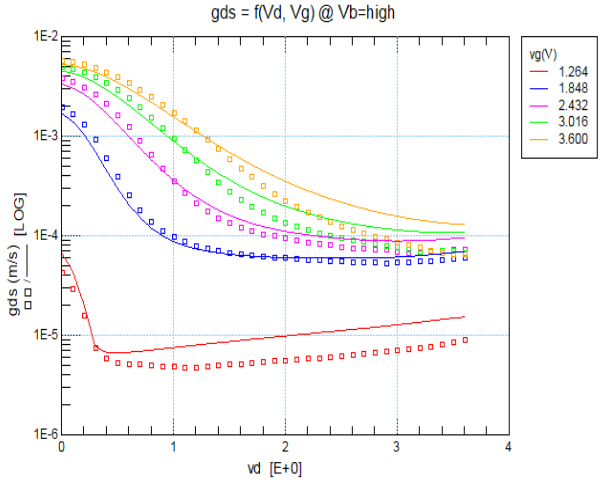




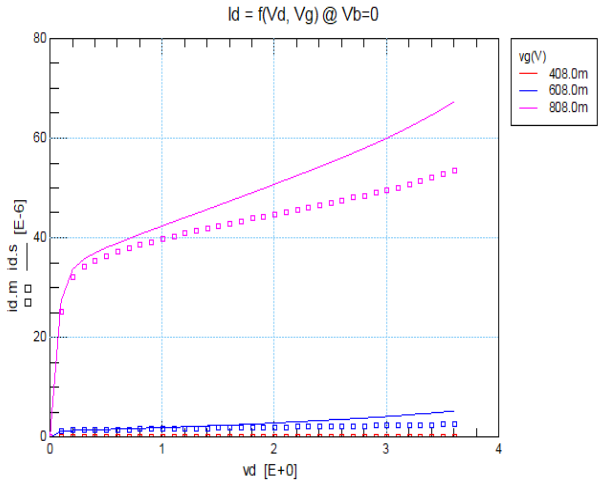
idvd_vbmin, Vb = -3.6V, T = 27°C

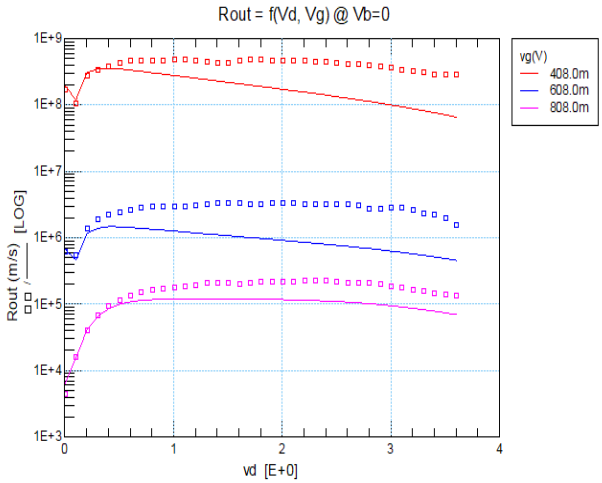


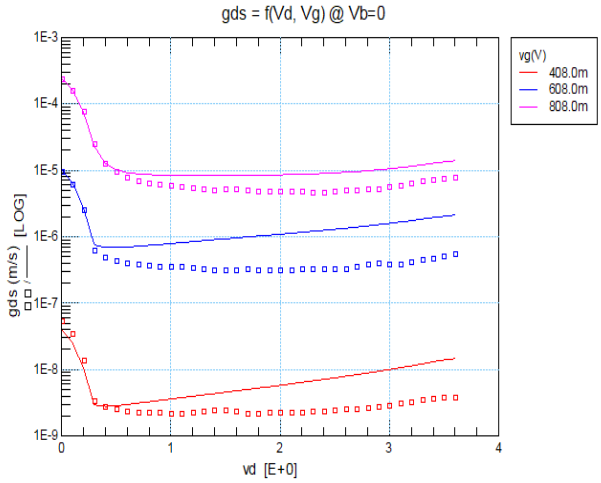




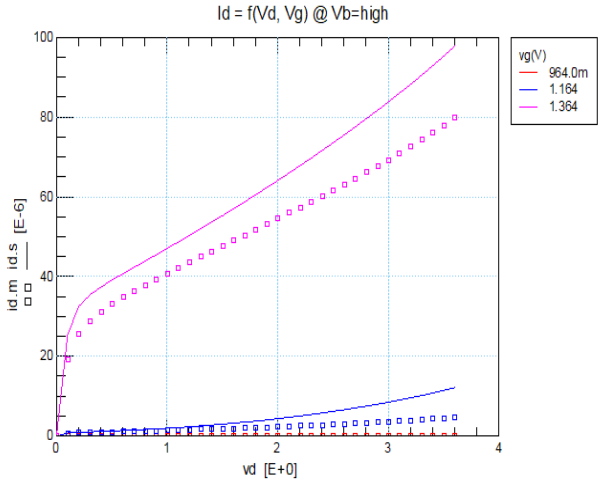
idvd_vth, Vb = 0V, T = 27°C

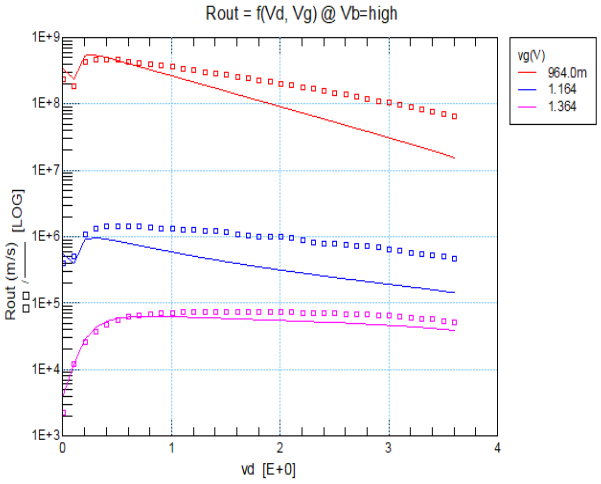


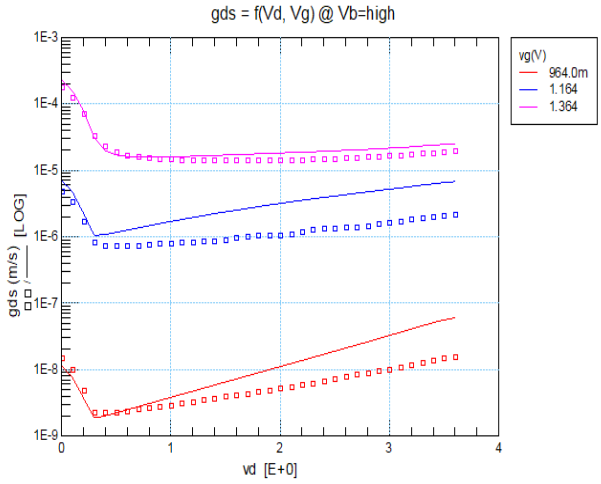




idvd_vth_vbmin, Vb = -3.6V, T = 27°C

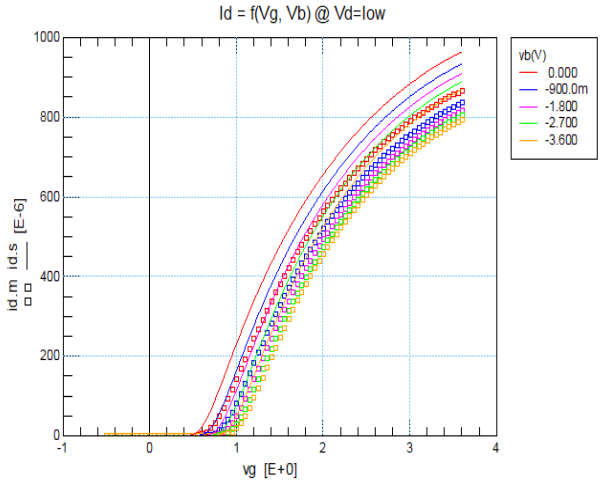


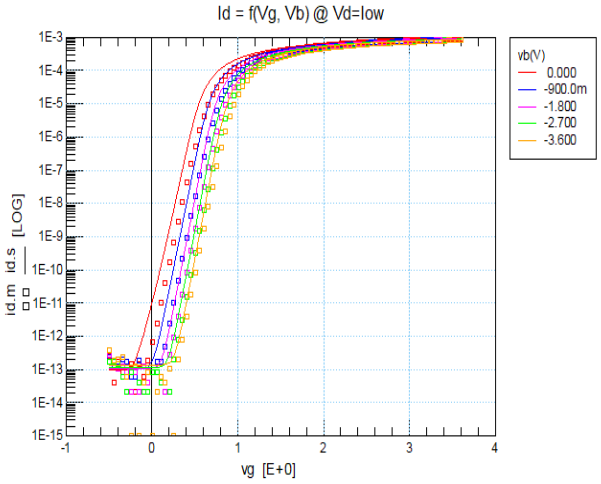


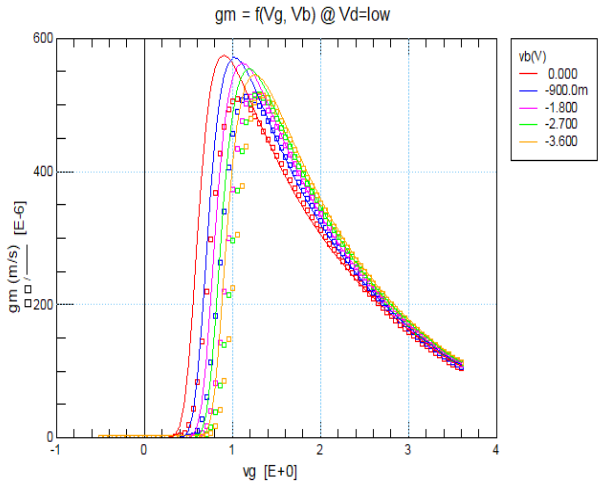


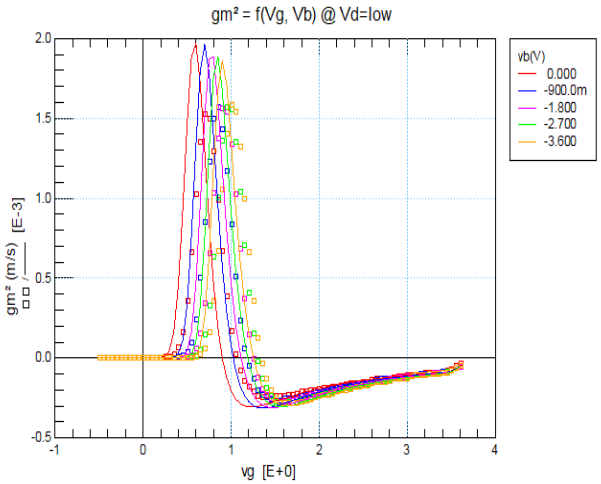
4.5 W10u0_L0u35_S556_3

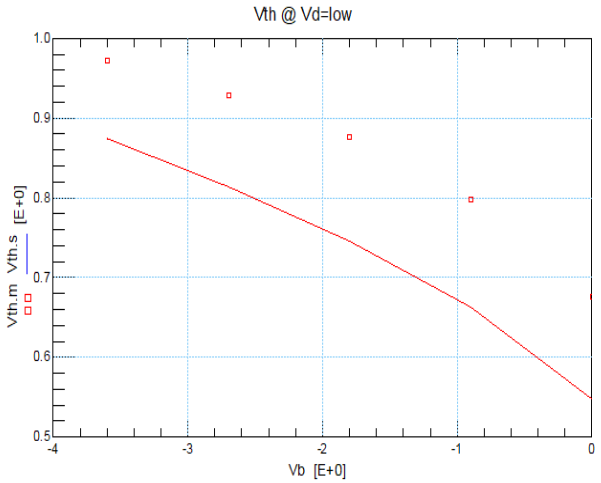
idvg, Vd = 0.1V, T = 27°C

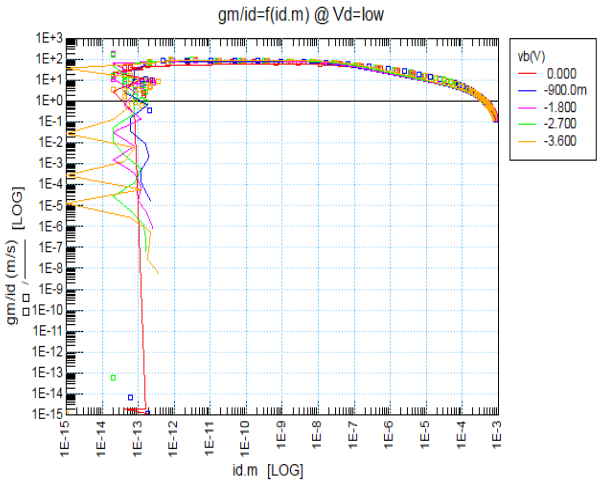


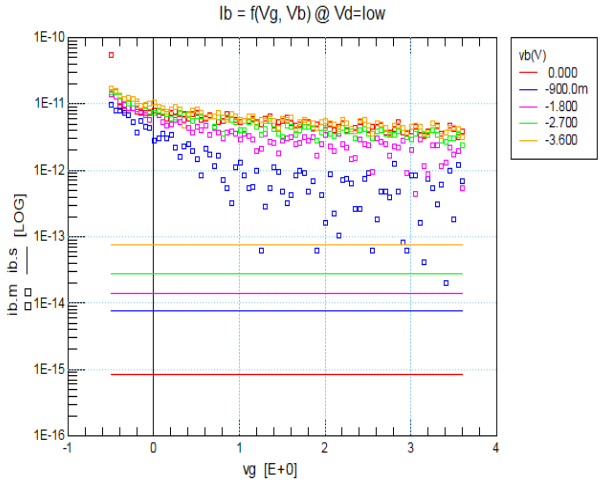


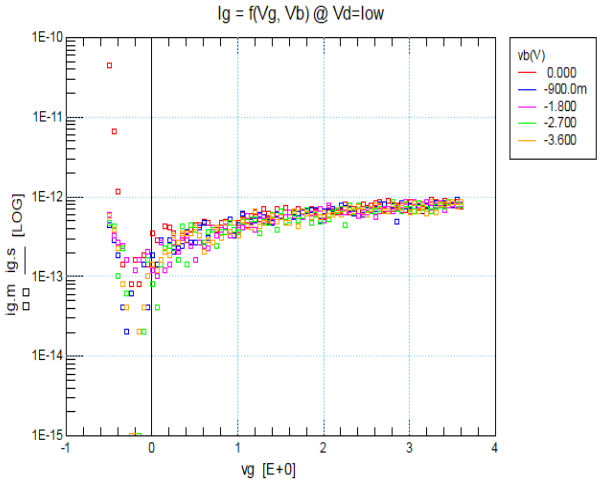




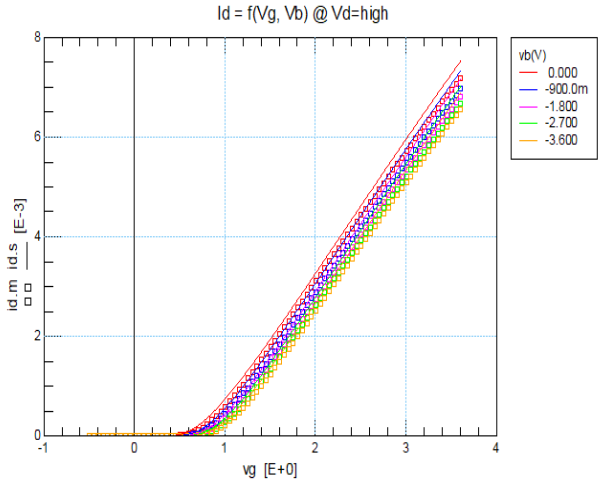


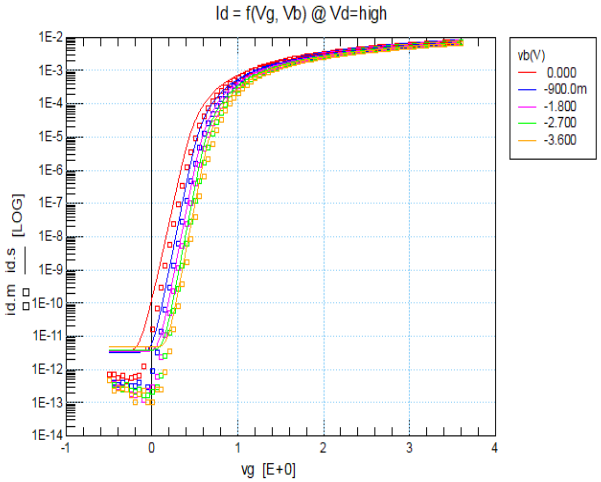


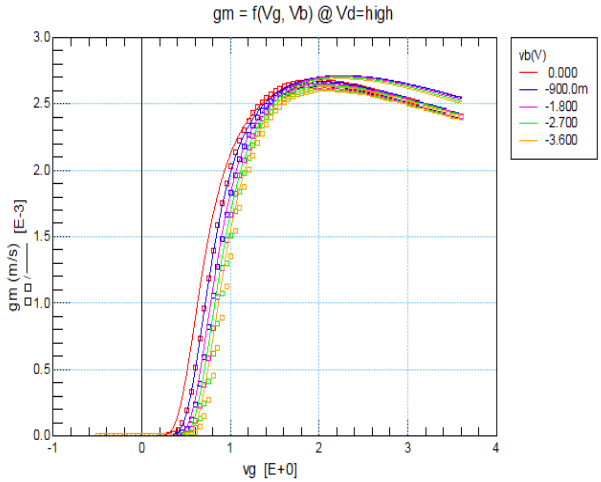


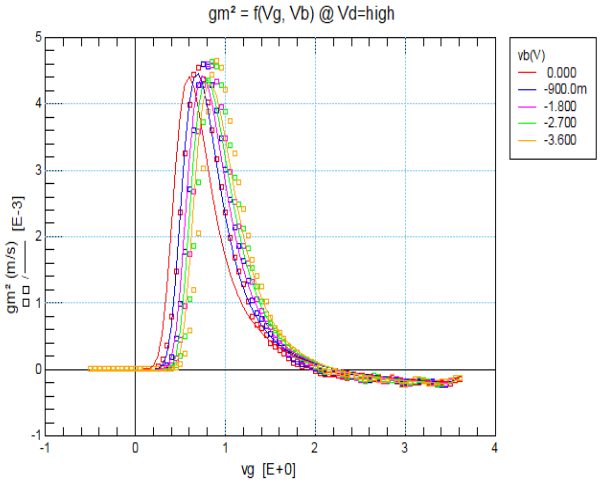


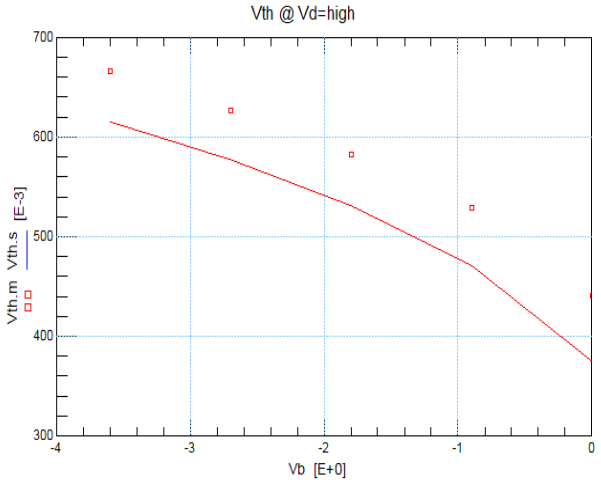
idvg, Vd = 3.3V, T = 27°C

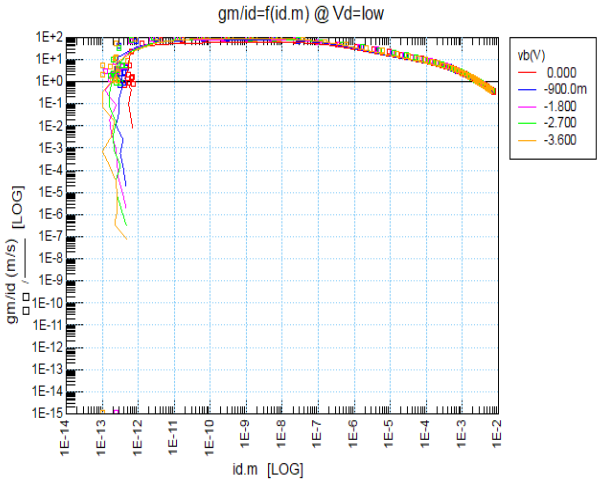


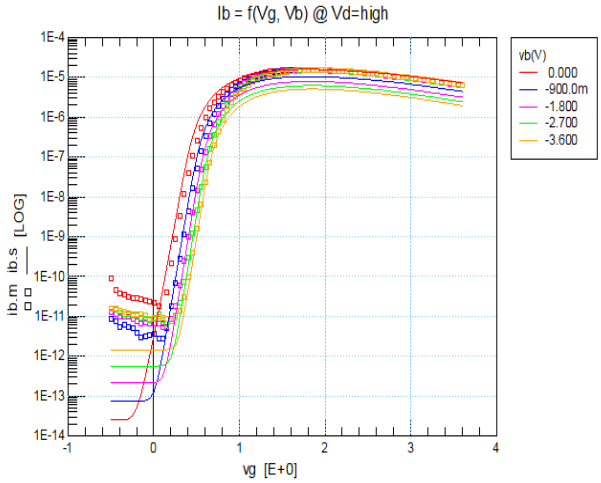


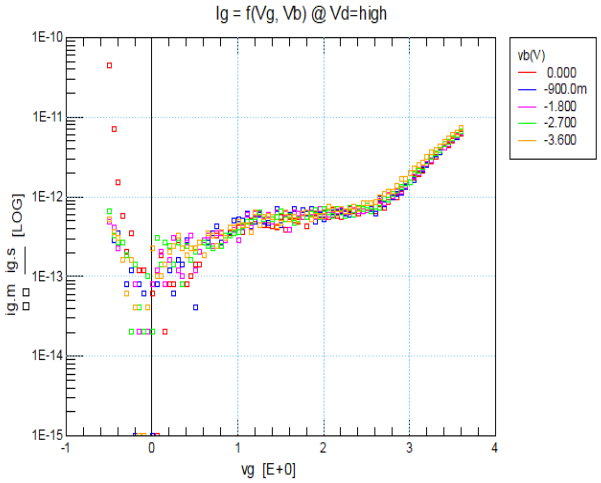




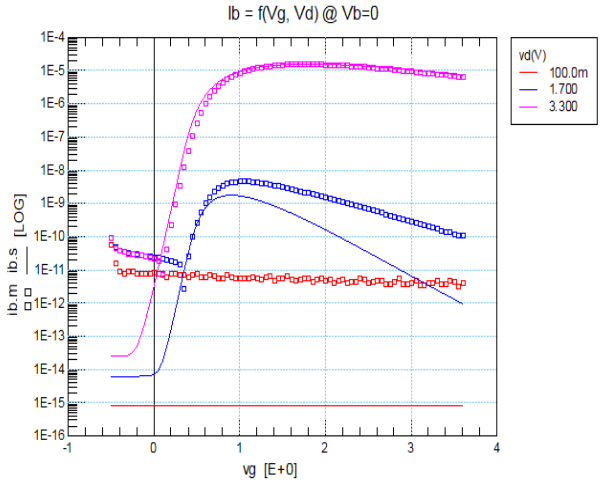


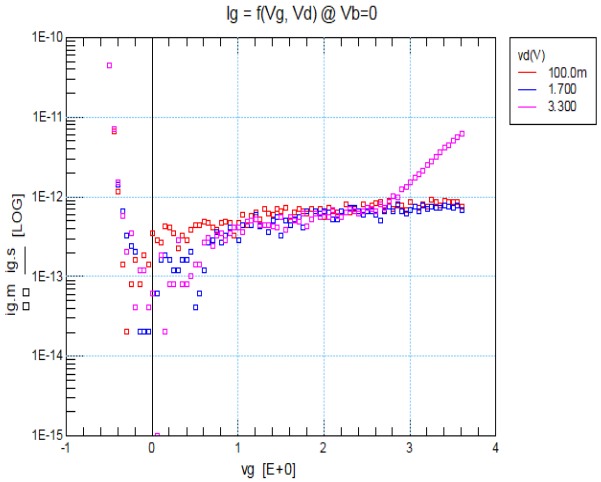




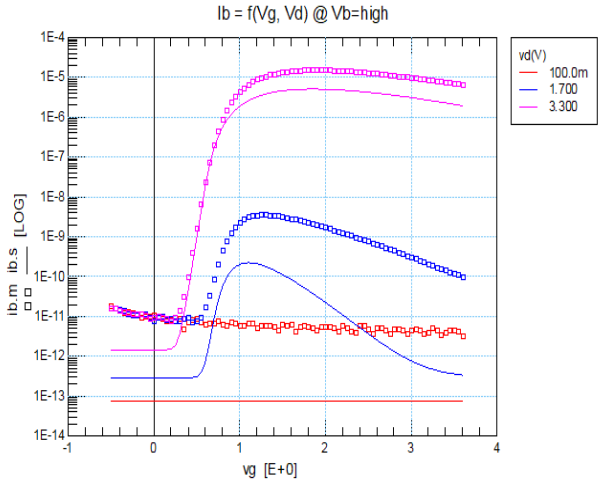


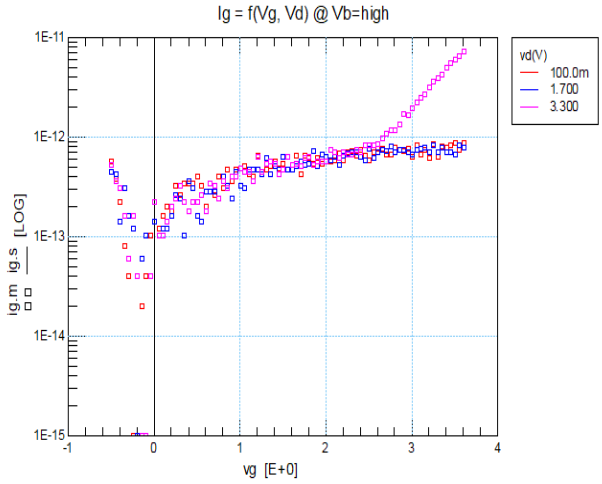
idvg, Ib,Ig, Vb = 0V, T = 27°C



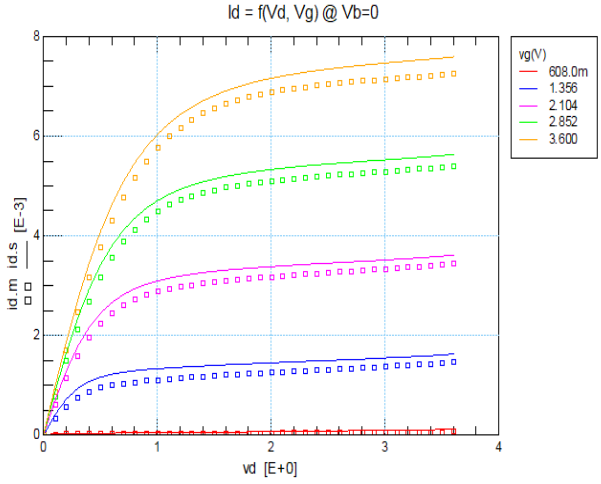


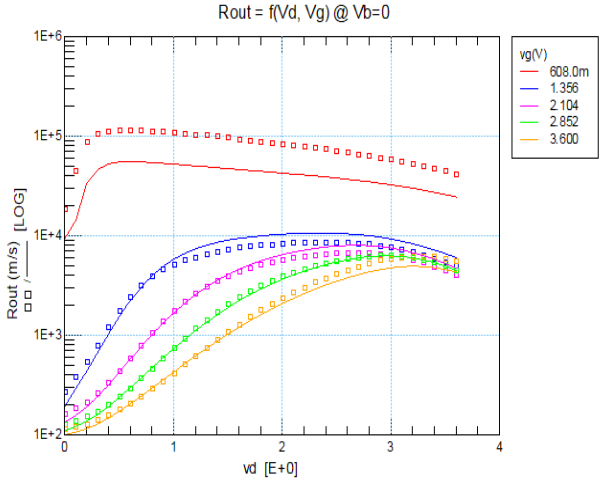
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

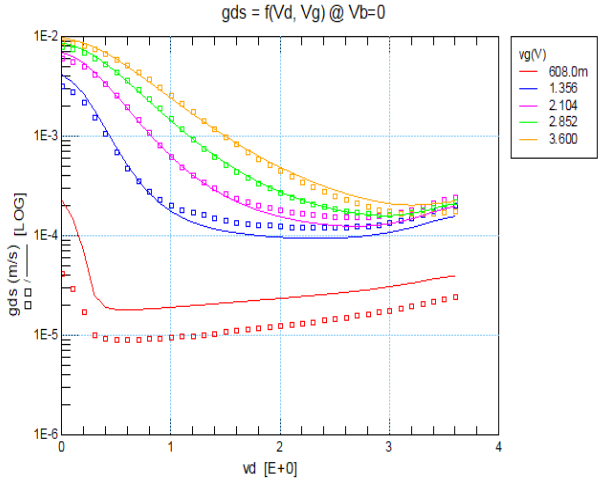




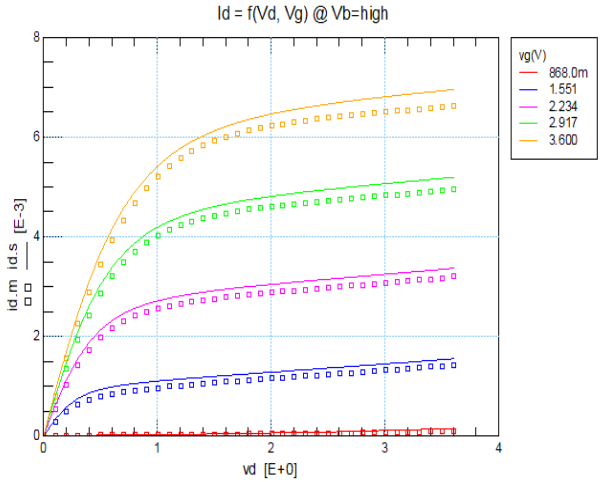
idvd, Vb = 0V, T = 27°C

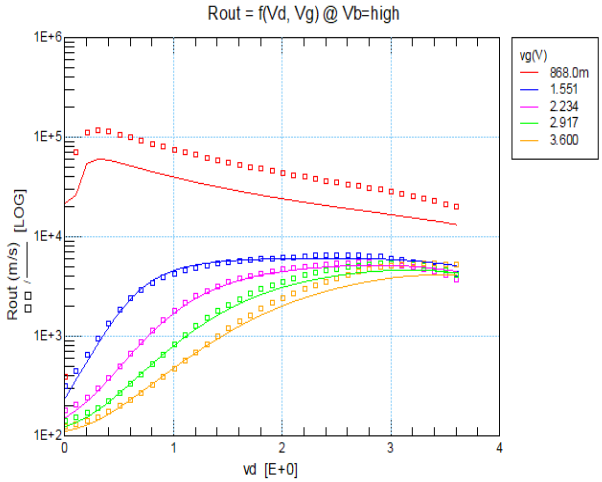


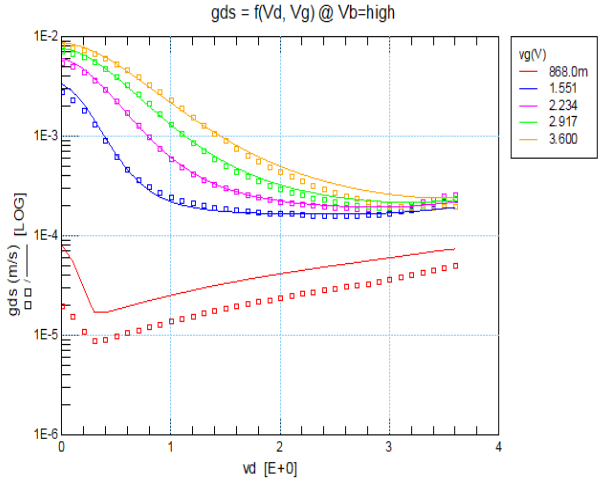




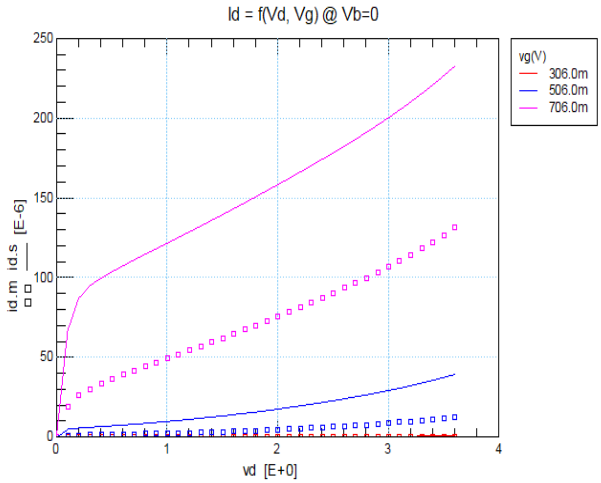
idvd_vbmin, Vb = -3.6V, T = 27°C

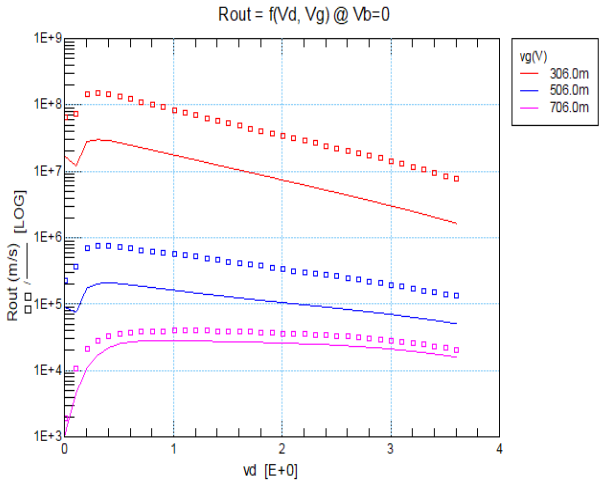


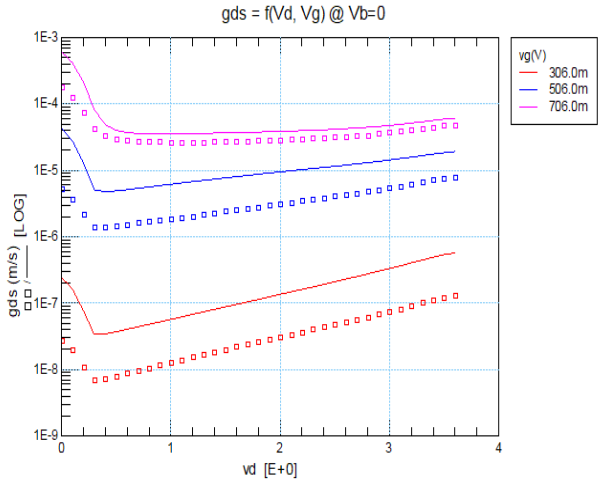




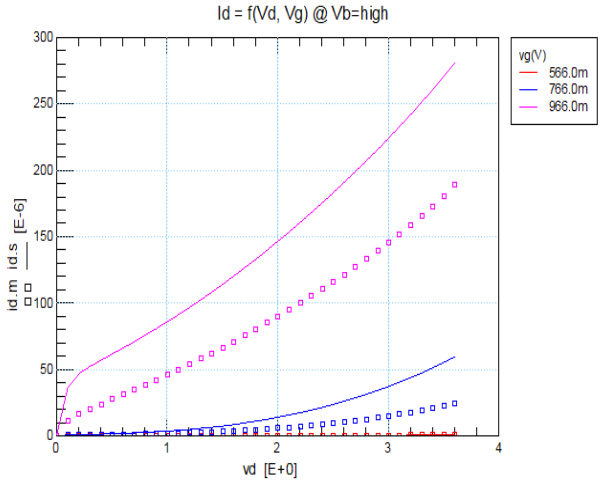
idvd_vth, Vb = 0V, T = 27°C

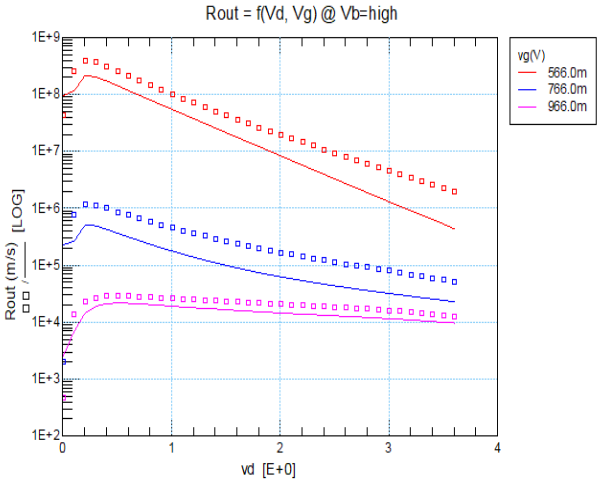


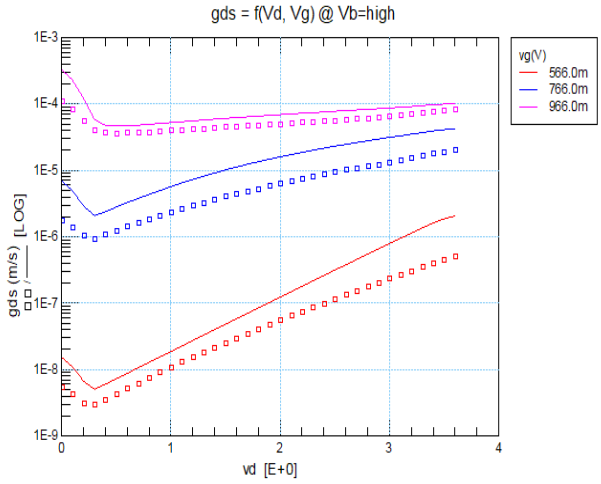




idvd_vth_vbmin, Vb = -3.6V, T = 27°C

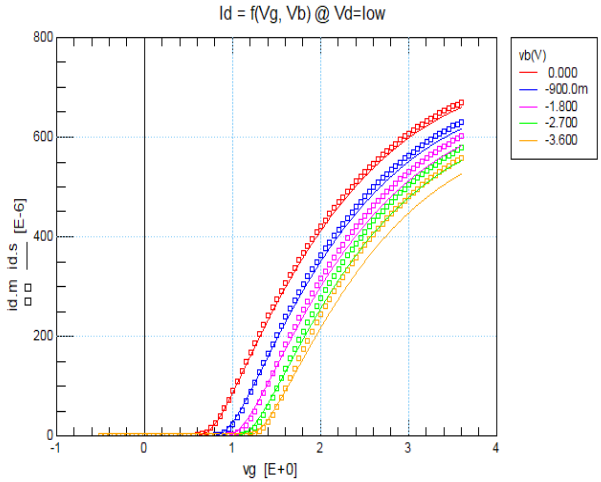


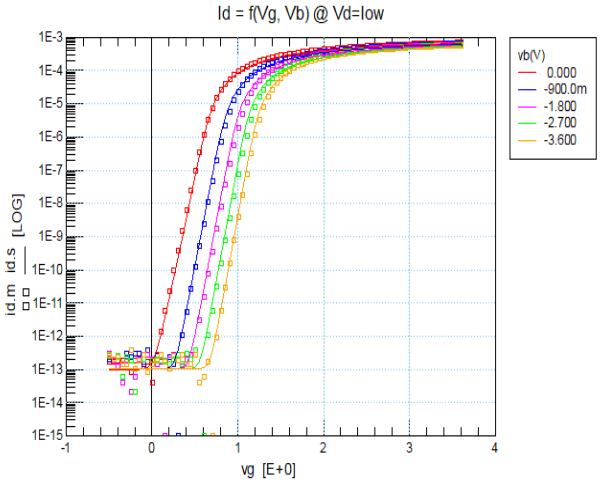


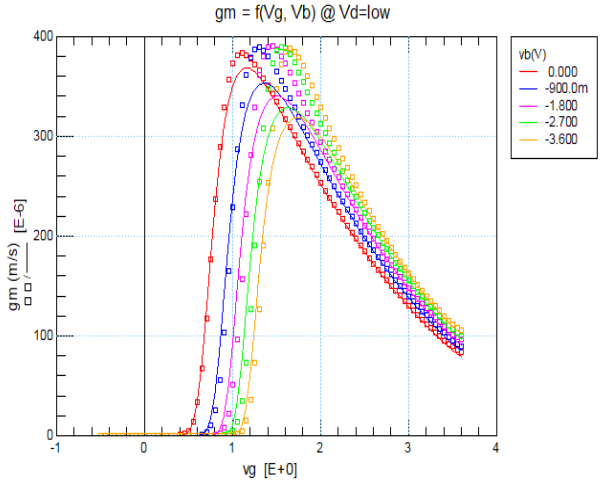


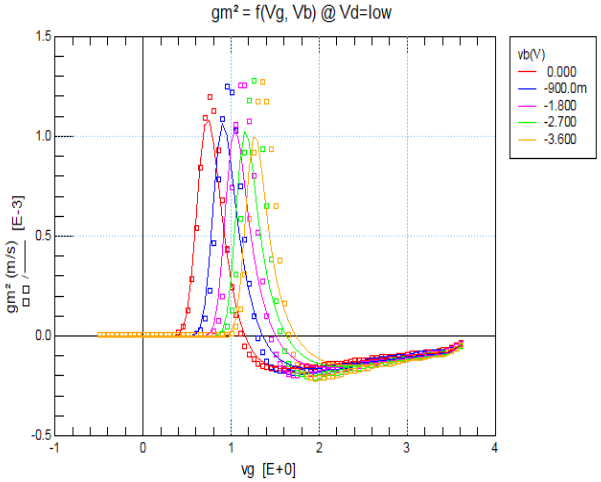
4.6 W10u0_L0u5_T356_S387_5

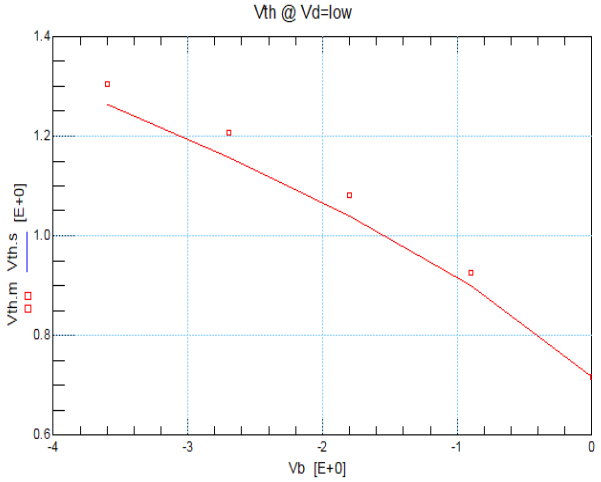
idvg, Vd = 0.1V, T = 27°C

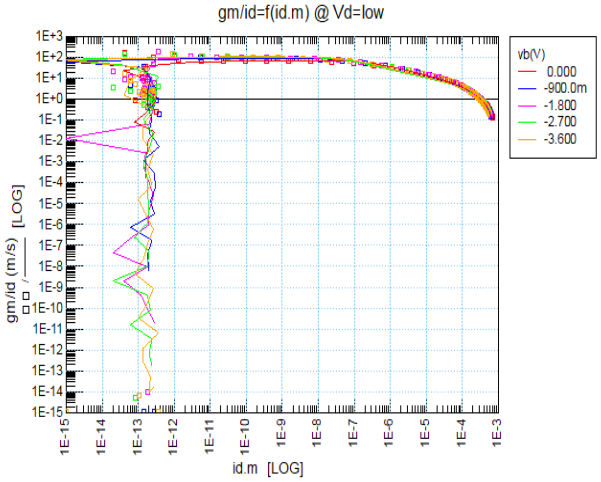


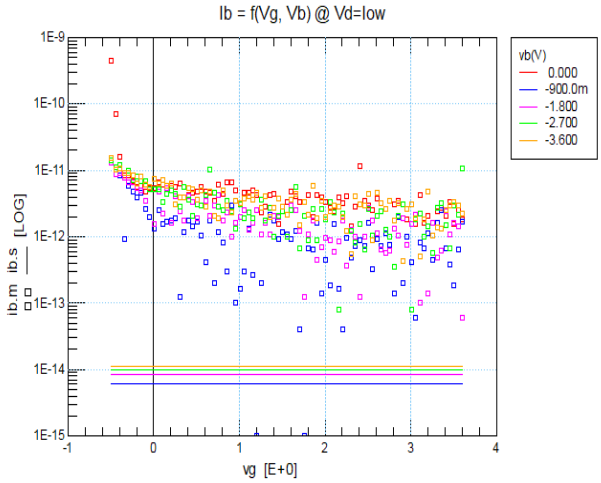


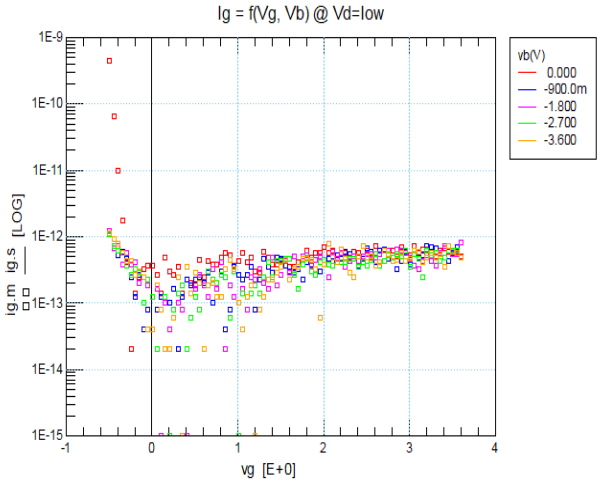




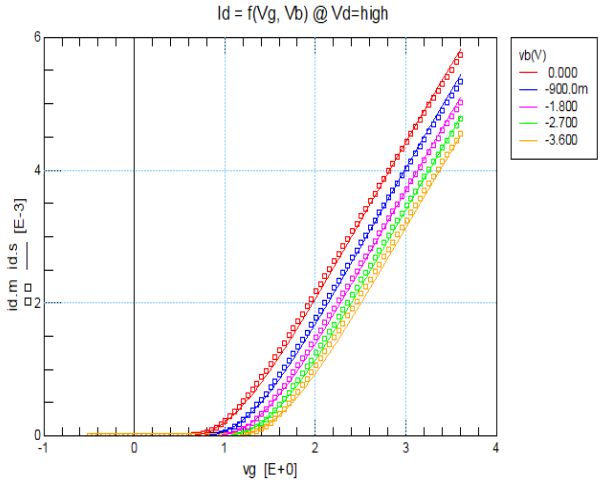


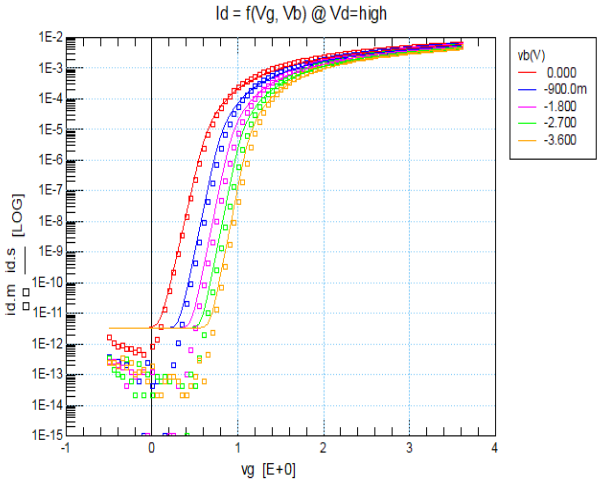


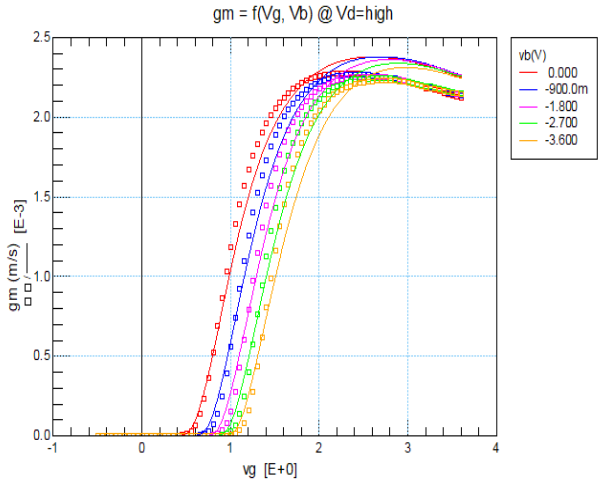


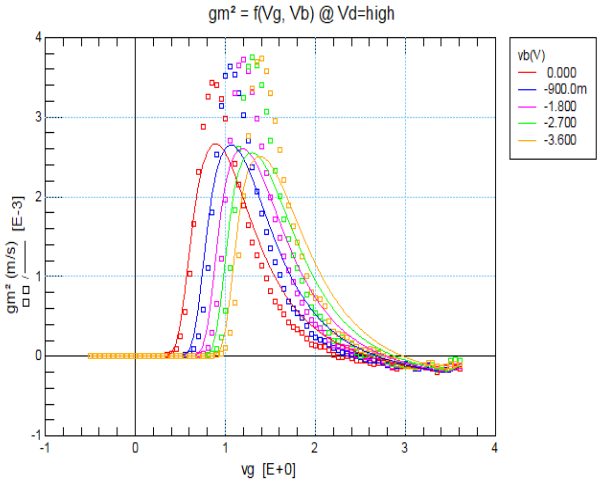


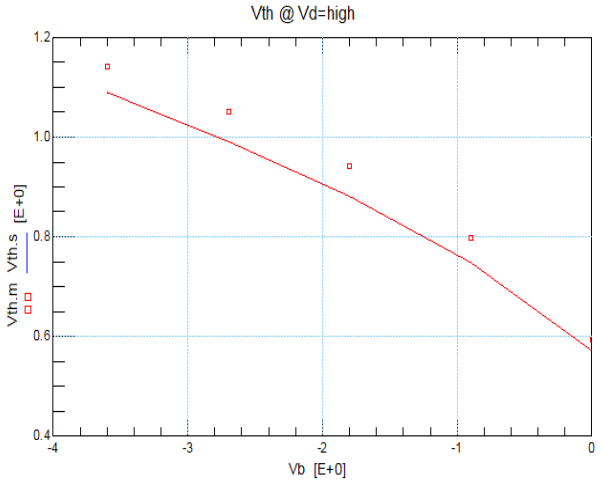
idvg, Vd = 3.3V, T = 27°C

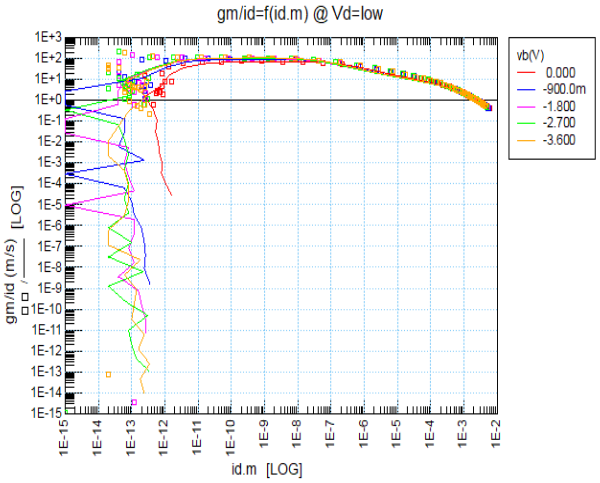


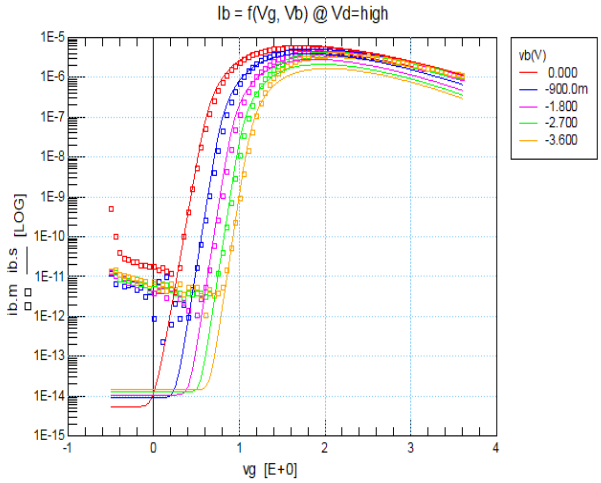


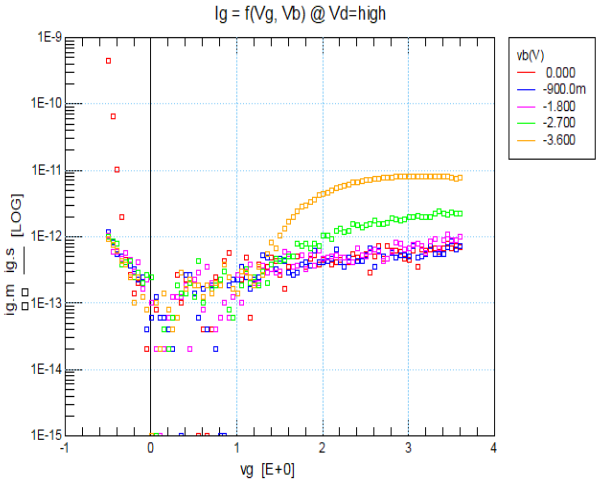




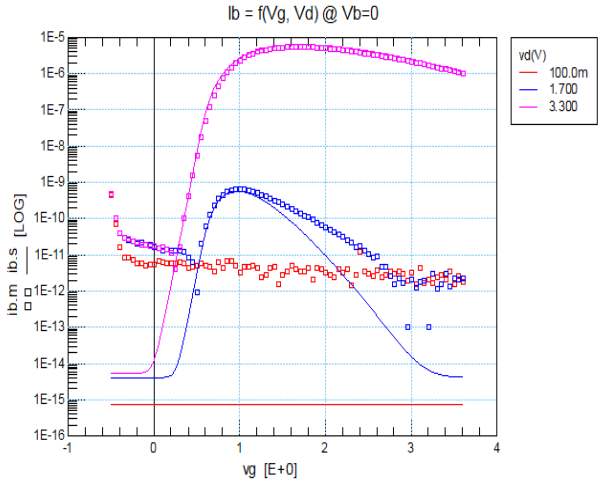


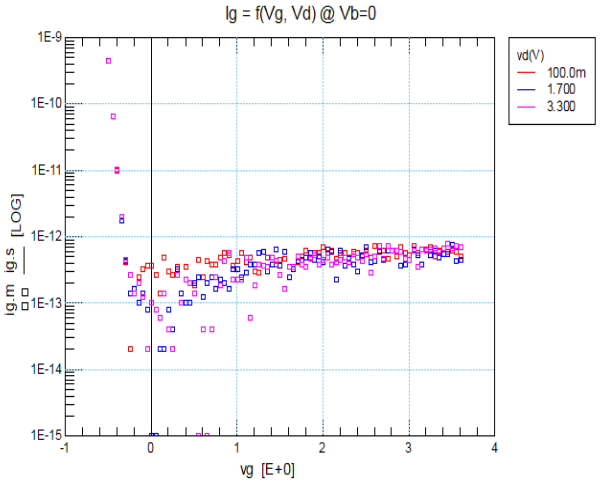




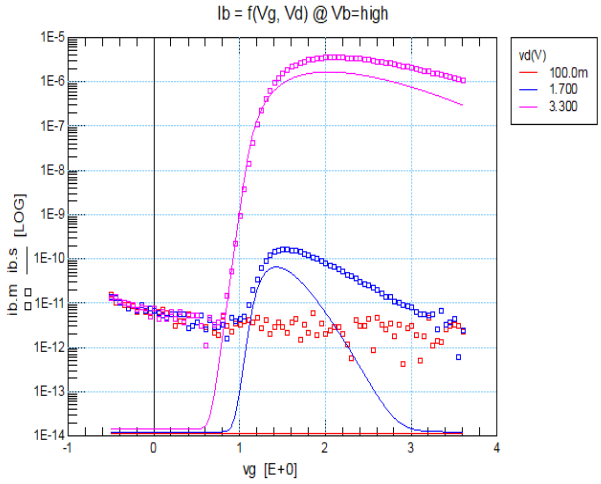


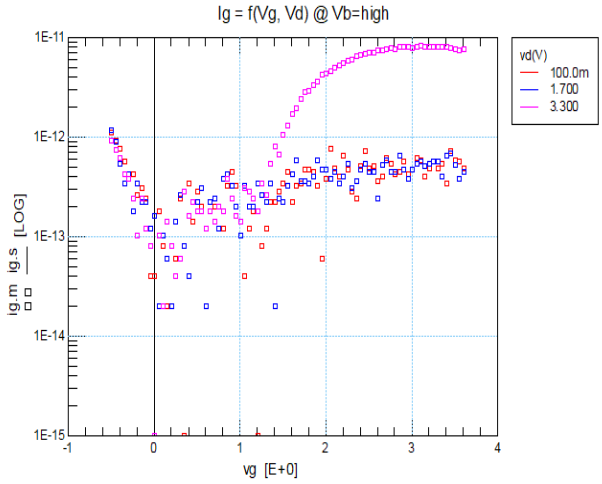
idvg, Ib,Ig, Vb = 0V, T = 27°C



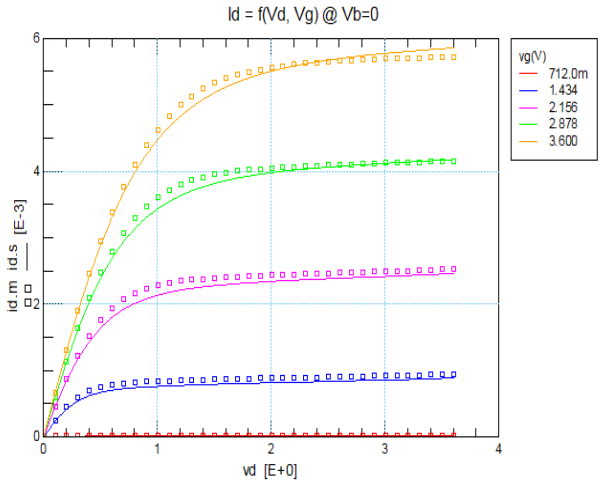


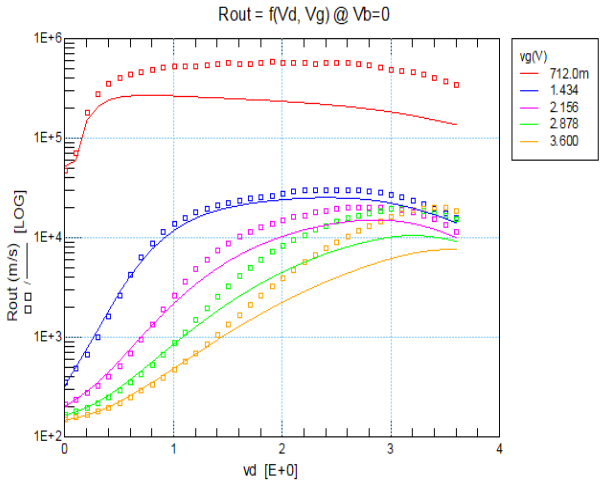
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

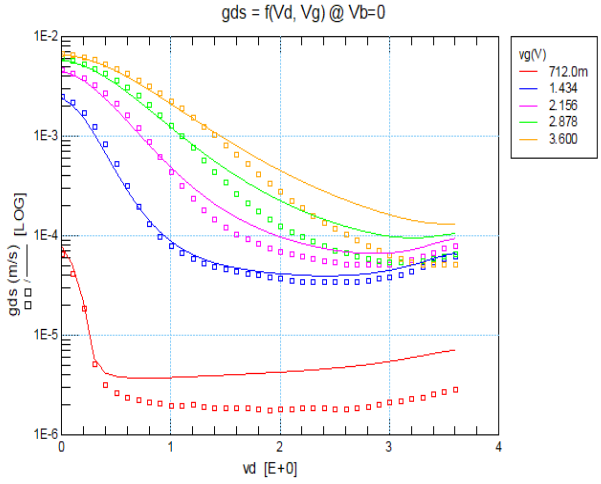




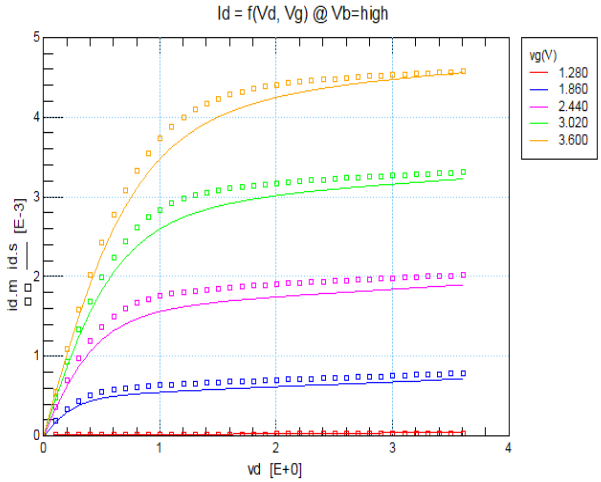
idvd, Vb = 0V, T = 27°C

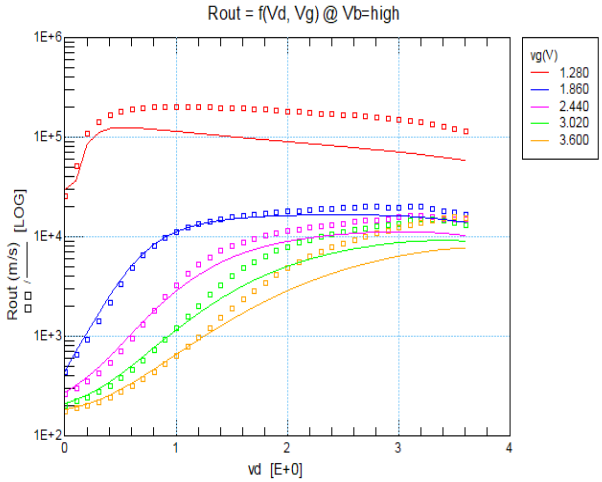


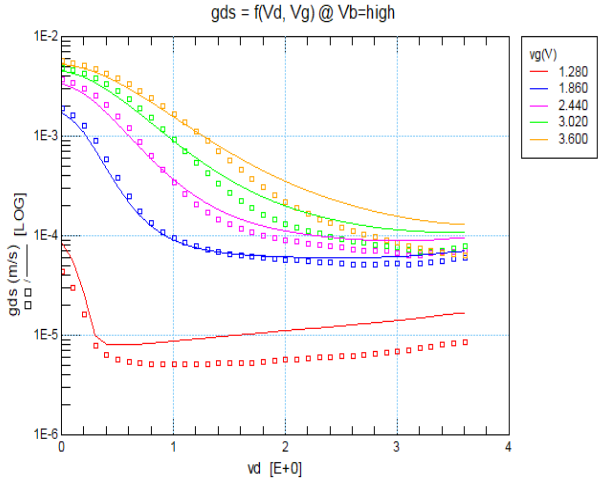




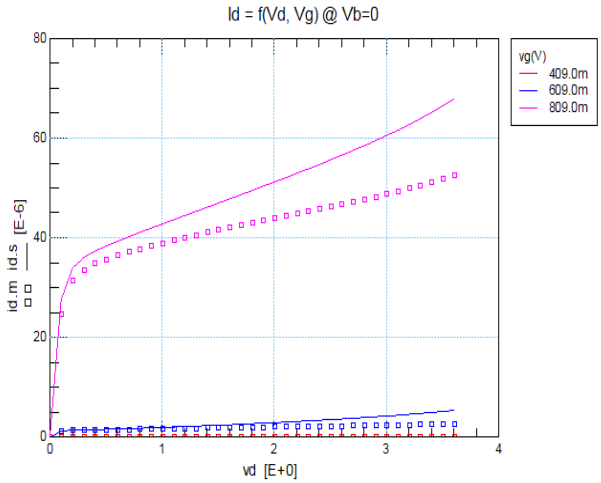
idvd_vbmin, Vb = -3.6V, T = 27°C

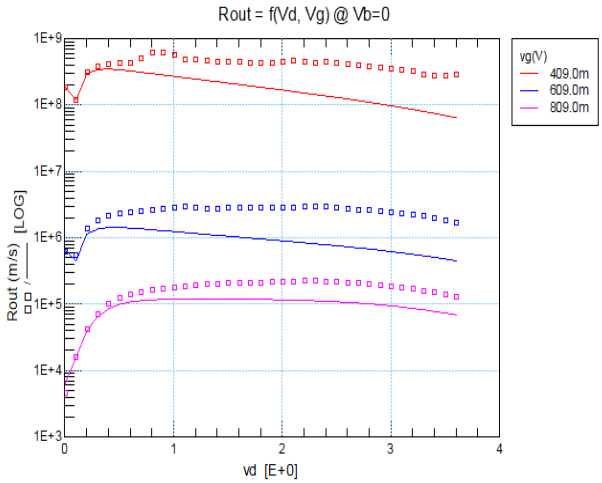


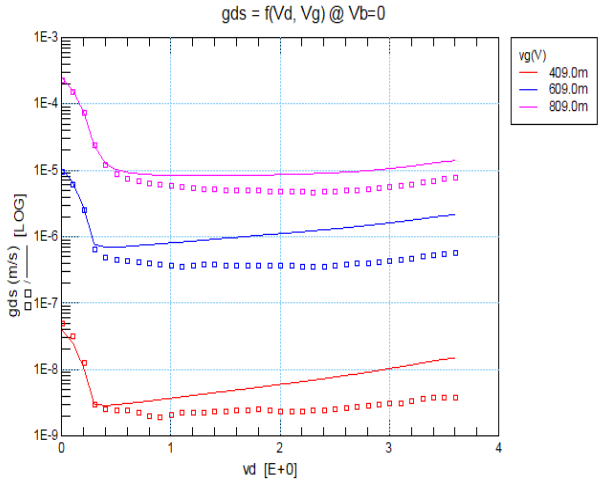




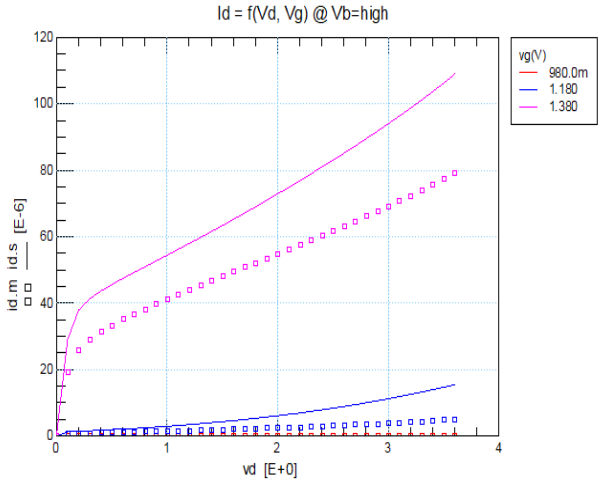
idvd_vth, Vb = 0V, T = 27°C

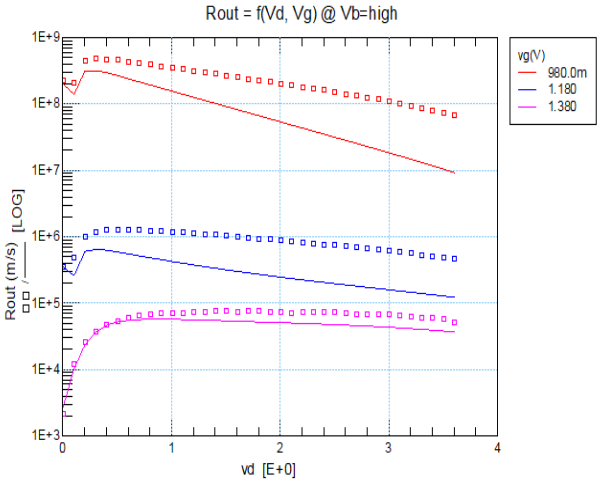


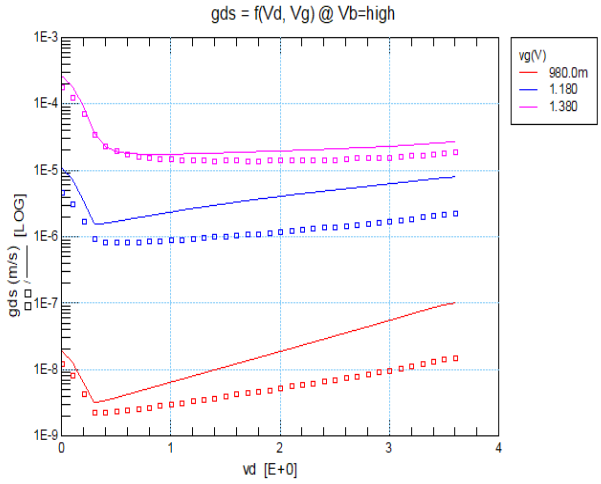




idvd_vth_vbmin, Vb = -3.6V, T = 27°C

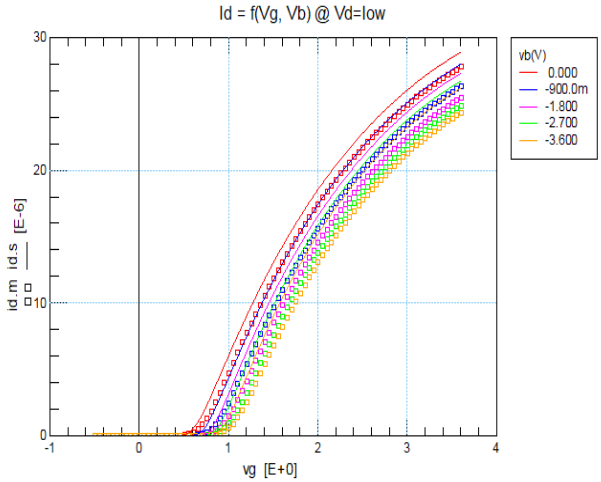


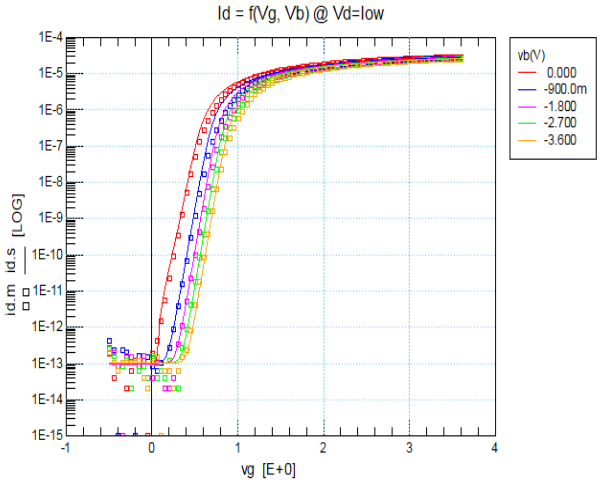


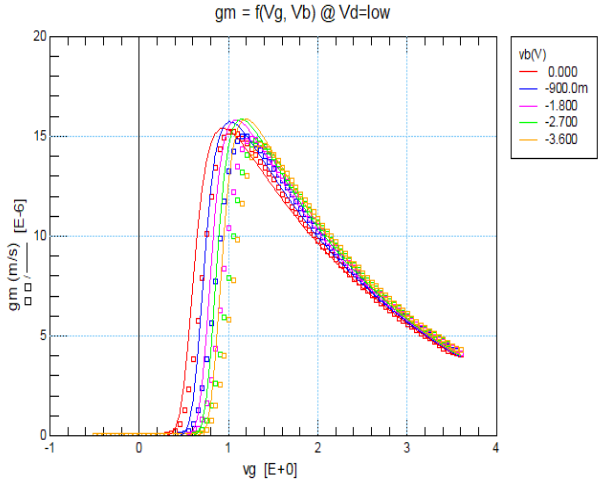


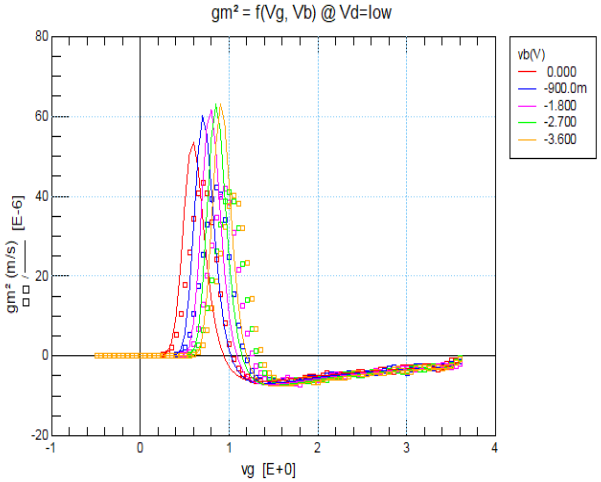
4.7 W0u3_L0u35_S560_3

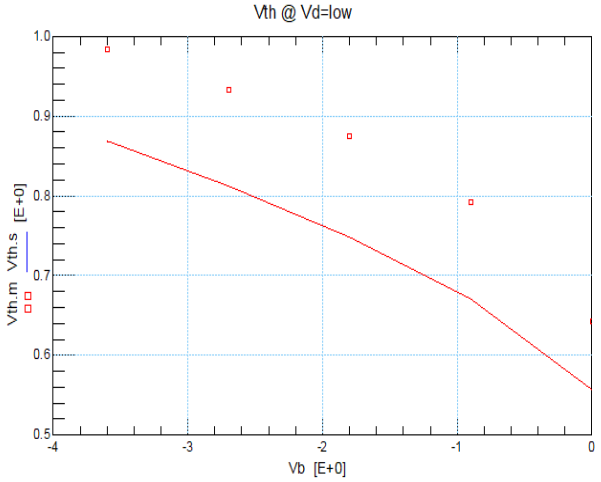
idvg, Vd = 0.1V, T = 27°C

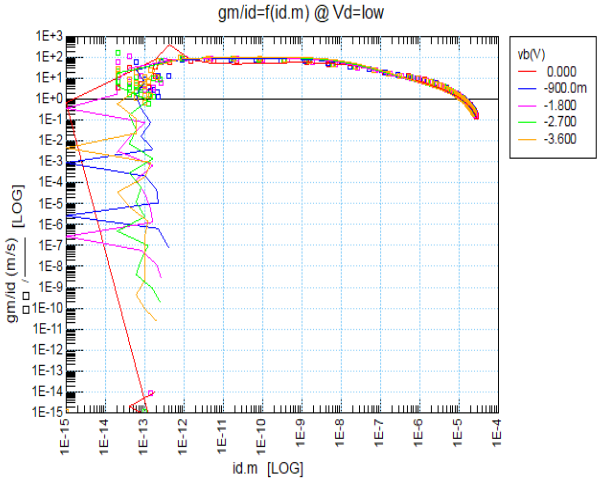


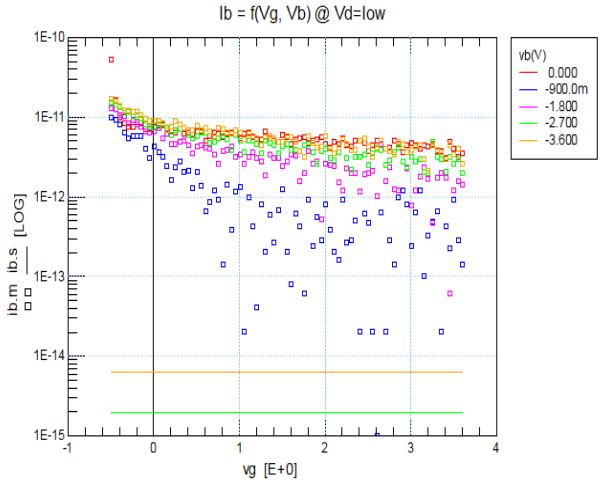


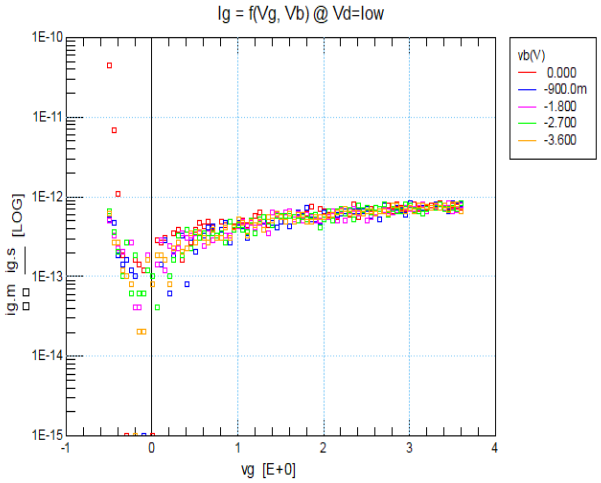




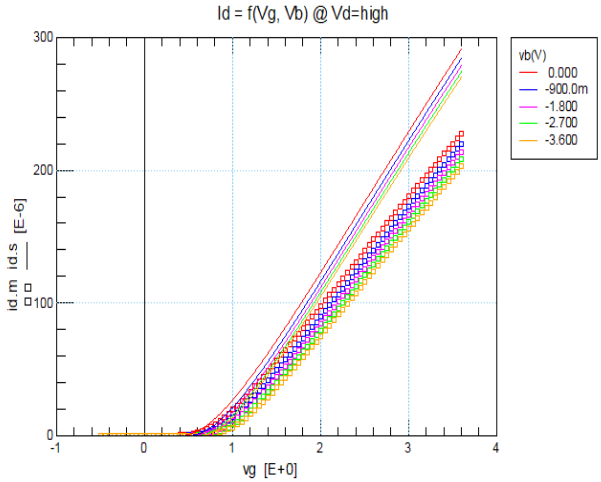


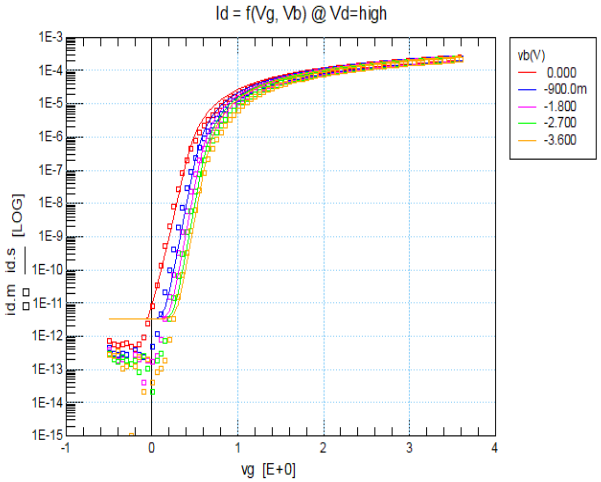


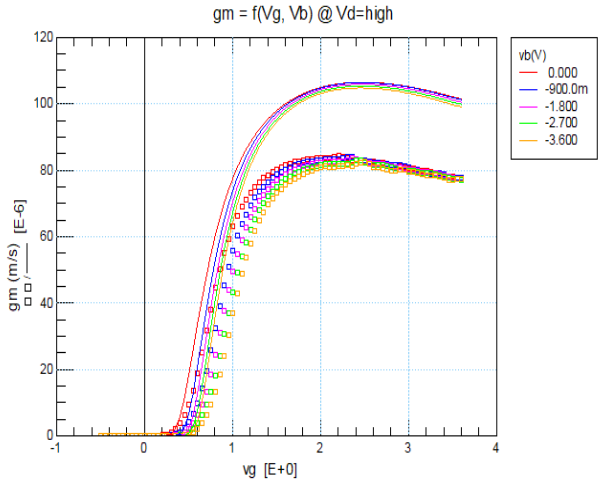


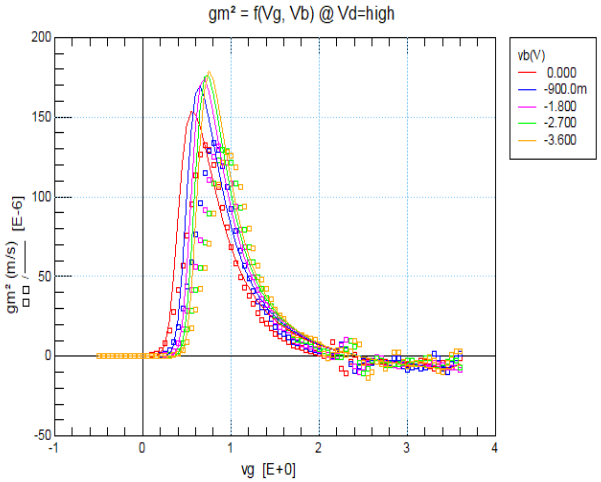


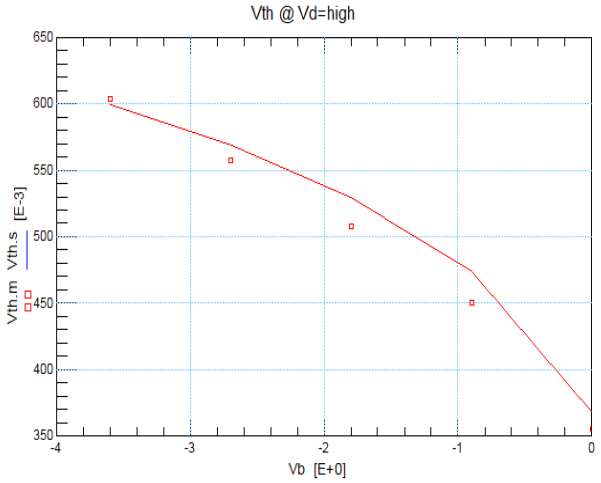
idvg, Vd = 3.3V, T = 27°C

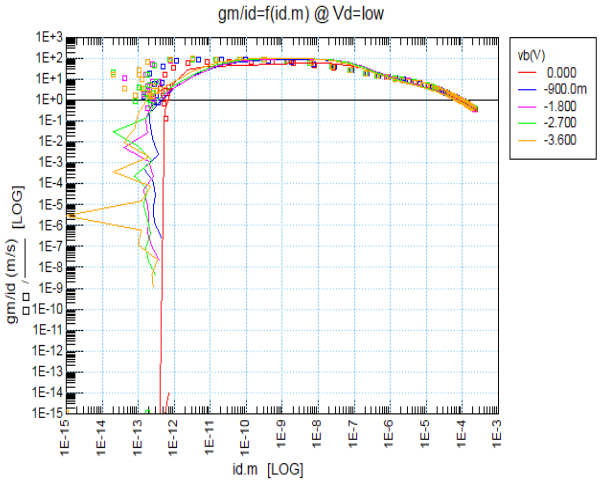


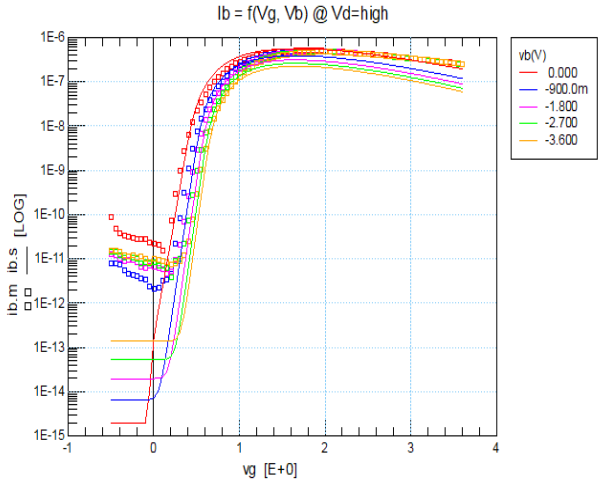


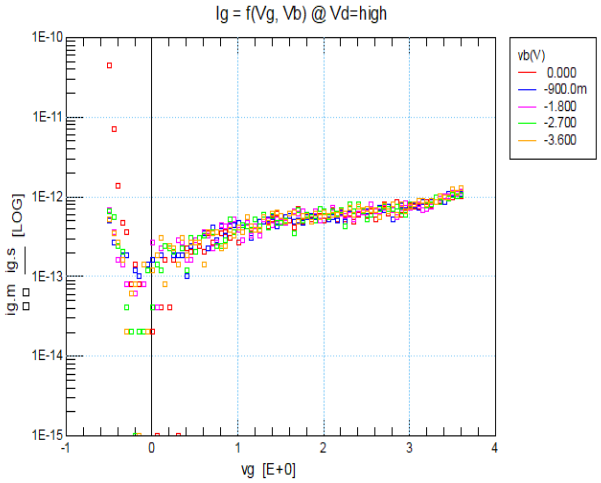




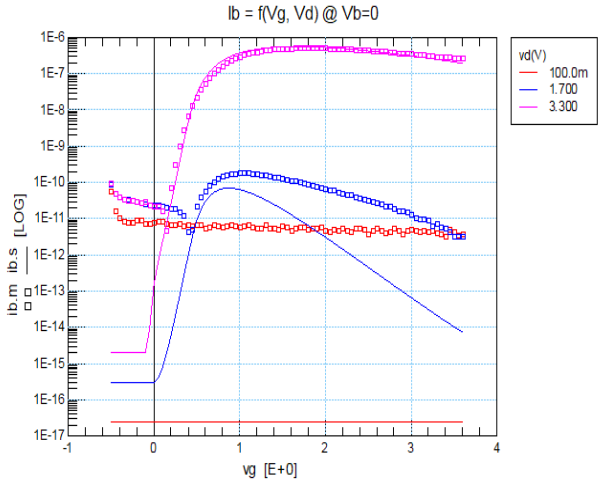


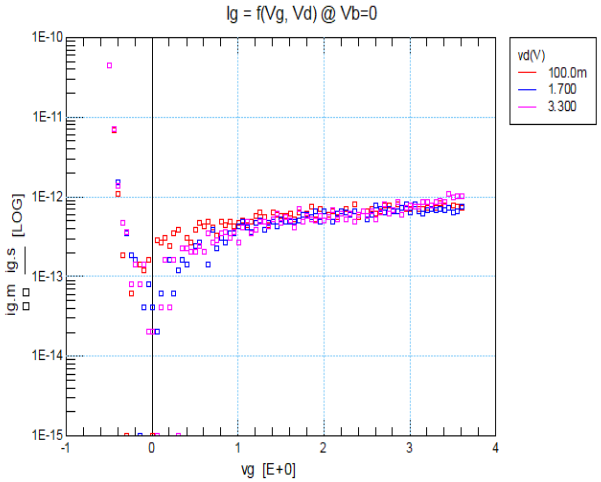




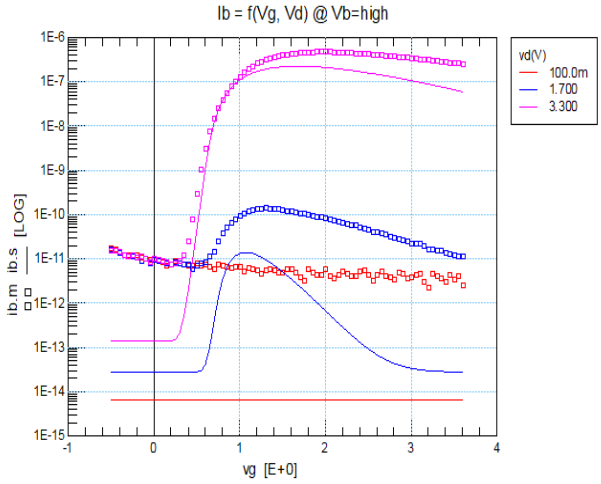


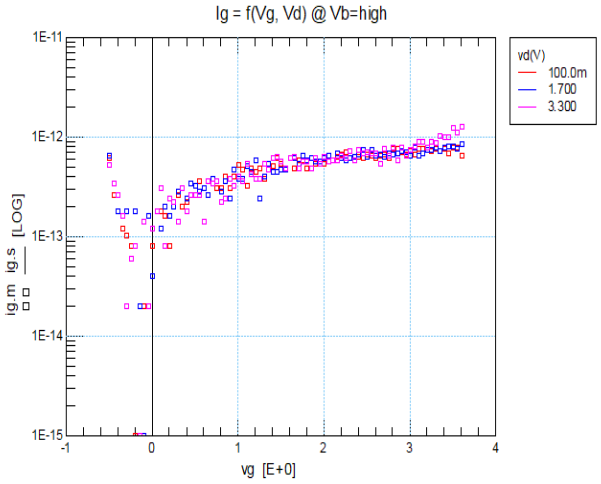
idvg, Ib,Ig, Vb = 0V, T = 27°C



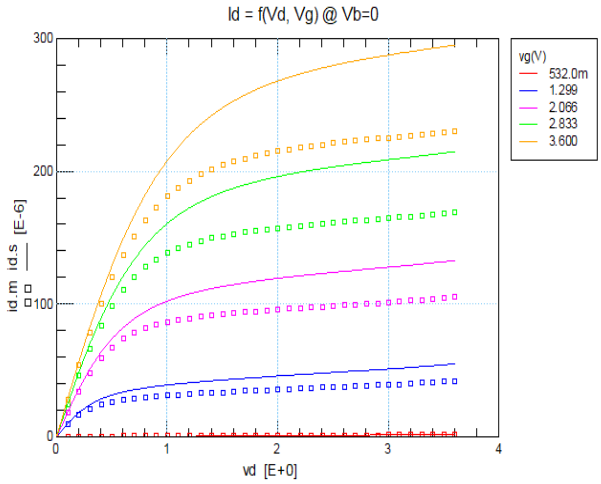


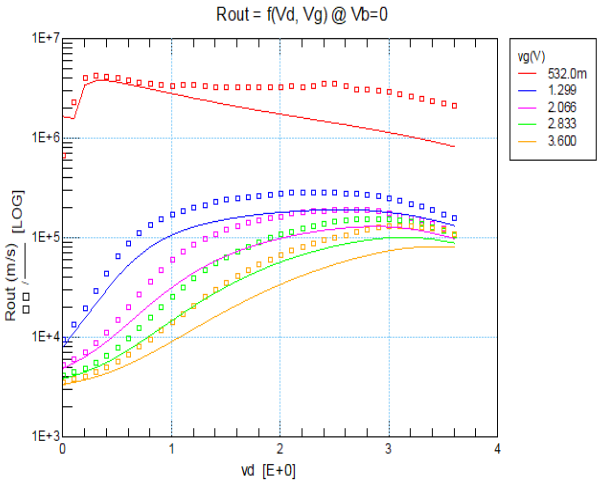
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

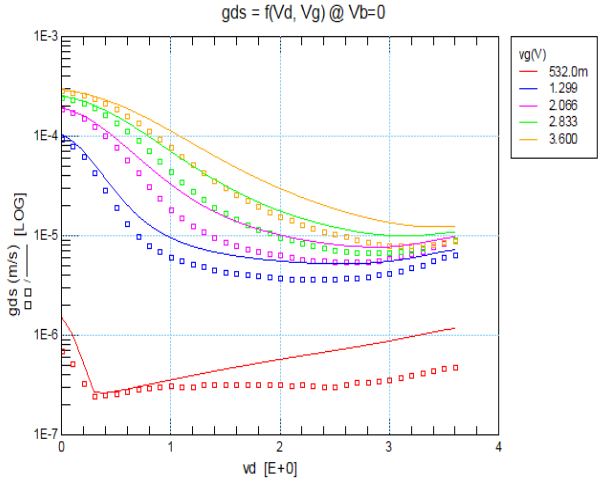




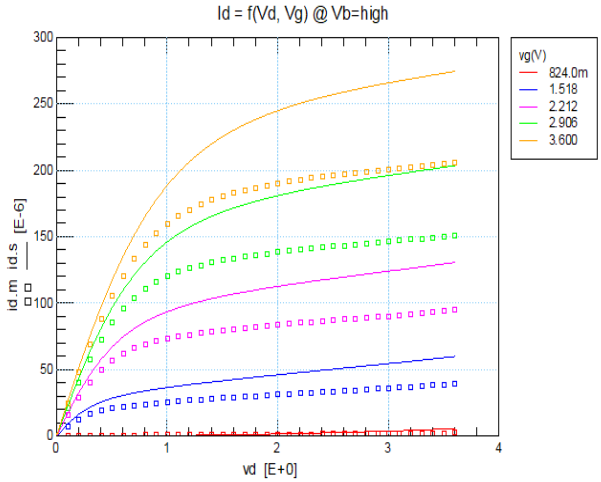
idvd, Vb = 0V, T = 27°C

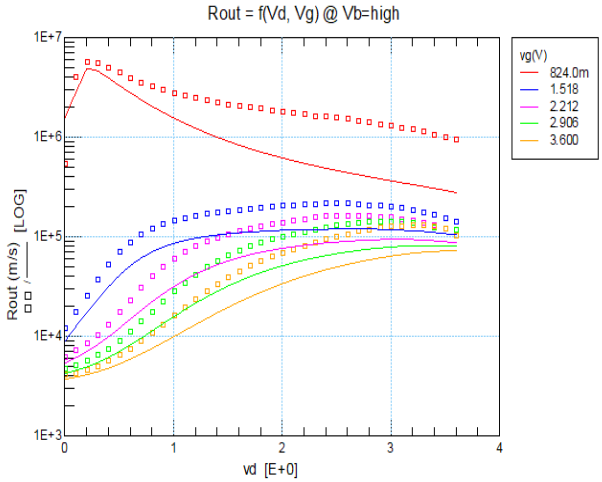


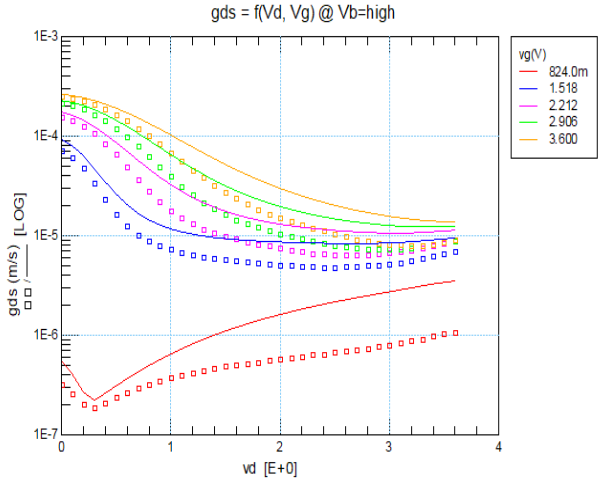




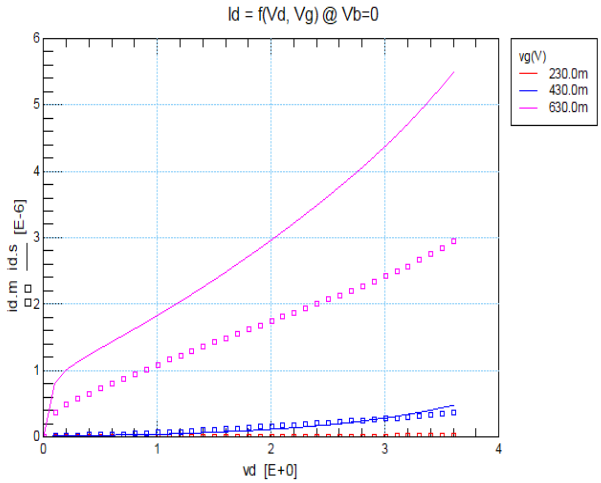
idvd_vbmin, Vb = -3.6V, T = 27°C

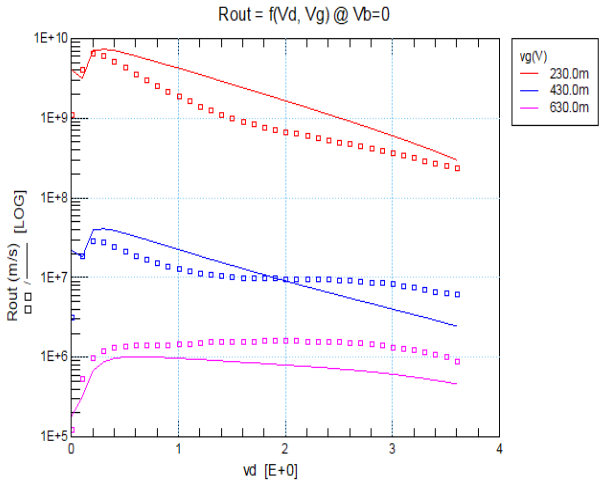


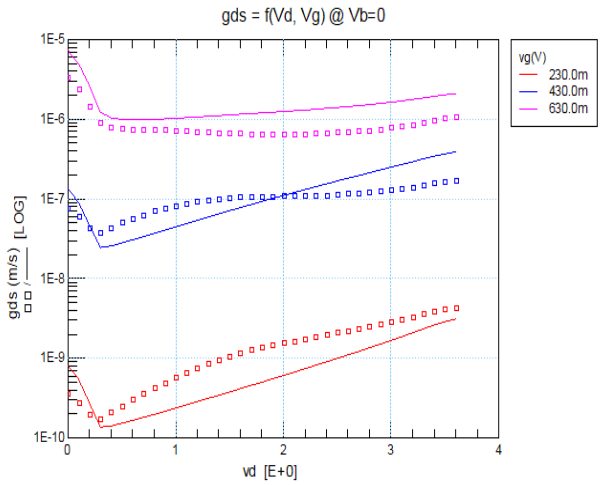




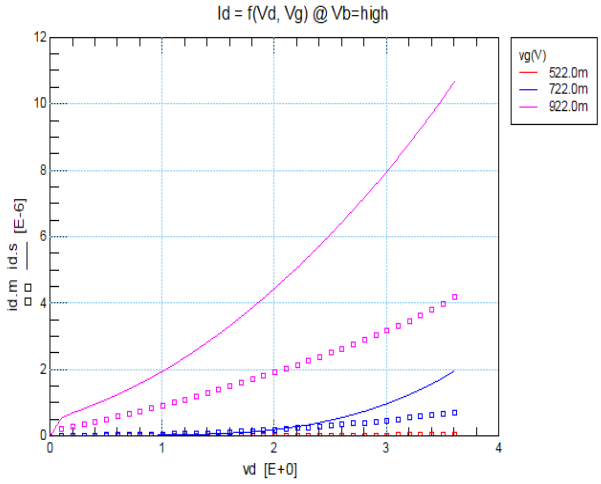
idvd_vth, Vb = 0V, T = 27°C

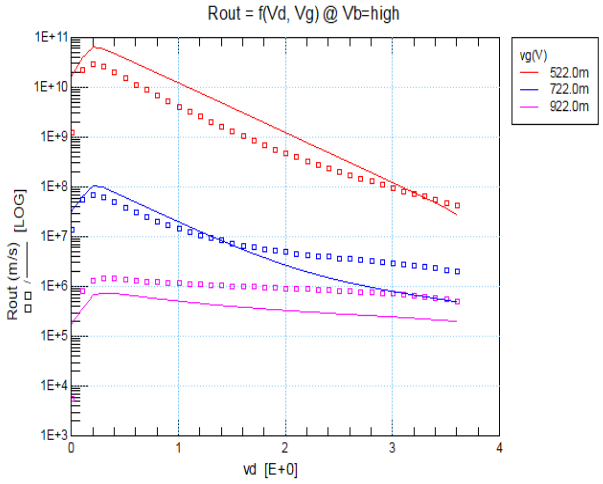


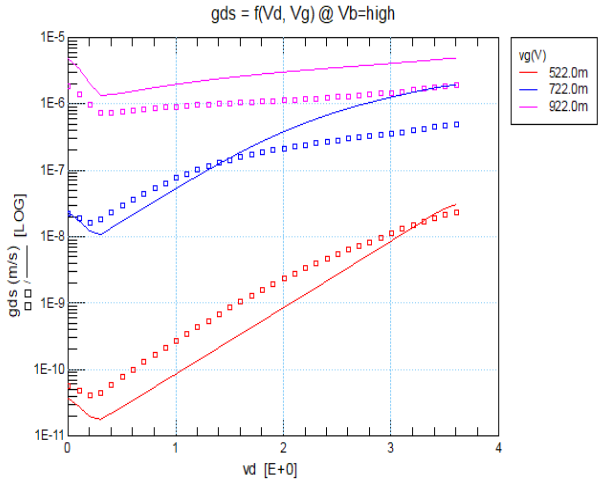




idvd_vth_vbmin, Vb = -3.6V, T = 27°C

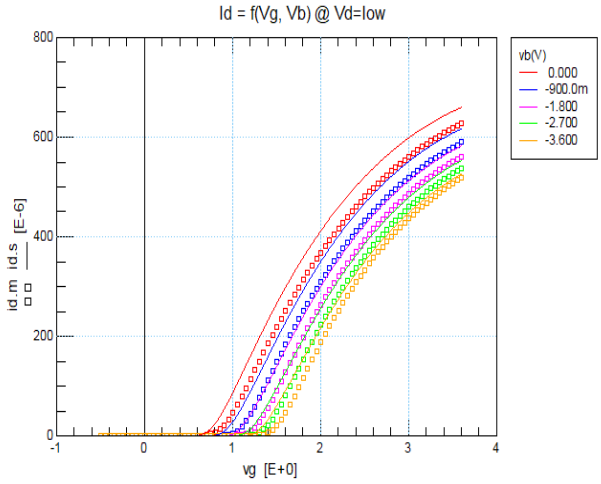


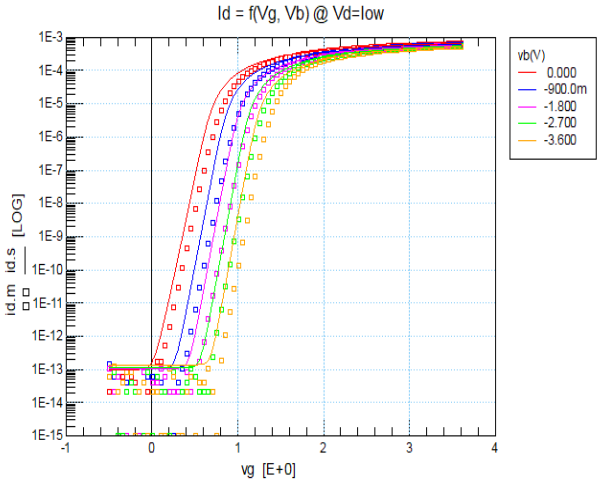


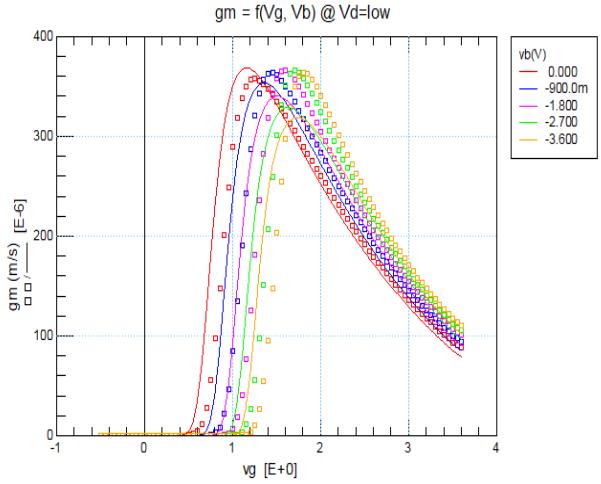


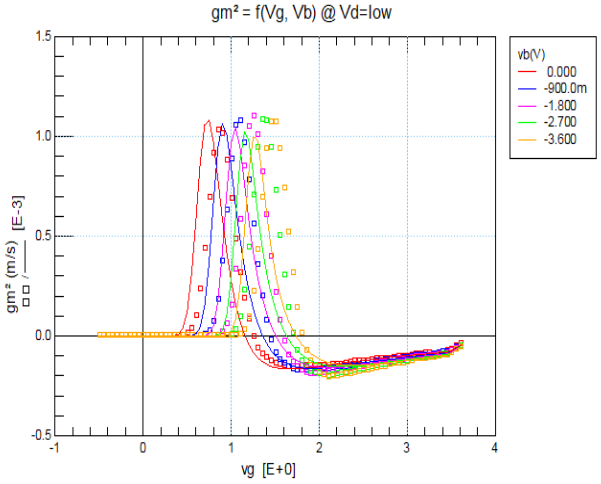
4.8 W10u0_L0u5_S556_4

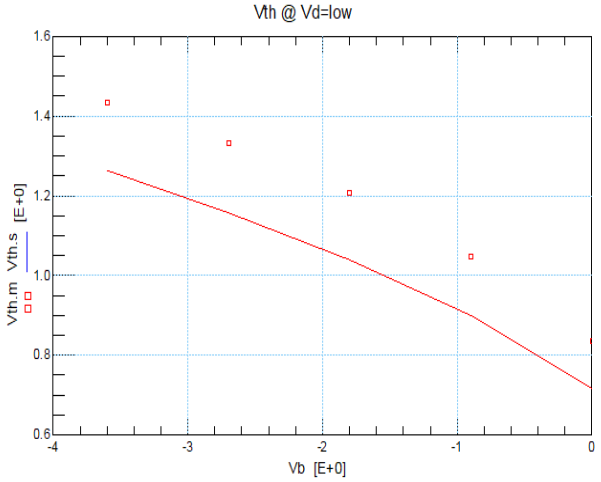
idvg, Vd = 0.1V, T = 27°C

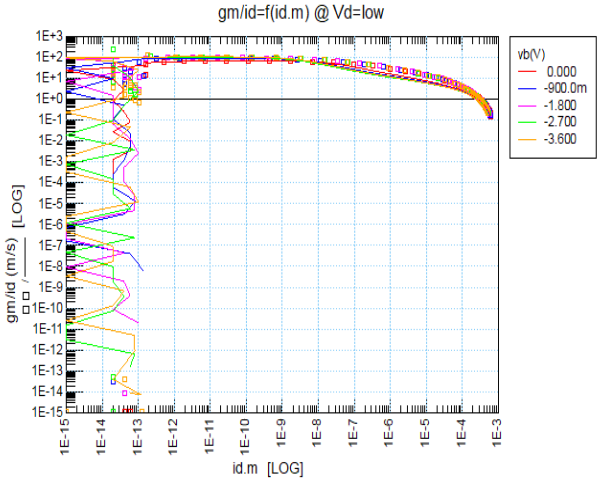


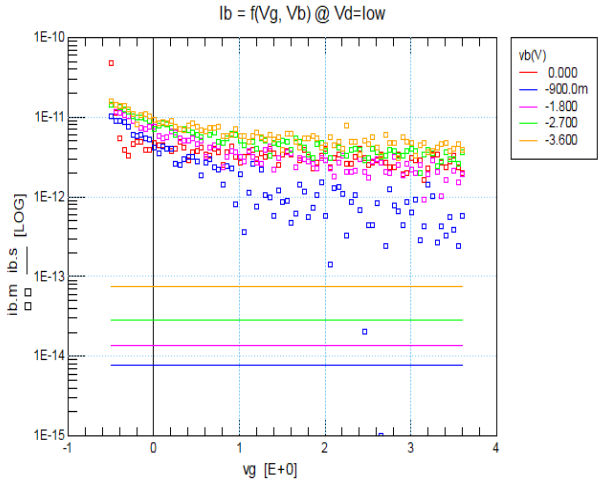


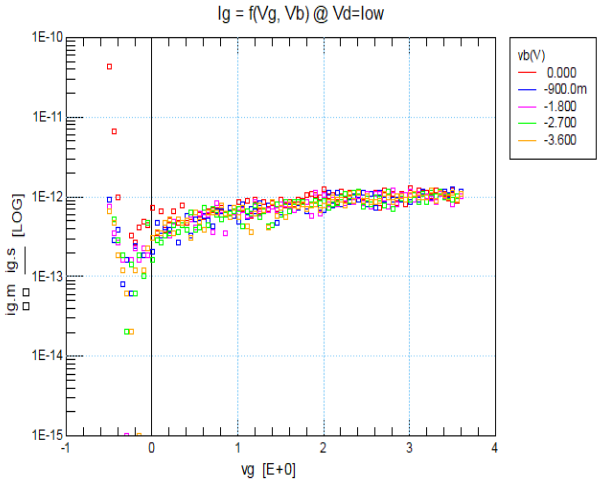




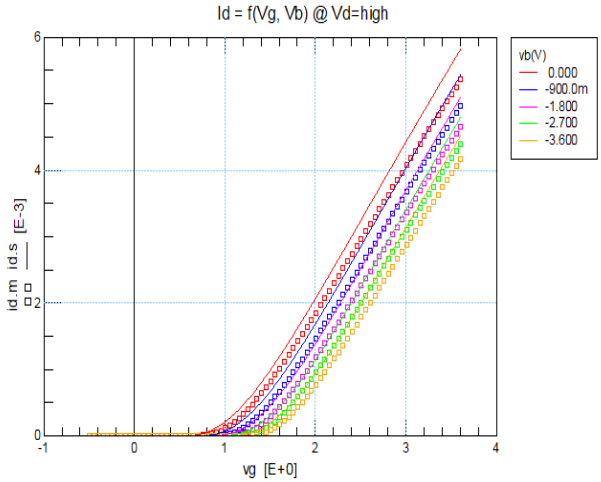


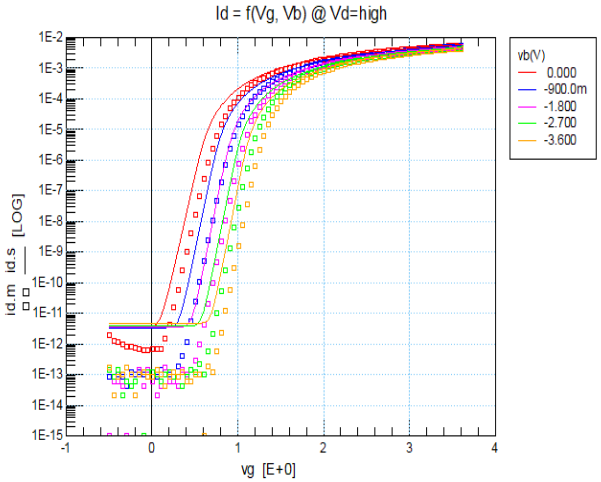


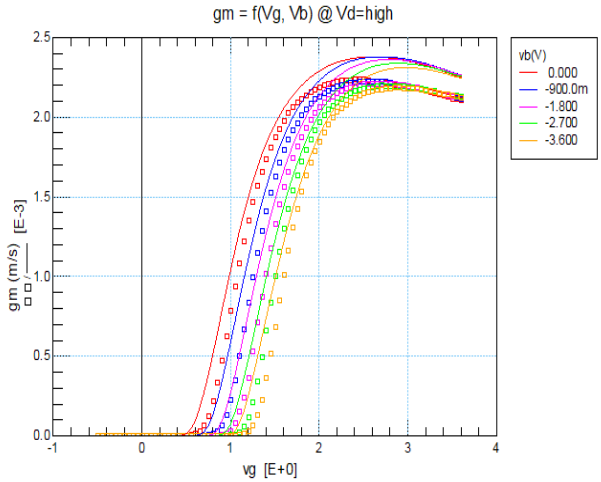


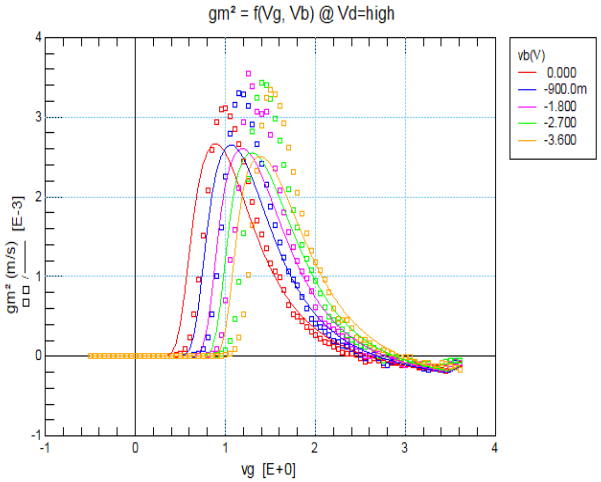


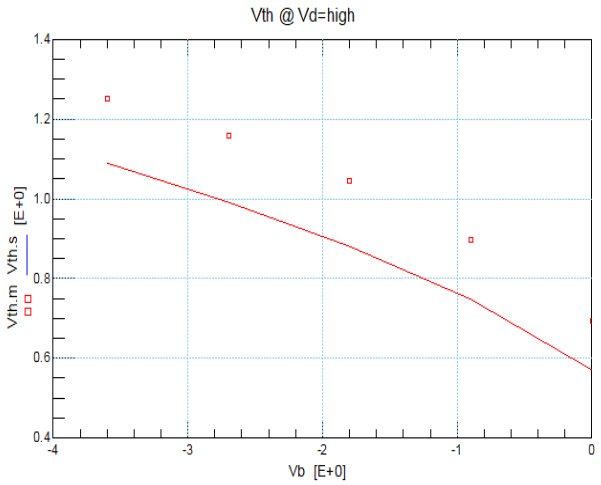
idvg, Vd = 3.3V, T = 27°C

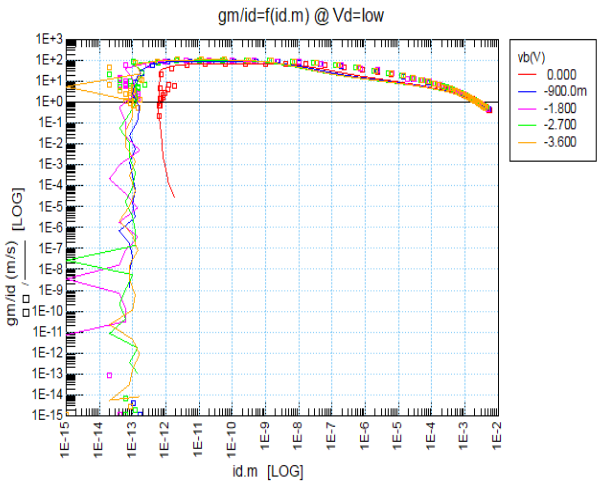


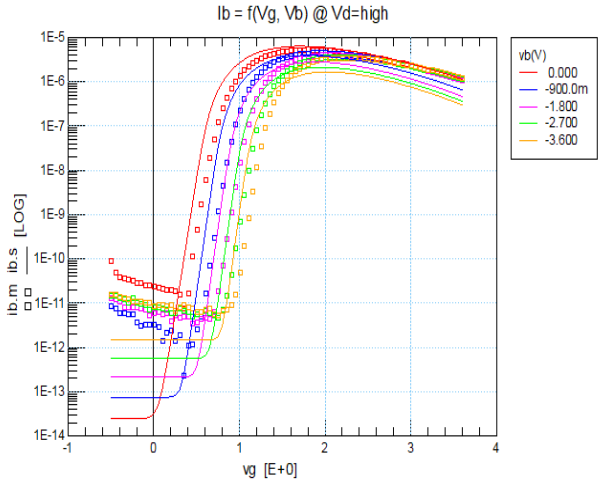


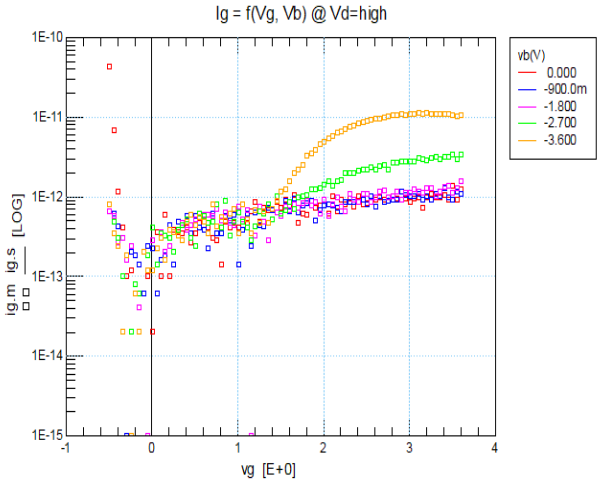




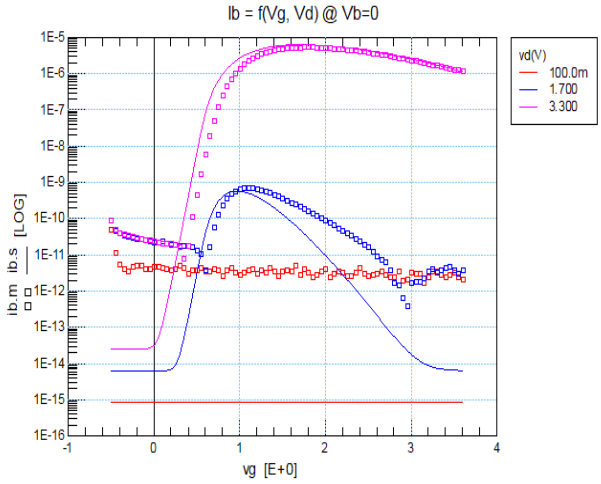


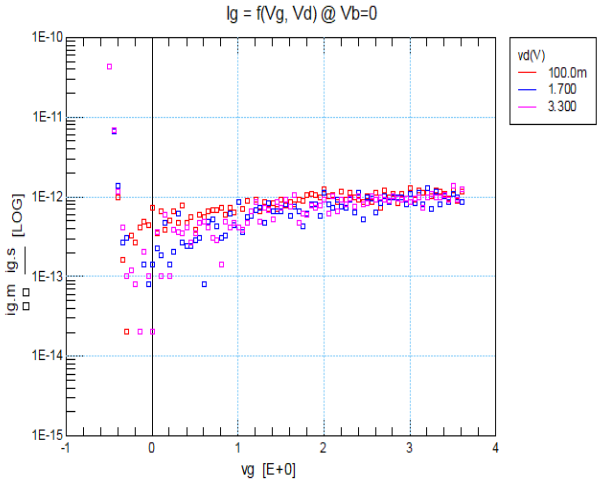




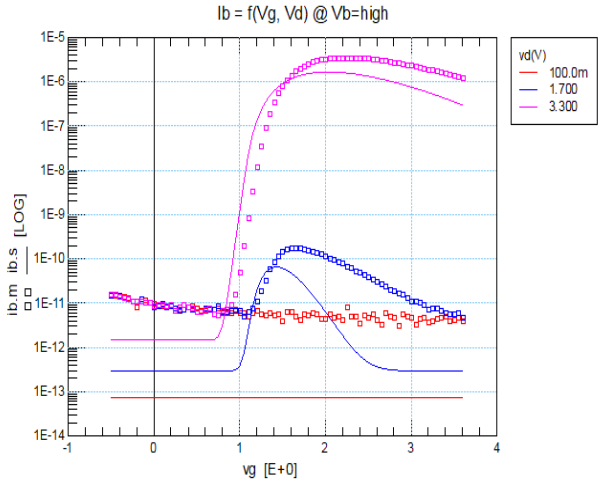


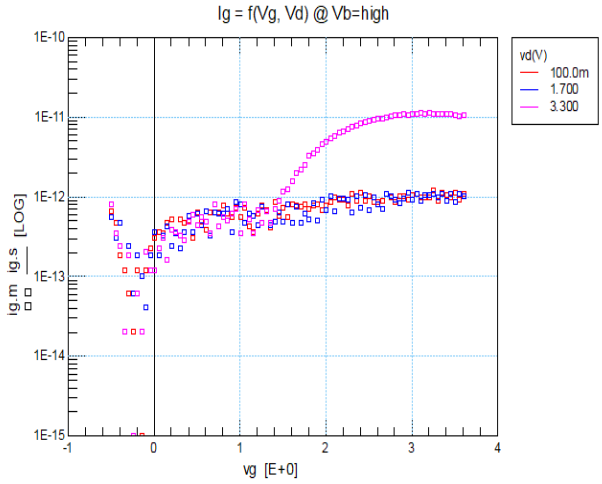
idvg, Ib,Ig, Vb = 0V, T = 27°C



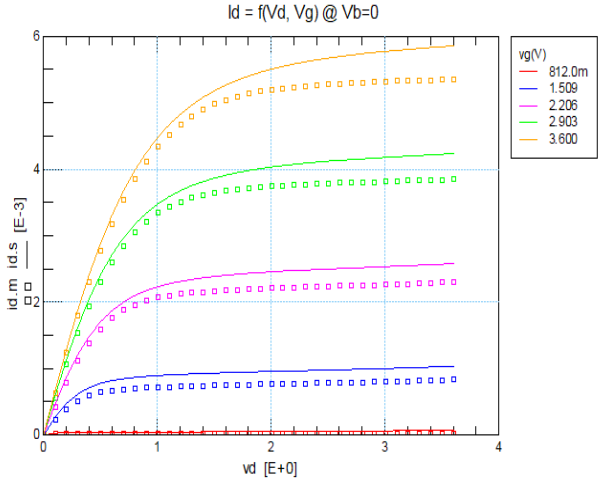


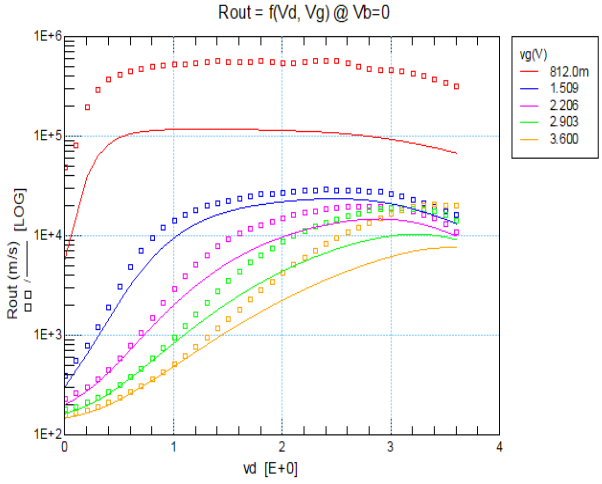
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

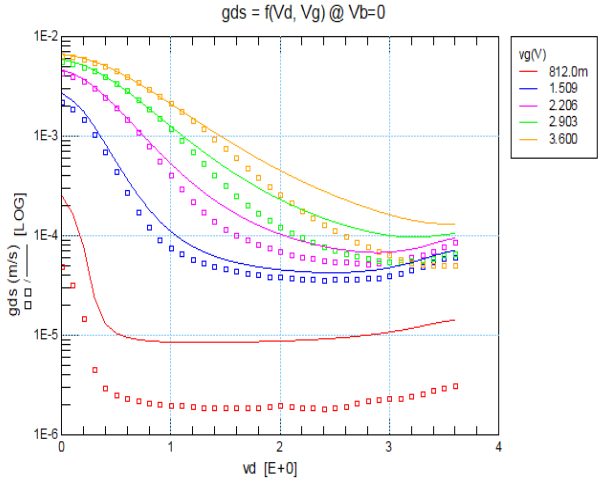




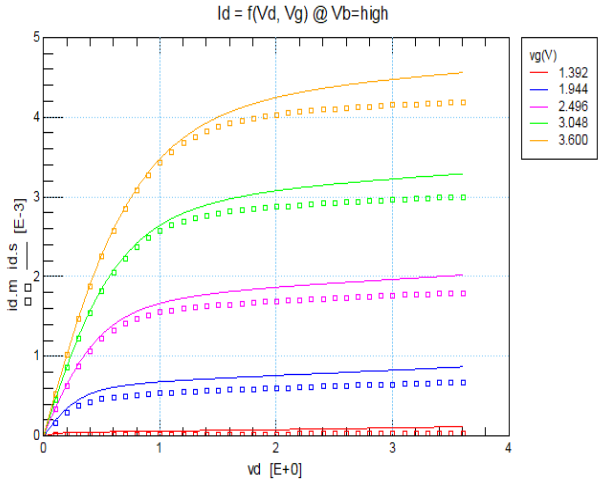
idvd, Vb = 0V, T = 27°C

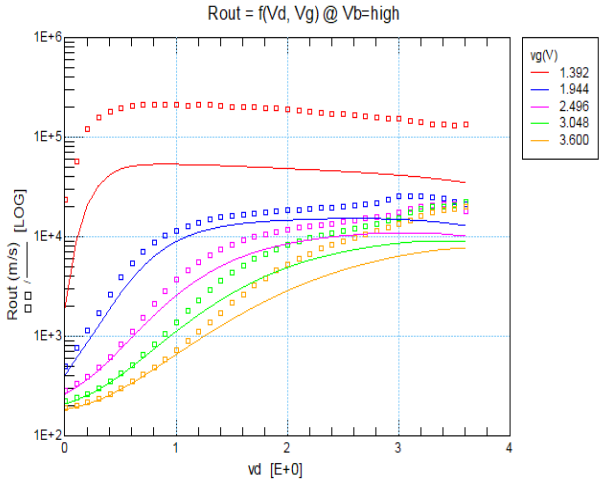


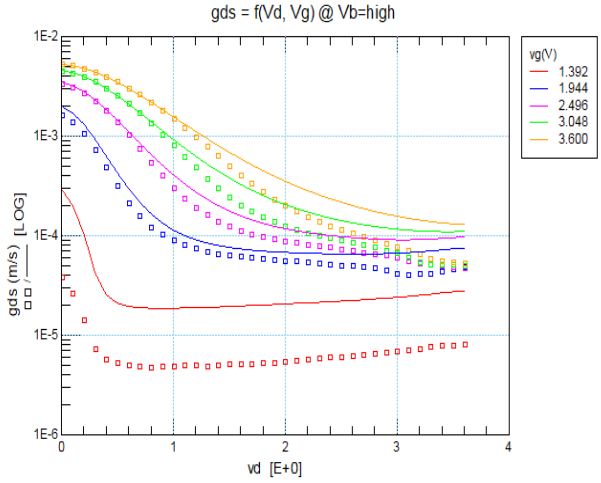




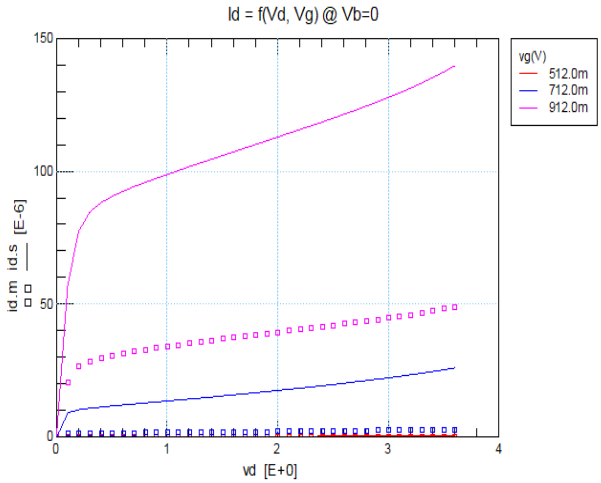
idvd_vbmin, Vb = -3.6V, T = 27°C

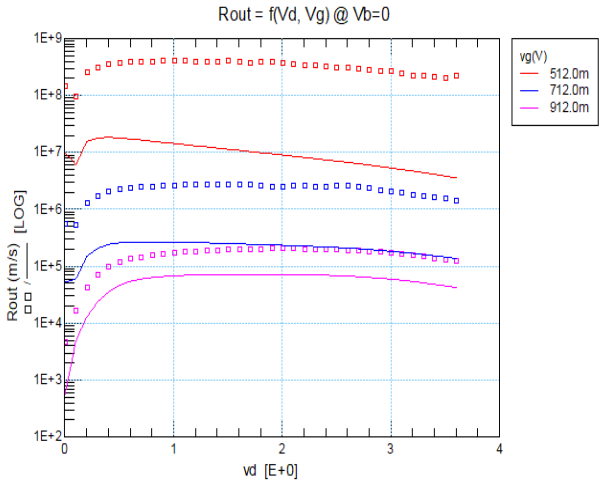


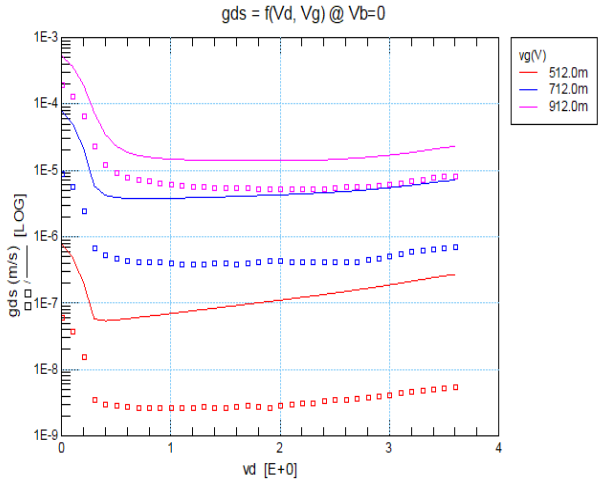




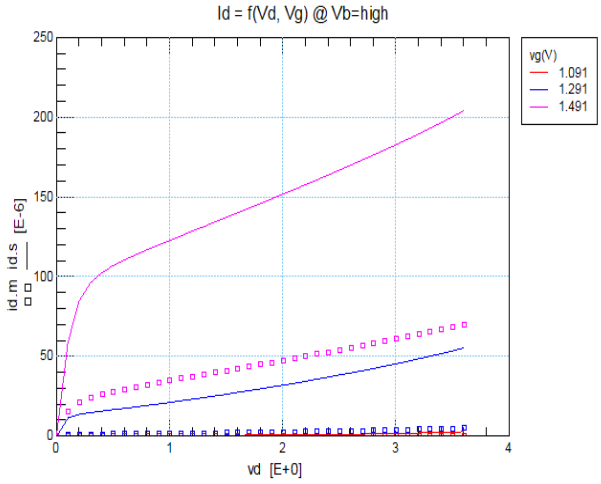
idvd_vth, Vb = 0V, T = 27°C

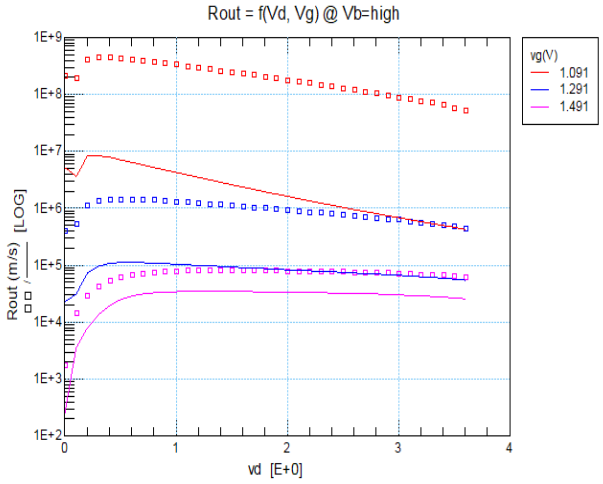


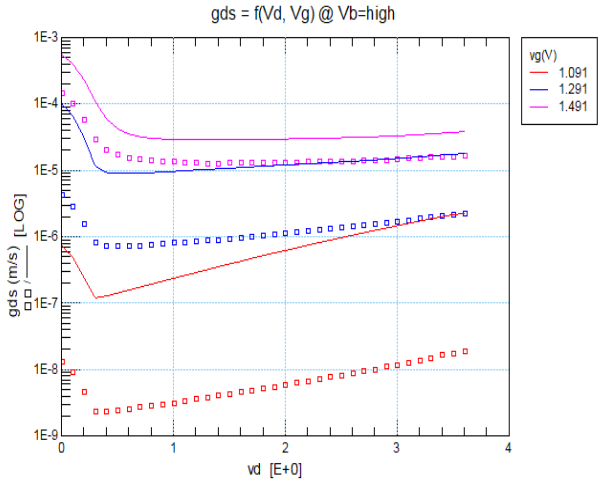




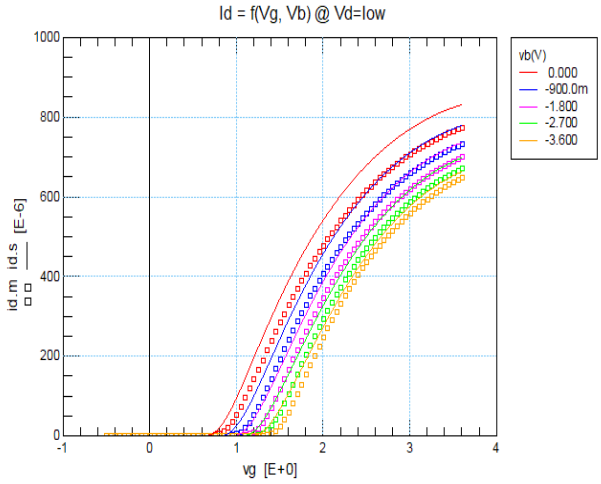
idvd_vth_vbmin, Vb = -3.6V, T = 27°C

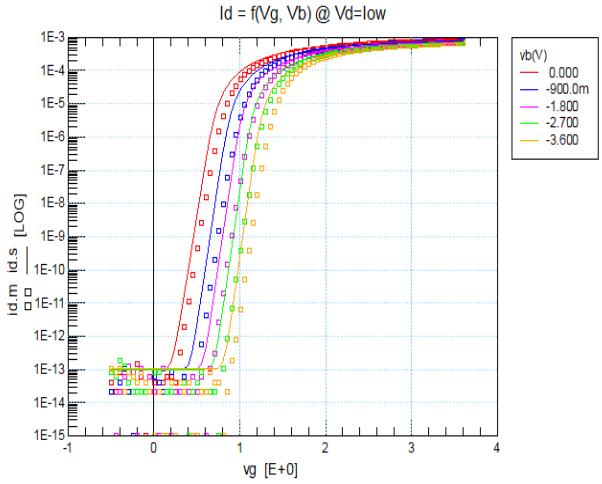


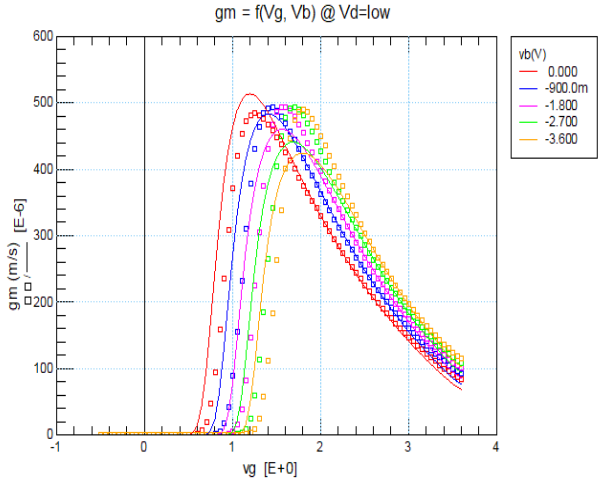


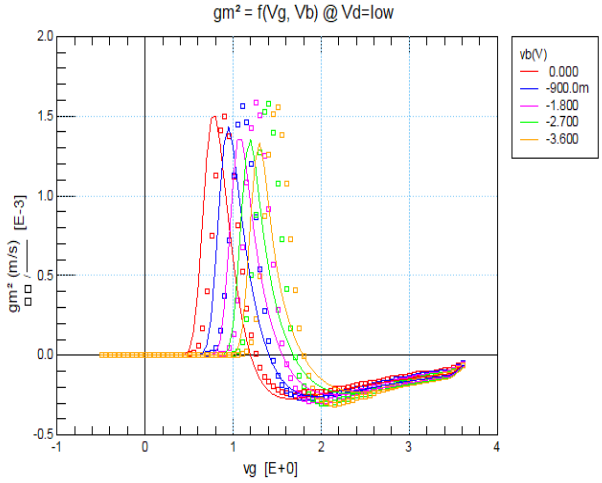


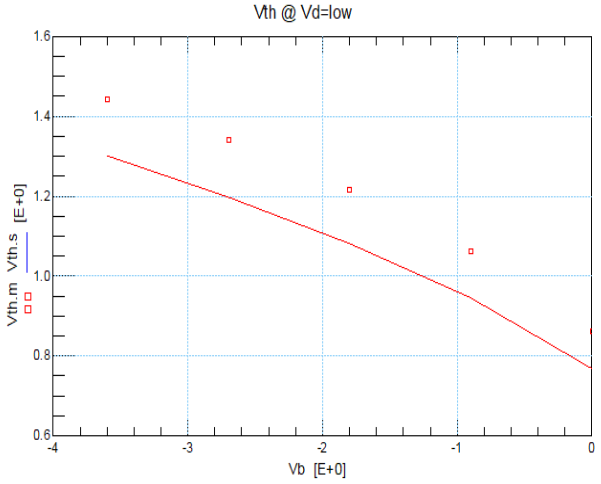
idvg, Vd = 0.1V, T = -40°C

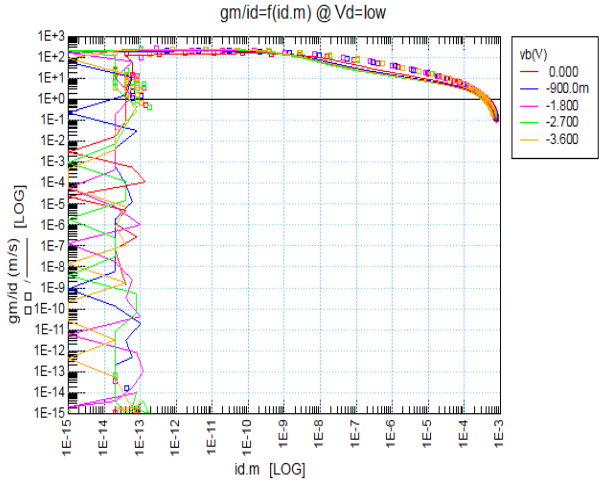


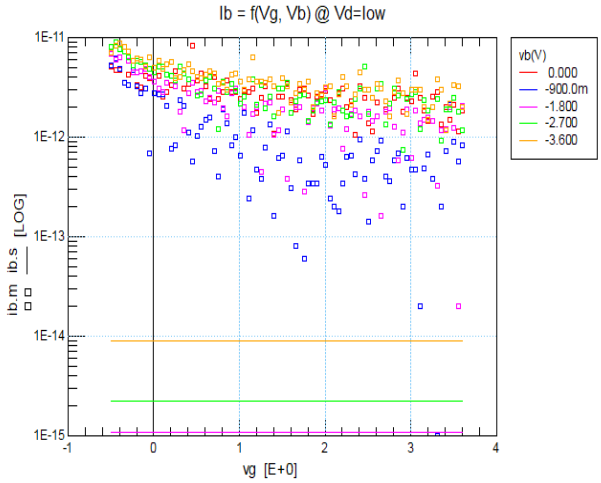


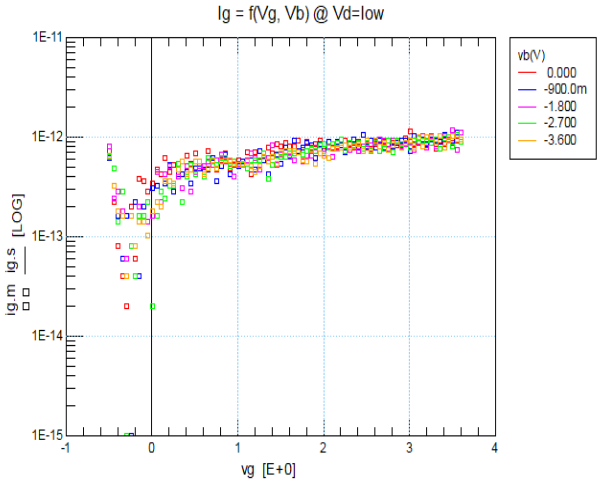




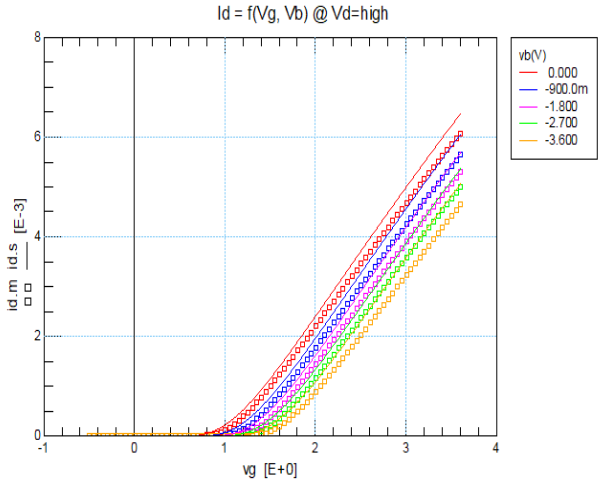


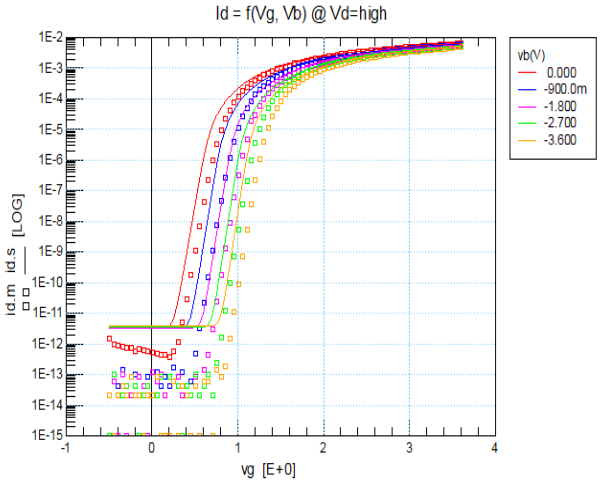


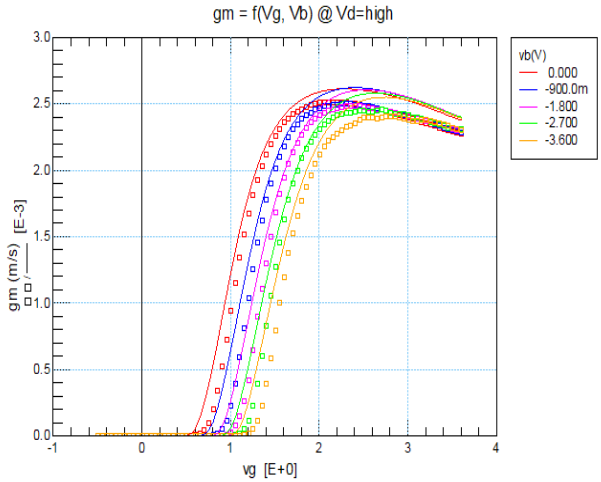


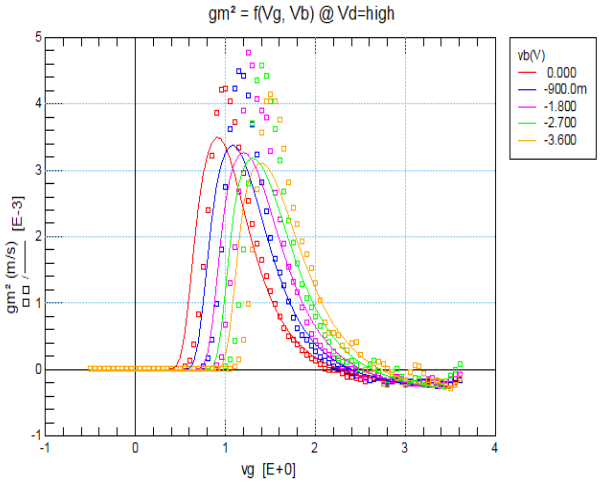


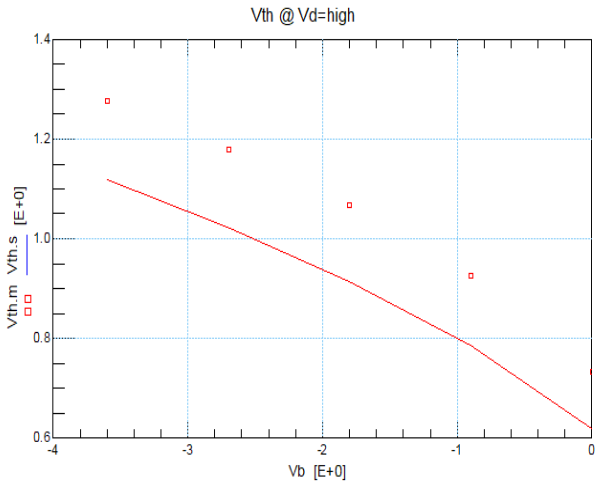
idvg, Vd = 3.3V, T = -40°C

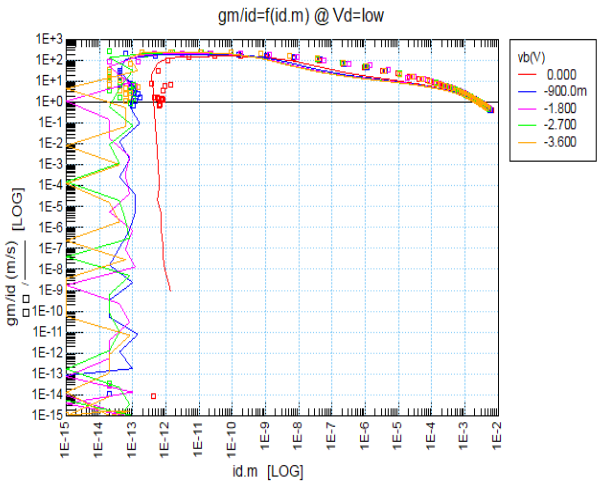


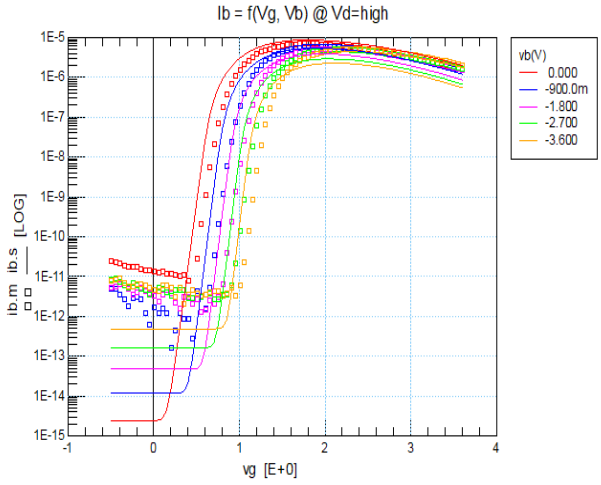


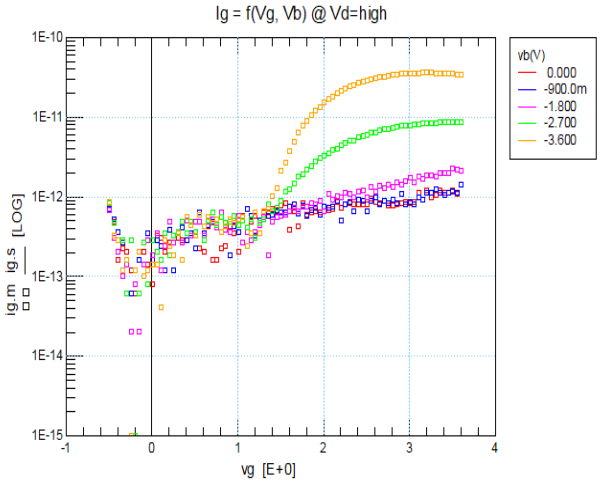




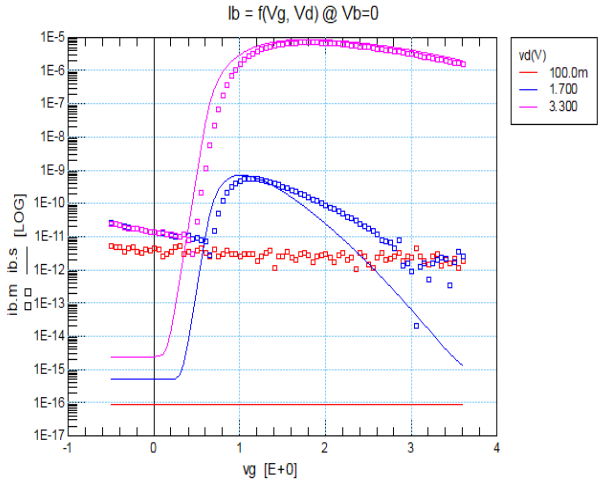


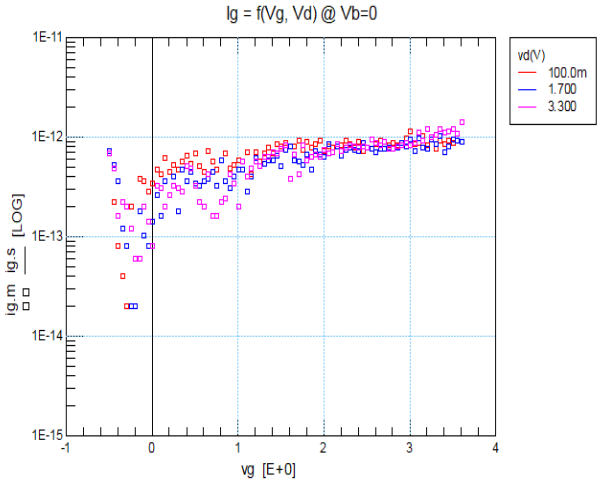




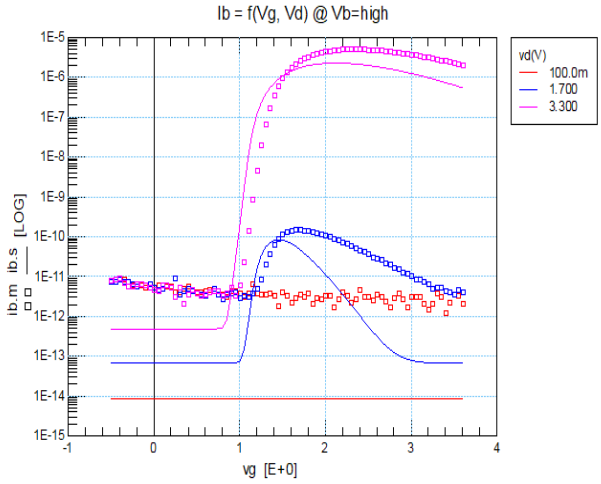


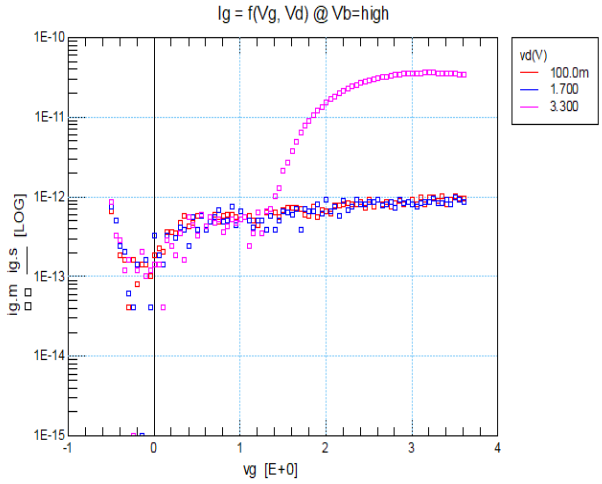
idvg, Ib,Ig, Vb = 0V, T = -40°C



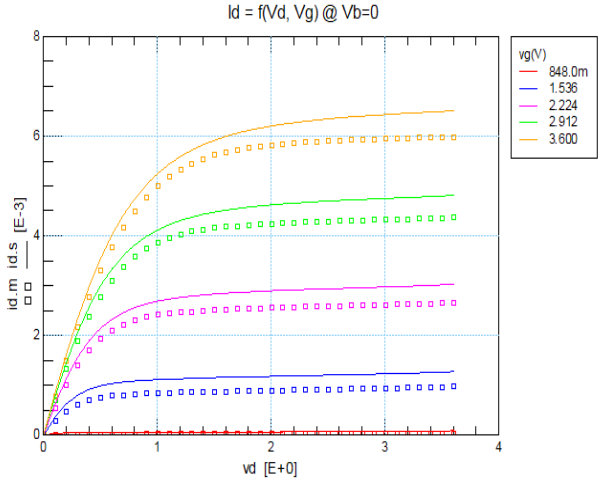


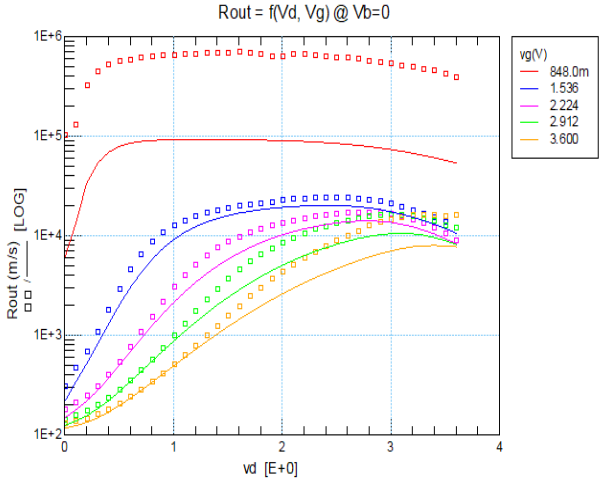
idvg, Ib,Ig, Vb = -3.6V, T = -40°C

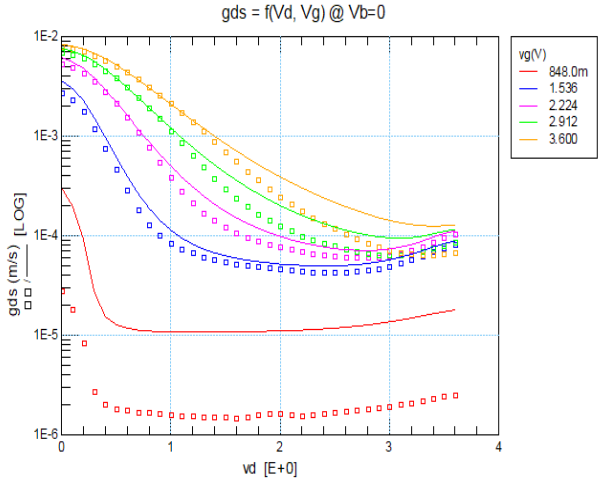




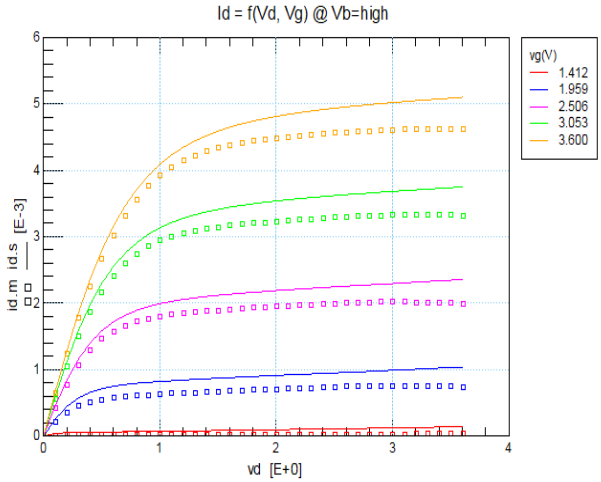
idvd, Vb = 0V, T = -40°C

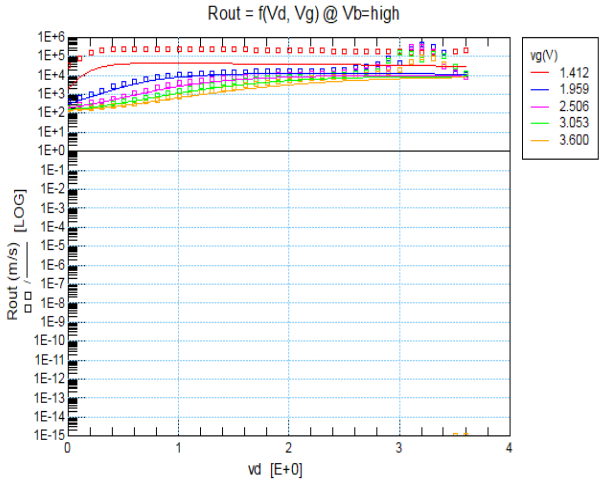


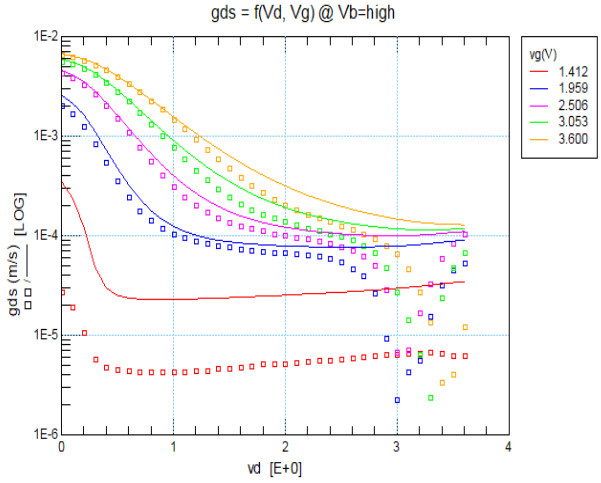




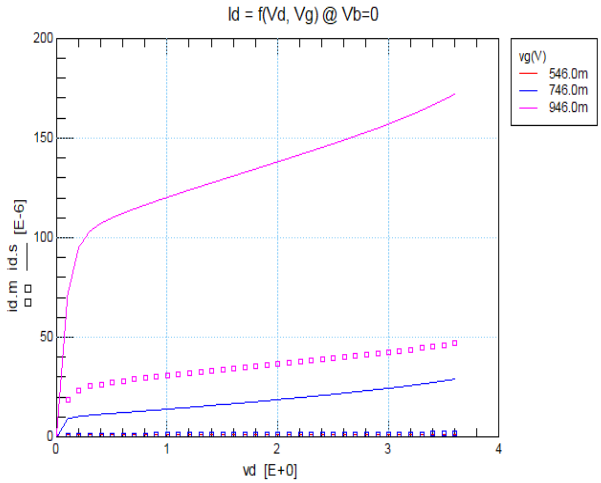
idvd_vbmin, Vb = -3.6V, T = -40°C

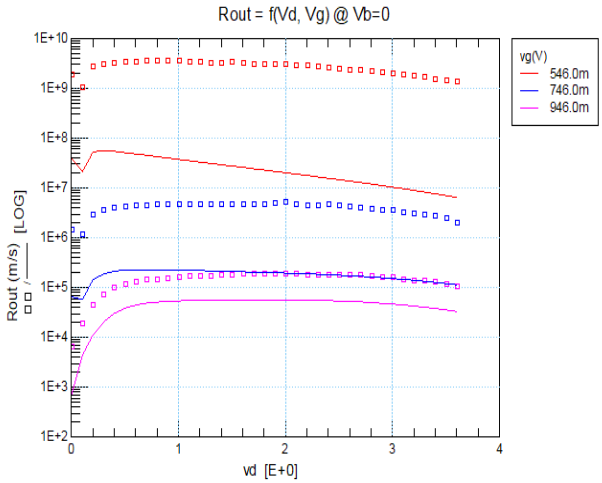


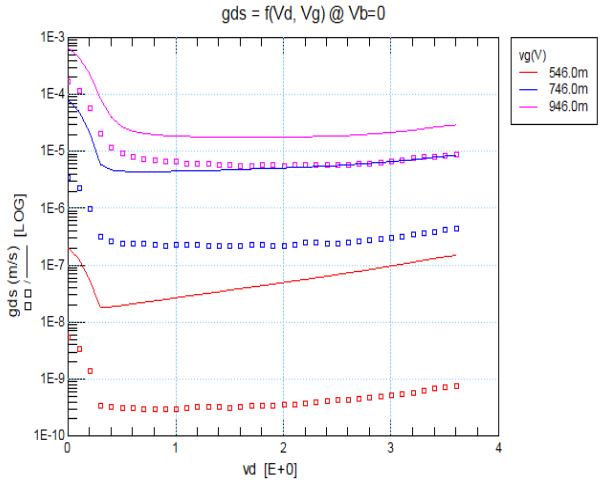




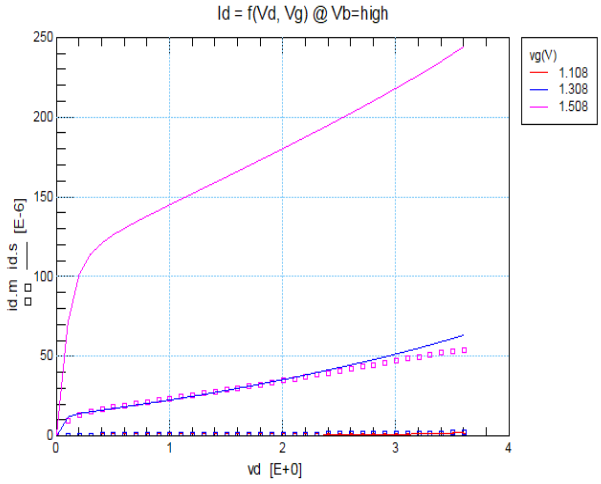
idvd_vth, Vb = 0V, T = -40°C

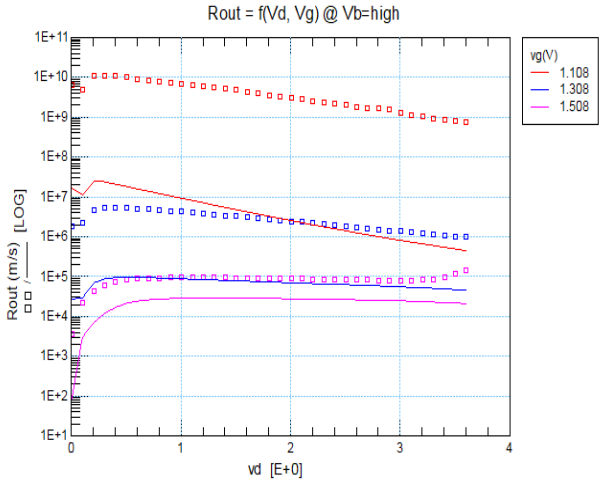


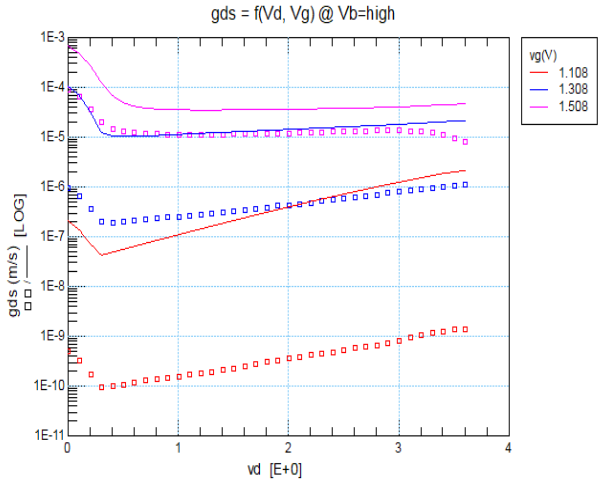




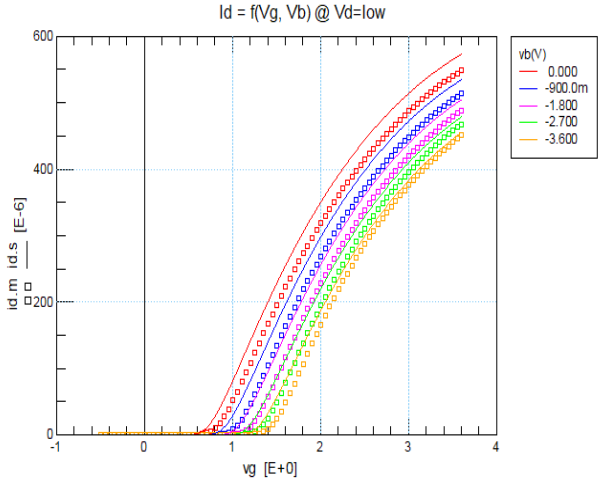
idvd_vth_vbmin, Vb = -3.6V, T = -40°C

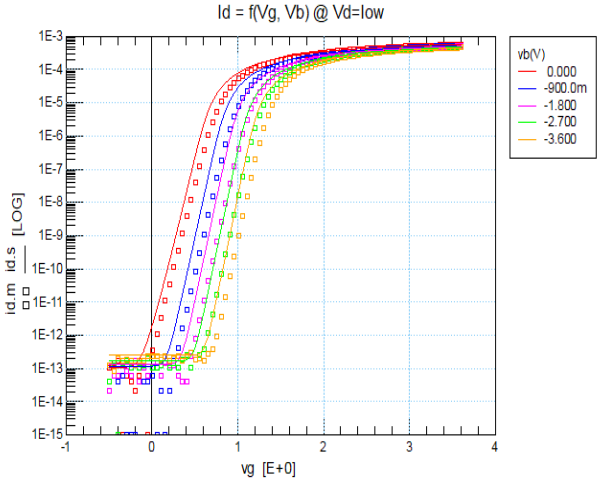


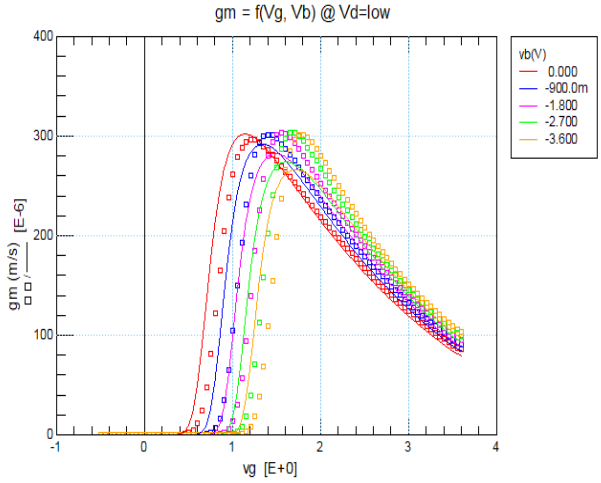


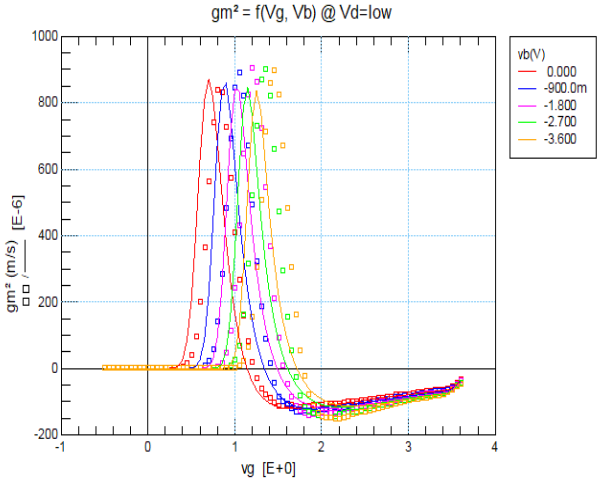


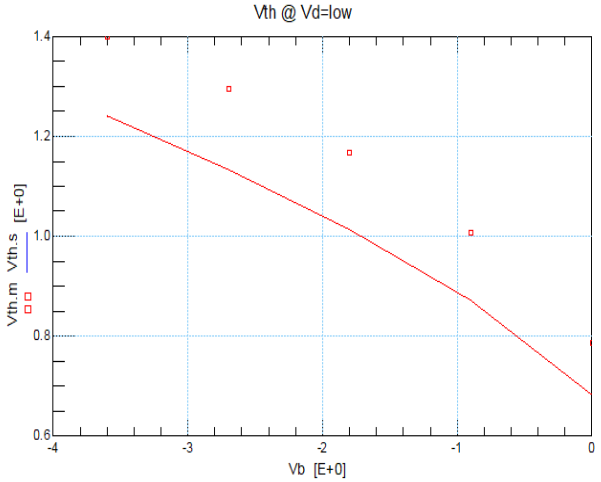
idvg, Vd = 0.1V, T = 70°C

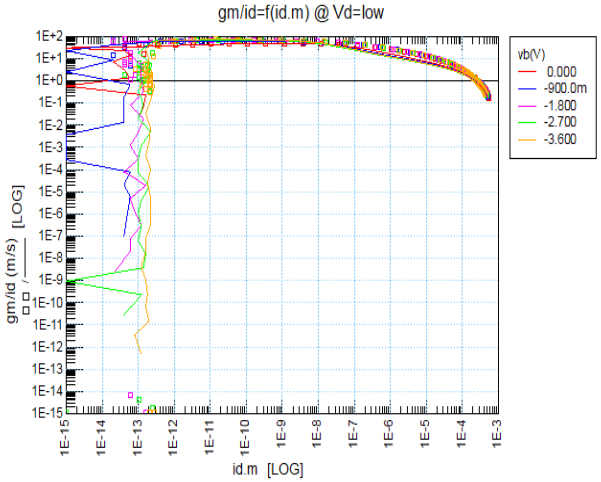


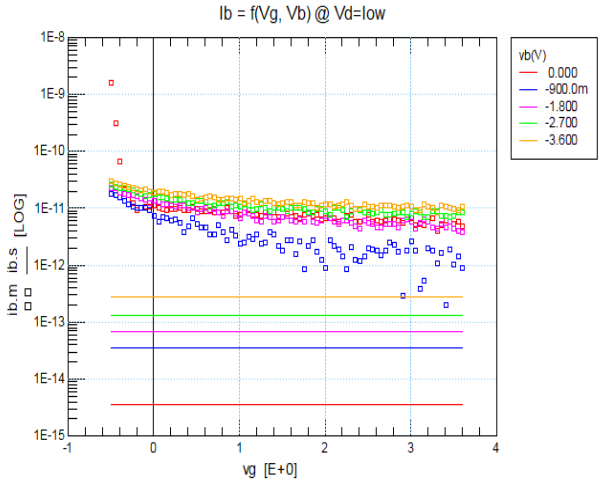


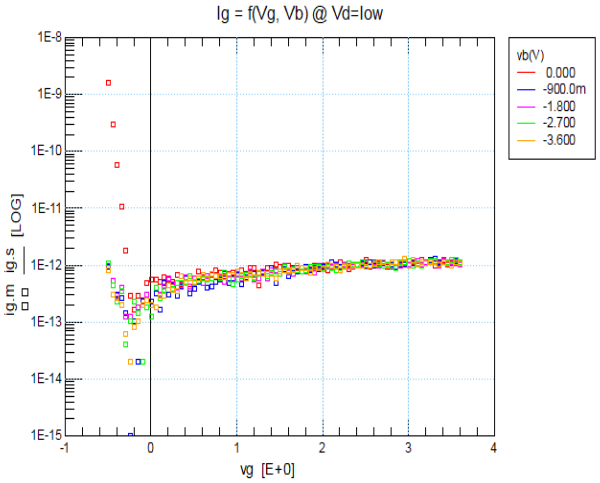




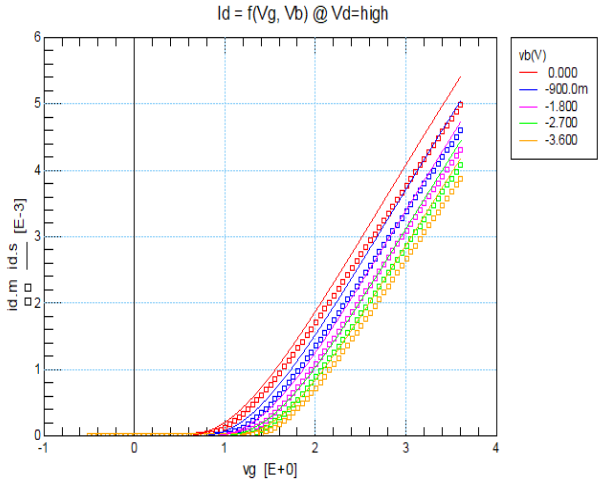


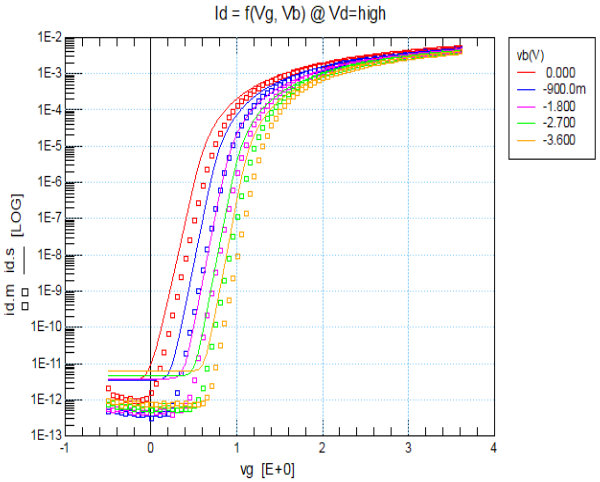


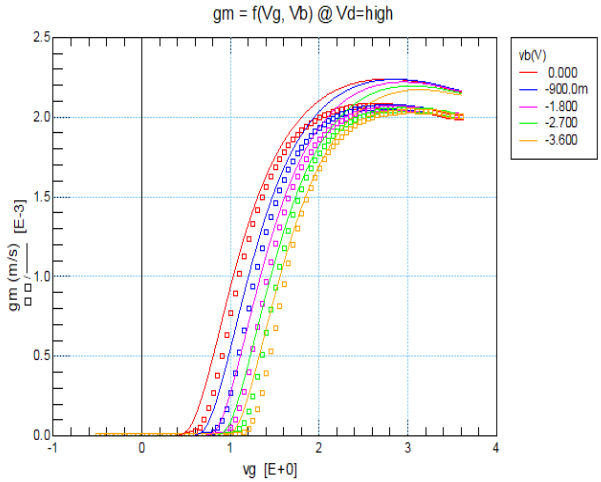


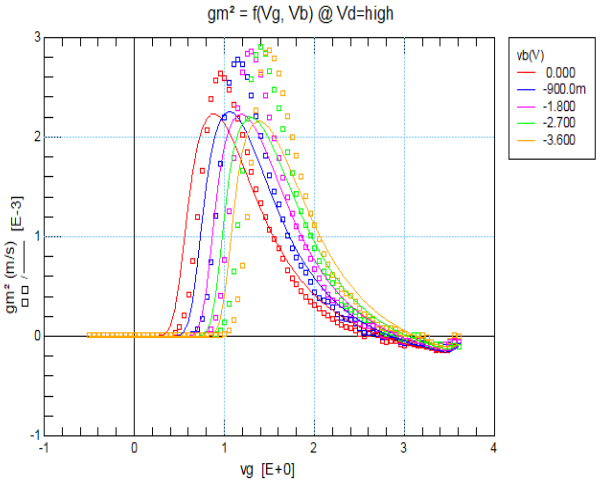


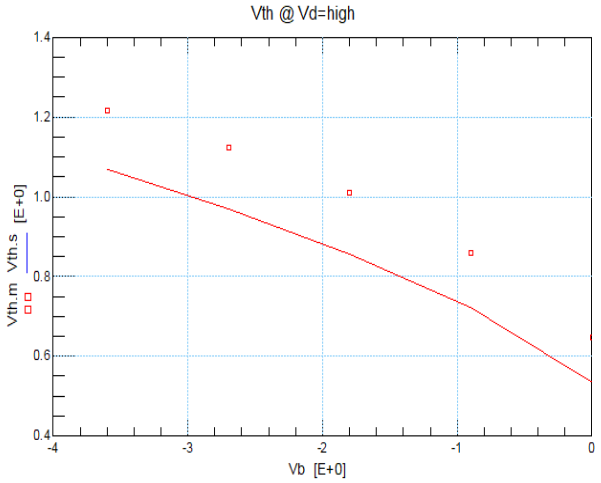
idvg, Vd = 3.3V, T = 70°C

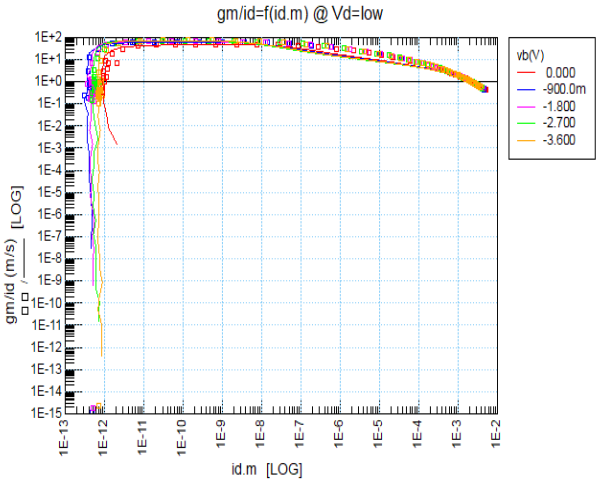


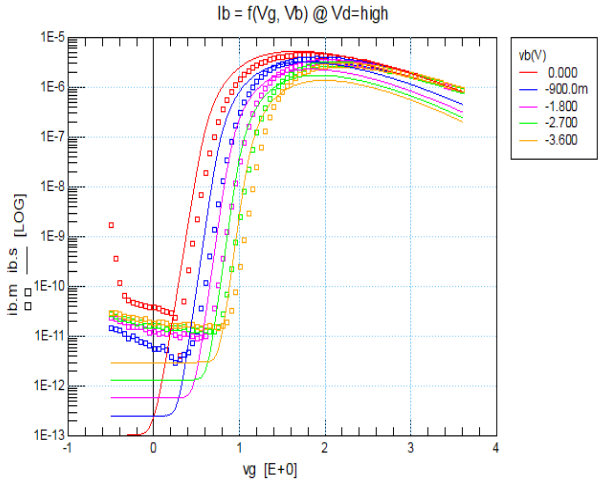


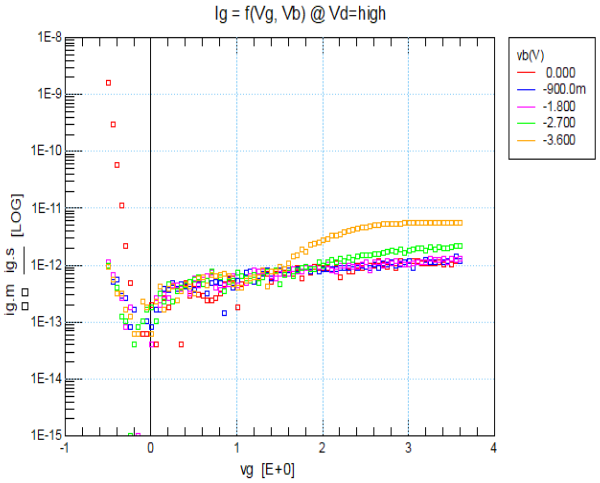




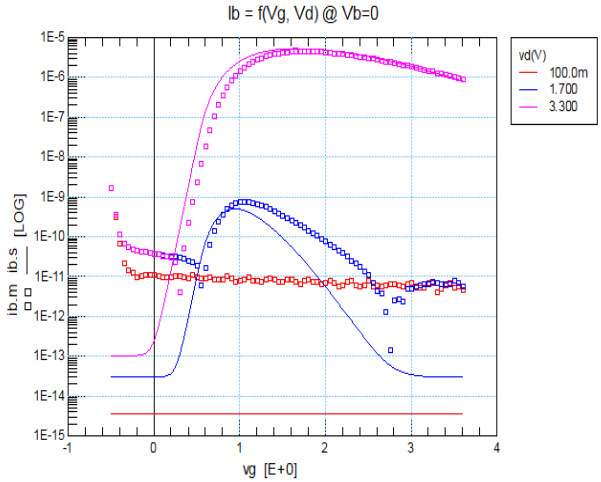


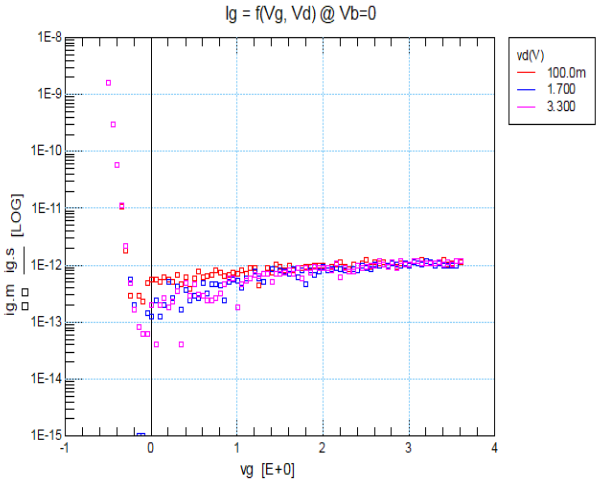




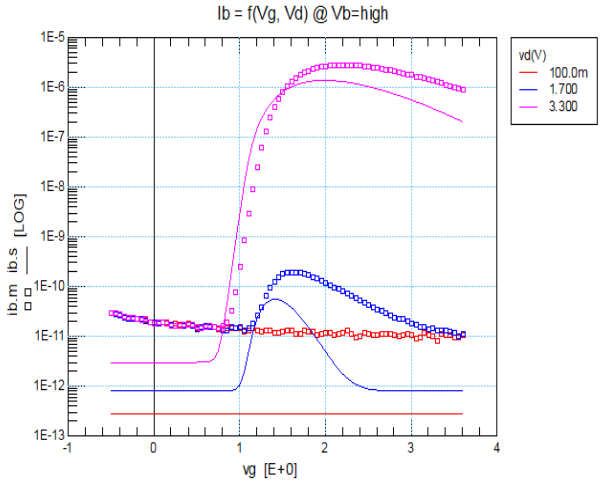


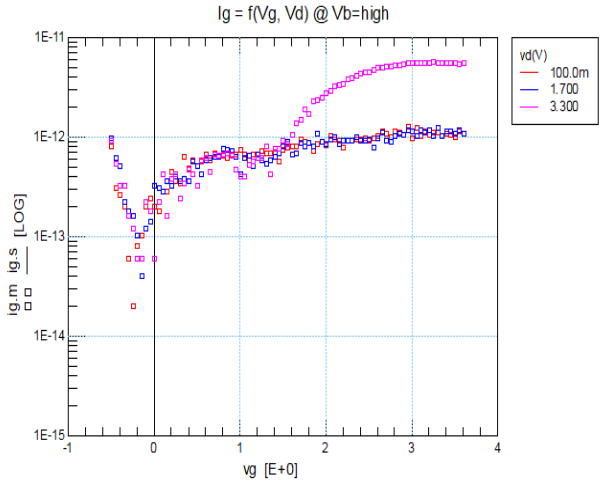
idvg, Ib,Ig, Vb = 0V, T = 70°C



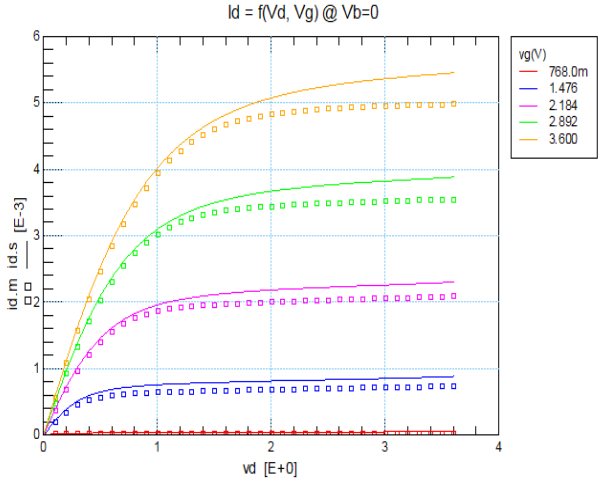


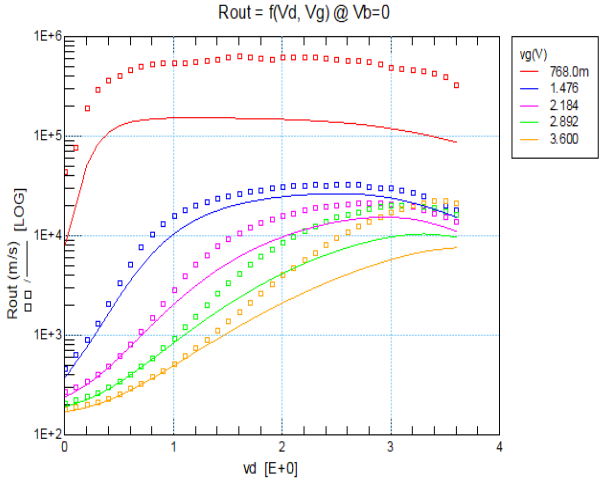
idvg, Ib,Ig, Vb = -3.6V, T = 70°C

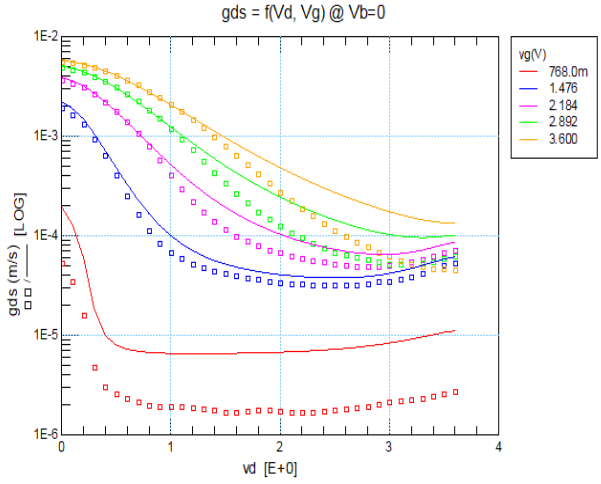




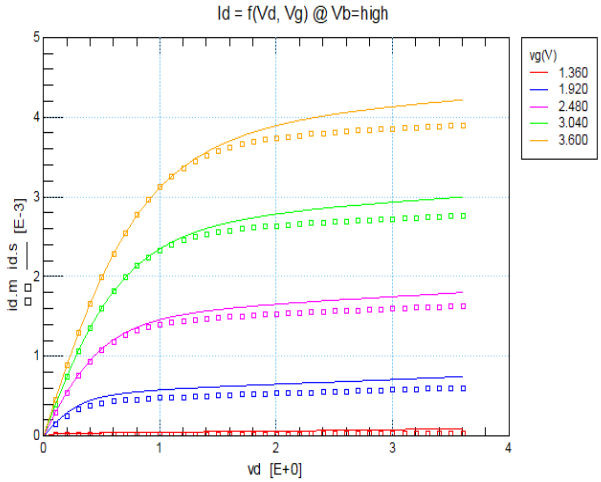
idvd, Vb = 0V, T = 70°C

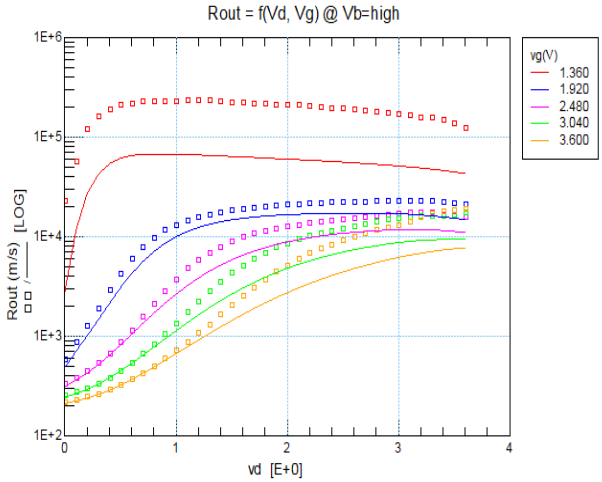


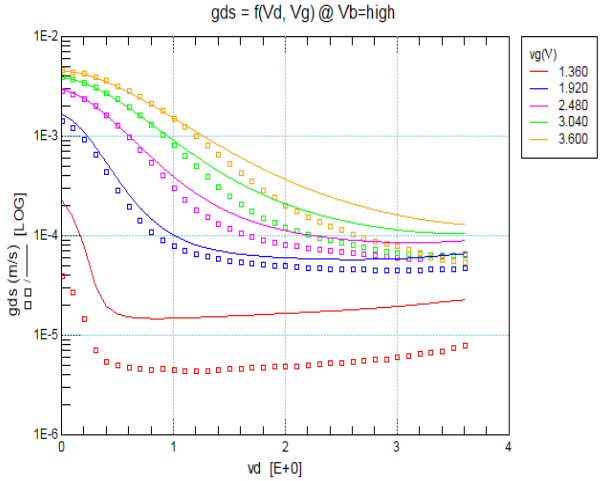




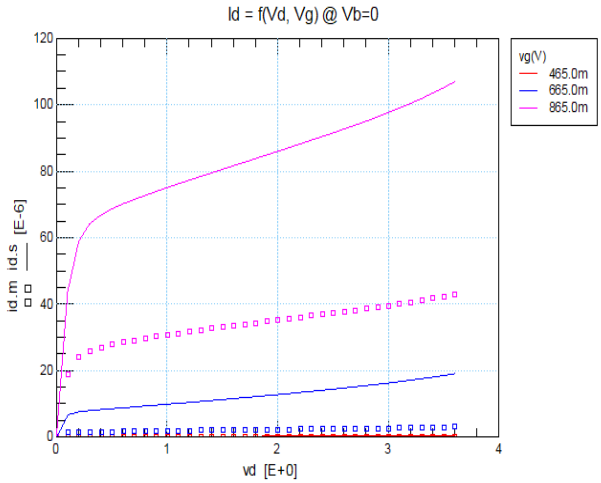
idvd_vbmin, Vb = -3.6V, T = 70°C

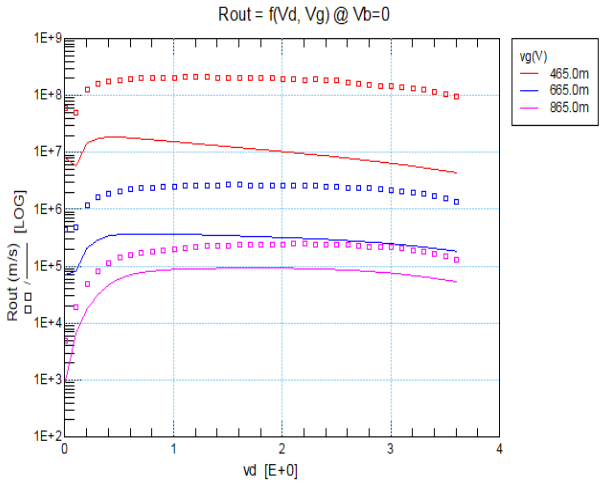


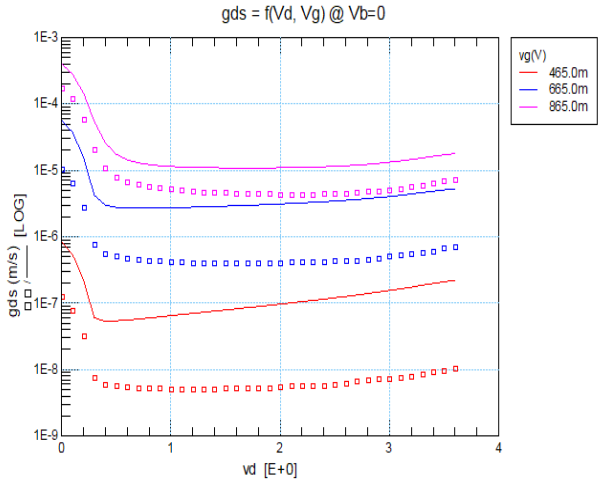




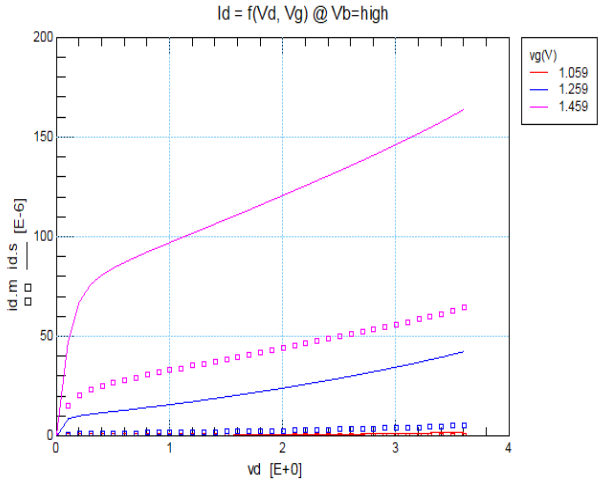
idvd_vth, Vb = 0V, T = 70°C

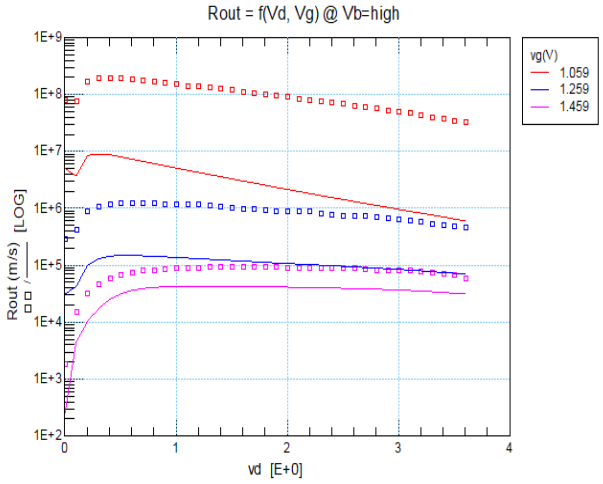


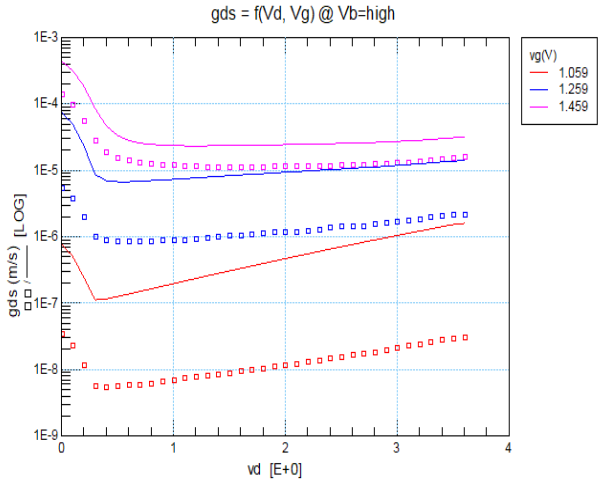




idvd_vth_vbmin, Vb = -3.6V, T = 70°C

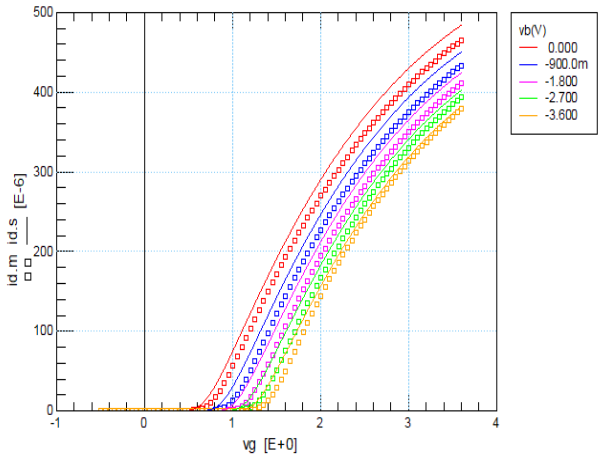




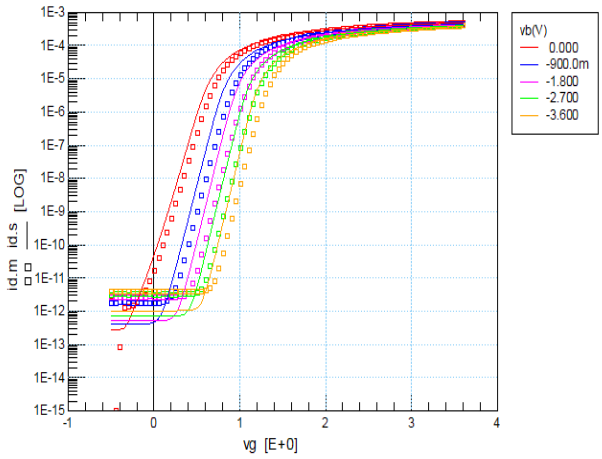


idvg, Vd = 0.1V, T = 125°C

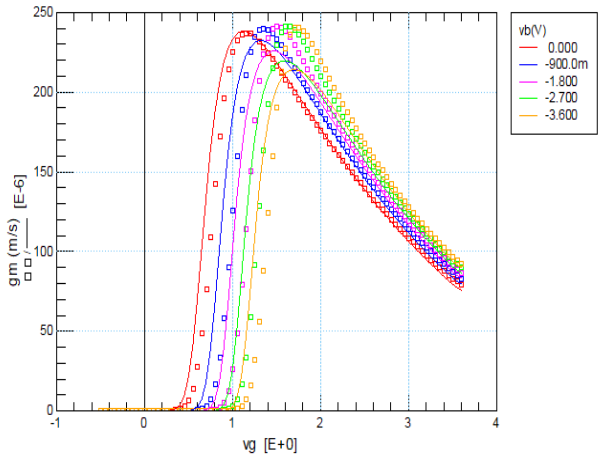
Id = f(Vg, Vb) @ Vd=low



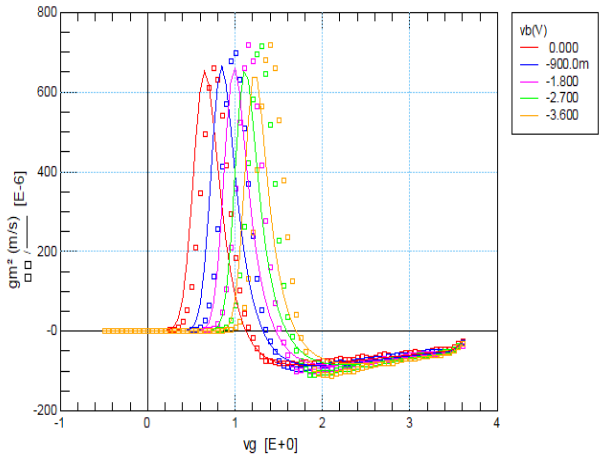
Id = f(Vg, Vb) @ Vd=low



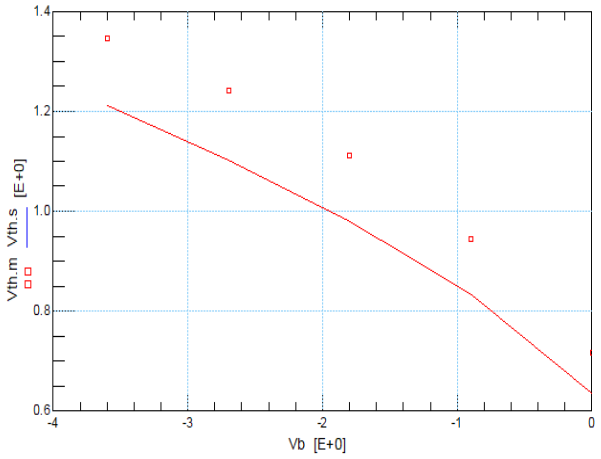
gm = f(Vg, Vb) @ Vd=low



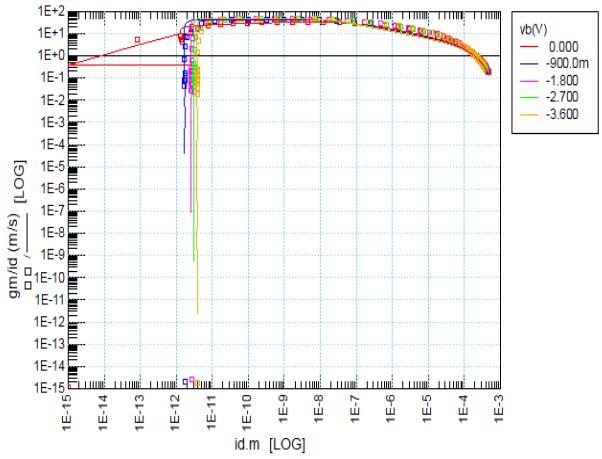
gm² = f(Vg, Vb) @ Vd=low

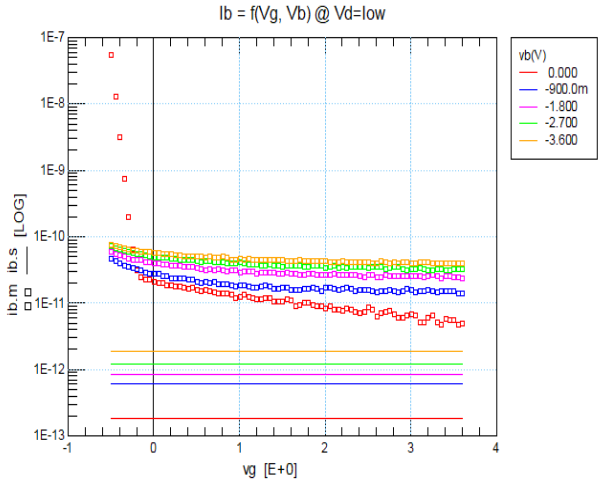


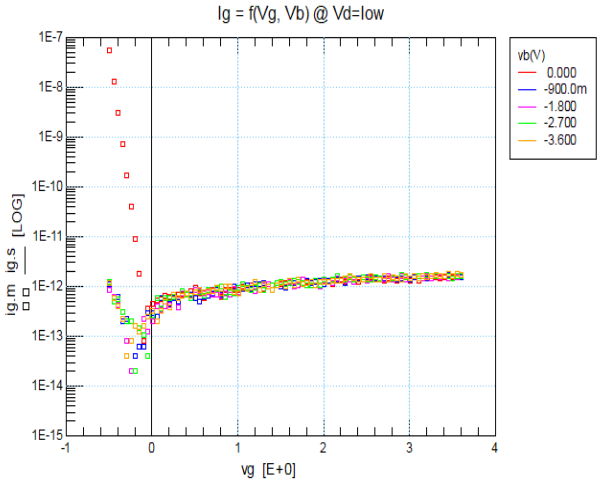
Vth @ Vd=low



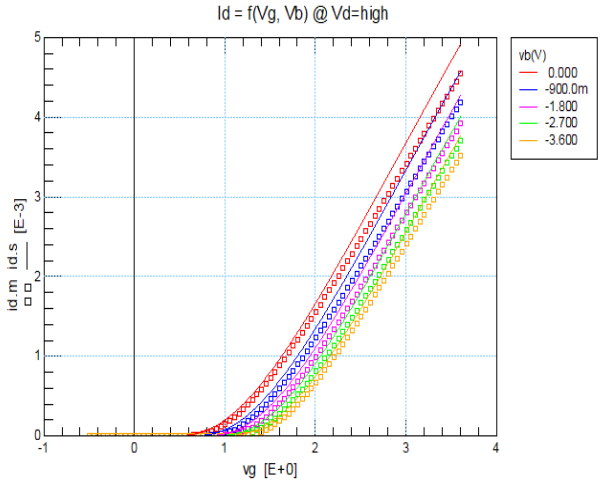
gm/id=f(id.m) @ Vd=low

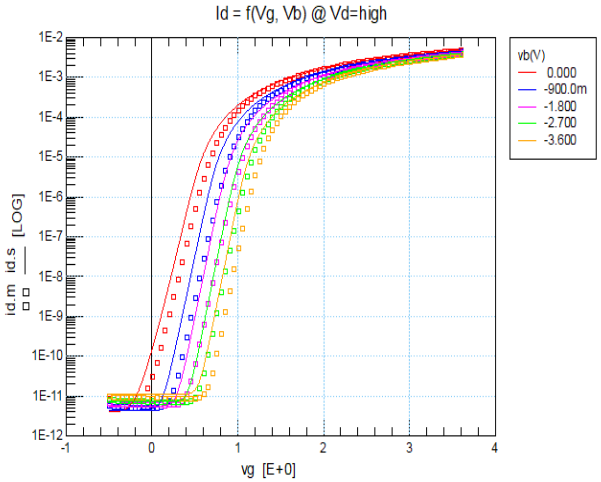


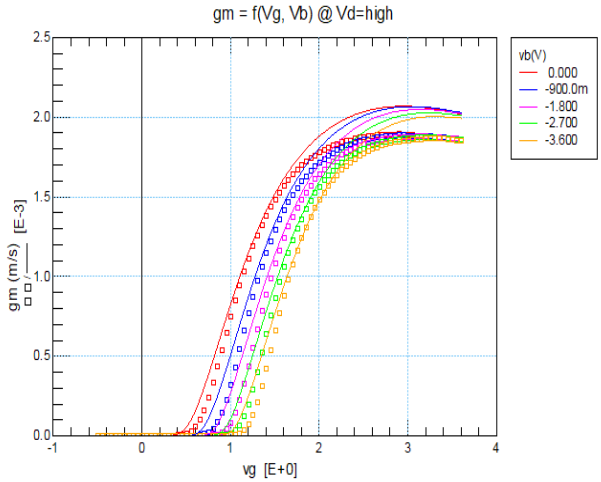


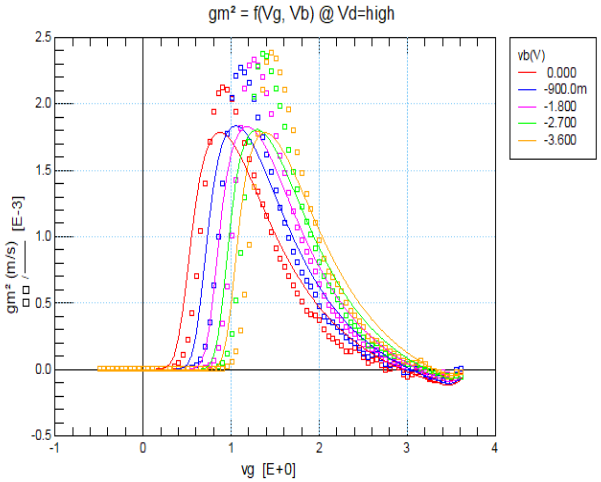


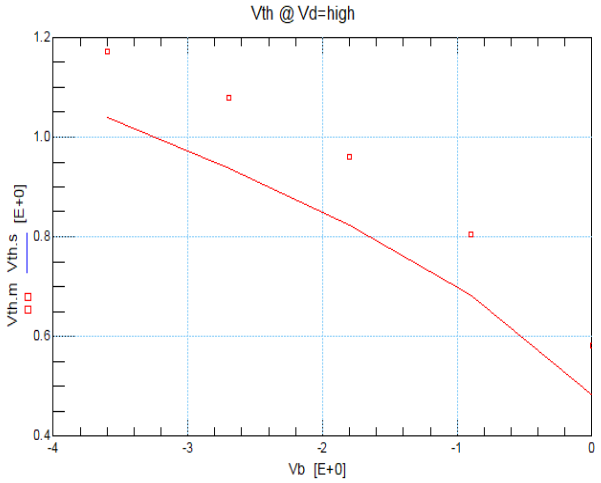
idvg, Vd = 3.3V, T = 125°C

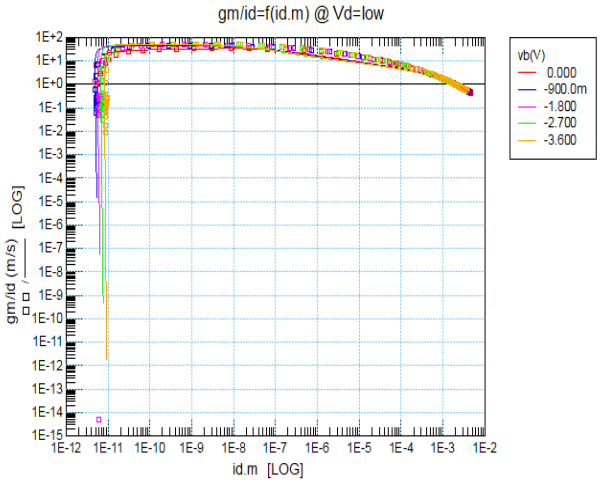


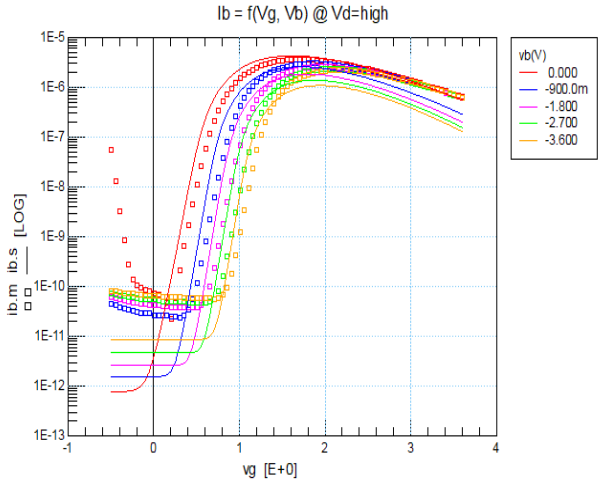


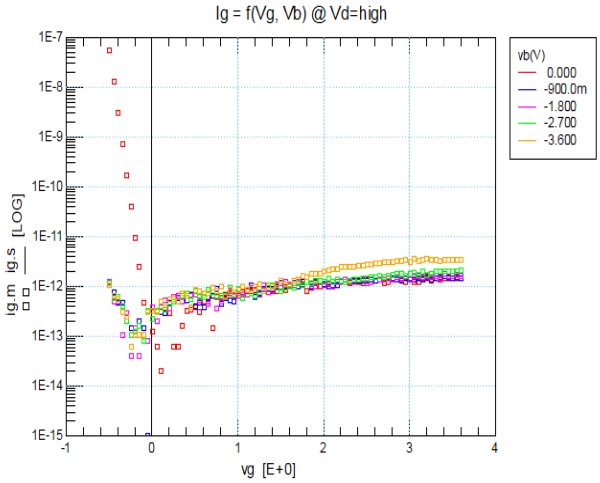




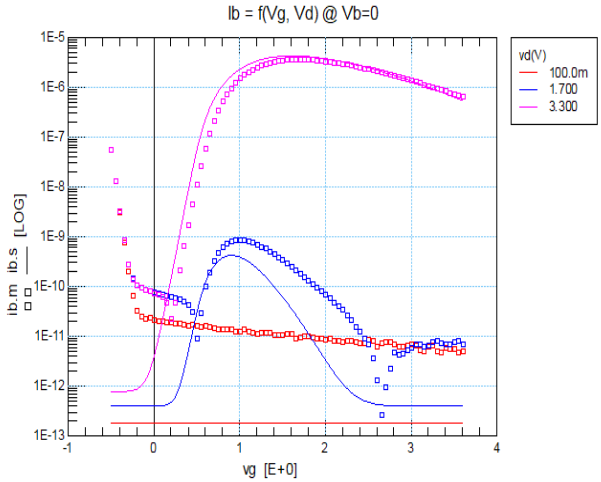


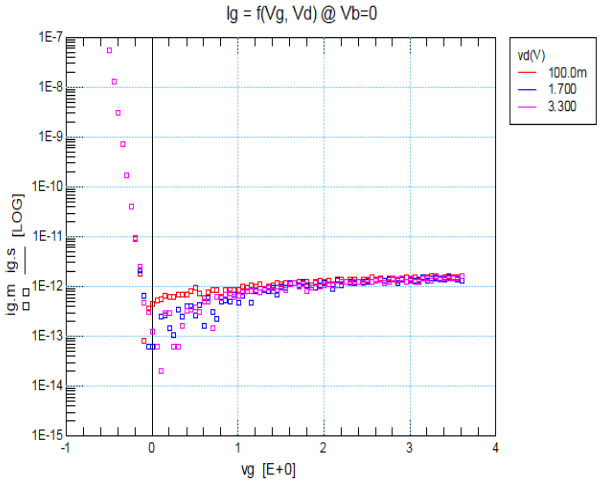




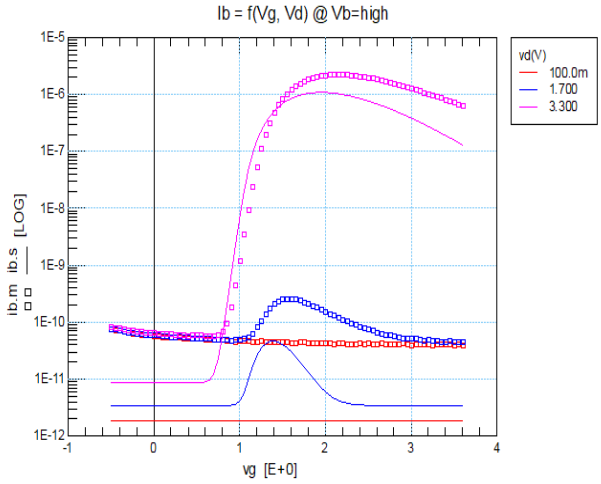


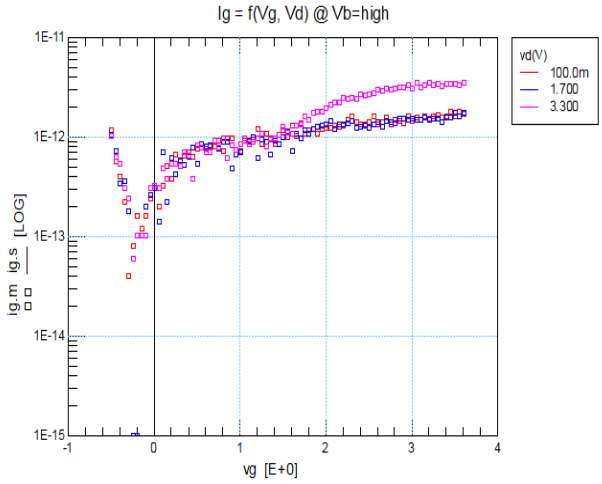
idvg, Ib,Ig, Vb = 0V, T = 125°C



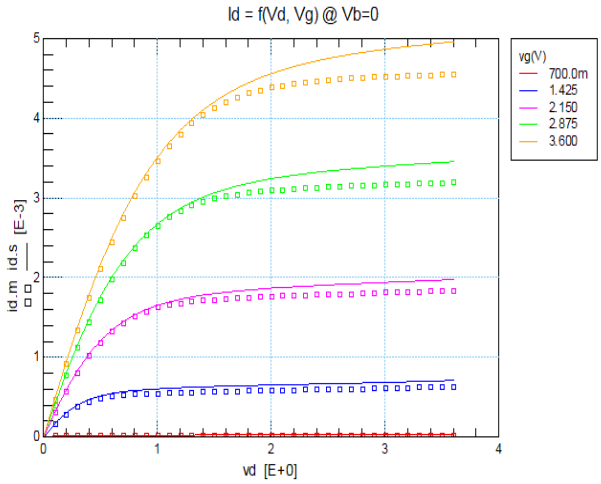


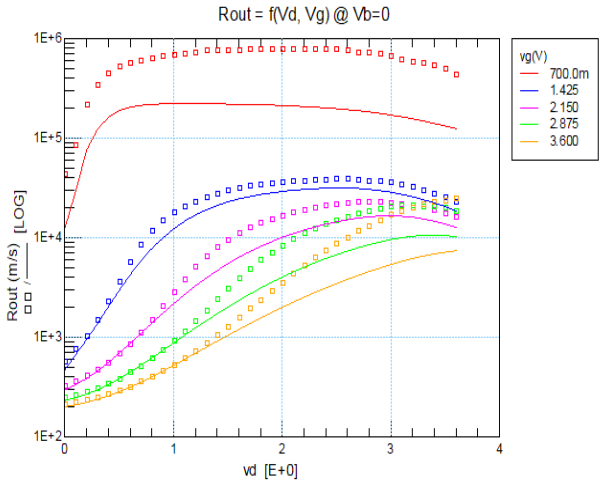
idvg, Ib,Ig, Vb = -3.6V, T = 125°C

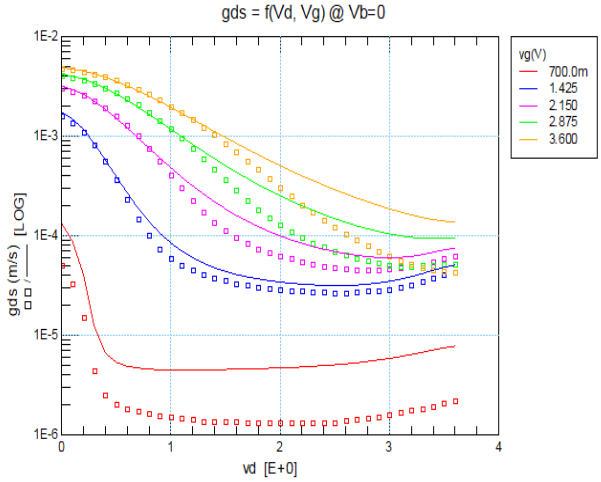




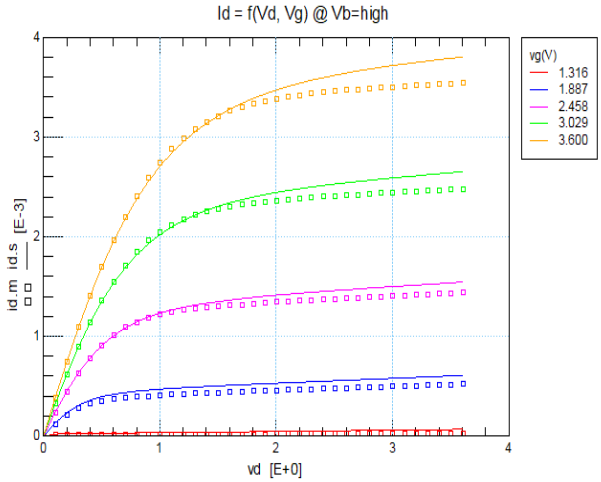
idvd, Vb = 0V, T = 125°C

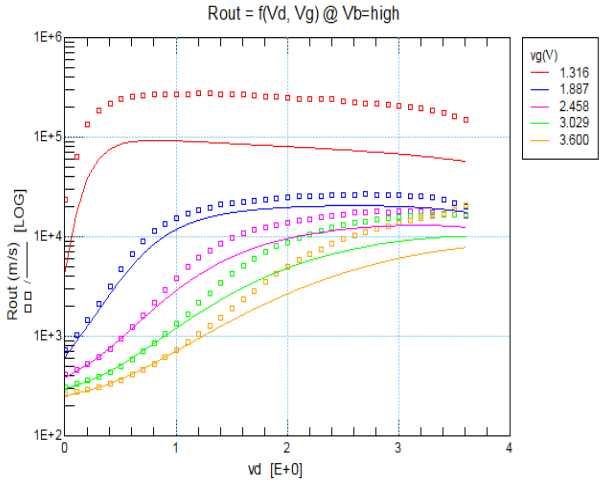


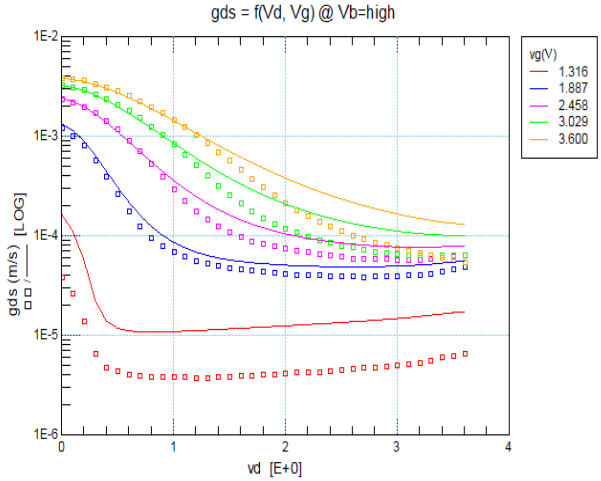




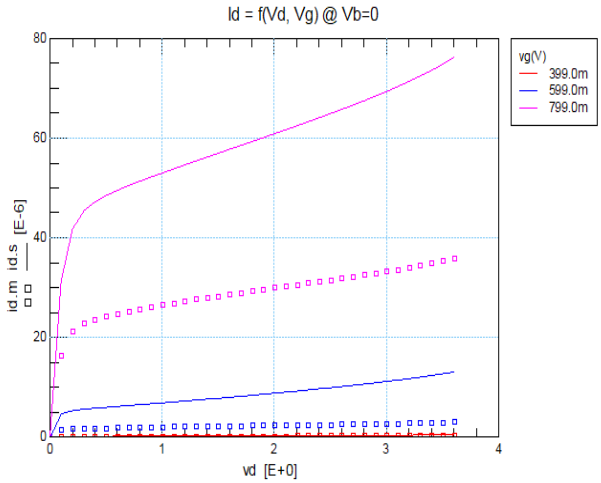
idvd_vbmin, Vb = -3.6V, T = 125°C

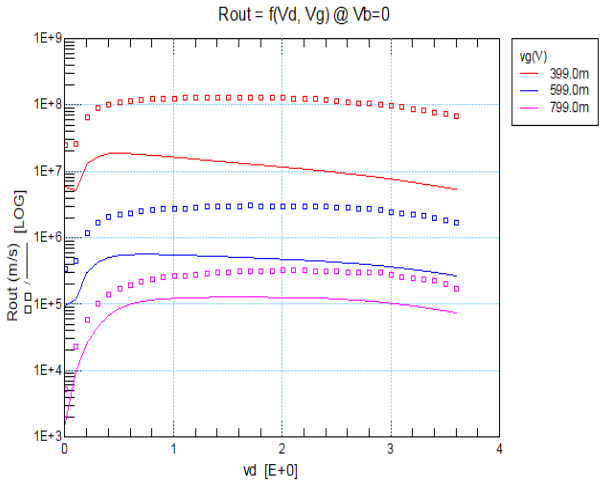


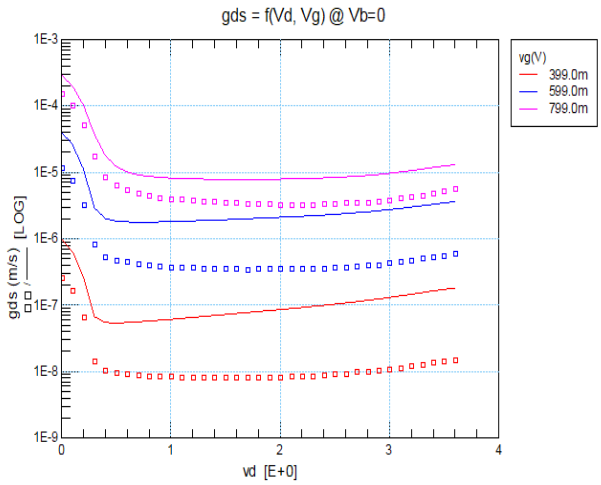




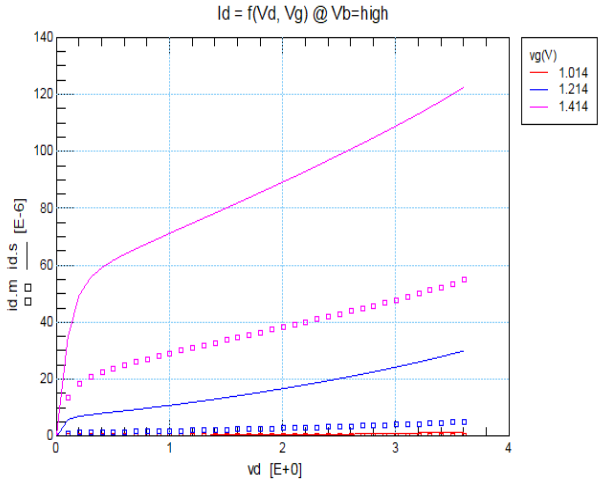
idvd_vth, Vb = 0V, T = 125°C

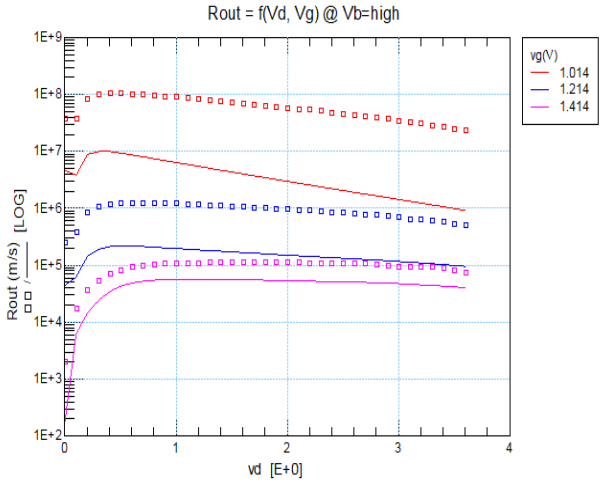


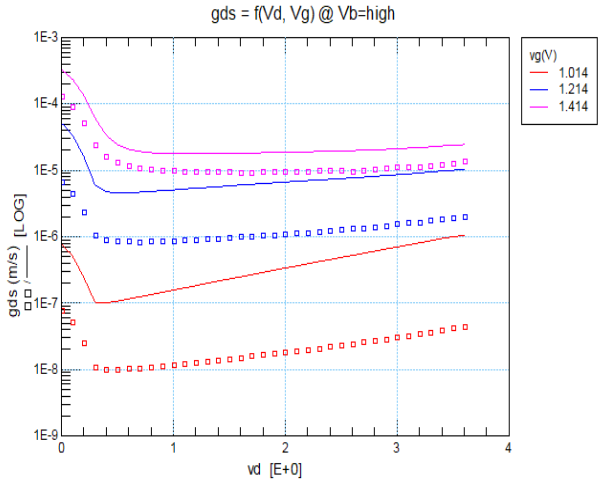




idvd_vth_vbmin, Vb = -3.6V, T = 125°C

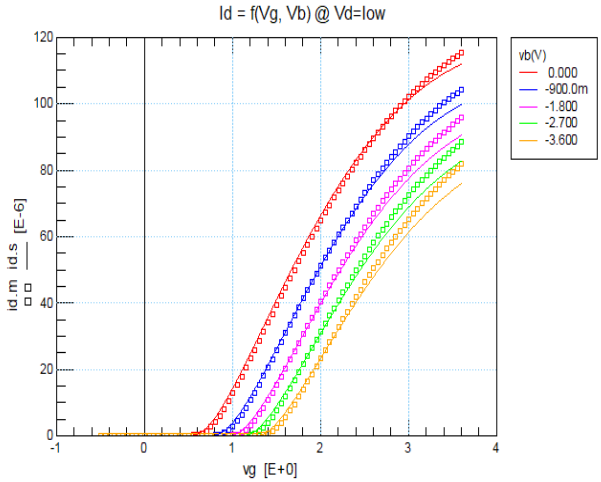


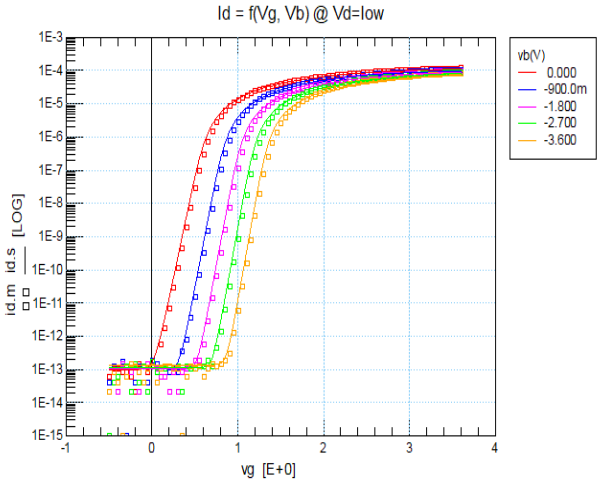


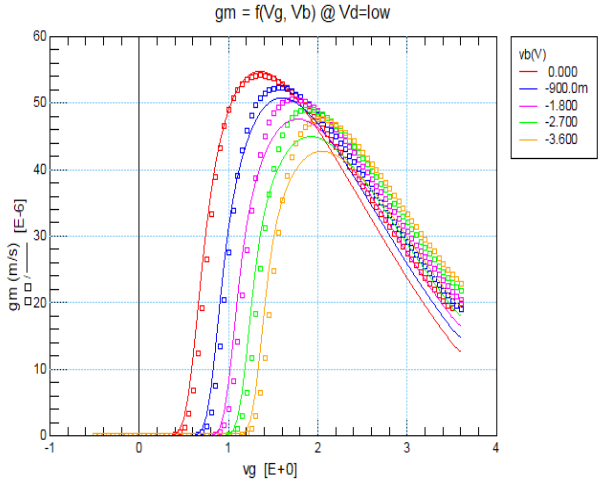


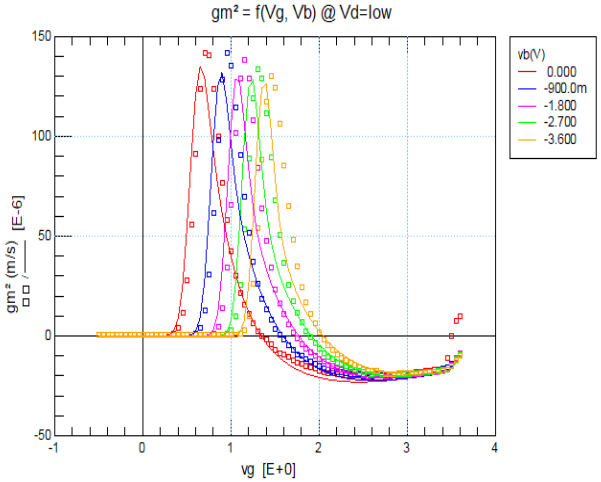
4.9 W10u0_L3u0_S557_1

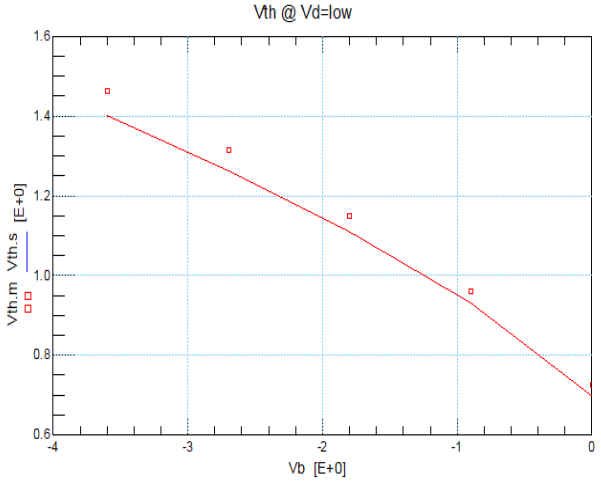
idvg, Vd = 0.1V, T = 27°C

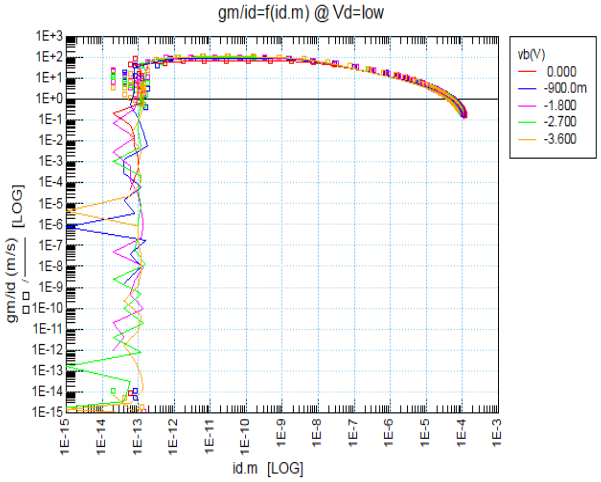


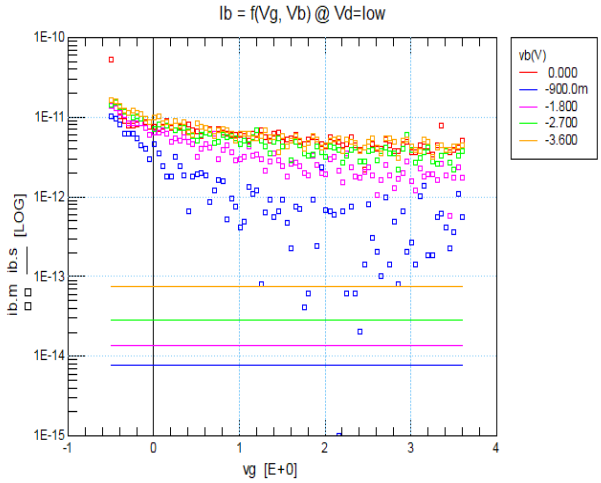


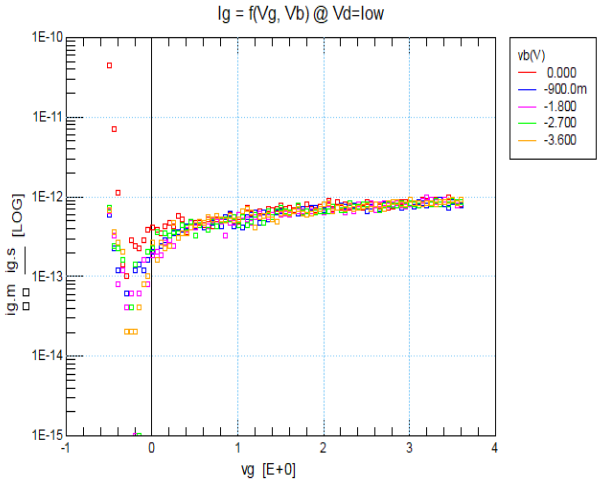




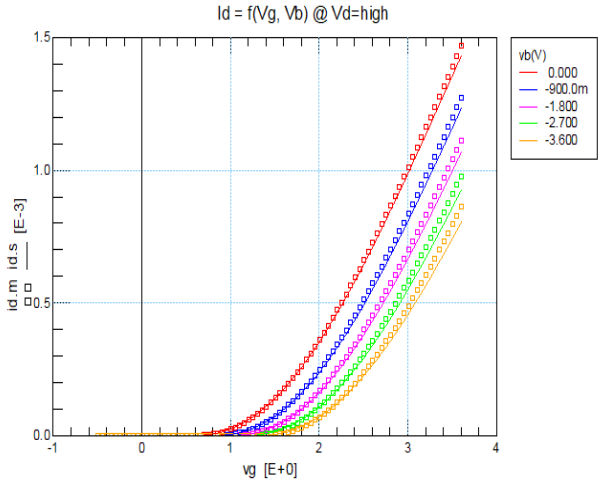


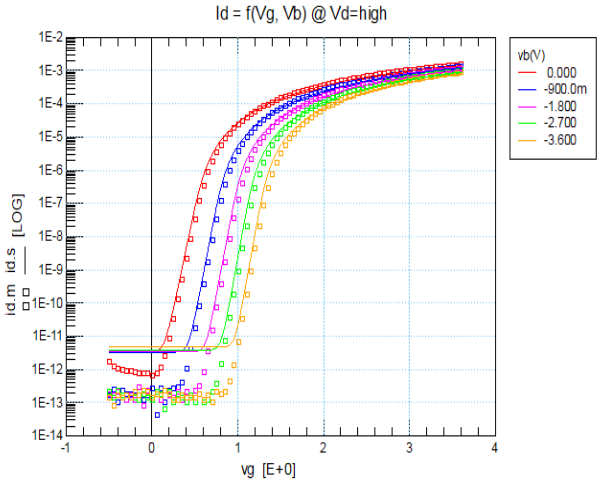


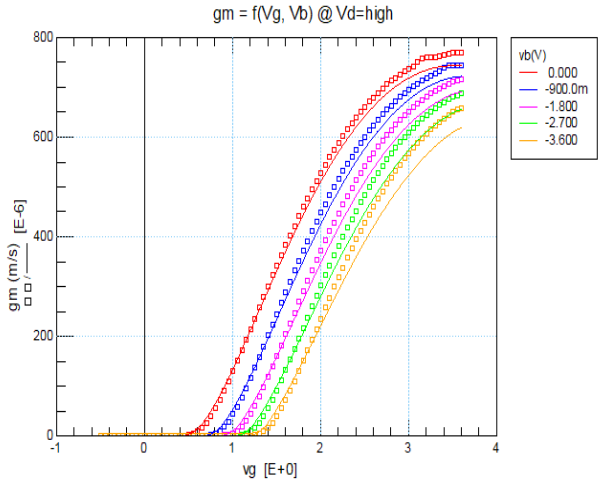


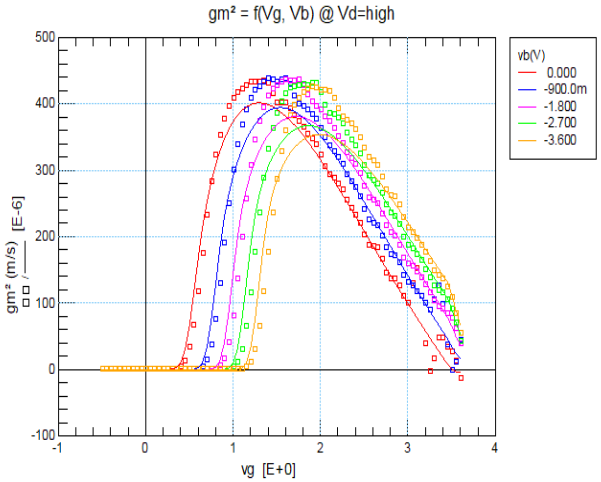


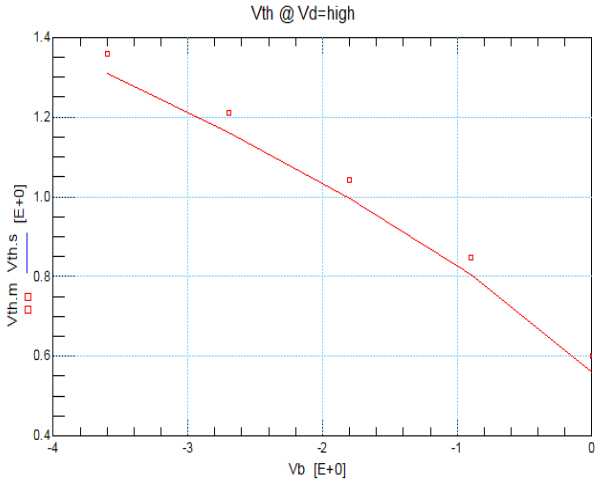
idvg, Vd = 3.3V, T = 27°C

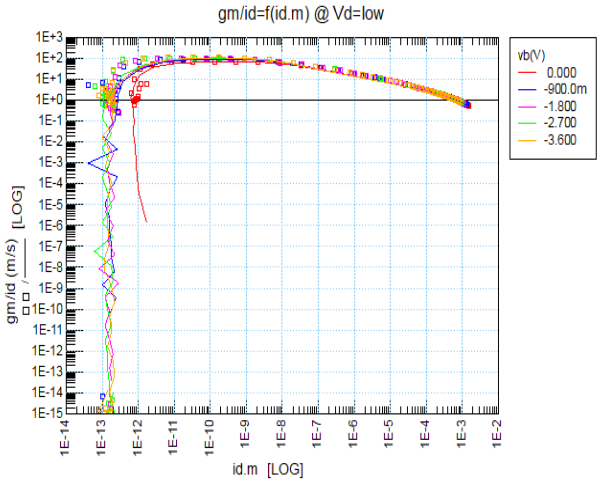


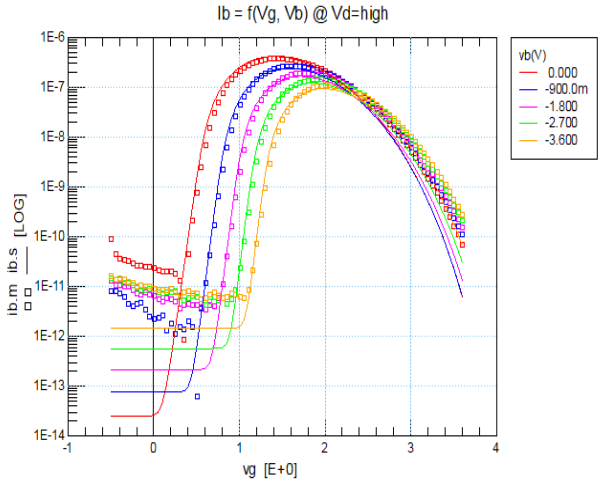


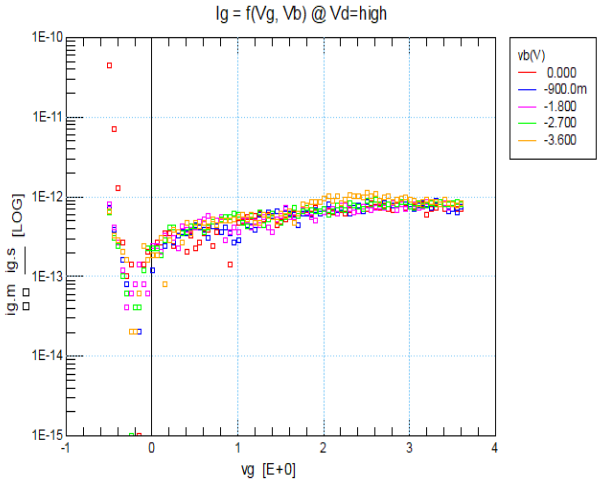




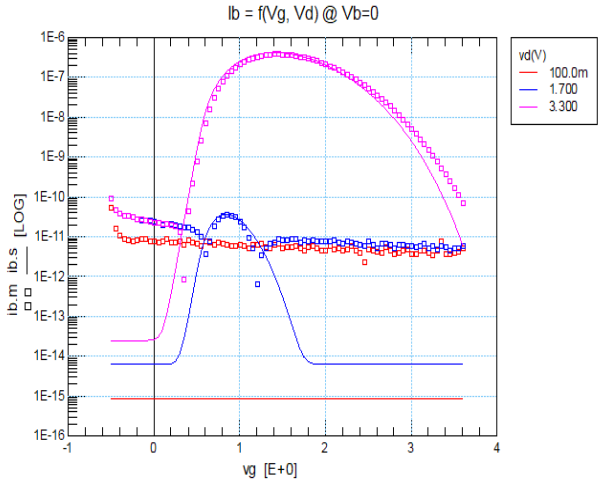


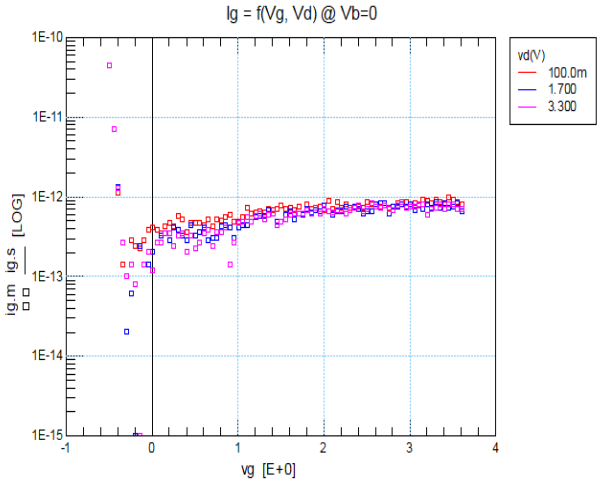




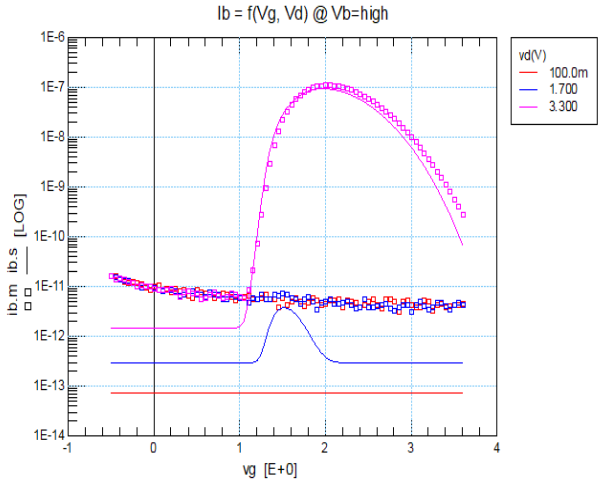


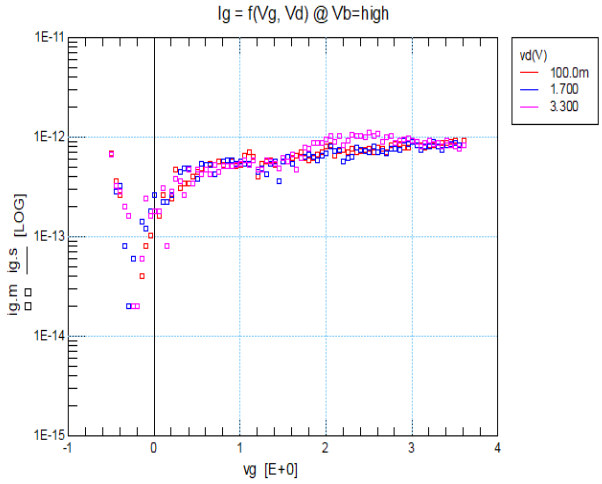
idvg, Ib,Ig, Vb = 0V, T = 27°C



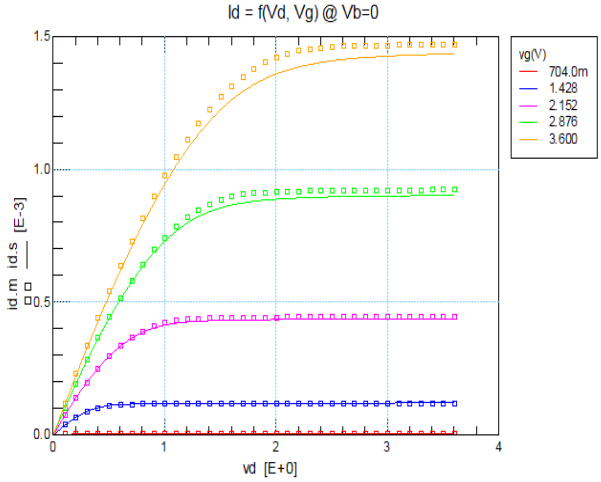


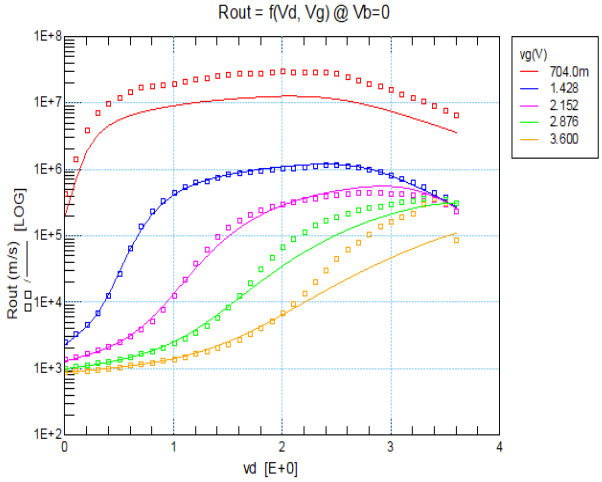
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

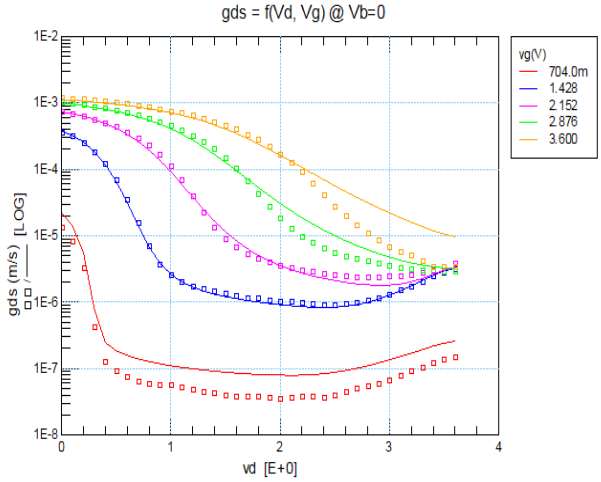




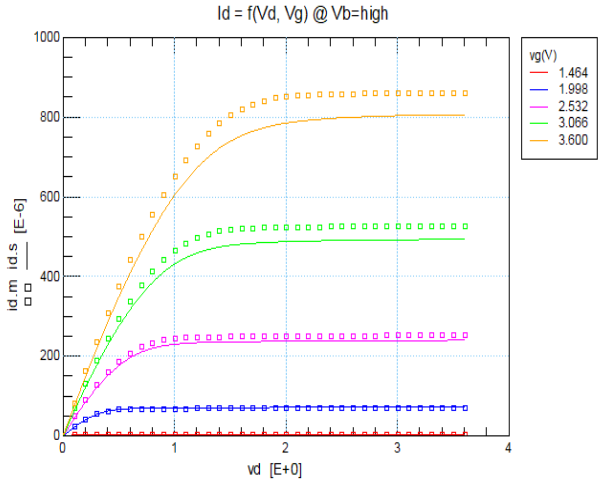
idvd, Vb = 0V, T = 27°C

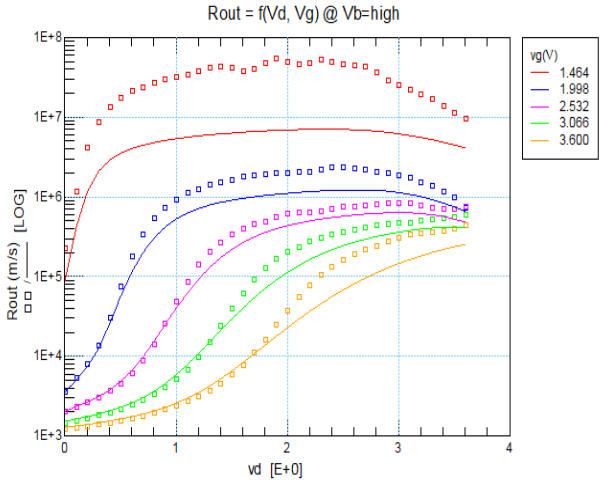


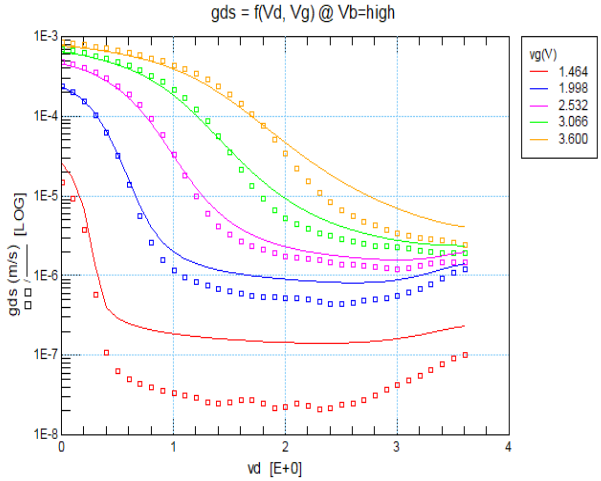




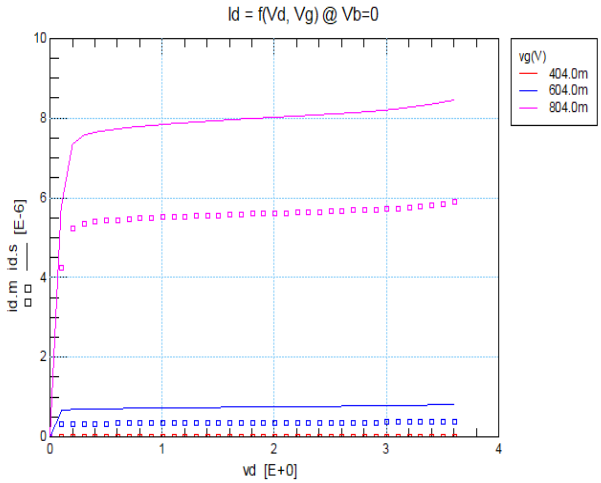
idvd_vbmin, Vb = -3.6V, T = 27°C

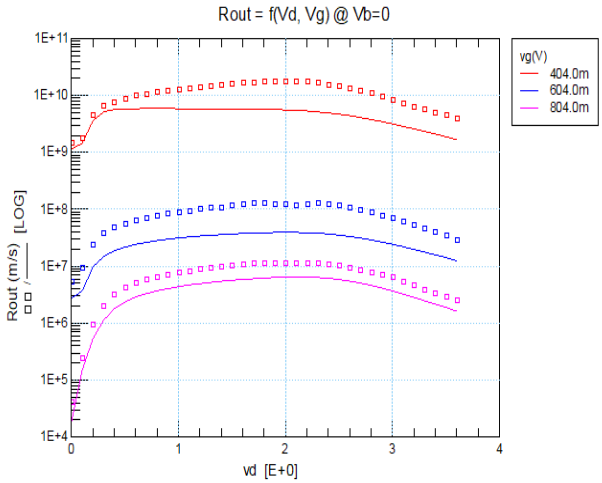


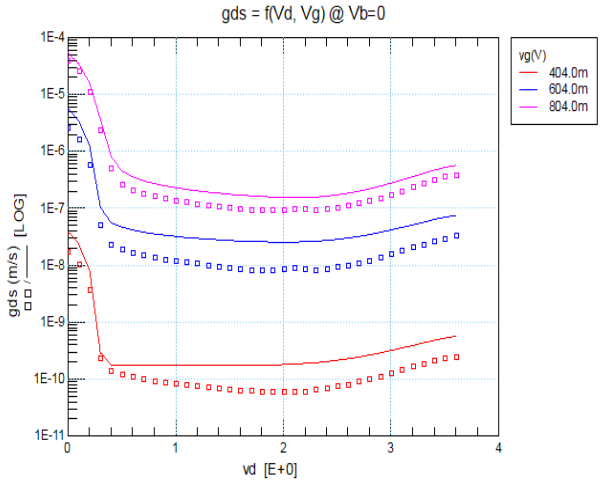




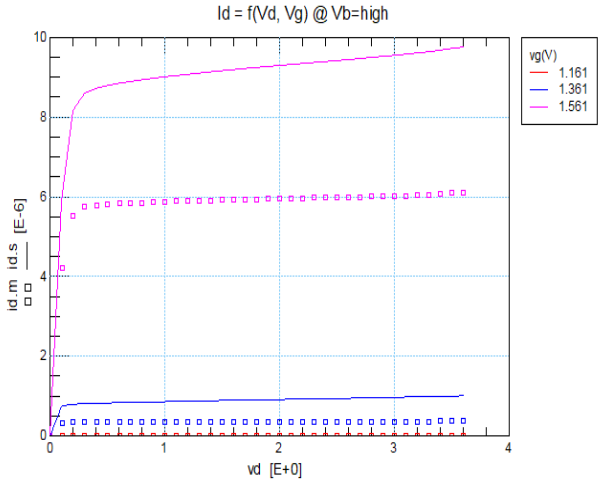
idvd_vth, Vb = 0V, T = 27°C

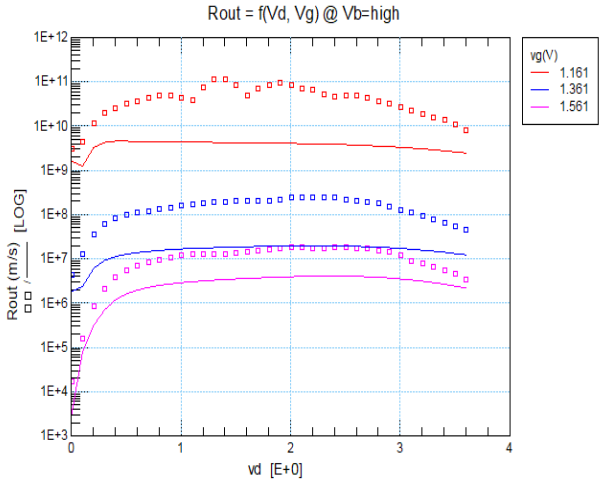


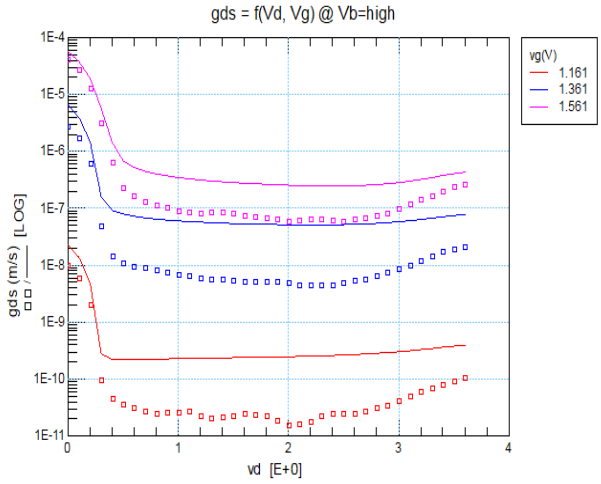




idvd_vth_vbmin, Vb = -3.6V, T = 27°C

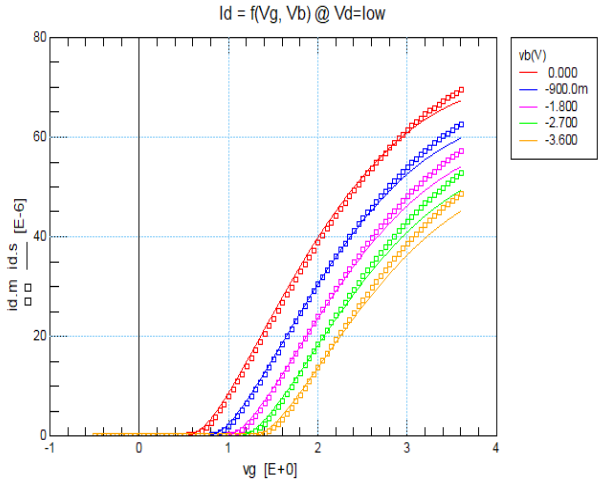


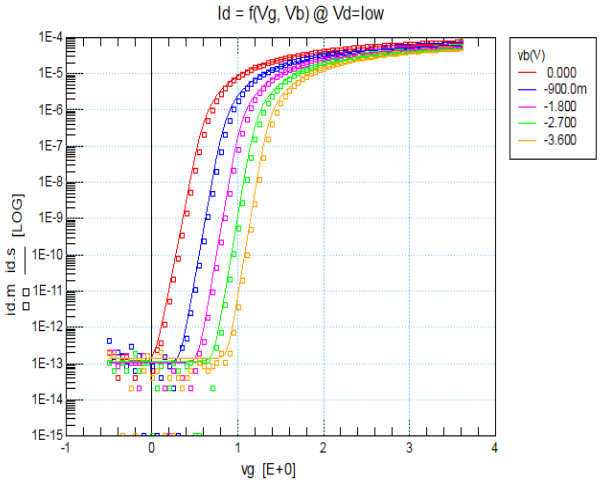


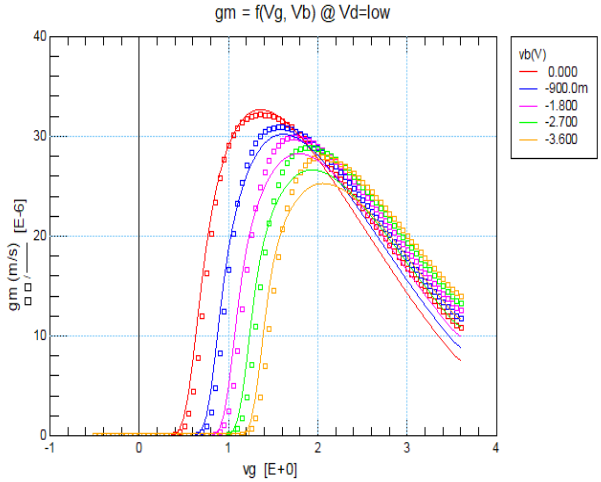


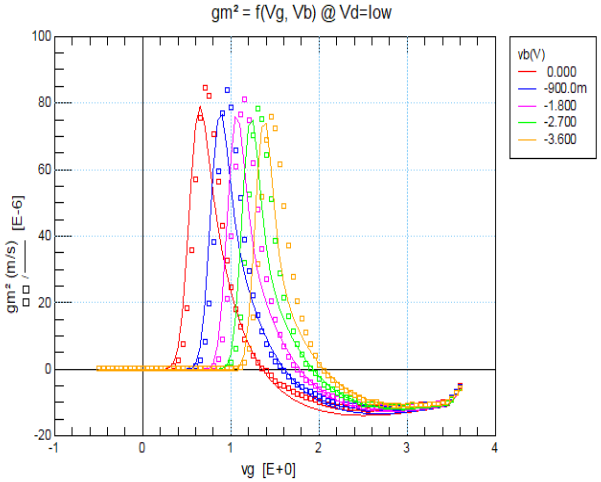
4.10 W10u0_L5u0_S557_2

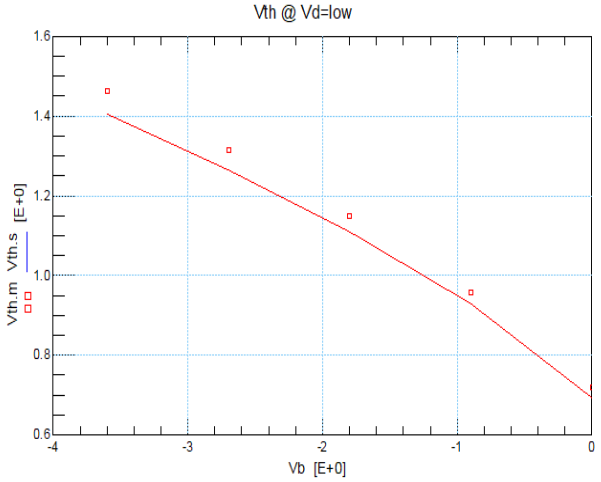
idvg, Vd = 0.1V, T = 27°C

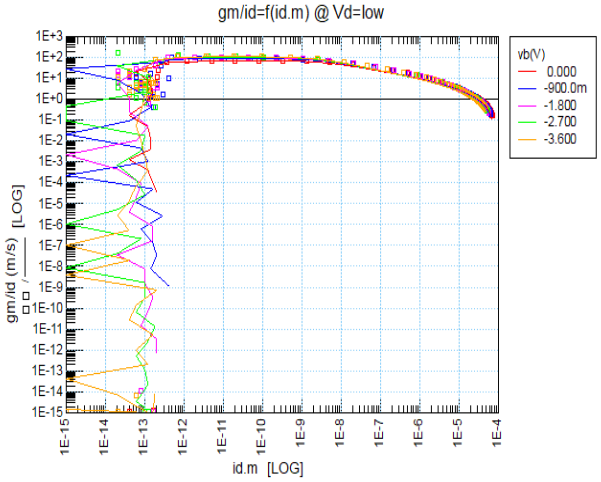


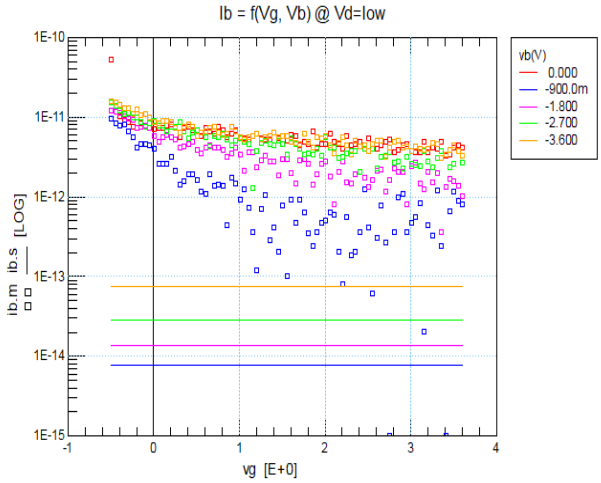


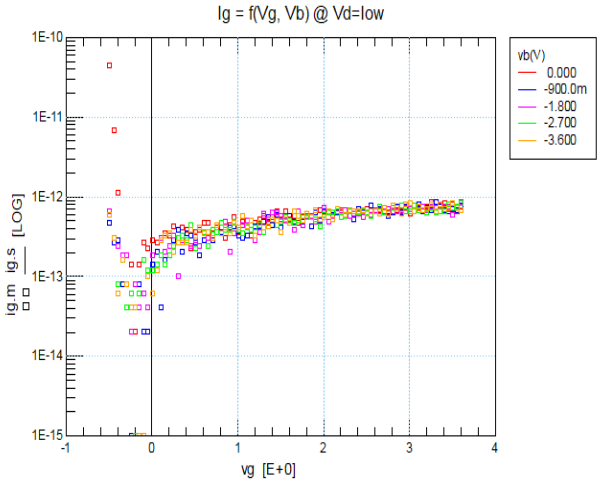




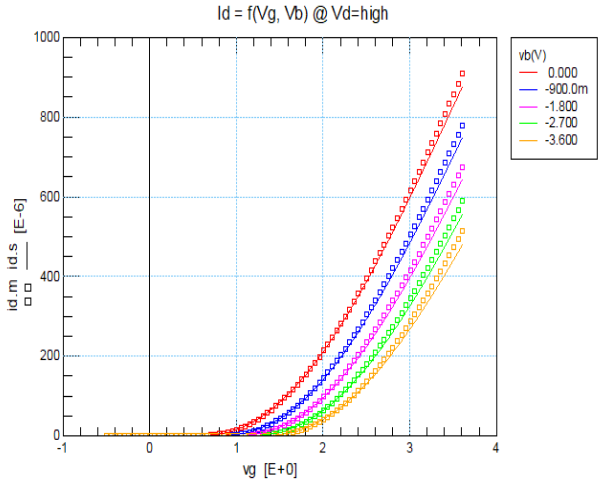


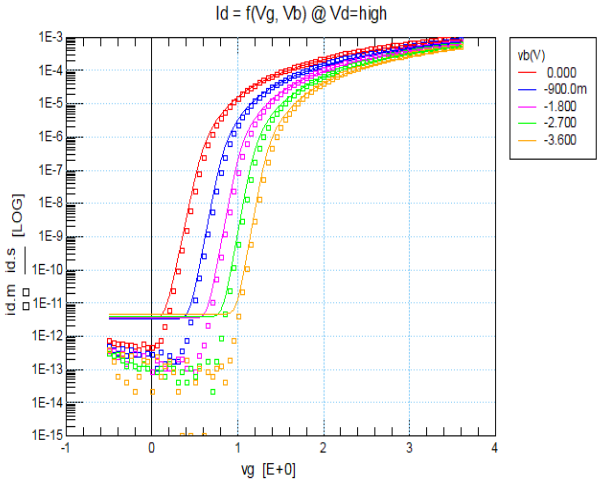


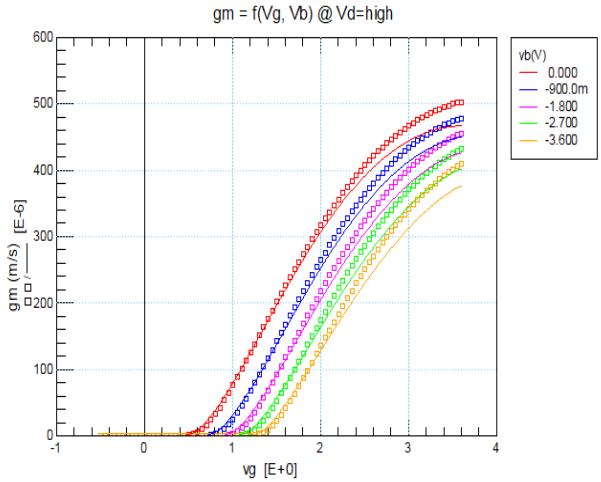


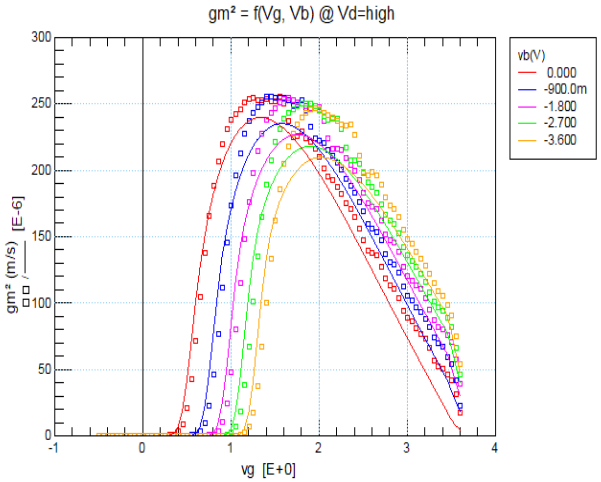


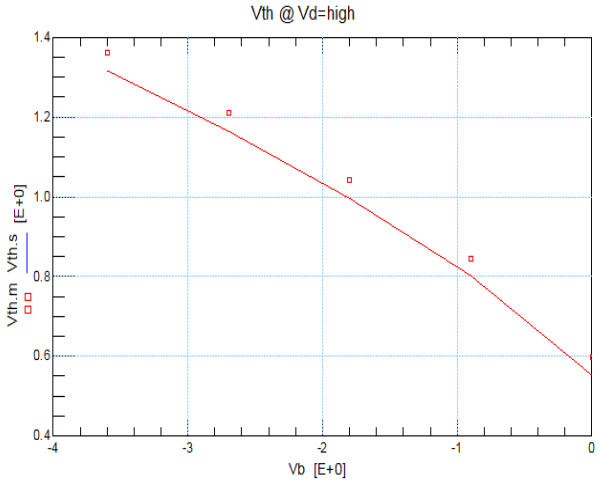
idvg, Vd = 3.3V, T = 27°C

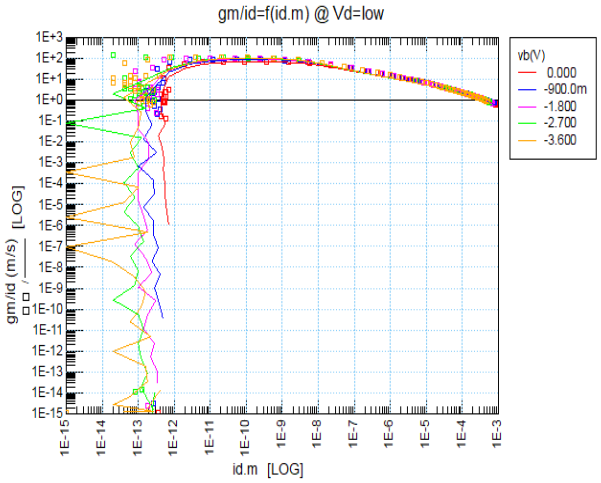


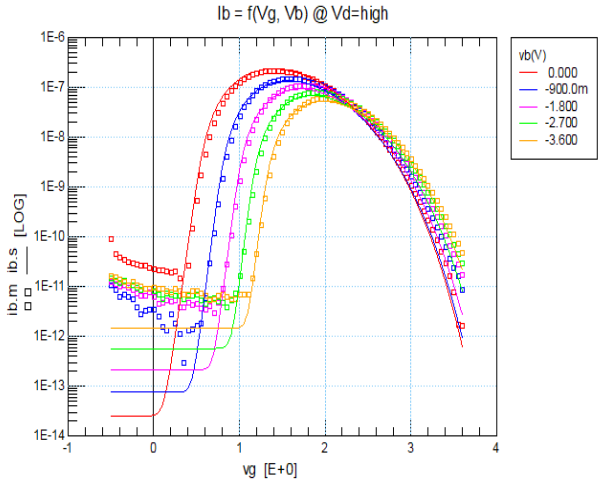


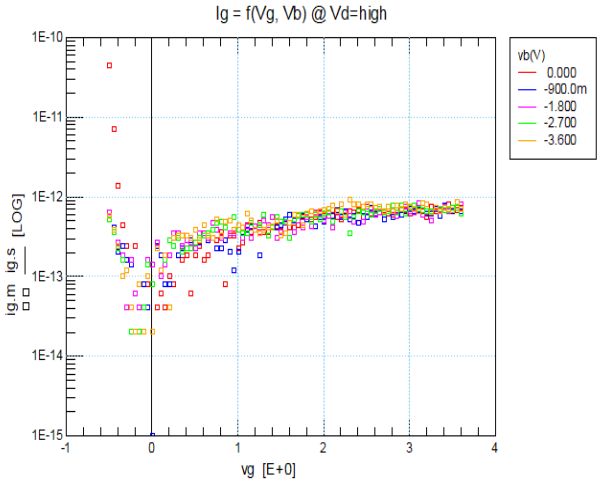




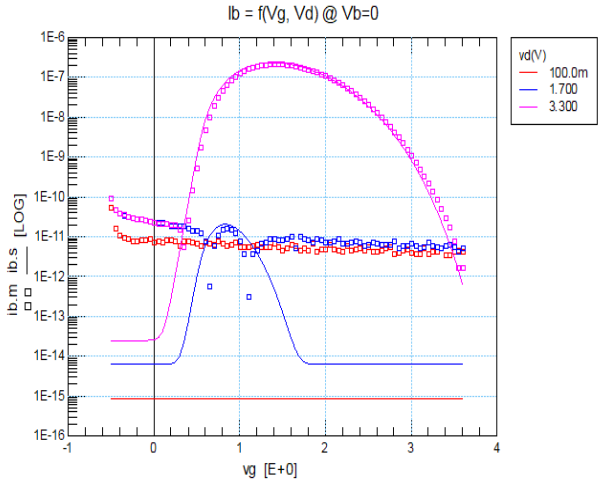


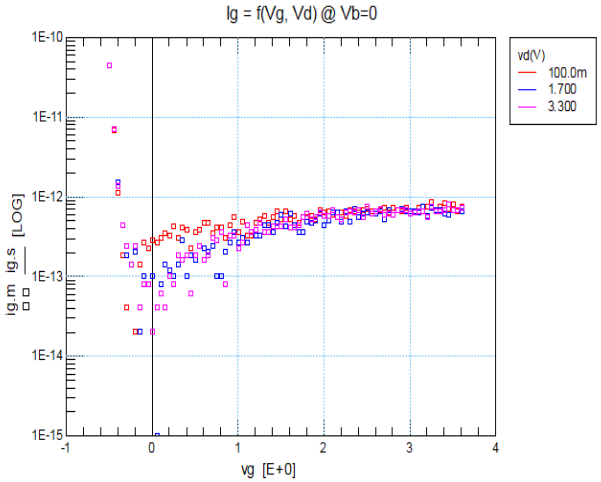




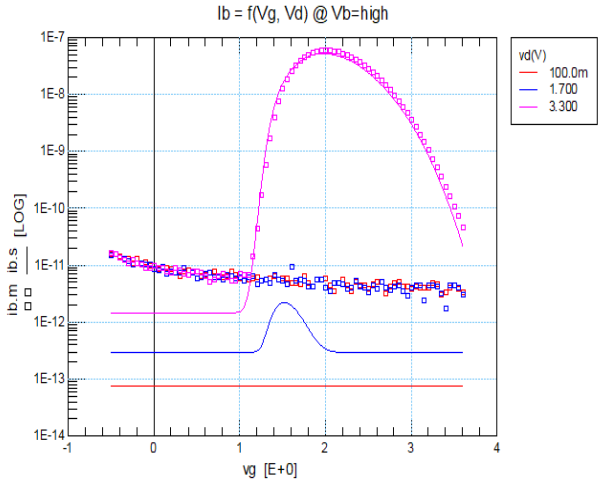


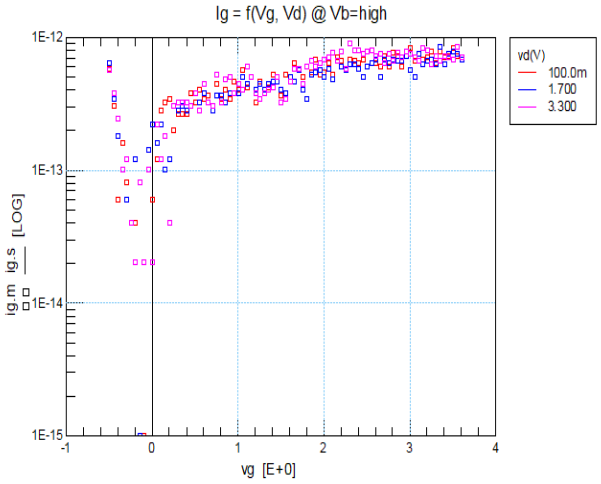
idvg, Ib,Ig, Vb = 0V, T = 27°C



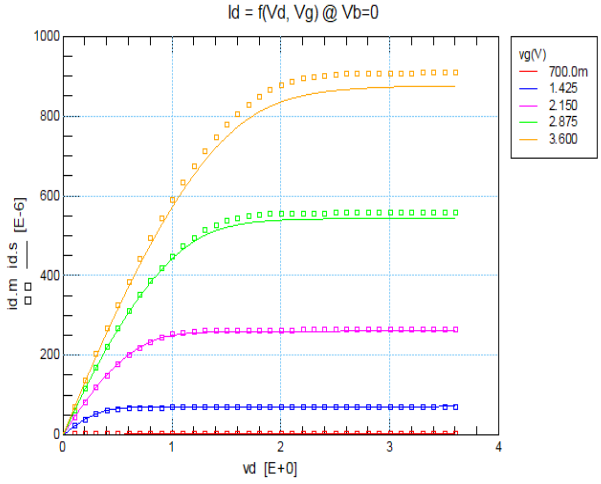


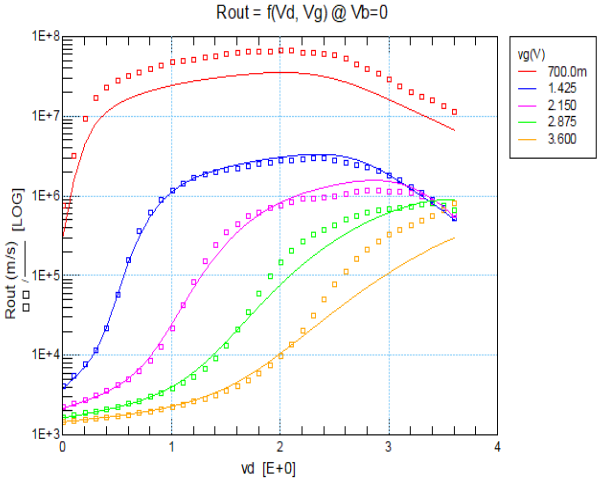
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

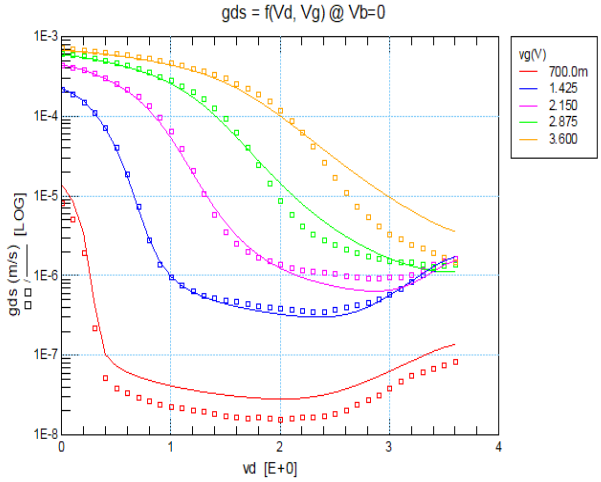




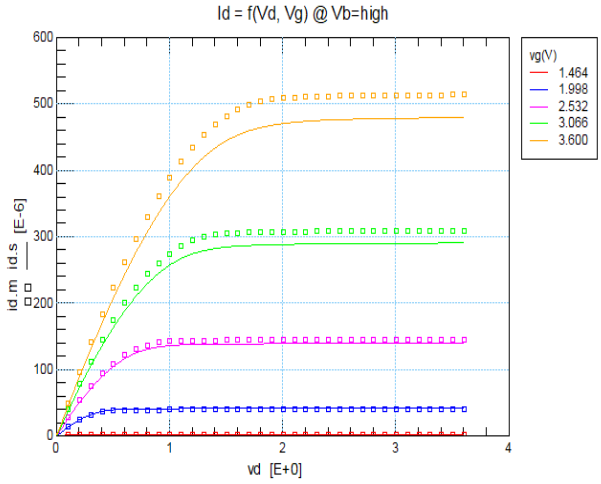
idvd, Vb = 0V, T = 27°C

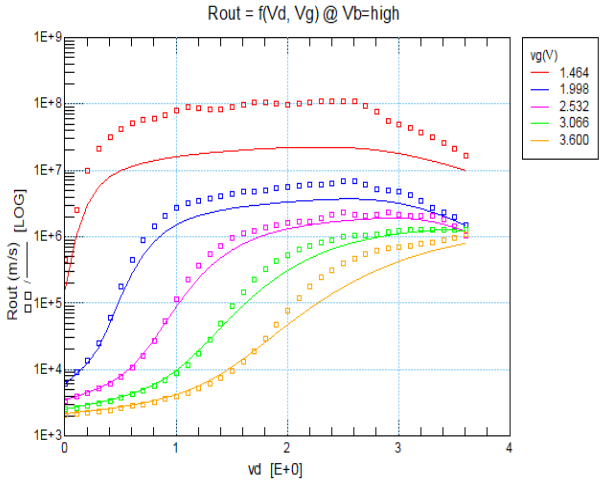


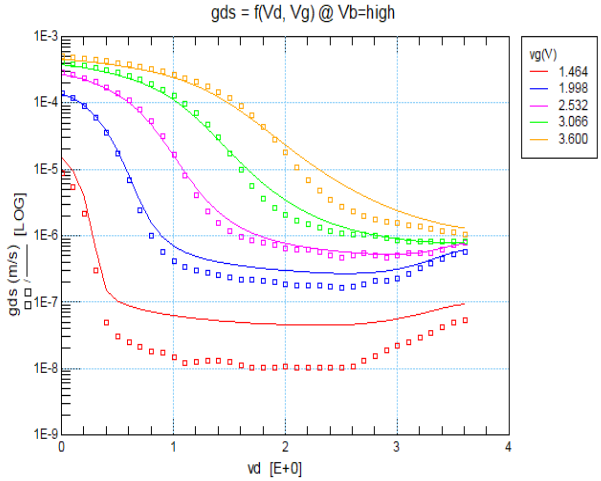




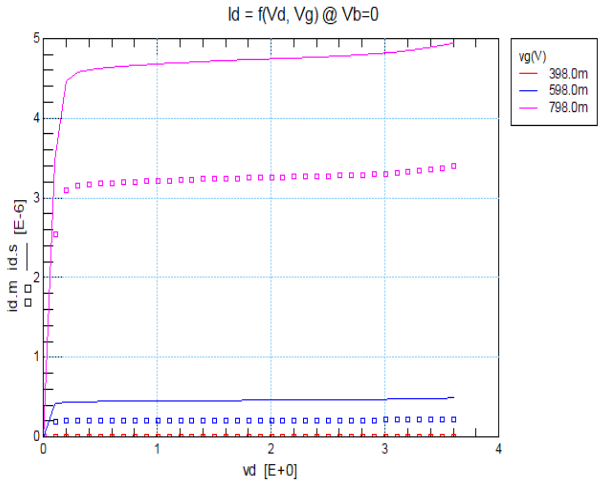
idvd_vbmin, Vb = -3.6V, T = 27°C

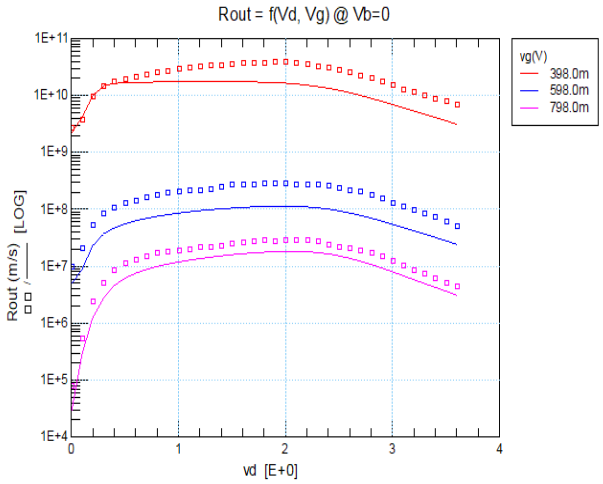


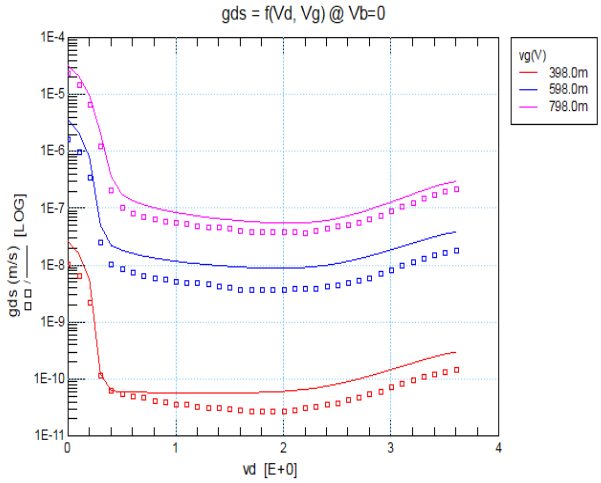




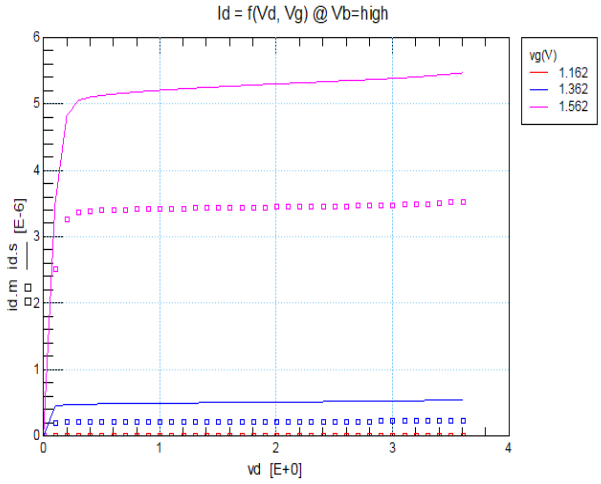
idvd_vth, Vb = 0V, T = 27°C

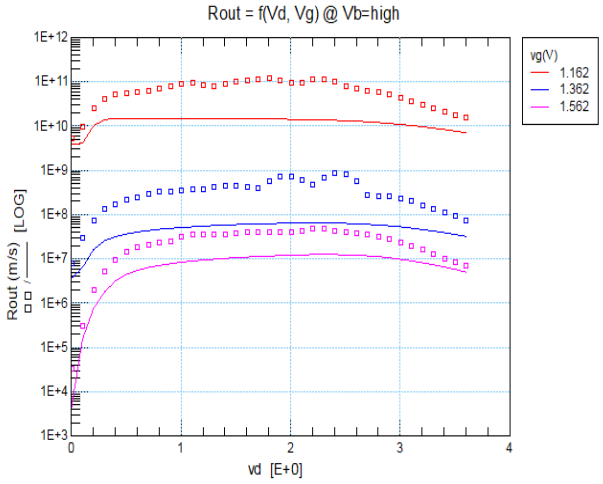


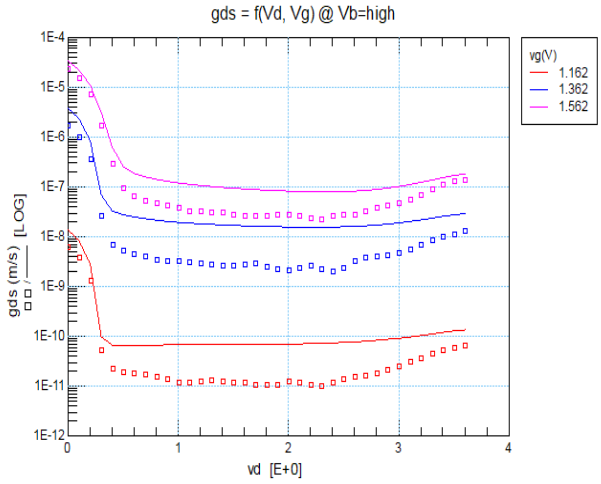




idvd_vth_vbmin, Vb = -3.6V, T = 27°C



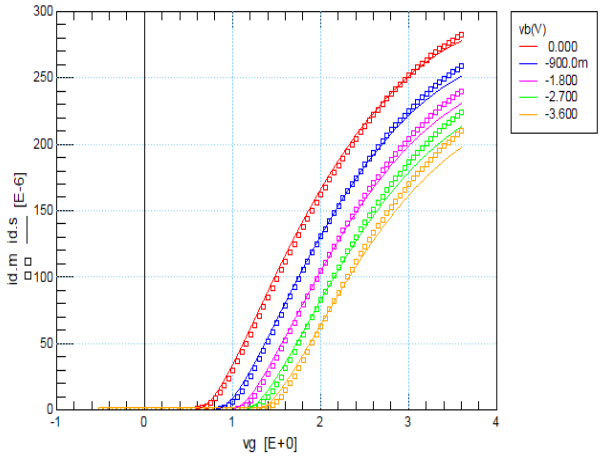




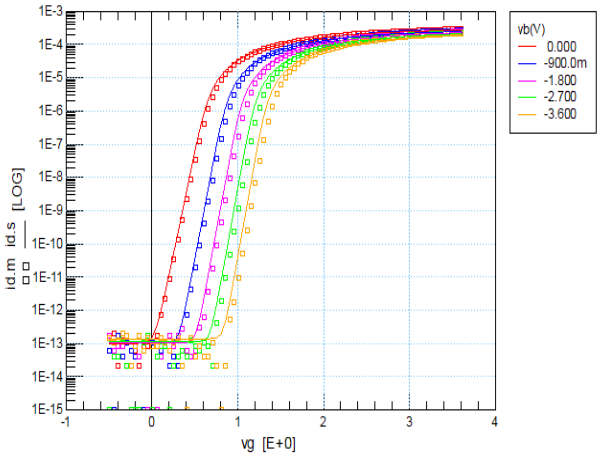
4.11 W10u0_L1u2_S556_5

idvg, Vd = 0.1V, T = 27°C

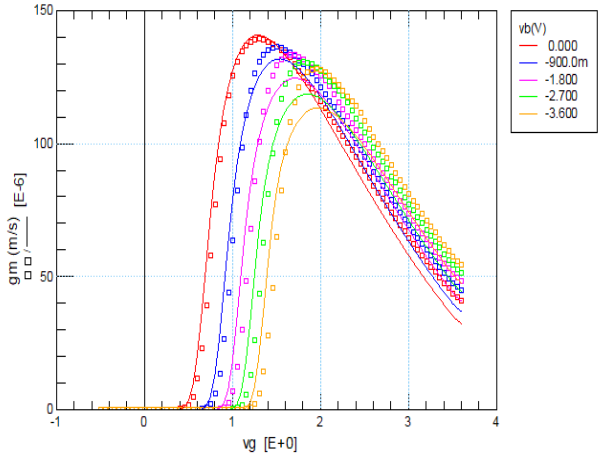
Id = f(Vg, Vb) @ Vd=low



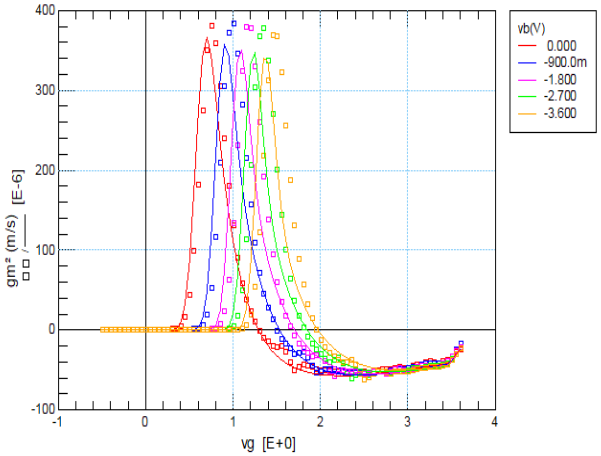
Id = f(Vg, Vb) @ Vd=low



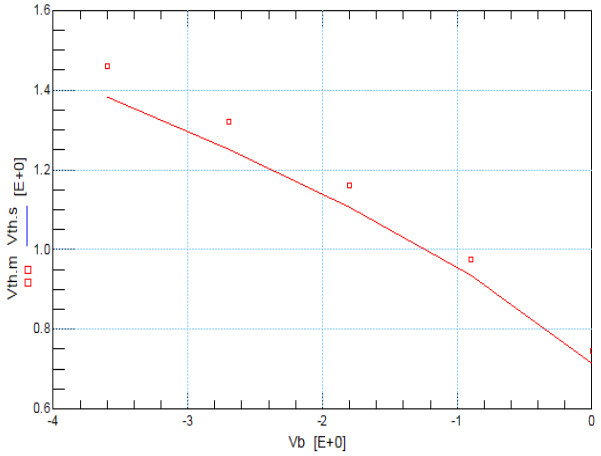
gm = f(Vg, Vb) @ Vd=low



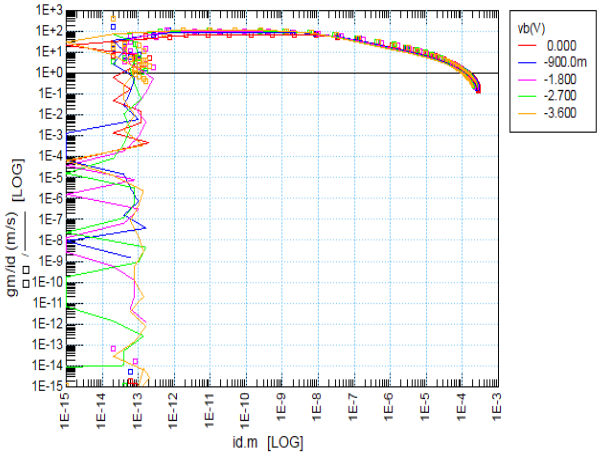
gm² = f(Vg, Vb) @ Vd=low

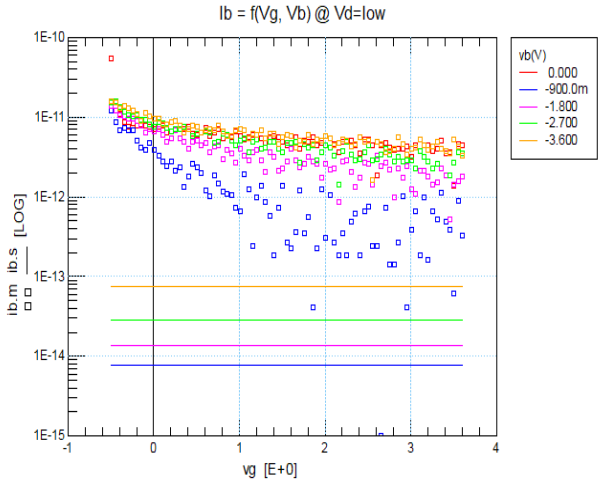


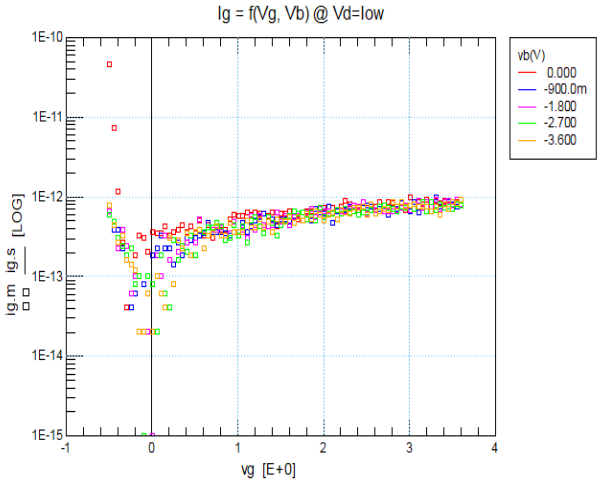
Vth @ Vd=low



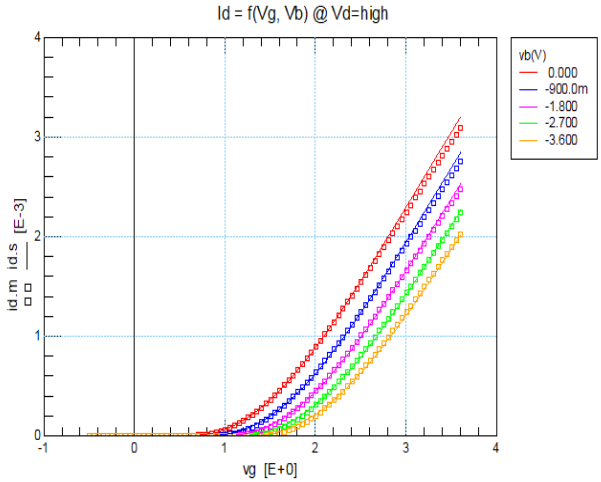
gm/id=f(id.m) @ Vd=low

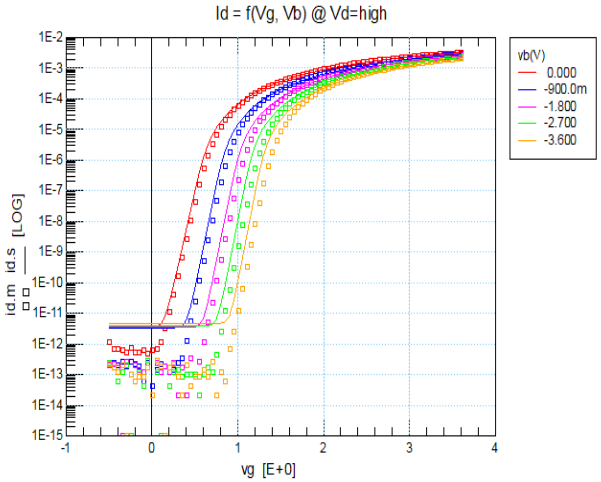


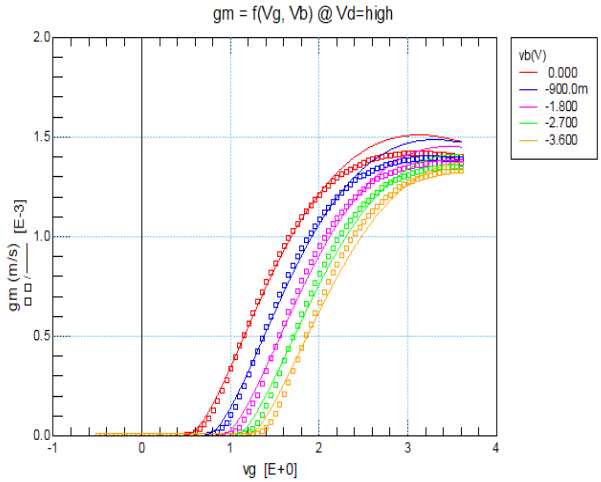


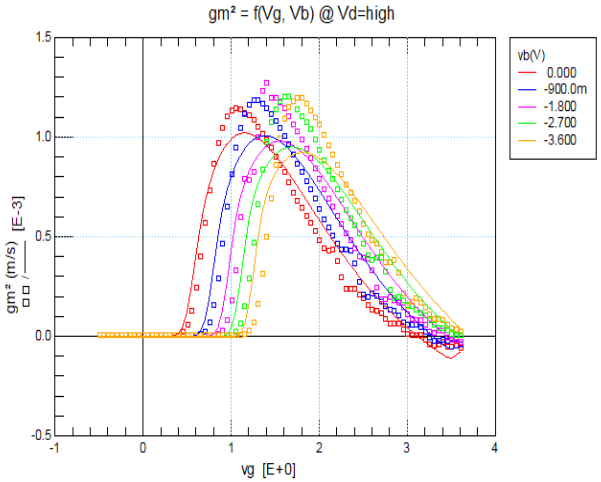


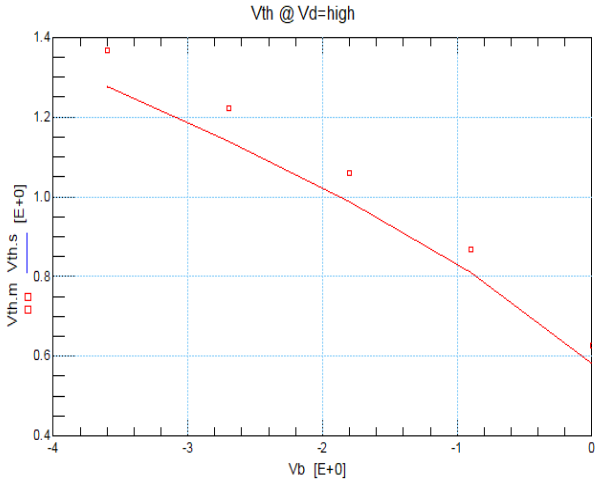
idvg, Vd = 3.3V, T = 27°C

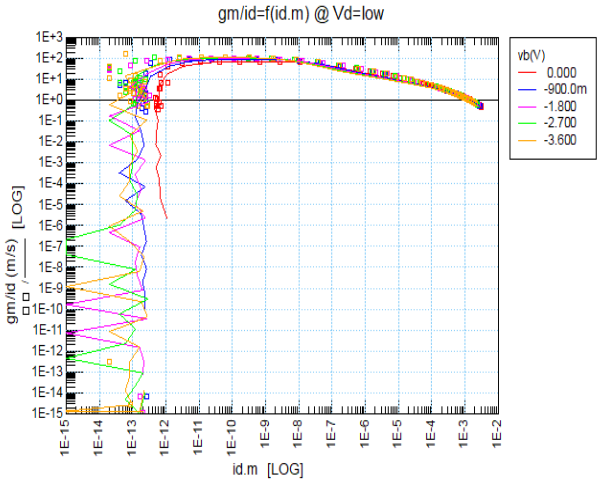


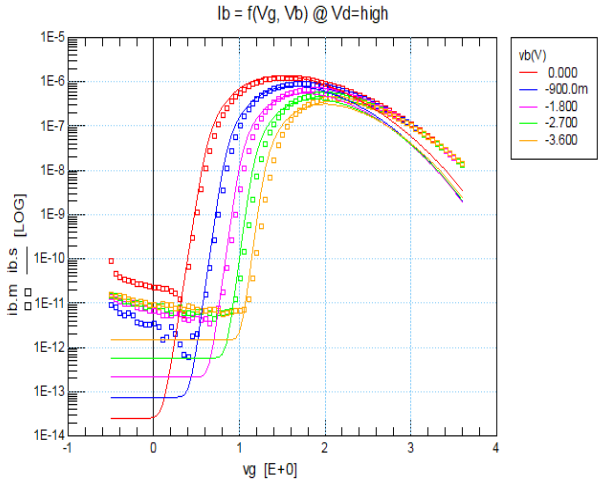


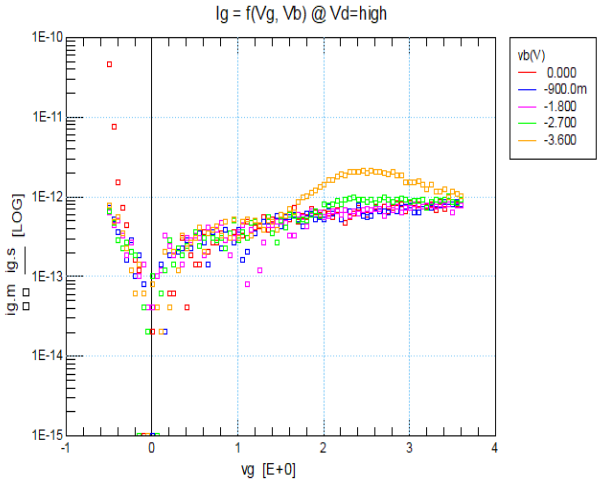




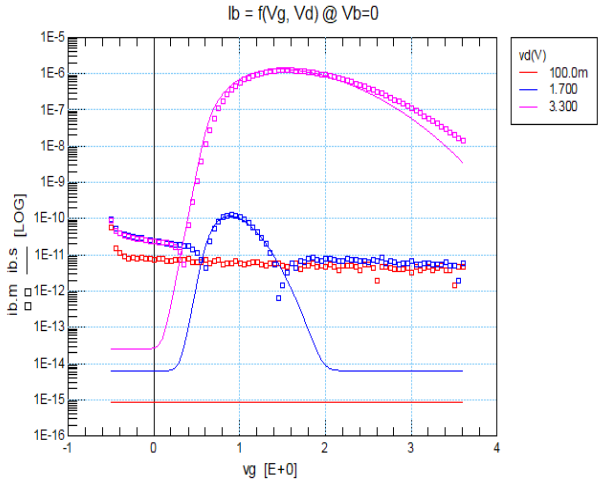


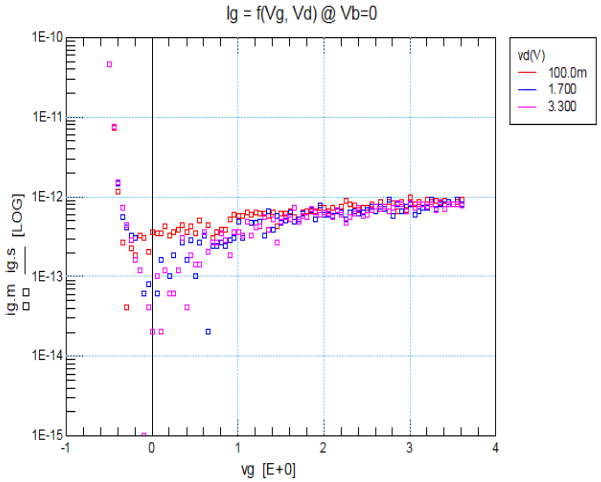




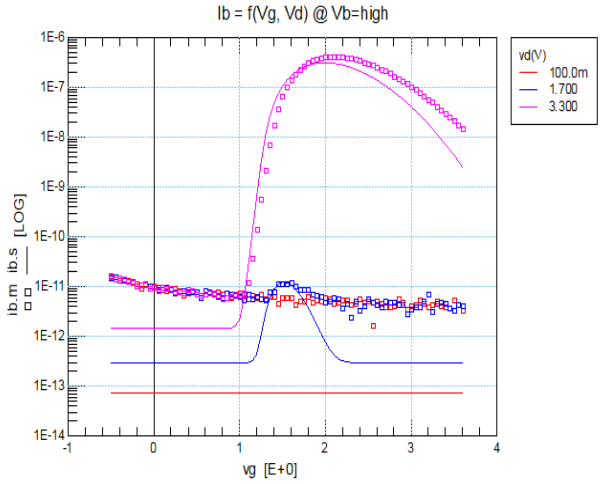


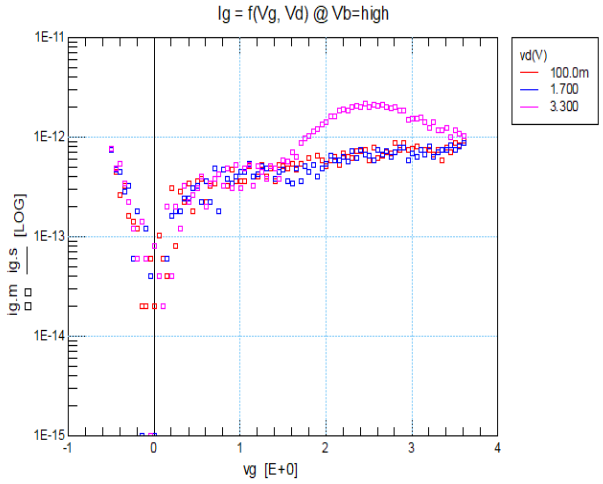
idvg, Ib,Ig, Vb = 0V, T = 27°C



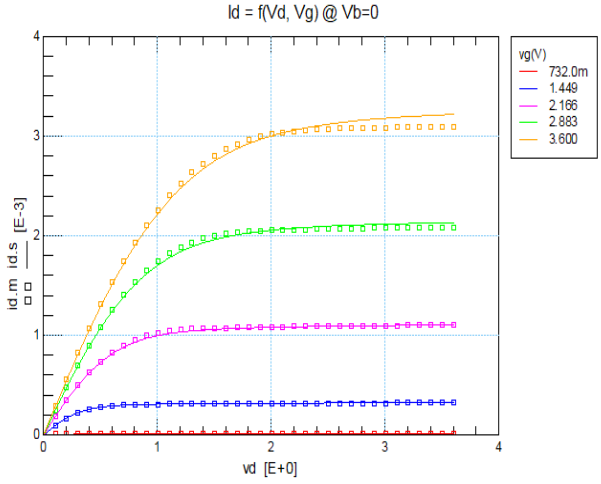


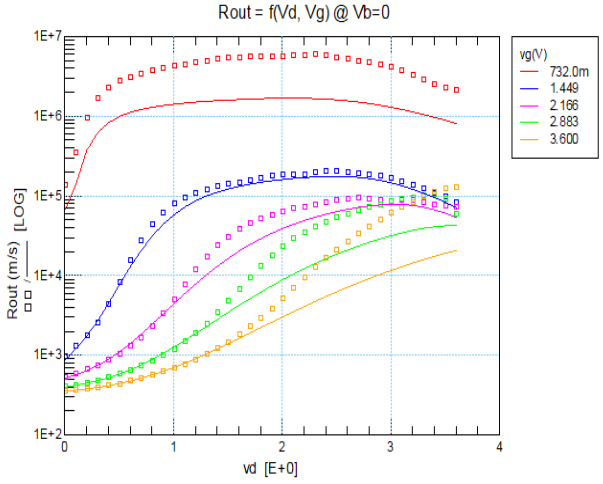
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

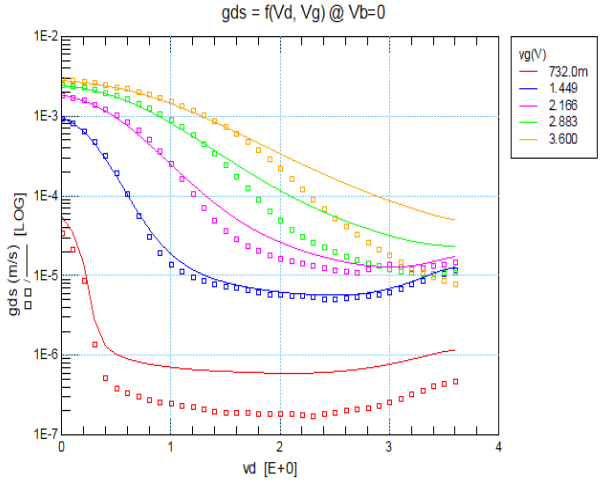




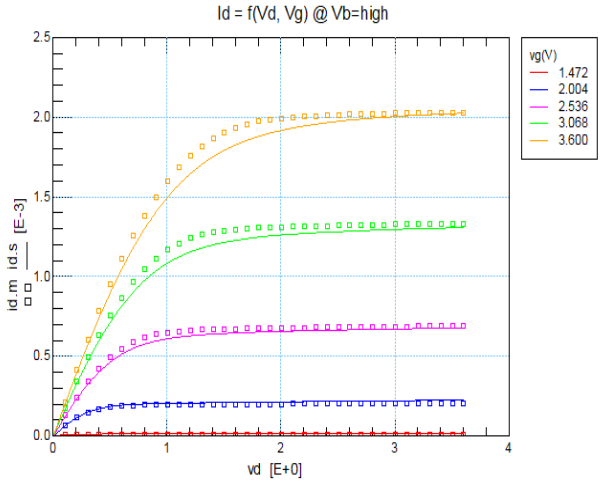
idvd, Vb = 0V, T = 27°C

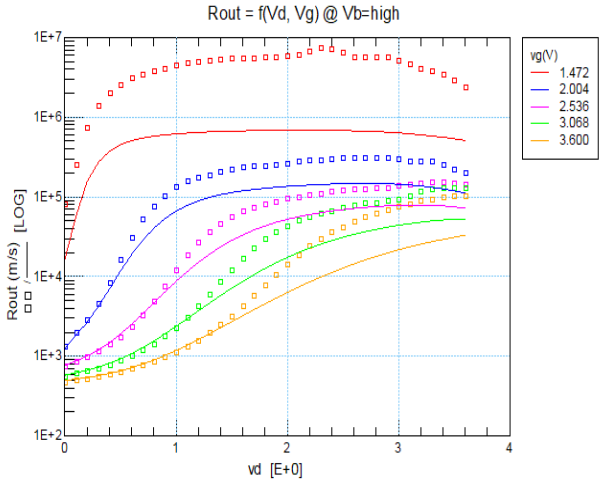


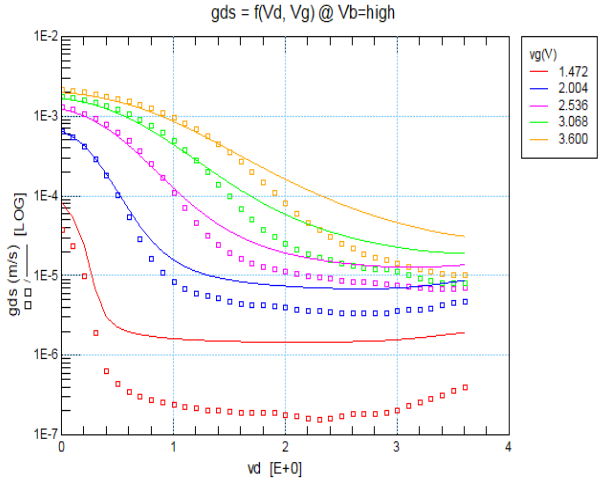




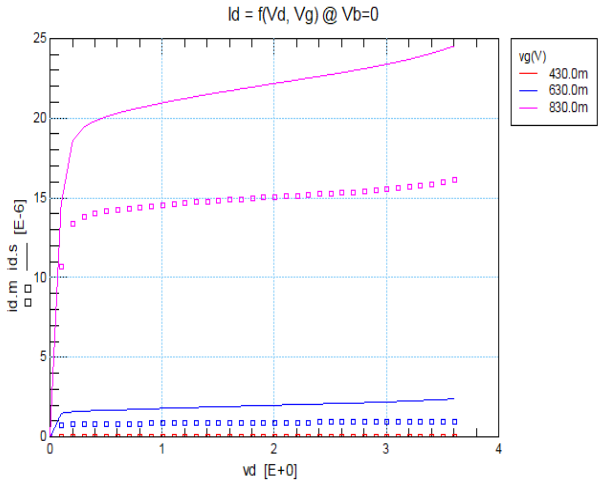
idvd_vbmin, Vb = -3.6V, T = 27°C

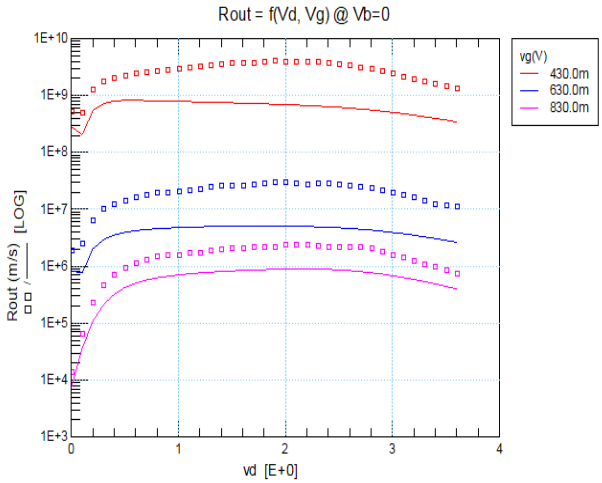


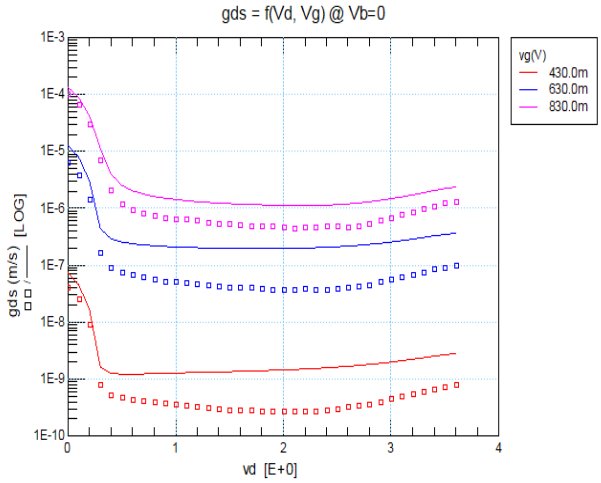




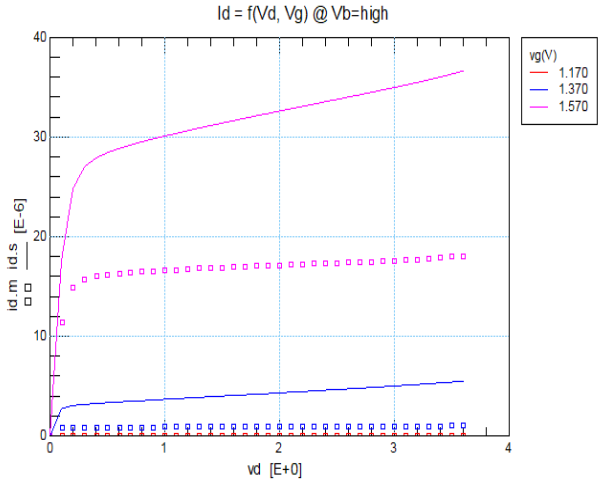
idvd_vth, Vb = 0V, T = 27°C

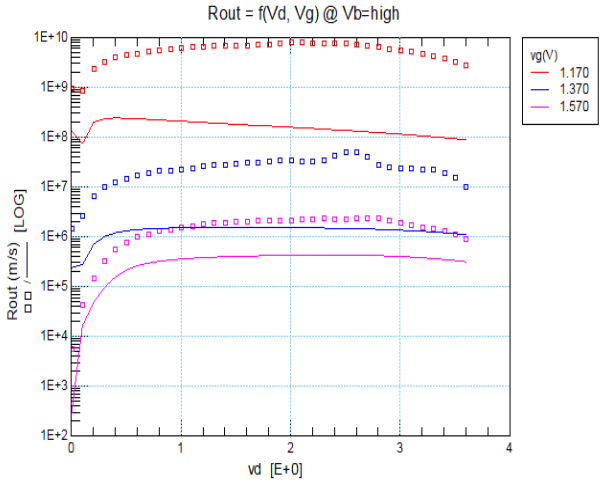


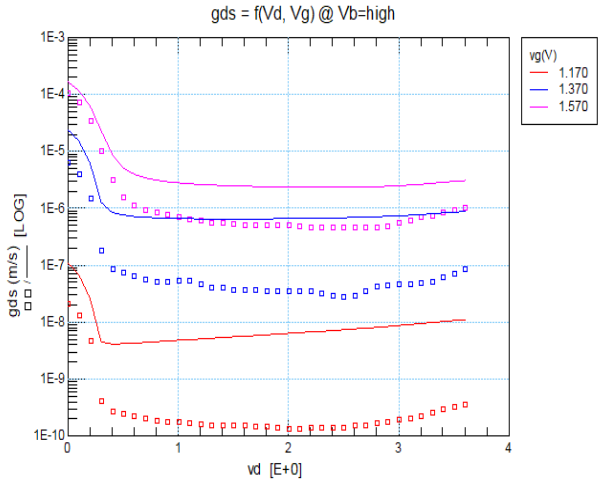




idvd_vth_vbmin, Vb = -3.6V, T = 27°C

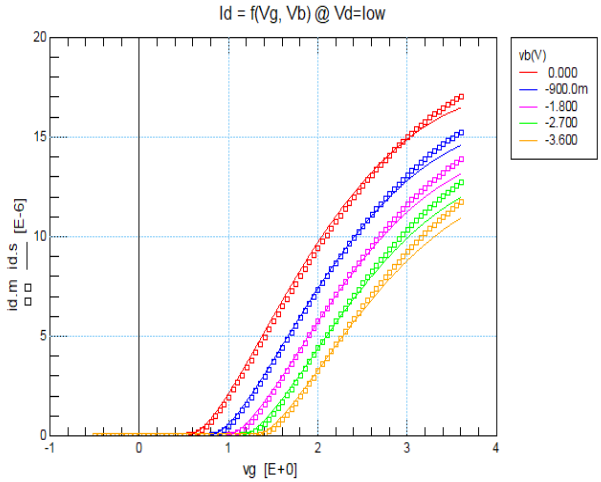


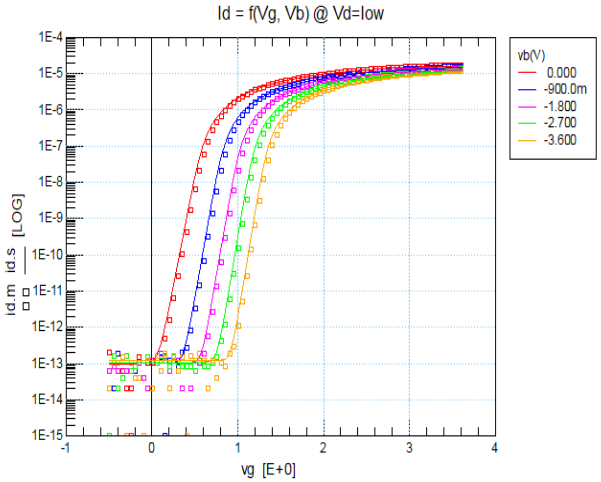


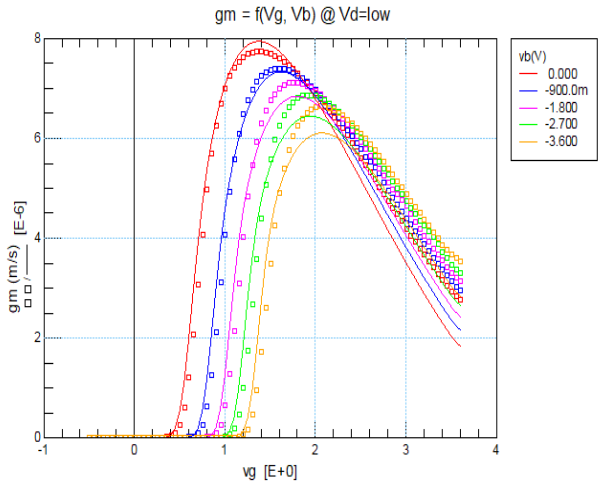


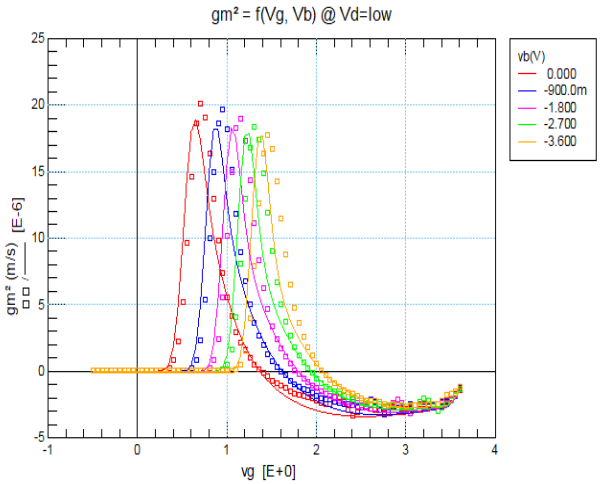
4.12 W05u0_L10u0_S558_1

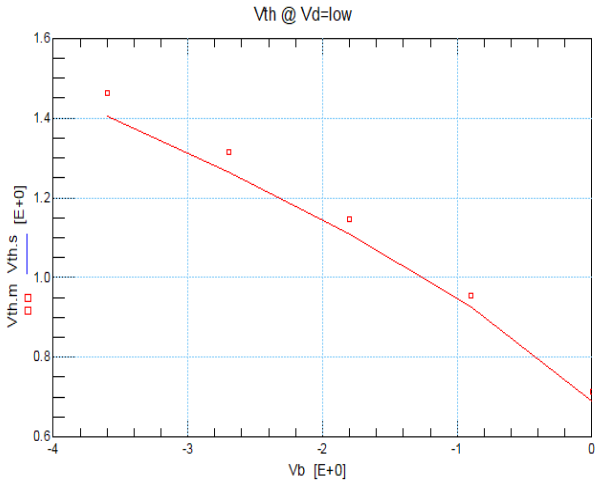
idvg, Vd = 0.1V, T = 27°C

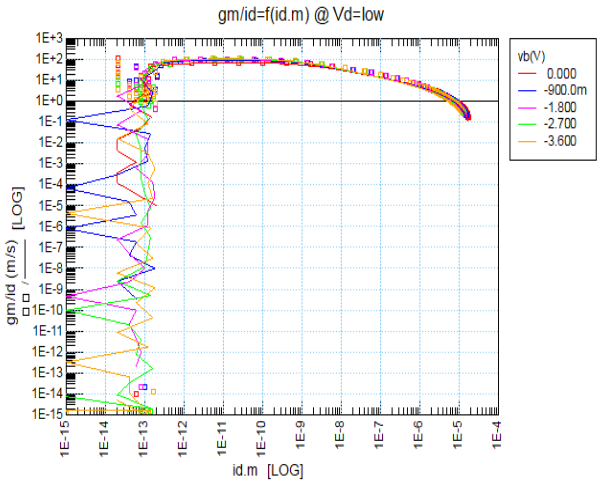


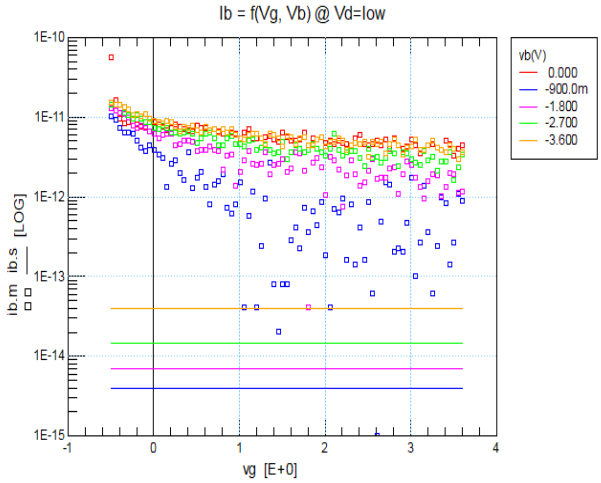


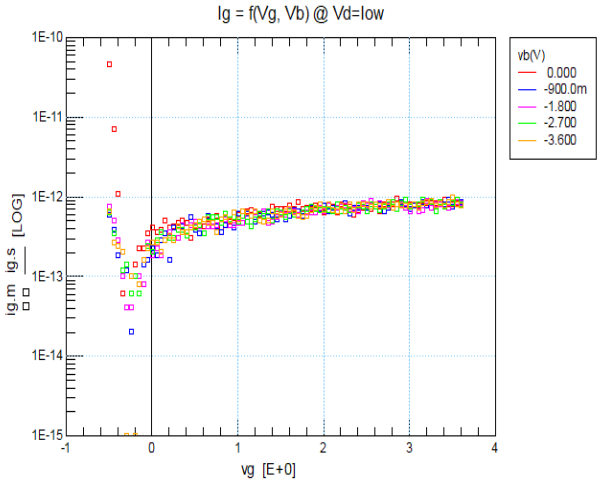




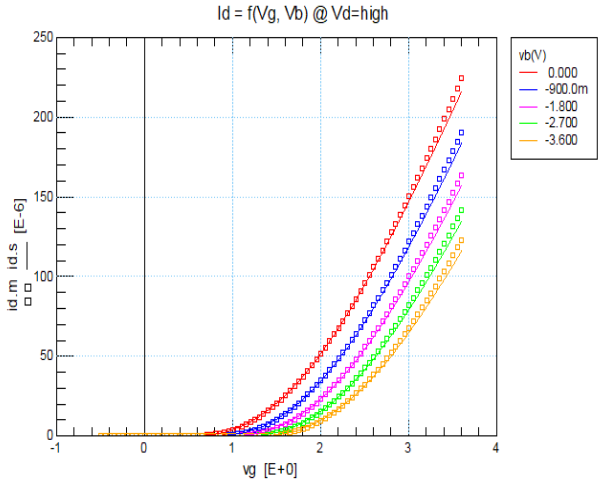


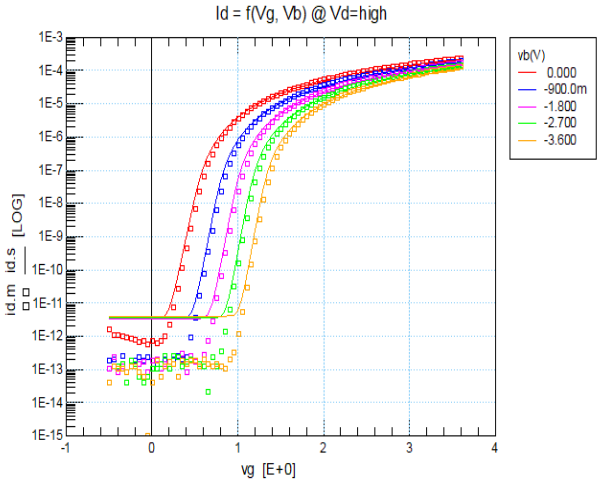


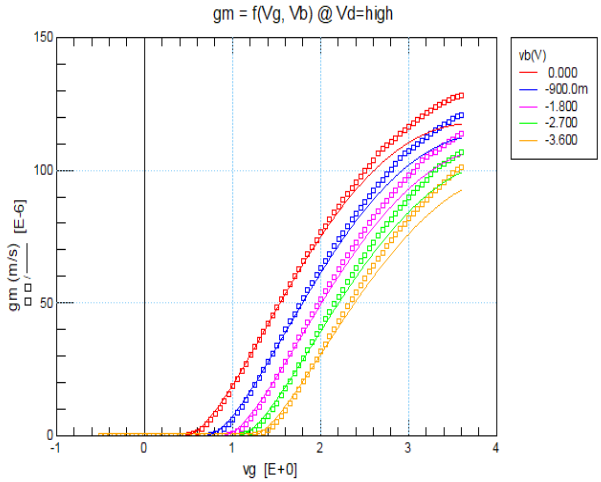


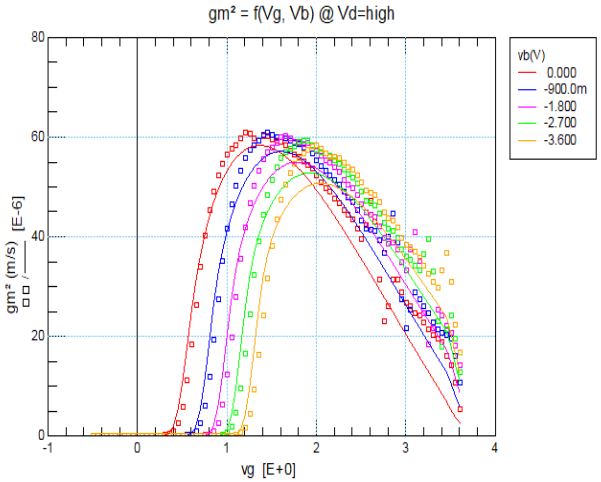


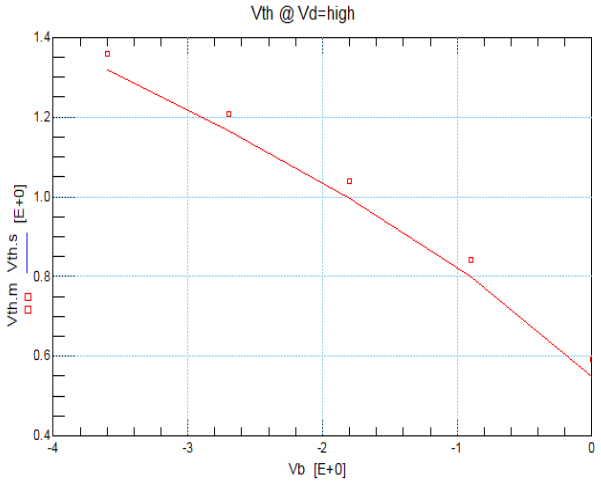
idvg, Vd = 3.3V, T = 27°C

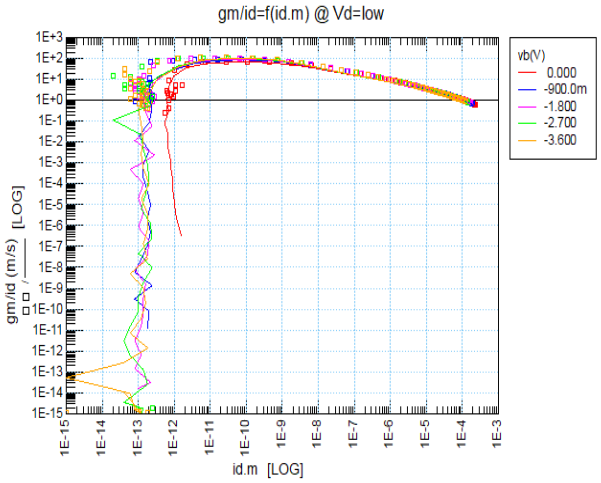


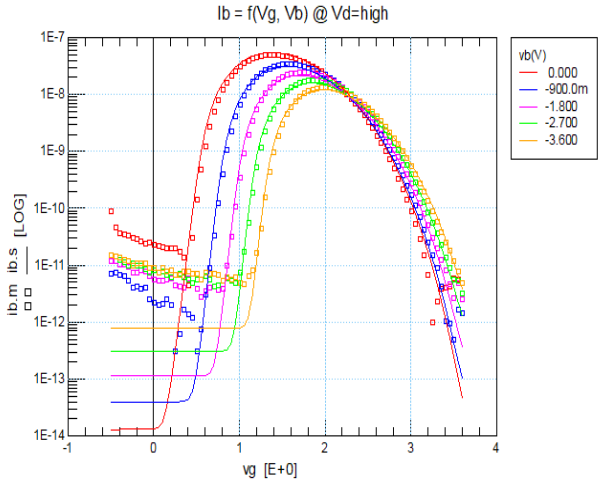


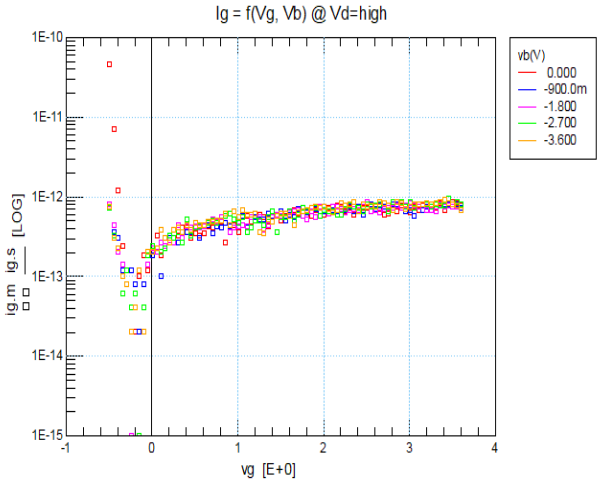




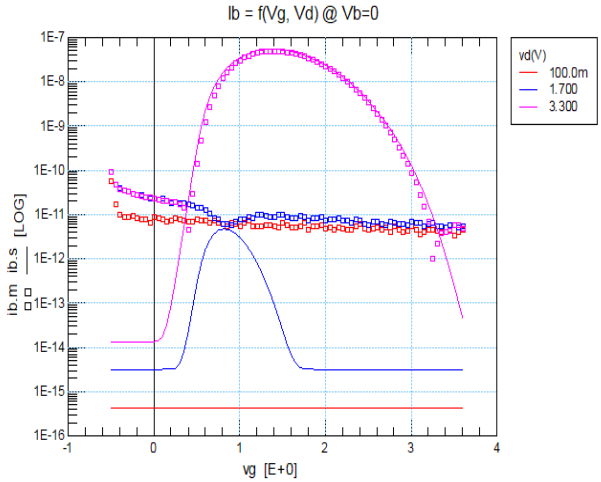


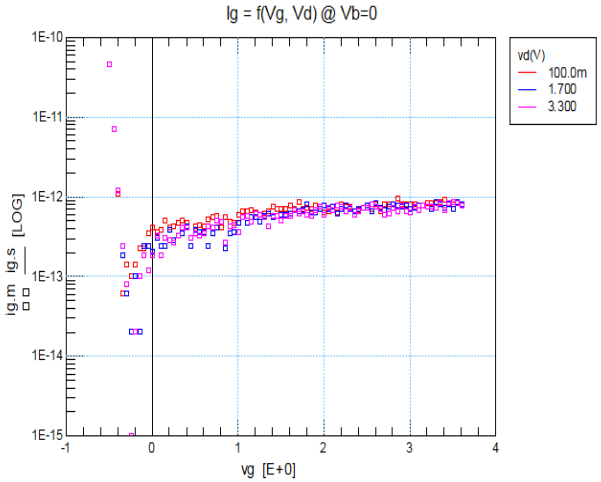




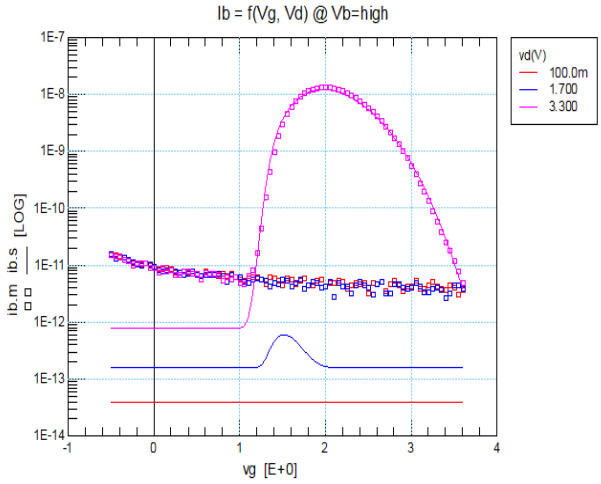


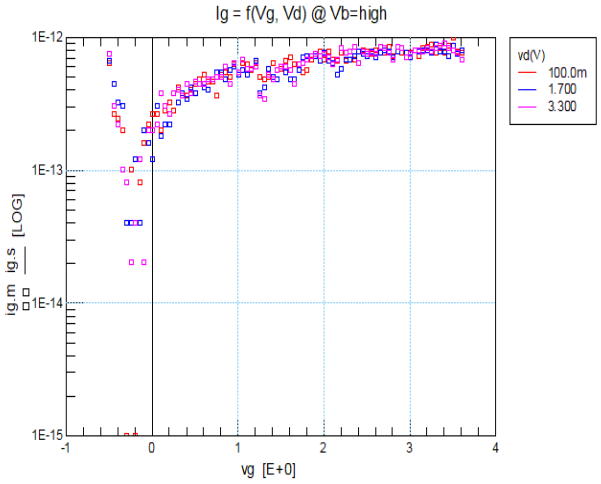
idvg, Ib,Ig, Vb = 0V, T = 27°C



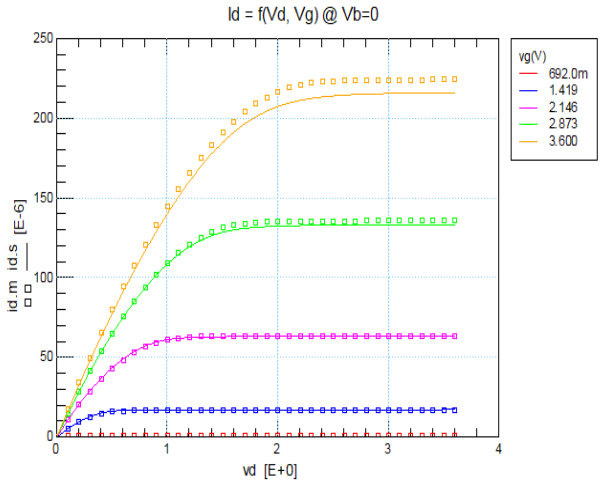


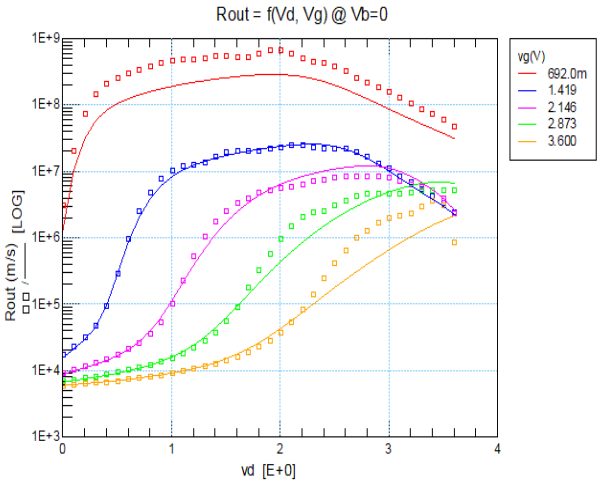
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

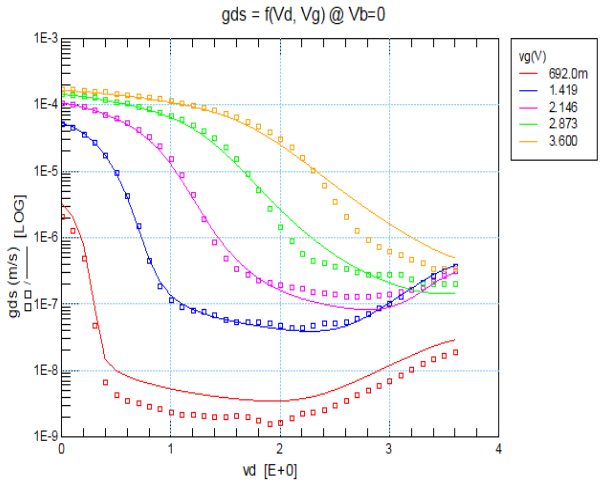




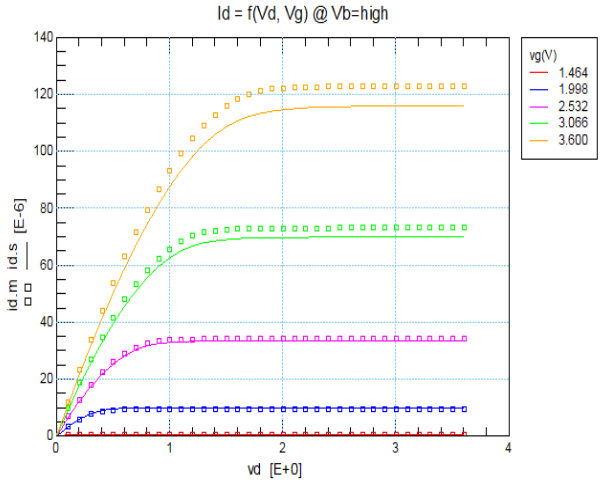
idvd, Vb = 0V, T = 27°C

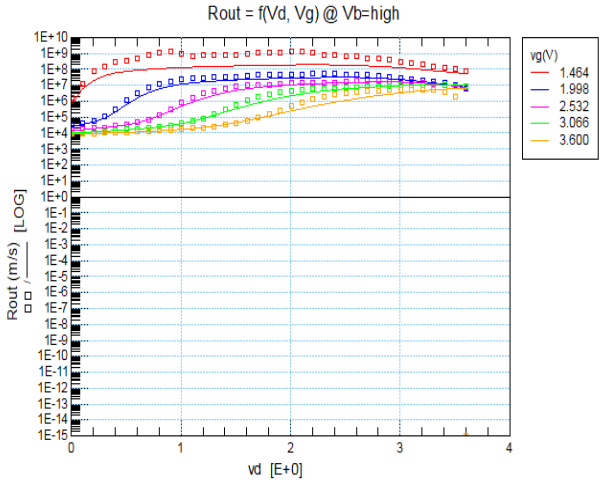


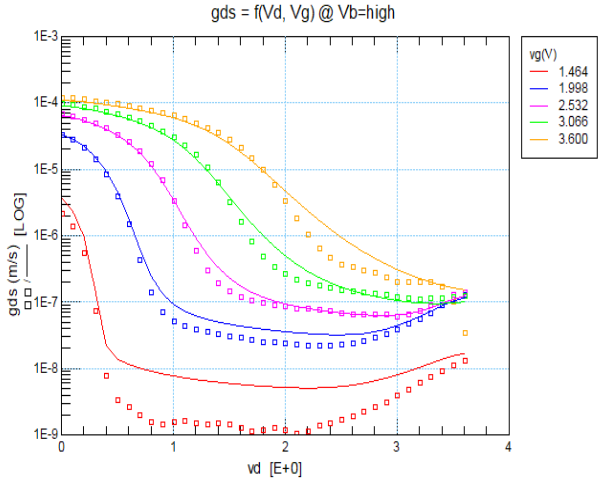




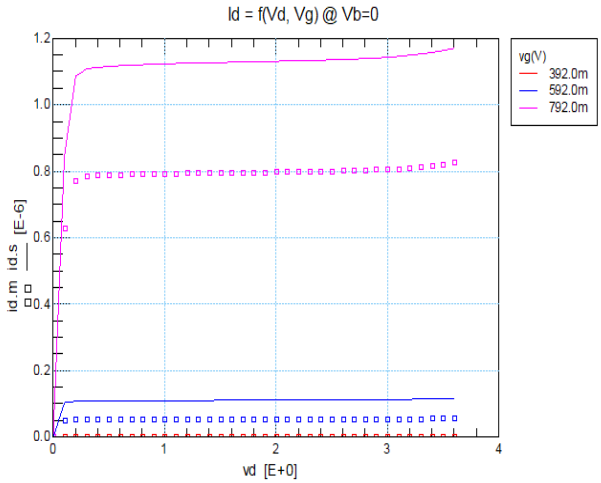
idvd_vbmin, Vb = -3.6V, T = 27°C

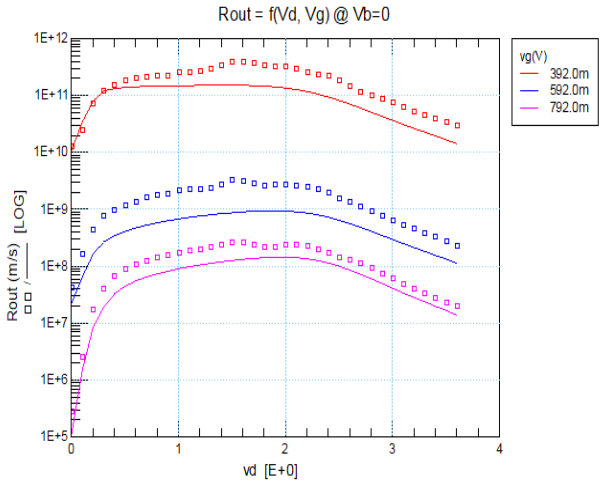


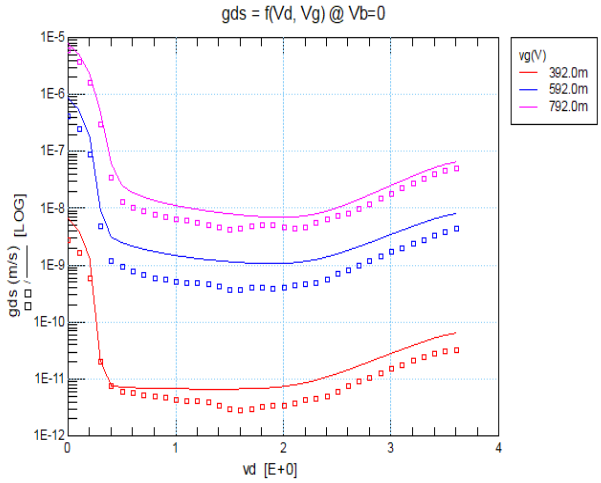




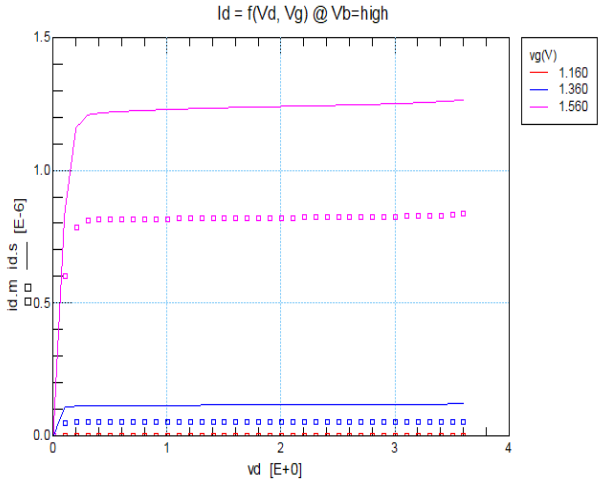
idvd_vth, Vb = 0V, T = 27°C

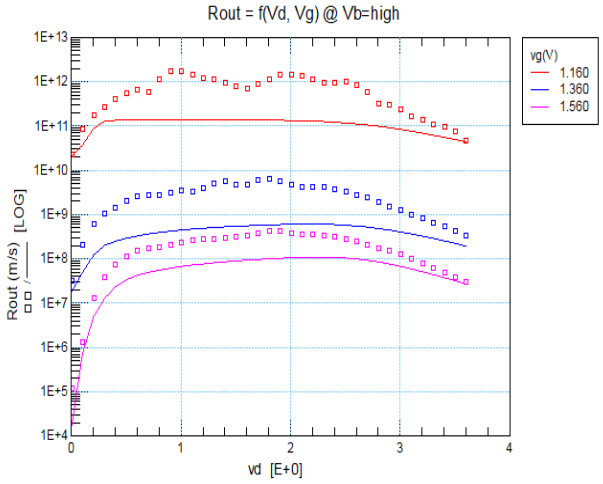


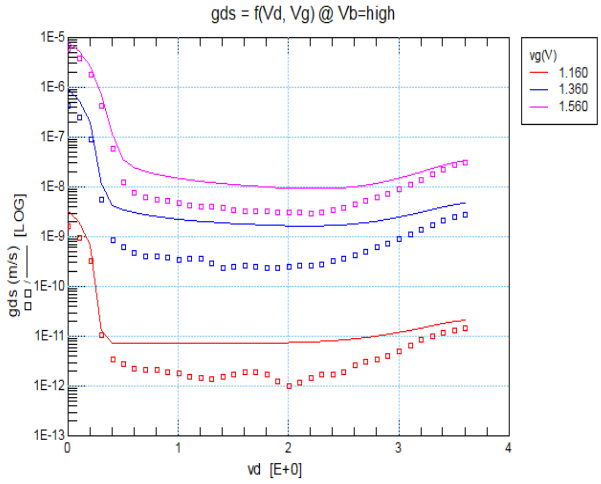




idvd_vth_vbmin, Vb = -3.6V, T = 27°C

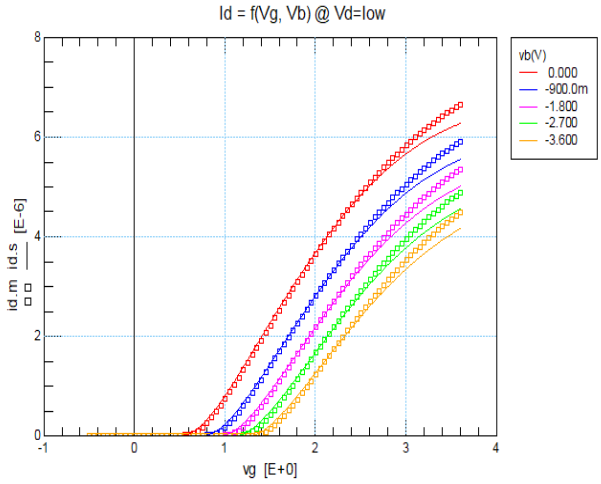


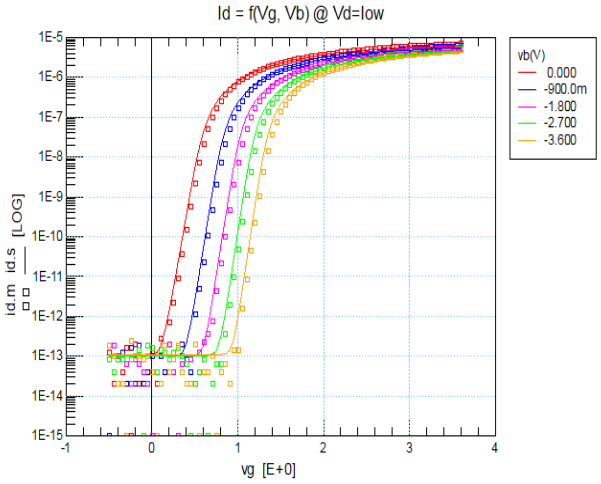


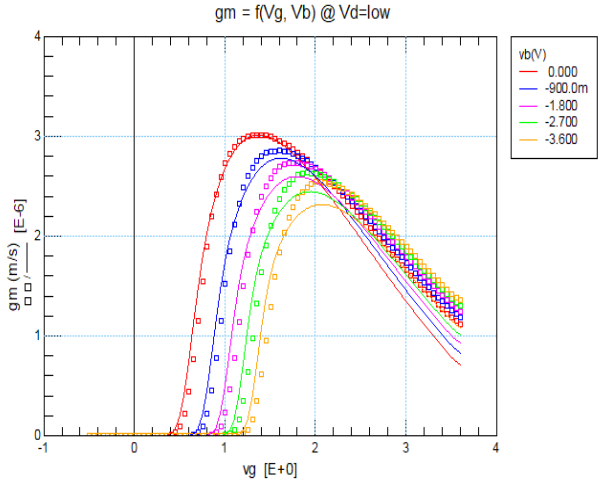


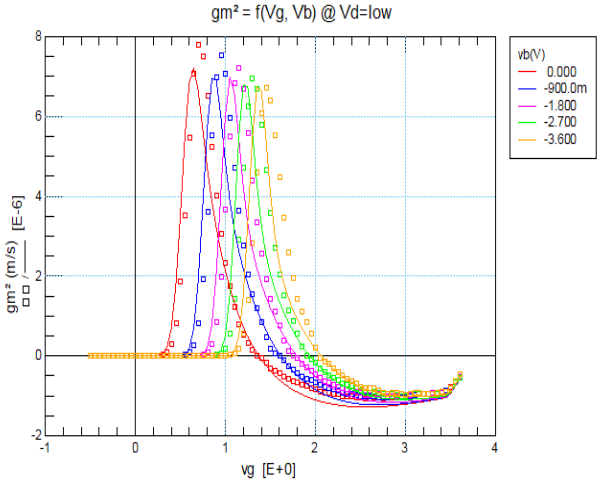
4.13 W02u0_L10u0_S558_5

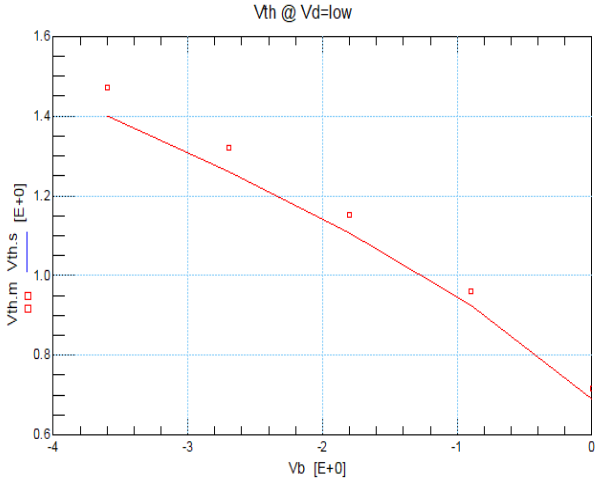
idvg, Vd = 0.1V, T = 27°C

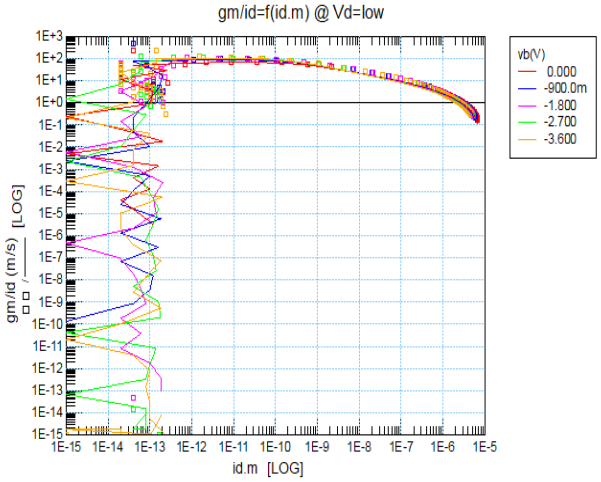


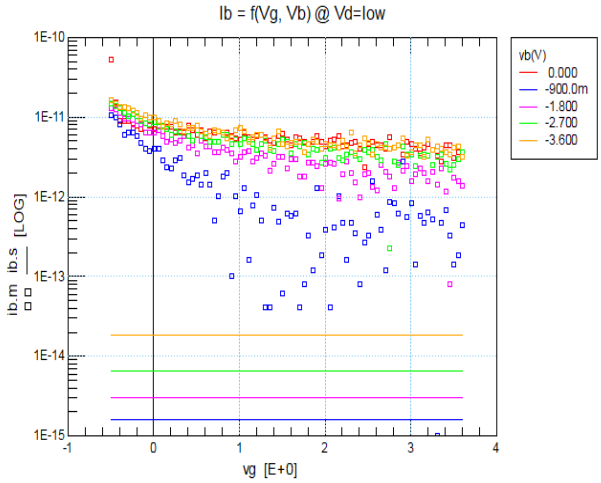


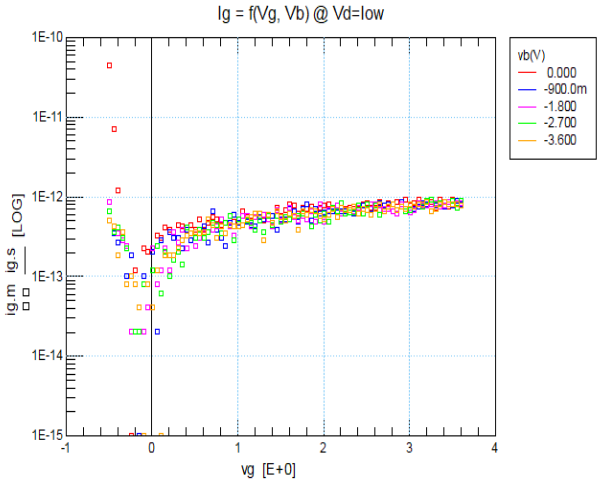




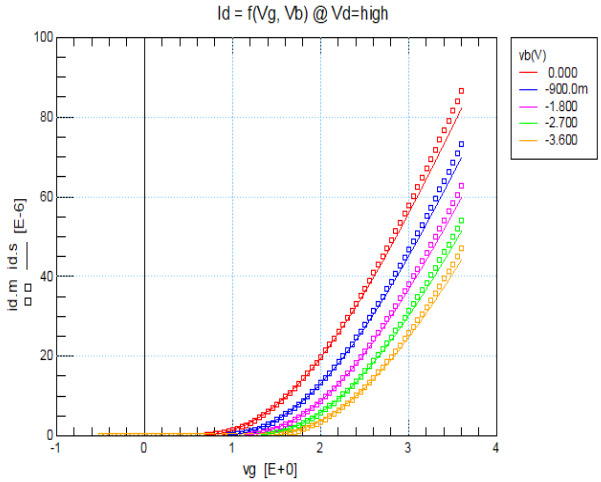


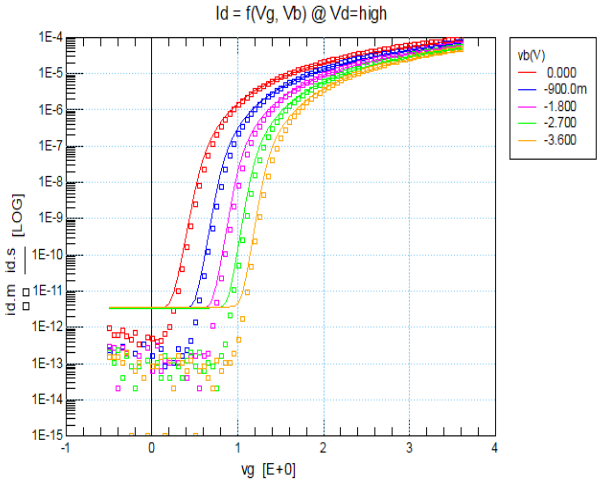


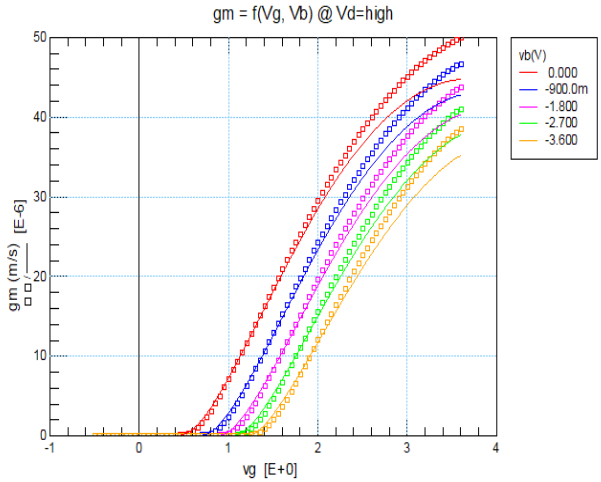


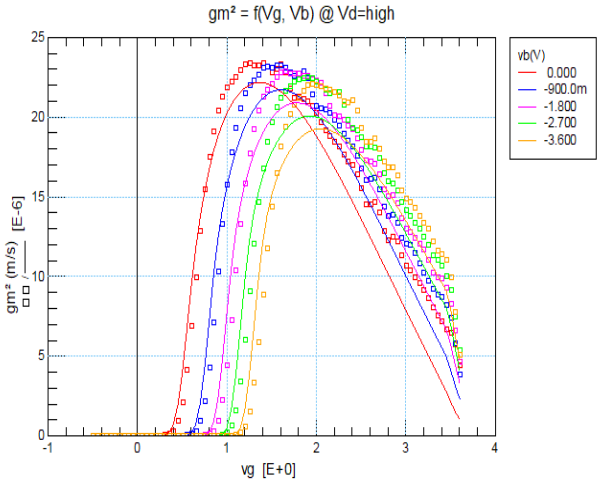


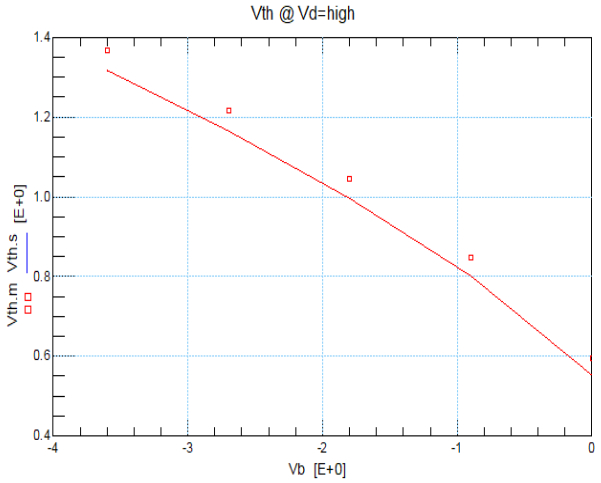
idvg, Vd = 3.3V, T = 27°C

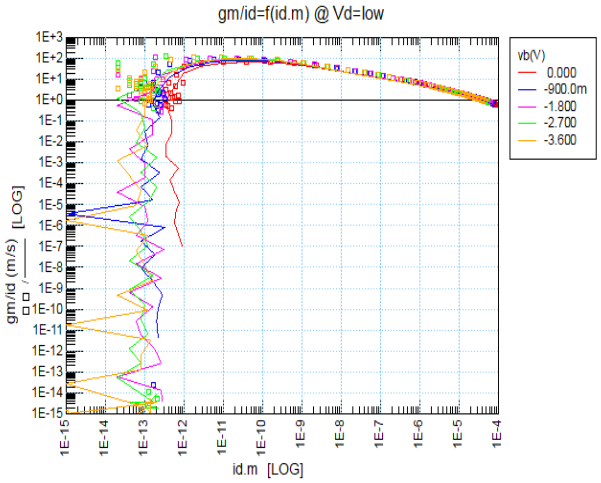


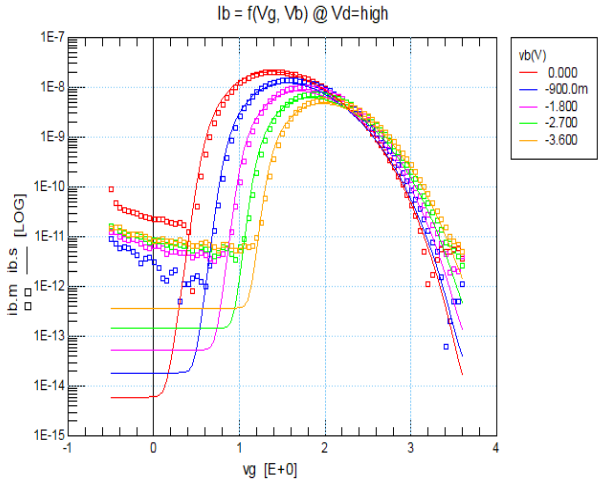


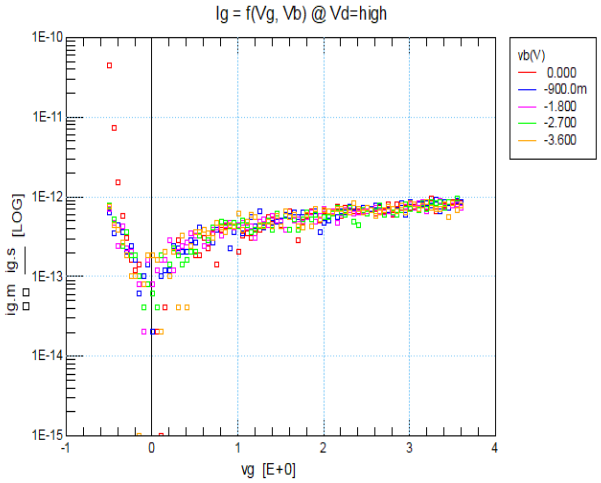




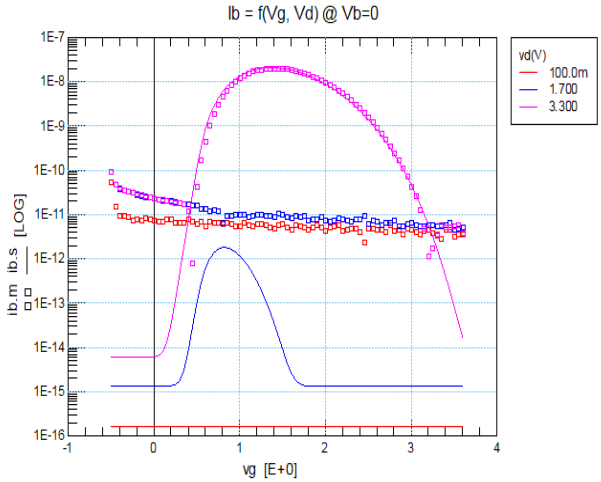


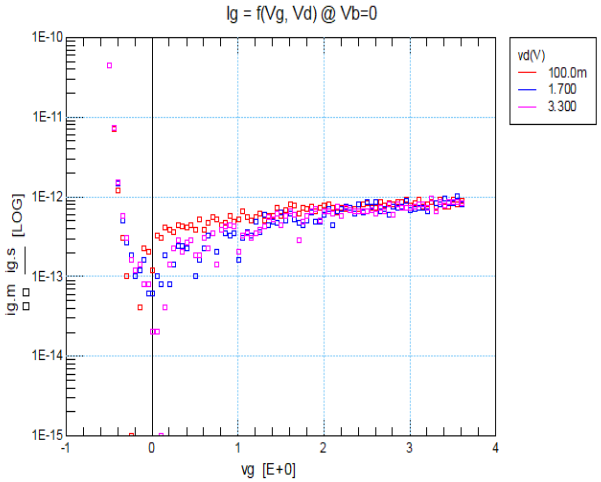




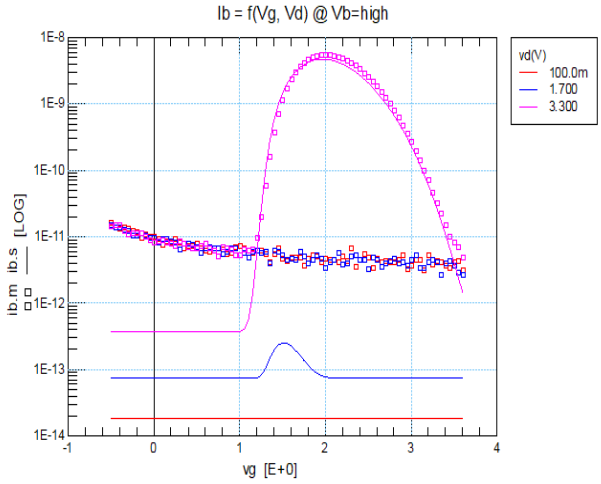


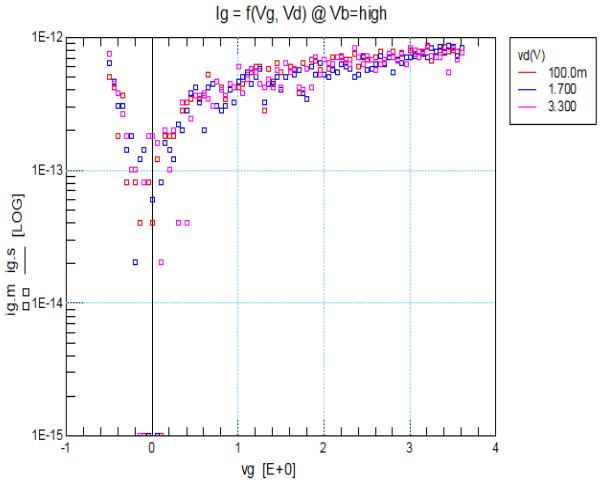
idvg, Ib,Ig, Vb = 0V, T = 27°C



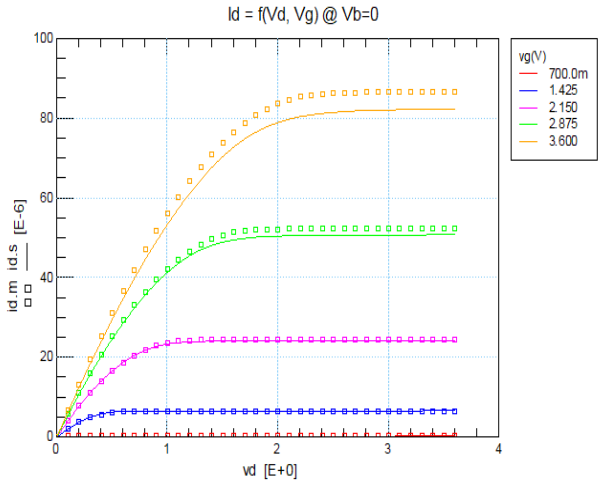


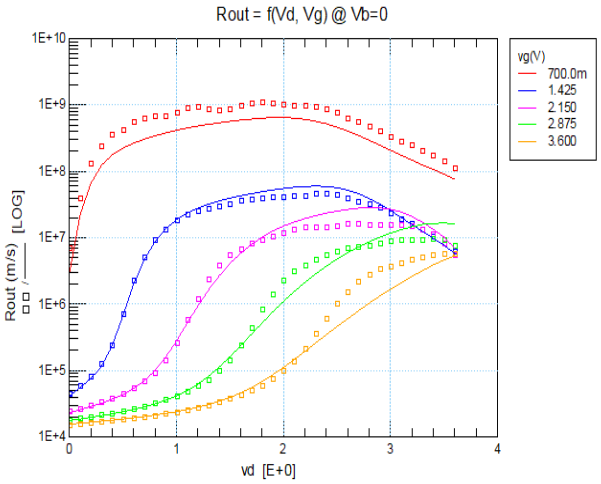
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

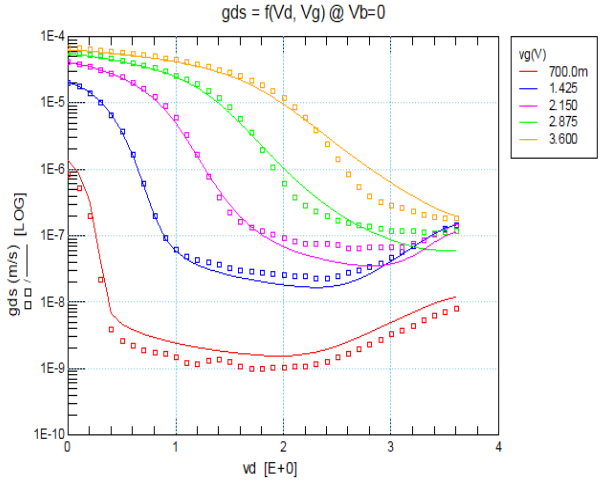




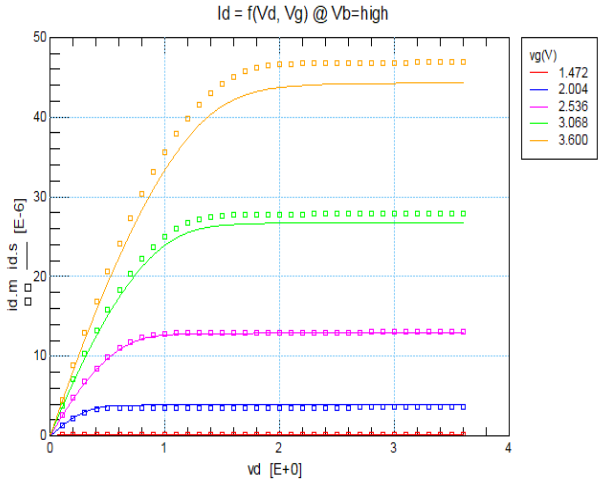
idvd, Vb = 0V, T = 27°C

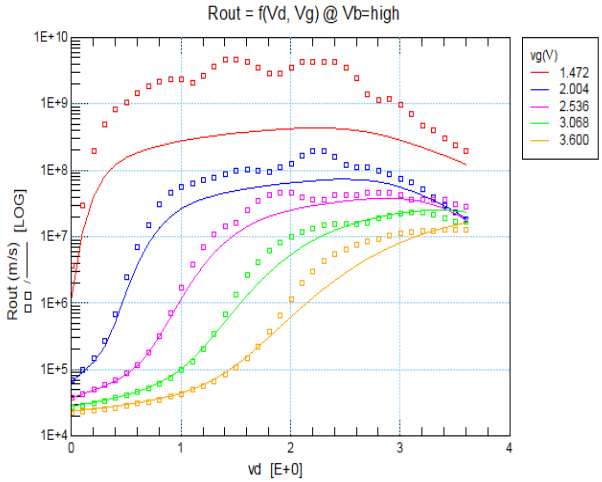


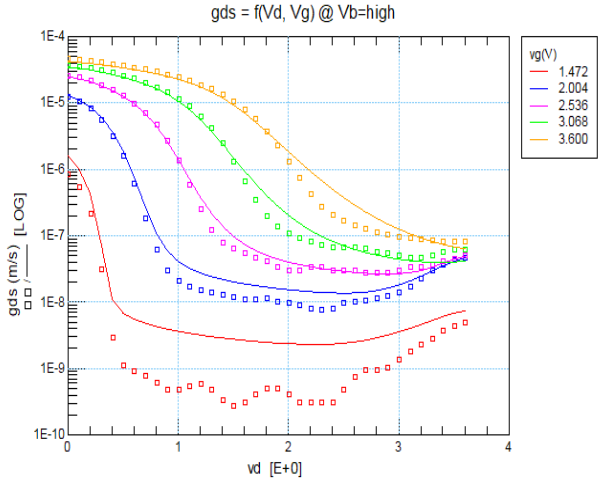




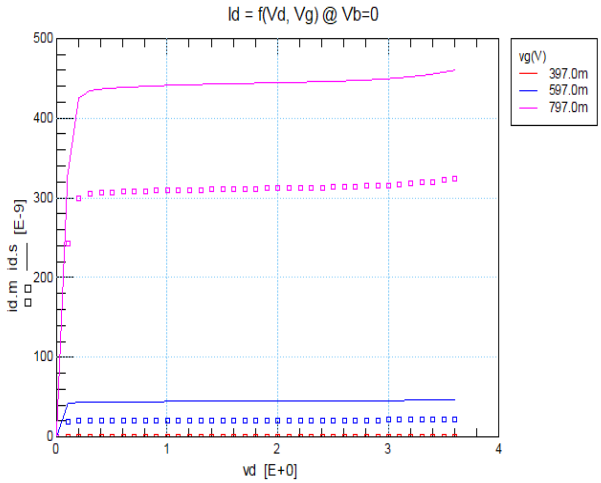
idvd_vbmin, Vb = -3.6V, T = 27°C

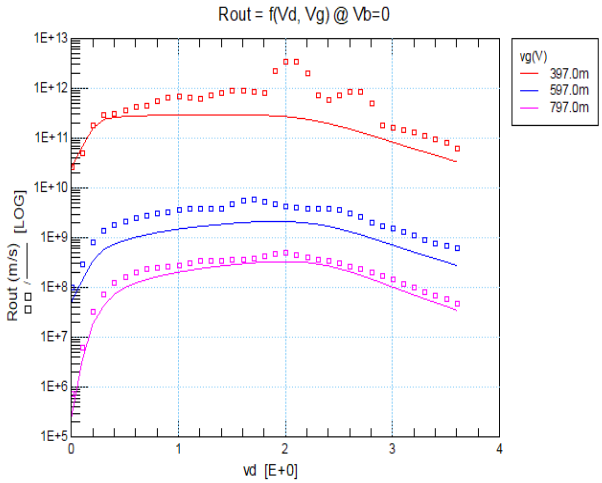


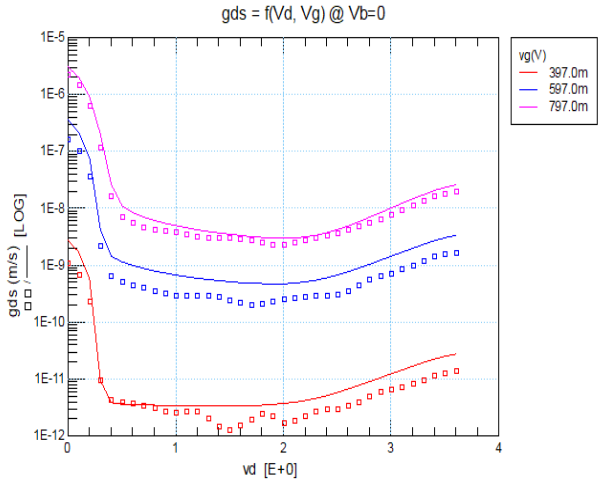




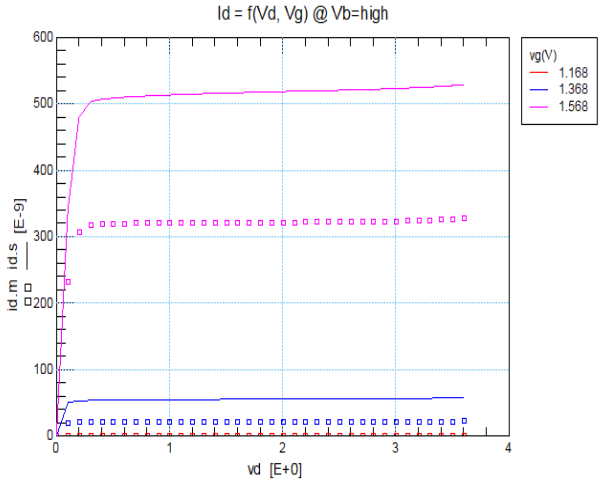
idvd_vth, Vb = 0V, T = 27°C

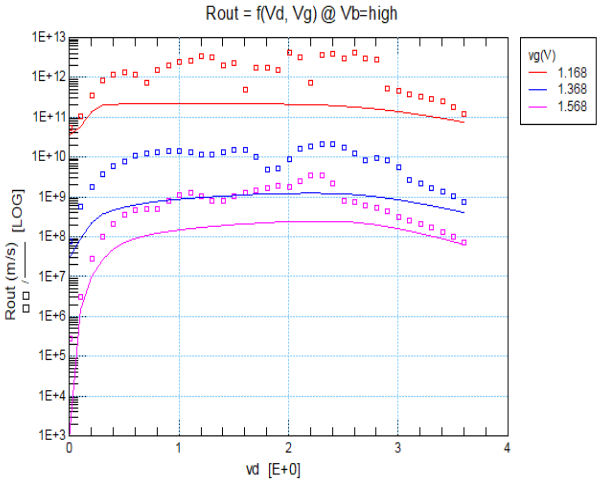


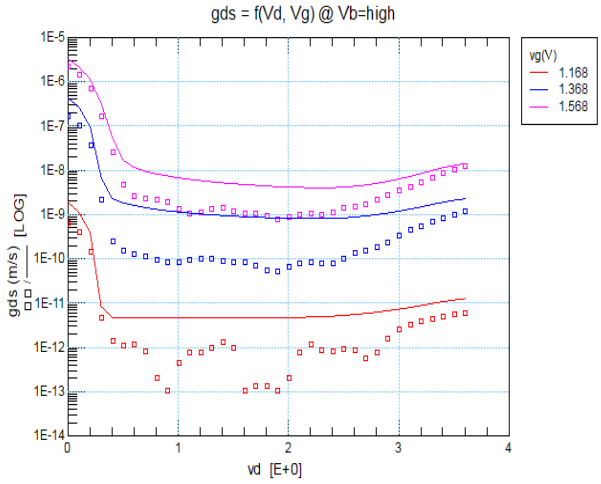




idvd_vth_vbmin, Vb = -3.6V, T = 27°C



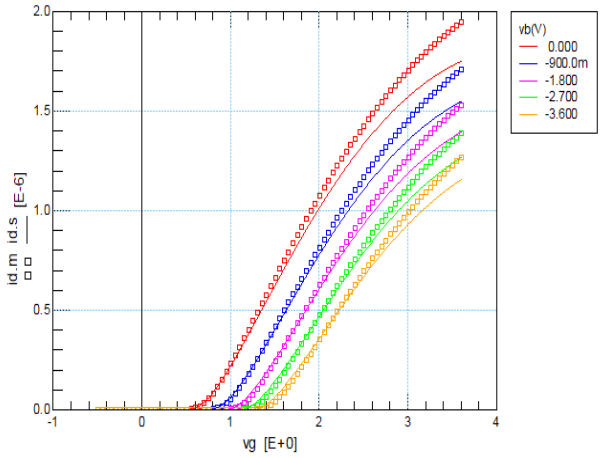




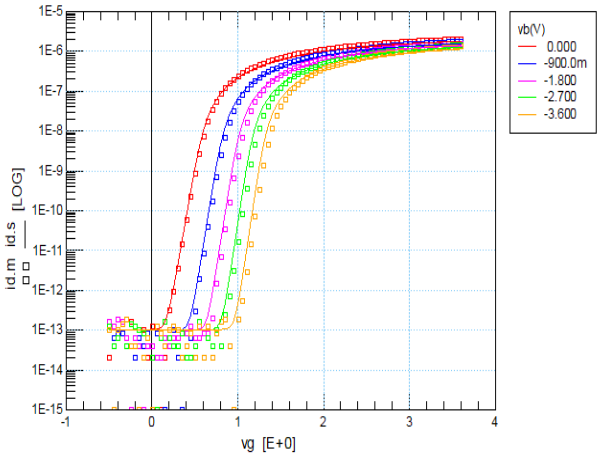
4.14 W0u6_L10u0_S559_5

idvg, Vd = 0.1V, T = 27°C

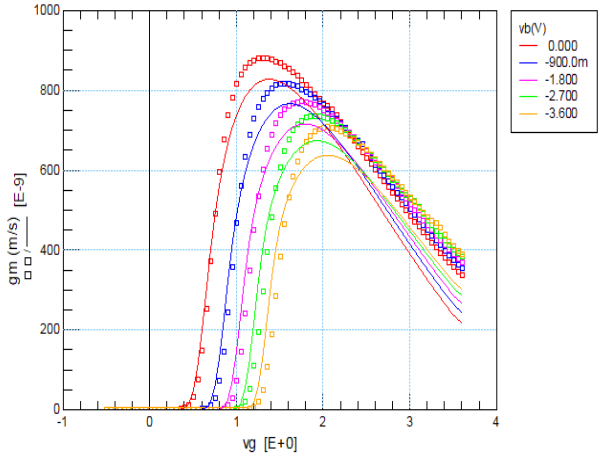
Id = f(Vg, Vb) @ Vd=low



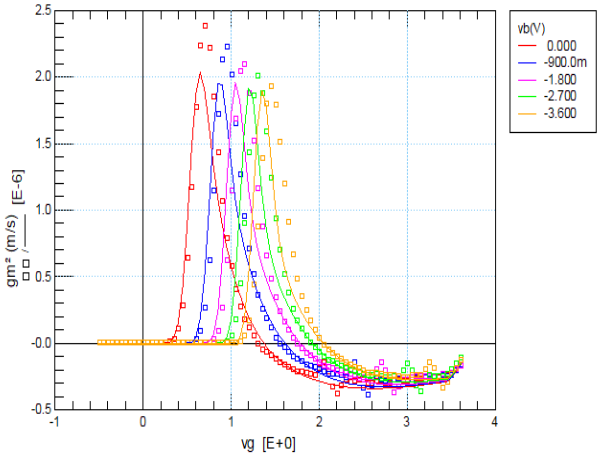
Id = f(Vg, Vb) @ Vd=low



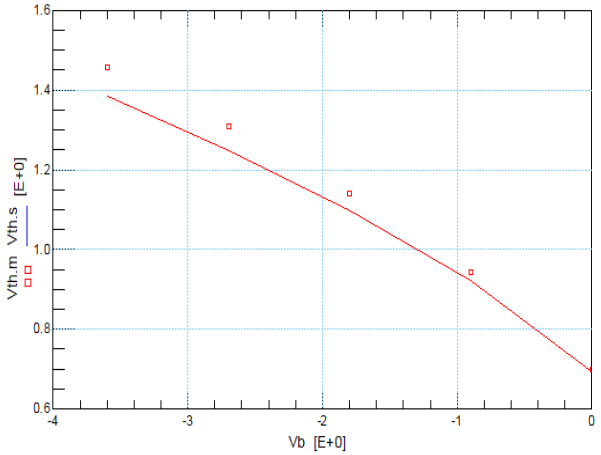
gm = f(Vg, Vb) @ Vd=low



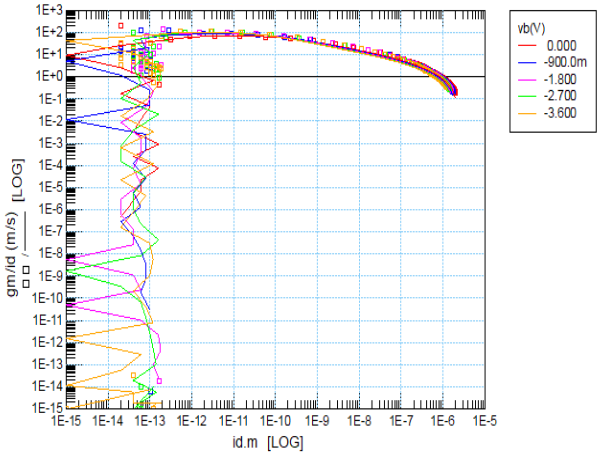
gm² = f(Vg, Vb) @ Vd=low

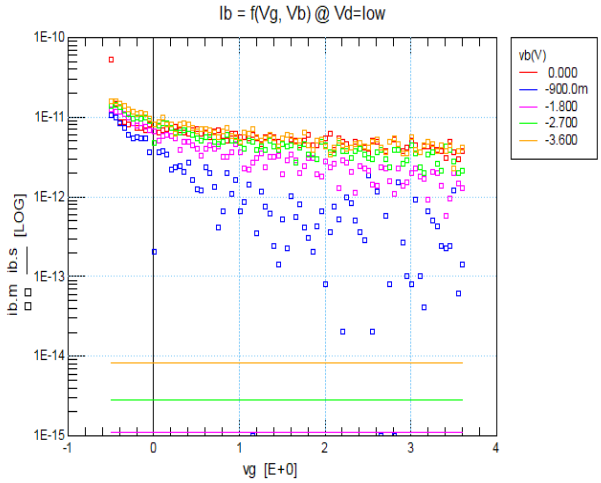


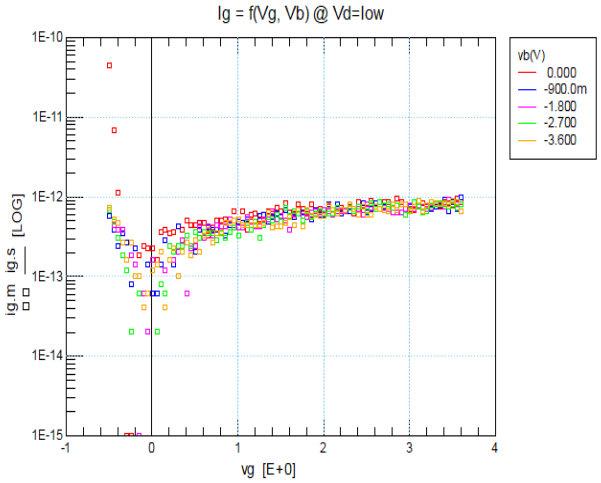
Vth @ Vd=low



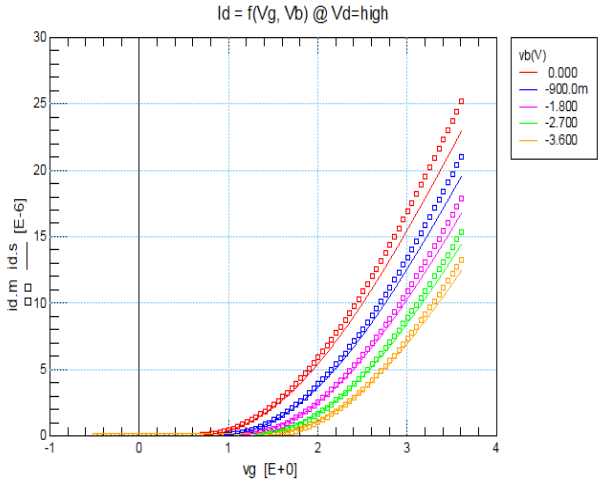
gm/id=f(id.m) @ Vd=low

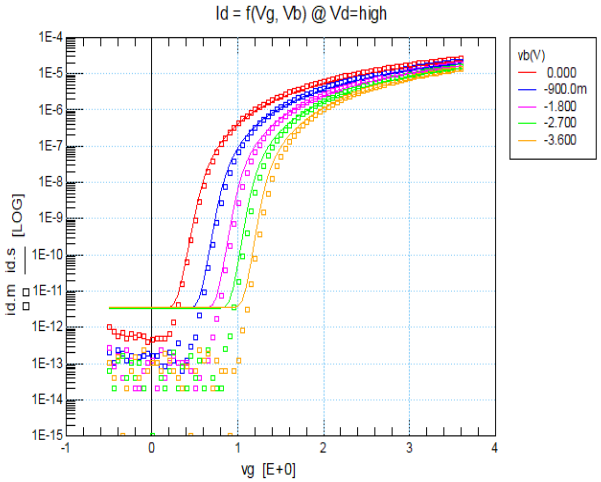


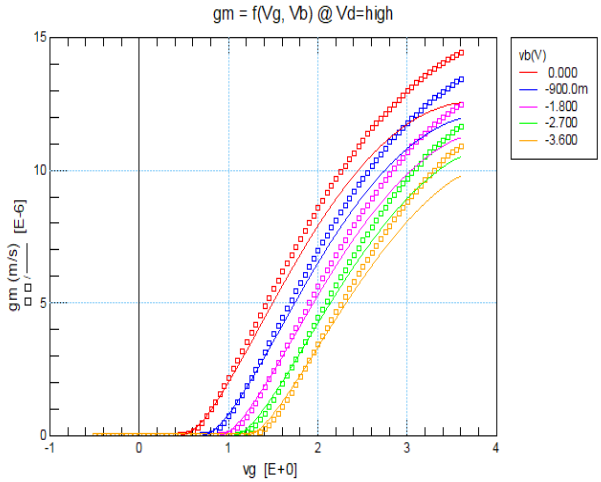


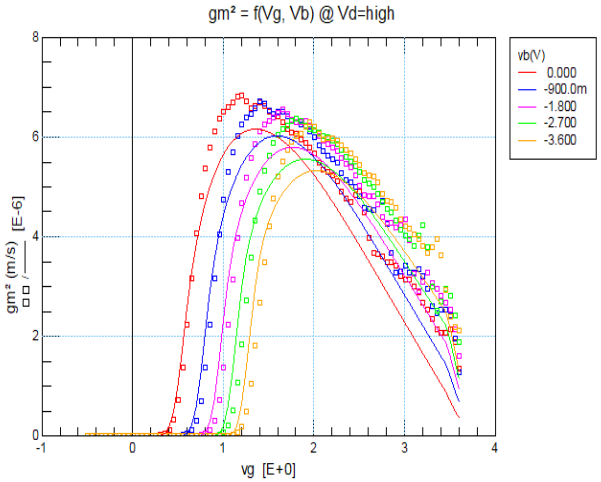


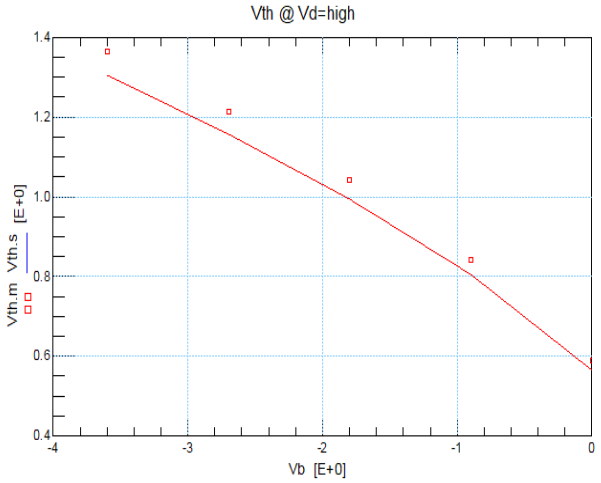
idvg, Vd = 3.3V, T = 27°C

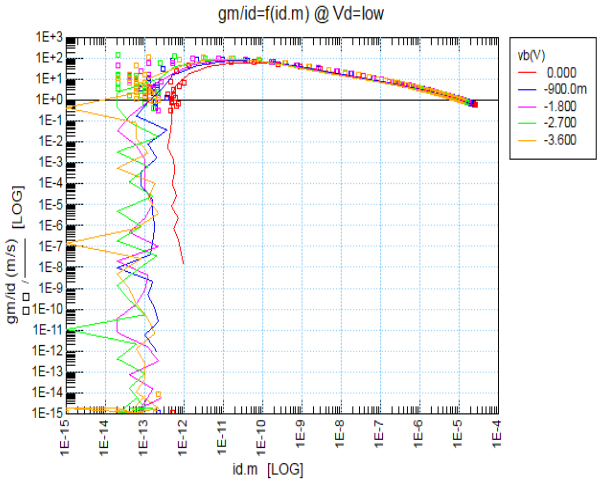


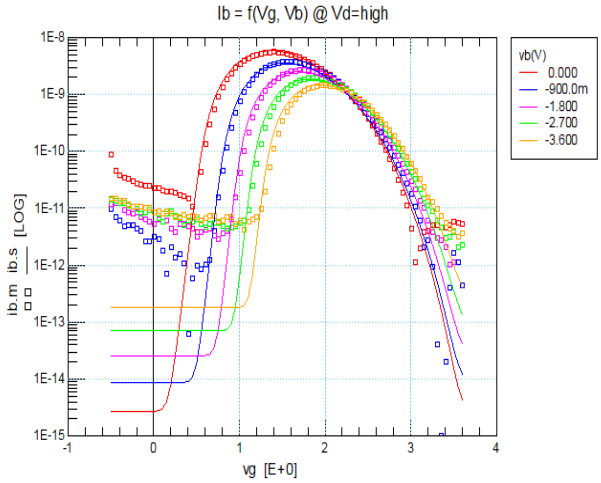


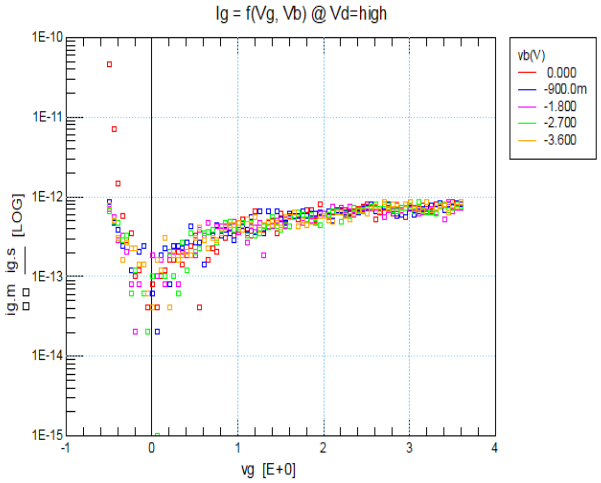




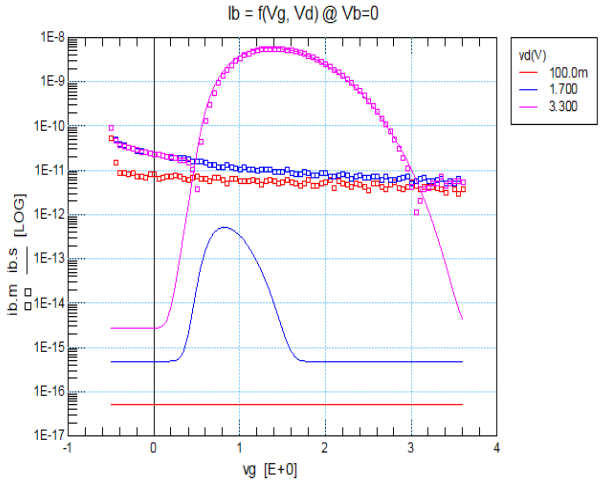


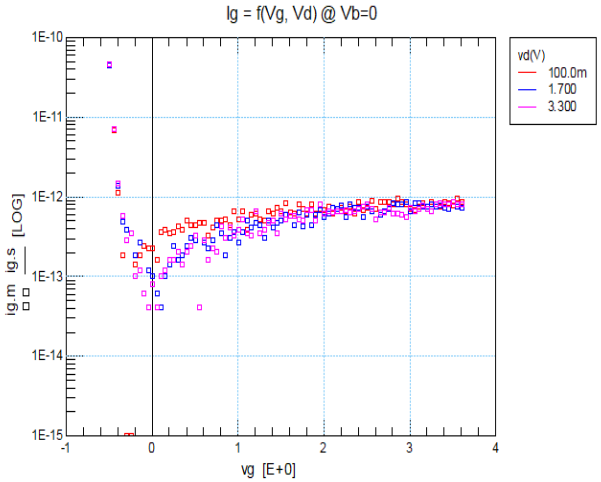




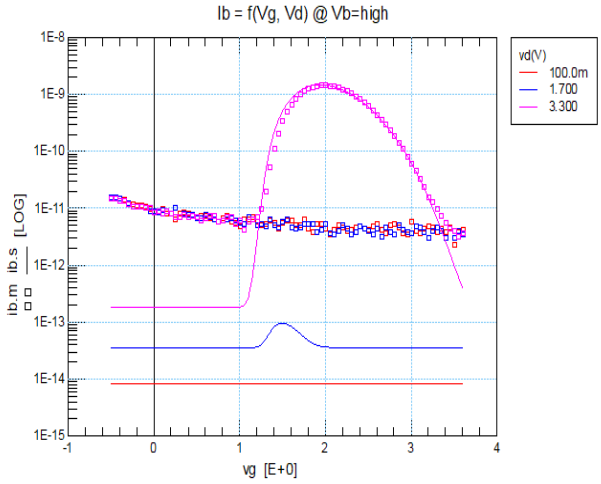


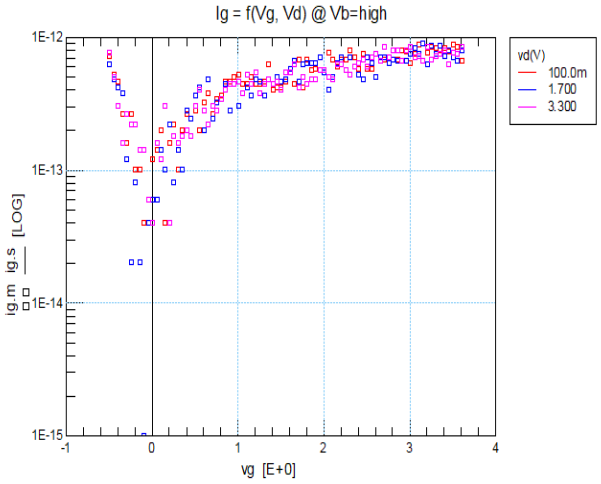
idvg, Ib,Ig, Vb = 0V, T = 27°C



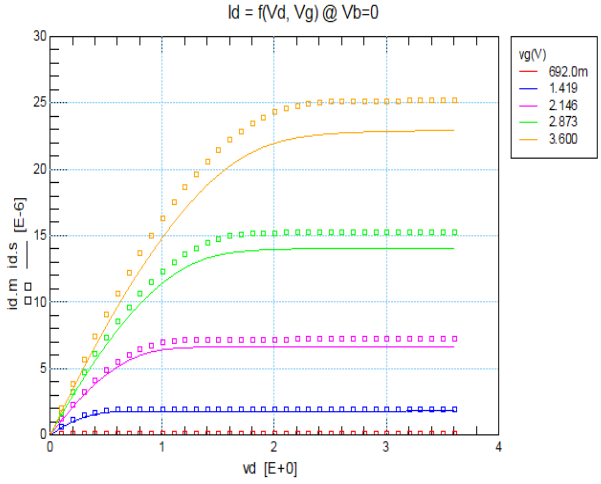


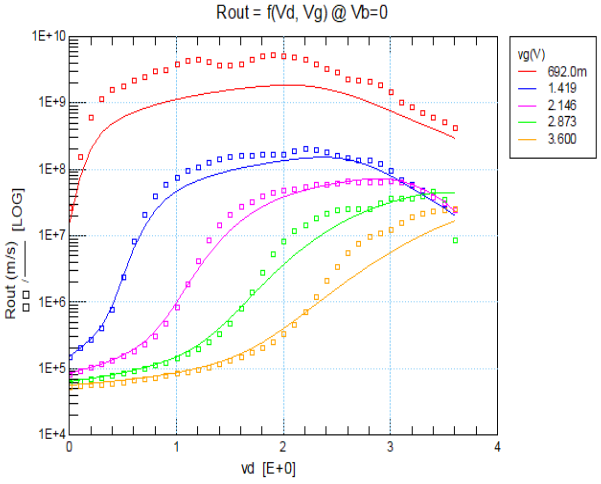
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

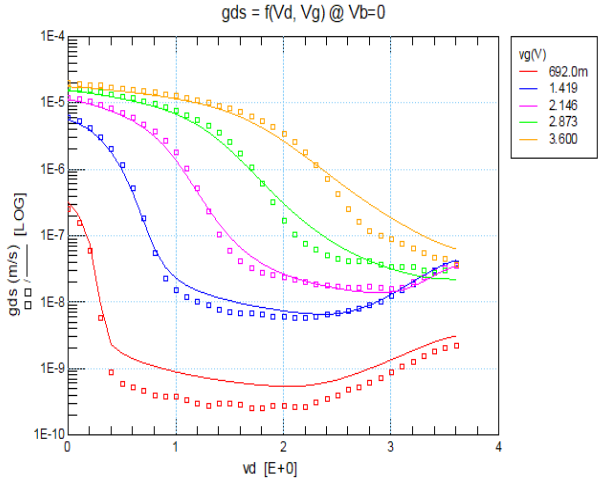




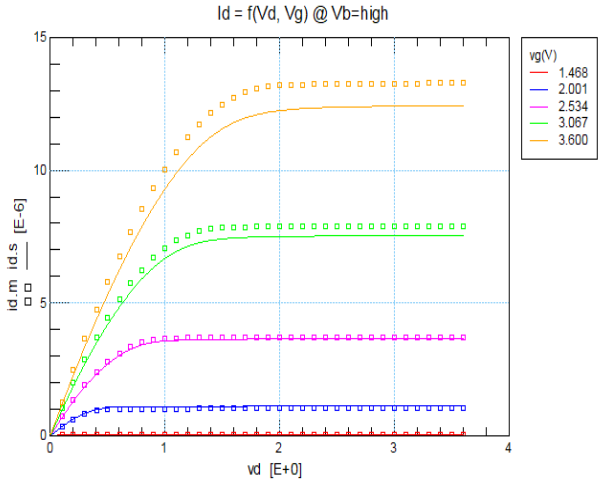
idvd, Vb = 0V, T = 27°C

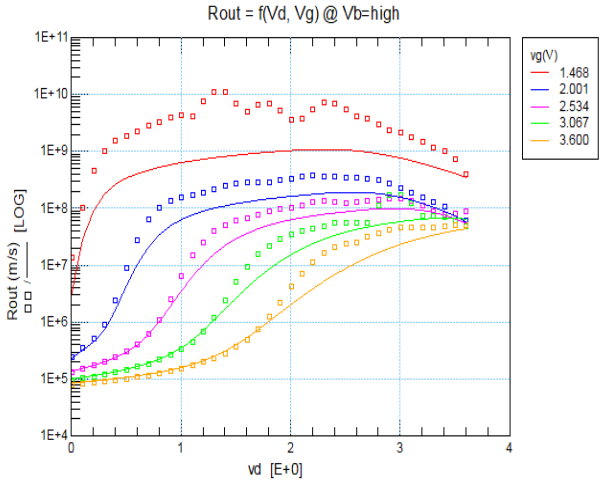


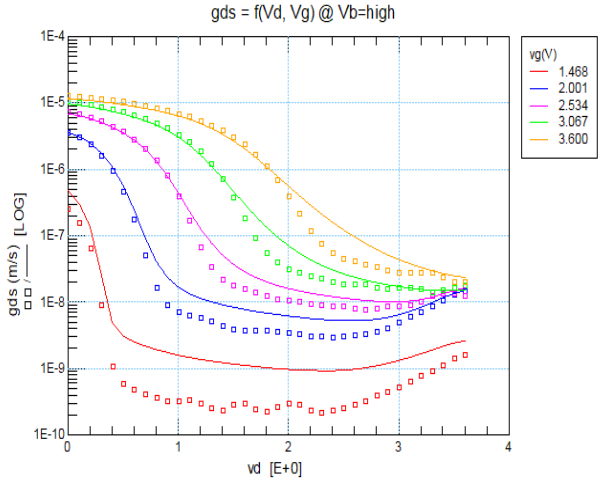




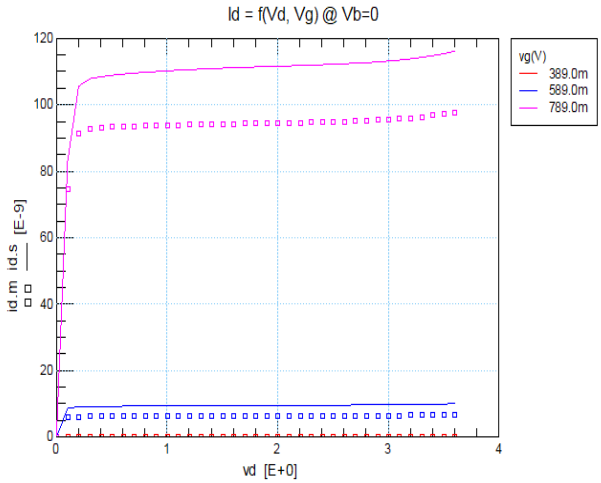
idvd_vbmin, Vb = -3.6V, T = 27°C

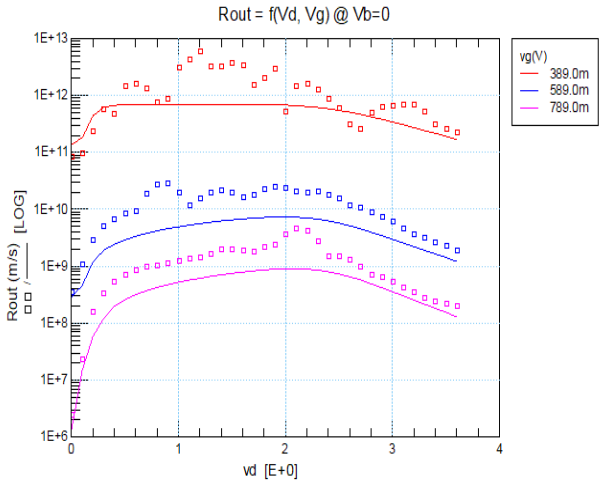


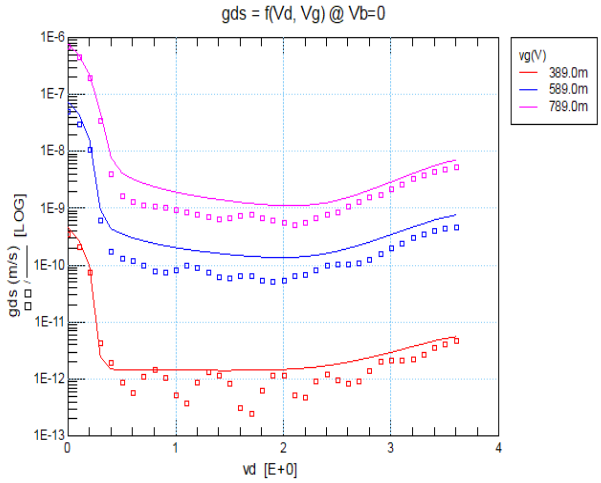




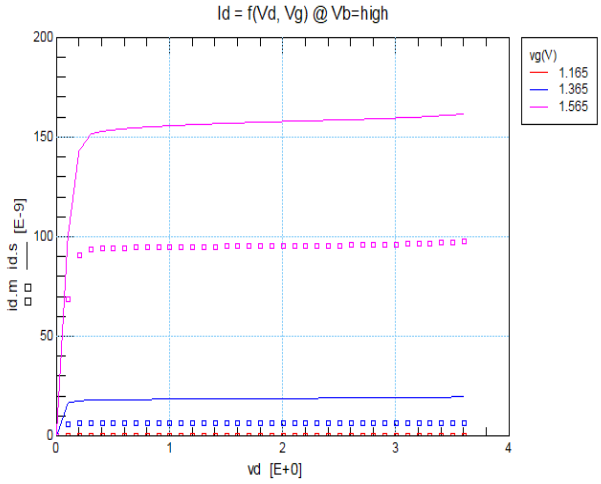
idvd_vth, Vb = 0V, T = 27°C

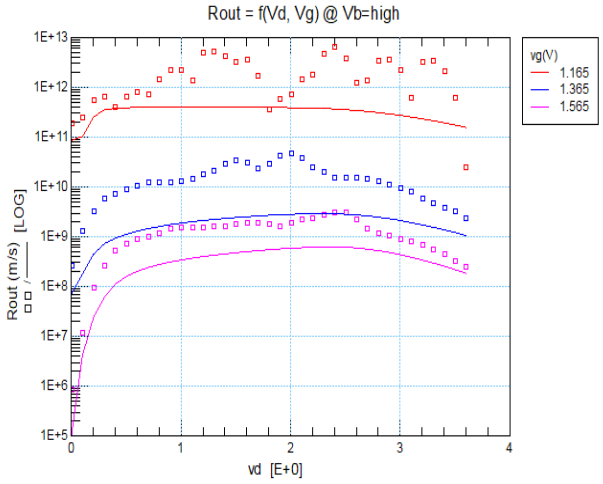


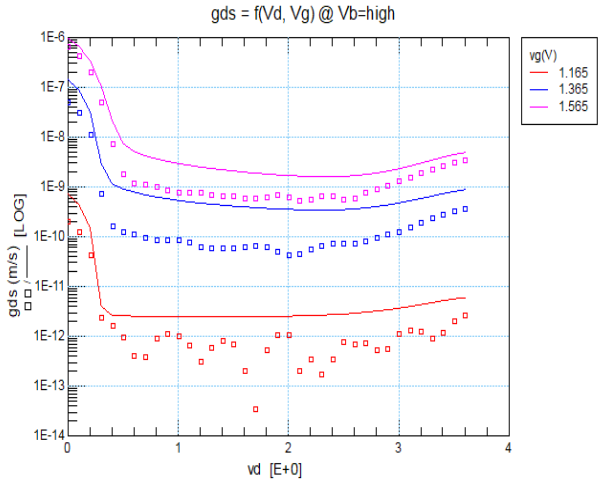




idvd_vth_vbmin, Vb = -3.6V, T = 27°C

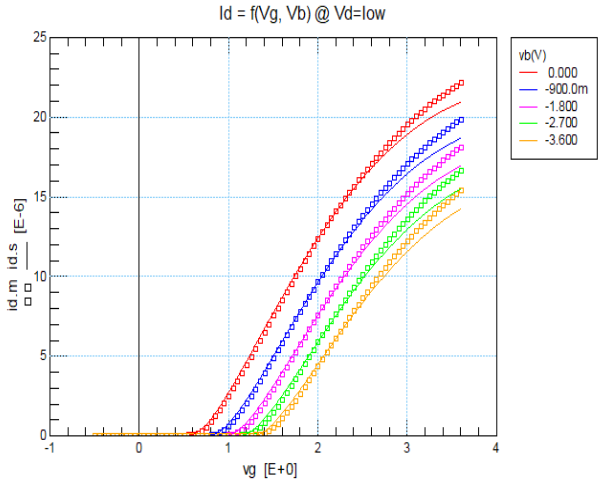


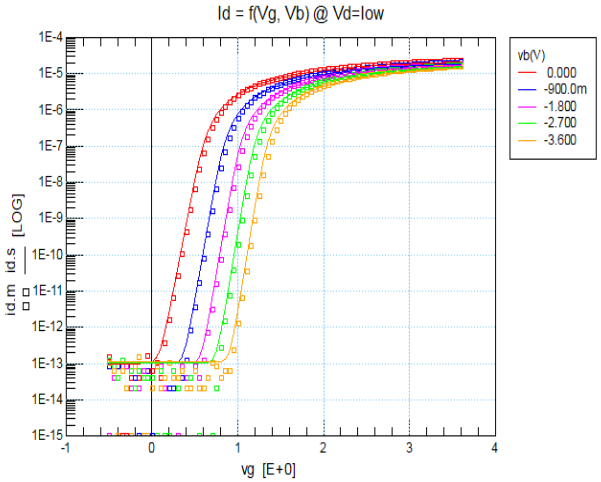


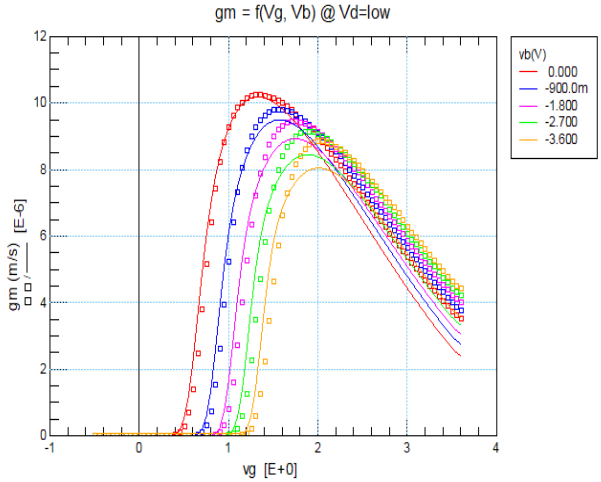


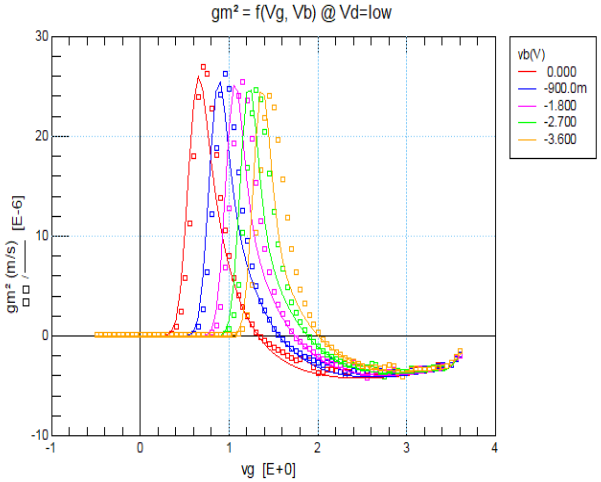
4.15W02u0_L3u0_S558_4

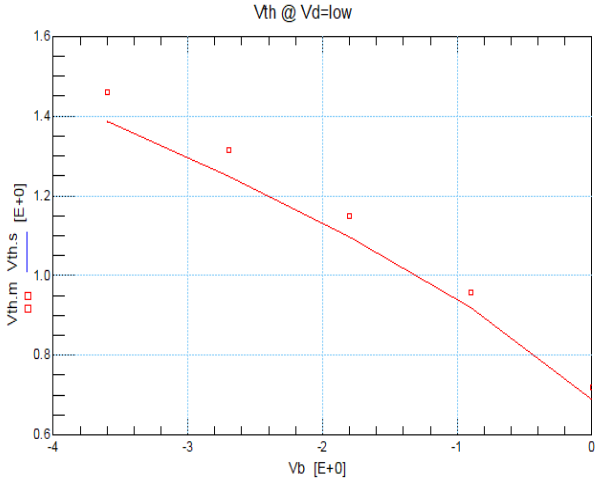
idvg, Vd = 0.1V, T = 27°C

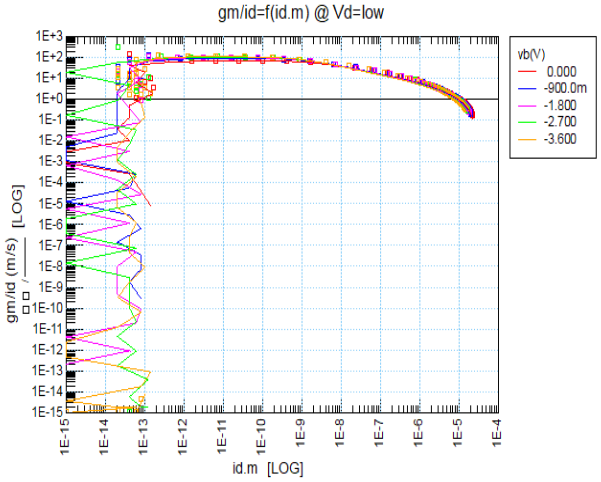


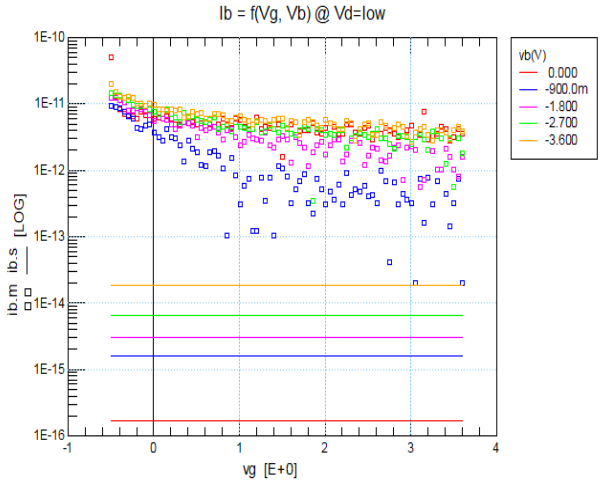


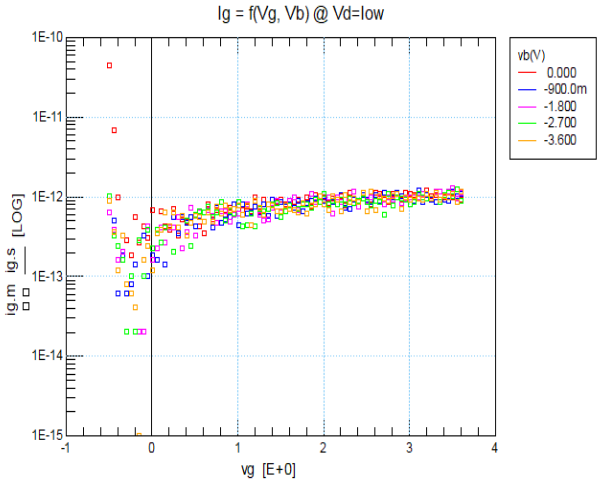




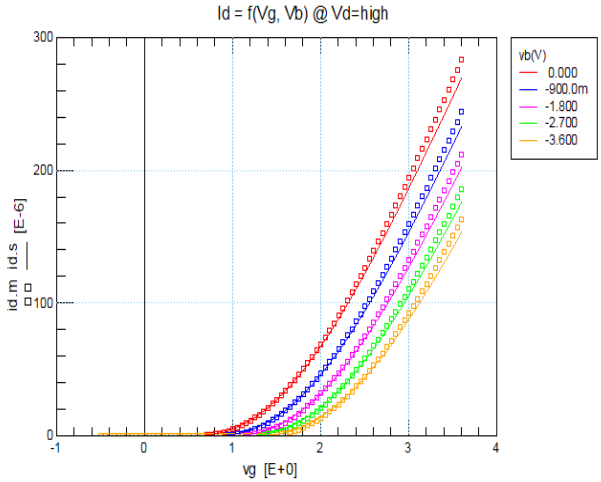


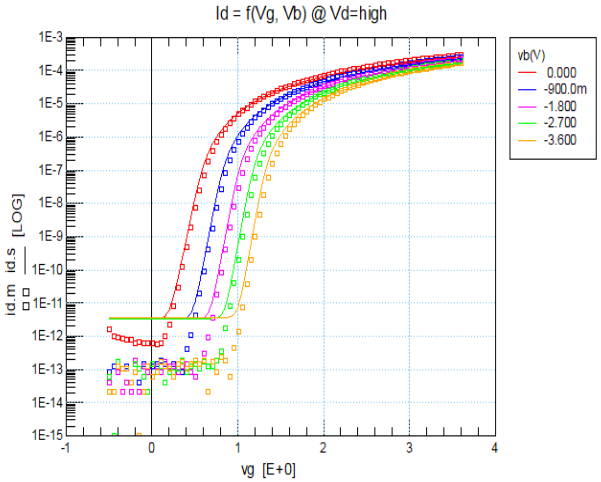


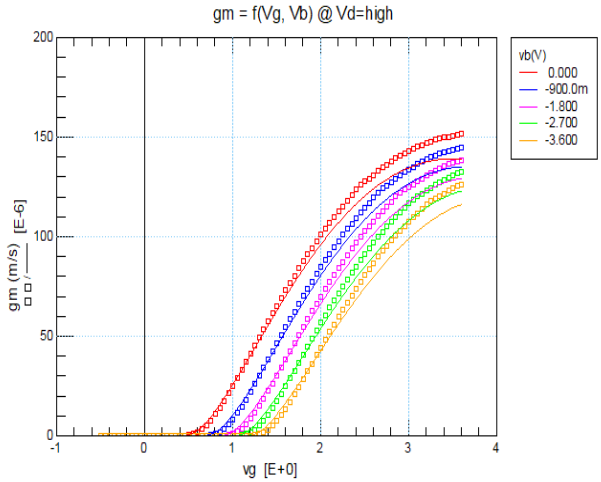


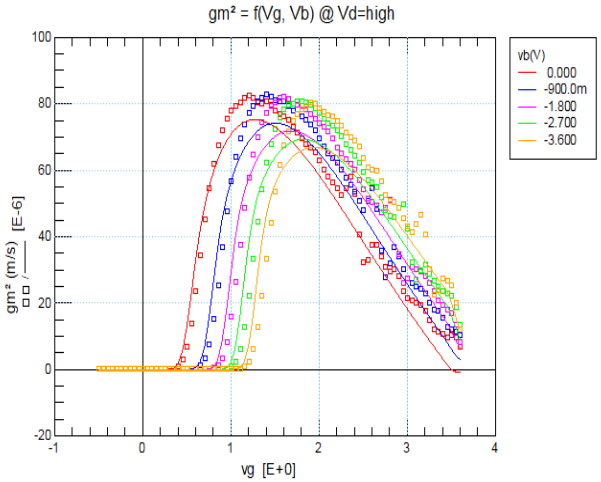


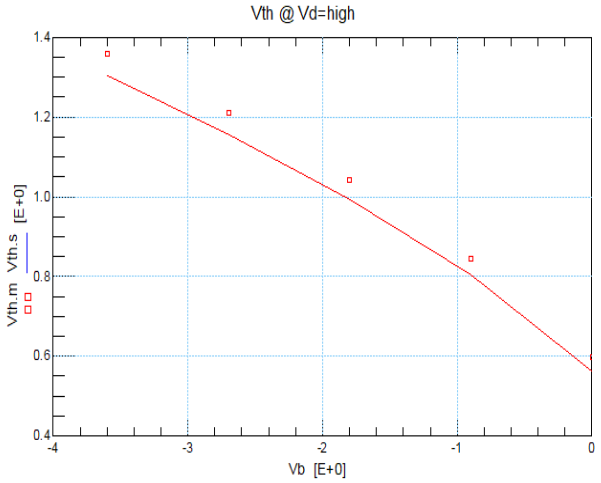
idvg, Vd = 3.3V, T = 27°C

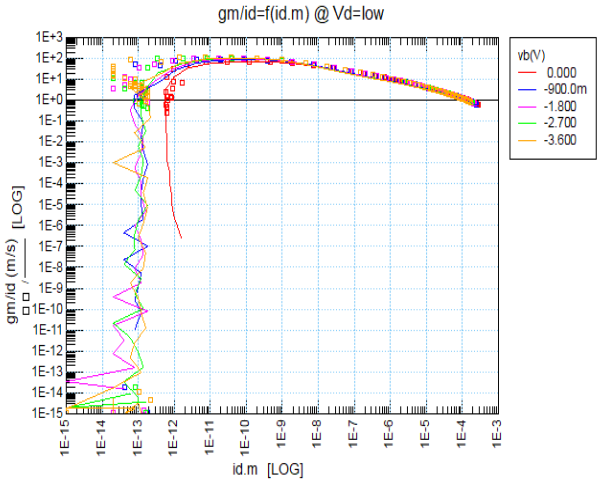


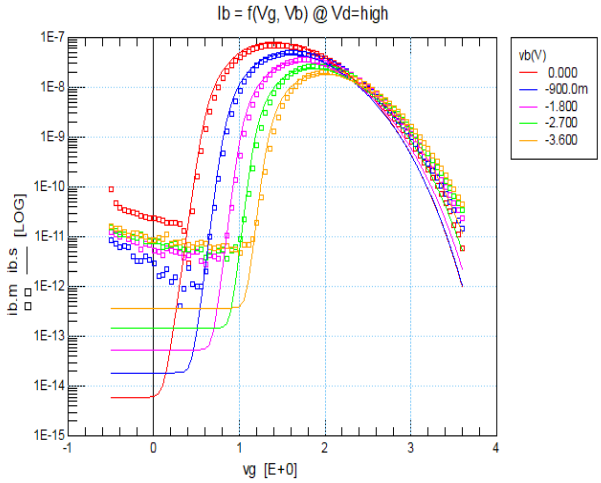


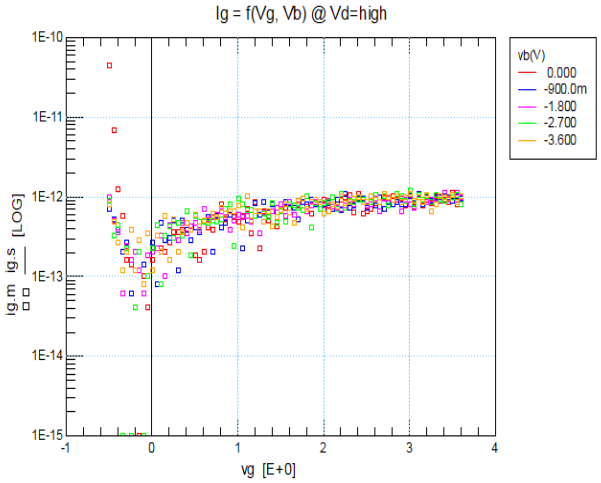




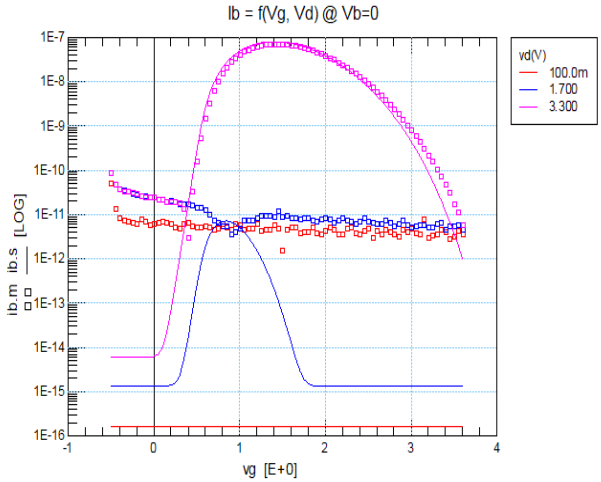


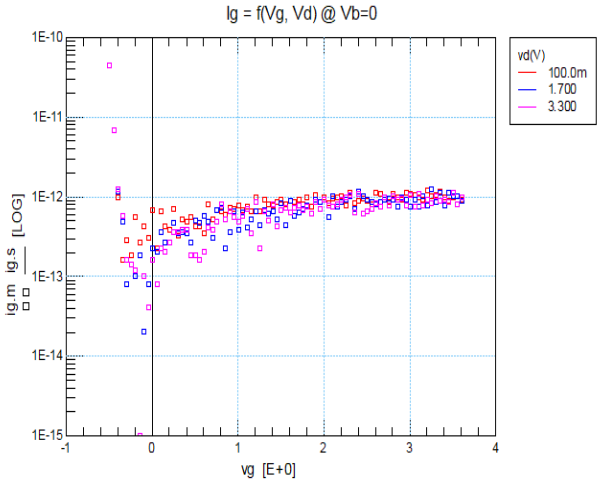




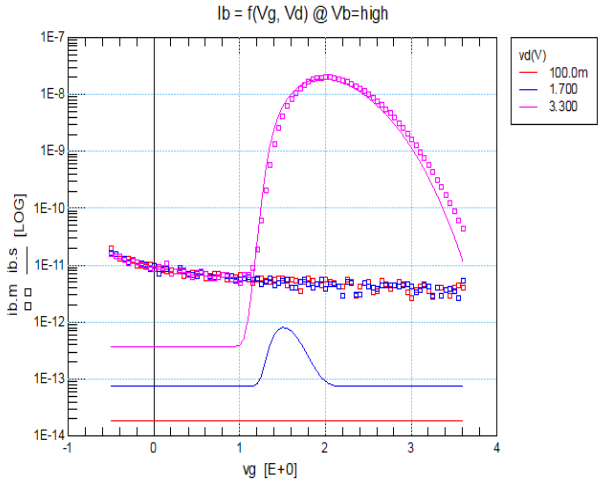


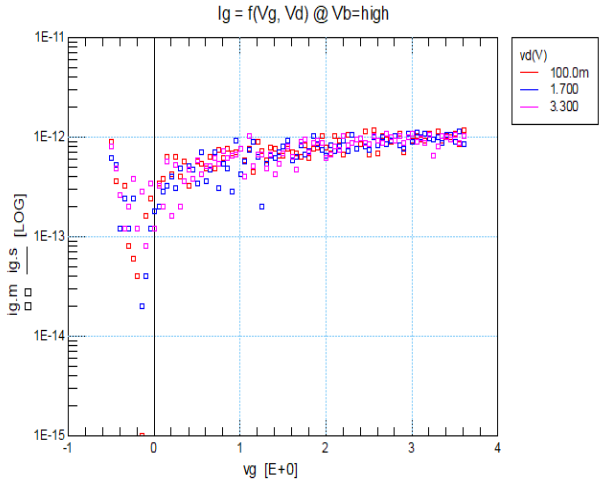
idvg, Ib,Ig, Vb = 0V, T = 27°C



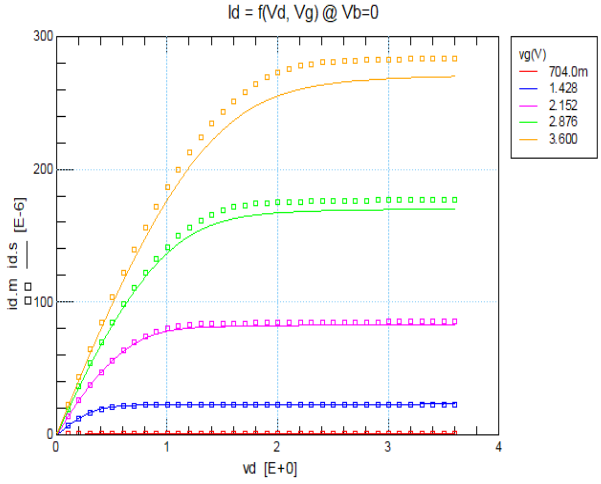


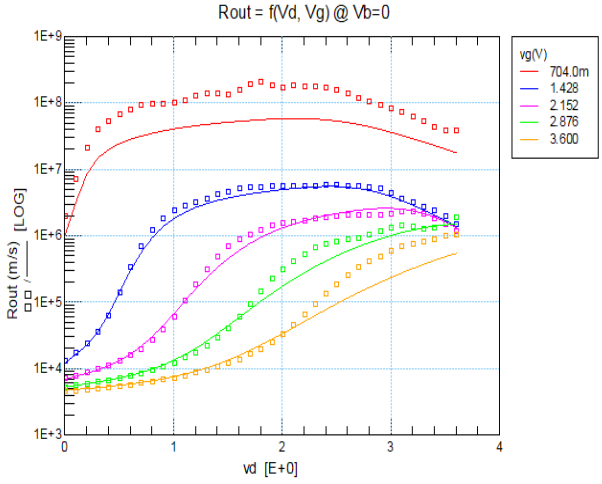
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

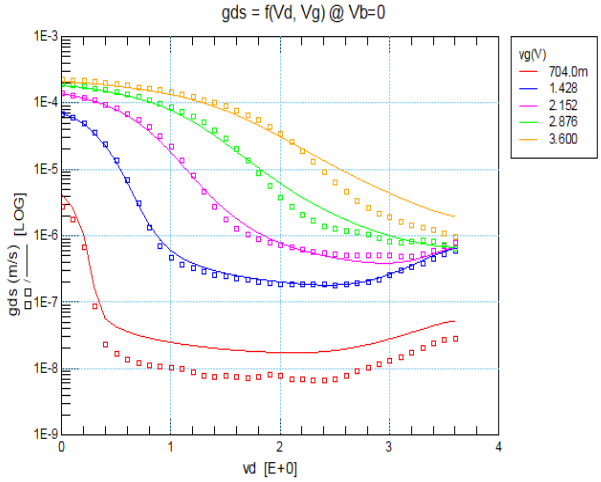




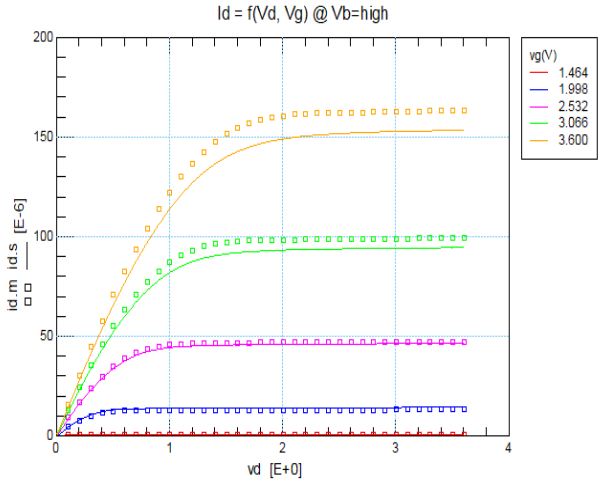
idvd, Vb = 0V, T = 27°C

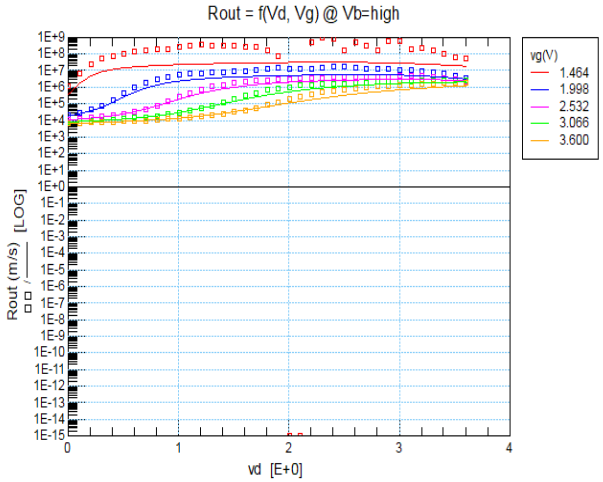


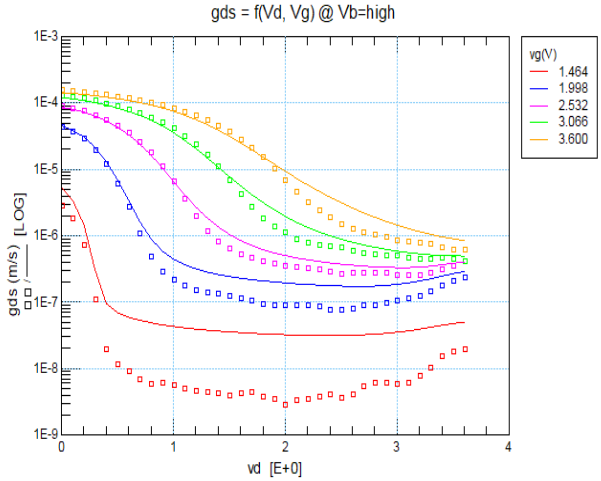




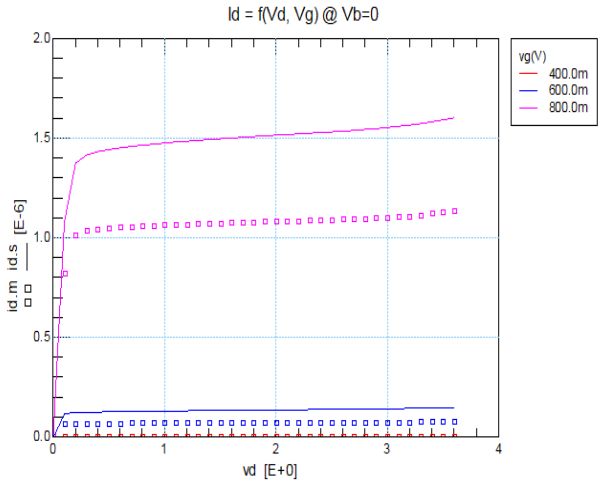
idvd_vbmin, Vb = -3.6V, T = 27°C

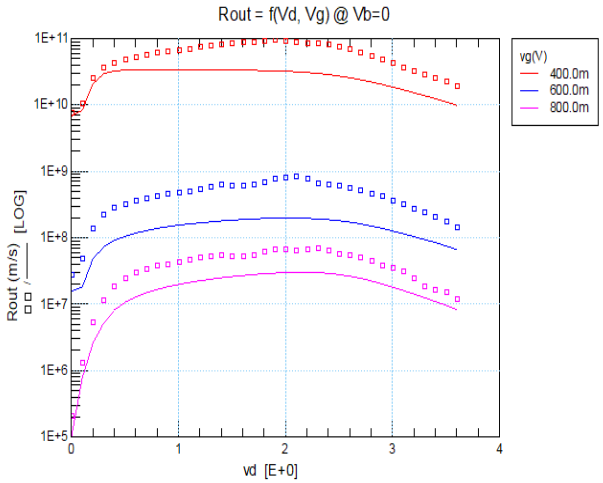


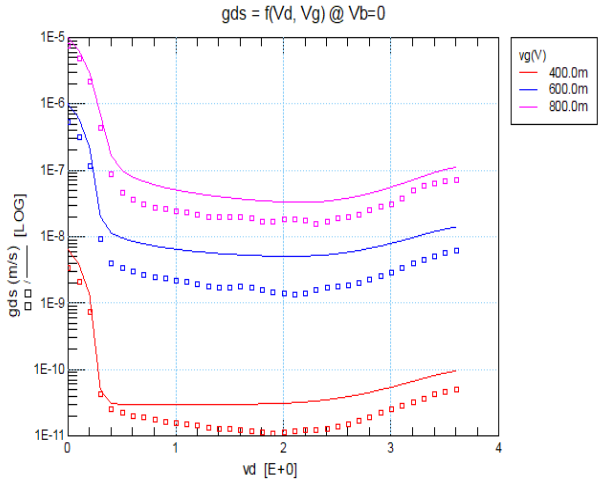




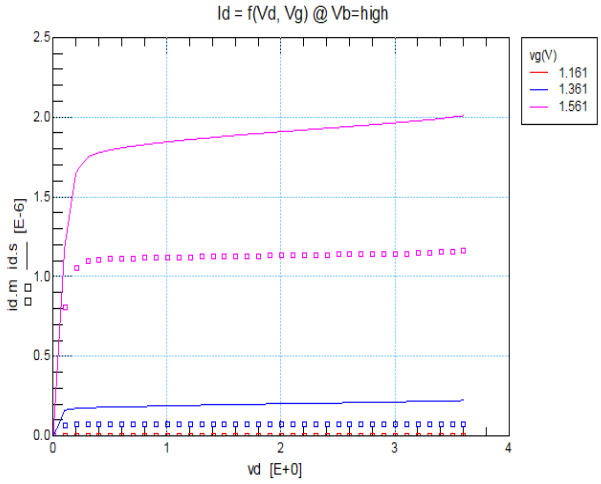
idvd_vth, Vb = 0V, T = 27°C

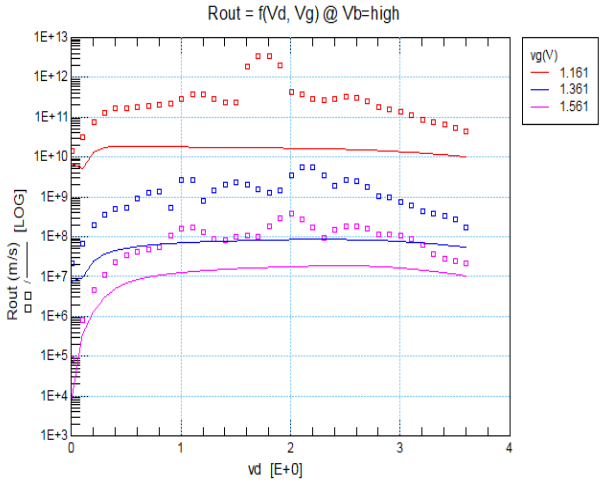


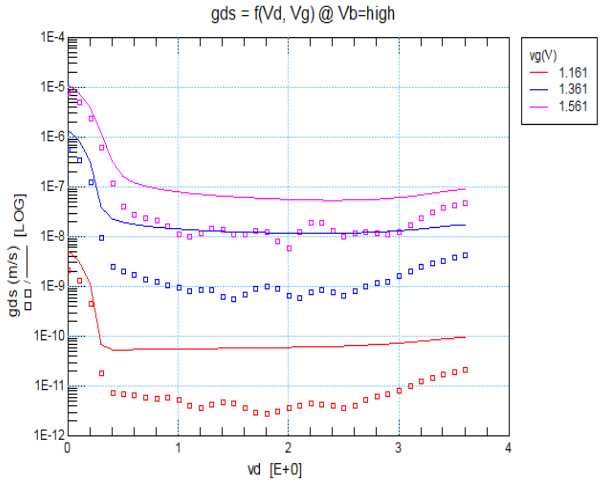




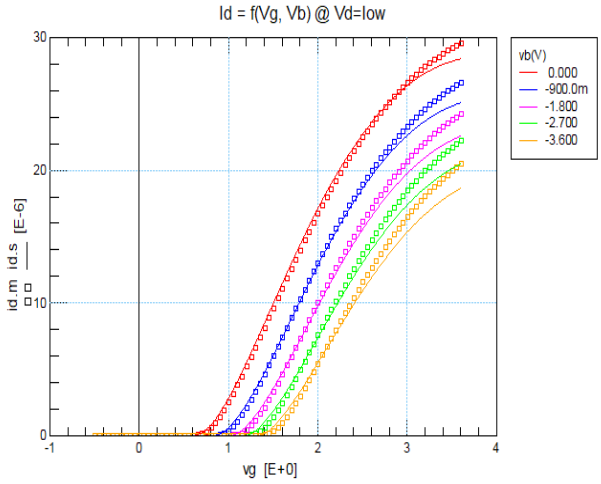
idvd_vth_vbmin, Vb = -3.6V, T = 27°C

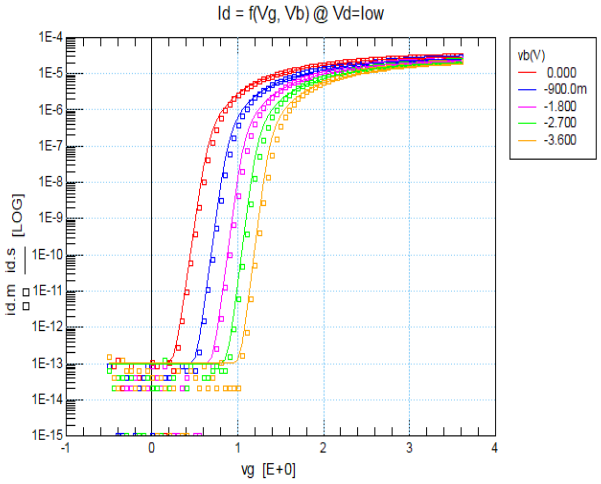


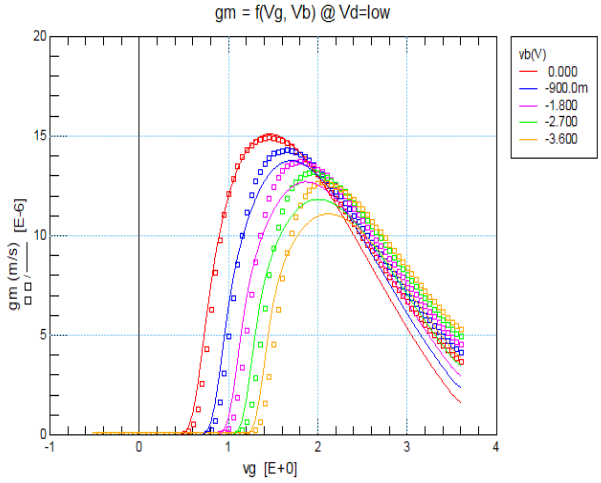


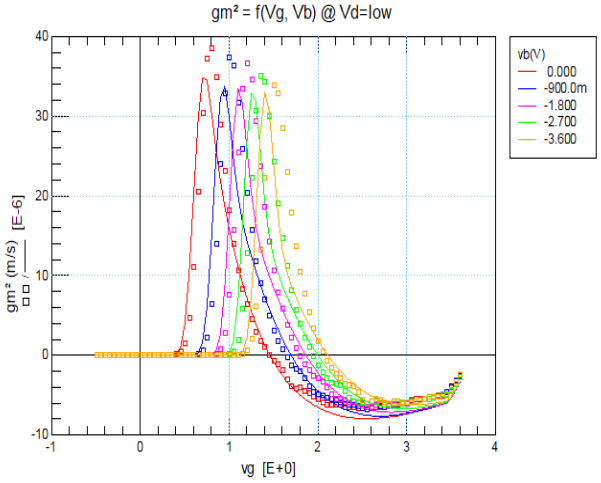


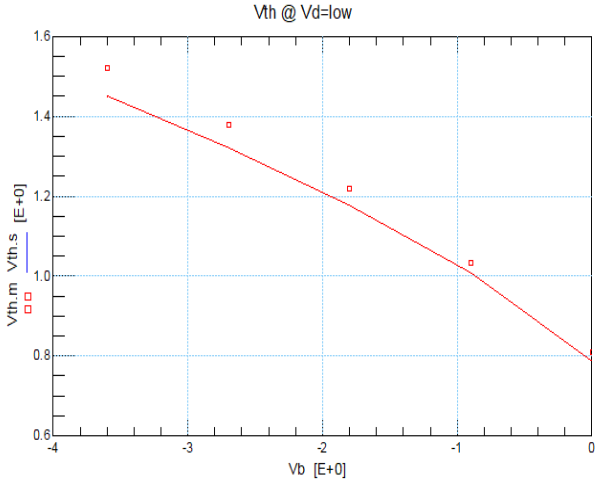
idvg, Vd = 0.1V, T = -40°C

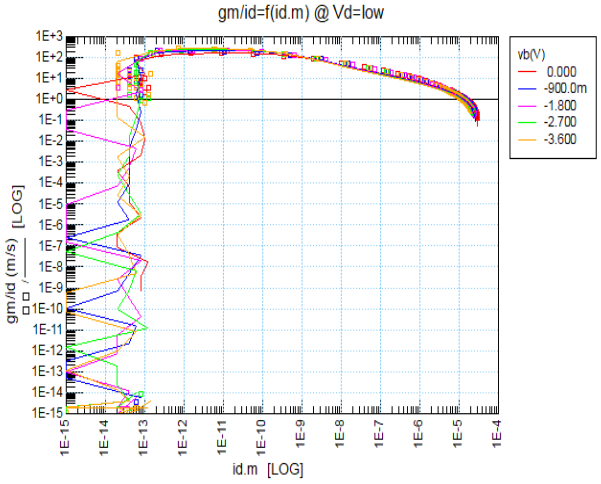


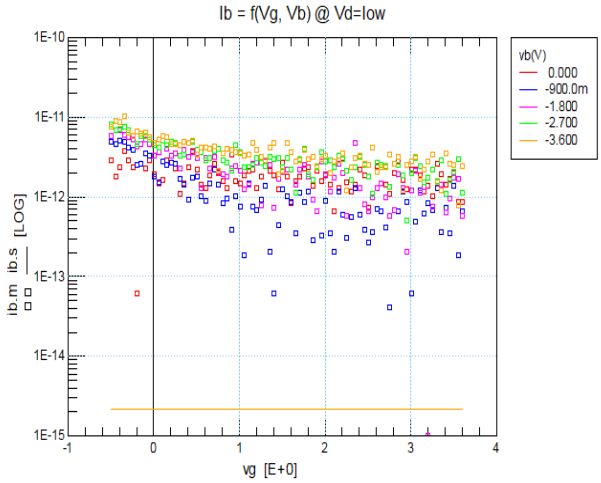


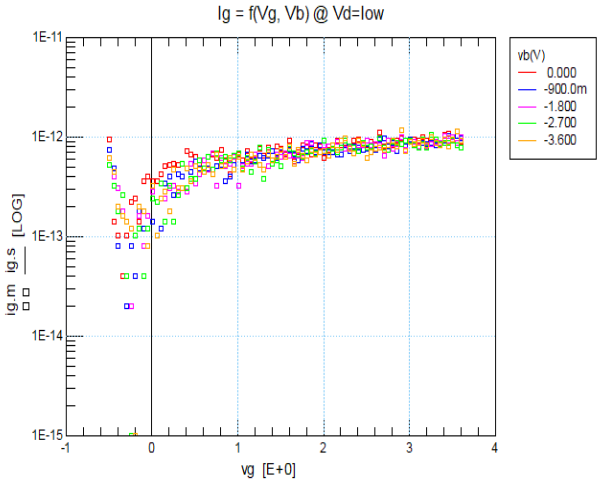




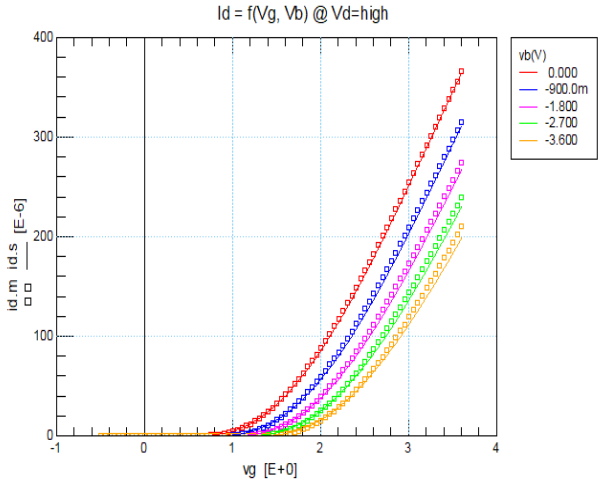


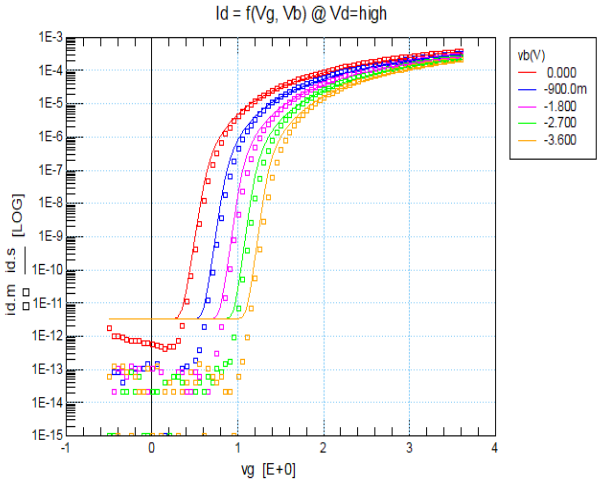


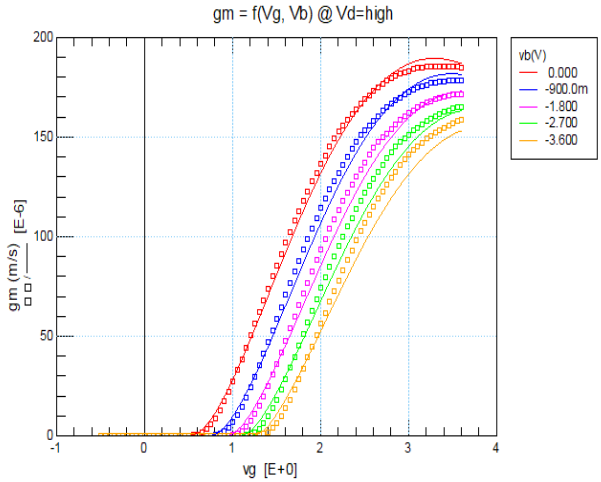


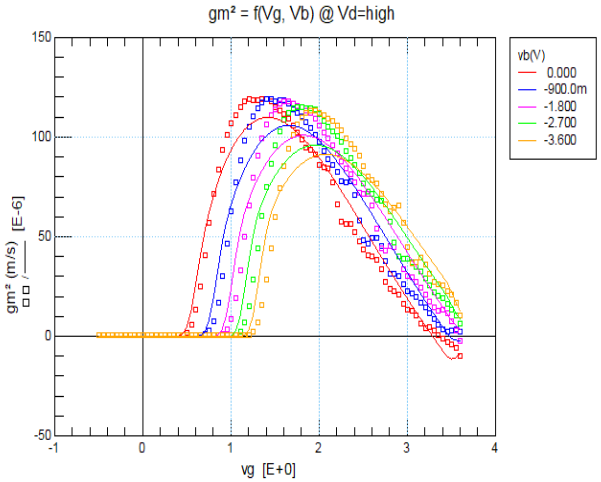


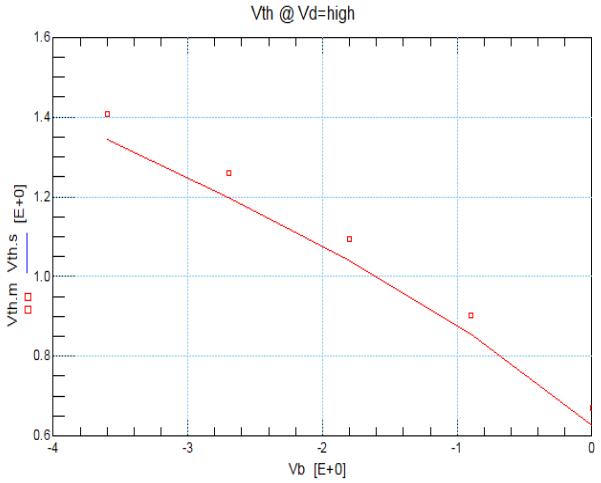
idvg, Vd = 3.3V, T = -40°C

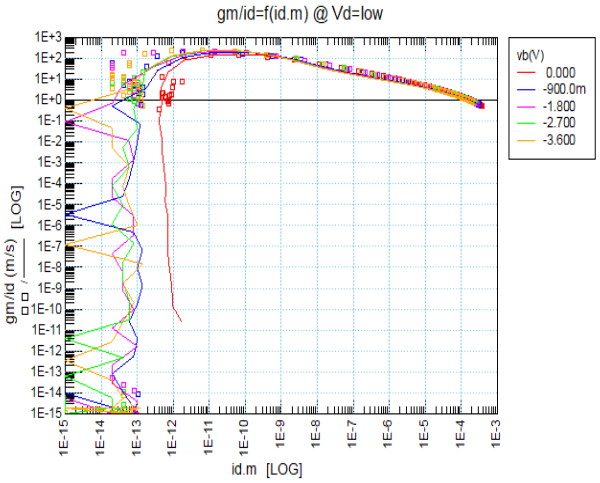


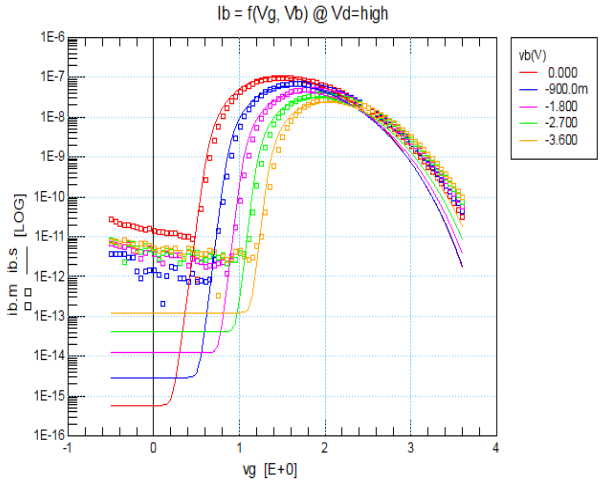


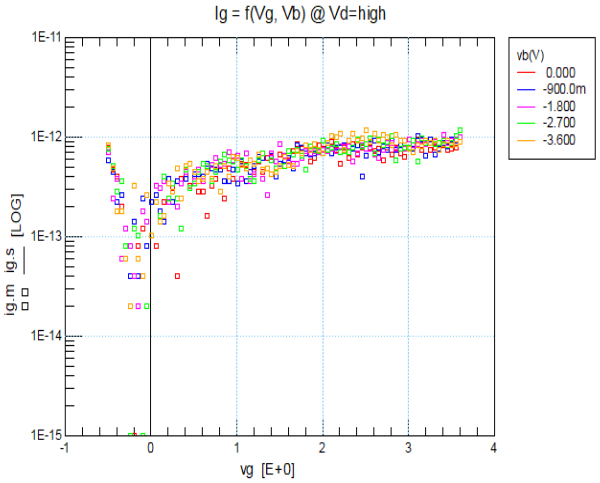




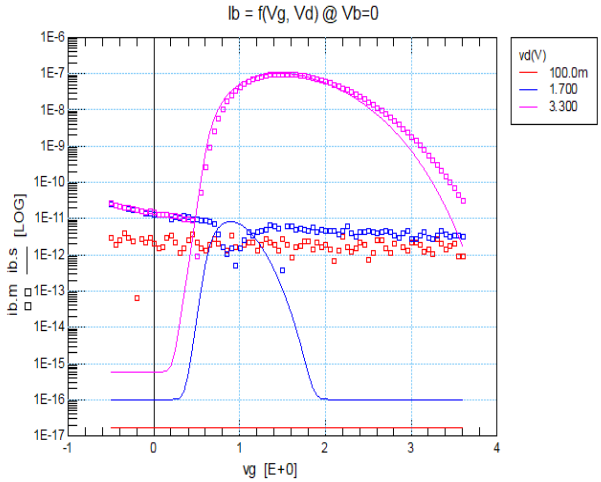


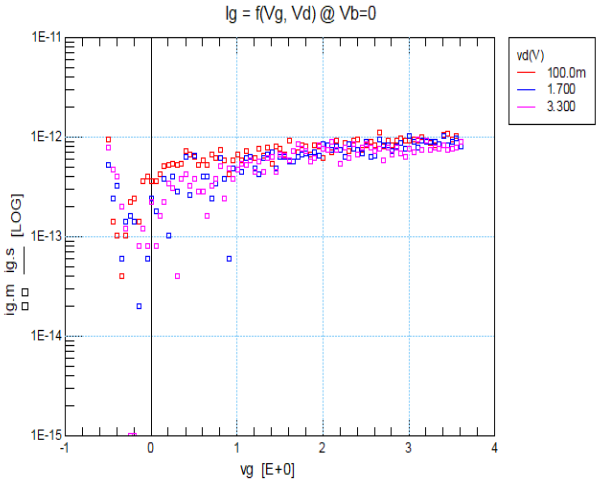




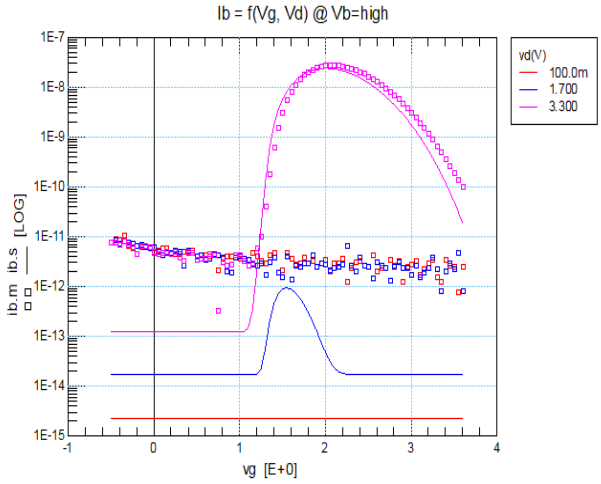


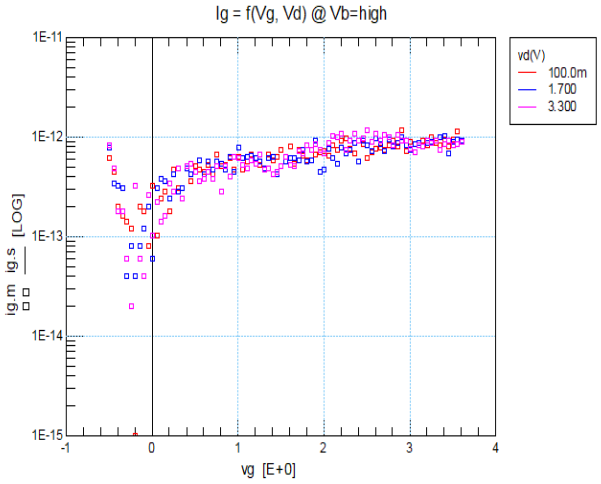
idvg, Ib,Ig, Vb = 0V, T = -40°C



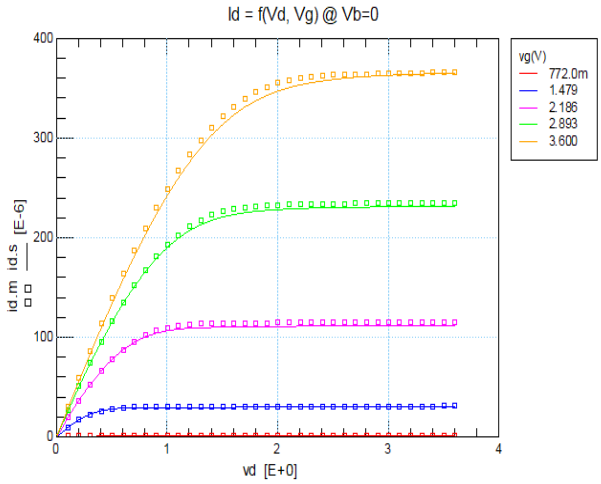


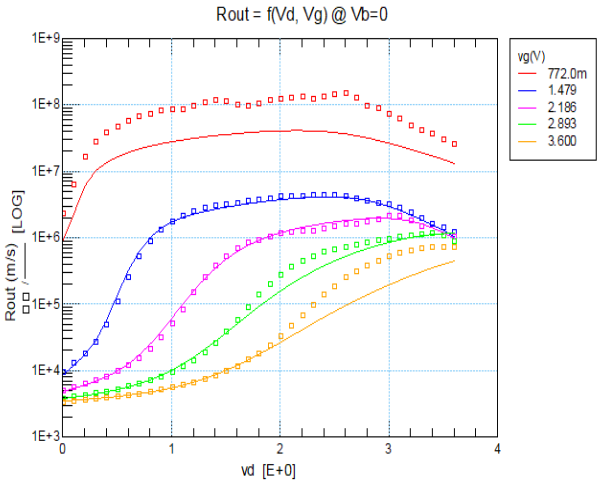
idvg, Ib,Ig, Vb = -3.6V, T = -40°C

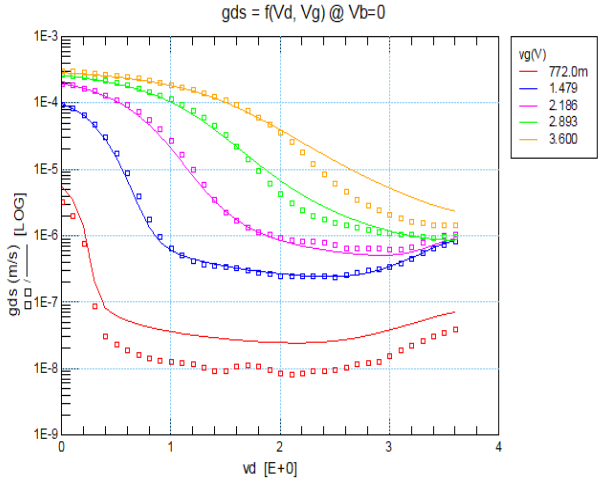




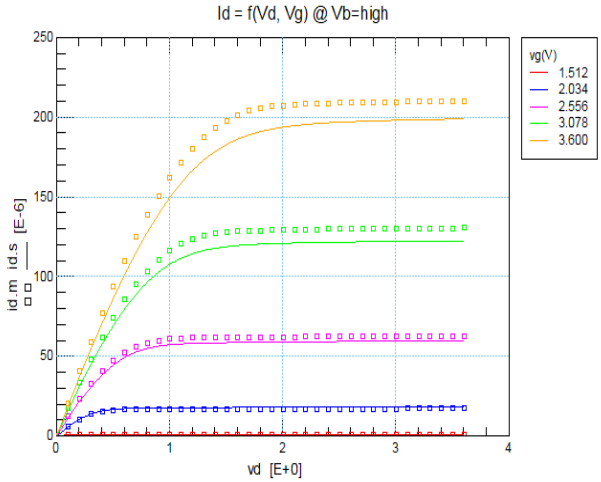
idvd, Vb = 0V, T = -40°C

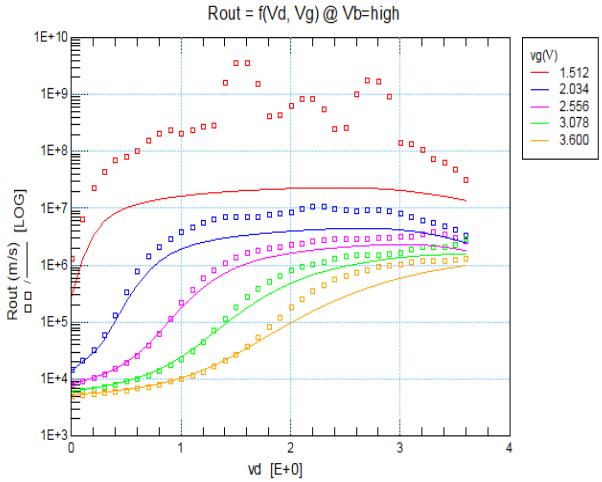


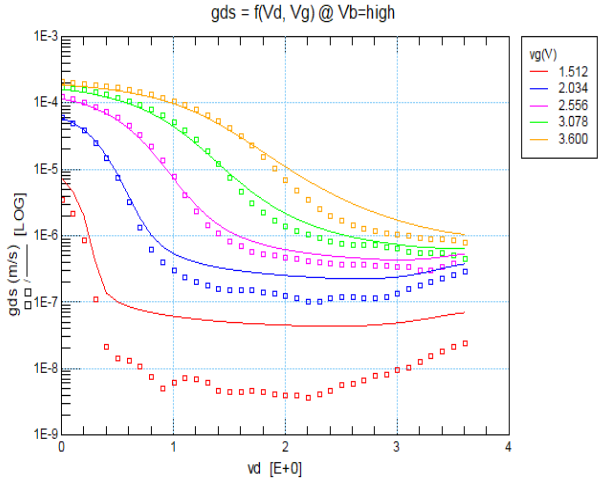




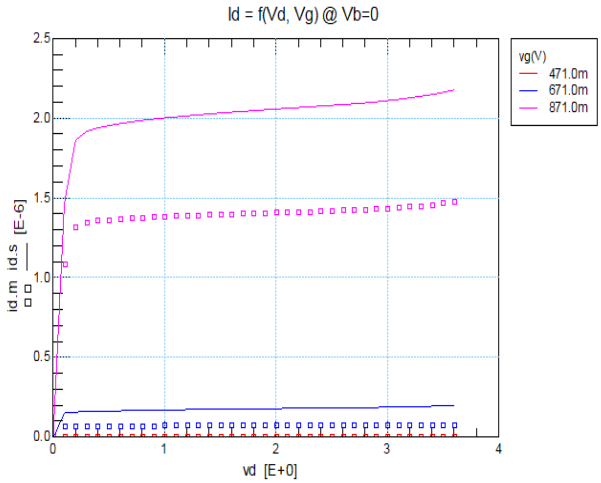
idvd_vbmin, Vb = -3.6V, T = -40°C

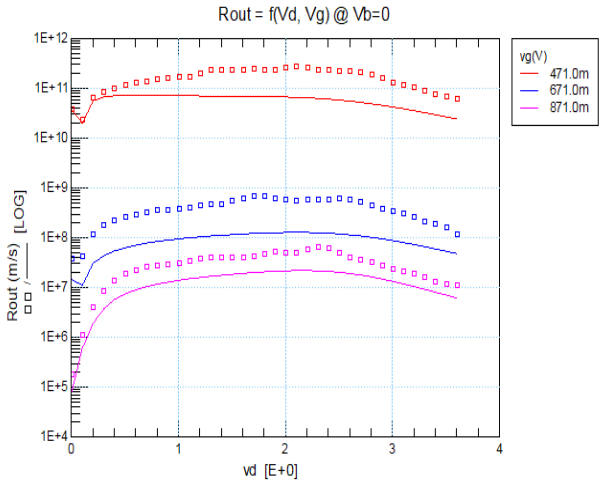


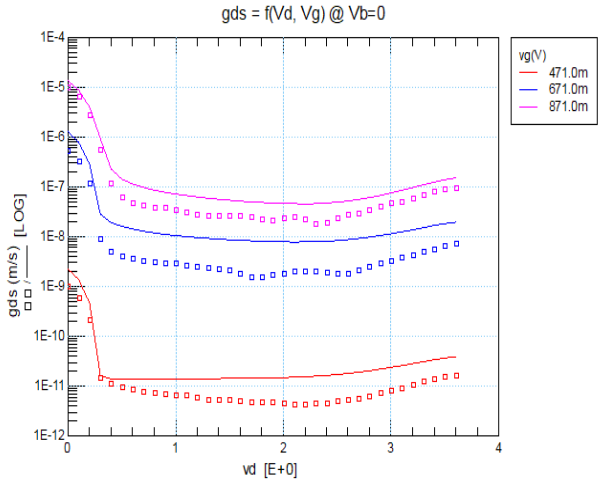




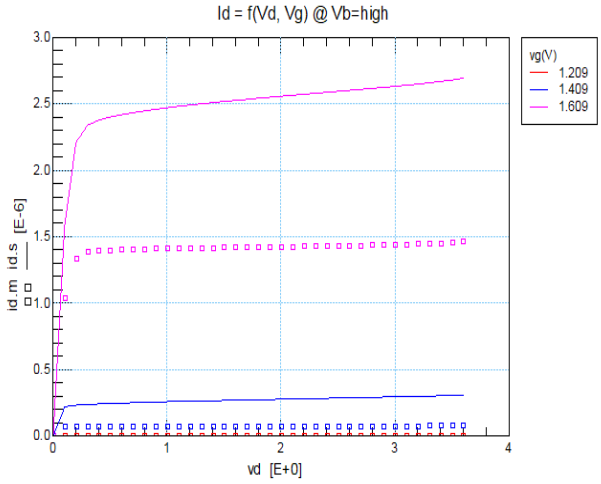
idvd_vth, Vb = 0V, T = -40°C

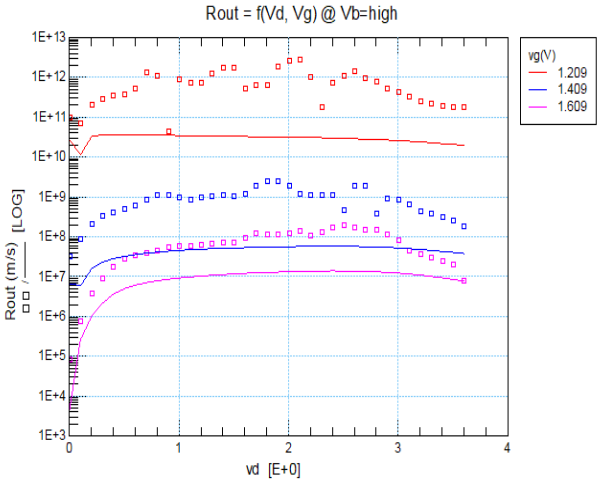


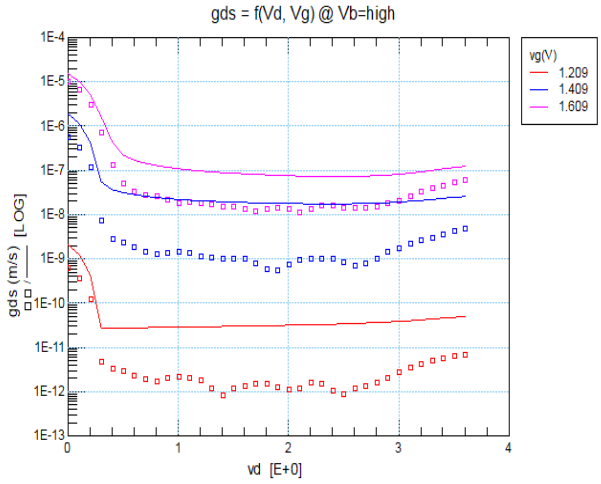




idvd_vth_vbmin, Vb = -3.6V, T = -40°C

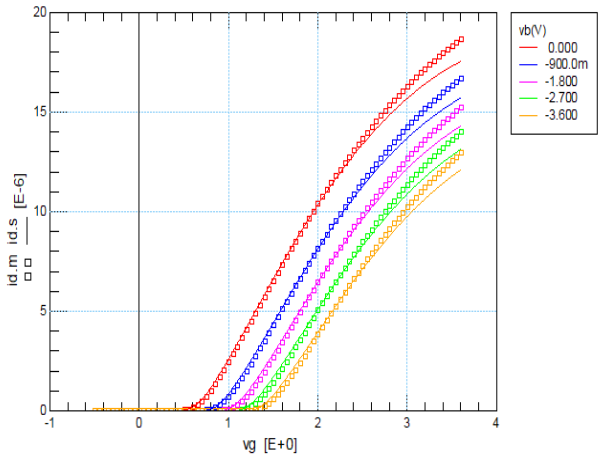




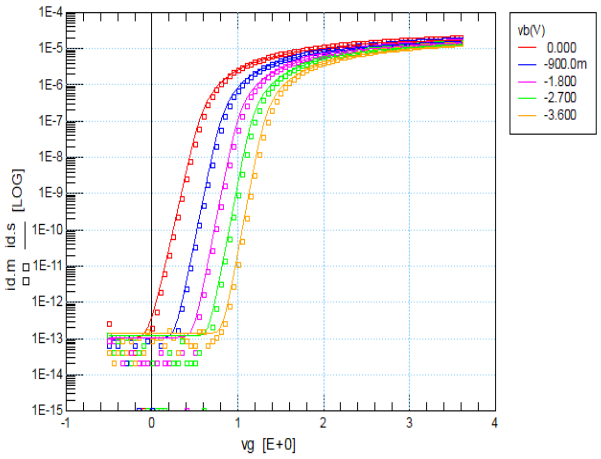


idvg, Vd = 0.1V, T = 70°C

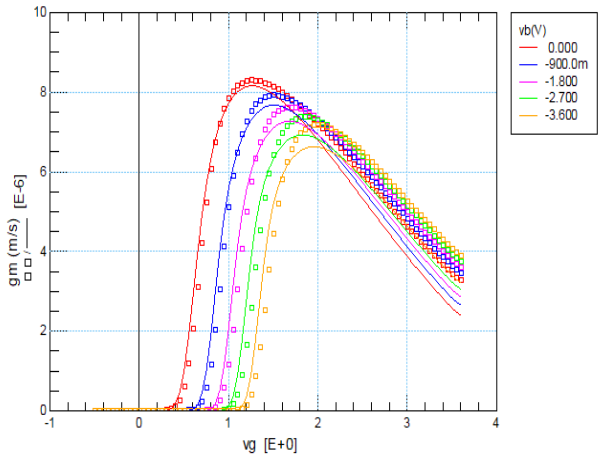
Id = f(Vg, Vb) @ Vd=low



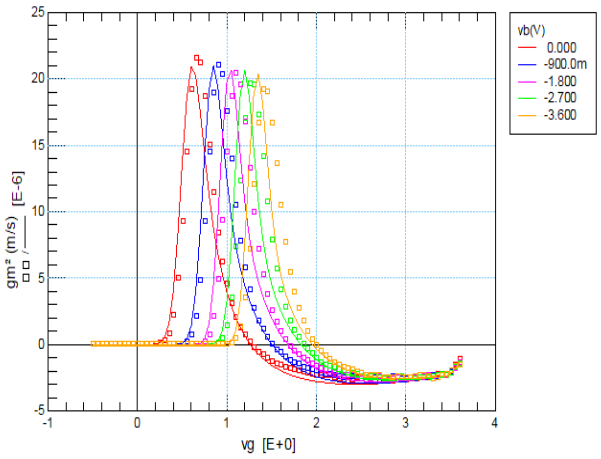
Id = f(Vg, Vb) @ Vd=low



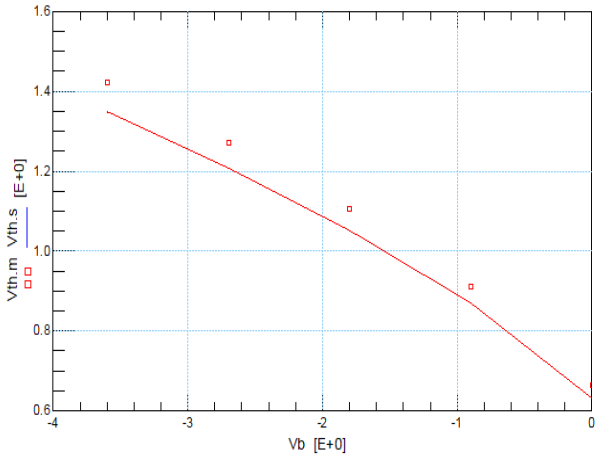
gm = f(Vg, Vb) @ Vd=low



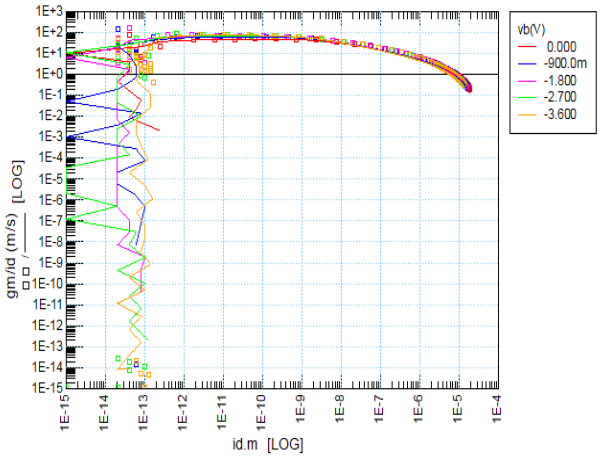
gm² = f(Vg, Vb) @ Vd=low

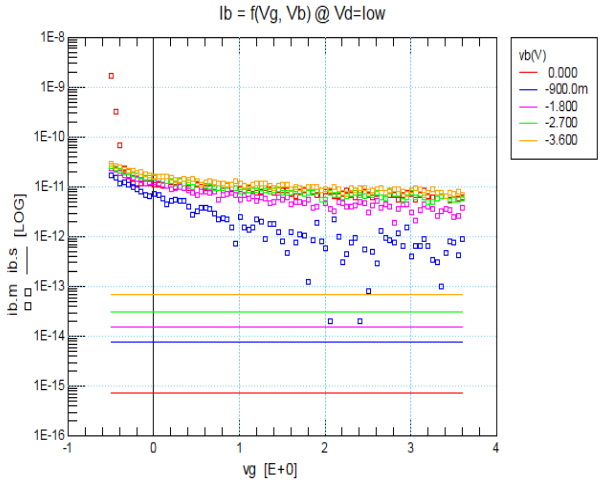


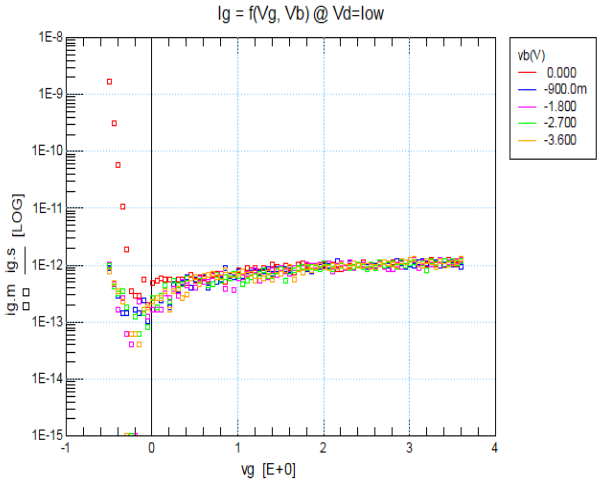
Vth @ Vd=low



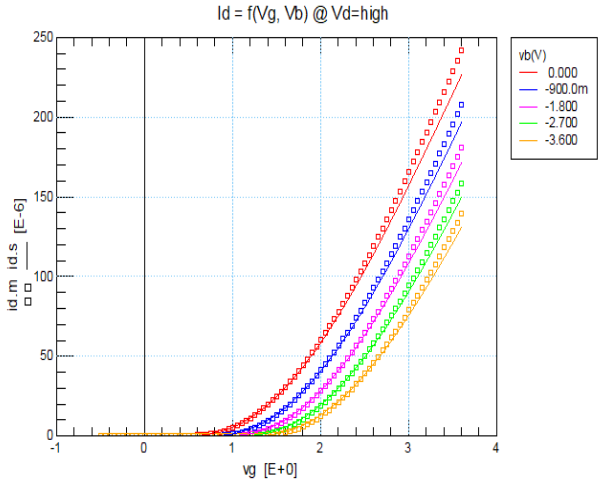
gm/id=f(id.m) @ Vd=low

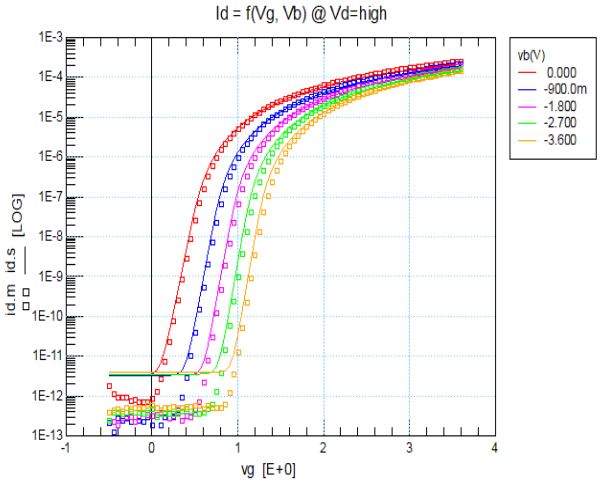


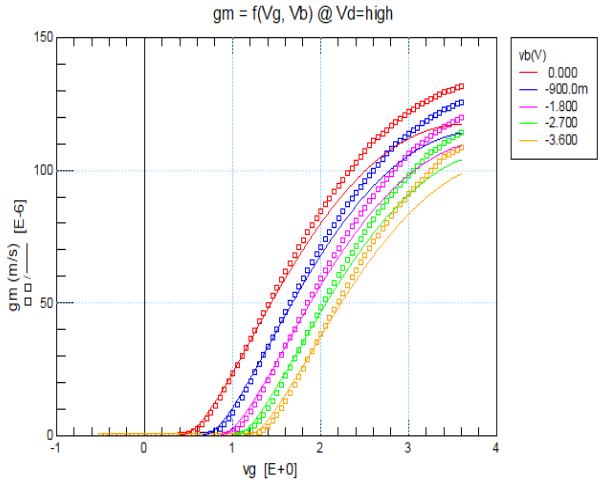


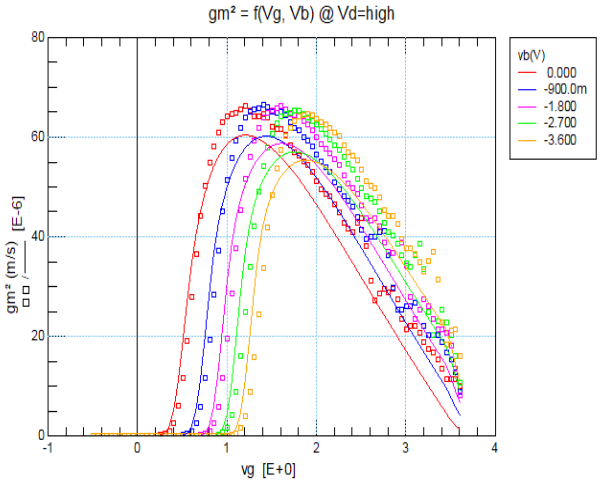


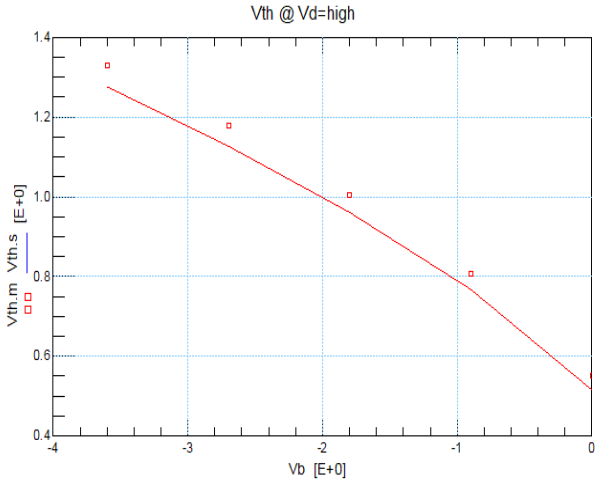
idvg, Vd = 3.3V, T = 70°C

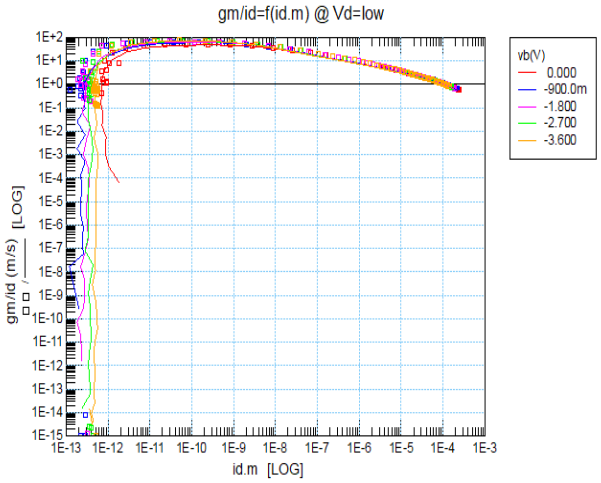


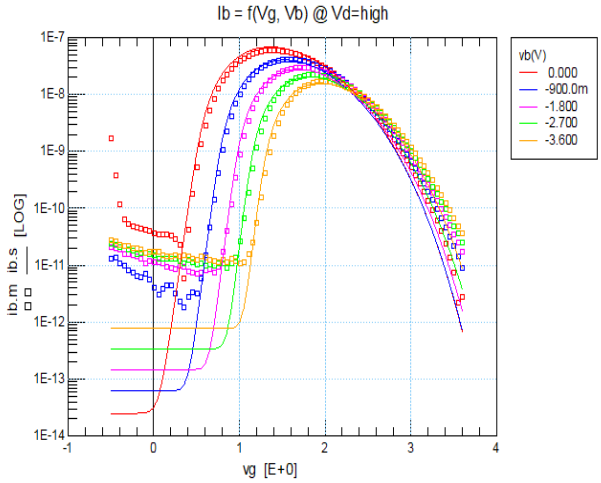


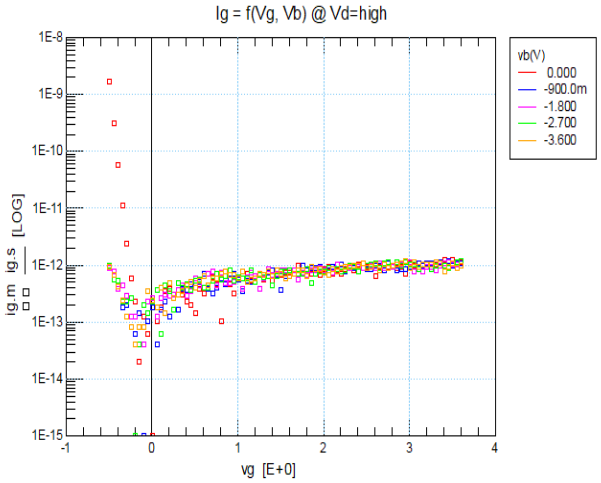




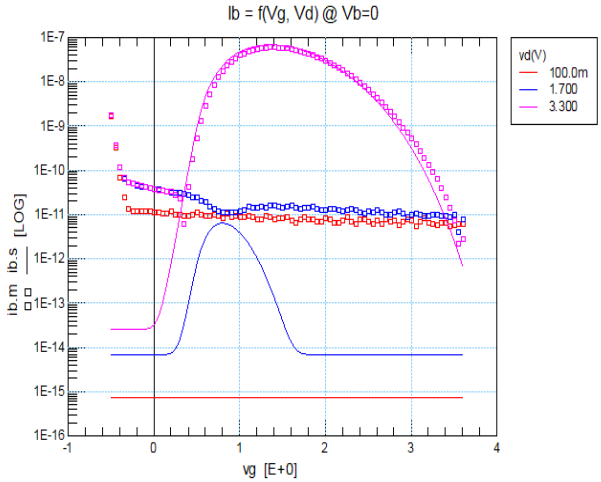


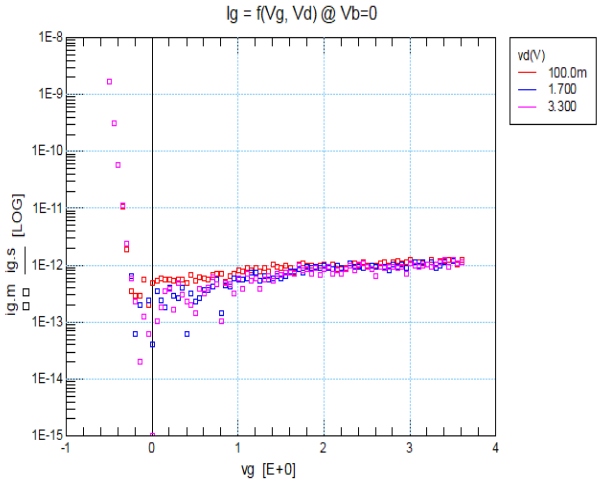




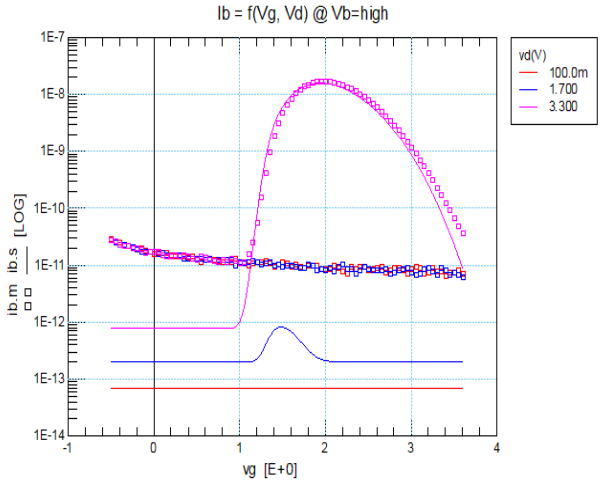


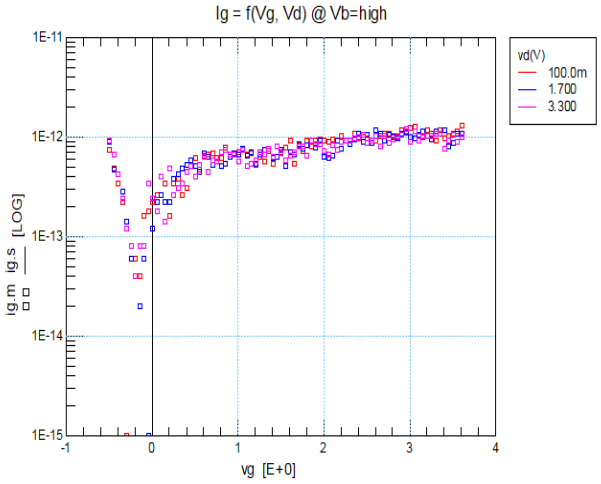
idvg, Ib,Ig, Vb = 0V, T = 70°C



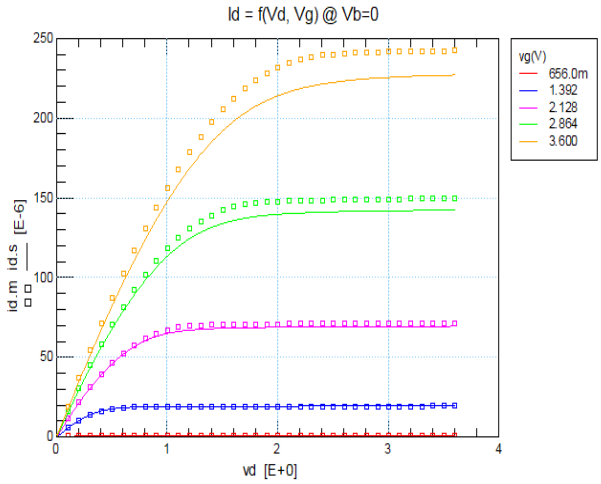


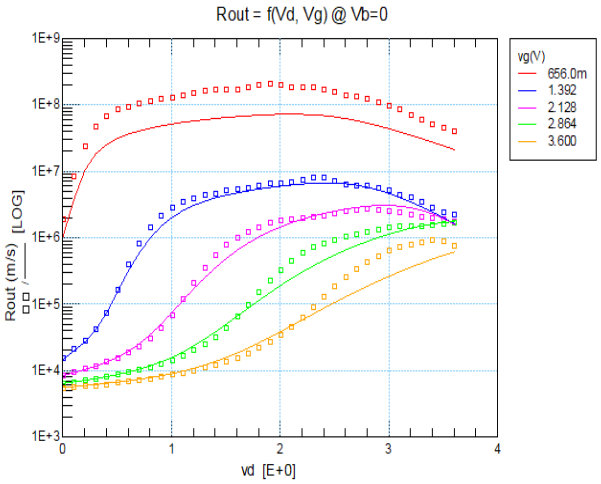
idvg, Ib,Ig, Vb = -3.6V, T = 70°C

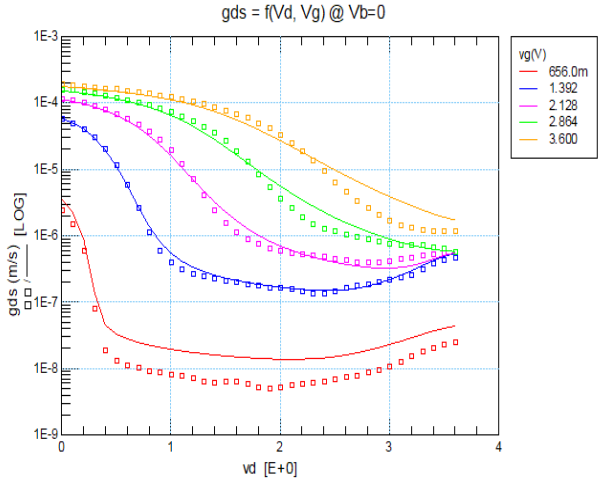




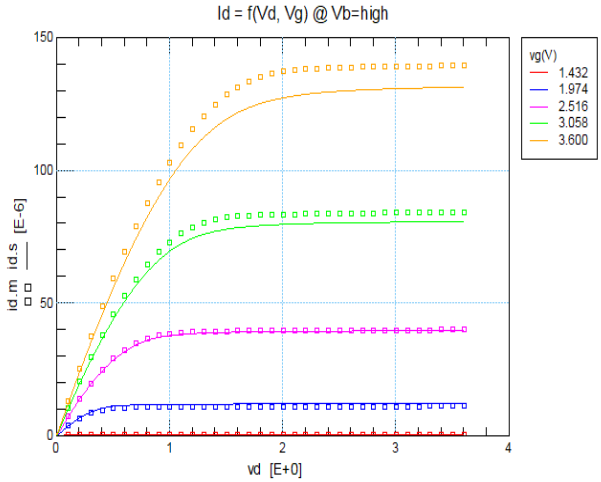
idvd, Vb = 0V, T = 70°C

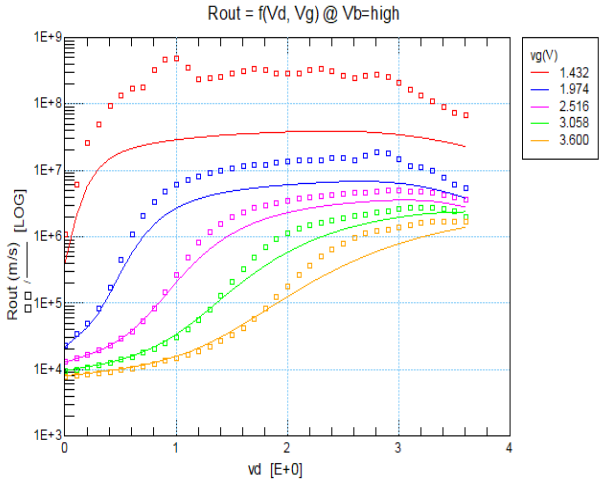


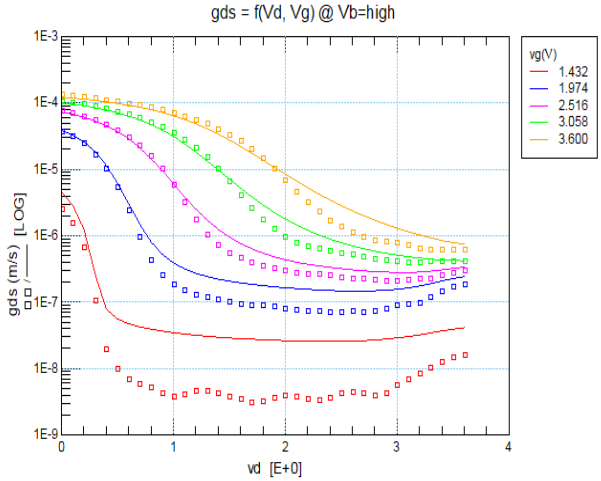




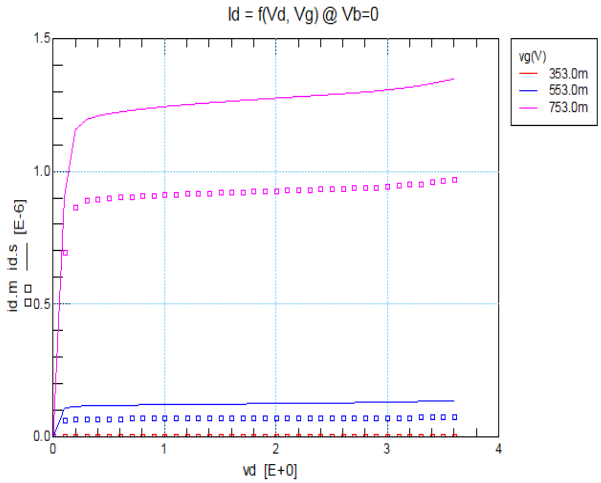
idvd_vbmin, Vb = -3.6V, T = 70°C

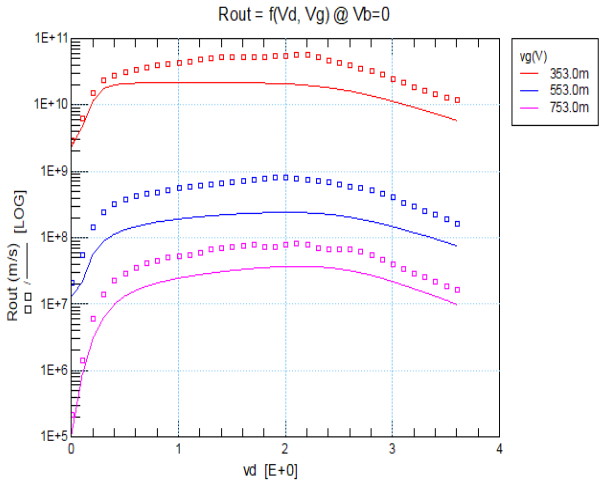


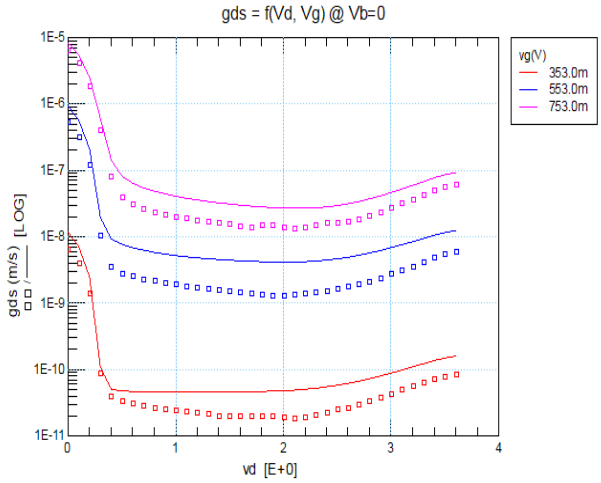




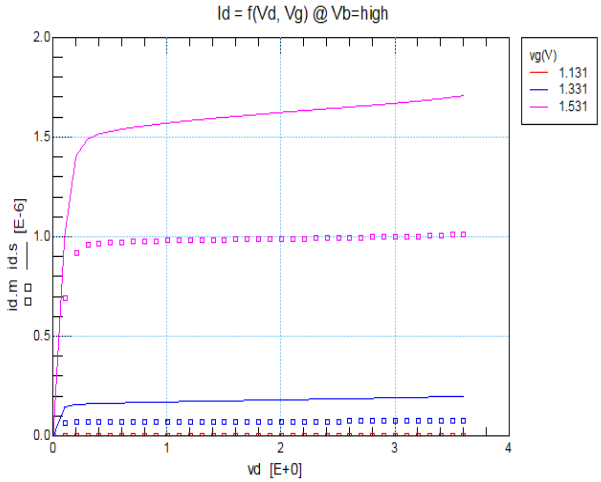
idvd_vth, Vb = 0V, T = 70°C

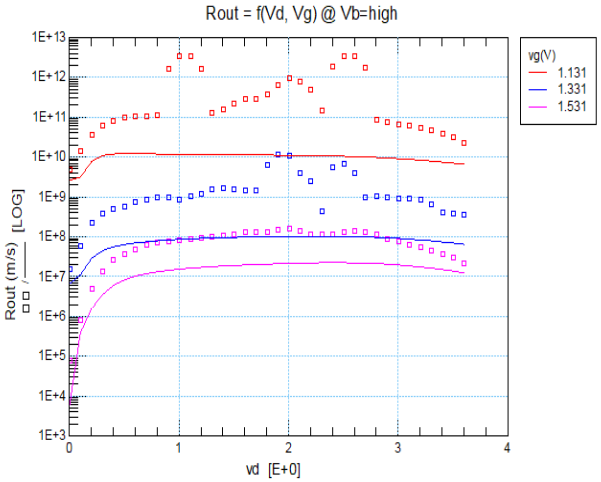


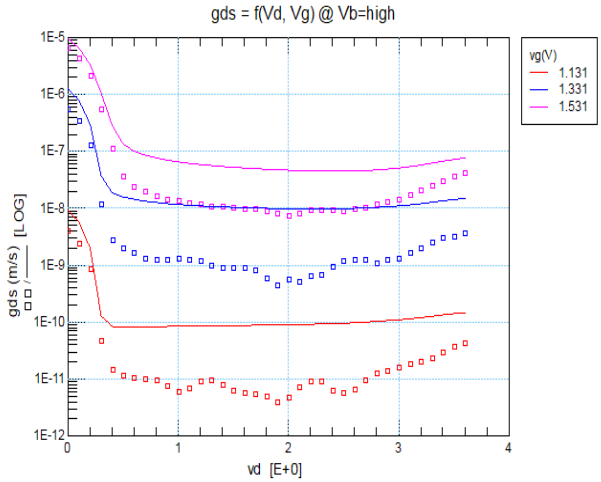




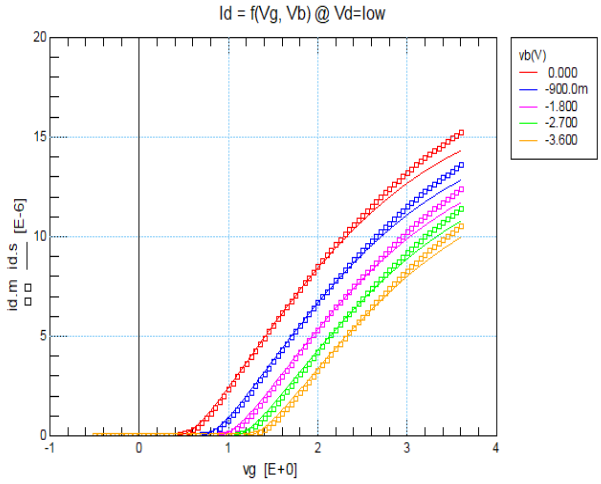
idvd_vth_vbmin, Vb = -3.6V, T = 70°C

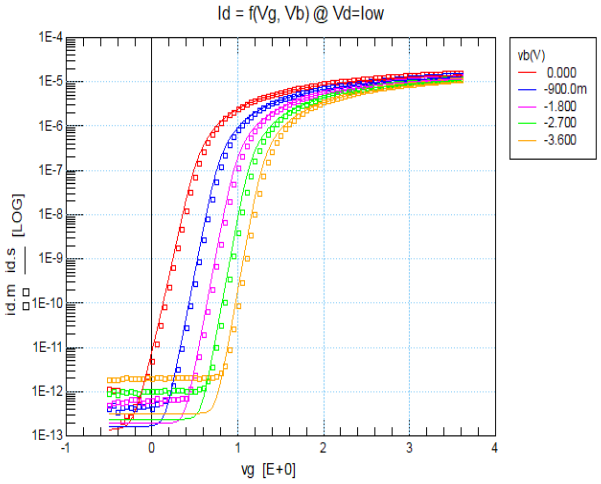


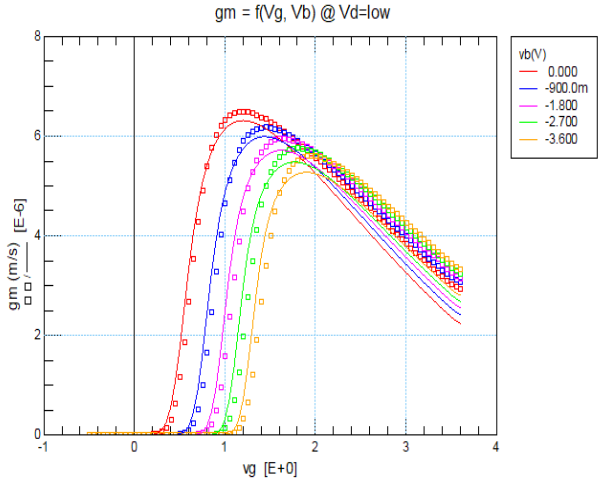


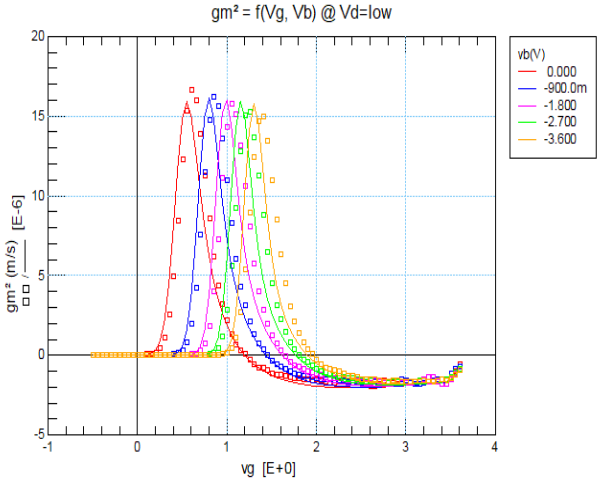


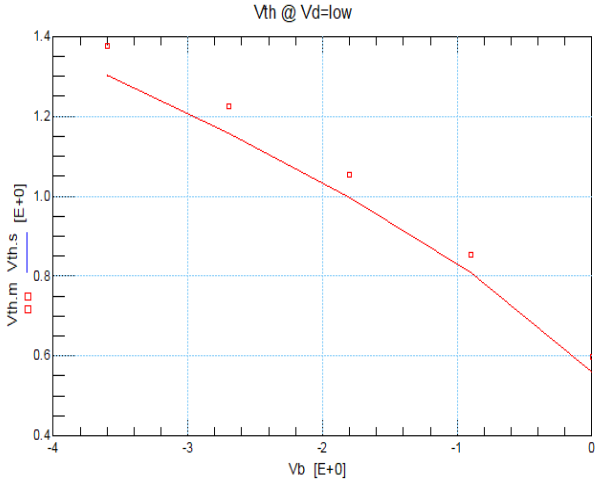
idvg, Vd = 0.1V, T = 125°C

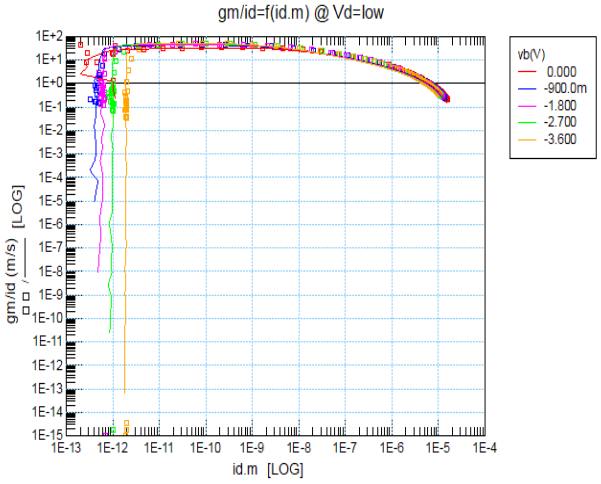


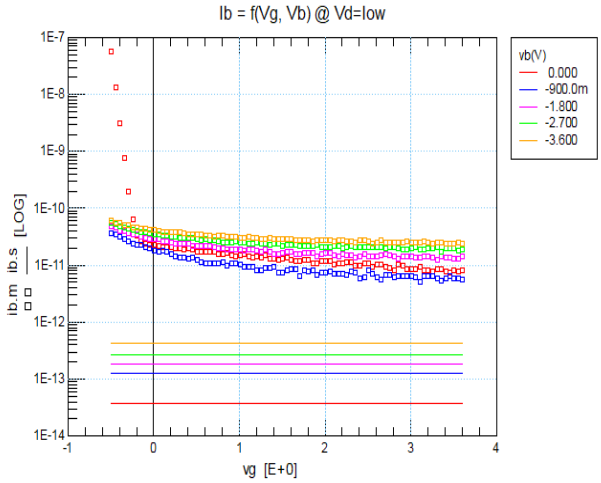


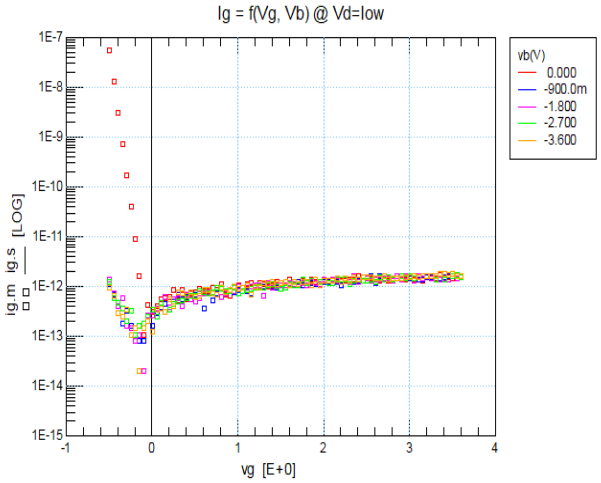




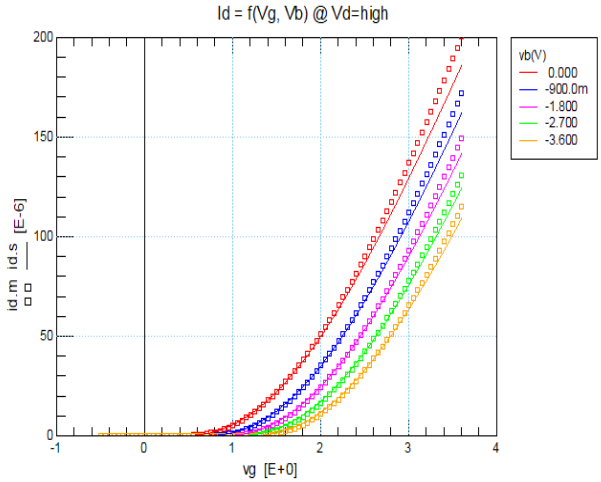


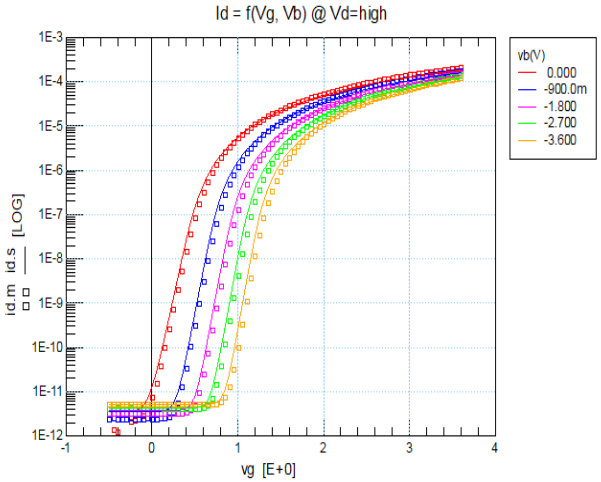


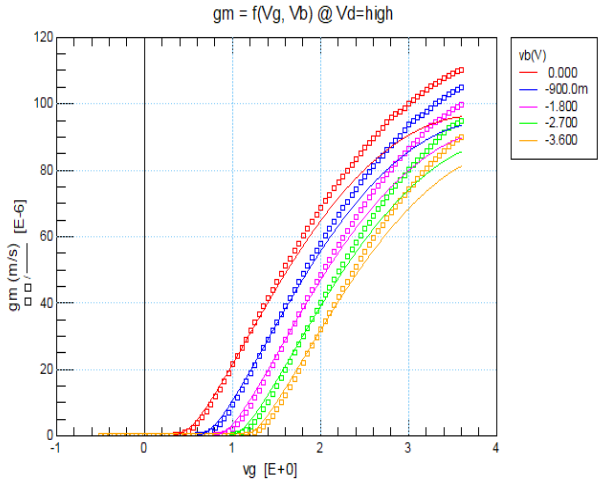


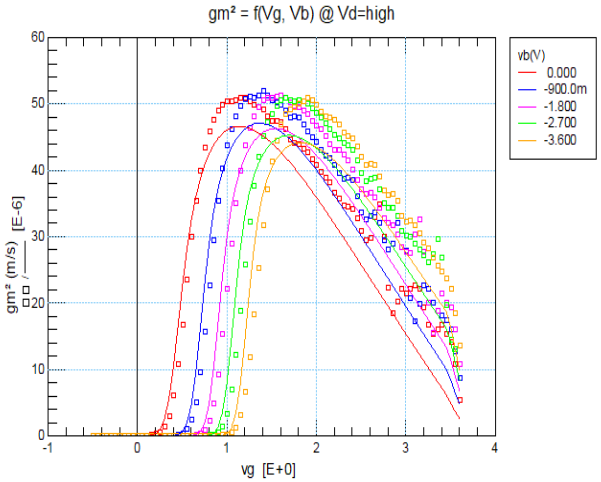


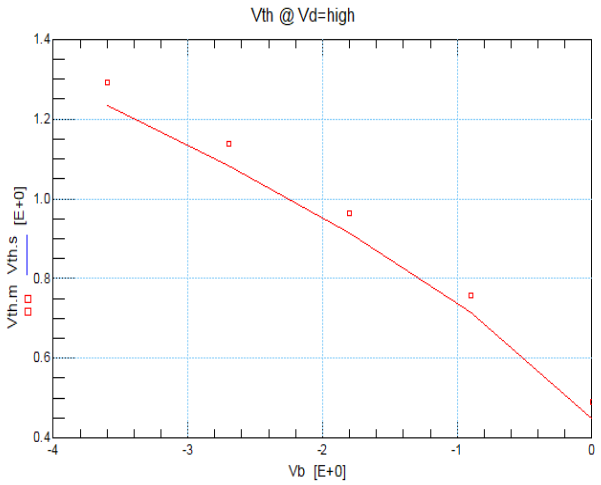
idvg, Vd = 3.3V, T = 125°C

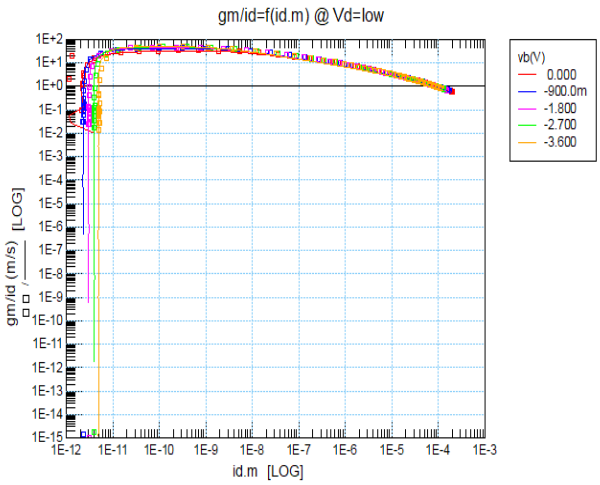


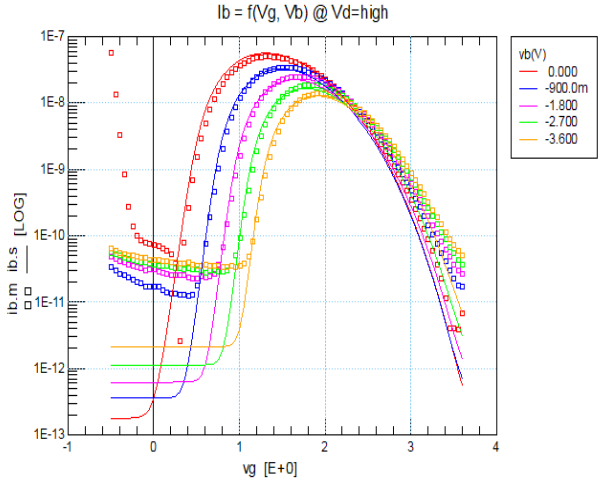


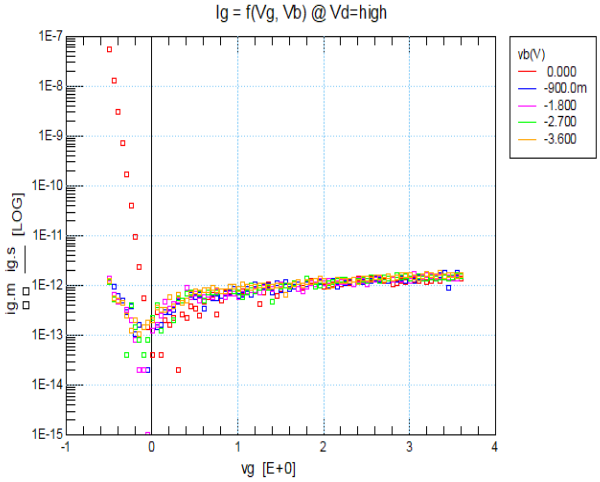




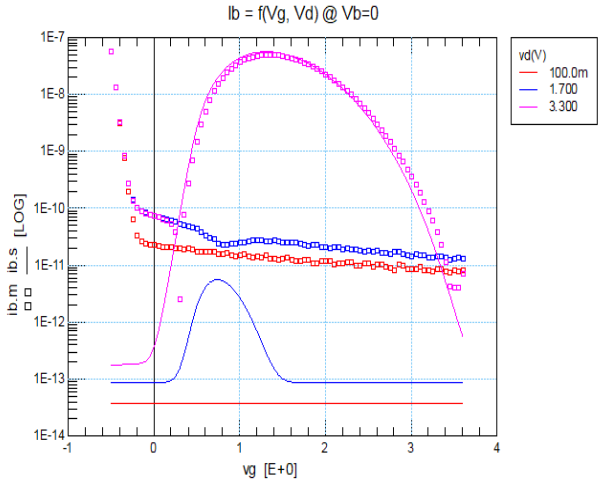


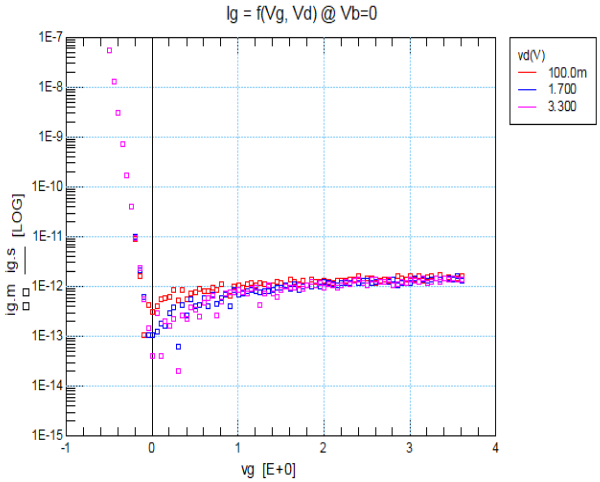




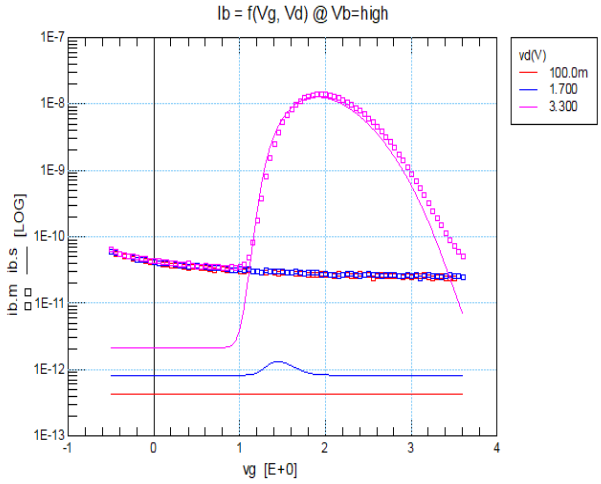


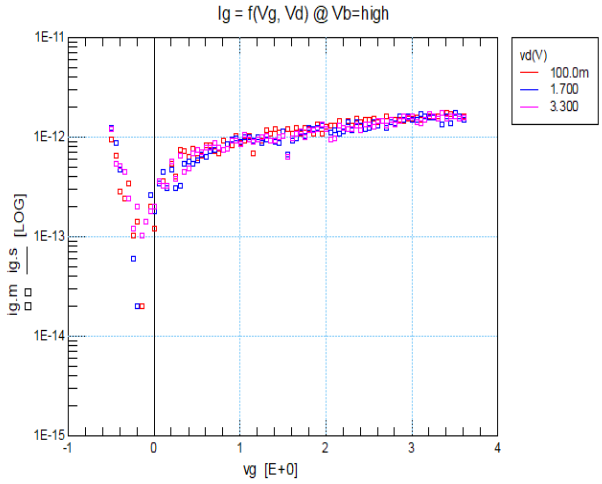
idvg, Ib,Ig, Vb = 0V, T = 125°C



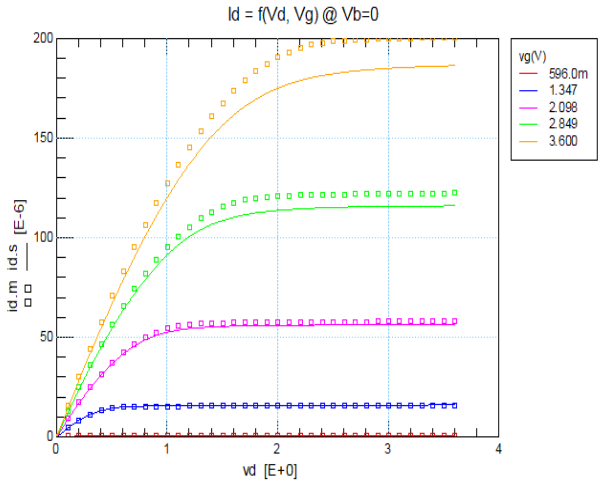


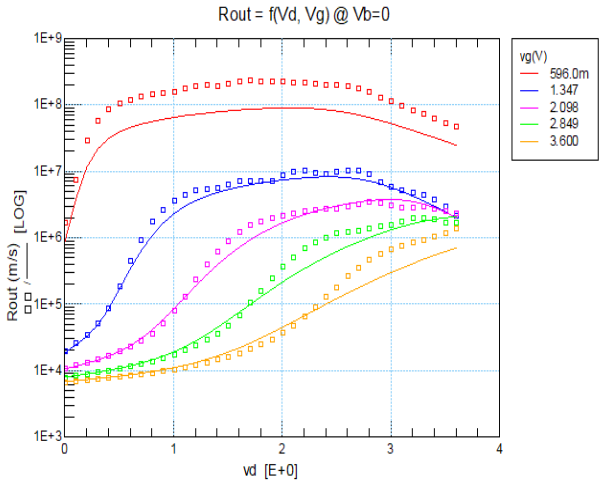
idvg, Ib,Ig, Vb = -3.6V, T = 125°C

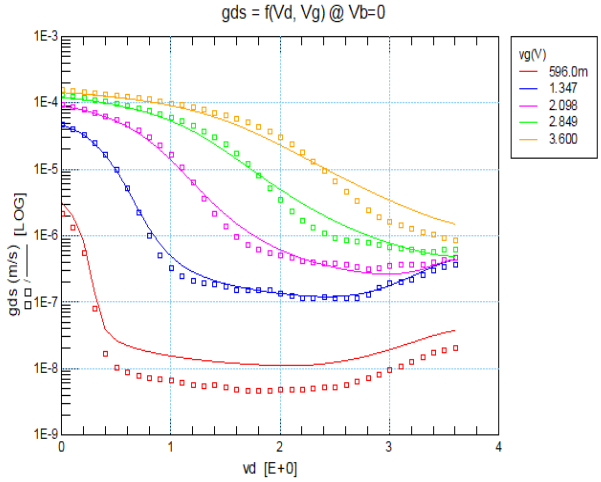




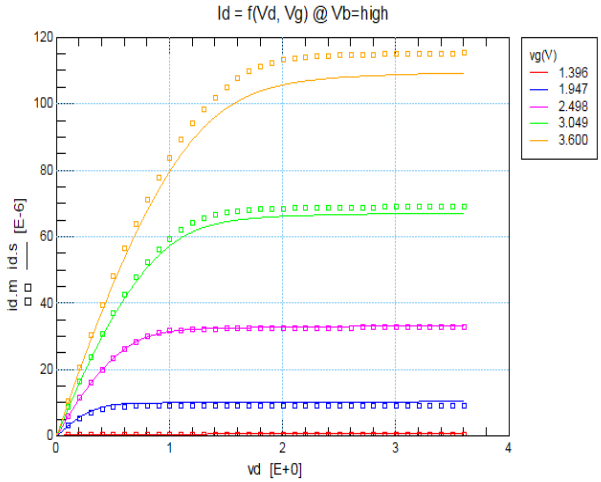
idvd, Vb = 0V, T = 125°C

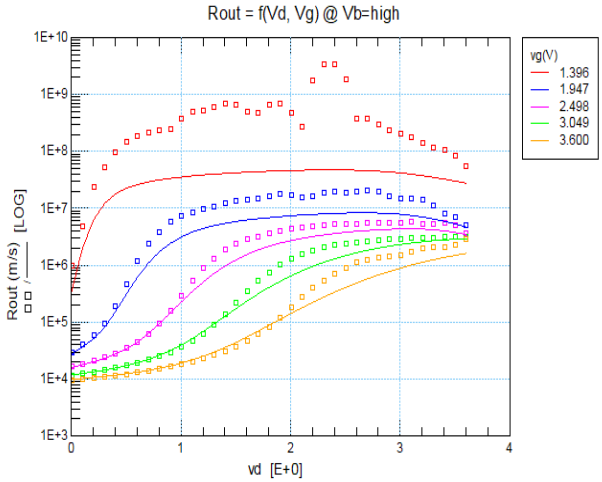


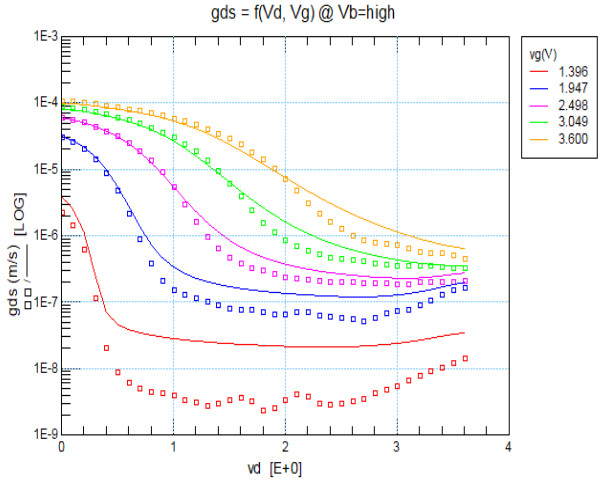




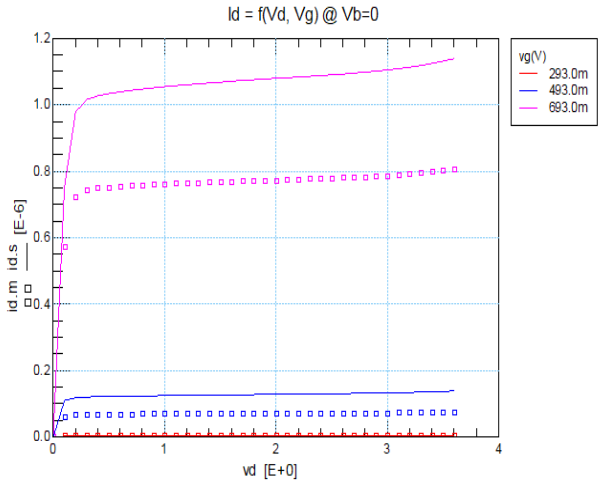
idvd_vbmin, Vb = -3.6V, T = 125°C

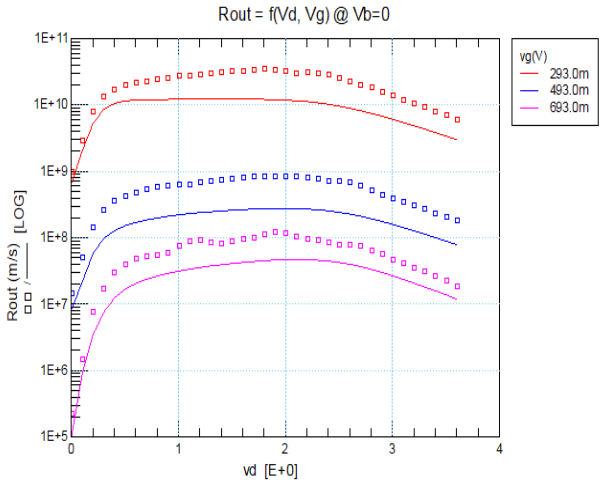


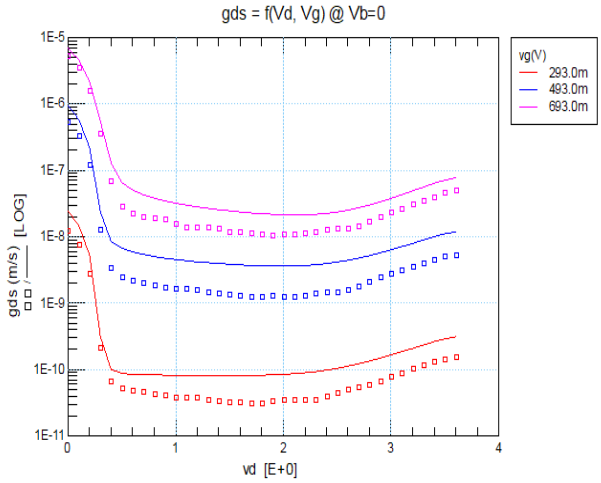




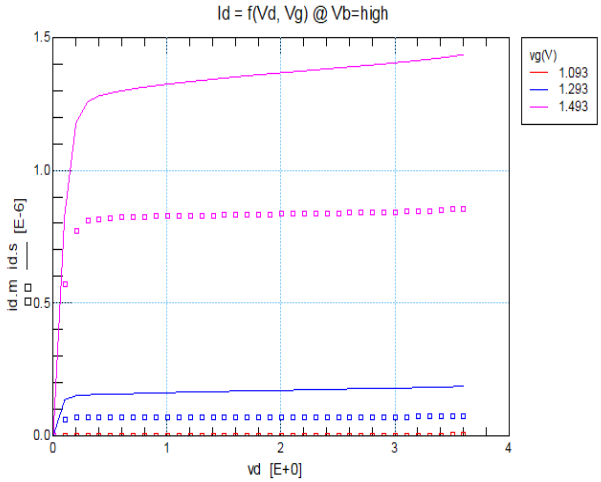
idvd_vth, Vb = 0V, T = 125°C

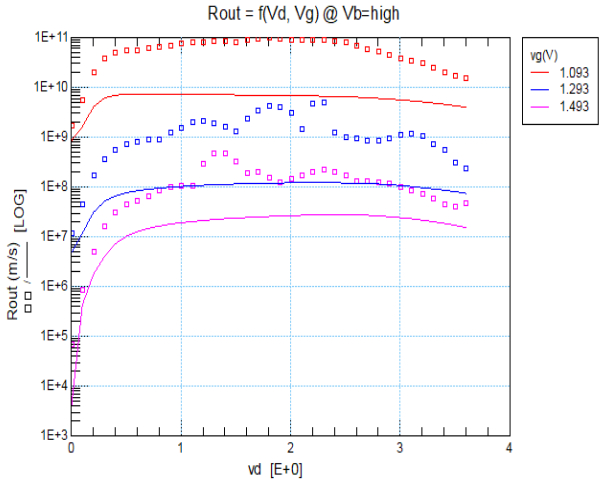


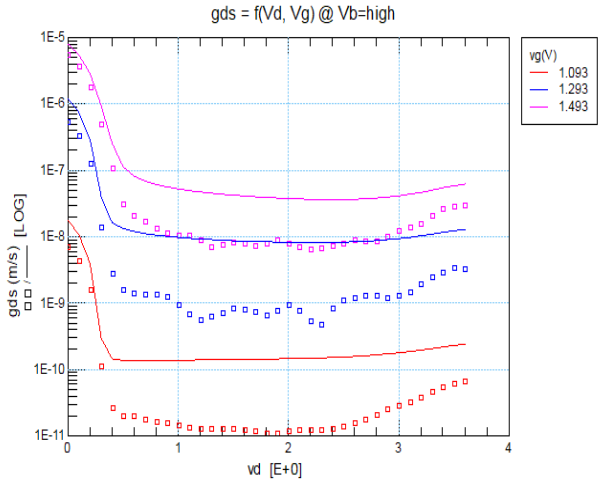




idvd_vth_vbmin, Vb = -3.6V, T = 125°C

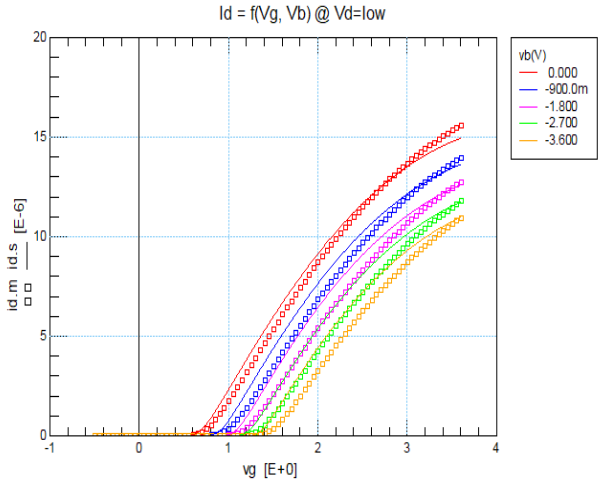


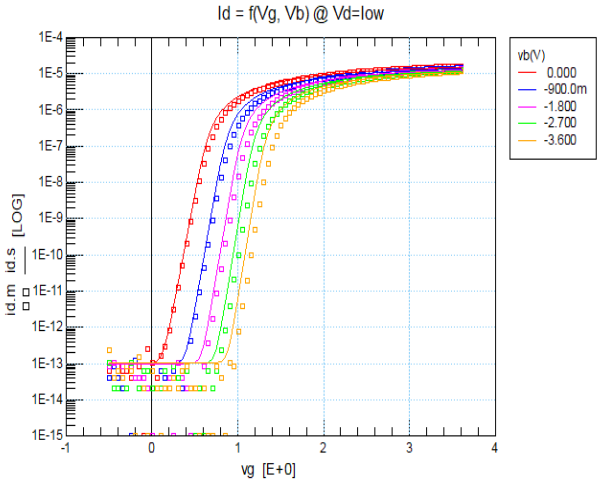


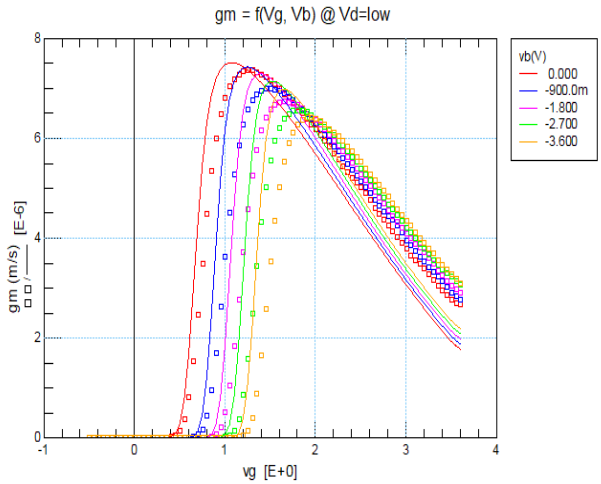


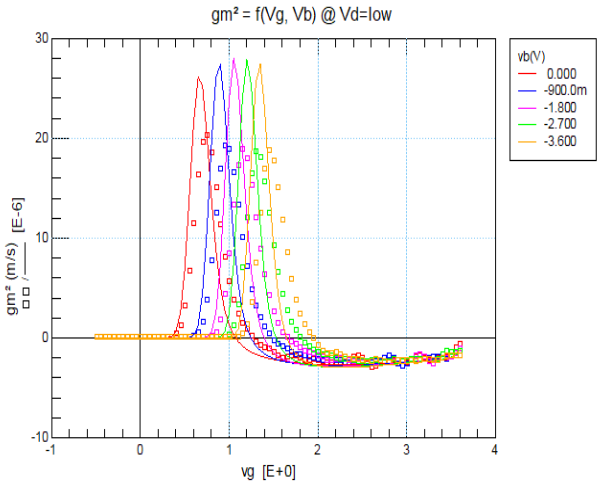
4.16W0u6_L1u2_S559_4

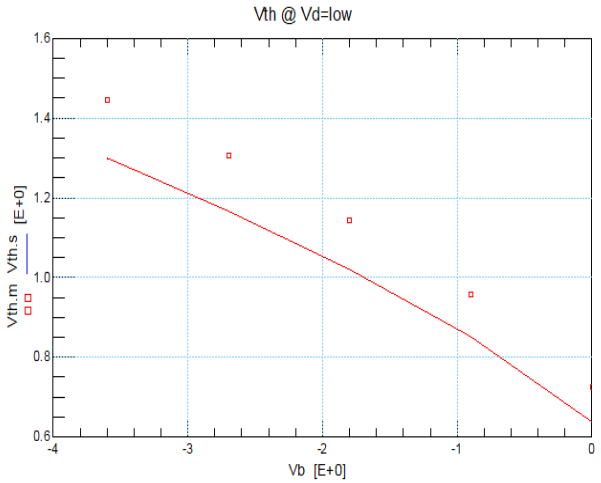
idvg, Vd = 0.1V, T = 27°C

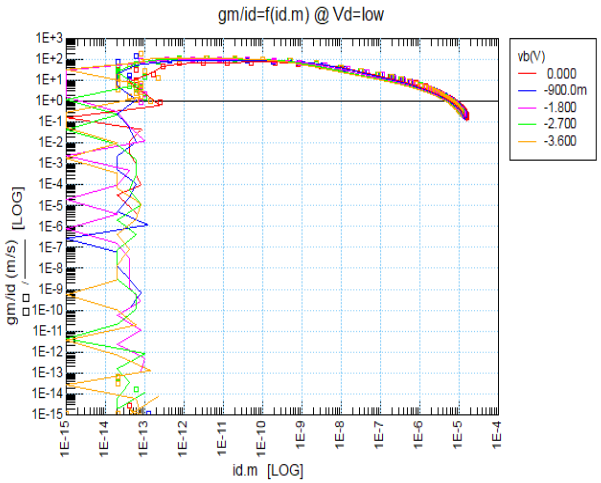


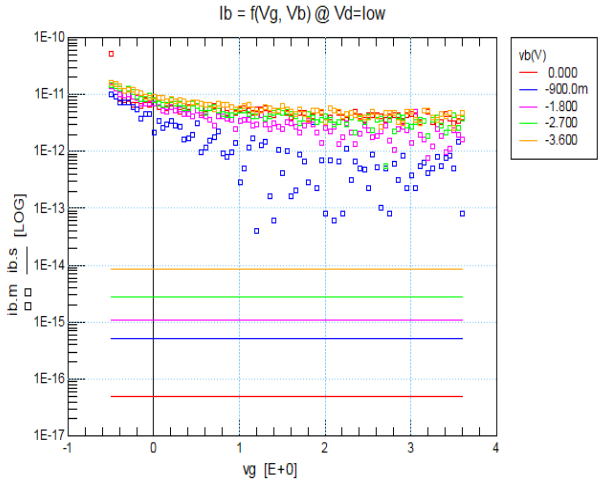


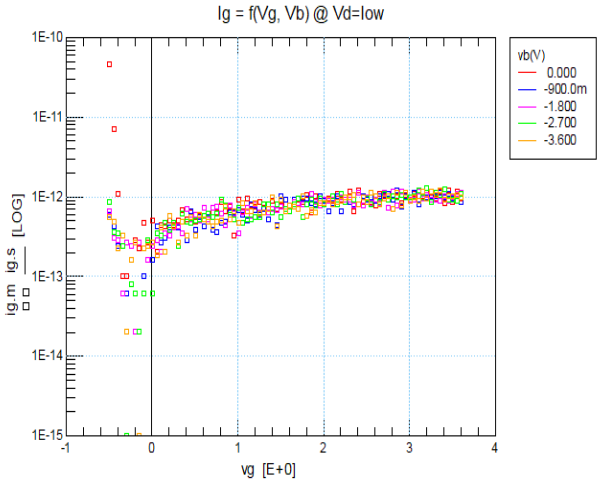




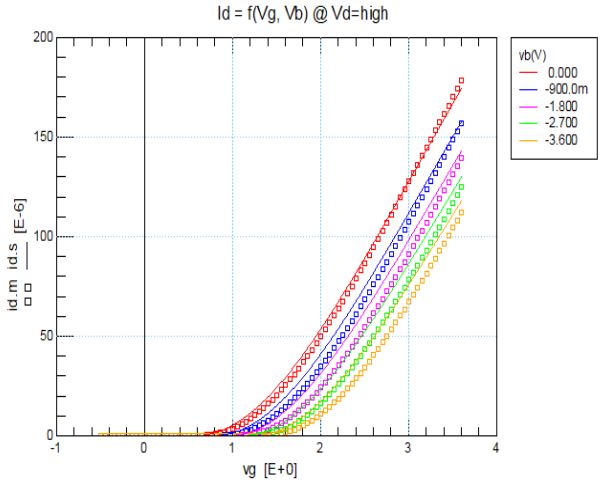


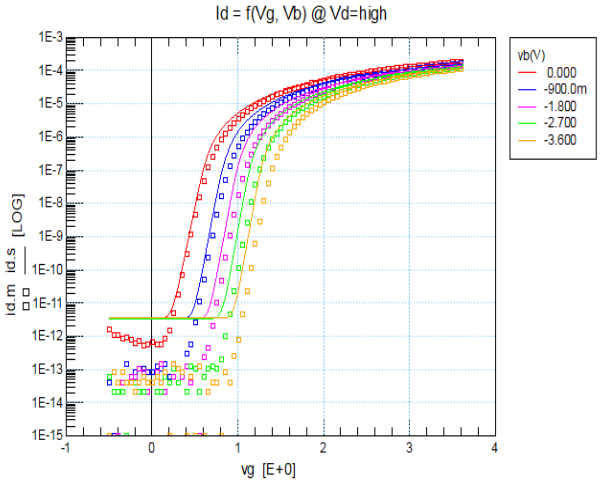


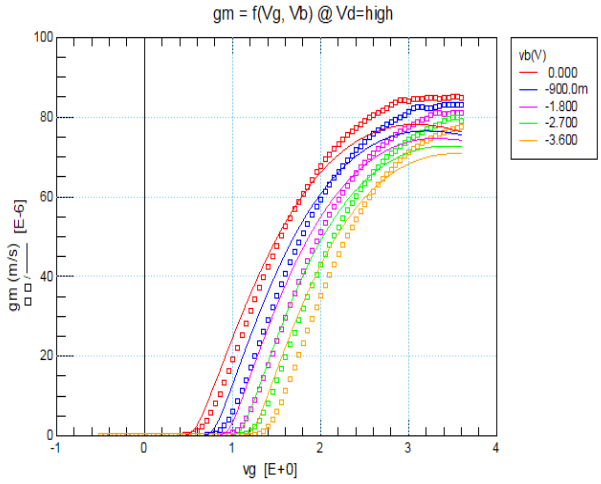


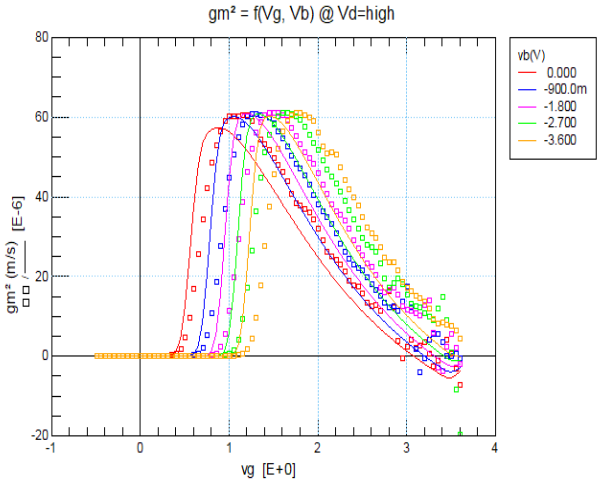


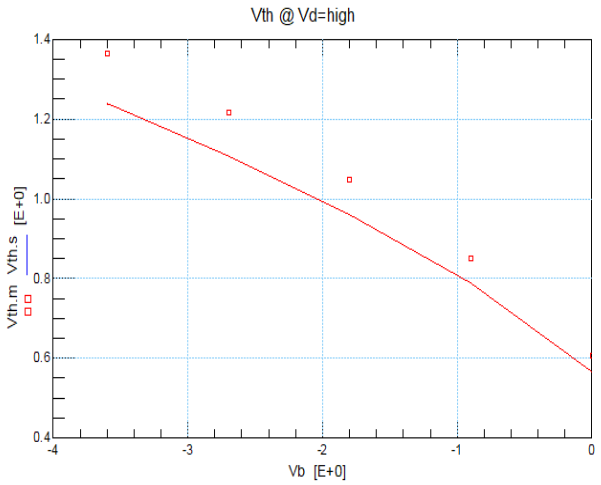
idvg, Vd = 3.3V, T = 27°C

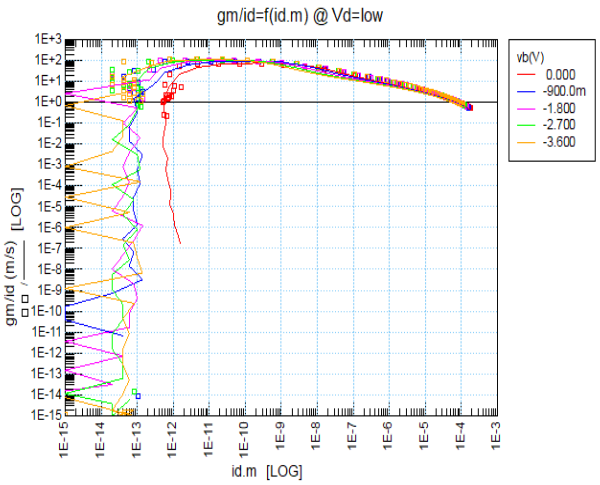


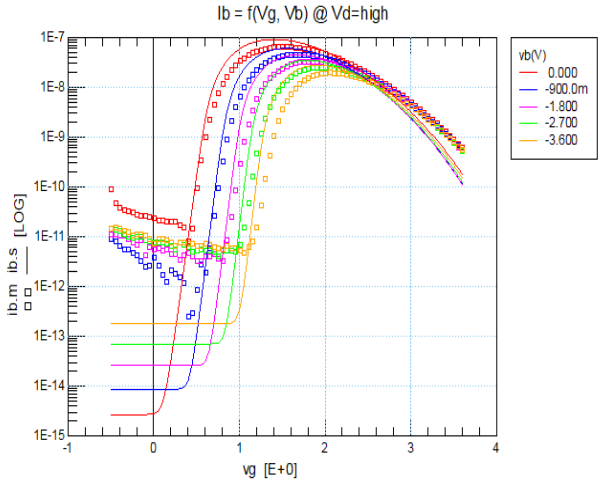


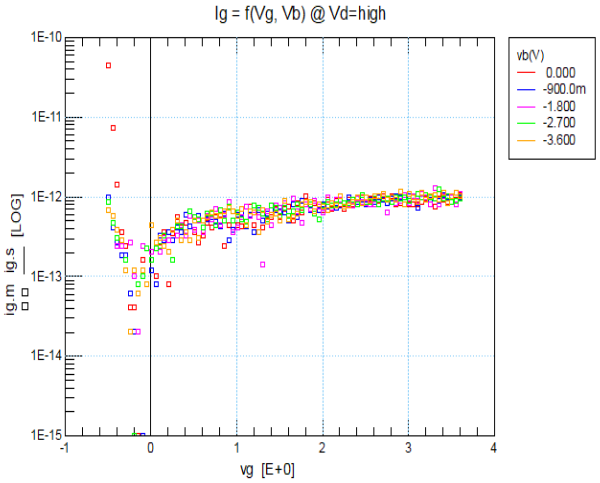




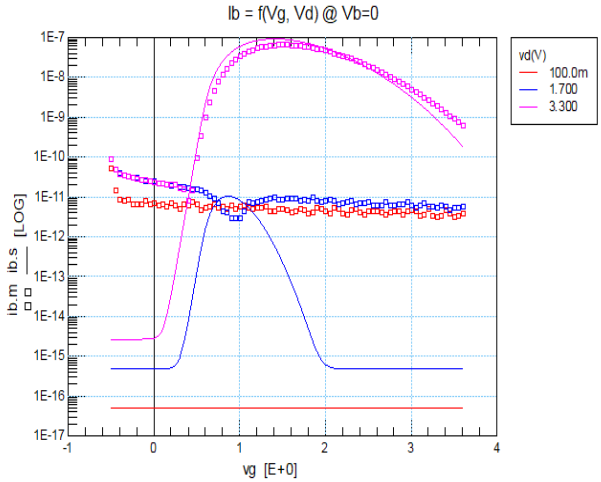


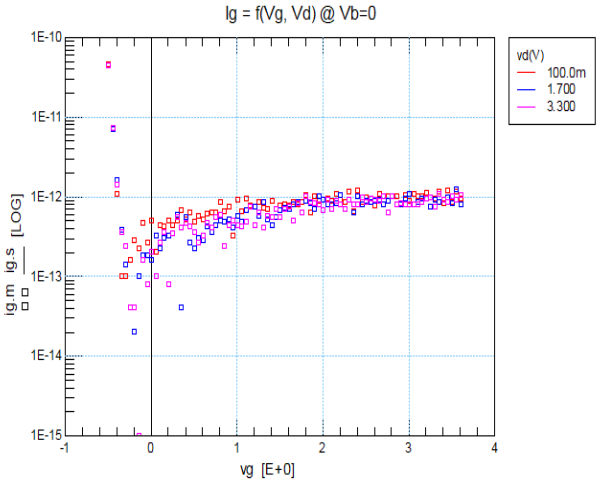




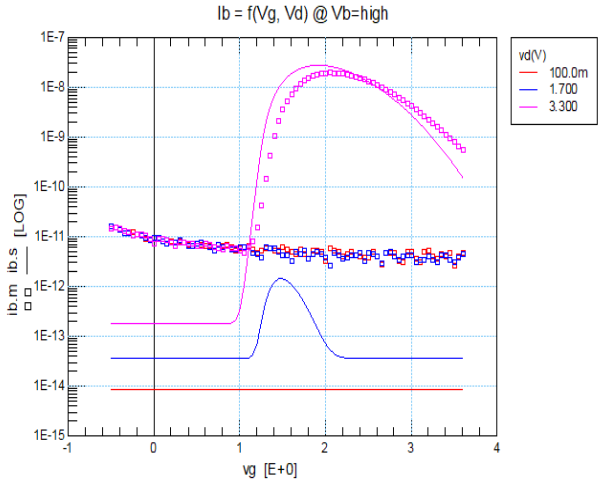


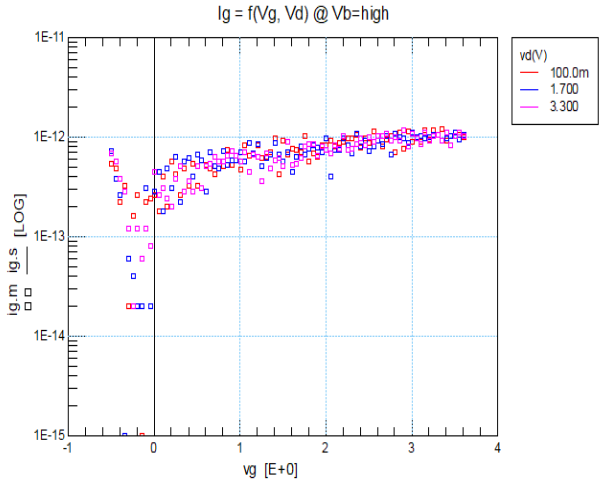
idvg, Ib,Ig, Vb = 0V, T = 27°C



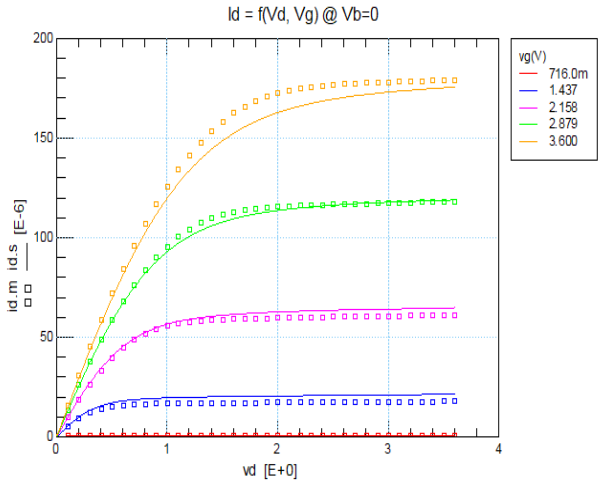


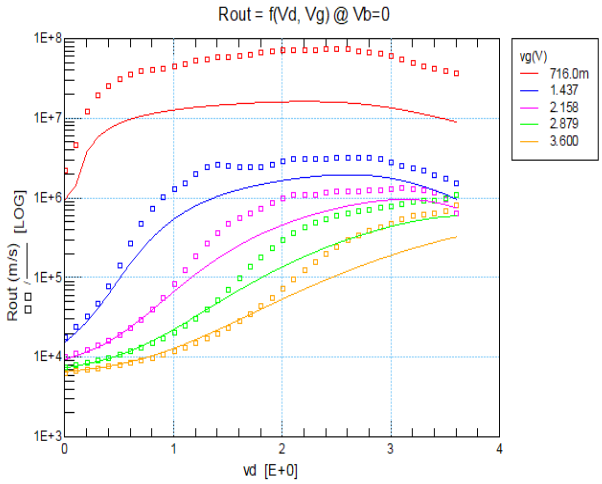
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

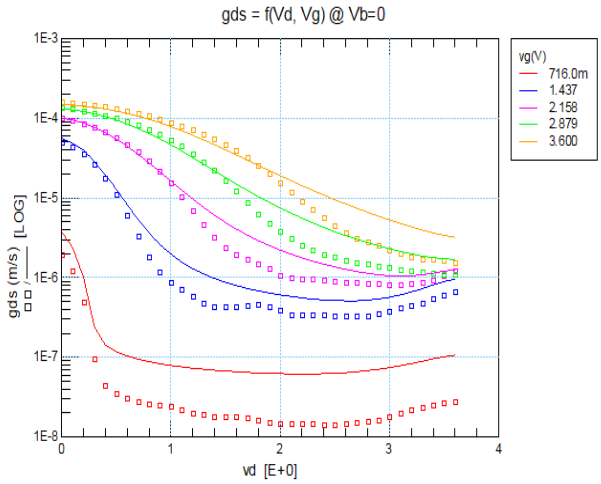




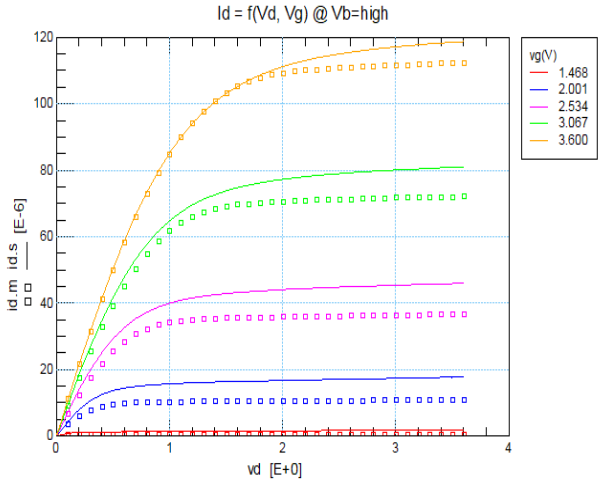
idvd, Vb = 0V, T = 27°C

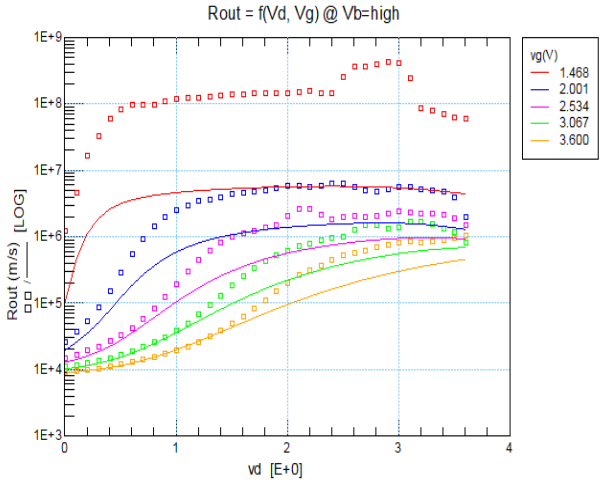


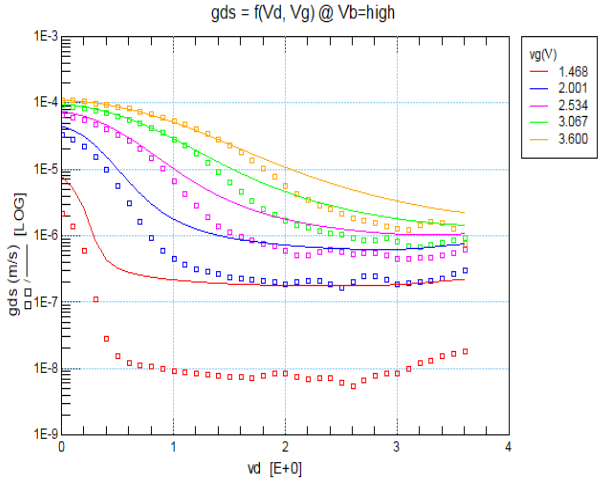




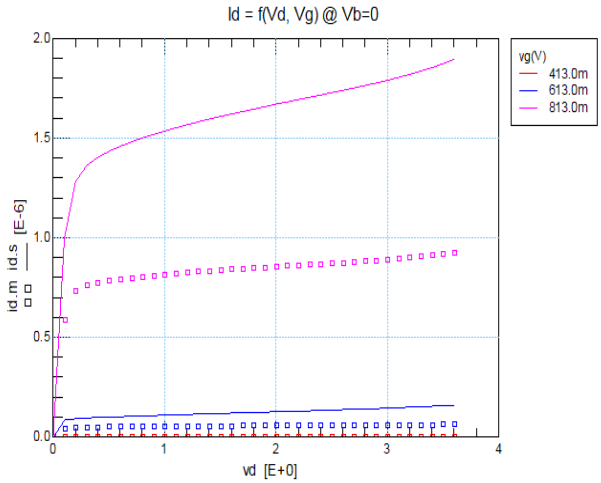
idvd_vbmin, Vb = -3.6V, T = 27°C

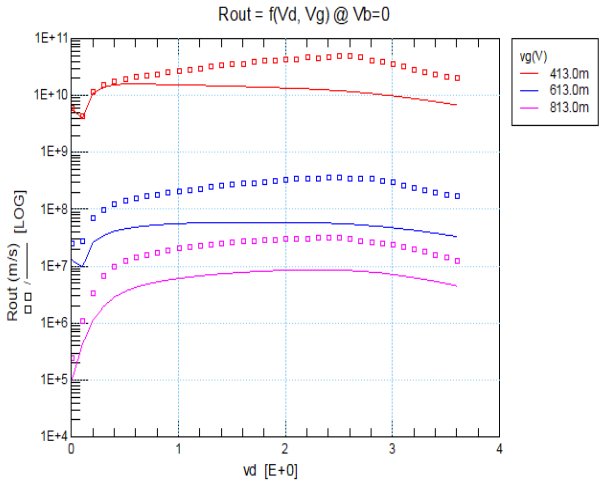


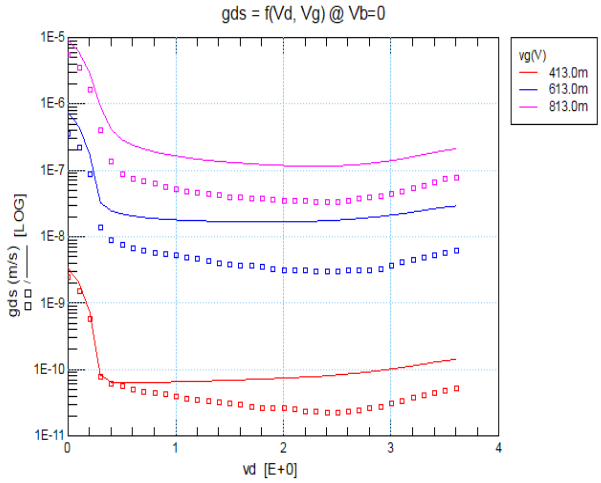




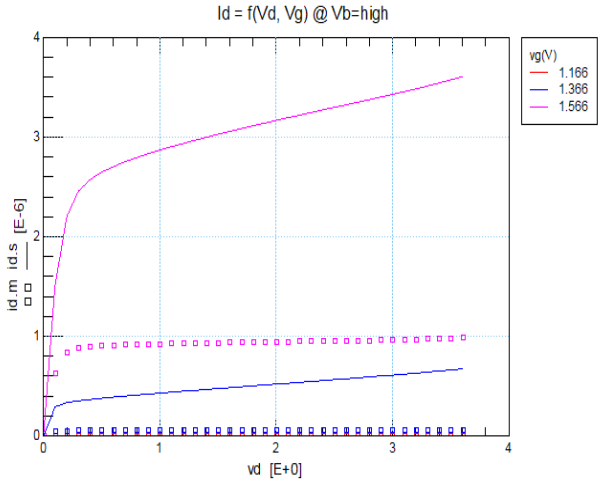
idvd_vth, Vb = 0V, T = 27°C

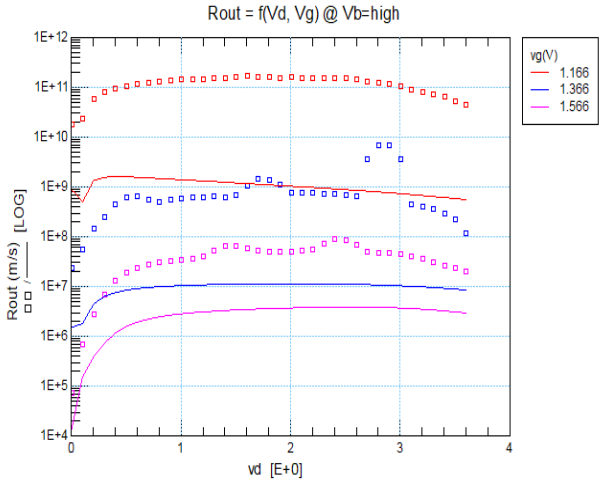


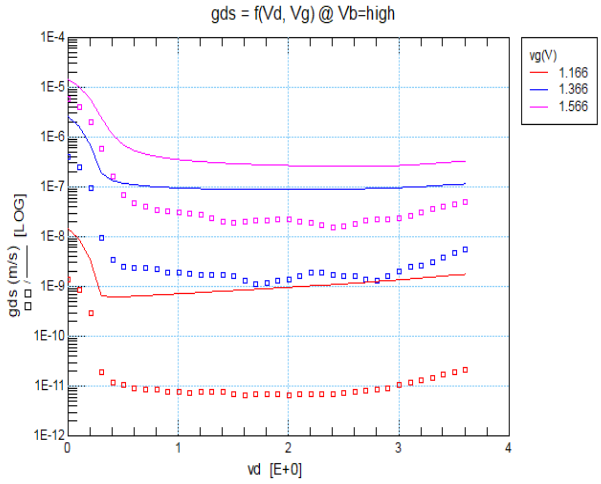




idvd_vth_vbmin, Vb = -3.6V, T = 27°C

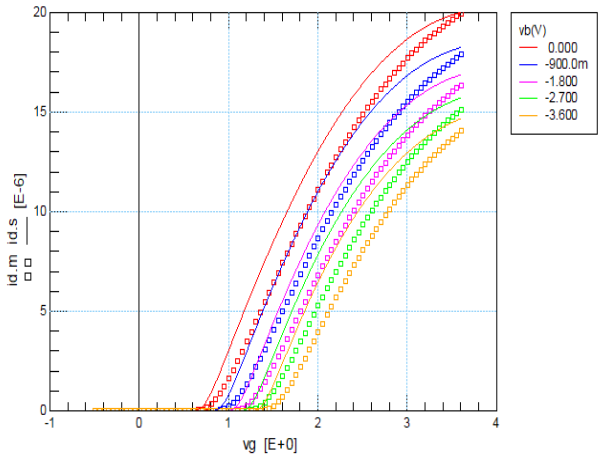




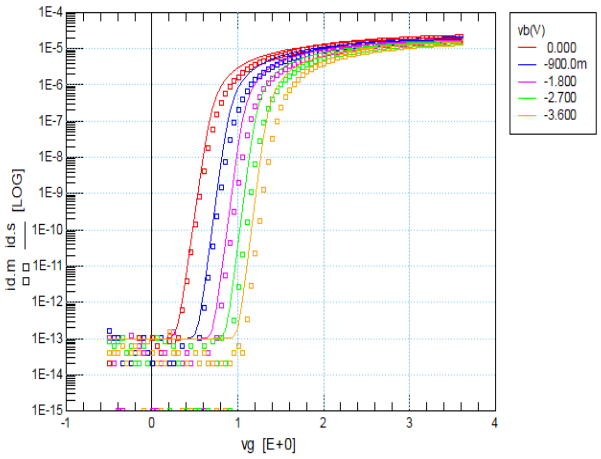


idvg, Vd = 0.1V, T = -40°C

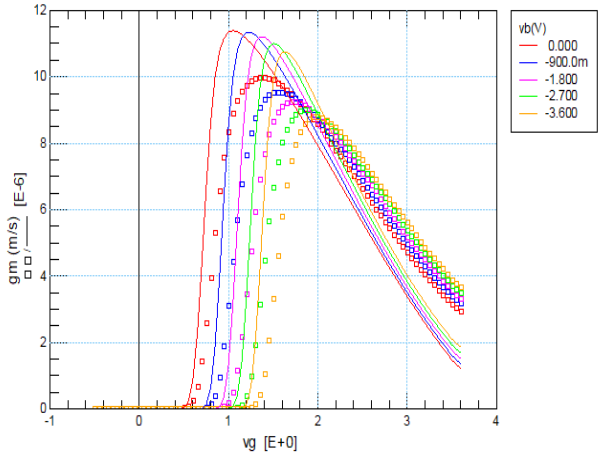
Id = f(Vg, Vb) @ Vd=low



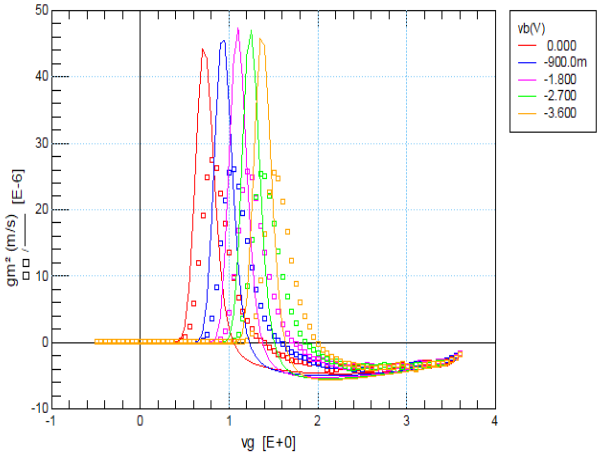
Id = f(Vg, Vb) @ Vd=low



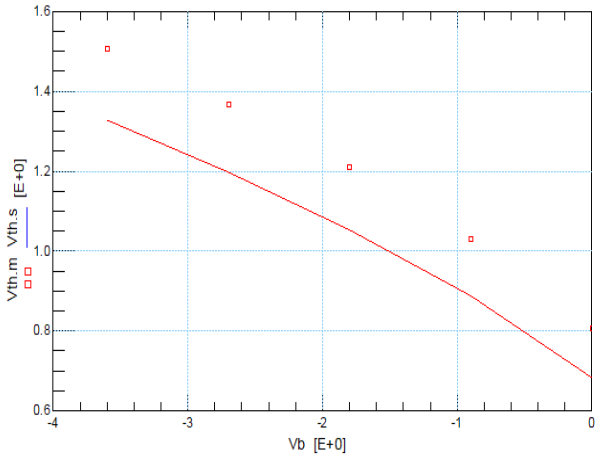
gm = f(Vg, Vb) @ Vd=low



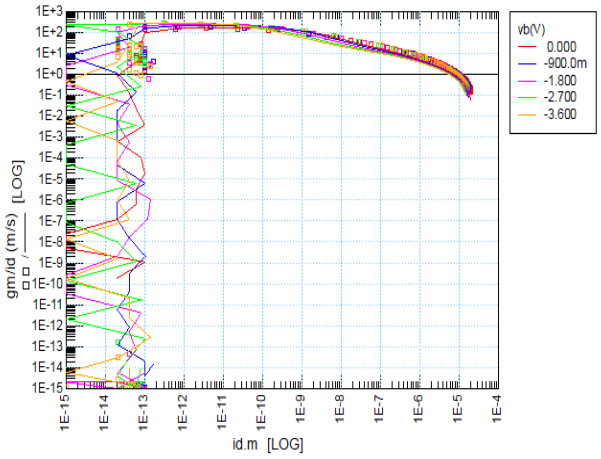
gm² = f(Vg, Vb) @ Vd=low

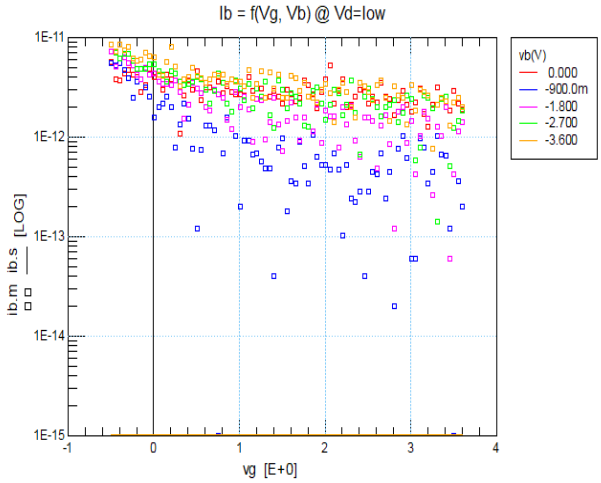


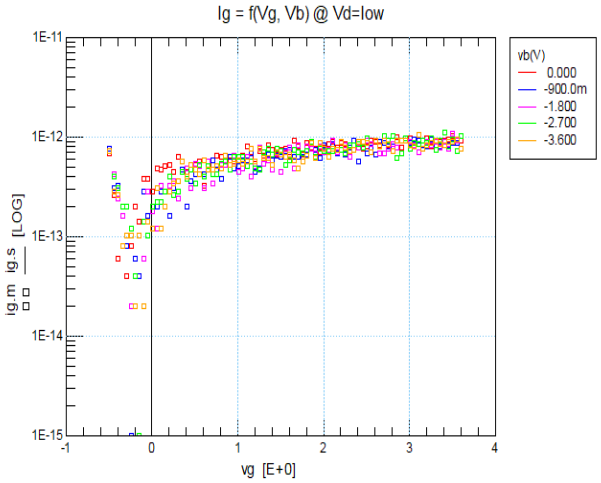
Vth @ Vd=low



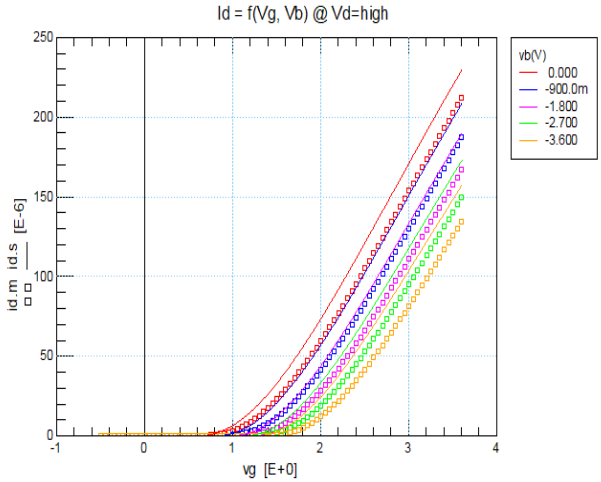
gm/id=f(id.m) @ Vd=low

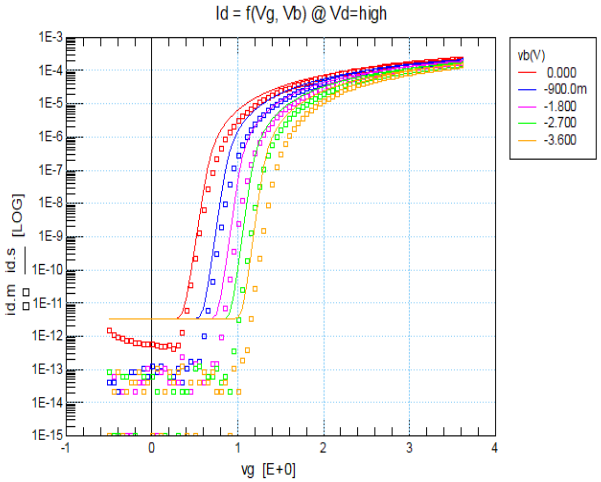


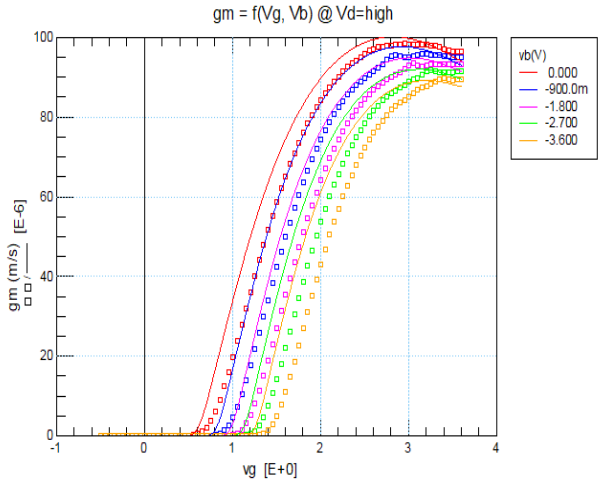


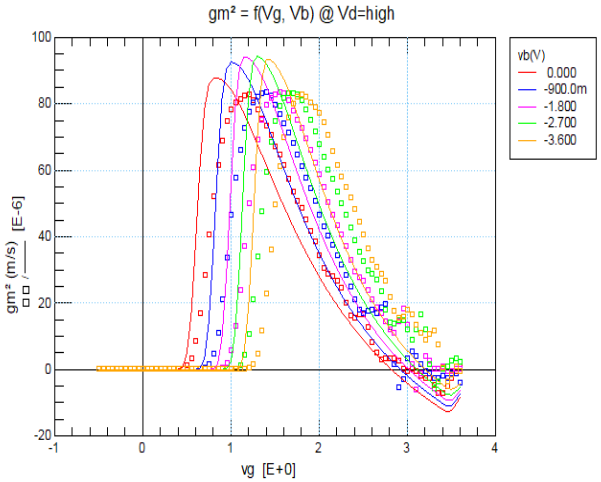


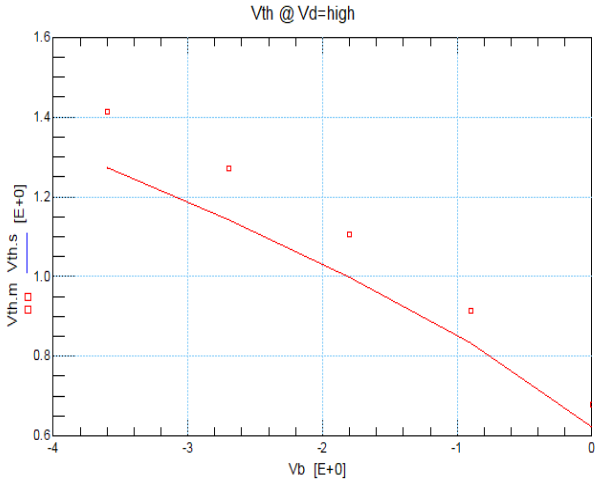
idvg, Vd = 3.3V, T = -40°C

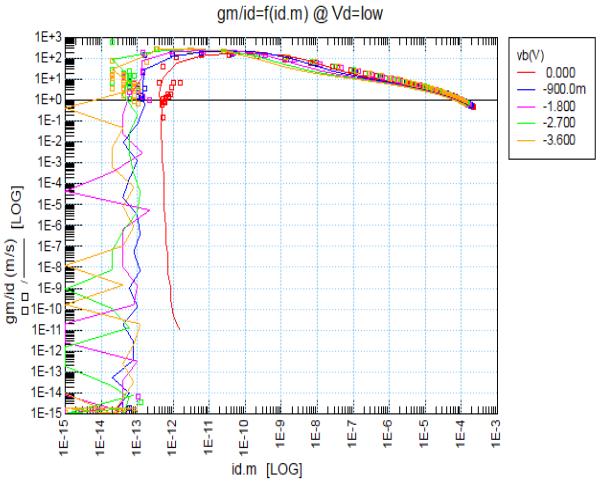


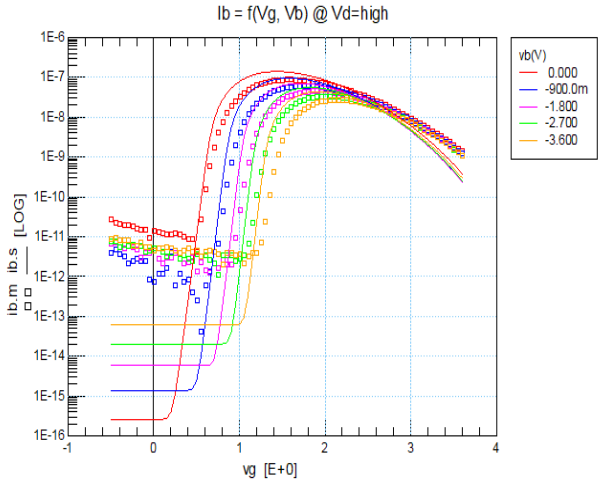


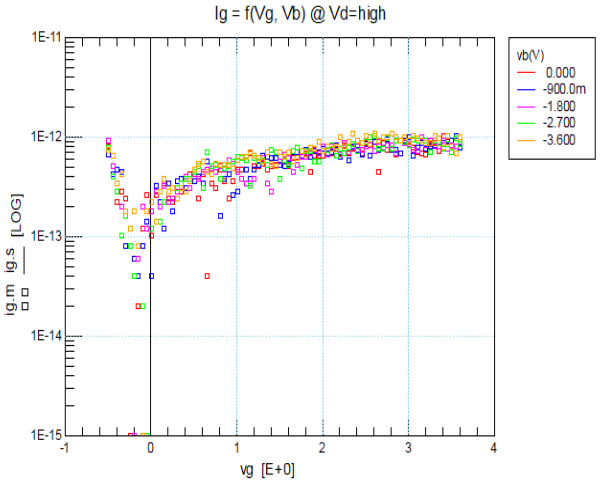




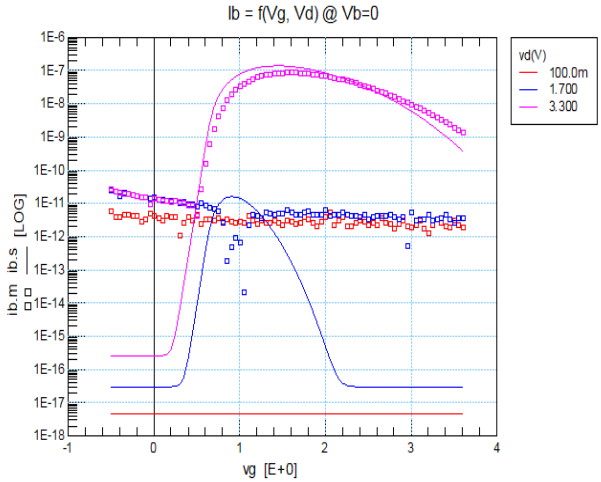


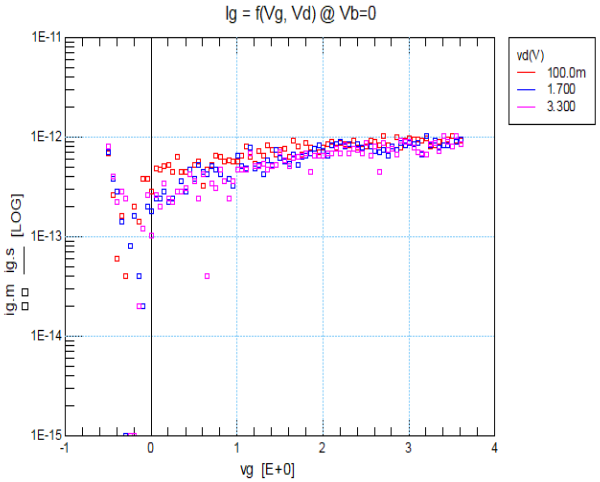




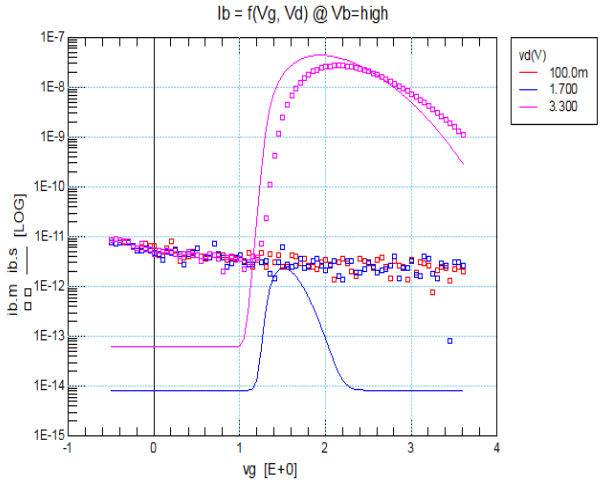


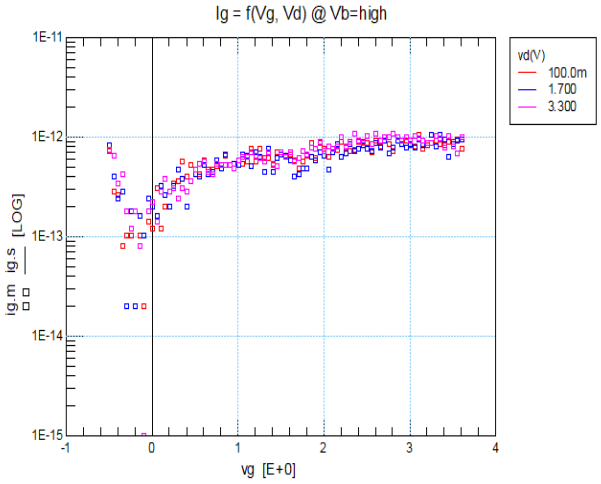
idvg, Ib,Ig, Vb = 0V, T = -40°C



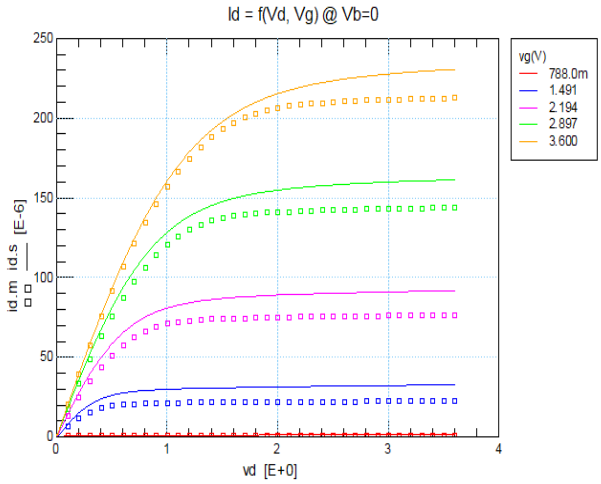


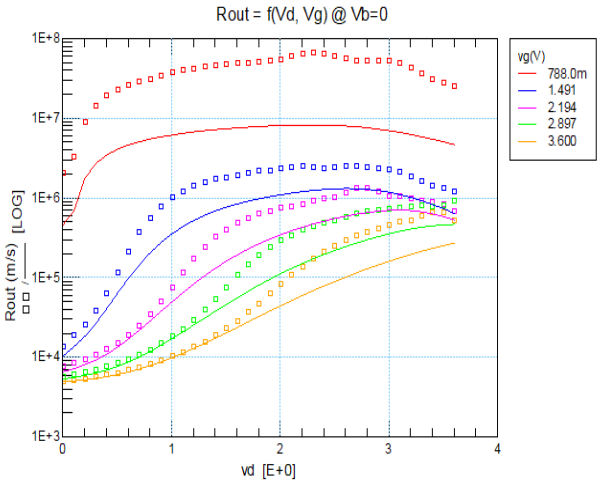
idvg, Ib,Ig, Vb = -3.6V, T = -40°C

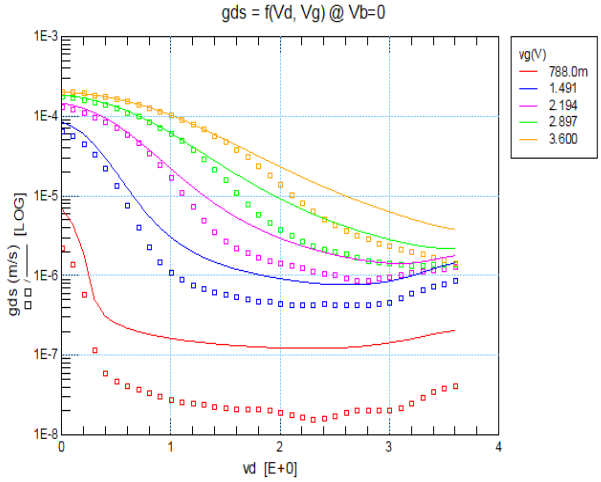




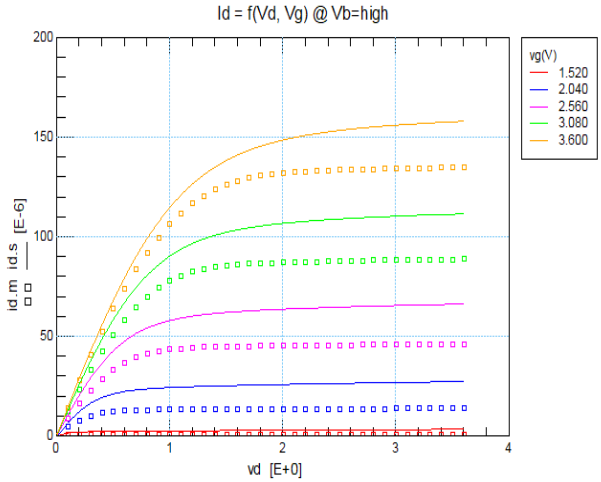
idvd, Vb = 0V, T = -40°C

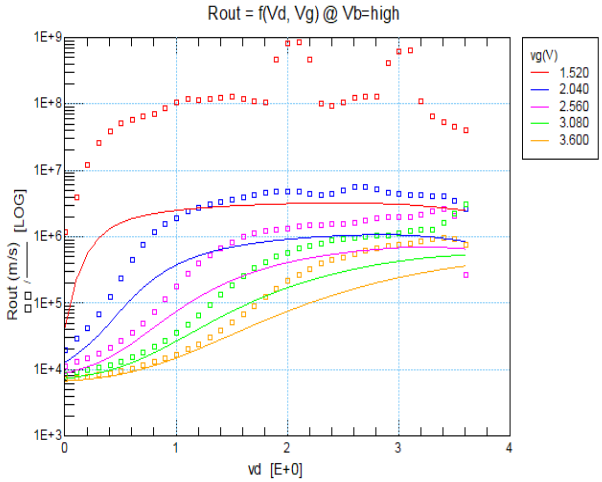


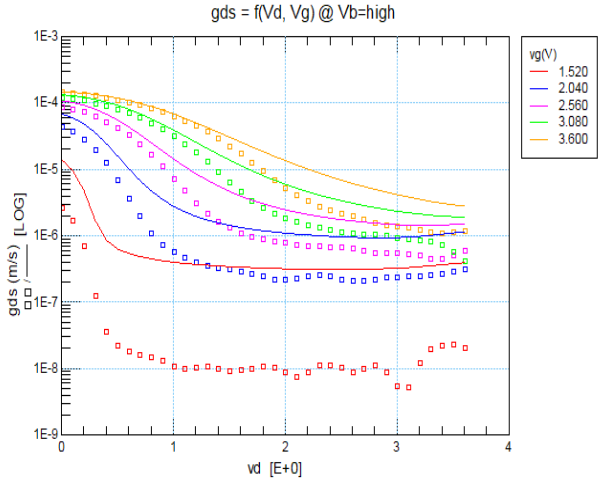




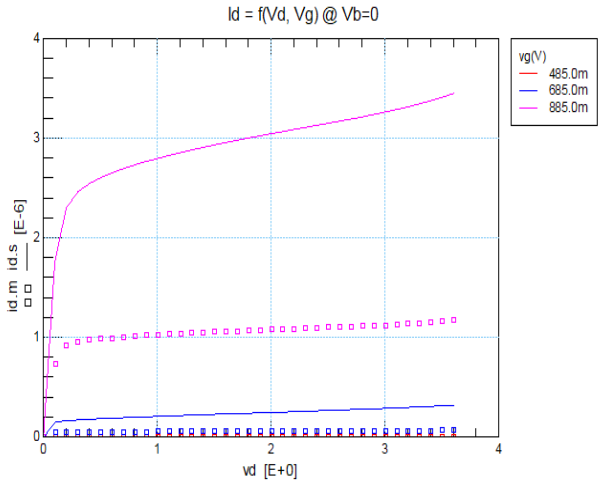
idvd_vbmin, Vb = -3.6V, T = -40°C

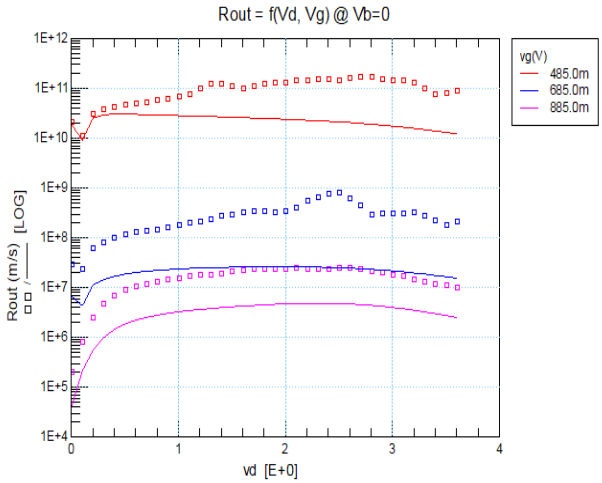


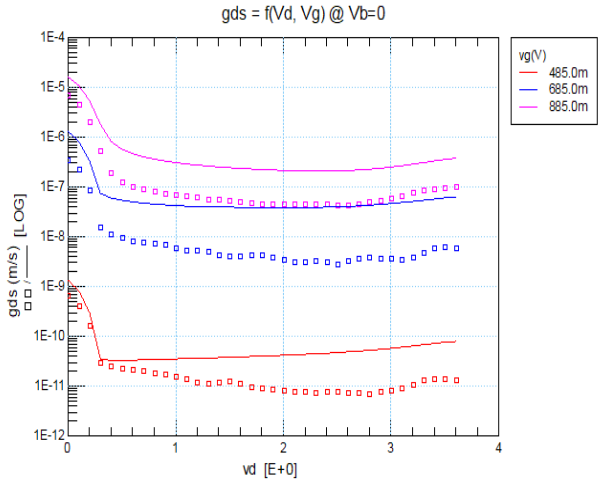




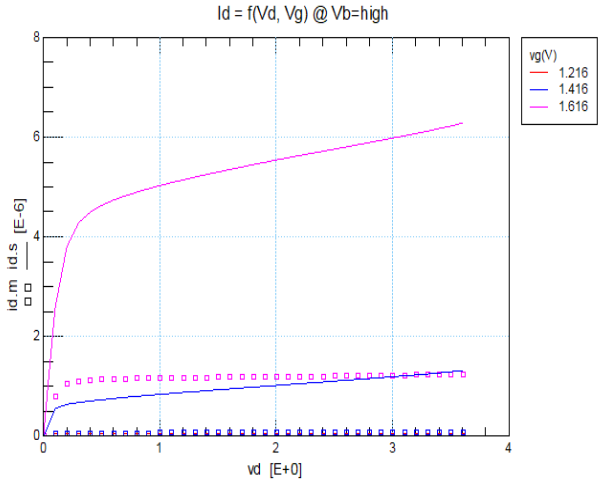
idvd_vth, Vb = 0V, T = -40°C

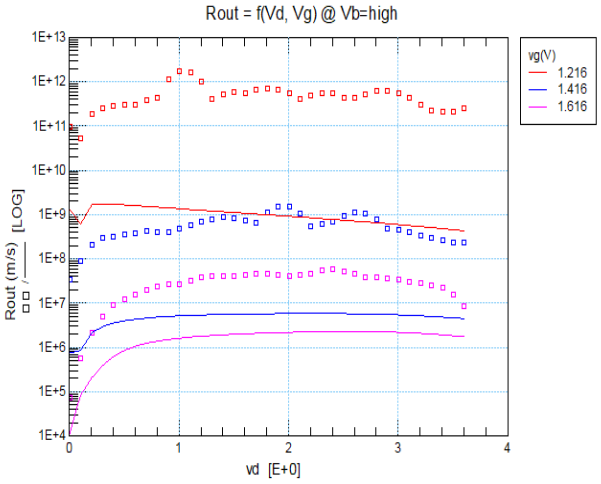


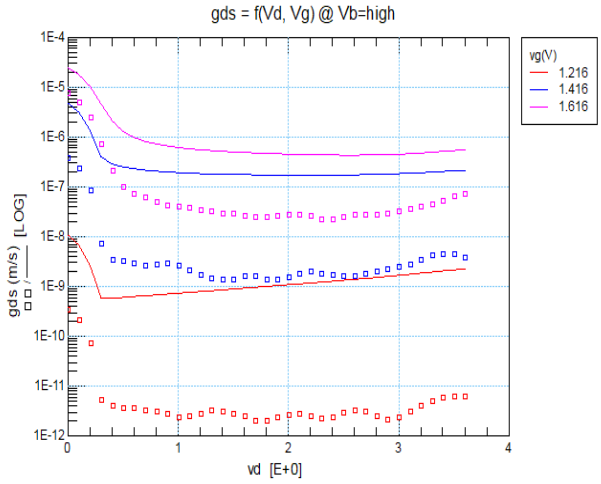




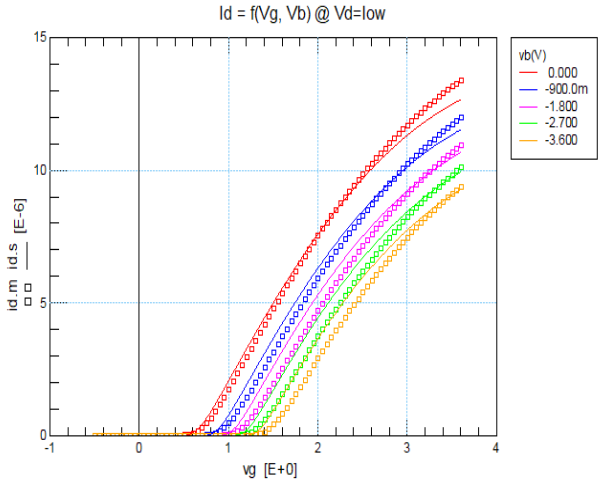
idvd_vth_vbmin, Vb = -3.6V, T = -40°C

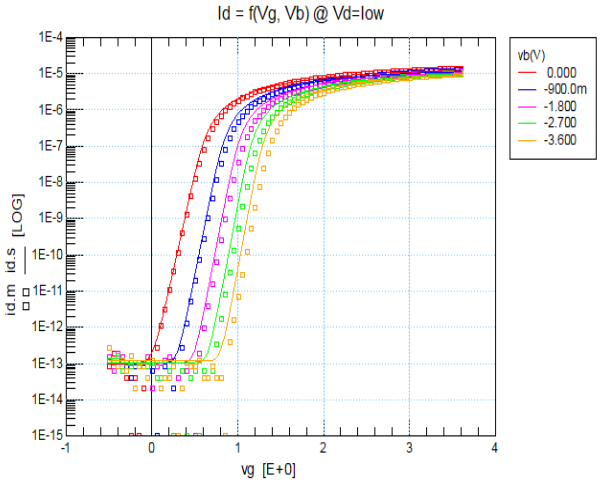


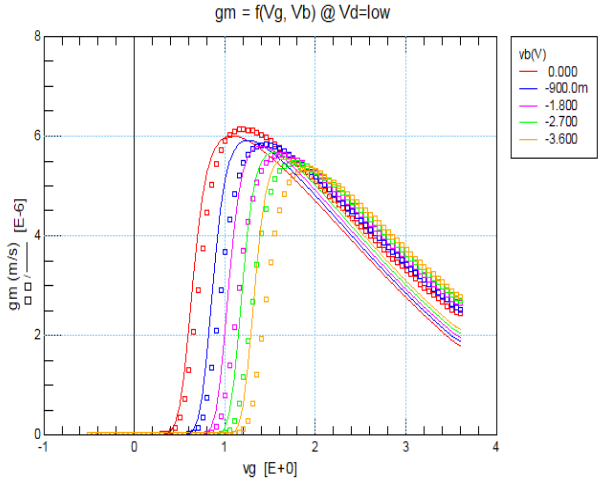


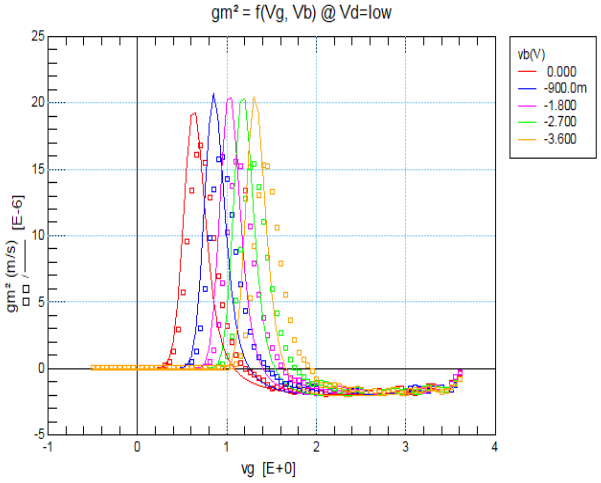


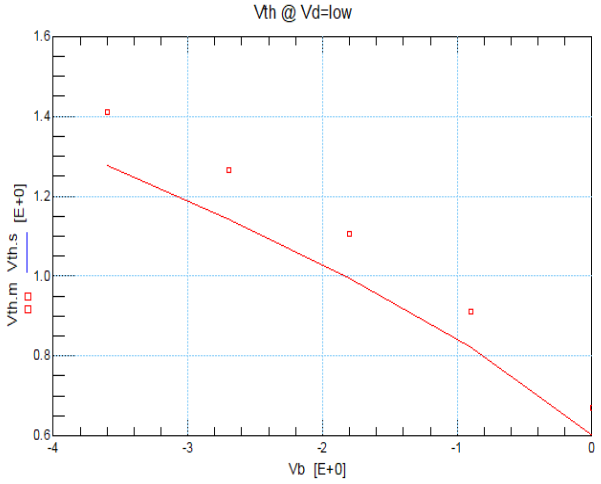
idvg, Vd = 0.1V, T = 70°C

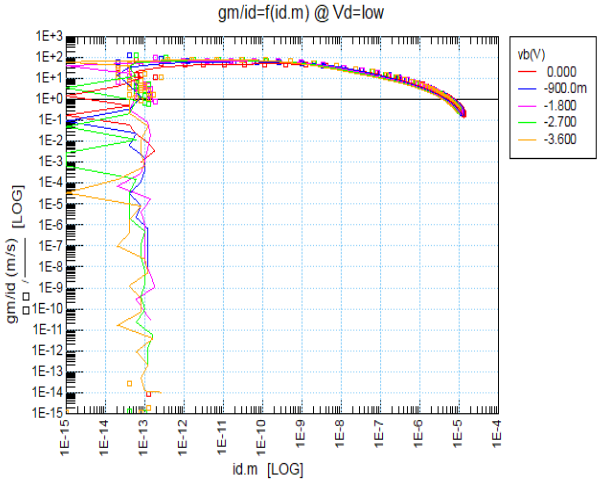


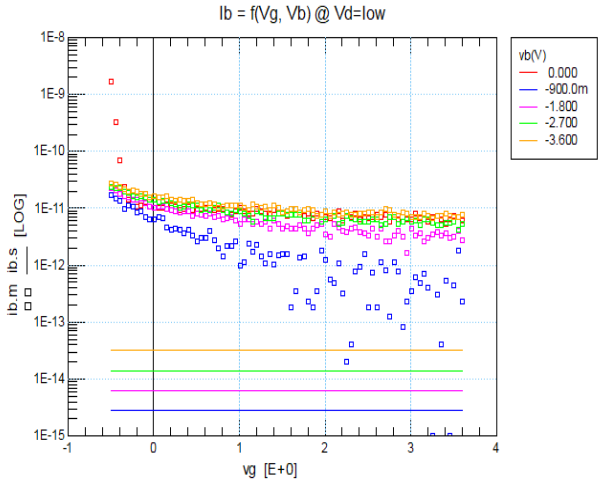


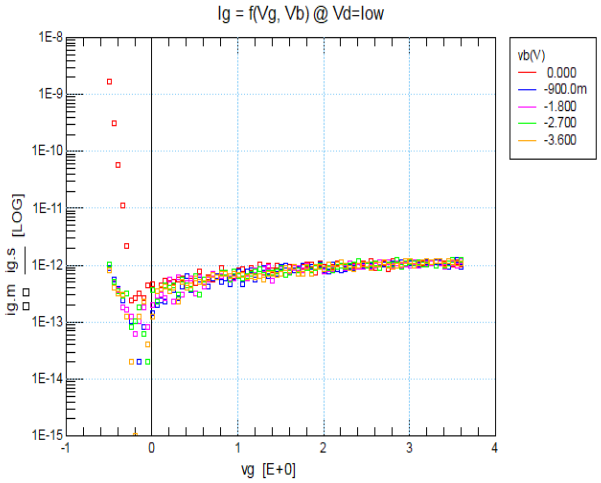




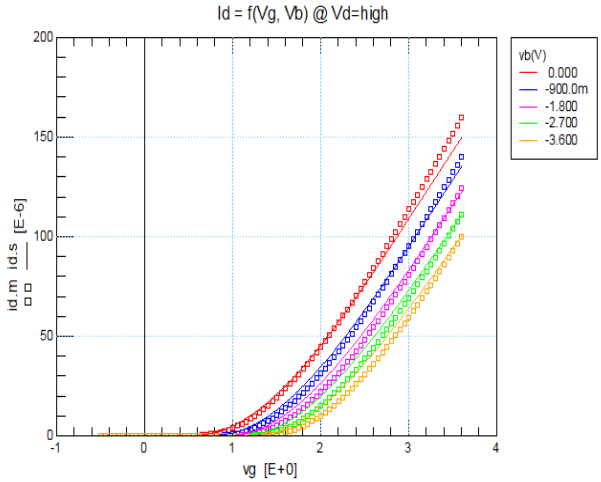


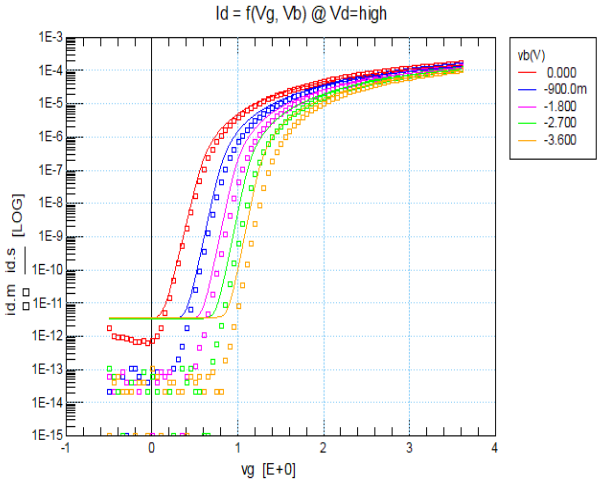


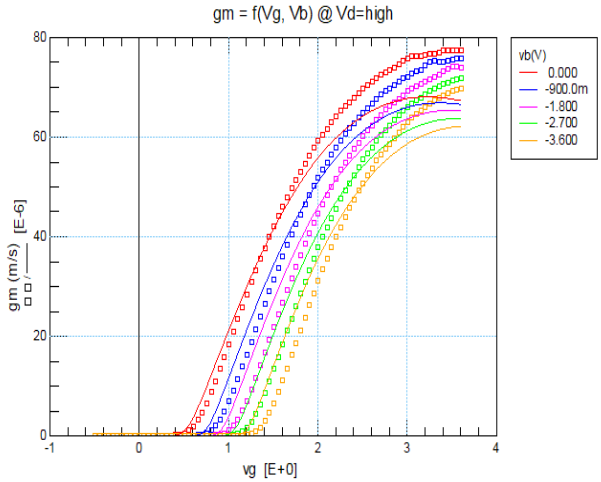


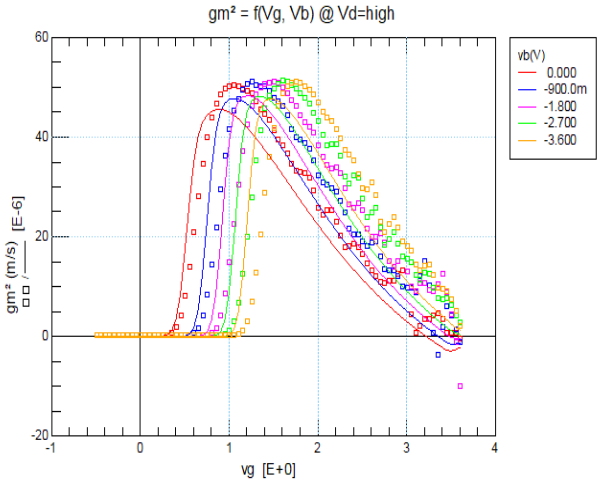


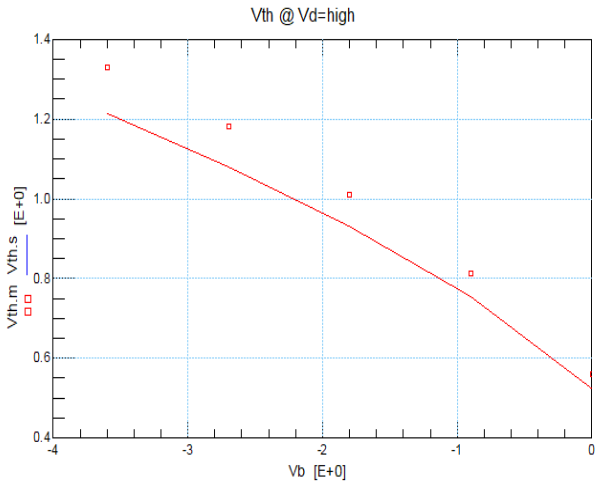
idvg, Vd = 3.3V, T = 70°C

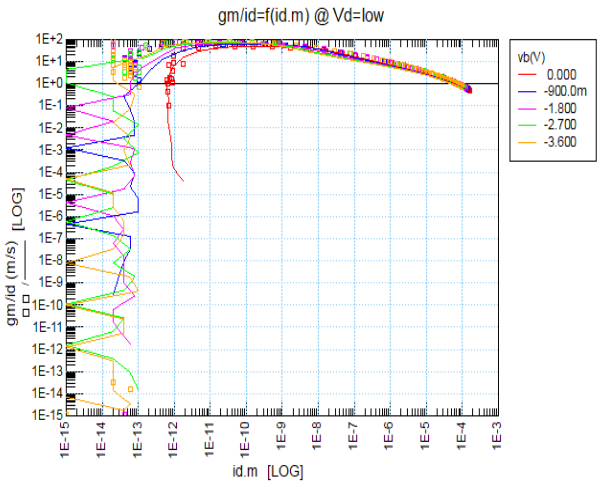


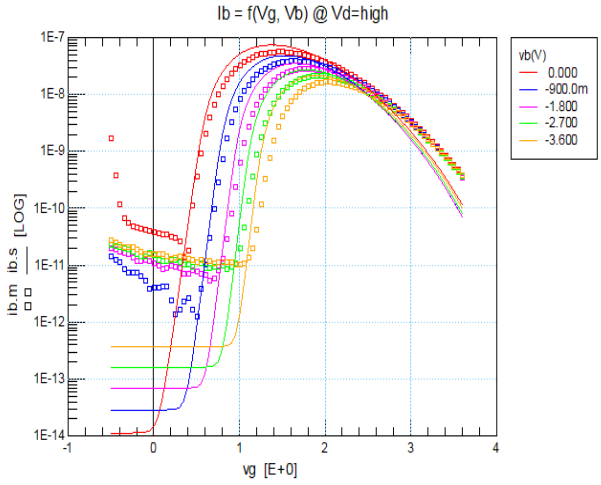


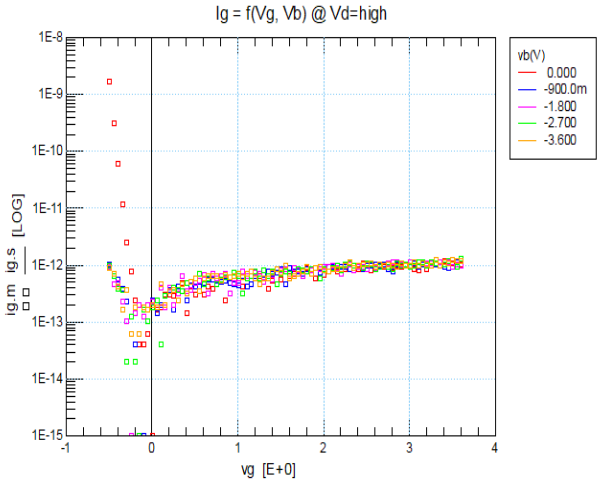




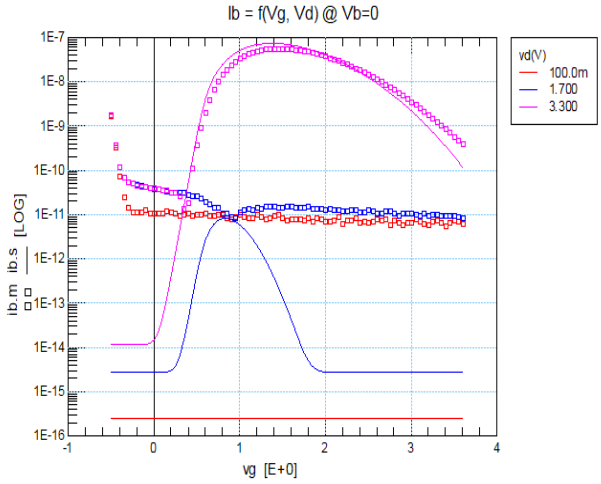


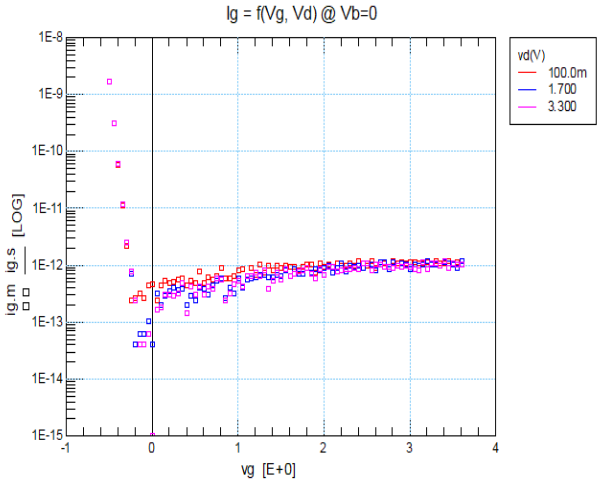




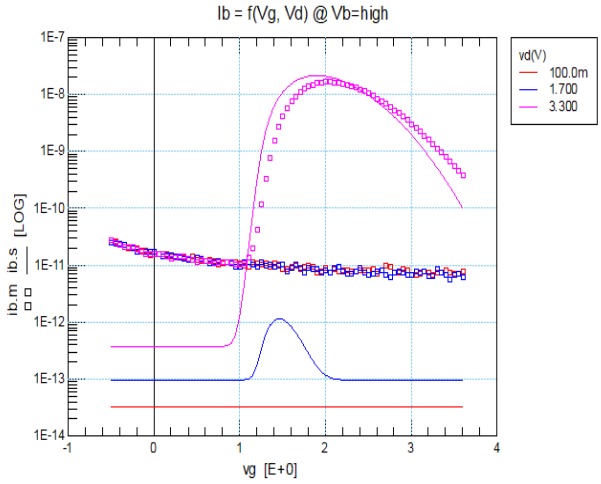


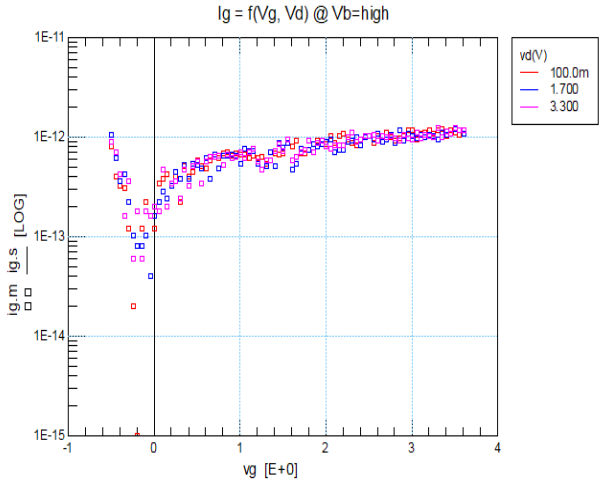
idvg, Ib,Ig, Vb = 0V, T = 70°C



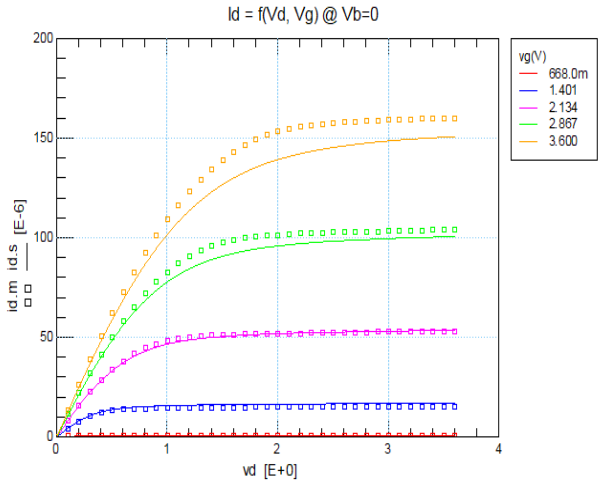


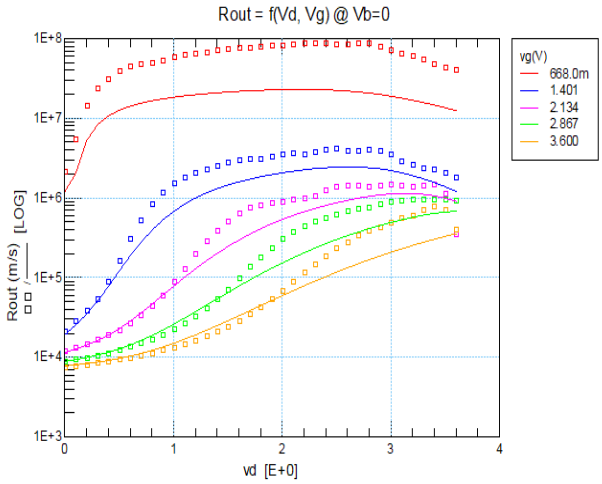
idvg, Ib,Ig, Vb = -3.6V, T = 70°C

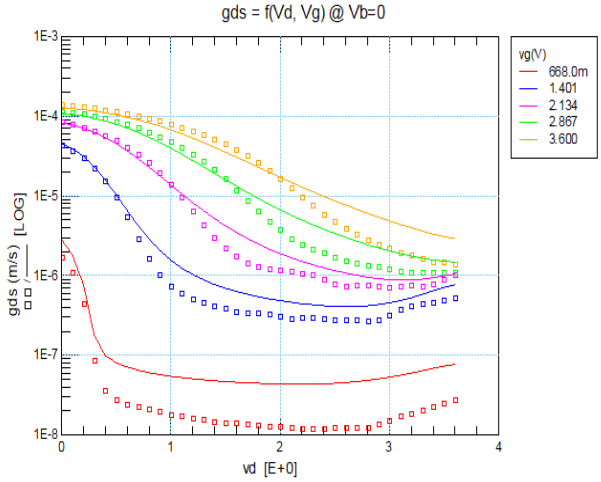




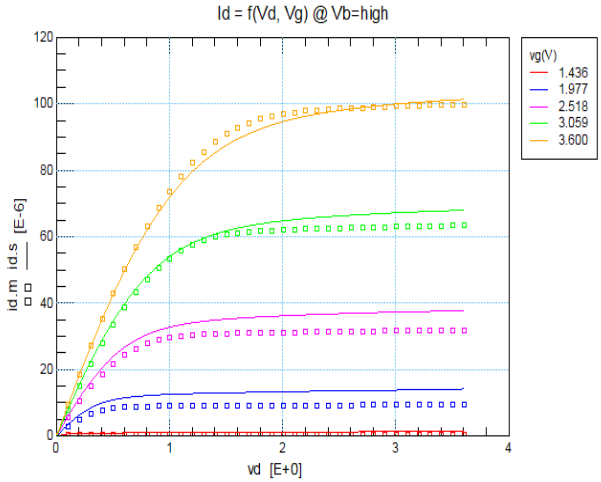
idvd, Vb = 0V, T = 70°C

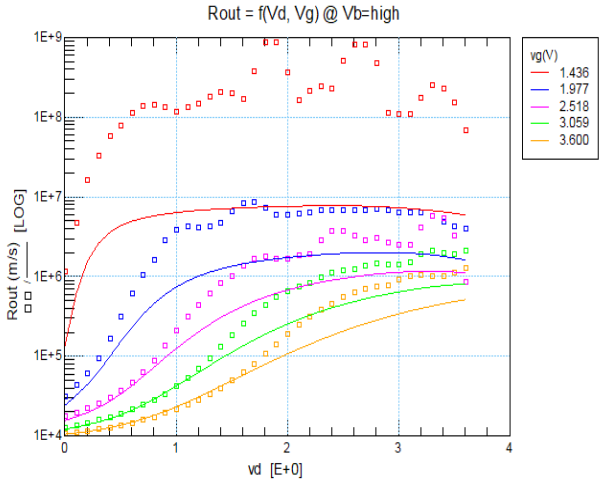


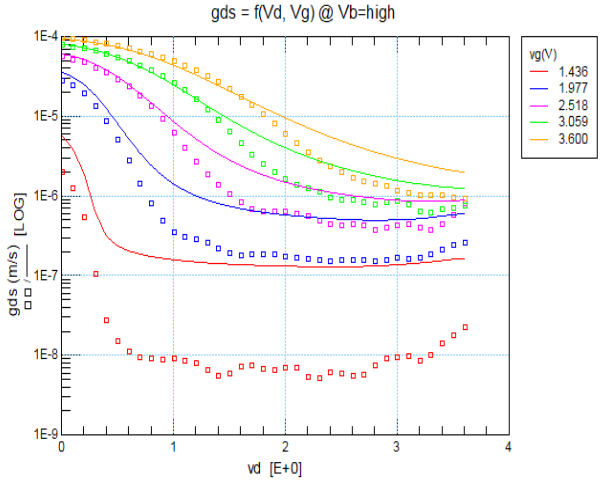




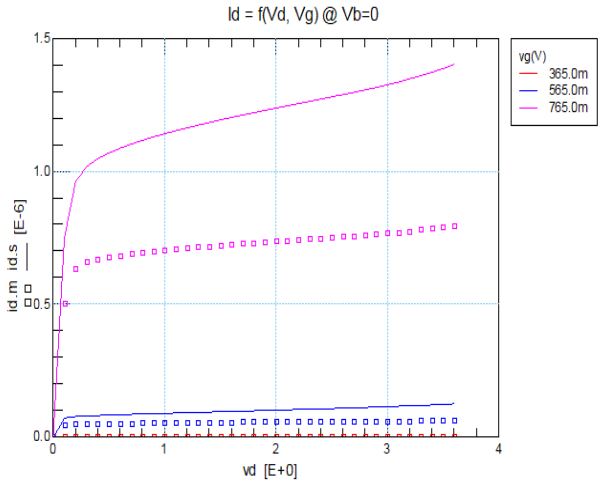
idvd_vbmin, Vb = -3.6V, T = 70°C

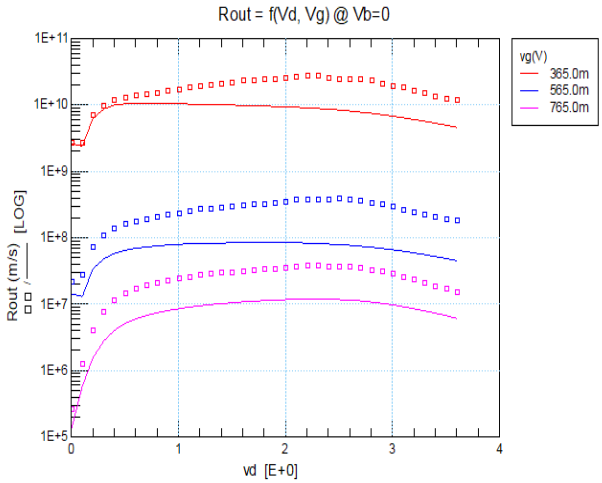


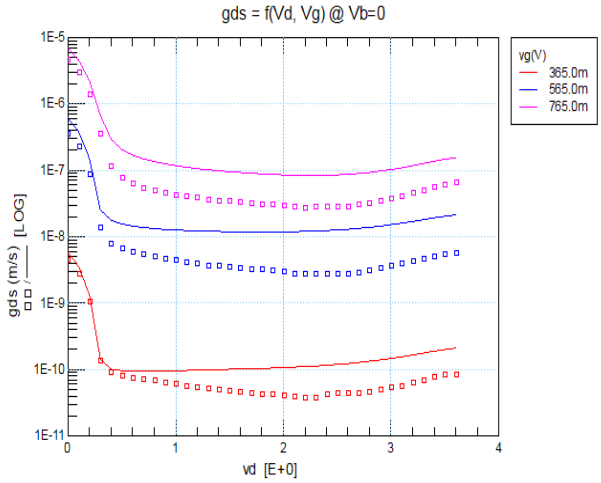




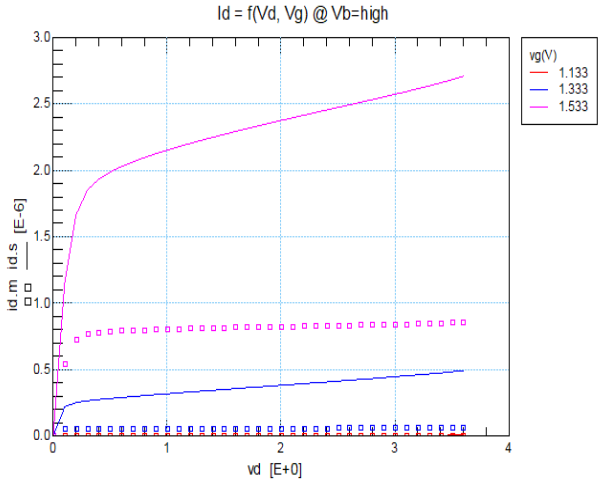
idvd_vth, Vb = 0V, T = 70°C

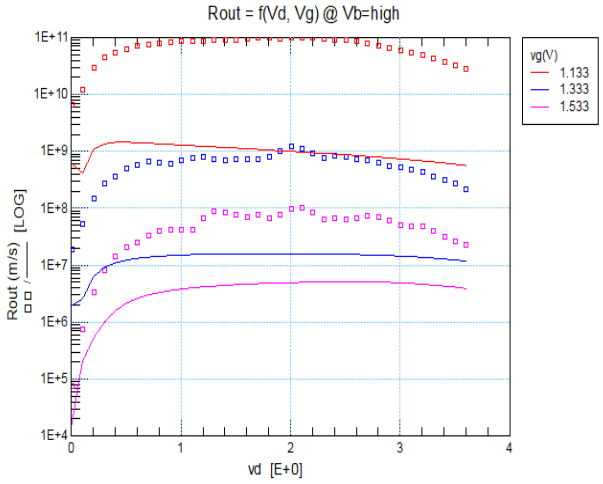


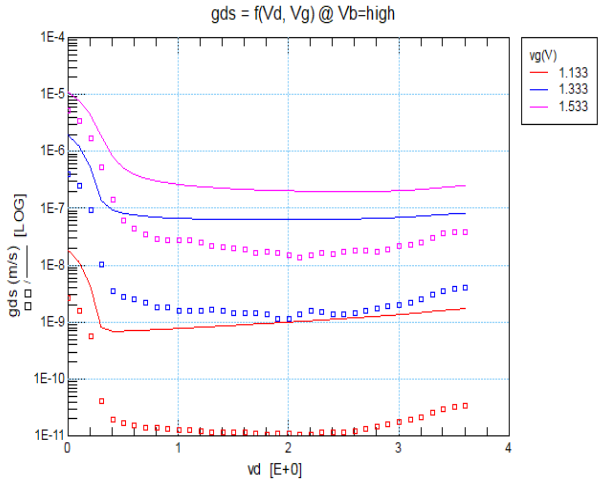




idvd_vth_vbmin, Vb = -3.6V, T = 70°C

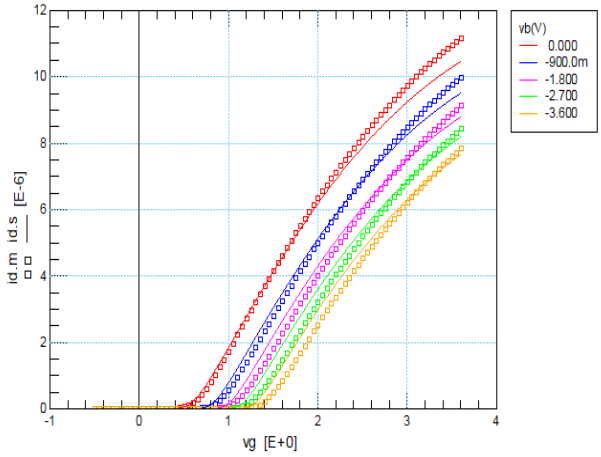




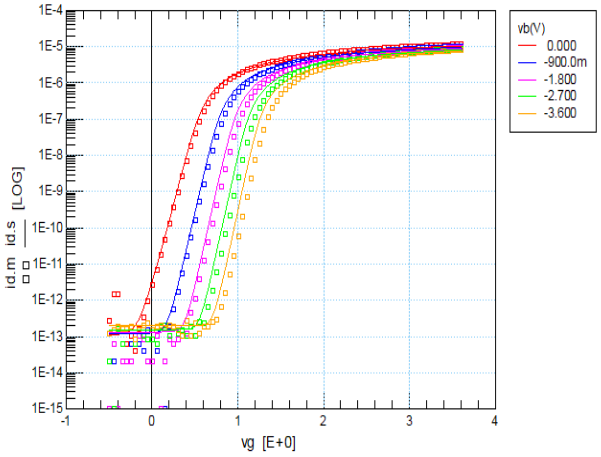


idvg, Vd = 0.1V, T = 125°C

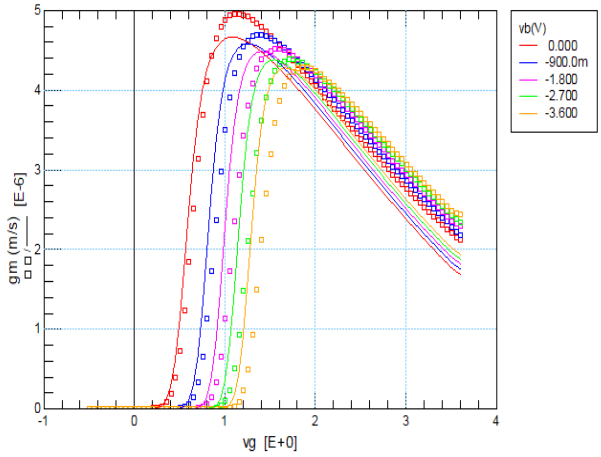
Id = f(Vg, Vb) @ Vd=low



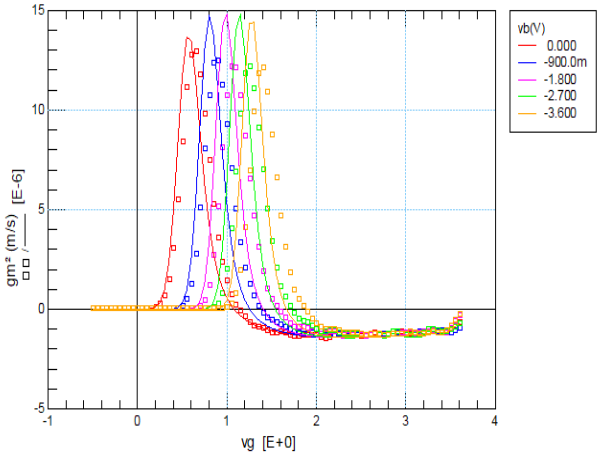
Id = f(Vg, Vb) @ Vd=low



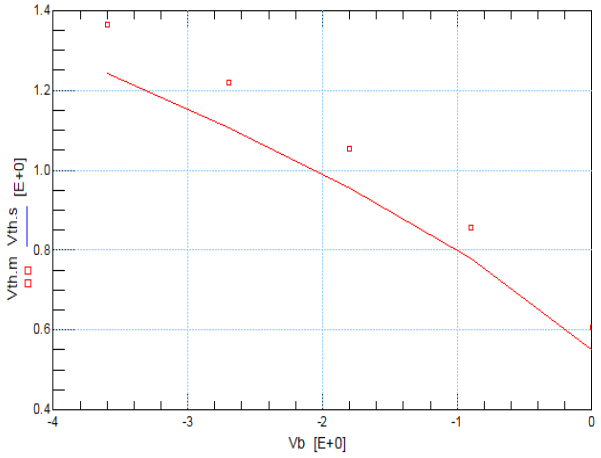
gm = f(Vg, Vb) @ Vd=low



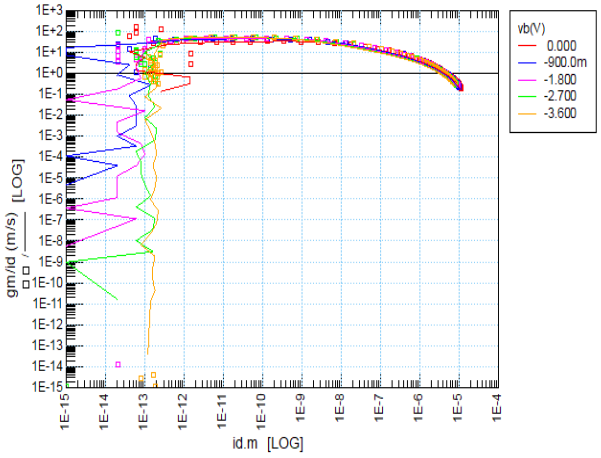
gm² = f(Vg, Vb) @ Vd=low

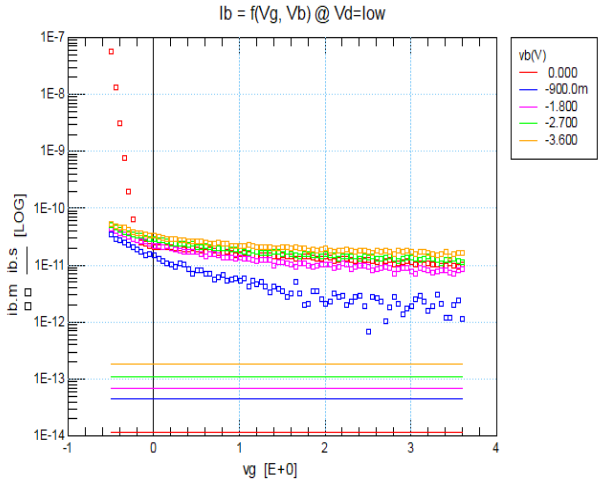


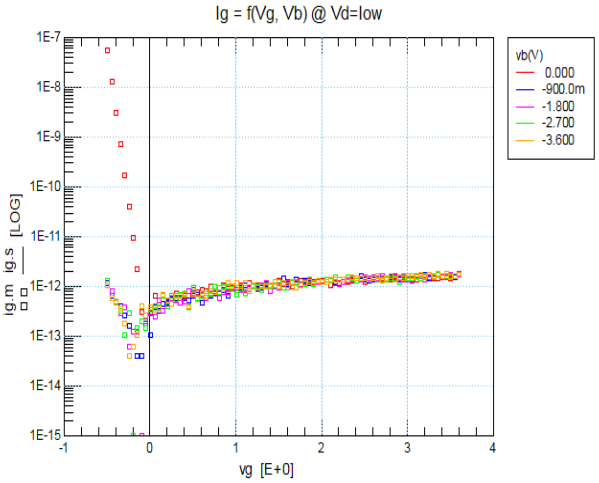
Vth @ Vd=low



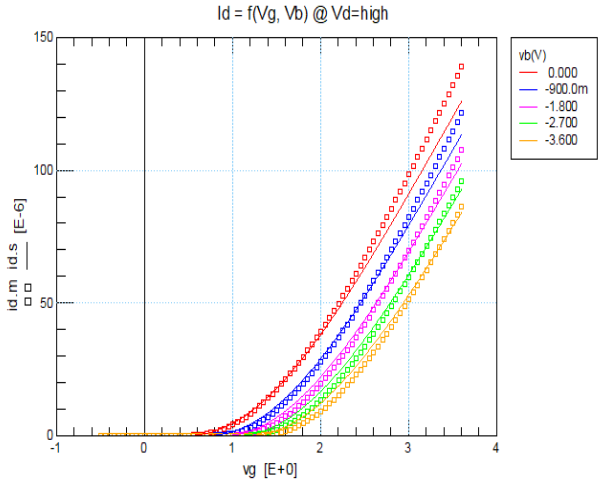
gm/id=f(id.m) @ Vd=low

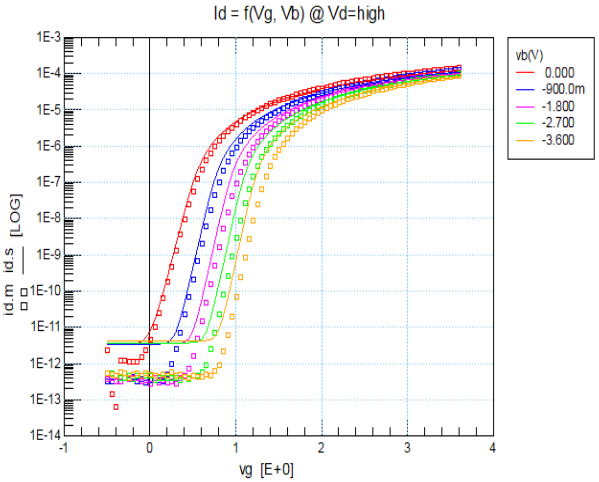


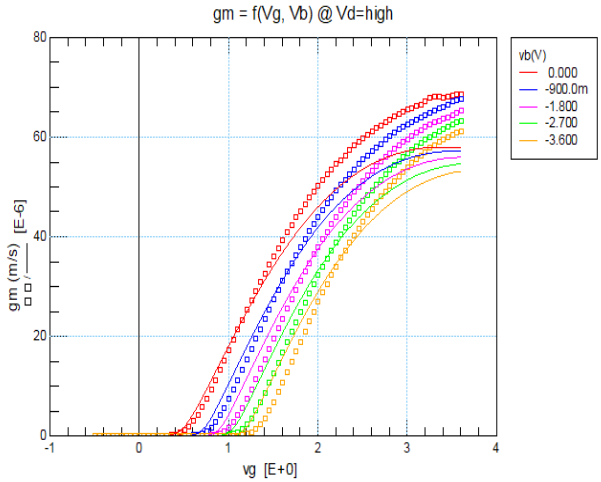


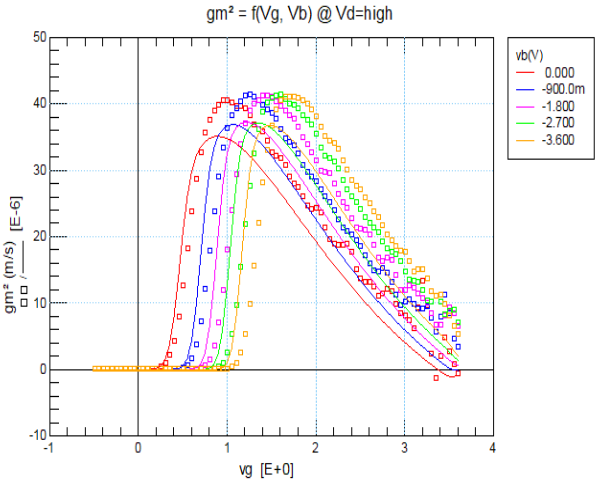


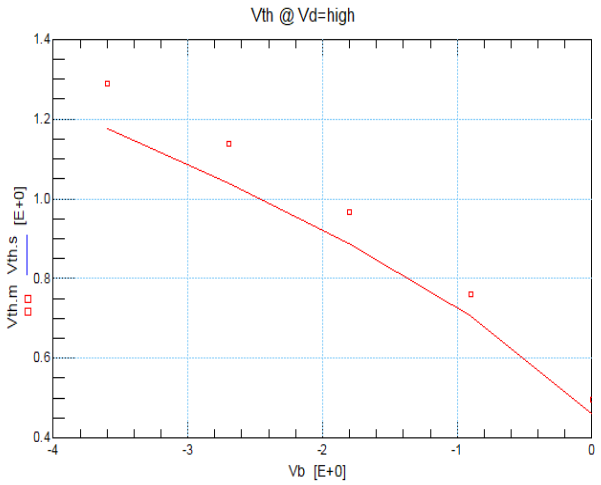
idvg, Vd = 3.3V, T = 125°C

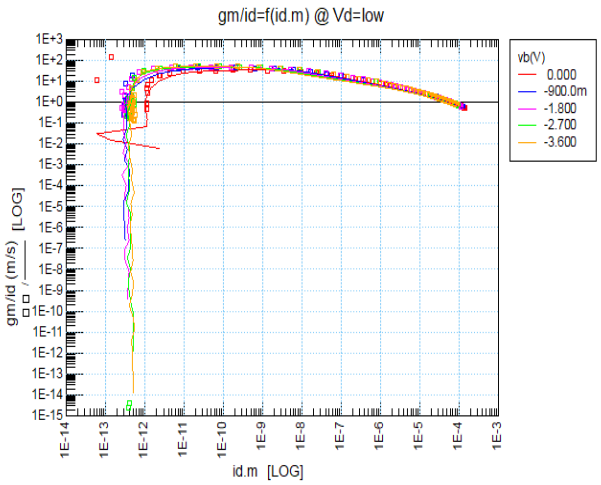


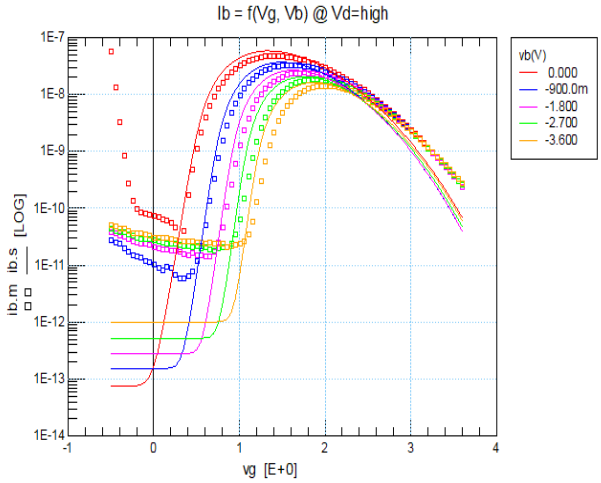


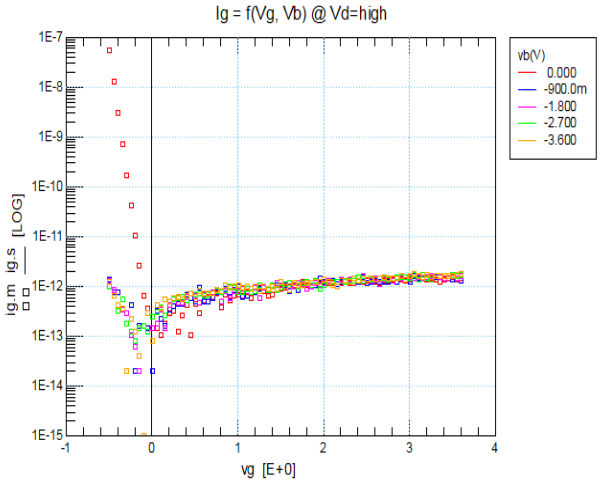




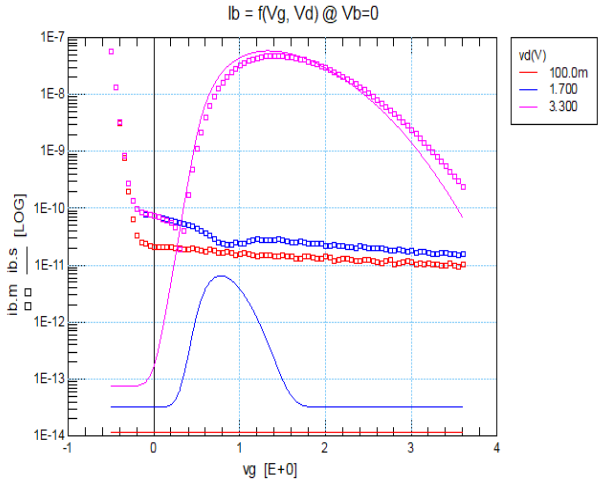


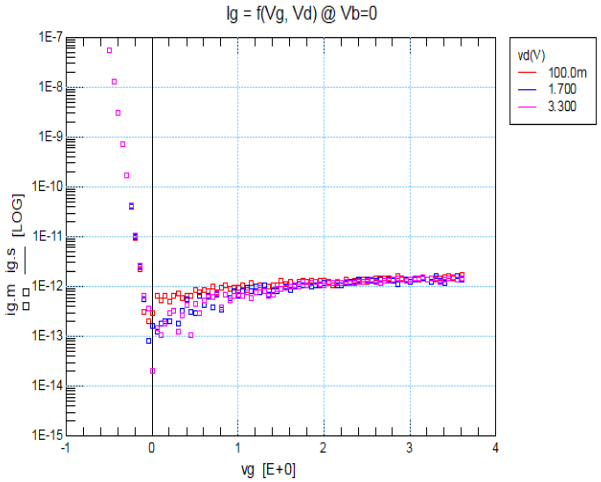




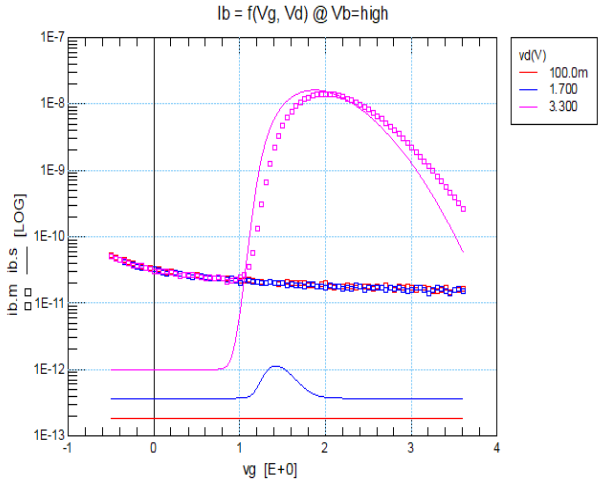


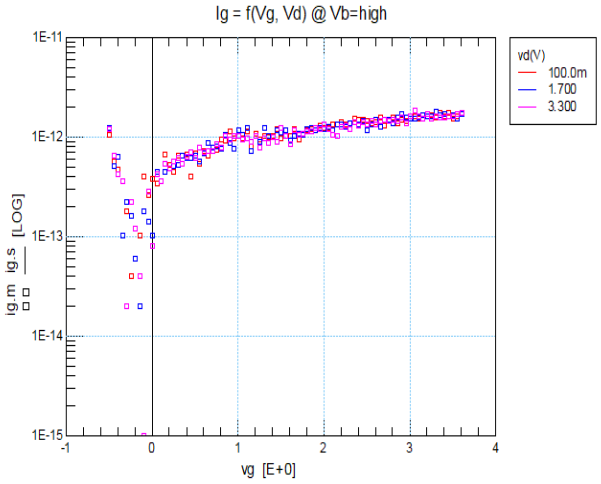
idvg, Ib,Ig, Vb = 0V, T = 125°C



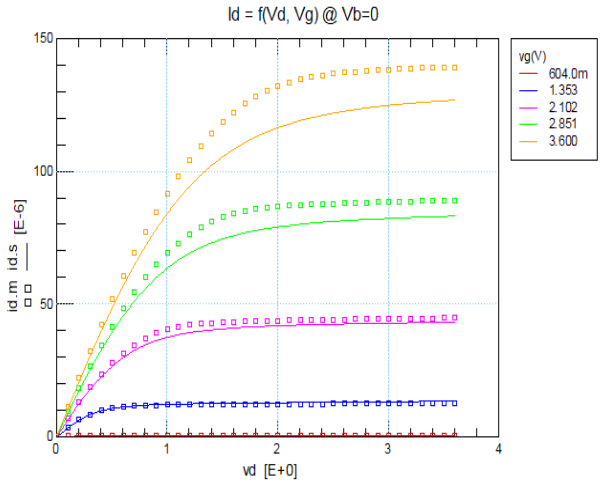


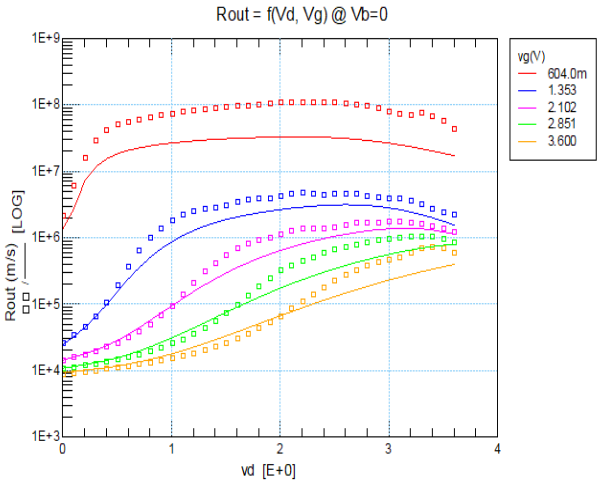
idvg, Ib,Ig, Vb = -3.6V, T = 125°C

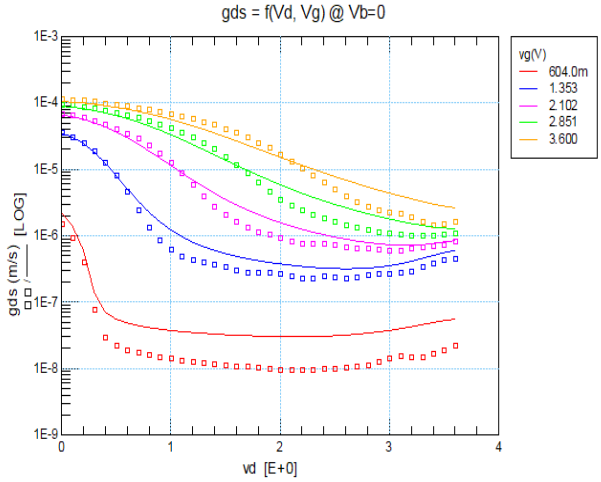




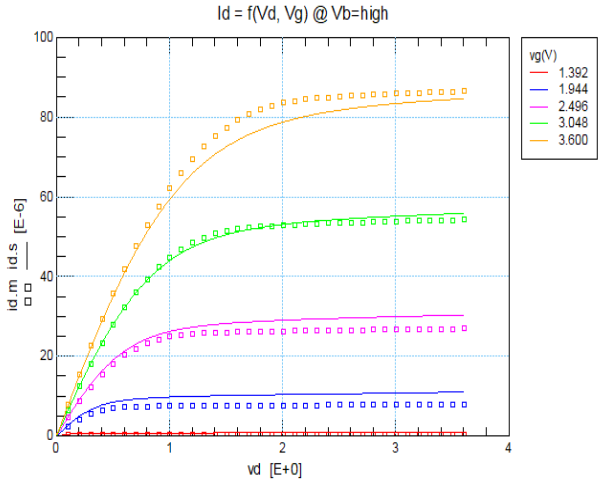
idvd, Vb = 0V, T = 125°C

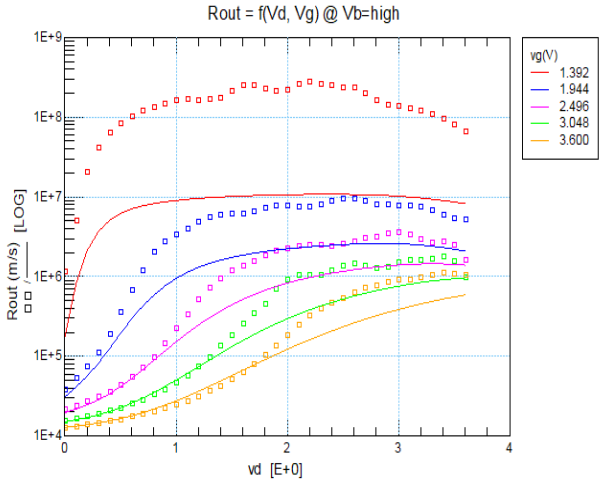


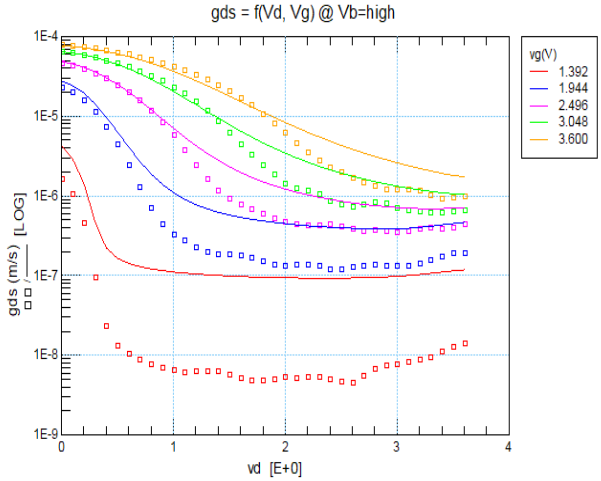




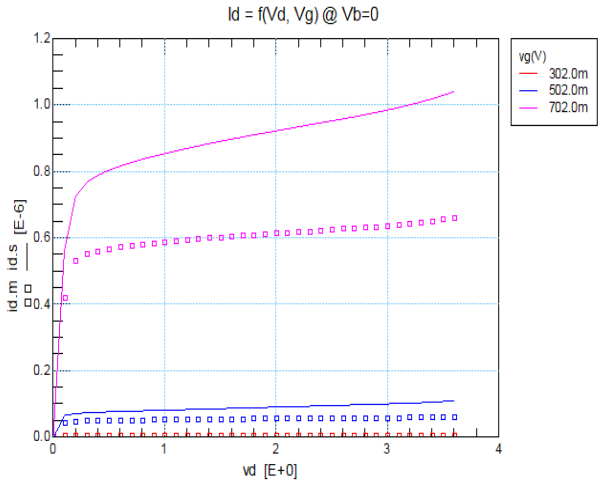
idvd_vbmin, Vb = -3.6V, T = 125°C

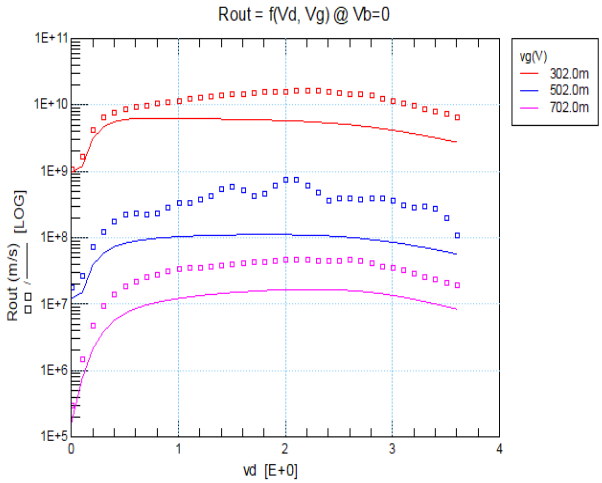


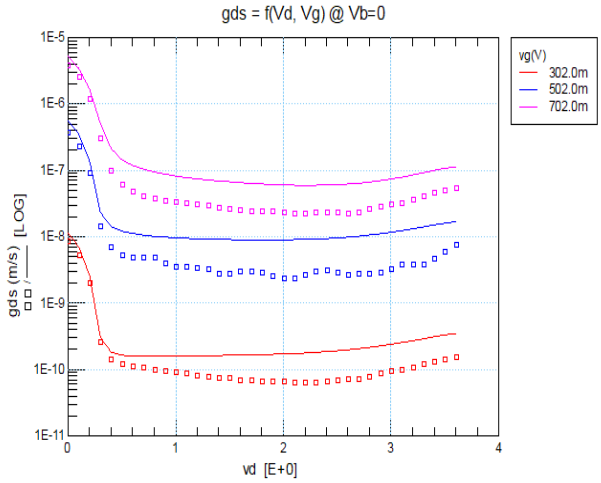




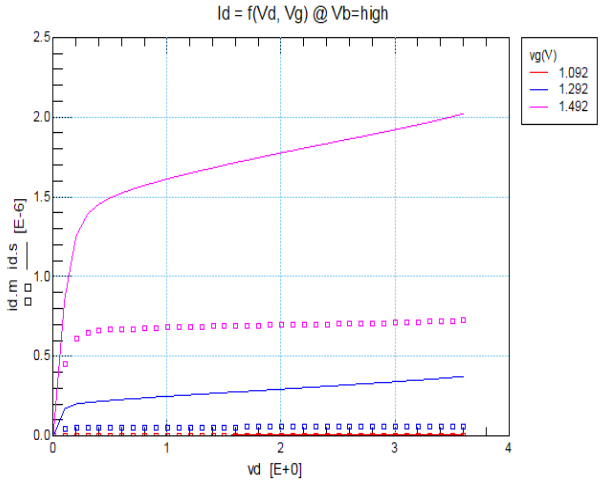
idvd_vth, Vb = 0V, T = 125°C

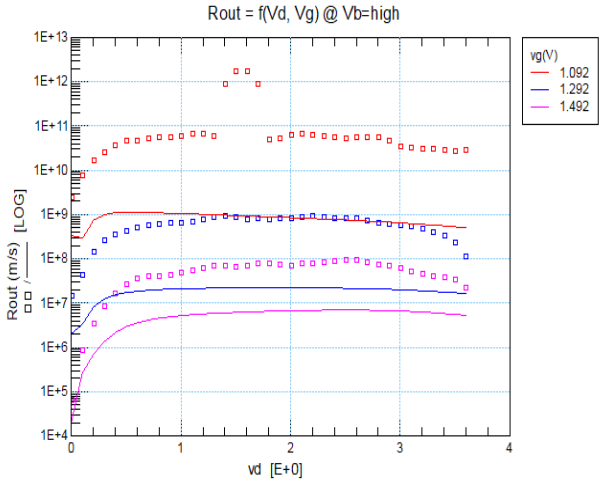


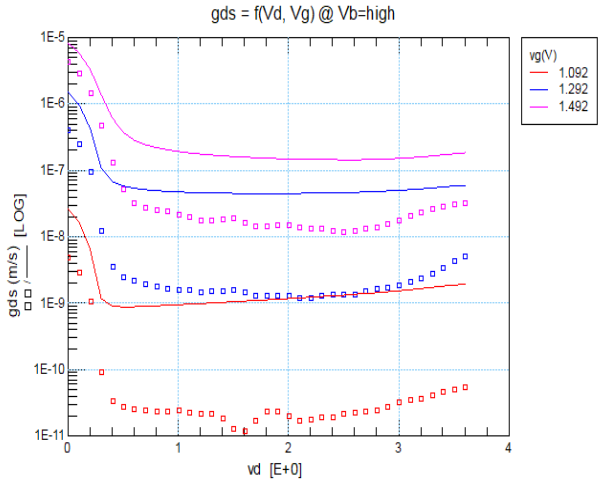




idvd_vth_vbmin, Vb = -3.6V, T = 125°C

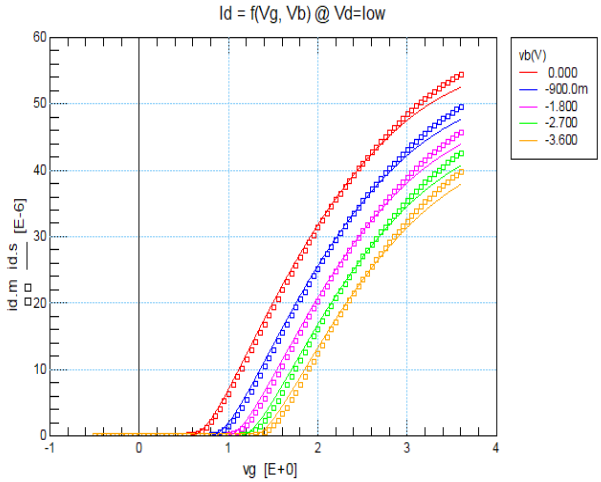


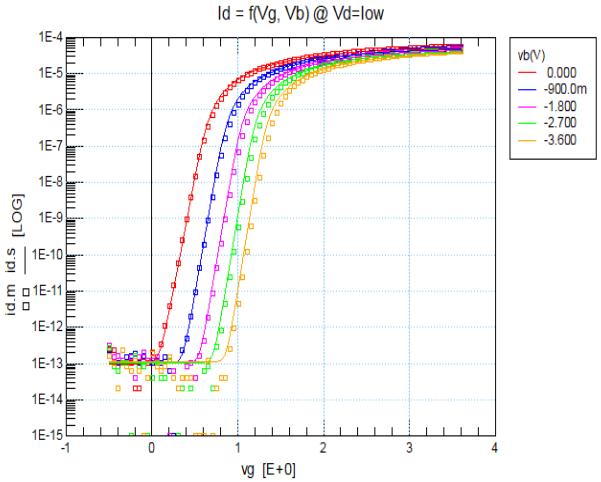


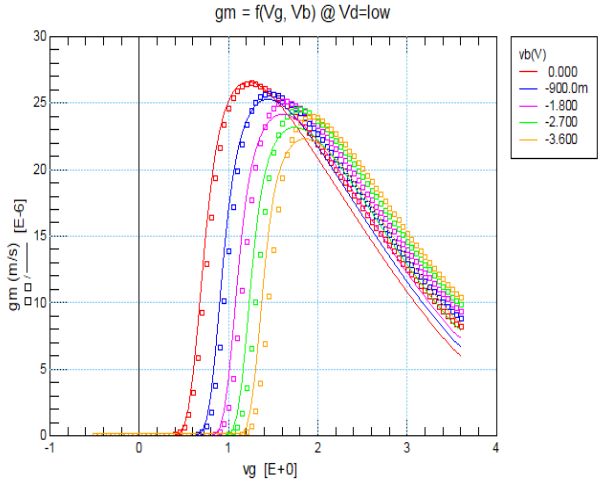


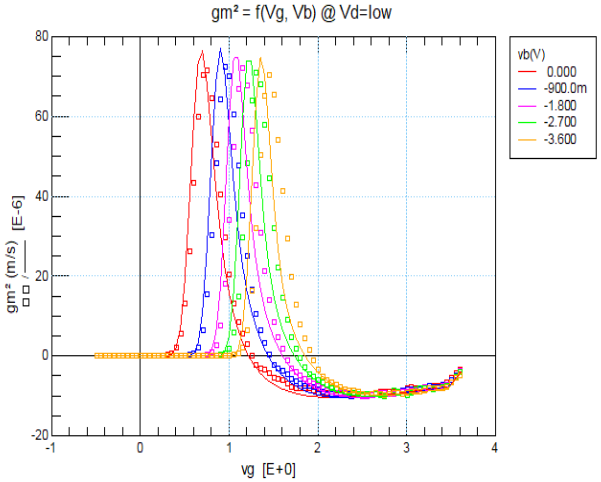
4.17W02u0_L1u2_S558_3

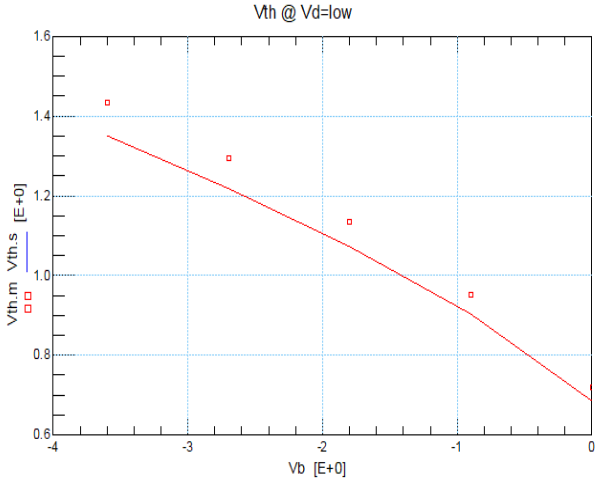
idvg, Vd = 0.1V, T = 27°C

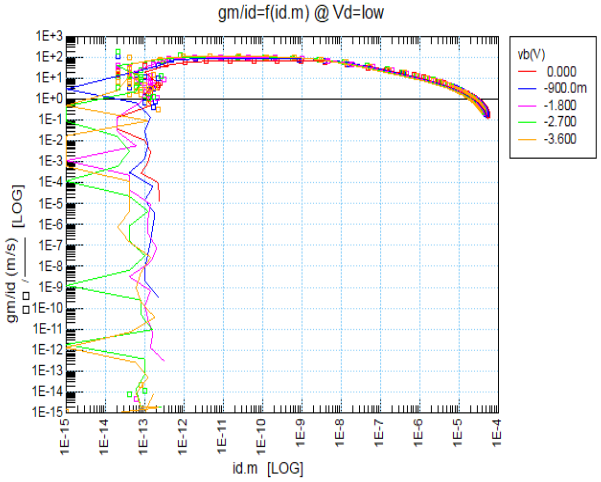


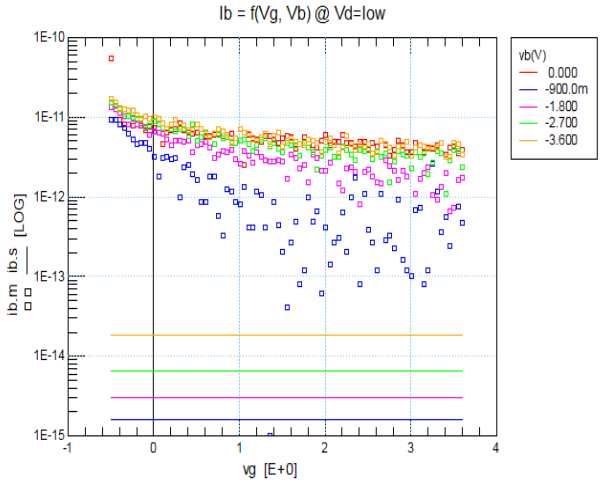


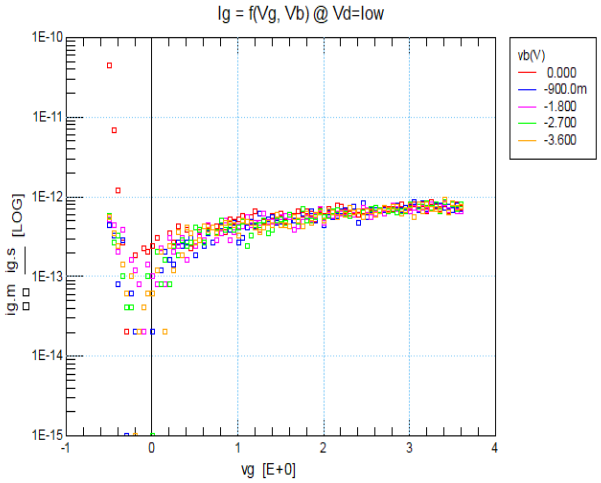




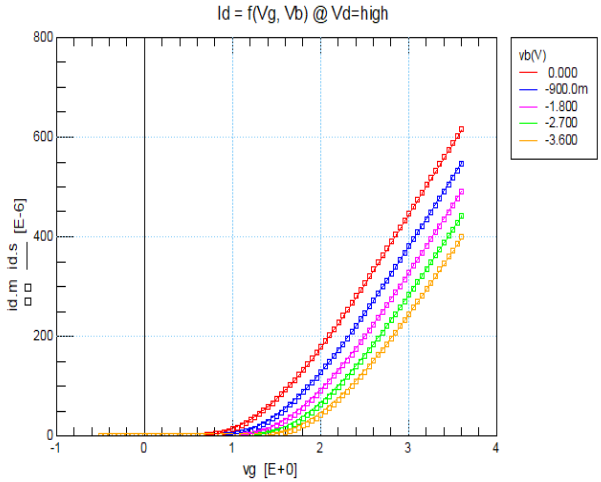


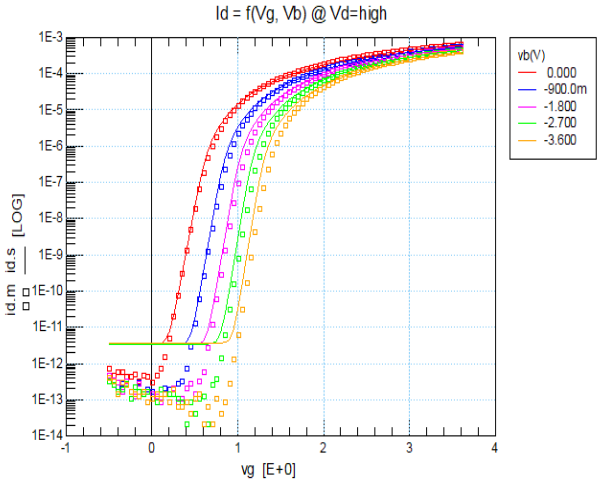


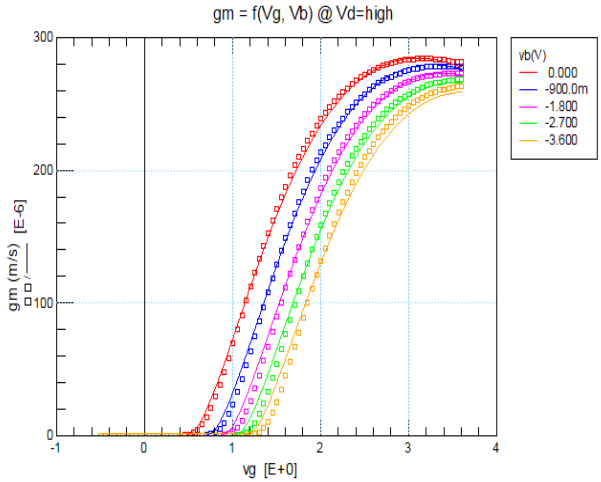


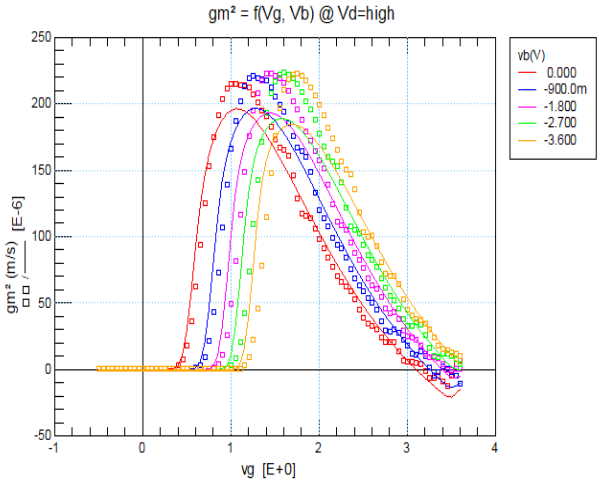


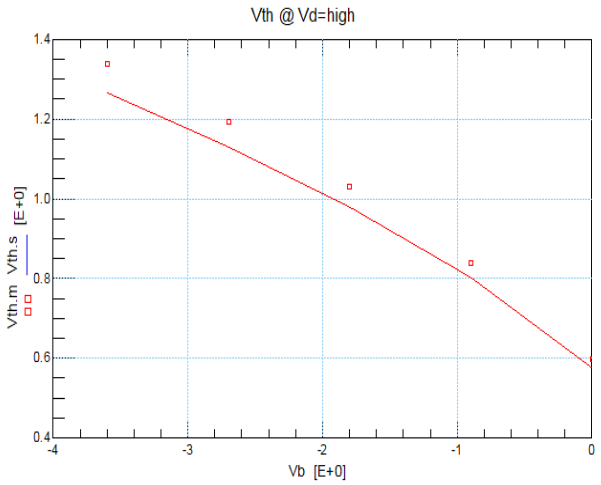
idvg, Vd = 3.3V, T = 27°C

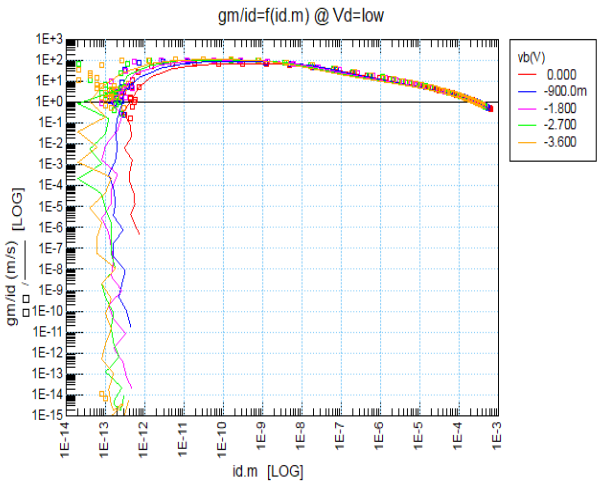


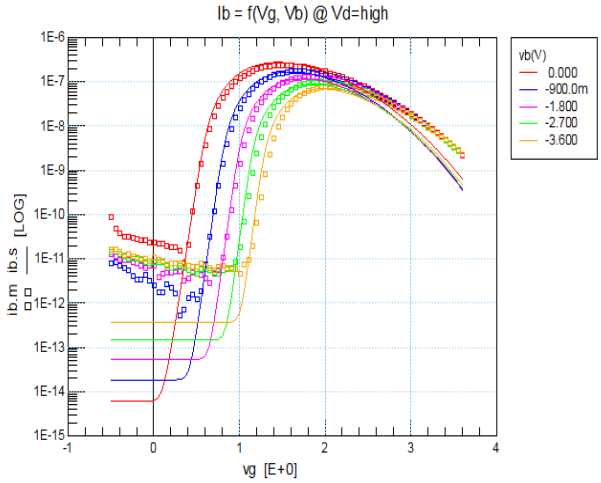


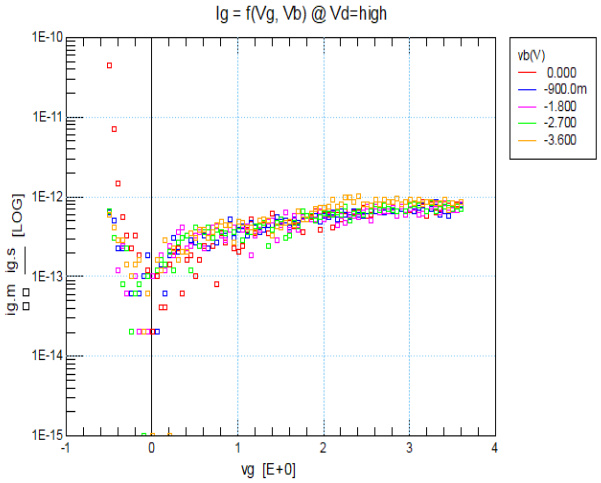




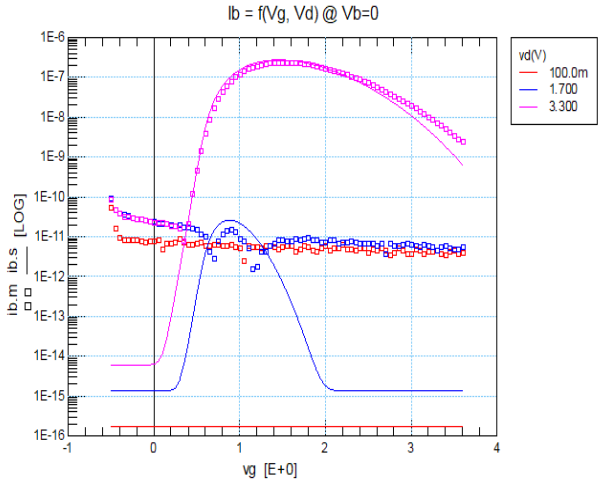


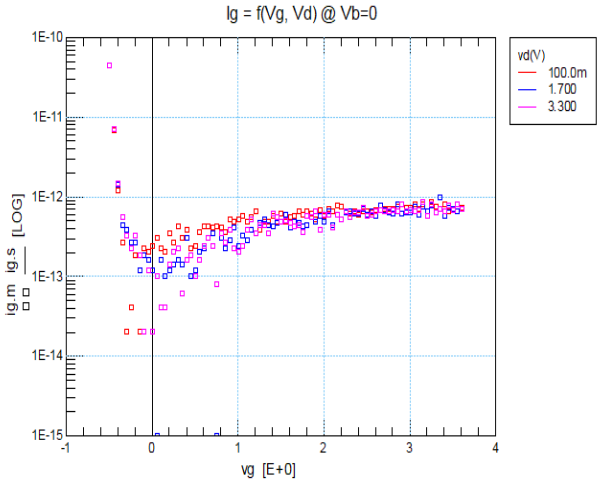




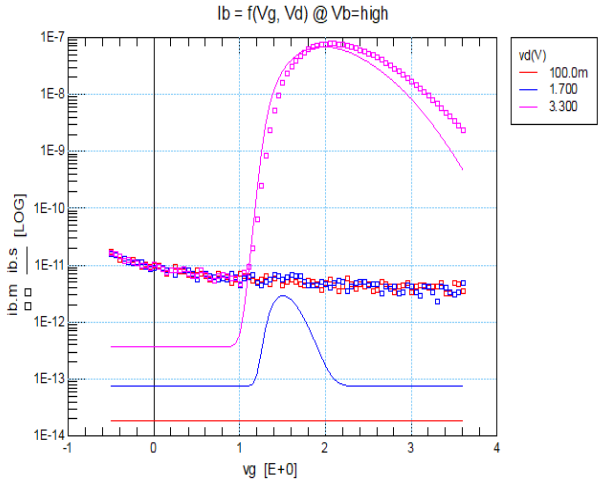


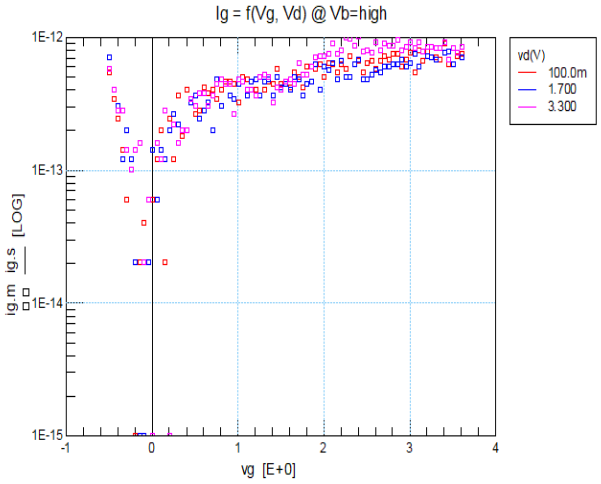
idvg, Ib,Ig, Vb = 0V, T = 27°C



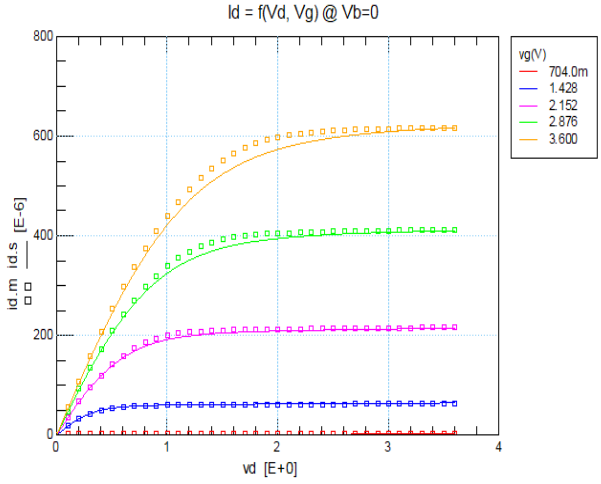


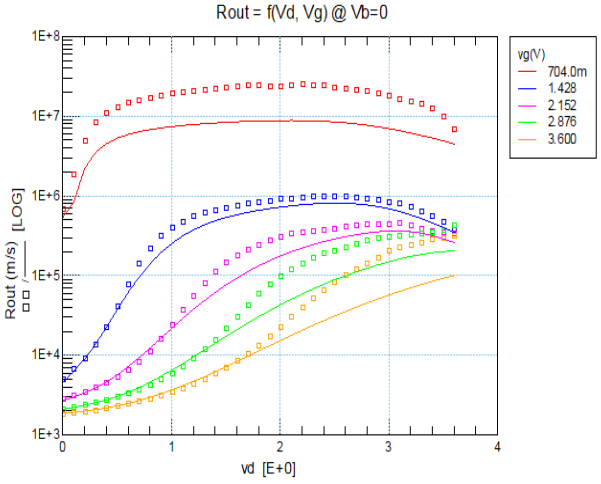
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

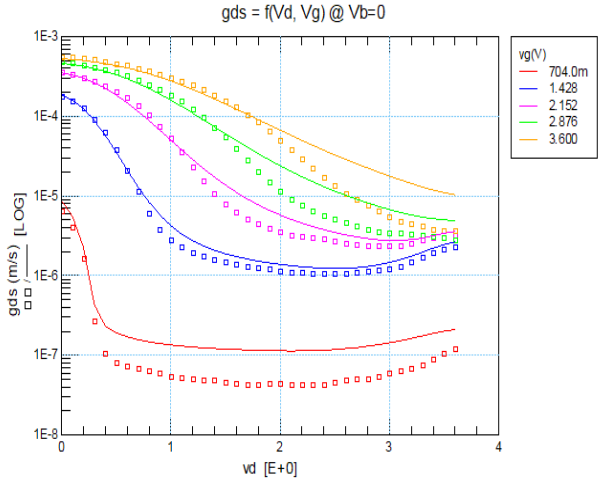




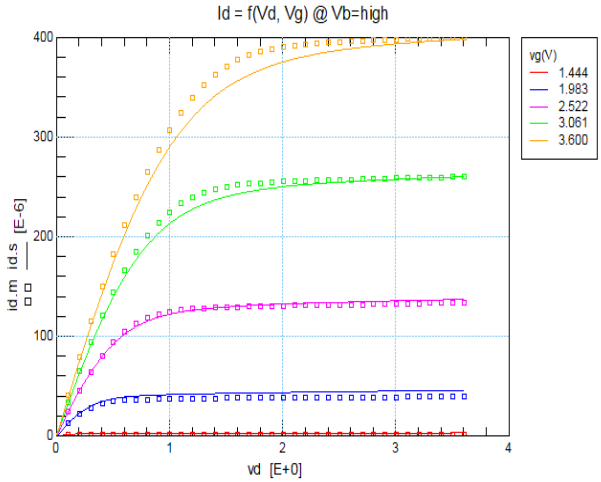
idvd, Vb = 0V, T = 27°C

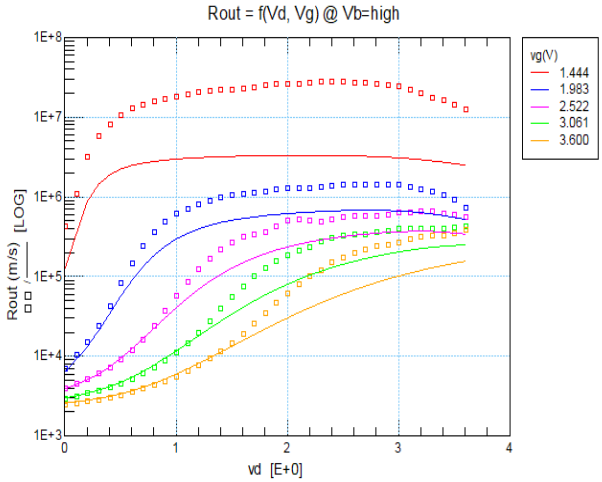


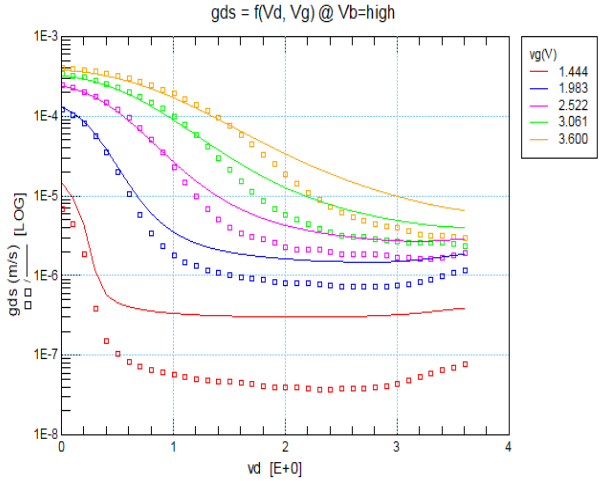




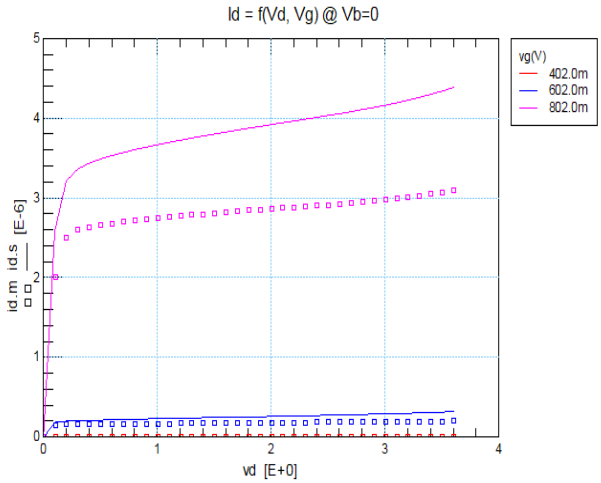
idvd_vbmin, Vb = -3.6V, T = 27°C

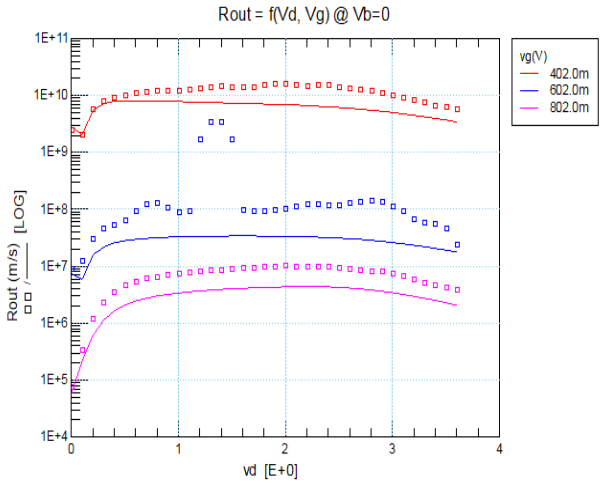


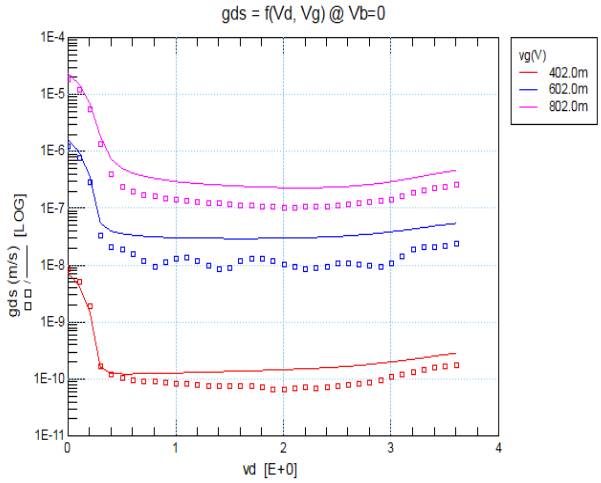




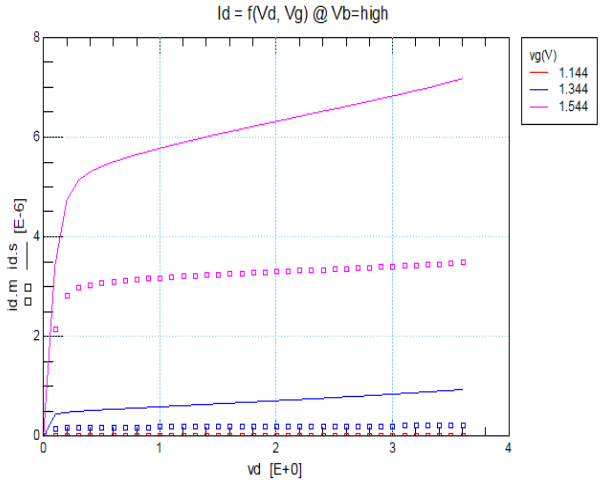
idvd_vth, Vb = 0V, T = 27°C

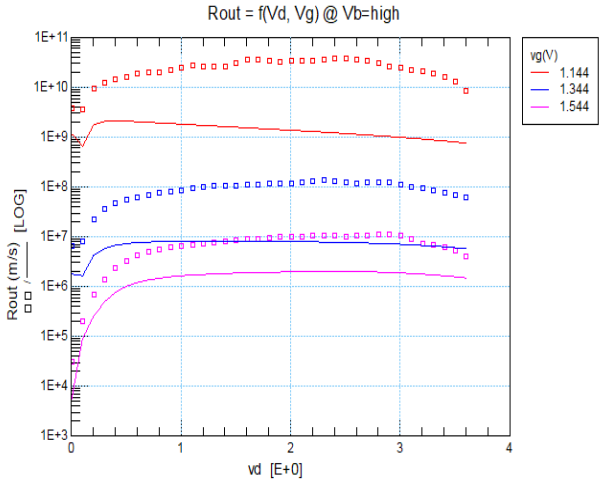


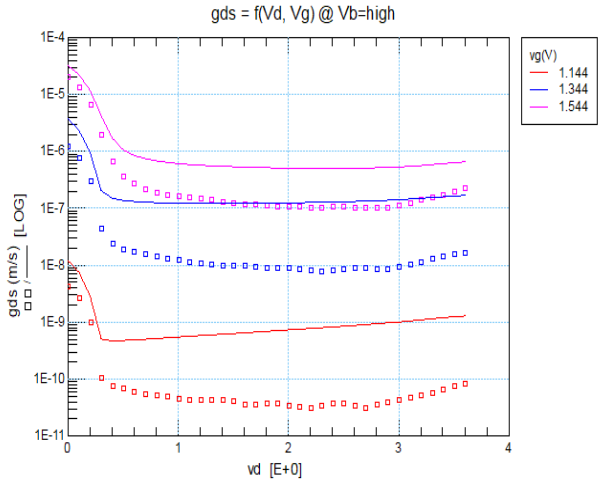




idvd_vth_vbmin, Vb = -3.6V, T = 27°C



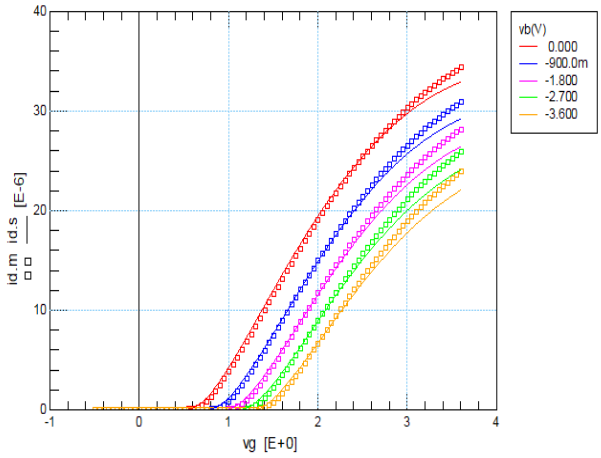




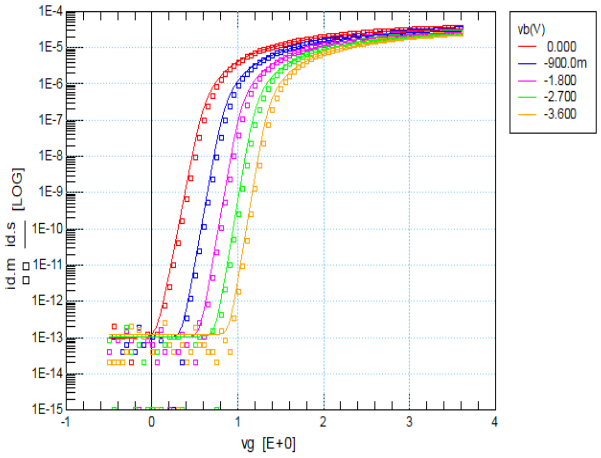
4.18 W05u0_L5u0_S557_5

idvg, Vd = 0.1V, T = 27°C

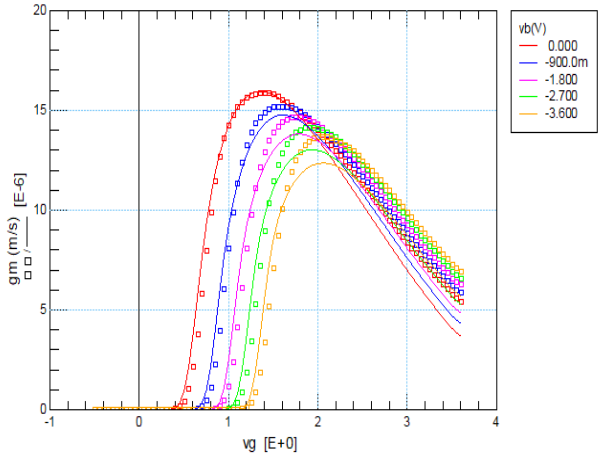
Id = f(Vg, Vb) @ Vd=low



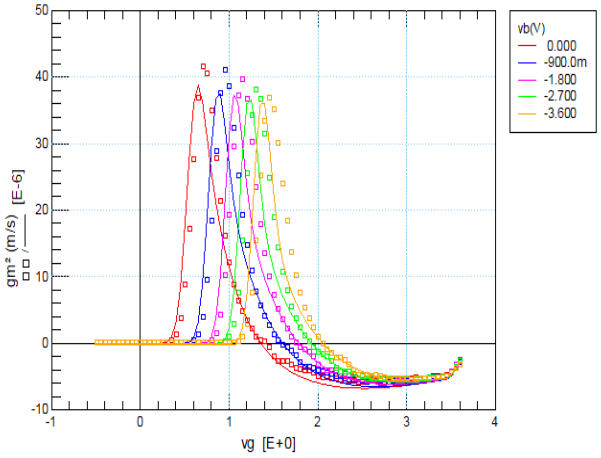
Id = f(Vg, Vb) @ Vd=low



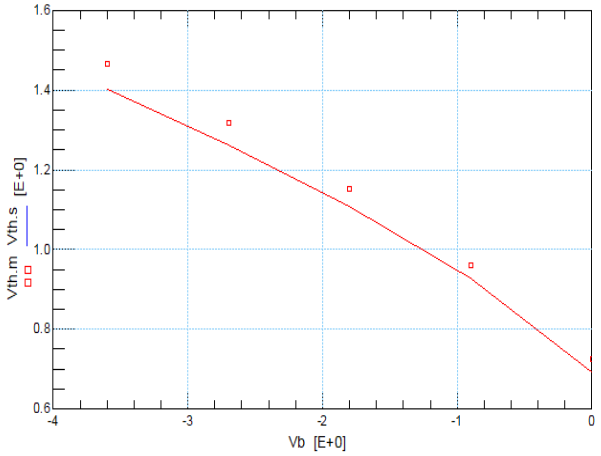
gm = f(Vg, Vb) @ Vd=low



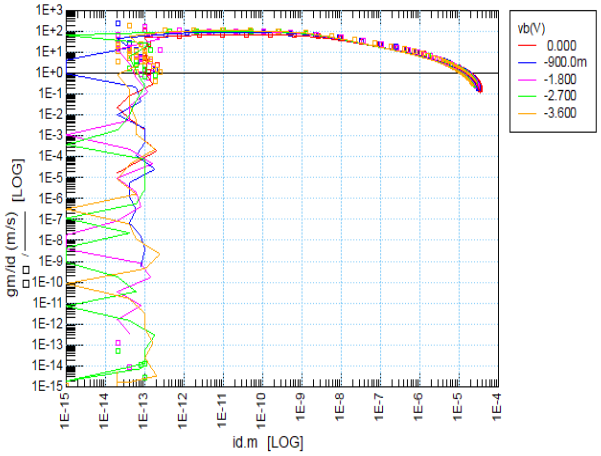
gm² = f(Vg, Vb) @ Vd=low

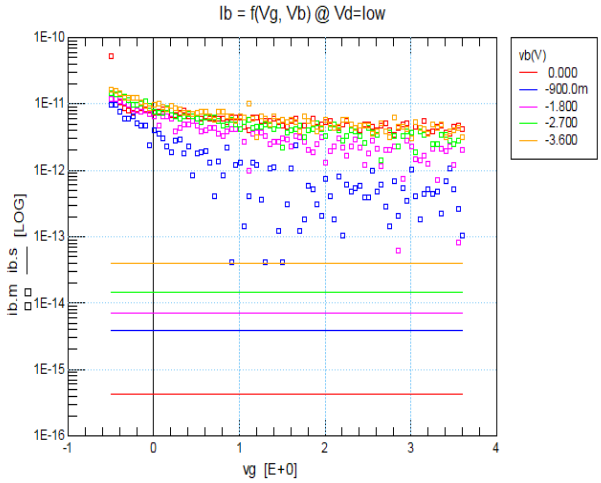


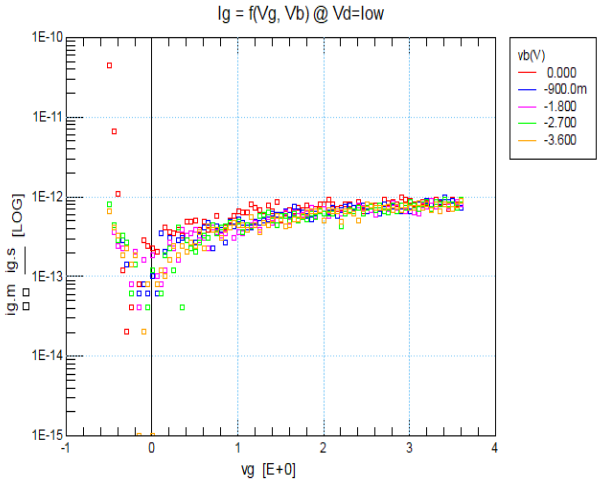
Vth @ Vd=low



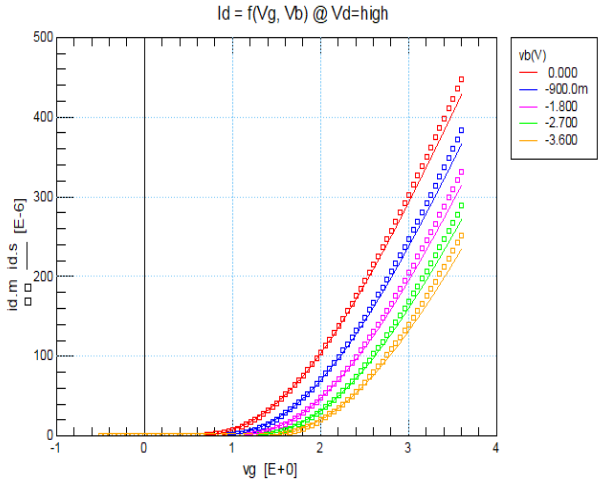
gm/id=f(id.m) @ Vd=low

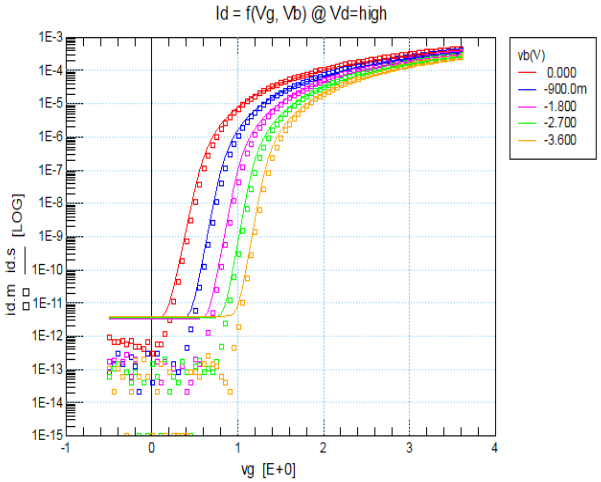


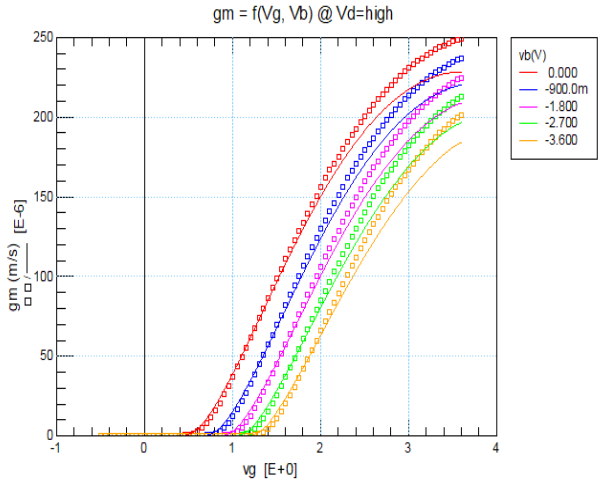


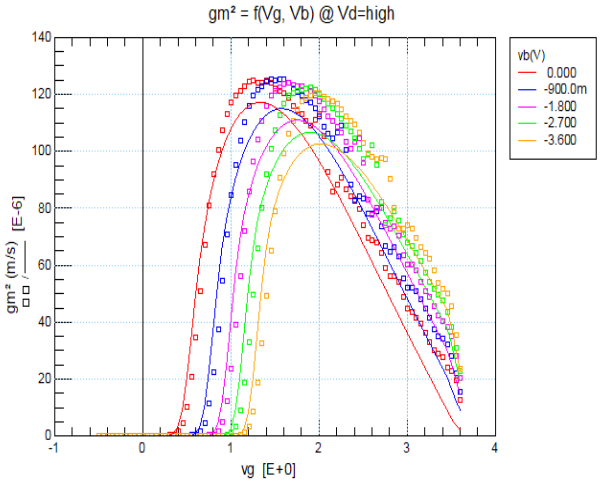


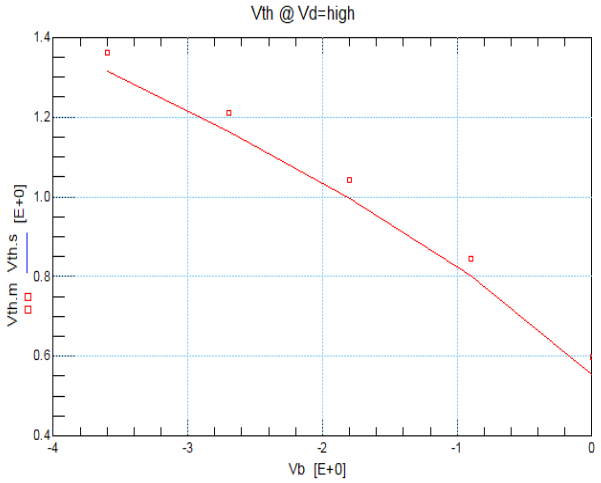
idvg, Vd = 3.3V, T = 27°C

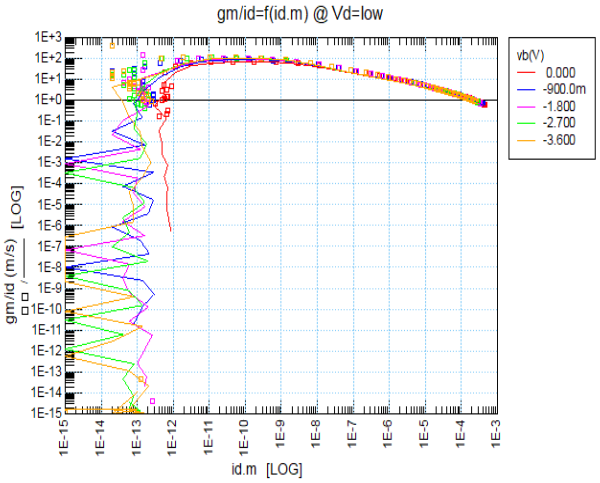


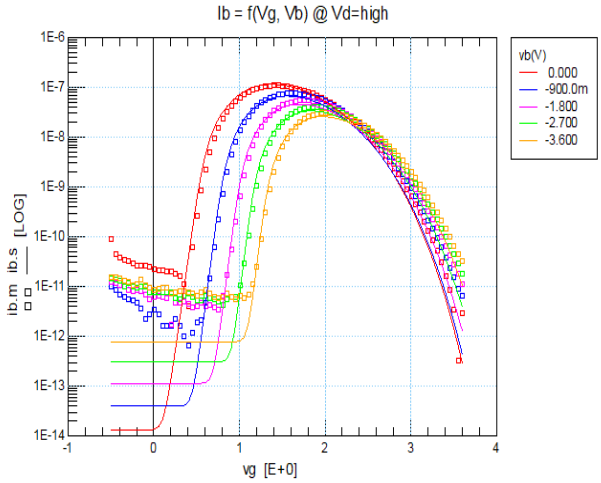


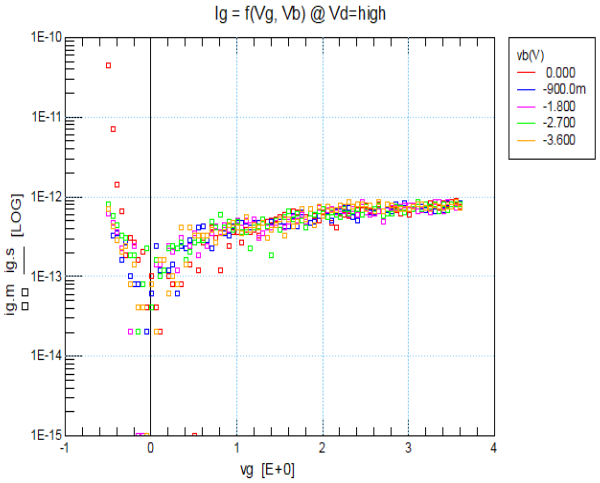




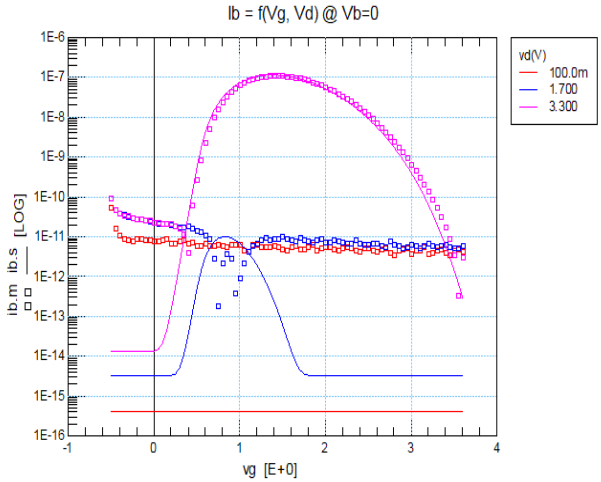


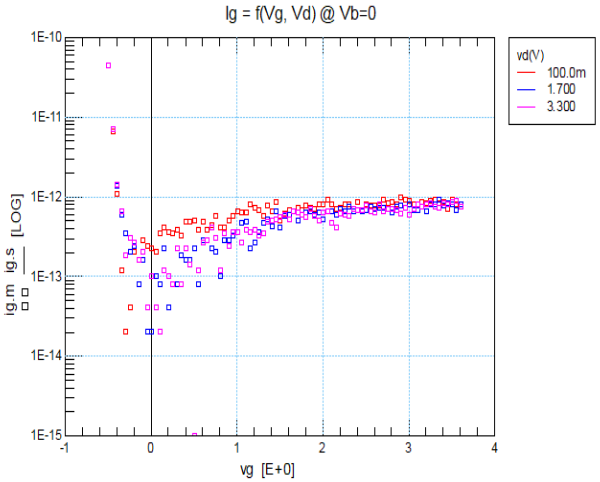




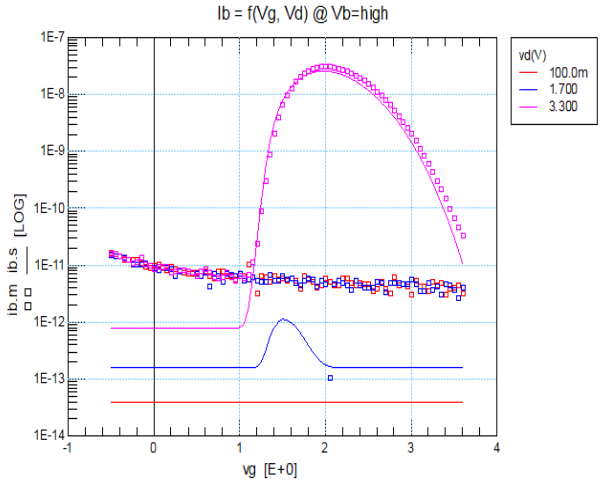


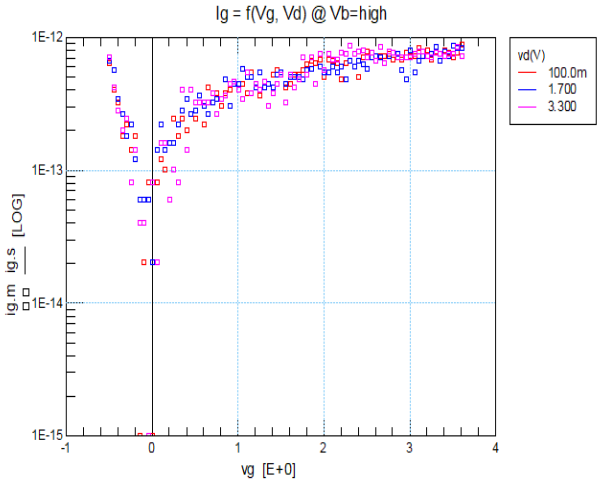
idvg, Ib,Ig, Vb = 0V, T = 27°C



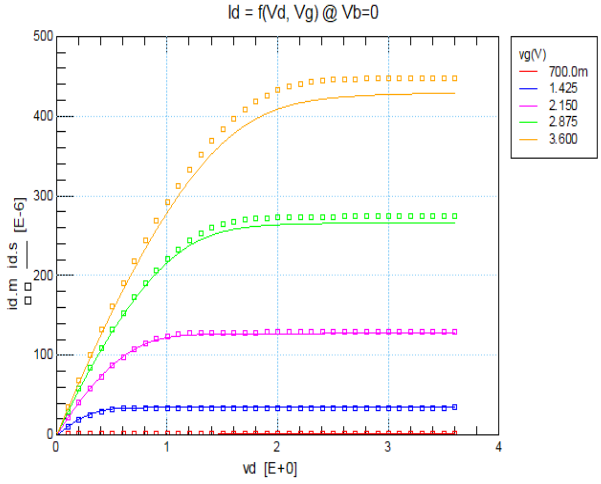


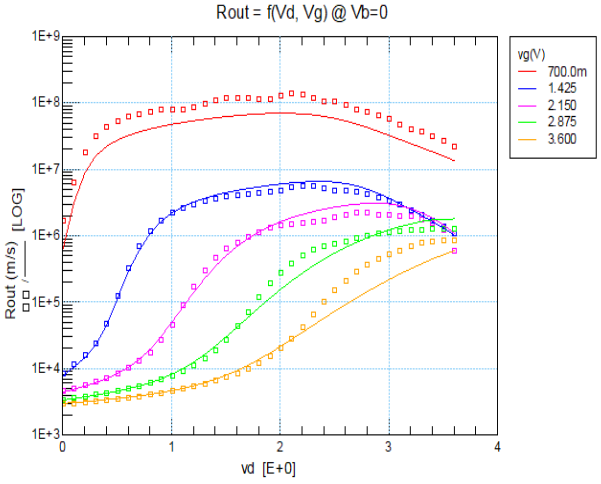
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

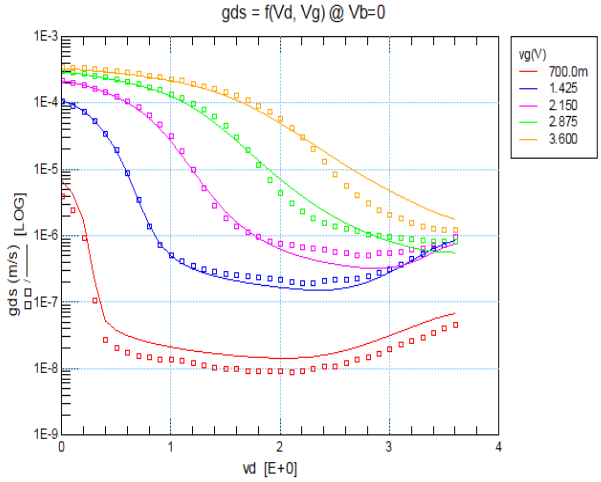




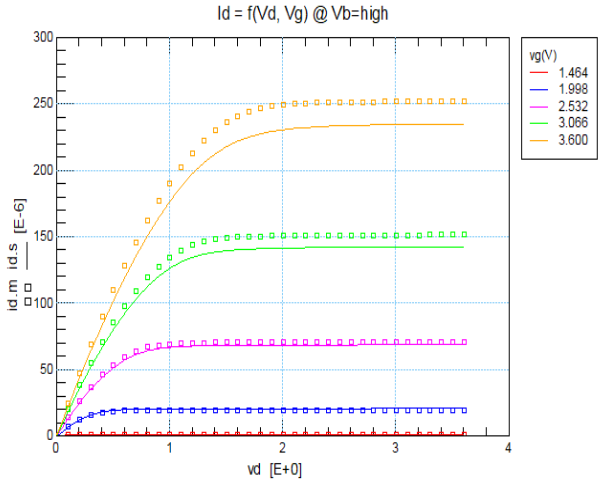
idvd, Vb = 0V, T = 27°C

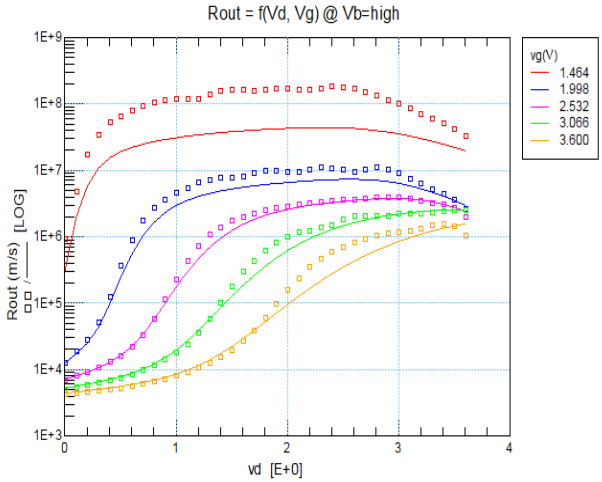


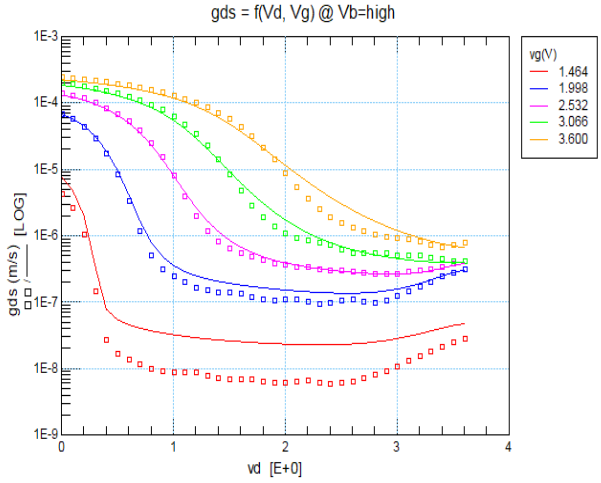




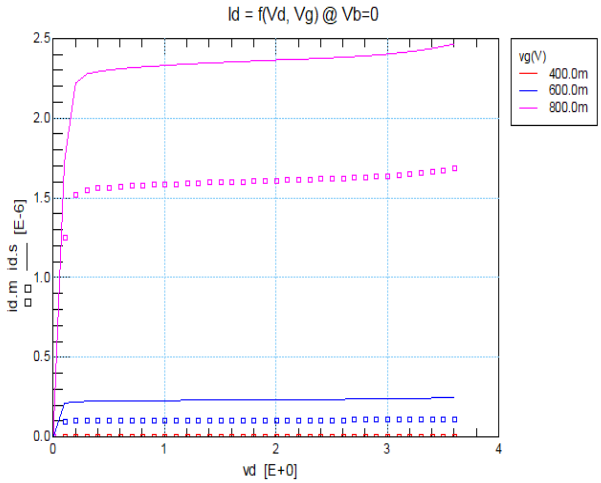
idvd_vbmin, Vb = -3.6V, T = 27°C

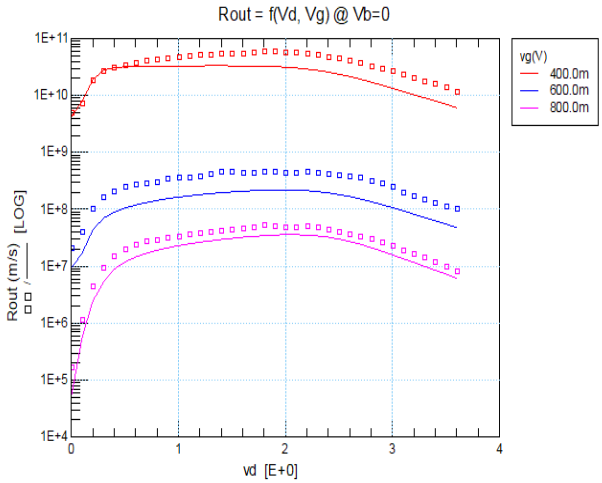


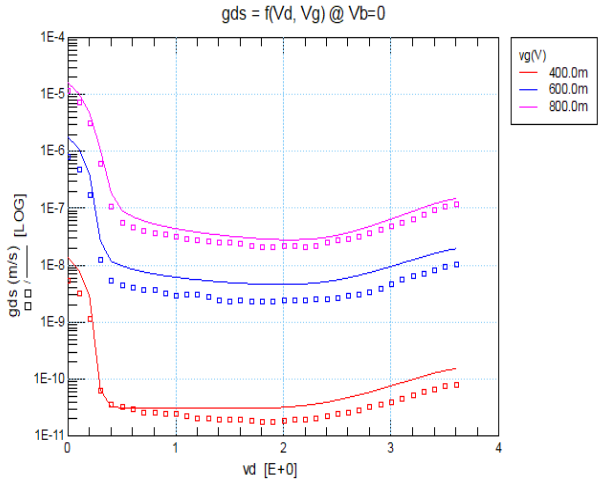




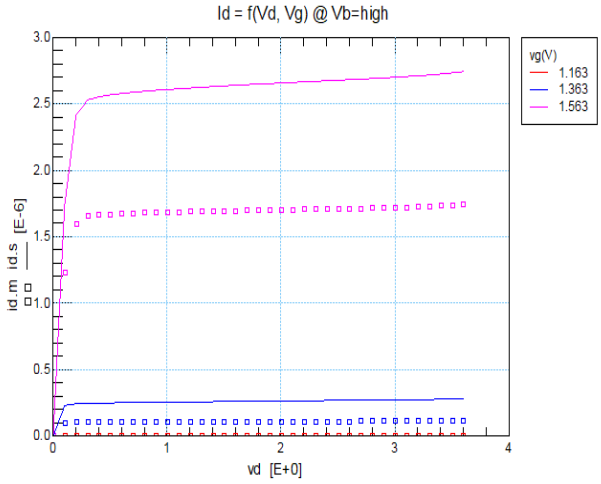
idvd_vth, Vb = 0V, T = 27°C

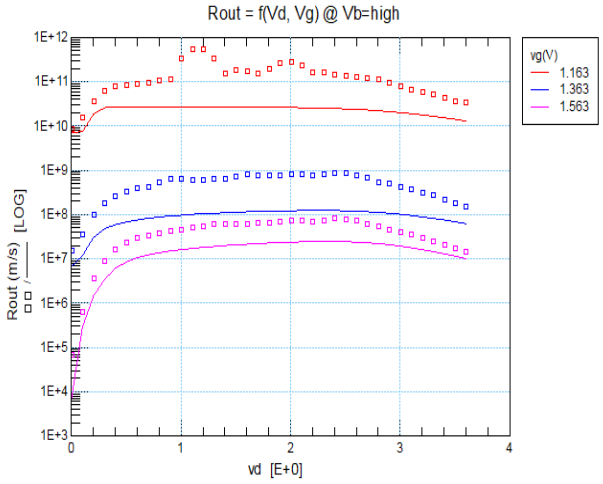


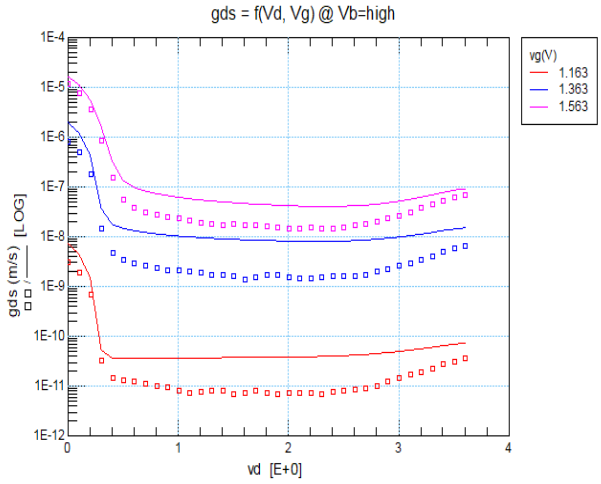




idvd_vth_vbmin, Vb = -3.6V, T = 27°C

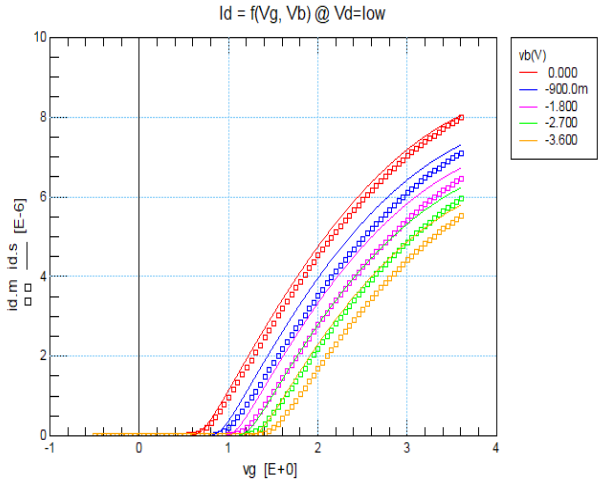


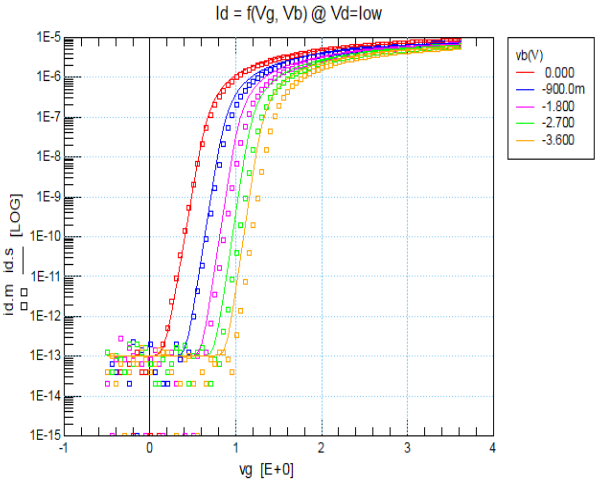


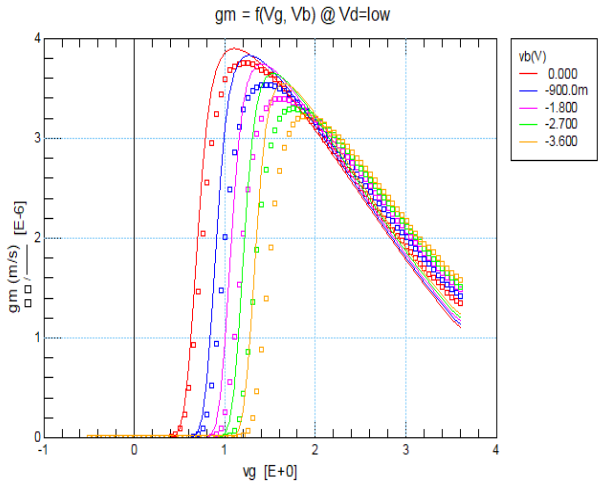


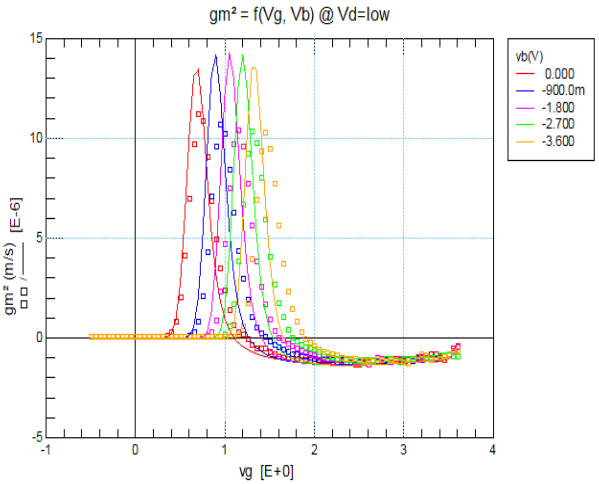
4.19W0u3_L1u2_S560_5

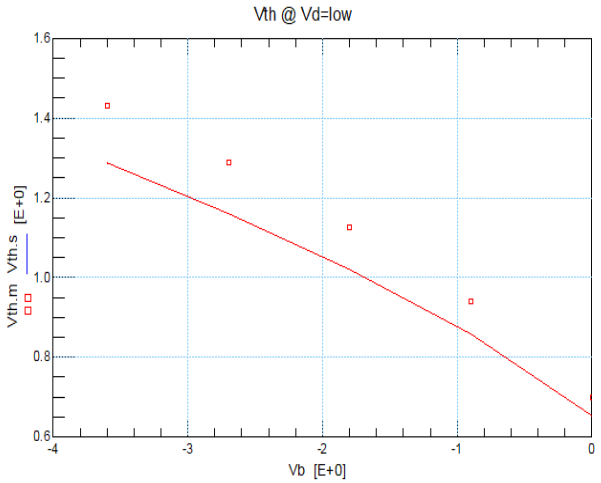
idvg, Vd = 0.1V, T = 27°C

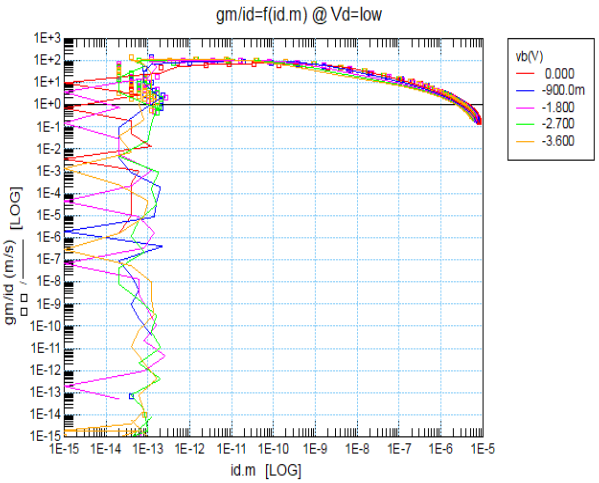


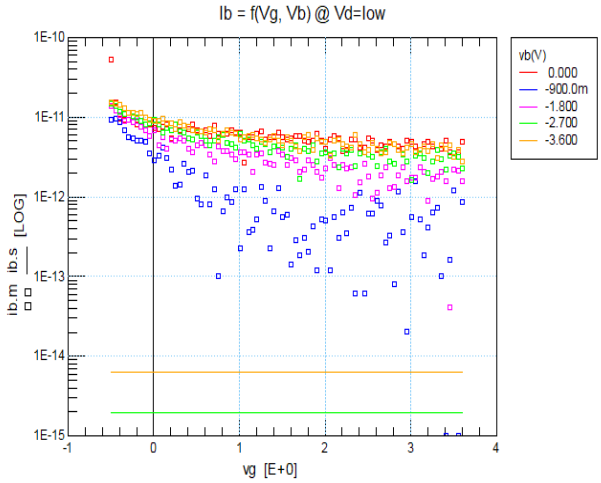


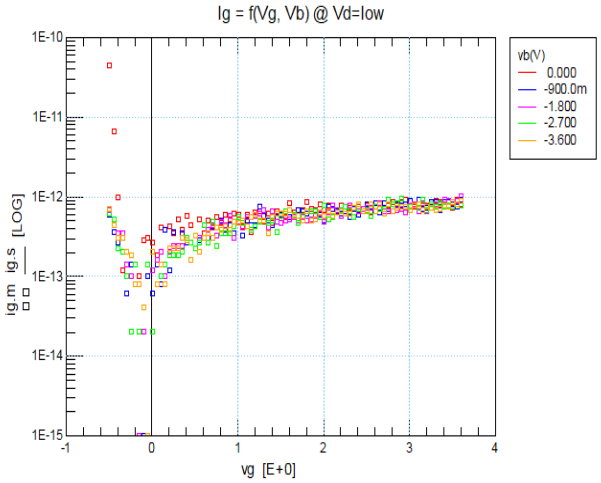




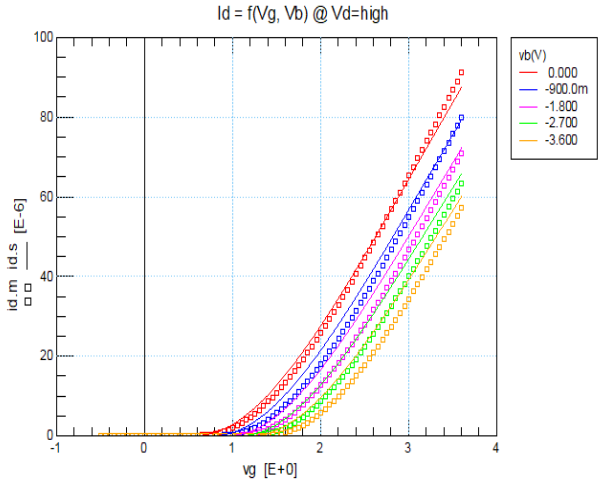


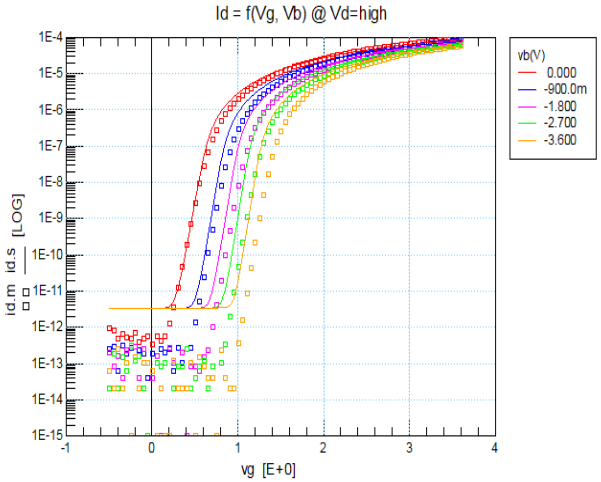


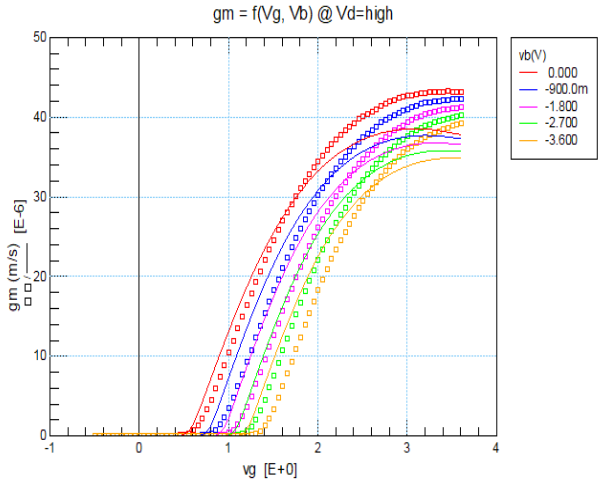


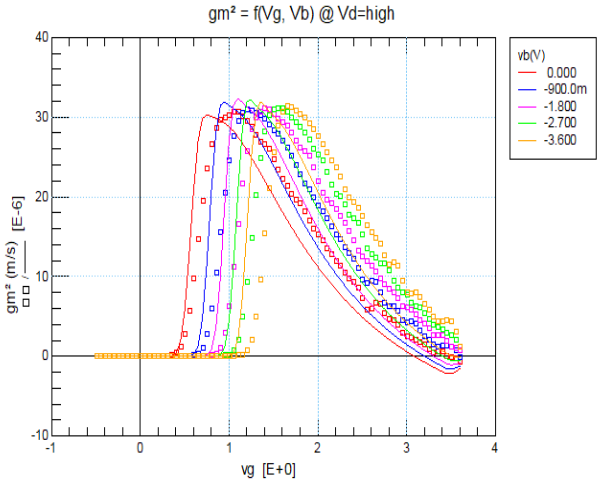


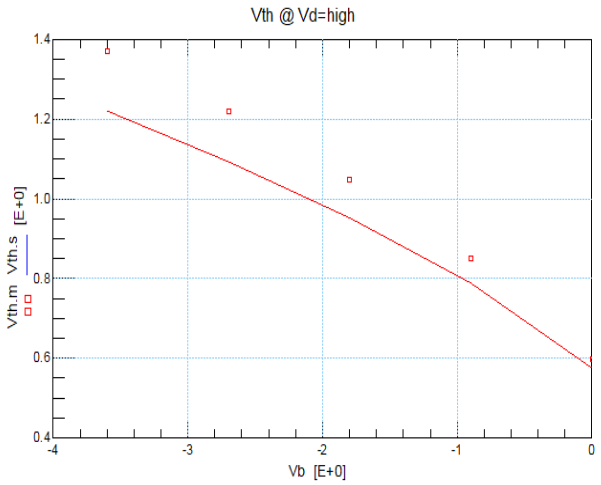
idvg, Vd = 3.3V, T = 27°C

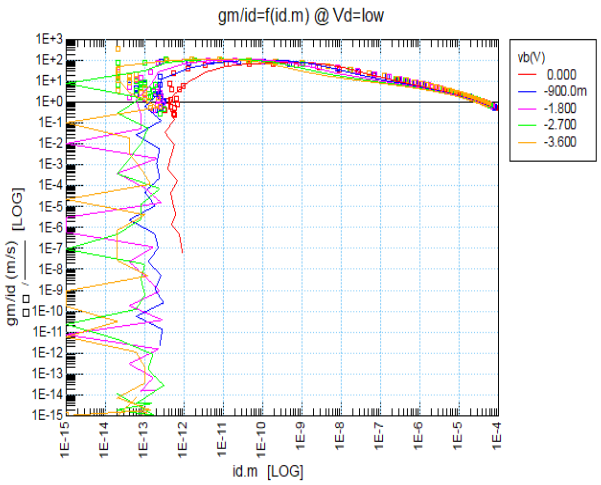


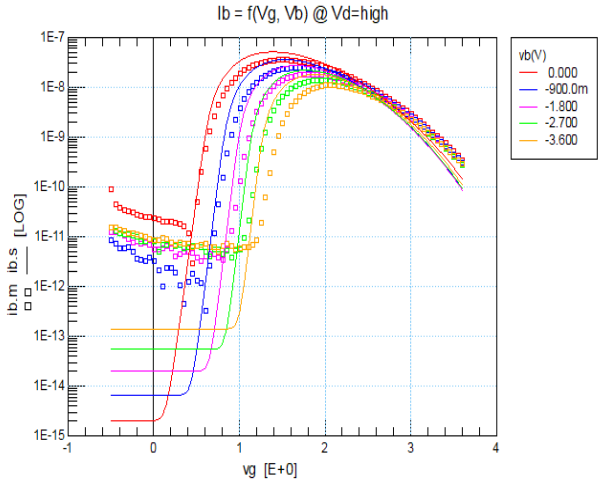


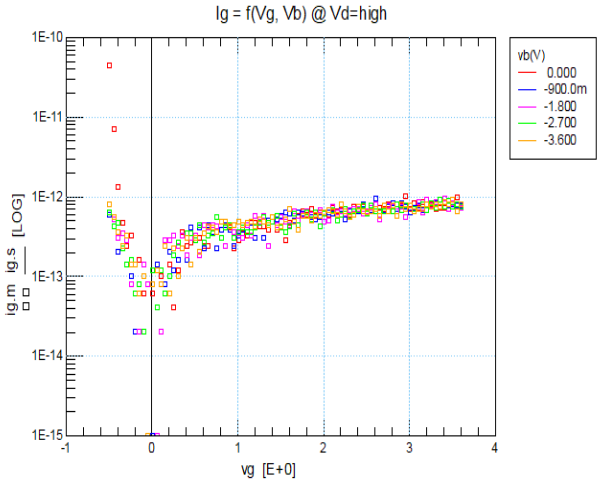




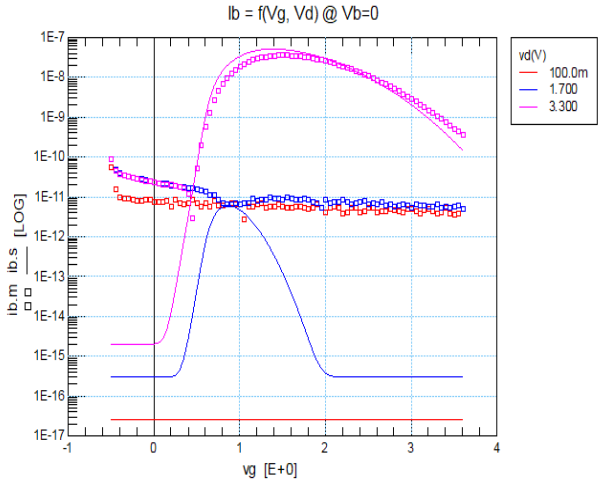


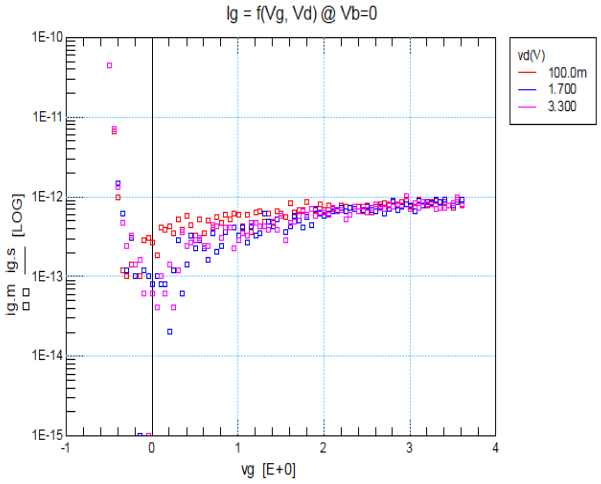




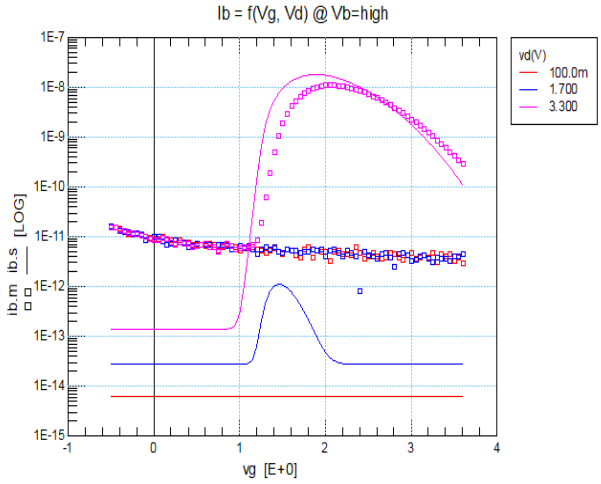


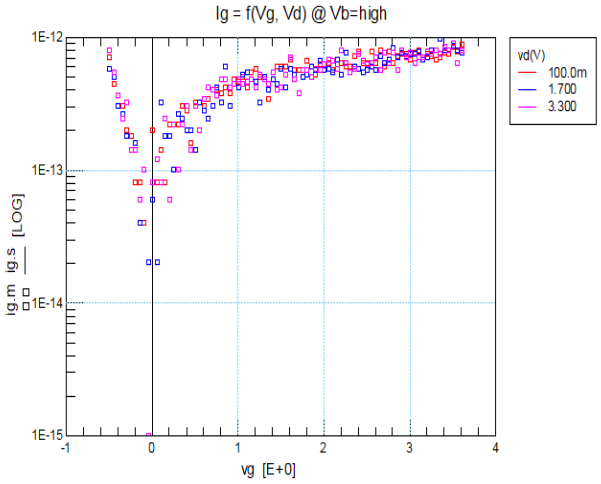
idvg, Ib,Ig, Vb = 0V, T = 27°C



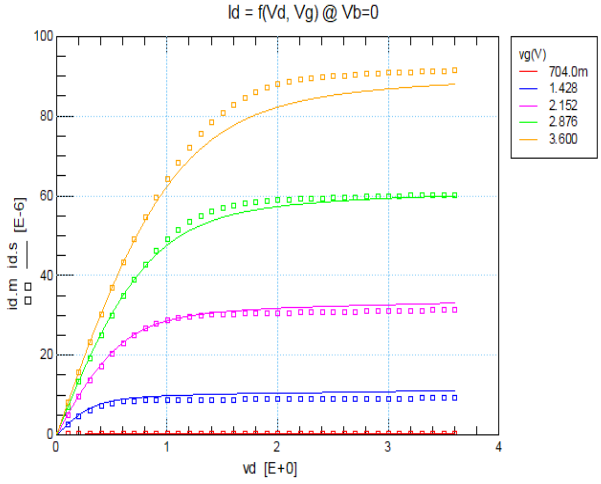


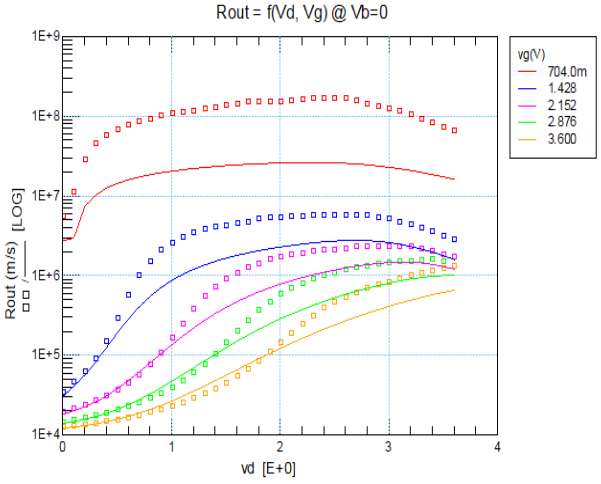
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

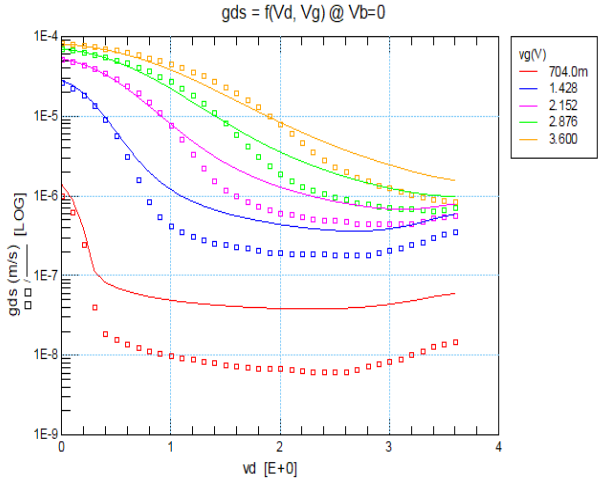




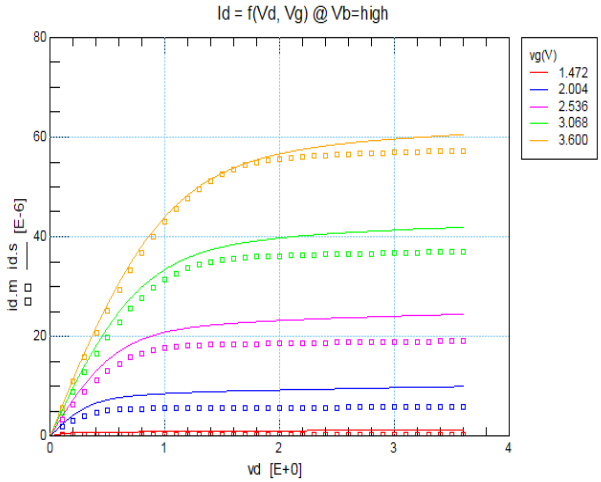
idvd, Vb = 0V, T = 27°C

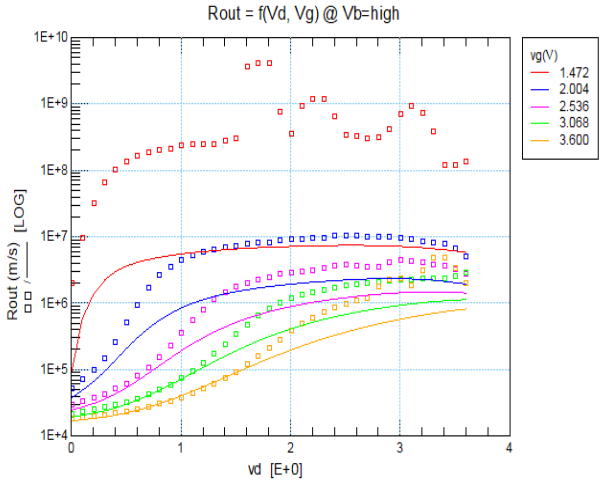


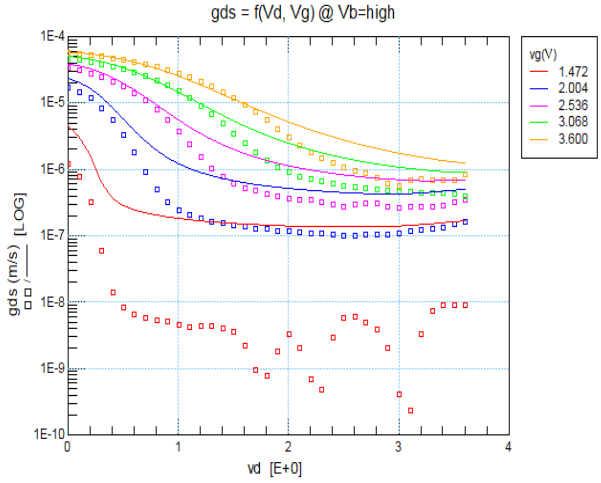




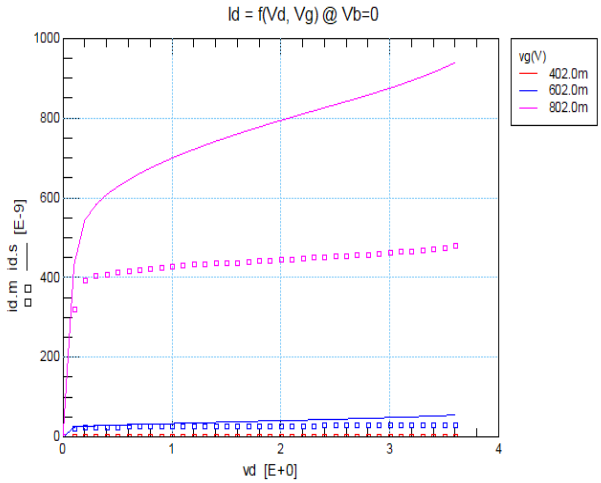
idvd_vbmin, Vb = -3.6V, T = 27°C

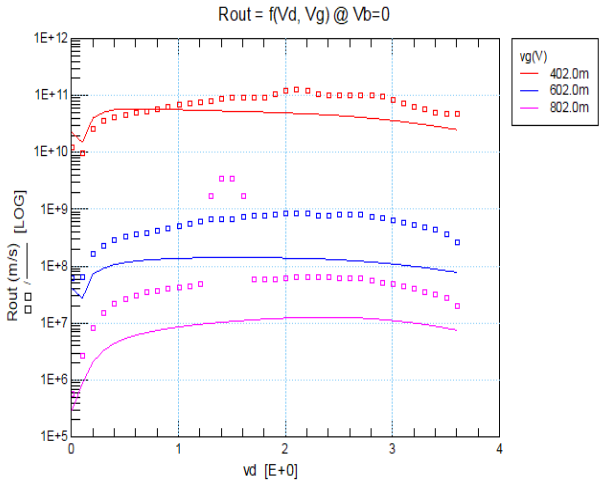


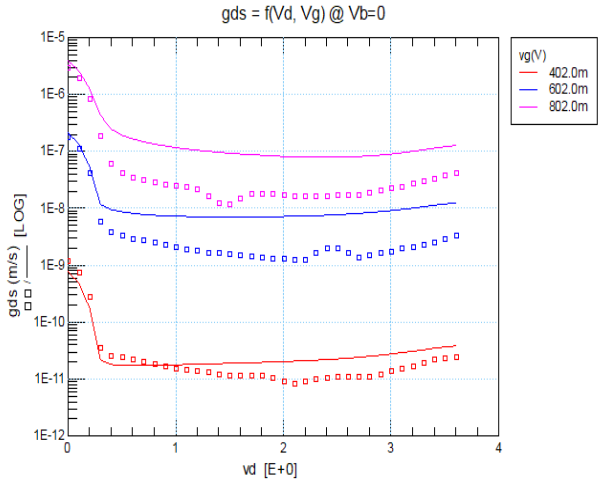




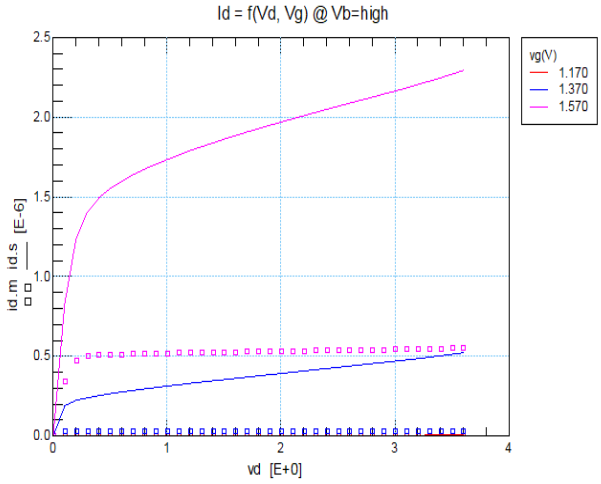
idvd_vth, Vb = 0V, T = 27°C

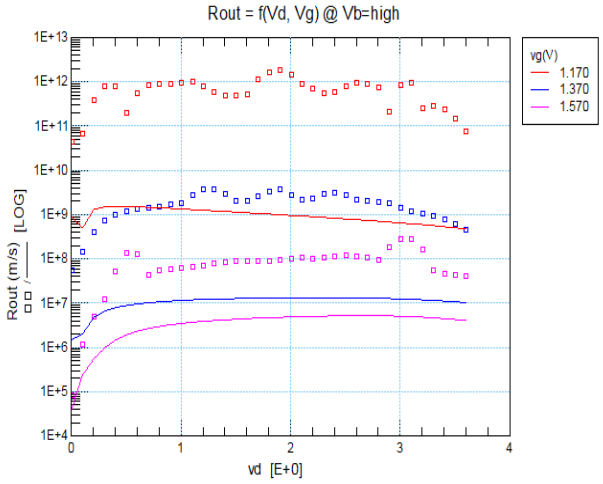


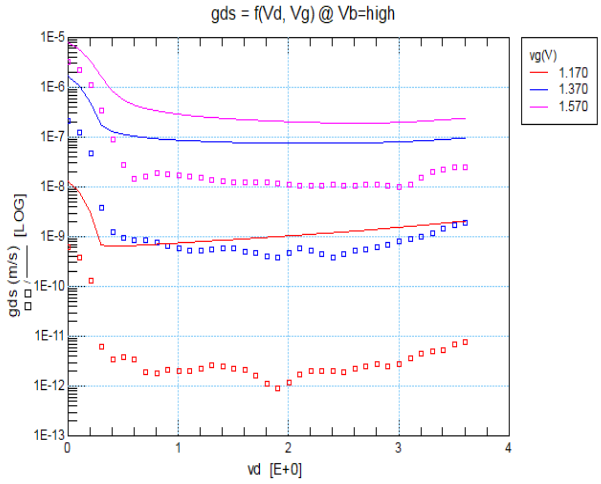




idvd_vth_vbmin, Vb = -3.6V, T = 27°C

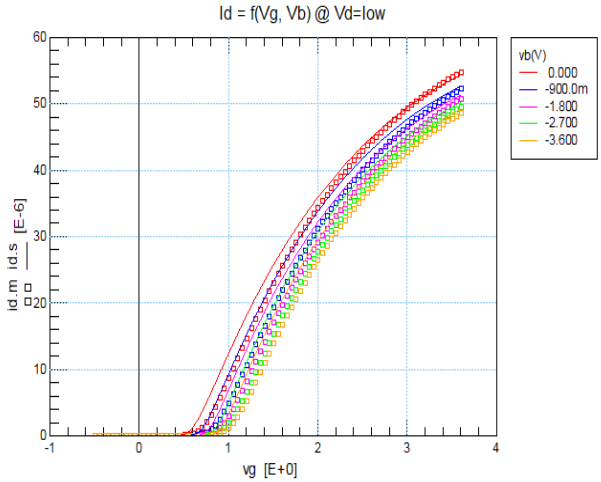


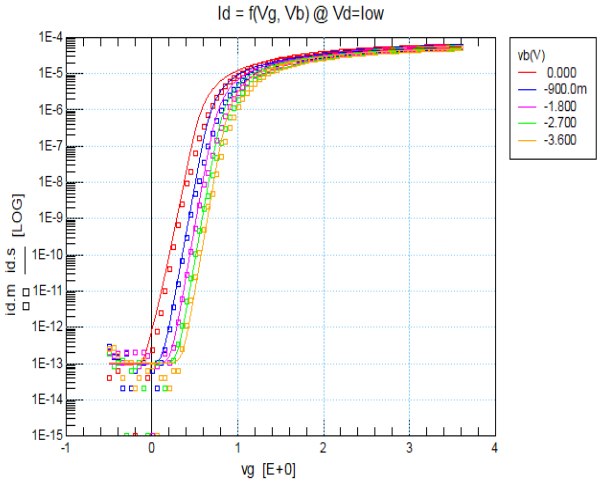


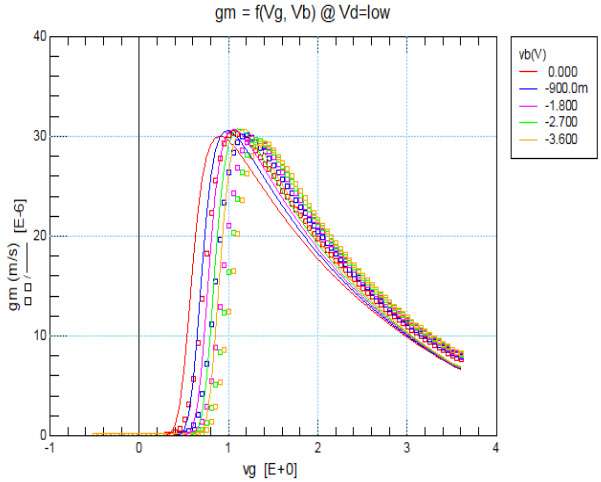


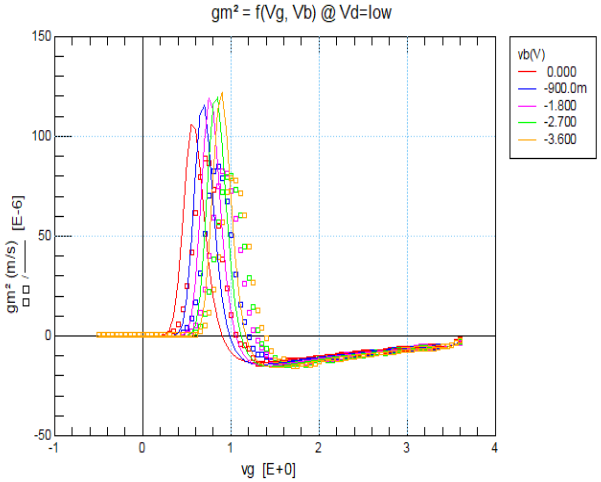
4.20W0u6_L0u35_S559_2

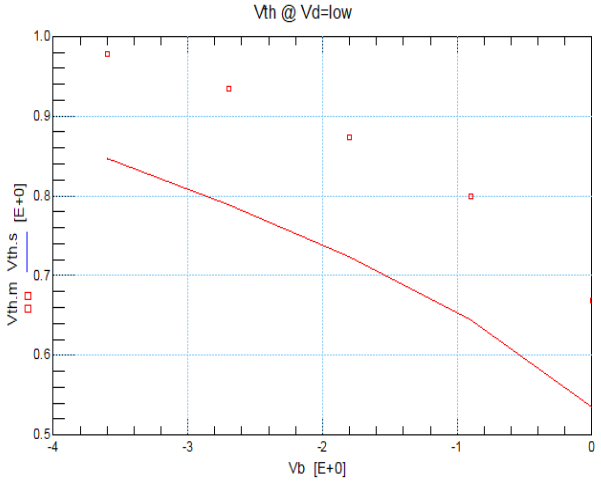
idvg, Vd = 0.1V, T = 27°C

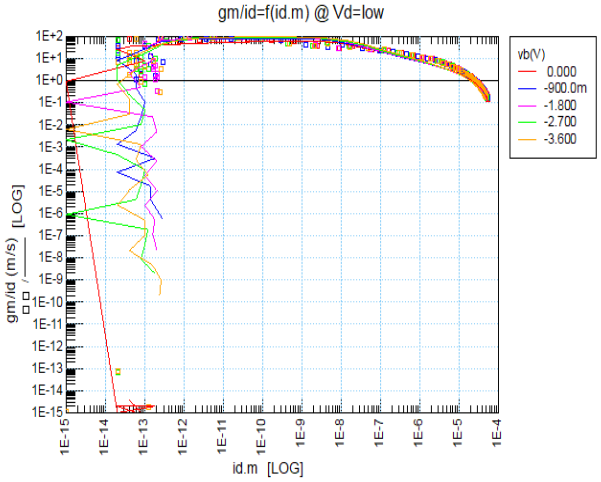


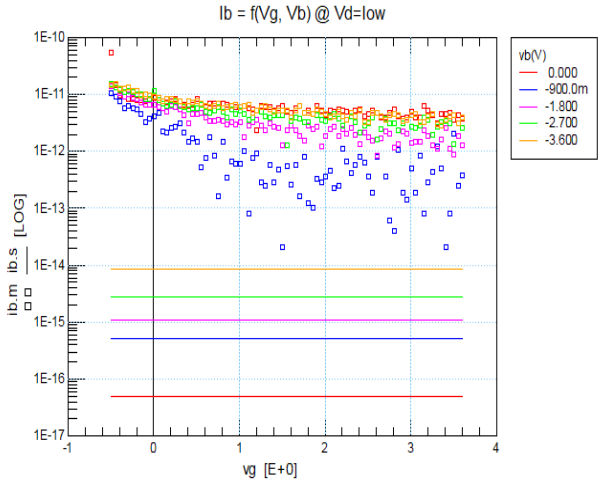


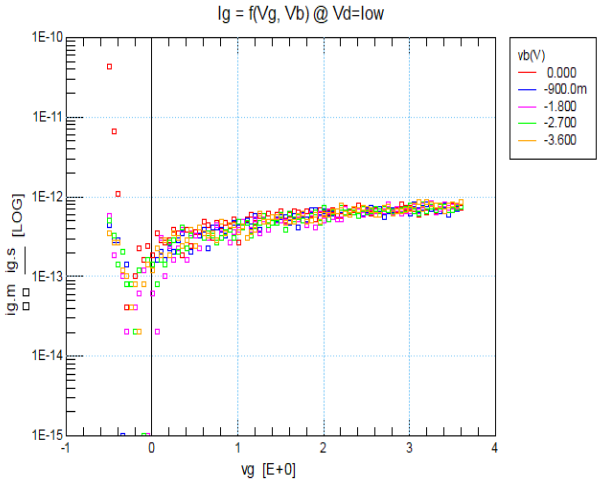




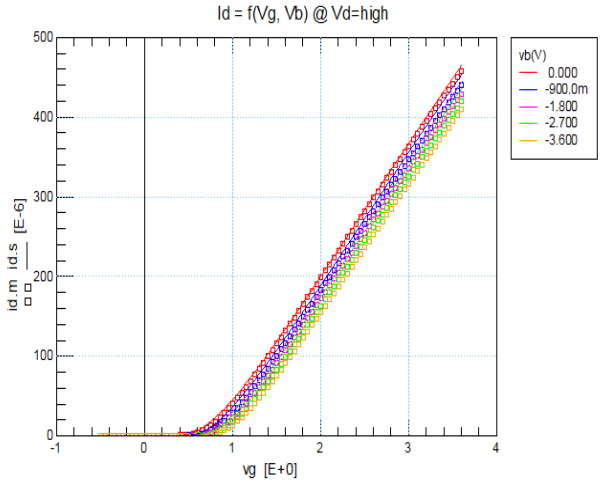


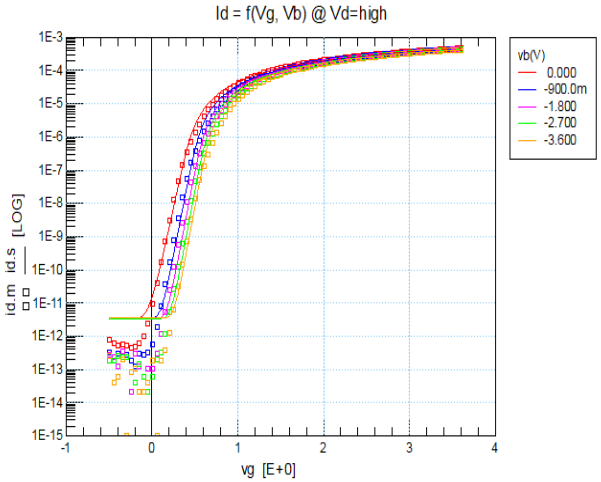


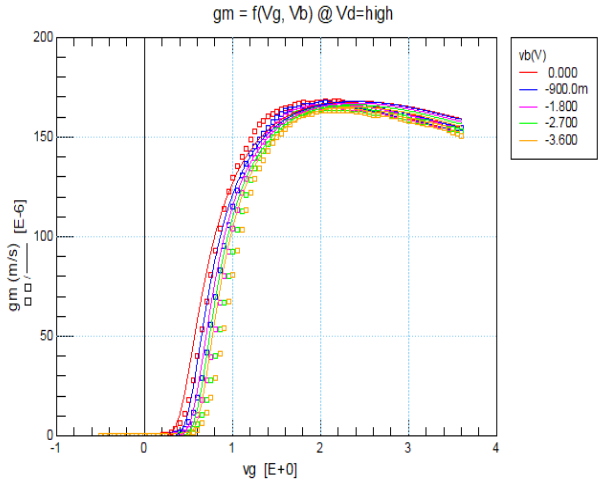


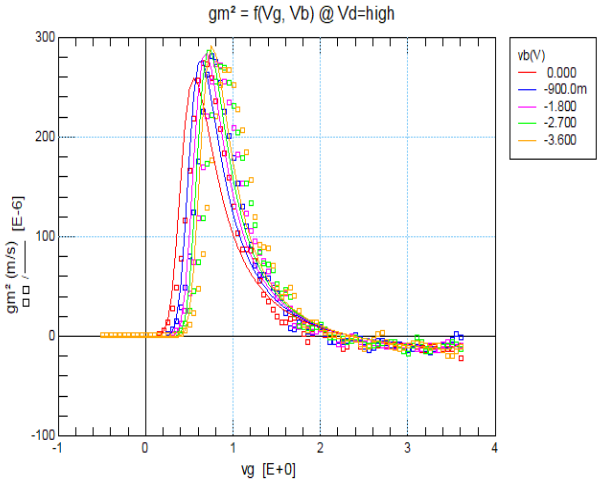


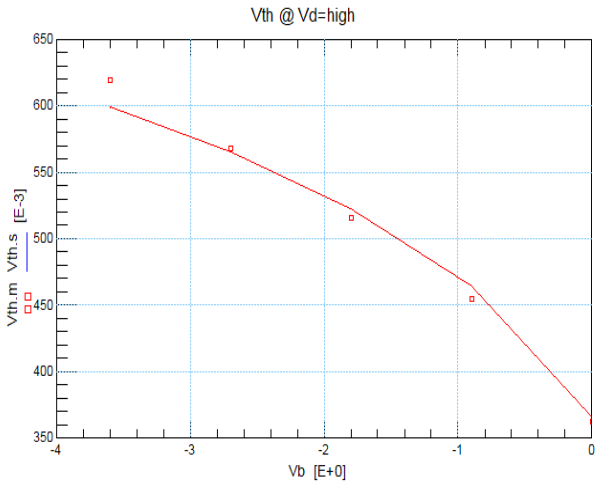
idvg, Vd = 3.3V, T = 27°C

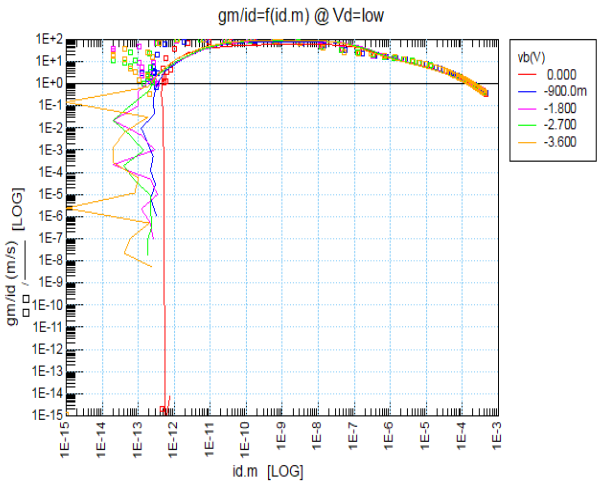


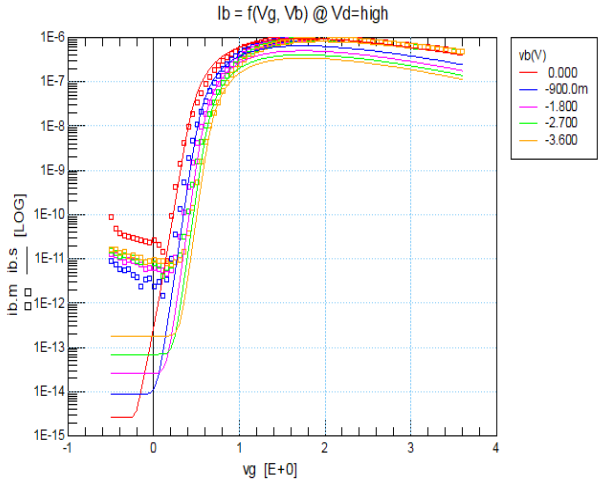


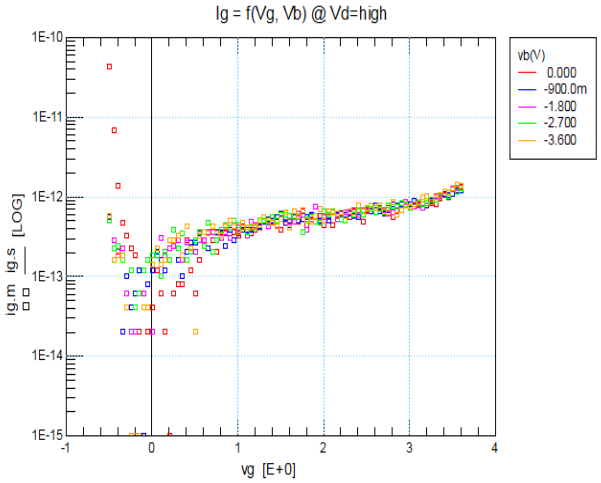




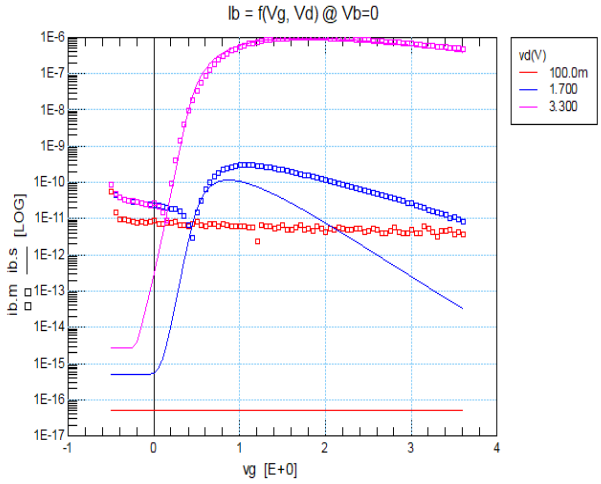


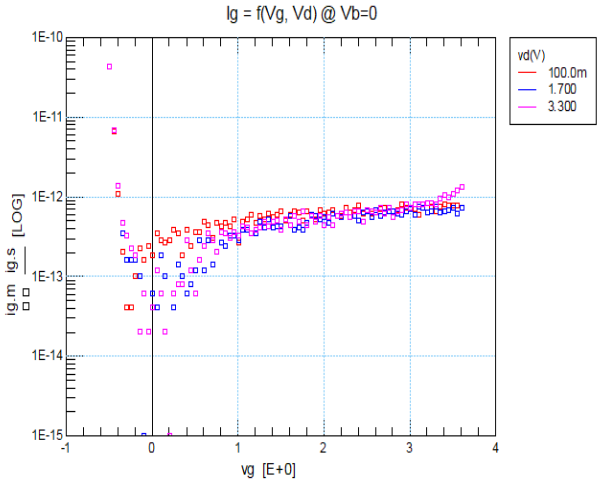




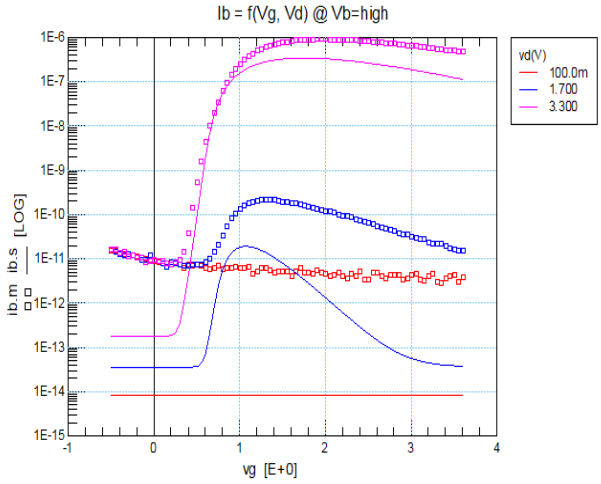


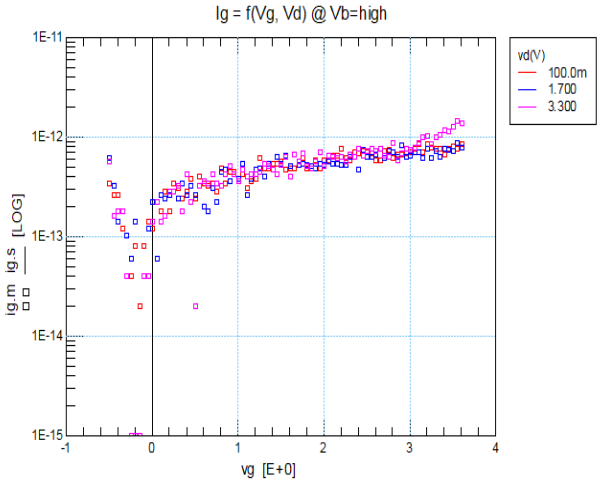
idvg, Ib,Ig, Vb = 0V, T = 27°C



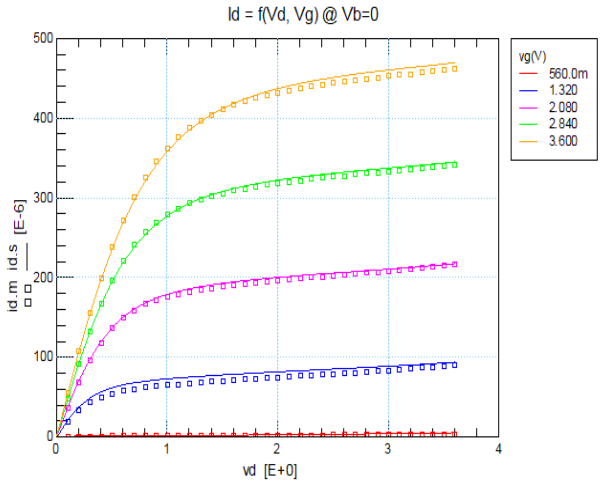


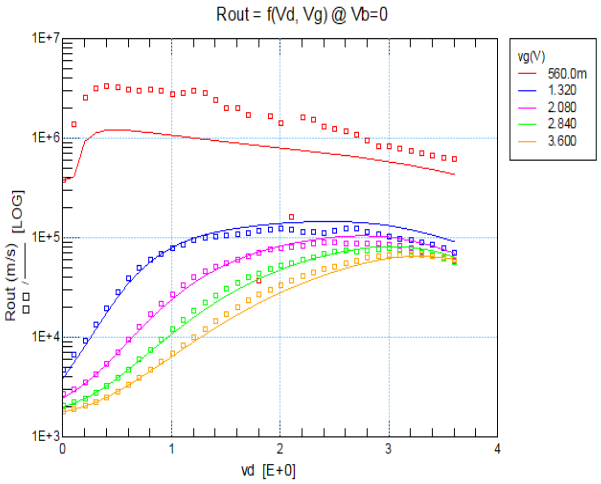
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

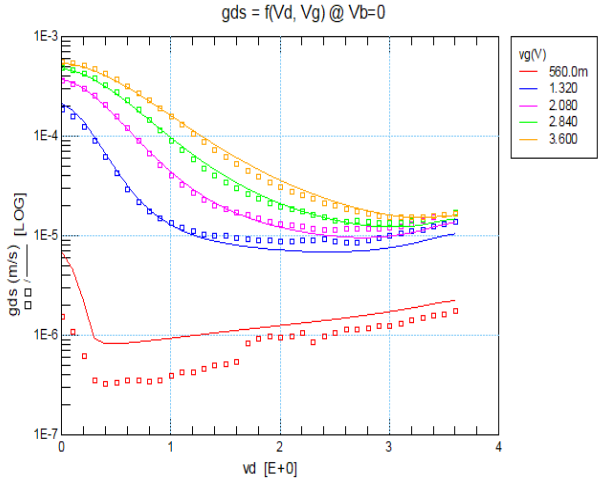




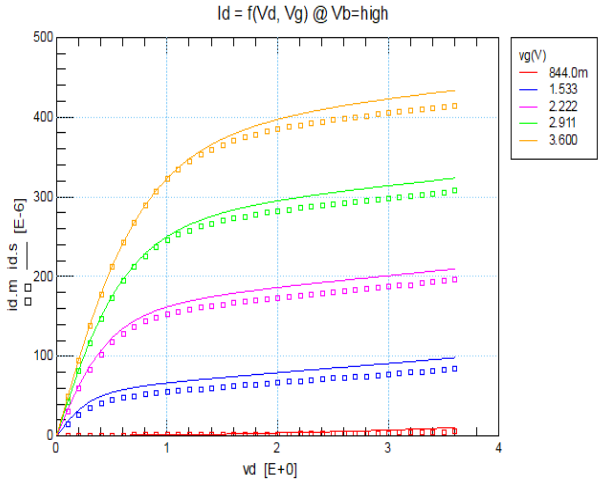
idvd, Vb = 0V, T = 27°C

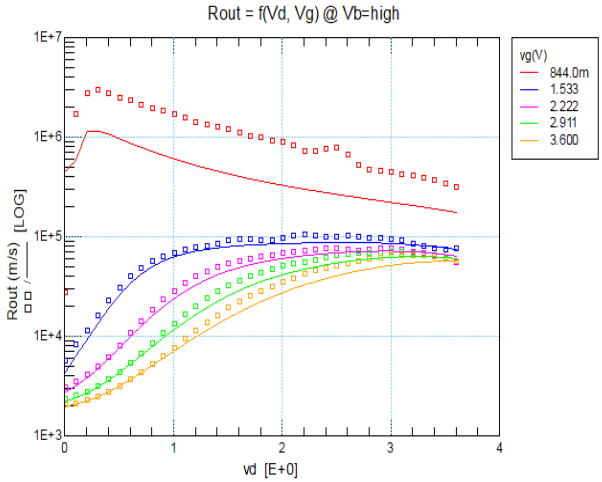


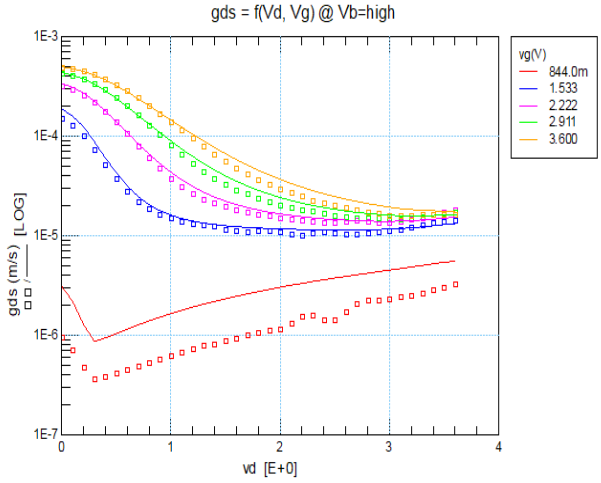




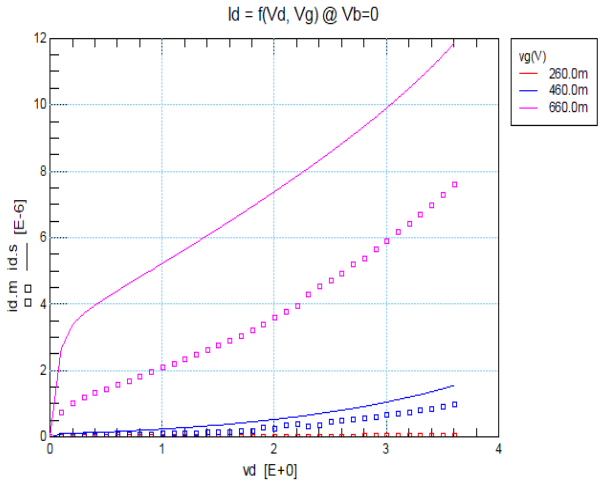
idvd_vbmin, Vb = -3.6V, T = 27°C

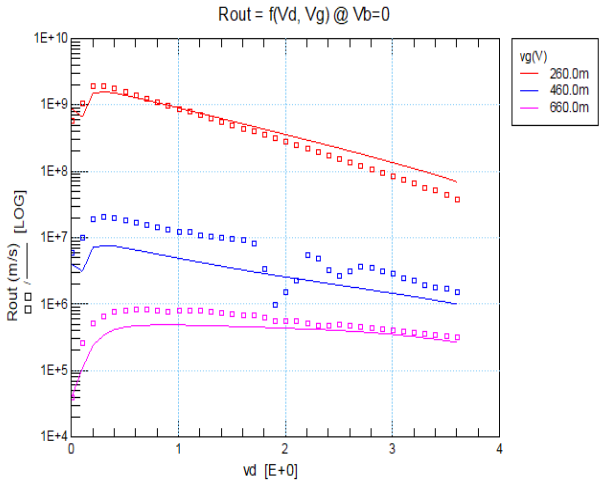


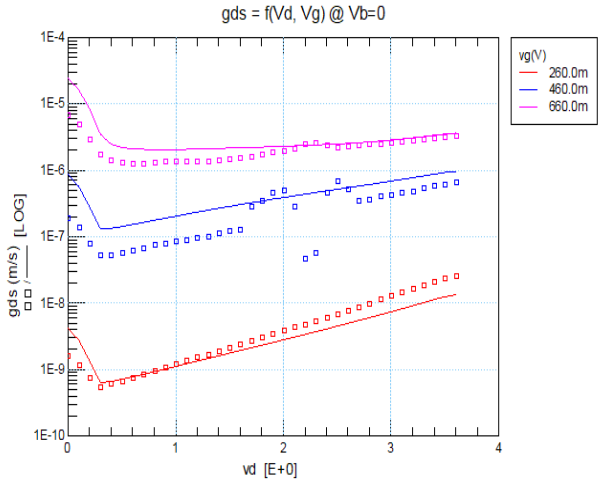




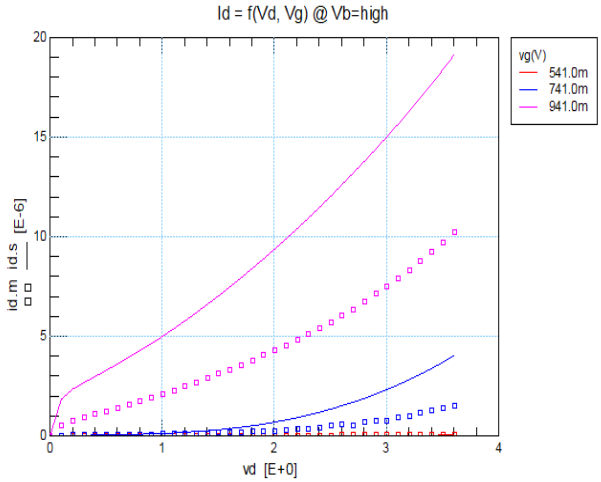
idvd_vth, Vb = 0V, T = 27°C

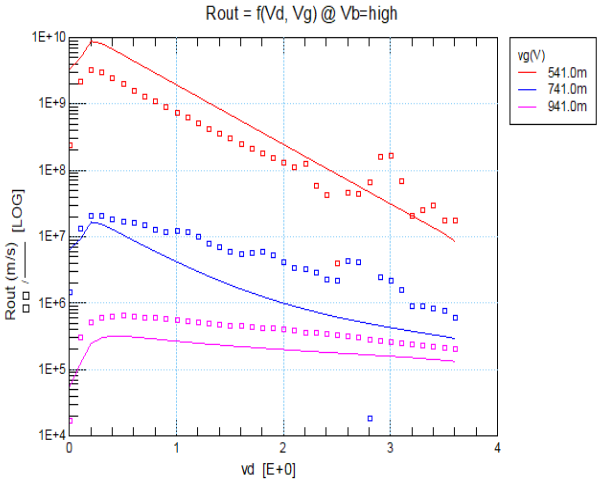


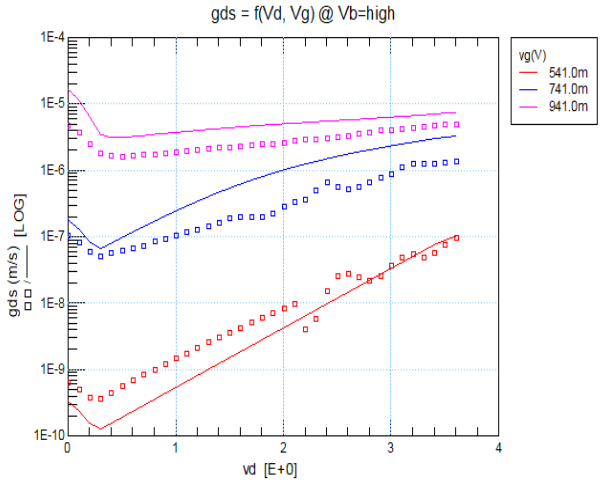




idvd_vth_vbmin, Vb = -3.6V, T = 27°C

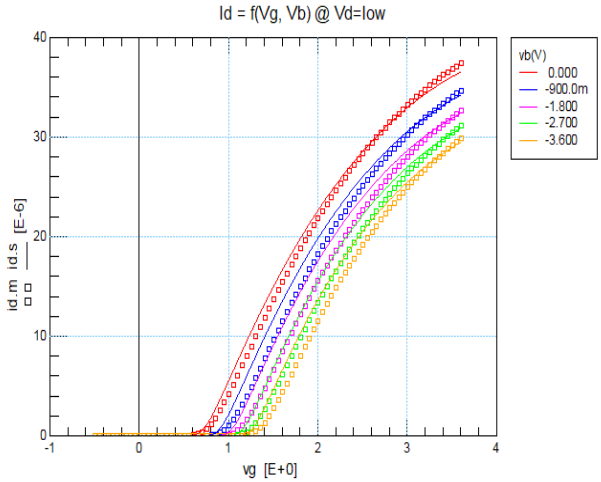


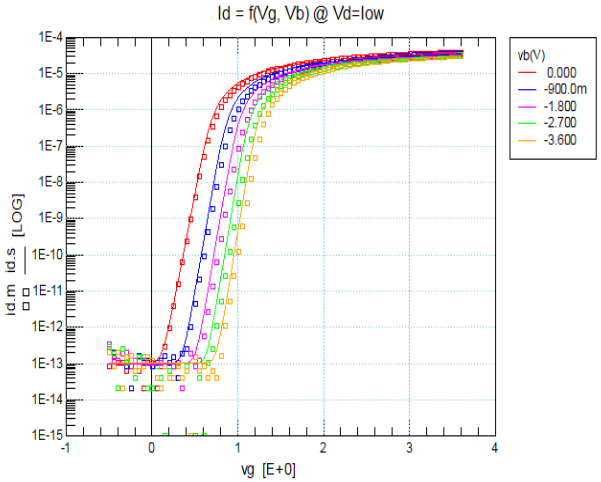


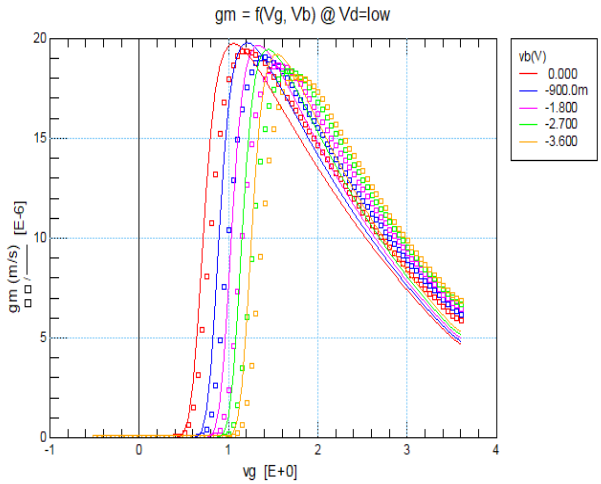


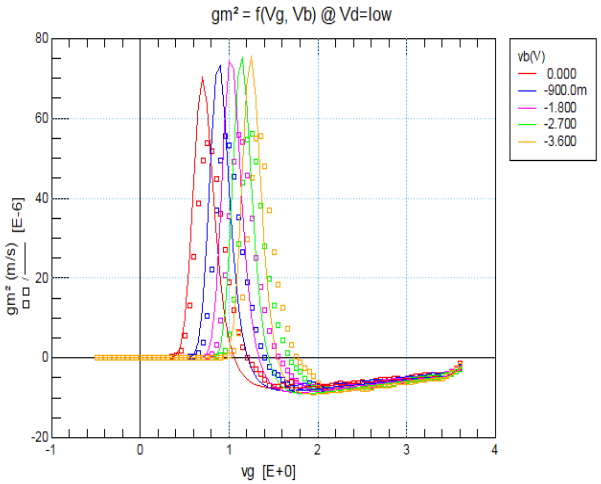
4.21 W0u6_L0u5_S559_3

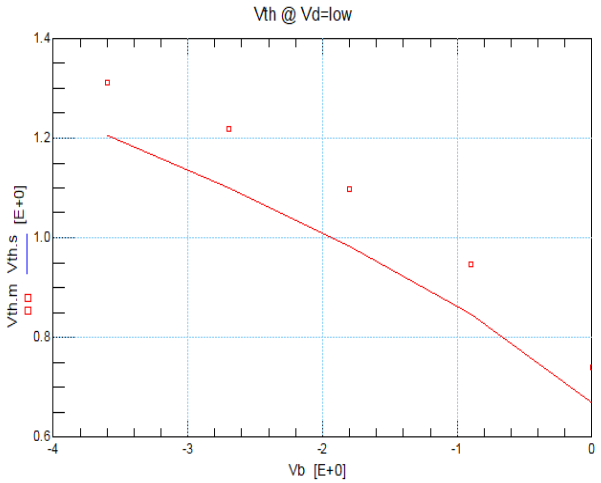
idvg, Vd = 0.1V, T = 27°C

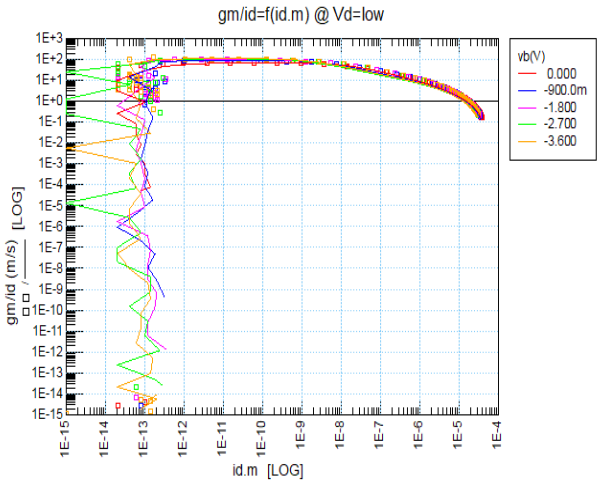


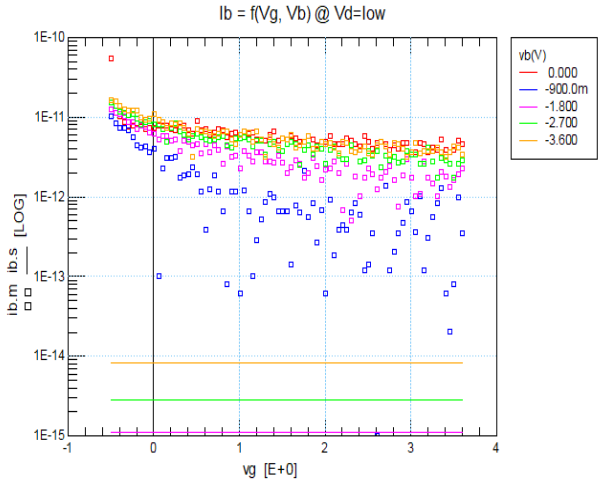


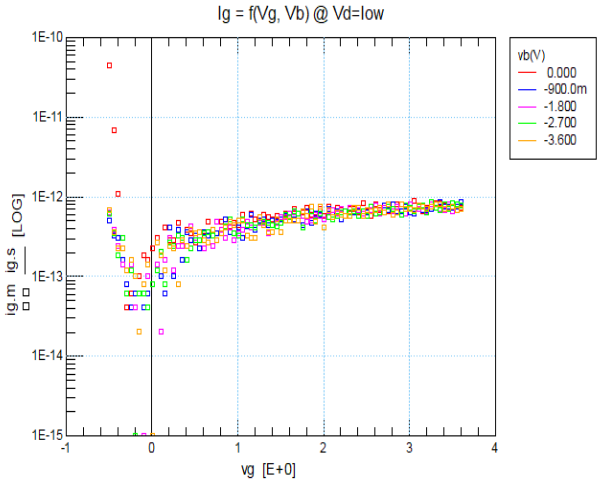




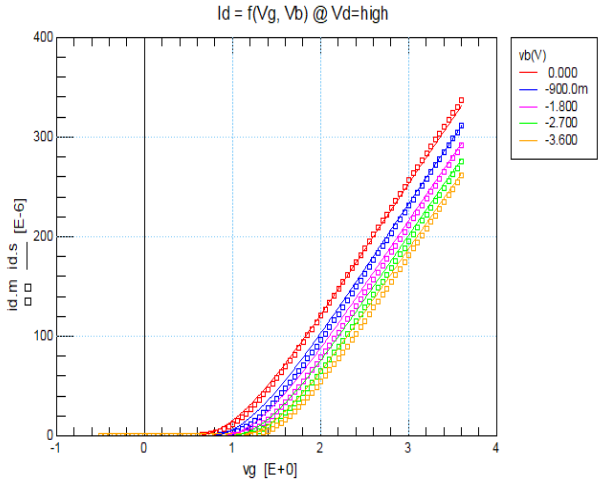


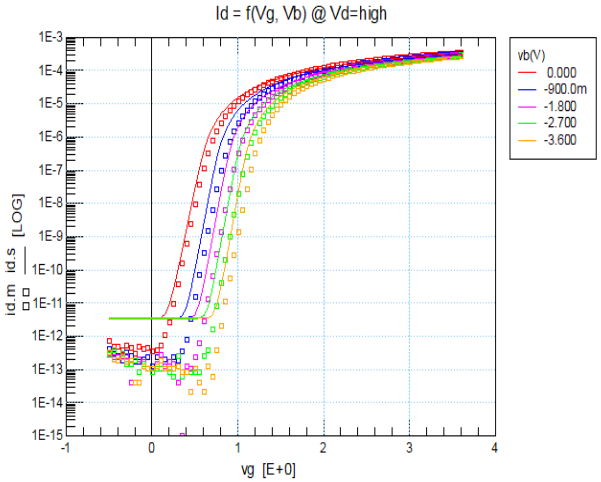


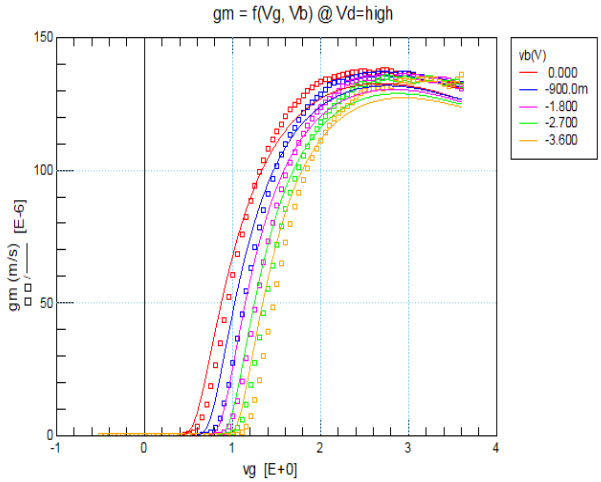


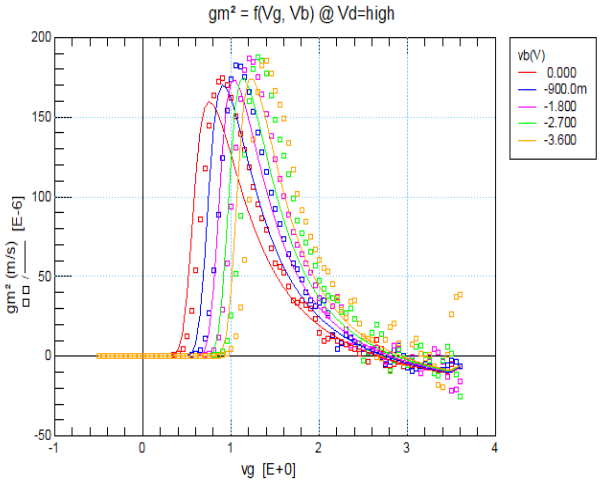


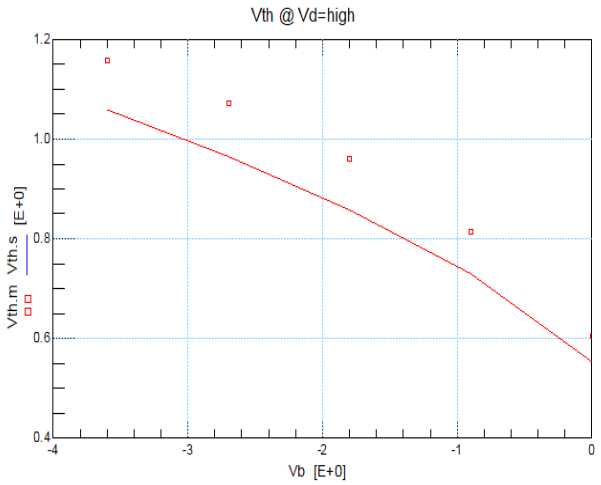
idvg, Vd = 3.3V, T = 27°C

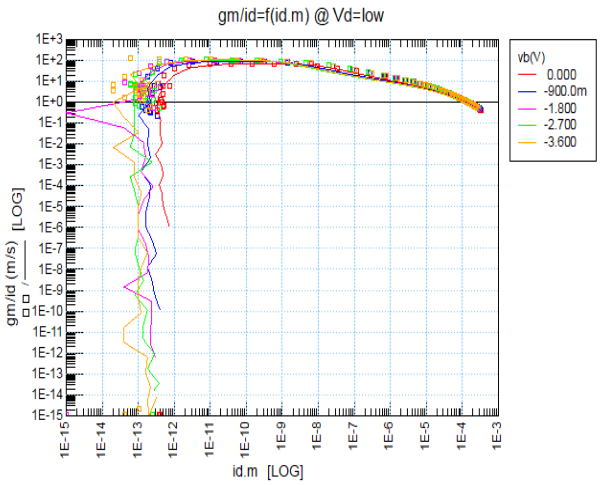


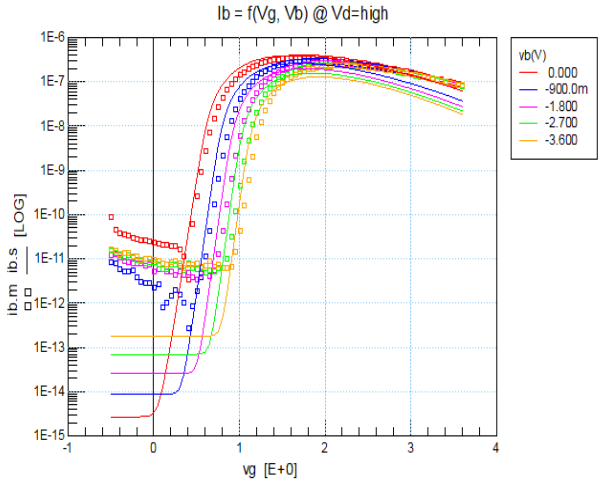


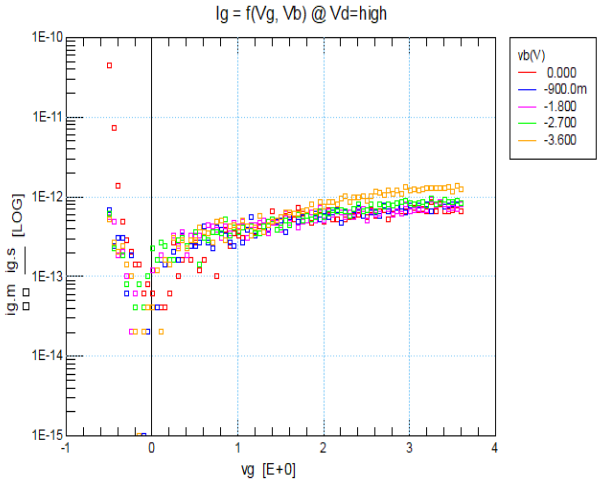




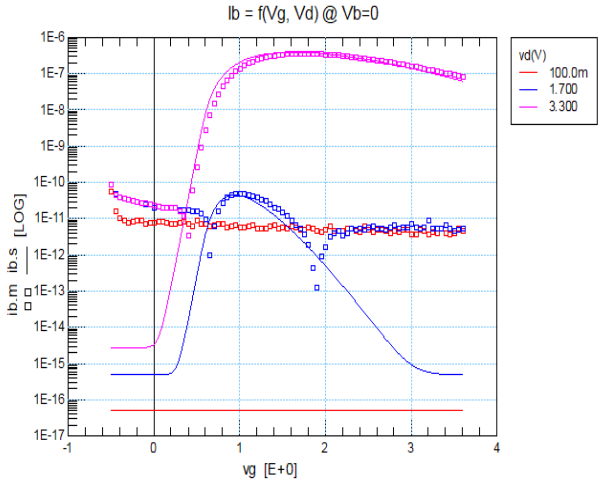


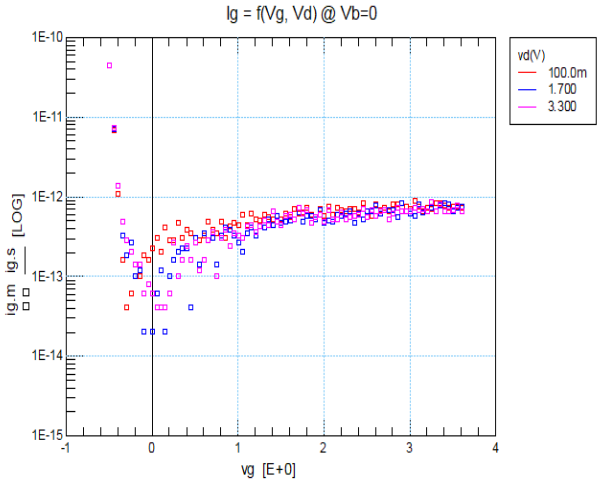




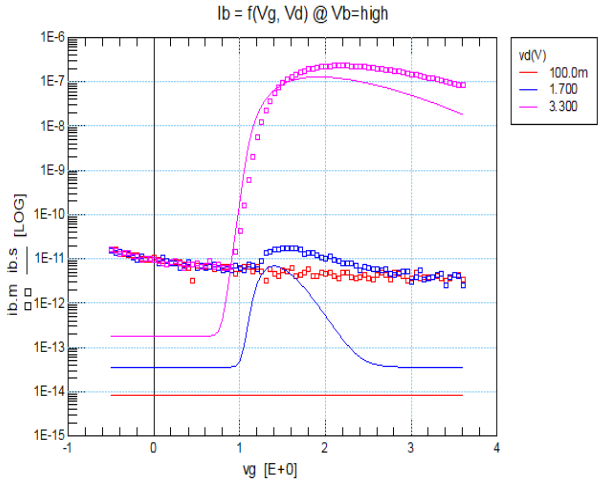


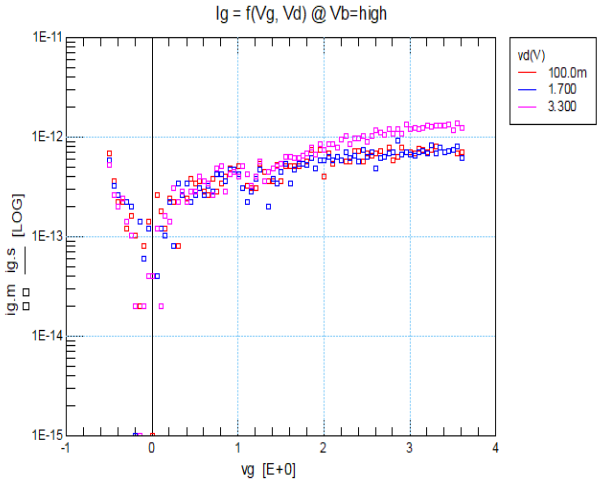
idvg, Ib,Ig, Vb = 0V, T = 27°C



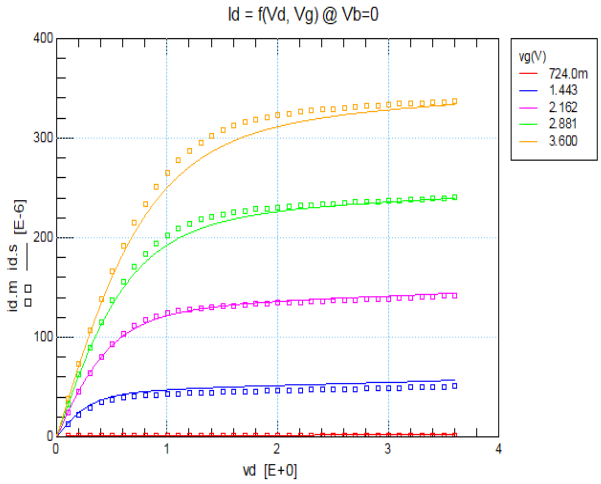


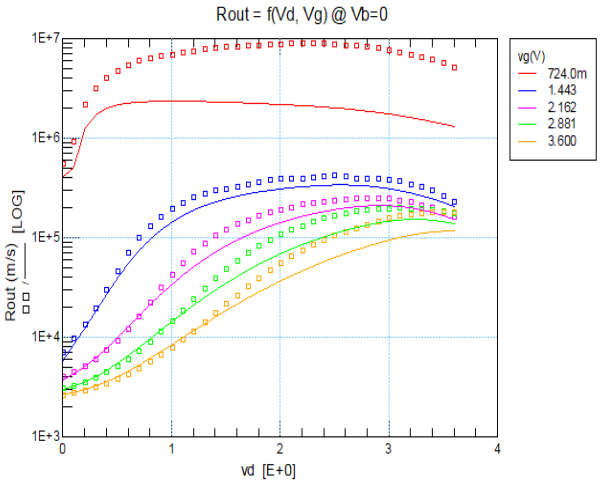
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

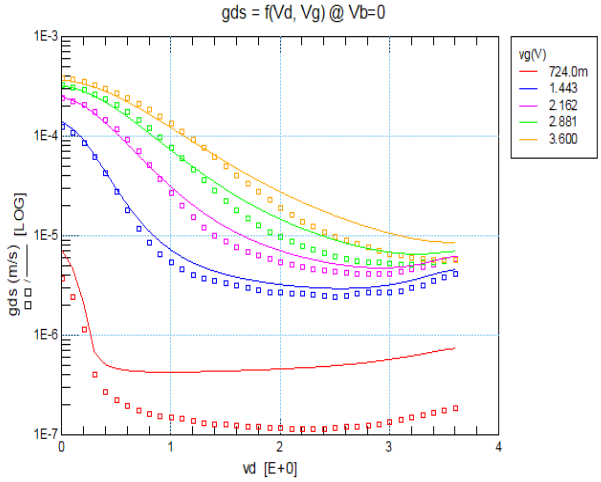




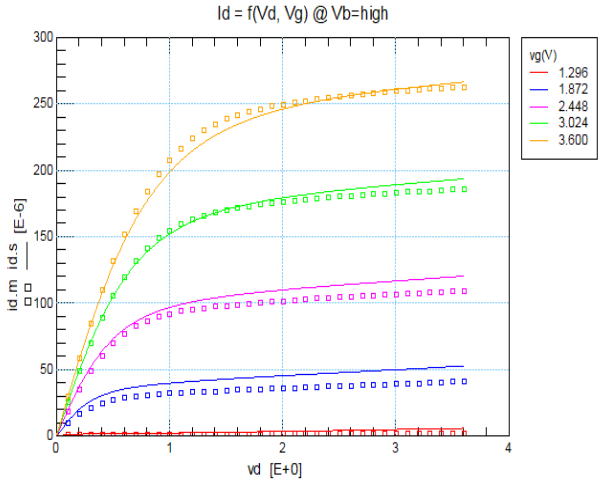
idvd, Vb = 0V, T = 27°C

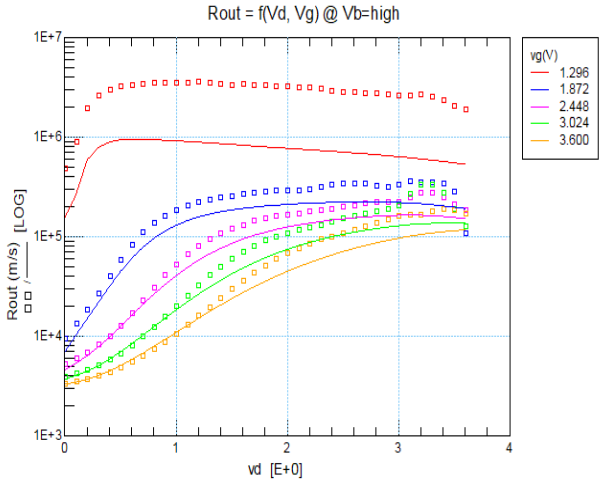


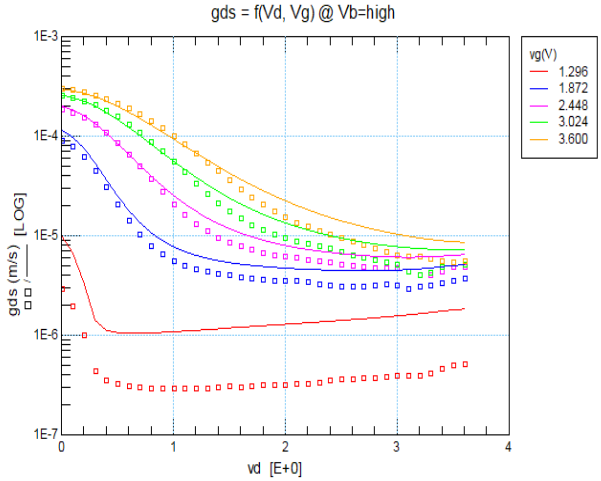




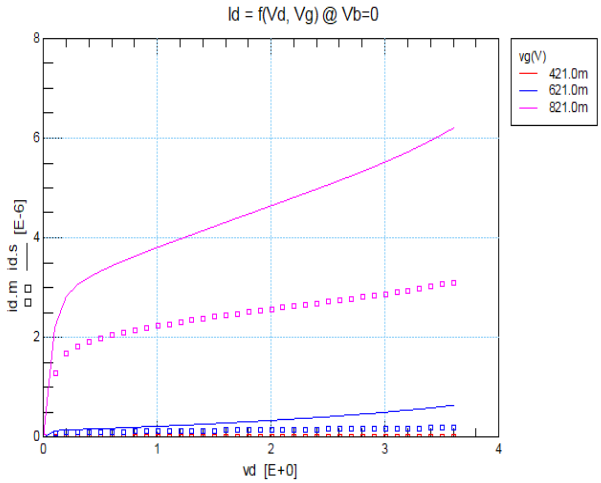
idvd_vbmin, Vb = -3.6V, T = 27°C

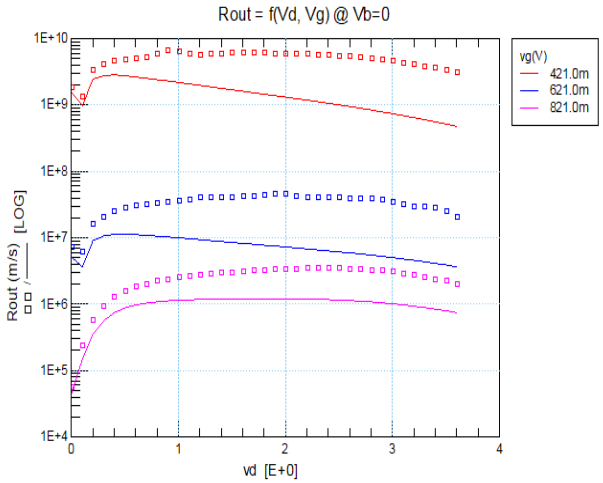


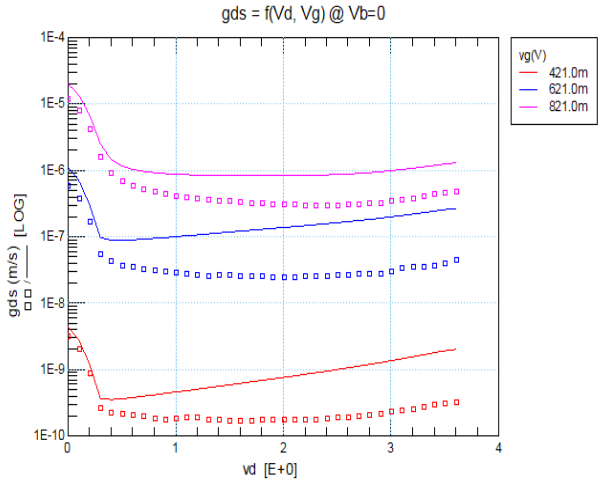




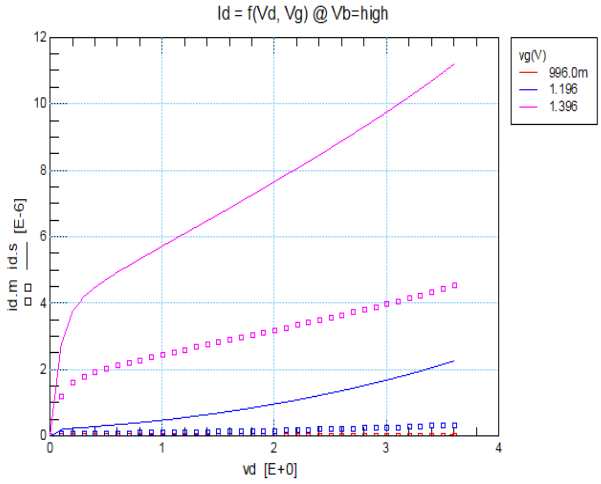
idvd_vth, Vb = 0V, T = 27°C

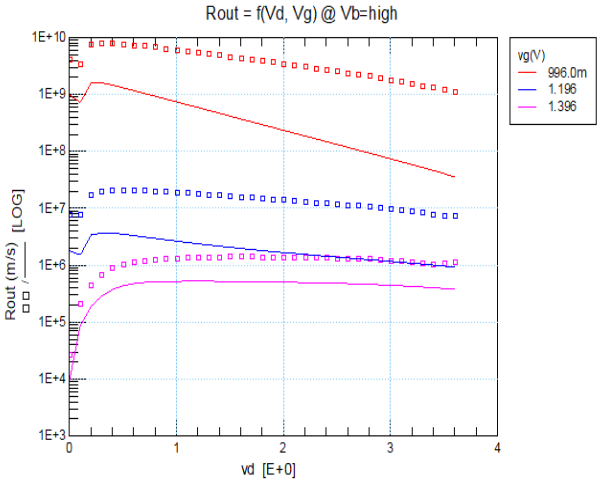


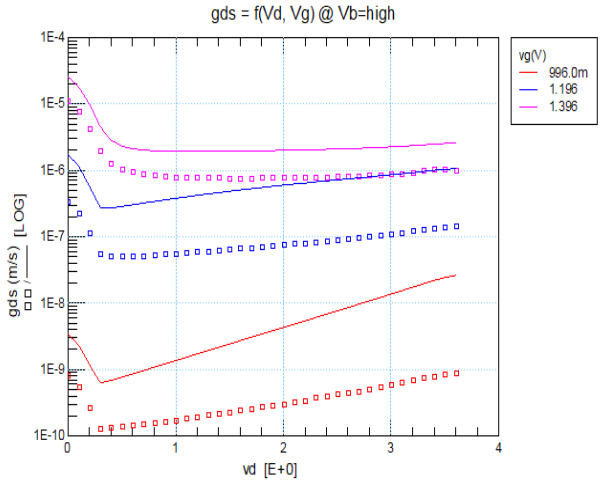




idvd_vth_vbmin, Vb = -3.6V, T = 27°C

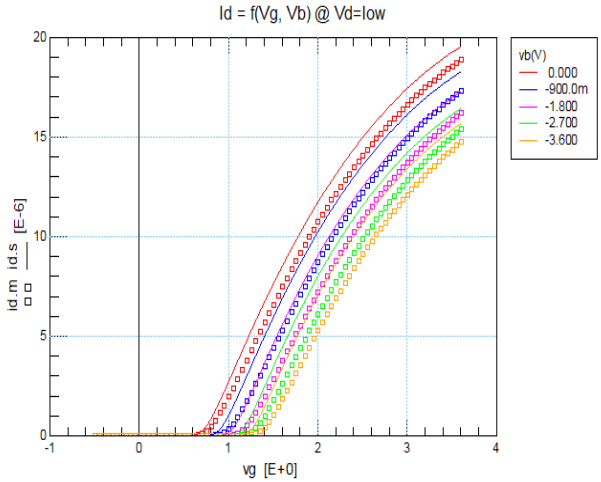


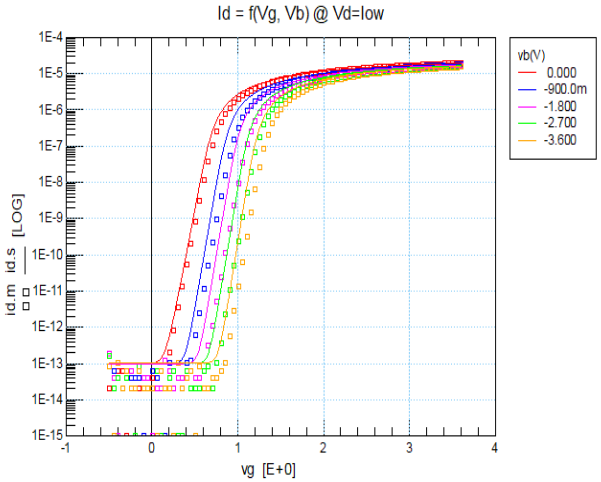


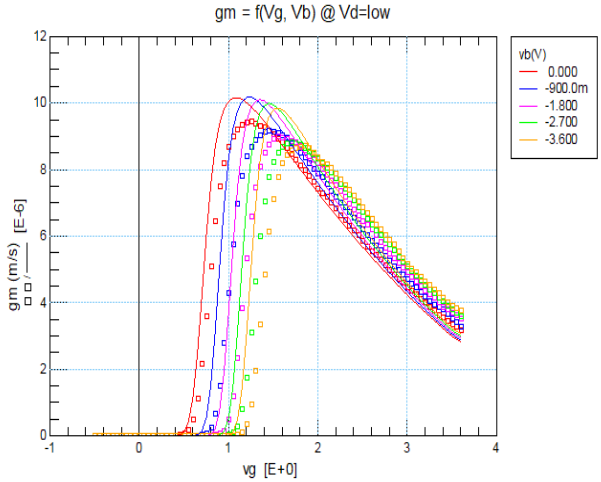


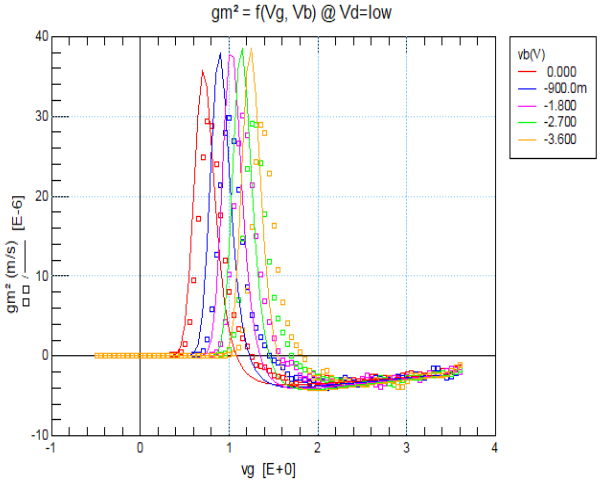
4.22 W0u3_L0u5_S560_4

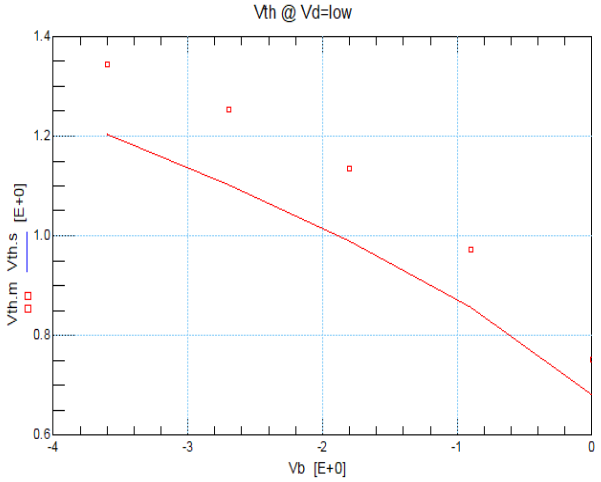
idvg, Vd = 0.1V, T = 27°C

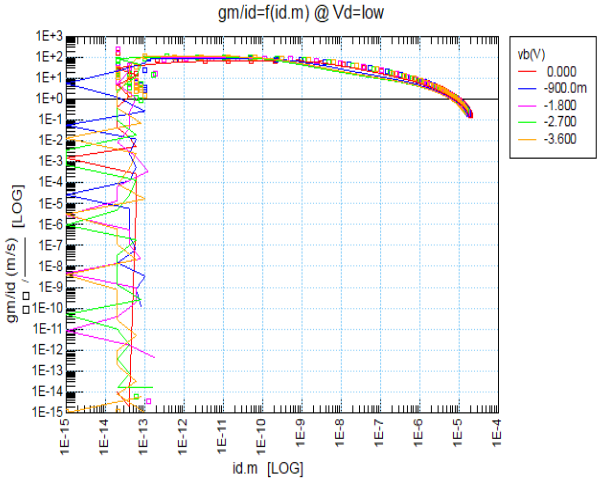


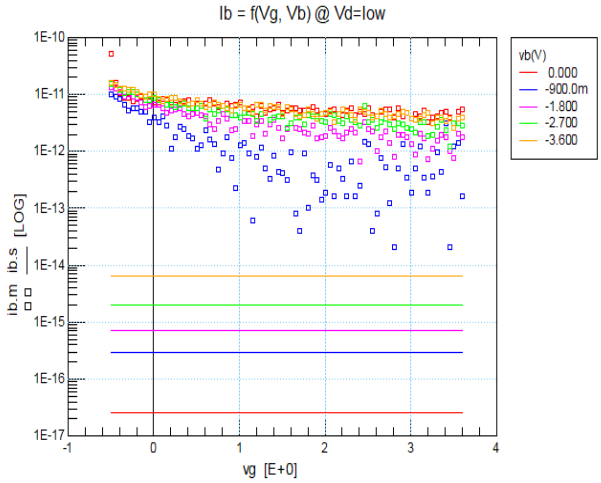


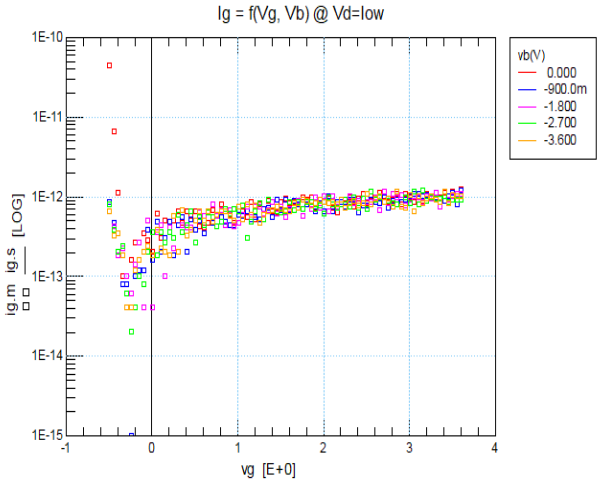




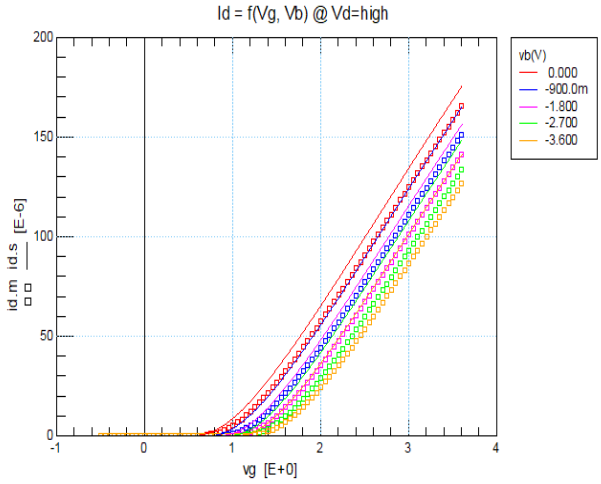


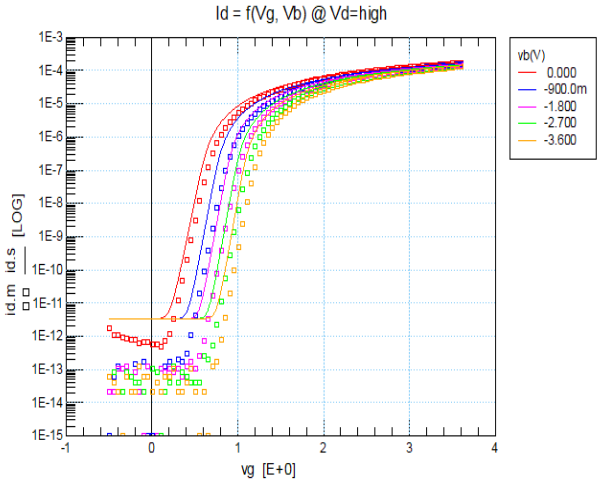


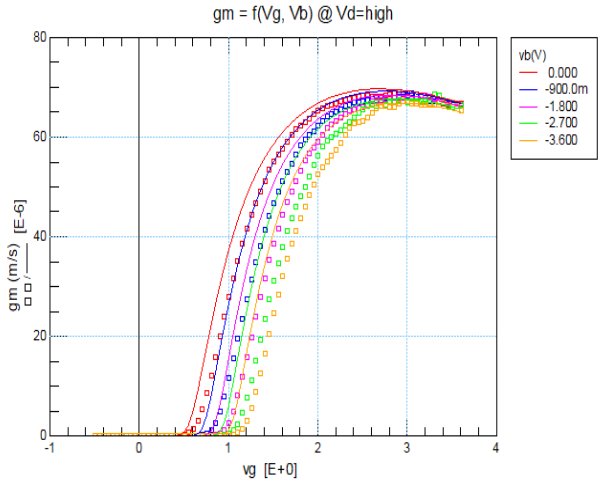


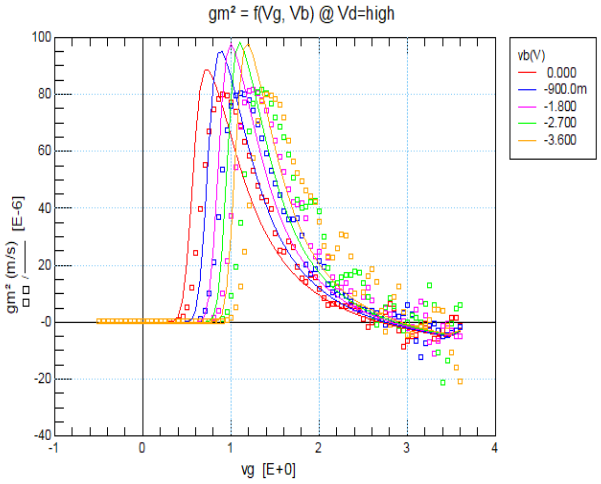


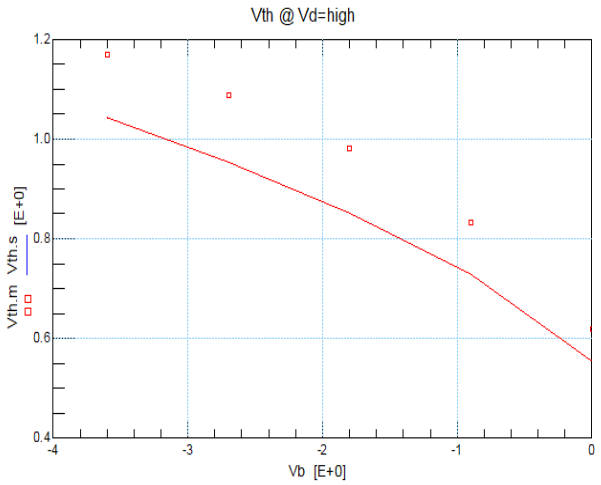
idvg, Vd = 3.3V, T = 27°C

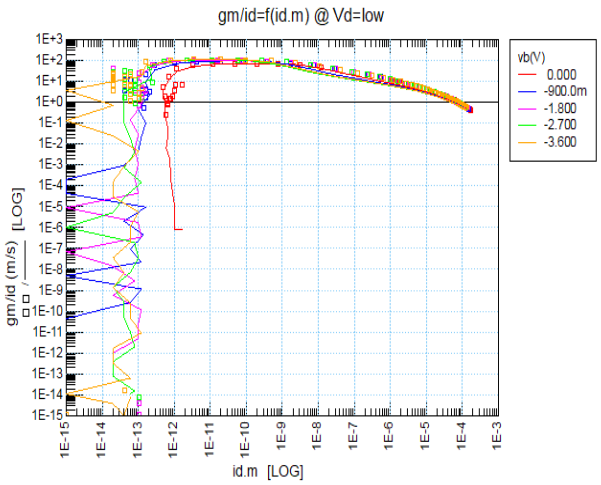


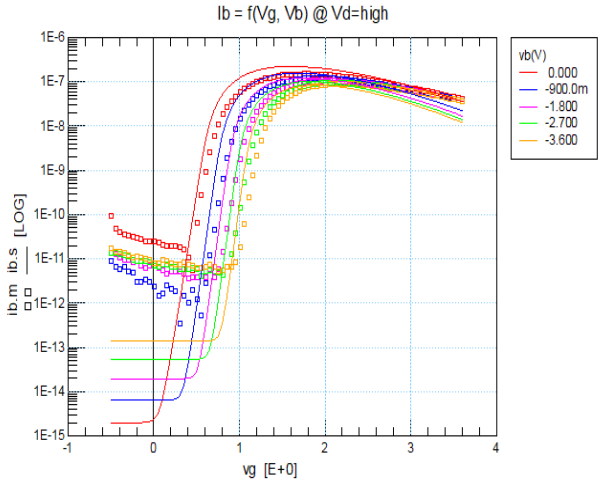


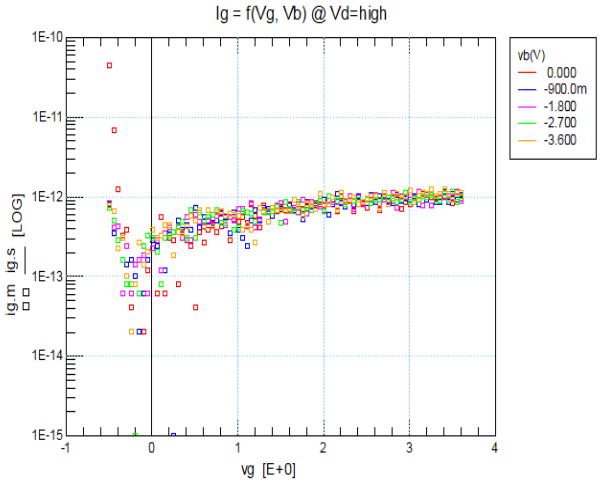




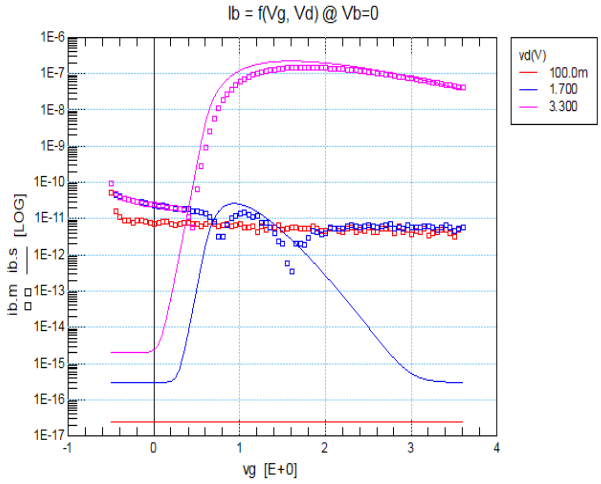


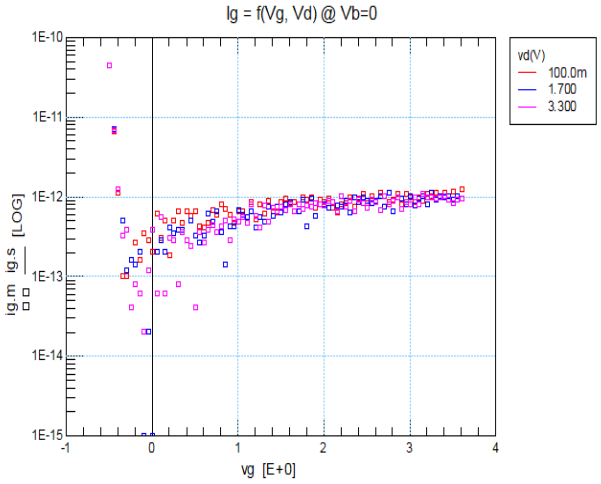




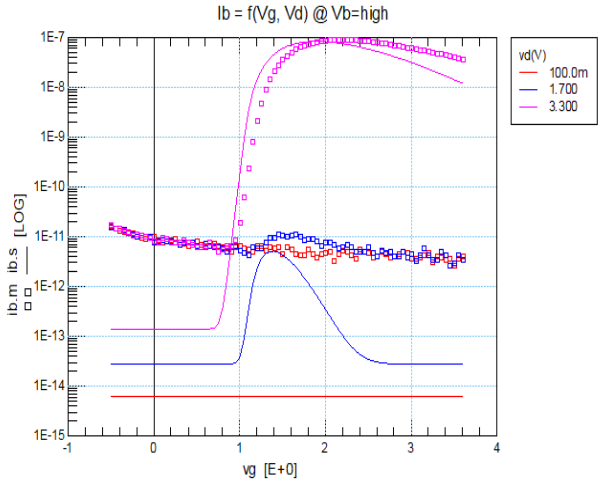


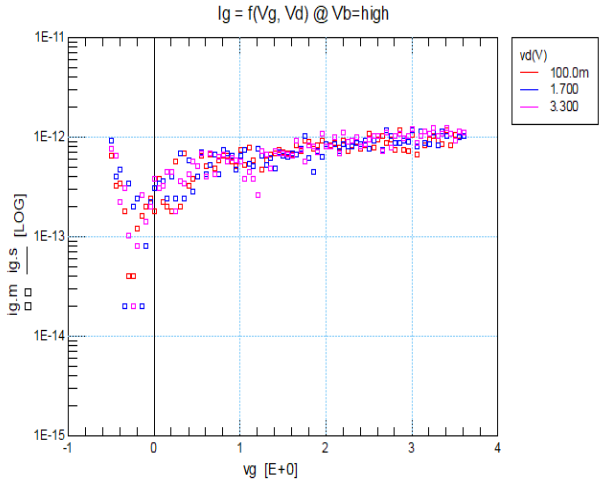
idvg, Ib,Ig, Vb = 0V, T = 27°C



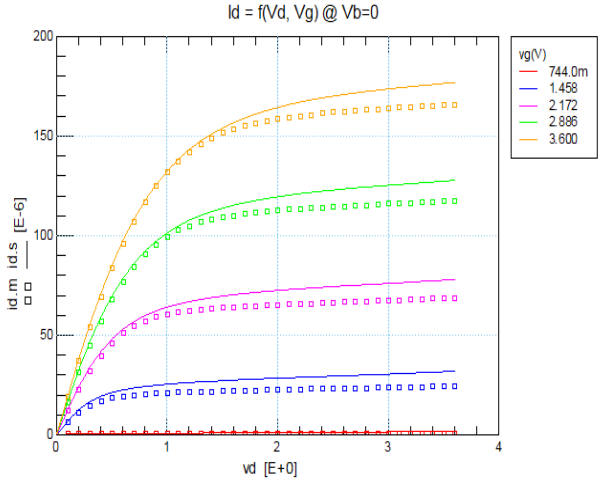


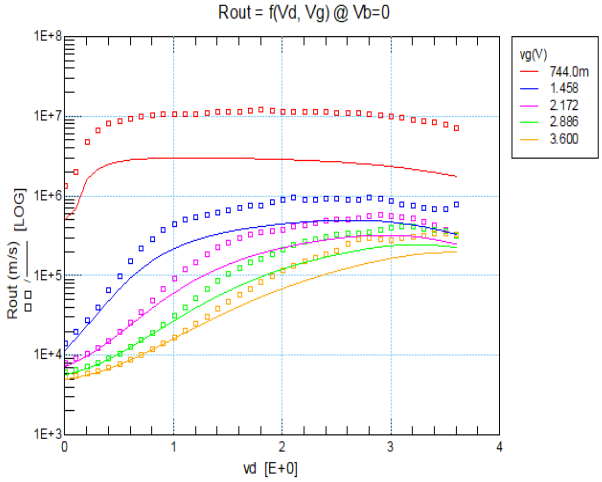
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

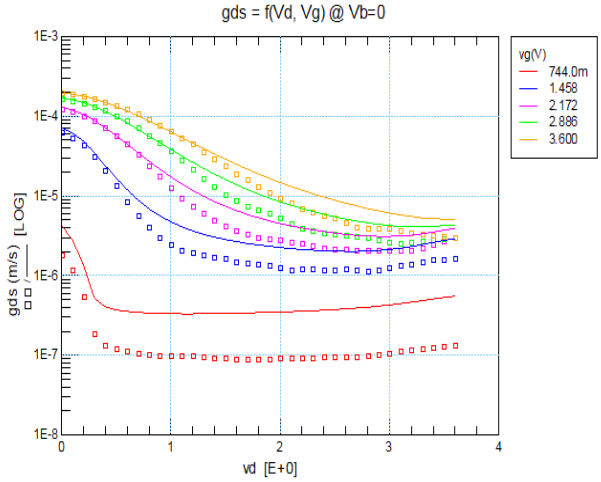




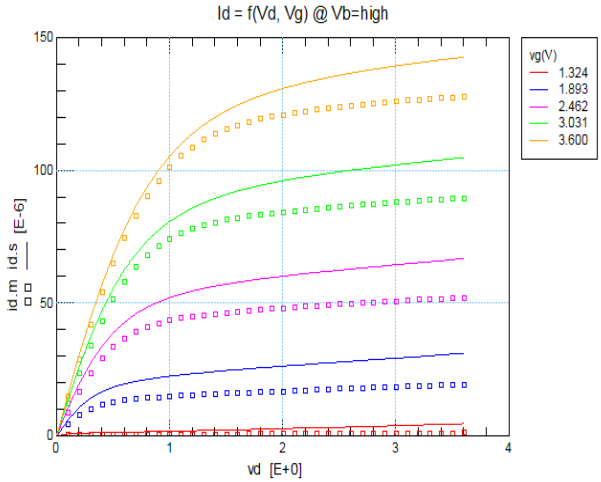
idvd, Vb = 0V, T = 27°C

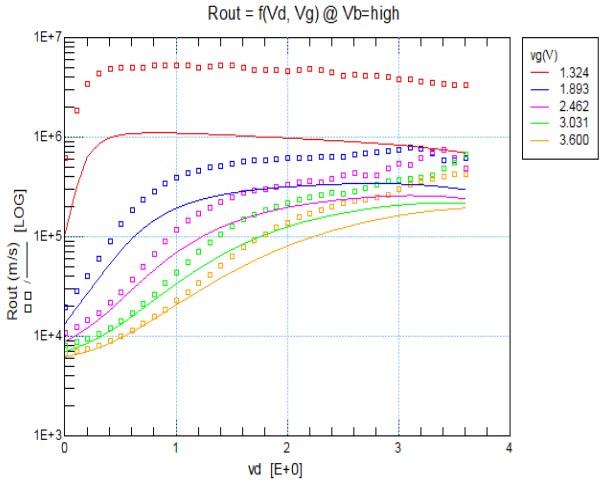


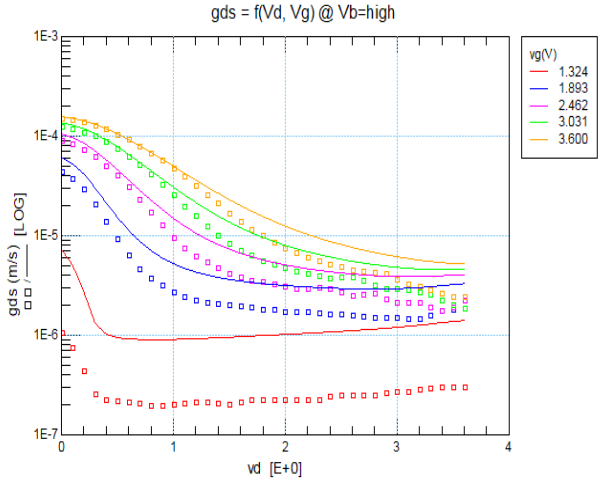




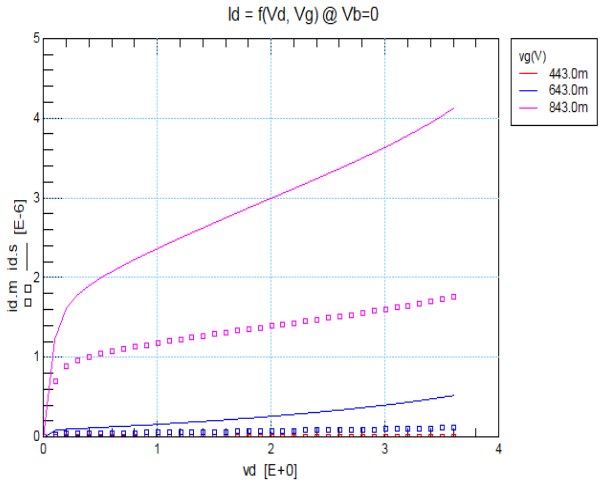
idvd_vbmin, Vb = -3.6V, T = 27°C

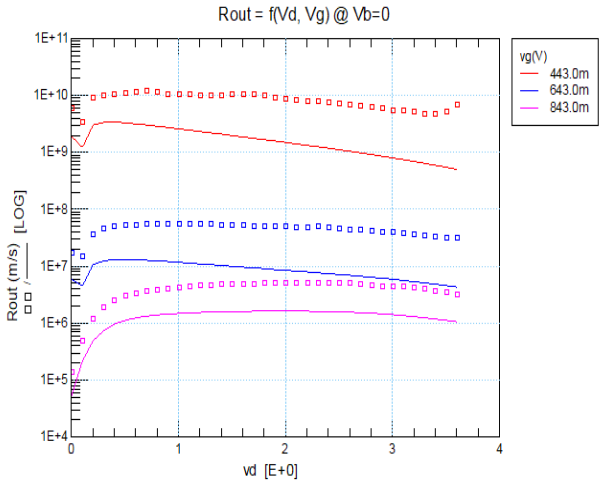


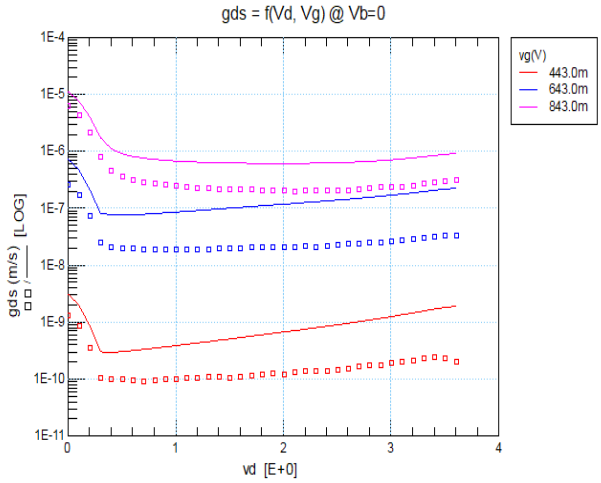




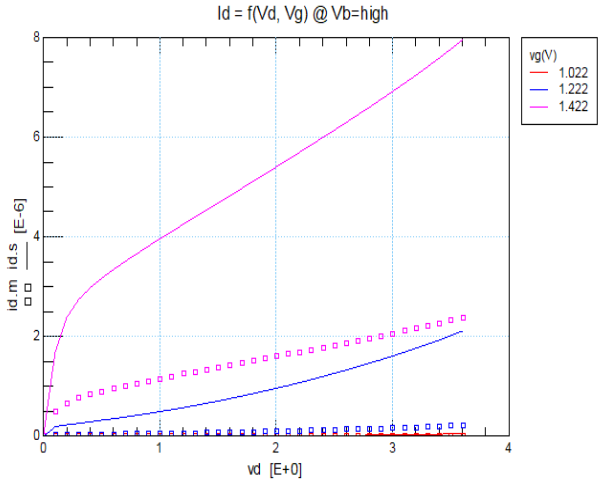
idvd_vth, Vb = 0V, T = 27°C

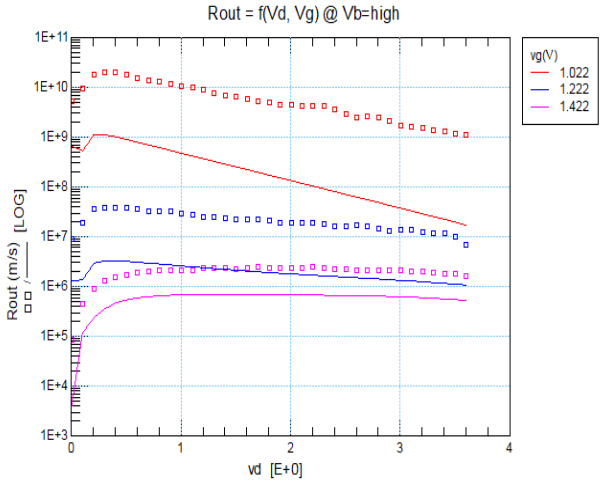


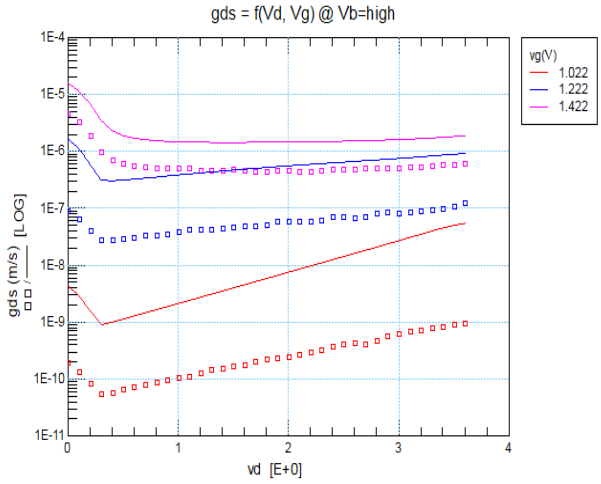




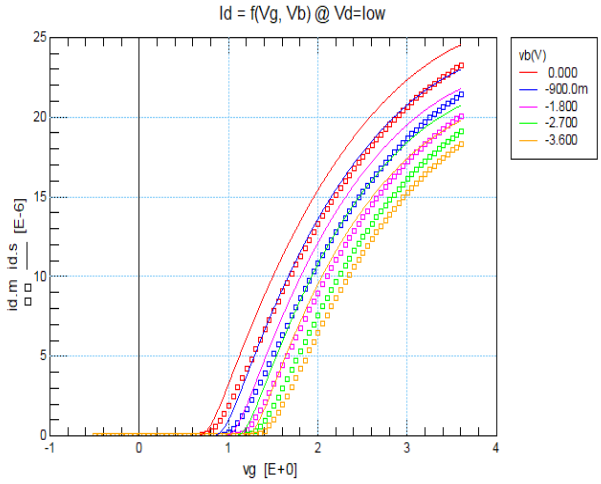
idvd_vth_vbmin, Vb = -3.6V, T = 27°C

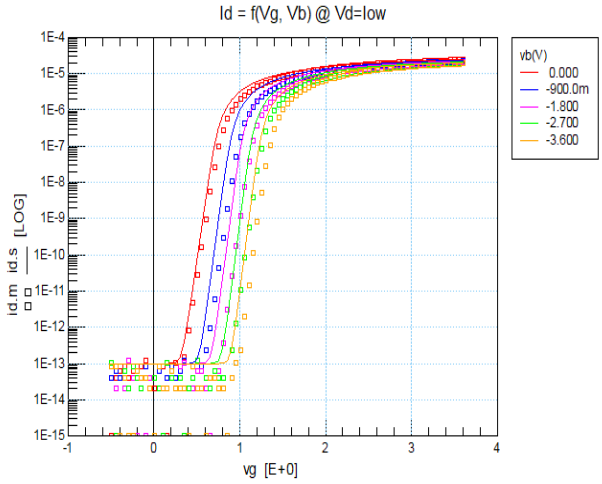


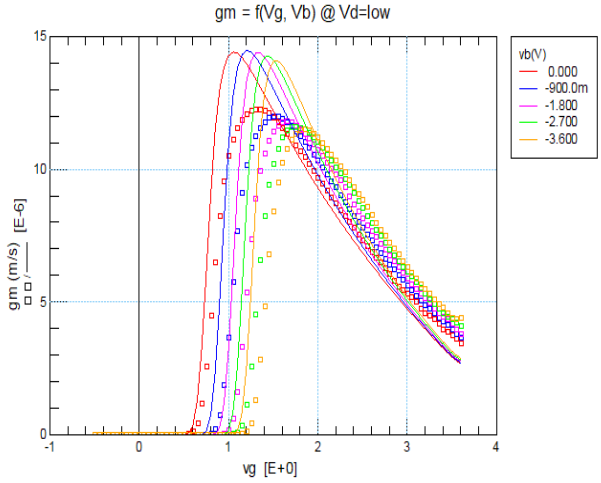


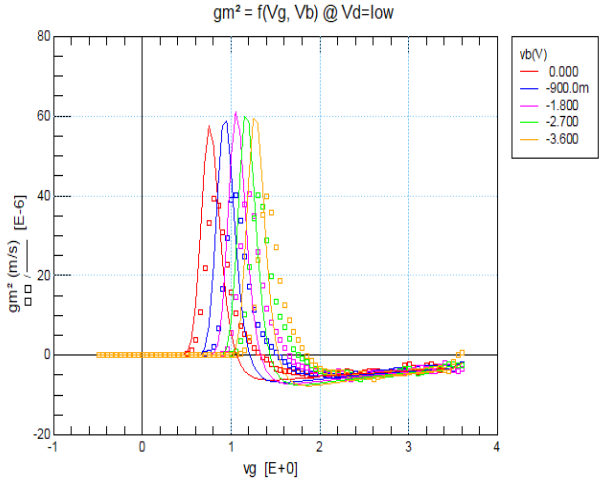


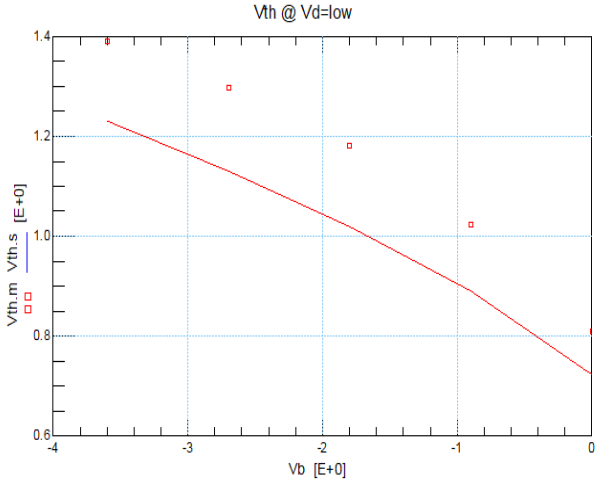
idvg, Vd = 0.1V, T = -40°C

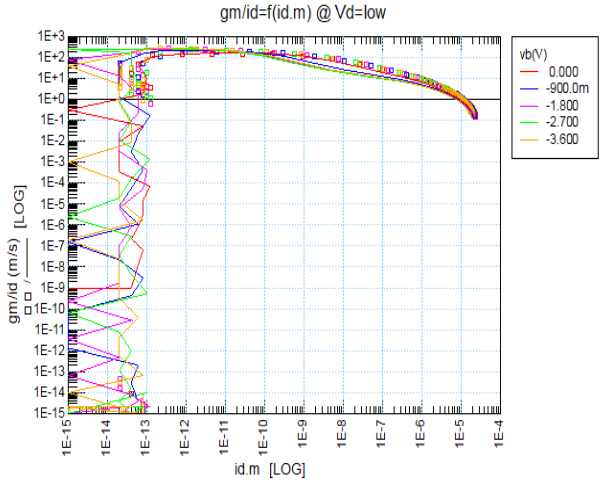


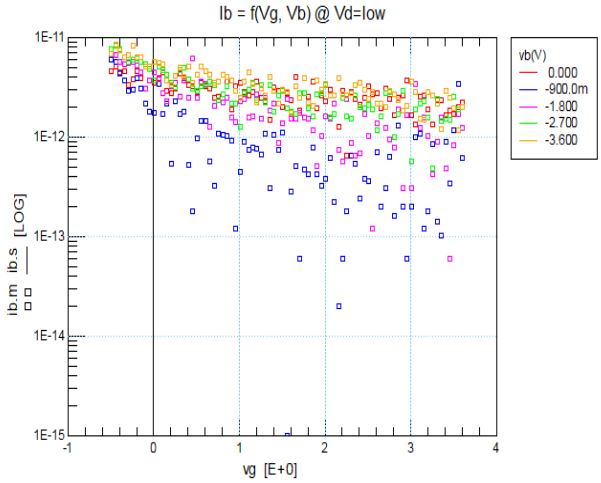


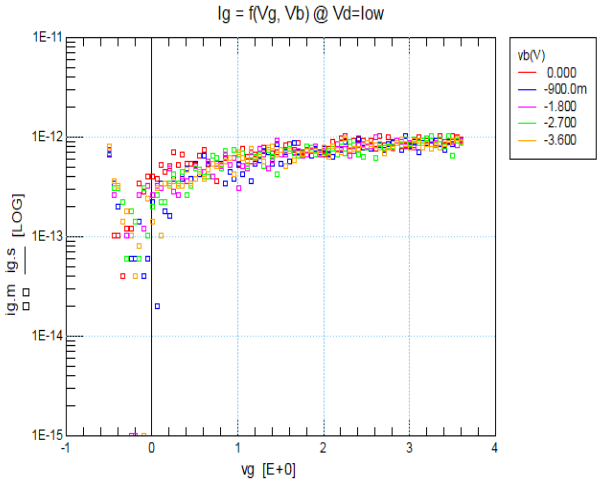




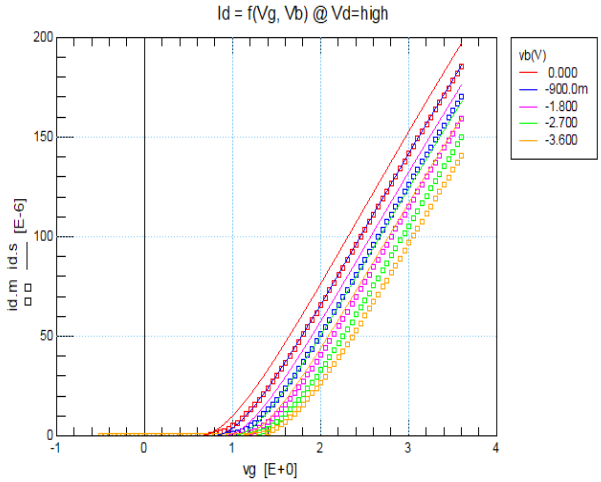


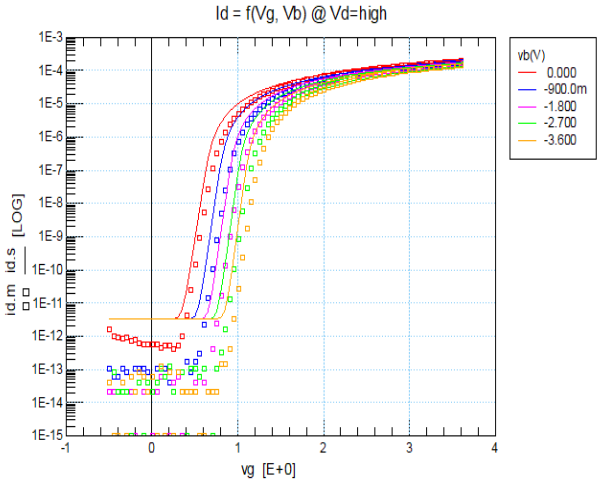


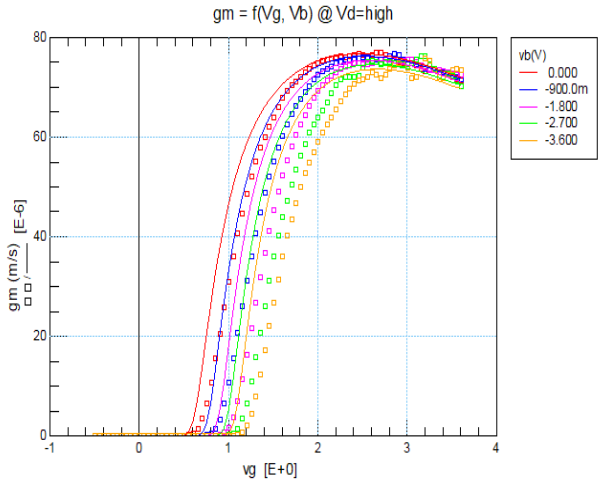


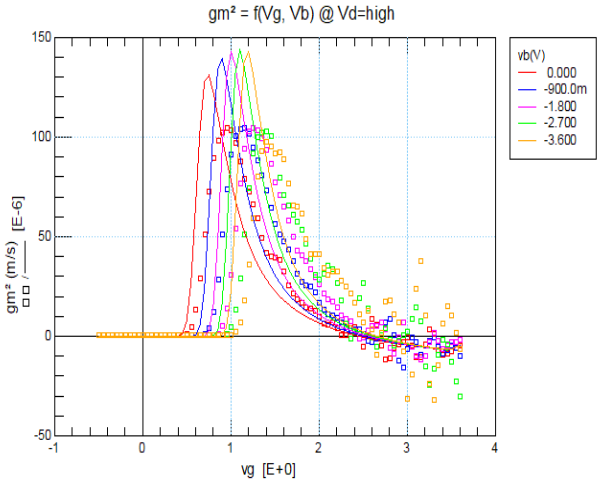


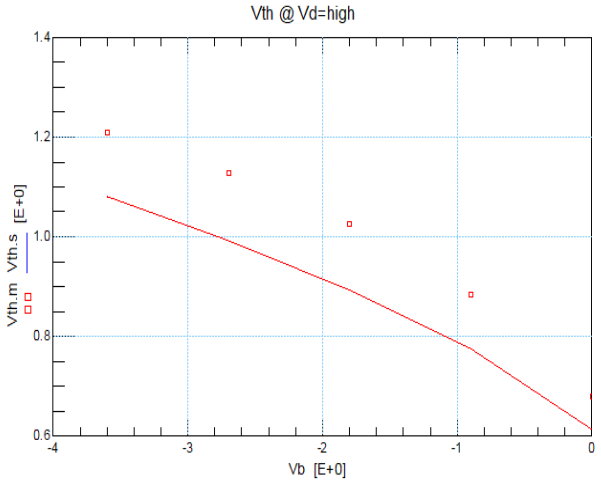
idvg, Vd = 3.3V, T = -40°C

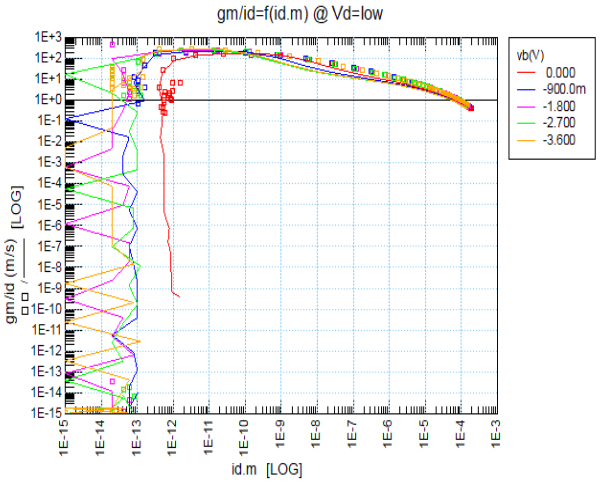


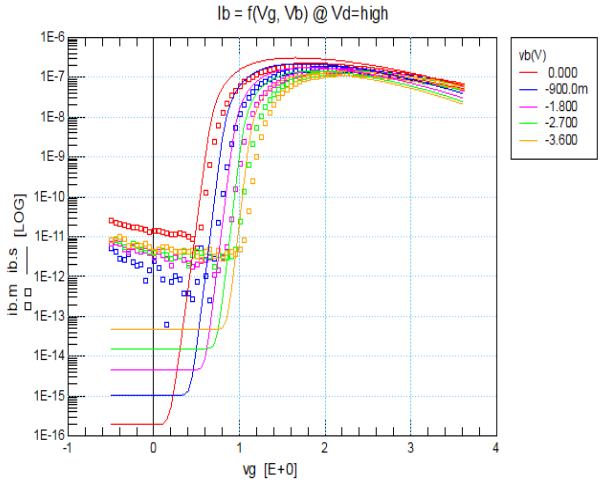


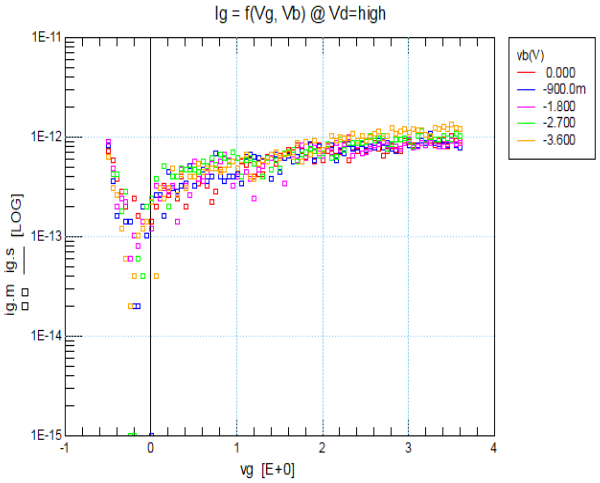




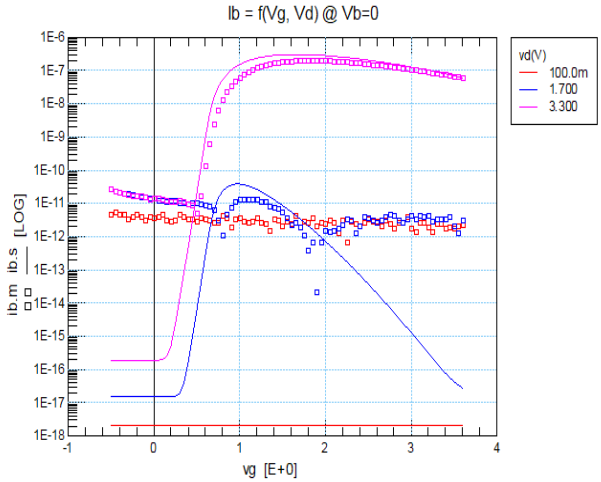


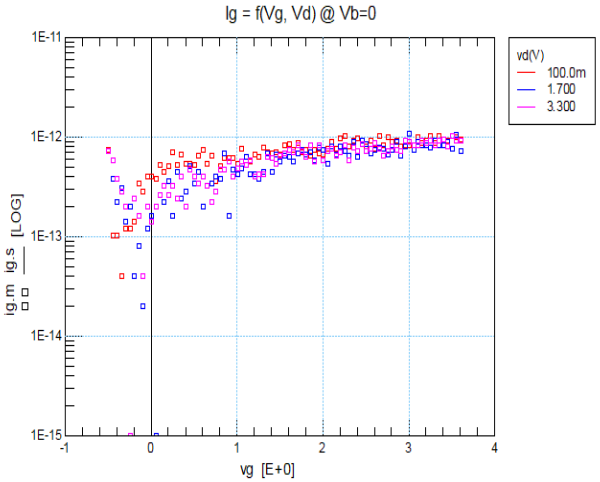




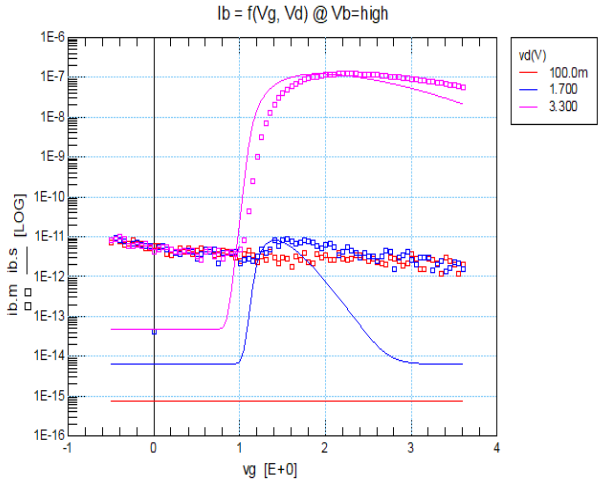


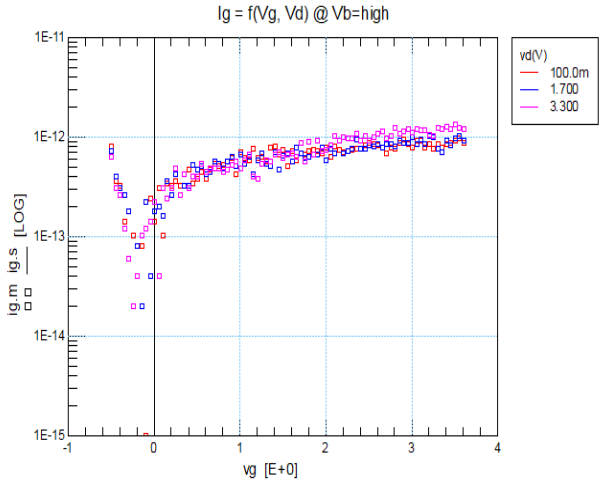
idvg, Ib,Ig, Vb = 0V, T = -40°C



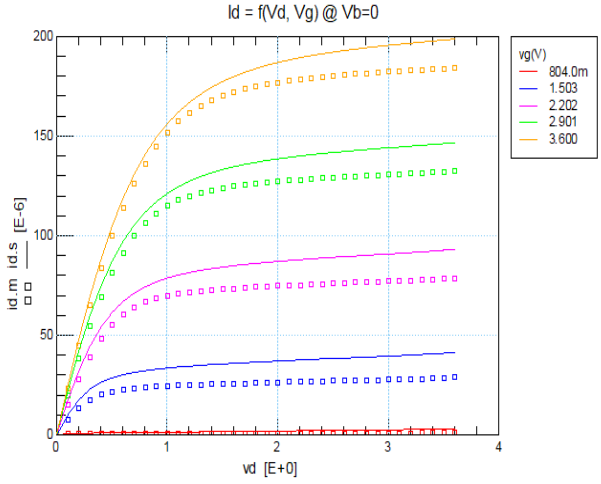


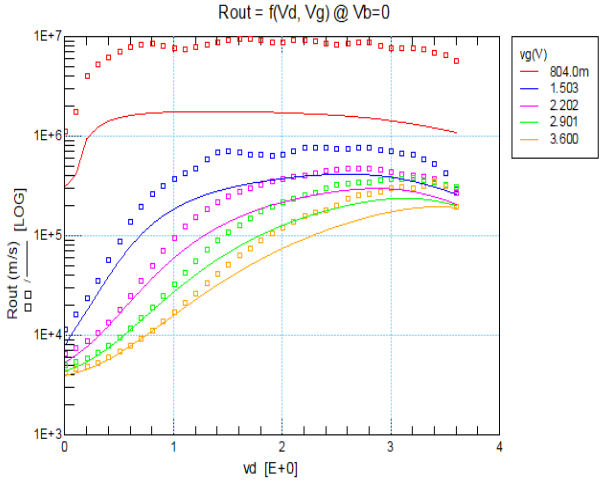
idvg, Ib,Ig, Vb = -3.6V, T = -40°C

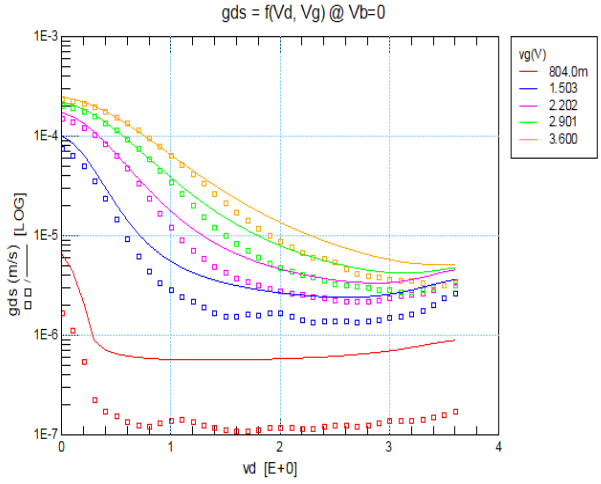




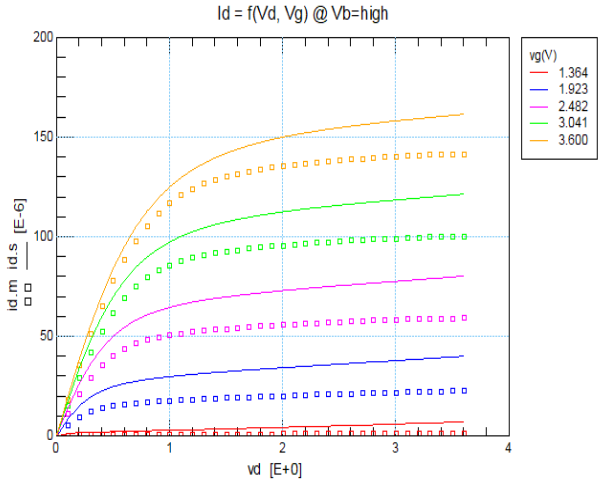
idvd, Vb = 0V, T = -40°C

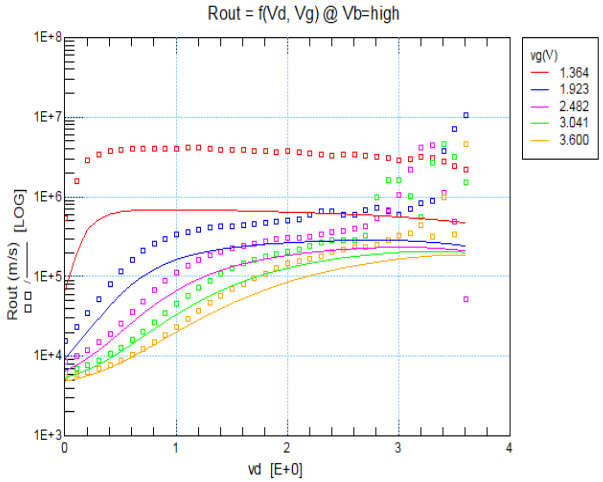


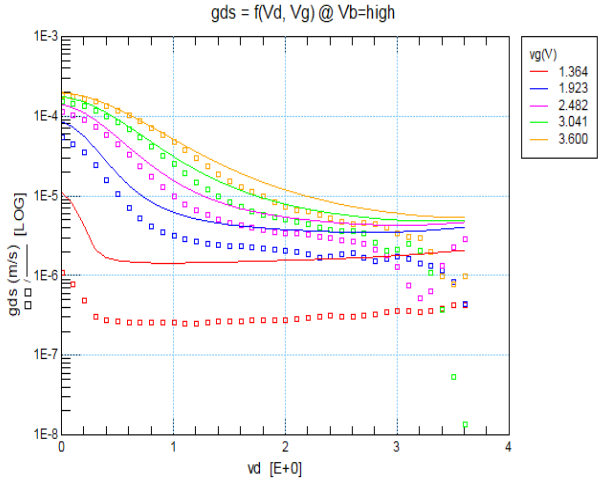




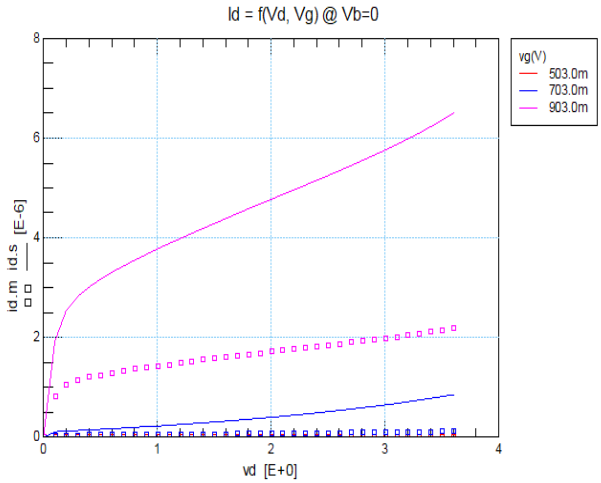
idvd_vbmin, Vb = -3.6V, T = -40°C

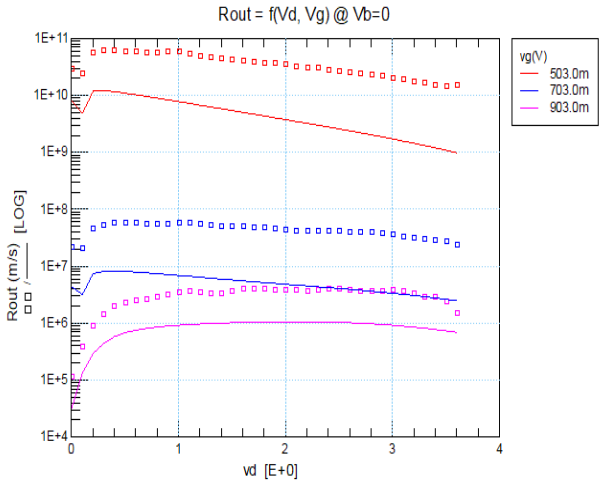


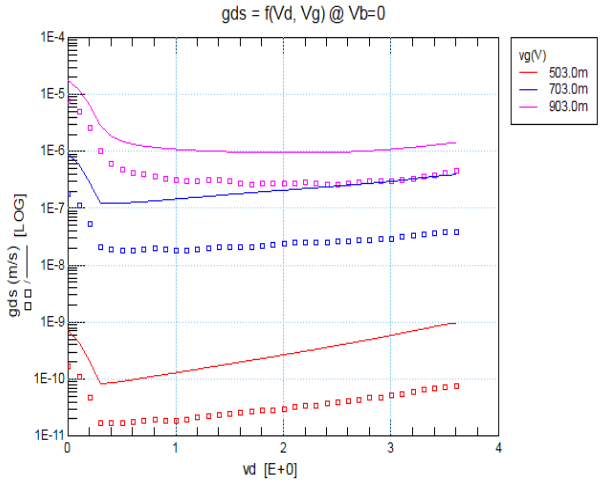




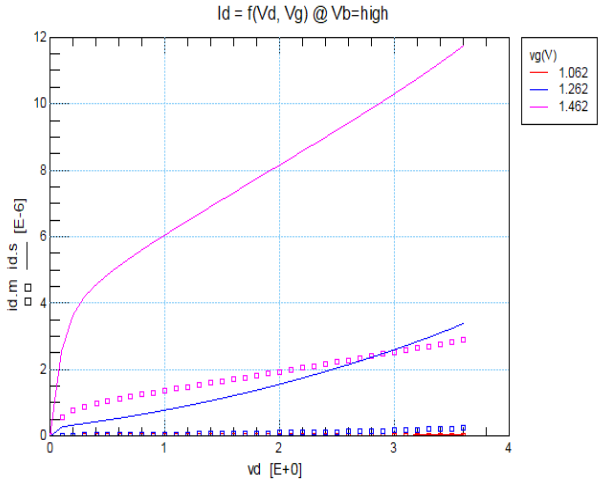
idvd_vth, Vb = 0V, T = -40°C

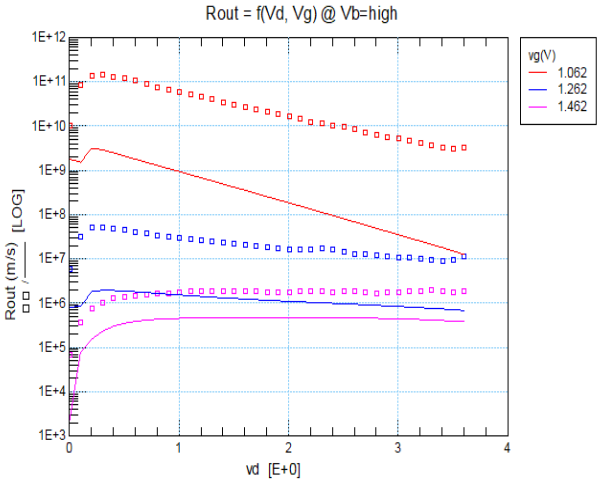


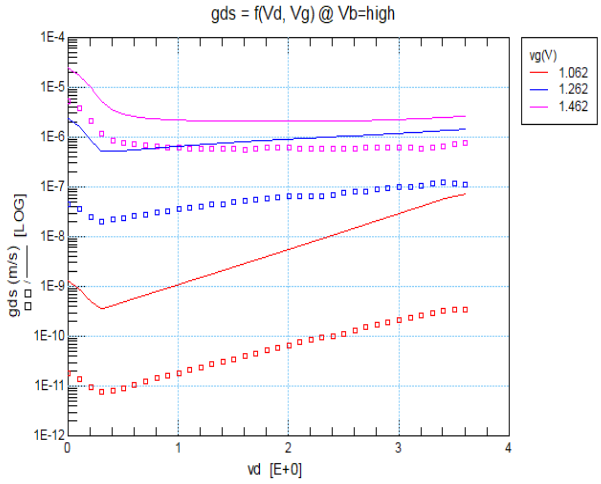




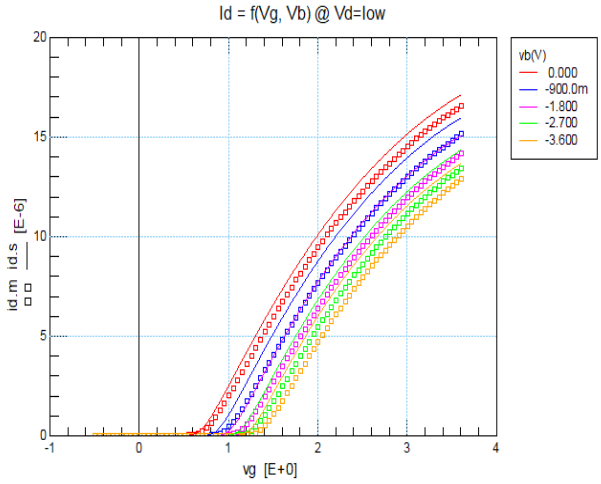
idvd_vth_vbmin, Vb = -3.6V, T = -40°C

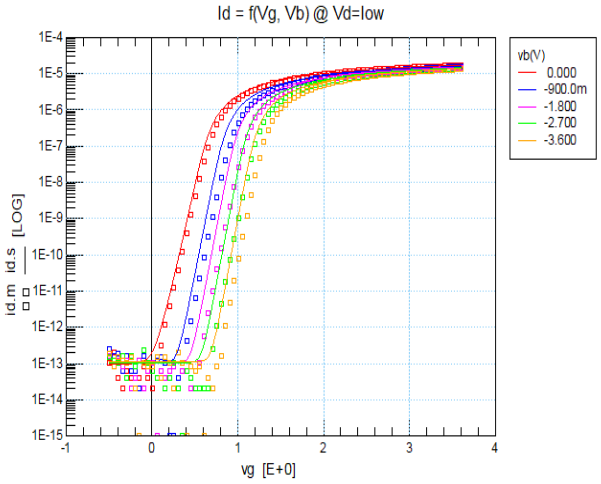


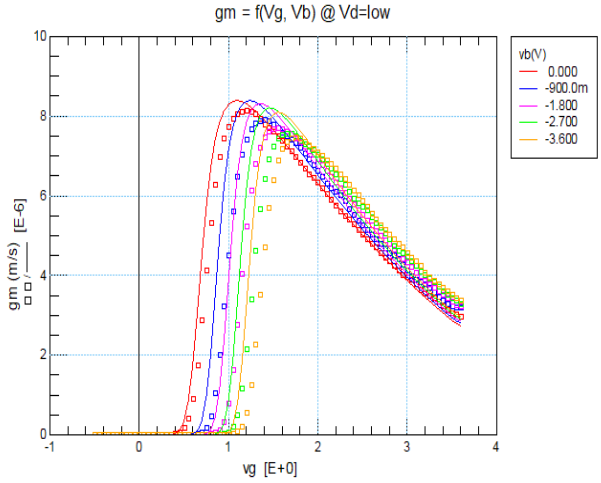


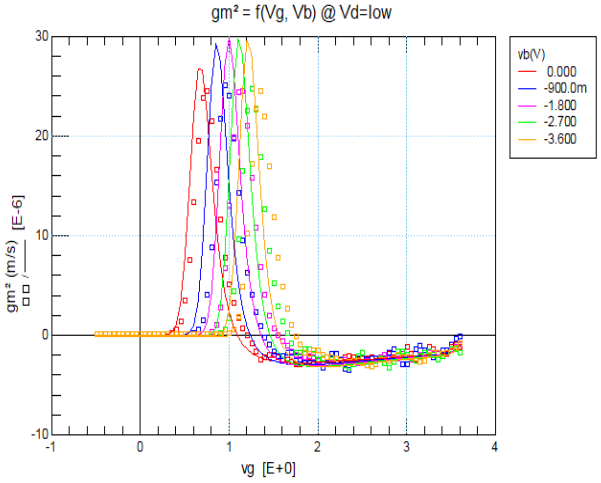


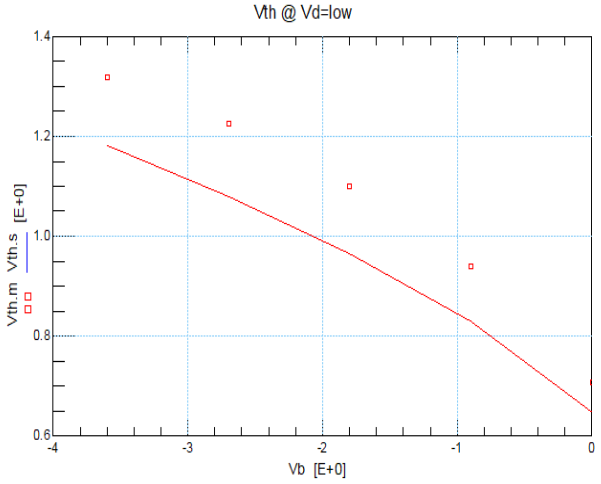
idvg, Vd = 0.1V, T = 70°C

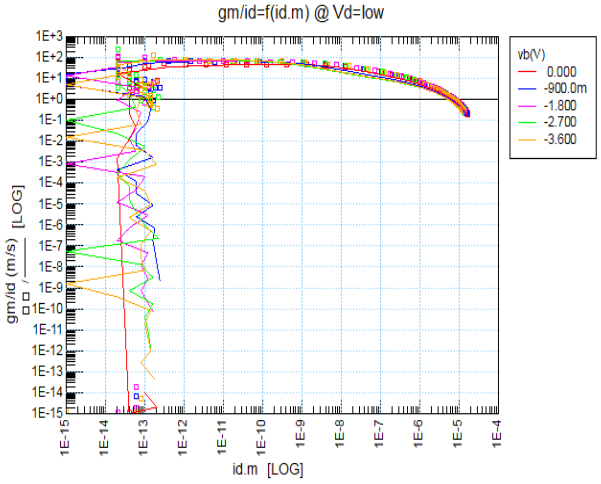


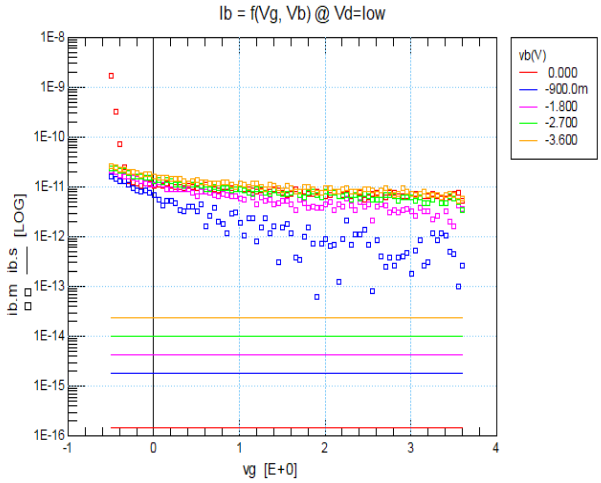


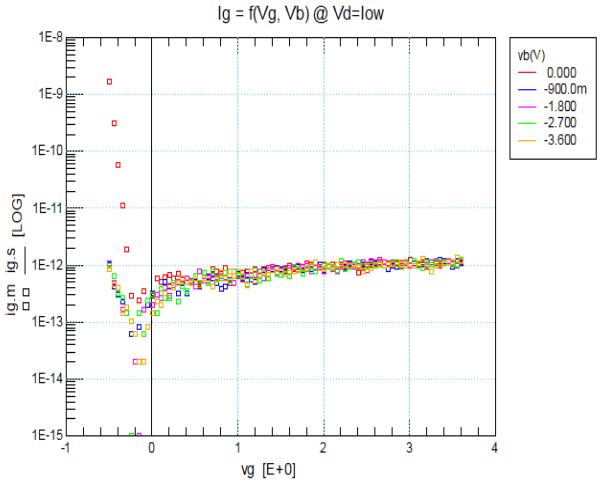




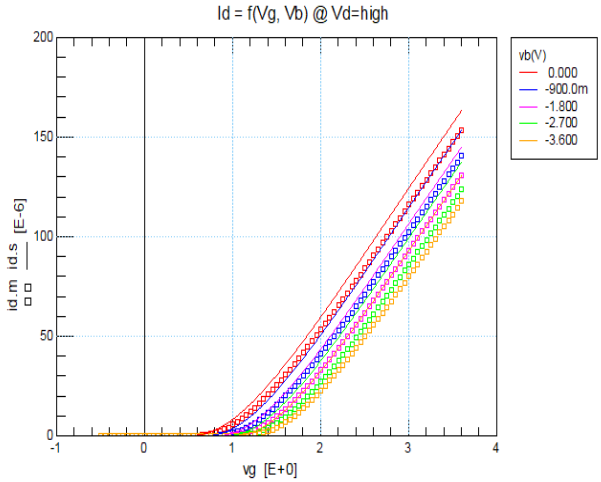


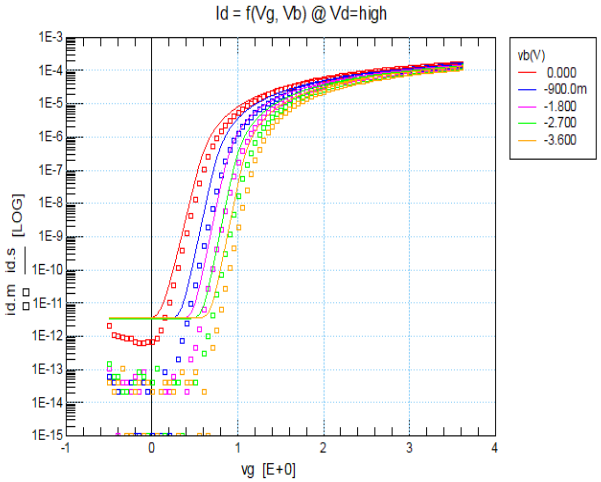


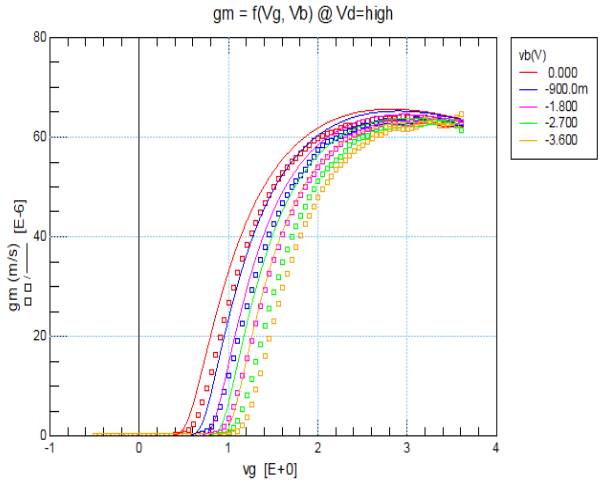


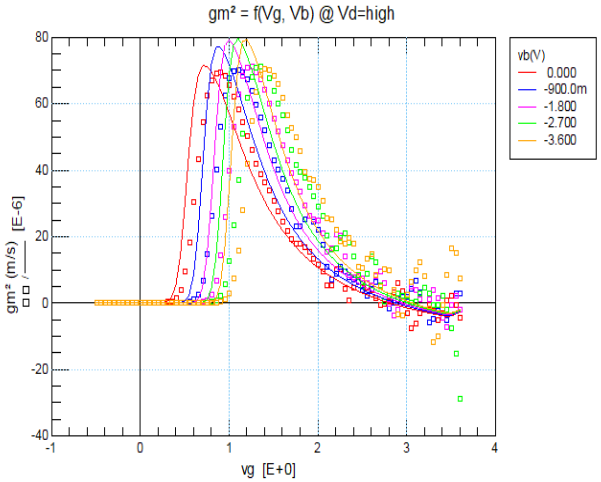


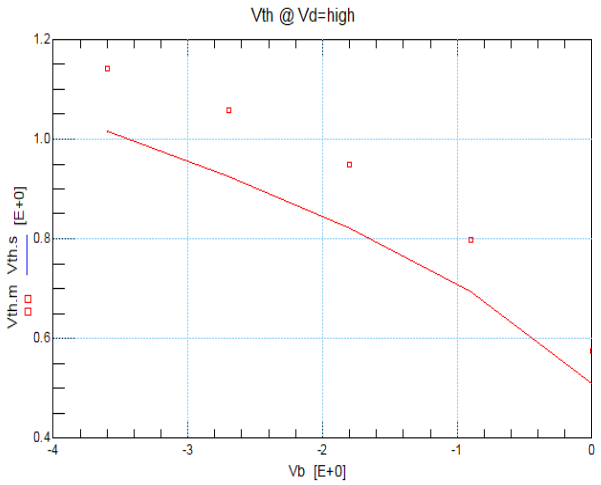
idvg, Vd = 3.3V, T = 70°C

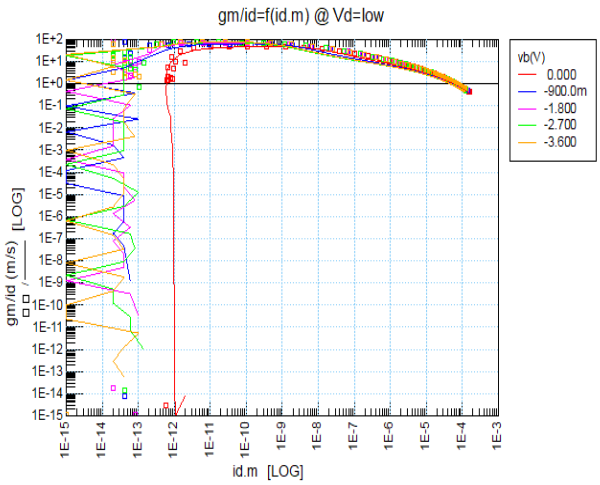


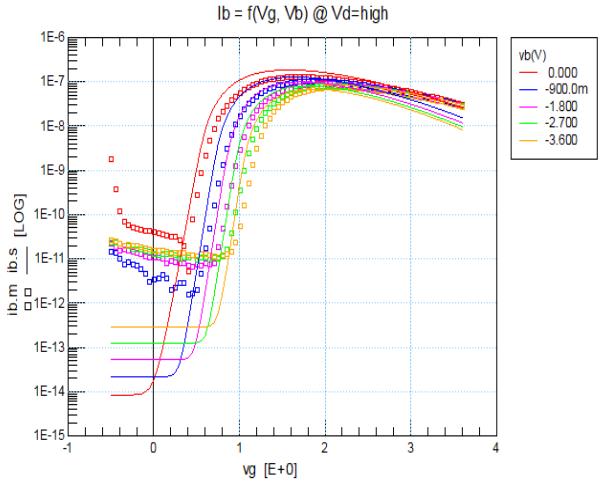


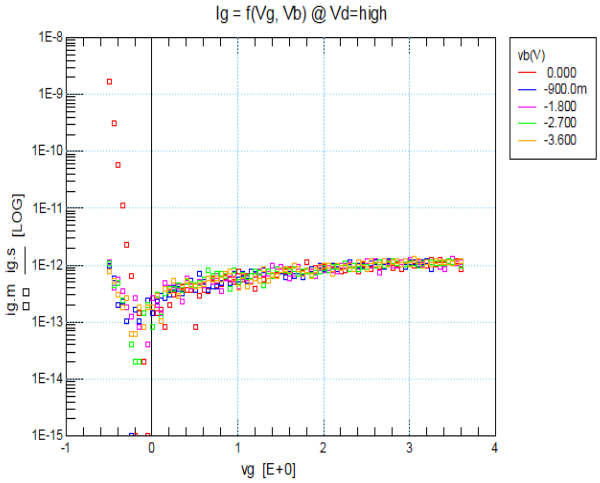




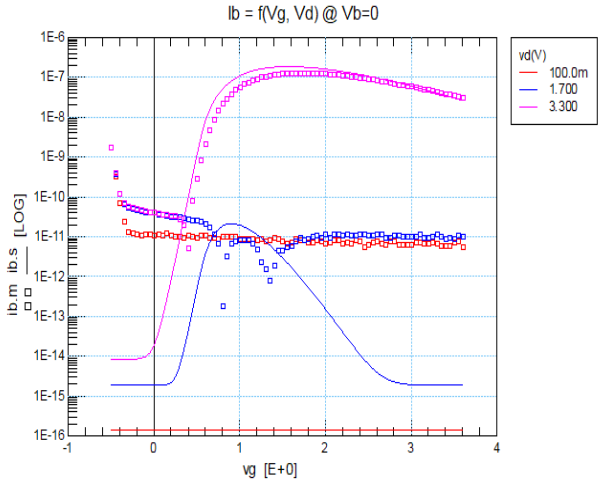


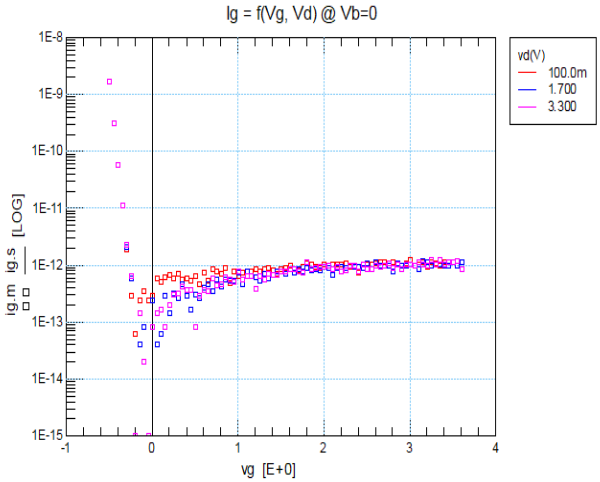




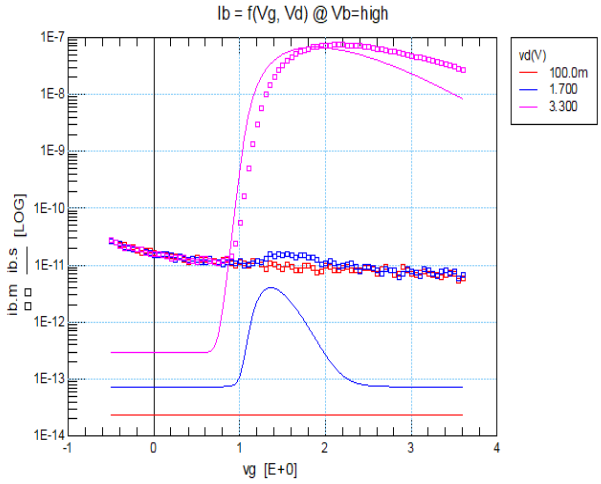


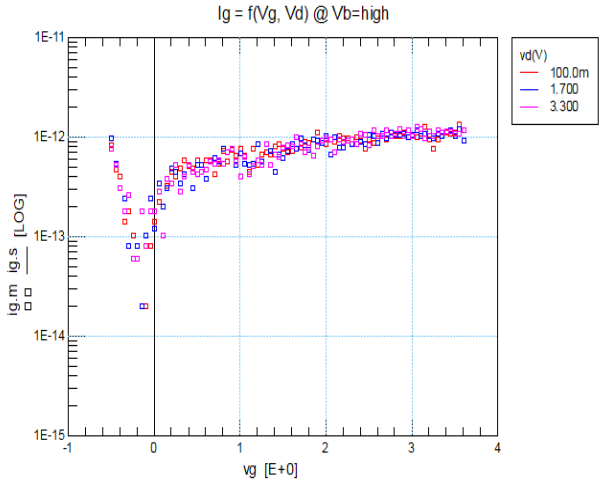
idvg, Ib,Ig, Vb = 0V, T = 70°C



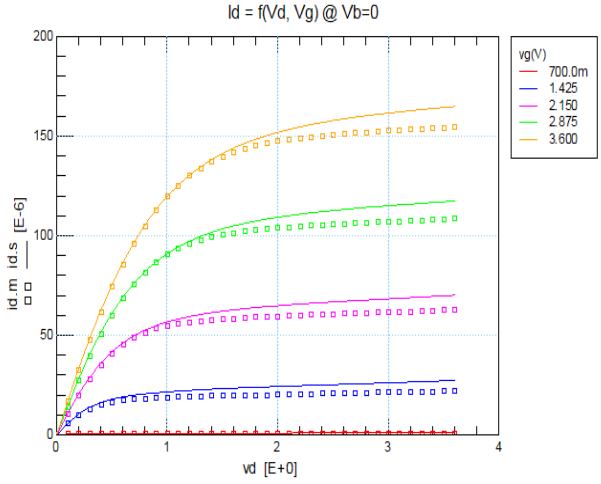


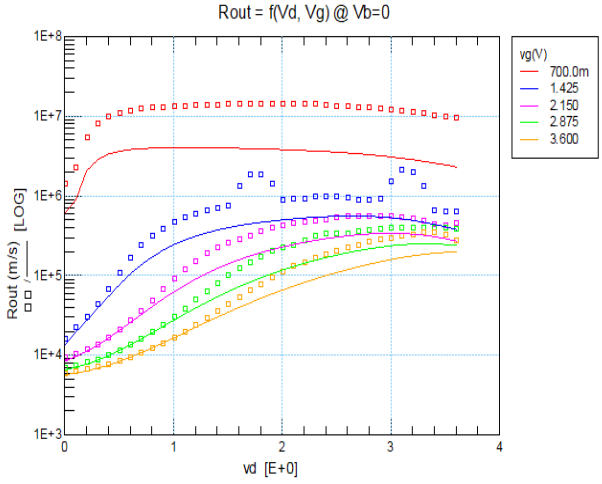
idvg, Ib,Ig, Vb = -3.6V, T = 70°C

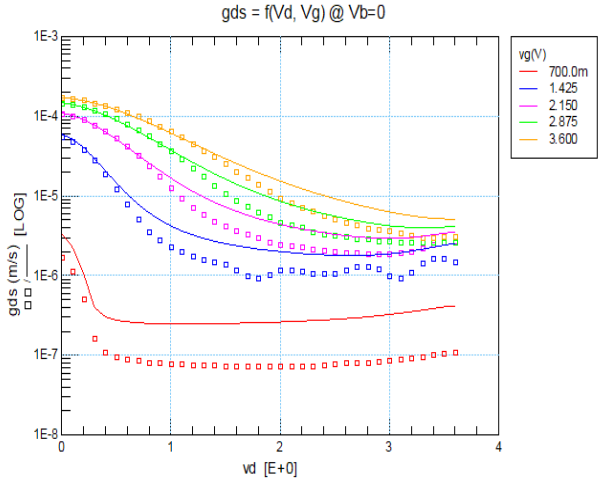




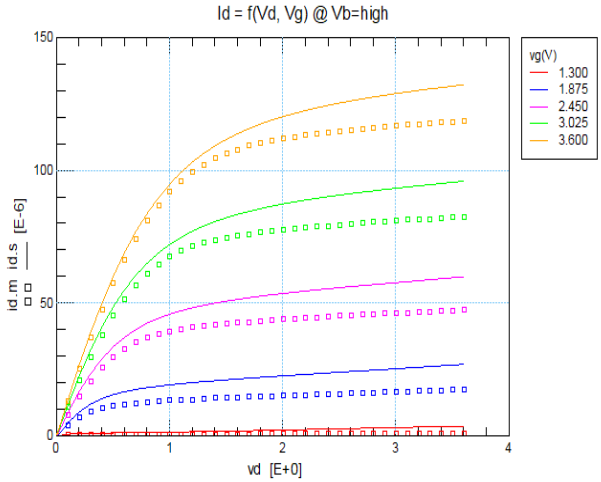
idvd, Vb = 0V, T = 70°C

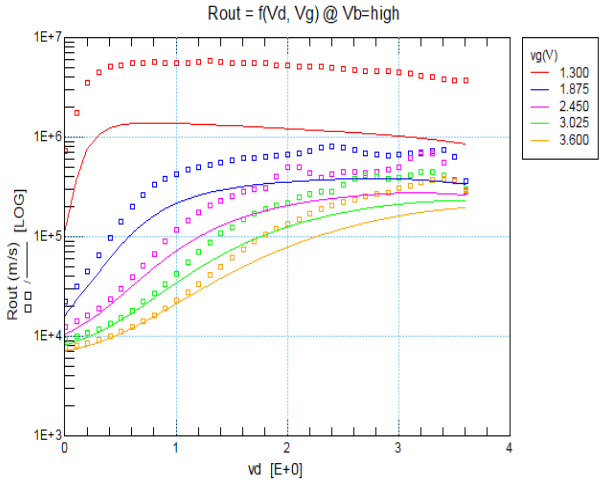


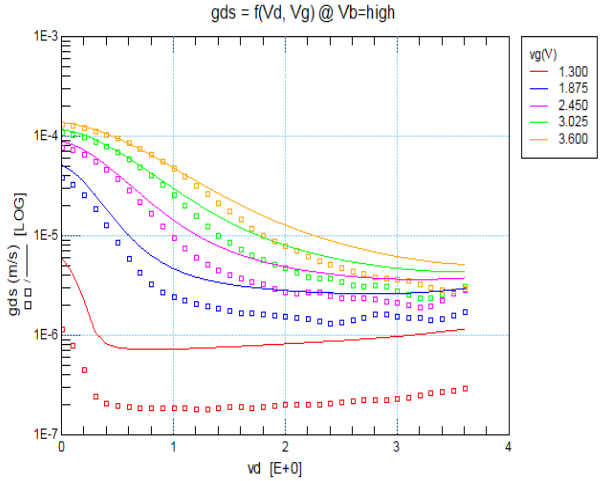




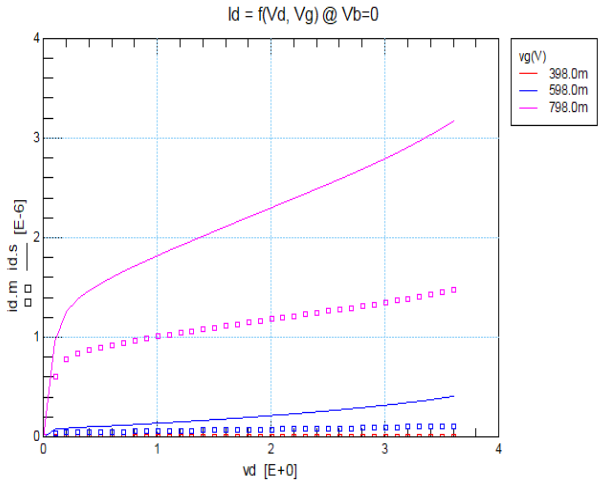
idvd_vbmin, Vb = -3.6V, T = 70°C

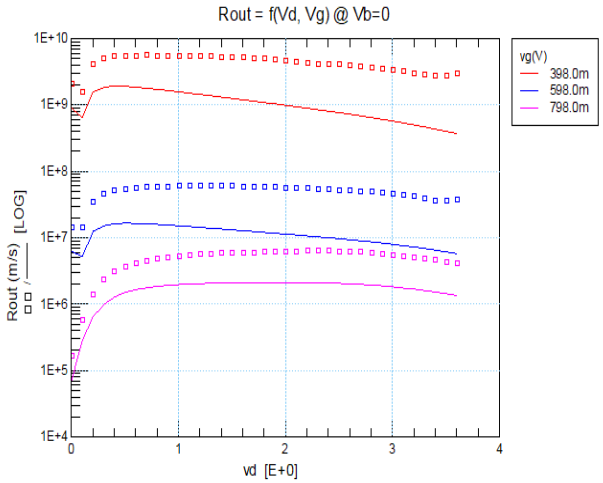


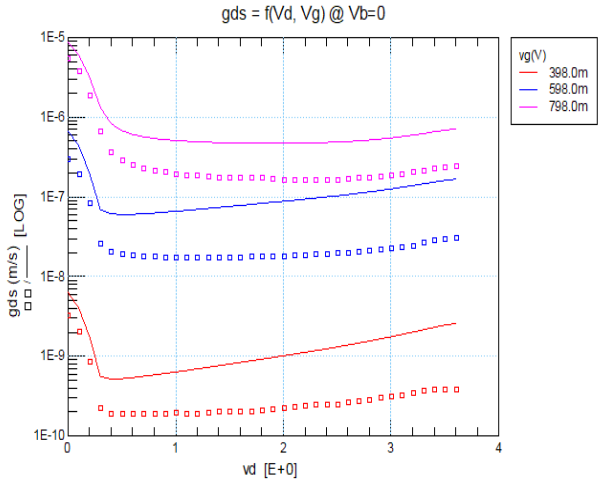




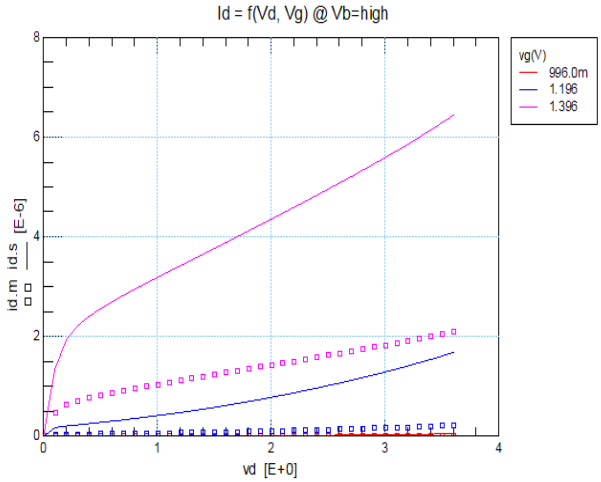
idvd_vth, Vb = 0V, T = 70°C

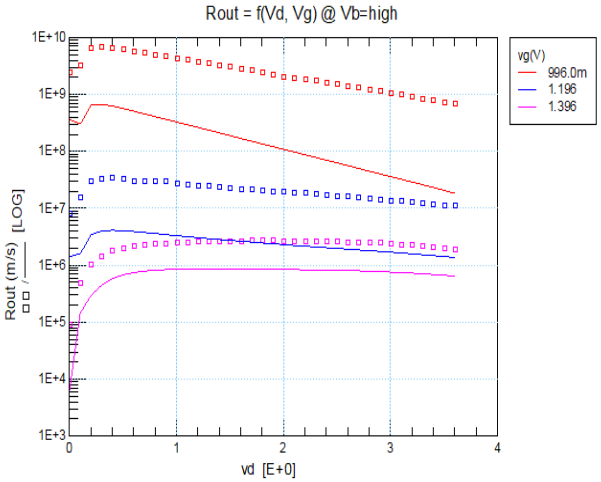


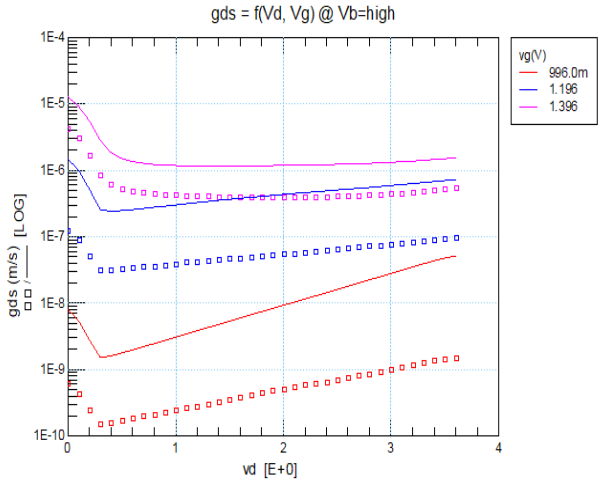




idvd_vth_vbmin, Vb = -3.6V, T = 70°C

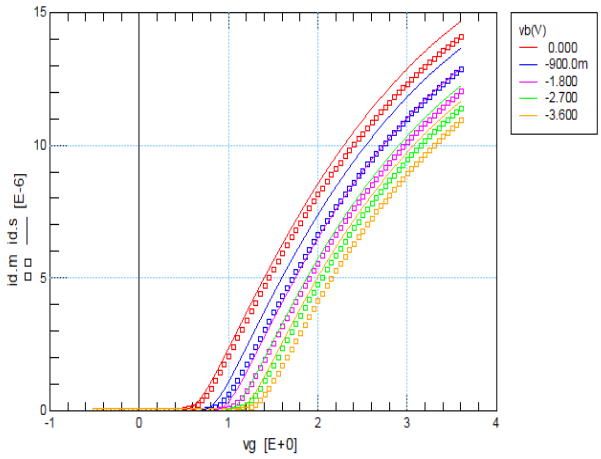




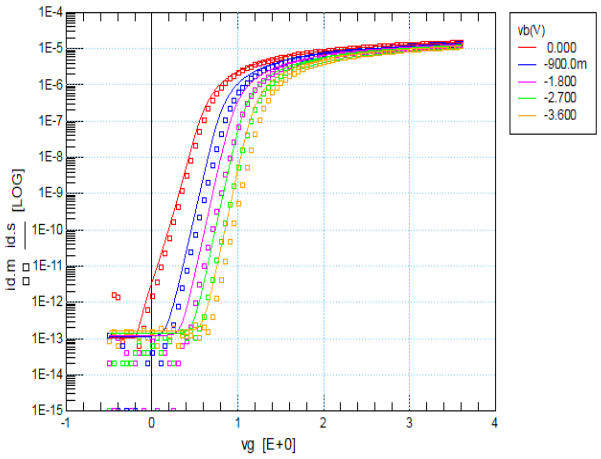


idvg, Vd = 0.1V, T = 125°C

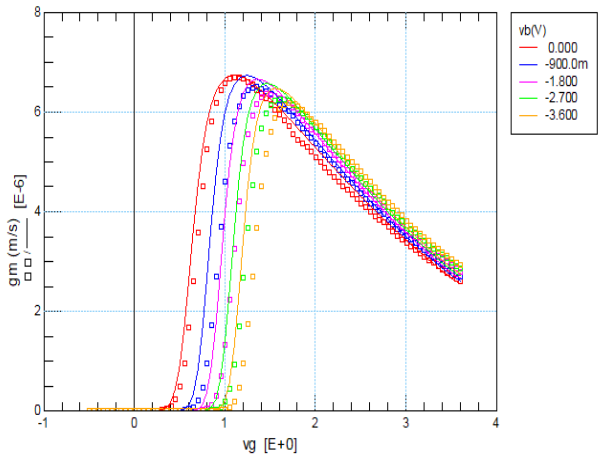
Id = f(Vg, Vb) @ Vd=low



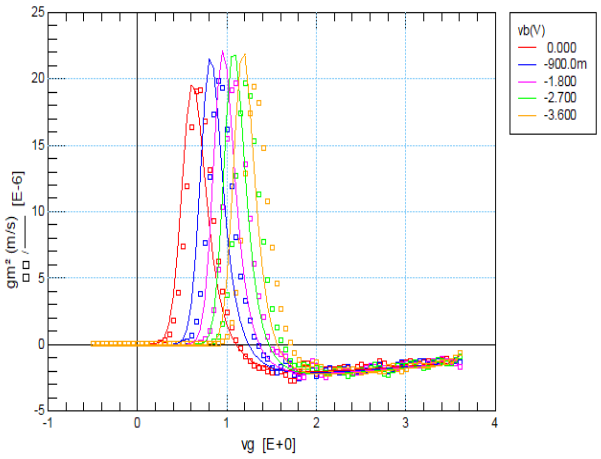
Id = f(Vg, Vb) @ Vd=low



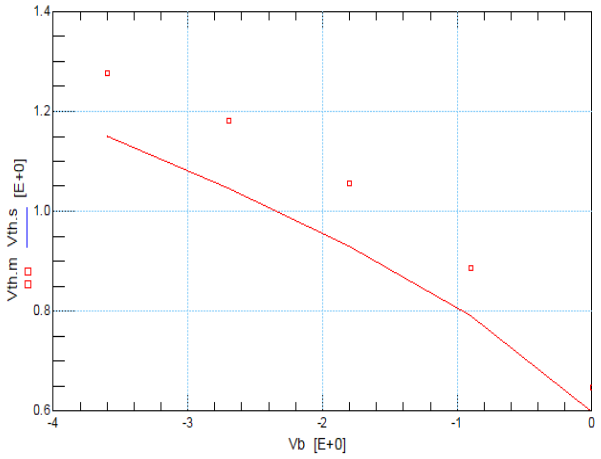
gm = f(Vg, Vb) @ Vd=low



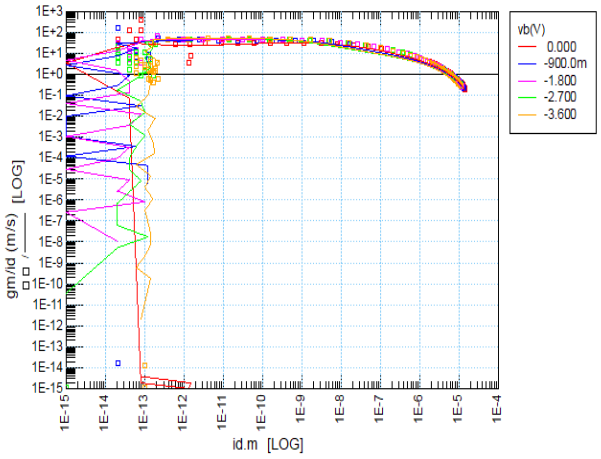
gm² = f(Vg, Vb) @ Vd=low

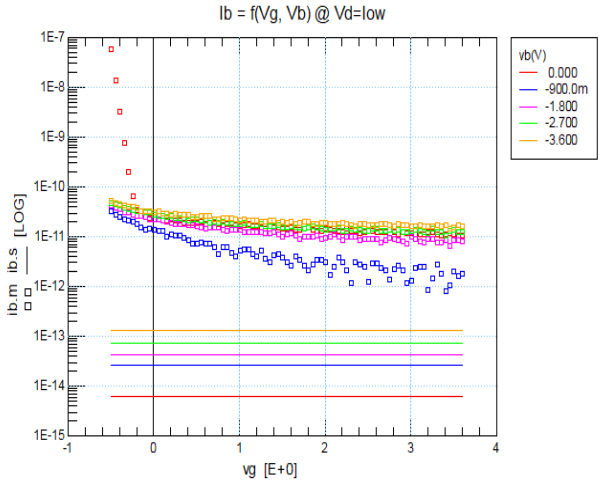


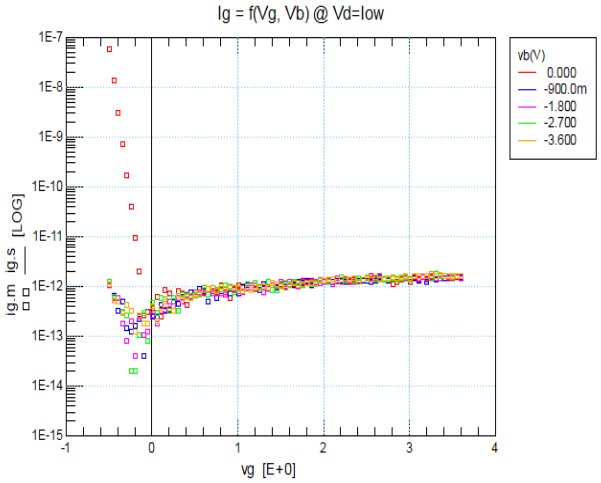
Vth @ Vd=low



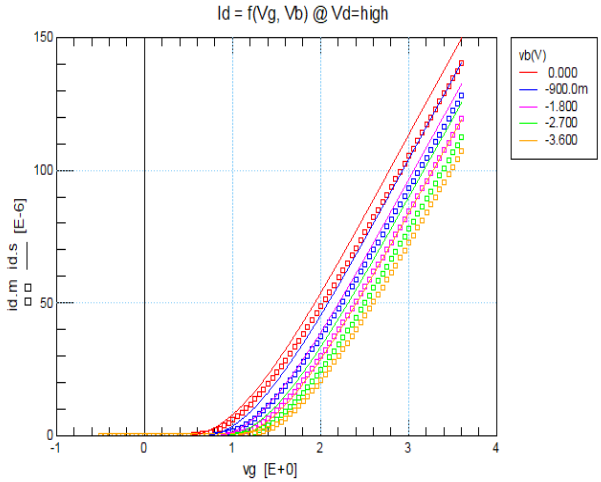
gm/id=f(id.m) @ Vd=low

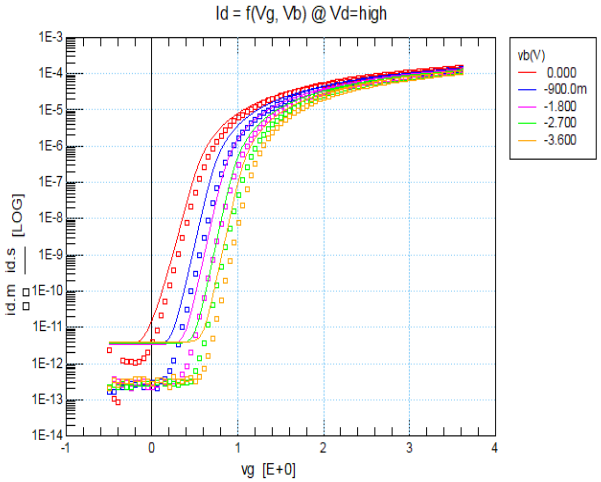


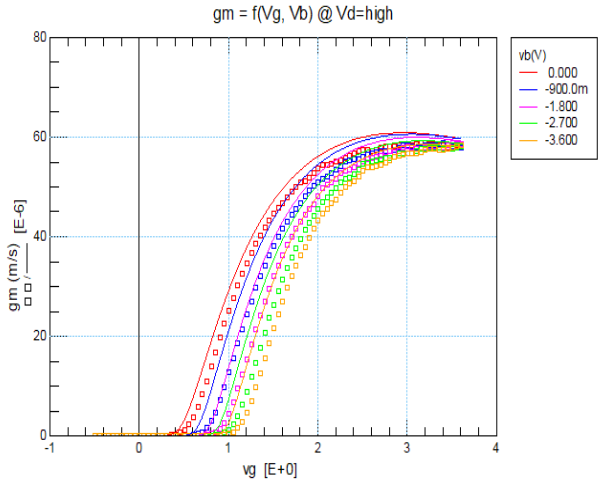


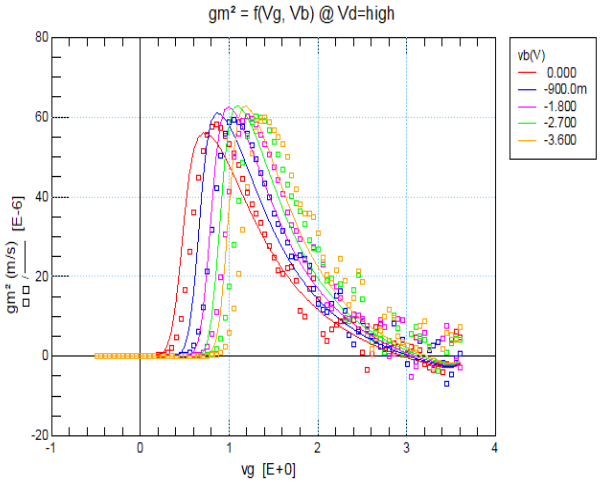


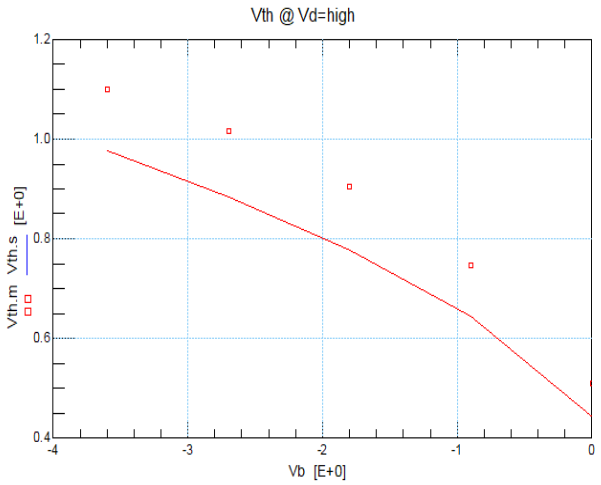
idvg, Vd = 3.3V, T = 125°C

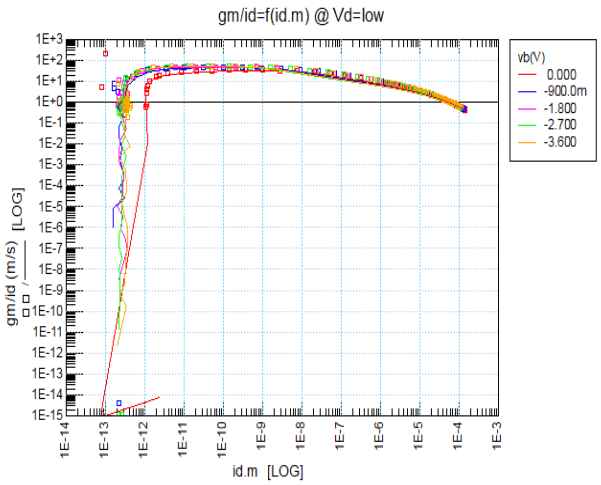


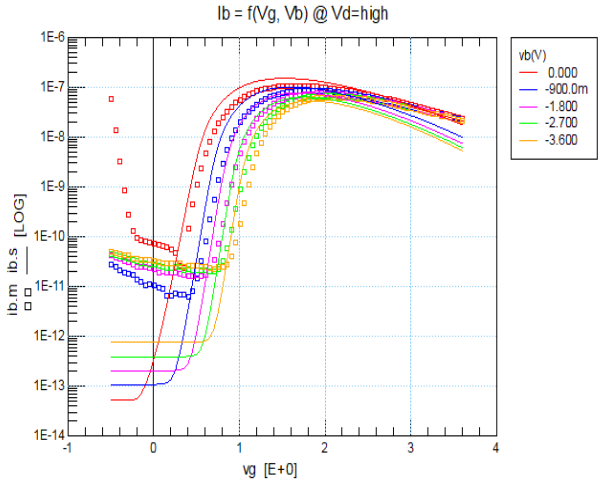


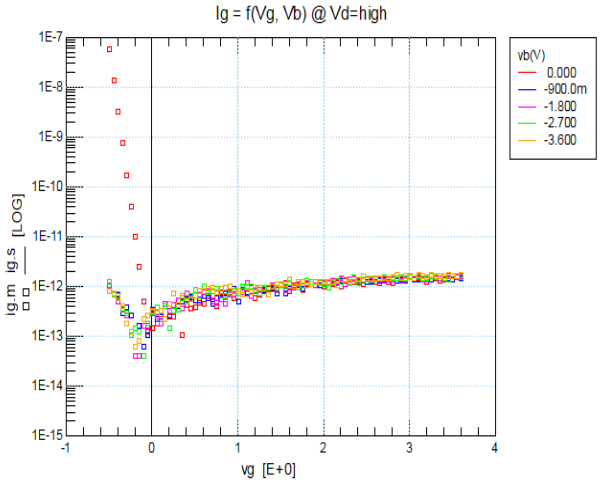




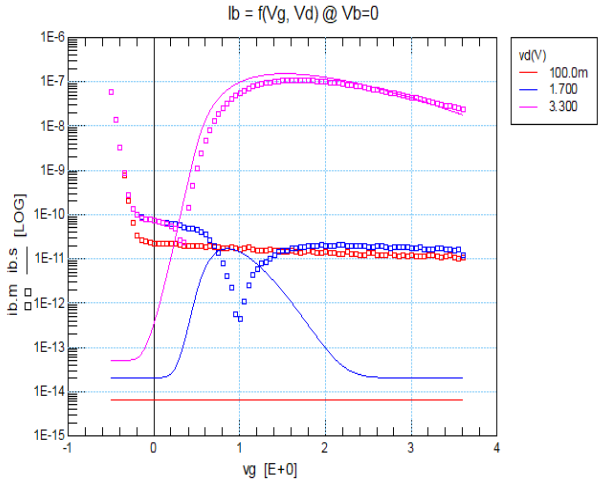


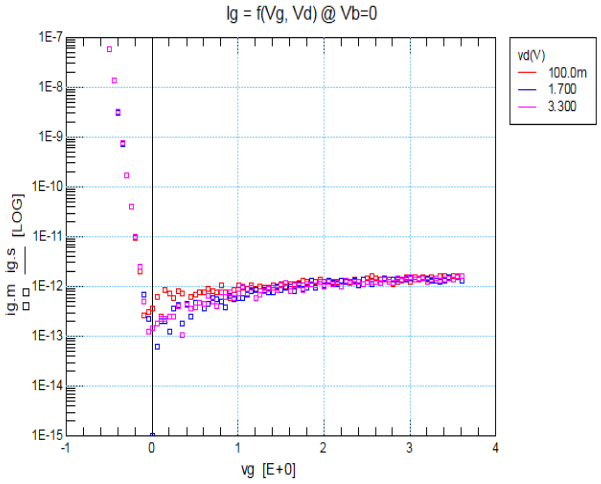




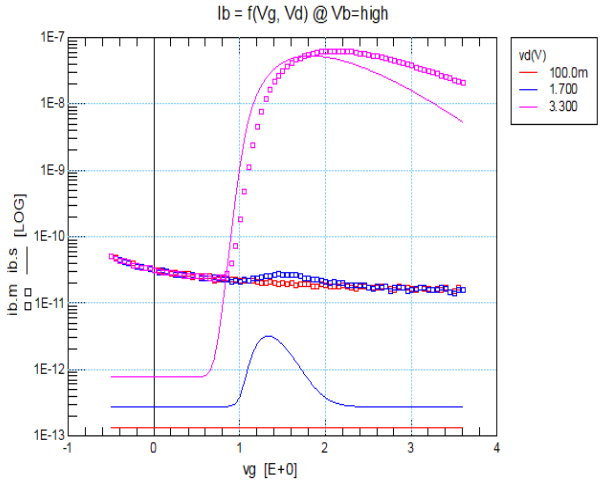


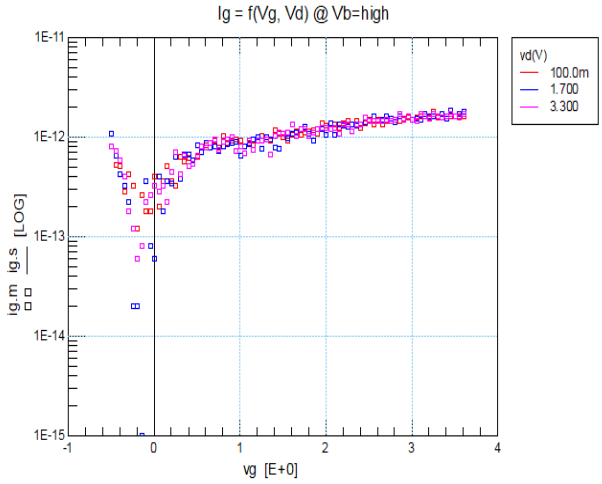
idvg, Ib,Ig, Vb = 0V, T = 125°C



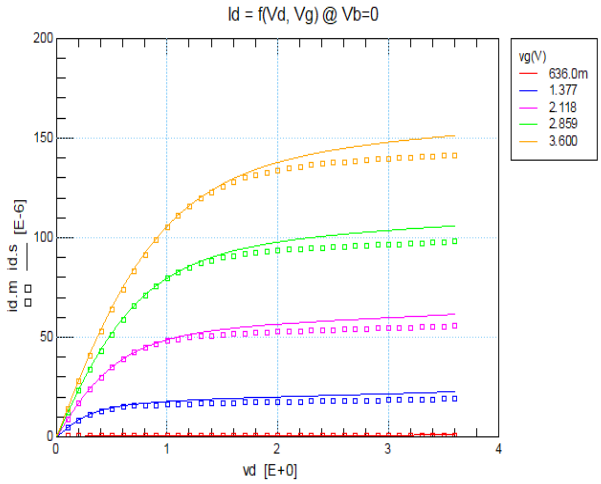


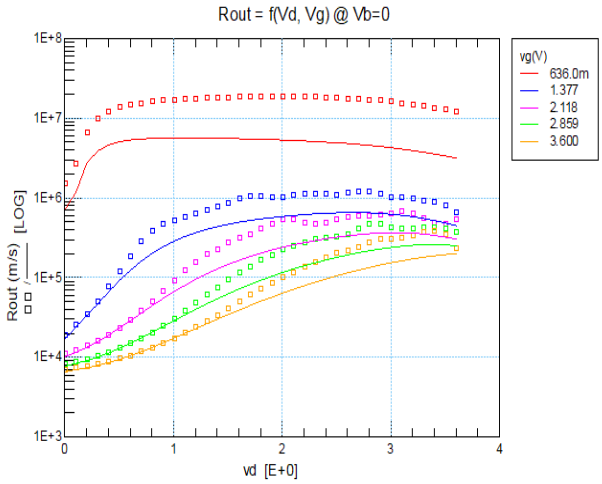
idvg, Ib,Ig, Vb = -3.6V, T = 125°C

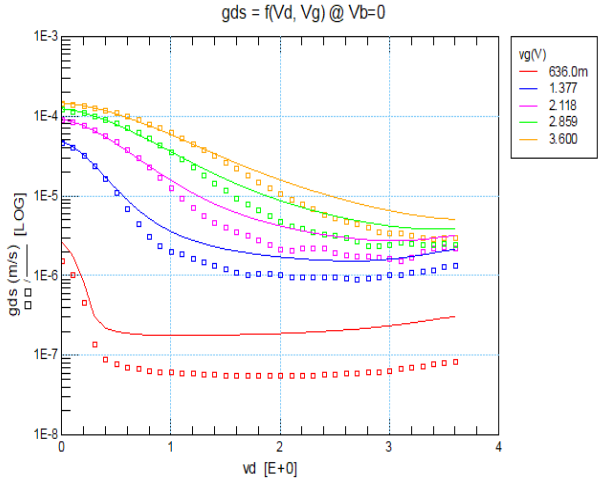




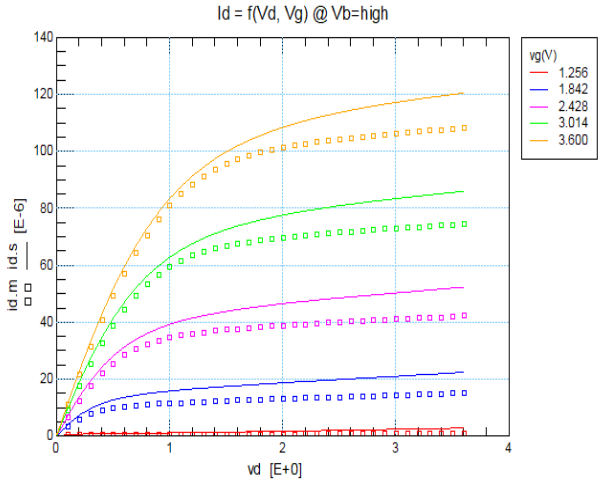
idvd, Vb = 0V, T = 125°C

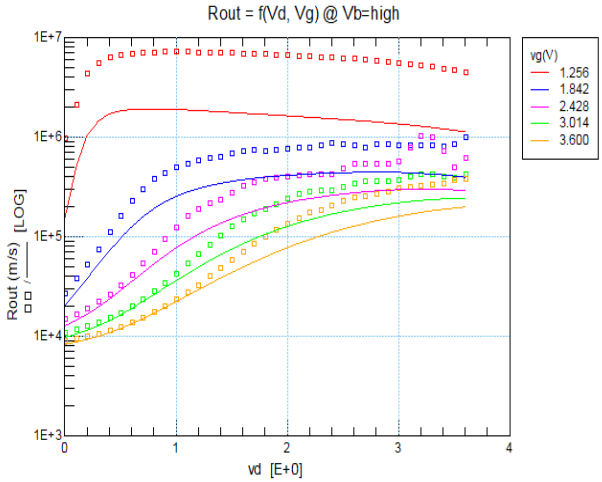


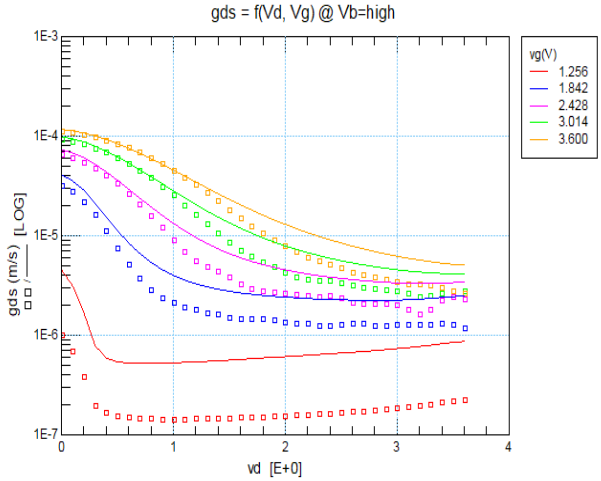




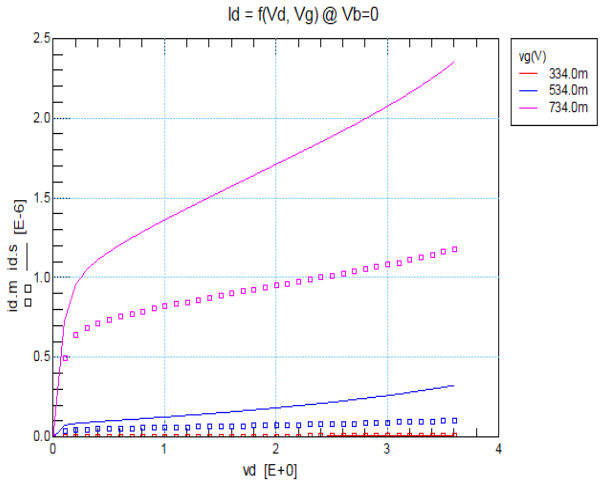
idvd_vbmin, Vb = -3.6V, T = 125°C

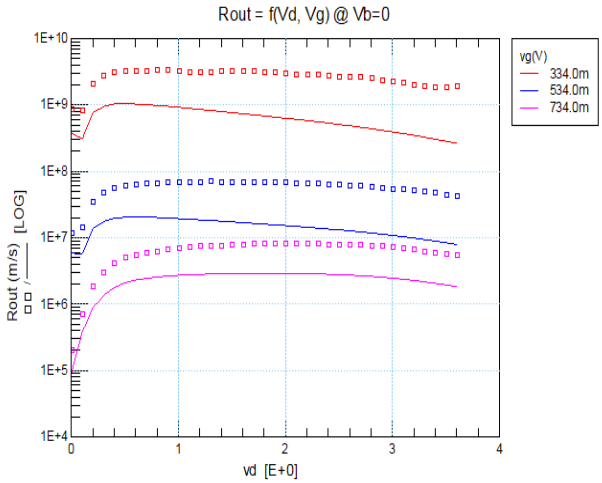


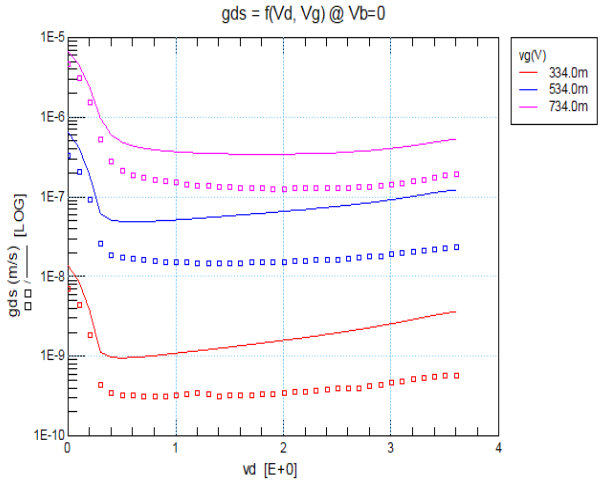




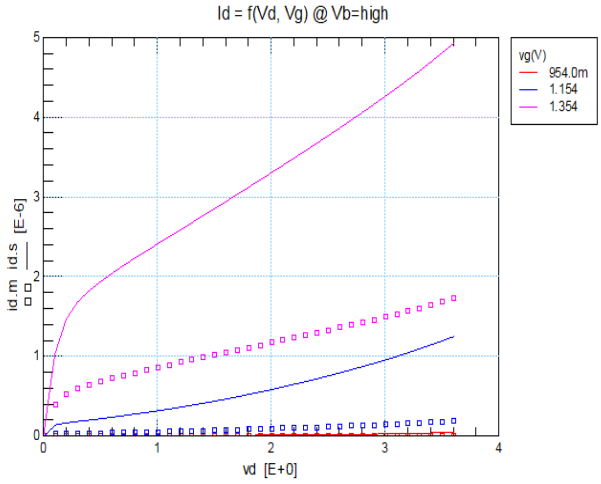
idvd_vth, Vb = 0V, T = 125°C

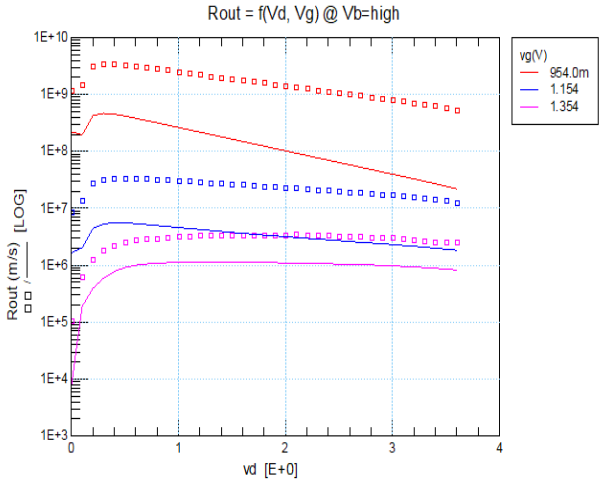


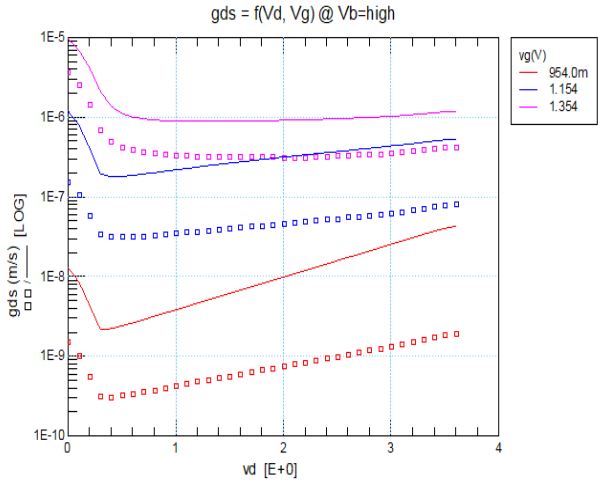




idvd_vth_vbmin, Vb = -3.6V, T = 125°C

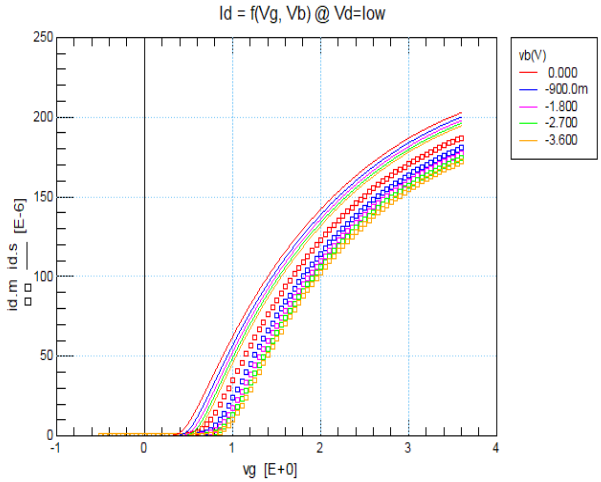


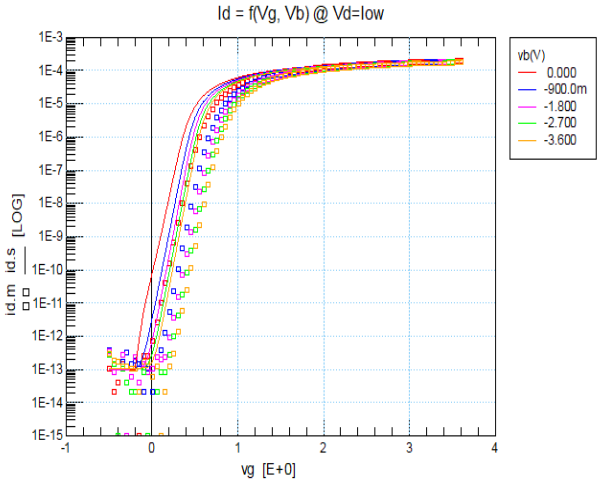


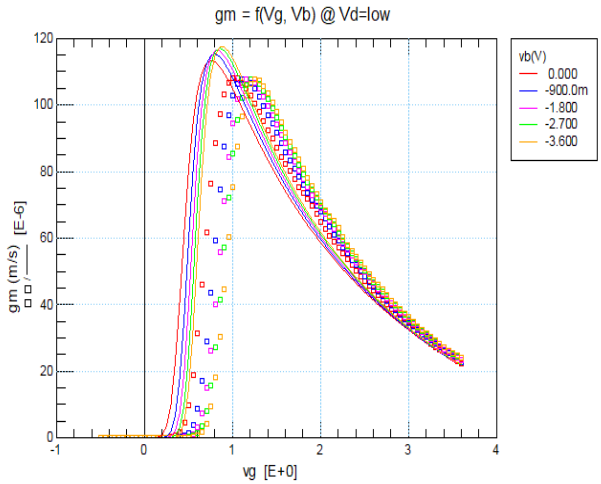


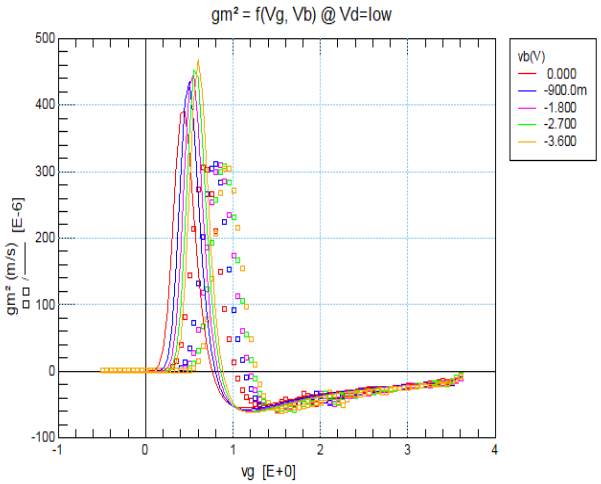
4.23 W02u0_L0u33_S558_2

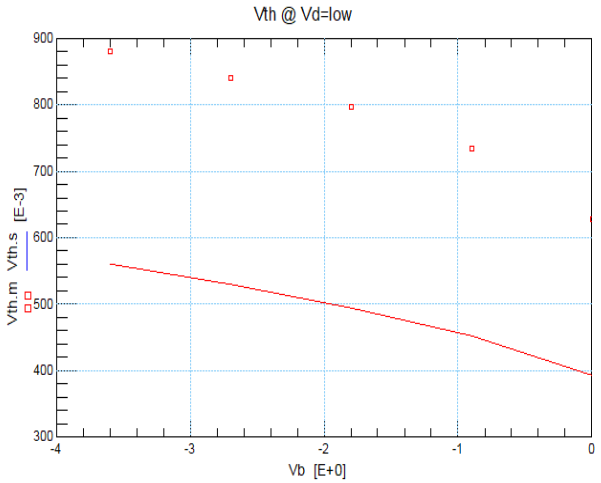
idvg, Vd = 0.1V, T = 27°C

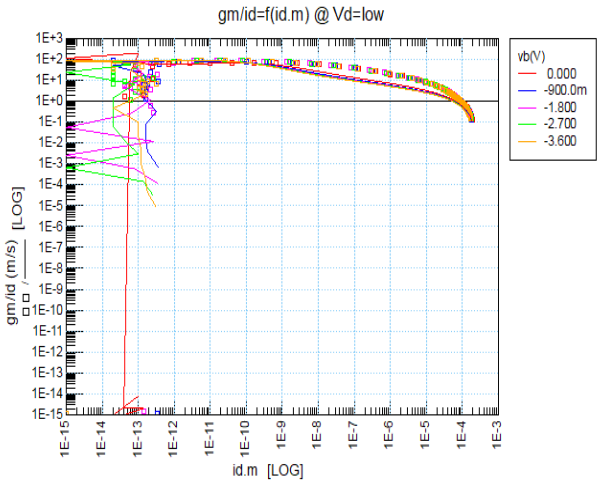


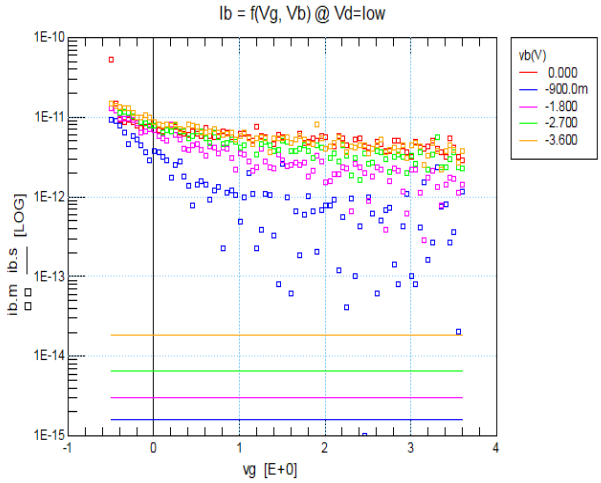


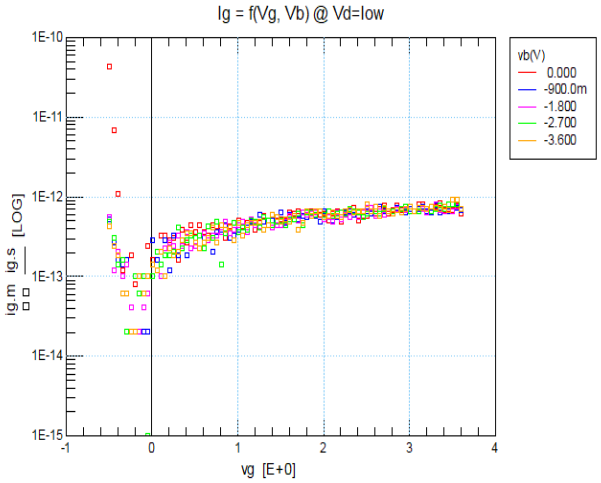




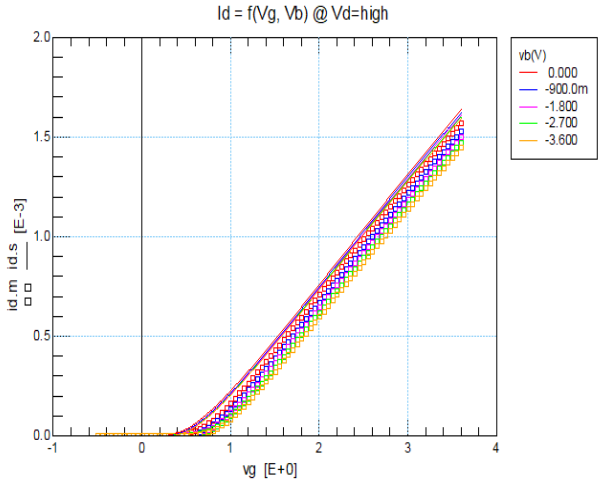


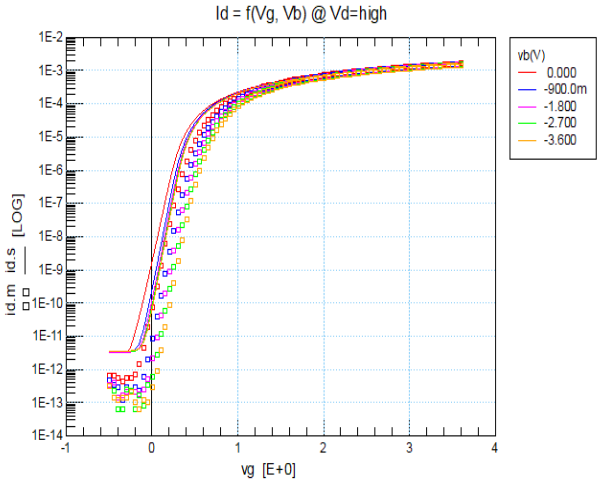


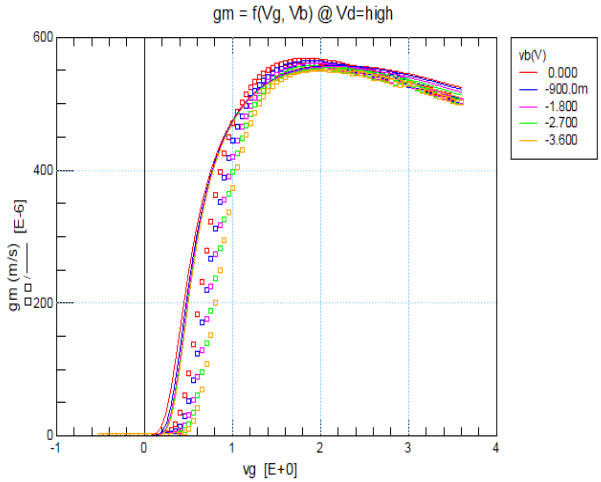


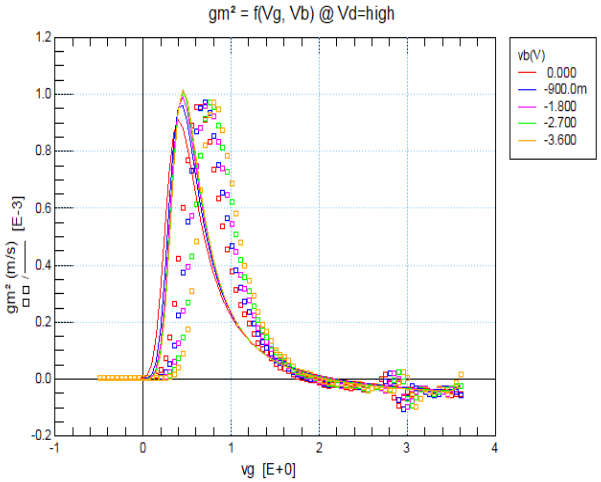


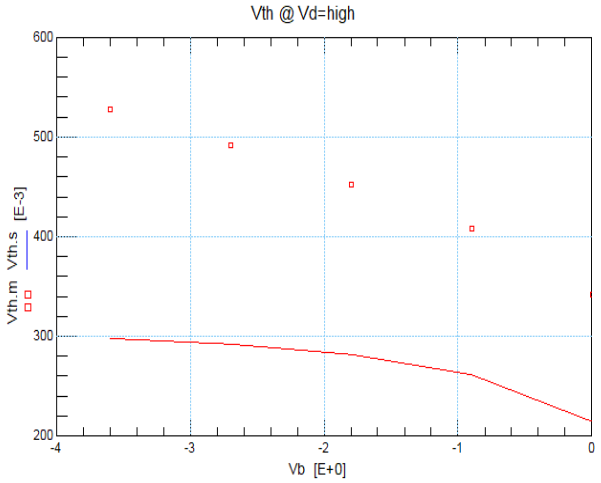
idvg, Vd = 3.3V, T = 27°C

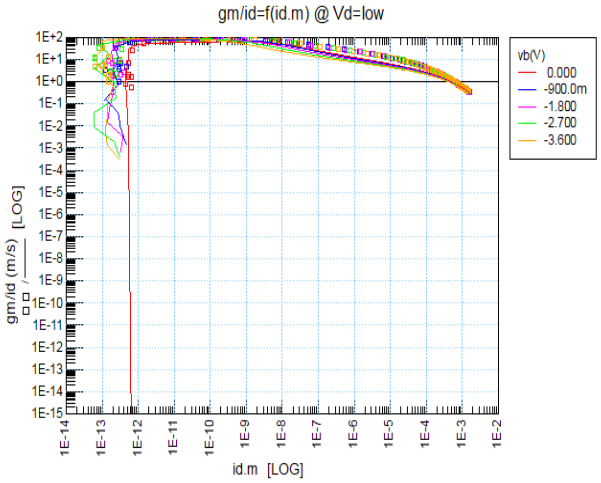


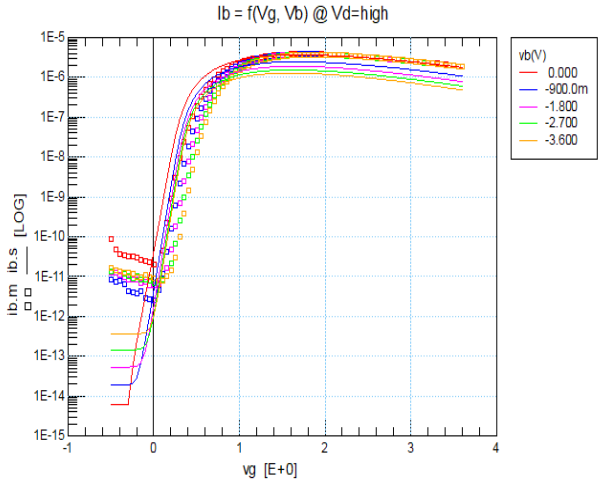


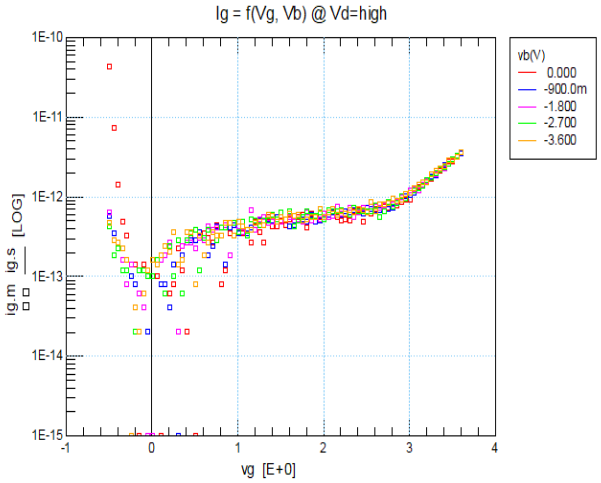




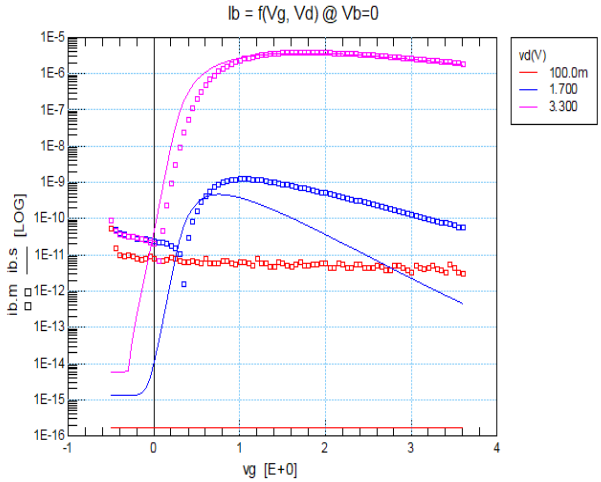


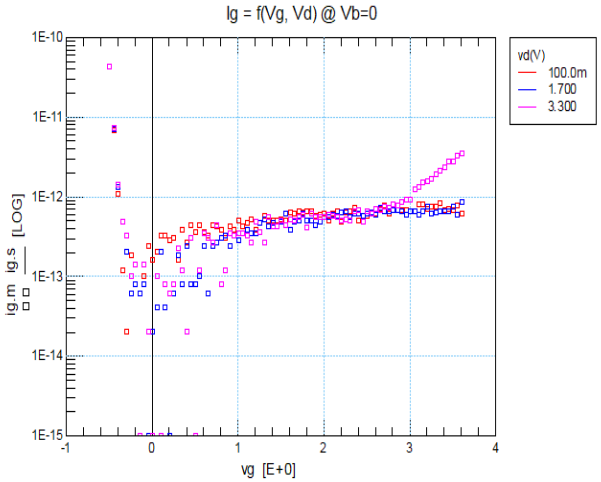




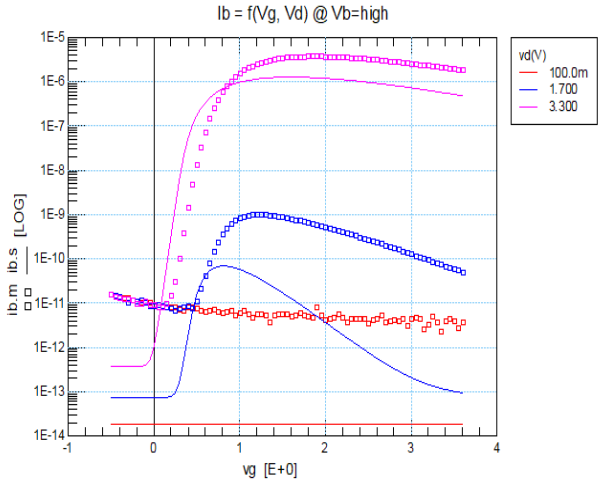


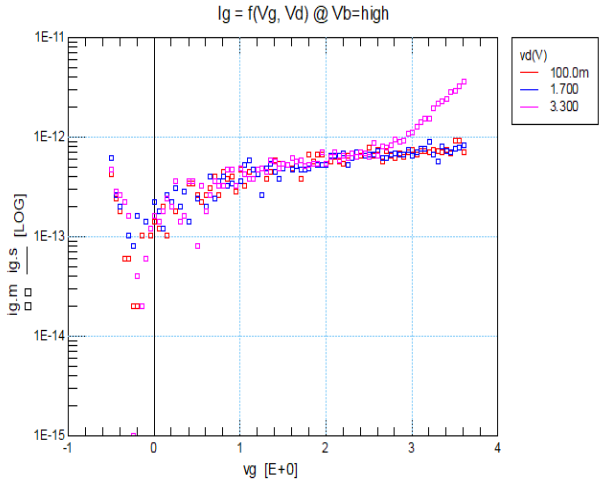
idvg, Ib,Ig, Vb = 0V, T = 27°C



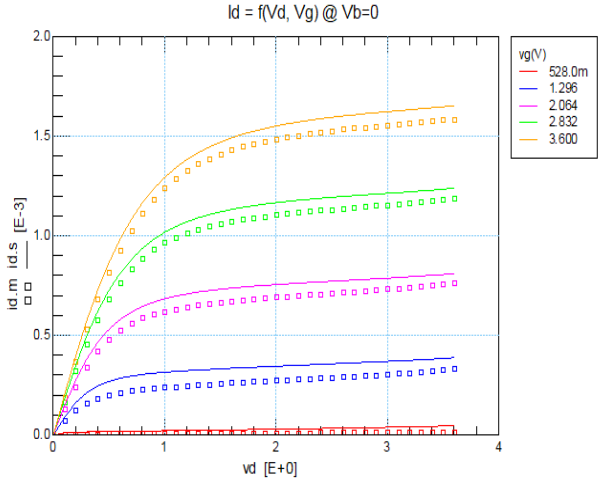


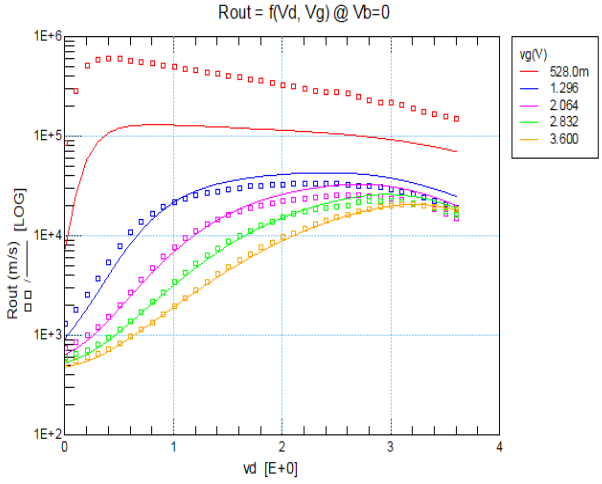
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

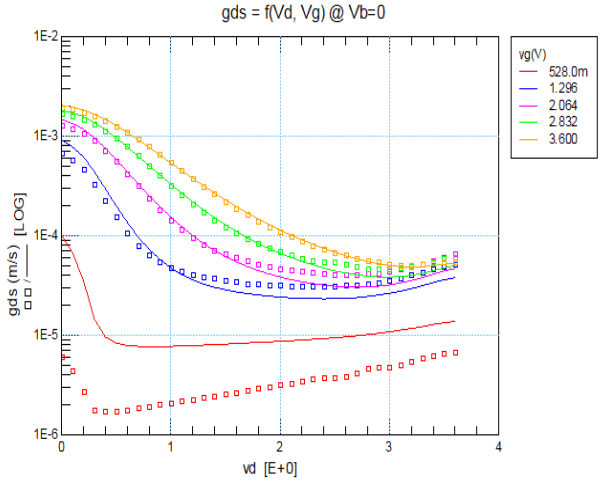




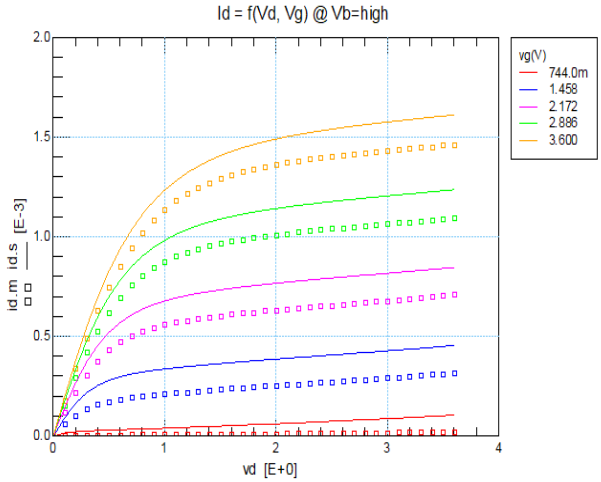
idvd, Vb = 0V, T = 27°C

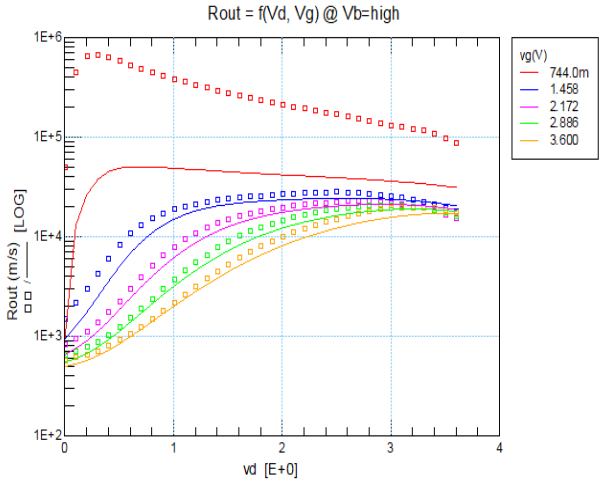


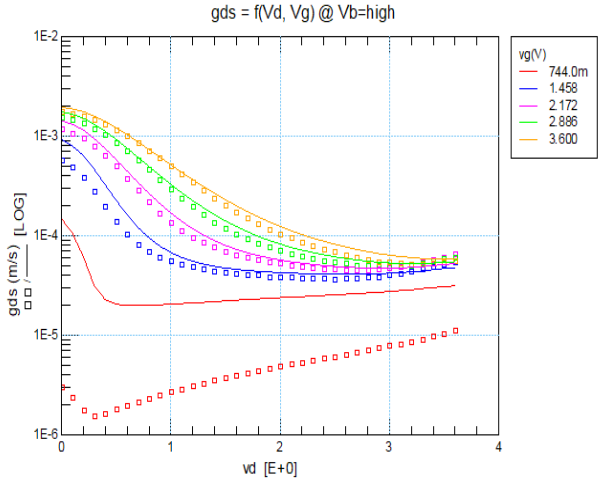




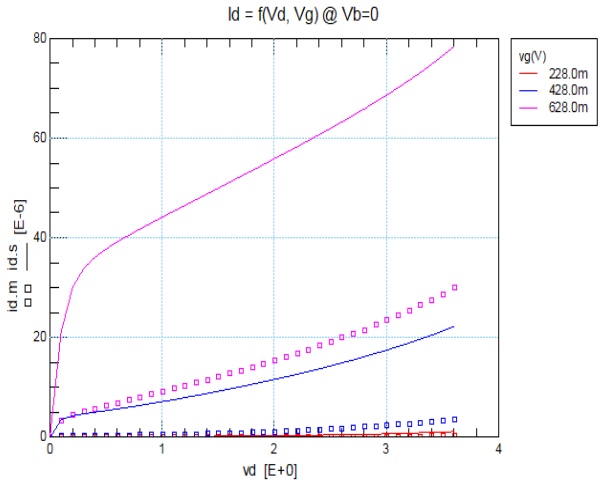
idvd_vbmin, Vb = -3.6V, T = 27°C

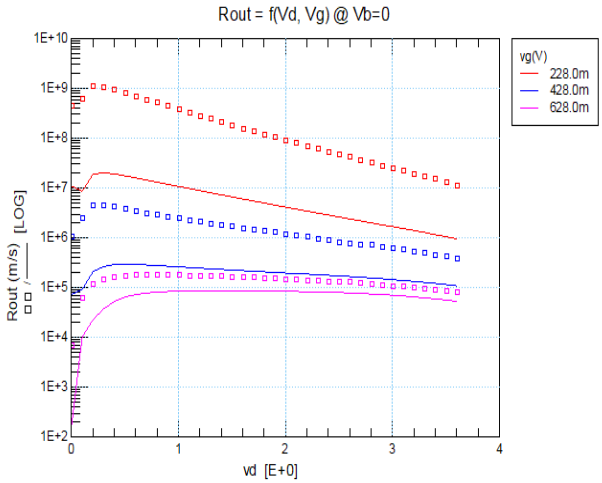


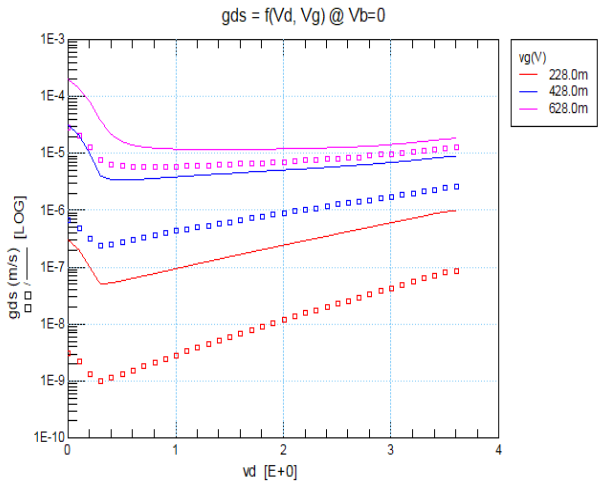




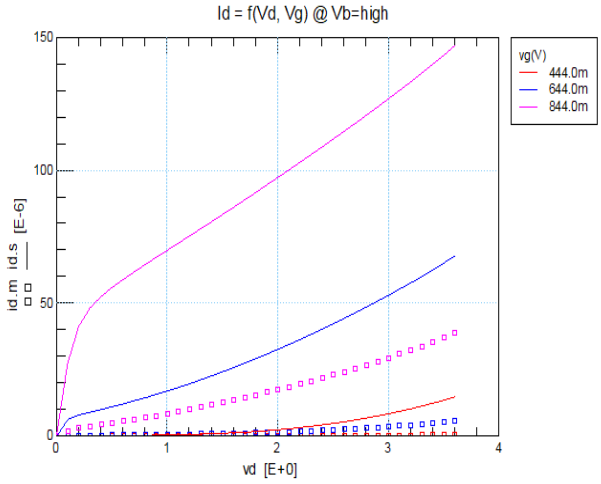
idvd_vth, Vb = 0V, T = 27°C

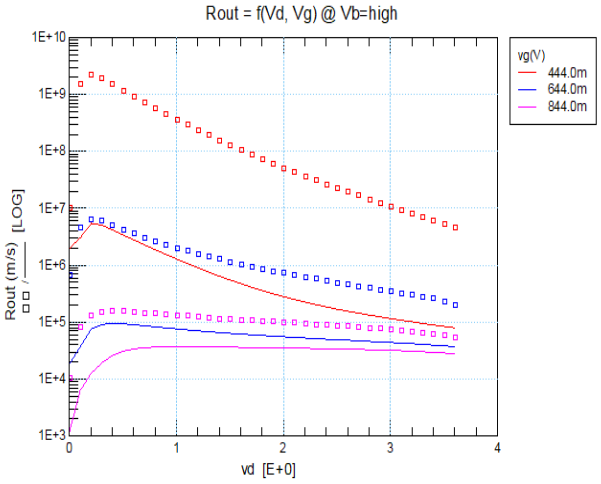


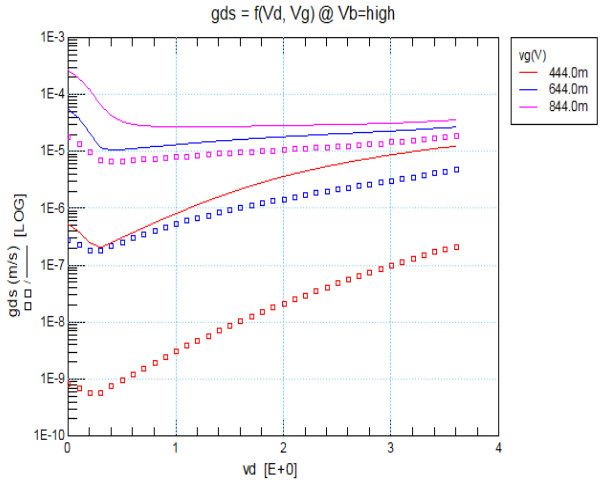




idvd_vth_vbmin, Vb = -3.6V, T = 27°C

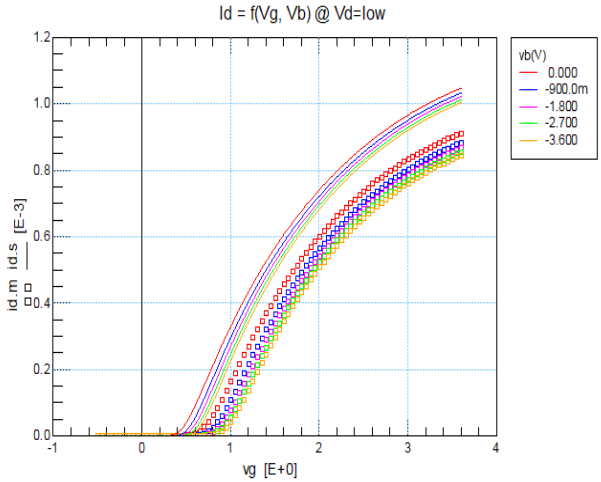


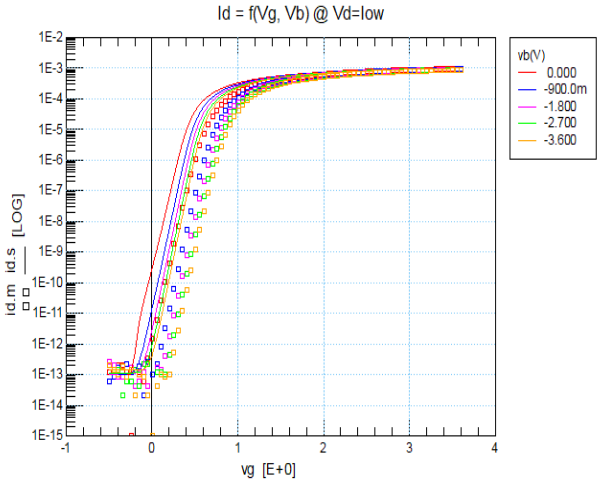


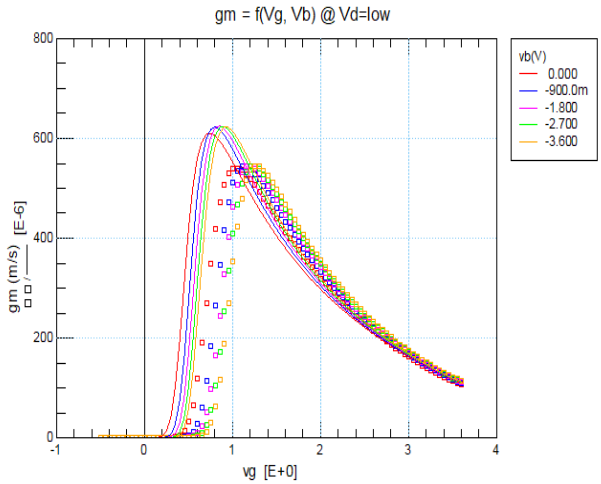


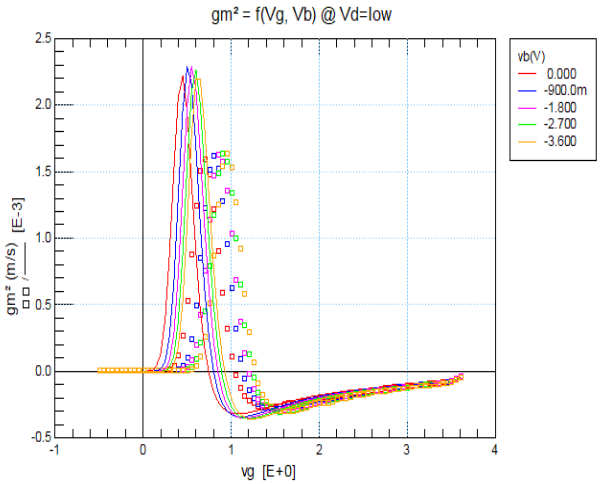
4.24W10u0_L0u33_S556_2

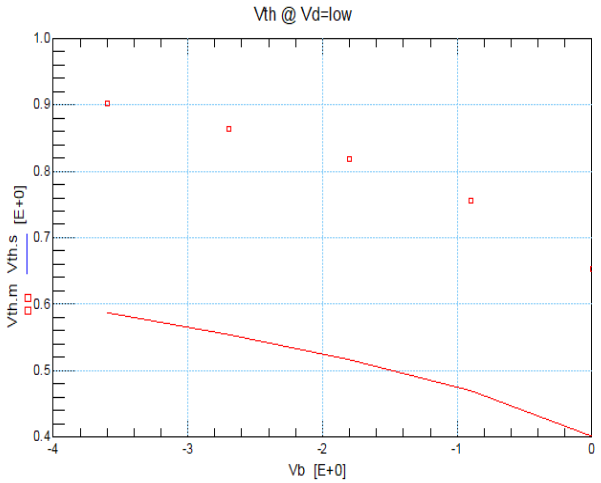
idvg, Vd = 0.1V, T = 27°C

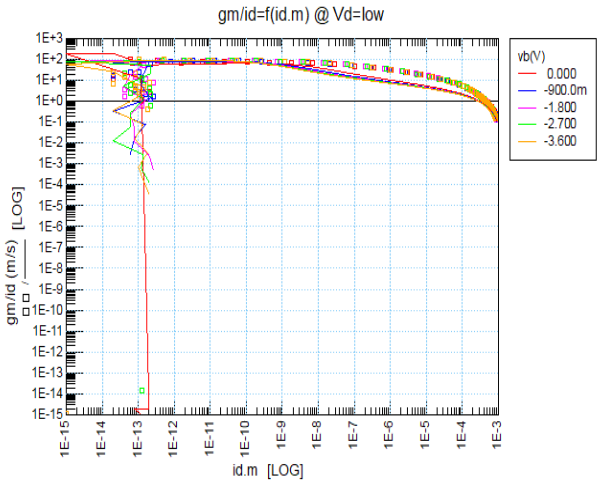


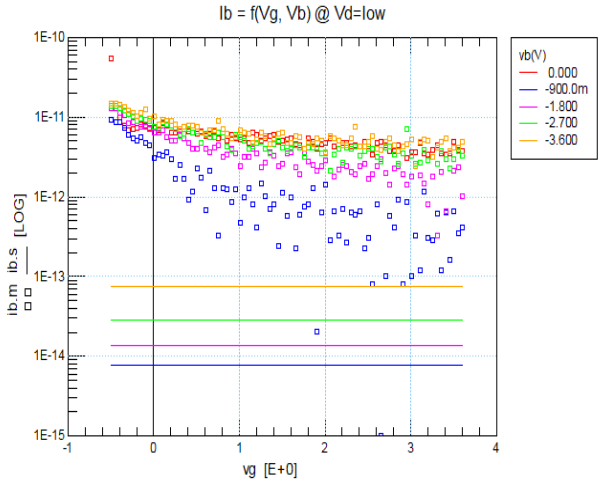


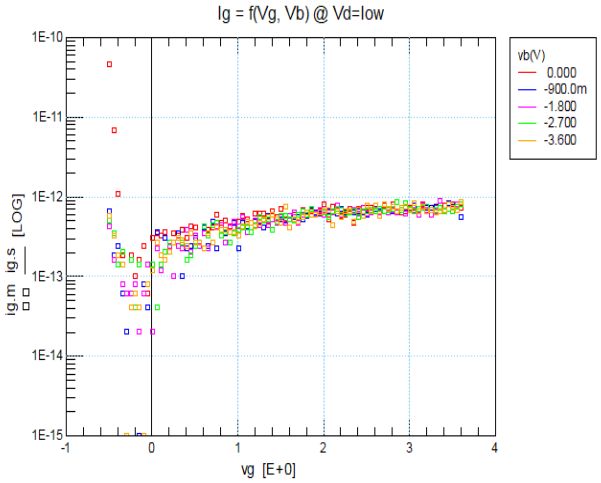




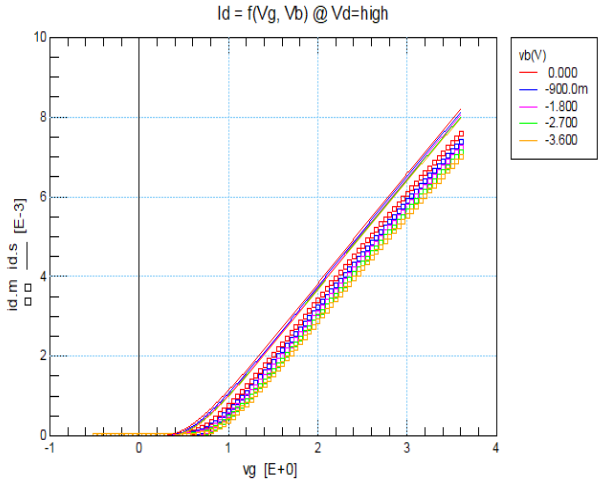


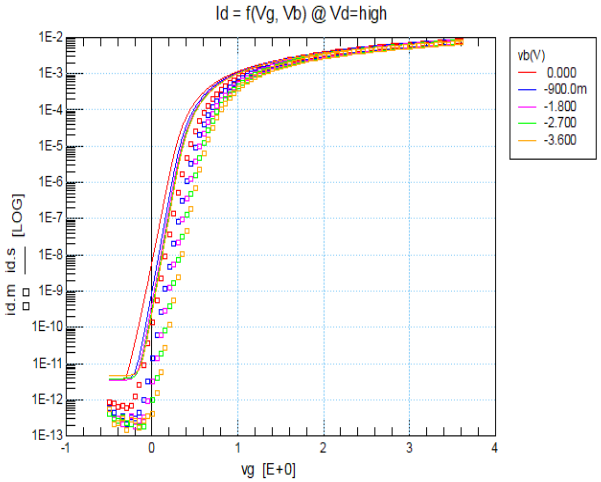


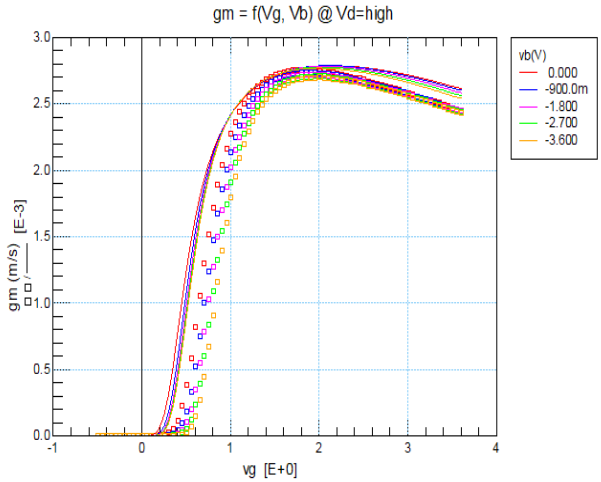


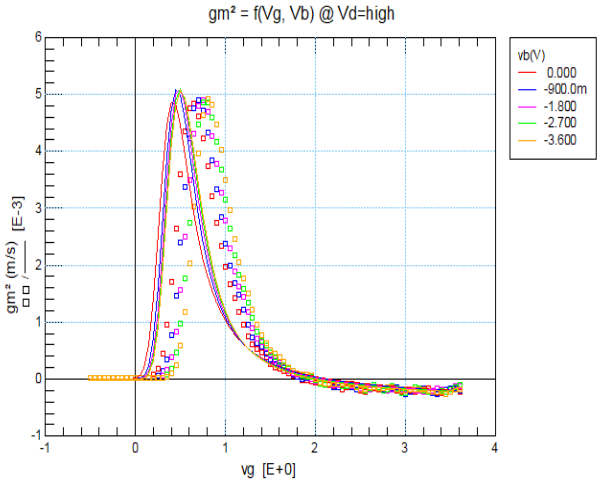


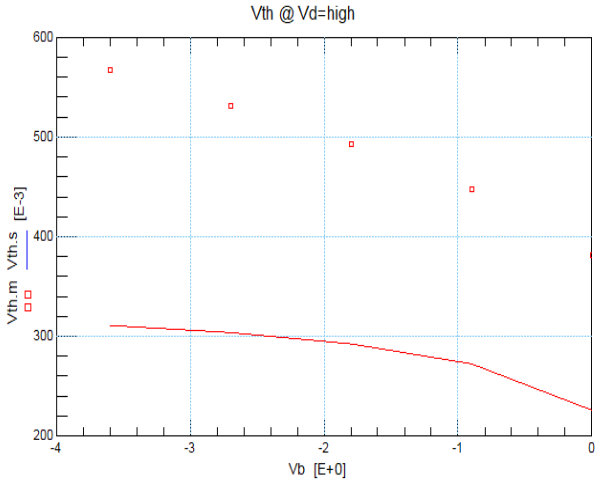
idvg, Vd = 3.3V, T = 27°C

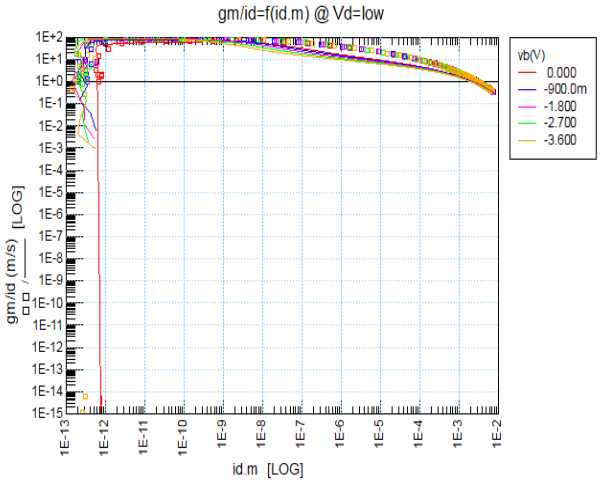


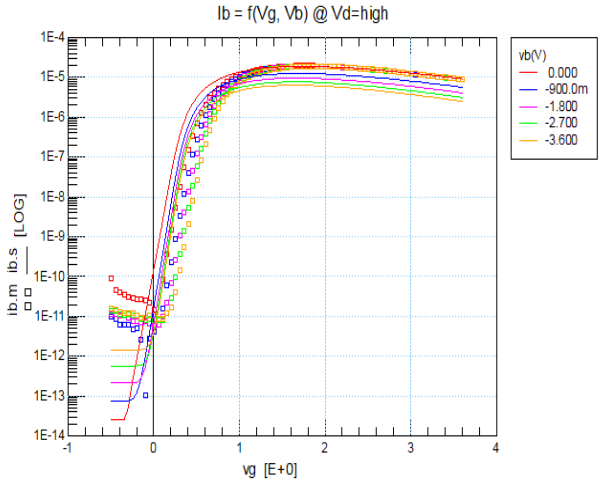


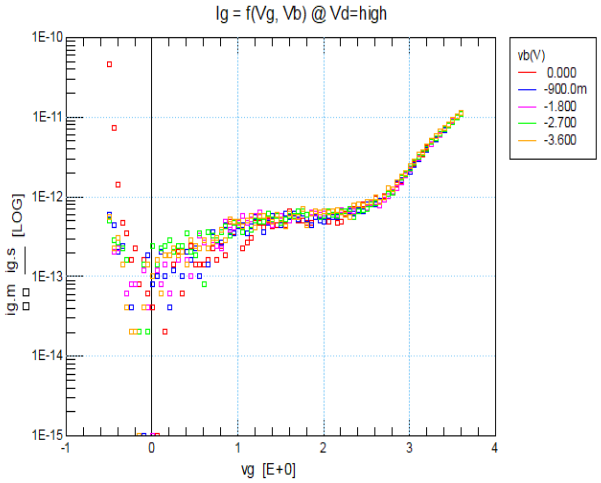




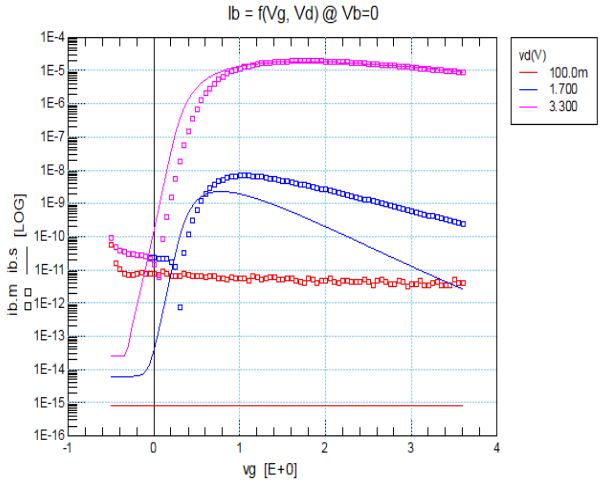


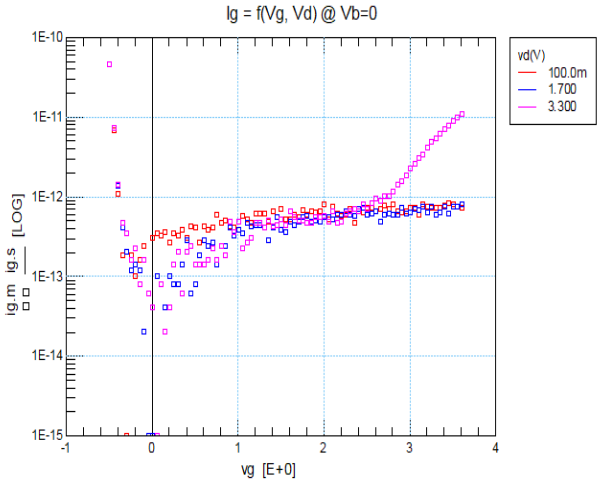




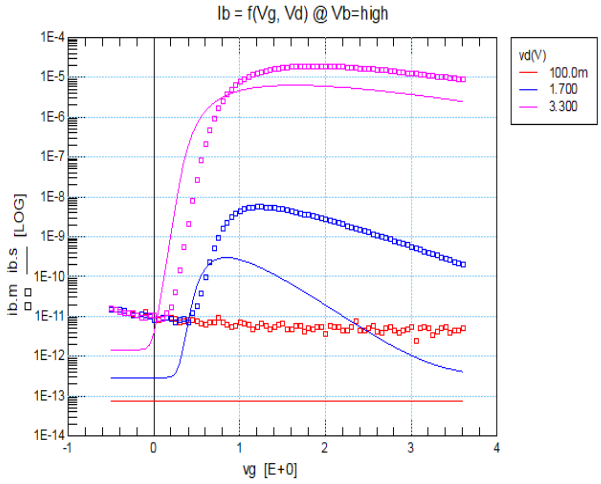


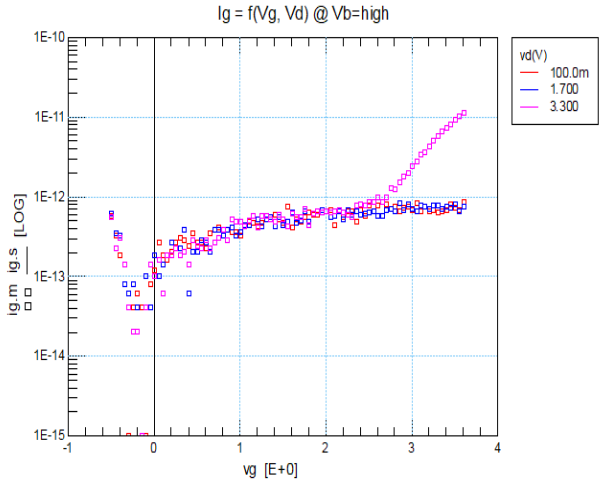
idvg, Ib,Ig, Vb = 0V, T = 27°C



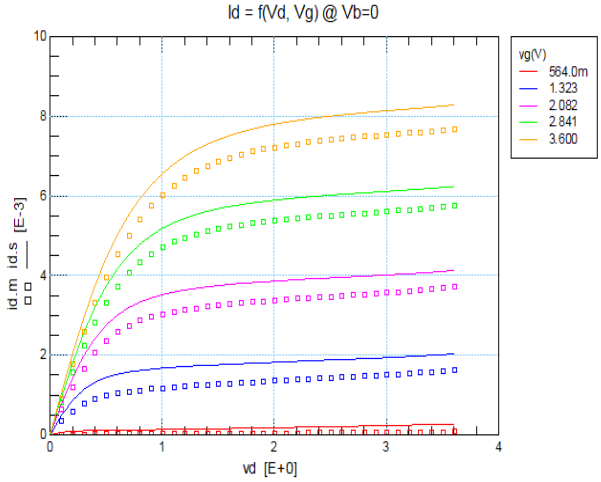


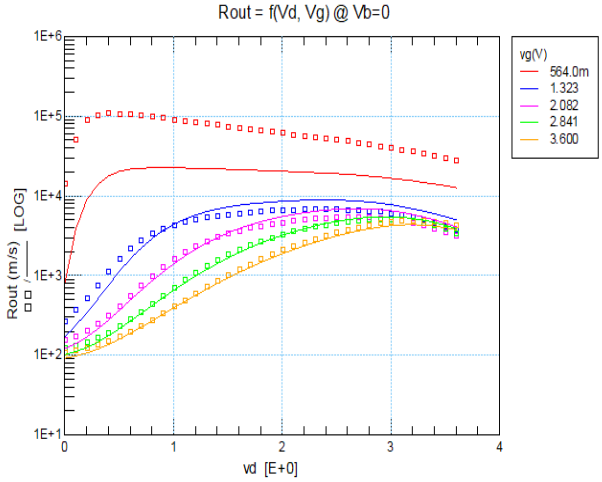
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

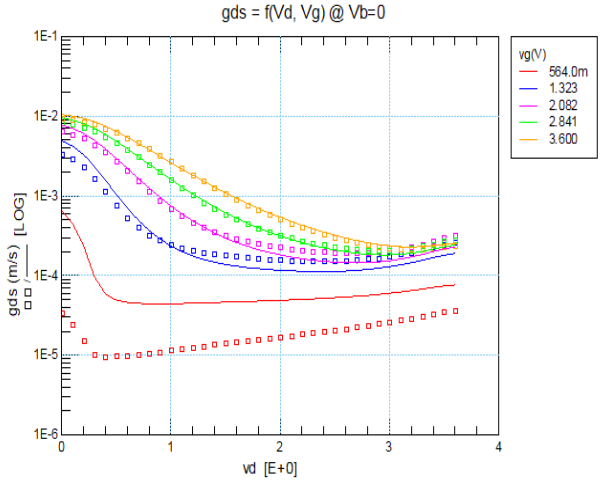




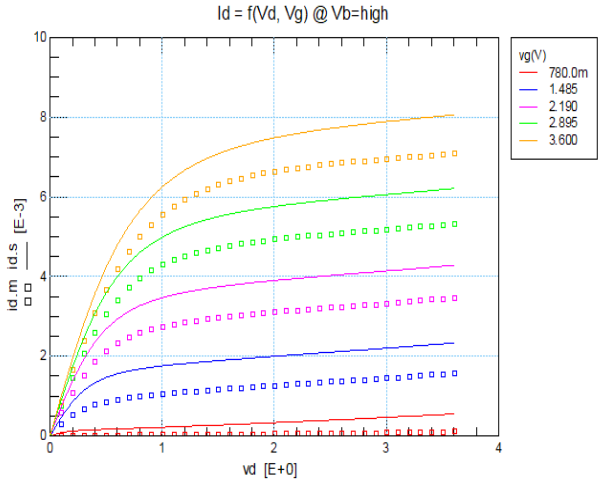
idvd, Vb = 0V, T = 27°C

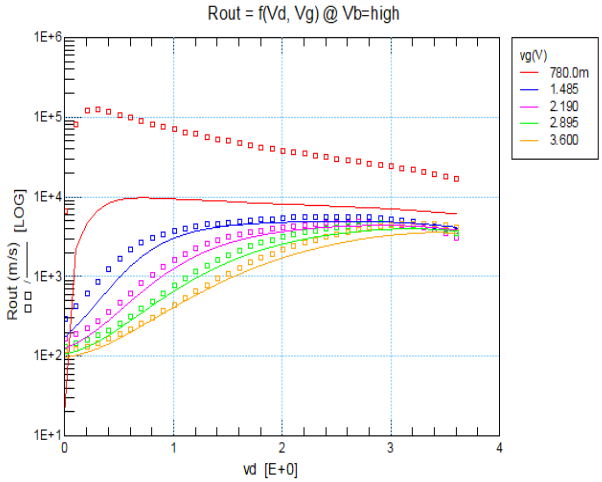


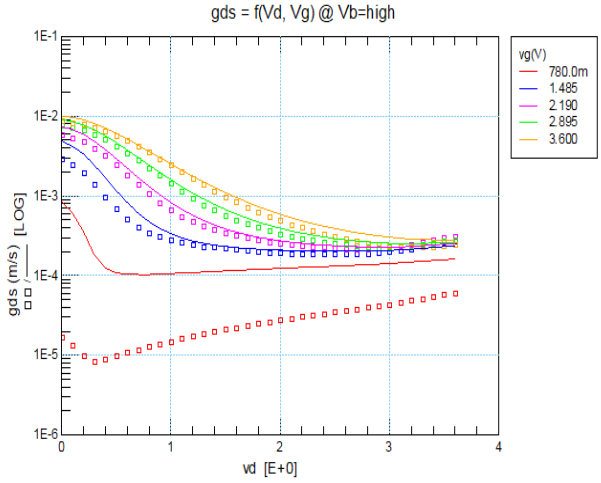




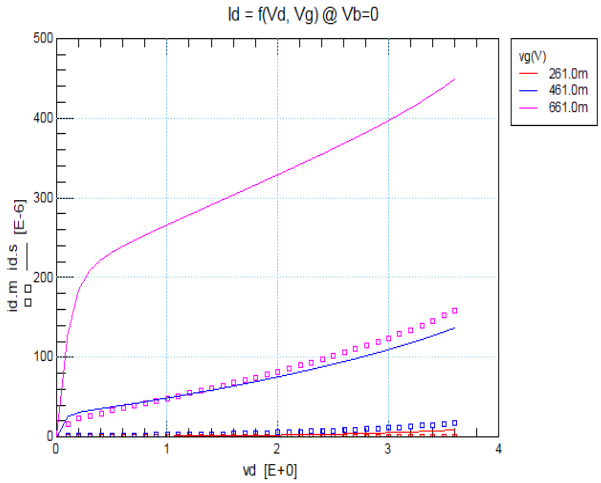
idvd_vbmin, Vb = -3.6V, T = 27°C

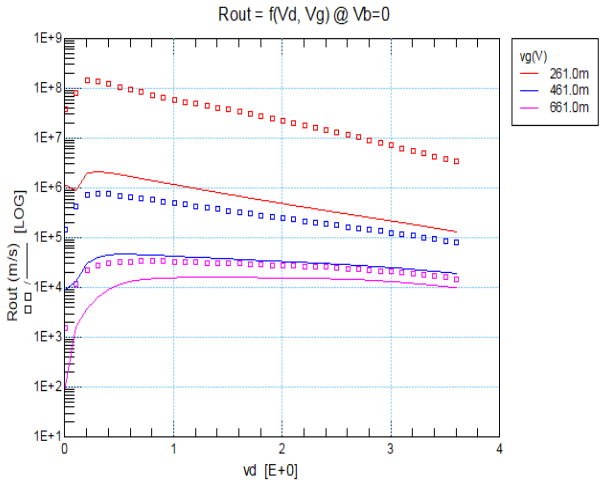


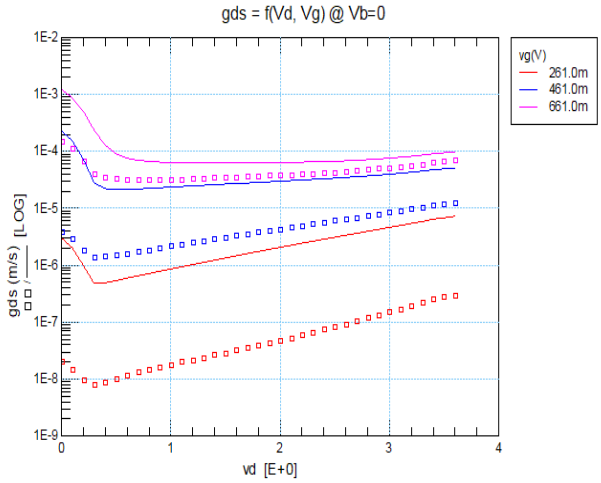




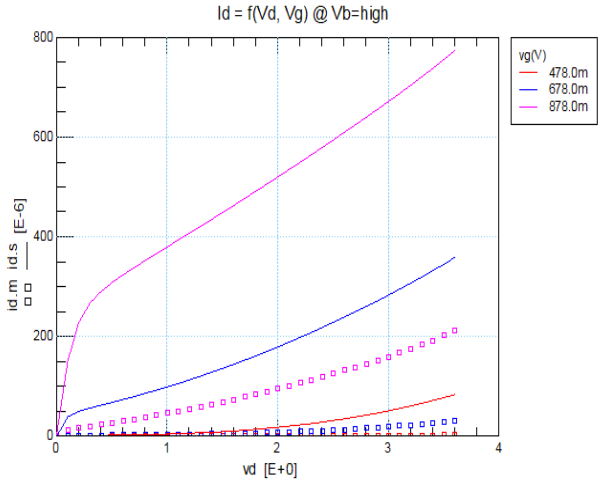
idvd_vth, Vb = 0V, T = 27°C

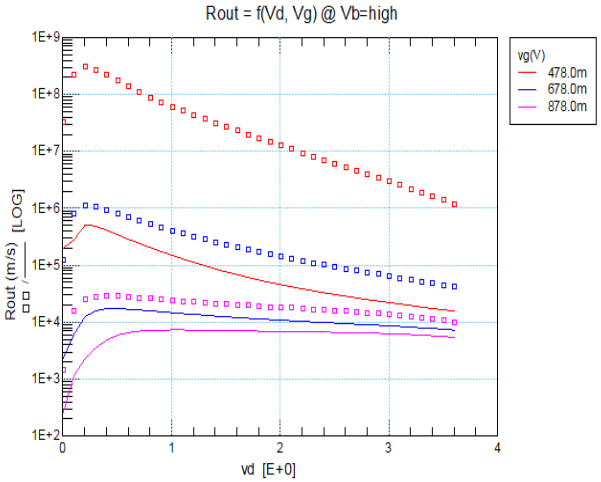


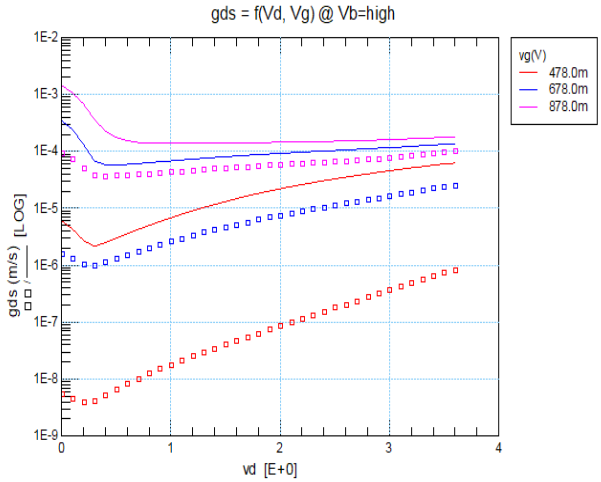




idvd_vth_vbmin, Vb = -3.6V, T = 27°C

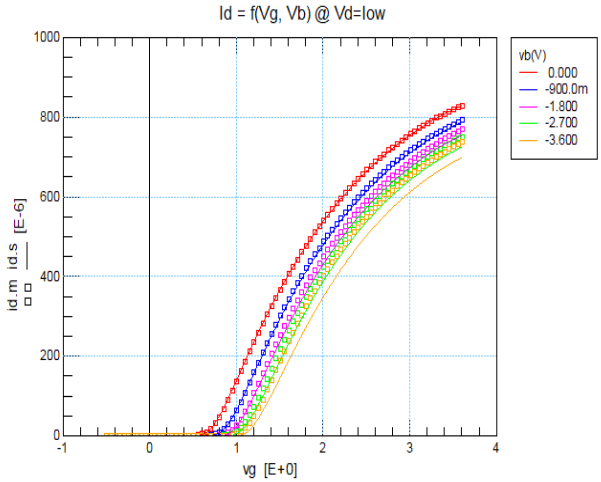


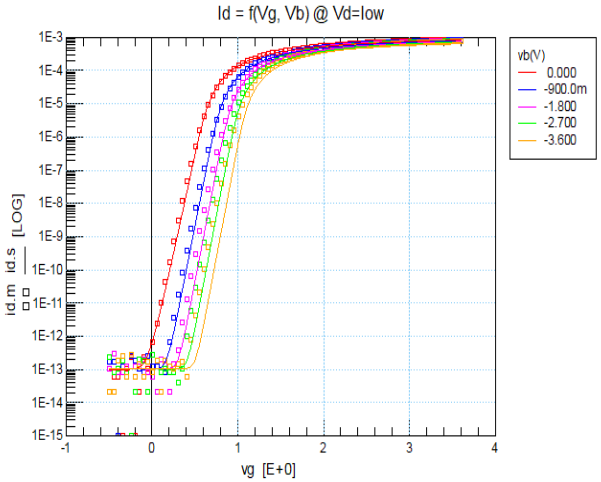


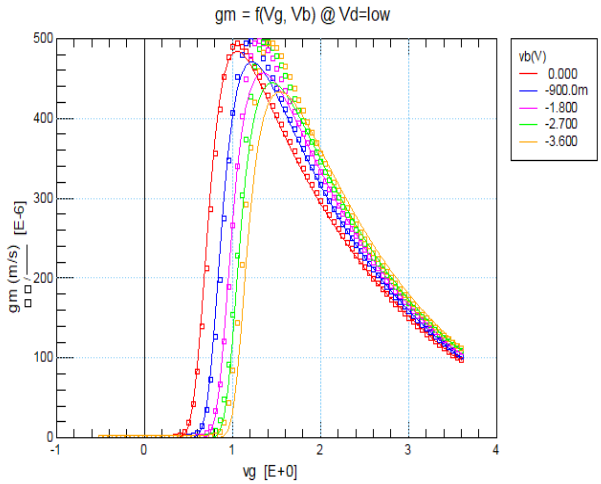


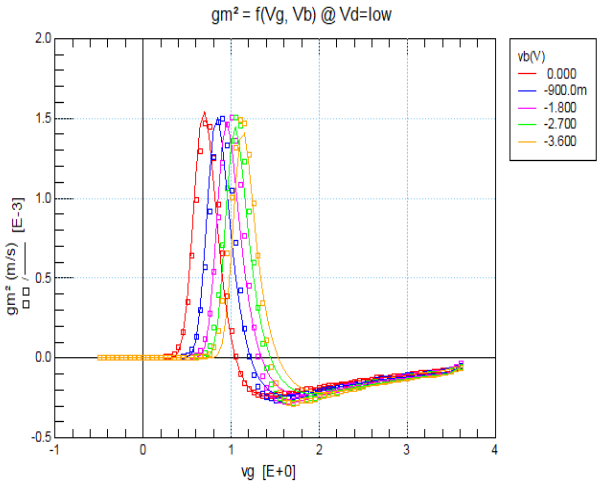
4.25W10u0_L0u4_T356_S387_4

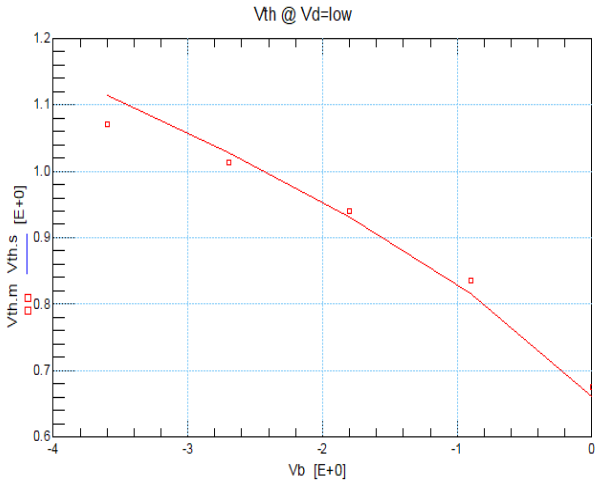
idvg, Vd = 0.1V, T = 27°C

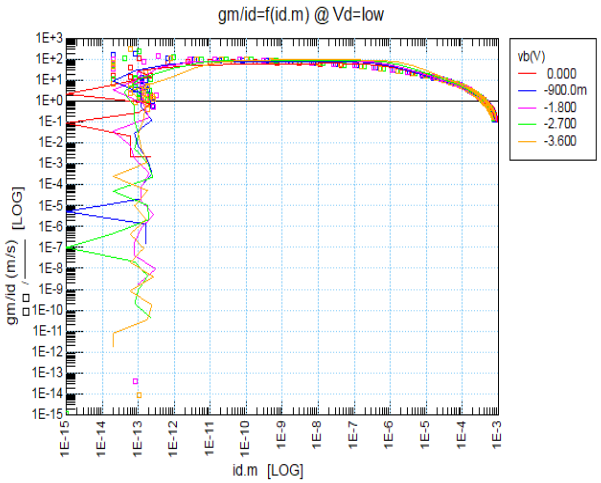


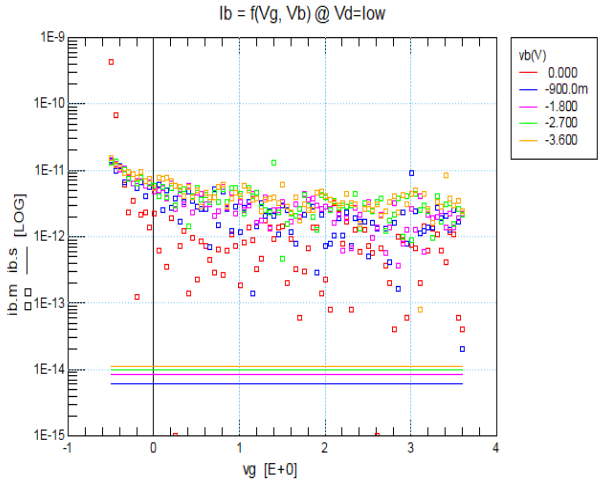


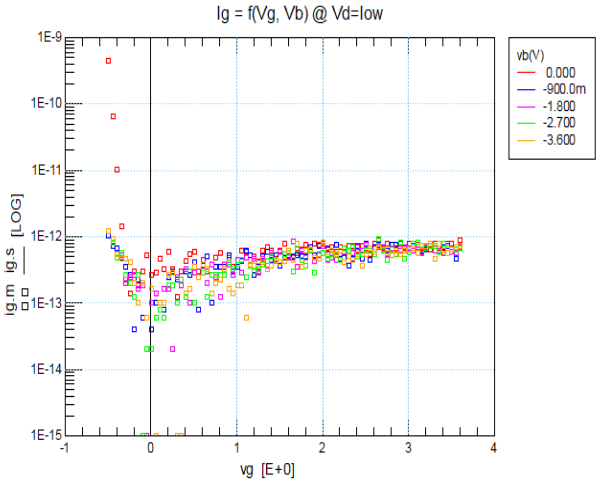




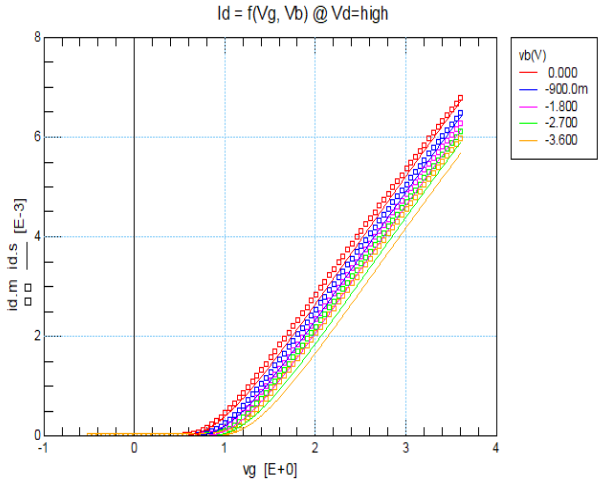


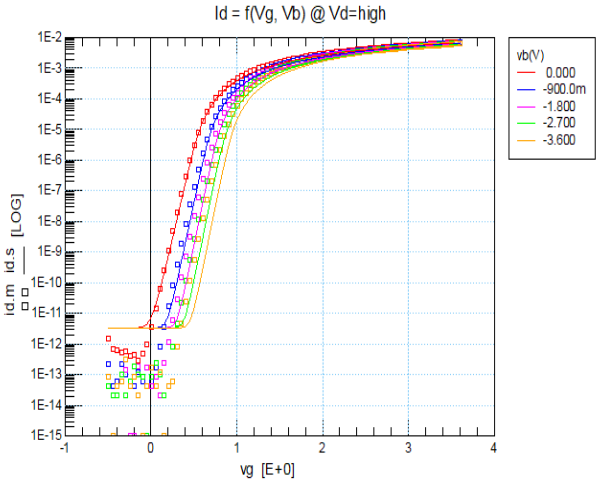


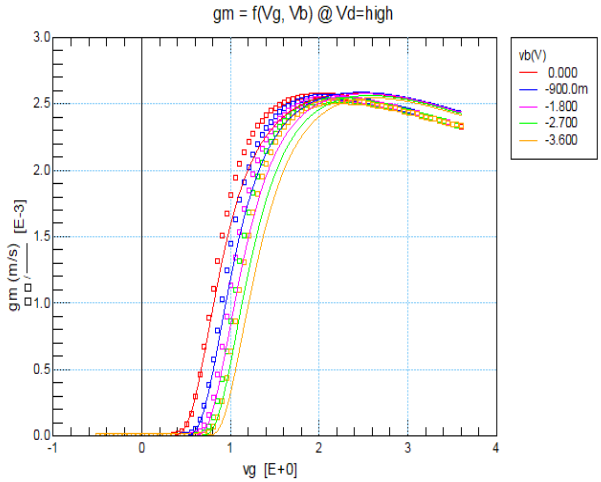


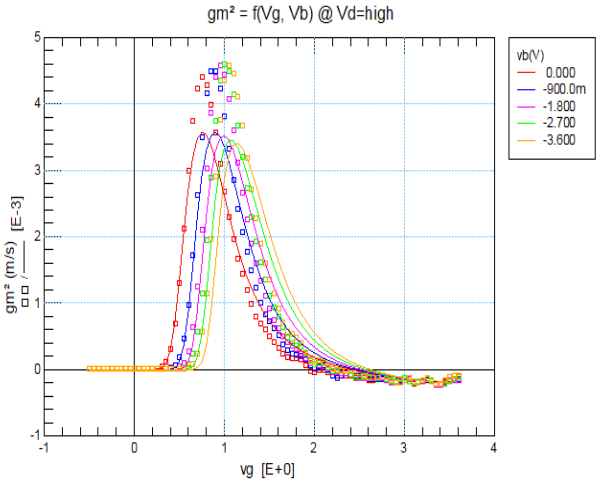


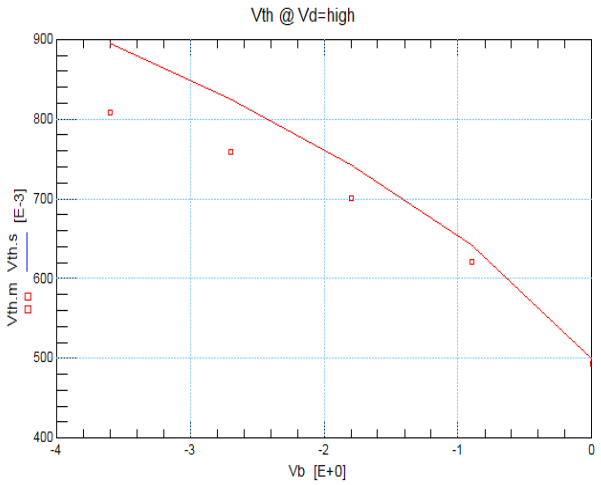
idvg, Vd = 3.3V, T = 27°C

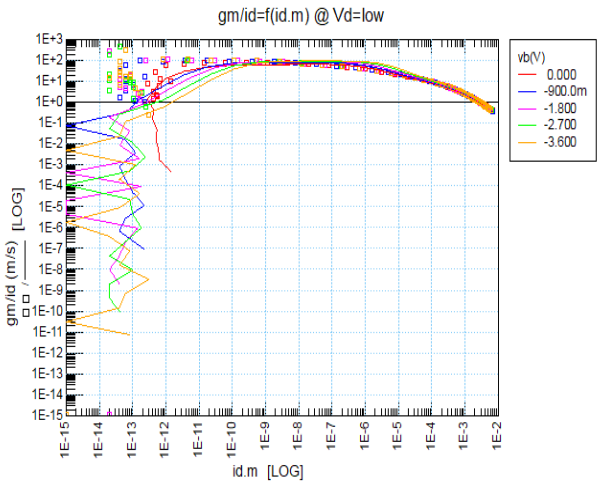


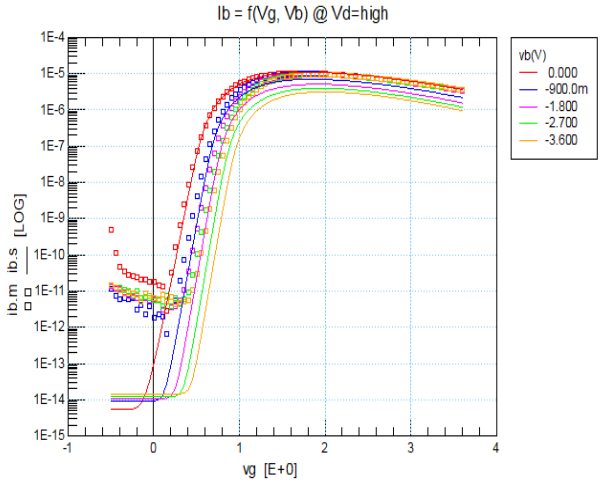


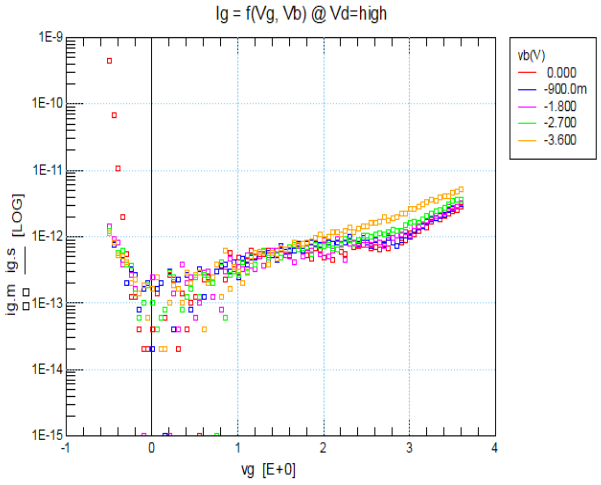




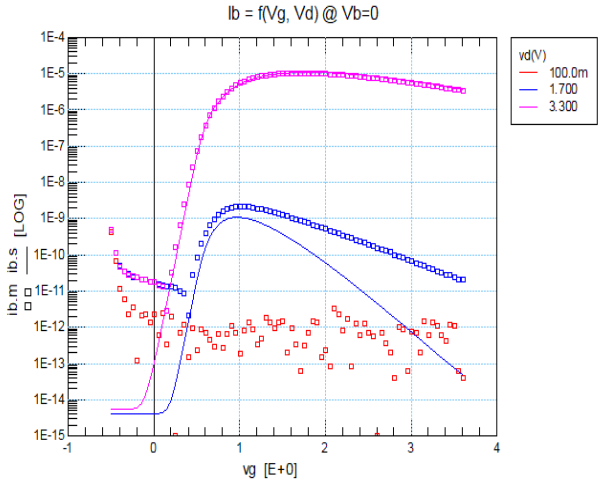


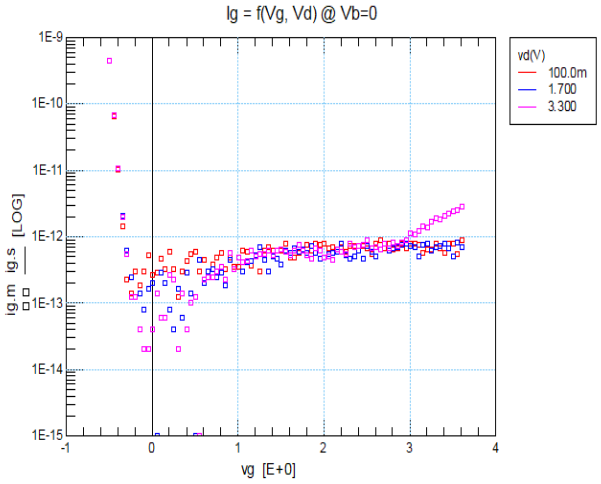




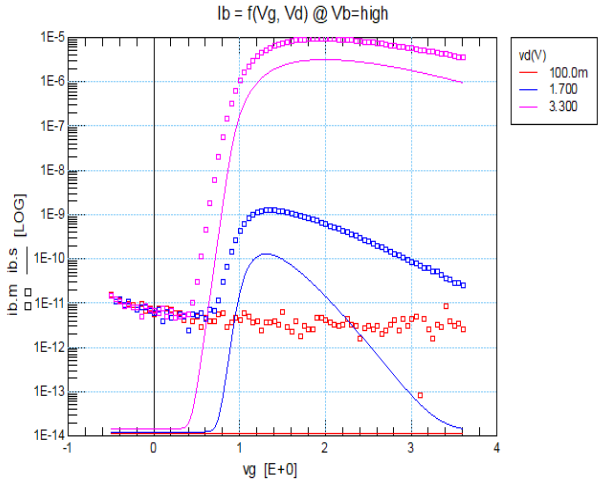


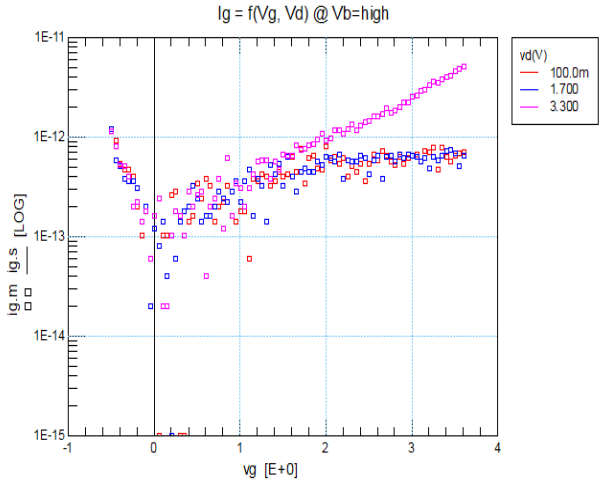
idvg, Ib,Ig, Vb = 0V, T = 27°C



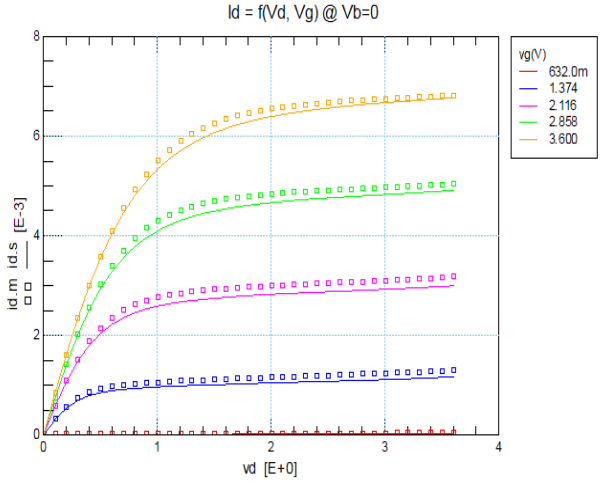


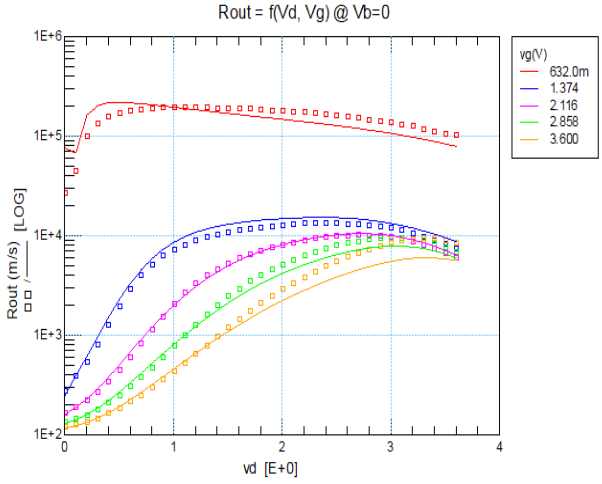
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

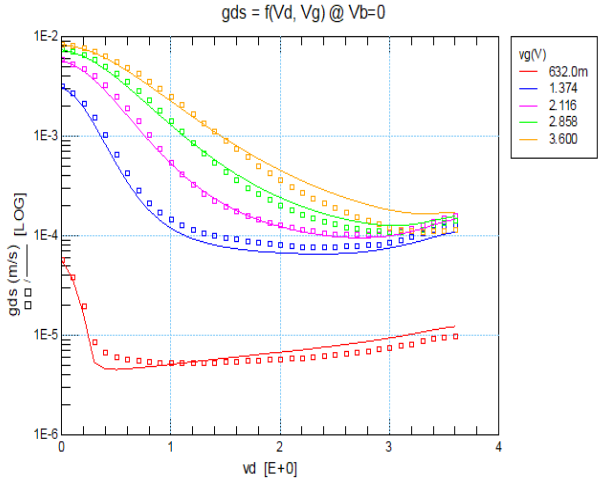




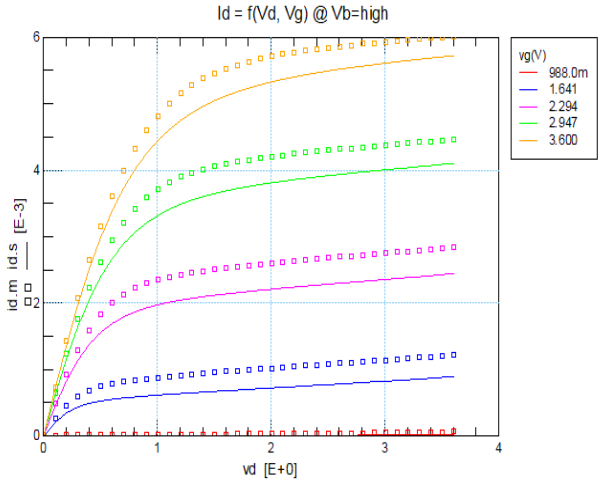
idvd, Vb = 0V, T = 27°C

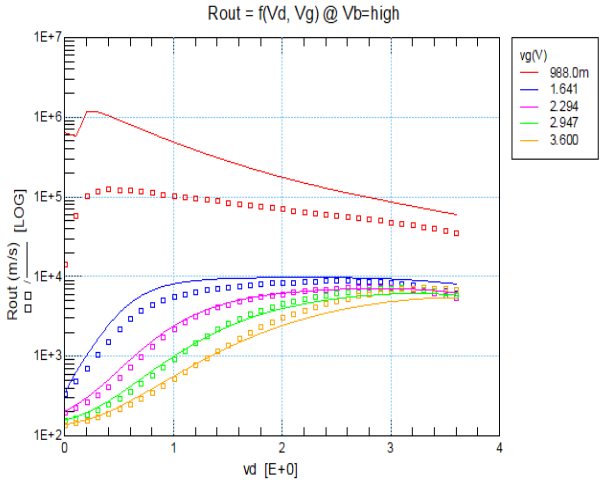


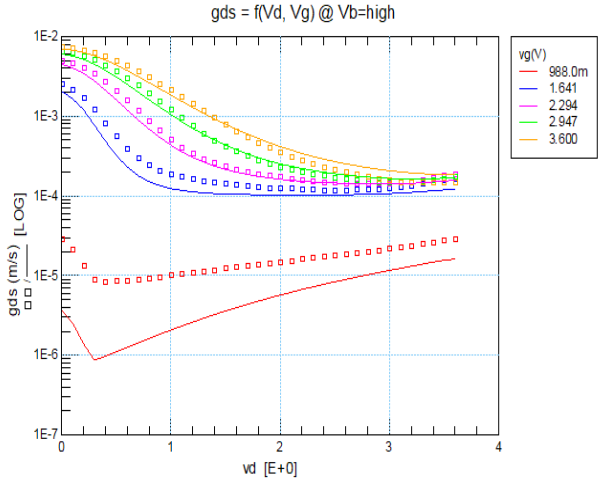




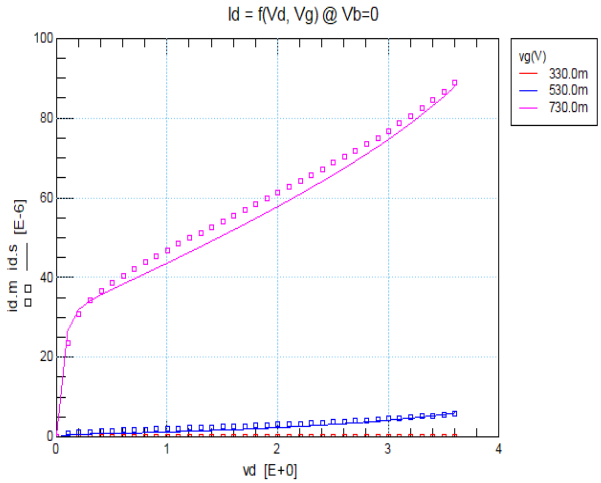
idvd_vbmin, Vb = -3.6V, T = 27°C

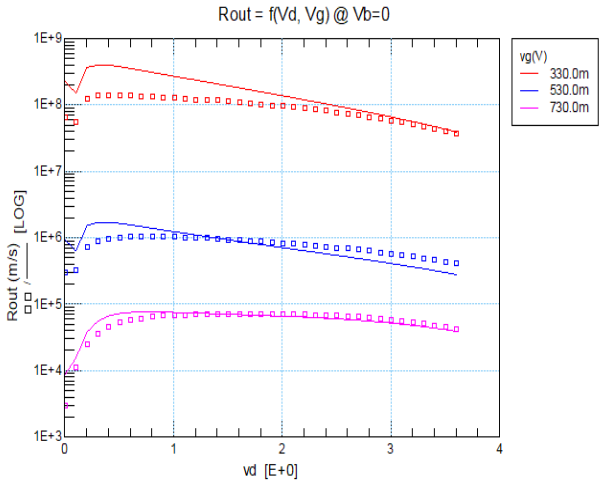


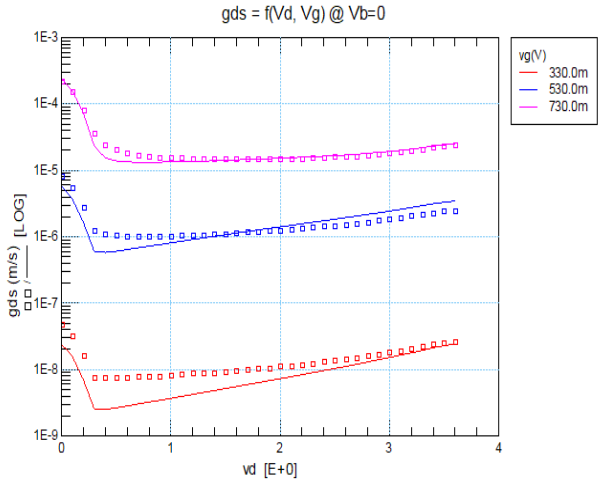




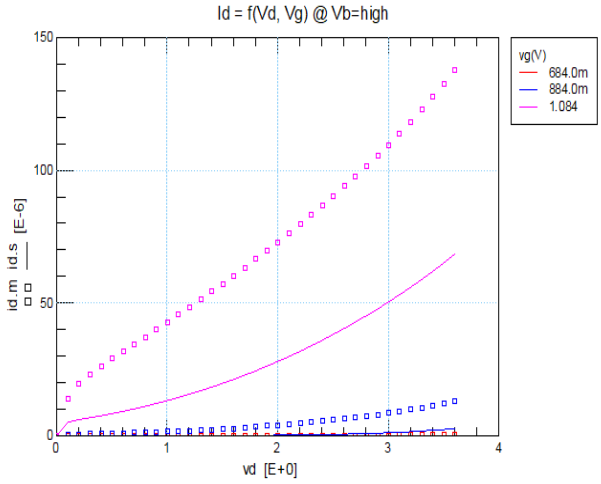
idvd_vth, Vb = 0V, T = 27°C

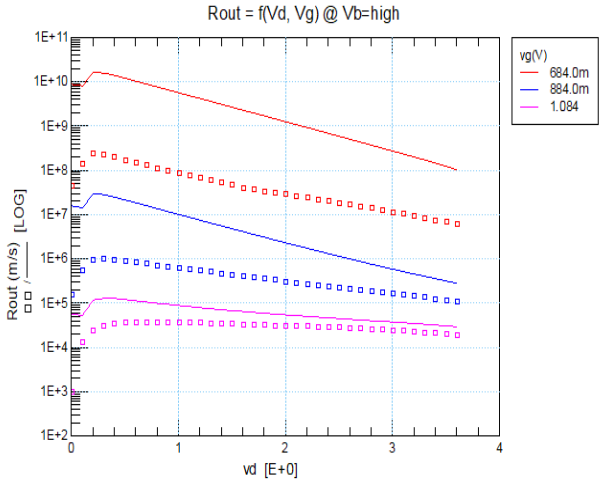


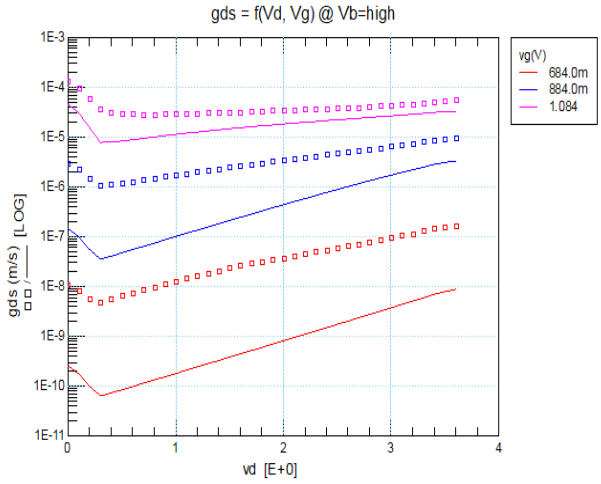




idvd_vth_vbmin, Vb = -3.6V, T = 27°C

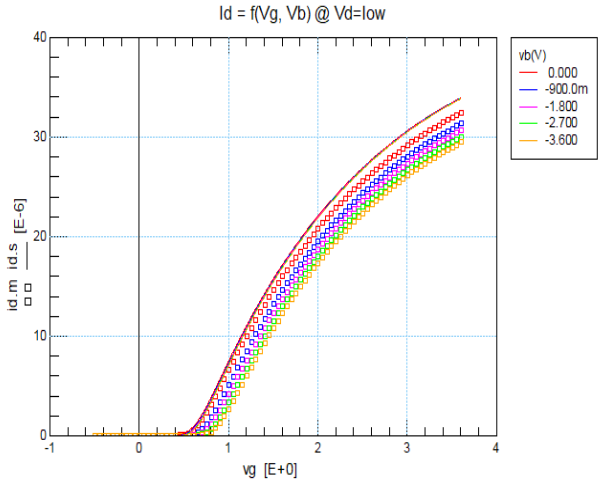


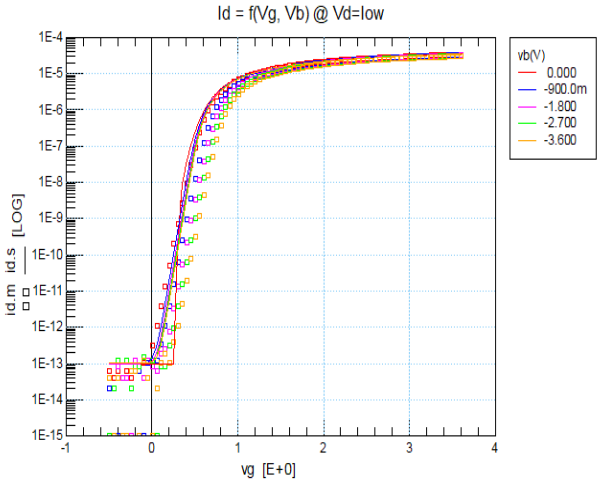


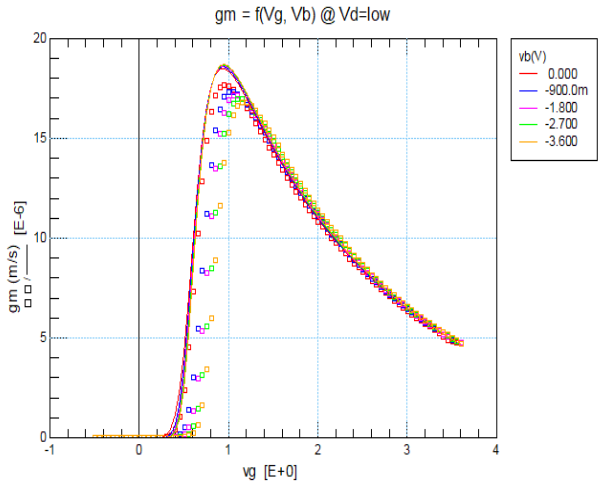


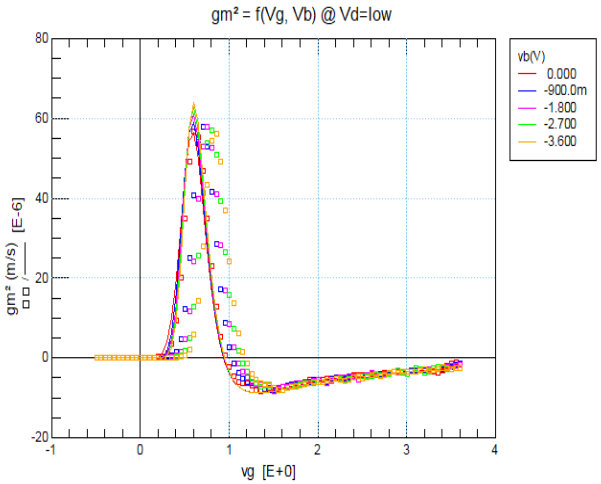
4.26 W0u3_L0u3_S560_1

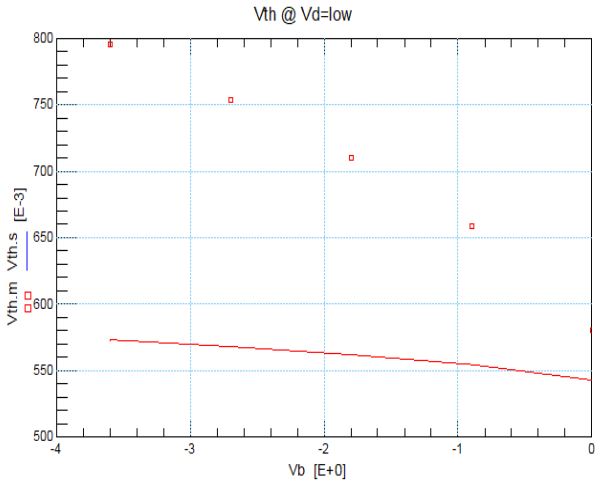
idvg, Vd = 0.1V, T = 27°C

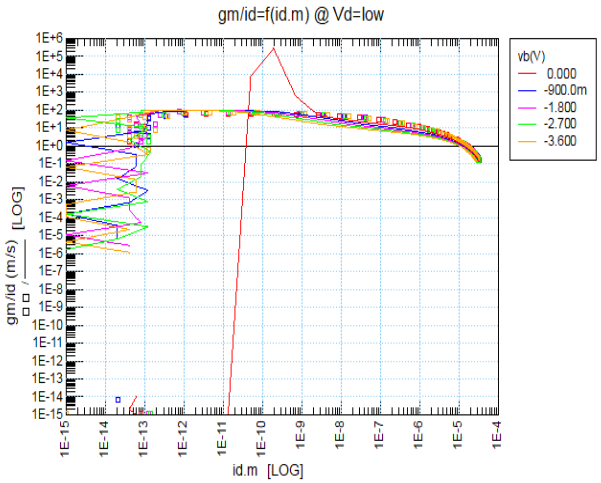


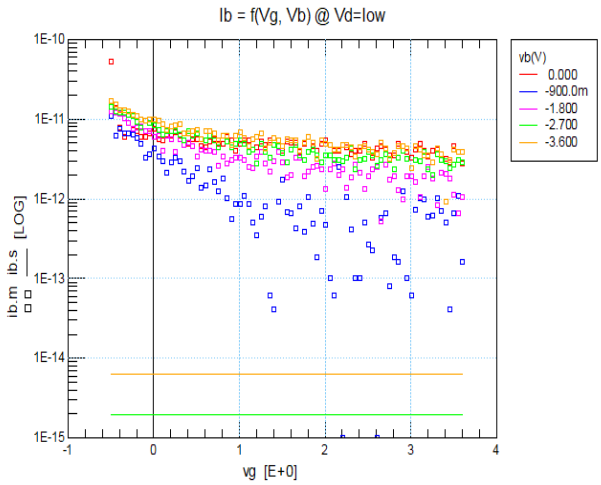


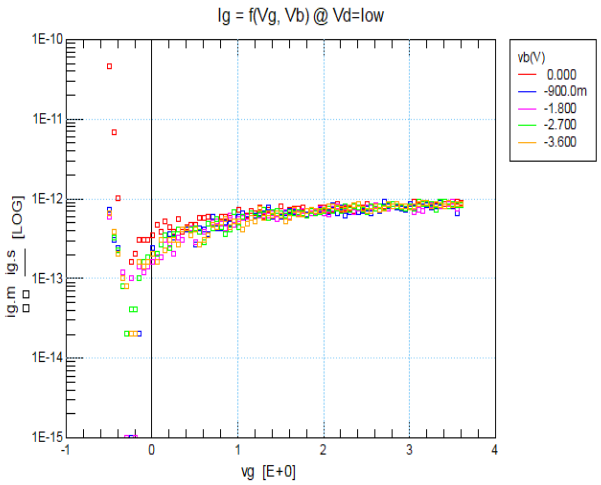




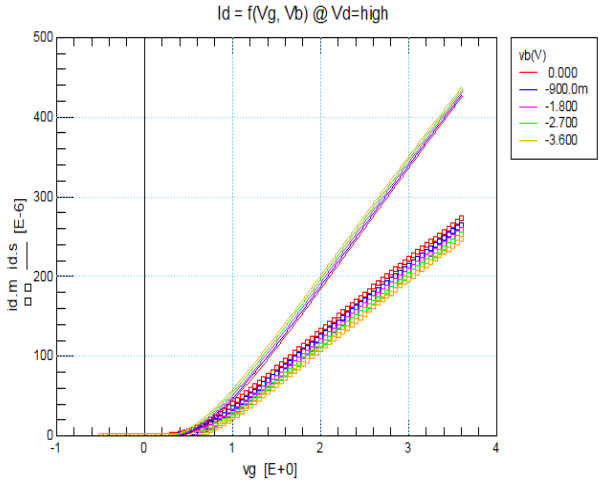


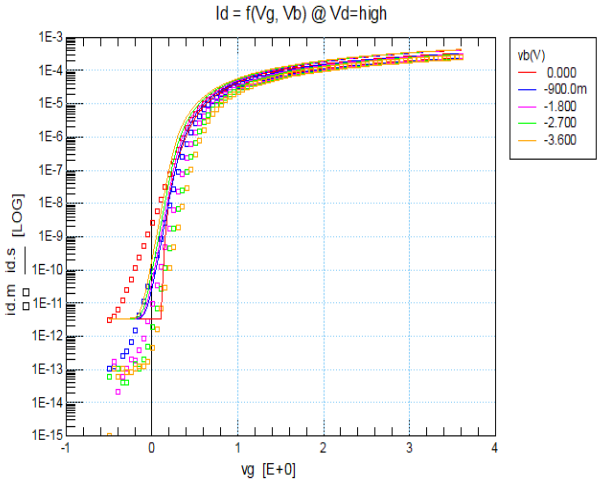


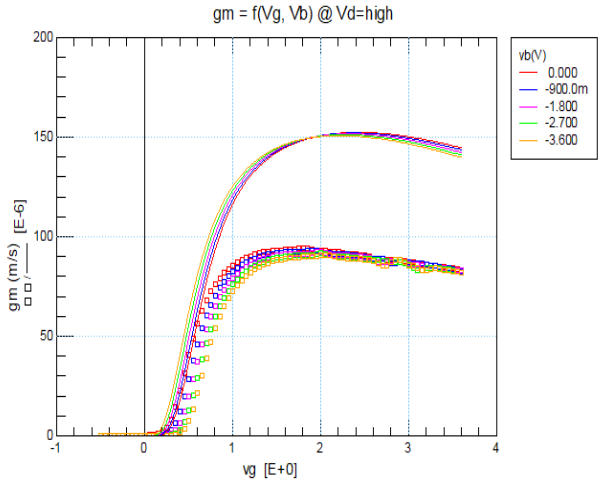


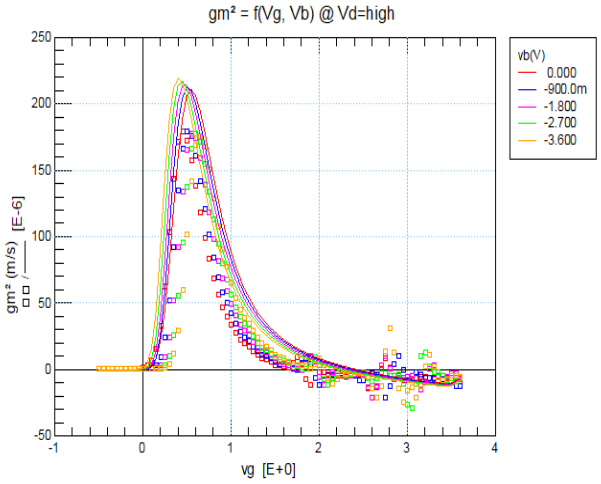


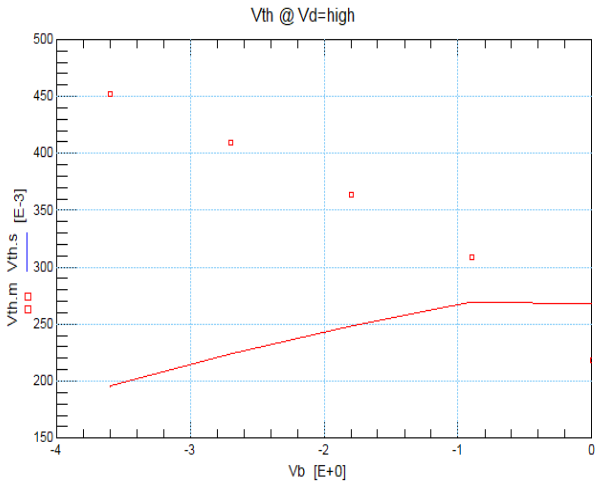
idvg, Vd = 3.3V, T = 27°C

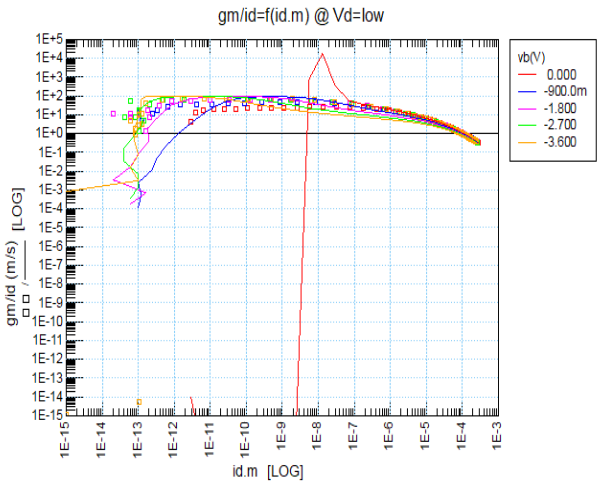


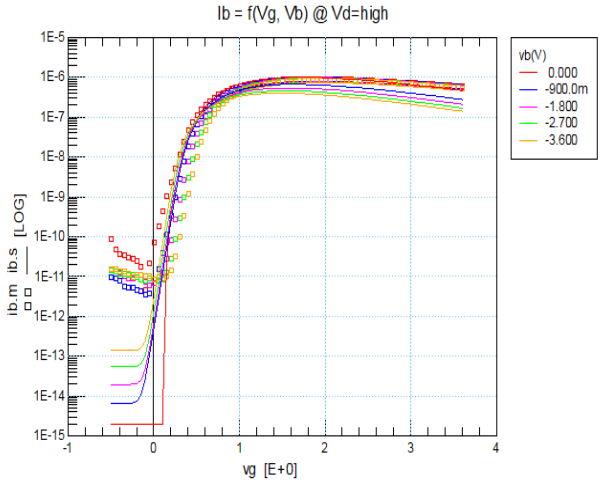


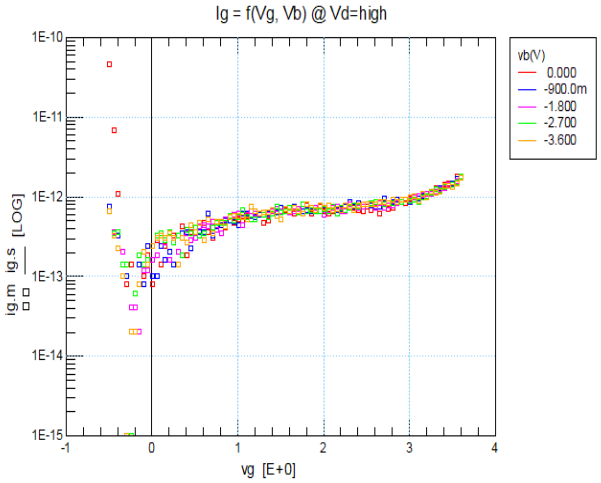




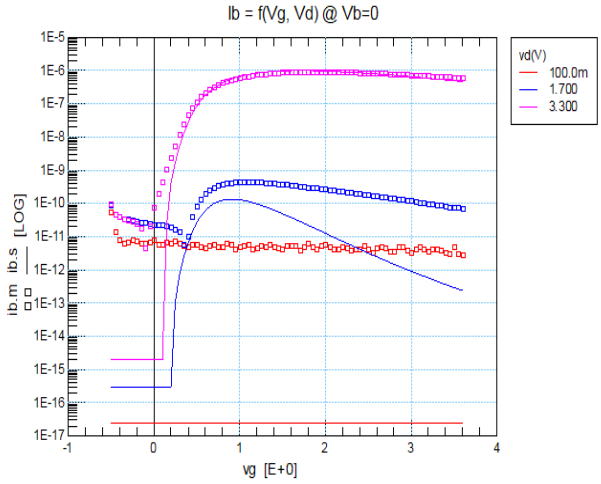


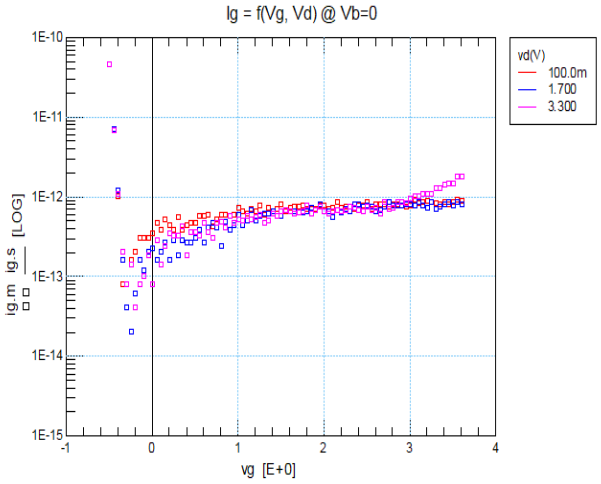




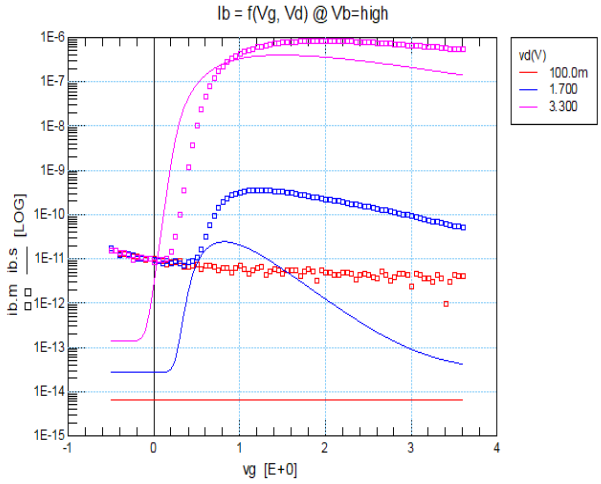


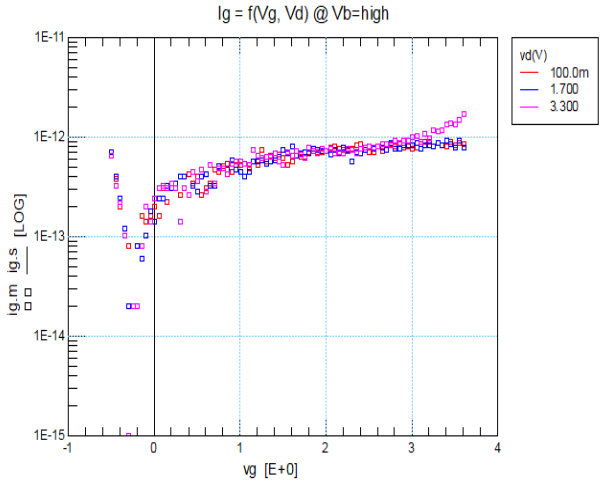
idvg, Ib,Ig, Vb = 0V, T = 27°C



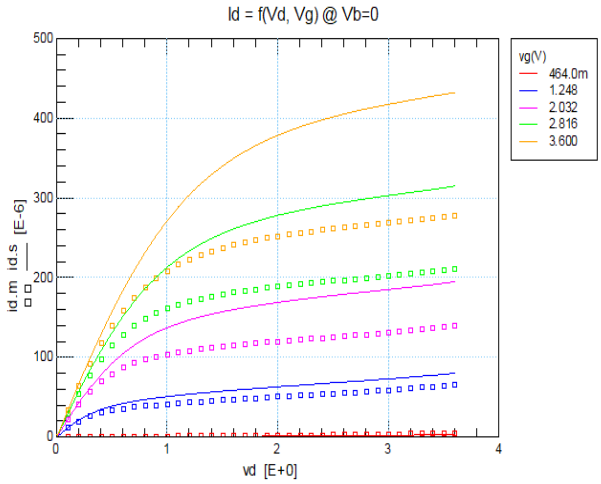


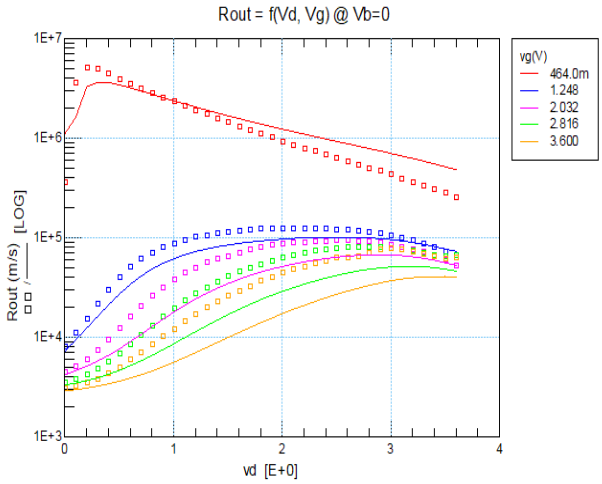
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

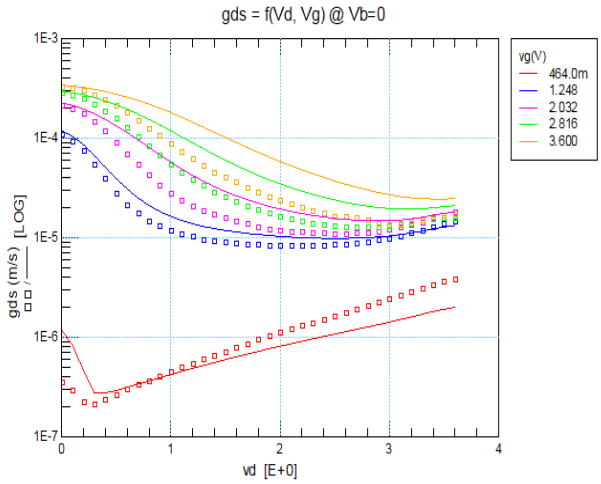




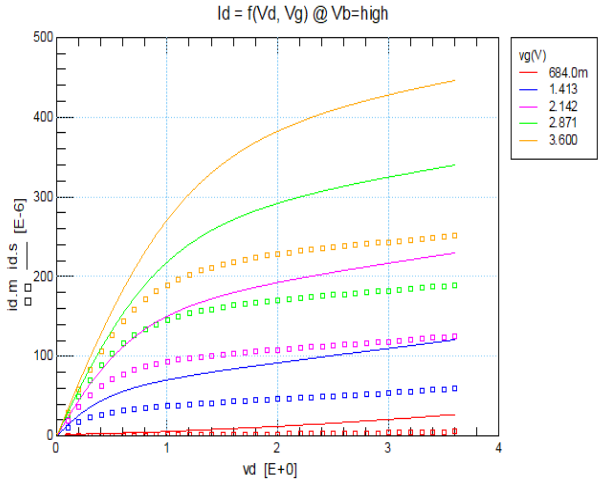
idvd, Vb = 0V, T = 27°C

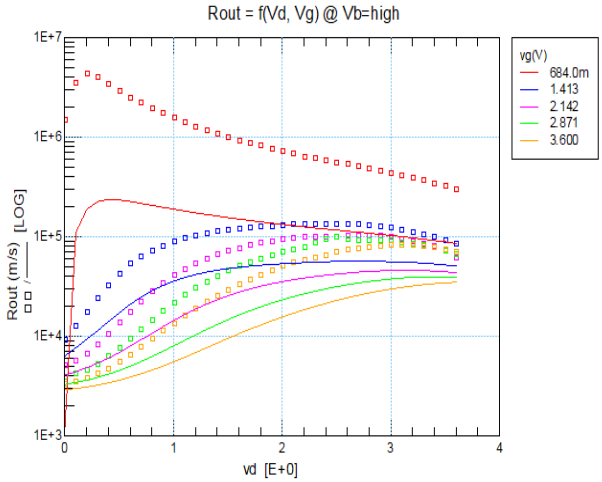


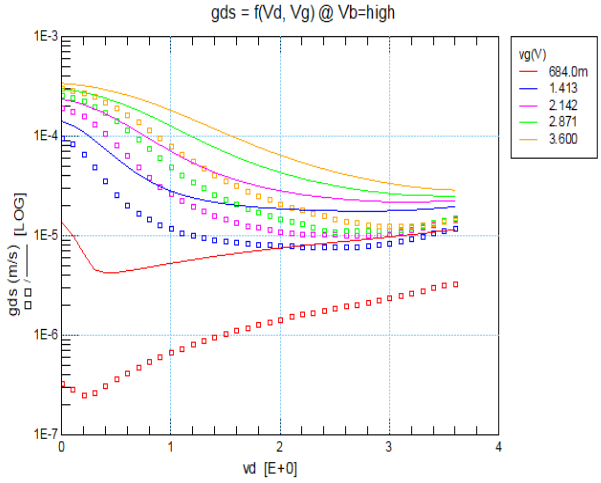




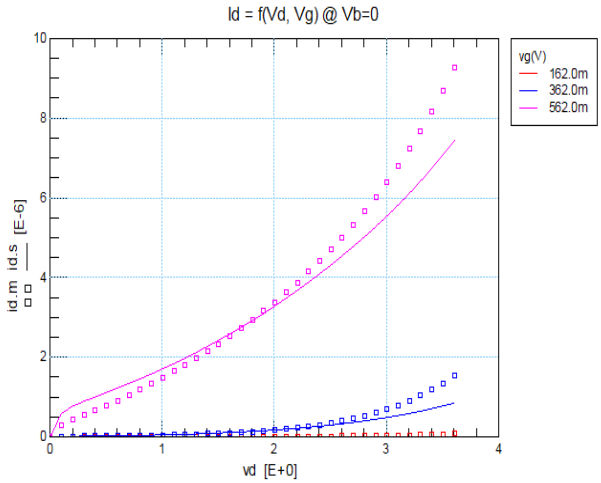
idvd_vbmin, Vb = -3.6V, T = 27°C

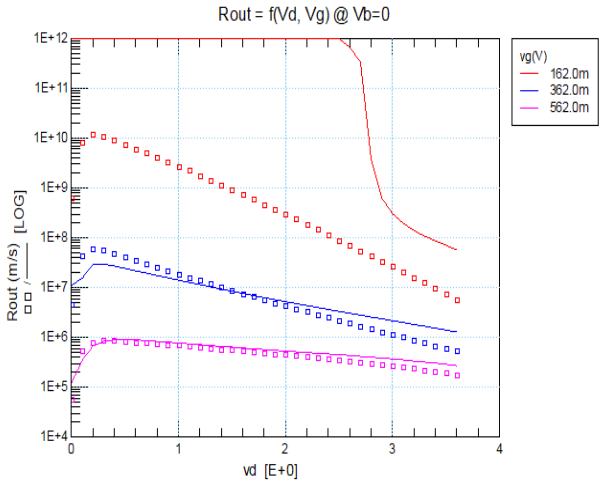


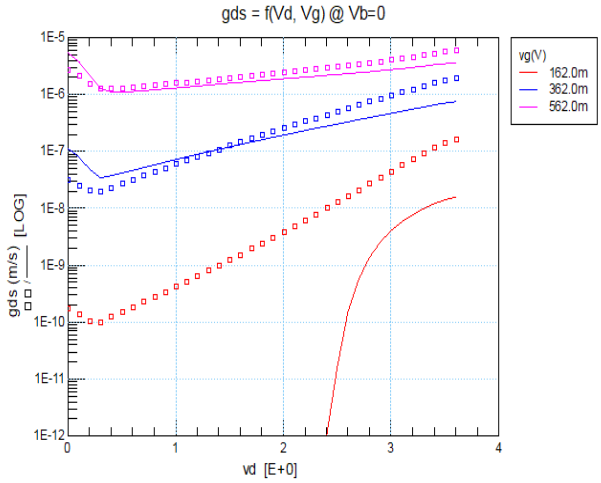




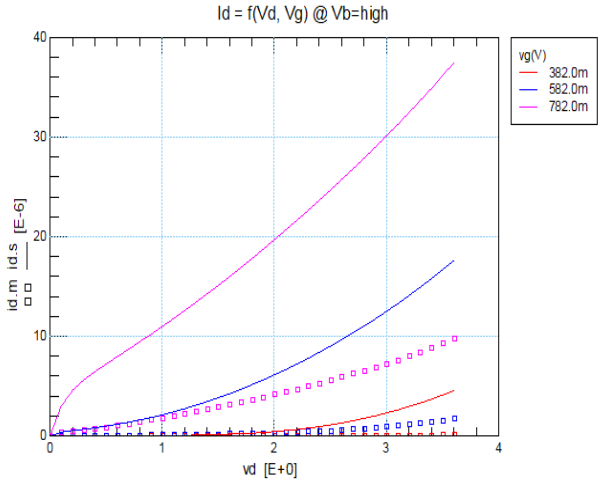
idvd_vth, Vb = 0V, T = 27°C

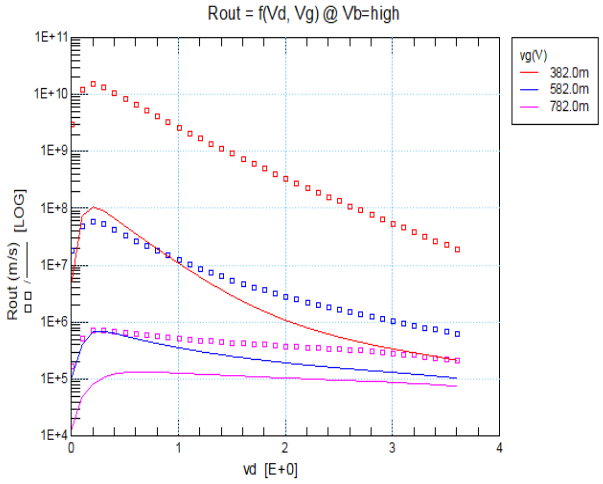


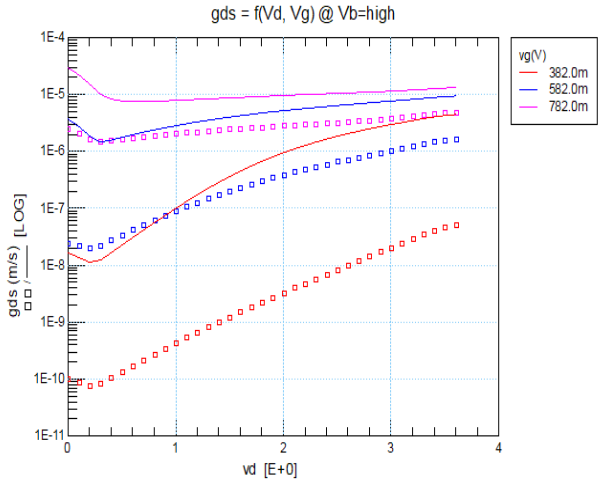




idvd_vth_vbmin, Vb = -3.6V, T = 27°C







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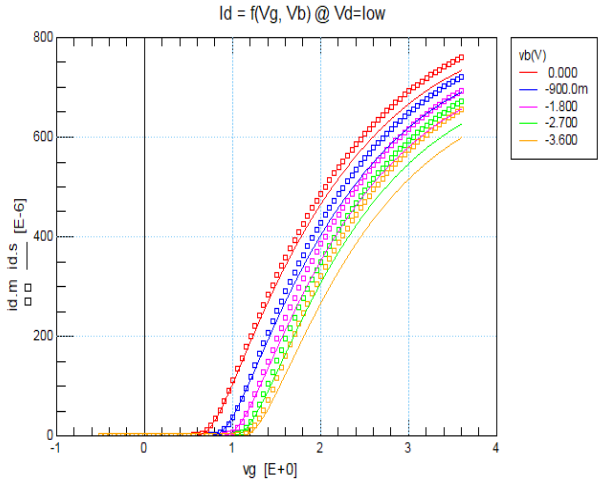
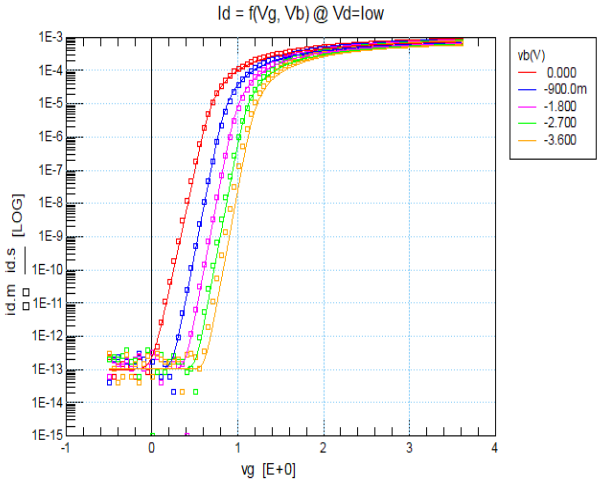
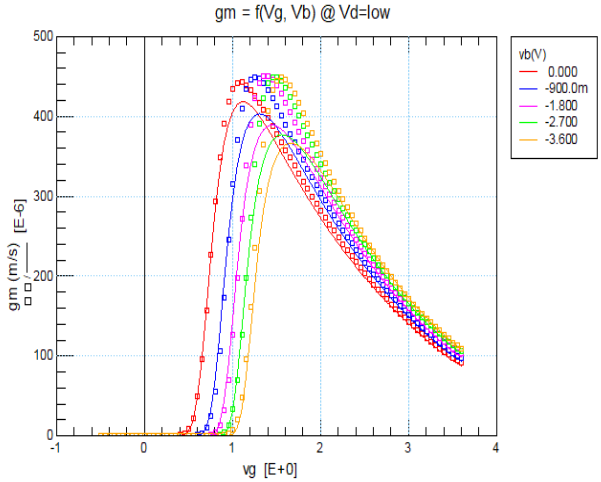
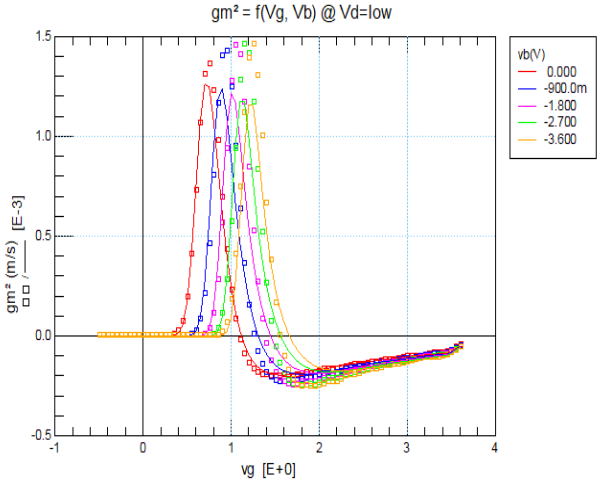
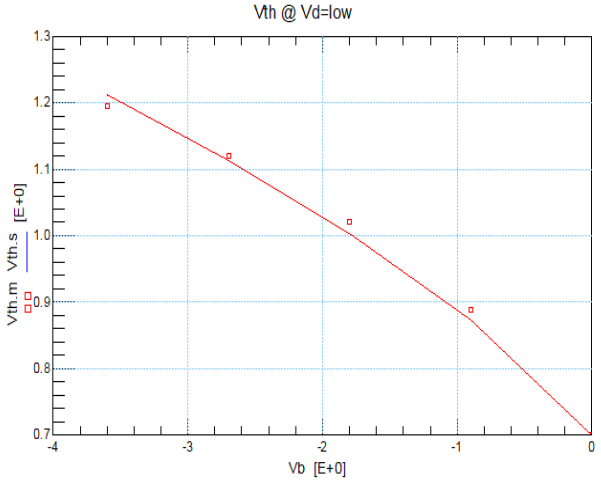
idvg, Vd = 0.1V, T = 27°C

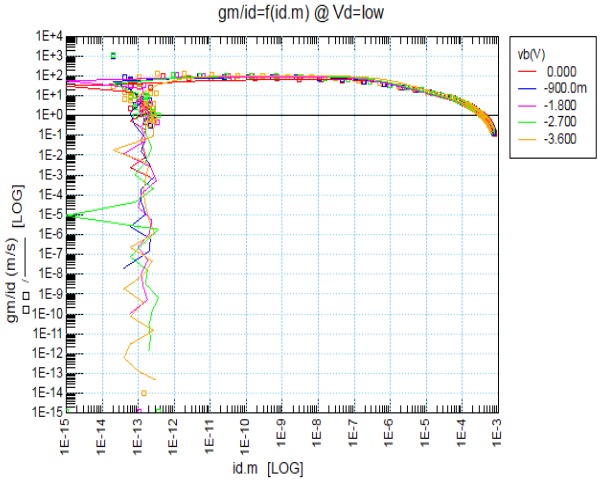


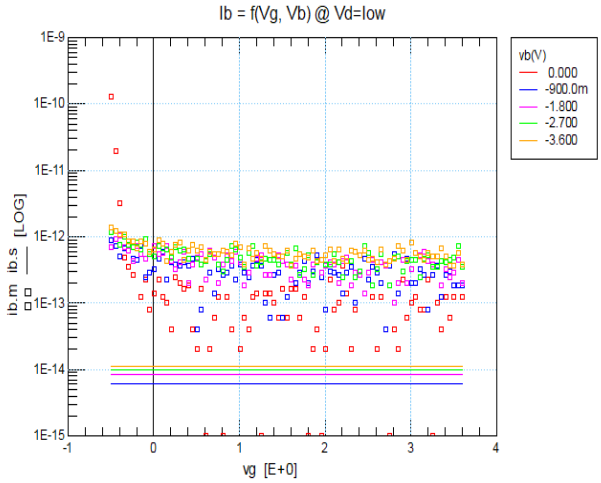


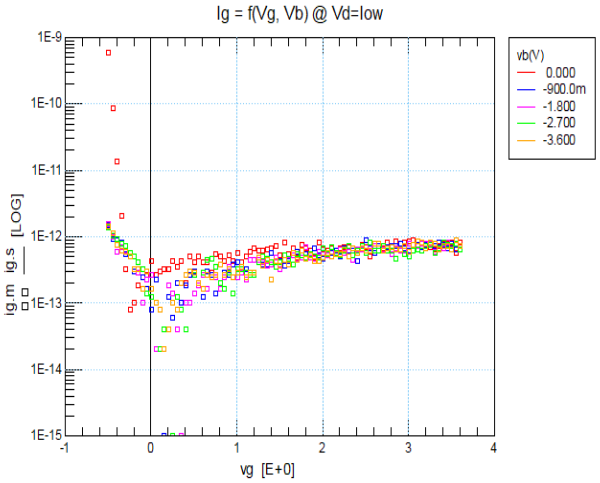




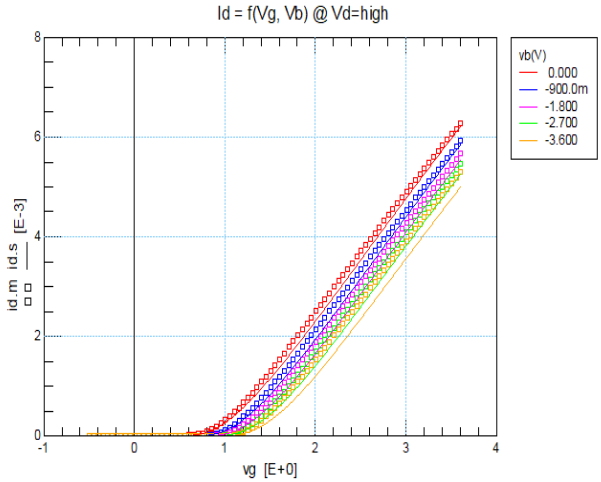


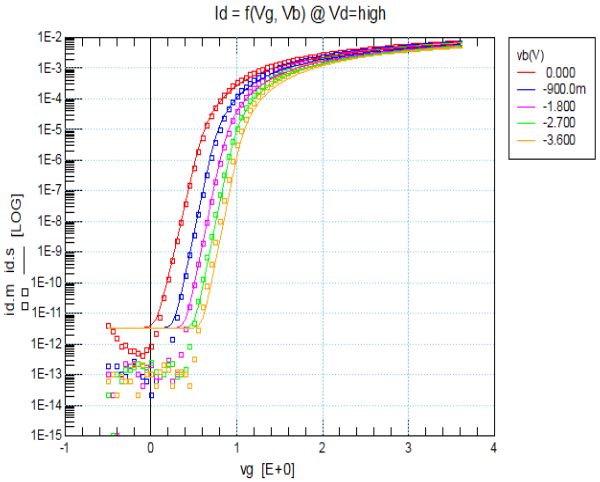


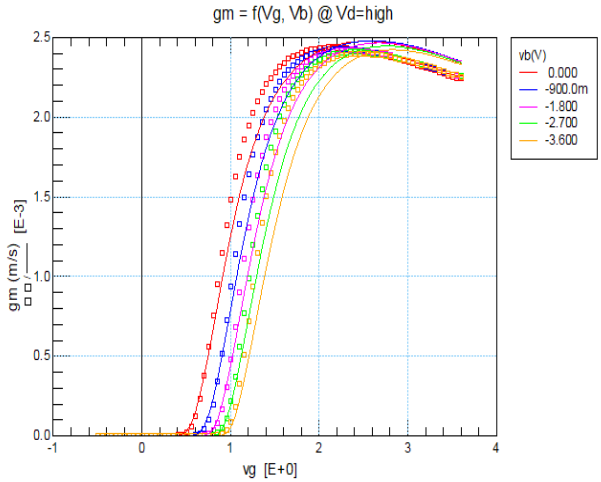


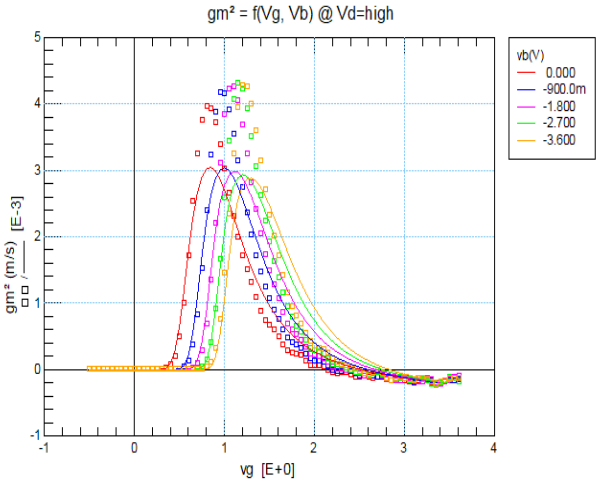


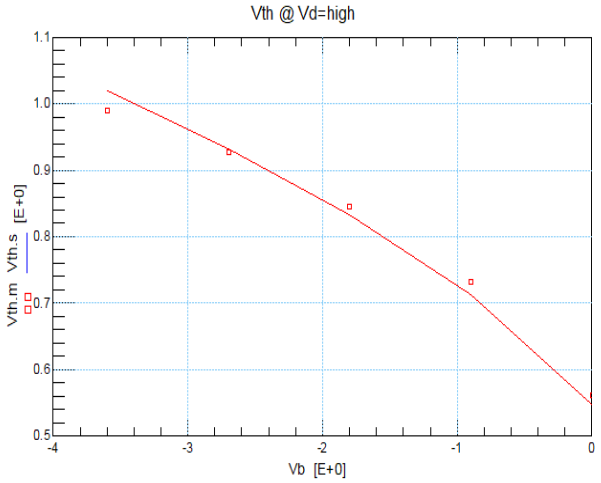
idvg, Vd = 3.3V, T = 27°C

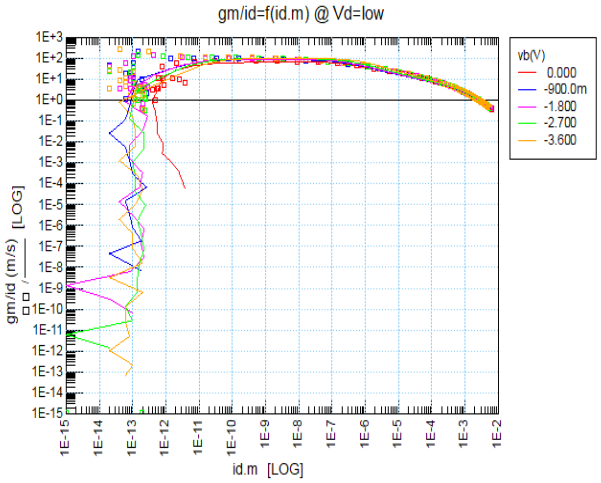


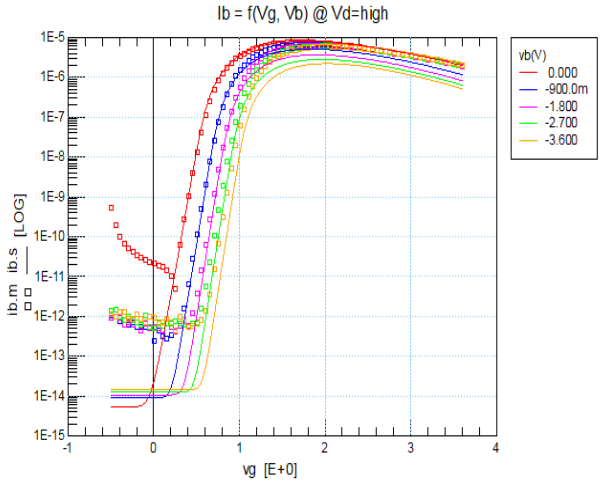


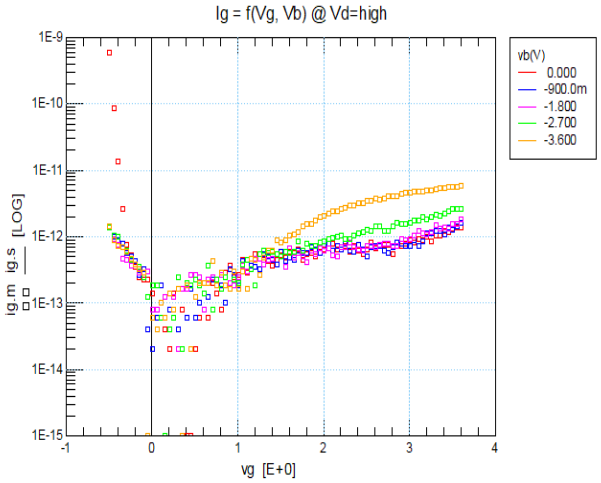




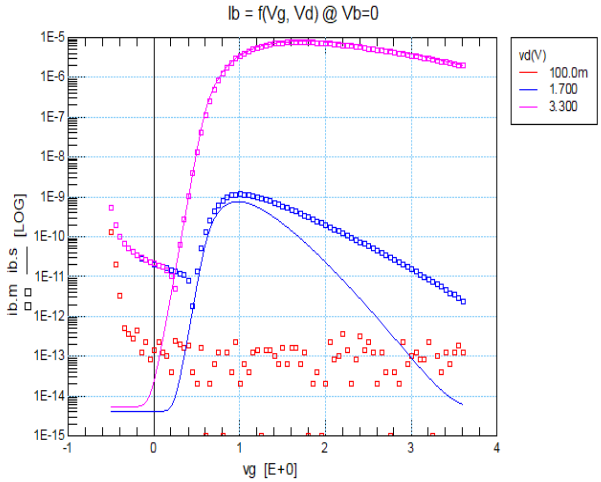


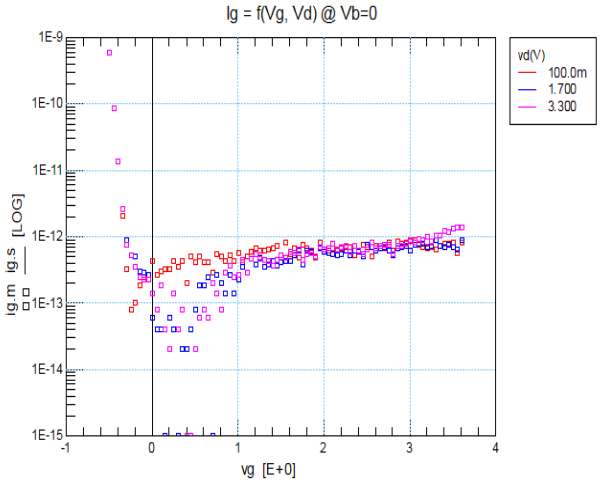




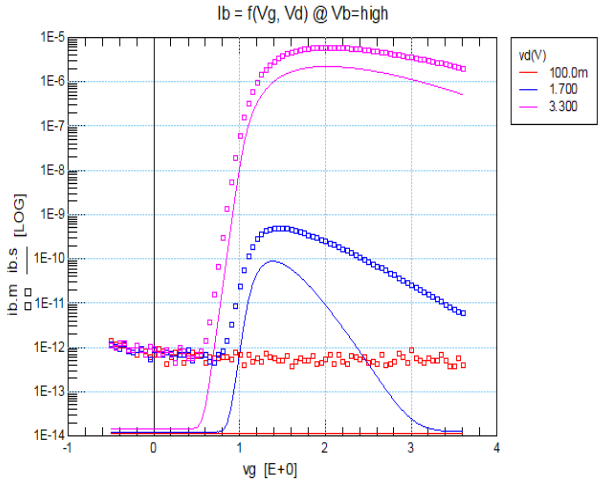


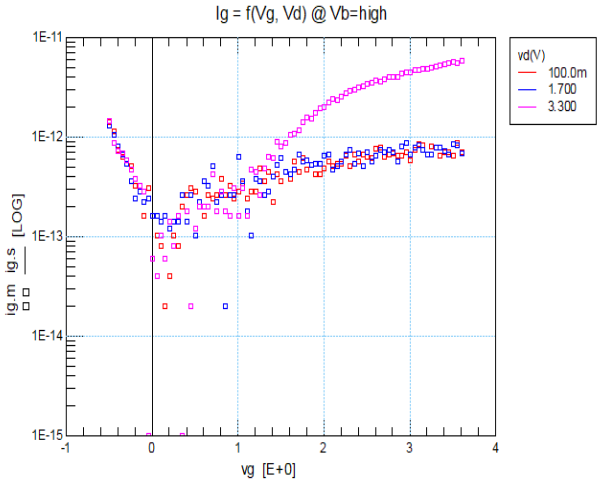
idvg, Ib,Ig, Vb = 0V, T = 27°C



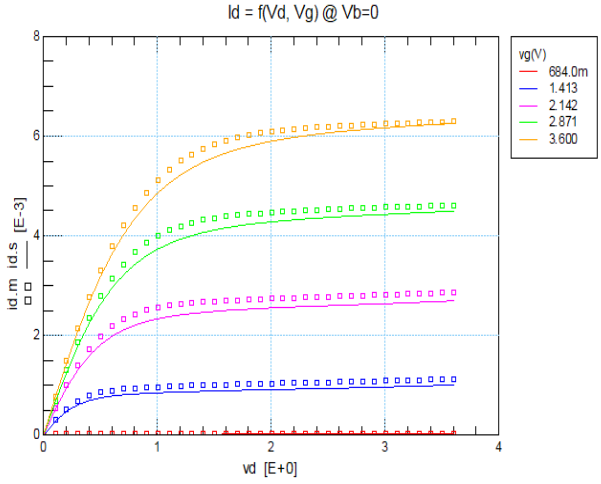


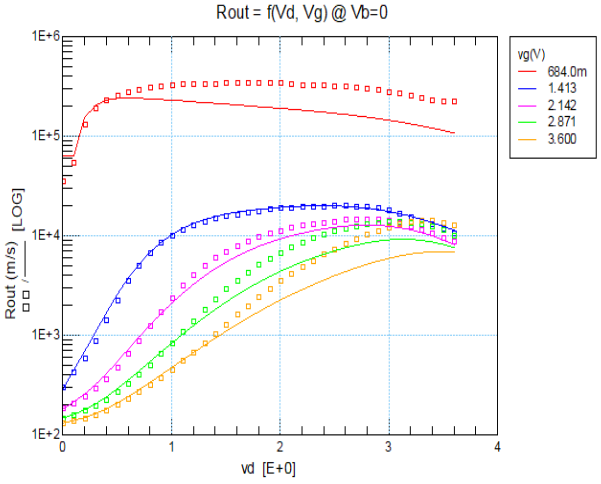
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

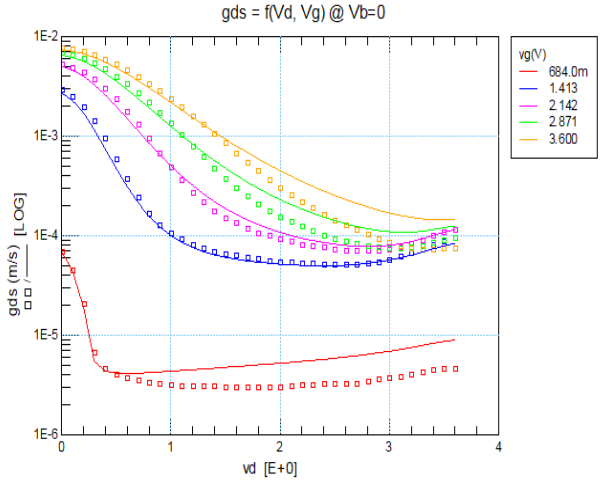




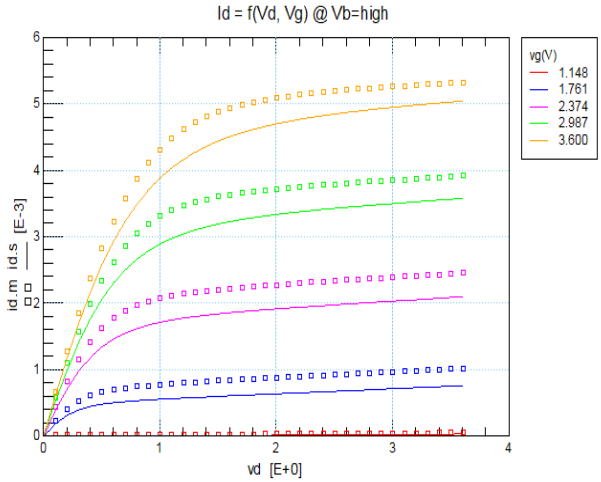
idvd, Vb = 0V, T = 27°C

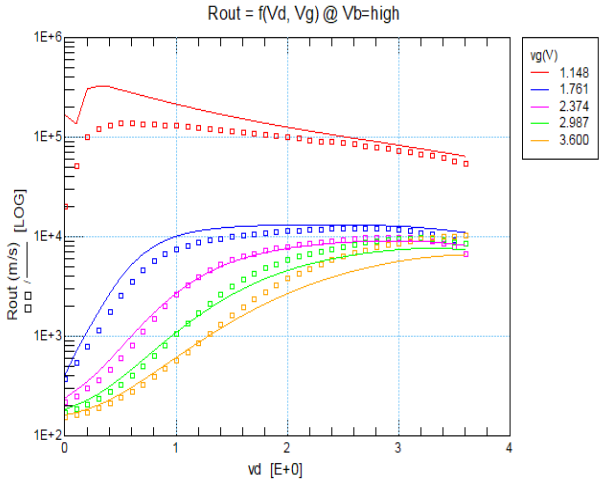


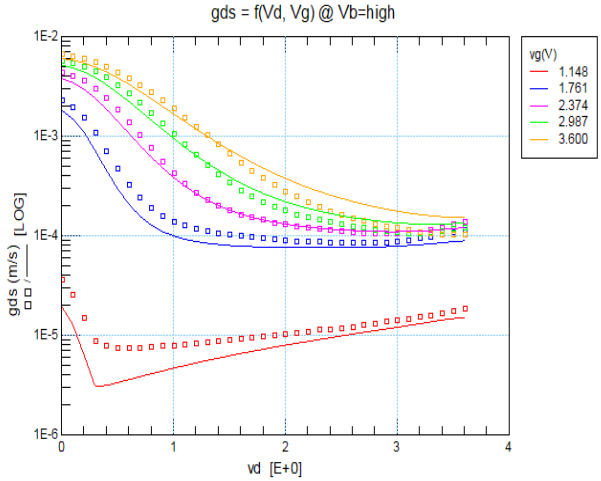




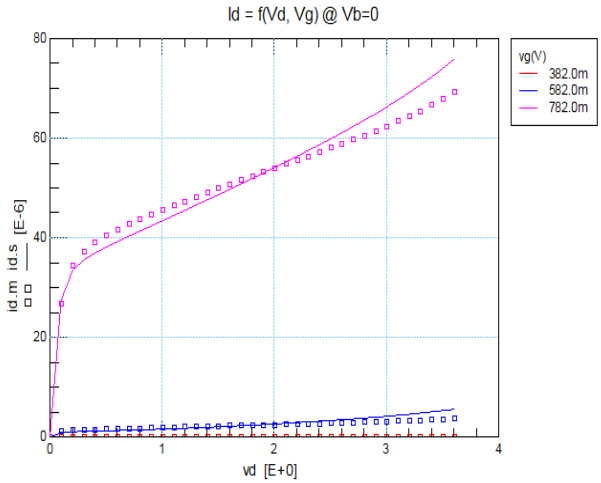
idvd_vbmin, Vb = -3.6V, T = 27°C

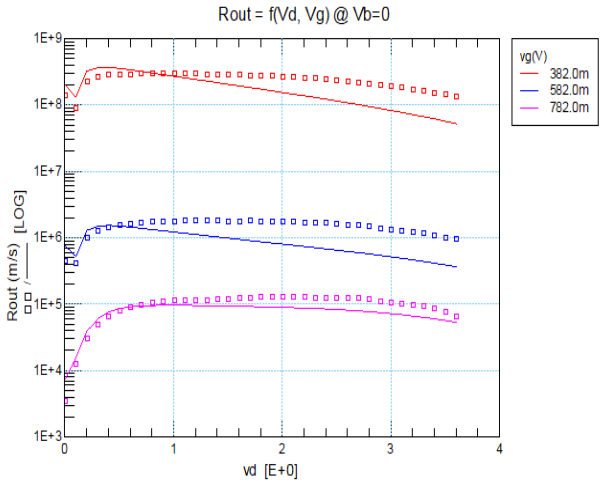


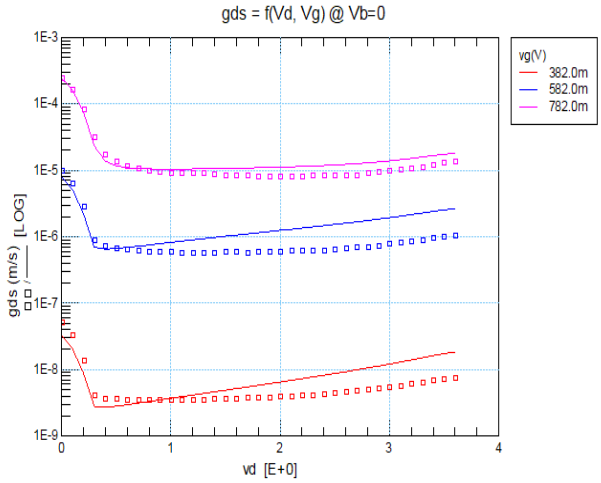




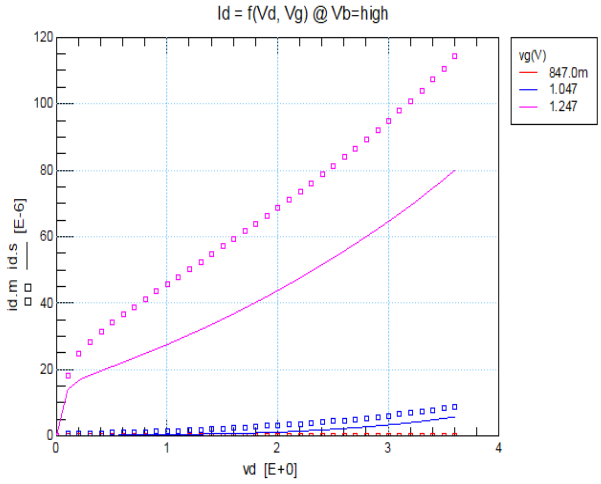
idvd_vth, Vb = 0V, T = 27°C

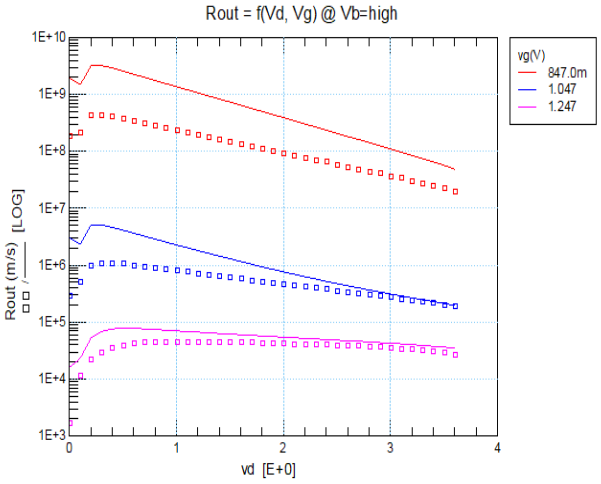


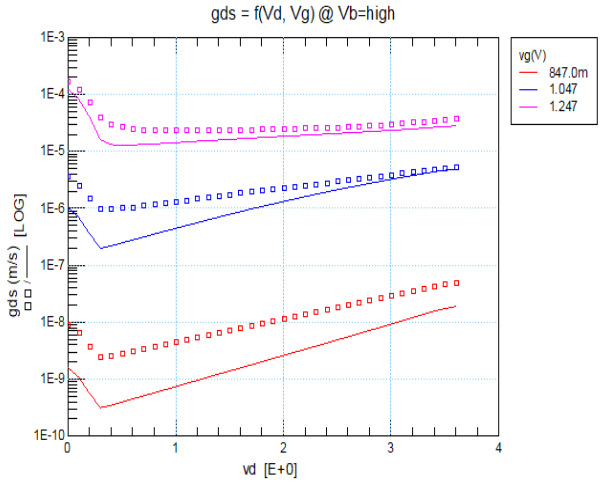




idvd_vth_vbmin, Vb = -3.6V, T = 27°C

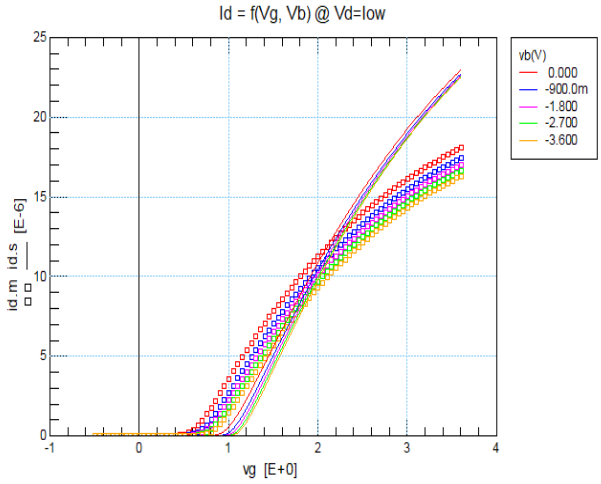


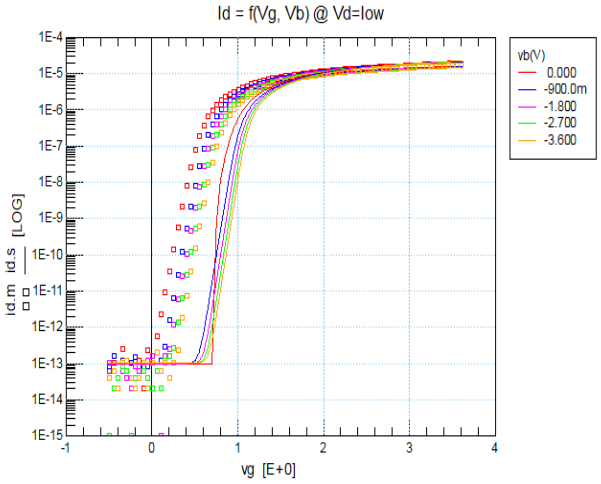


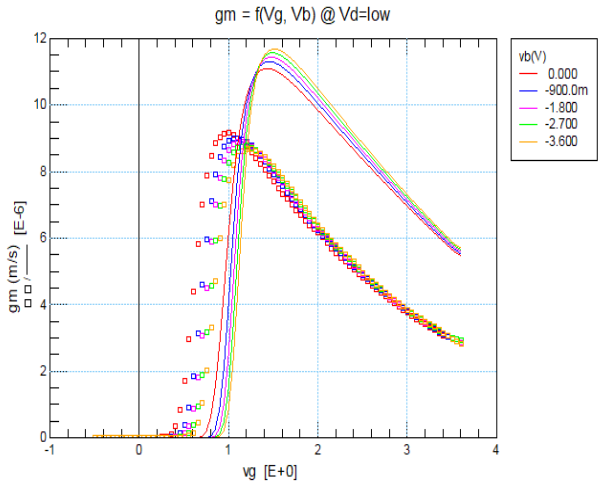


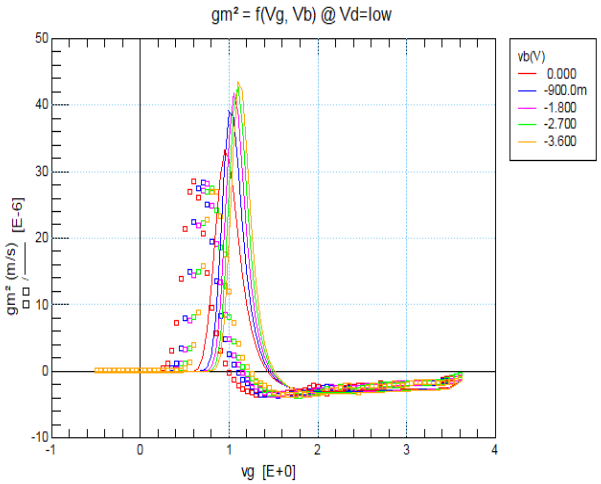
4.28W0u15_L0u33_S547_5

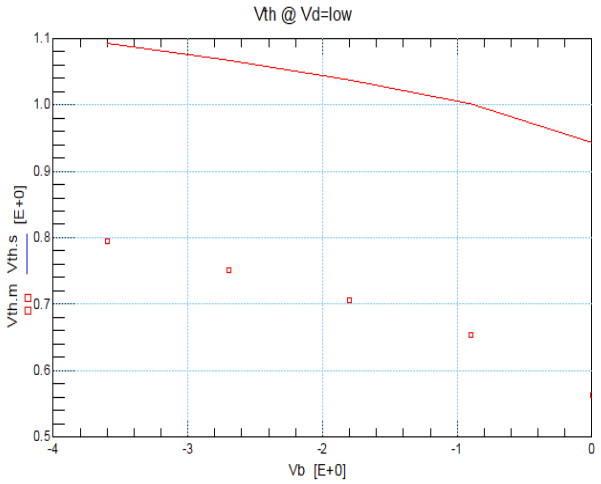
idvg, Vd = 0.1V, T = 27°C

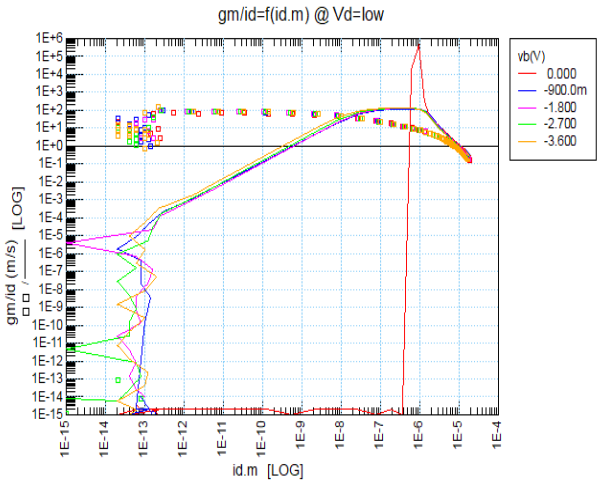


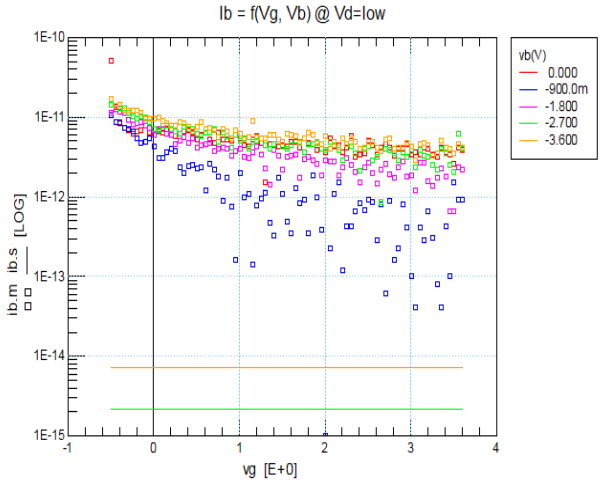


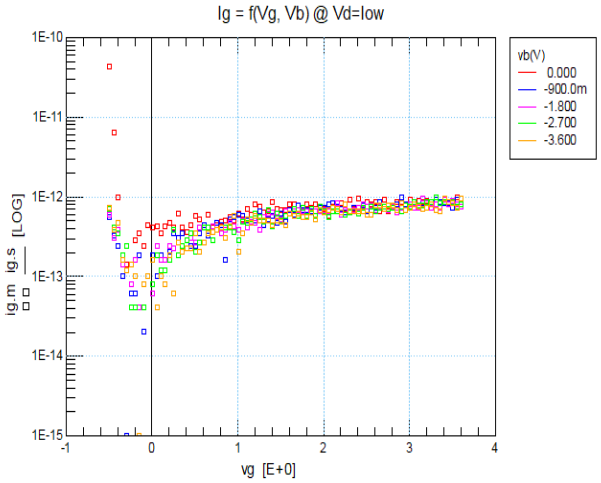




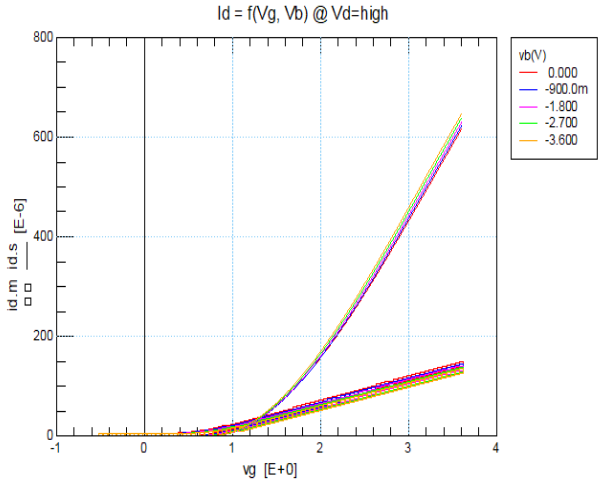


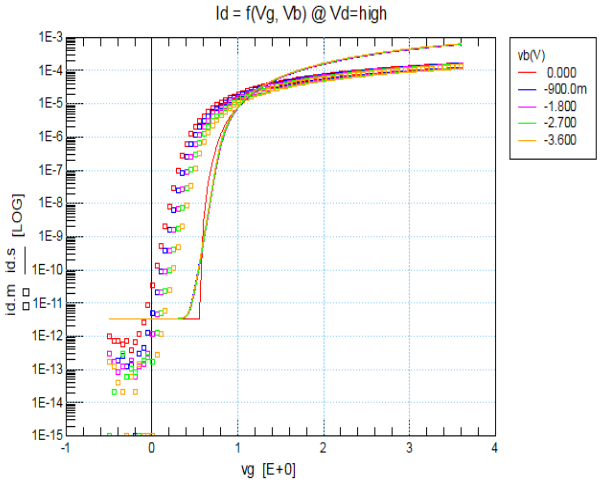


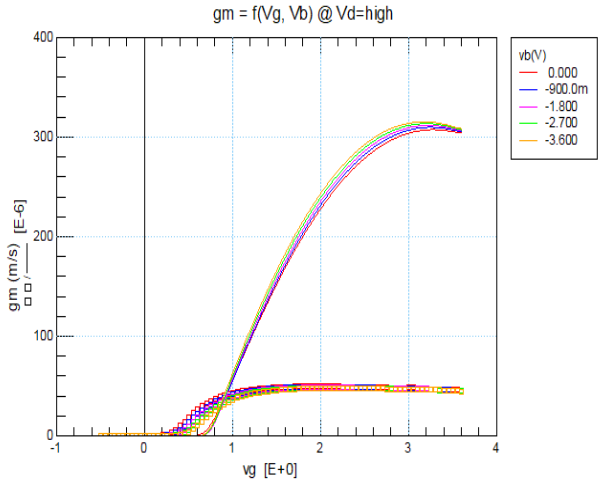


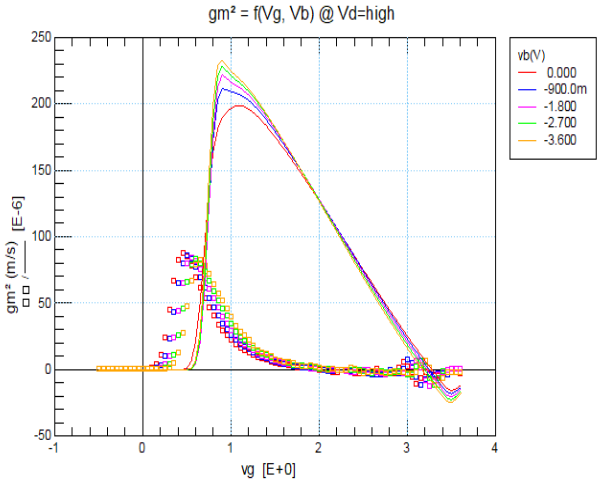


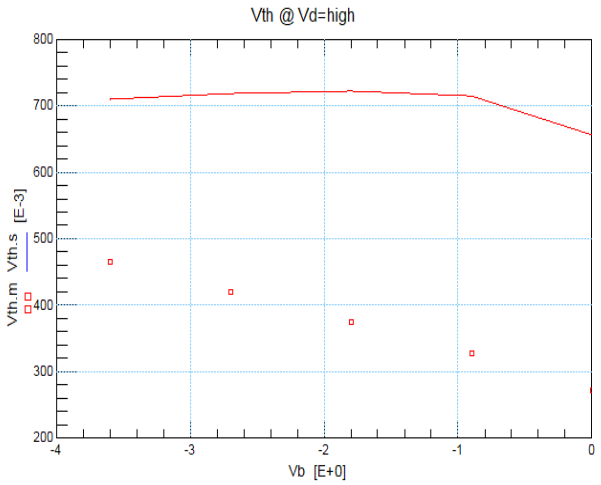
idvg, Vd = 3.3V, T = 27°C

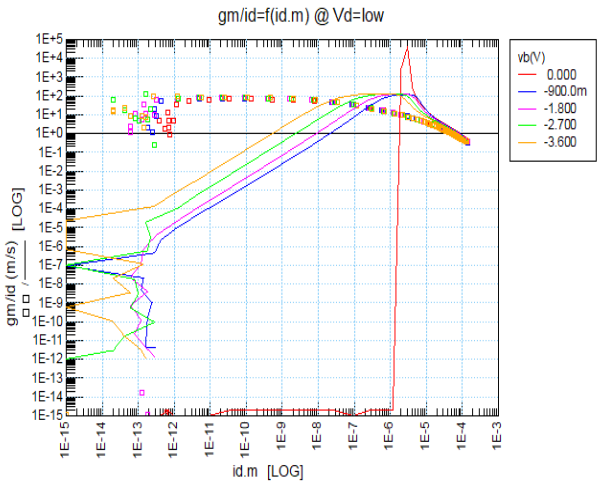


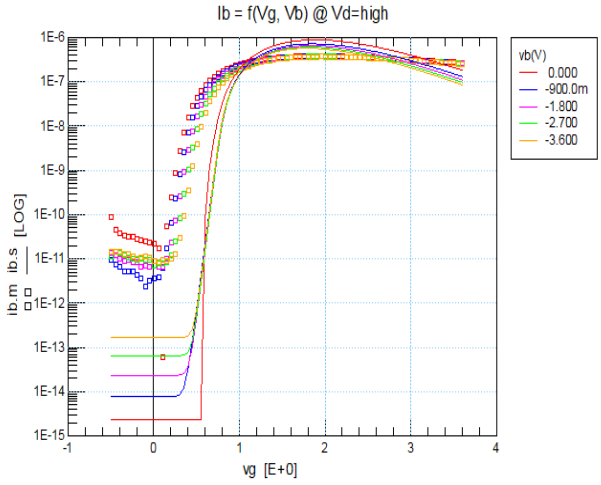


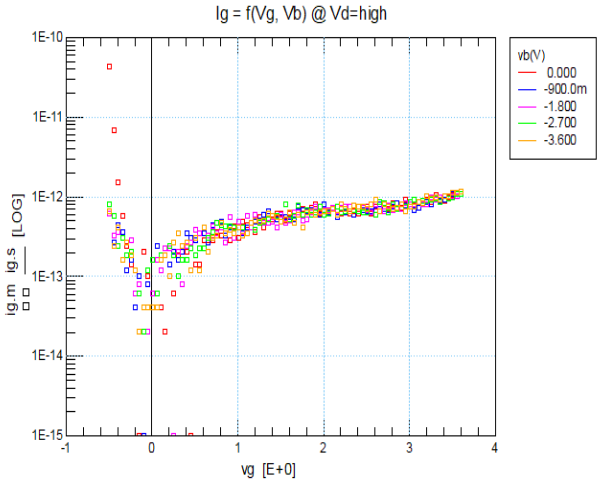




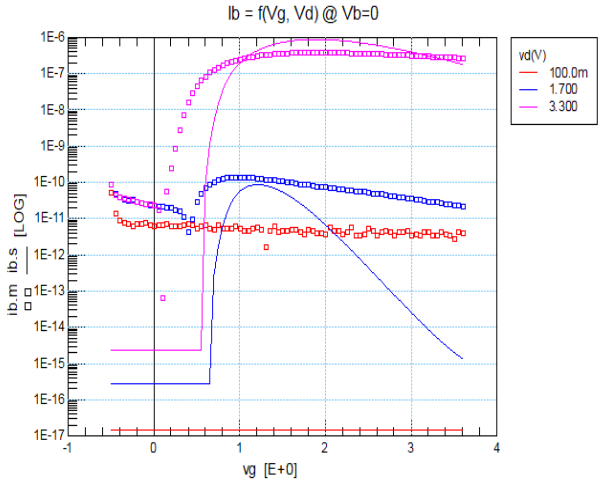


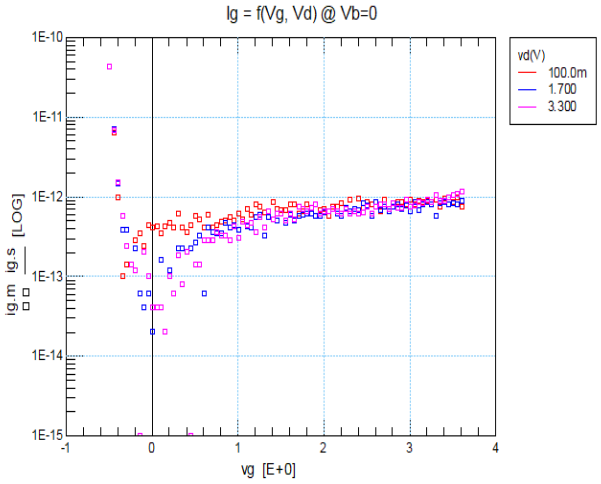




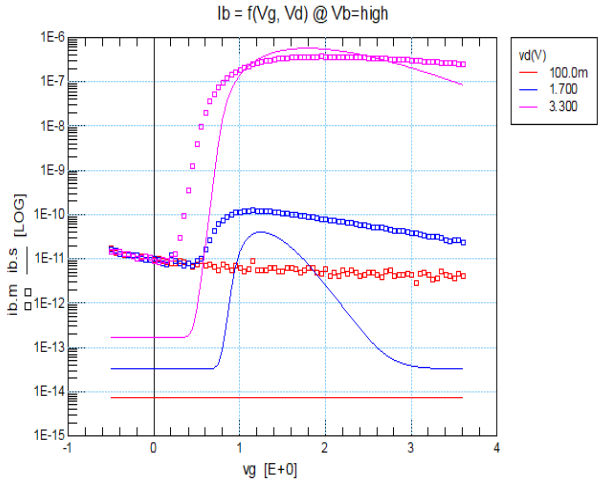


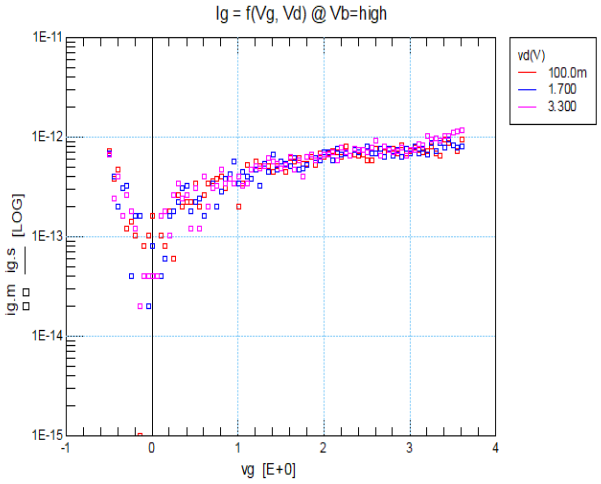
idvg, Ib,Ig, Vb = 0V, T = 27°C



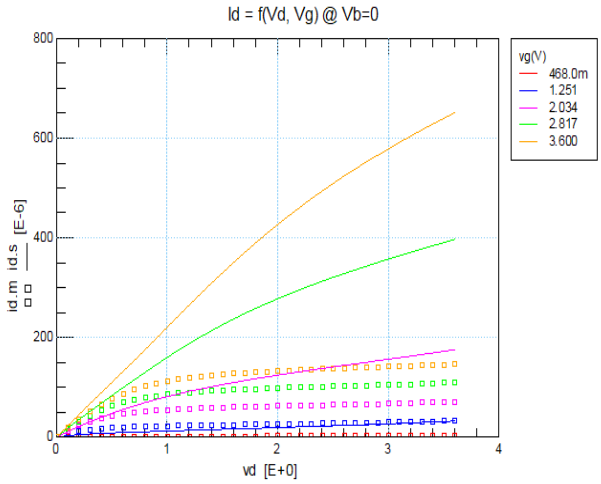


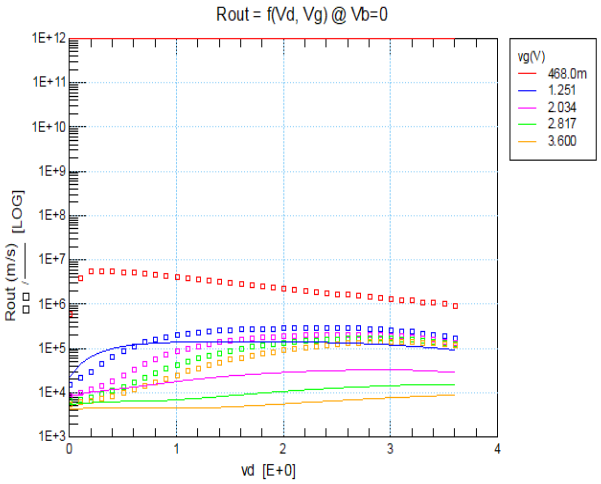
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

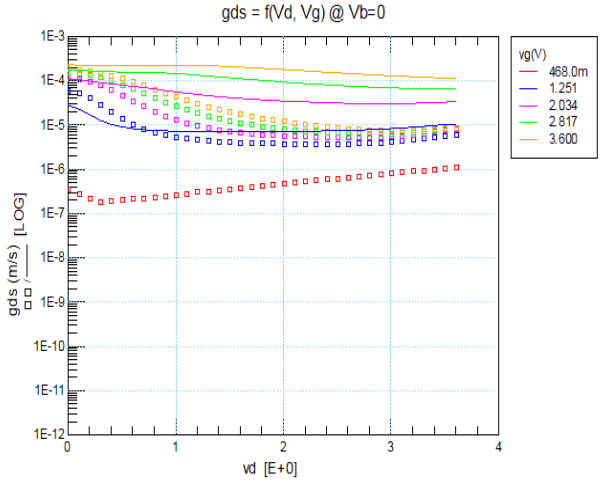




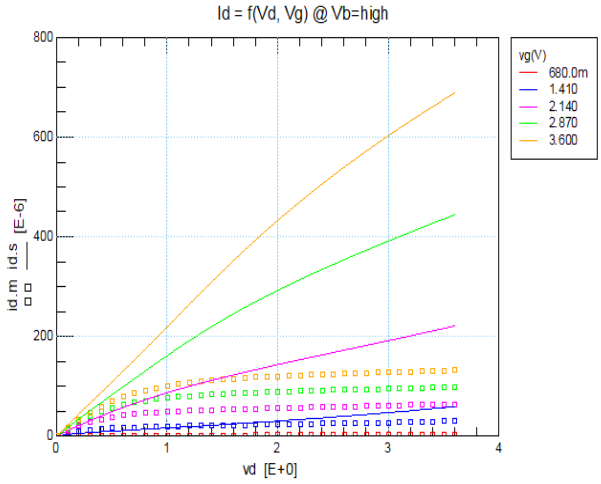
idvd, Vb = 0V, T = 27°C

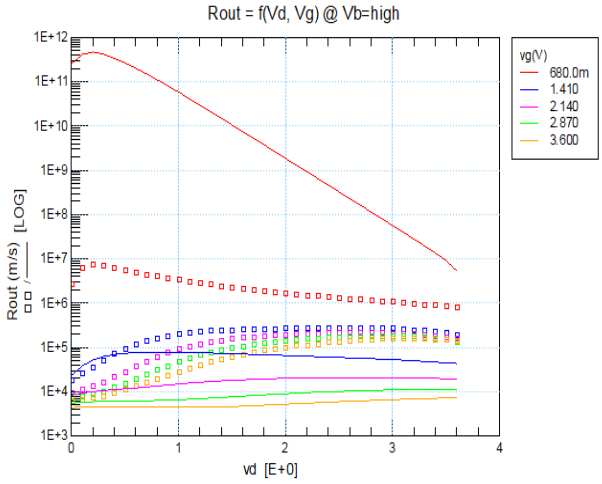


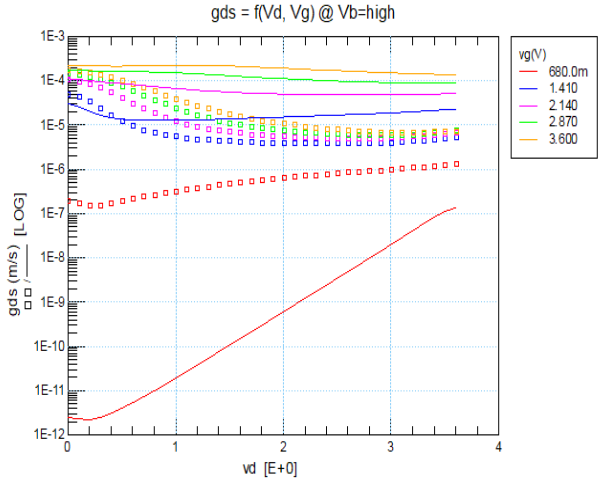




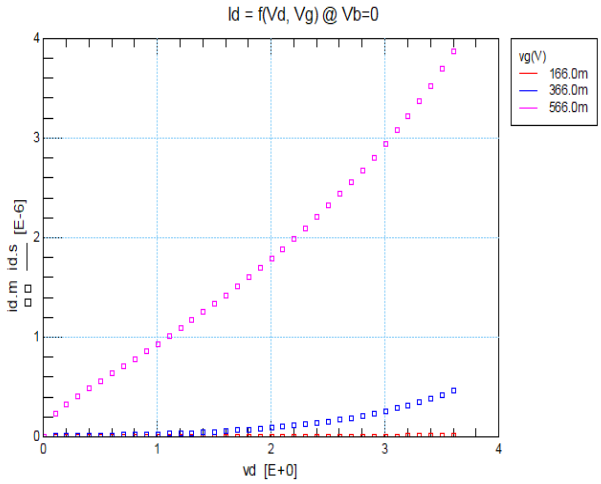
idvd_vbmin, Vb = -3.6V, T = 27°C

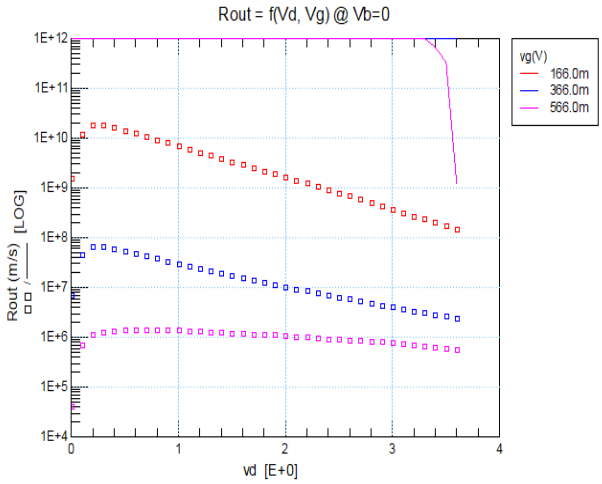


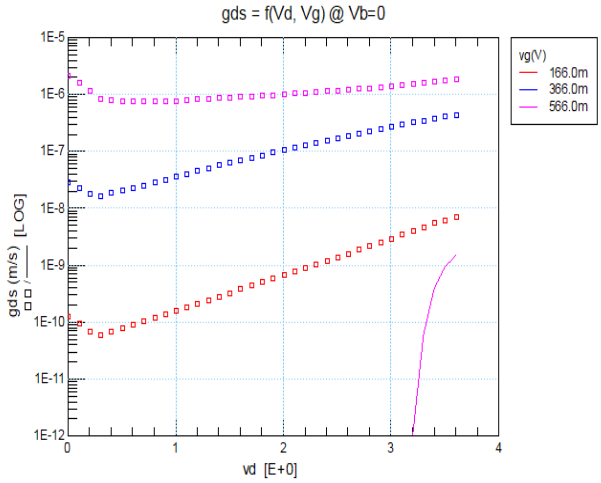




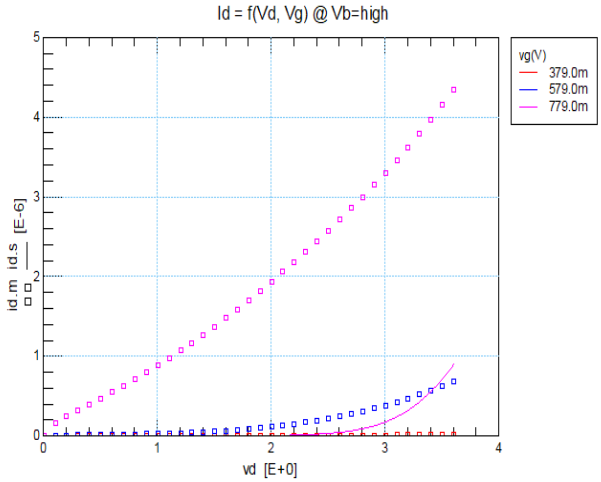
idvd_vth, Vb = 0V, T = 27°C

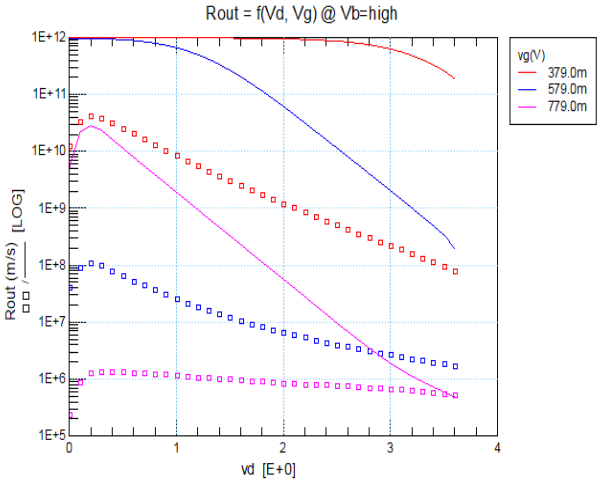


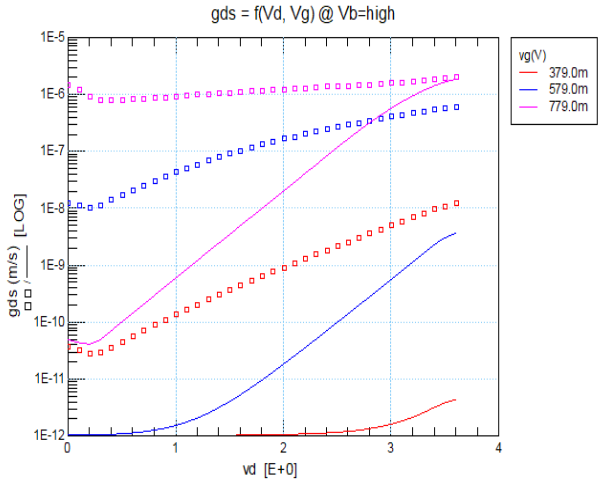




idvd_vth_vbmin, Vb = -3.6V, T = 27°C

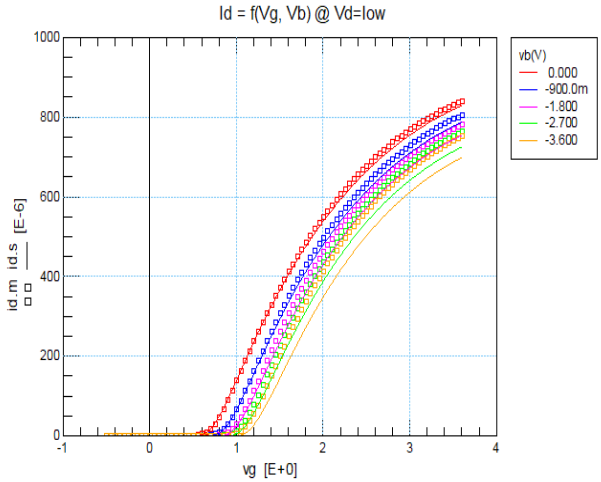


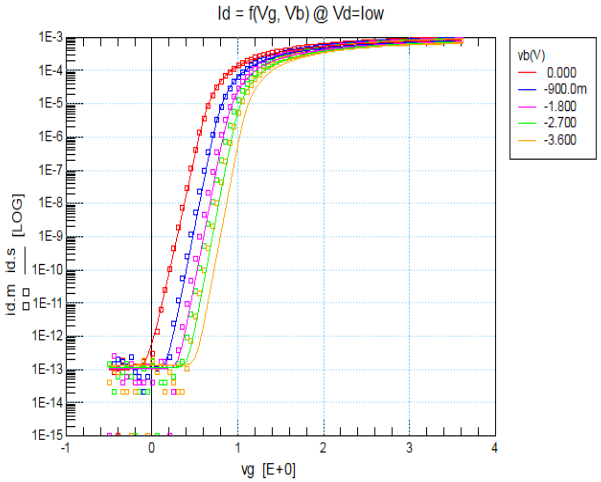


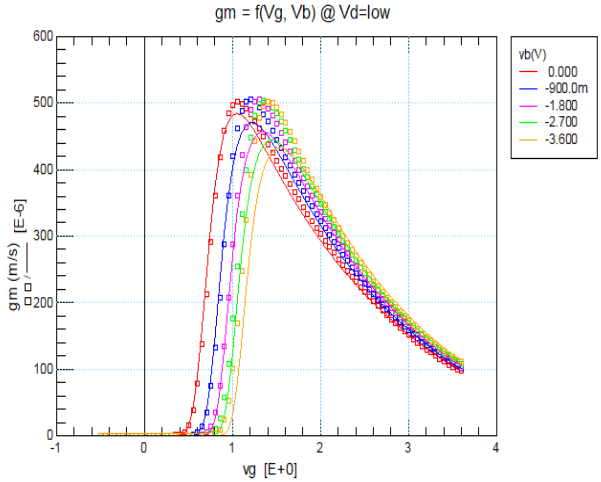


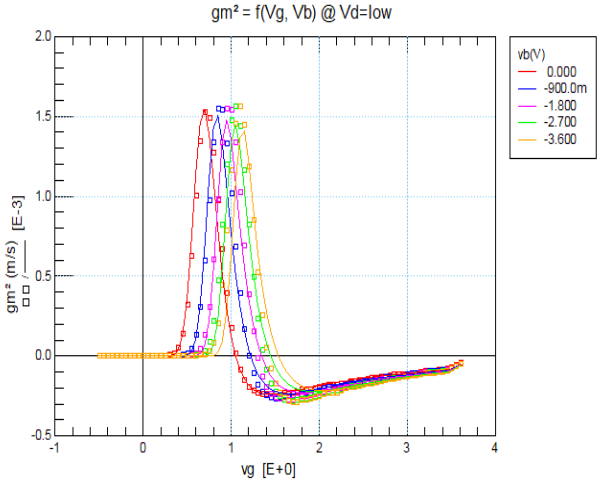
4.29W10u0_L0u4_S387_4

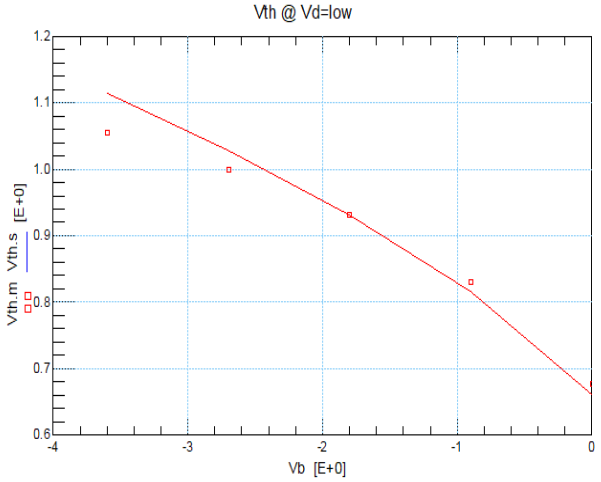
idvg, Vd = 0.1V, T = 27°C

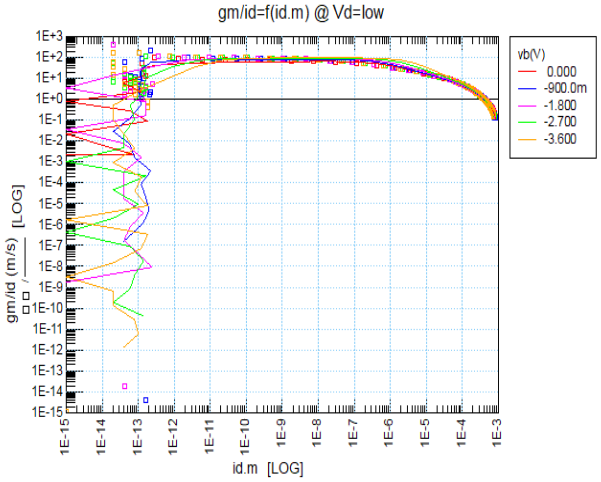


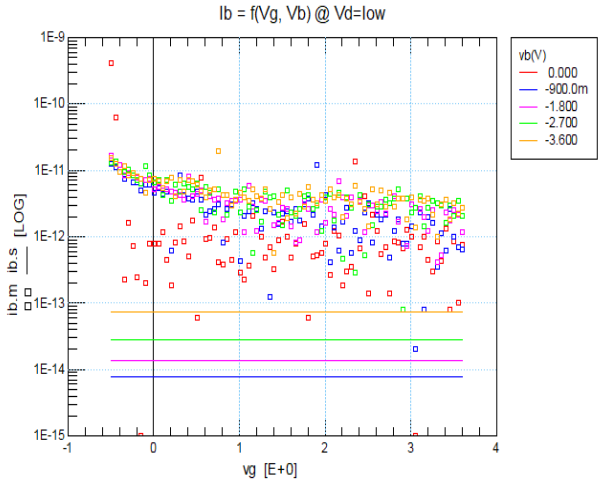


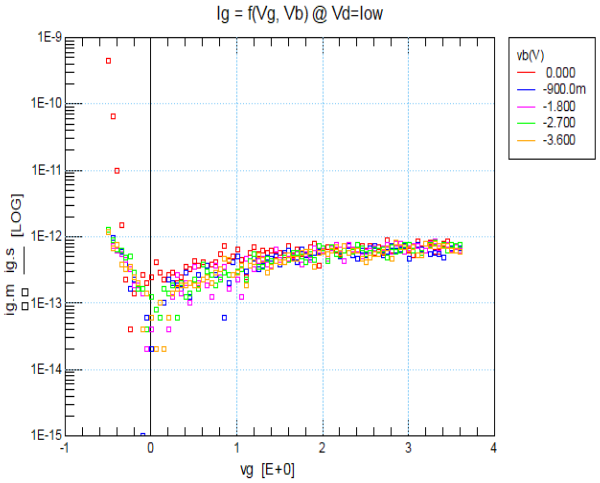




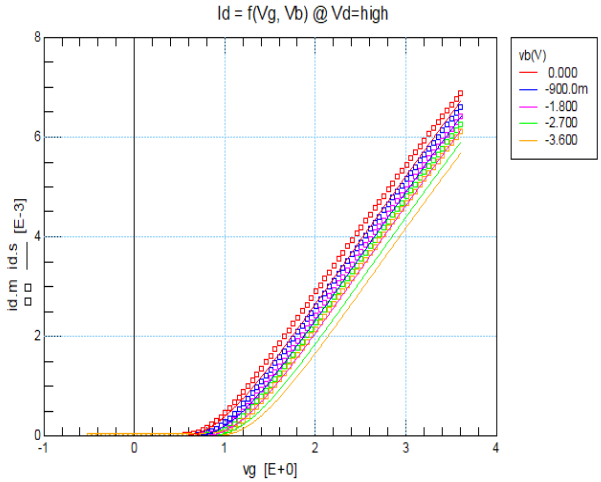


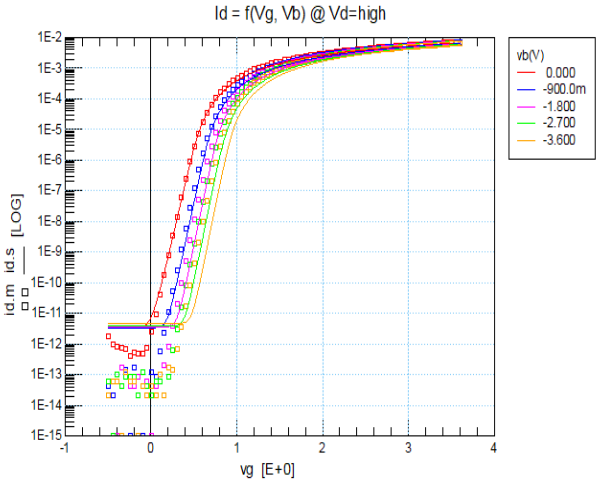


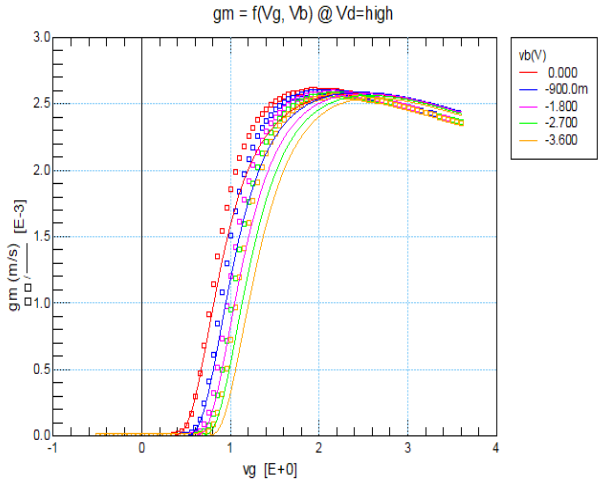


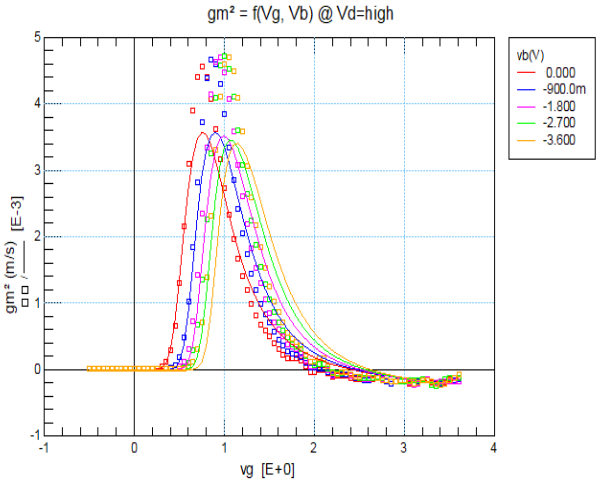


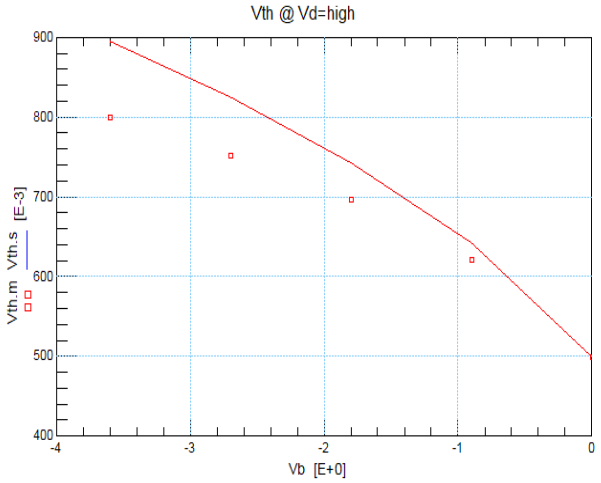
idvg, Vd = 3.3V, T = 27°C

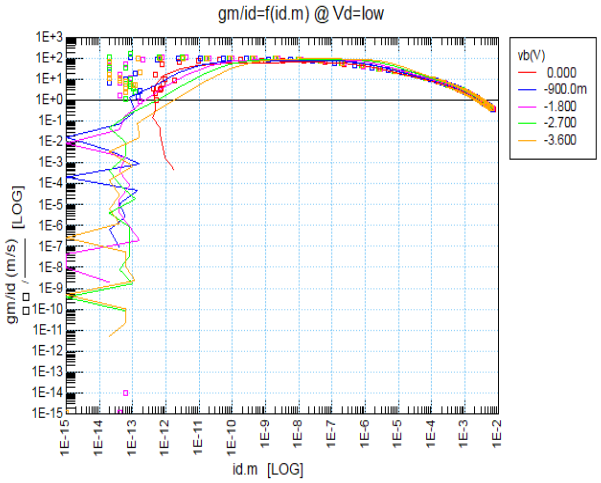


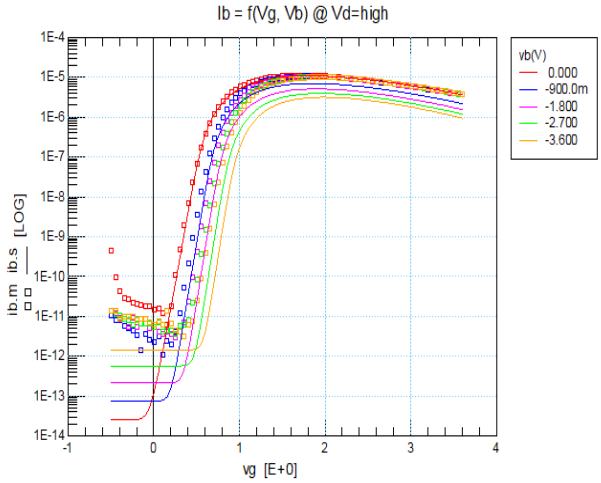


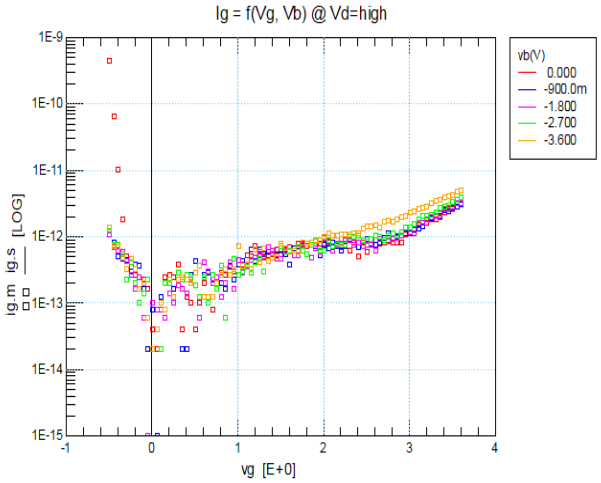




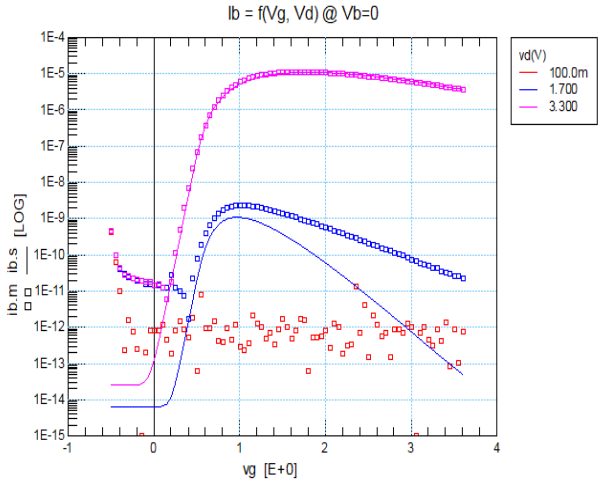


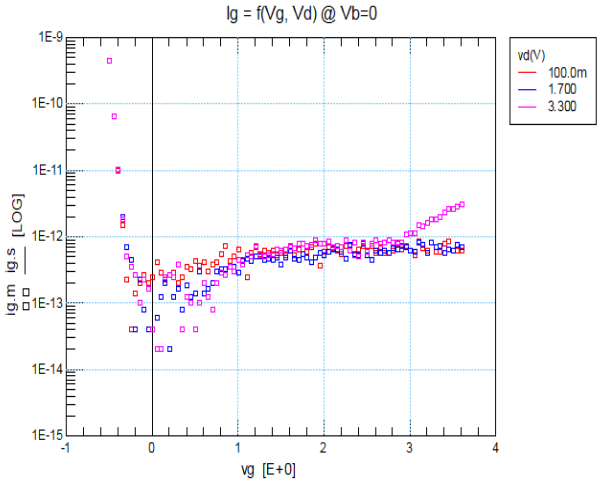




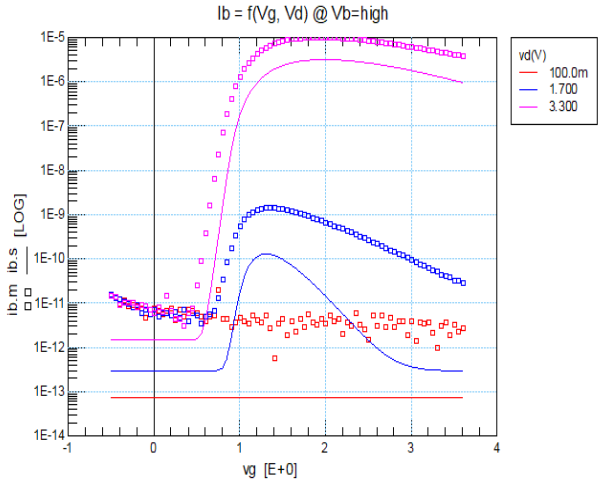


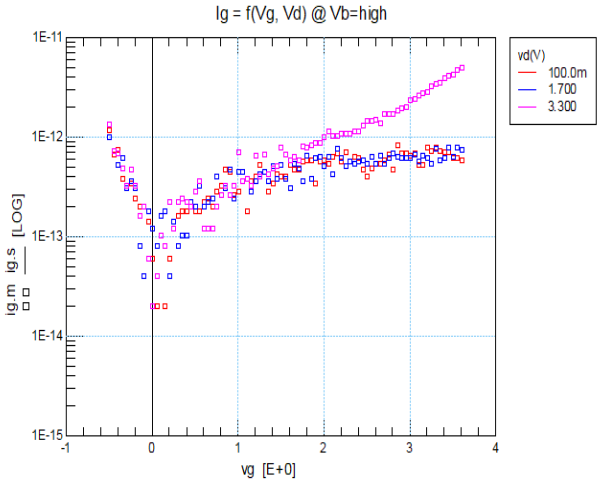
idvg, Ib,Ig, Vb = 0V, T = 27°C



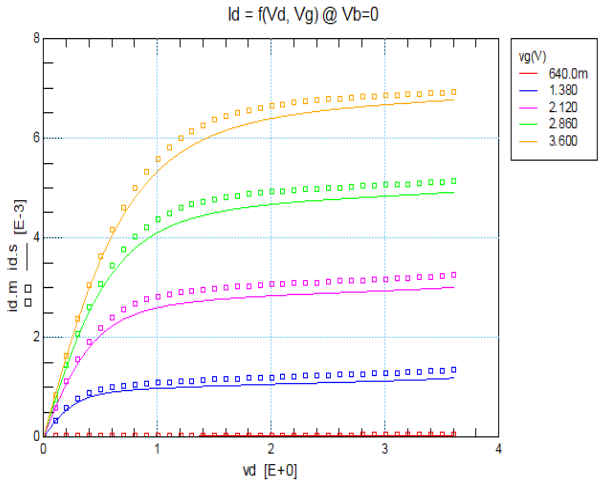


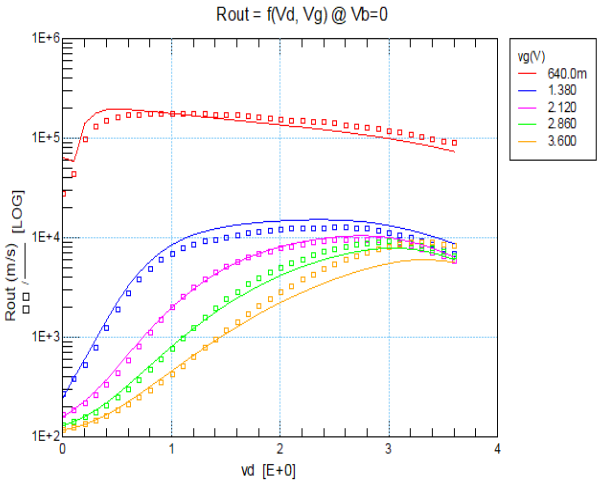
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

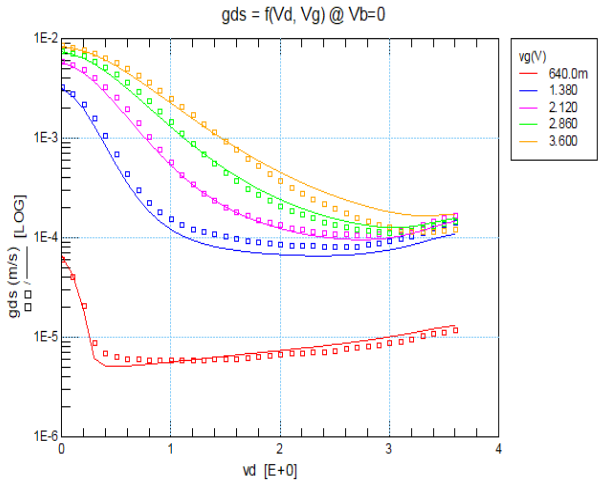




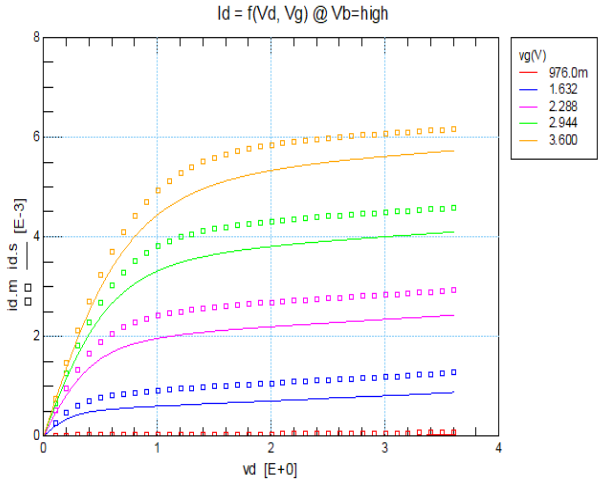
idvd, Vb = 0V, T = 27°C

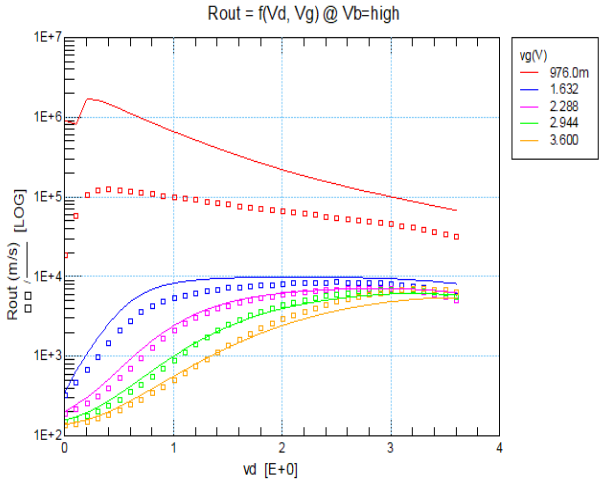


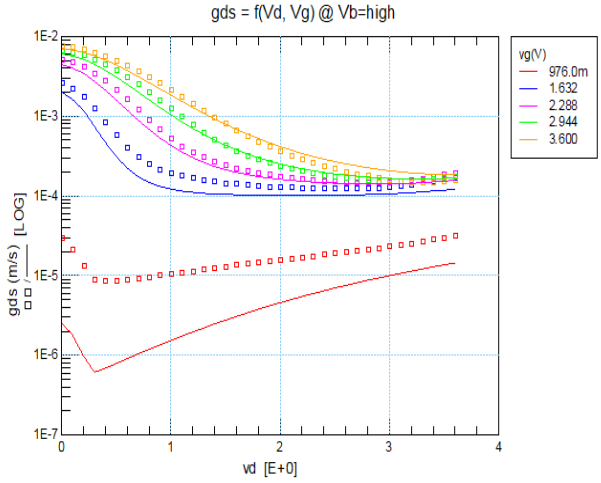




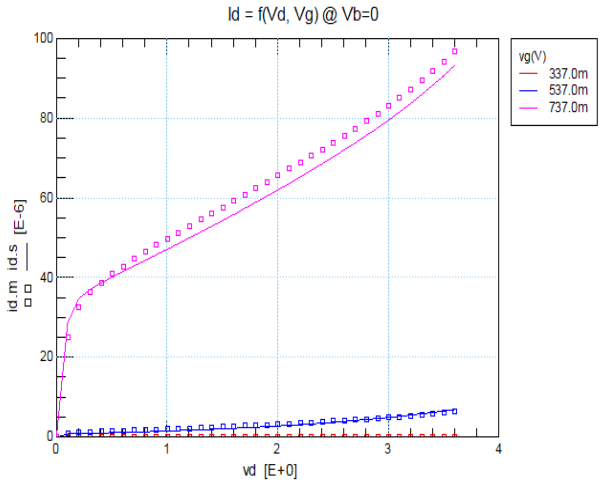
idvd_vbmin, Vb = -3.6V, T = 27°C

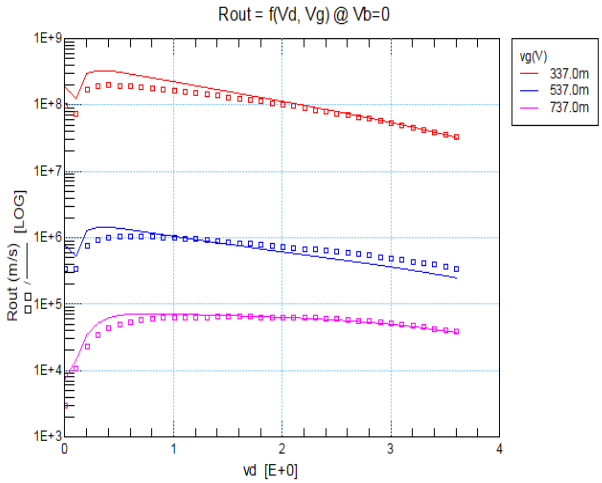


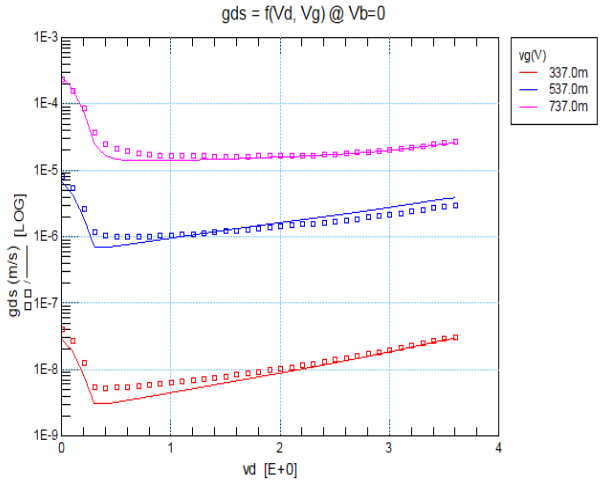




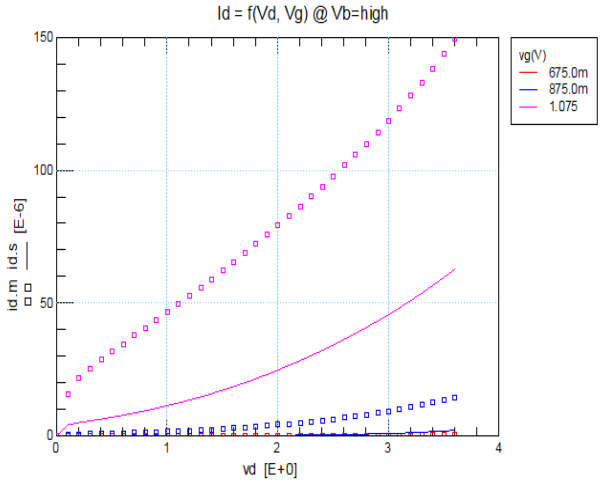
idvd_vth, Vb = 0V, T = 27°C

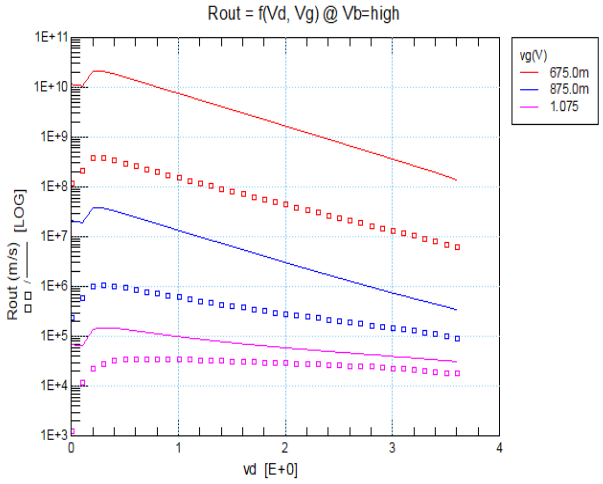


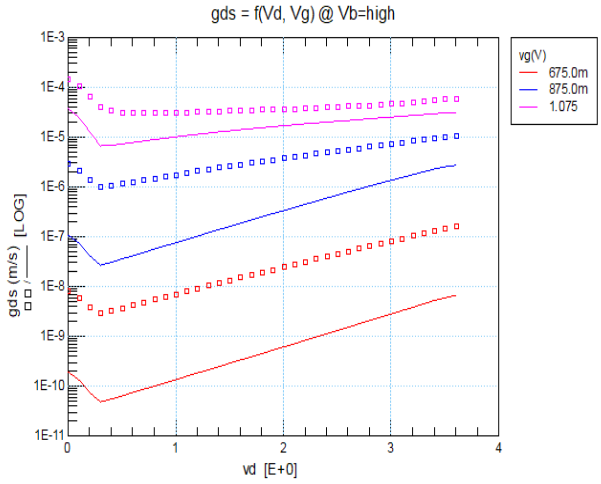




idvd_vth_vbmin, Vb = -3.6V, T = 27°C



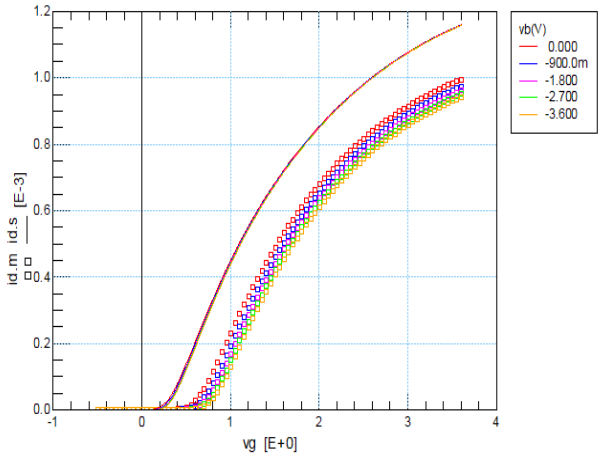




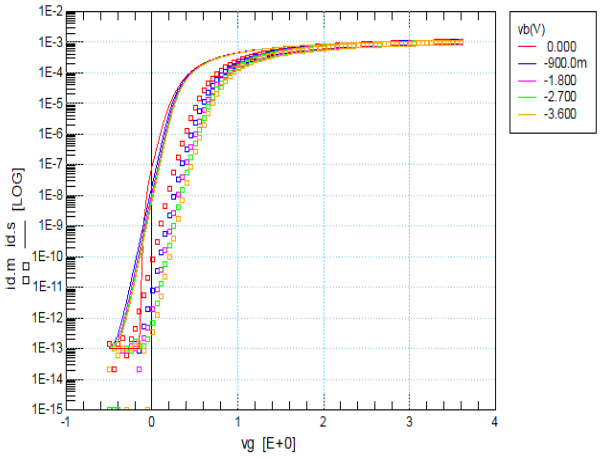
4.30 W10u0_L0u3_S556_1

idvg, Vd = 0.1V, T = 27°C

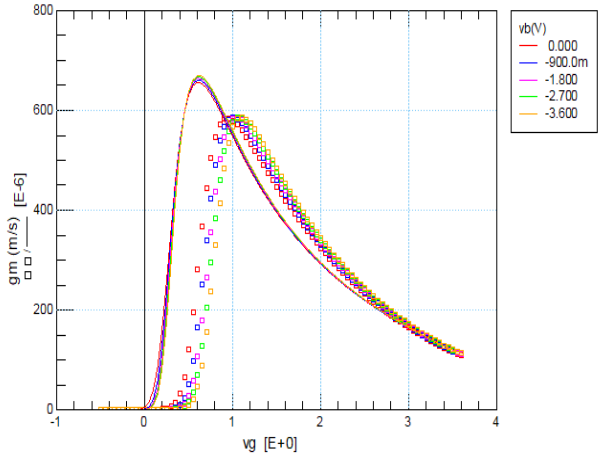
Id = f(Vg, Vb) @ Vd=low



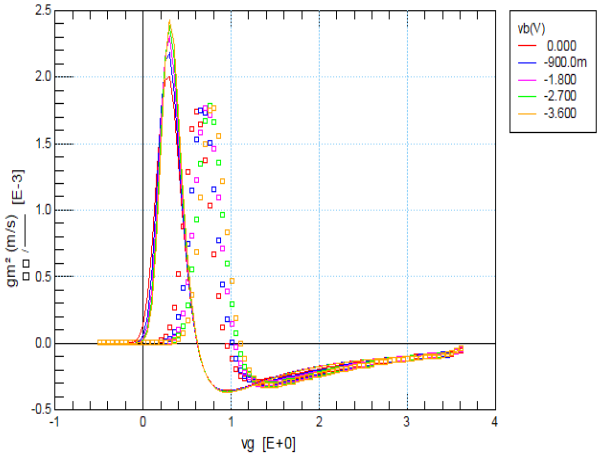
Id = f(Vg, Vb) @ Vd=low



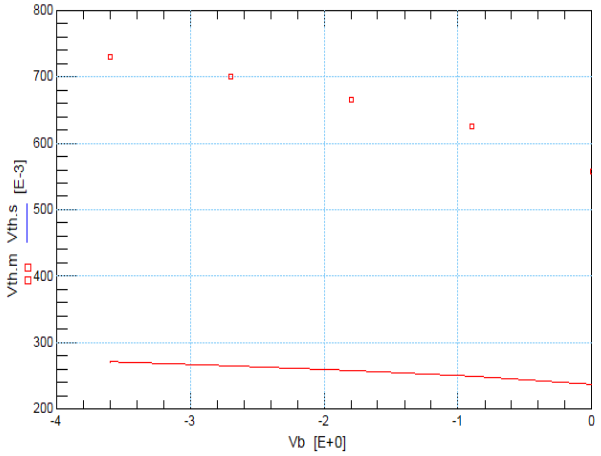
gm = f(Vg, Vb) @ Vd=low



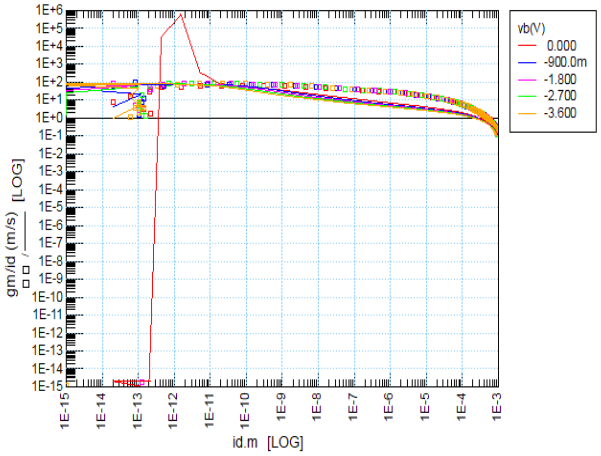
gm² = f(Vg, Vb) @ Vd=low

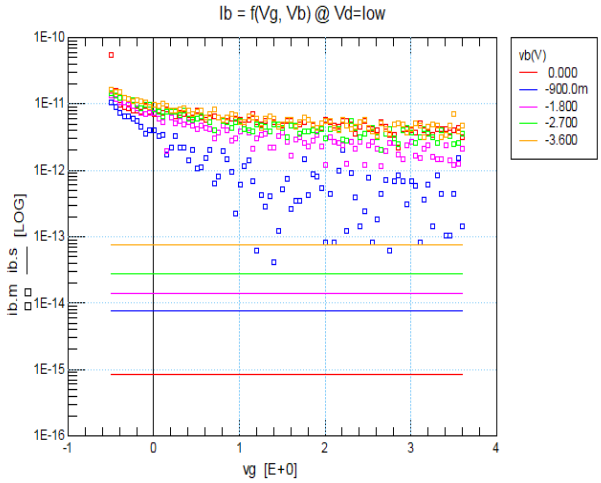


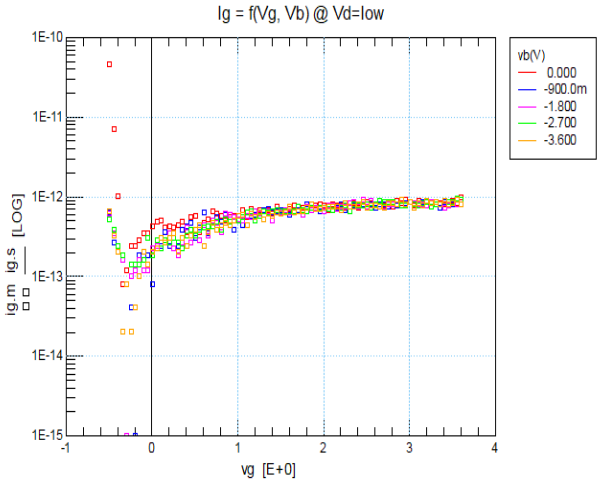
Vth @ Vd=low



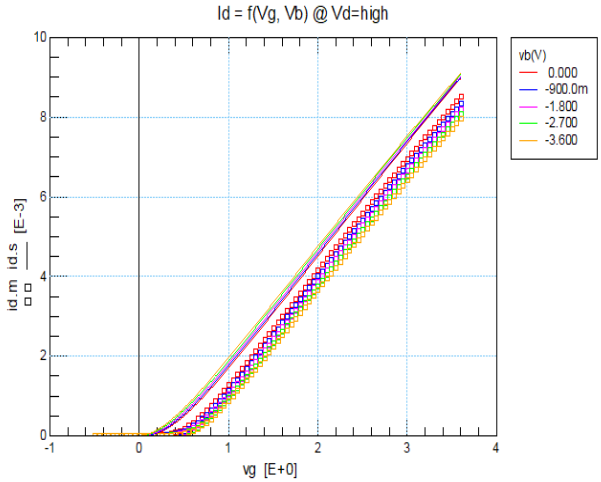
gm/id=f(id.m) @ Vd=low

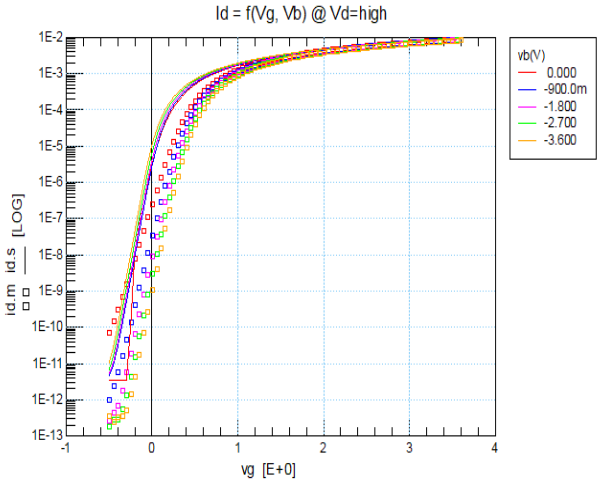


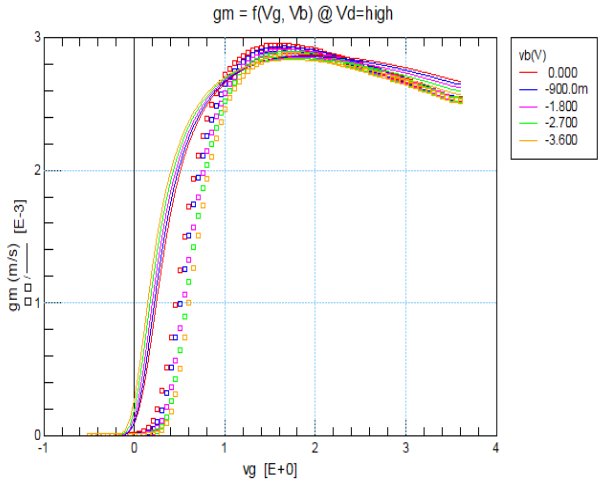


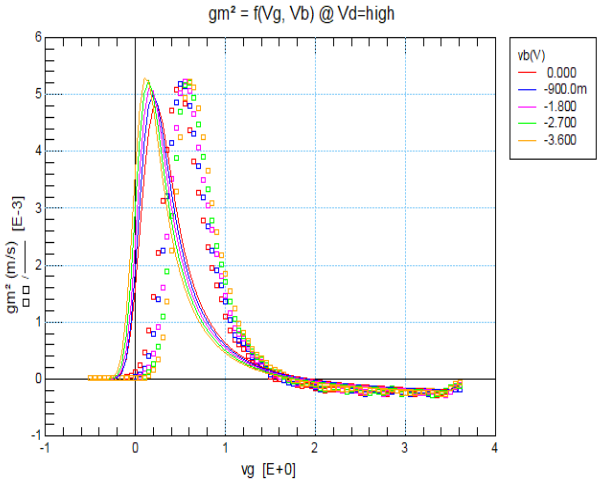


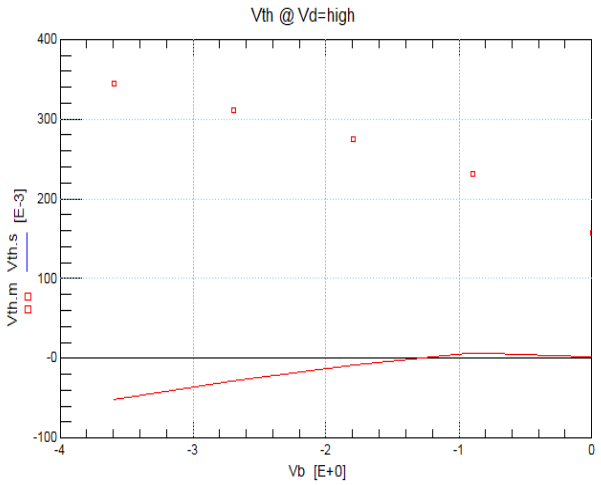
idvg, Vd = 3.3V, T = 27°C

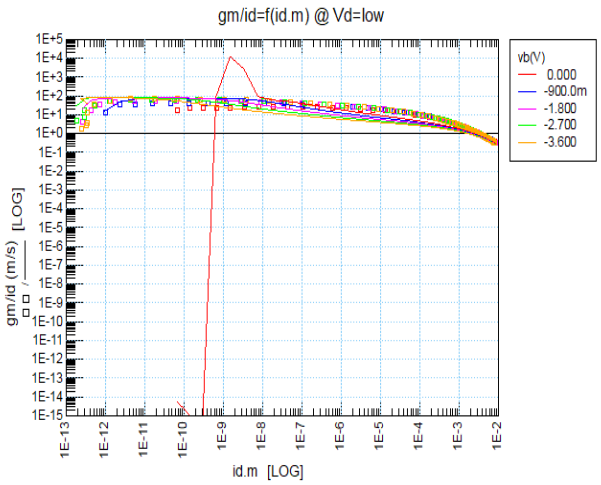


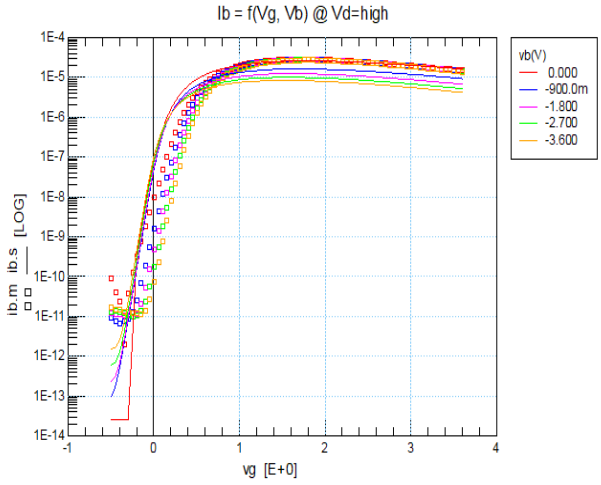


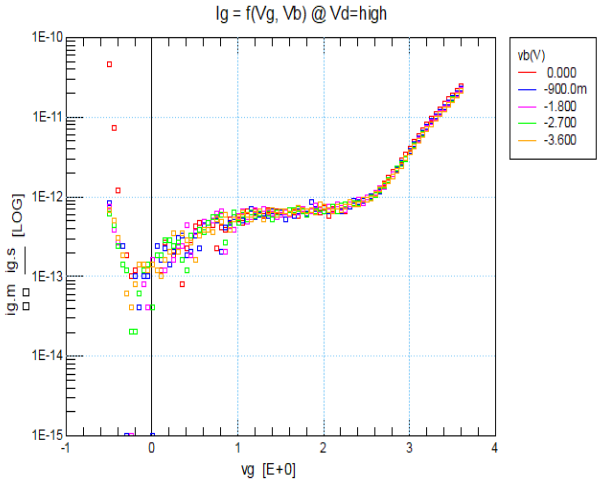




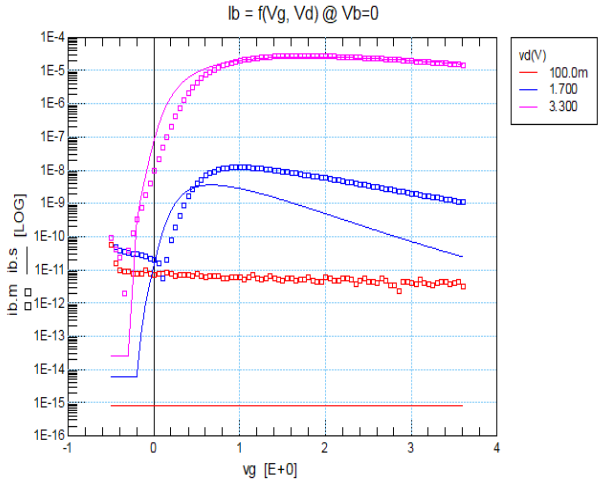


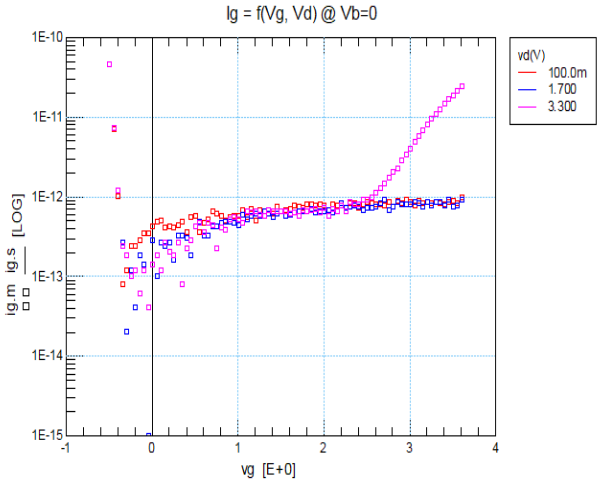




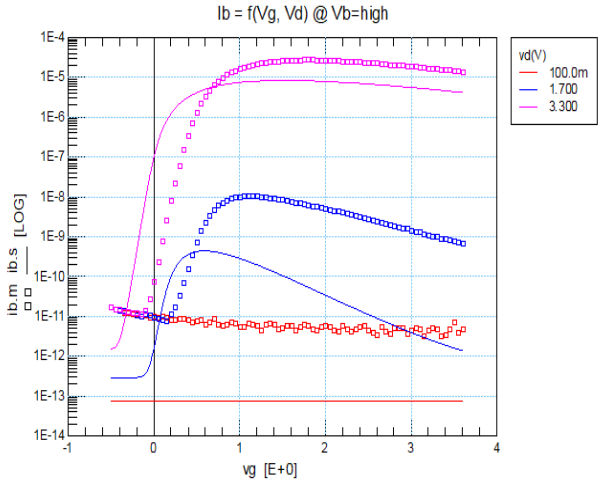


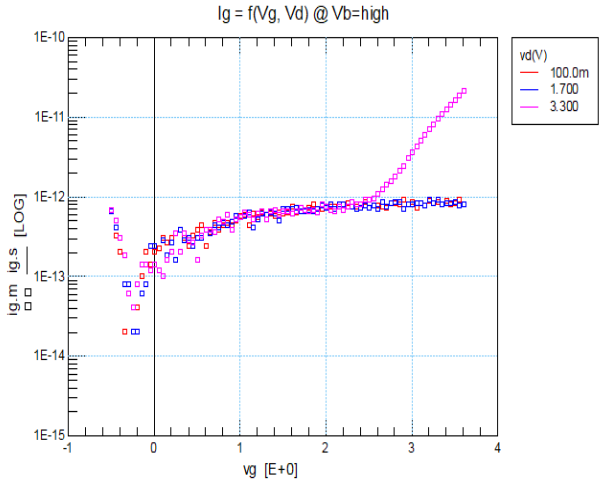
idvg, Ib,Ig, Vb = 0V, T = 27°C



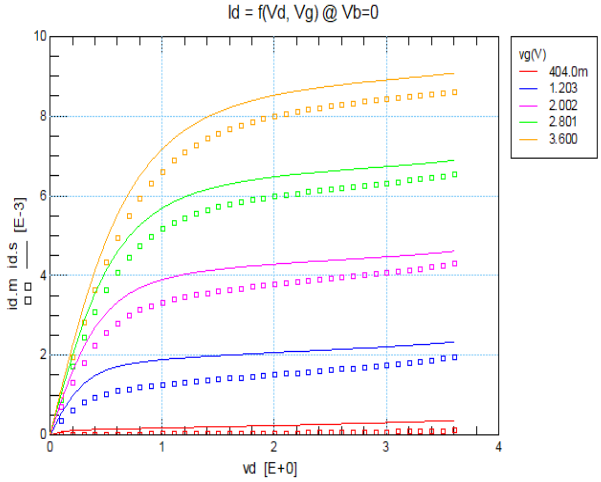


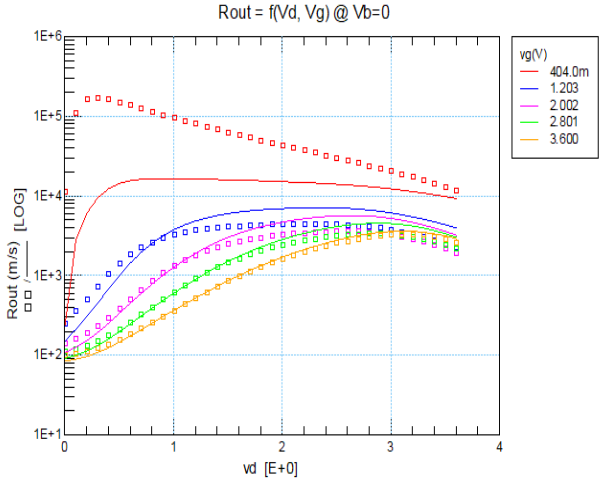
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

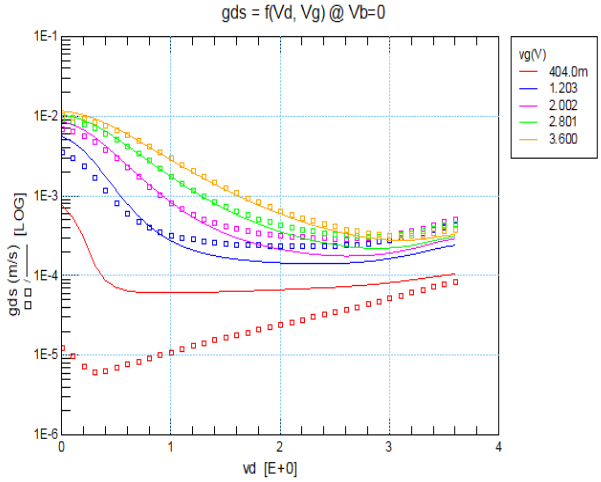




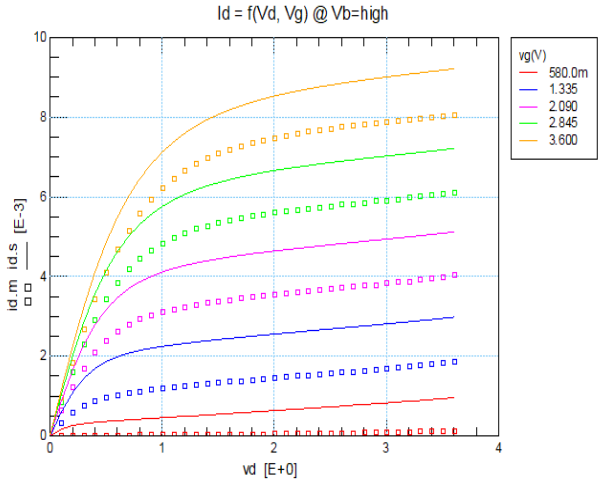
idvd, Vb = 0V, T = 27°C

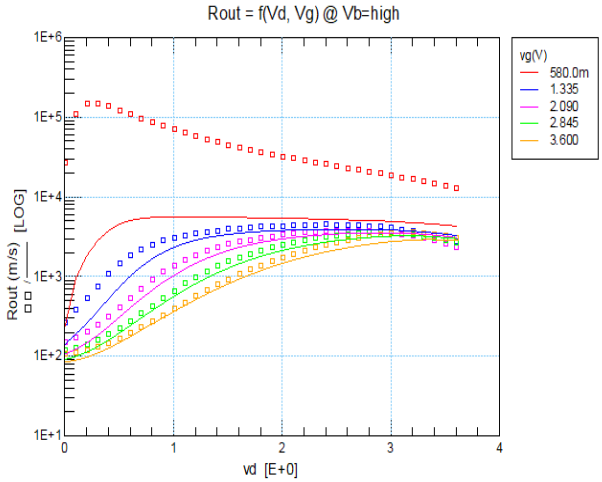


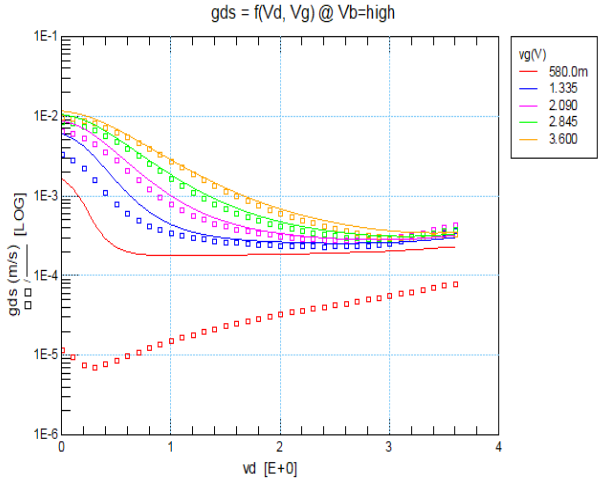




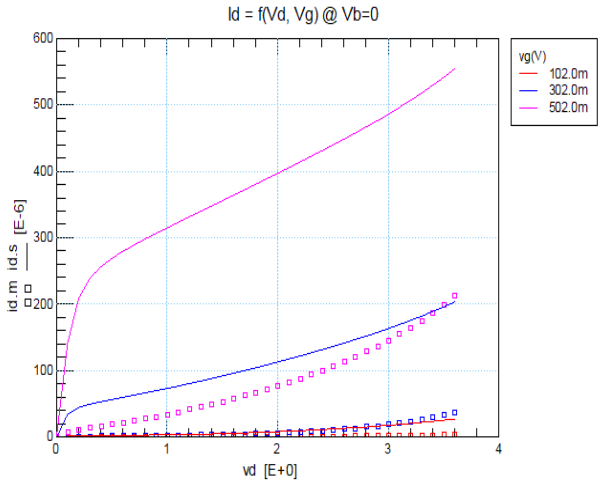
idvd_vbmin, Vb = -3.6V, T = 27°C

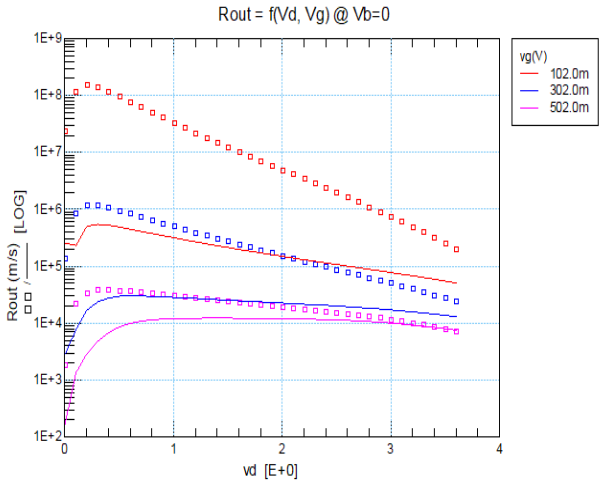


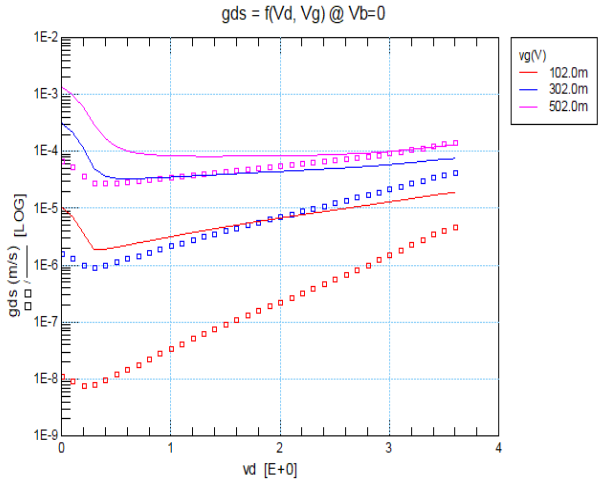




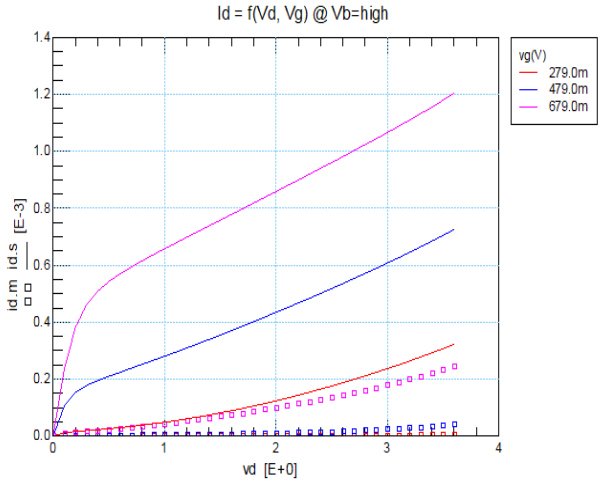
idvd_vth, Vb = 0V, T = 27°C

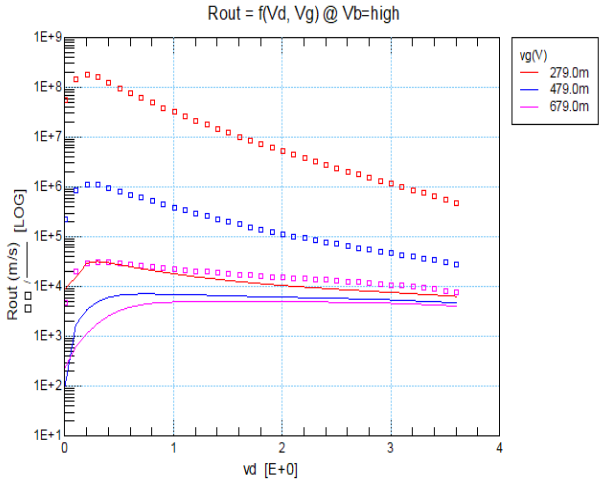


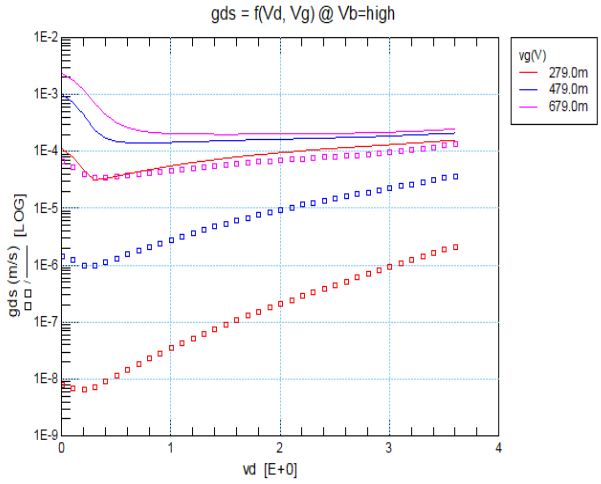




idvd_vth_vbmin, Vb = -3.6V, T = 27°C

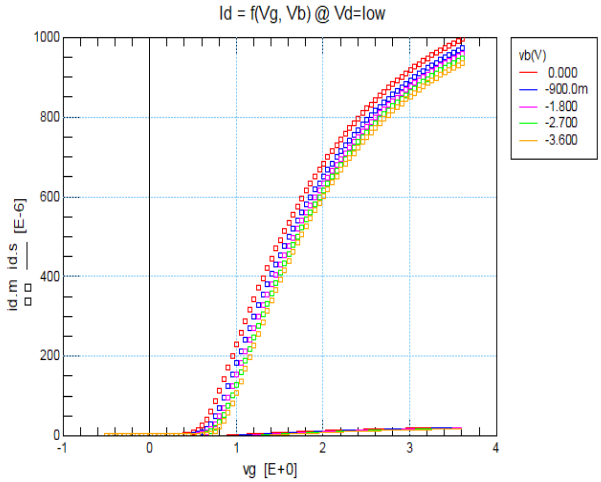


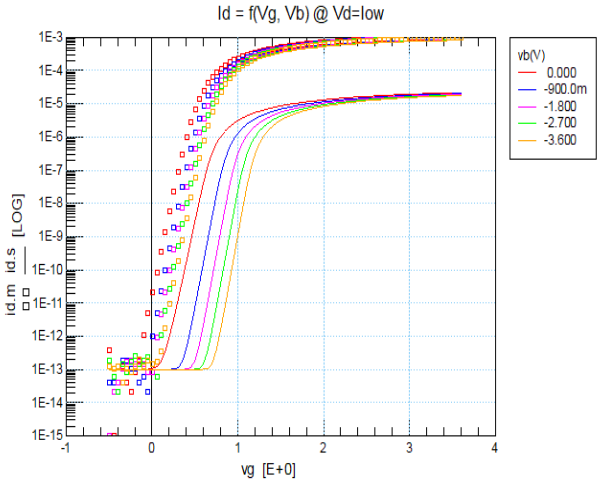


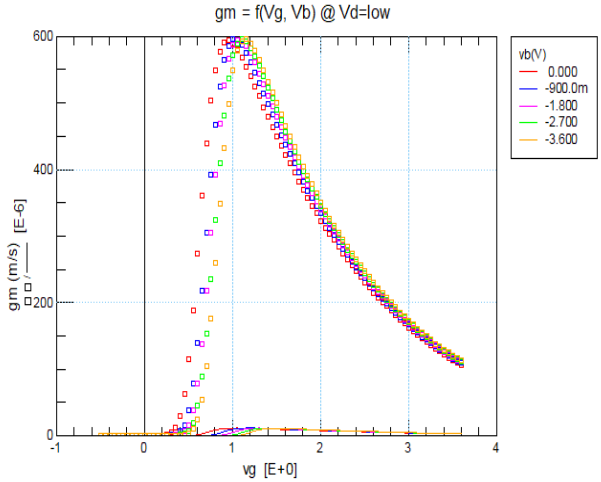


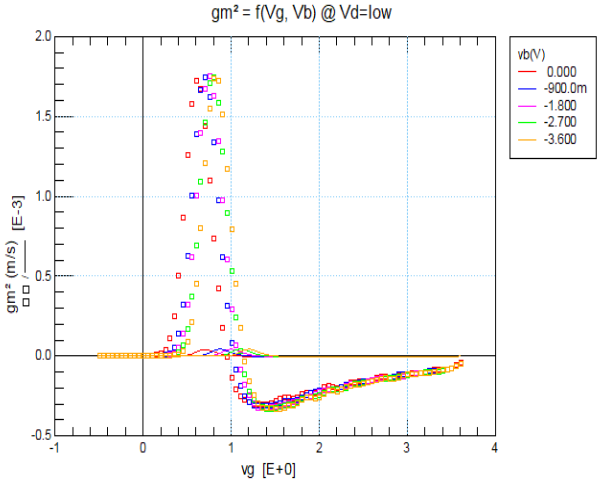
4.31 W0u3_L0u45_T356_S384M_3

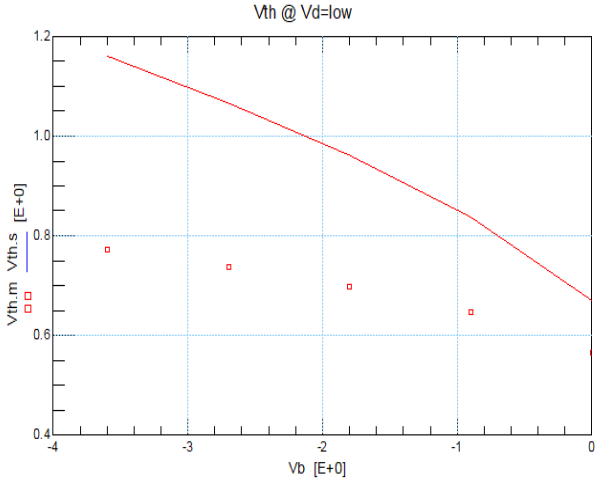
idvg, Vd = 0.1V, T = 27°C

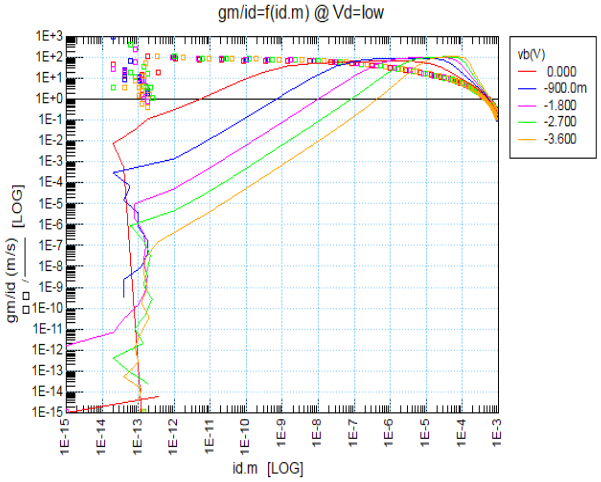


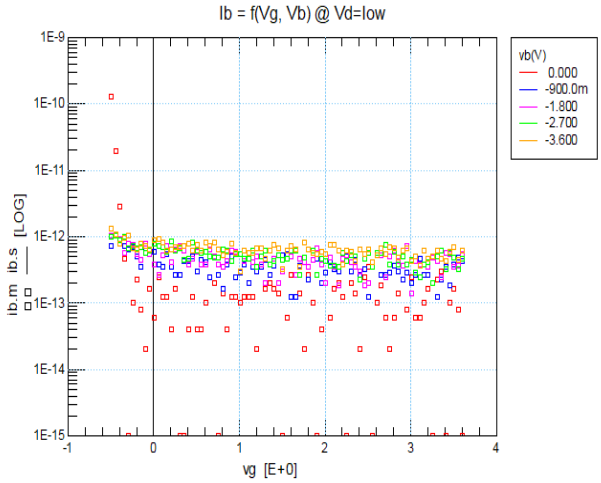


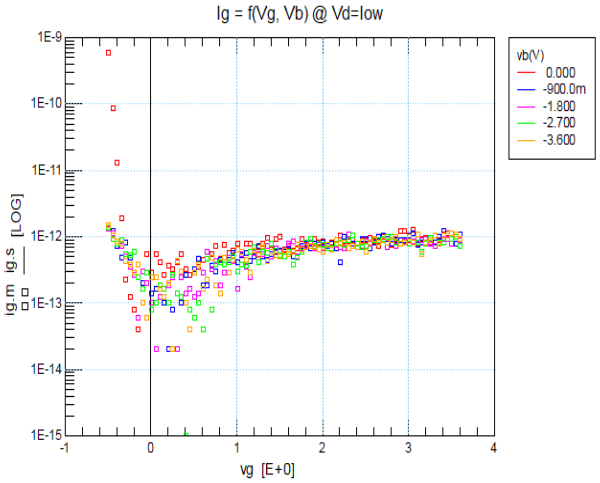




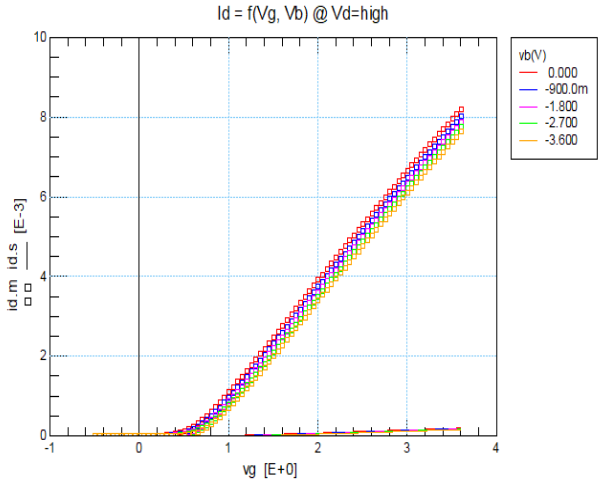


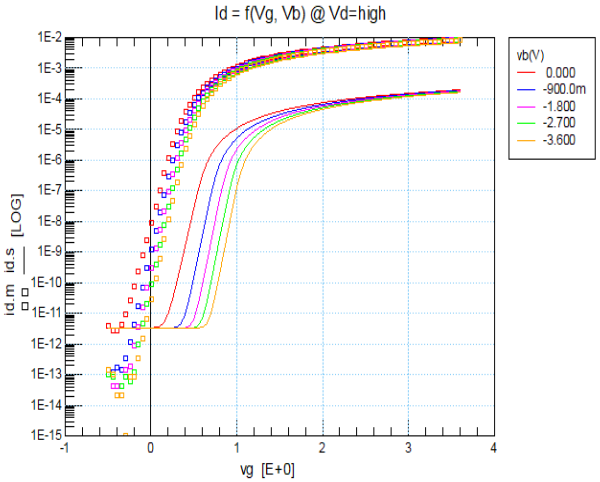


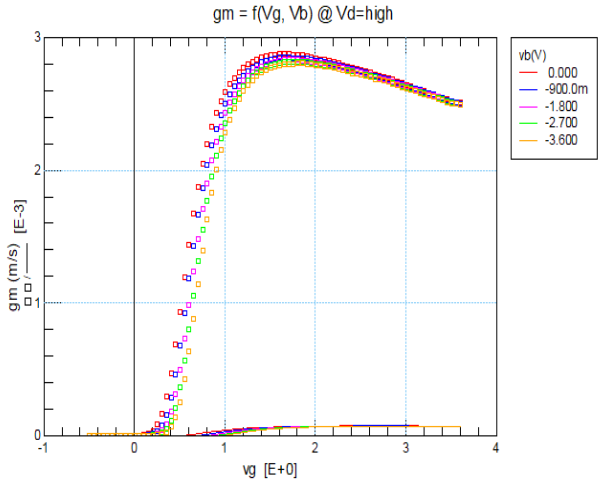


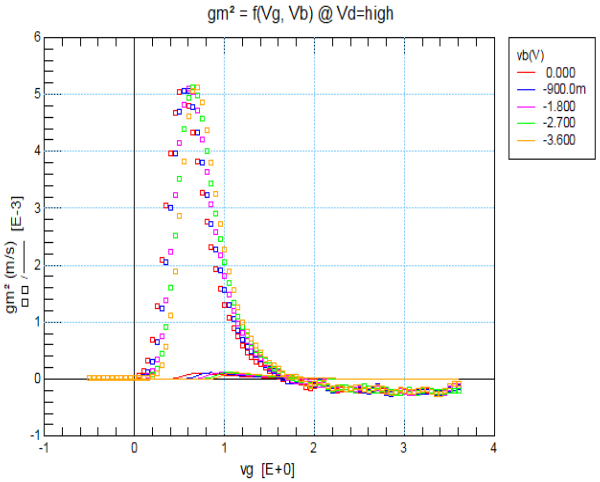


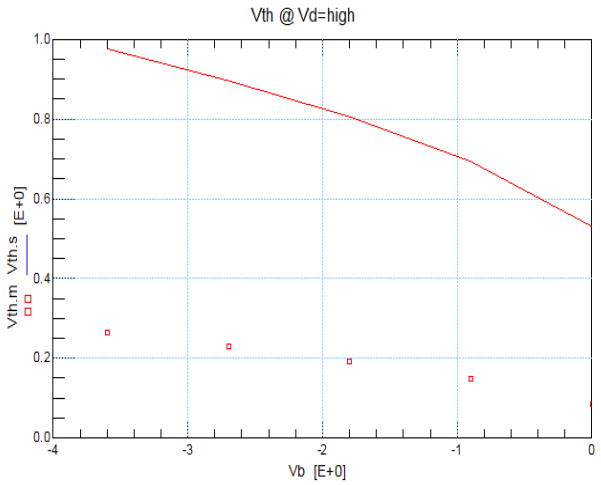
idvg, Vd = 3.3V, T = 27°C

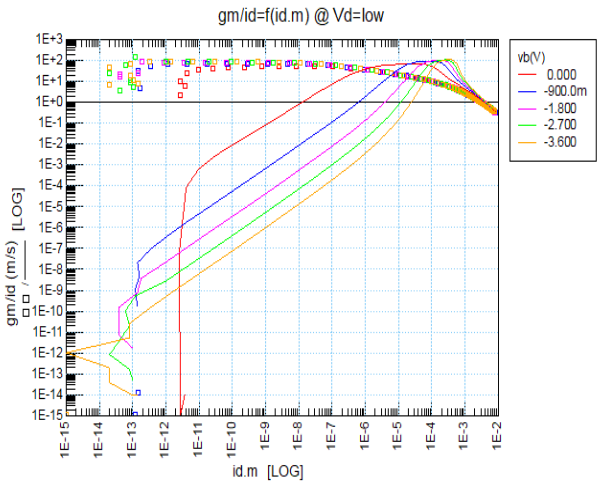


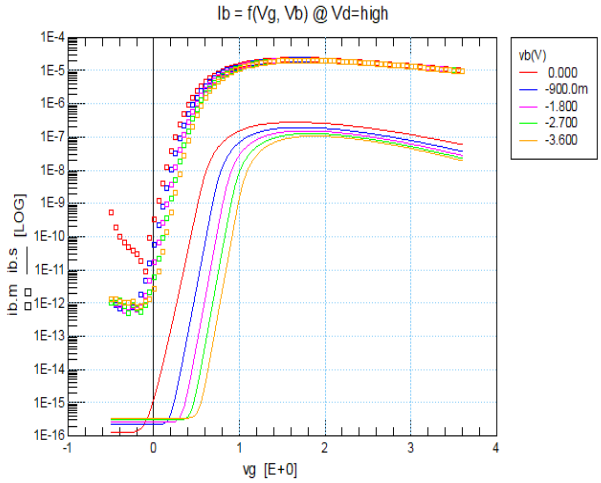


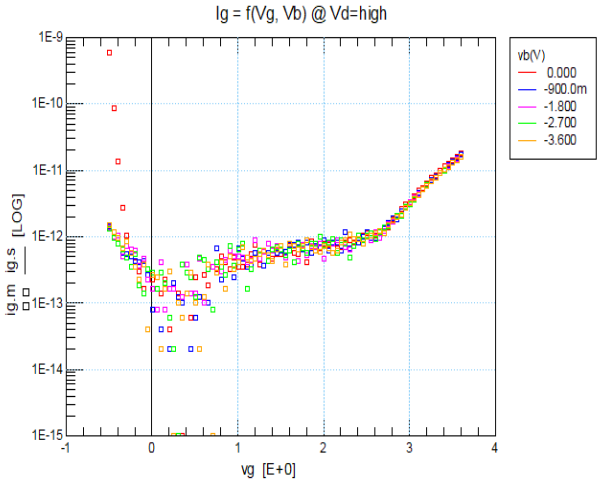




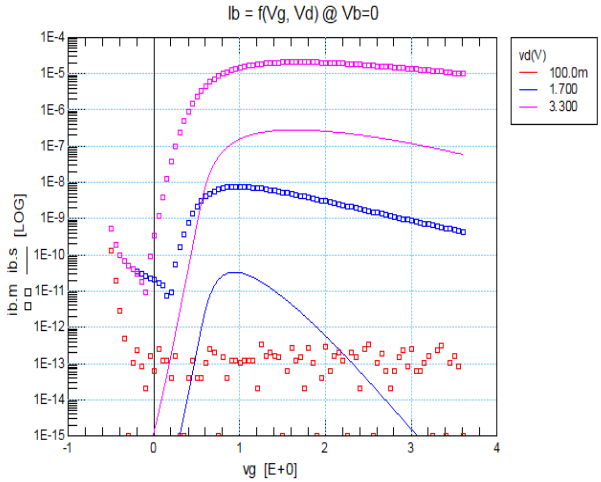


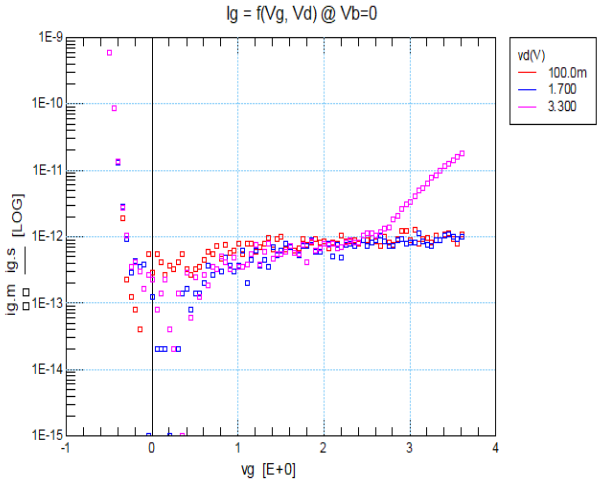




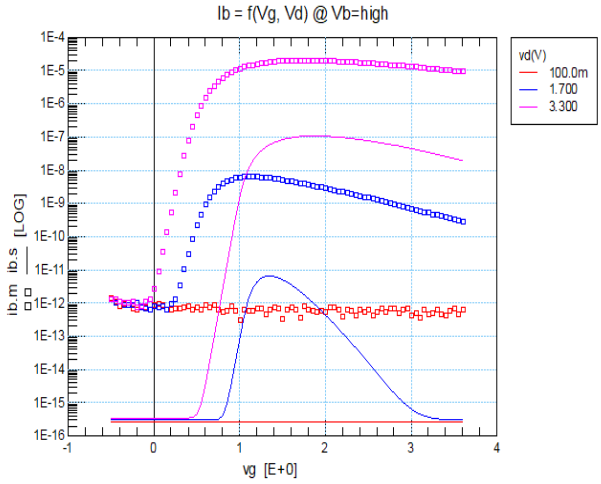


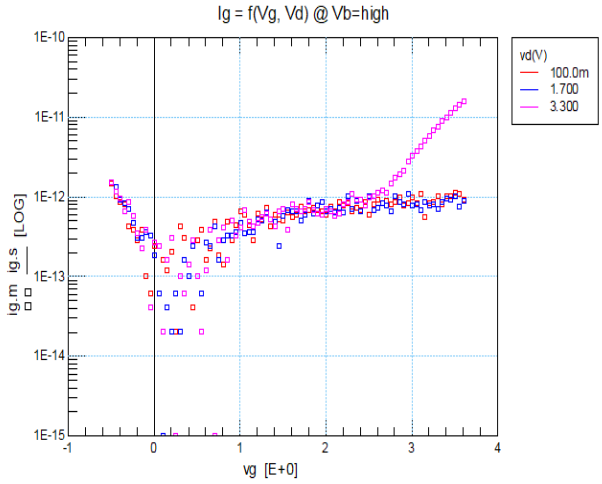
idvg, Ib,Ig, Vb = 0V, T = 27°C



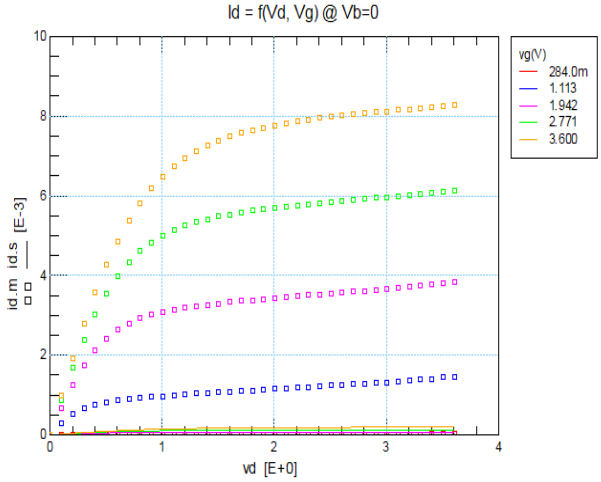


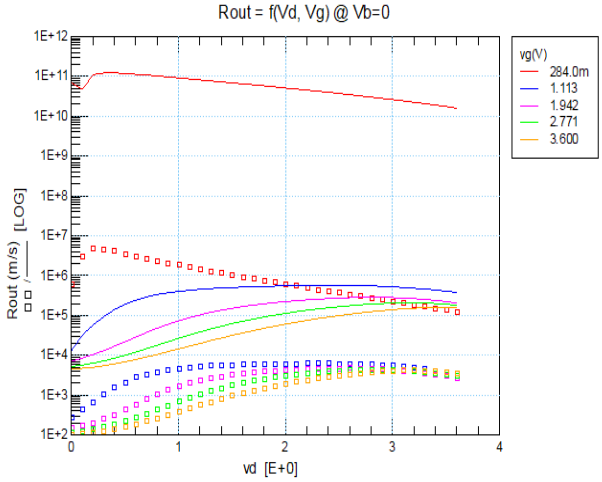
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

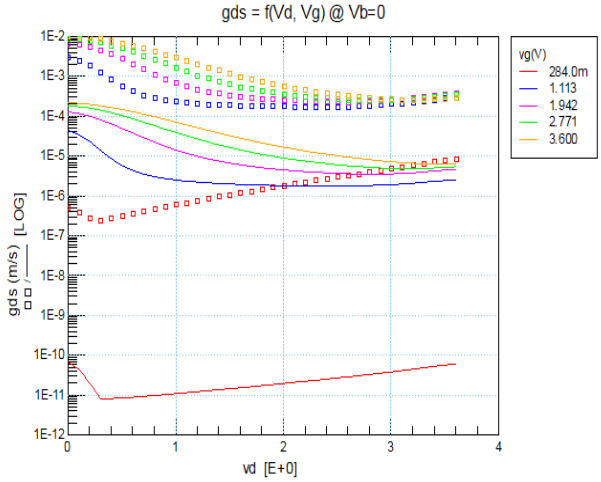




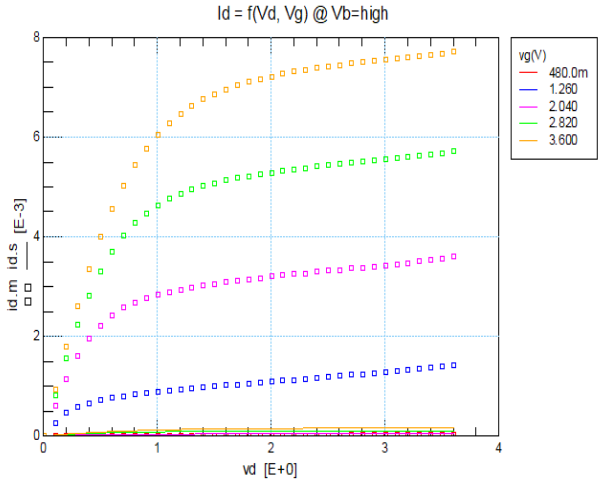
idvd, Vb = 0V, T = 27°C

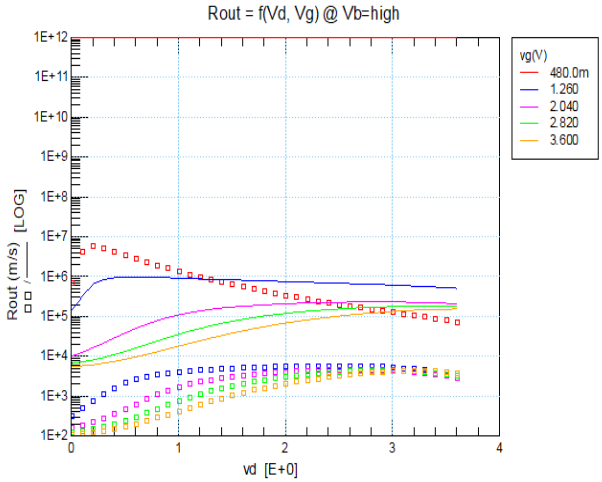


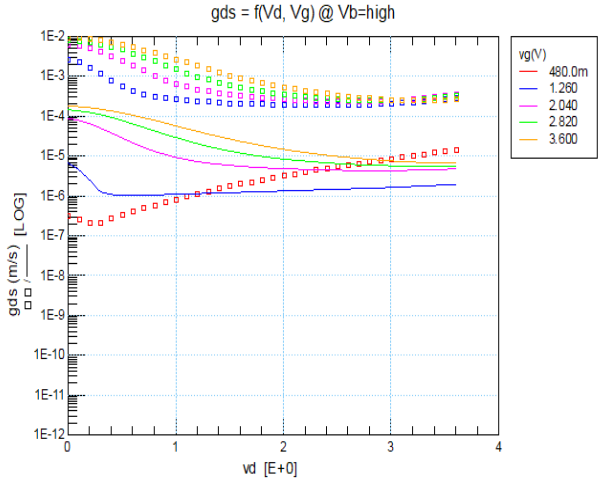




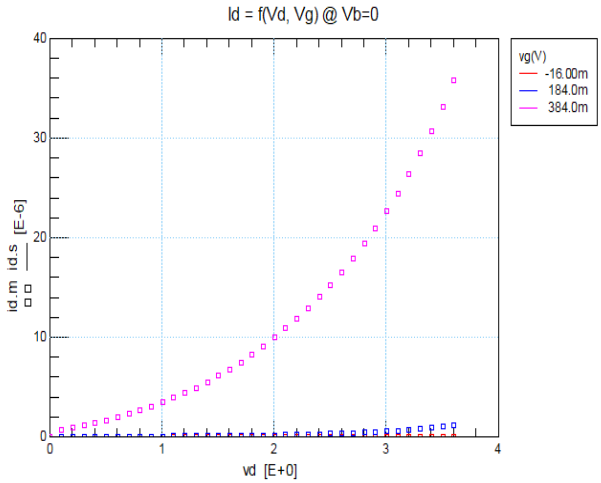
idvd_vbmin, Vb = -3.6V, T = 27°C

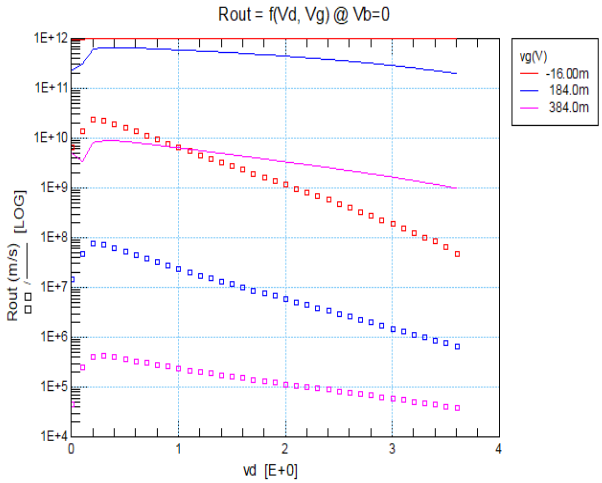


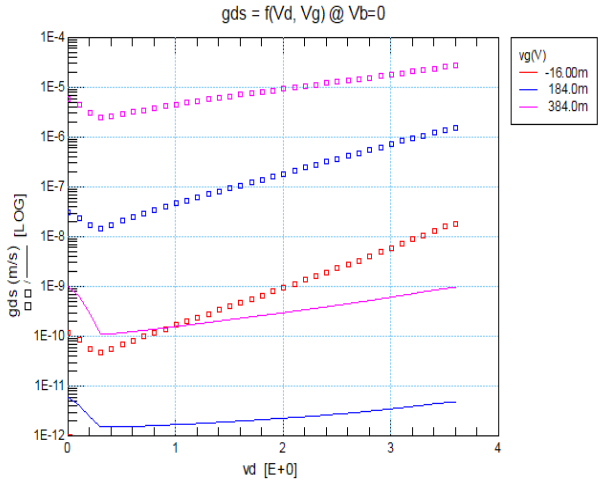




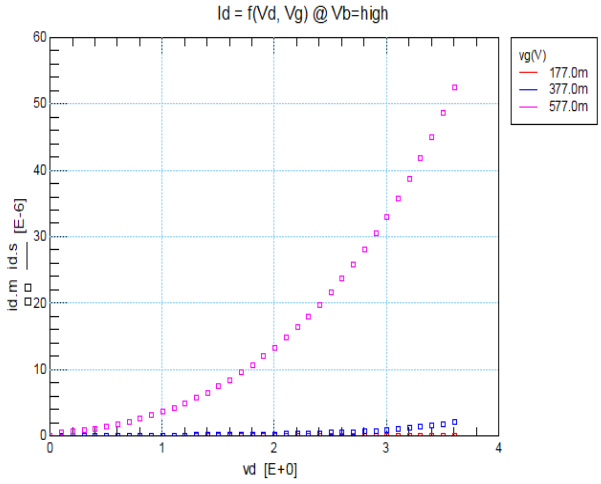
idvd_vth, Vb = 0V, T = 27°C

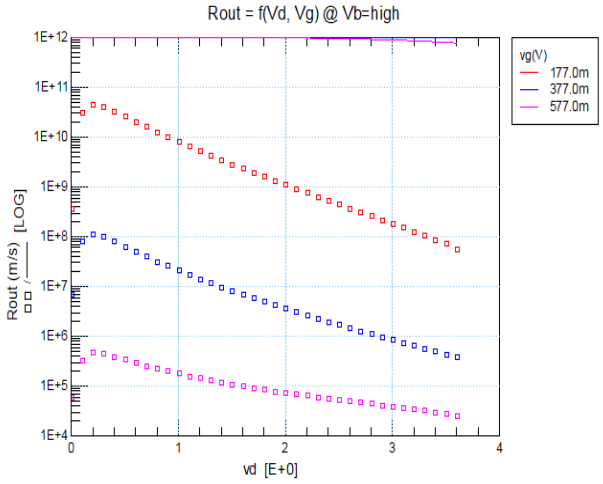


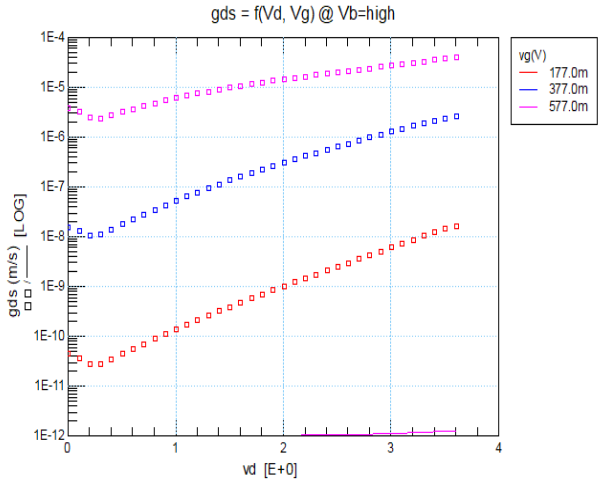




idvd_vth_vbmin, Vb = -3.6V, T = 27°C

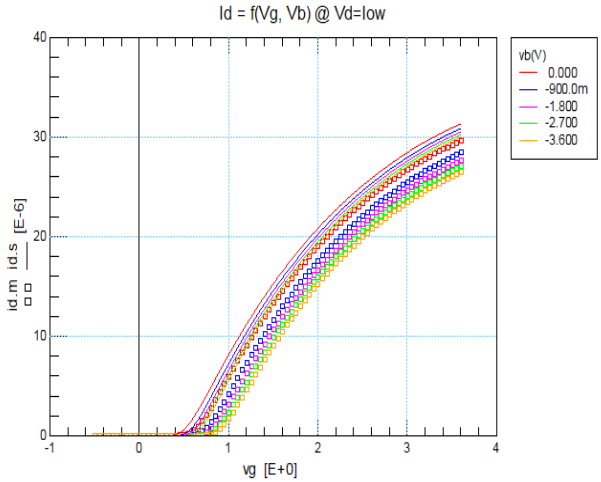


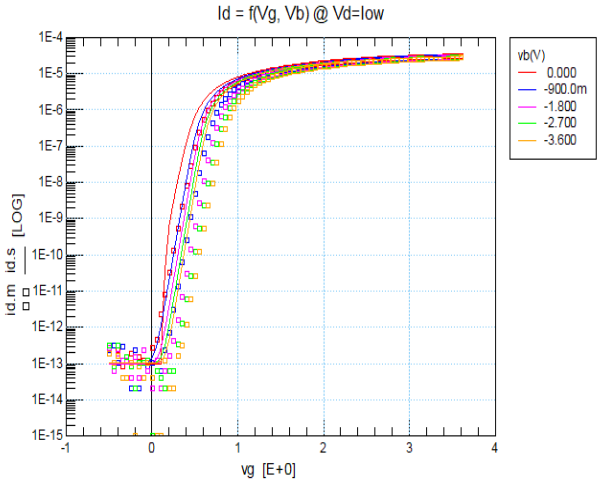


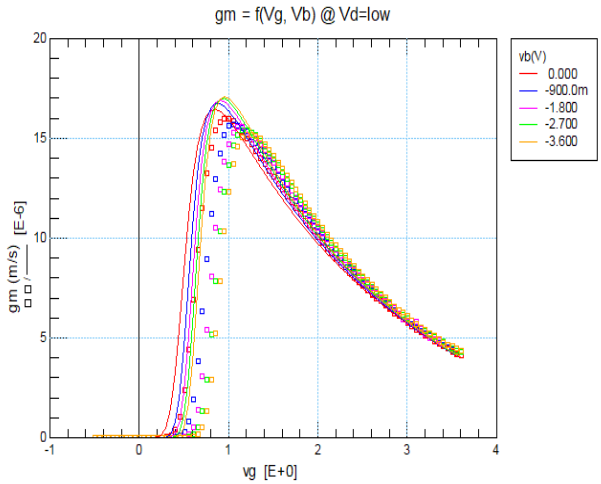


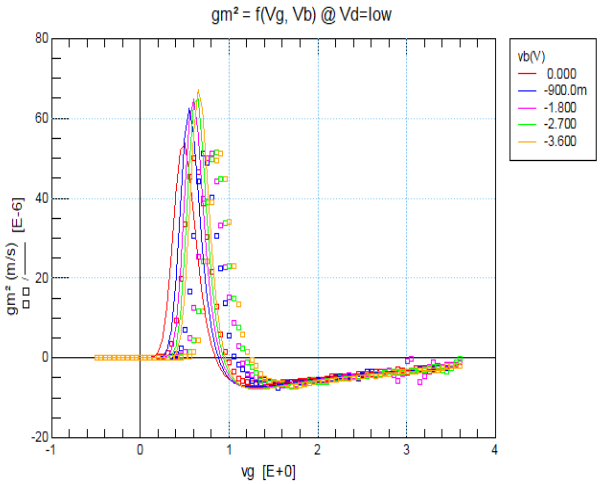
4.32 W0u3_L0u33_S560_2

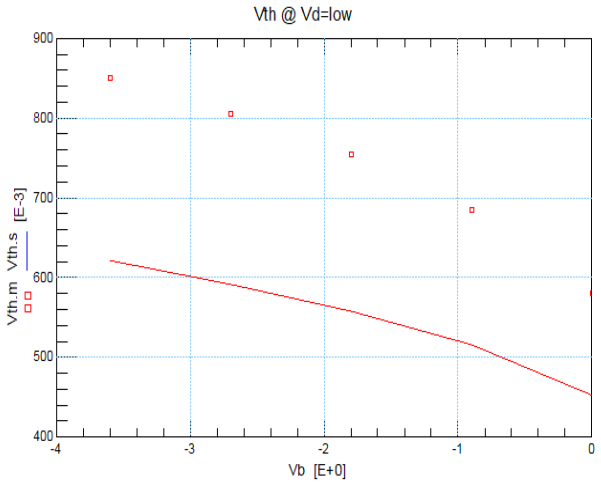
idvg, Vd = 0.1V, T = 27°C

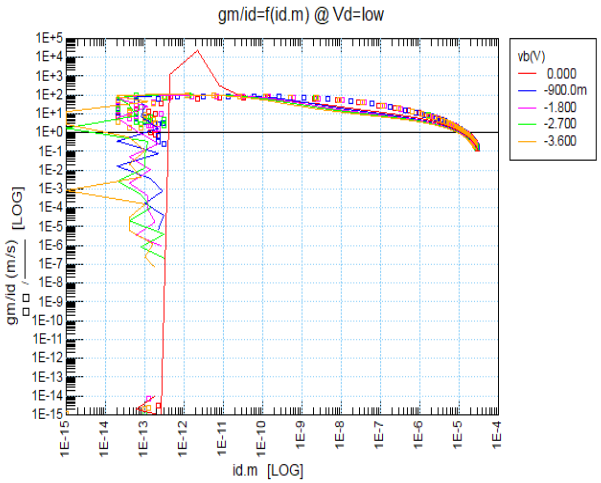


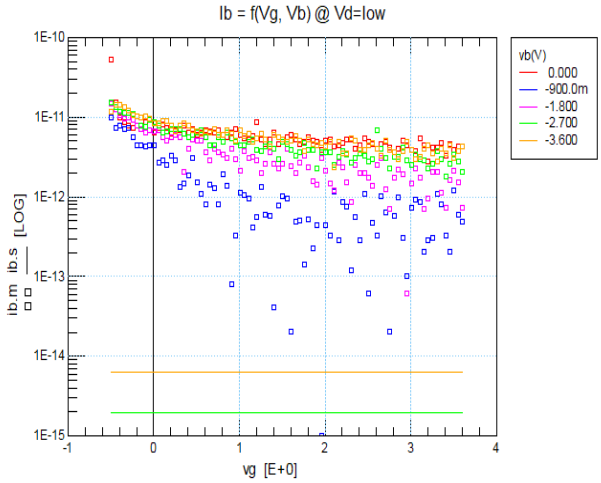


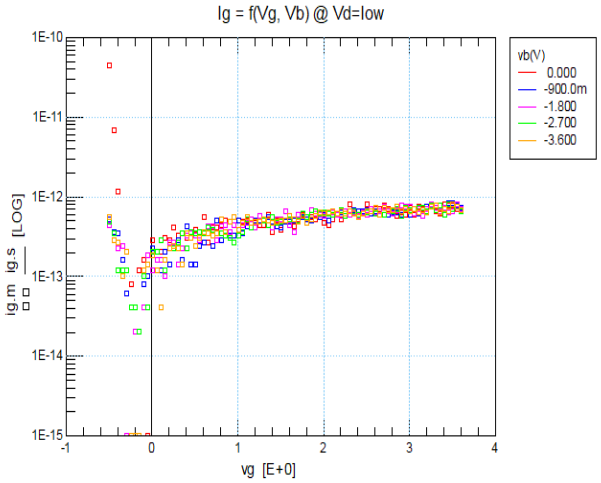




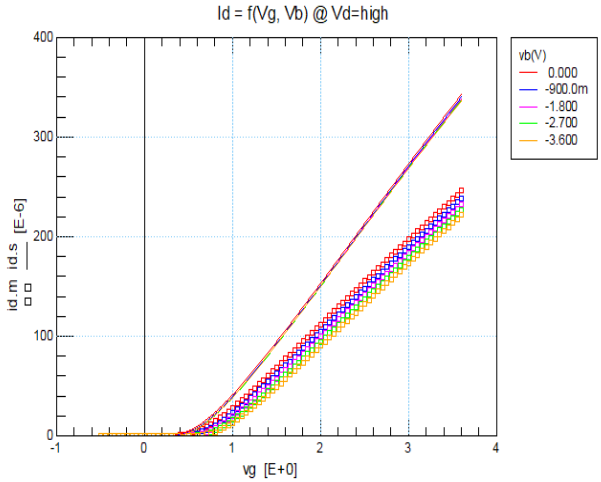


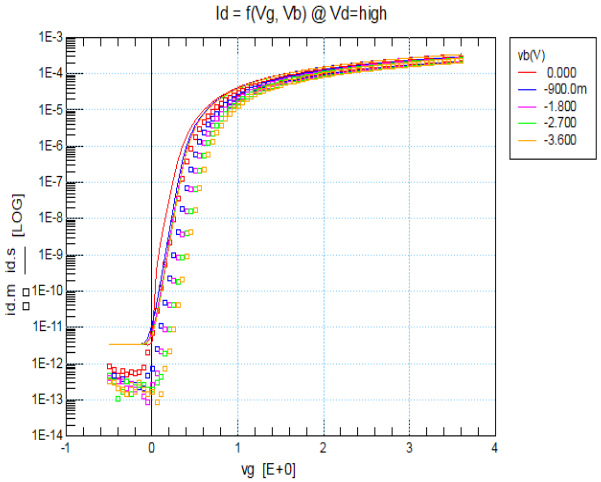


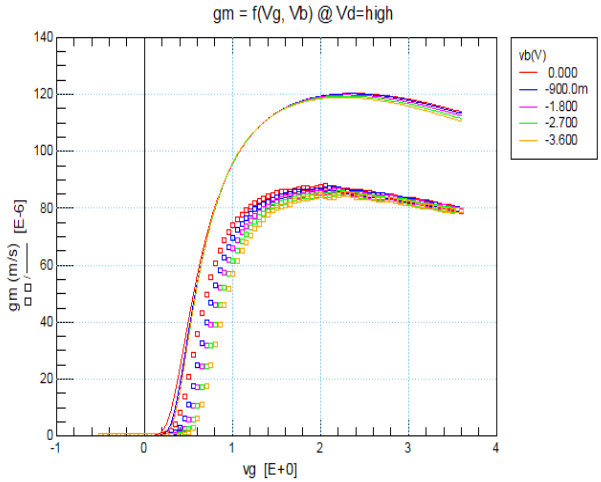


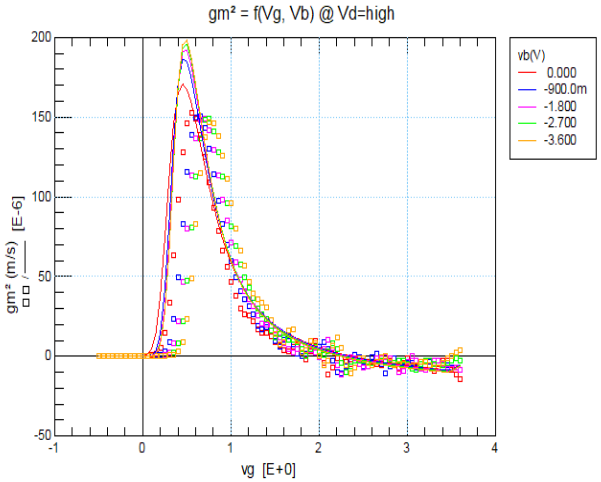


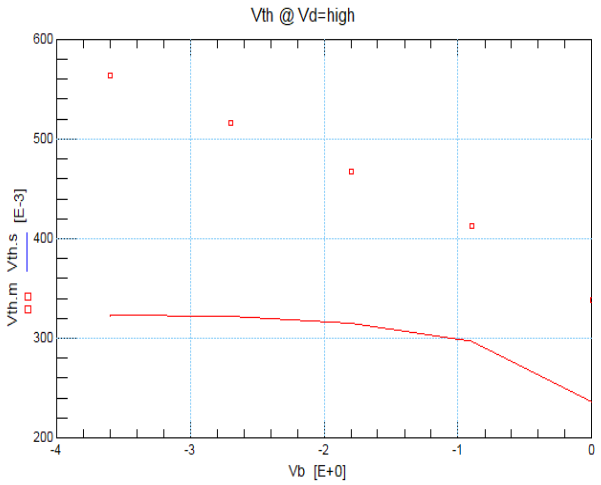
idvg, Vd = 3.3V, T = 27°C

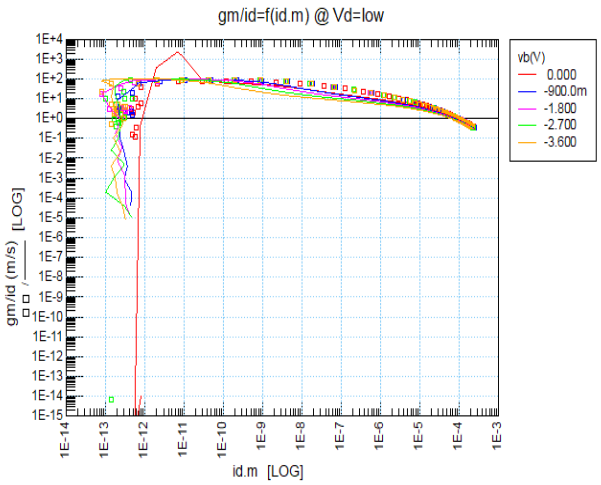


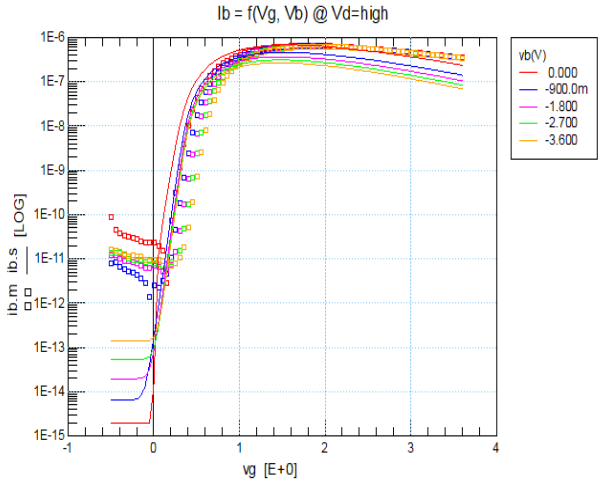


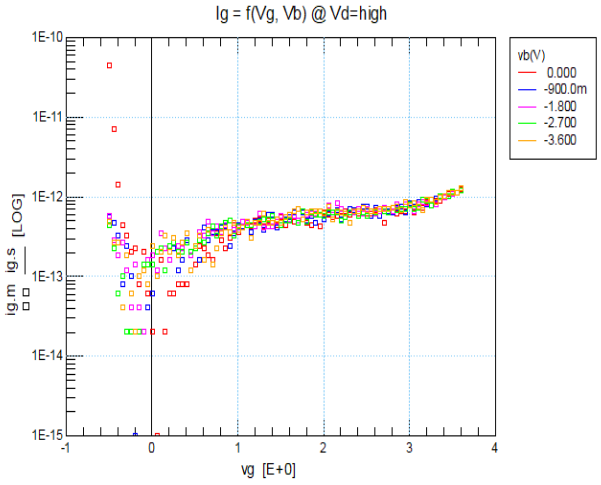




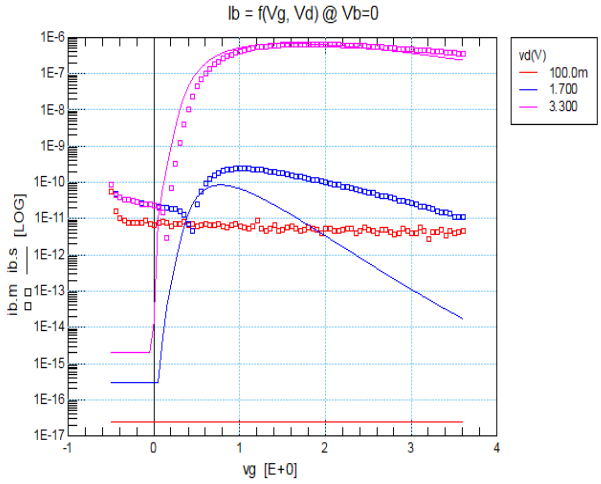


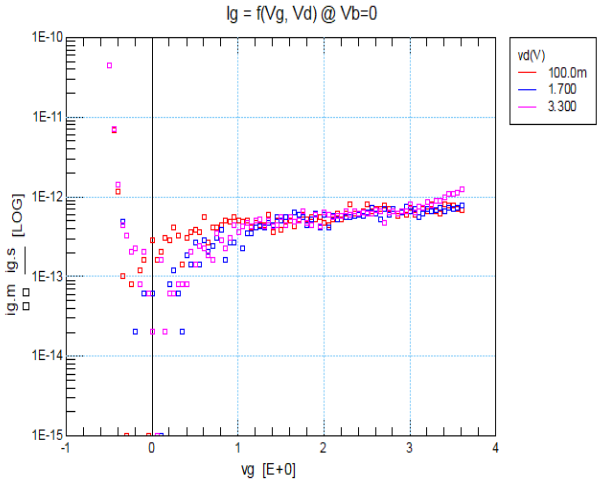




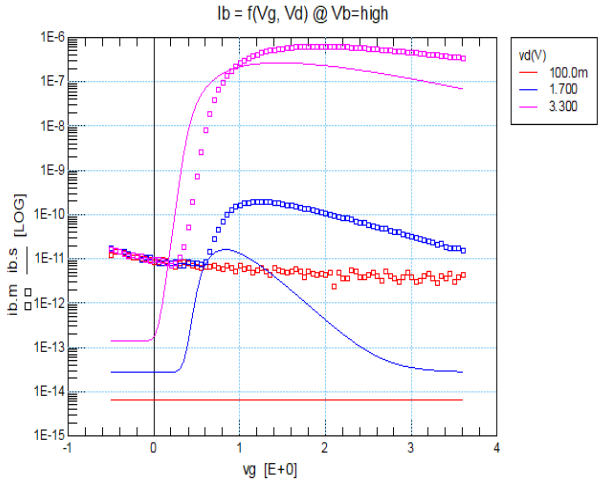


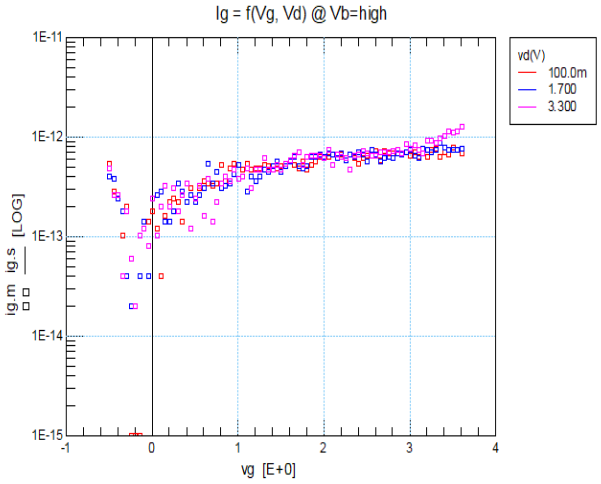
idvg, Ib,Ig, Vb = 0V, T = 27°C



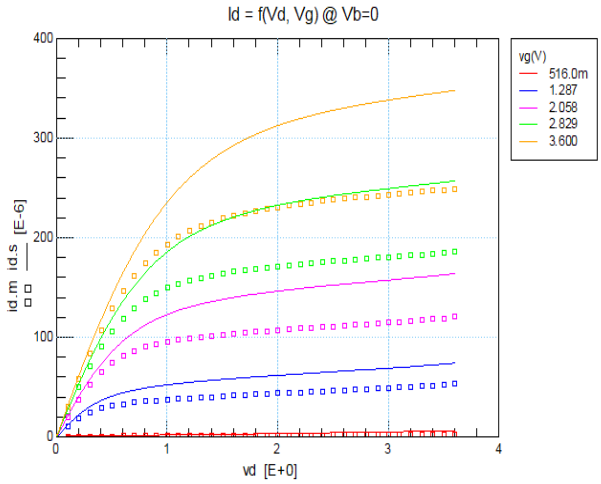


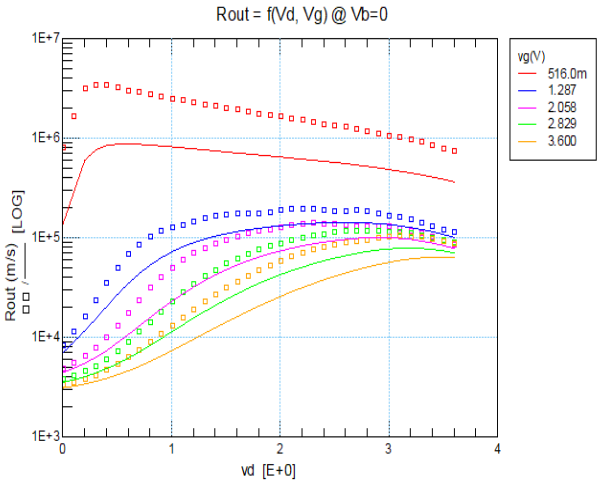
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

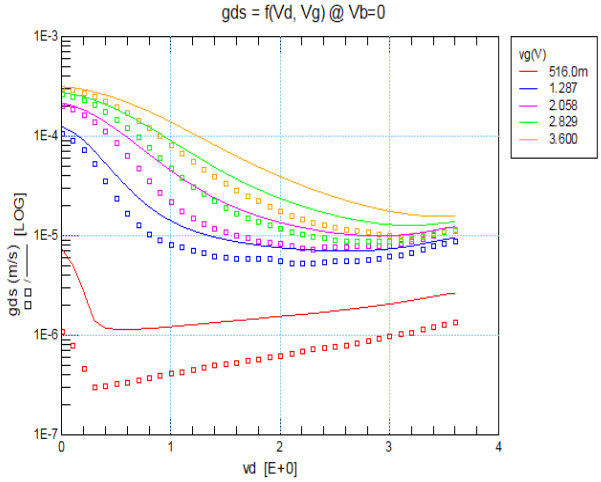




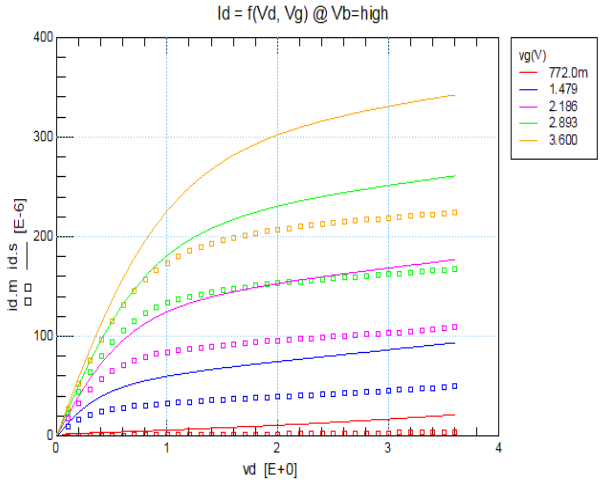
idvd, Vb = 0V, T = 27°C

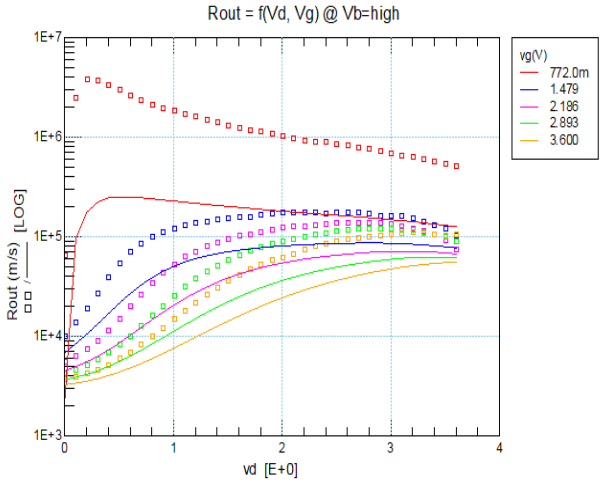


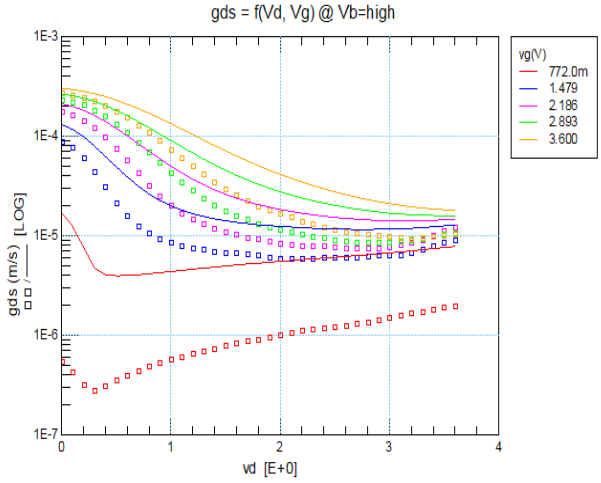




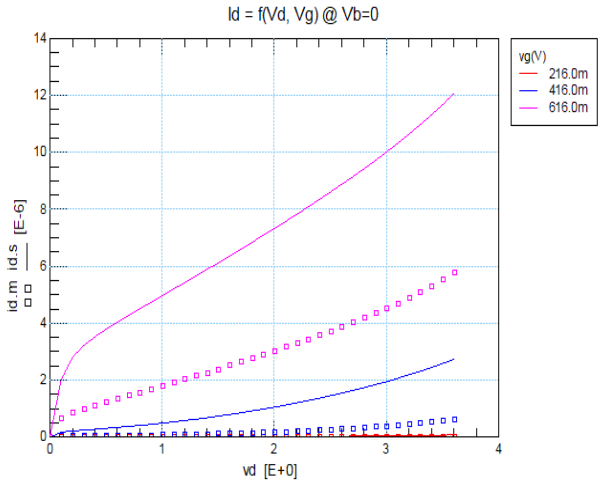
idvd_vbmin, Vb = -3.6V, T = 27°C

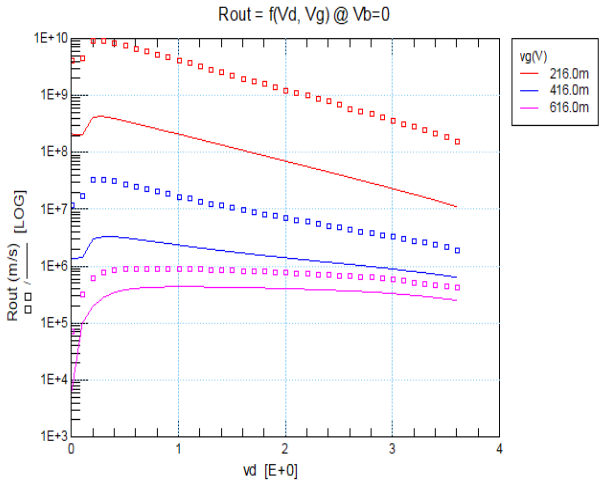


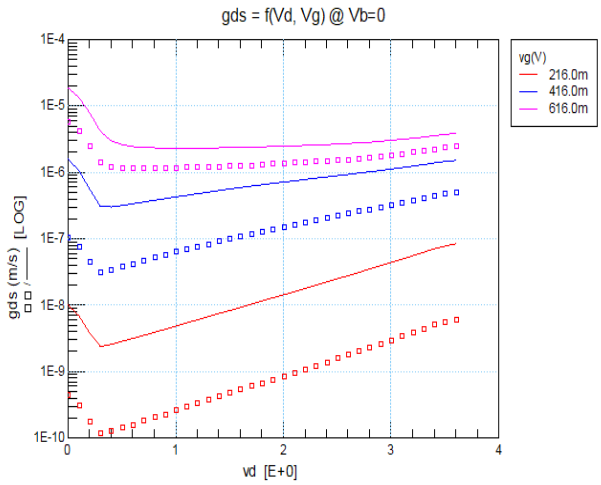




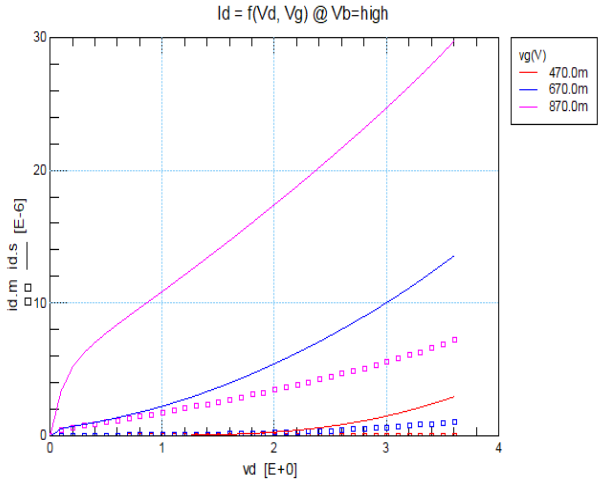
idvd_vth, Vb = 0V, T = 27°C

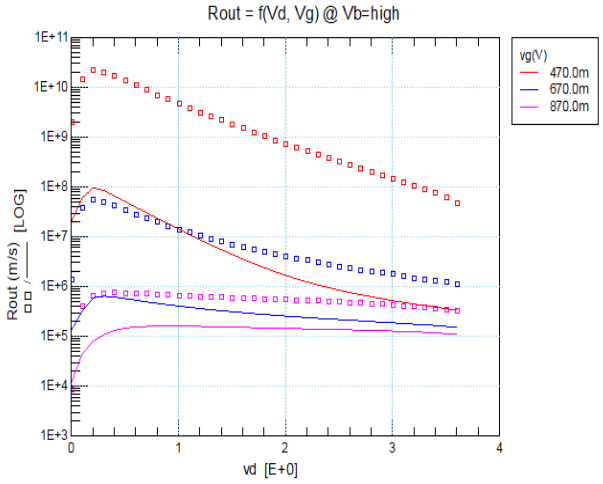


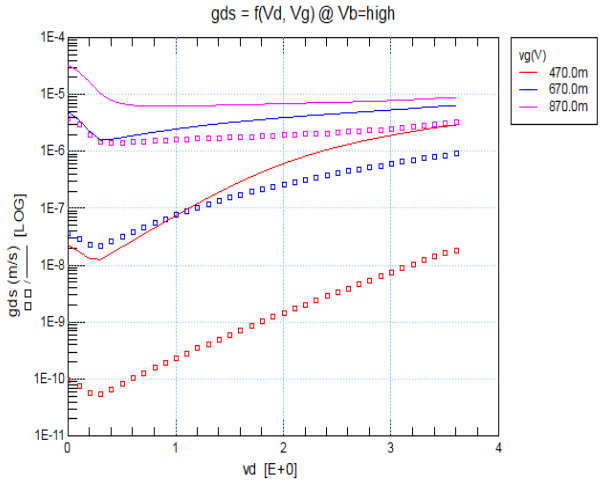




idvd_vth_vbmin, Vb = -3.6V, T = 27°C

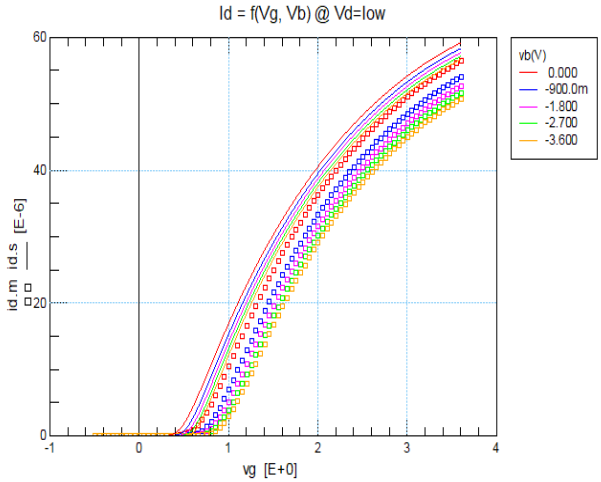


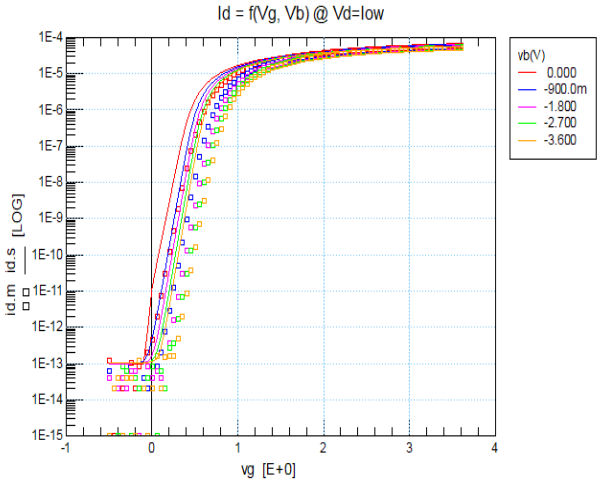


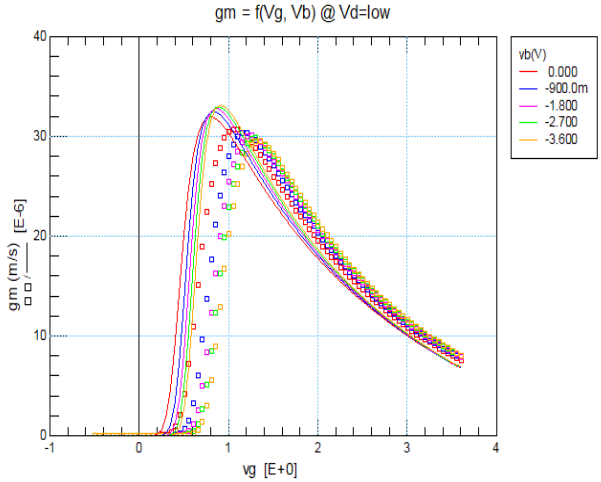


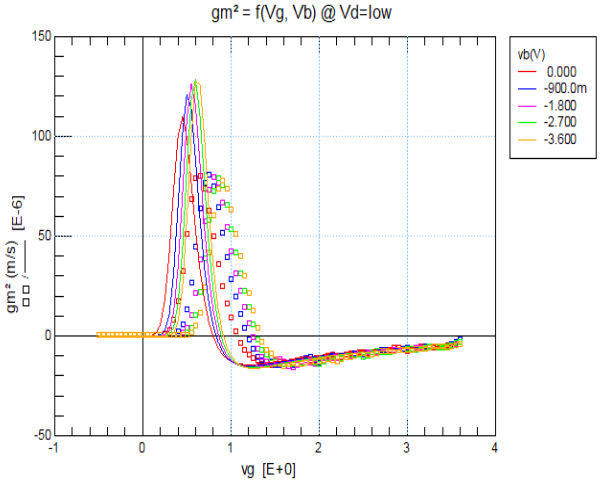
4.33 W0u6_L0u33_S559_1

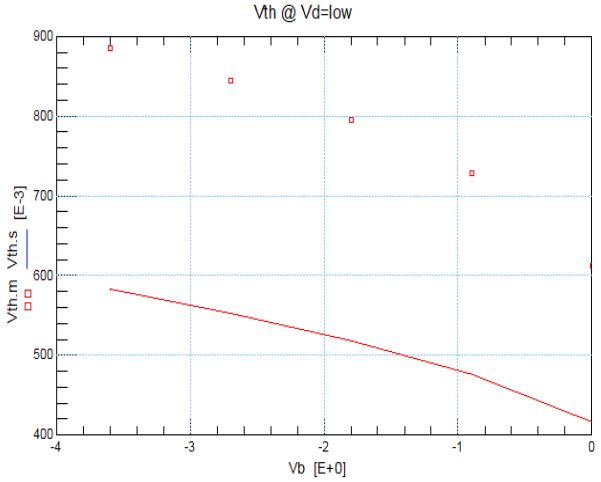
idvg, Vd = 0.1V, T = 27°C

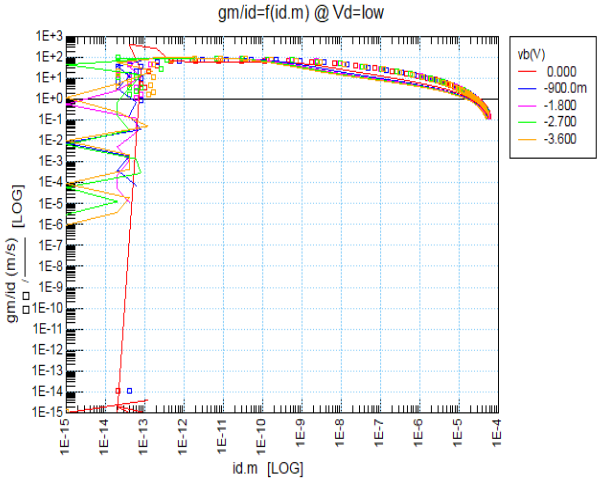


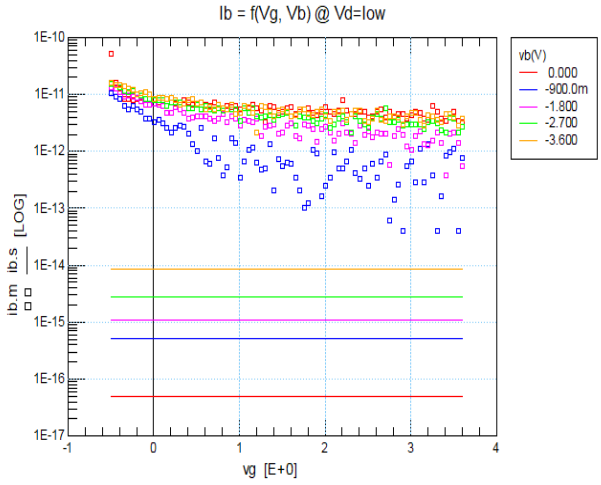


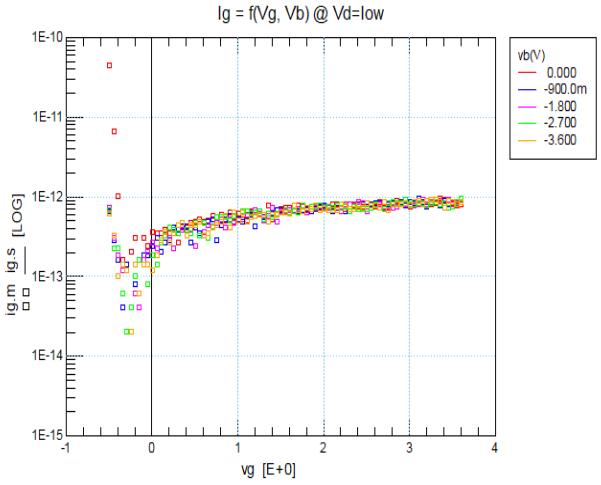




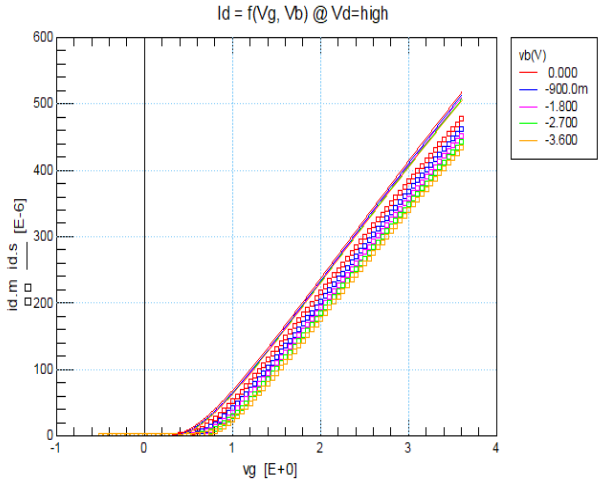


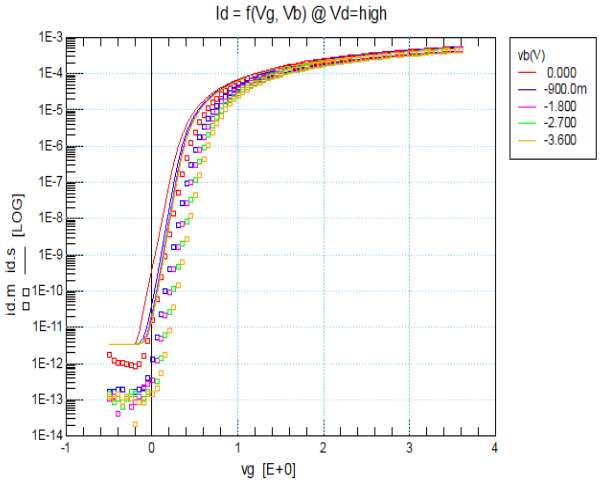


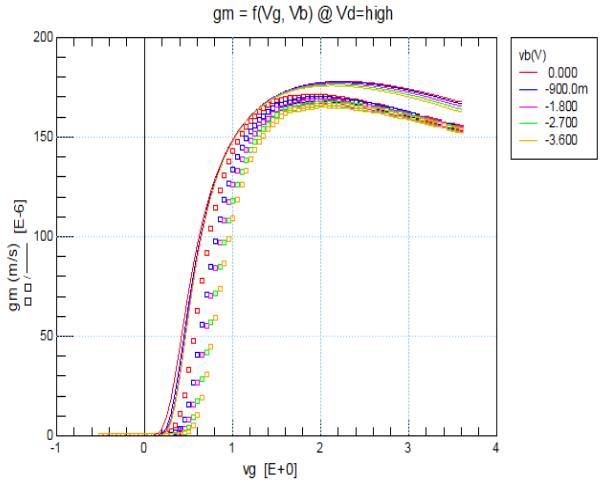


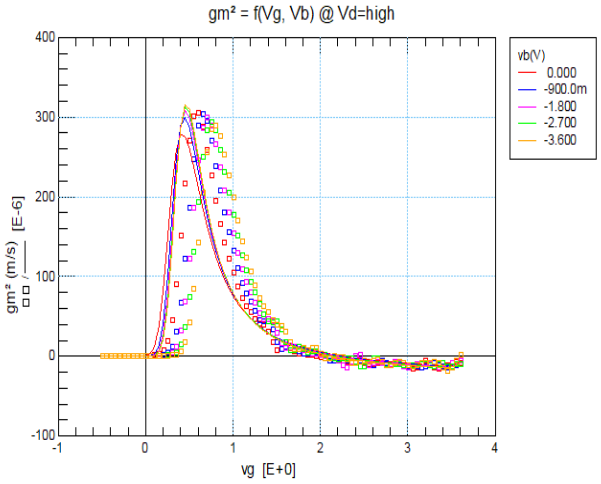


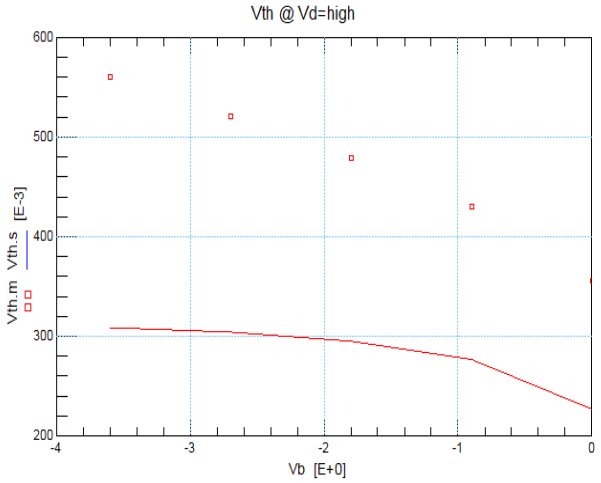
idvg, Vd = 3.3V, T = 27°C

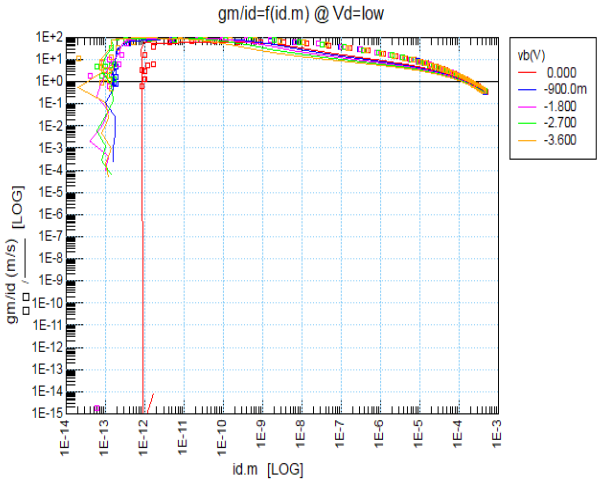


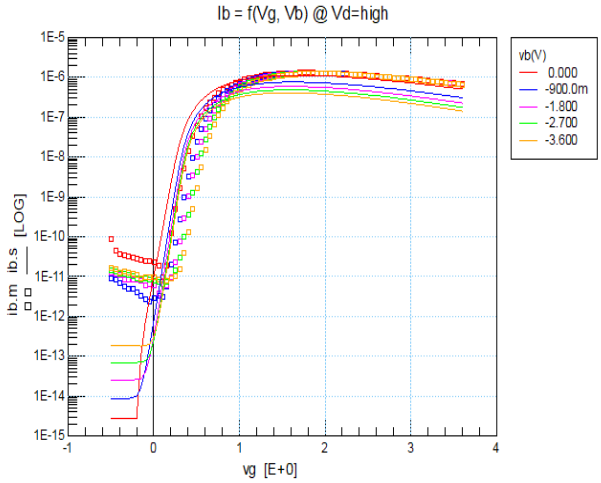


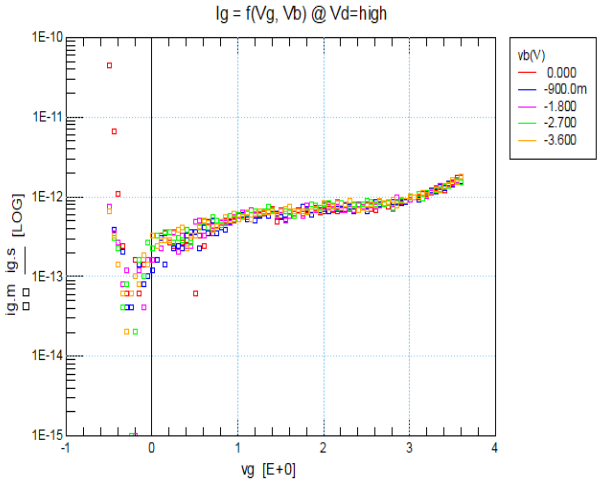




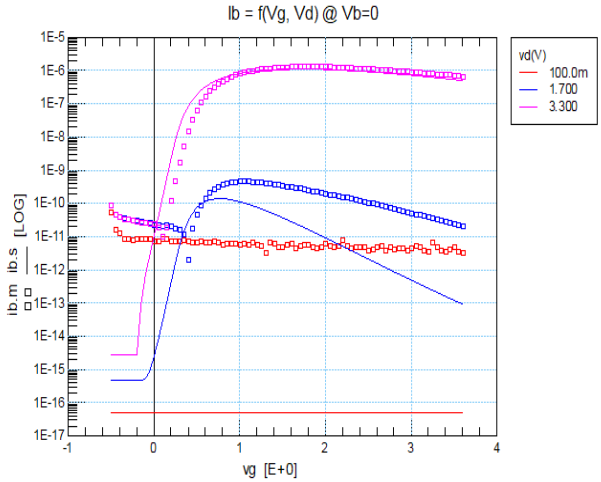


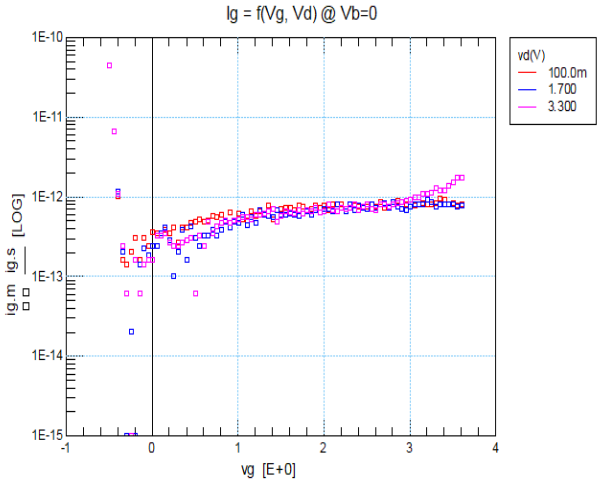




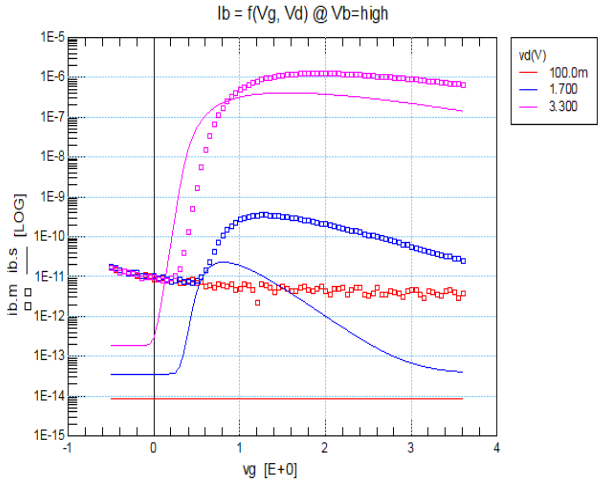


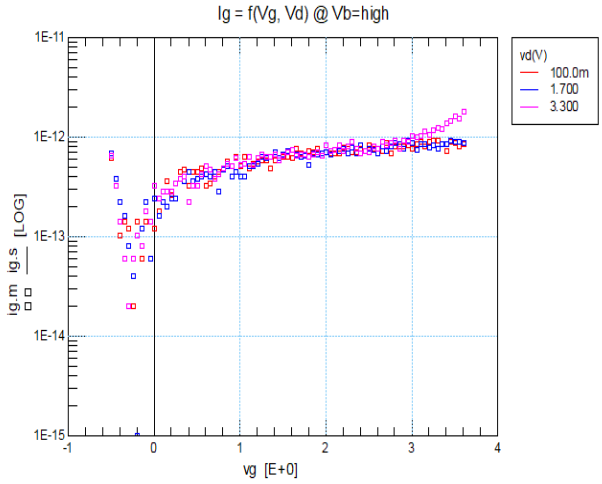
idvg, Ib,Ig, Vb = 0V, T = 27°C



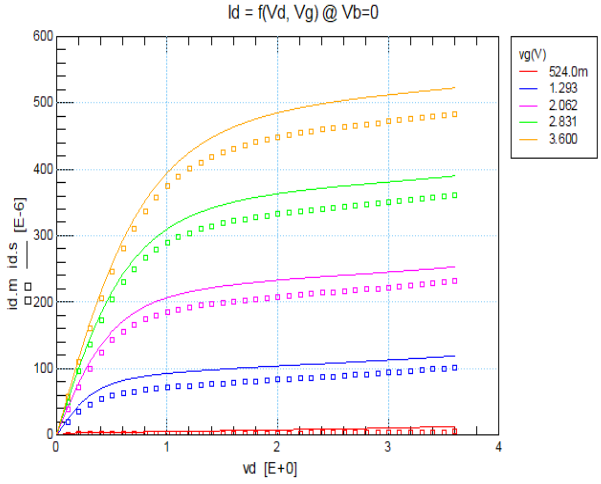


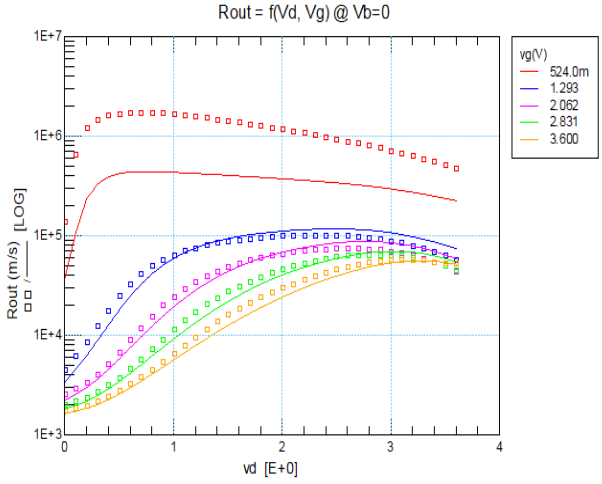
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

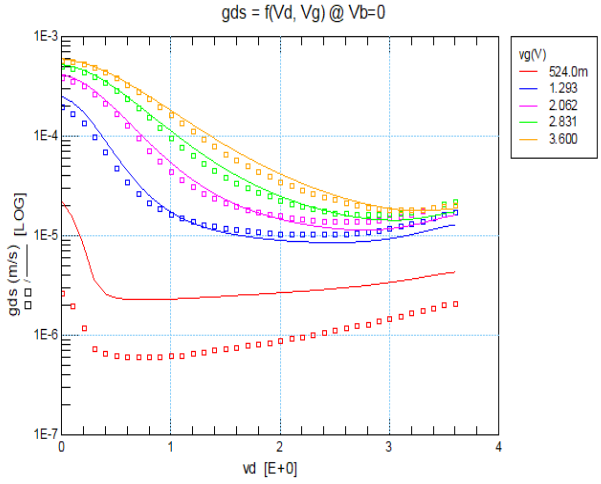




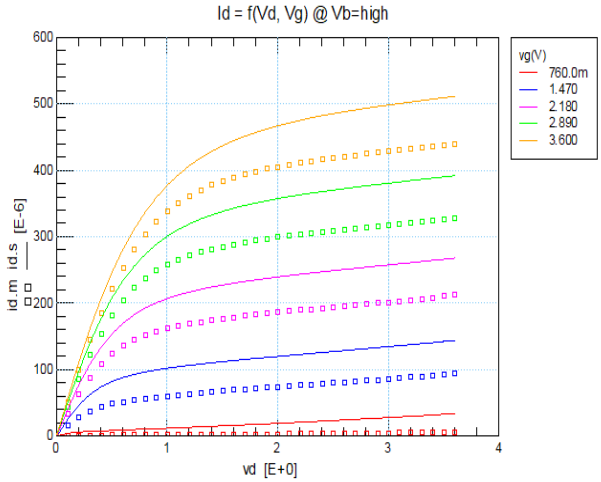
idvd, Vb = 0V, T = 27°C

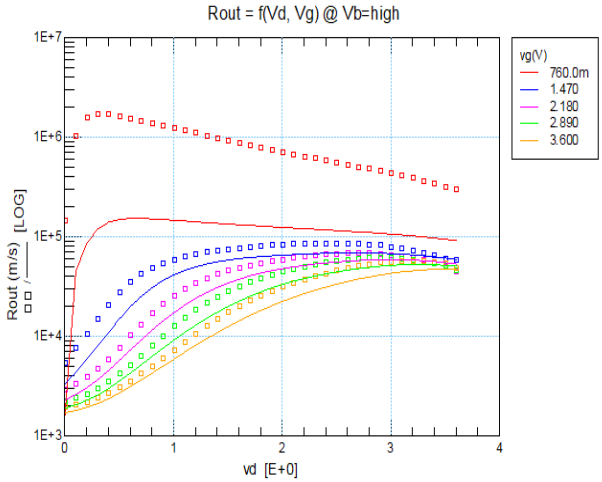


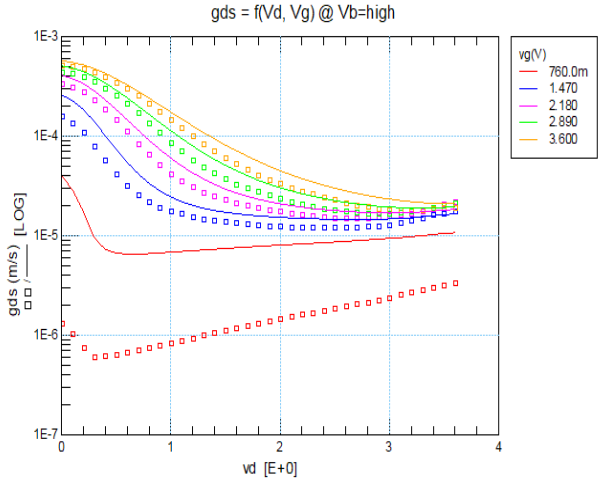




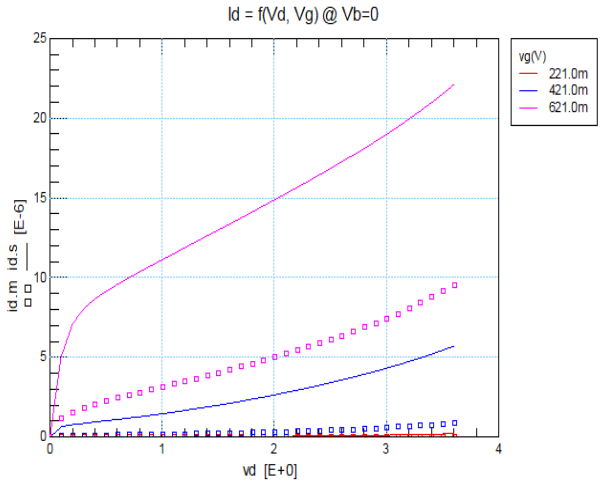
idvd_vbmin, Vb = -3.6V, T = 27°C

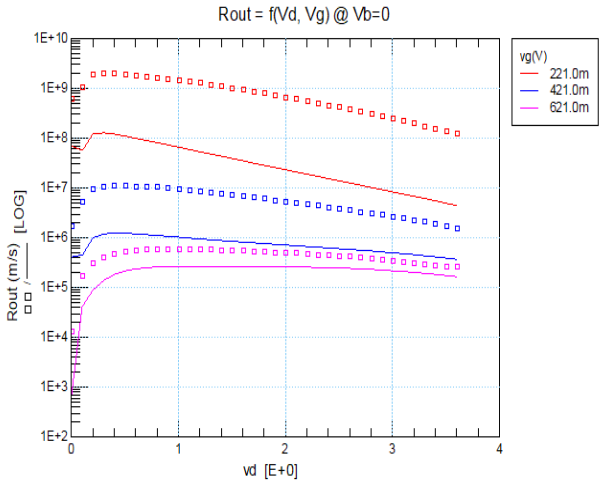


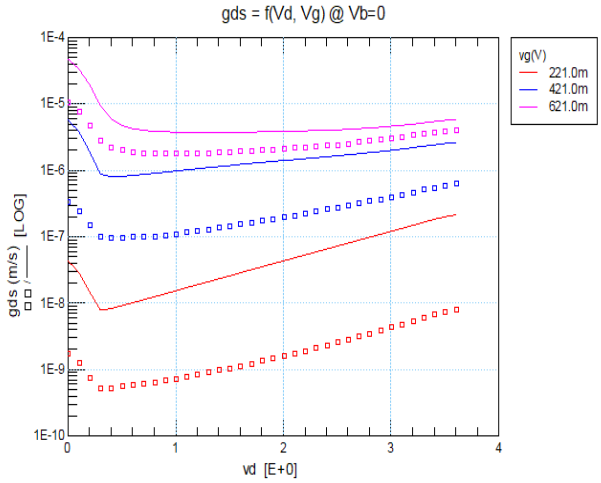




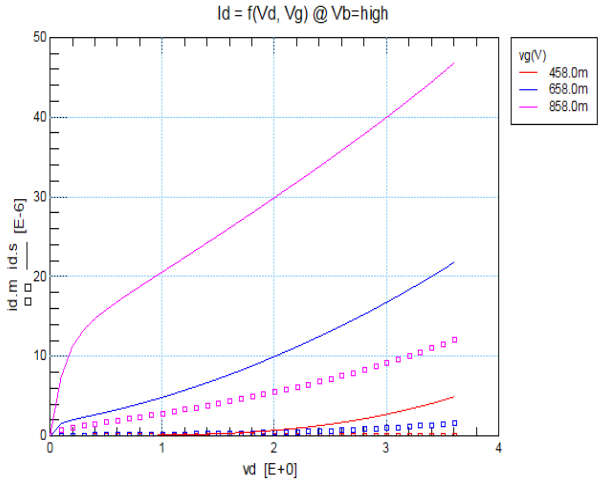
idvd_vth, Vb = 0V, T = 27°C

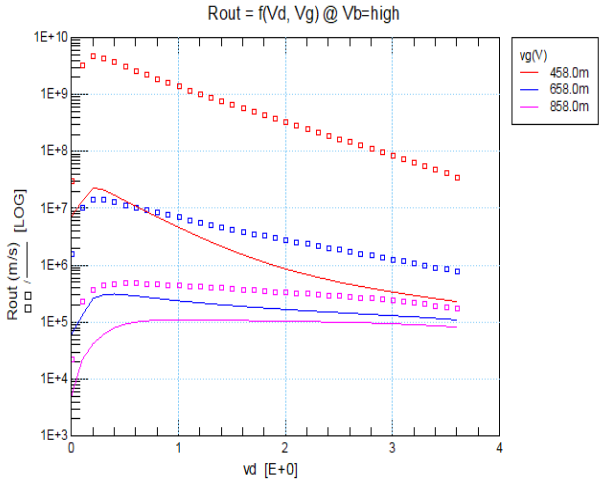


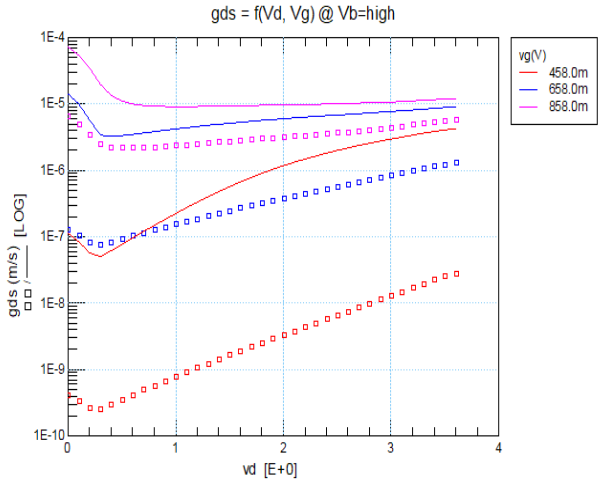




idvd_vth_vbmin, Vb = -3.6V, T = 27°C

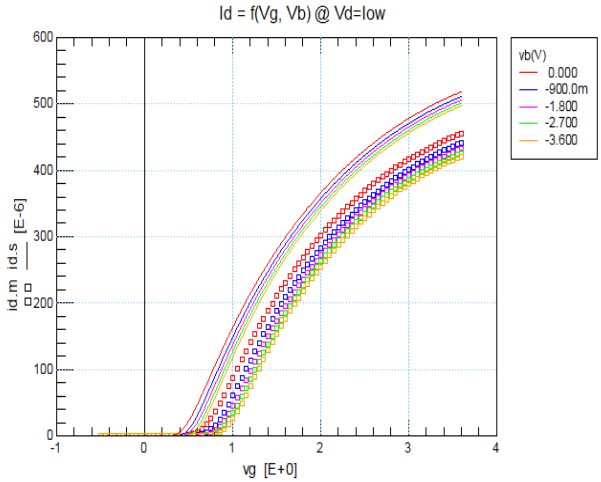


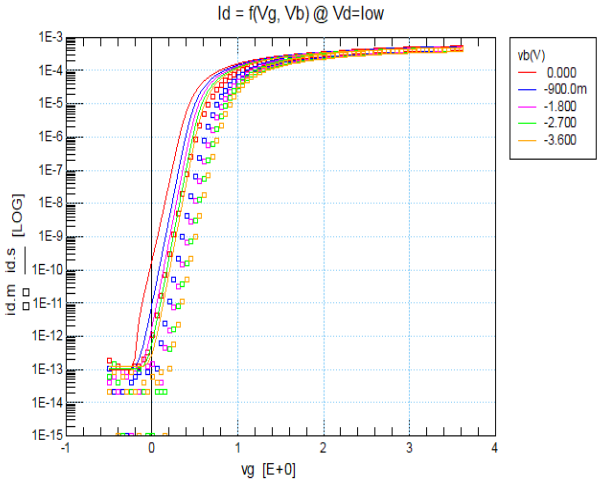


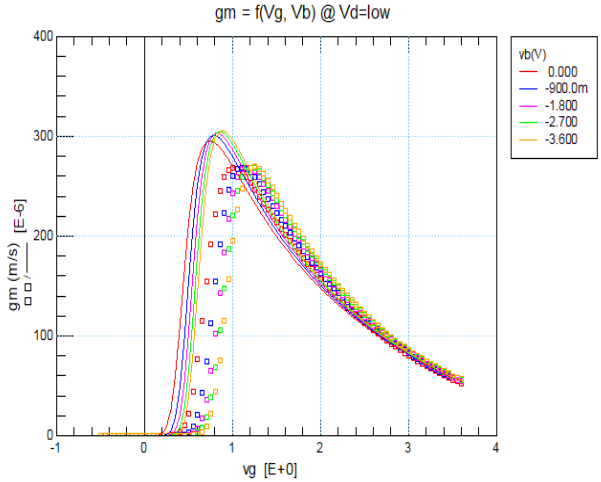


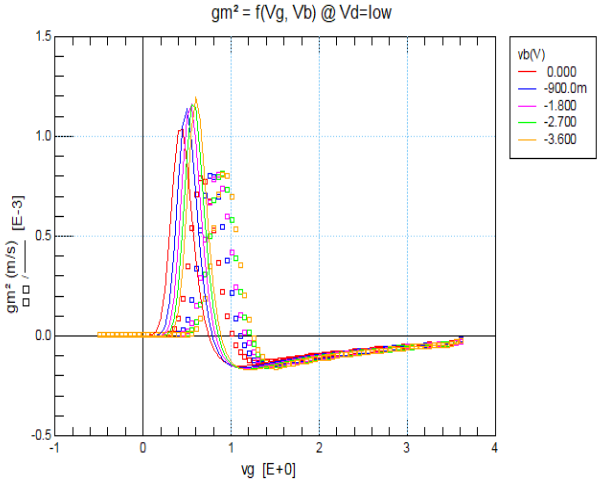
4.34 W05u0_L0u33_S557_4

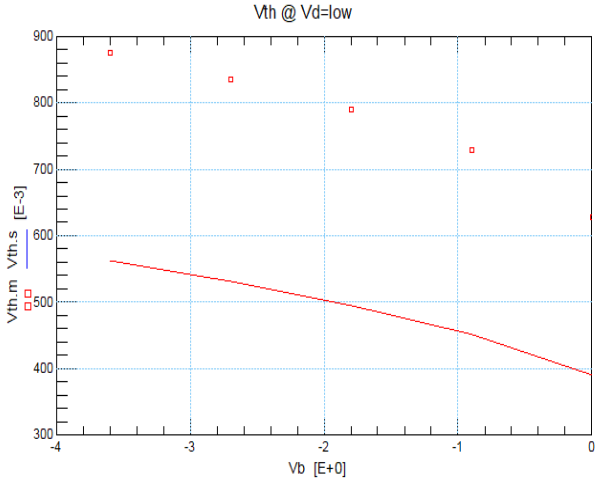
idvg, Vd = 0.1V, T = 27°C

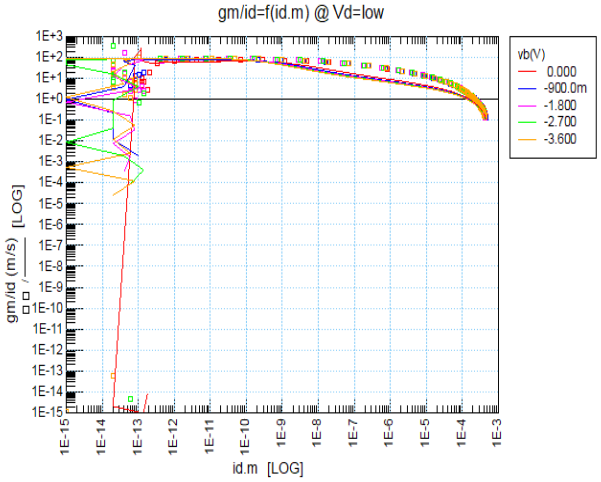


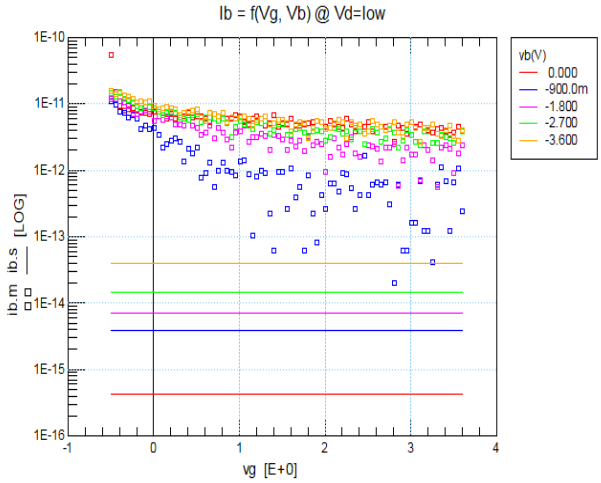


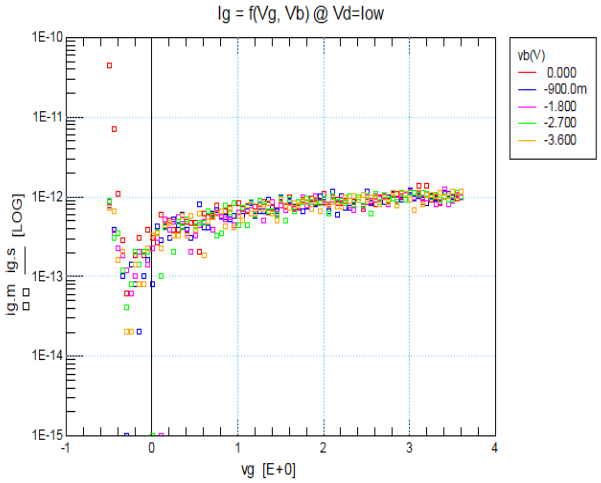




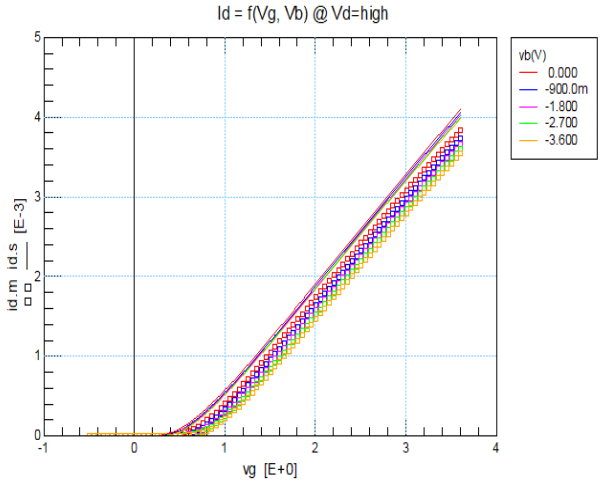


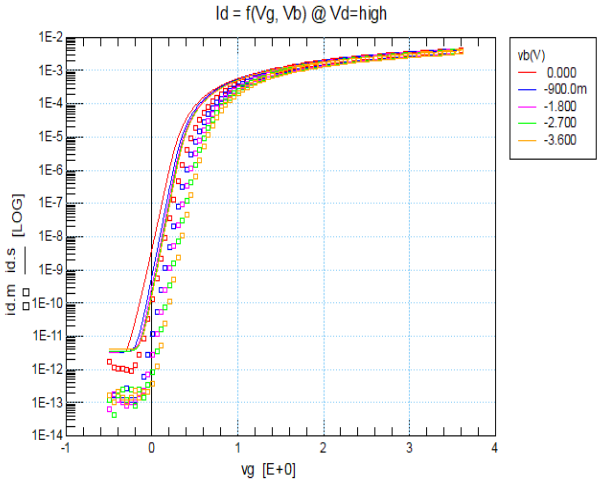


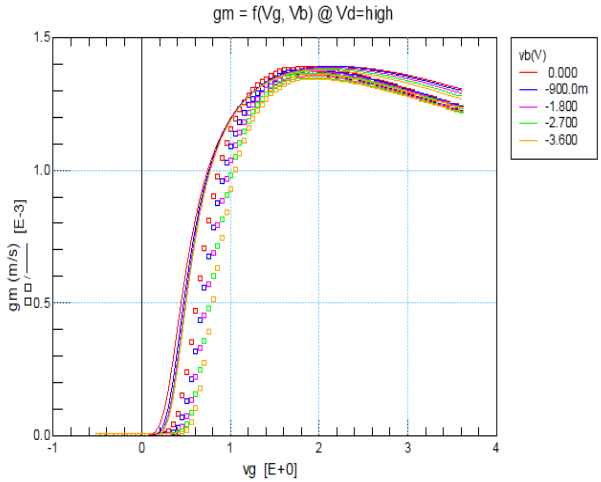


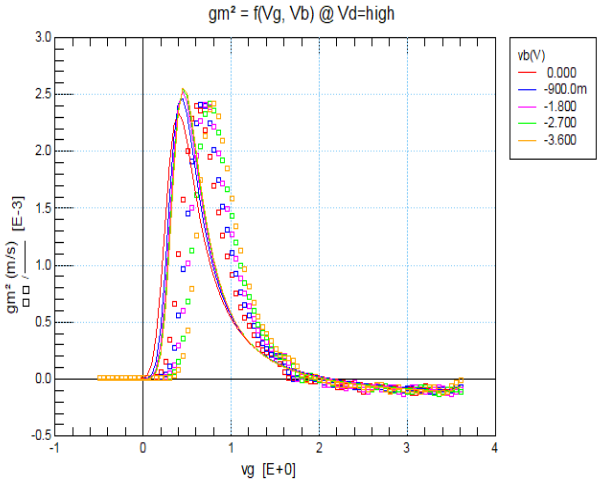


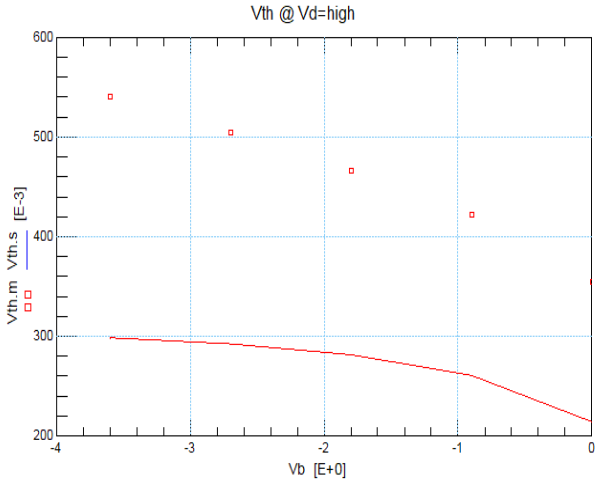
idvg, Vd = 3.3V, T = 27°C

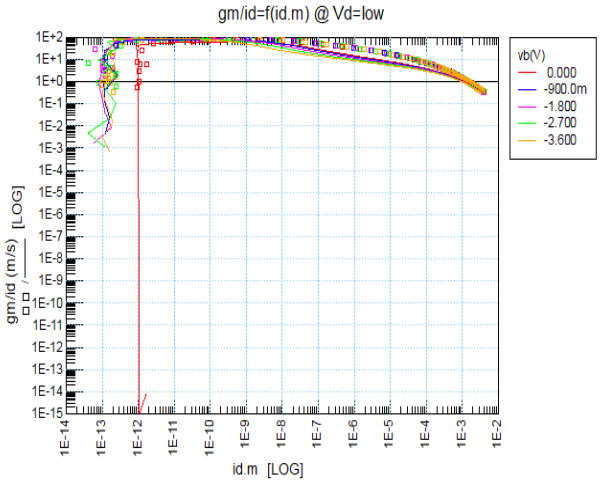


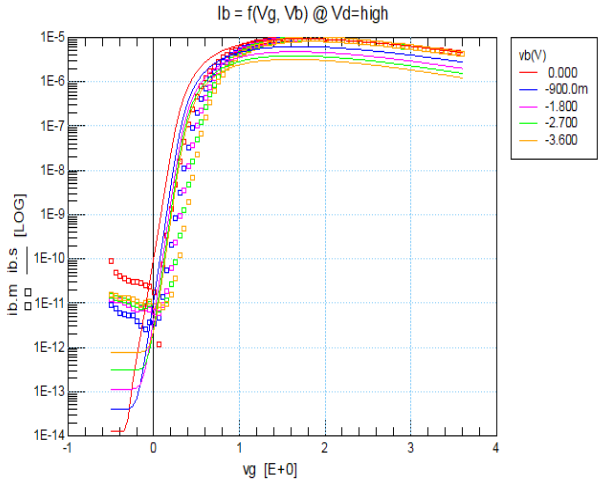


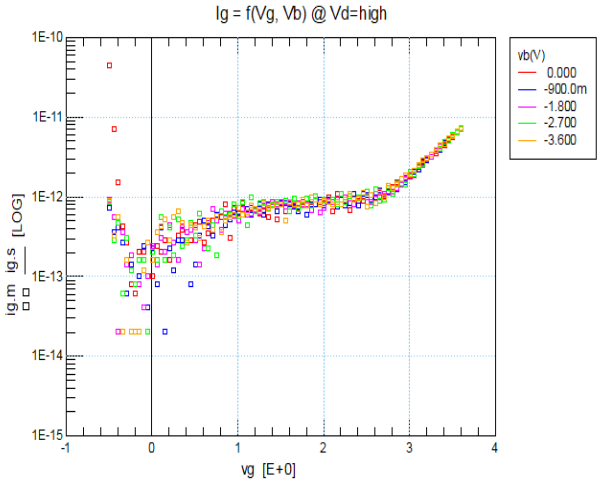




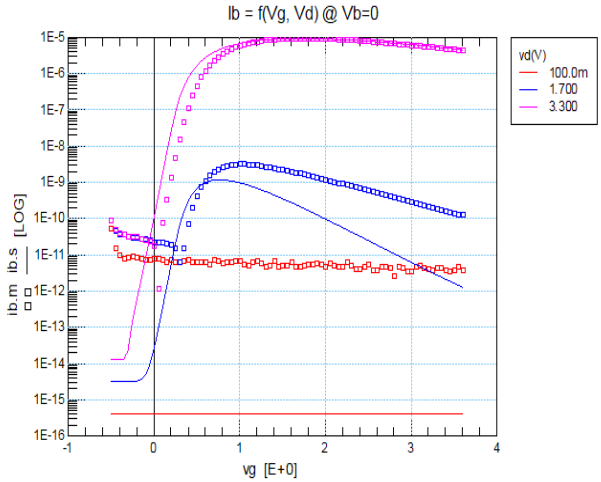


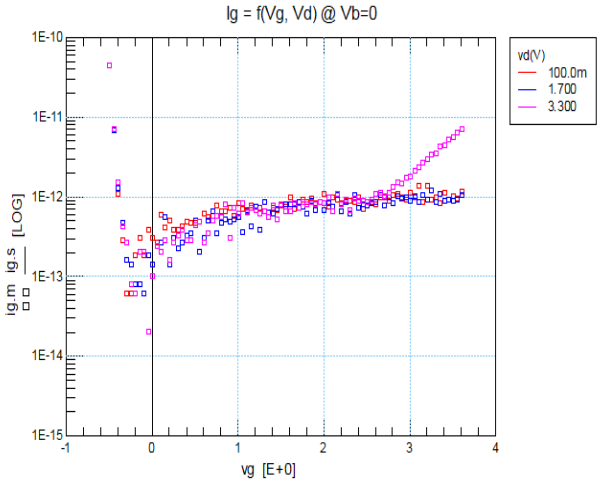




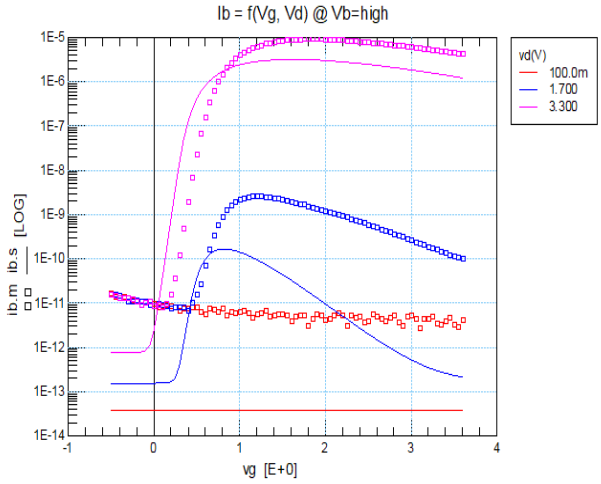


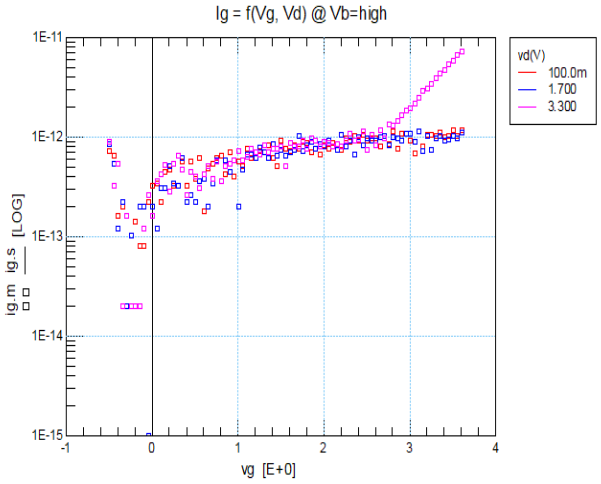
idvg, Ib,Ig, Vb = 0V, T = 27°C



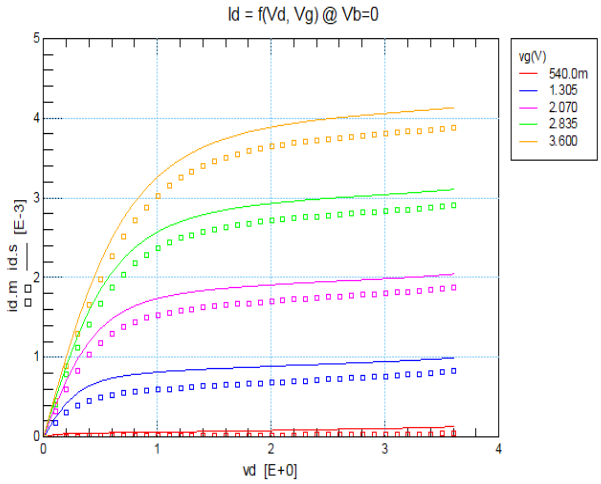


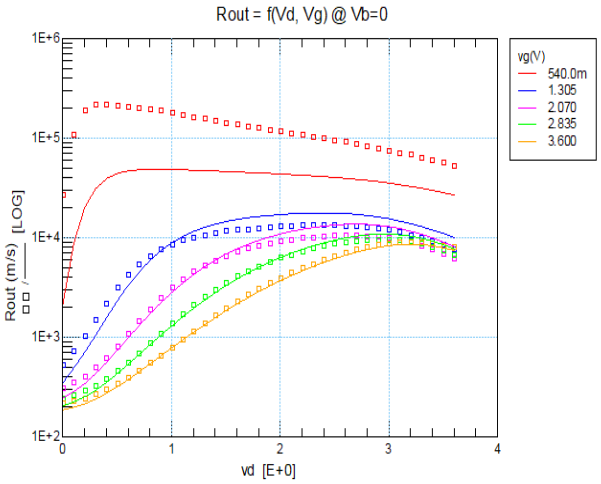
idvg, Ib,Ig, Vb = -3.6V, T = 27°C

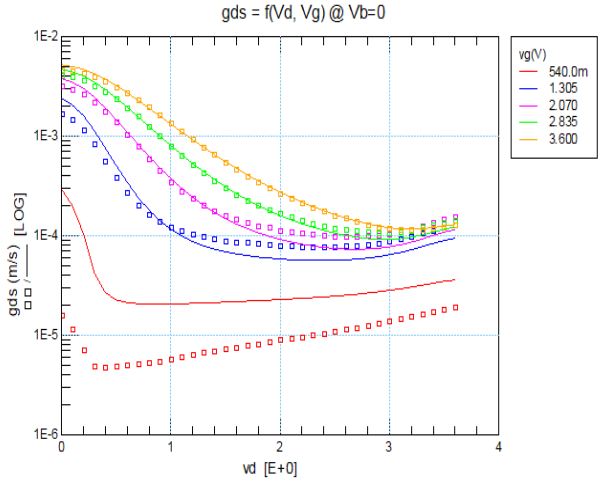




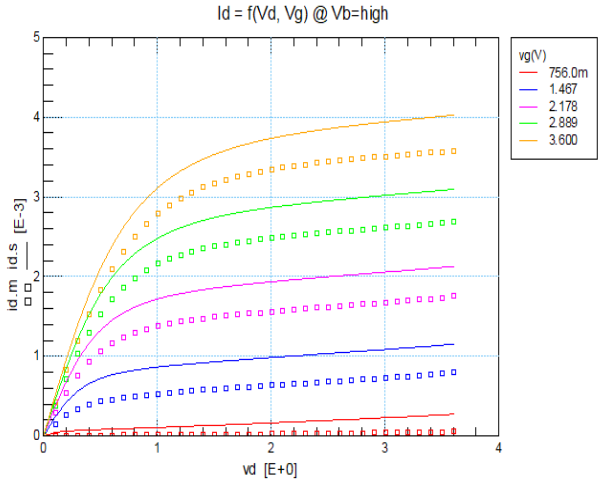
idvd, Vb = 0V, T = 27°C

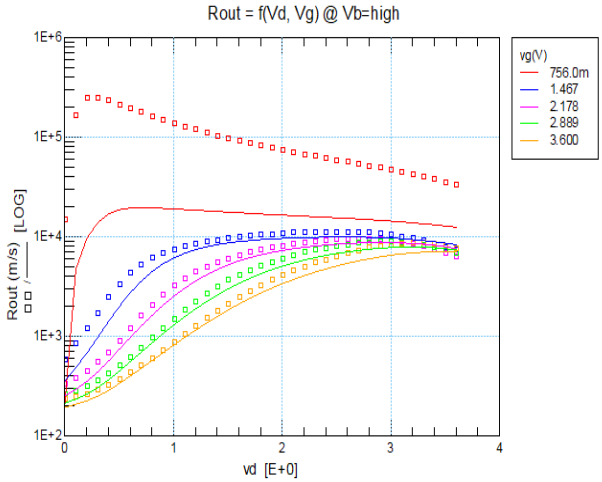


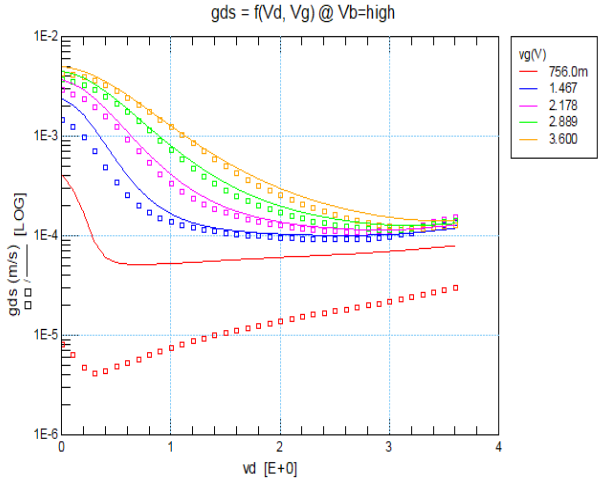




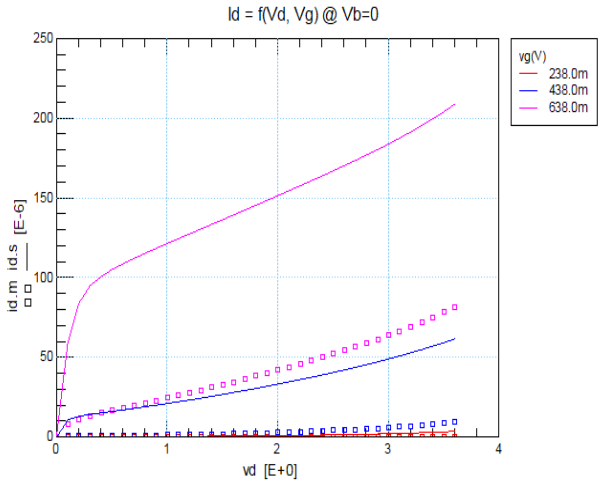
idvd_vbmin, Vb = -3.6V, T = 27°C

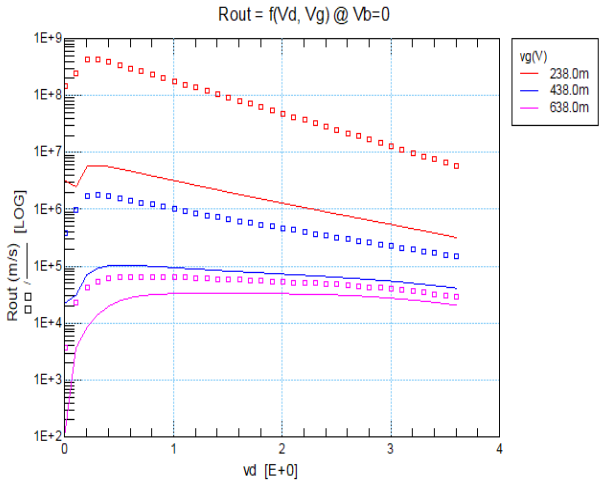


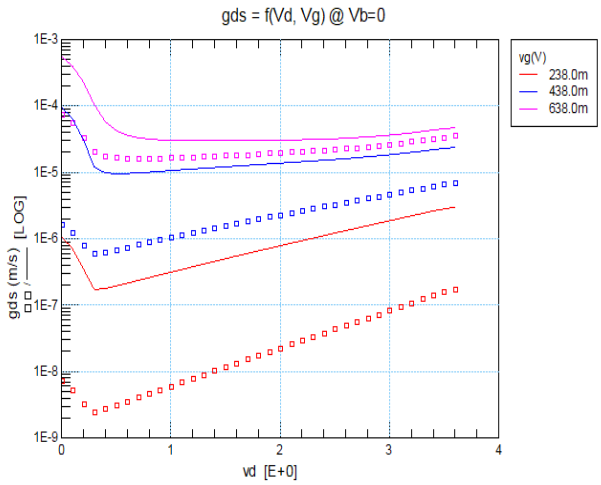




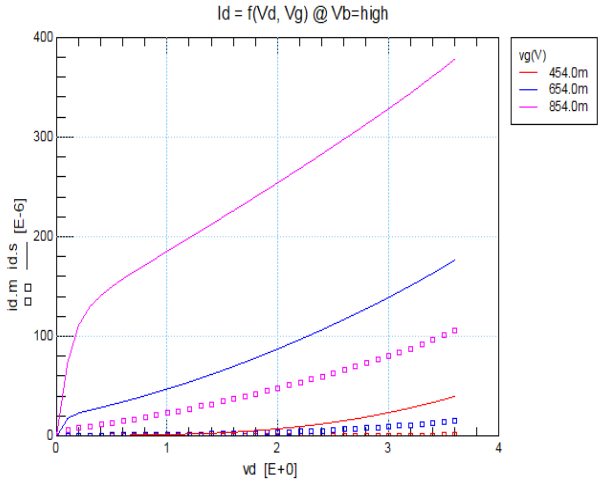
idvd_vth, Vb = 0V, T = 27°C

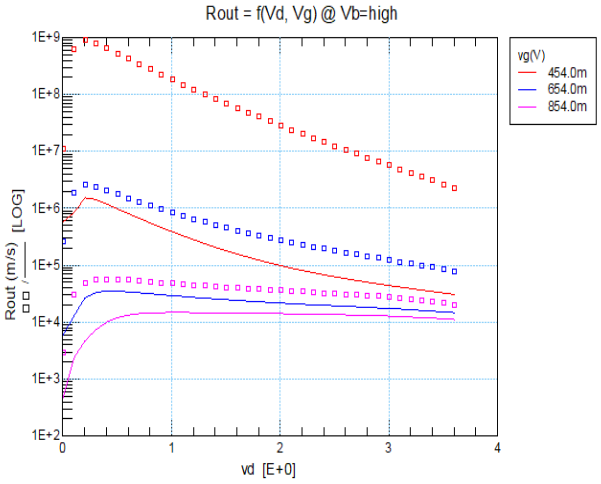


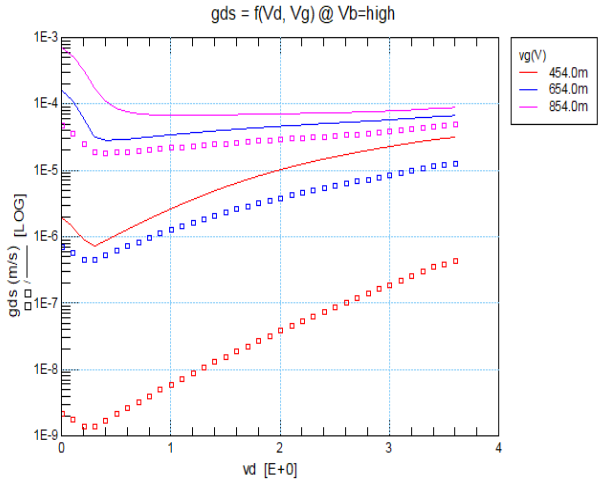




idvd_vth_vbmin, Vb = -3.6V, T = 27°C



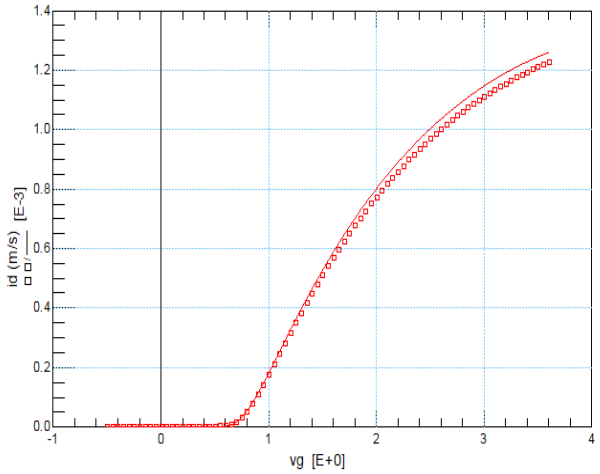




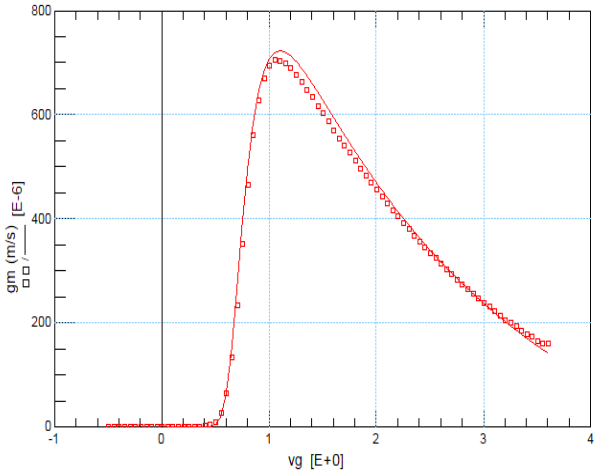
5 RF Transistors

5.1 NF04_WF05u0_L_0u5_post_90

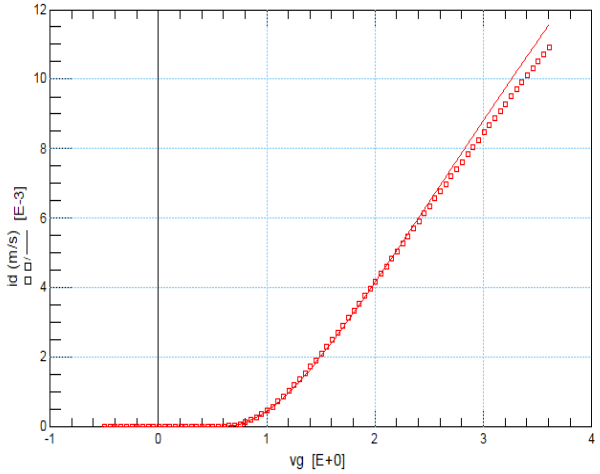
IdVg, T = 27°C



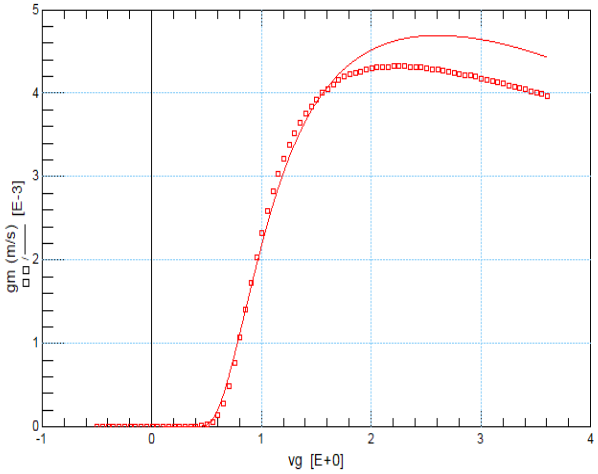
Id = f(Vg) @ Vd = 0.1V



gm = f(Vg) @ Vd = 0.1V

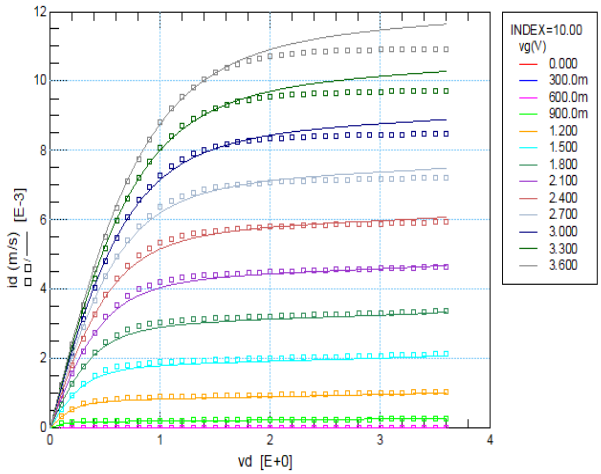


Id = f(Vg) @ Vd = 3.3V

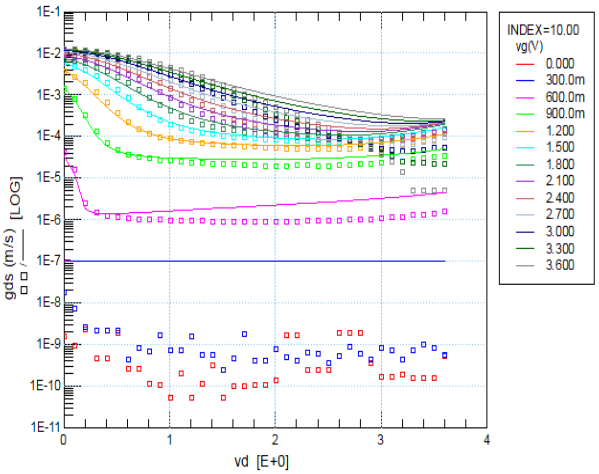


gm = f(Vg) @ Vd = 3.3V

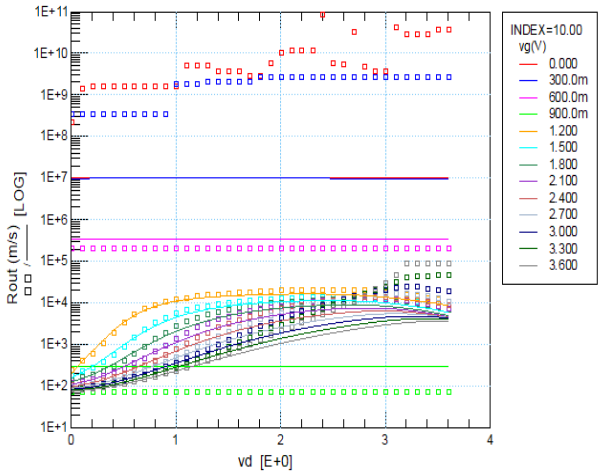
IdVd, T = 27°C



Id = f(Vd)

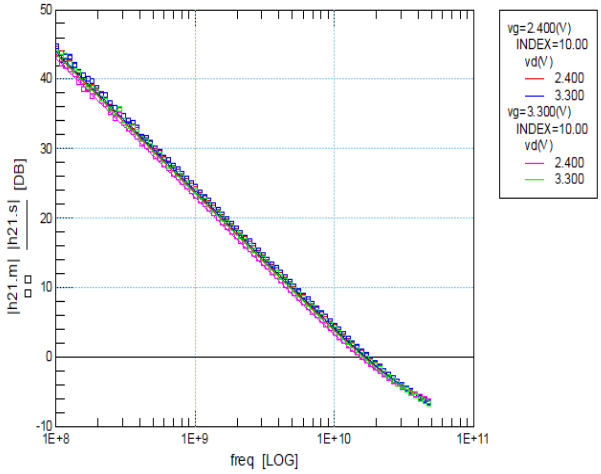


gds = f(Vd)

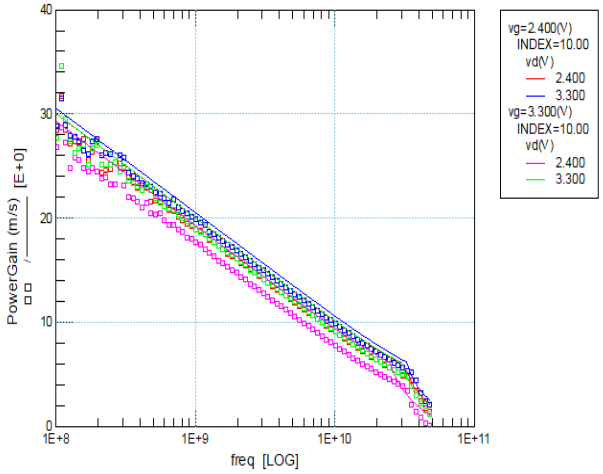


Rout = f(Vd)

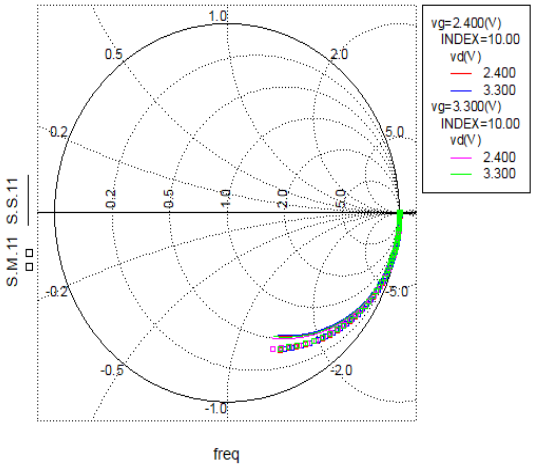
S-Parameter, T = 27°C



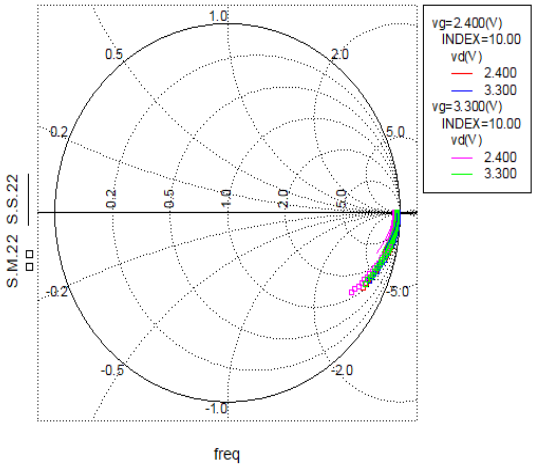
|h21| = f(Vg,Vd)



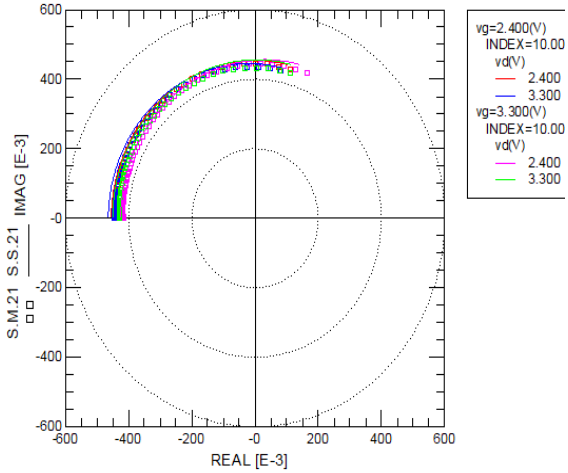
MSG/MAG = f(Vg,Vd)



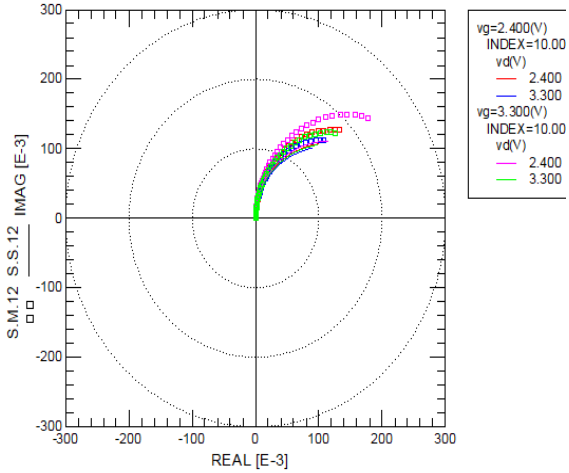
S11 = f(Vg,Vd)



S22 = f(Vg,Vd)

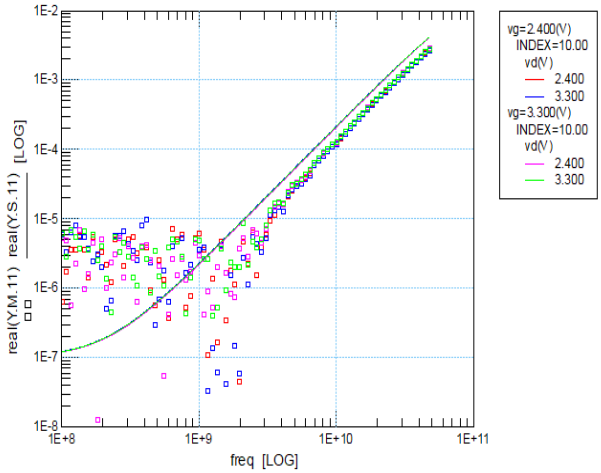


S21 = f(Vg,Vd)

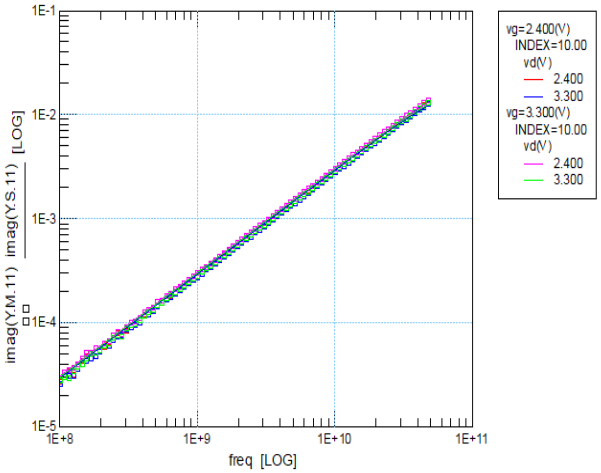


S12 = f(Vg,Vd)

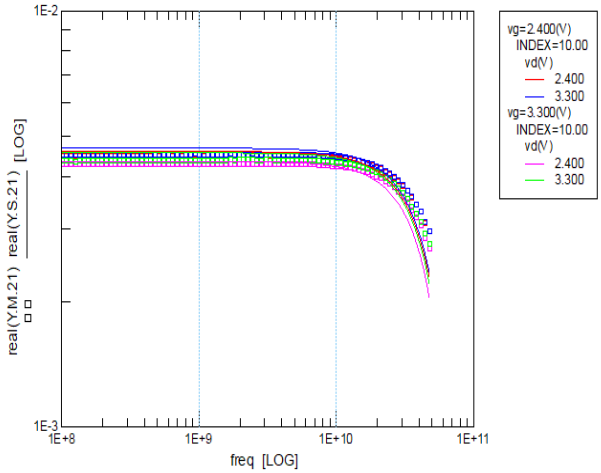
Y-Parameter, T = 27°C



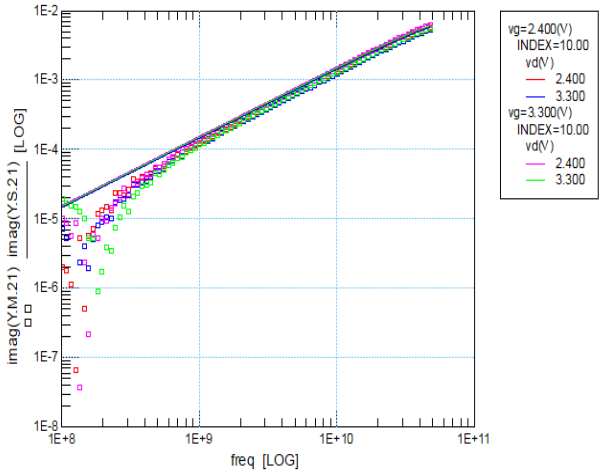
re(Y11) = f(Vg,Vd)



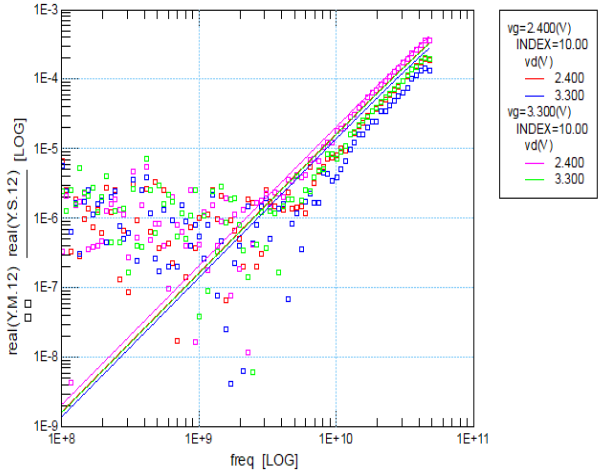
im(Y11) = f(Vg,Vd)



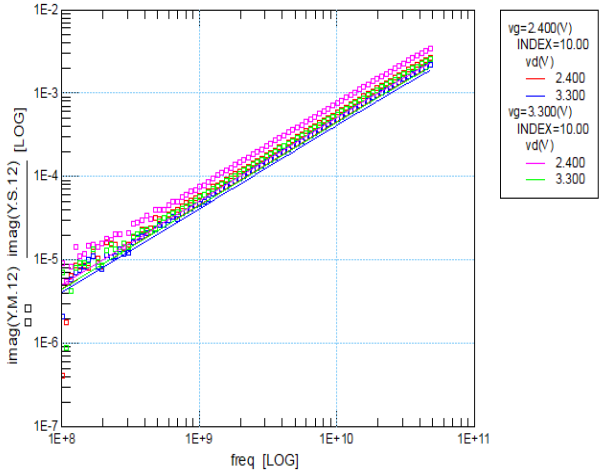
re(Y21) = f(Vg,Vd)



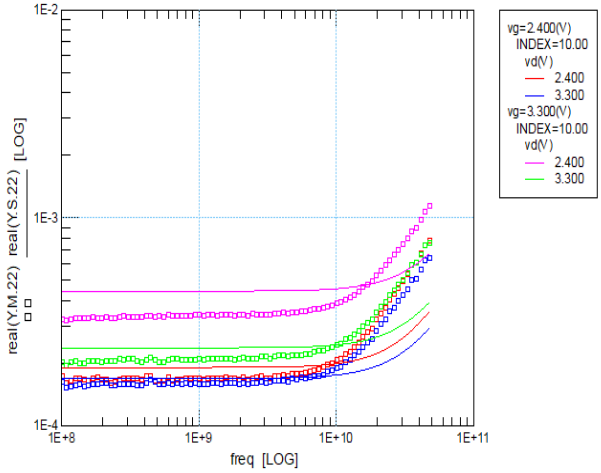
im(Y21) = f(Vg,Vd)



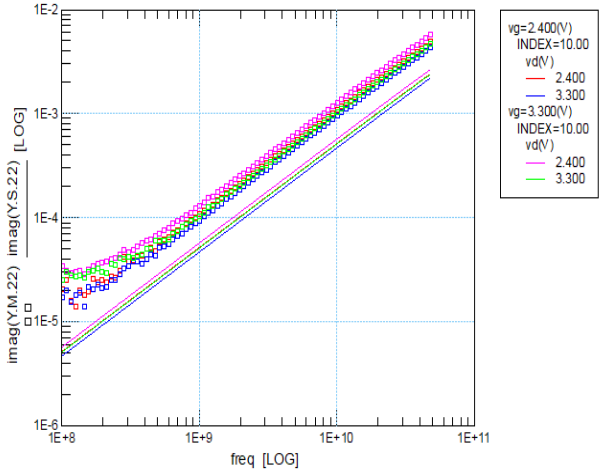
re(Y12) = f(Vg,Vd)



im(Y12) = f(Vg,Vd)

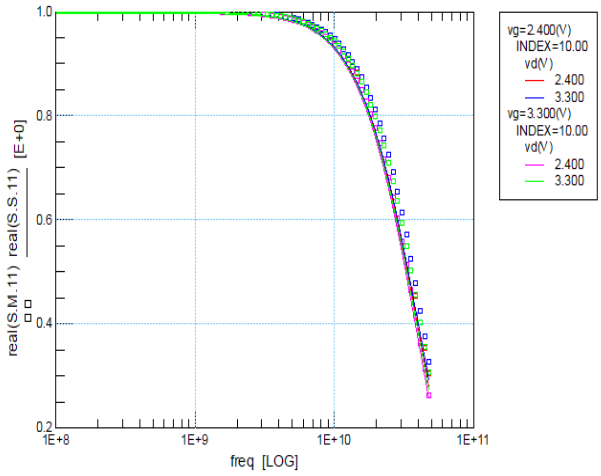


re(Y22) = f(Vg,Vd)

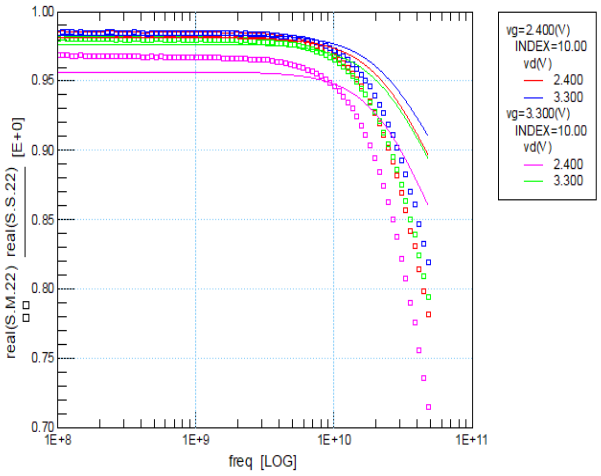


im(Y22) = f(Vg,Vd)

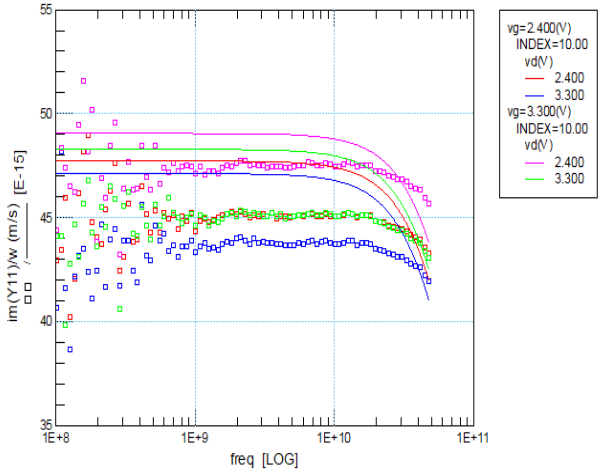
Verify Plots, T = 27°C



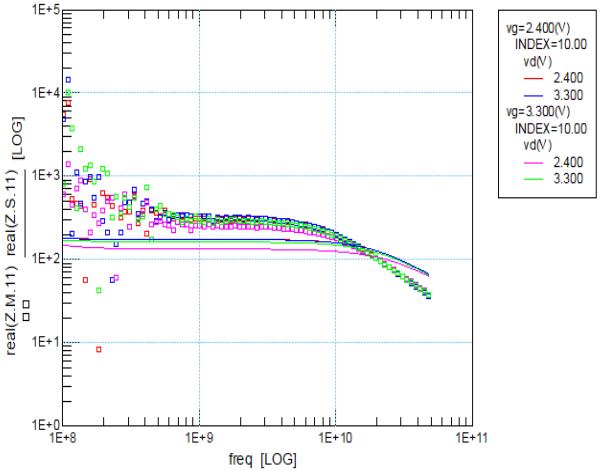
re(S11) = f(Vg,Vd)



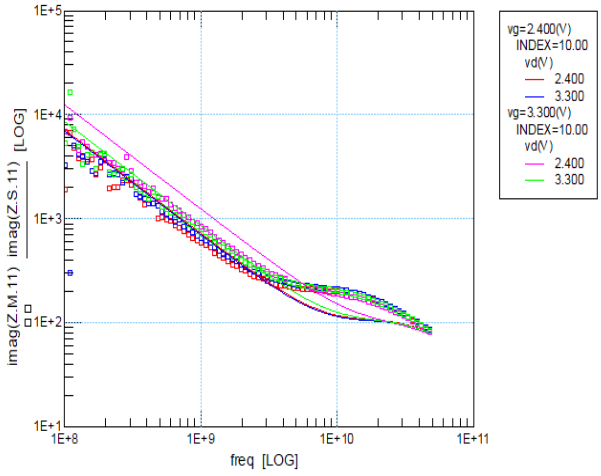
re(S22) = f(Vg,Vd)



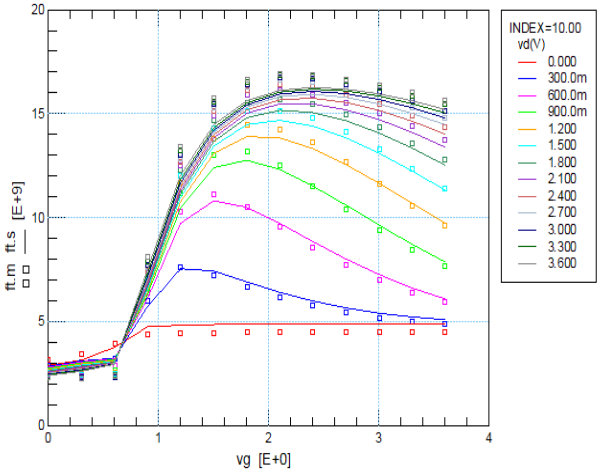
im(Y11) divided by ω



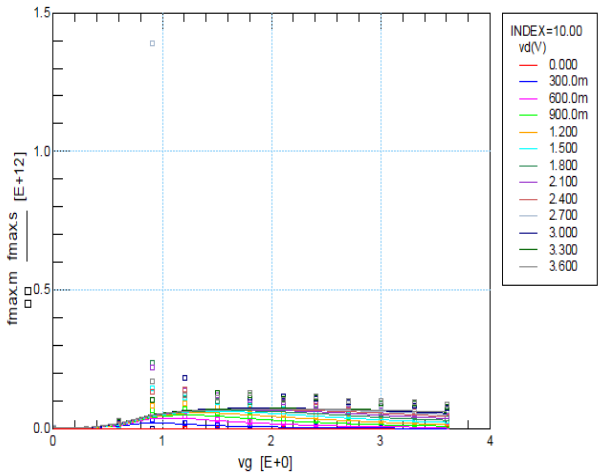
re(Z11) = f(Vg,Vd)



im(Z11) = f(Vg,Vd)



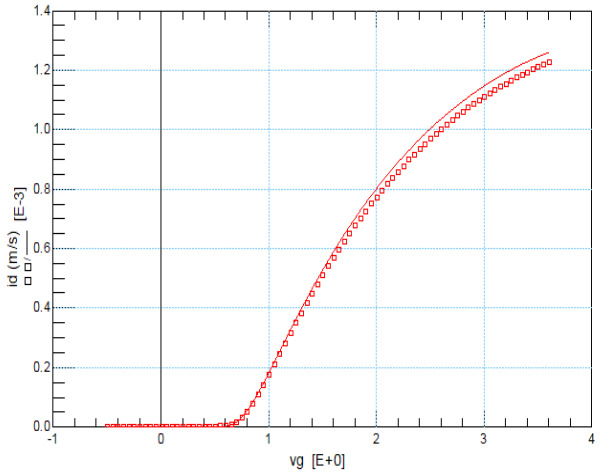
fT_vg = f(Vg,Vd)



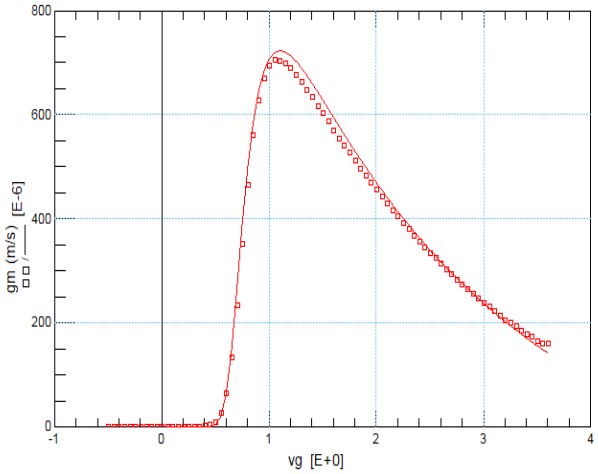
fmax_vg = f(Vg,Vd)

5.2 NF04_WF05u0_L_0u5_S453A_90

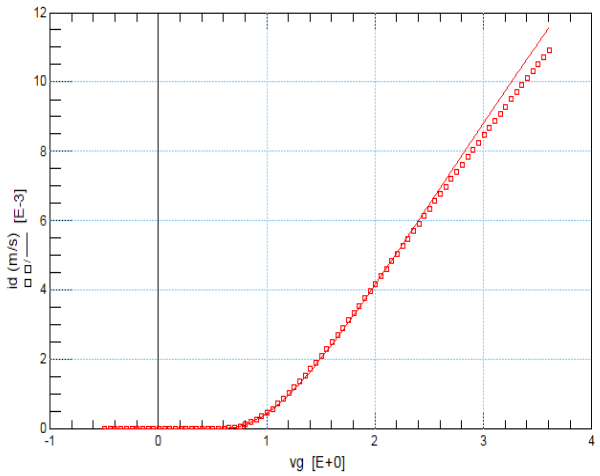
IdVg, T = 27°C



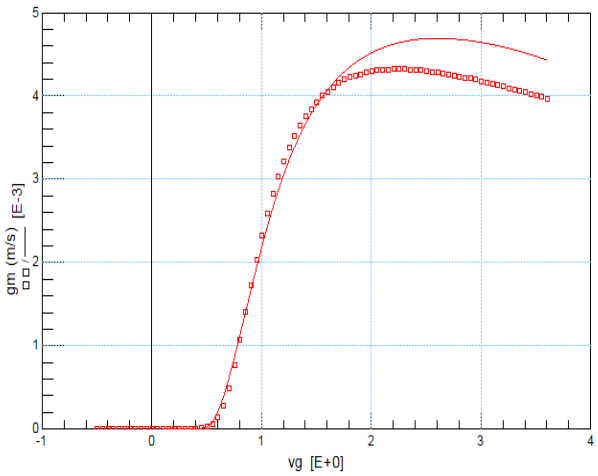
Id = f(Vg) @ Vd = 0.1V



gm = f(Vg) @ Vd = 0.1V

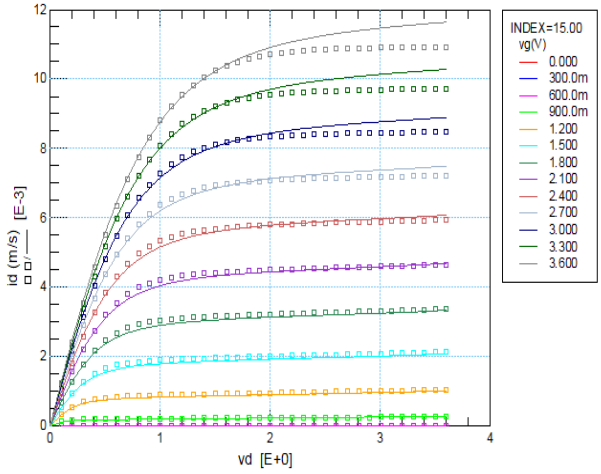


Id = f(Vg) @ Vd = 3.3V

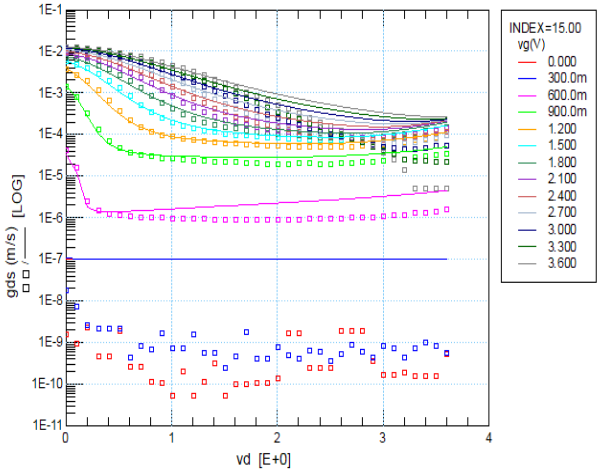


gm = f(Vg) @ Vd = 3.3V

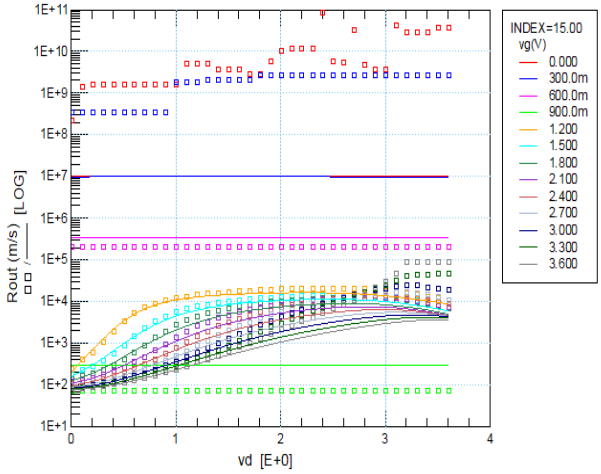
IdVd, T = 27°C



Id = f(Vd)

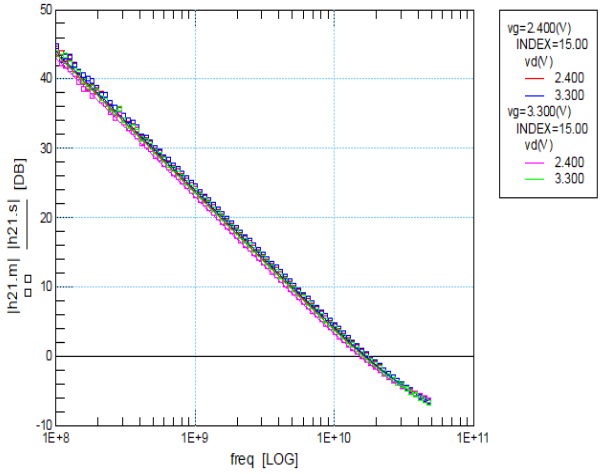


gds = f(Vd)

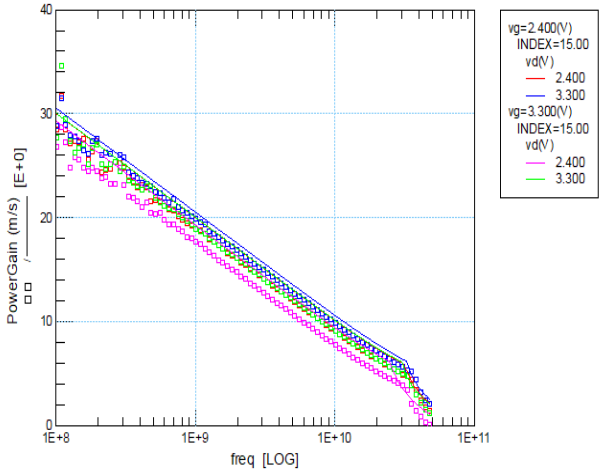


Rout = f(Vd)

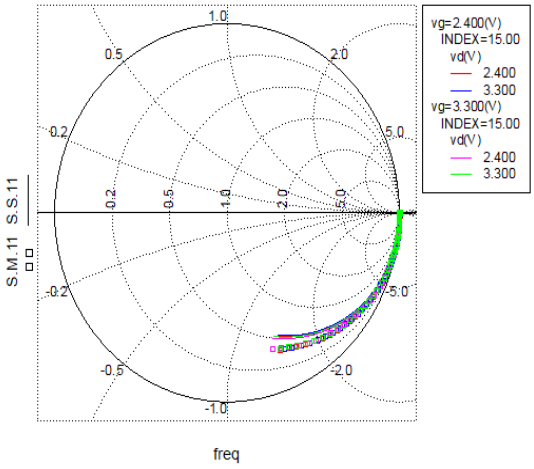
S-Parameter, T = 27°C



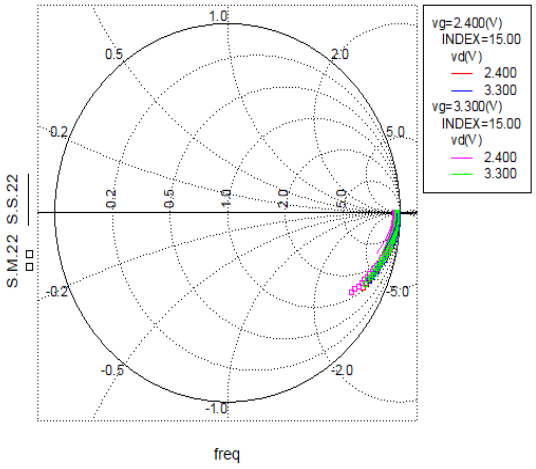
|h21| = f(Vg,Vd)



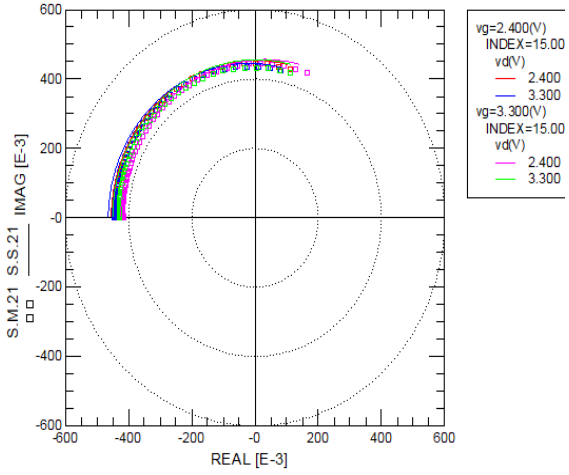
MSG/MAG = f(Vg,Vd)



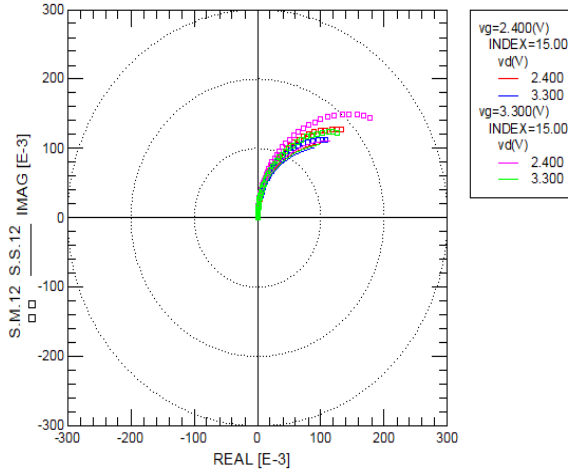
S11 = f(Vg,Vd)



S22 = f(Vg,Vd)

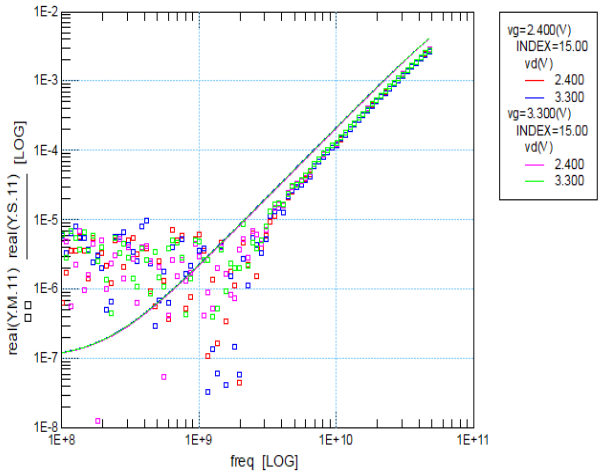


S21 = f(Vg,Vd)

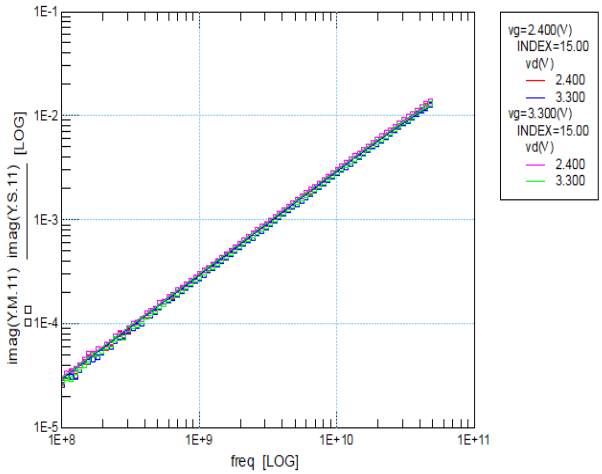


S12 = f(Vg,Vd)

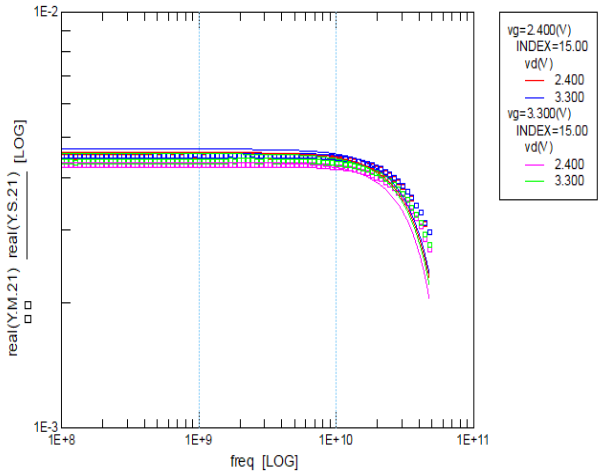
Y-Parameter, T = 27°C



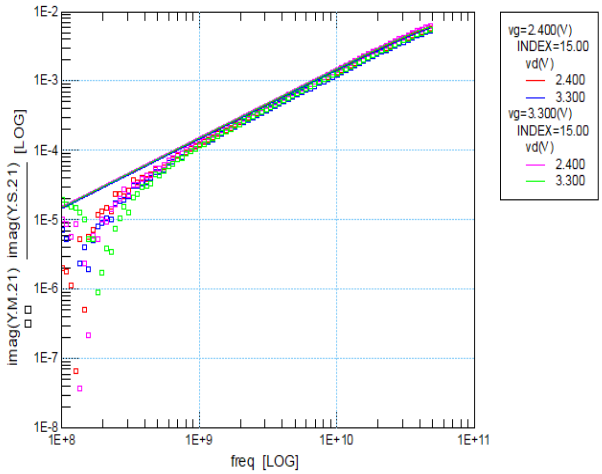
re(Y11) = f(Vg,Vd)



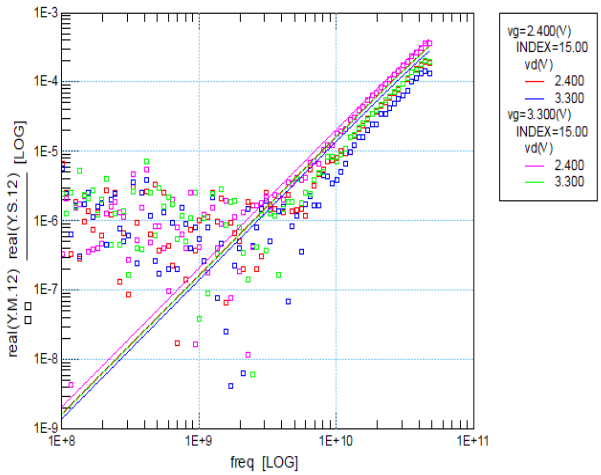
im(Y11) = f(Vg,Vd)



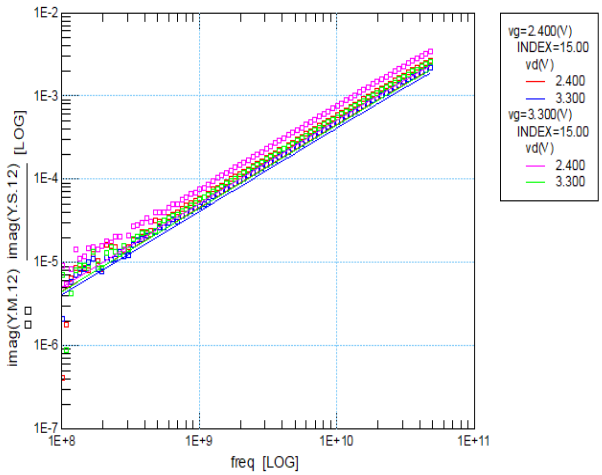
re(Y21) = f(Vg,Vd)



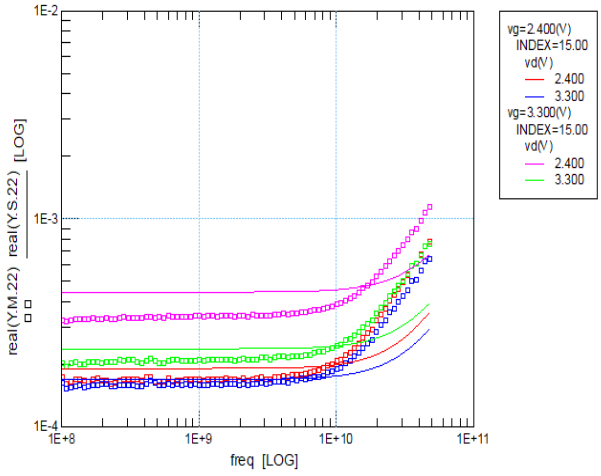
im(Y21) = f(Vg,Vd)



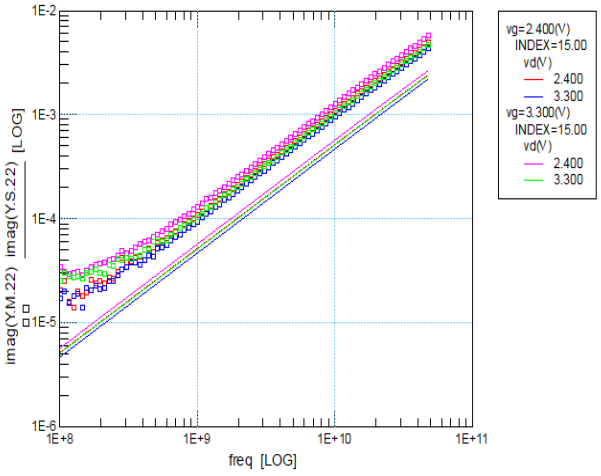
re(Y12) = f(Vg,Vd)



im(Y12) = f(Vg,Vd)

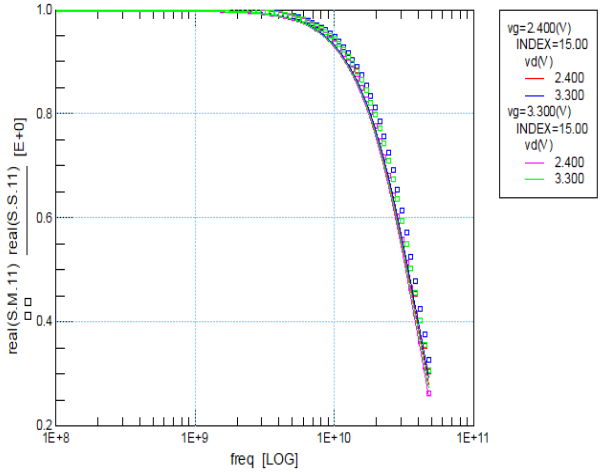


re(Y22) = f(Vg,Vd)

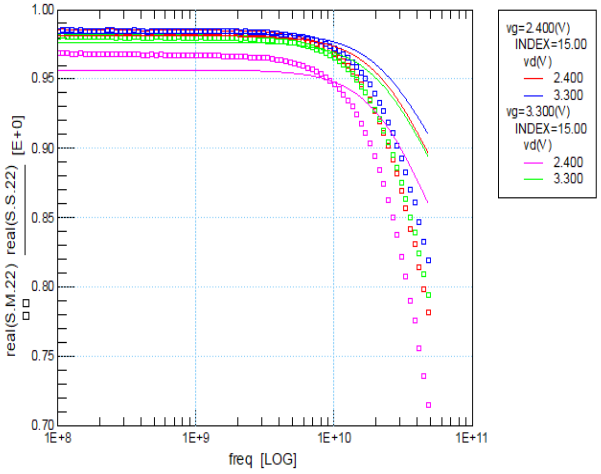


im(Y22) = f(Vg,Vd)

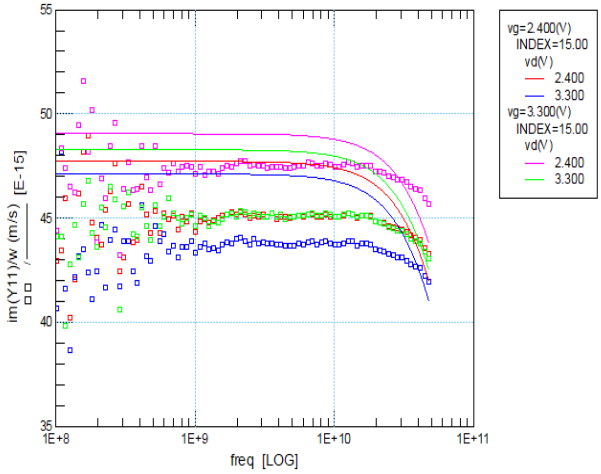
Verify Plots, T = 27°C



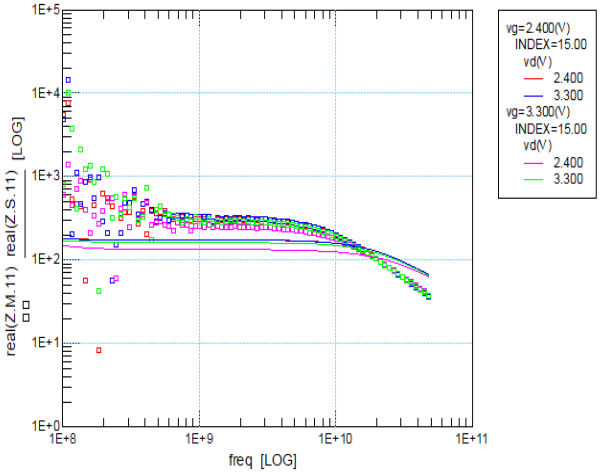
re(S11) = f(Vg,Vd)



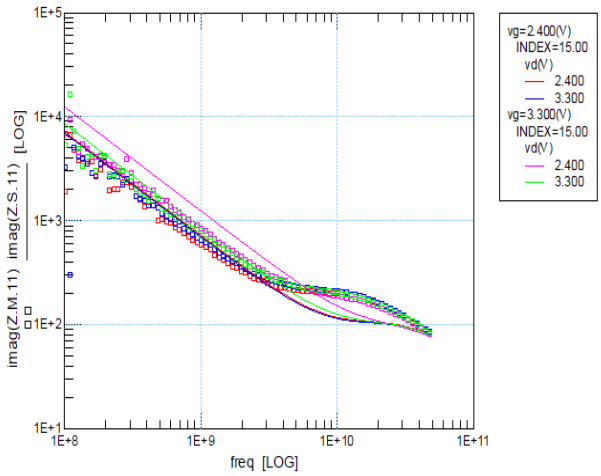
re(S22) = f(Vg,Vd)



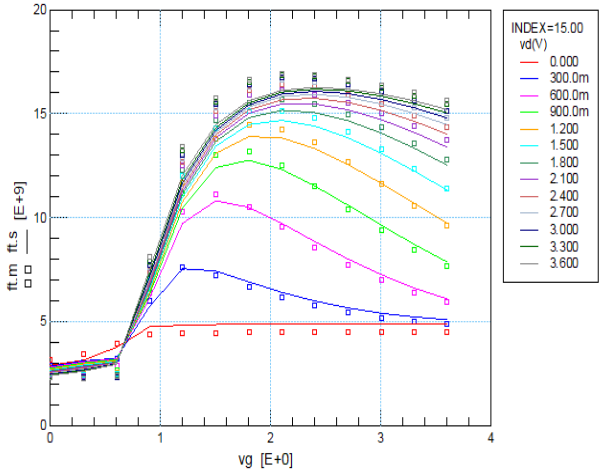
im(Y11) divided by ω



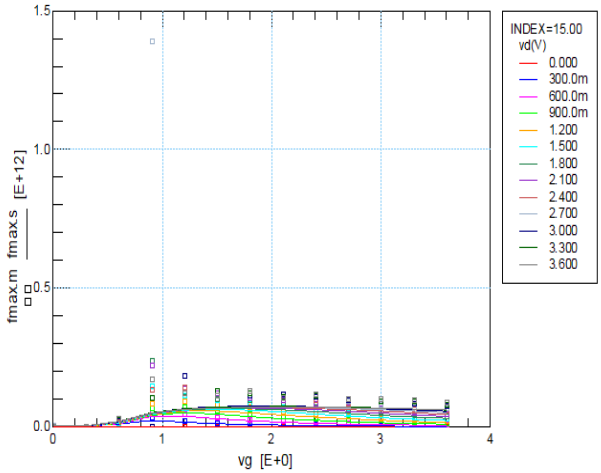
re(Z11) = f(Vg,Vd)



im(Z11) = f(Vg,Vd)



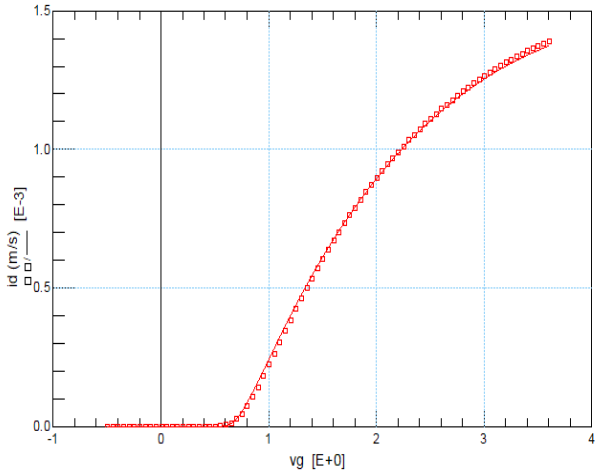
fT_vg = f(Vg,Vd)



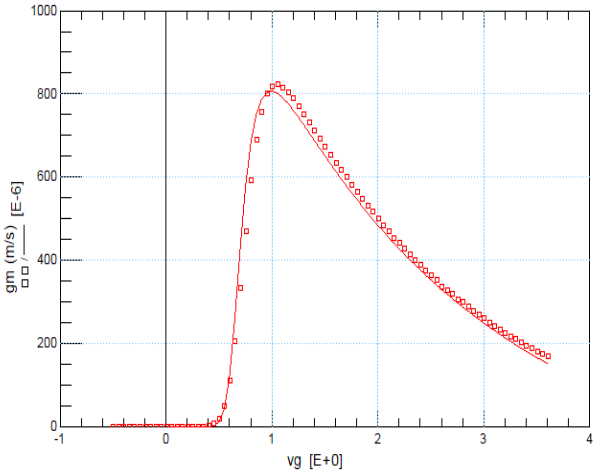
fmax_vg = f(Vg,Vd)

5.3 NF08_WF02u5_L_0u45_S453A_A1

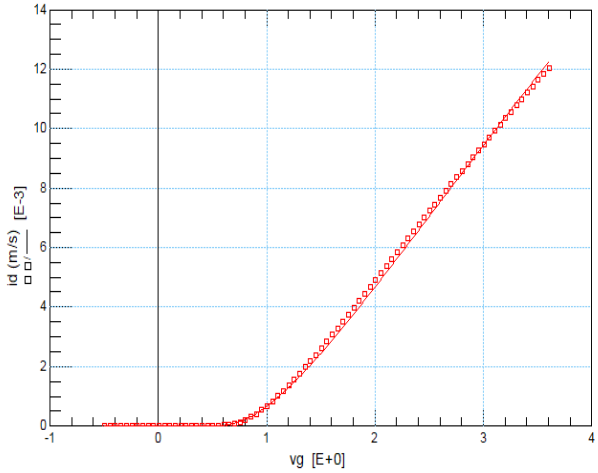
IdVg, T = 27°C



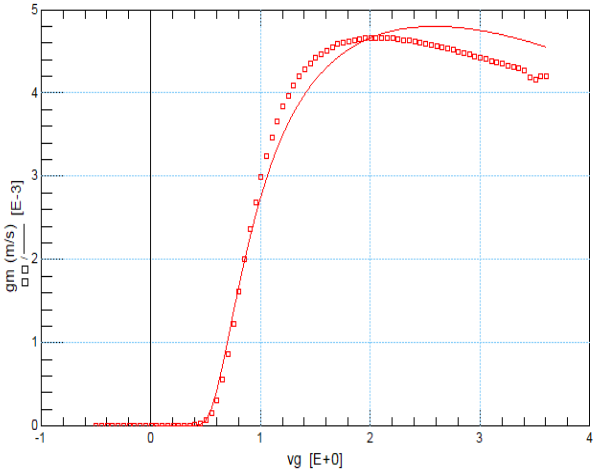
Id = f(Vg) @ Vd = 0.1V



gm = f(Vg) @ Vd = 0.1V

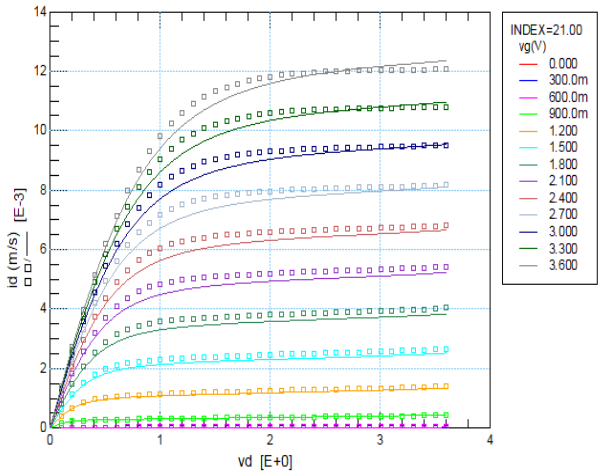


Id = f(Vg) @ Vd = 3.3V

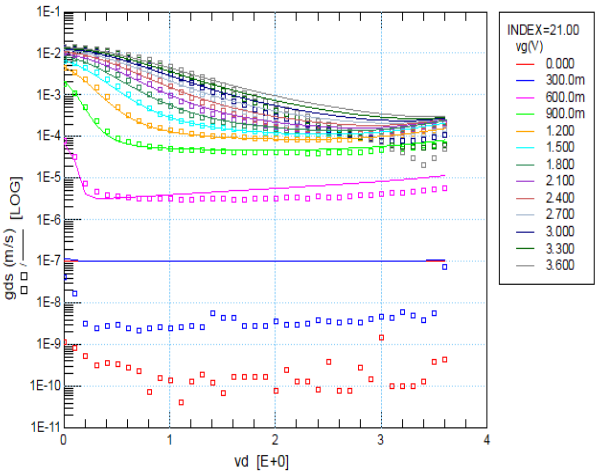


gm = f(Vg) @ Vd = 3.3V

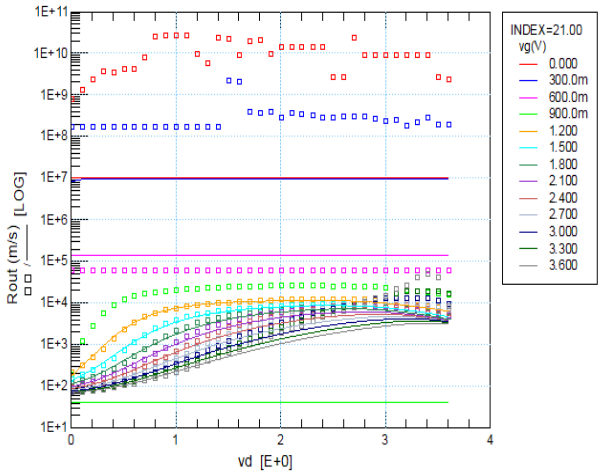
IdVd, T = 27°C



Id = f(Vd)

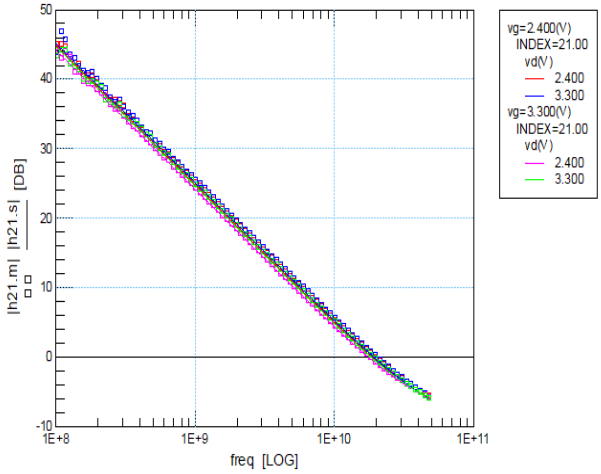


gds = f(Vd)

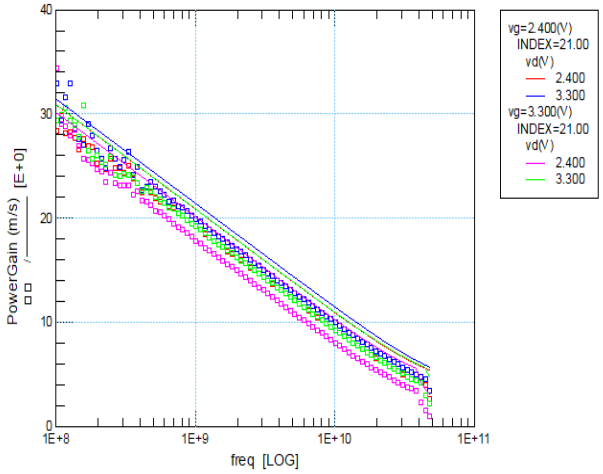


Rout = f(Vd)

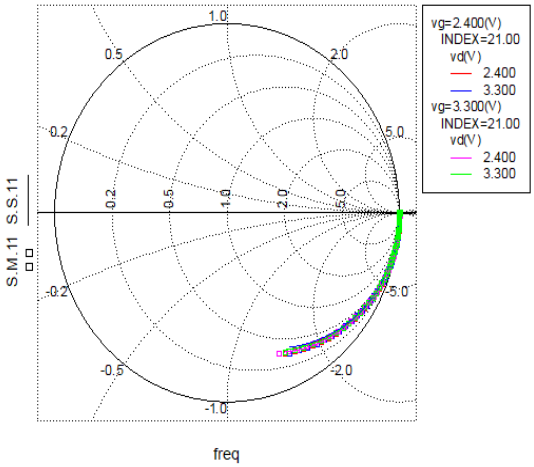
S-Parameter, T = 27°C



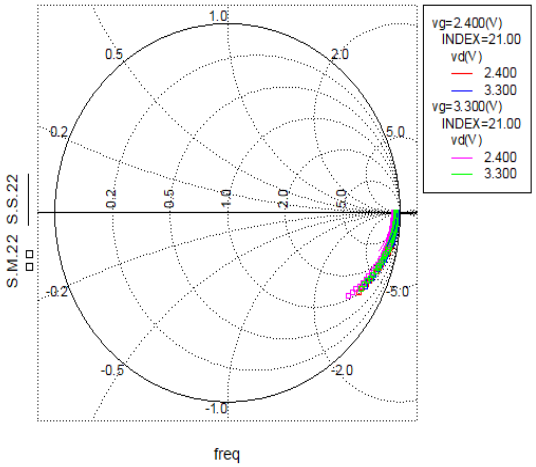
|h21| = f(Vg,Vd)



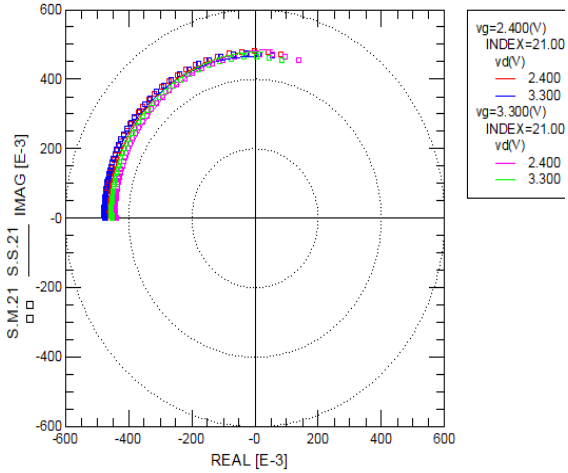
MSG/MAG = f(Vg,Vd)



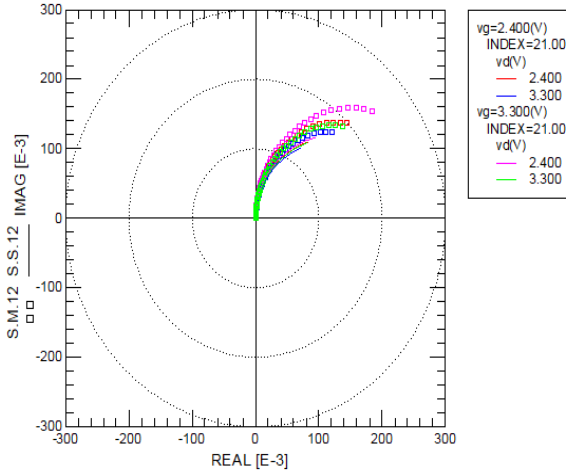
S11 = f(Vg,Vd)



S22 = f(Vg,Vd)

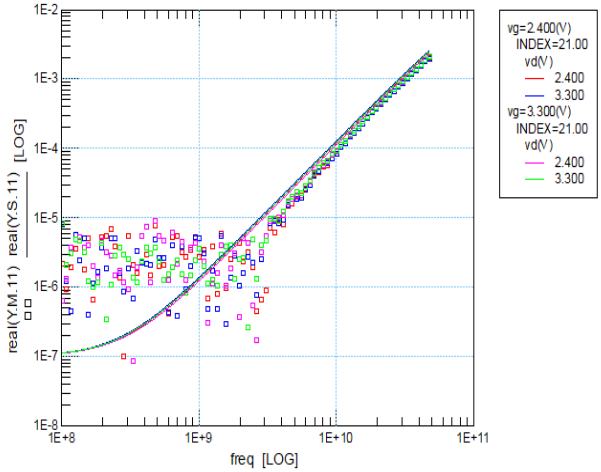


S21 = f(Vg,Vd)

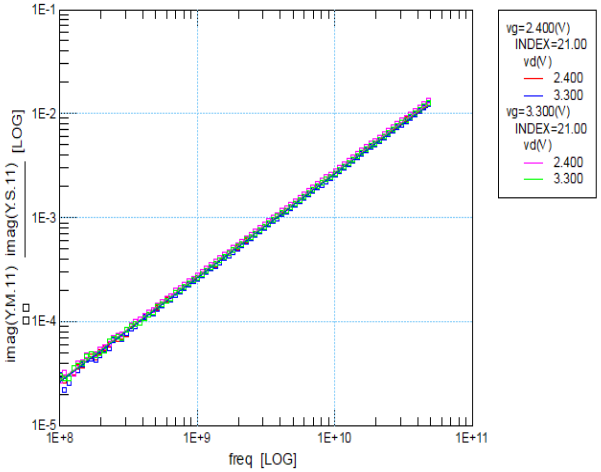


S12 = f(Vg,Vd)

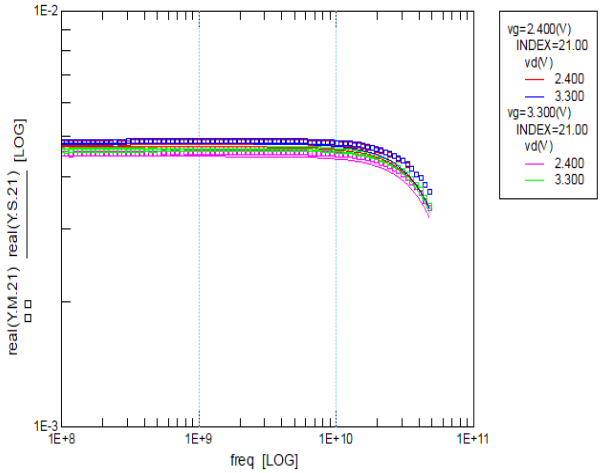
Y-Parameter, T = 27°C



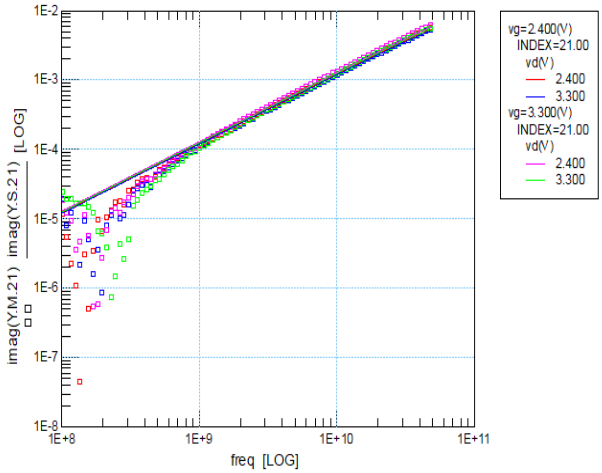
re(Y11) = f(Vg,Vd)



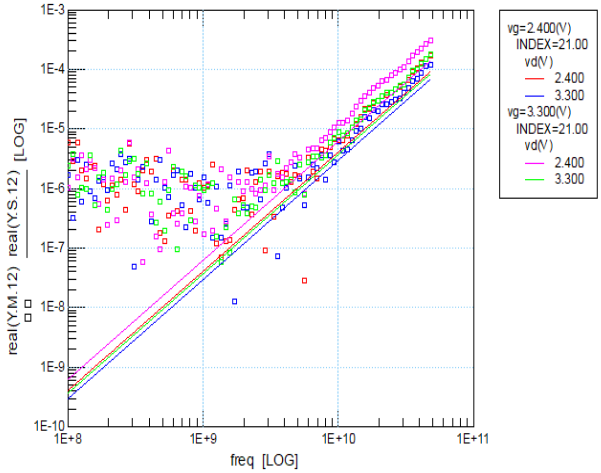
im(Y11) = f(Vg,Vd)



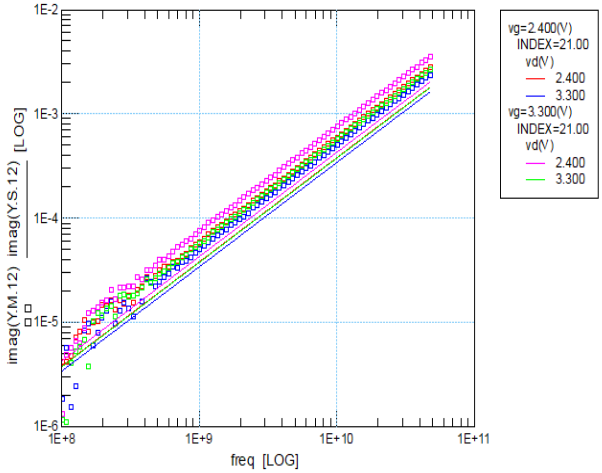
re(Y21) = f(Vg,Vd)



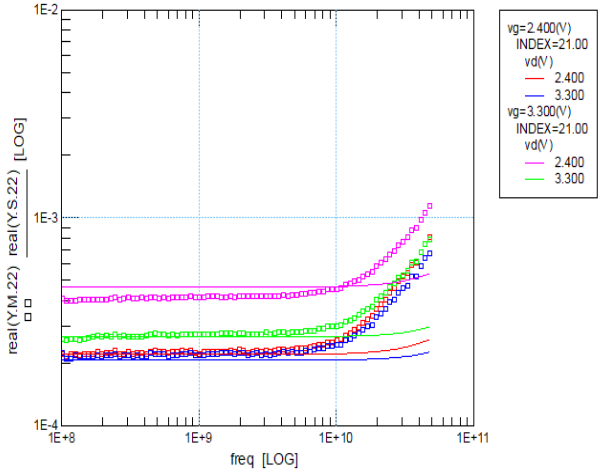
im(Y21) = f(Vg,Vd)



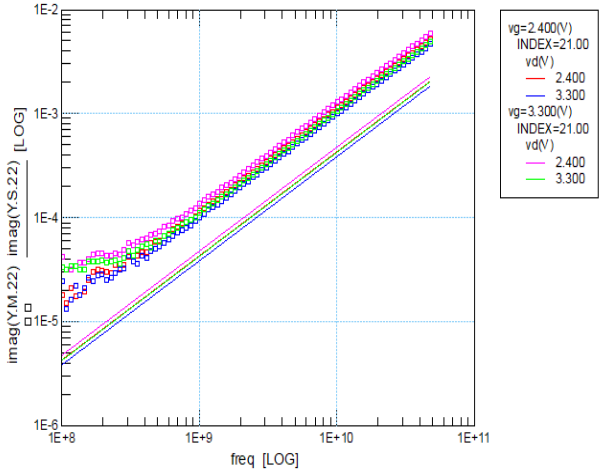
re(Y12) = f(Vg,Vd)



im(Y12) = f(Vg,Vd)

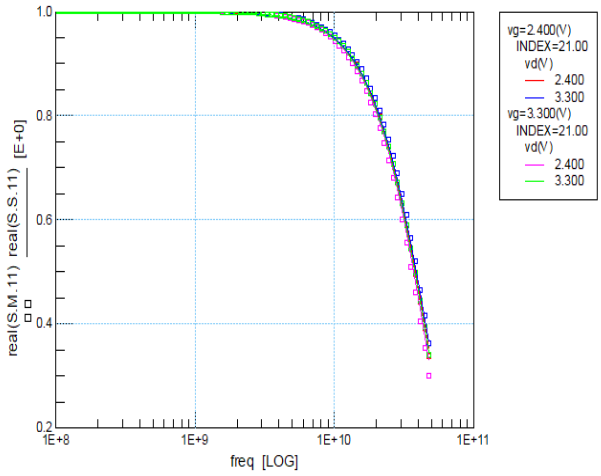


re(Y22) = f(Vg,Vd)

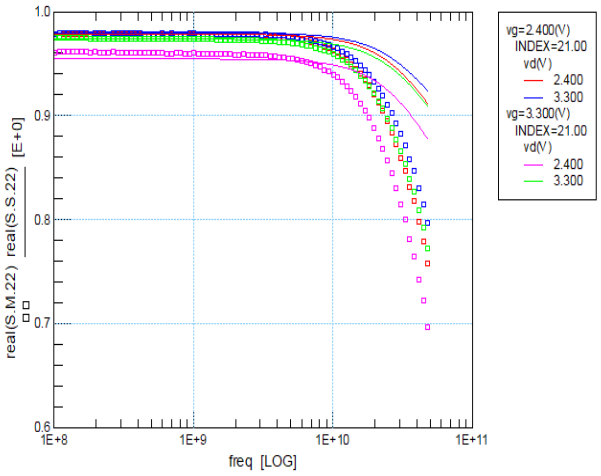


im(Y22) = f(Vg,Vd)

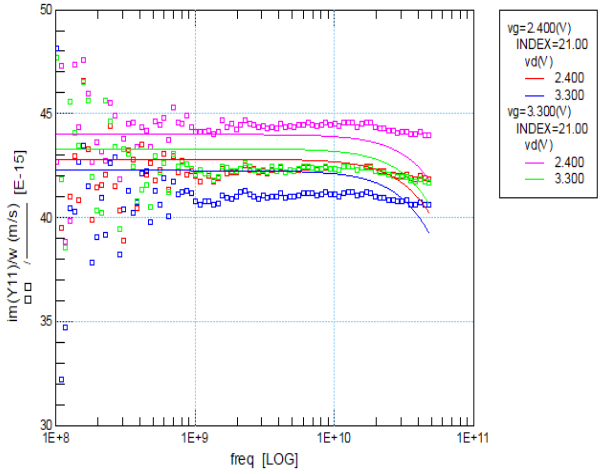
Verify Plots, T = 27°C



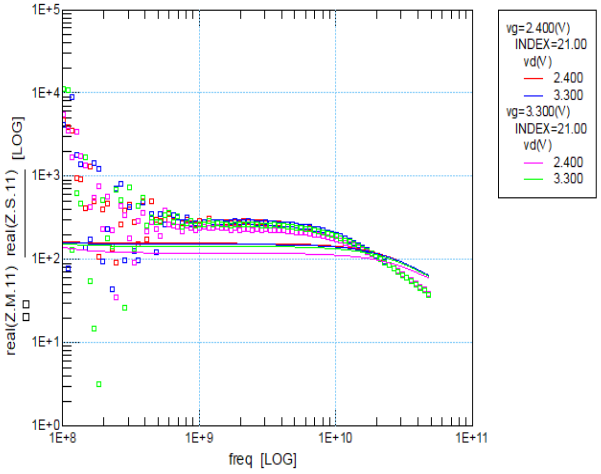
re(S11) = f(Vg,Vd)



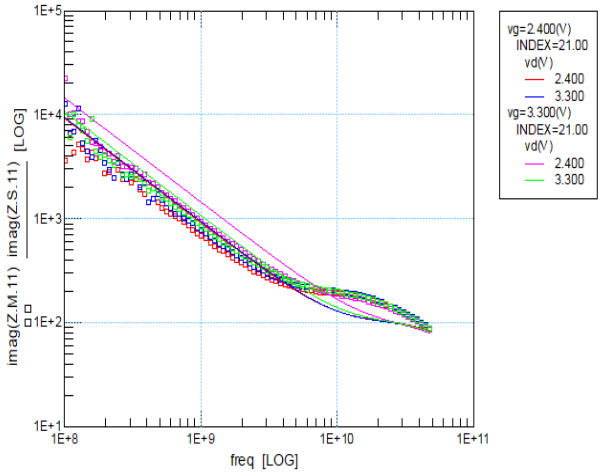
re(S22) = f(Vg,Vd)



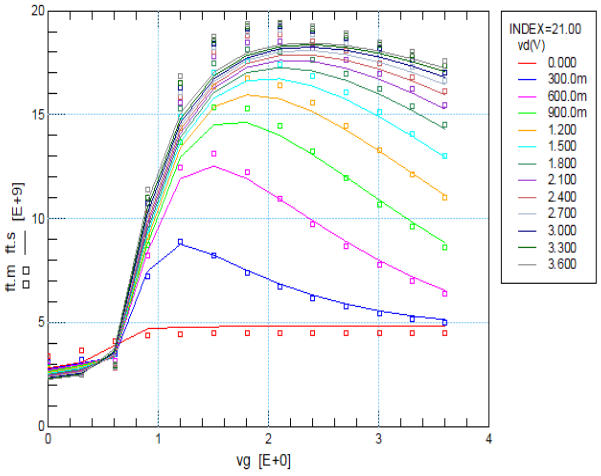
im(Y11) divided by ω



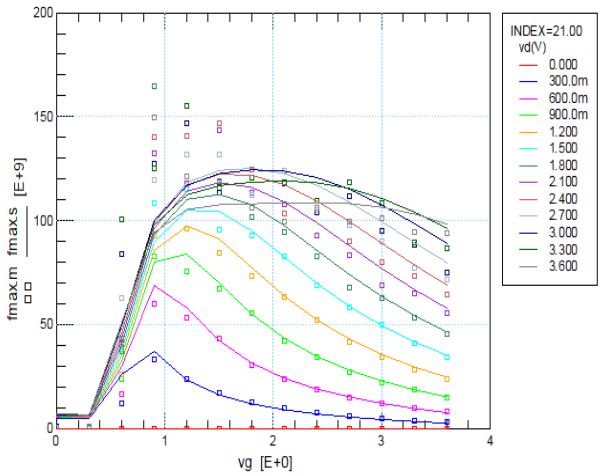
re(Z11) = f(Vg,Vd)



im(Z11) = f(Vg,Vd)



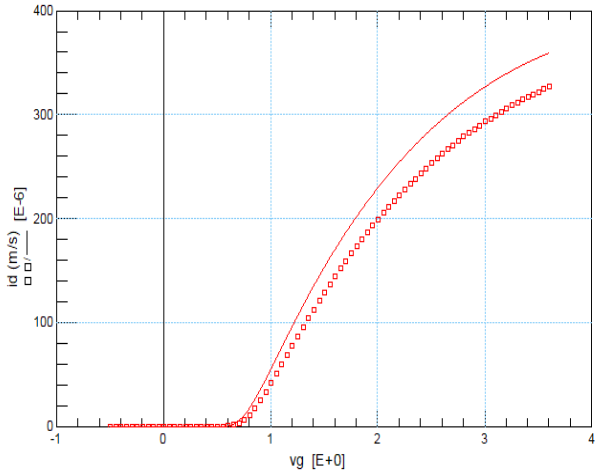
fT_vg = f(Vg,Vd)



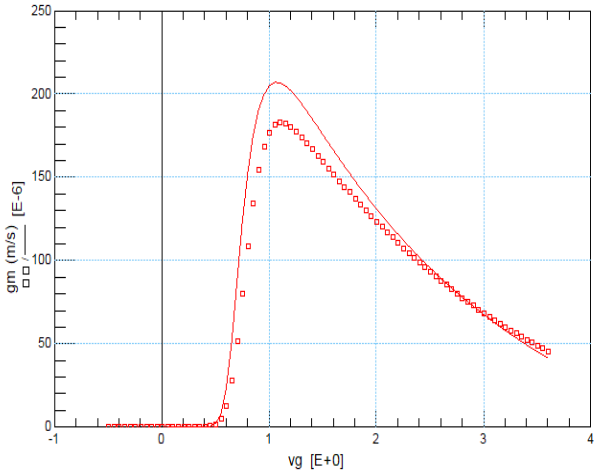
fmax_vg = f(Vg,Vd)

5.4 NF01_WF05u0_L_0u45_S453A_C1

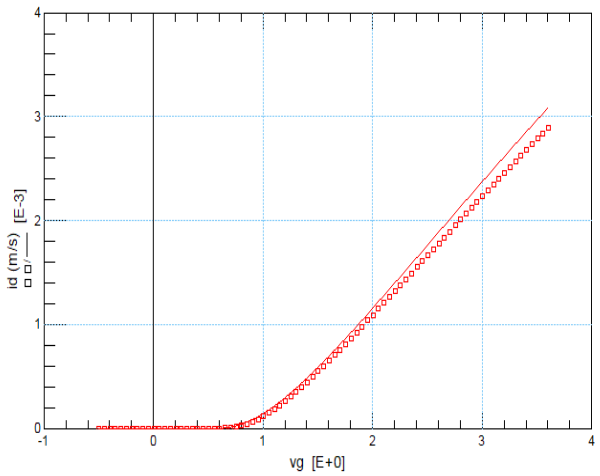
IdVg, T = 27°C



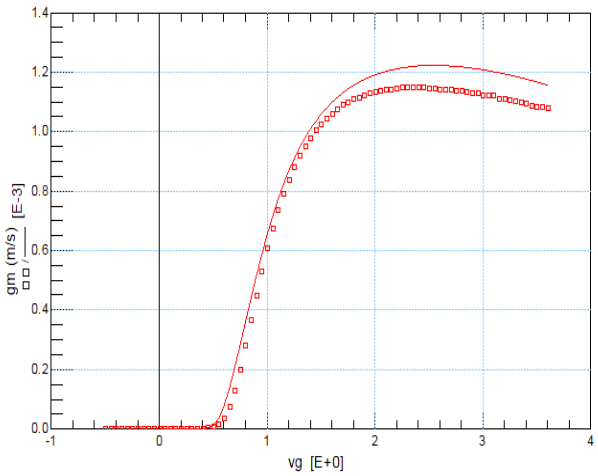
Id = f(Vg) @ Vd = 0.1V



gm = f(Vg) @ Vd = 0.1V

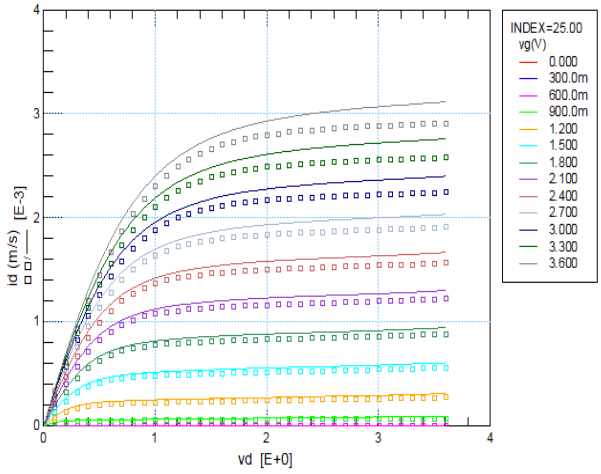


Id = f(Vg) @ Vd = 3.3V

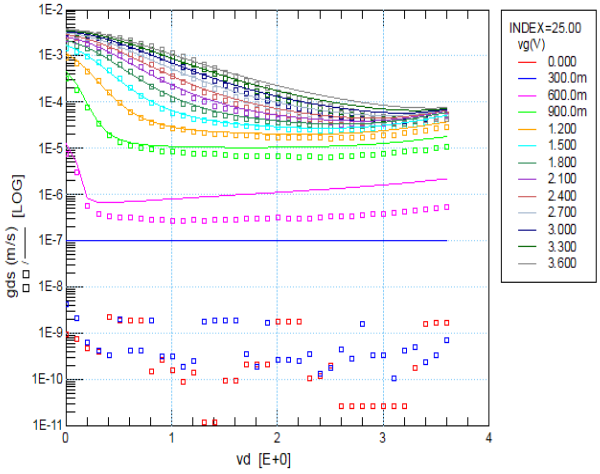


gm = f(Vg) @ Vd = 3.3V

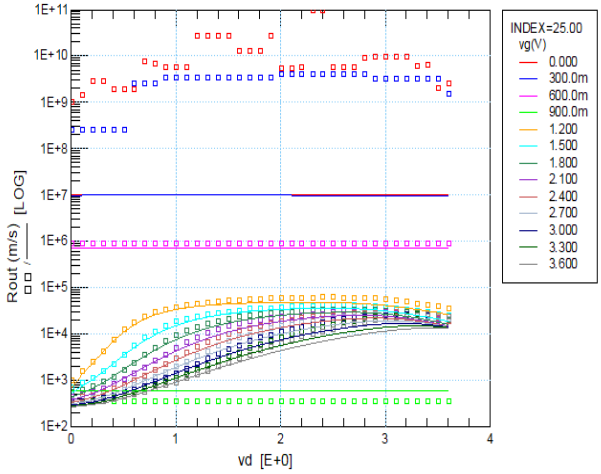
IdVd, T = 27°C



Id = f(Vd)

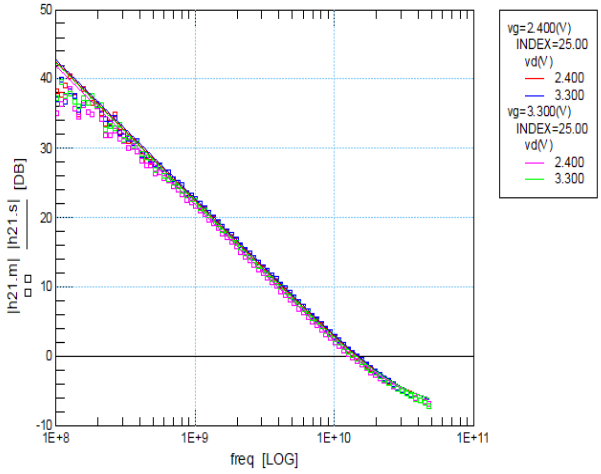


gds = f(Vd)

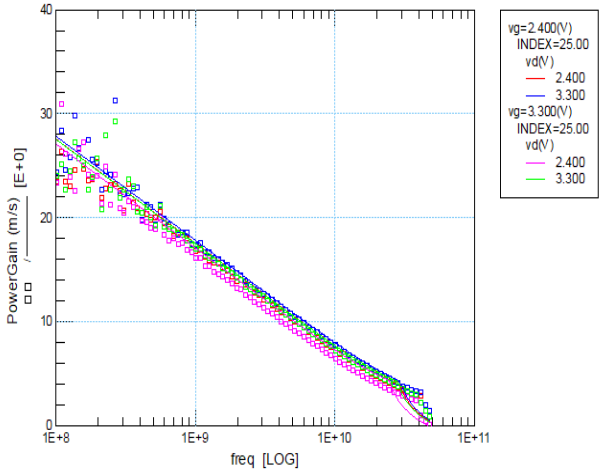


Rout = f(Vd)

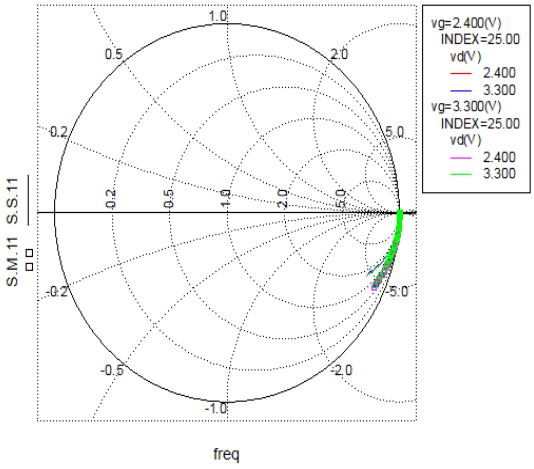
S-Parameter, T = 27°C



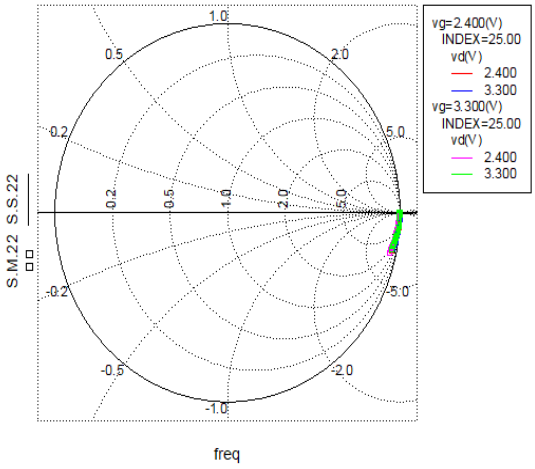
|h21| = f(Vg,Vd)



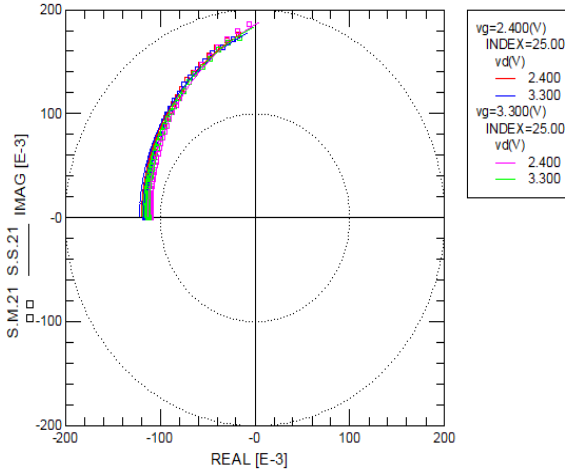
MSG/MAG = f(Vg,Vd)



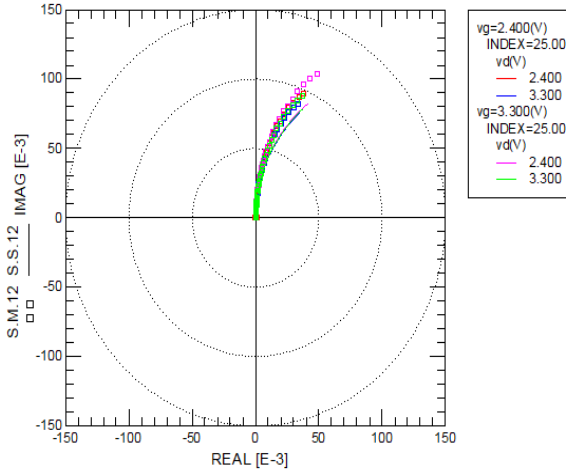
S11 = f(Vg,Vd)



S22 = f(Vg,Vd)

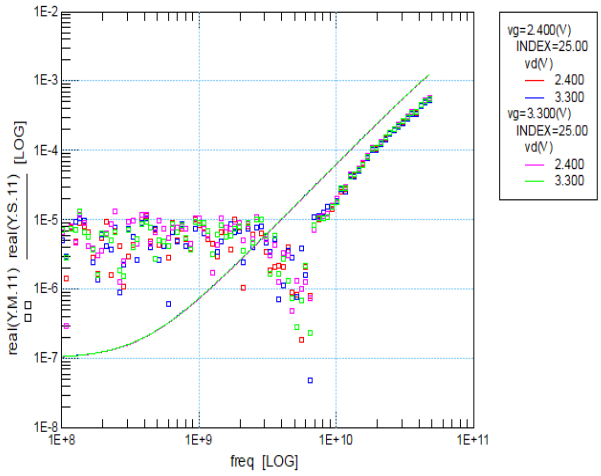


S21 = f(Vg,Vd)

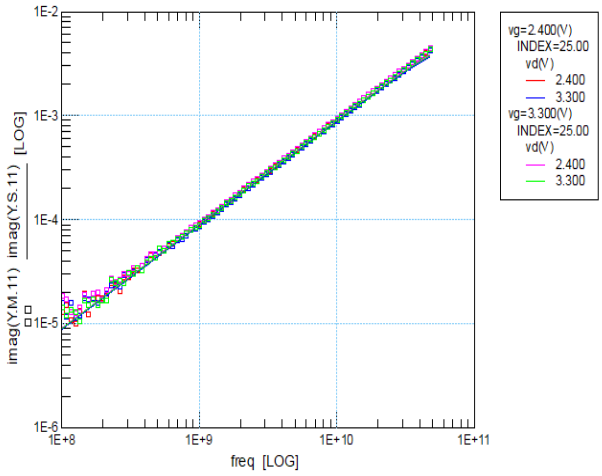


S12 = f(Vg,Vd)

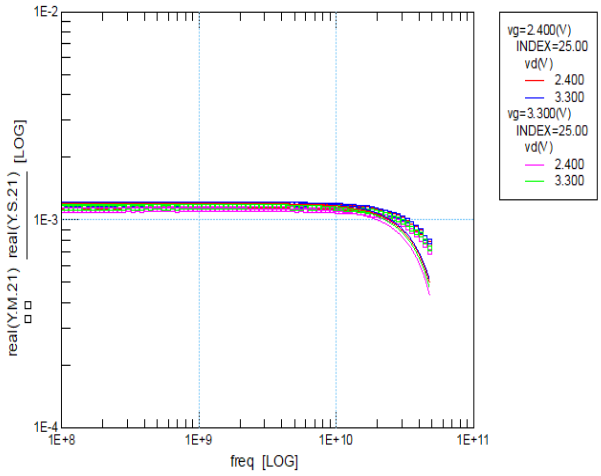
Y-Parameter, T = 27°C



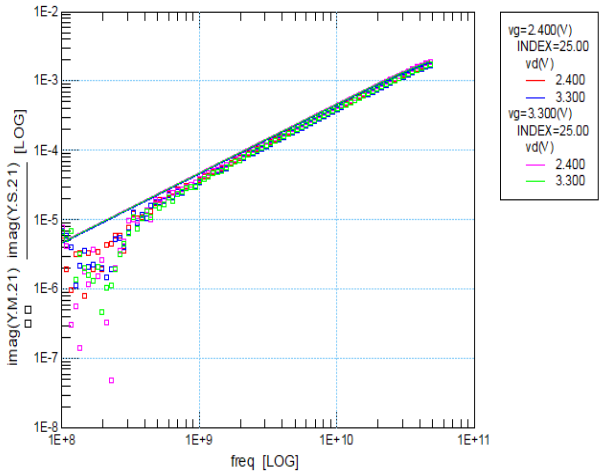
re(Y11) = f(Vg,Vd)



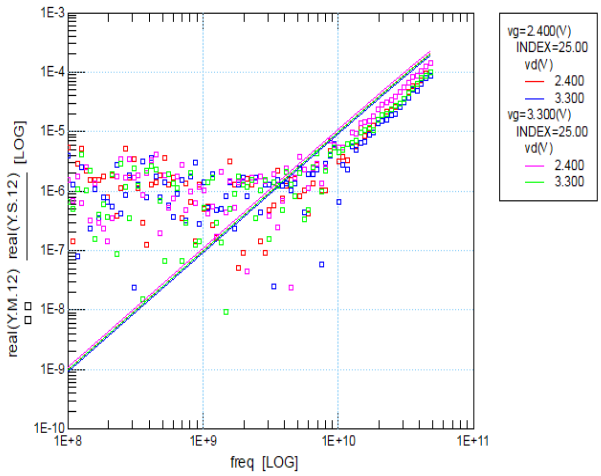
im(Y11) = f(Vg,Vd)



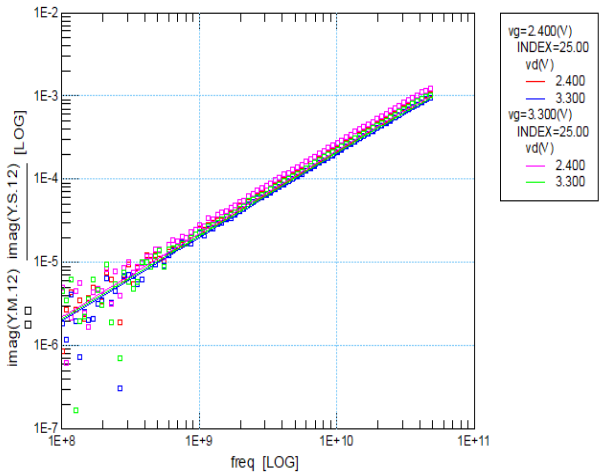
re(Y21) = f(Vg,Vd)



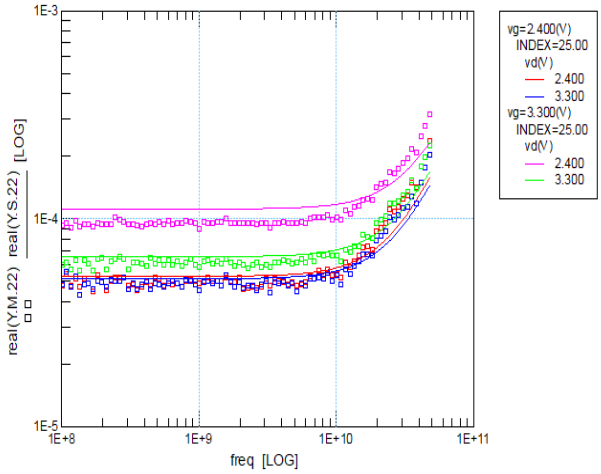
im(Y21) = f(Vg,Vd)



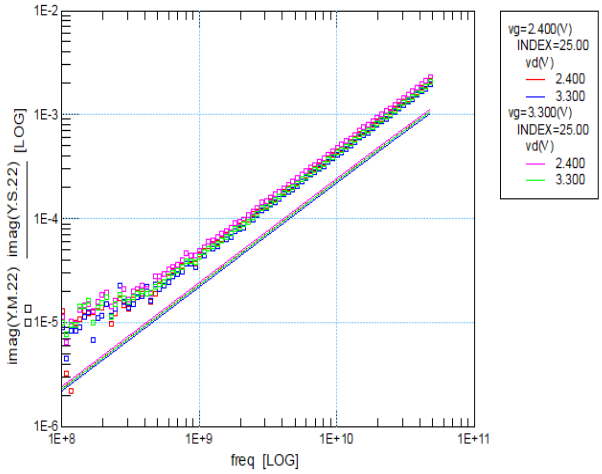
re(Y12) = f(Vg,Vd)



im(Y12) = f(Vg,Vd)

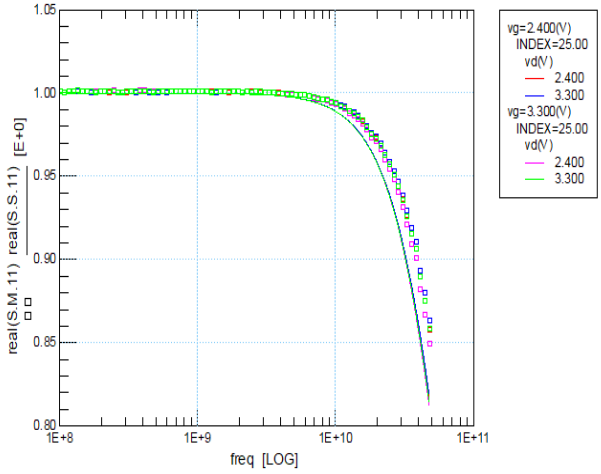


re(Y22) = f(Vg,Vd)

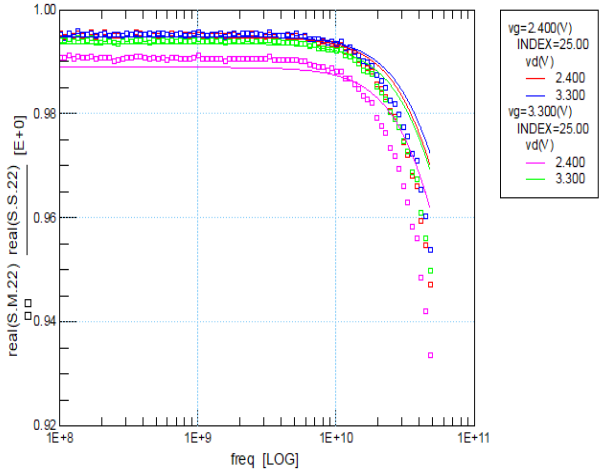


im(Y22) = f(Vg,Vd)

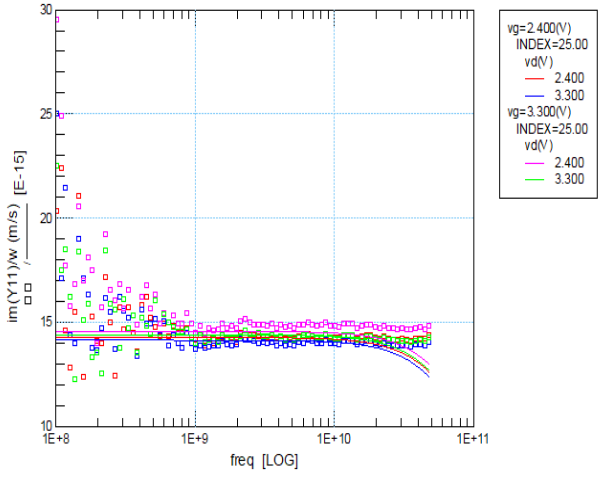
Verify Plots, T = 27°C



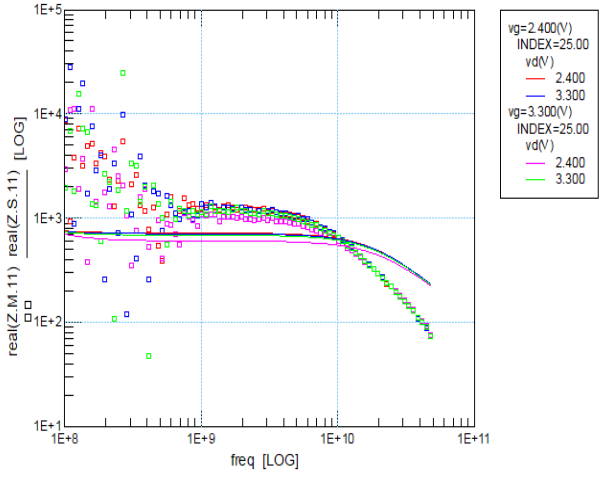
re(S11) = f(Vg,Vd)



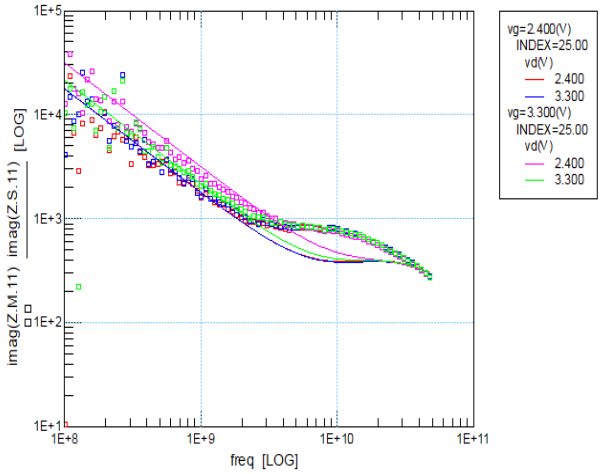
re(S22) = f(Vg,Vd)



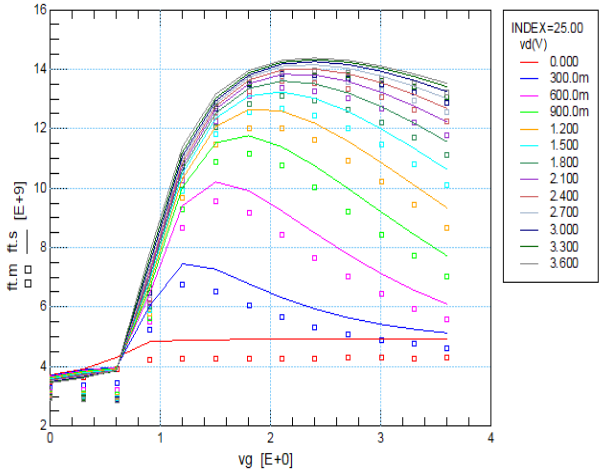
im(Y11) divided by ω



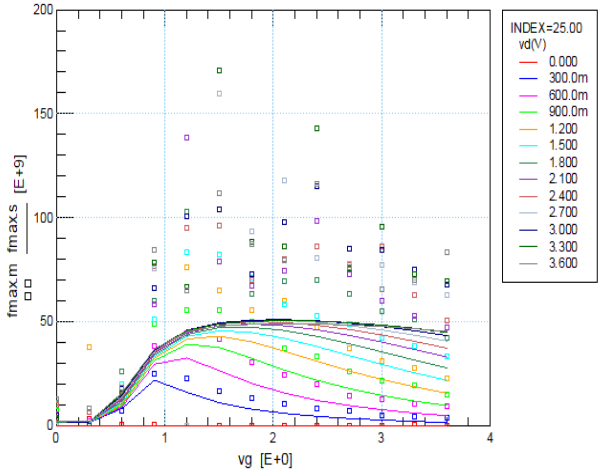
re(Z11) = f(Vg,Vd)



im(Z11) = f(Vg,Vd)



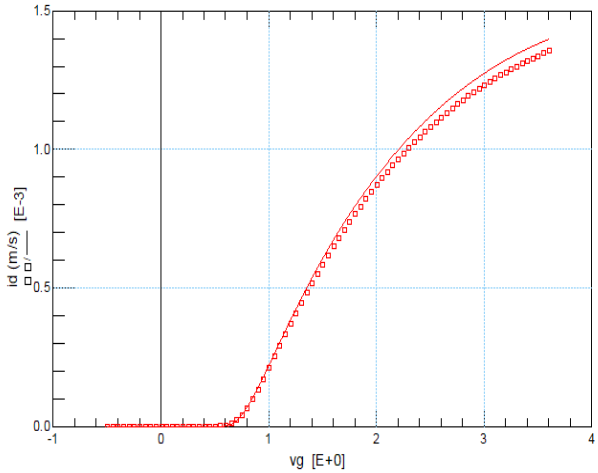
fT_vg = f(Vg,Vd)



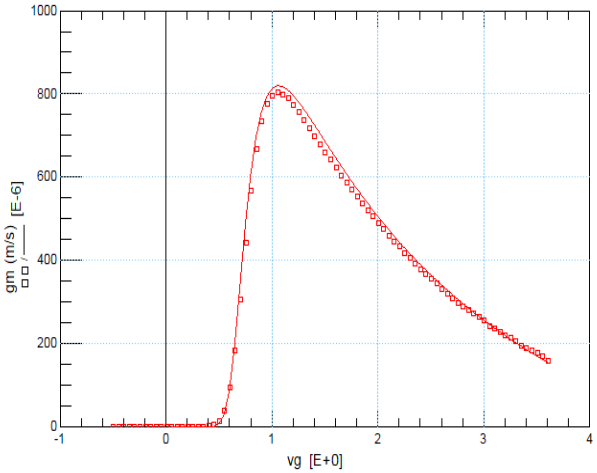
fmax_vg = f(Vg,Vd)

5.5 NF04_WF05u0_L_0u45_S453A_83

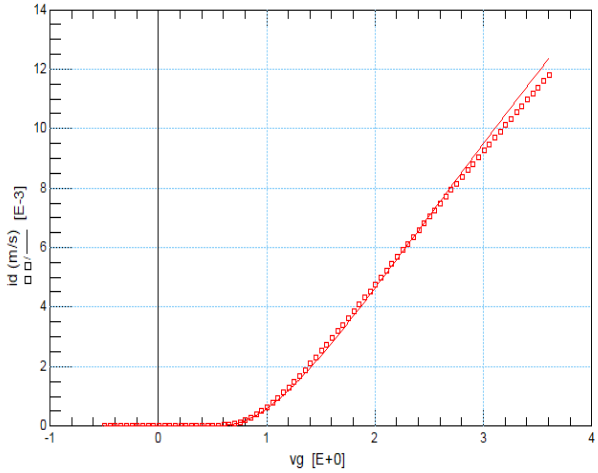
IdVg, T = 27°C



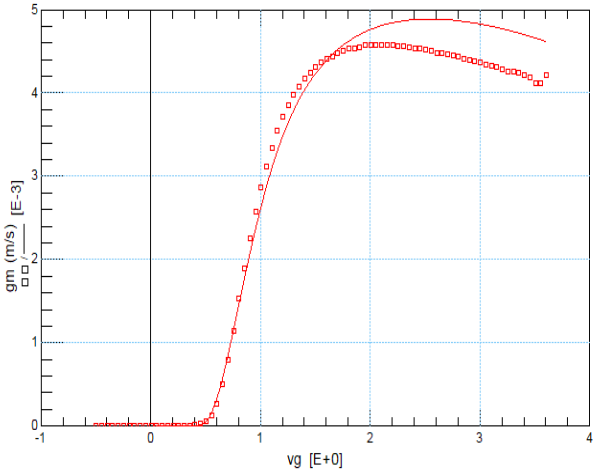
Id = f(Vg) @ Vd = 0.1V



gm = f(Vg) @ Vd = 0.1V

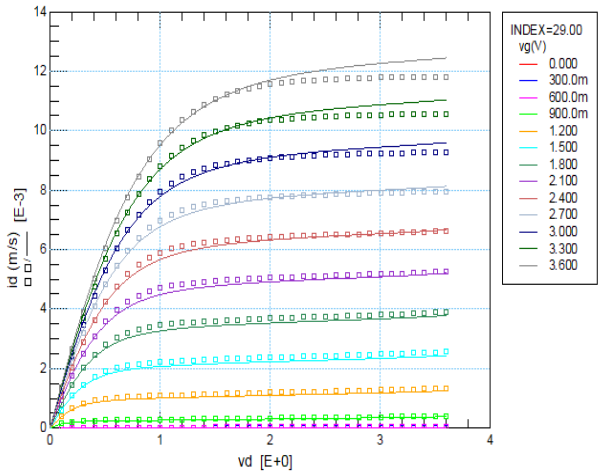


Id = f(Vg) @ Vd = 3.3V

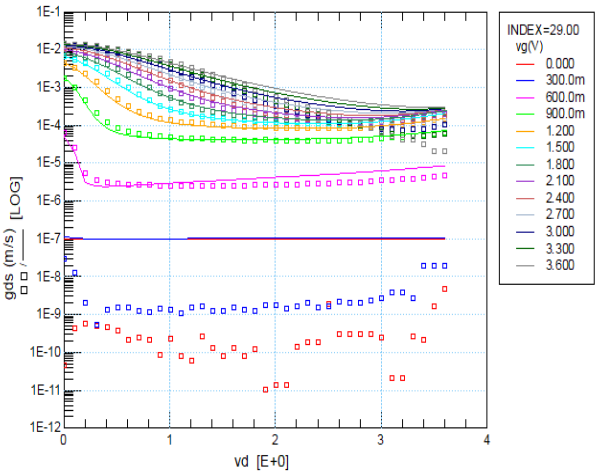


gm = f(Vg) @ Vd = 3.3V

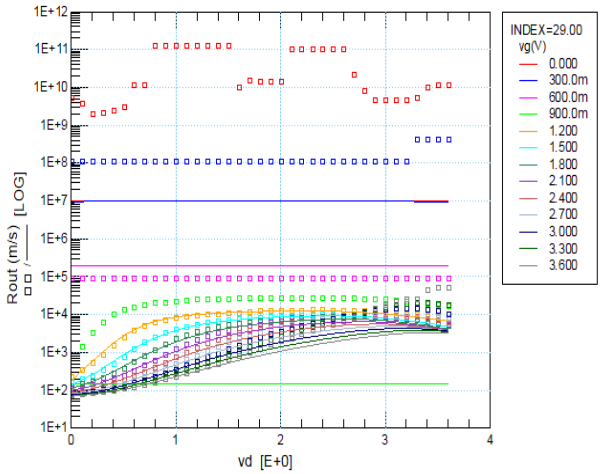
IdVd, T = 27°C



Id = f(Vd)

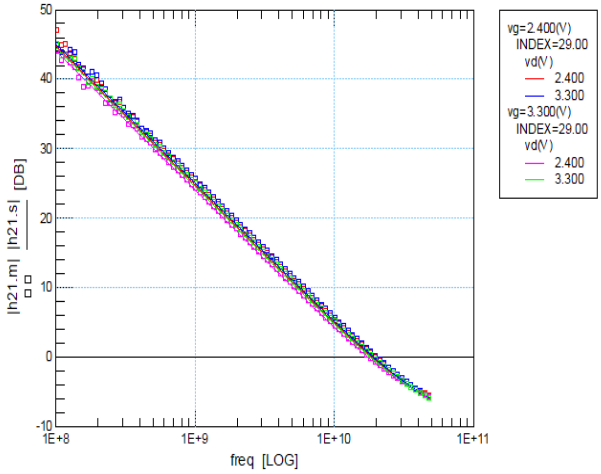


gds = f(Vd)

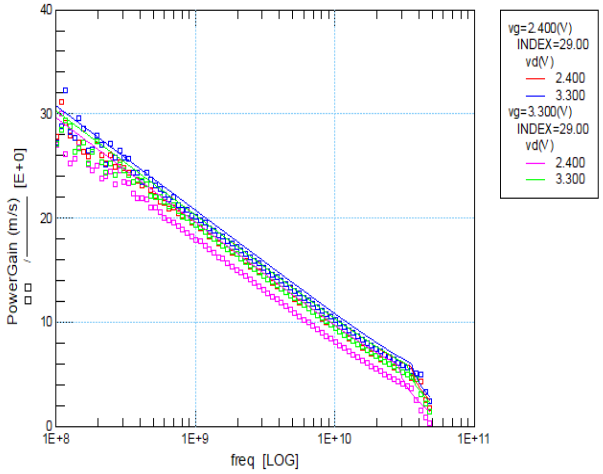


Rout = f(Vd)

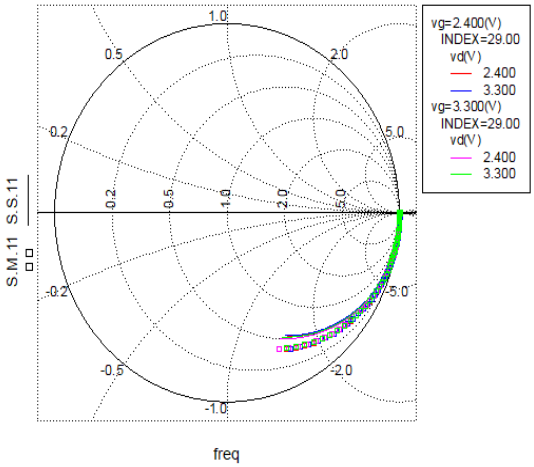
S-Parameter, T = 27°C



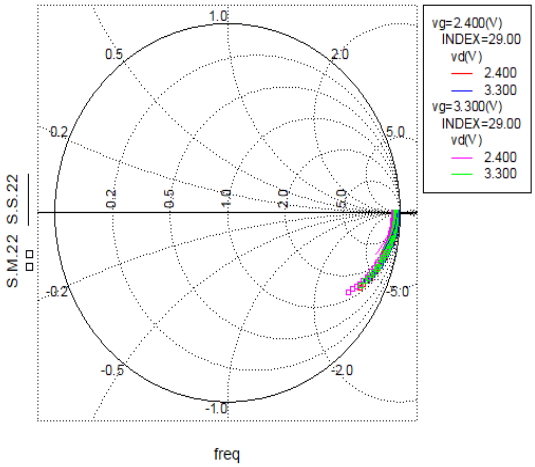
|h21| = f(Vg,Vd)



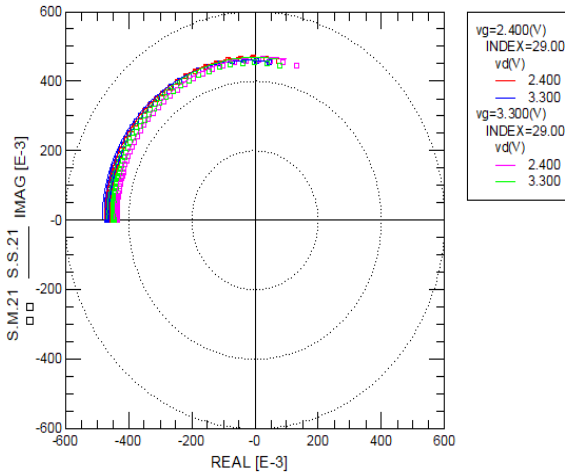
MSG/MAG = f(Vg,Vd)



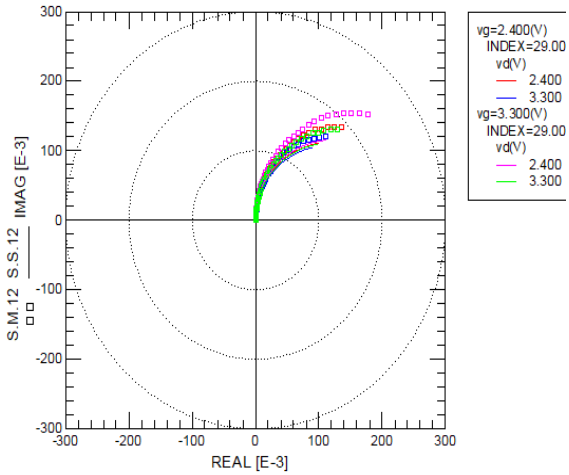
S11 = f(Vg,Vd)



S22 = f(Vg,Vd)

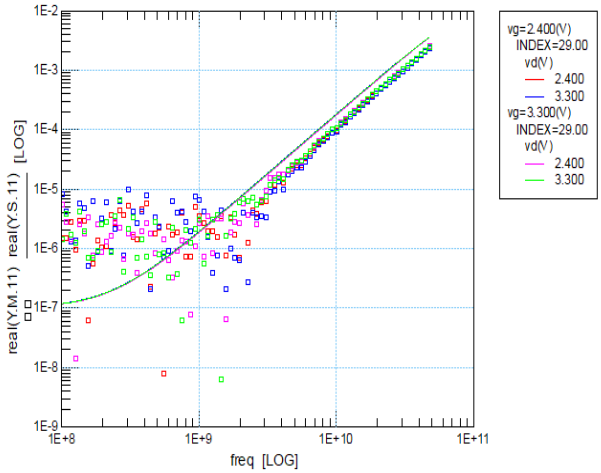


S21 = f(Vg,Vd)

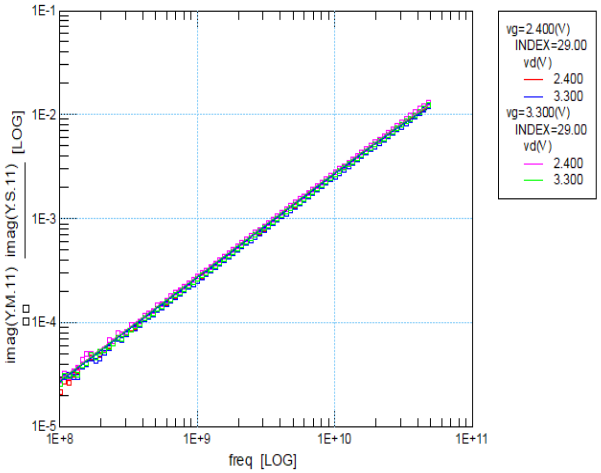


S12 = f(Vg,Vd)

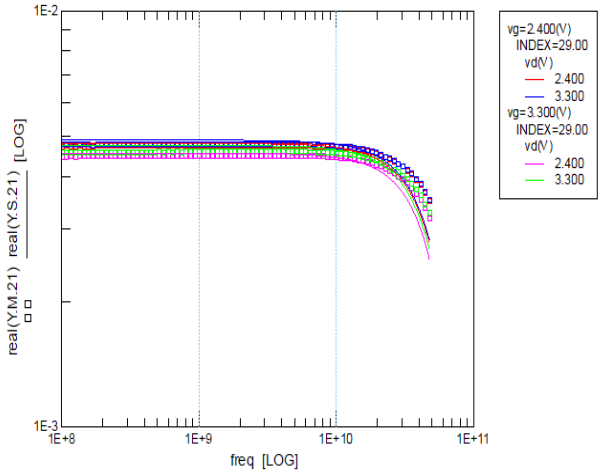
Y-Parameter, T = 27°C



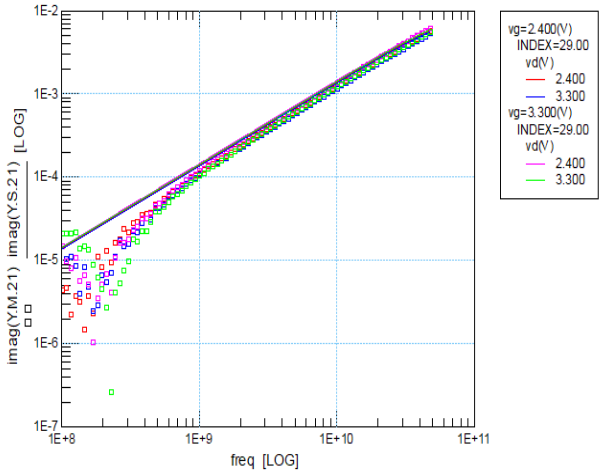
re(Y11) = f(Vg,Vd)



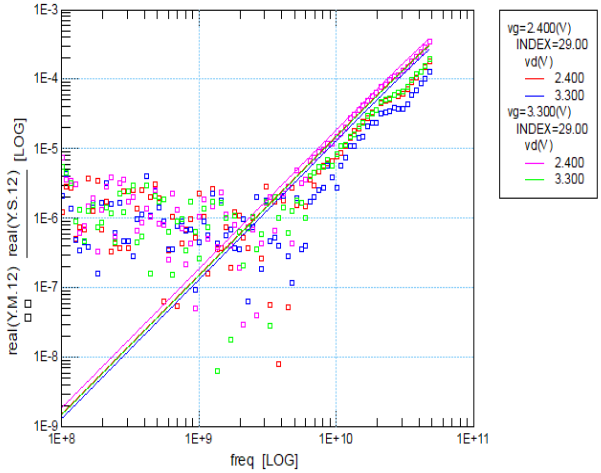
im(Y11) = f(Vg,Vd)



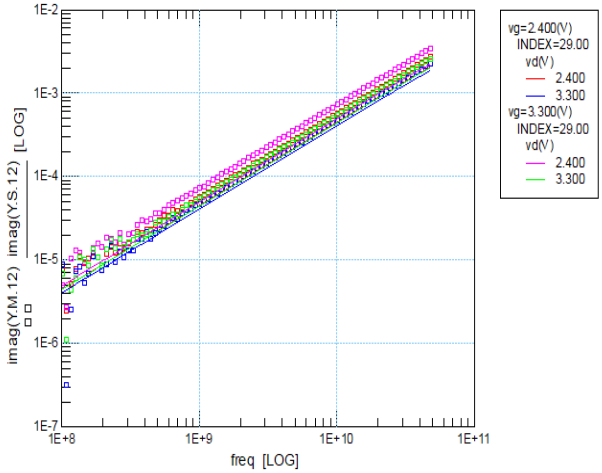
re(Y21) = f(Vg,Vd)



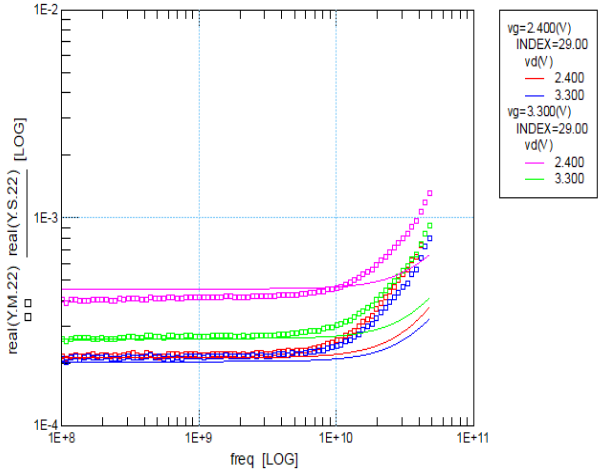
im(Y21) = f(Vg,Vd)



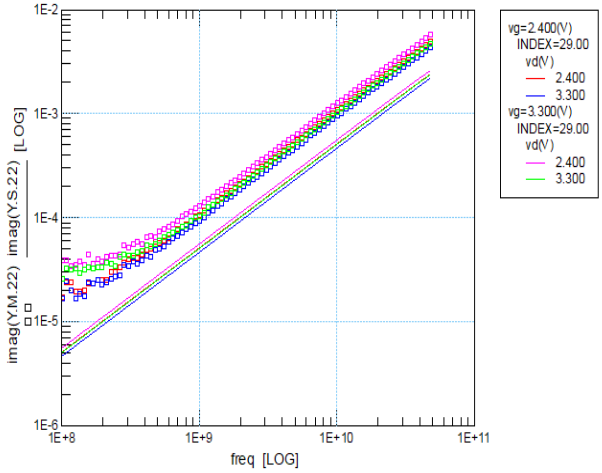
re(Y12) = f(Vg,Vd)



im(Y12) = f(Vg,Vd)

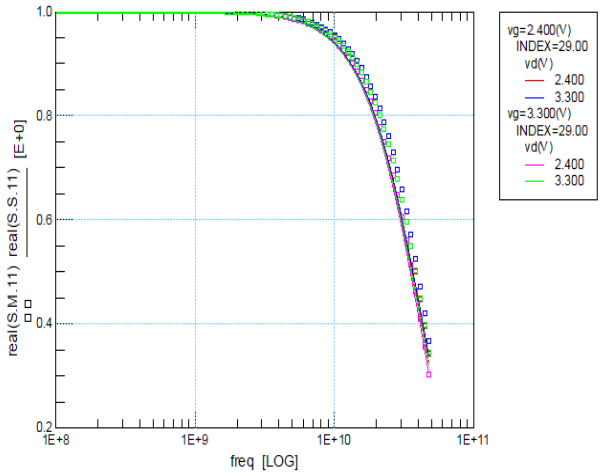


re(Y22) = f(Vg,Vd)

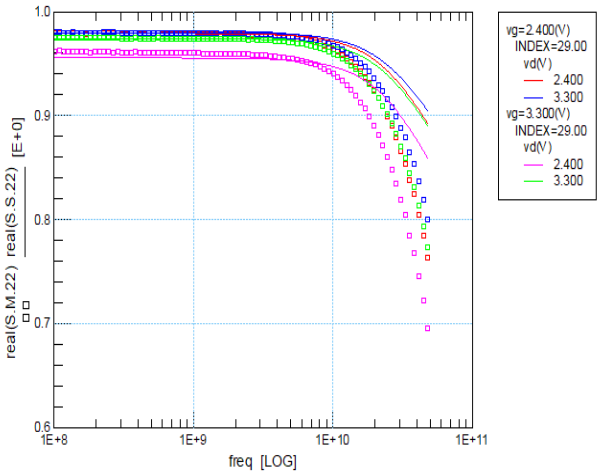


im(Y22) = f(Vg,Vd)

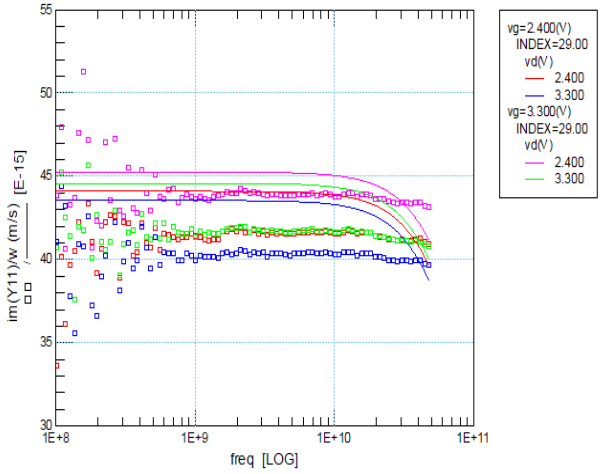
Verify Plots, T = 27°C



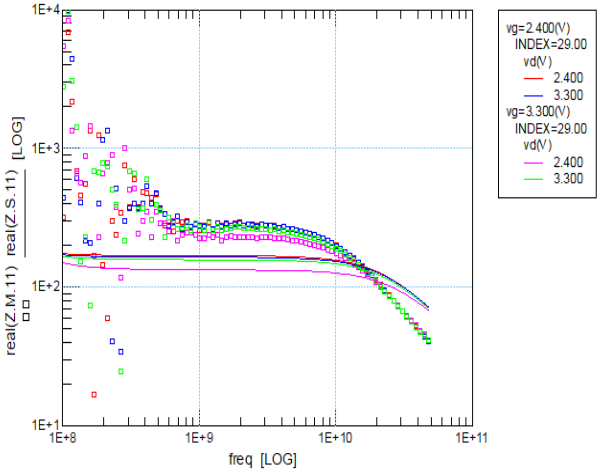
re(S11) = f(Vg,Vd)



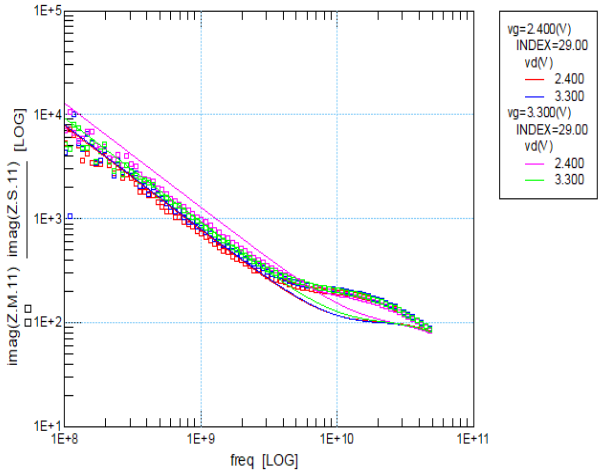
re(S22) = f(Vg,Vd)



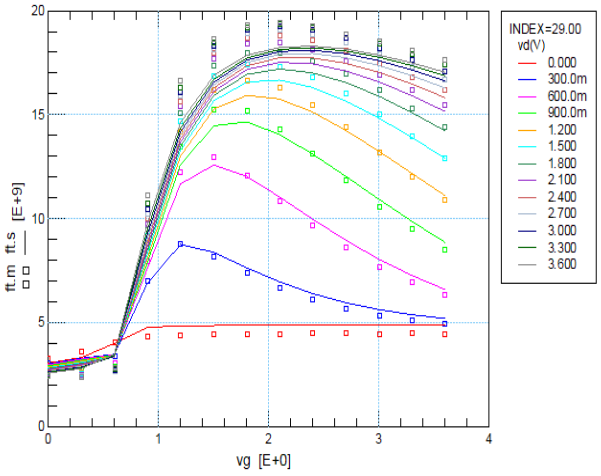
im(Y11) divided by ω



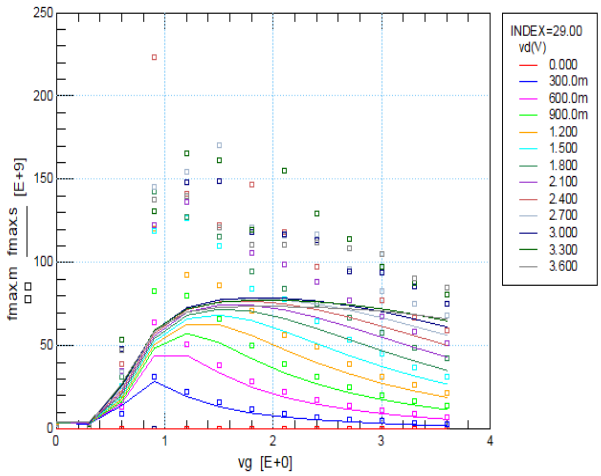
re(Z11) = f(Vg,Vd)



im(Z11) = f(Vg,Vd)



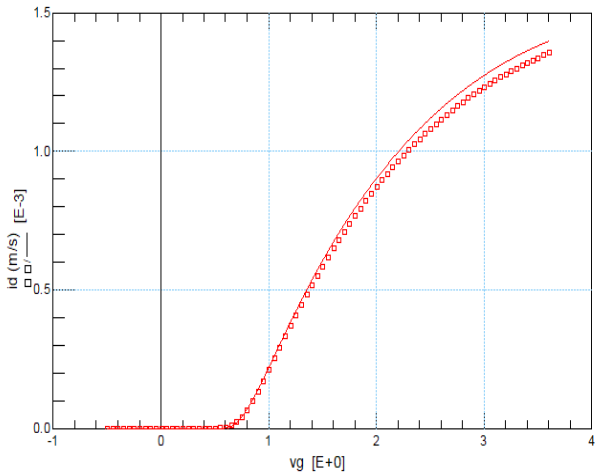
fT_vg = f(Vg,Vd)



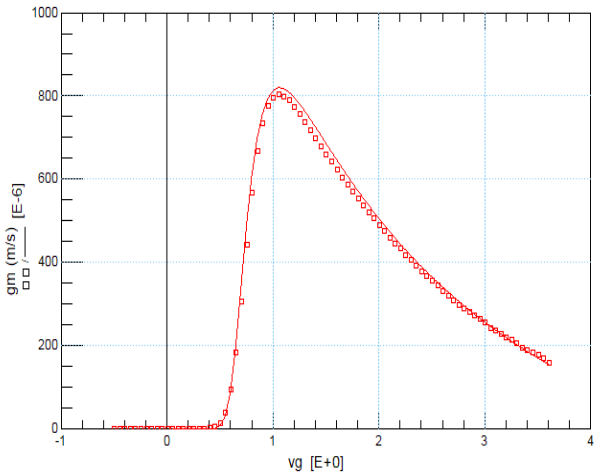
fmax_vg = f(Vg,Vd)

5.6 NF04_WF05u0_L_0u45_post_83

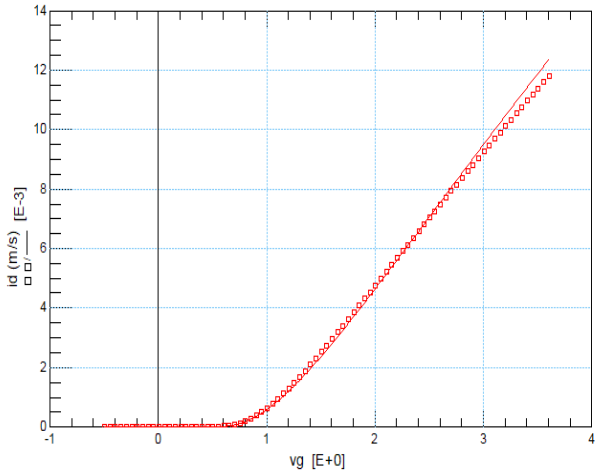
IdVg, T = 27°C



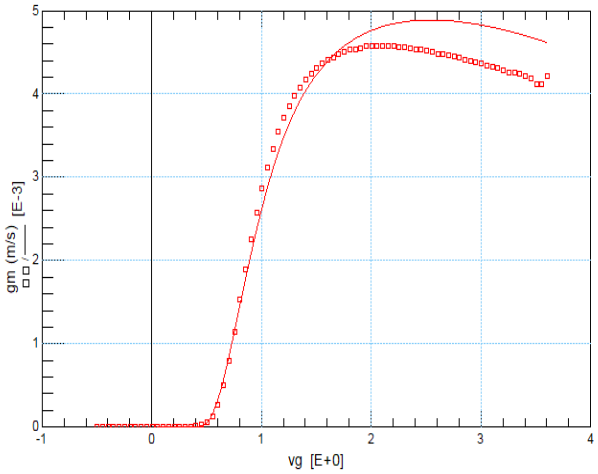
Id = f(Vg) @ Vd = 0.1V



gm = f(Vg) @ Vd = 0.1V

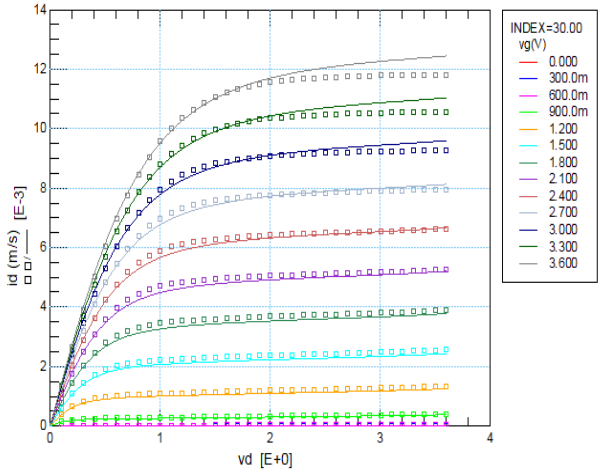


Id = f(Vg) @ Vd = 3.3V

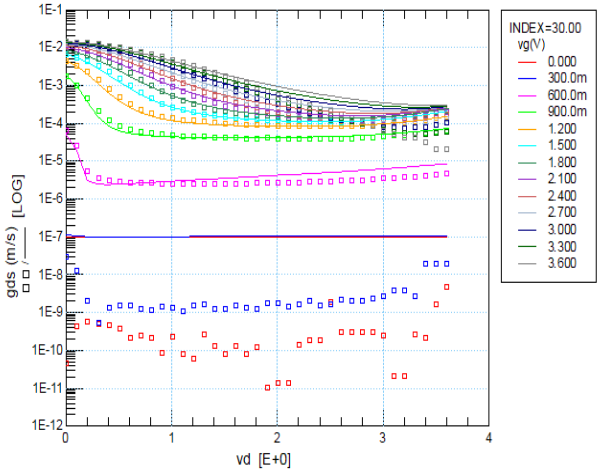


gm = f(Vg) @ Vd = 3.3V

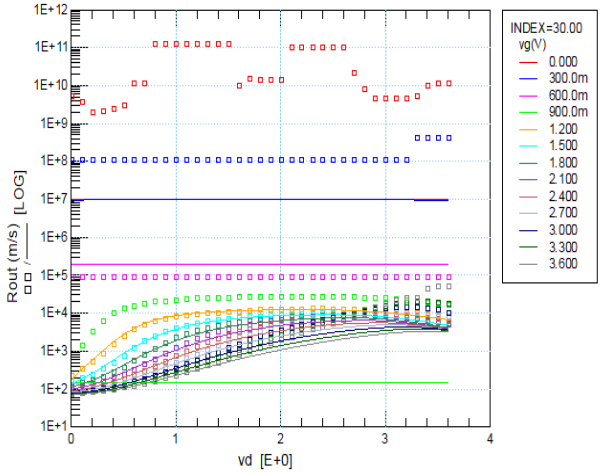
IdVd, T = 27°C



Id = f(Vd)

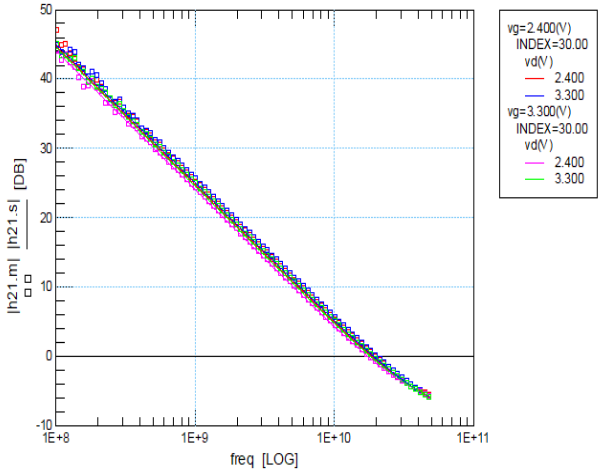


gds = f(Vd)

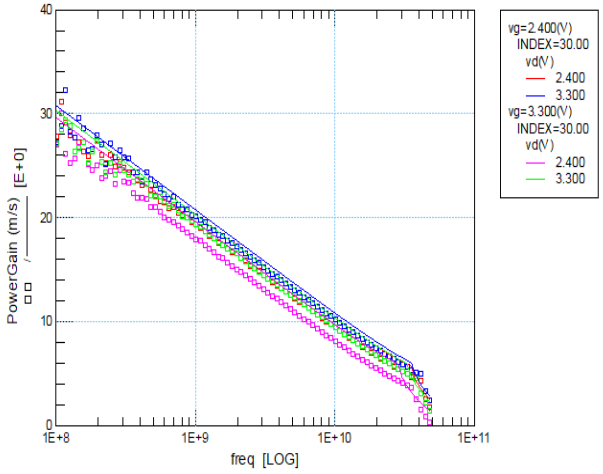


Rout = f(Vd)

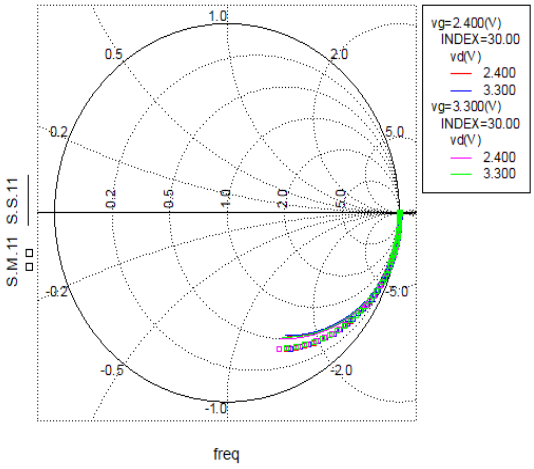
S-Parameter, T = 27°C



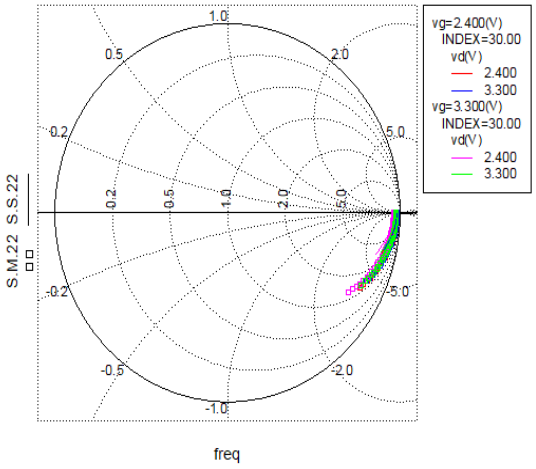
|h21| = f(Vg,Vd)



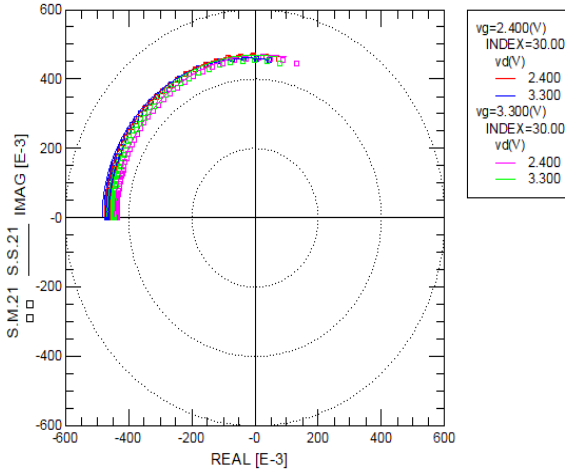
MSG/MAG = f(Vg,Vd)



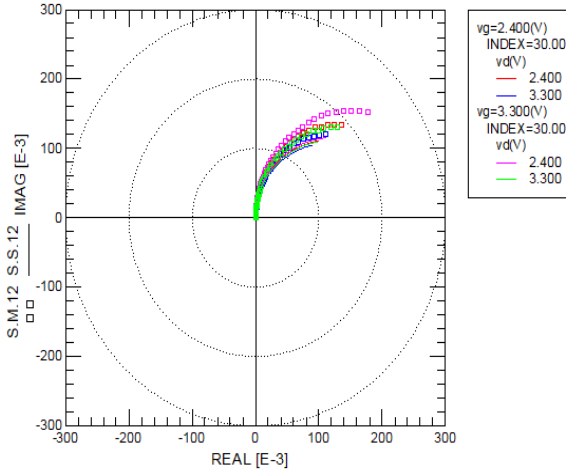
S11 = f(Vg,Vd)



S22 = f(Vg,Vd)

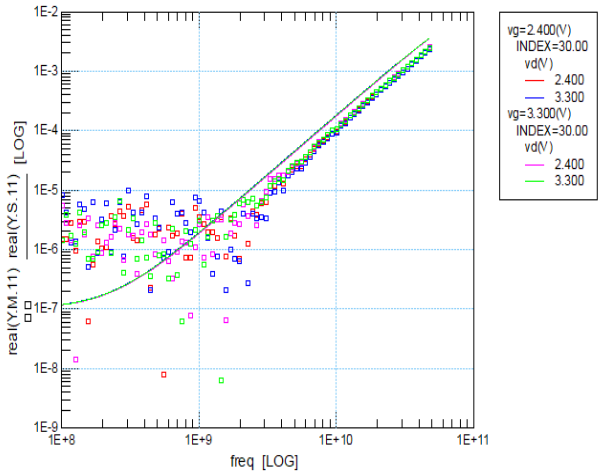


S21 = f(Vg,Vd)

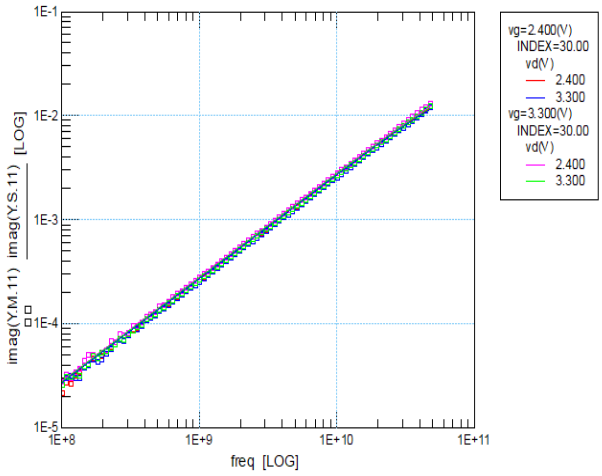


S12 = f(Vg,Vd)

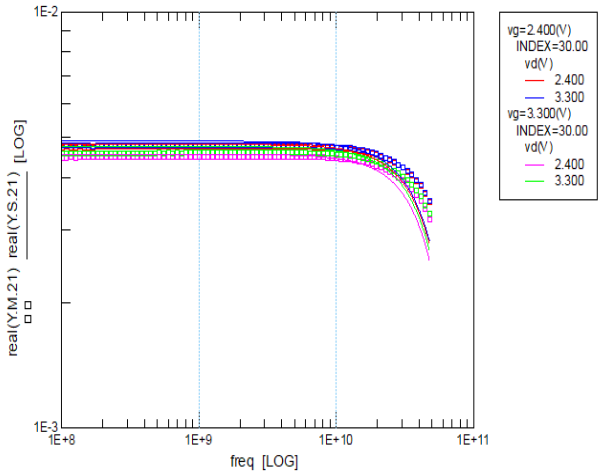
Y-Parameter, T = 27°C



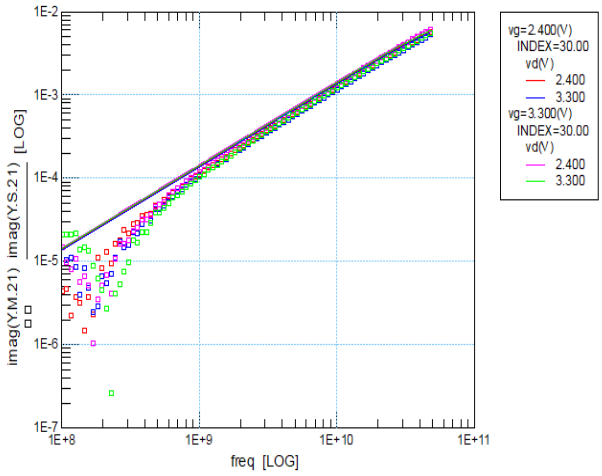
re(Y11) = f(Vg,Vd)



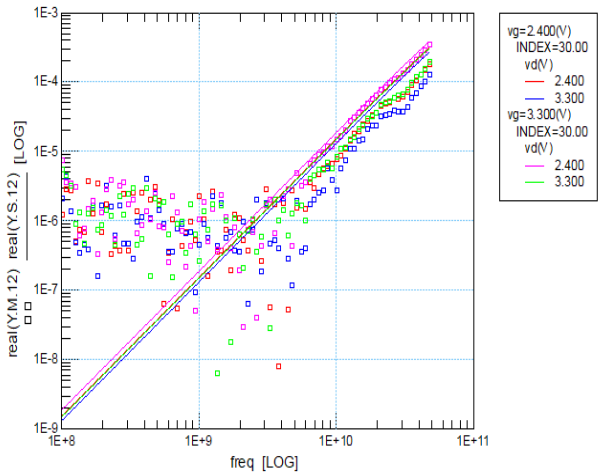
im(Y11) = f(Vg,Vd)



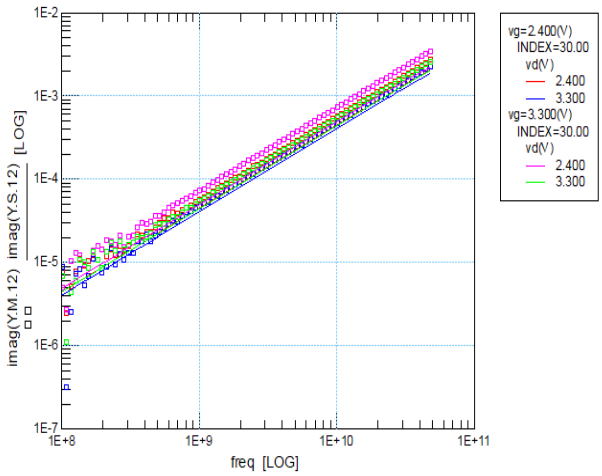
re(Y21) = f(Vg,Vd)



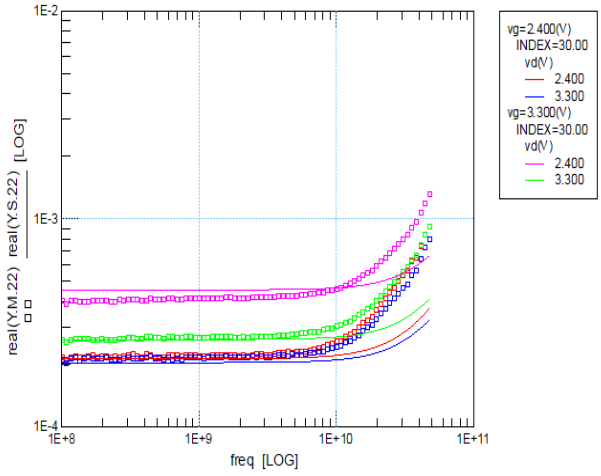
im(Y21) = f(Vg,Vd)



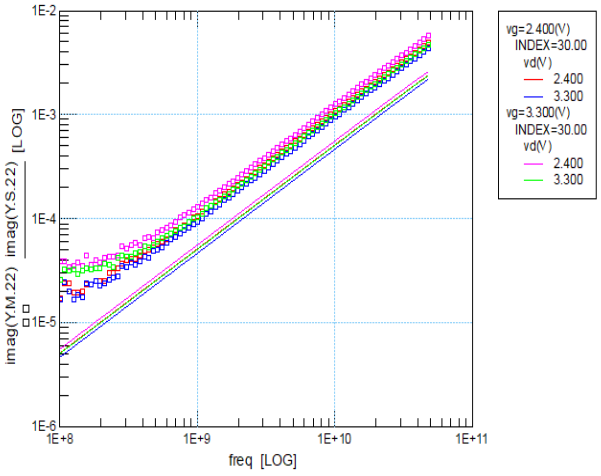
re(Y12) = f(Vg,Vd)



im(Y12) = f(Vg,Vd)

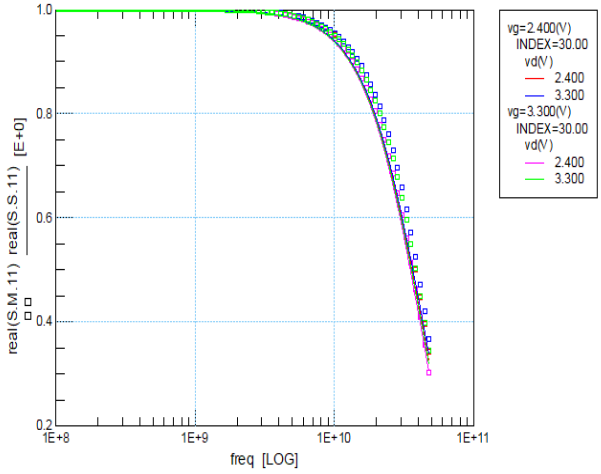


re(Y22) = f(Vg,Vd)

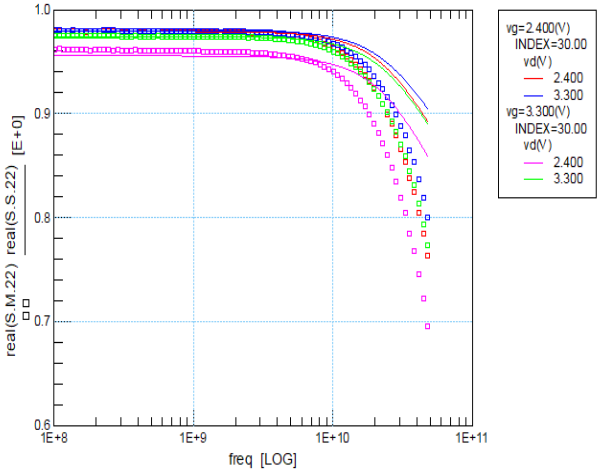


im(Y22) = f(Vg,Vd)

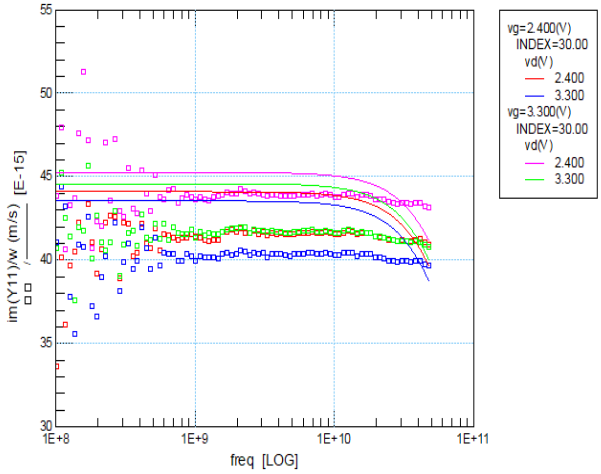
Verify Plots, T = 27°C



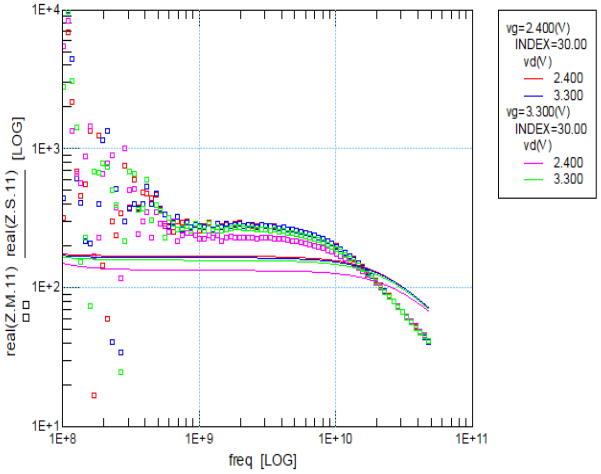
re(S11) = f(Vg,Vd)



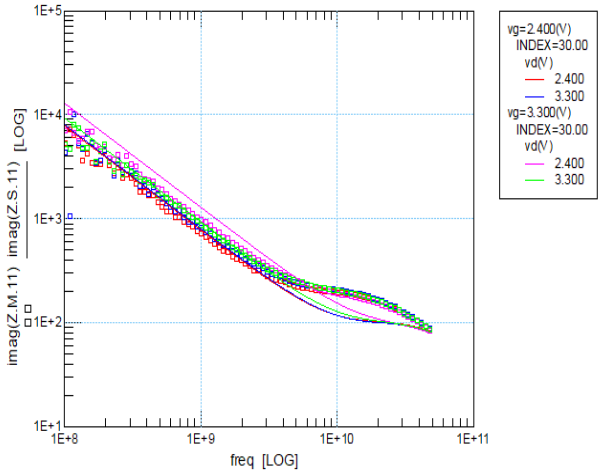
re(S22) = f(Vg,Vd)



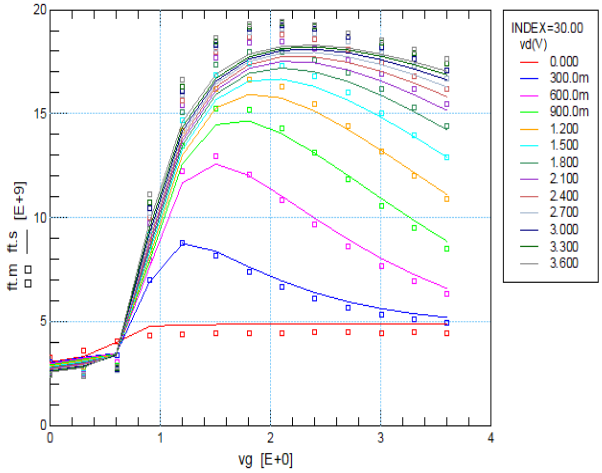
im(Y11) divided by ω



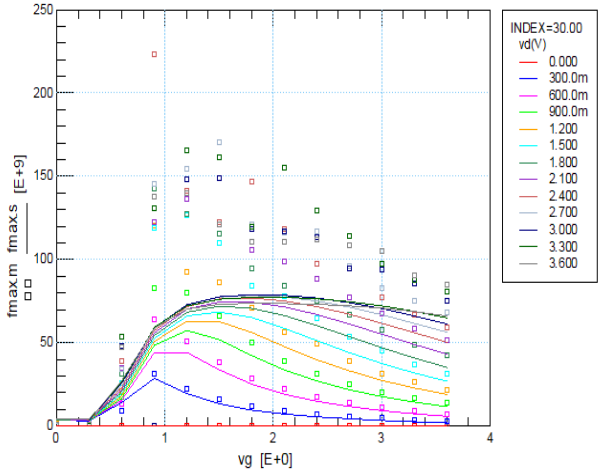
re(Z11) = f(Vg,Vd)



im(Z11) = f(Vg,Vd)



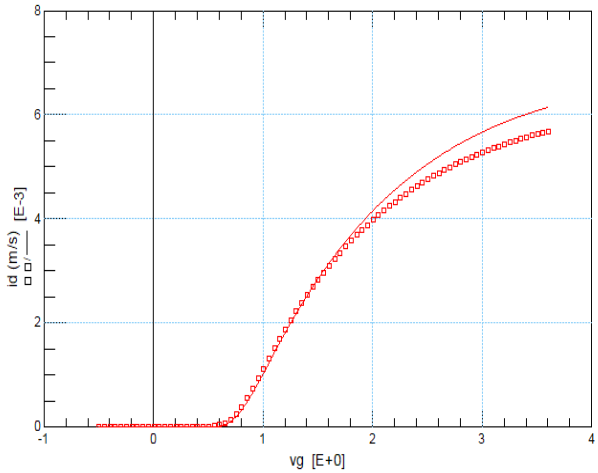
fT_vg = f(Vg,Vd)



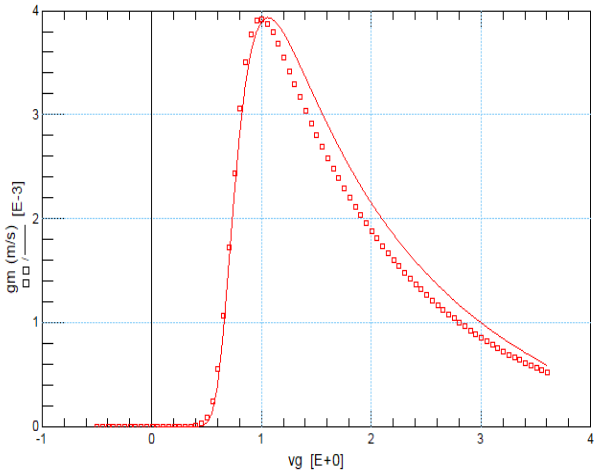
fmax_vg = f(Vg,Vd)

5.7 NF10_WF10u0_L_0u45_S453A_80

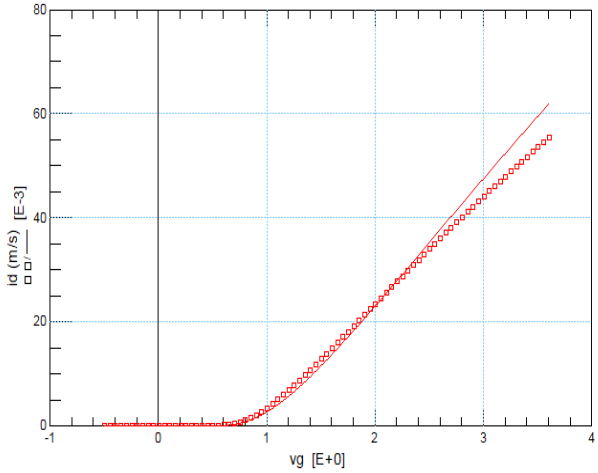
IdVg, T = 27°C



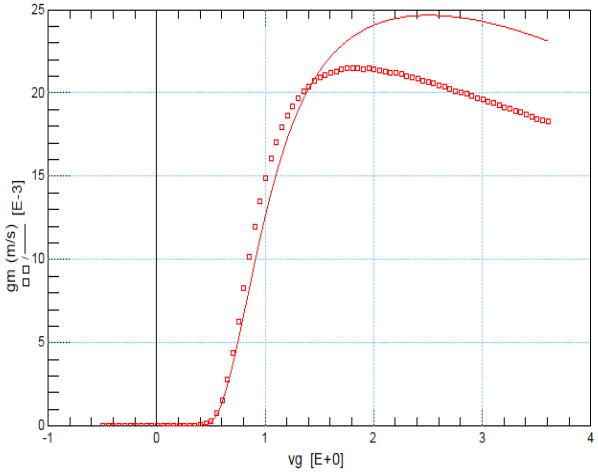
Id = f(Vg) @ Vd = 0.1V



gm = f(Vg) @ Vd = 0.1V

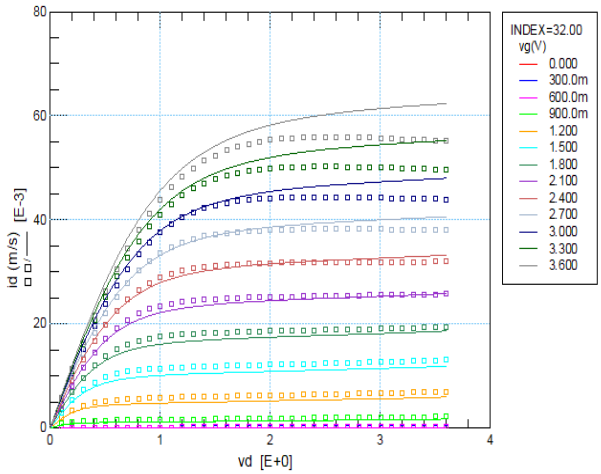


Id = f(Vg) @ Vd = 3.3V

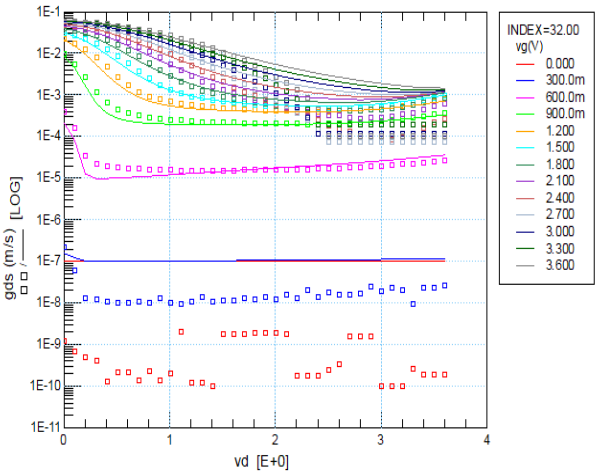


gm = f(Vg) @ Vd = 3.3V

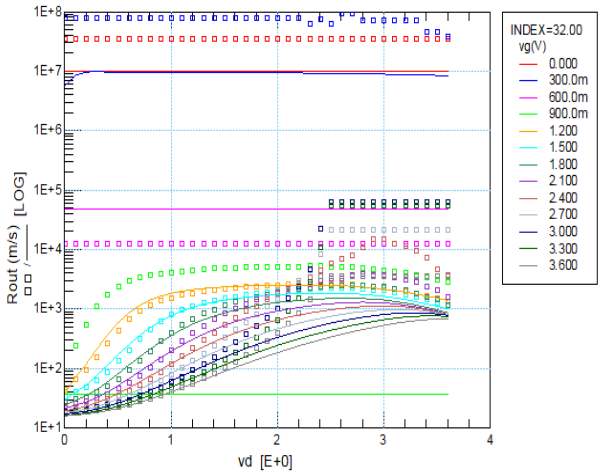
IdVd, T = 27°C



Id = f(Vd)

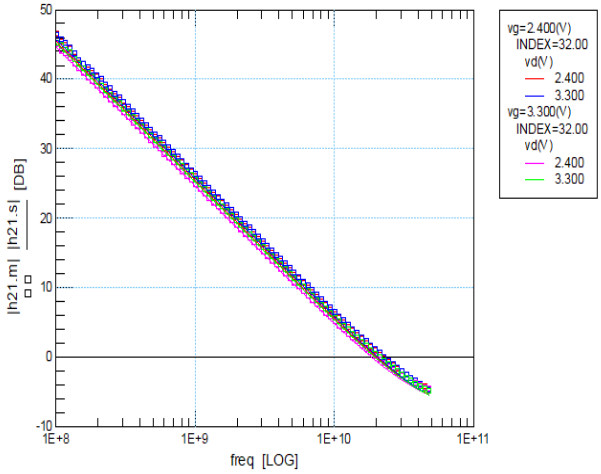


gds = f(Vd)

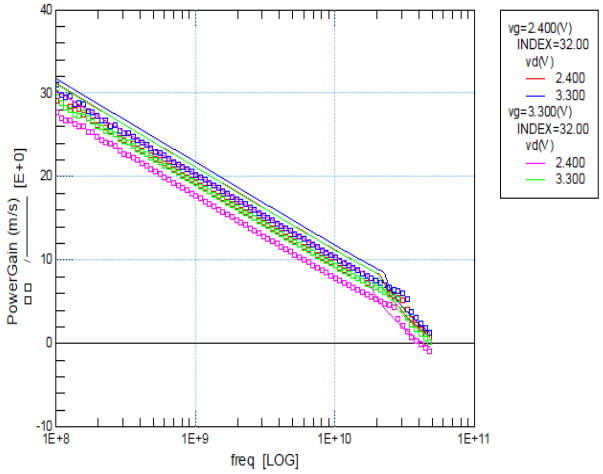


Rout = f(Vd)

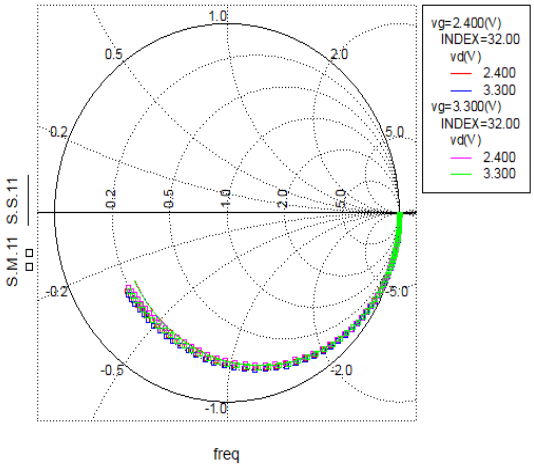
S-Parameter, T = 27°C



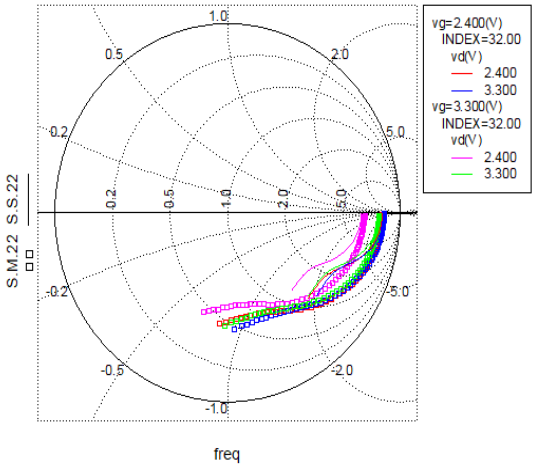
|h21| = f(Vg,Vd)



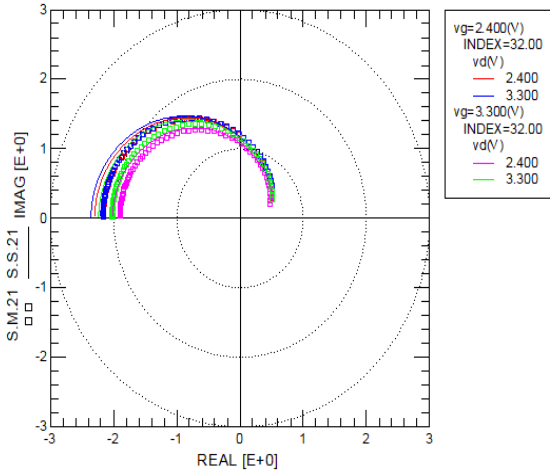
MSG/MAG = f(Vg,Vd)



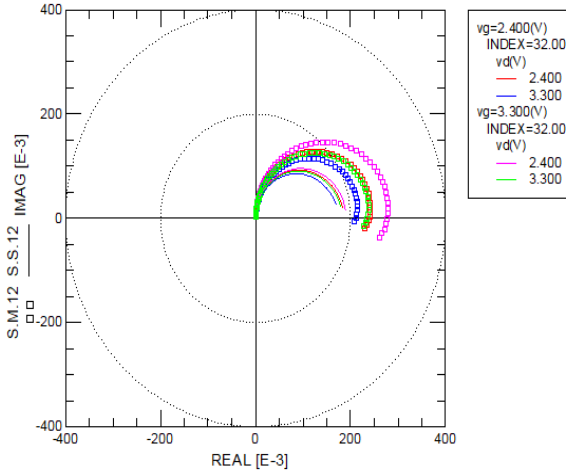
S11 = f(Vg,Vd)



S22 = f(Vg,Vd)

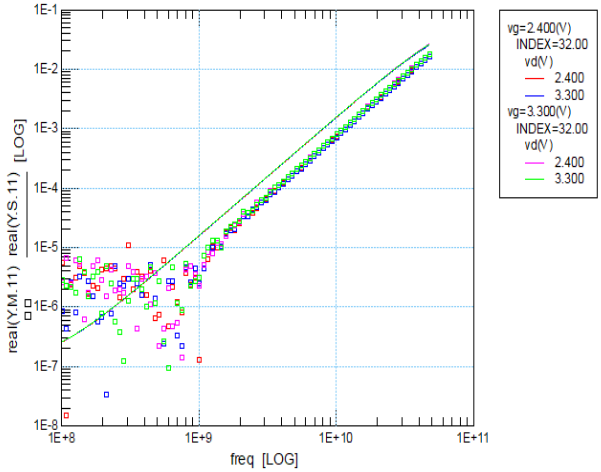


S21 = f(Vg,Vd)

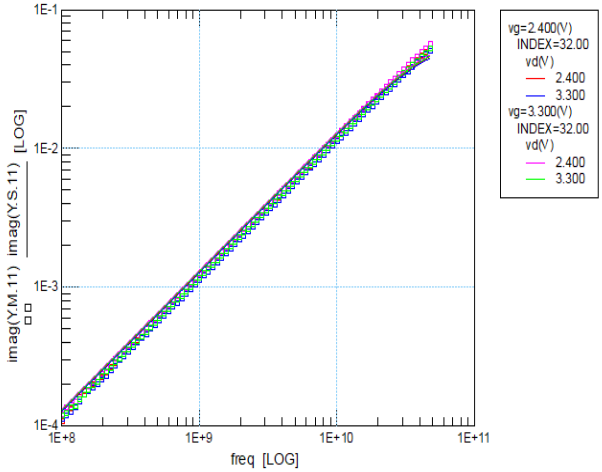


S12 = f(Vg,Vd)

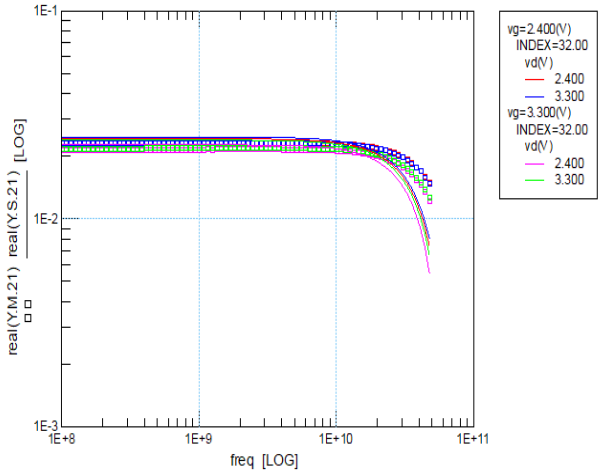
Y-Parameter, T = 27°C



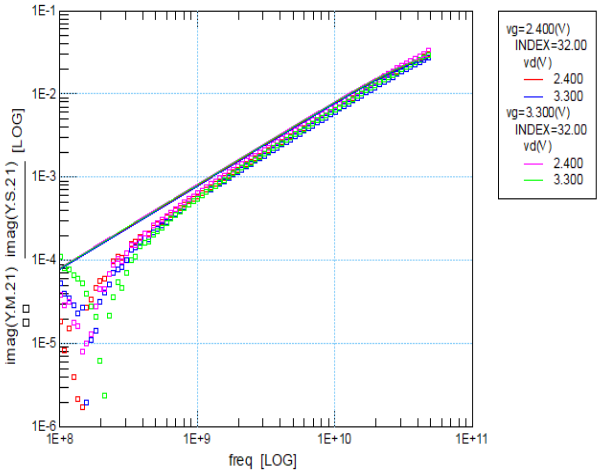
re(Y11) = f(Vg,Vd)



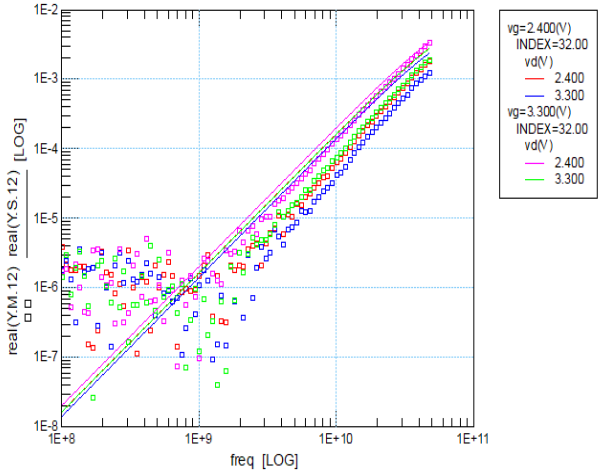
im(Y11) = f(Vg,Vd)



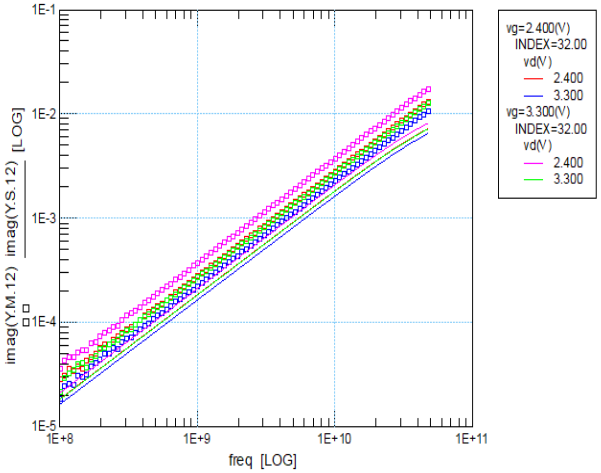
re(Y21) = f(Vg,Vd)



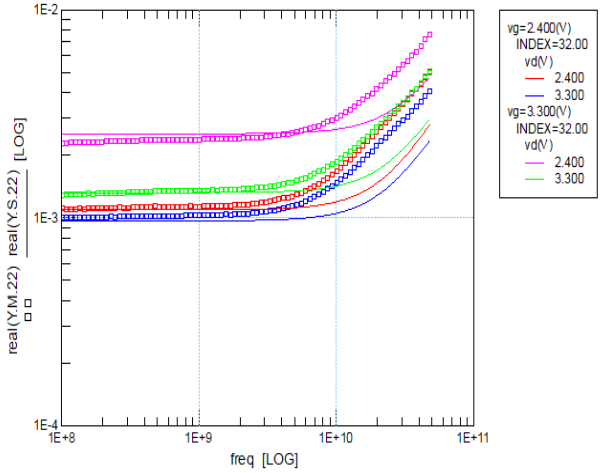
im(Y21) = f(Vg,Vd)



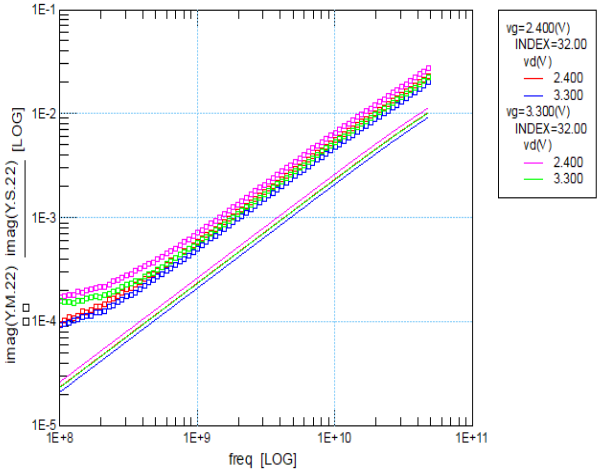
re(Y12) = f(Vg,Vd)



im(Y12) = f(Vg,Vd)

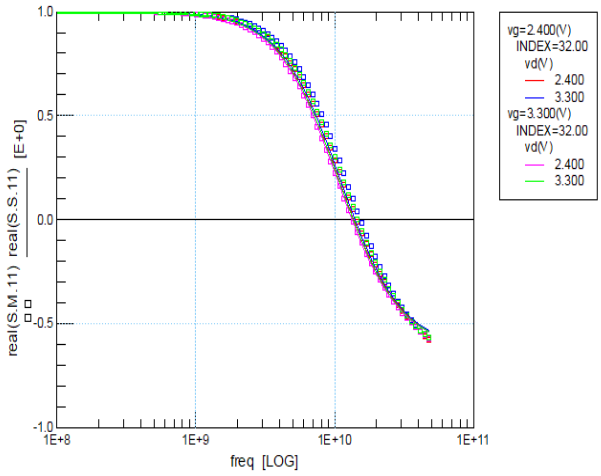


re(Y22) = f(Vg,Vd)

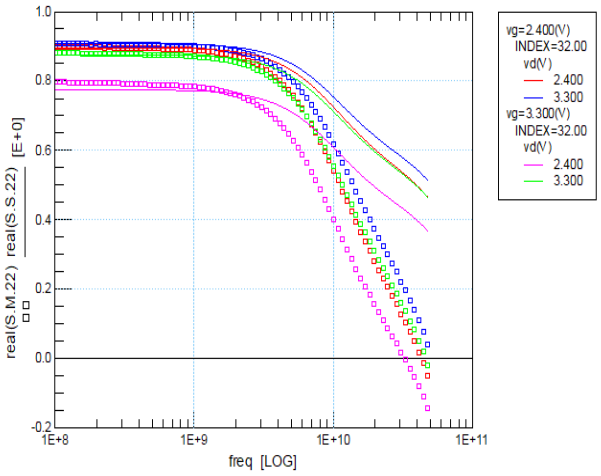


im(Y22) = f(Vg,Vd)

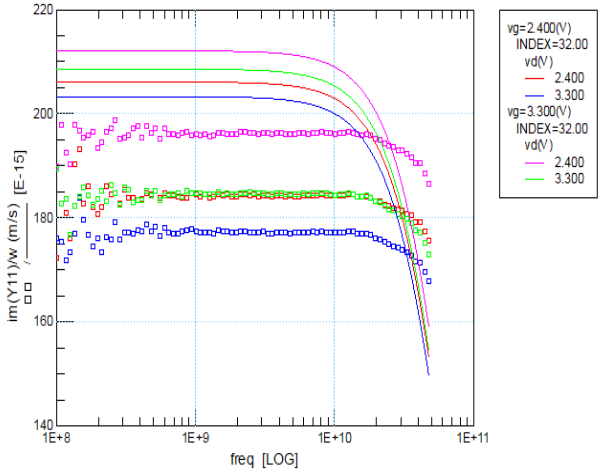
Verify Plots, T = 27°C



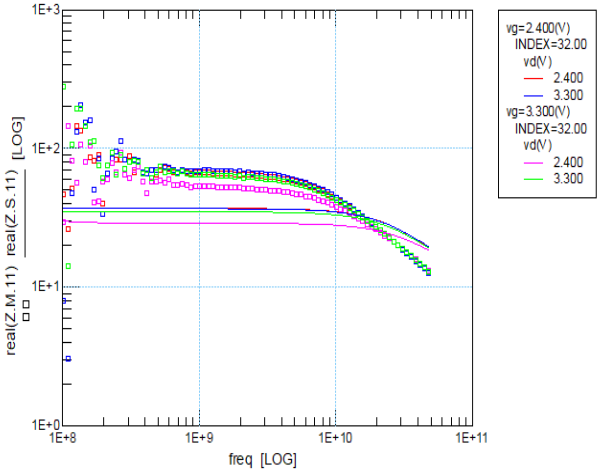
re(S11) = f(Vg,Vd)



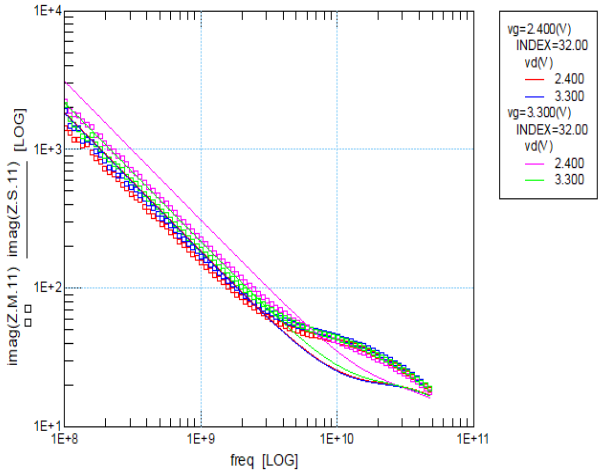
re(S22) = f(Vg,Vd)



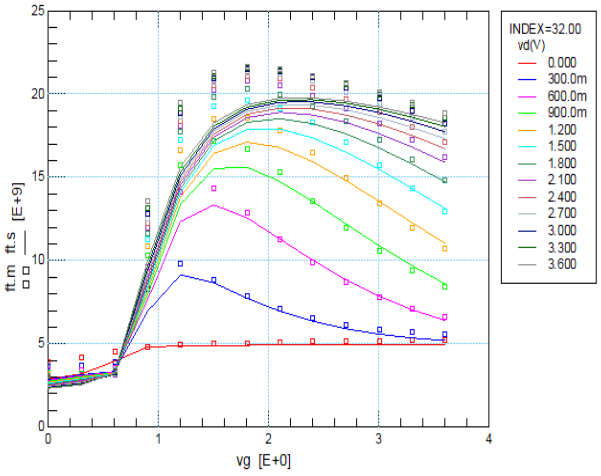
im(Y11) divided by ω



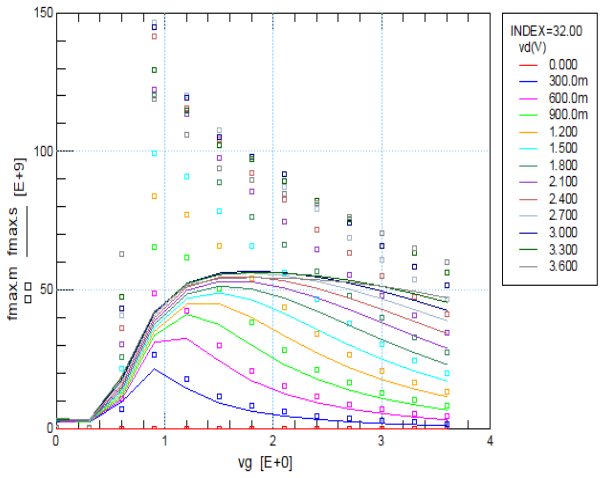
re(Z11) = f(Vg,Vd)



im(Z11) = f(Vg,Vd)



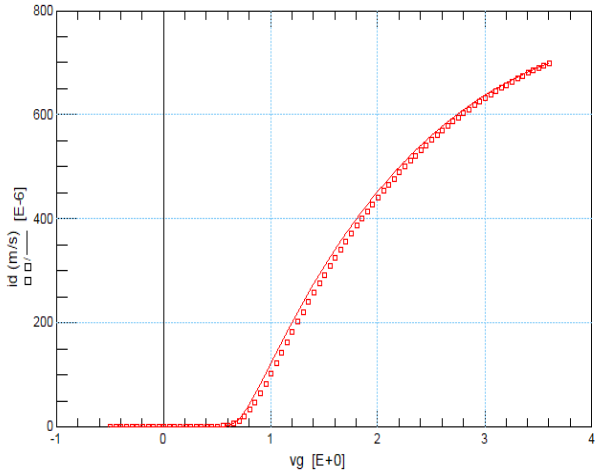
fT_vg = f(Vg,Vd)



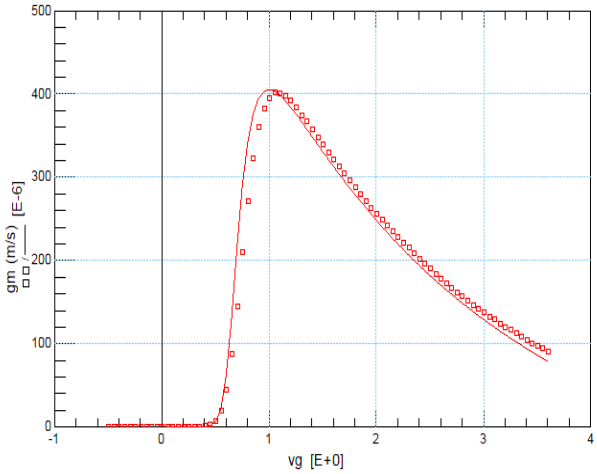
fmax_vg = f(Vg,Vd)

5.8 NF04_WF02u5_L_0u45_S453A_91

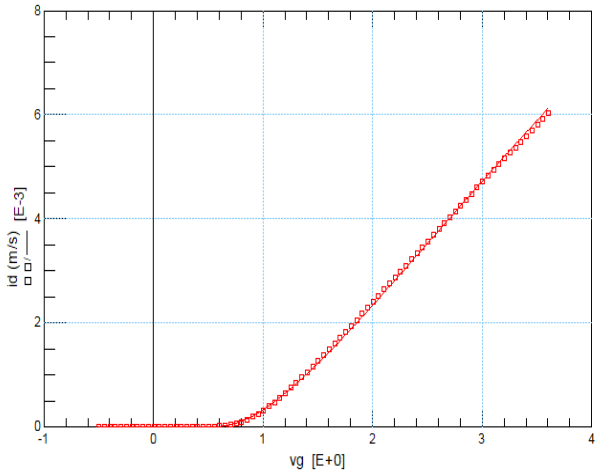
IdVg, T = 27°C



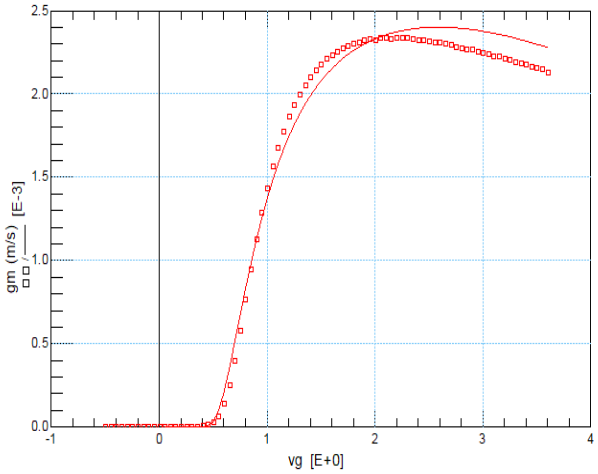
Id = f(Vg) @ Vd = 0.1V



gm = f(Vg) @ Vd = 0.1V

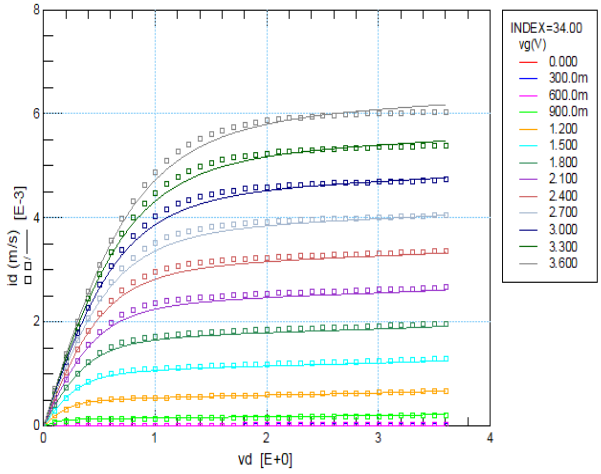


Id = f(Vg) @ Vd = 3.3V

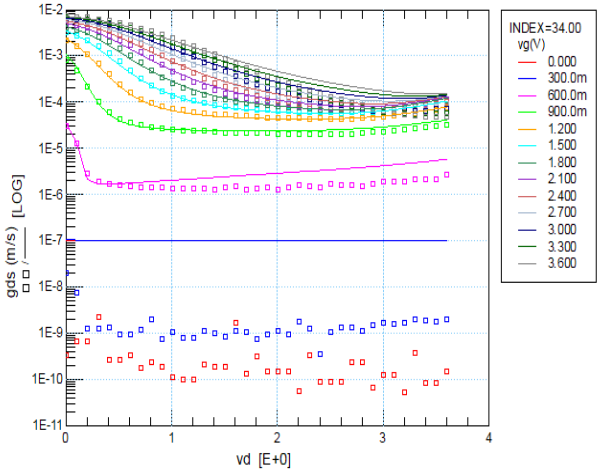


gm = f(Vg) @ Vd = 3.3V

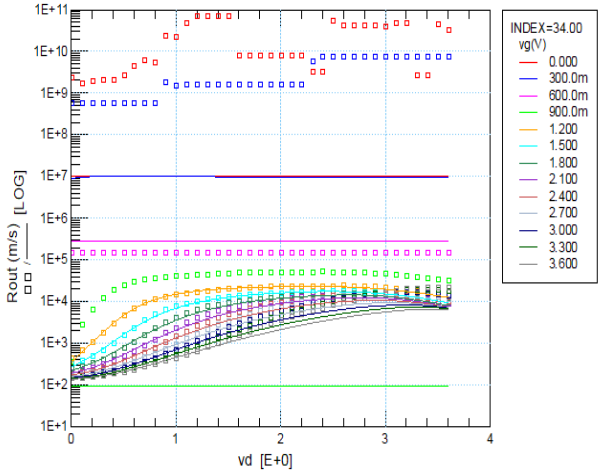
IdVd, T = 27°C



Id = f(Vd)

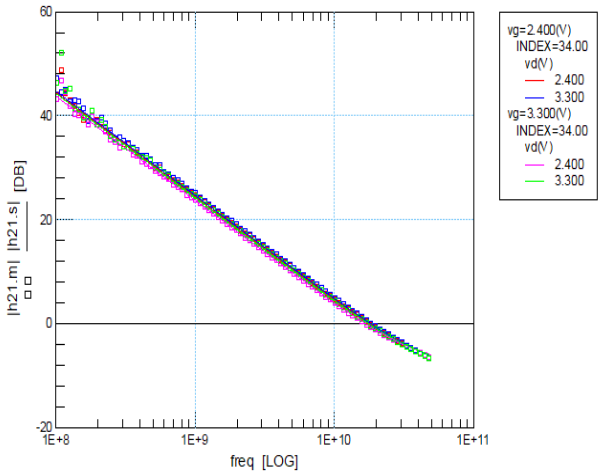


gds = f(Vd)

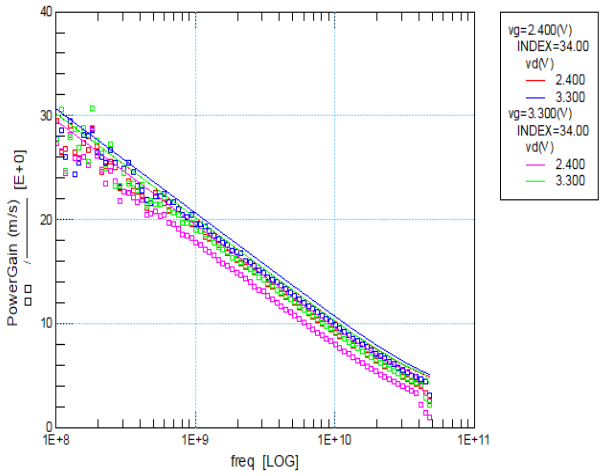


Rout = f(Vd)

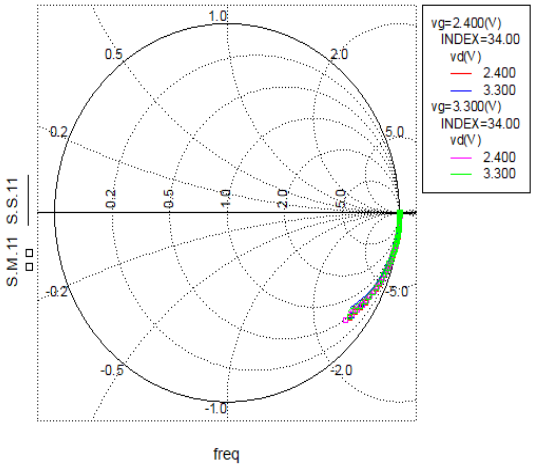
S-Parameter, T = 27°C



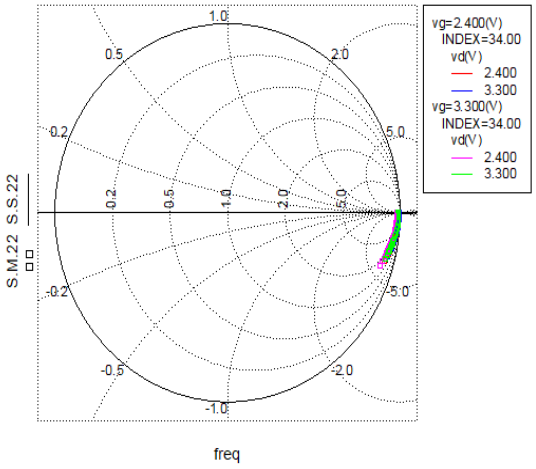
|h21| = f(Vg,Vd)



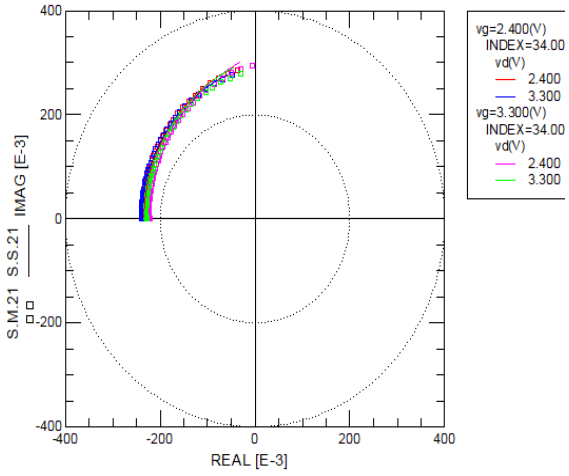
MSG/MAG = f(Vg,Vd)



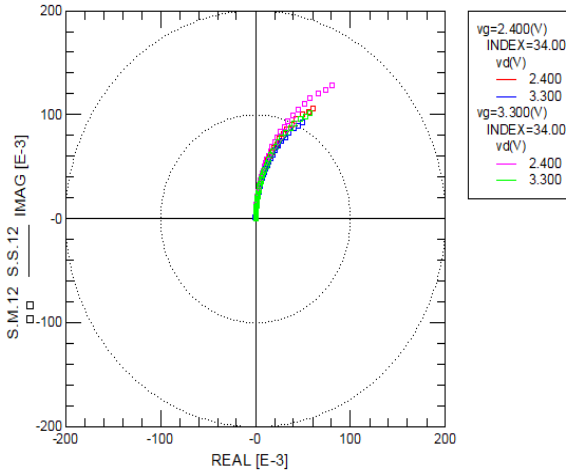
S11 = f(Vg,Vd)



S22 = f(Vg,Vd)

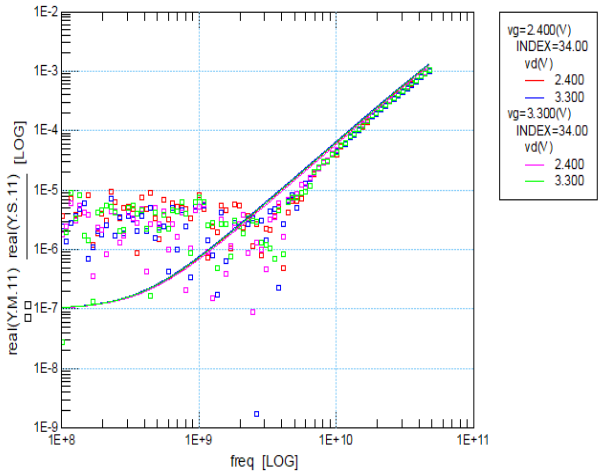


S21 = f(Vg,Vd)

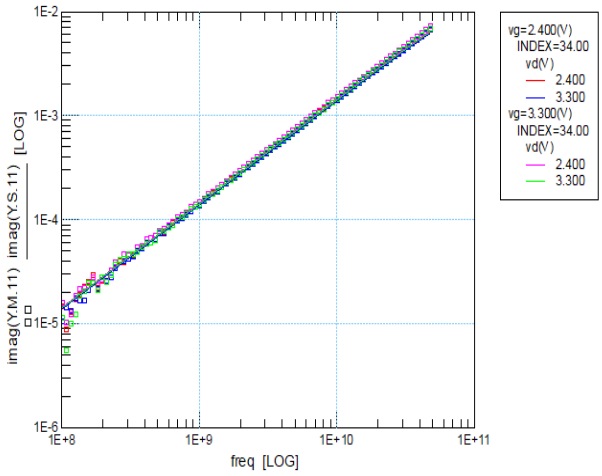


S12 = f(Vg,Vd)

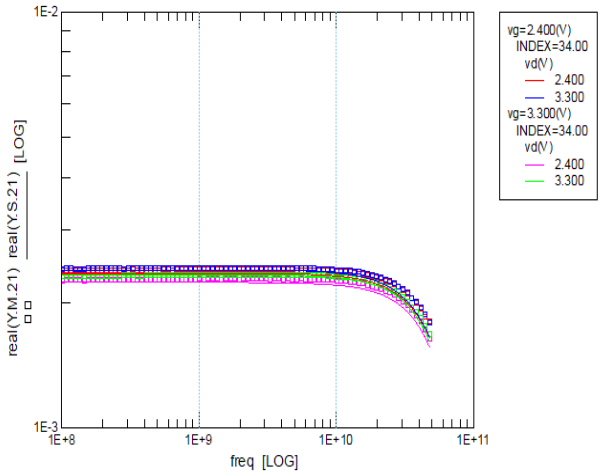
Y-Parameter, T = 27°C



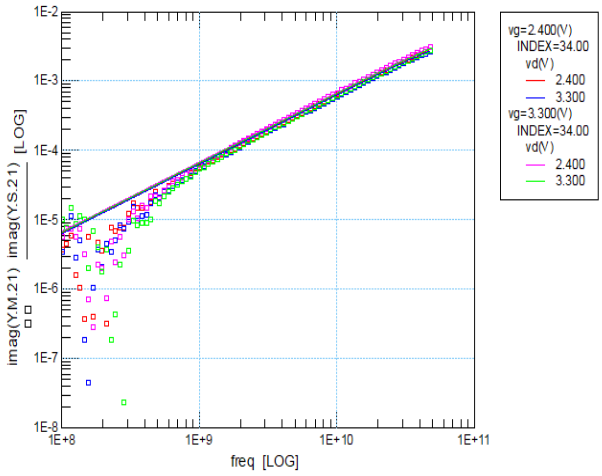
re(Y11) = f(Vg,Vd)



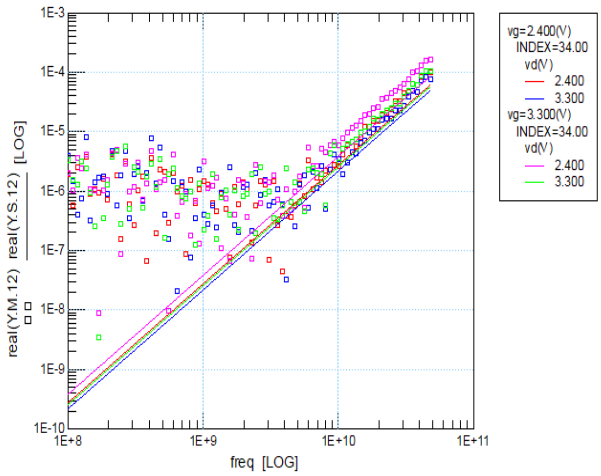
im(Y11) = f(Vg,Vd)



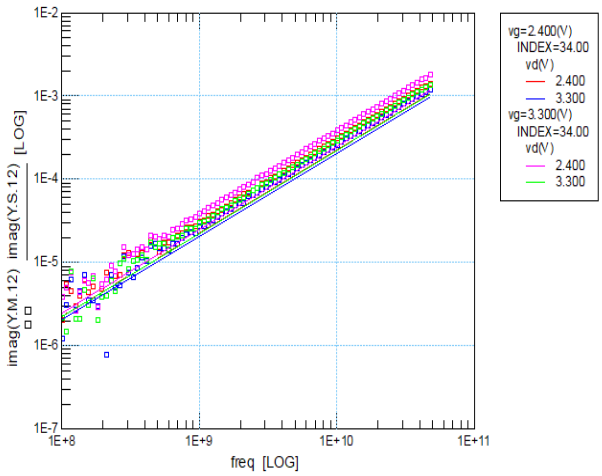
re(Y21) = f(Vg,Vd)



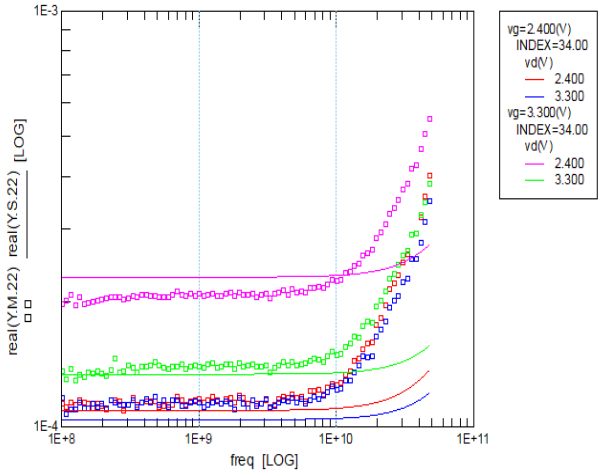
im(Y21) = f(Vg,Vd)



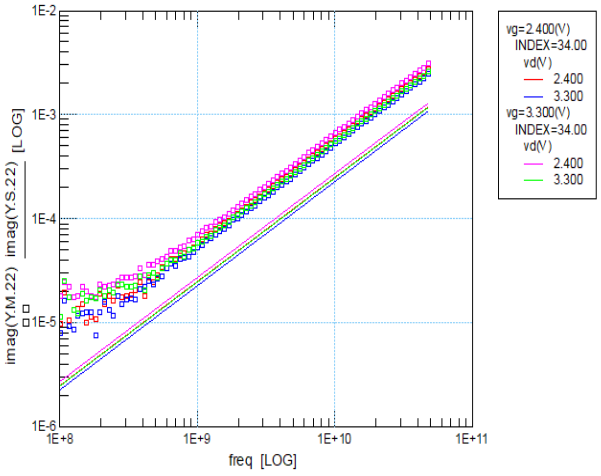
re(Y12) = f(Vg,Vd)



im(Y12) = f(Vg,Vd)

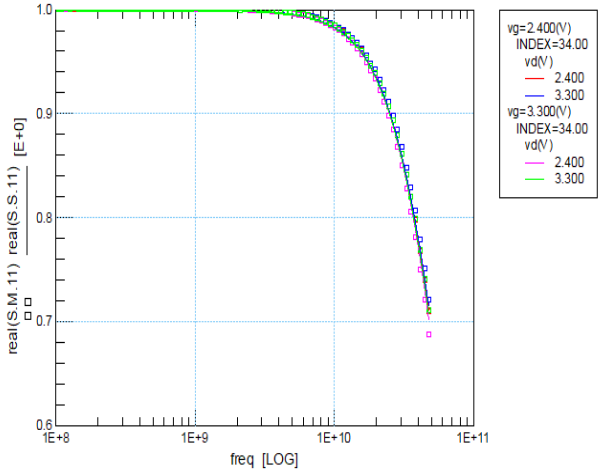


re(Y22) = f(Vg,Vd)

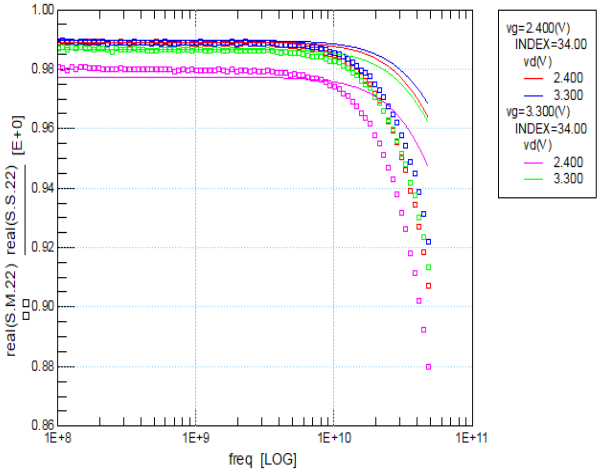


im(Y22) = f(Vg,Vd)

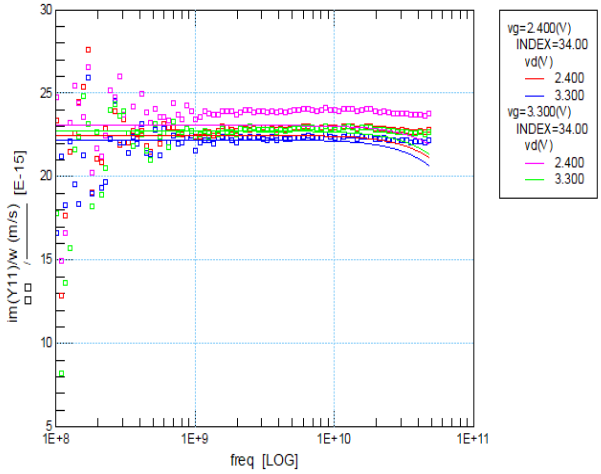
Verify Plots, T = 27°C



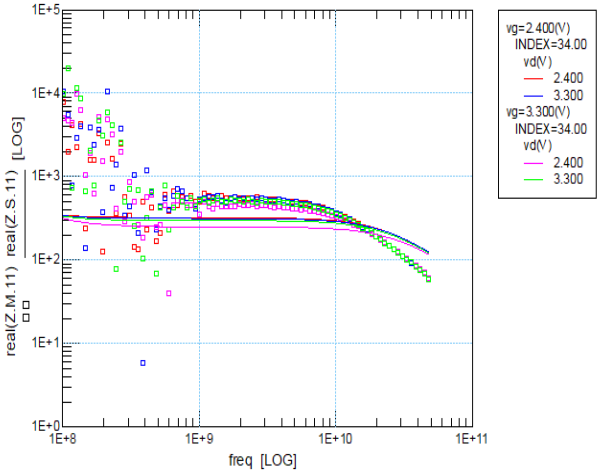
re(S11) = f(Vg,Vd)



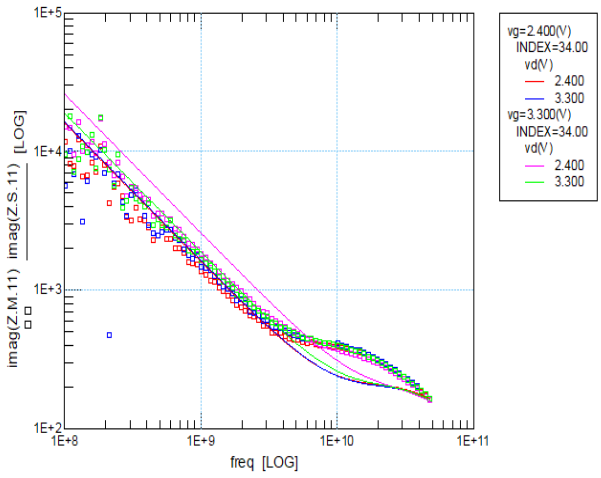
re(S22) = f(Vg,Vd)



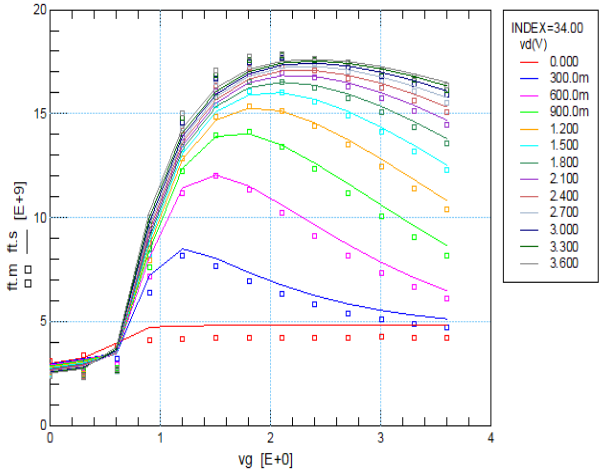
im(Y11) divided by ω



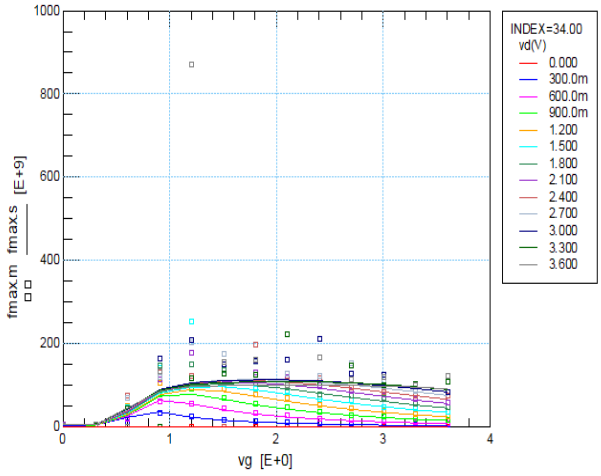
re(Z11) = f(Vg,Vd)



im(Z11) = f(Vg,Vd)



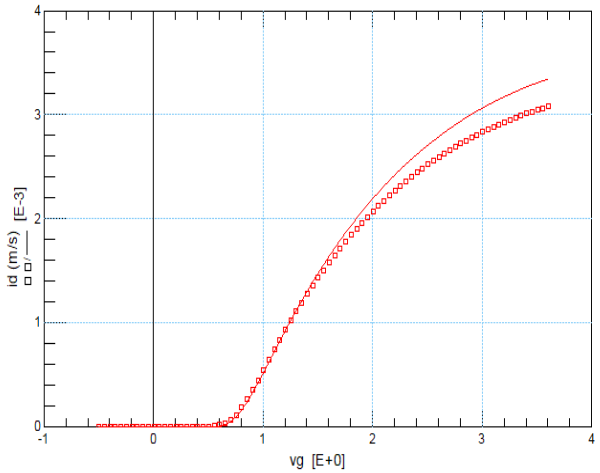
fT_vg = f(Vg,Vd)



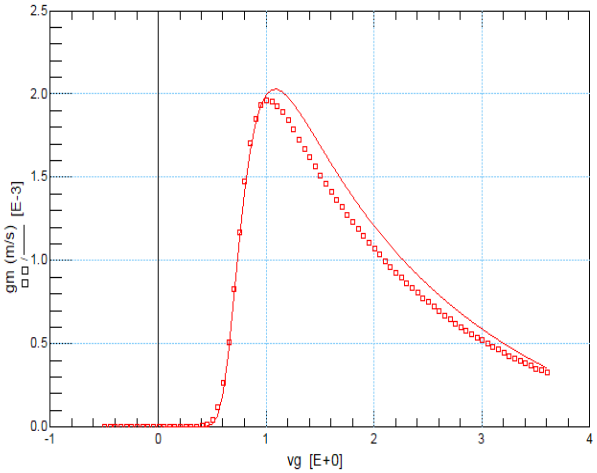
fmax_vg = f(Vg,Vd)

5.9 NF05_WF10u0_L_0u45_S453A_A4

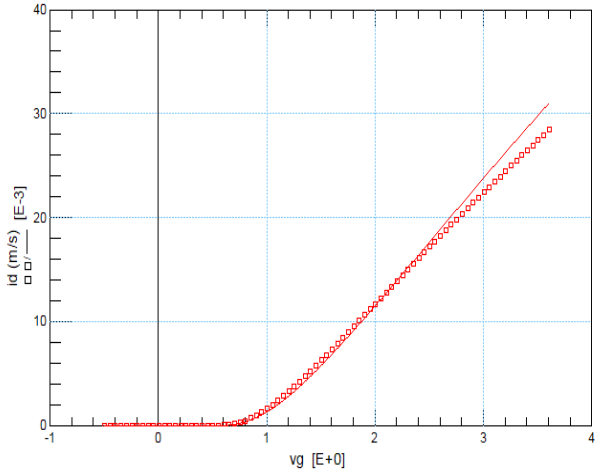
IdVg, T = 27°C



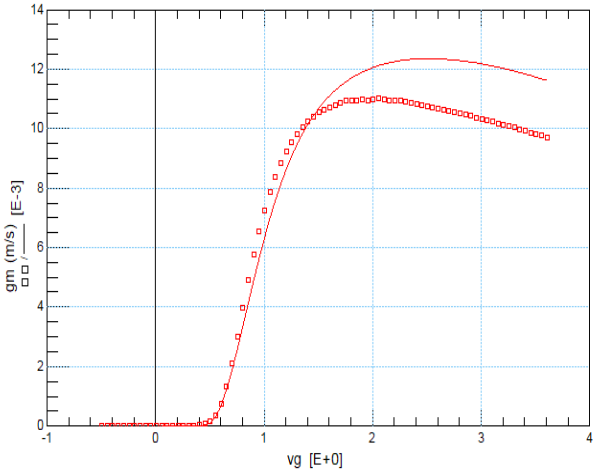
Id = f(Vg) @ Vd = 0.1V



gm = f(Vg) @ Vd = 0.1V

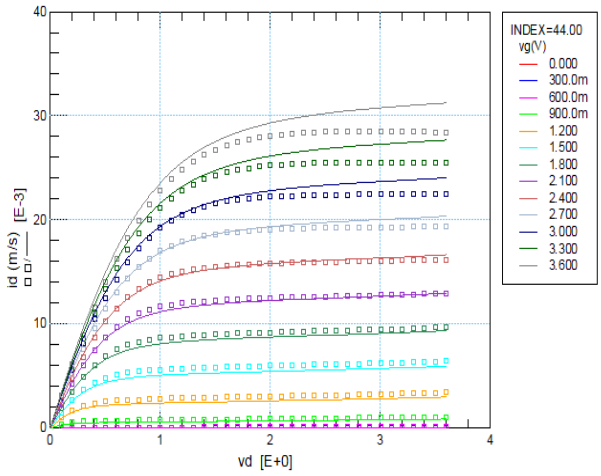


Id = f(Vg) @ Vd = 3.3V

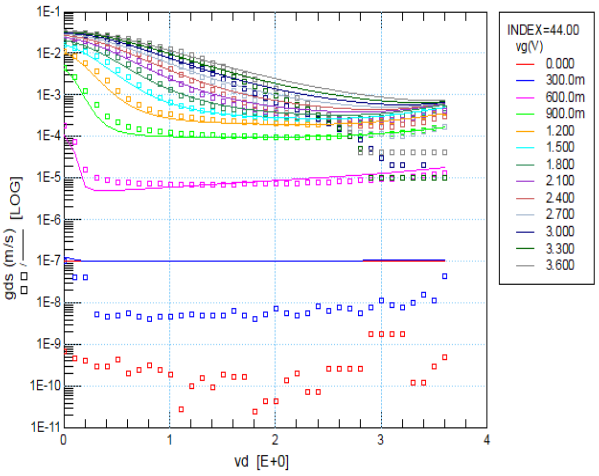


gm = f(Vg) @ Vd = 3.3V

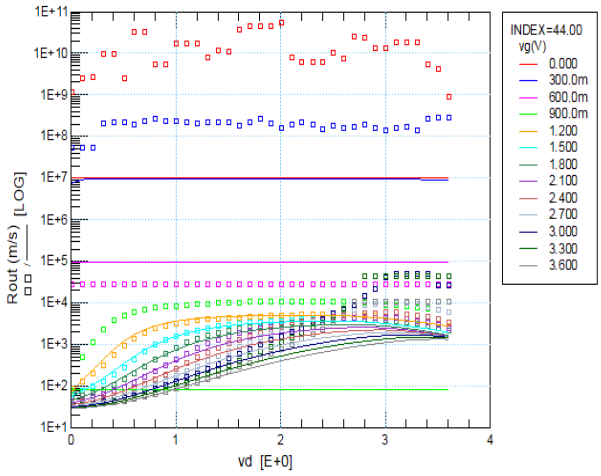
IdVd, T = 27°C



Id = f(Vd)

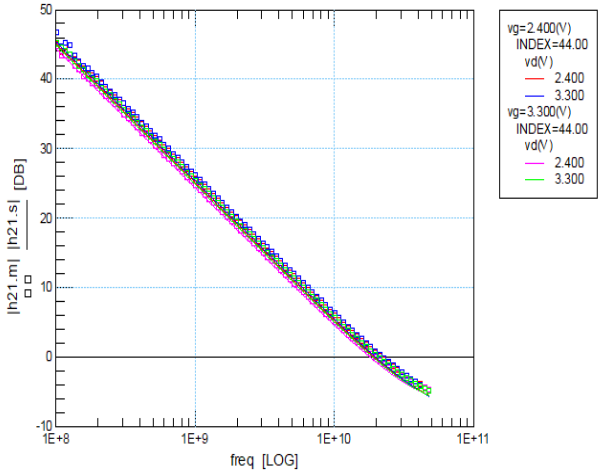


gds = f(Vd)

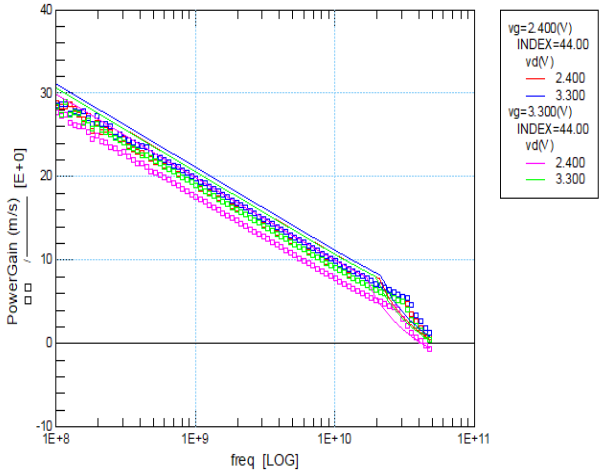


Rout = f(Vd)

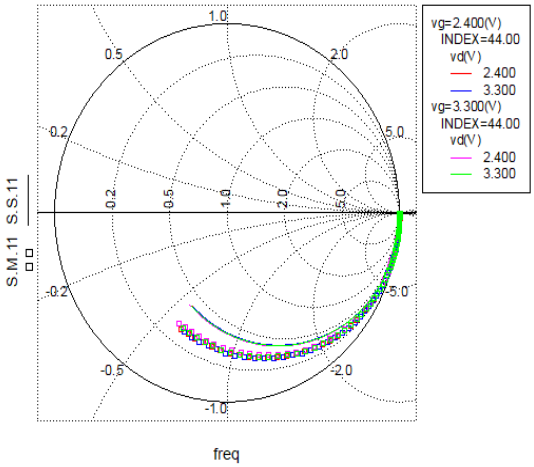
S-Parameter, T = 27°C



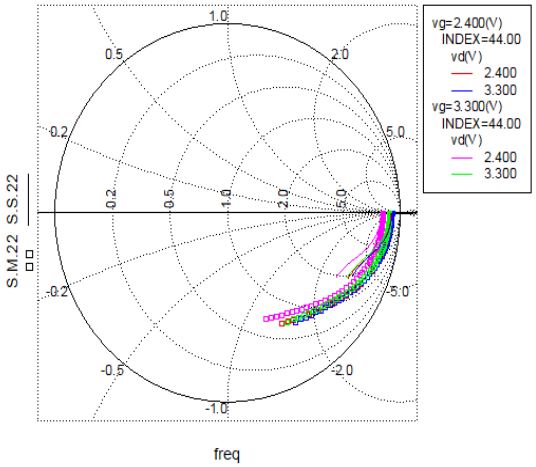
|h21| = f(Vg,Vd)



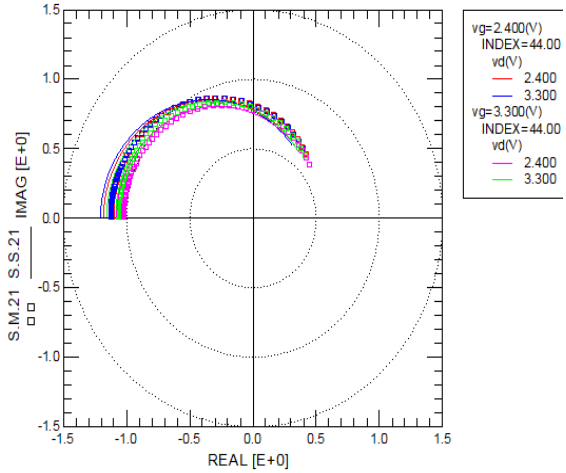
MSG/MAG = f(Vg,Vd)



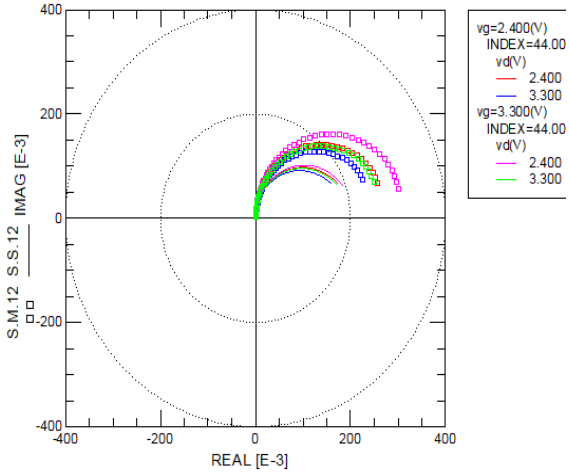
S11 = f(Vg,Vd)



S22 = f(Vg,Vd)

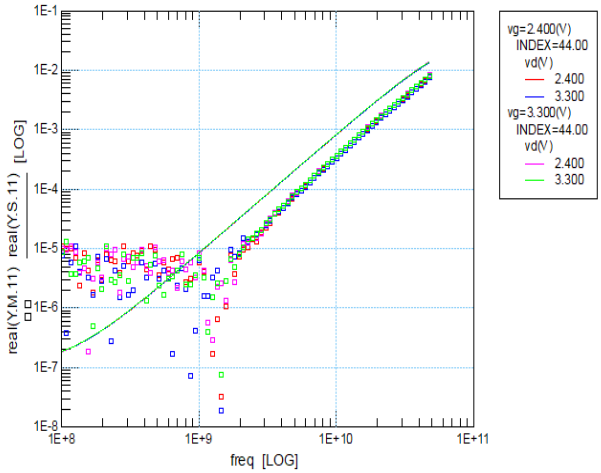


S21 = f(Vg,Vd)

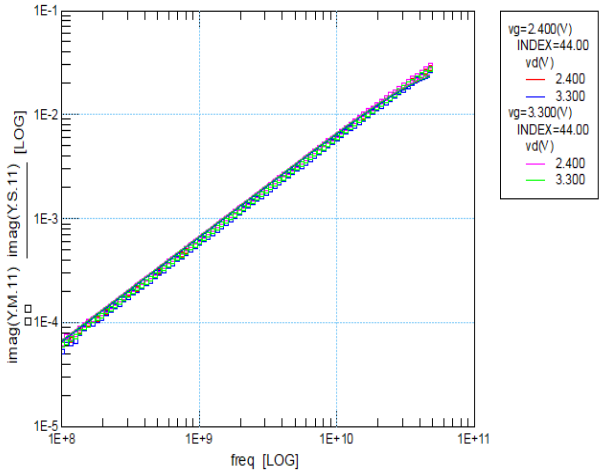


S12 = f(Vg,Vd)

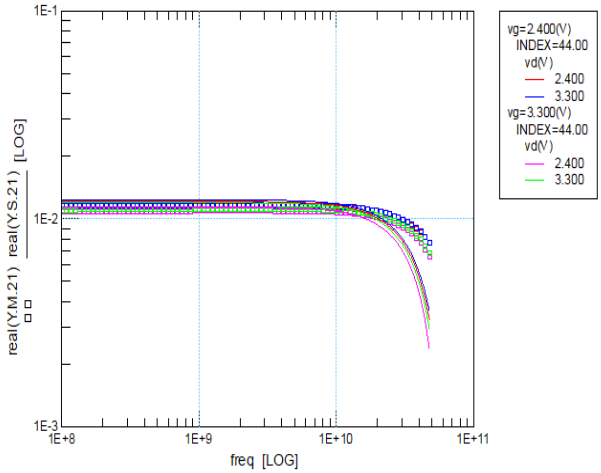
Y-Parameter, T = 27°C



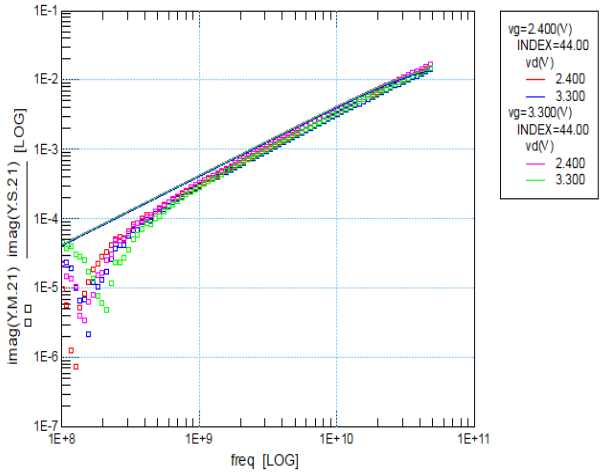
re(Y11) = f(Vg,Vd)



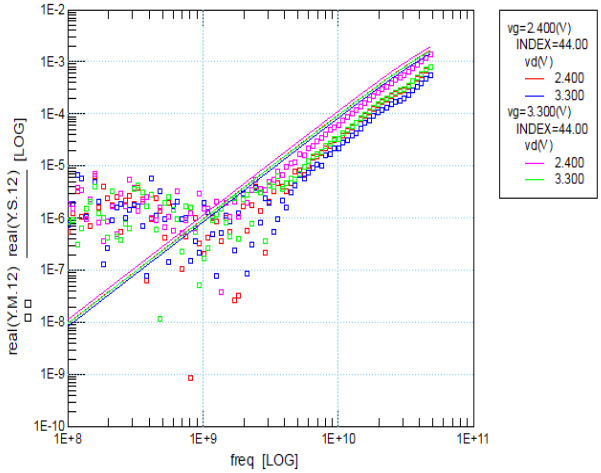
im(Y11) = f(Vg,Vd)



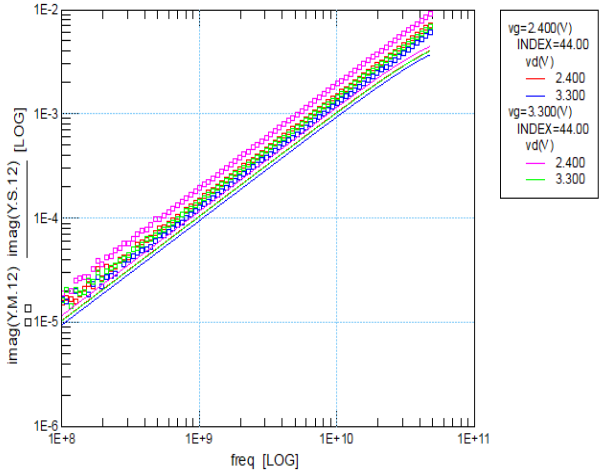
re(Y21) = f(Vg,Vd)



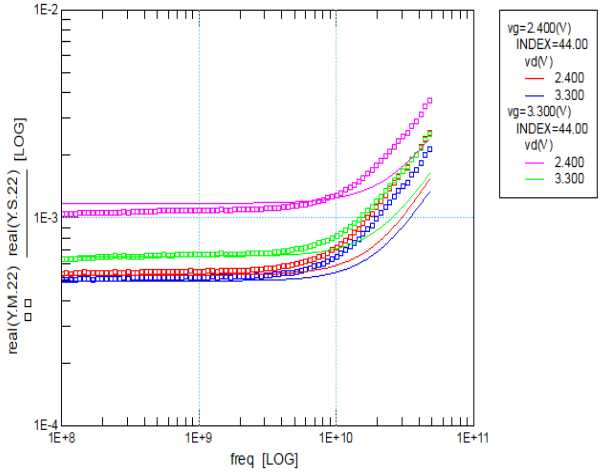
im(Y21) = f(Vg,Vd)



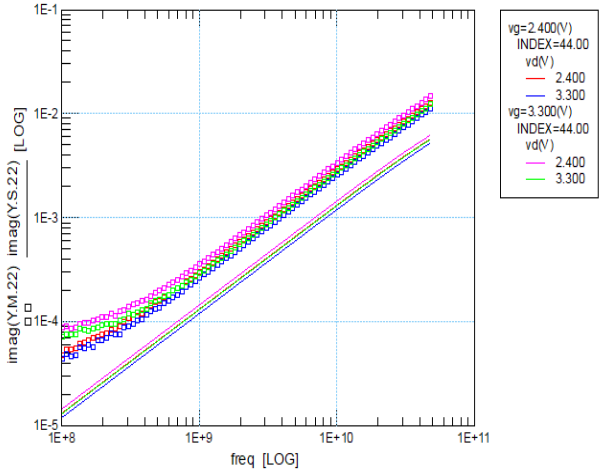
re(Y12) = f(Vg,Vd)



im(Y12) = f(Vg,Vd)

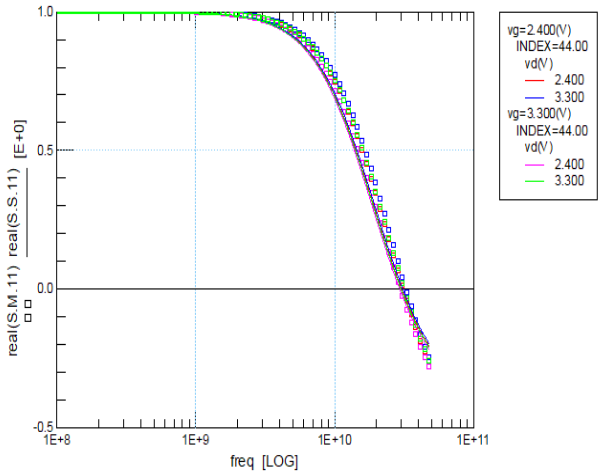


re(Y22) = f(Vg,Vd)

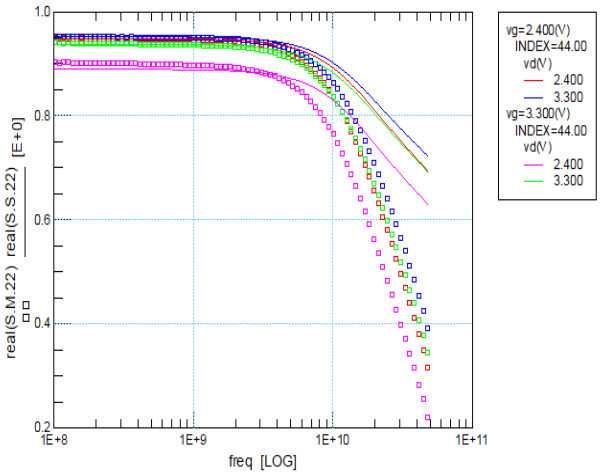


im(Y22) = f(Vg,Vd)

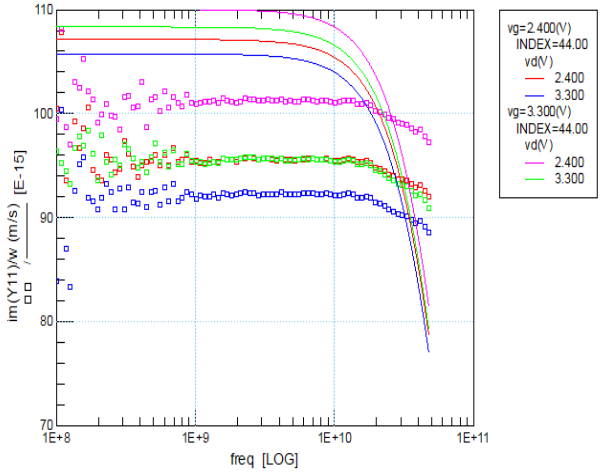
Verify Plots, T = 27°C



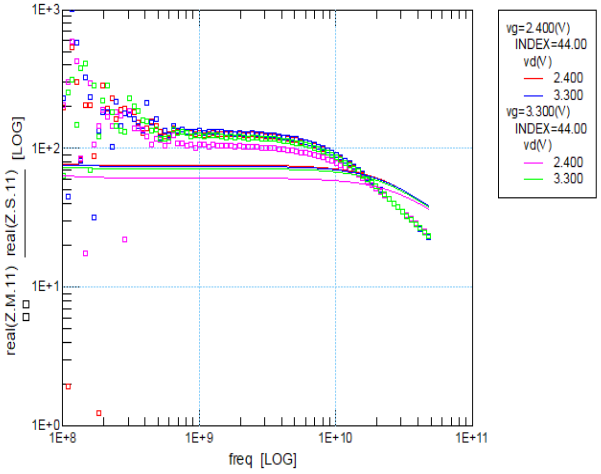
re(S11) = f(Vg,Vd)



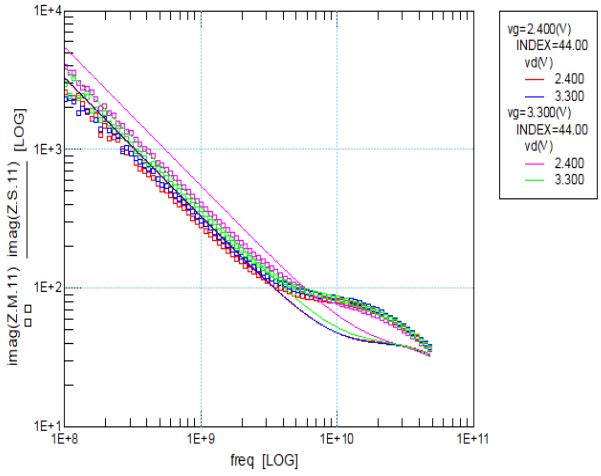
re(S22) = f(Vg,Vd)



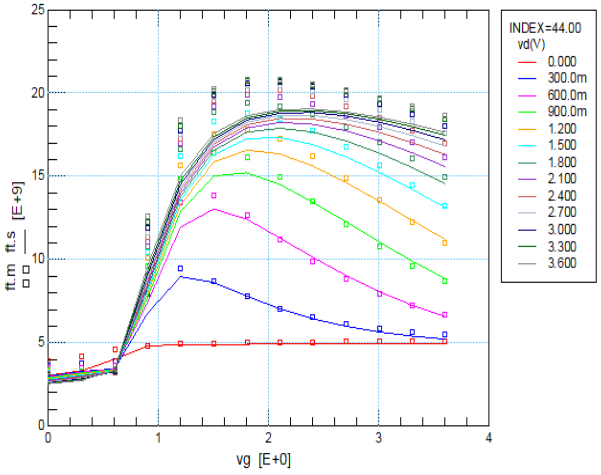
im(Y11) divided by ω



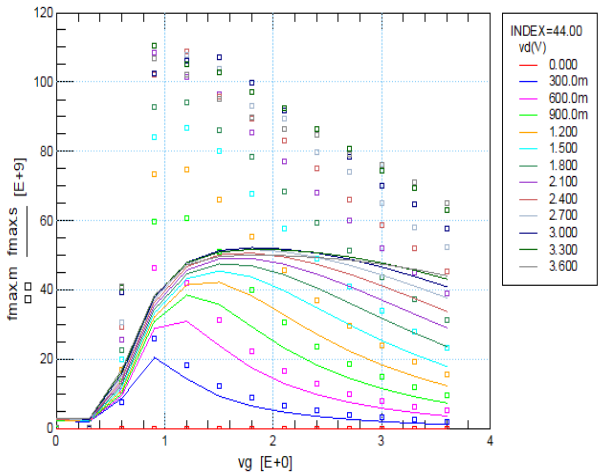
re(Z11) = f(Vg,Vd)



im(Z11) = f(Vg,Vd)



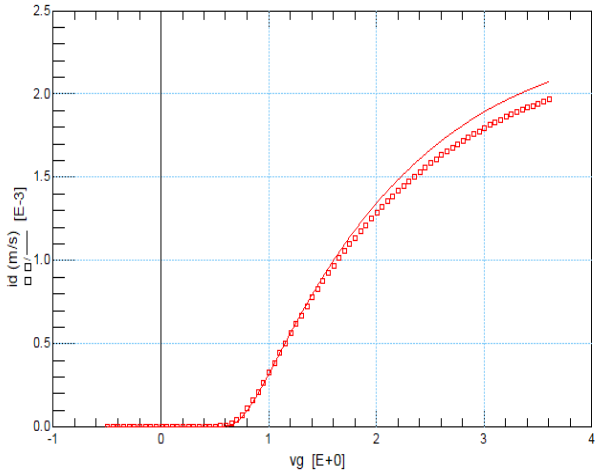
fT_vg = f(Vg,Vd)



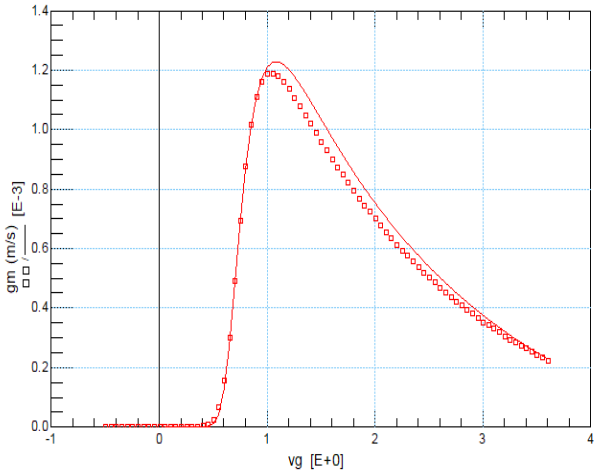
fmax_vg = f(Vg,Vd)

5.10 NF04_WF07u5_L_0u45_S453A_94

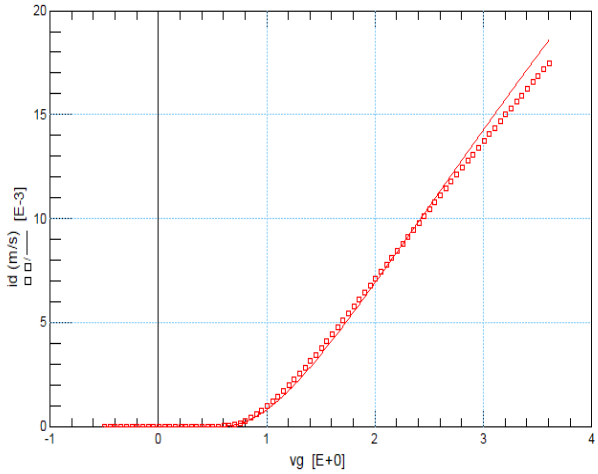
IdVg, T = 27°C



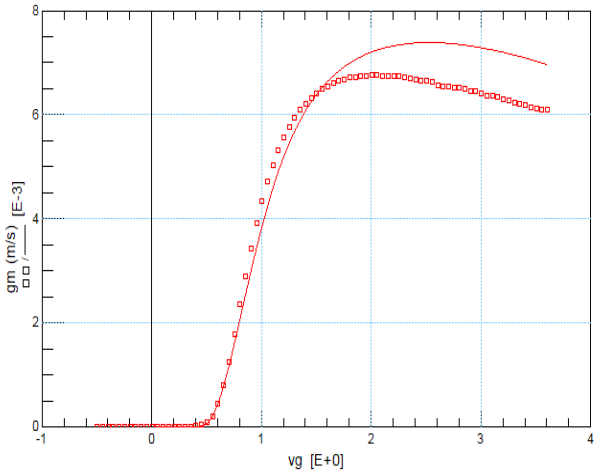
Id = f(Vg) @ Vd = 0.1V



gm = f(Vg) @ Vd = 0.1V

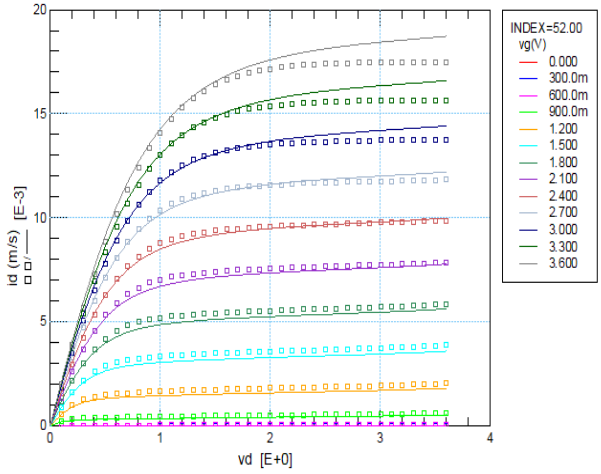


Id = f(Vg) @ Vd = 3.3V

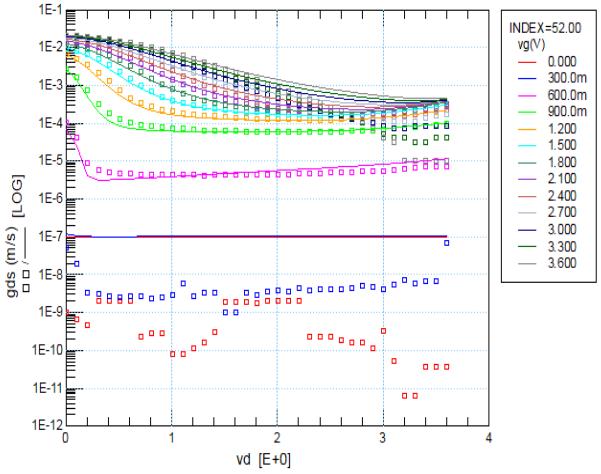


gm = f(Vg) @ Vd = 3.3V

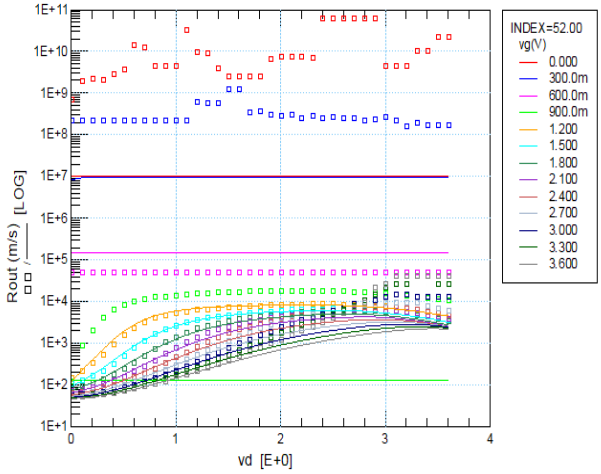
IdVd, T = 27°C



Id = f(Vd)

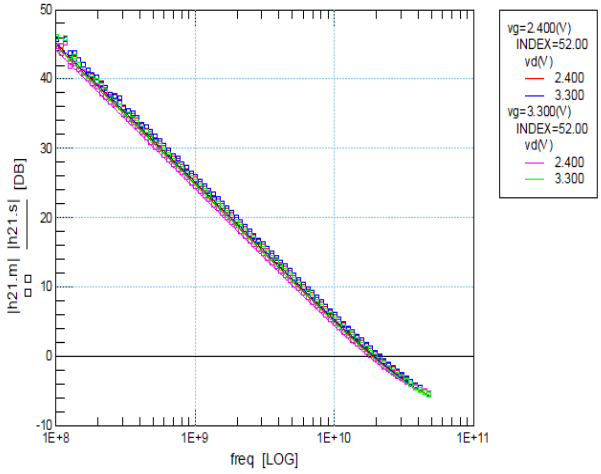


gds = f(Vd)

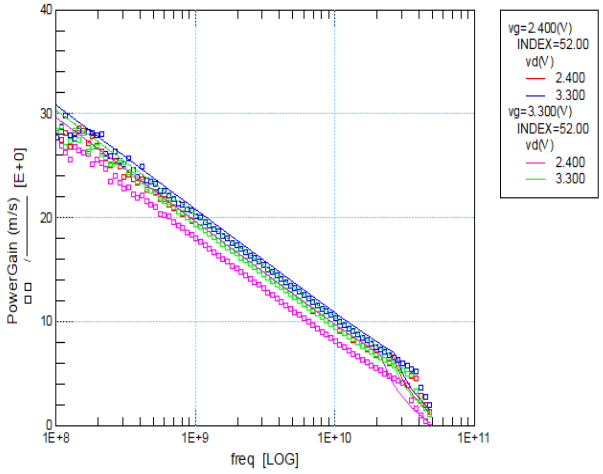


Rout = f(Vd)

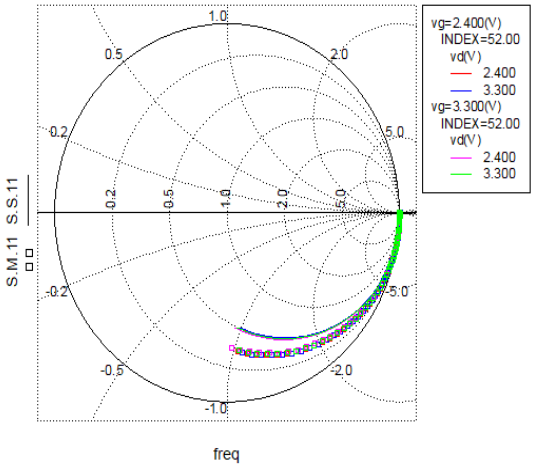
S-Parameter, T = 27°C



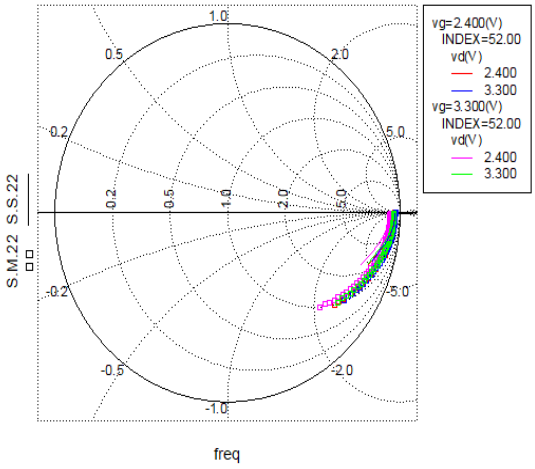
|h21| = f(Vg,Vd)



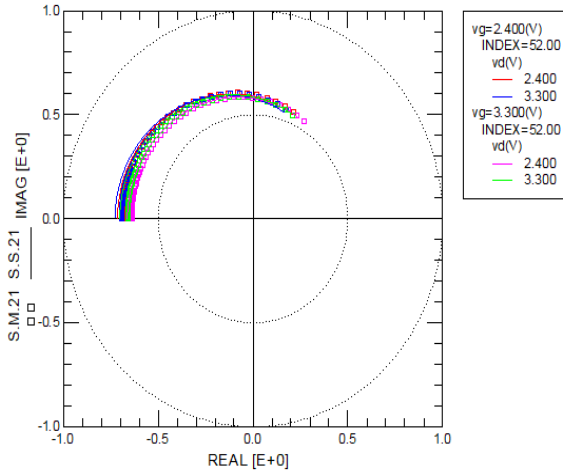
MSG/MAG = f(Vg,Vd)



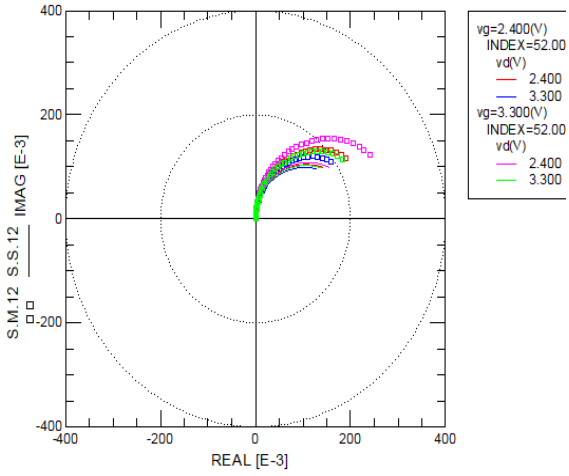
S11 = f(Vg,Vd)



S22 = f(Vg,Vd)

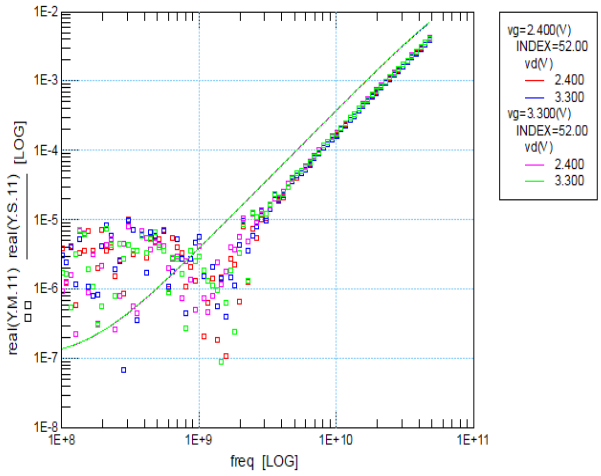


S21 = f(Vg,Vd)

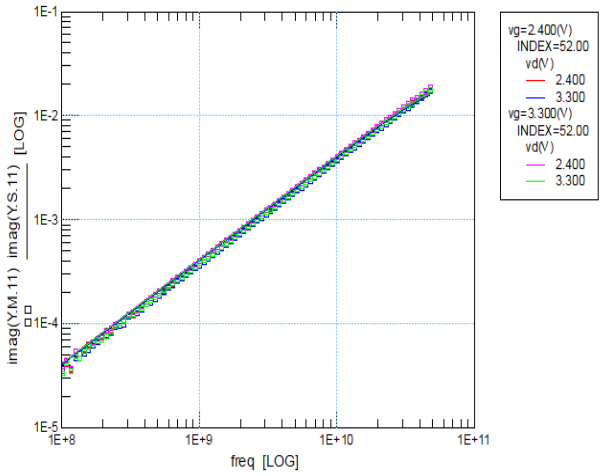


S12 = f(Vg,Vd)

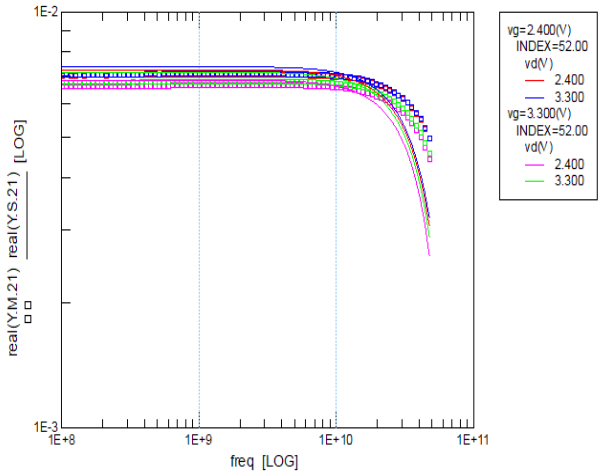
Y-Parameter, T = 27°C



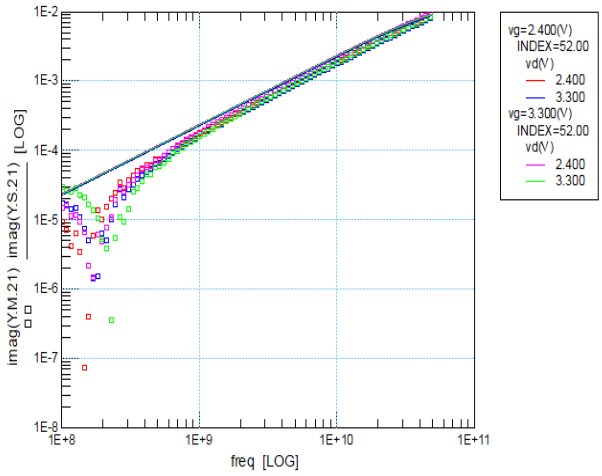
re(Y11) = f(Vg,Vd)



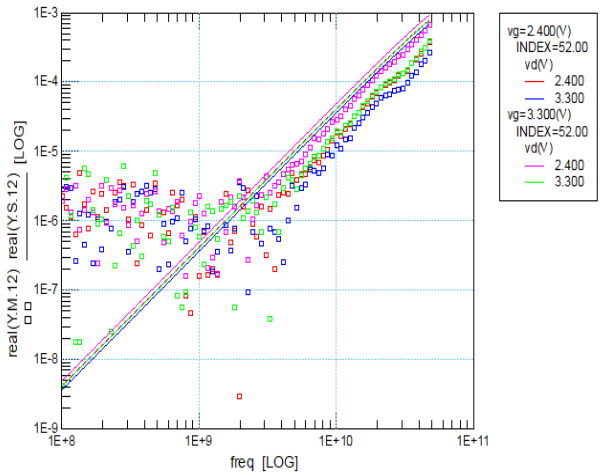
im(Y11) = f(Vg,Vd)



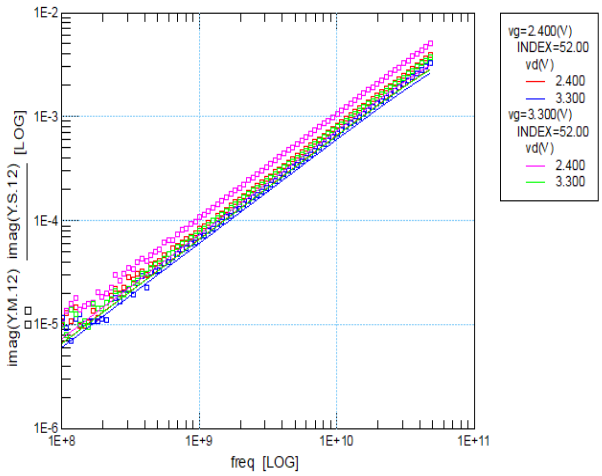
re(Y21) = f(Vg,Vd)



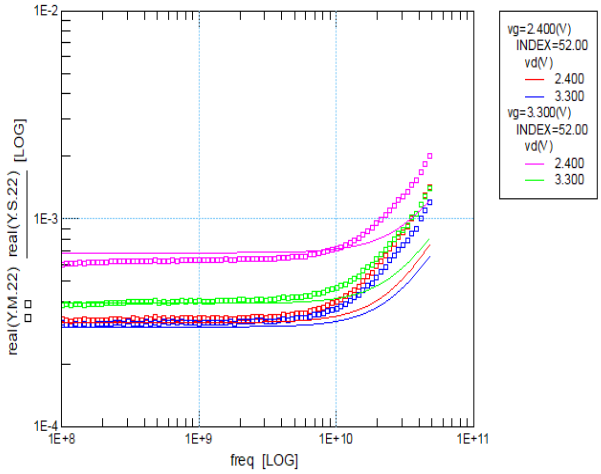
im(Y21) = f(Vg,Vd)



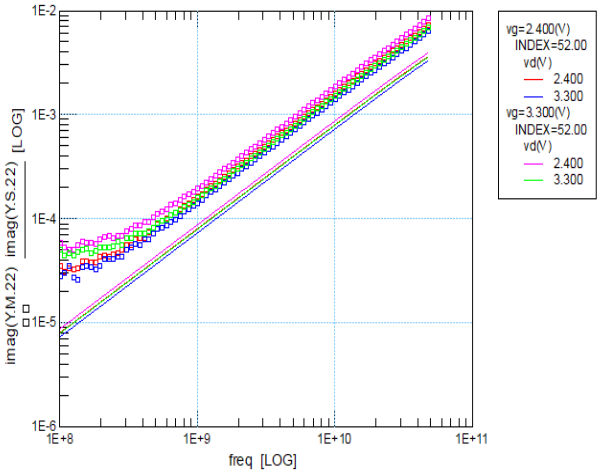
re(Y12) = f(Vg,Vd)



im(Y12) = f(Vg,Vd)

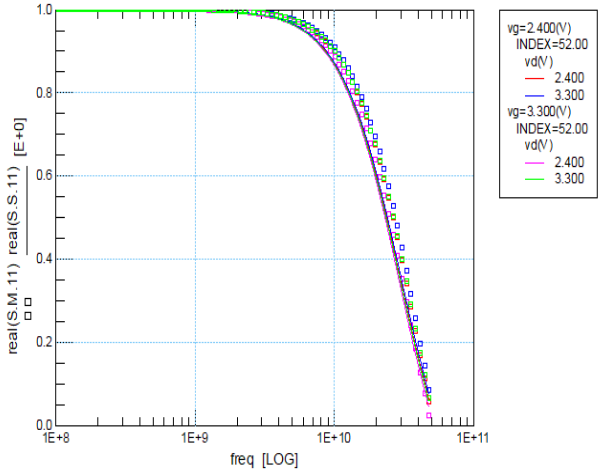


re(Y22) = f(Vg,Vd)

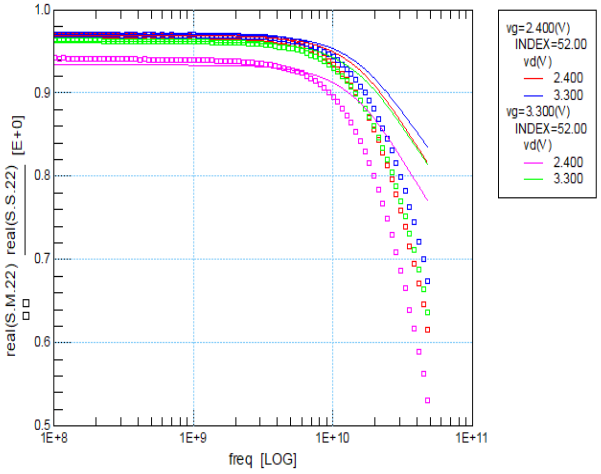


im(Y22) = f(Vg,Vd)

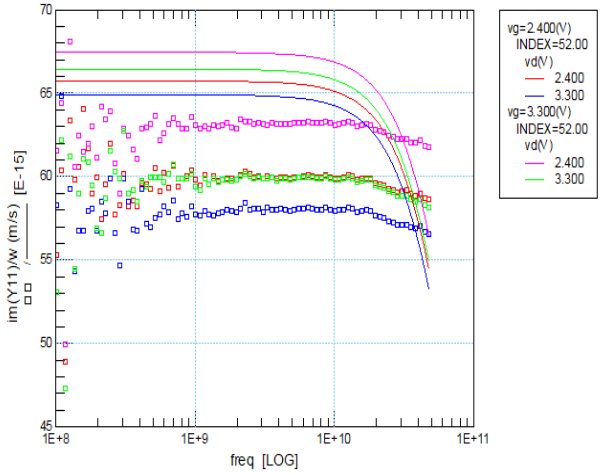
Verify Plots, T = 27°C



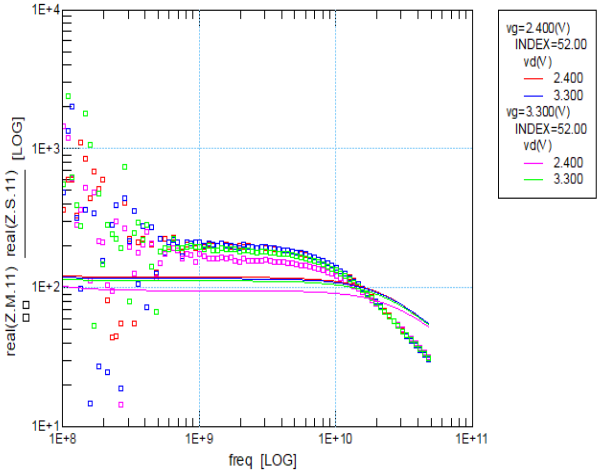
re(S11) = f(Vg,Vd)



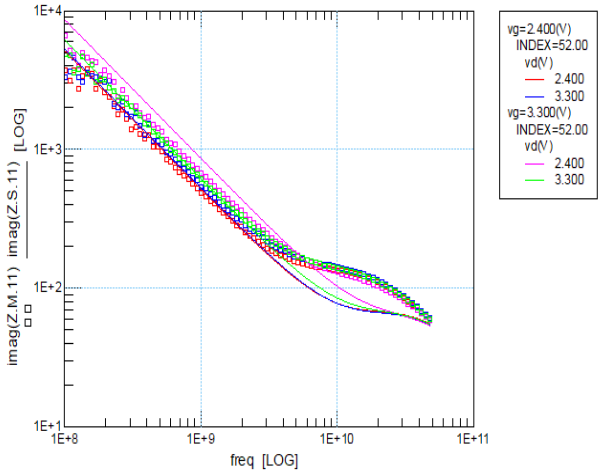
re(S22) = f(Vg,Vd)



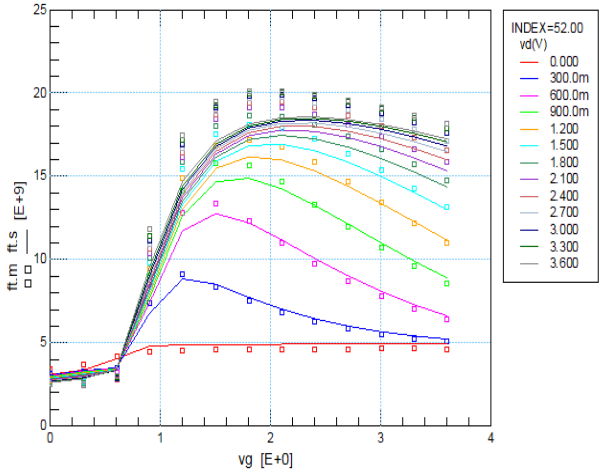
im(Y11) divided by ω



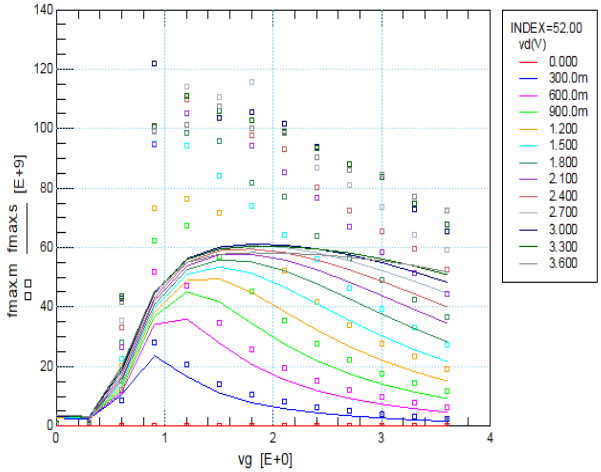
re(Z11) = f(Vg,Vd)



im(Z11) = f(Vg,Vd)



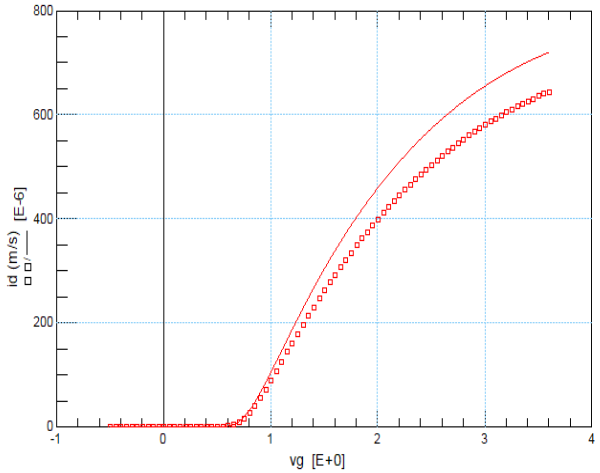
fT_vg = f(Vg,Vd)



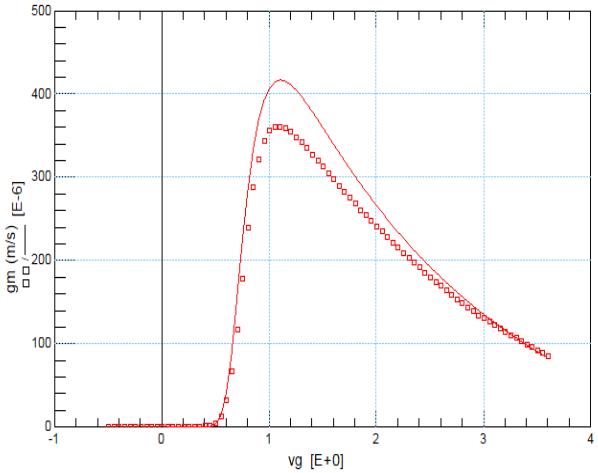
fmax_vg = f(Vg,Vd)

5.11 NF01_WF10u0_L_0u45_S453A_B1

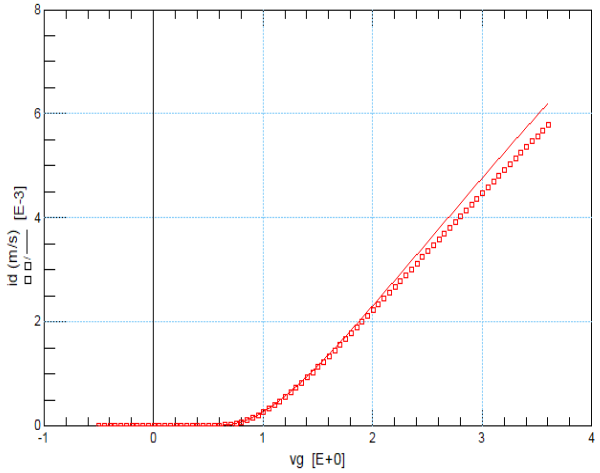
IdVg, T = 27°C



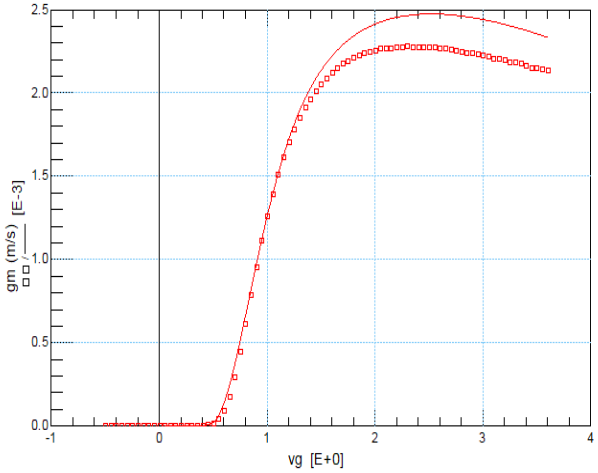
Id = f(Vg) @ Vd = 0.1V



gm = f(Vg) @ Vd = 0.1V

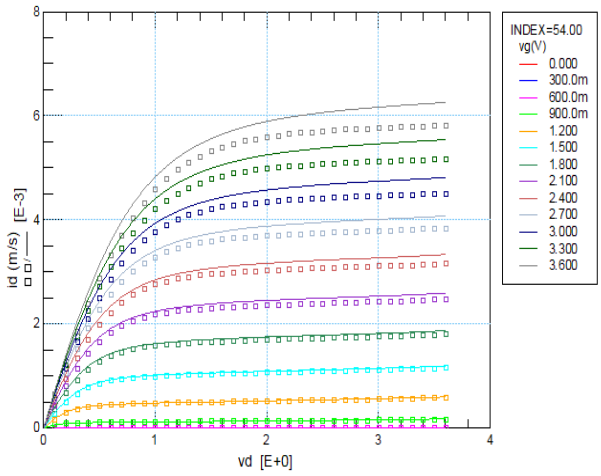


Id = f(Vg) @ Vd = 3.3V

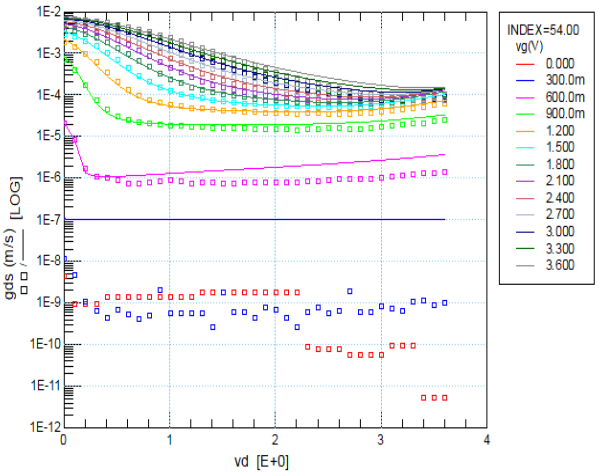


gm = f(Vg) @ Vd = 3.3V

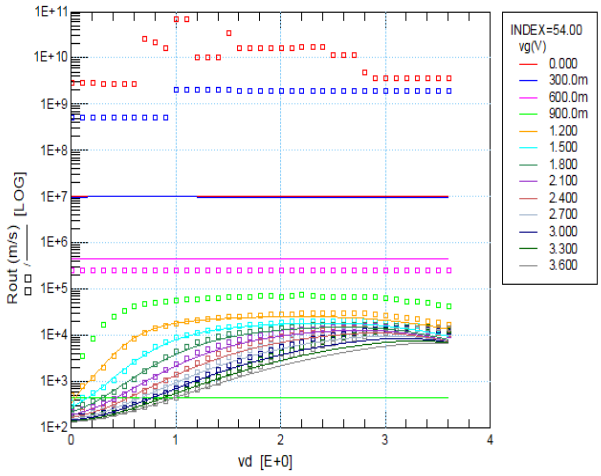
IdVd, T = 27°C



Id = f(Vd)

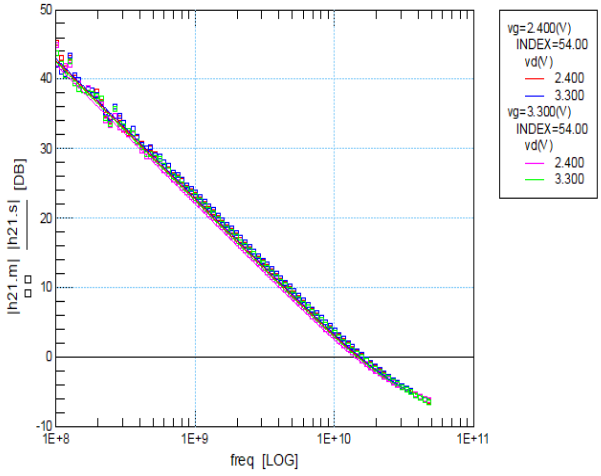


gds = f(Vd)

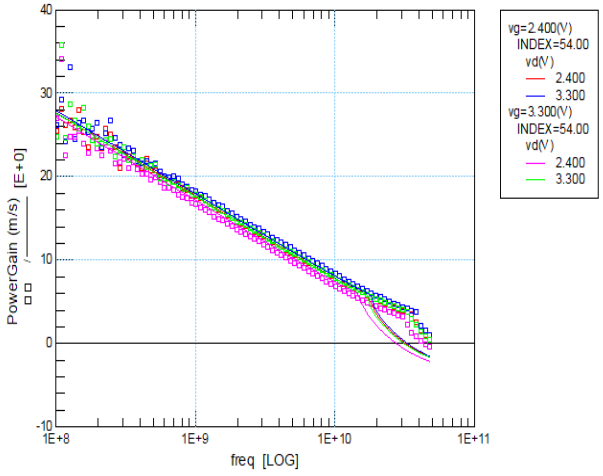


Rout = f(Vd)

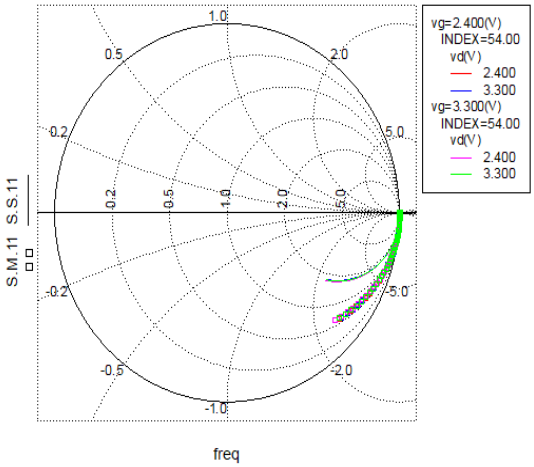
S-Parameter, T = 27°C



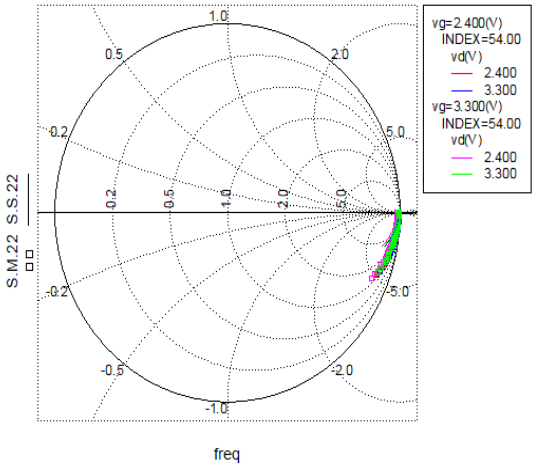
|h21| = f(Vg,Vd)



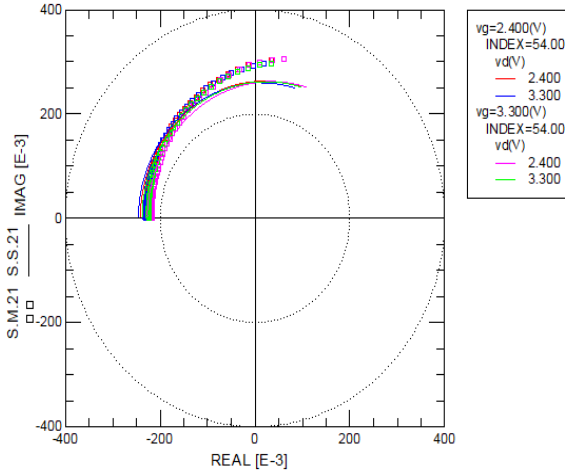
MSG/MAG = f(Vg,Vd)



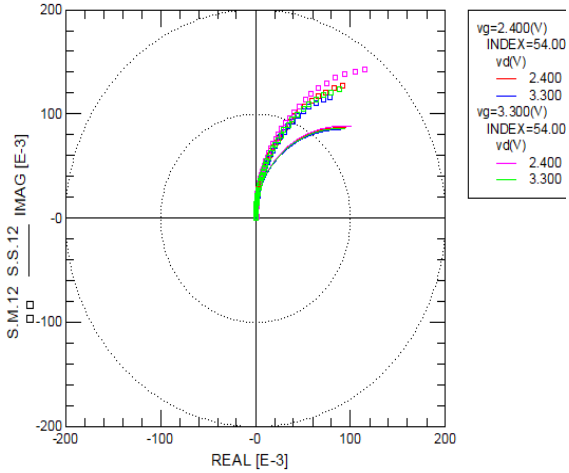
S11 = f(Vg,Vd)



S22 = f(Vg,Vd)

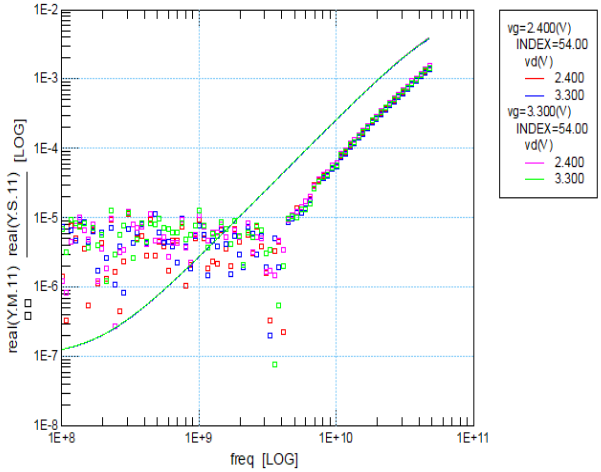


S21 = f(Vg,Vd)

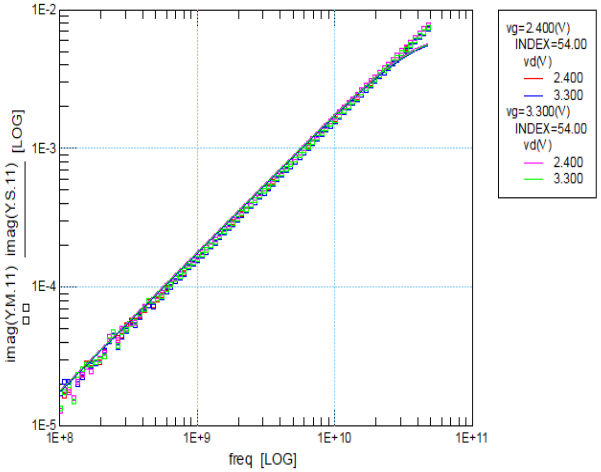


S12 = f(Vg,Vd)

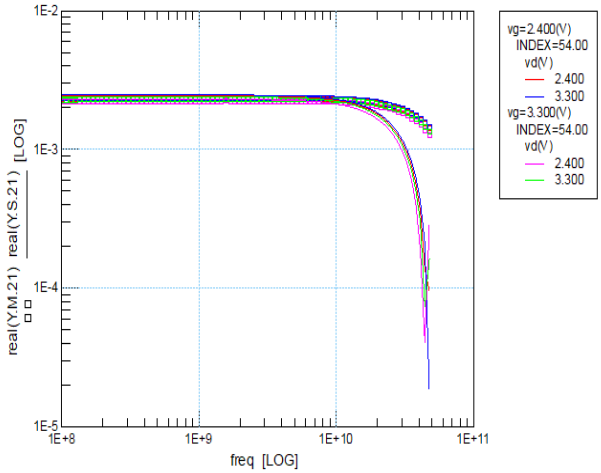
Y-Parameter, T = 27°C



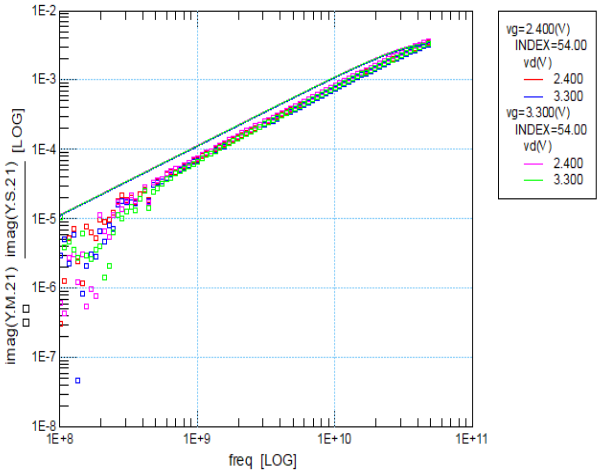
re(Y11) = f(Vg,Vd)



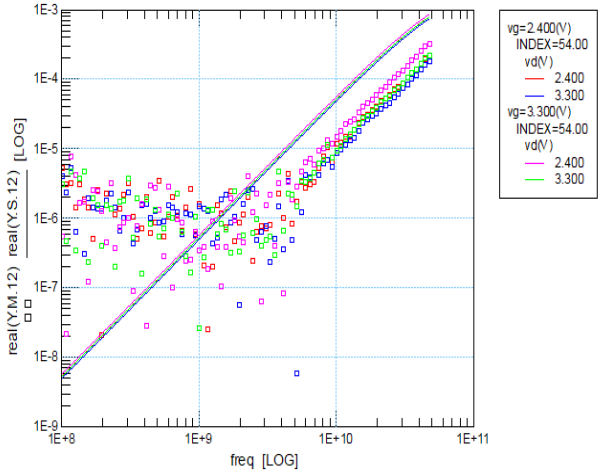
im(Y11) = f(Vg,Vd)



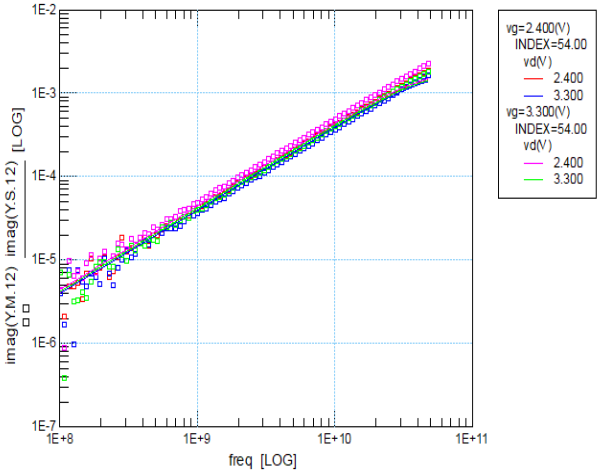
re(Y21) = f(Vg,Vd)



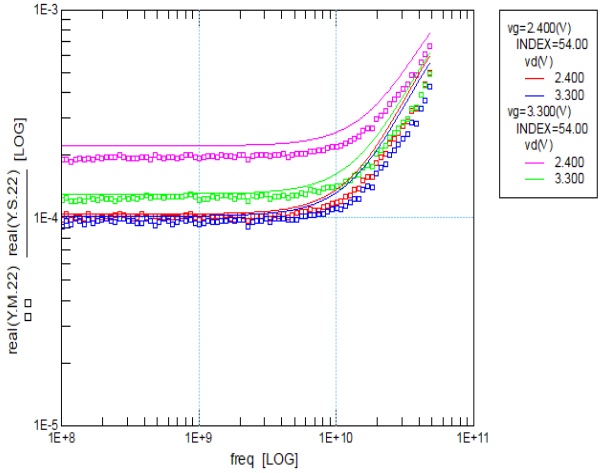
im(Y21) = f(Vg,Vd)



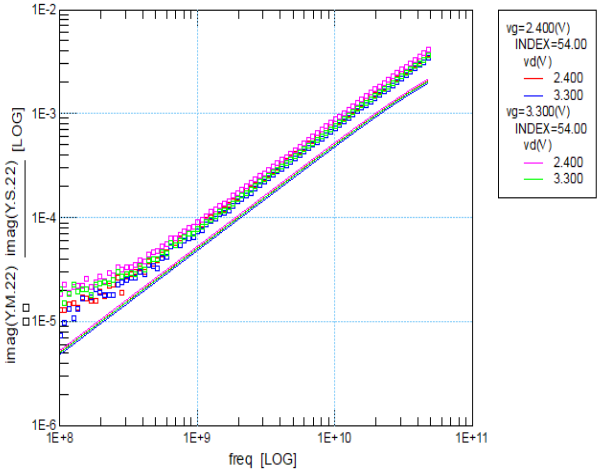
re(Y12) = f(Vg,Vd)



im(Y12) = f(Vg,Vd)

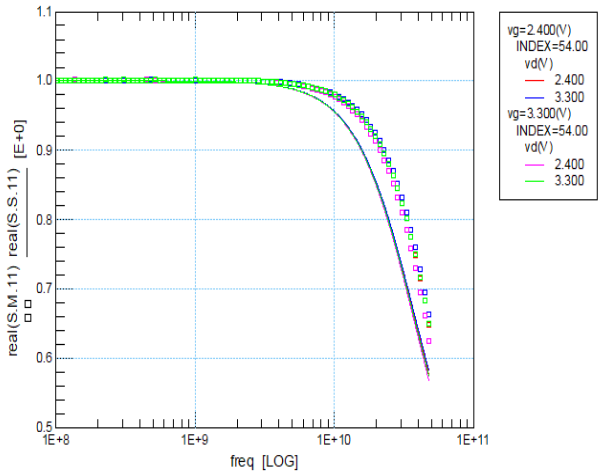


re(Y22) = f(Vg,Vd)

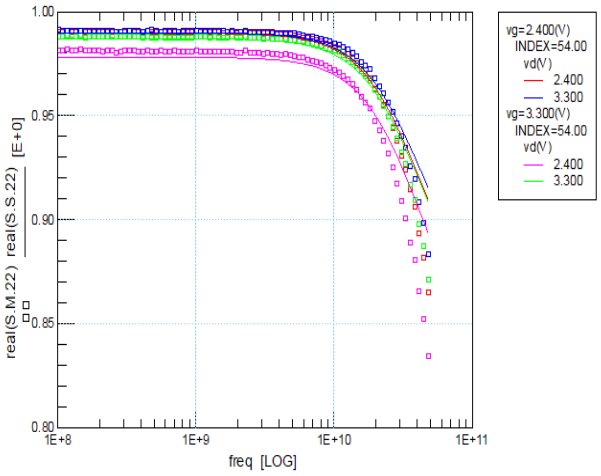


im(Y22) = f(Vg,Vd)

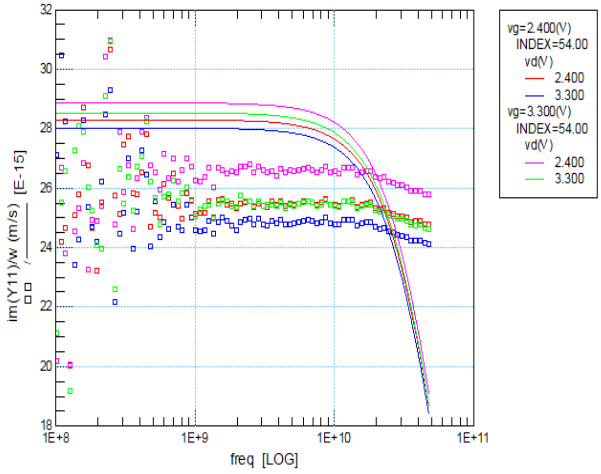
Verify Plots, T = 27°C



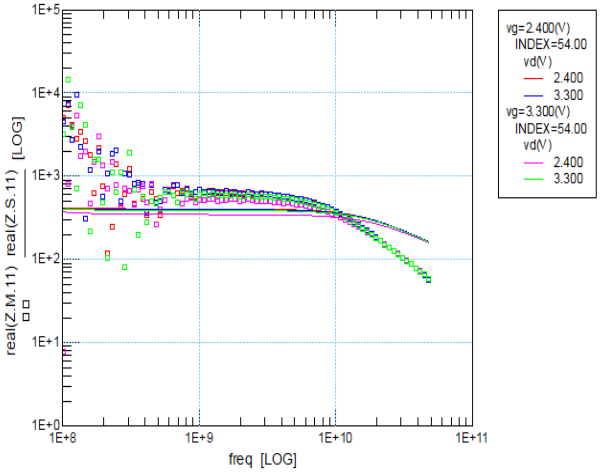
re(S11) = f(Vg,Vd)



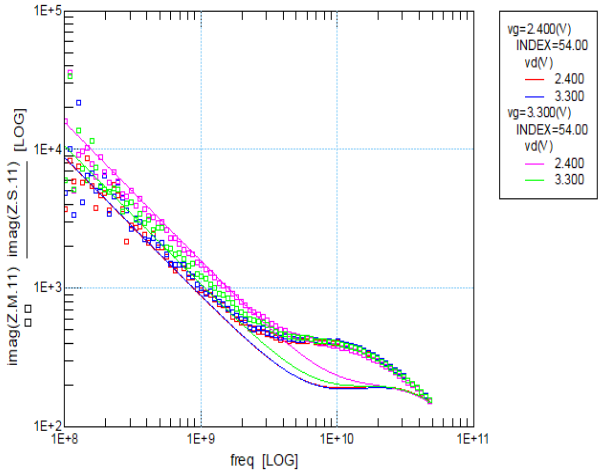
re(S22) = f(Vg,Vd)



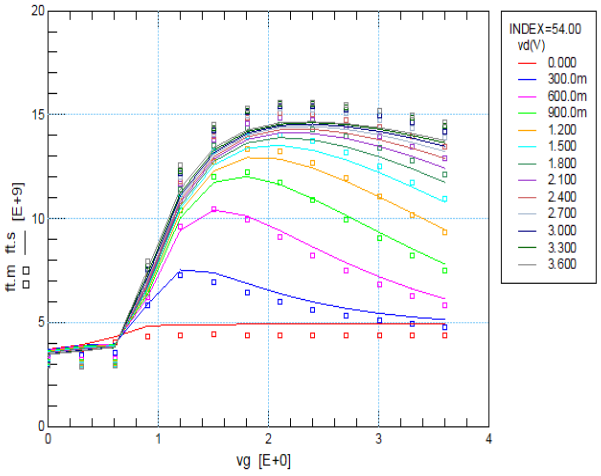
im(Y11) divided by ω



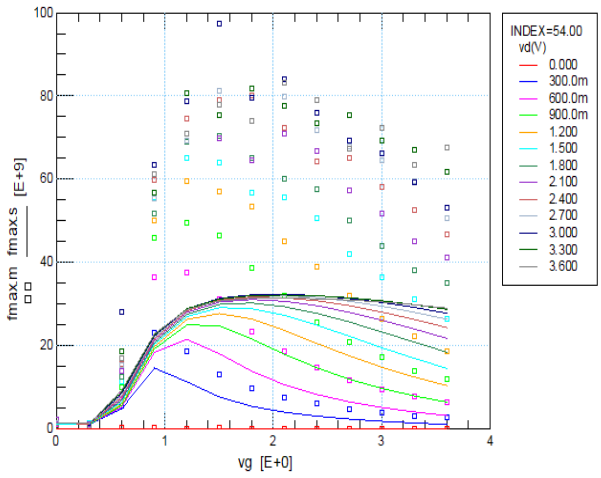
re(Z11) = f(Vg,Vd)



im(Z11) = f(Vg,Vd)



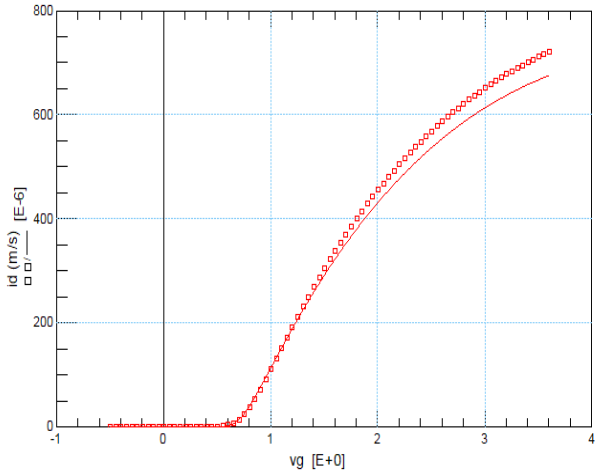
fT_vg = f(Vg,Vd)



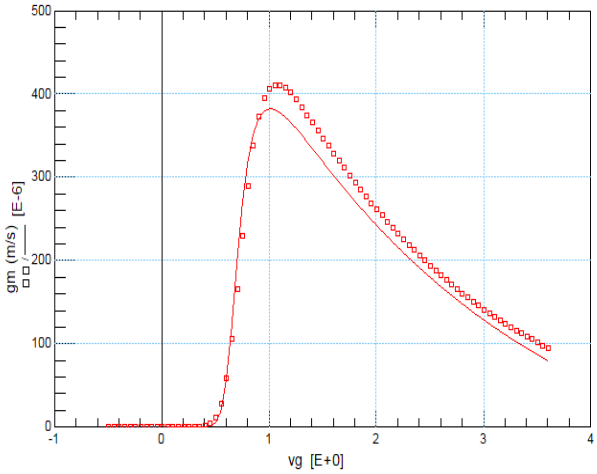
fmax_vg = f(Vg,Vd)

5.12 NF10_WF01u0_L_0u45_S453A_B4

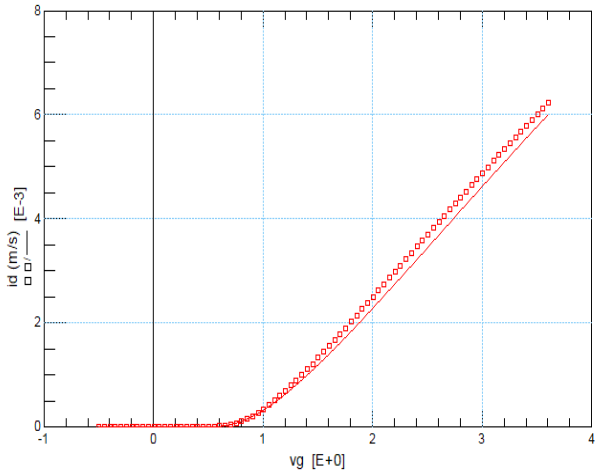
IdVg, T = 27°C



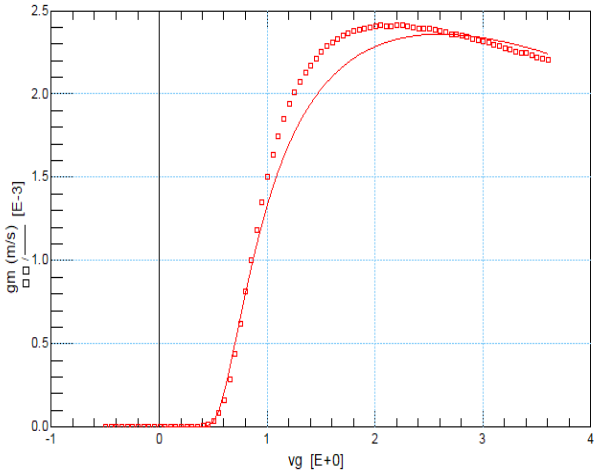
Id = f(Vg) @ Vd = 0.1V



gm = f(Vg) @ Vd = 0.1V

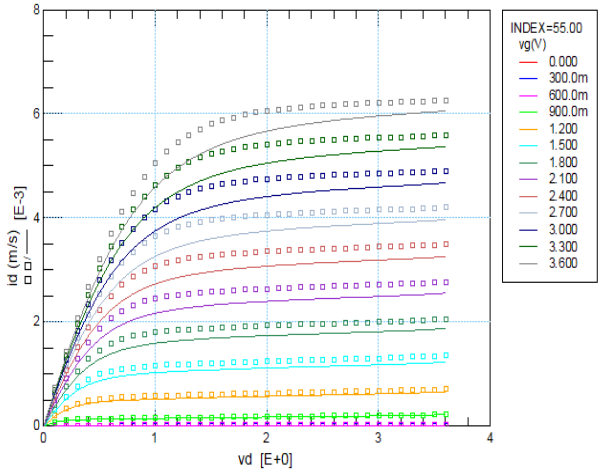


Id = f(Vg) @ Vd = 3.3V

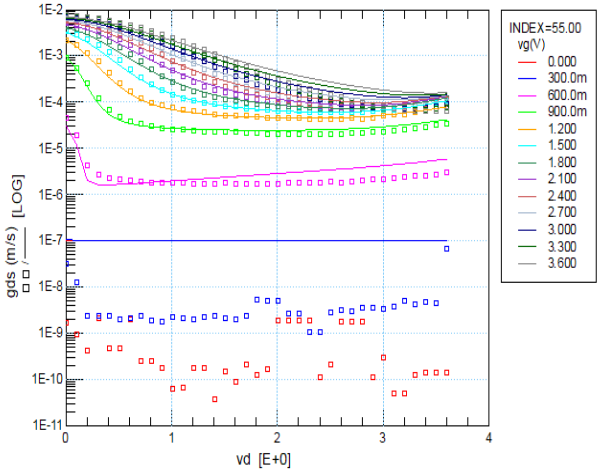


gm = f(Vg) @ Vd = 3.3V

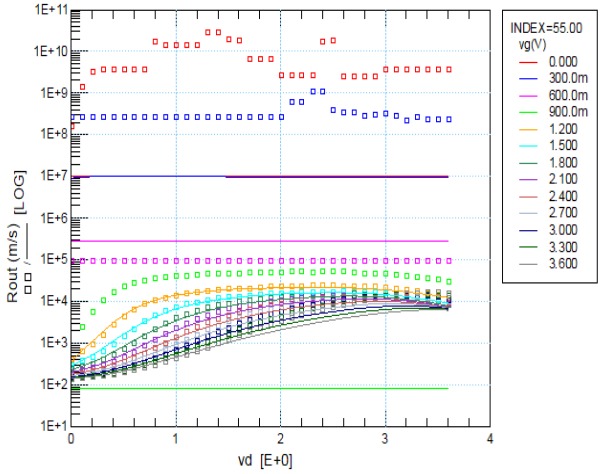
IdVd, T = 27°C



Id = f(Vd)

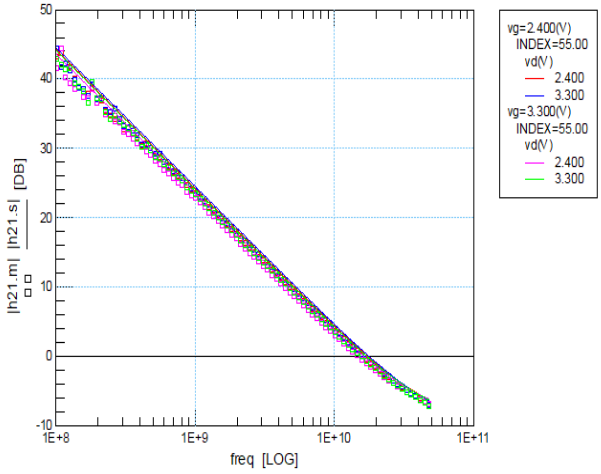


gds = f(Vd)

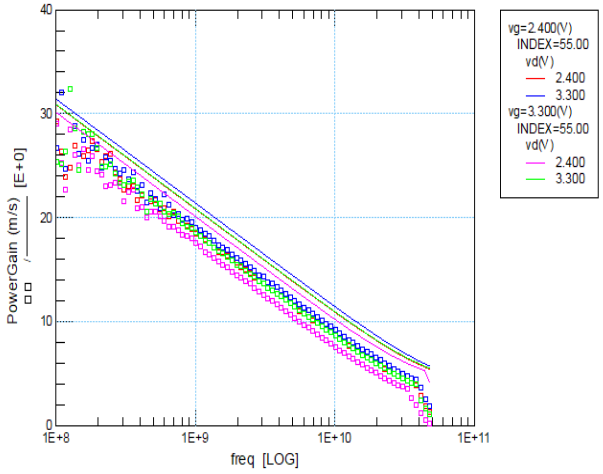


Rout = f(Vd)

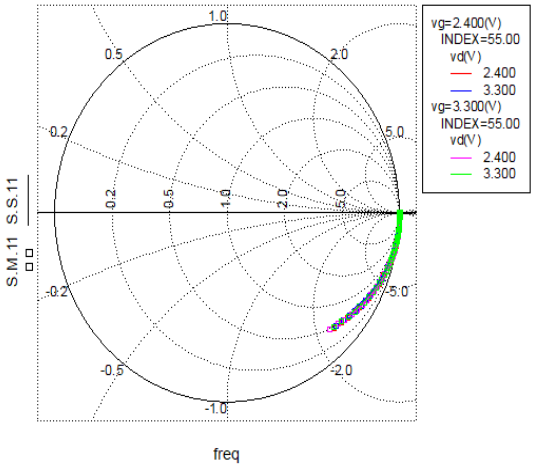
S-Parameter, T = 27°C



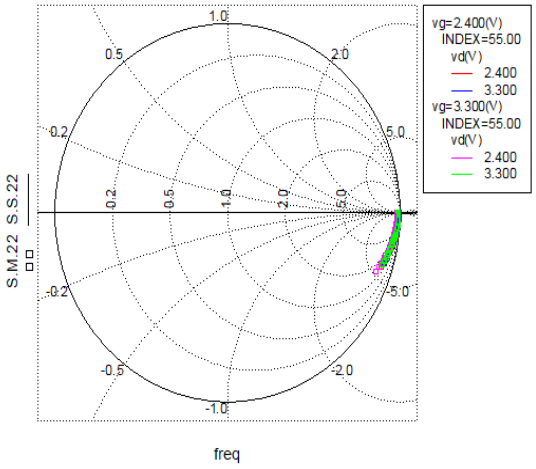
|h21| = f(Vg,Vd)



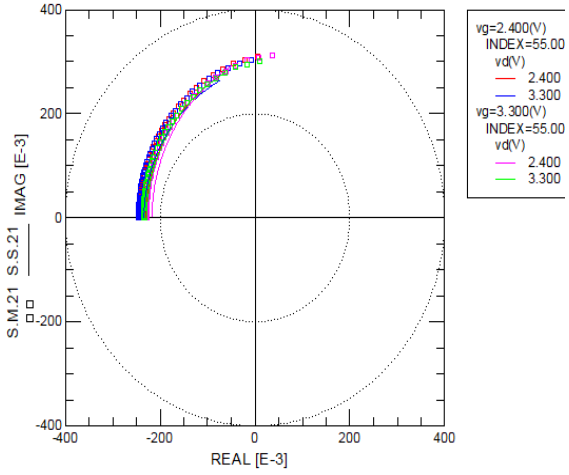
MSG/MAG = f(Vg,Vd)



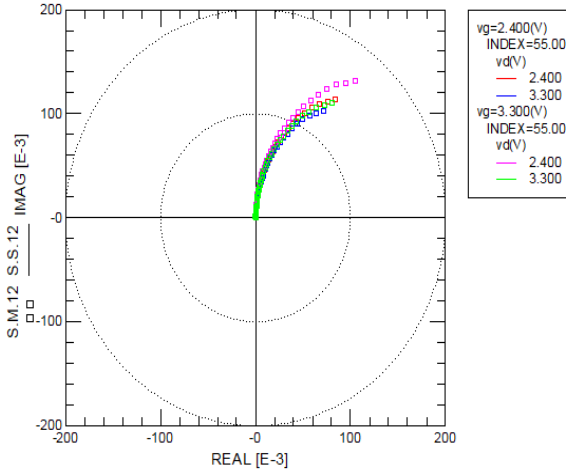
S11 = f(Vg,Vd)



S22 = f(Vg,Vd)

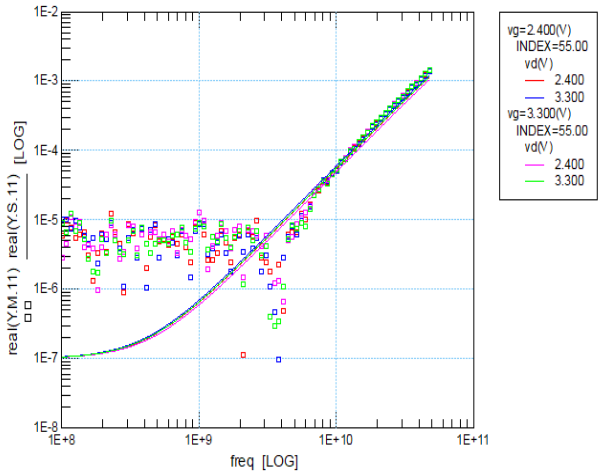


S21 = f(Vg,Vd)

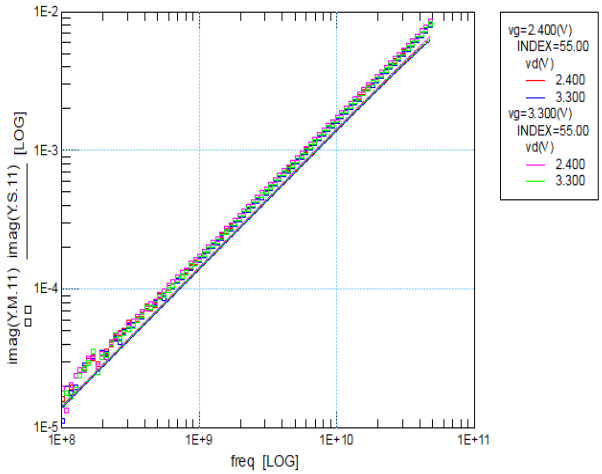


S12 = f(Vg,Vd)

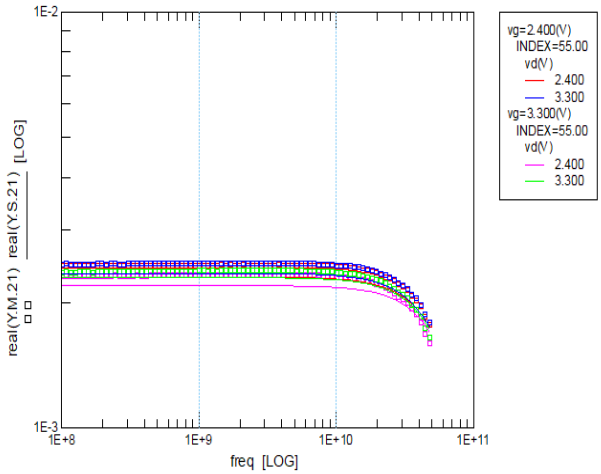
Y-Parameter, T = 27°C



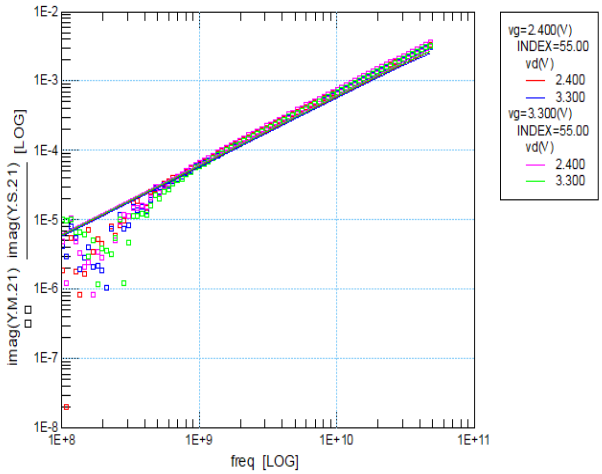
re(Y11) = f(Vg,Vd)



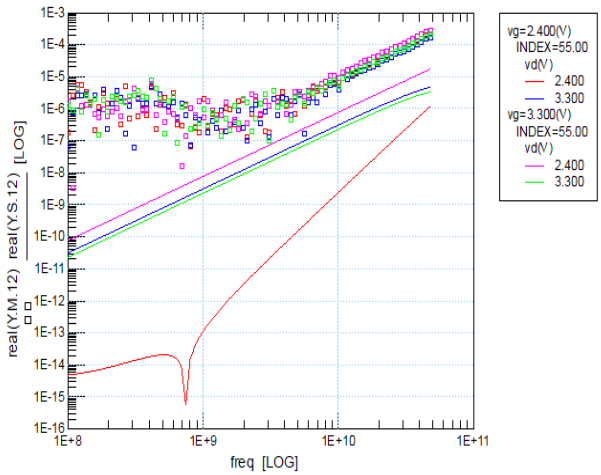
im(Y11) = f(Vg,Vd)



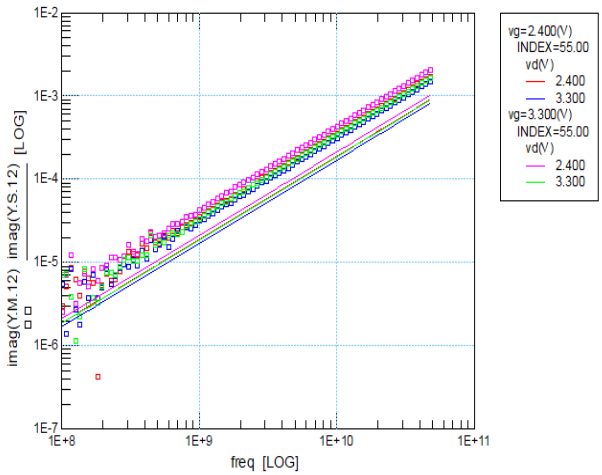
re(Y21) = f(Vg,Vd)



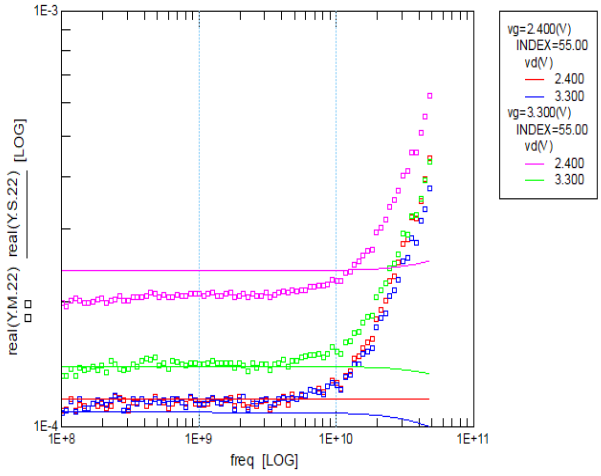
im(Y21) = f(Vg,Vd)



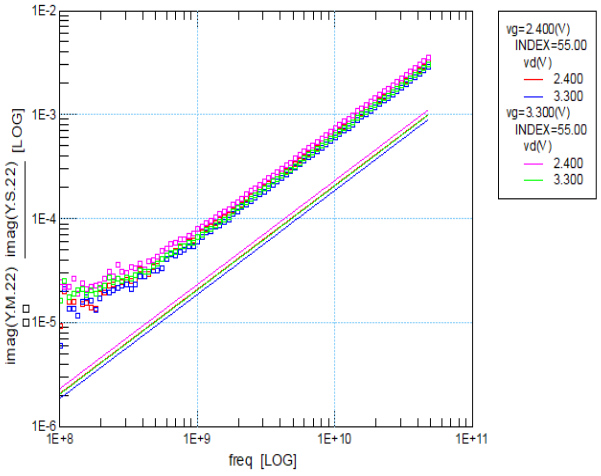
re(Y12) = f(Vg,Vd)



im(Y12) = f(Vg,Vd)

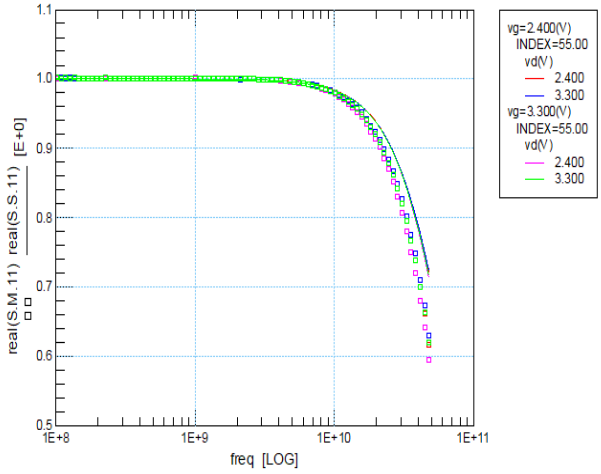


re(Y22) = f(Vg,Vd)

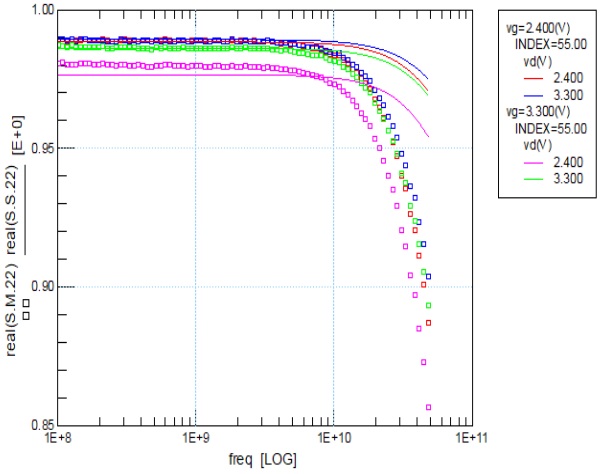


im(Y22) = f(Vg,Vd)

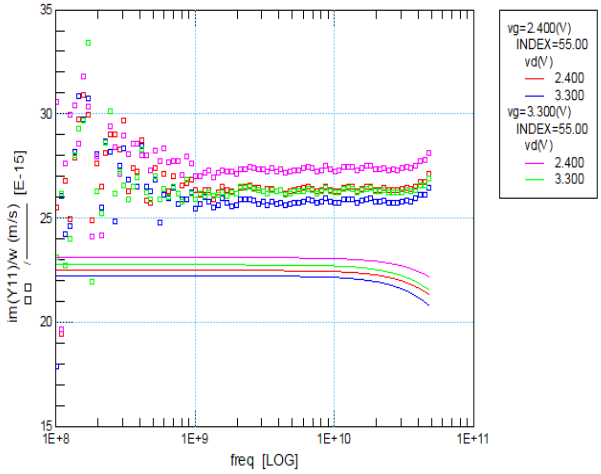
Verify Plots, T = 27°C



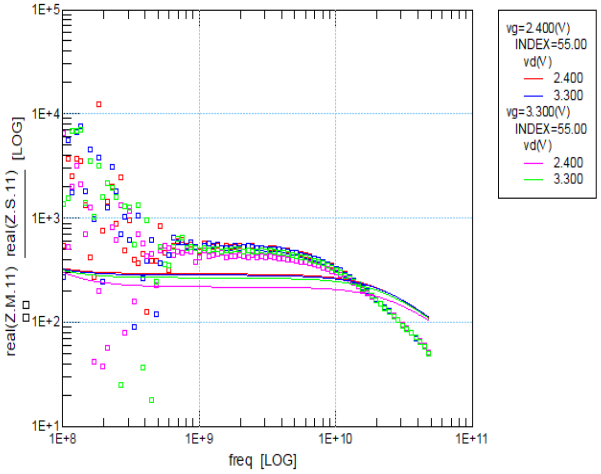
re(S11) = f(Vg,Vd)



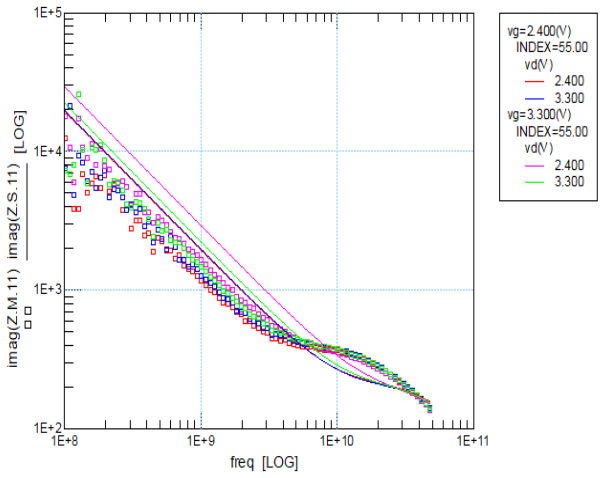
re(S22) = f(Vg,Vd)



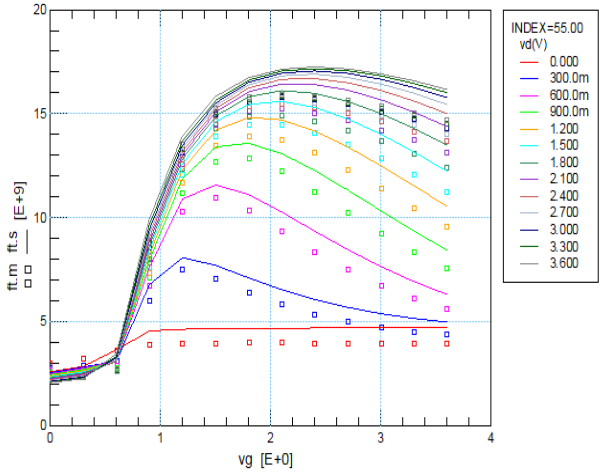
im(Y11) divided by ω



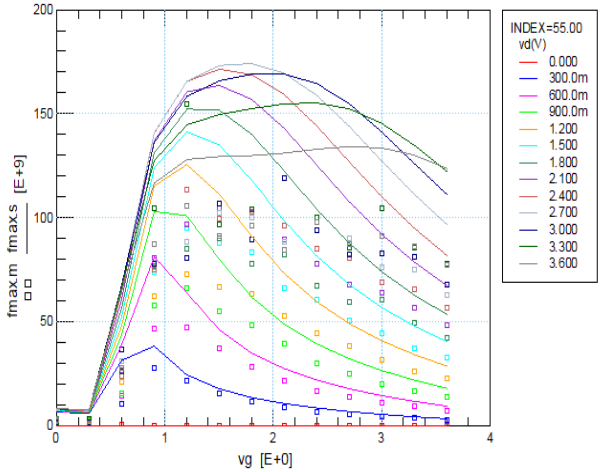
re(Z11) = f(Vg,Vd)



im(Z11) = f(Vg,Vd)



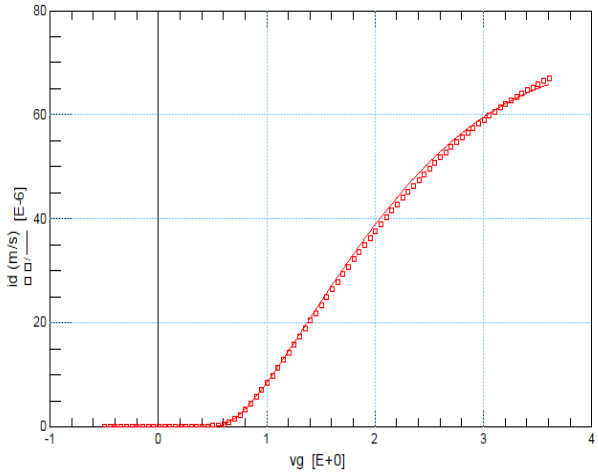
fT_vg = f(Vg,Vd)



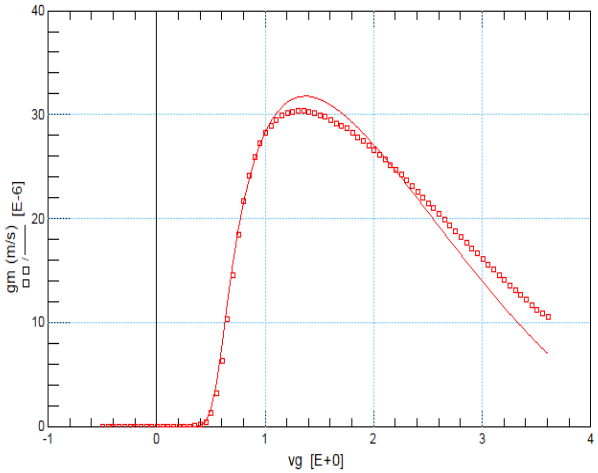
fmax_vg = f(Vg,Vd)

5.13 NF04_WF05u0_L_10u0_S453A_14

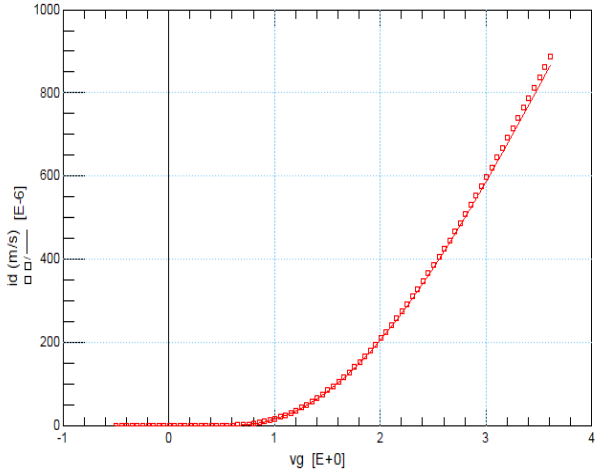
IdVg, T = 27°C



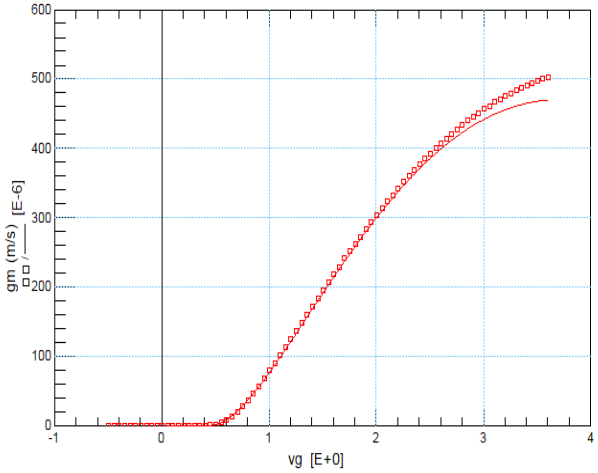
Id = f(Vg) @ Vd = 0.1V



gm = f(Vg) @ Vd = 0.1V

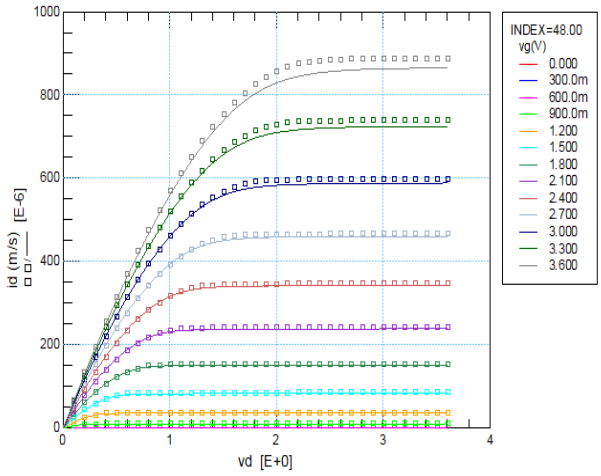


Id = f(Vg) @ Vd = 3.3V

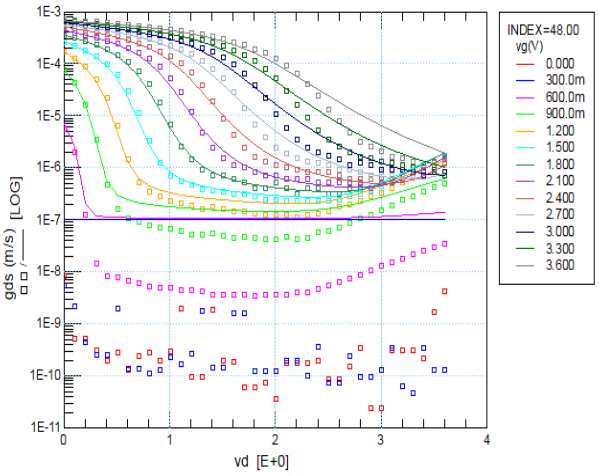


gm = f(Vg) @ Vd = 3.3V

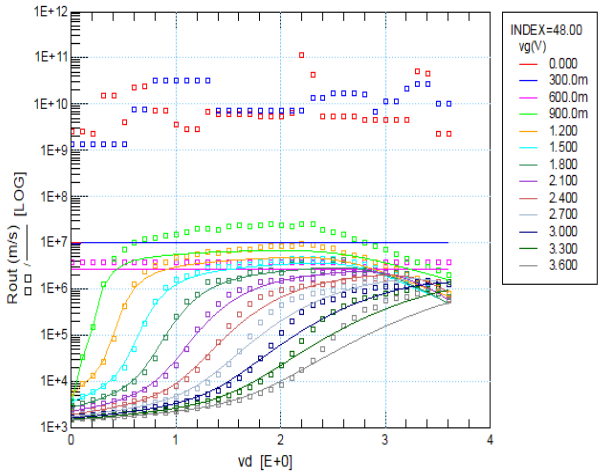
IdVd, T = 27°C



Id = f(Vd)

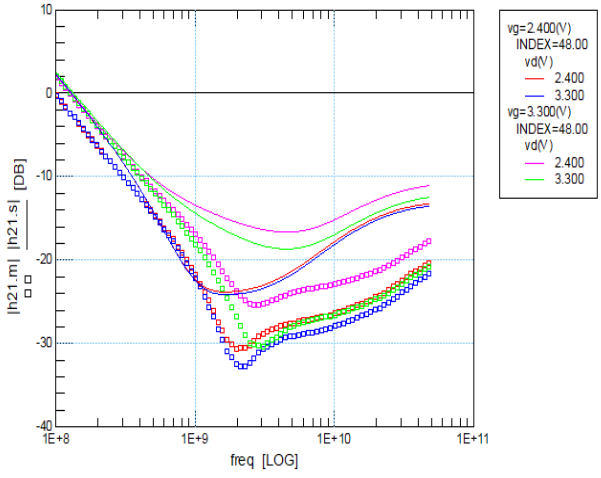


gds = f(Vd)

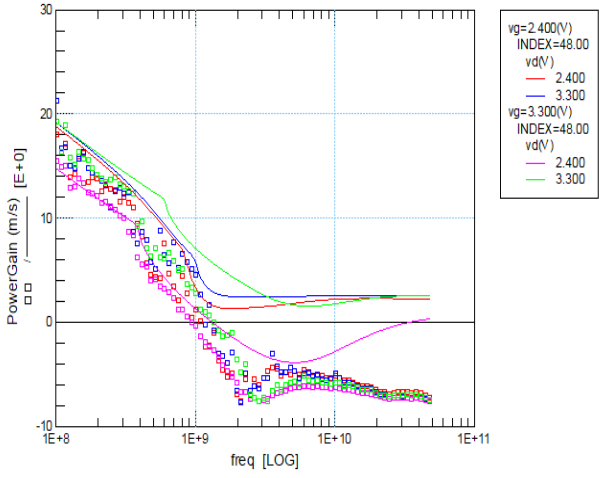


Rout = f(Vd)

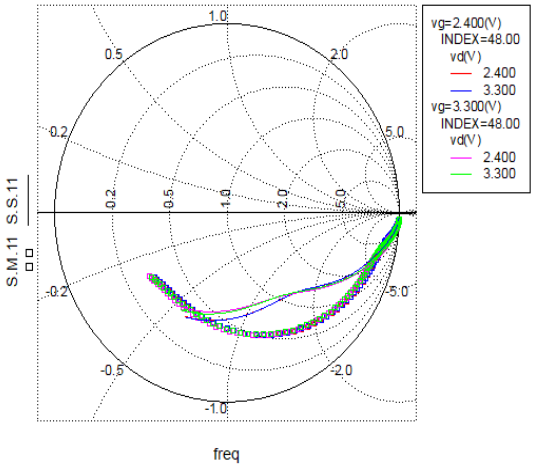
S-Parameter, T = 27°C



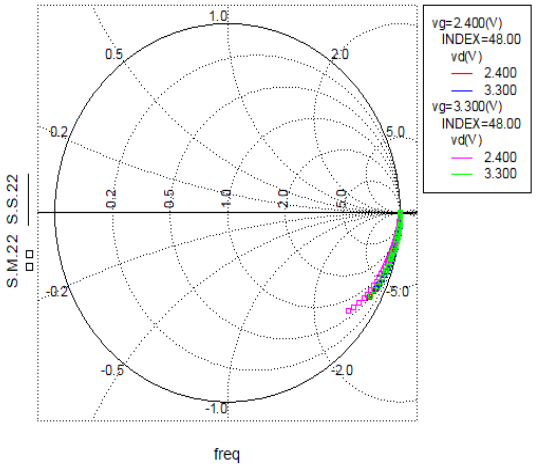
|h21| = f(Vg,Vd)



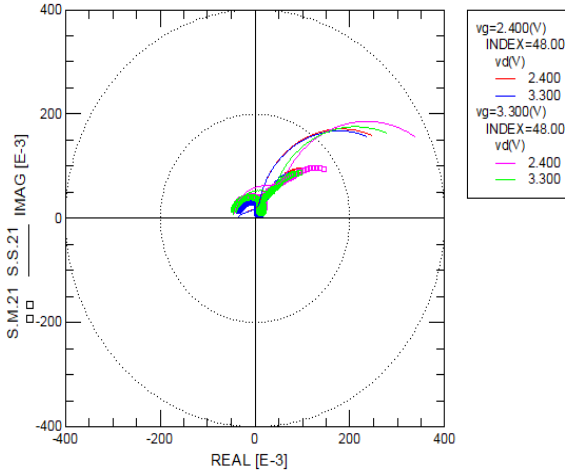
MSG/MAG = f(Vg,Vd)



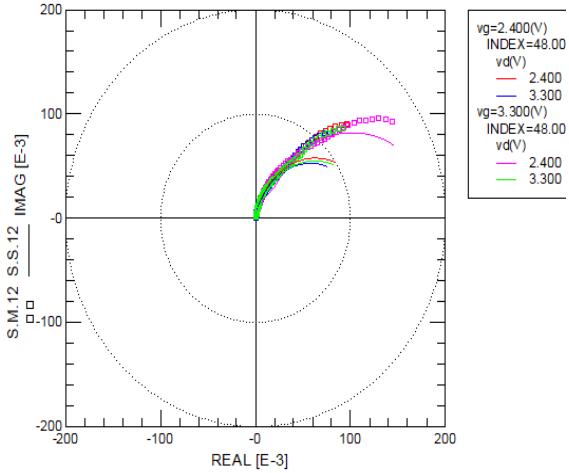
S11 = f(Vg,Vd)



S22 = f(Vg,Vd)

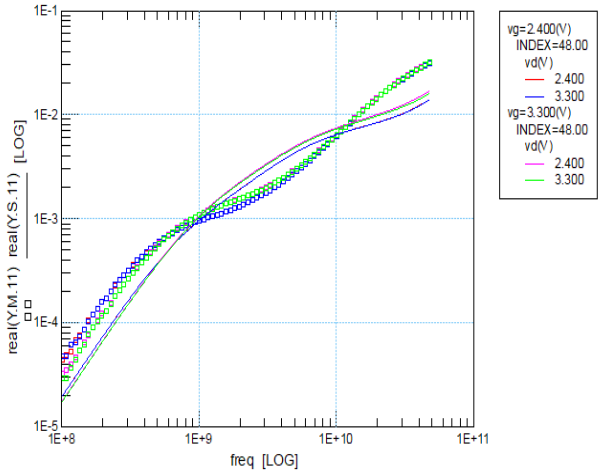


S21 = f(Vg,Vd)

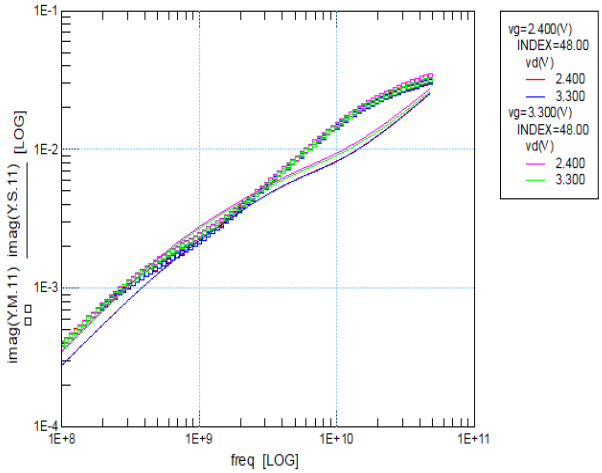


S12 = f(Vg,Vd)

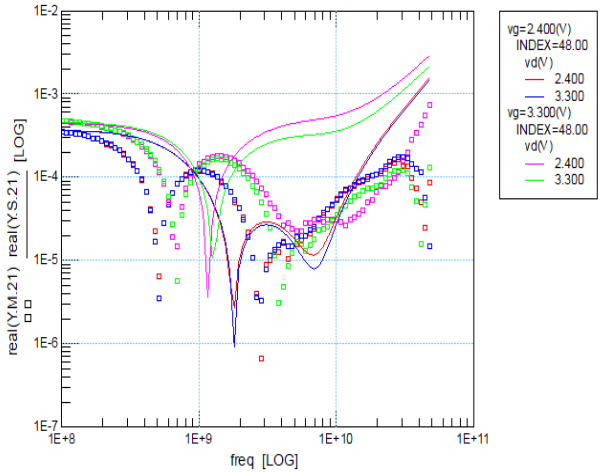
Y-Parameter, T = 27°C



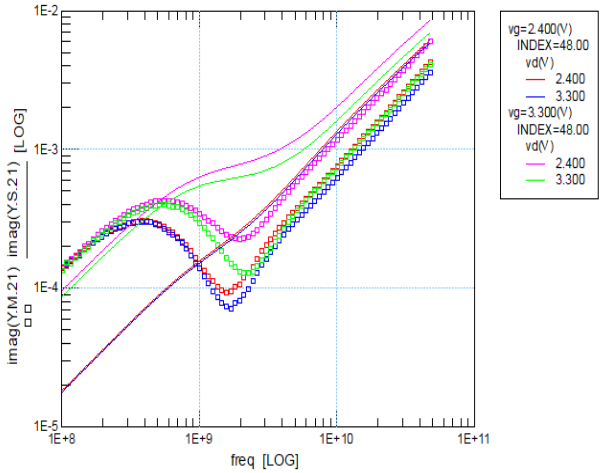
re(Y11) = f(Vg,Vd)



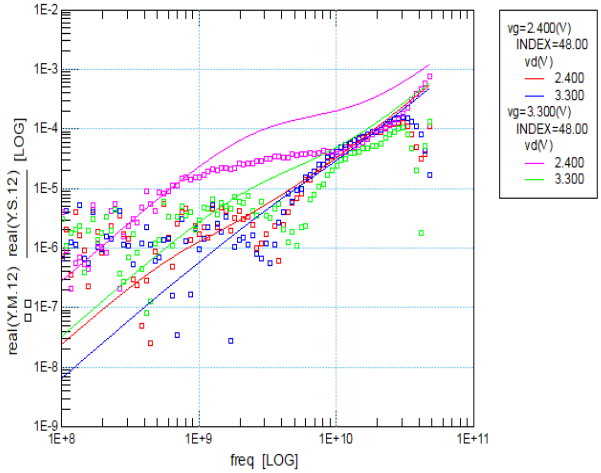
im(Y11) = f(Vg,Vd)



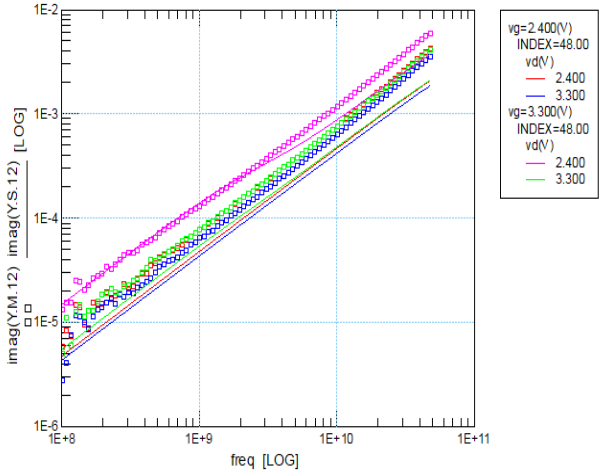
re(Y21) = f(Vg,Vd)



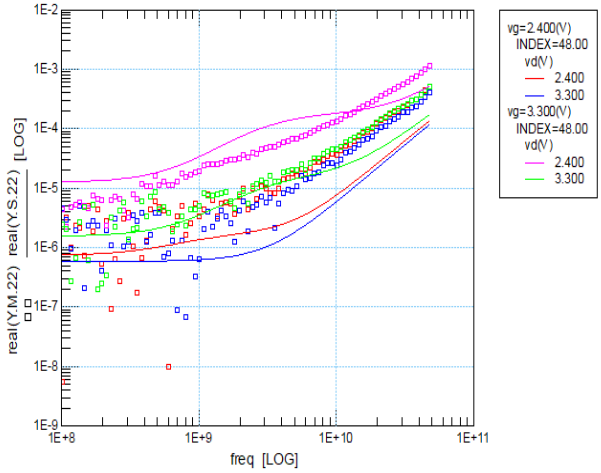
im(Y21) = f(Vg,Vd)



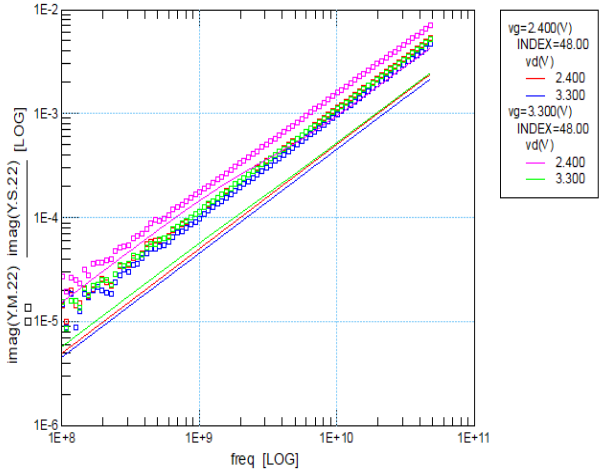
re(Y12) = f(Vg,Vd)



im(Y12) = f(Vg,Vd)

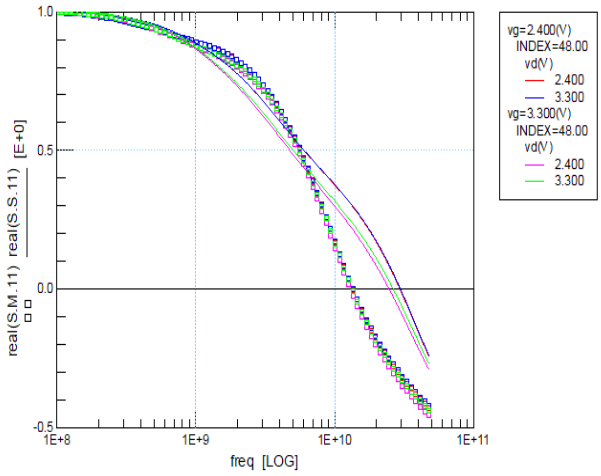


re(Y22) = f(Vg,Vd)

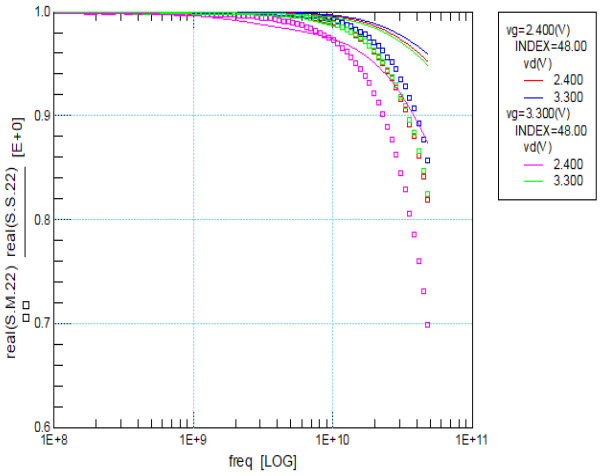


im(Y22) = f(Vg,Vd)

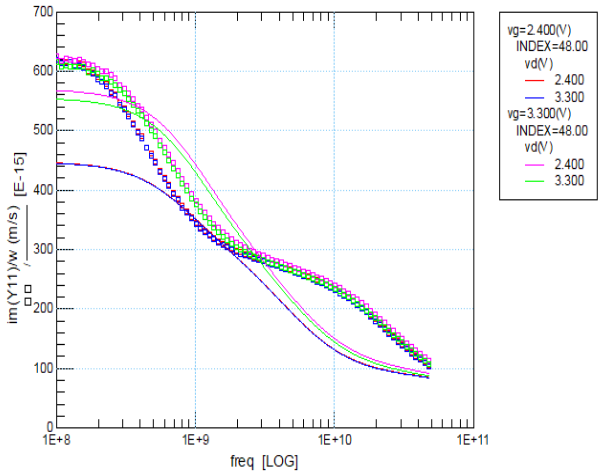
Verify Plots, T = 27°C



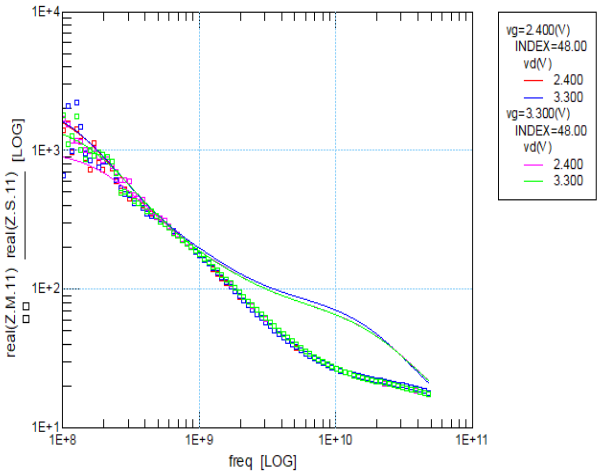
re(S11) = f(Vg,Vd)



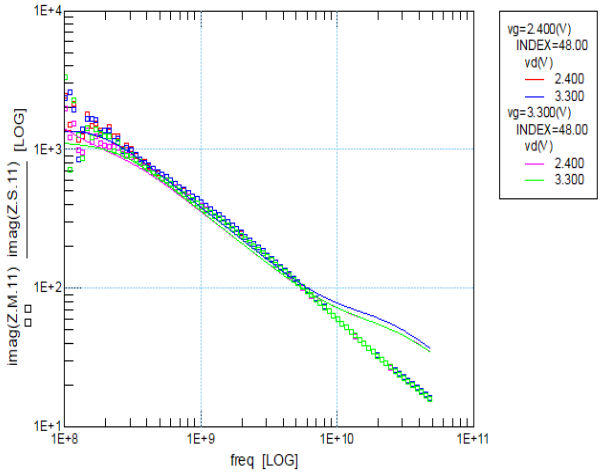
re(S22) = f(Vg,Vd)



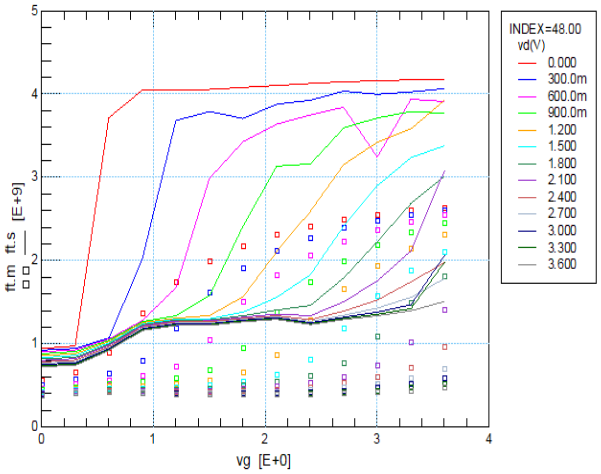
im(Y11) divided by ω



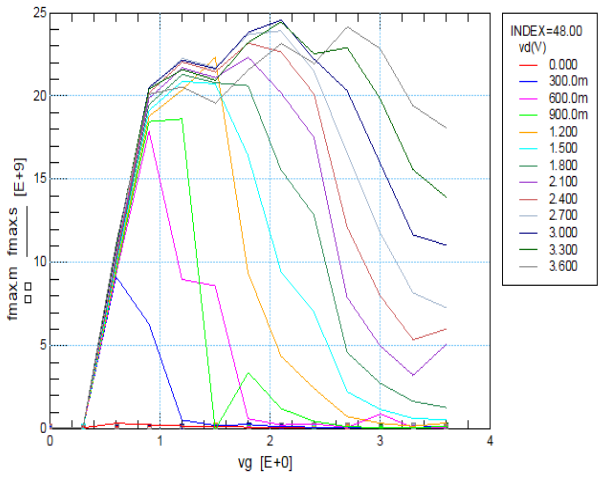
re(Z11) = f(Vg,Vd)



im(Z11) = f(Vg,Vd)



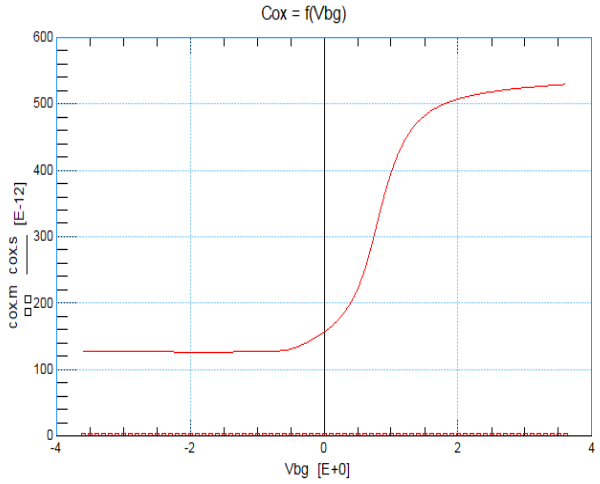
fT_vg = f(Vg,Vd)



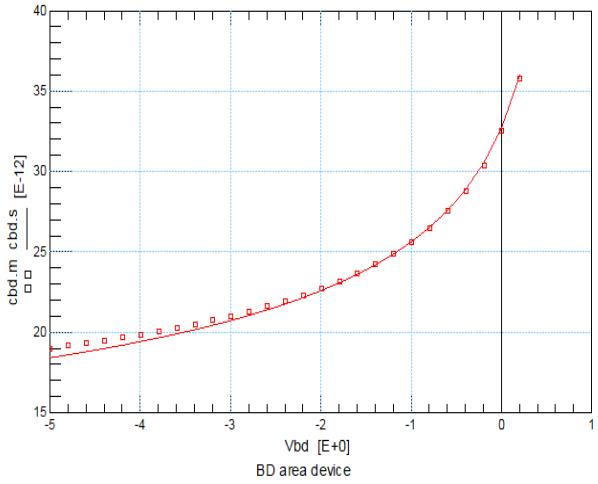
fmax_vg = f(Vg,Vd)

6 Capacitances

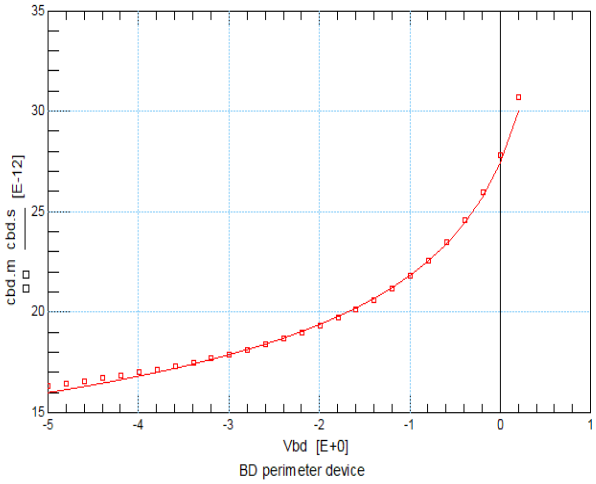
6.1 Gate Oxide Capacitance



6.2 Junction Capacitance Bulk - Drain

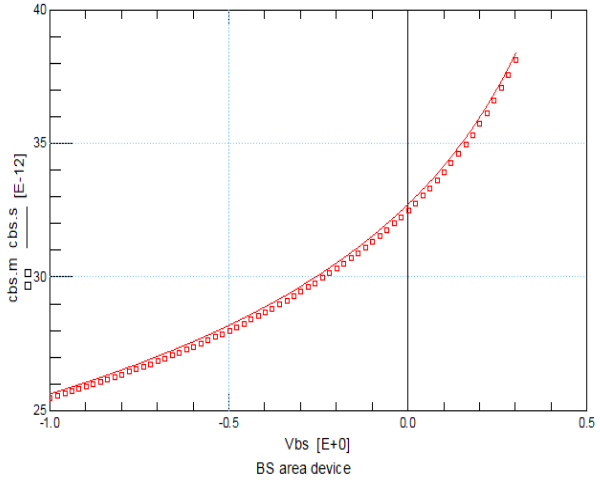


Area

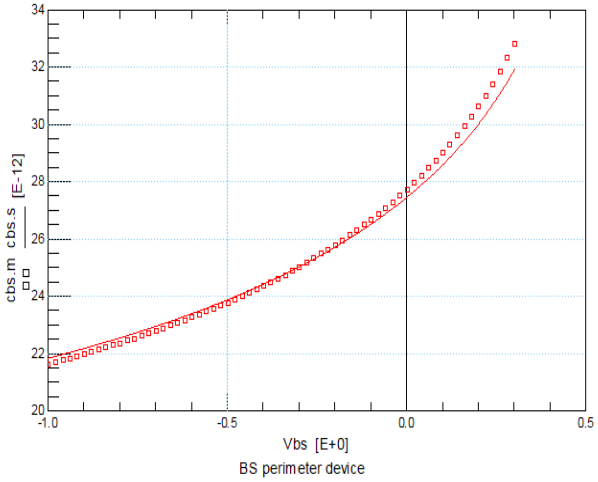


Perimeter

6.3 Junction Capacitance Bulk - Source

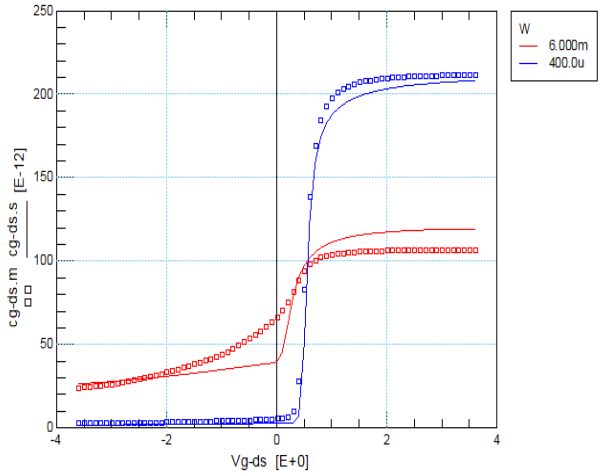


Area

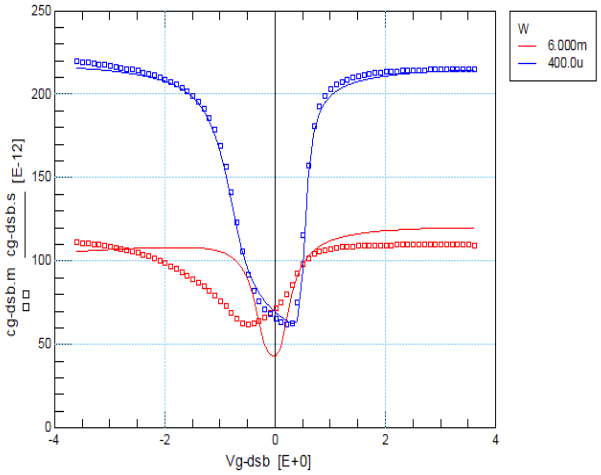


Perimeter

6.4 Overlap Capacitance



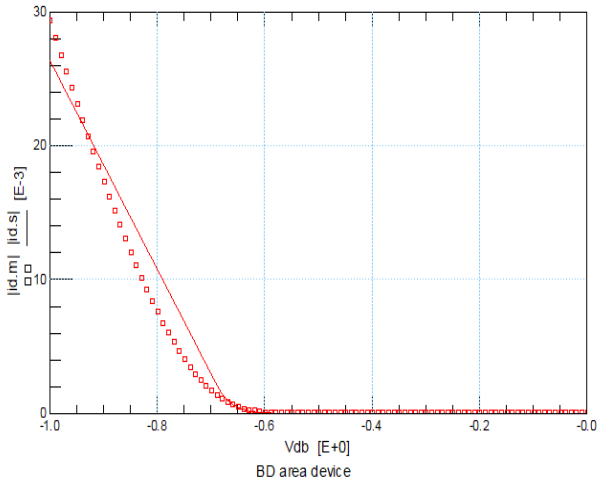
Gate - Drain,Source



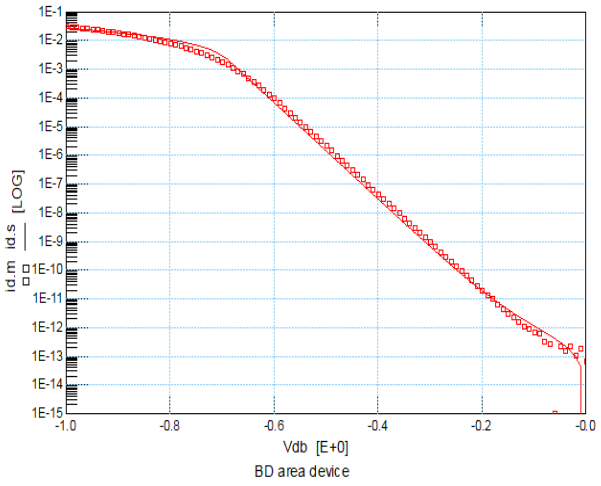
Gate - Drain,Source,Bulk

7 Diodes

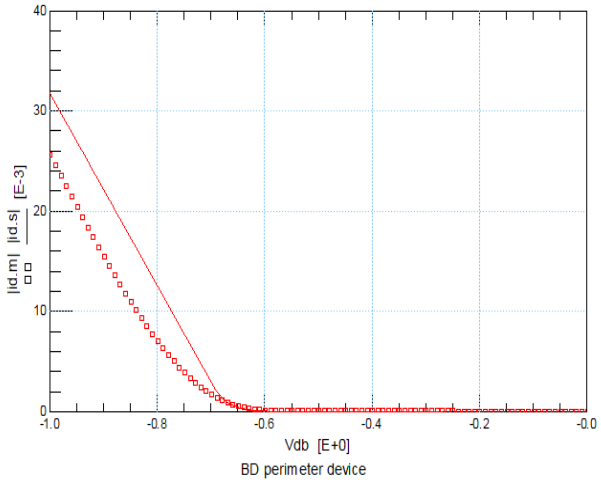
7.1 Diode Bulk - Drain



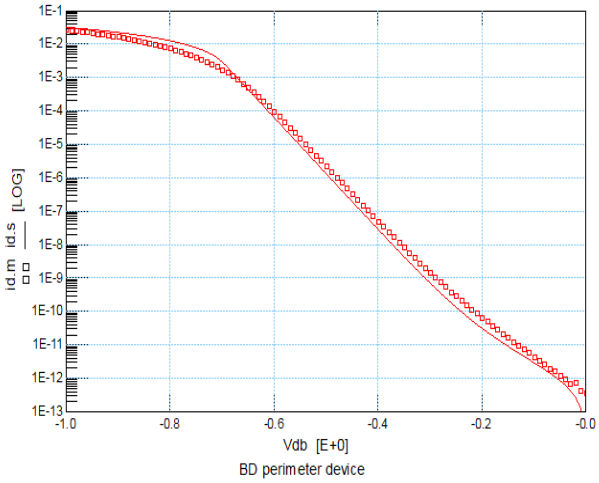
Area



Area

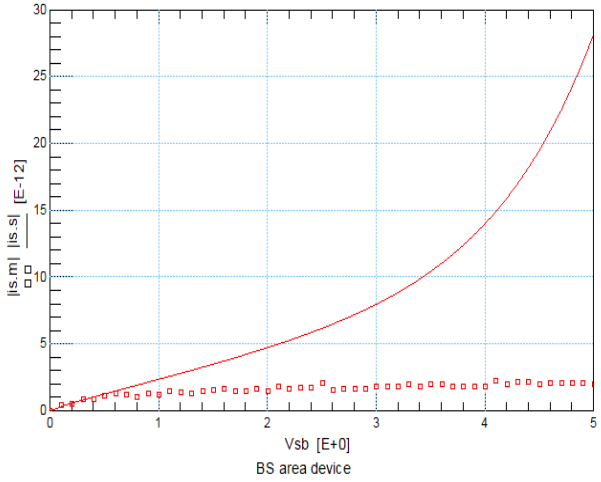


Perimeter

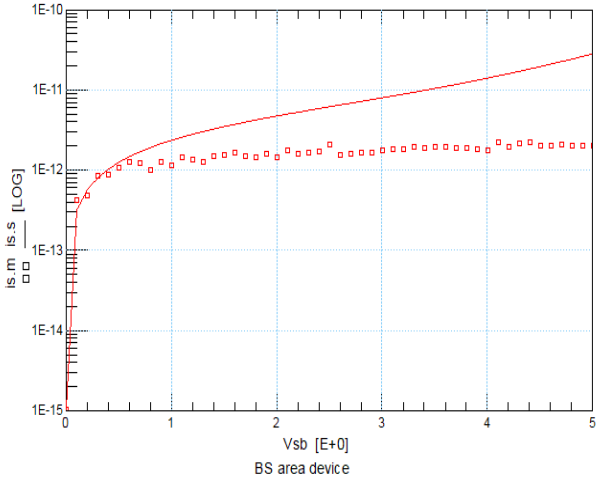


Perimeter

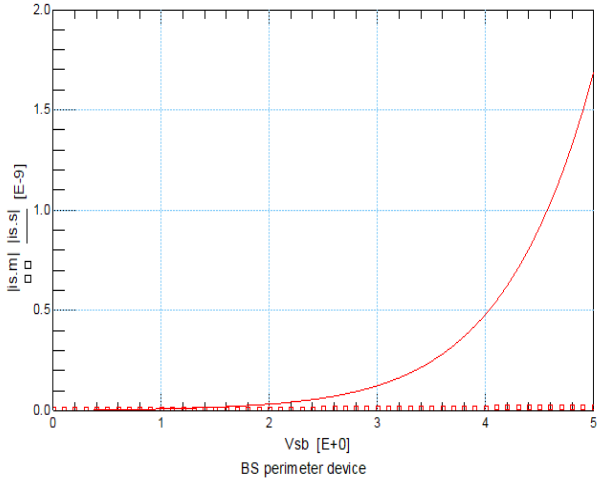
7.2 Diode Bulk - Source



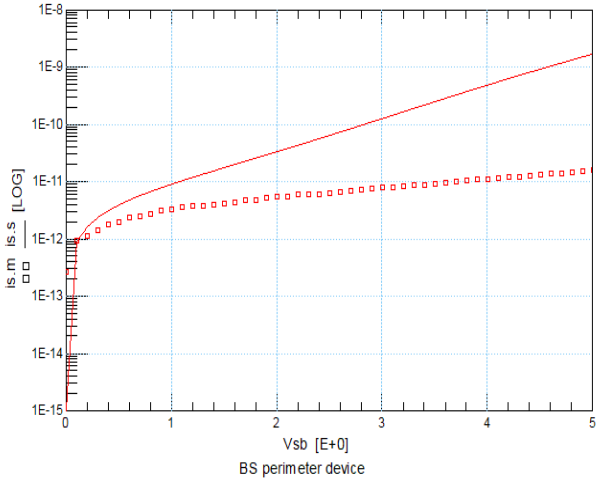
Area



Area



Perimeter

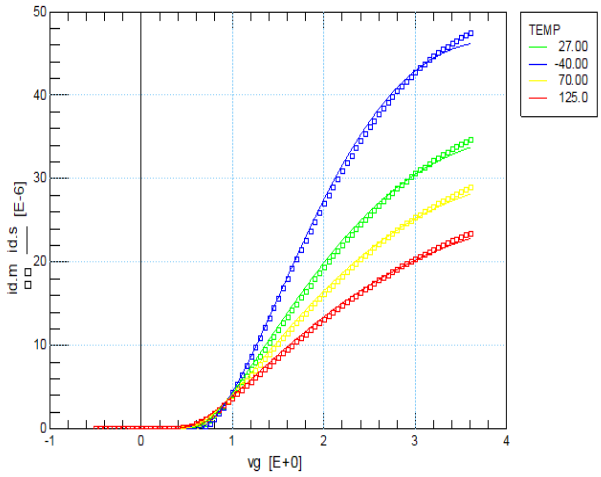


Perimeter

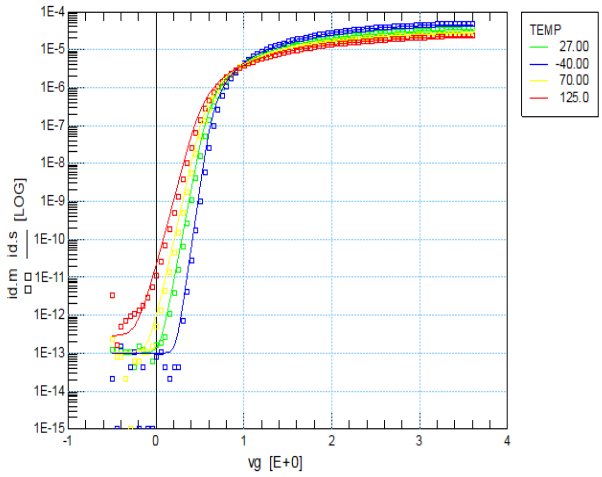
8 Temperature Measurement

8.1 Transistors

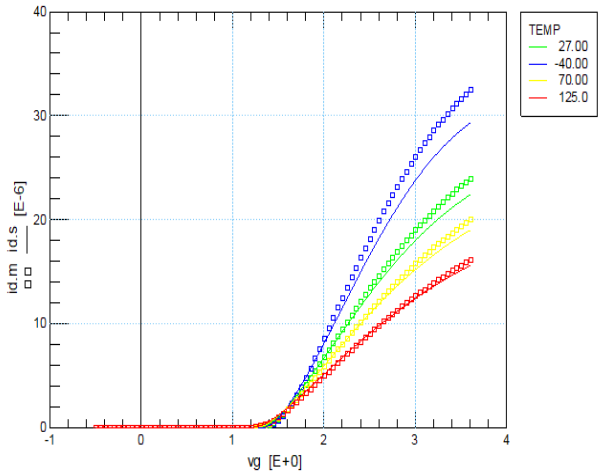
8.1.1 W10u0_L10u0_S557_3



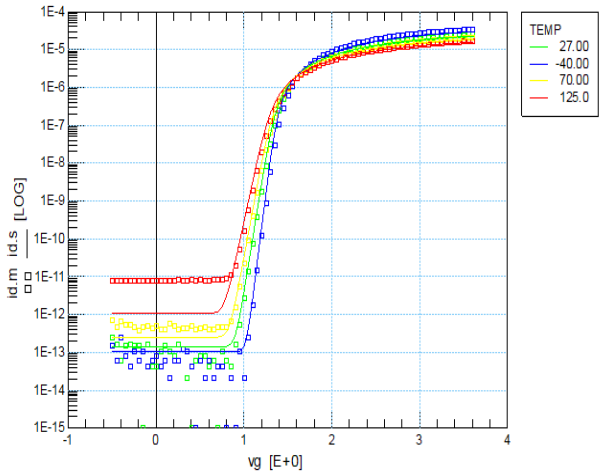
Id = f (vg, T) @ low Vds, Vbs = 0



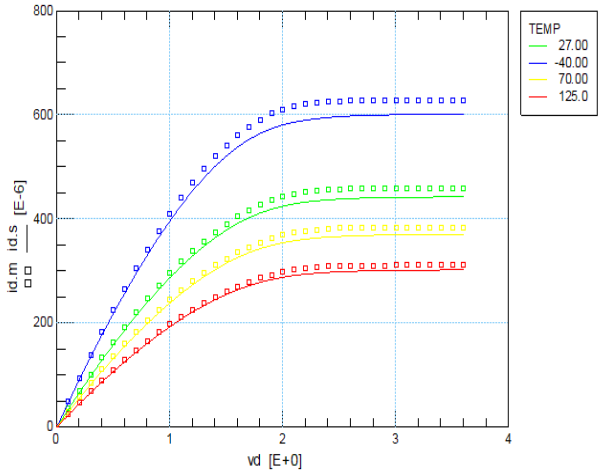
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max

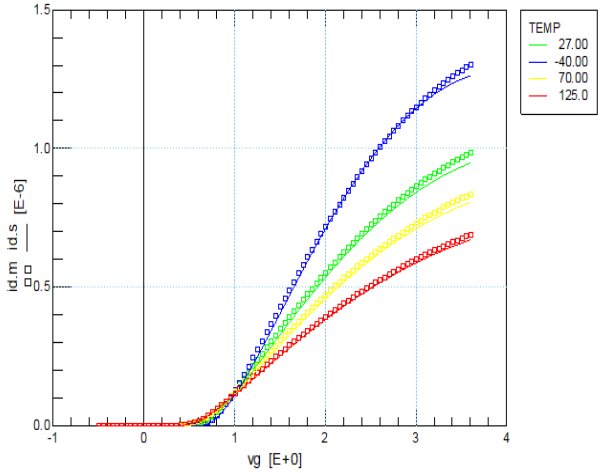


log(Id) = f (vg, T) @ low Vds, Vbs = max

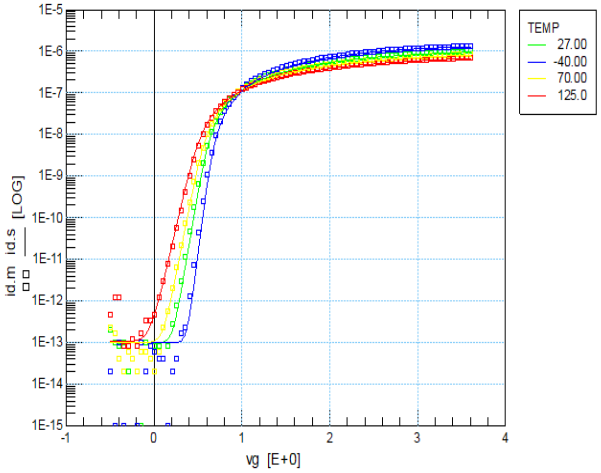


Id = f (Vd, Vg, T) @ Vbs = 0

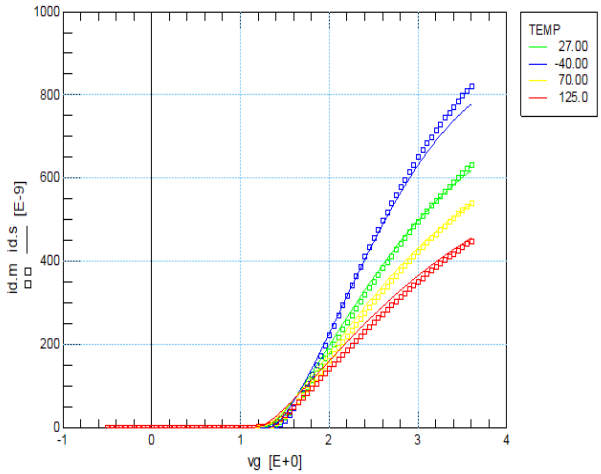
8.1.2 W0u3_L10u0_S547_4



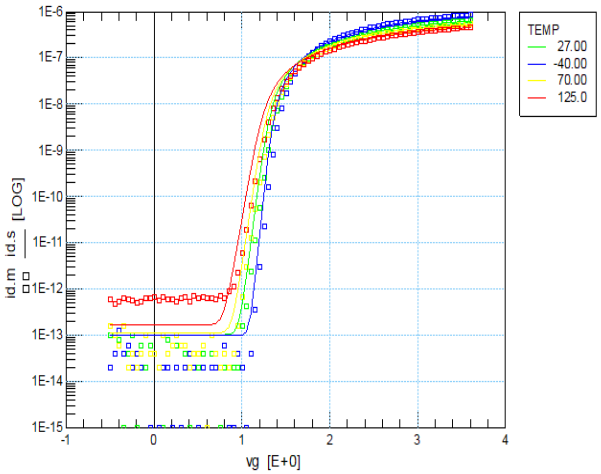
Id = f (vg, T) @ low Vds, Vbs = 0



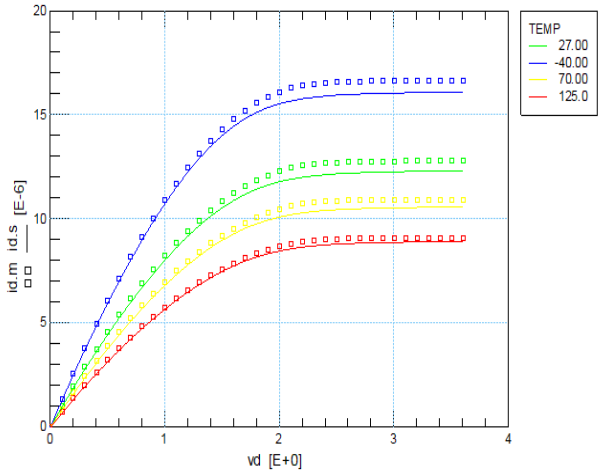
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max

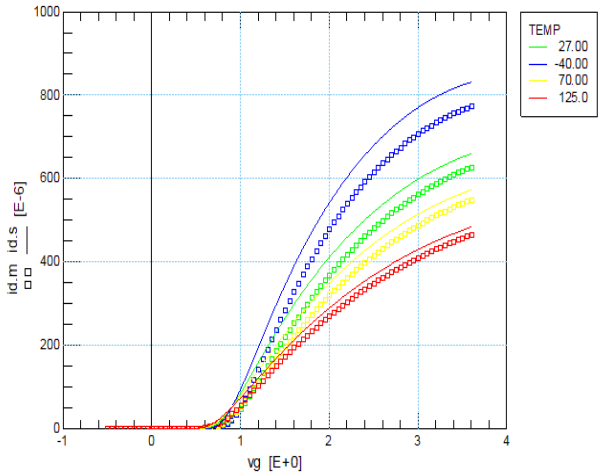


log(Id) = f (vg, T) @ low Vds, Vbs = max

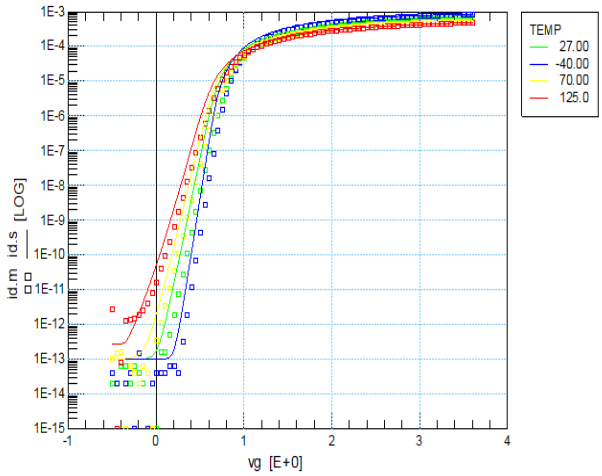


Id = f (Vd, Vg, T) @ Vbs = 0

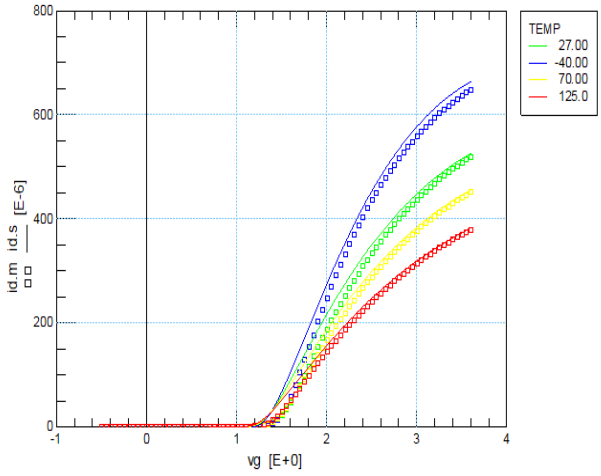
8.1.3 W10u0_L0u5_S556_4



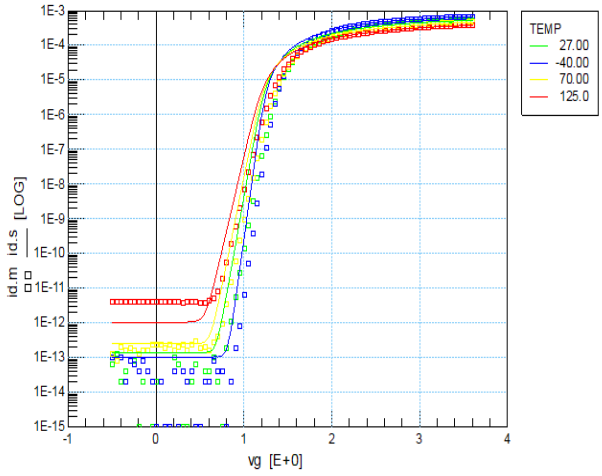
Id = f (vg, T) @ low Vds, Vbs = 0



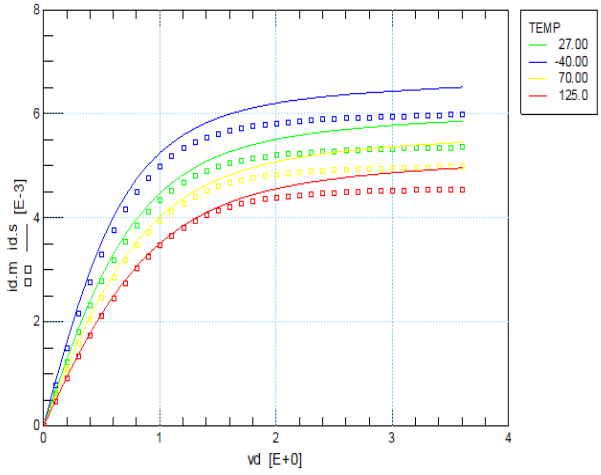
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max

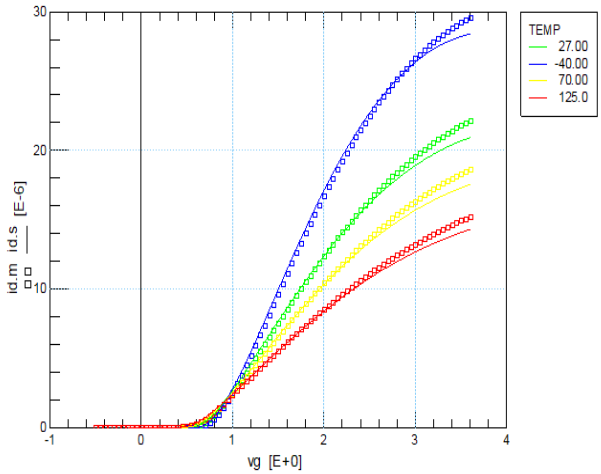


log(Id) = f (vg, T) @ low Vds, Vbs = max

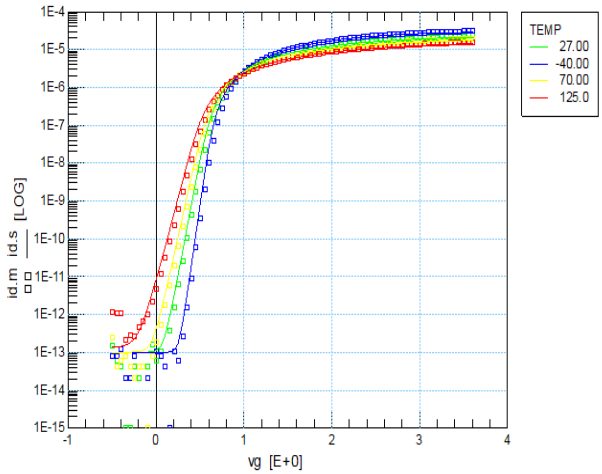


Id = f (Vd, Vg, T) @ Vbs = 0

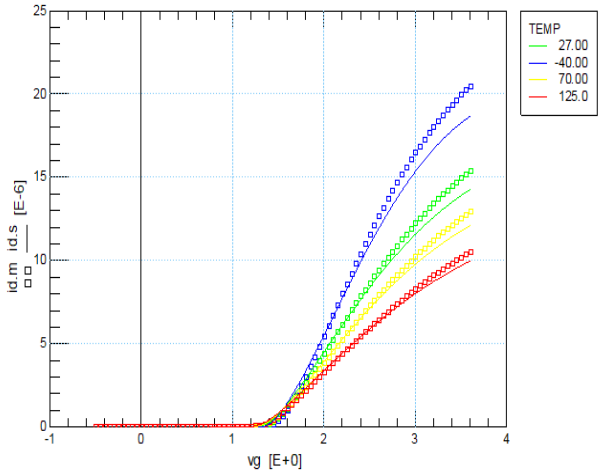
8.1.4 W02u0_L3u0_S558_4



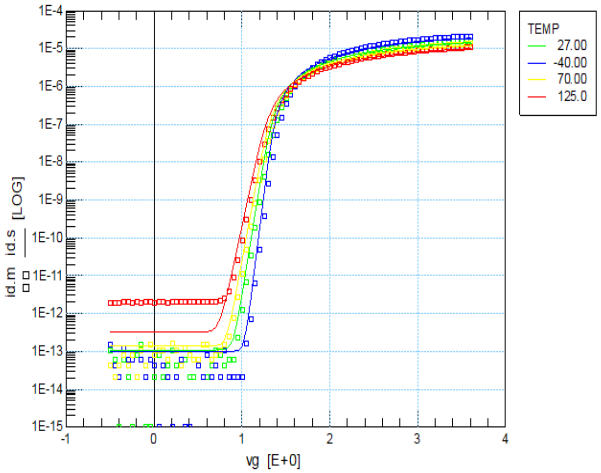
Id = f (vg, T) @ low Vds, Vbs = 0



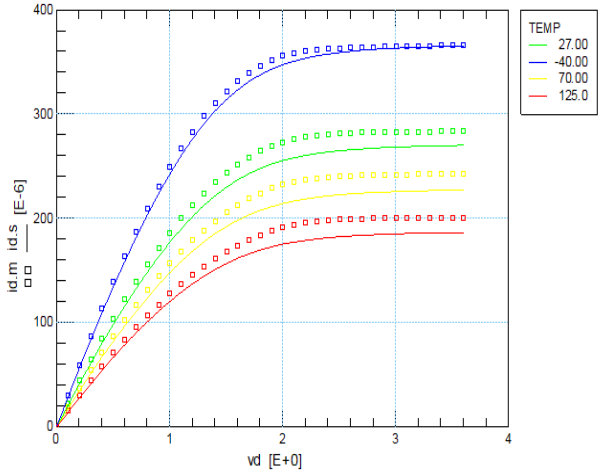
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max

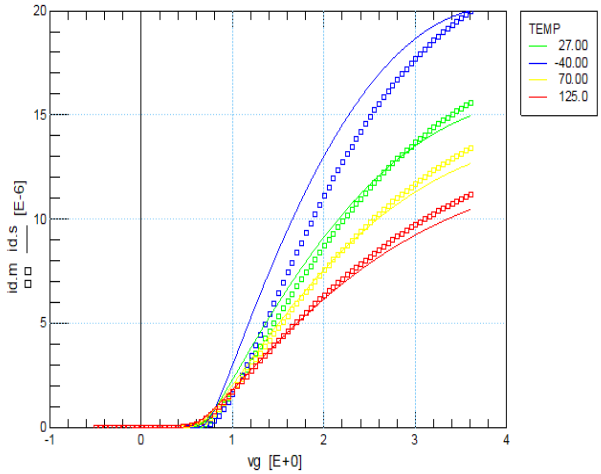


log(Id) = f (vg, T) @ low Vds, Vbs = max

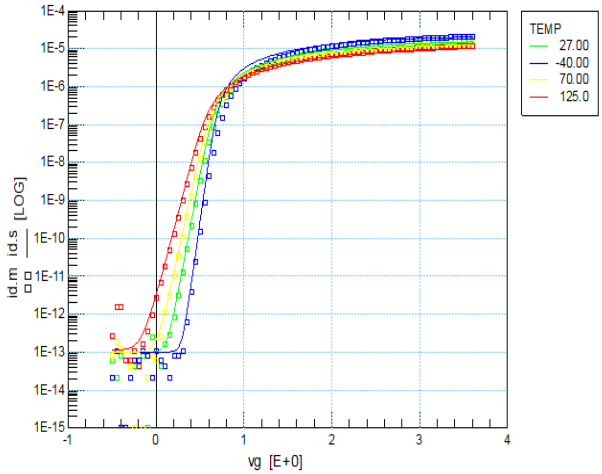


Id = f (Vd, Vg, T) @ Vbs = 0

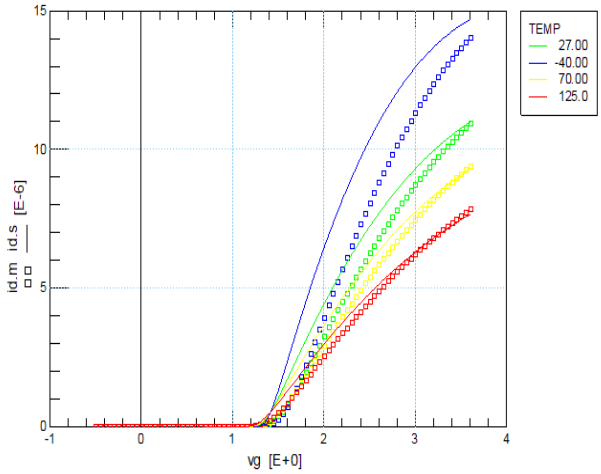
8.1.5 W0u6_L1u2_S559_4



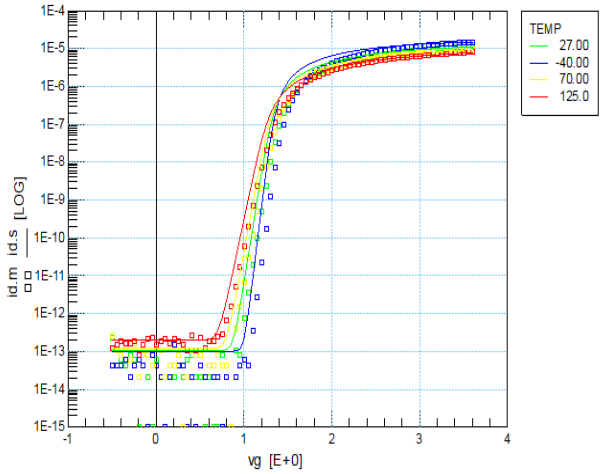
Id = f (vg, T) @ low Vds, Vbs = 0



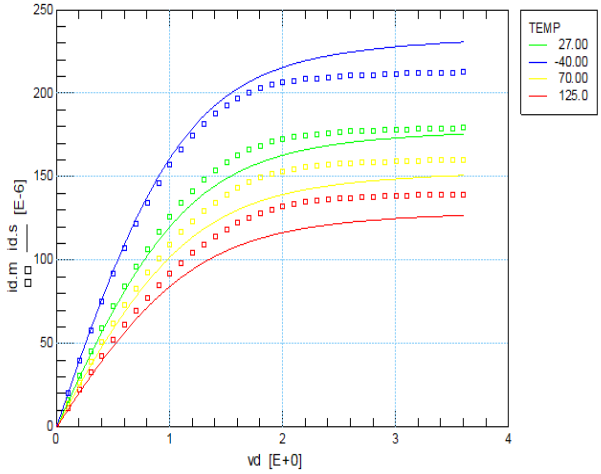
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max

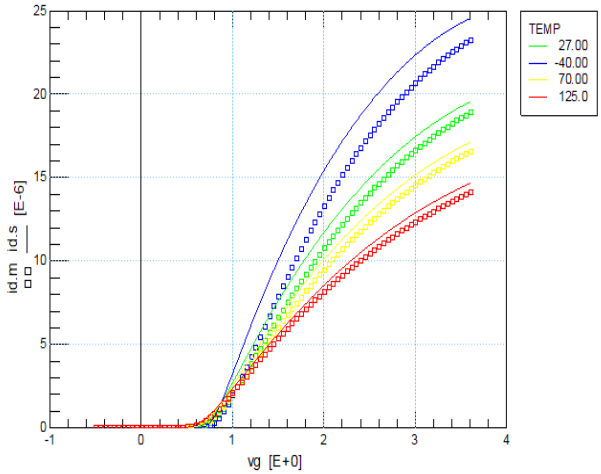


log(Id) = f (vg, T) @ low Vds, Vbs = max

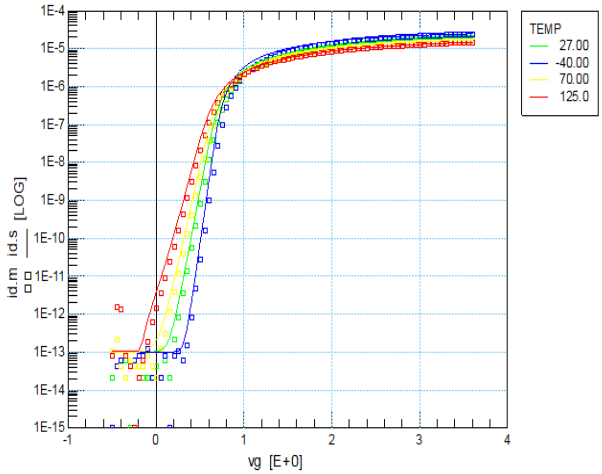


Id = f (Vd, Vg, T) @ Vbs = 0

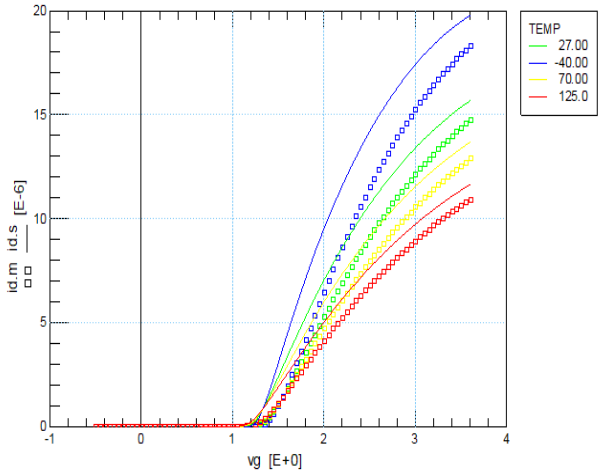
8.1.6 W0u3_L0u5_S560_4



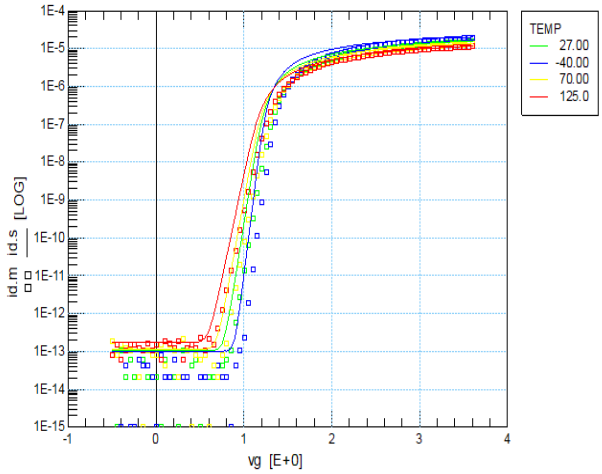
Id = f (vg, T) @ low Vds, Vbs = 0



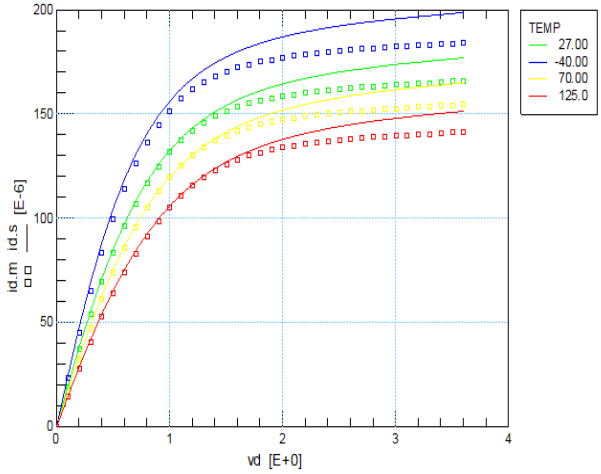
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max



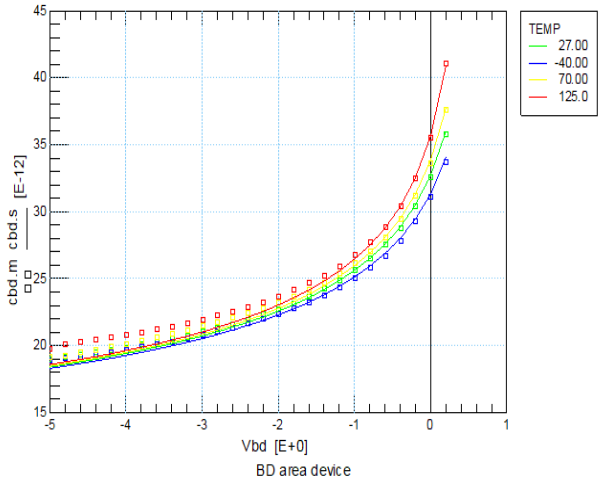
log(Id) = f (vg, T) @ low Vds, Vbs = max



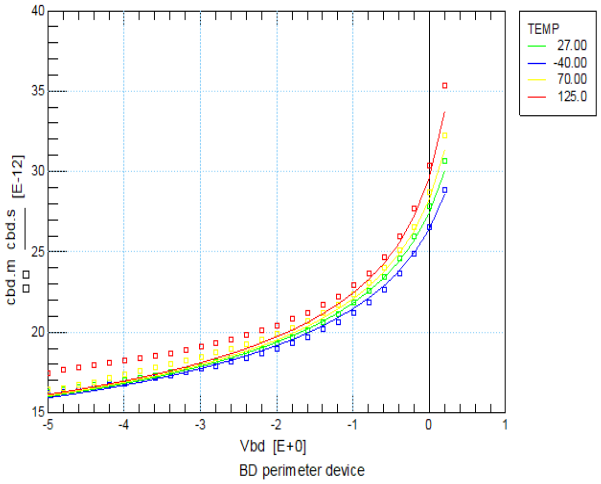
Id = f (Vd, Vg, T) @ Vbs = 0

8.2 Junction Capacitance

8.2.1 Bulk - Drain

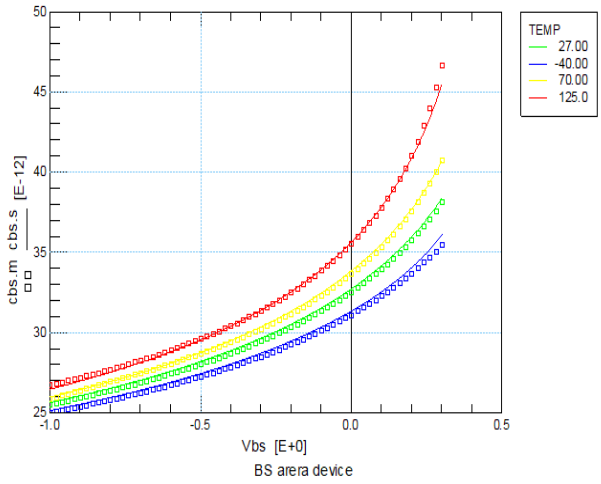


Area

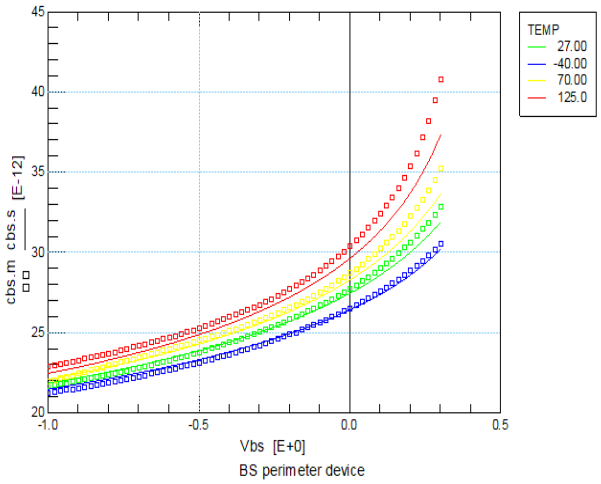


Perimeter

8.2.2 Bulk - Source



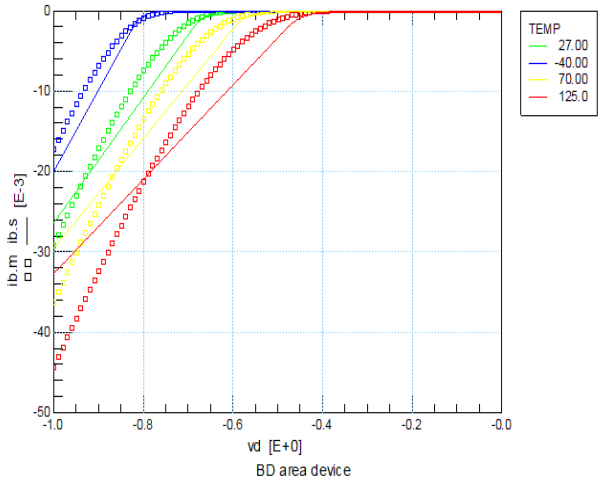
Area



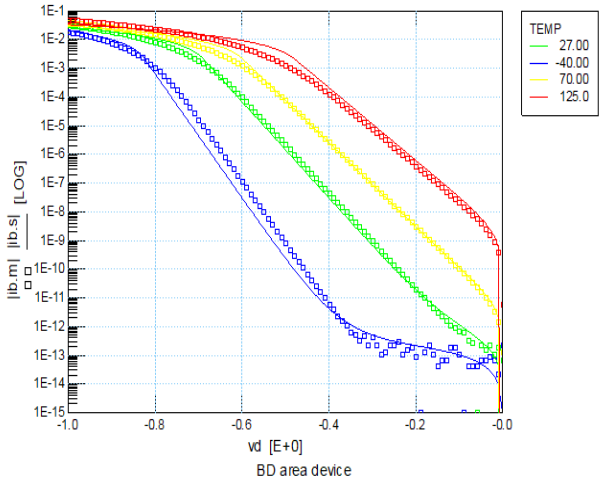
Perimeter

8.3 Parasitic Diodes

8.3.1 Bulk - Drain



Area



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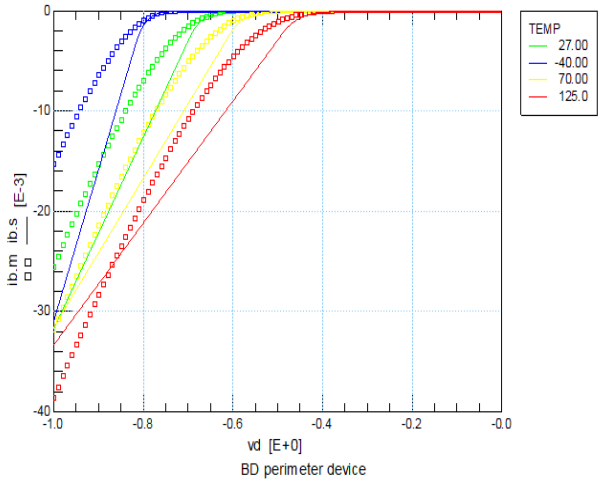
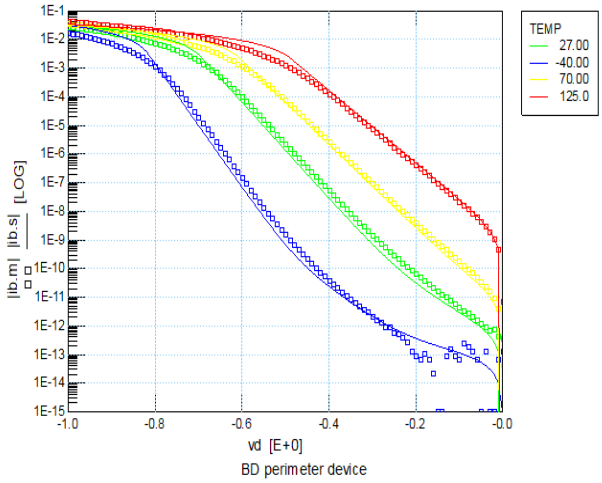
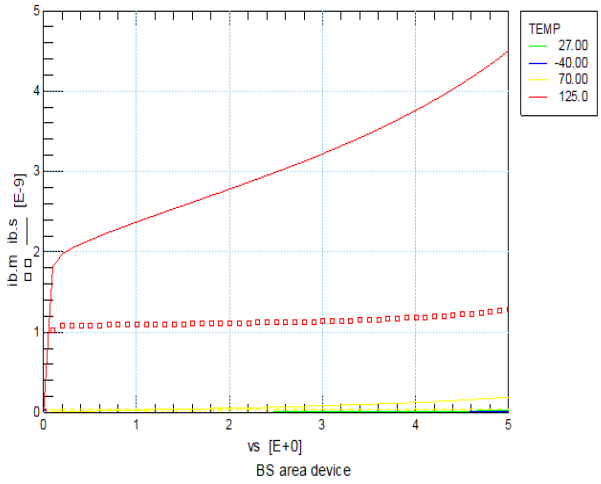


Perimeter



Perimeter

8.3.2 Bulk - Source



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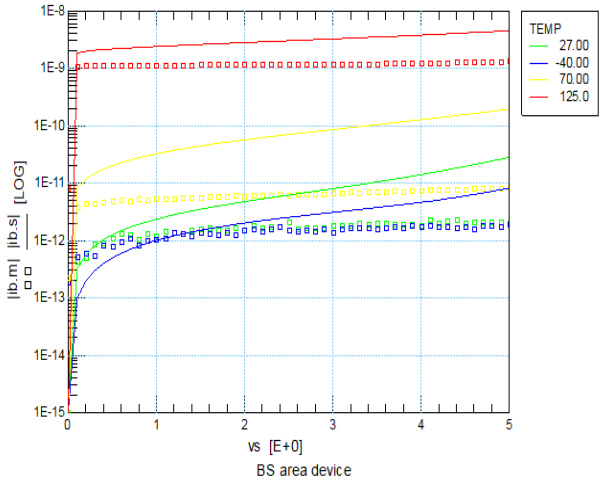

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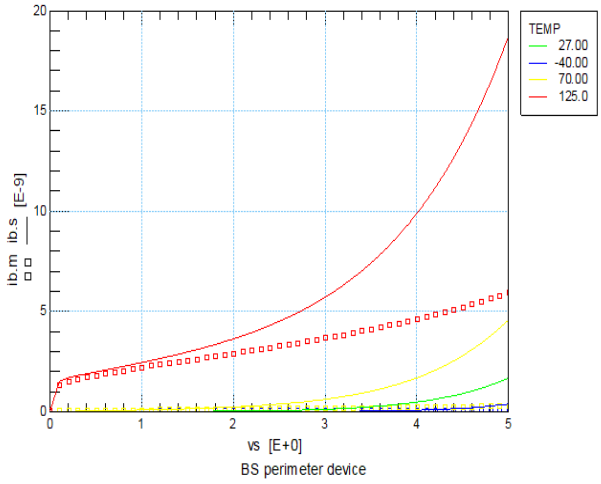
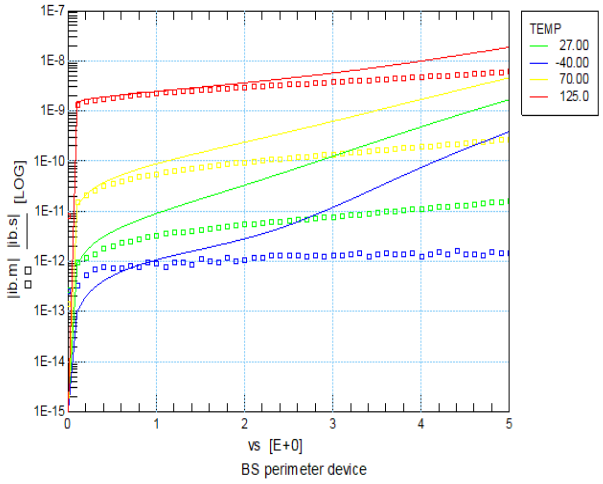


Perimeter



Perimeter

9 Flicker Noise

Flicker noise model parameters are taken from the SG13 simulation models.

9.1 Measurement Conditions

Not defined as no flicker noise measurements are yet done on SG13G2 devices.

9.2 Devices

Not defined as no flicker noise measurements are yet done on SG13G2 devices.

10 Change History

Revision	Device	Changes
190606	All	Initial version
190730	All	Adjust temperature and RF behavior
190924	Summarized Results	Legend added